Reverse Engineering para principiantes

Denis Yurichev

Ingeniería inversa para principiantes

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Versión de este texto (25 de junio de 2026).

La versión más reciente de este texto (y la edición en ruso) está disponible en beginners.re. La versión en formato A4 también está disponible allí.

Existe también una versión LITE (versión introductoria abreviada), pensada para quienes quieren una introducción rápida a los conceptos básicos de la ingeniería inversa: beginners.re

También puedes seguirme en Twitter para enterarte de las actualizaciones de este texto: `@yurichev`[1] o suscribirte a la lista de correo[2].

La portada fue realizada por Andy Nechaevsky: facebook.

[1] twitter.com/yurichev
[2] yurichev.com

Contenido abreviado

Índice general

[3]Instruction Set Architecture

[4]General Purpose Registers

Prefacio

Temas tratados en profundidad

x86/x64, ARM/ARM64, MIPS, Java/JVM.

Temas mencionados

Oracle RDBMS (81 on page 1648), Itanium (93 on page 1769), (78 on page 1465), LD_PRELOAD (67.2 on page 1334), , ELF[5], (68.2 on page 1347), x86-64 (26.1 on page 815), (68.4 on page 1408), (66 on page 1325), TLS[6], (PIC[7]) (67.1 on page 1328), profile-guided optimization (95.1 on page 1777), C++ STL (51.4 on page 1073), OpenMP (92 on page 1757), SEH (68.3 on page 1359).

[5] Formato de archivo ejecutable usado ampliamente en sistemas *NIX, incluido Linux

[6] Thread Local Storage

[7] Position Independent Code: 67.1 on page 1328

Sobre el autor

dennis(a)yurichev.com, **dennis.yurichev**.

Ingeniería inversa para principiantes

- «It's very well done .. and for free .. amazing.»[8] Daniel Bilar, Siege Technologies, LLC.
- «… excellent and free»[9] Pete Finnigan, Oracle RDBMS.
- «… book is interesting, great job!» Michael Sikorski, *Practical Malware Analysis: The Hands-On Guide to Dissecting Malicious Software*.
- «… my compliments for the very nice tutorial!» Herbert Bos, Vrije Universiteit Amsterdam, *Modern Ope-*

[8] twitter.com/daniel_bilar/status/436578617221742593
[9] twitter.com/petefinnigan/status/400551705797869568

rating Systems (4th Edition).

- «… It is amazing and unbelievable.» Luis Rocha, CISSP / ISSAP, Technical Manager, Network & Information Security at Verizon Business.
- «Thanks for the great work and your book.» Joris van de Vis, SAP Netweaver & Security.
- «… reasonable intro to some of the techniques.»[10] Mike Stay, Federal Law Enforcement Training Center, Georgia, US.
- «I love this book! I have several students reading it at the moment, plan to use it in graduate course.»[11] , Research Assistant Professor Dartmouth College
- «Dennis @Yurichev has published an impressive (and free!) book on reverse engineering»[12] Tanel Poder, .
- «This book is some kind of Wikipedia to beginners…» Archer, Chinese Translator, IT Security Researcher.

Agradecimientos

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[10]reddit
[11]twitter.com/sergeybratus/status/505590326560833536
[12]twitter.com/TanelPoder/status/524668104065159169

: Archer.

: Byungho Min.

: , , , “Logxen” , Yuan Jochen Kang, Mal Malakov, Lewis Porter, Jarle Thorsen.

2 * Oleg Vygovsky (50+100 UAH), Daniel Bilar ($50), James Truscott ($4.5), Luis Rocha ($63), Joris van de Vis ($127), Richard S Shultz ($20), Jang Minchang ($20), Shade Atlas (5 AUD), Yao Xiao ($10), Pawel Szczur (40 CHF), Justin Simms ($20), Shawn the R0ck ($27), Ki Chan Ahn ($50), Triop AB (100 SEK), Ange Albertini (€10+50), Sergey Lukianov (300 RUR), Ludvig Gislason (200 SEK), Gérard Labadie (€40), Sergey Volchkov (10 AUD), Vankayala Vigneswararao ($50), Philippe Teuwen ($4), Martin Haeberli ($10), Victor Cazacov (€5), Tobias Sturzenegger (10 CHF), Sonny Thai ($15), Bayna AlZaabi ($75), Redfive B.V. (€25), Joona Oskari Heikkilä (€5), Marshall Bishop ($50), Nicolas Werner (€12), Jeremy Brown ($100), Alexandre Borges ($25), Vladimir Dikovski (€50), Jiarui Hong (100.00 SEK), Jim Di (500 RUR), Tan Vincent ($30), Sri Harsha Kandrakota (10 AUD), Pillay Harish (10 SGD), Timur Valiev (230 RUR), Carlos Garcia Prado (€10), Salikov Alexander (500 RUR), Oliver Whitehouse (30 GBP), Katy Moe ($14), Maxim Dyakonov ($3), Sebastian Aguilera (€20), Hans-Martin Münch (€15), Jarle Thorsen (100 NOK), Vitaly Osipov ($100), Yuri Romanov (1000 RUR), Aliaksandr Autayeu (€10), Tudor Azoitei ($40), Z0vsky (€10), Yu Dai ($10).

mini-FAQ

Q: ¿Por qué debería aprenderse el lenguaje ensamblador hoy en día?
A: A menos que seas desarrollador de un OS[13], probablemente no necesites programar en ensamblador: los compiladores modernos optimizan mucho mejor que los humanos[14]. Además, las CPU[15] modernas son dispositivos muy complejos y saber ensamblador difícilmente ayuda a comprender su funcionamiento interno. Aun así, hay al menos dos áreas donde comprender bien el ensamblador resulta útil: 1) el análisis de seguridad y malware; 2) entender mejor tu código compilado mientras depuras. Por ello, este libro está pensado para quienes quieren entender el ensamblador más que programar en él, de ahí la cantidad de ejemplos de salidas de compiladores.

Q: Hice clic en un hipervínculo dentro de un PDF, ¿cómo vuelvo atrás?
A: En Adobe Acrobat Reader pulsa Alt+Flecha Izquierda.

Q: ¡Tu libro es enorme! ¿Hay algo más corto?
A: Hay una versión abreviada (lite) aquí: `http://beginners.re/#lite`.

Q: No estoy seguro de si debería aprender ingeniería inversa o no.

[13] Sistema Operativo
[14] Un texto muy bueno sobre este tema: [Fog13b]
[15] Central processing unit

A: Quizá el tiempo medio para familiarizarse con la versión abreviada LITE sea de 1 a 2 meses.

Q: ¿Puedo imprimir este libro? ¿Usarlo para enseñar?
A: Por supuesto; por eso el libro está bajo licencia Creative Commons. También puedes compilar tu propia versión del libro; lee aquí para saber más.

Q: Quiero traducir tu libro a otro idioma.
A: Lee mi nota para traductores.

Q: ? ¿Cómo conseguir trabajo como ingeniero inverso?
A: En reddit dedicado a RE[16]aparecen hilos de contratación de vez en cuando (2013 Q3, 2014). .Échales un ojo. En el subred «netsec» hay un hilo parecido: 2014 Q2.

Q: ¿Dónde estudiar en Ucrania?
A: NTUU «KPI»: «Аналіз програмного коду та бінарних вразлив
optativas.

Q: Tengo una pregunta...
A: Envíamela por correo (dennis(a)yurichev.com).

. . : beginners.re.

Sobre la traducción al coreano

.

[16]reddit.com/r/ReverseEngineering/

Byungho Min (twitter/tais9).

: facebook/andydinka.

Parte I

Patrones de código

> Todo es relativo
>
> Autor desconocido

Cuando el autor de este libro comenzó a aprender C y, más tarde, C++, él solía escribir pequeños trozos de código, compilarlos, y luego ver los resultados en lenguaje assembly. Esto lo hizo muy fácil para él entender lo que estaba pasando en el código que había escrito. [17]. Él lo hizo tantas veces que la relación entre el código C/C++ y lo que el compilador producido se imprimió profundamente en su mente. És fácil imaginar al instante un esbozo de la aparencia y función del código C. Quizás esta técnica podría ser útil para otra persona.

En ciertas partes, se han empleado aquí compiladores muy antiguas, con el fin de obtener lo mas corta (o simple) posible snippet.

Ejercicios

Cuando el autor de este libro estudió la lenguaje assembly, también con frecuencia compilaba pequeñas funciones en C, y reescribia gradualmente en assembly, tratando de hacer el código lo más pequeño posible   Probablemente no vale la pena hacer esto en escenarios reales actualmente, porque es dificil competir con los compiladores modernos

[17] De hecho, todavia lo hace cuando no puede entender lo que hace una determinada pieza de código.

en términos de eficiencia. Es, sin embargo, una muy buena manera de obtener una mejor compreensión de la assembly

Siéntase libre, por lo tanto, para tomar cualquier código de este libro y tratar de hacerlo más pequeño. Sin embargo, no se olvide de probar lo que has escrito.

Níveles de optimización y la información de depuración

El código fuente puede ser compilado por diferentes compiladores com varios niveles de optimización. Un compilador típico tiene alredor de tres de esos niveles, donde el nivel cero significa desactivar la optimización. La optimización también puede dirigirse hacia el tamaño del código o la velocidad de código.

Un compilador sin optimización es más rápido y produce código más inteligible (aunque más grande), mientras un compilador con optimización es más lento y trata de producir un código que corre más rápido (pero no necesariamente más compacto).

Además de los niveles y dirección de la otimización, el compilador puede incluir informaciones de depuración en el archivo resultante, produciendo así código para fácil depuración.

Una de los características importantes del código de ´debug' és que puede contener enlaces entre cada línea del código fuente y las direcciones de código de máquina respectivos. Compiladores con optimización, por otro lado, tienden a producir una salida donde líneas enteras de código fuente pueden ser optimizados al punto de ser eliminados y por consiguiente no estar presentes en el código de máquina resultante.

Ingenieros Inversos pueden encontrar ambas versiones, simplesmente porque alguns desarrolladores activan los flags de optimización del compilador, y otros no activan. Debido a esto, vamos a tratar de trabajar con ejemplos de ambas versiones de debug y release del código resaltado en este libro, cuando sea posible.

CAPÍTULO 1. UNA BREVE INTRODUCCIÓN A LA CPU

Capítulo 1

Una breve introducción a la CPU

La CPU es el dispositivo que ejecuta el código de máquina que constituye un programa.

Un breve glosario:

Instrucción : Una primitiva CPU comando. Los ejemplos más simples incluyen: mover datos entre registros, trabajar con la memoria, operaciones aritméticas primitivas. Como regla general, cada CPU tiene su propio conjunto de instrucciones (ISA).

: Código que la CPU procesa directamente. Cada instrucción generalmente se codifica por vários bytes.

 CAPÍTULO 1. UNA BREVE INTRODUCCIÓN A LA CPU

Lenguaje assembly : Código mnemónico y algunas extensiones como macros que destinados a hacer la vida del programador más fácil.

Registros de la CPU : Cada CPU tiene un conjunto fijo de registros de propósito general (GPR). ≈ 8 en x86, ≈ 16 en x86-64, ≈ 16 en ARM. La forma más fácil de entender un registro es pensar en ello como una variable temporal sin tipo. Imagine si estuviera trabajando con una PL[1] de alto nivel y sólo podría utilizar ocho variables de 32-bit (o de 64-bit). Sin embargo mucho se puede hacer usando sólo estos!

Uno podría perguntarse por qué es necessário que haya diferencia entre el código de la máquina y una lenguaje de programación de alto nivel. La respuesta está en el hecho de que los seres humanos y CPUs no son iguales. És mucho más fácil para los humanos utilizar un PL de alto nivel como C/C++, Java, Python, etc., pero és más fácil para una CPU utilizar un nivel mucho más bajo de abstración. Tal vez sería posible inventar una CPU que podría ejecutar código de PL de alto nivel, pero sería muchas veces más compleja que las CPUs que conocemos hoy. En uma manera similar, es muy incómodo para los seres humanos escribir en lenguaje assembly, debido a que es tan bajo nivel y difícil escribir sin hacer una gran cantidade de errores molestos. El programa que convierte el código de PL de alto nivel en assembly se llama *compiler*.

[1] Lenguaje de programación

CAPÍTULO 1. UNA BREVE INTRODUCCIÓN A LA CPU

1.1 Algunas palabras sobre diferentes ISAs

El ISA x86 siempre ha tenido opcodes de tamaño variable, de modo que cuanco llegó la era de 64-bit, las extensiones x64 no impactan el ISA de manera muy significativa. De hecho, el ISA x86 aún contiene una gran cantidade de instrucciones que primero aparecieron en CPU 8086 16-bit, pero aún se encuentran en las CPUs de hoy.

ARM és una CPU RISC[2] diseñado con la idea de opcodes con tamaño constante, que tenía algunas ventajas en el pasado. En el principio, todas las instrucciones ARM fueron codificados en 4 bytes[3]. Esto actualmente se conoce como «ARM mode».

Entonces se llegó a la conclusión que no era tan económico como se imaginó al princípio. En realidad, la mayoría de las instrucciones de CPU utilizados [4] en aplicaciones del mundo real pueden ser codificados utilizando menos información. Por lo tanto añadieron otra ISA, llamado Thumb, donde cada instrucción fue codificada en sólo 2 bytes. Esto se conoce como «Thumb mode». No obstante, no todas

[2]Reduced instruction set computing

[3]Dicho sea de paso, las instrucciones de longitud fija son muy útiles porque se puede calcular la dirección de instrucción siguiente (o anterior) sin esfuerzo. Esta característica se discutirá en la sección de el operador switch() (13.2.2 on page 312).

[4]Son estos MOV/PUSH/CALL/Jcc

CAPÍTULO 1. UNA BREVE INTRODUCCIÓN A LA CPU
las instrucciones ARM pueden ser codificadas en apenas 2 bytes, entonces el conjunto de instrucciones Thumb es algo limitada. Es importante destacar que el código compilado para el modo ARM y para el modo Thumb pueden, por supuesto, coexistir dentro de un solo programa.

Los creadores de ARM concluyeron que se podría extender el Thumb, dando origem al Thumb-2, que apareció en el ARMv7. Thumb-2 sigue utilizando instrucciones de 2 bytes, pero tiene algunas nuevas instrucciones que tienen el tamaño de 4 bytes. Hay una idea errónea de que Thumb-2 es una mezcla de ARM y Thumb. Esto es incorrecto. Más bien, se extendió Thumb-2 para apoyar plenamente todas las características de processador por lo que podría competir con el modo ARMun objetivo que se logró con claridad, ya que la mayoria de aplicacciones para iPod/iPhone/iPad son compmilados para el conjunto de instrucciones del Thumb-2 (la verdade es, en gran parte debido al hecho de que Xcode hace esto por defecto). Más tarde, el ARM 64-bit salió. Este ISA tiene opcodes de 4 bytes, y descarta la necesidade de cualquier modo Thumb adicional. Pero, los requisitos de 64-bit afectaron la ISA, resultando en ahora tenermos tres conjuntos de instrucciones ARM: ARM mode, Thumb mode (incluyendo Thumb-2) y ARM64. Estos ISAs se intersectan parcialmente, pero puede ser más bien decir que son ISAs diferentes, en lugar de variaciones de lo mismo. Por lo tanto, nos gustaría intentar añadir fragmentos de código de los tres ISAs del ARM en este libro.

Hay, por cierto, muchos otros RISC ISAs con opcodes de tamaño fijo de 32-bit, tales como MIPS, PowerPC y Alpha

CAPÍTULO 1. UNA BREVE INTRODUCCIÓN A LA CPU

AXP.

 CAPÍTULO 2. LA FUNCIÓN MÁS SIMPLE

Capítulo 2

La función más simple

La función más simple posible es, probablemente, aquella que simplemente devuelve un valor constante:

: Aquí está:

Listing 2.1: Código C/C++

```
int f()
{
        return 123;
};
```

¡Compilémosla!

 CAPÍTULO 2. LA FUNCIÓN MÁS SIMPLE

2.1 x86

: Esto es lo que producen los compiladores GCC y MSVC con optimización en la plataforma x86:

Listing 2.2: Con optimización GCC/MSVC (salida de assembly)

```
f:
        mov     eax, 123
        ret
```

Solo hay dos instrucciones: la primera coloca el valor 123 en el registro EAX, que por convención se usa para almacenar el valor de retorno, y la segunda es RET, que devuelve la ejecución a la caller. La función llamadora tomará el resultado del registro EAX.

2.2 ARM

Hay algunas diferencias en la plataforma ARM:

Listing 2.3: Con optimización Keil 6/2013 (Modo ARM) ASM Output

```
f PROC
        MOV      r0,#0x7b ; 123
        BX       lr
        ENDP
```

ARM utiliza el registro R0 para devolver los resultados de las funciones, por lo que 123 se copia en R0.

 CAPÍTULO 2. LA FUNCIÓN MÁS SIMPLE

En ARM la dirección de retorno no se guarda en la pila local, sino en el registro de enlace (LR[1]). Por ello, la instrucción BX LR salta a esa dirección, lo que equivale a devolver la ejecución a la caller.

Vale la pena señalar que MOV es un nombre equívoco para la instrucción tanto en x86 como en las ISA ARM. En realidad, los datos no se *mueven*, sino que se *copian*.

2.3 MIPS

Hay dos convenciones para nombrar registros en MIPS. por número (de $0 a $31) o por seudónimo ($V0, $A0, Spanish text placeholder). La salida de ensamblador de GCC a continuación muestra los registros por número:

Listing 2.4: Con optimización GCC 4.4.5 (salida de assembly)

```
        j       $31
        li      $2,123                  # 0↙
    ↘ x7b
```

...: ... mientras que IDA[2] los muestrapor sus seudónimos:

Listing 2.5: Con optimización GCC 4.4.5 (IDA)

```
                jr      $ra
                li      $v0, 0x7B
```

[1]Link Register

[2] Desensamblador y depurador interactivo desarrollado por Hex-Rays

 CAPÍTULO 2. LA FUNCIÓN MÁS SIMPLE

Así que el registro $2 (o $V0) se usa para almacenar el valor de retorno de la función. LI "Load Immediate" . significa "Load Immediate" y es el equivalente a MOV en MIPS.

La otra instrucción es la de salto (J o JR), que devuelve la ejecución a la caller saltando a la dirección contenida en el registro $31 (o $RA). Este registro es análogo al LR en ARM.

"branch delay slot". Quizá te preguntes por qué la instrucción de carga (LI) y la instrucción de salto (J o JR) están intercambiadas. Esto se debe a una característica de las arquitecturas RISC llamada "branch delay slot". En realidad, no necesitamos entrar en estos detalles. Solo hay que recordar que en MIPS, la instrucción que sigue a una instrucción de salto se ejecuta *antes* que la propia instrucción de salto/ramificación. Como consecuencia, las instrucciones de salto siempre intercambian su lugar con la instrucción que debe ejecutarse previamente.

2.3.1 Una nota sobre los nombres de instrucciones/registros en MIPS

En el mundo MIPS, los nombres de registros e instrucciones tradicionalmente se escriben en minúsculas. Sin embargo, para mantener la coherencia, usaremos mayúsculas, ya que es la convención seguida por las demás ISA presentadas en este libro.

 CAPÍTULO 3. HELLO, WORLD!

Capítulo 3

Hello, world!

Continuemos usando el famoso ejemplo del libro “The C programming Language”[Ker88]:

```
#include <stdio.h>

int main()
{
    printf("hello, world\n");
    return 0;
}
```

3.1 x86

3.1.1 MSVC

Compilémoslo en MSVC 2010:

```
cl 1.cpp /Fa1.asm
```

(La opción /Fa le indica al compilador que genere un archivo de listado en ensamblador)

Listing 3.1: MSVC 2010

```
CONST   SEGMENT
$SG3830 DB      'hello, world', 0AH, 00H
CONST   ENDS
PUBLIC  _main
EXTRN   _printf:PROC
; Function compile flags: /Odtp
_TEXT   SEGMENT
_main   PROC
        push    ebp
        mov     ebp, esp
        push    OFFSET $SG3830
        call    _printf
        add     esp, 4
        xor     eax, eax
        pop     ebp
        ret     0
_main   ENDP
_TEXT   ENDS
```

MSVC genera los listados de ensamblador en sintaxis Intel. La diferencia entre la sintaxis Intel y la sintaxis AT&T se discutirá en 3.1.3 on page 21.

El compilador generó el archivo `1.obj`, el cual será enlazado en `1.exe`. En nuestro caso, el archivo contiene dos segmentos: CONST (para constantes de datos) y _TEXT (para código).

La cadena `hello, world` en C/C++ tiene el tipo `const char[]` [Str13, p176, 7.3.2]; sin embargo, no tiene un nombre propio. El compilador necesita manejar la cadena de alguna manera, por lo que le define el nombre interno `$SG3830`.

Por eso el ejemplo podría reescribirse de la siguiente manera:

```
#include <stdio.h>

const char $SG3830[]="hello, world\n";

int main()
{
    printf($SG3830);
    return 0;
}
```

.Volvamos al listado en ensamblador. Como podemos ver, la cadena termina con un byte cero, lo cual es estándar para las cadenas en C/C++. : 57.1.1 on page 1273.Más sobre las cadenas en C: 57.1.1 on page 1273.

: `main()`.En el segmento de código, _TEXT, solo hay una

 CAPÍTULO 3. HELLO, WORLD!
función por ahora: `main()`. [1].La función `main()` comienza con el código de prólogo y termina con el de epílogo (como casi cualquier función)[2].

: `CALL _printf`.Después del prólogo de la función vemos la llamada a la función `printf()`: `CALL _printf`. Antes de la llamada, la dirección de la cadena (o un puntero a ella) que contiene nuestro saludo se coloca en la pila con la ayuda de la instrucción PUSH.

Cuando la función `printf()` devuelve el control a la función `main()`, la dirección de la cadena (o un puntero a ella) todavía está en la pila. Como ya no lo necesitamos, el stack pointer (el registro ESP) debe ser corregido.

`ADD ESP, 4` significa sumar 4 al valor del registro ESP. ¿Por qué 4? Como este es un programa de 32 bits, necesitamos exactamente 4 bytes para pasar la dirección a través de la pila. Si fuera código x64, necesitaríamos 8 bytes. `ADD ESP, 4` es efectivamente equivalente a `POP registro` pero sin usar ningún registro[3].

Algunos compiladores (como Intel C++ Compiler) para el mismo propósito pueden emitir POP ECX en lugar de ADD (por ejemplo, tal patrón se puede observar en el código de Oracle RDBMS compilado con el compilador Intel C++). Esta instrucción tiene casi el mismo efecto, pero el contenido del registro ECX se sobrescribirá. El compilador Intel C++

[1] (4 on page 61).
[2] Puedes leer más sobre esto en la sección sobre el prólogo y epílogo de la función (4 on page 61).
[3] Sin embargo, las banderas de la CPU se modifican

probablemente usa `POP ECX` ya que el opcode de esta instrucción es más corto que `ADD ESP, x` (1 byte para `POP` frente a 3 para `ADD`).

Aquí hay un ejemplo de uso de `POP` en lugar de `ADD` en Oracle RDBMS:

Listing 3.2: Oracle RDBMS 10.2 Linux (app.o)

```
.text:0800029A                 push    ebx
.text:0800029B                 call    ↙
    ↘ qksfroChild
.text:080002A0                 pop     ecx
```

Después de llamar a `printf()`, el código original en C/C++ contiene la instrucción `return 0` —devolver 0 como resultado de la función `main()`. En el código generado esto se implementa mediante la instrucción `XOR EAX, EAX`. `XOR` es en realidad solo un «O exclusivo»[4] , pero los compiladores a menudo lo usan en lugar de `MOV EAX, 0` nuevamente porque es un opcode ligeramente más corto (2 bytes para `XOR` frente a 5 para `MOV`).

Algunos compiladores emiten `SUB EAX, EAX`, lo que significa *restar el valor de* `EAX` *del valor de* `EAX`, lo que en cualquier caso da como resultado cero.

La última instrucción `RET` devuelve el control a la caller. Usualmente, este es el código de la CRT[5] de C/C++, el cual a su vez devuelve el control al OS.

[4] wikipedia
[5] C runtime library : 68.1 on page 1340

3.1.2 GCC

Ahora intentemos compilar el mismo código C/C++ con el compilador GCC 4.4.1 en Linux: `gcc 1.c -o 1` A continuación, con la ayuda del desensamblador IDA, veamos cómo se creó la función `main()`. IDA, Al igual que MSVC, IDA utiliza la sintaxis Intel[6].

Listing 3.3: código en IDA

```
main                proc near

var_10              = dword ptr -10h

                    push    ebp
                    mov     ebp, esp
                    and     esp, 0FFFFFFF0h
                    sub     esp, 10h
                    mov     eax, offset ⤦
    ↳ aHelloWorld ; "hello, world\n"
                    mov     [esp+10h+var_10], ⤦
    ↳ eax
                    call    _printf
                    mov     eax, 0
                    leave
                    retn
main                endp
```

El resultado es casi el mismo. La dirección de la cadena `hello, world` (almacenada en el segmento de datos) se

[6]También podríamos hacer que GCC genere listados de ensamblador en sintaxis Intel aplicando las opciones `-S -masm=intel`.

 CAPÍTULO 3. HELLO, WORLD!

carga primero en el registro EAX y luego se guarda en la pila. Además, el prólogo de la función contiene `AND ESP, 0FFFFFFF0h` —esta instrucción alinea el valor del registro ESP en un límite de 16 bytes. Esto hace que todos los valores en la pila se alineen de la misma manera (la CPU funciona mejor si los valores con los que trata están ubicados en memoria en direcciones alineadas en límites de 4 o 16 bytes)[7].

`SUB ESP, 10h` asigna 16 bytes en la pila. Aunque, como podemos ver más adelante, aquí solo son necesarios 4.

Esto se debe a que el tamaño de la pila asignada también está alineado en un límite de 16 bytes.

La dirección de la cadena (o un puntero a ella) se almacena directamente en la pila sin utilizar la instrucción PUSH. *var_10* —es una variable local y también un argumento para `printf()`. Lee sobre esto más abajo.

Luego se llama a la función `printf()`.

A diferencia de MSVC, cuando GCC compila sin la optimización activada, emite `MOV EAX, 0` en lugar de un opcode más corto.

La última instrucción, LEAVE —es el equivalente del par de instrucciones `MOV ESP, EBP` y `POP EBP` —en otras palabras, esta instrucción restablece el stack pointer (ESP) y restaura el registro EBP a su estado inicial. Esto es necesario ya que modificamos los valores de estos registros (ESP

---

[7] Spanish text placeholder

y EBP) al principio de la función (ejecutando `MOV EBP, ESP` / `AND ESP, …`).

3.1.3 GCC: Sintaxis AT&T

Veamos cómo se puede representar esto en la sintaxis AT&T del lenguaje ensamblador. Esta sintaxis es mucho más popular en el mundo UNIX.

Listing 3.4: compilemos en GCC 4.7.3

```
gcc -S 1_1.c
```

Obtenemos esto:

Listing 3.5: GCC 4.7.3

```
        .file   "1_1.c"
        .section        .rodata
.LC0:
        .string "hello, world\n"
        .text
        .globl  main
        .type   main, @function
main:
.LFB0:
        .cfi_startproc
        pushl   %ebp
        .cfi_def_cfa_offset 8
        .cfi_offset 5, -8
        movl    %esp, %ebp
        .cfi_def_cfa_register 5
        andl    $-16, %esp
```

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```
        subl    $16, %esp
        movl    $.LC0, (%esp)
        call    printf
        movl    $0, %eax
        leave
        .cfi_restore 5
        .cfi_def_cfa 4, 4
        ret
        .cfi_endproc
.LFE0:
        .size   main, .-main
        .ident  "GCC: (Ubuntu/Linaro 4.7.3-1↙
    ↳ ubuntu1) 4.7.3"
        .section        .note.GNU-stack,"",↙
    ↳ @progbits
```

El listado contiene muchas macros (que comienzan con un punto). Estas no son interesantes para nosotros en este momento. Por ahora, en aras de la simplificación, podemos ignorarlas (excepto la macro *.string* que codifica una secuencia de caracteres terminada en nulo al igual que una cadena de C). Entonces veremos esto[8]:

Listing 3.6: GCC 4.7.3

```
.LC0:
        .string "hello, world\n"
main:
        pushl   %ebp
        movl    %esp, %ebp
```

[8]Esta opción de GCC se puede utilizar para eliminar las macros «innecesarias»: *-fno-asynchronous-unwind-tables*

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```
        andl    $-16, %esp
        subl    $16, %esp
        movl    $.LC0, (%esp)
        call    printf
        movl    $0, %eax
        leave
        ret
```

Algunas de las principales diferencias entre la sintaxis Intel y la sintaxis AT&T son:

- Los operandos de origen y destino se escriben en orden opuesto.

  En la sintaxis Intel: <instrucción> <operando de destino> <operando de origen>.

  En la sintaxis AT&T: <instrucción> <operando de origen> <operando de destino>.

  : Aquí hay una manera de memorizar fácilmente la diferencia: cuando tratas con la sintaxis Intel, puedes imaginar que hay un signo de igualdad (=) entre los operandos, y cuando tratas con la sintaxis AT&T, imagina que hay una flecha hacia la derecha (→) [9].

- AT&T: Antes de los nombres de los registros se debe escribir un signo de porcentaje (%) y antes de los

[9] Por cierto, en algunas funciones estándar de C (por ejemplo, memcpy(), strcpy()) los argumentos se listan de la misma manera que en la sintaxis Intel: primero el puntero al bloque de memoria de destino y luego el puntero al bloque de memoria de origen.

números un signo de dólar ($). Se utilizan paréntesis en lugar de corchetes.

- AT&T: Se añade un sufijo a las instrucciones para definir el tamaño del operando:
  - q — quad (64 bits)
  - l — long (32 bits)
  - w — word (16 bits)
  - b — byte (8 bits)

Volvamos al resultado compilado: es idéntico a lo que vimos en IDA. : 0FFFFFFF0h $-16.Con una sutil diferencia: 0FFFFFFF0h se presenta como $-16. : 16 0x10 Es la misma cosa: 16 en el sistema decimal es 0x10 .en hexadecimal. -0x10 0xFFFFFFF0 ().-0x10 es igual a 0xFFFFFFF0 (para un tipo de datos de 32 bits).

.Una cosa más: el valor de retorno se establece en 0 usando el MOV habitual, no XOR. MOV.MOV simplemente carga el valor en un registro. Su nombre es erróneo (los datos no se mueven sino que se copian). En otras arquitecturas, esta instrucción se llama «LOAD» o «STORE» o algo similar.

3.2 x86-64

3.2.1 MSVCx86-64

Probemos también con MSVC de 64 bits:

Listing 3.7: MSVC 2012 x64

```
$SG2989 DB      'hello, world', 0AH, 00H

main    PROC
        sub     rsp, 40
        lea     rcx, OFFSET FLAT:$SG2989
        call    printf
        xor     eax, eax
        add     rsp, 40
        ret     0
main    ENDP
```

.En x86-64, todos los registros se extendieron a 64 bits y ahora sus nombres tienen el prefijo R- Con el fin de utilizar menos la pila (en otras palabras, para acceder menos a la memoria externa y a la caché), existe una forma muy popular de pasar los argumentos de las funciones a través de los registros (fastcall: 64.3 on page 1301). .Es decir, una parte de los argumentos de la función se pasa en los registros y el resto a través de la pila. RCX, RDX, R8, R9.En Win64, los primeros 4 argumentos de la función se pasan en los registros RCX, RDX, R8, R9. .Eso es lo que vemos aquí: el puntero a la cadena para `printf()` ahora no se pasa en la pila, sino en el registro RCX.

.Los punteros ahora son de 64 bits, por lo que se pasan en los registros de 64 bits (que tienen el prefijo R-). .Sin embargo, para mantener la compatibilidad hacia atrás, sigue siendo posible acceder a las partes de 32 bits, utilizando el prefijo E-.

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RAX/EAX/AX/AL registro se ve en x86-64:

Spanish text placeholder		
RAX^{x64}		
	EAX	
	AX	
	AH	AL

La función `main()` devuelve un valor de tipo *int*, el cual en C/C++, para una mejor compatibilidad hacia atrás y portabilidad, sigue siendo de 32 bits; es por eso que al final de la función se limpia el registro EAX (es decir, la parte de 32 bits del registro) en lugar de RAX.

.También se puede ver que se asignan 40 bytes en la pila local. Esto se llama «shadow space», del cual hablaremos más adelante: 8.2.1 on page 193.

3.2.2 GCCx86-64

Probemos también con GCC en Linux de 64 bits:

Listing 3.8: GCC 4.4.6 x64

```
.string "hello, world\n"
main:
        sub     rsp, 8
        mov     edi, OFFSET FLAT:.LC0 ; "hello, world\n"
        xor     eax, eax  ;
        call    printf
        xor     eax, eax
```

```
        add     rsp, 8
        ret
```

[Mit13].En Linux, *BSD y Mac OS X para x86-64 también se adopta el método de pasar los argumentos de las funciones a través de los registros RDI, RSI, RDX, RCX, R8, R9.Los primeros 6 argumentos se pasan en los registros RDI, RSI, RDX, RCX, R8, R9, y el resto a través de la pila.

.Así que el puntero a la cadena se pasa en EDI (la parte de 32 bits del registro). , RDI?Pero ¿por qué no usar la parte de 64 bits, RDI?

[Int13].Es importante tener en cuenta que en modo de 64 bits todas las instrucciones MOV que escriben algo en la parte inferior de 32 bits de un registro, también limpian los 32 bits superiores MOV EAX, 011223344h .Es decir, la instrucción MOV EAX, 011223344h escribe un valor en RAX correctamente, ya que los bits superiores se limpiarán.

Si abrimos el archivo objeto compilado (.o), también podemos ver todos los opcodes de las instrucciones[10]:

Listing 3.9: GCC 4.4.6 x64

```
.text:00000000004004D0                              main ↙
    ↘  proc near
.text:00000000004004D0 48 83 EC 08               sub ↙
    ↘     rsp, 8
```

[10]Options → Disassembly → Number of opcode bytesEsto se debe habilitar en Options → Disassembly → Number of opcode bytes

```
.text:00000000004004D4 BF E8 05 40 00   mov ↙
    ↘     edi, offset format ; "hello, world↙
    ↘ \n"
.text:00000000004004D9 31 C0                xor ↙
    ↘     eax, eax
.text:00000000004004DB E8 D8 FE FF FF   call↙
    ↘     _printf
.text:00000000004004E0 31 C0                xor ↙
    ↘     eax, eax
.text:00000000004004E2 48 83 C4 08          add ↙
    ↘     rsp, 8
.text:00000000004004E6 C3                   retn
.text:00000000004004E6                      main↙
    ↘   endp
```

.Como podemos ver, la instrucción que escribe en EDI en 0x4004D4 ocupa 5 bytes. La misma instrucción que escribe un valor de 64 bits en RDI ocupa 7 bytes. .Aparentemente, GCC está tratando de ahorrar algo de espacio. Además, puede estar seguro de que el segmento de datos que contiene la cadena no se asignará en direcciones superiores a 4GiB.

.También vemos que el registro EAX se limpió antes de la llamada a la función `printf()`. Esto se hace porque, según el estándar de transferencia de argumentos en *NIX para x86-64, en EAX se pasa el número de registros vectoriales utilizados: «with variable arguments passes information about the number of vector registers used» [Mit13].

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3.3 GCCuna cosa más

const (3.1.1 on page 16), :El hecho de que una cadena de C *anónima* sea de tipo *const* (3.1.1 on page 16), y el hecho de que las cadenas de C asignadas en el segmento de constantes tengan garantizada su inmutabilidad, tiene una consecuencia interesante: .el compilador puede usar una parte específica de la cadena.

Probemos este ejemplo:

```
#include <stdio.h>

int f1()
{
        printf ("world\n");
}

int f2()
{
        printf ("hello world\n");
}

int main()
{
        f1();
        f2();
}
```

:Los compiladores comunes de C/C++ (incluido MSVC) asignan dos cadenas, pero veamos qué hace GCC 4.8.1:

Listing 3.10: GCC 4.8.1 + IDAlistado en IDA

```
f1                  proc near

s                   = dword ptr -1Ch

                    sub     esp, 1Ch
                    mov     [esp+1Ch+s], offset ⤦
    ↳ s ; "world\n"
                    call    _puts
                    add     esp, 1Ch
                    retn
f1                  endp

f2                  proc near

s                   = dword ptr -1Ch

                    sub     esp, 1Ch
                    mov     [esp+1Ch+s], offset ⤦
    ↳ aHello ; "hello "
                    call    _puts
                    add     esp, 1Ch
                    retn
f2                  endp

aHello              db 'hello '
s                   db 'world',0xa,0
```

, .En efecto: cuando imprimimos la cadena «hello world», estas dos palabras se colocan en memoria de forma adyacente, y `puts()`, al ser llamada desde la función f2(), no es consciente de que esta cadena está dividida. .De hecho, no está dividida; está dividida solo «virtualmente» en este listado.

Cuando `puts()` es llamada desde f1(), utiliza la cadena «world» . más el byte cero. `puts()`!¡`puts()` no es consciente de que hay algo antes de esta cadena!

Este ingenioso truco suele ser utilizado al menos por GCC y puede ahorrar algo de memoria.

3.4 ARM

Para mis experimentos con procesadores ARM, se usaron varios compiladores:

- Popular en el entorno embebido Keil Release 6/2013.
- Apple Xcode 4.6.3 IDE (con LLVM-GCC 4.2 como compilador[11]).
- GCC 4.9 (Linaro) (para ARM64), disponible como ejecutables para win32 en http://go.yurichev.com/17325.

En todo este libro, salvo que se indique lo contrario, se usa ARM de 32 bits (incluidos los modos Thumb y Thumb-2). Cuando hablamos de ARM de 64 bits aquí, lo llamamos ARM64.

[11]Es así: Apple Xcode 4.6.3 usa GCC de código abierto como compilador de frontend y LLVM como generador de código

3.4.1 Sin optimización Keil 6/2013 (Modo ARM)

Para empezar, compilamos nuestro ejemplo en Keil:

```
armcc.exe --arm --c90 -O0 1.c
```

El compilador *armcc* produce listados en ensamblador con sintaxis Intel , pero incluye macros de alto nivel relacionados con ARM[12], pero nos importa más ver las instrucciones tal cual, así que veamos el resultado compilado en IDA.

Listing 3.11: Sin optimización Keil 6/2013 (Modo ARM) IDA

```
.text:00000000             main
.text:00000000 10 40 2D E9   STMFD   SP!, {↙
    ↳ R4,LR}
.text:00000004 1E 0E 8F E2   ADR     R0, ↙
    ↳ aHelloWorld ; "hello, world"
.text:00000008 15 19 00 EB   BL      ↙
    ↳ __2printf
.text:0000000C 00 00 A0 E3   MOV     R0, #0
.text:00000010 10 80 BD E8   LDMFD   SP!, {↙
    ↳ R4,PC}

.text:000001EC 68 65 6C 6C+aHelloWorld  DCB ↙
    ↳ "hello, world",0    ; DATA XREF: main↙
    ↳ +4
```

En el ejemplo anterior se ve fácilmente que cada instrucción mide 4 bytes. Efectivamente, compilamos el código para modo ARM, no Thumb.

[12] p. ej., muestra instrucciones PUSH/POP, ausentes en el modo ARM

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La primera instrucción, `STMFD SP!, {R4,LR}`[13], funciona como una PUSH de x86, escribiendo los valores de dos registros (R4 y LR) en la pila. `PUSH {r4,lr}` de hecho muestra la instrucción. . Pero eso no es del todo preciso: PUSH solo está disponible en modo Thumb; para evitar confusiones, lo veremos en IDA.

Esta instrucción primero decrements el SP[15] para que apunte a un lugar libre de la pila y después guarda los valores de R4 y LR en la dirección indicada por el SP modificado.

. Esta instrucción (como PUSH en modo Thumb) puede guardar varios registros a la vez, lo cual puede ser muy útil. .Por cierto, esto no existe en x86. También puede notarse que STMFD es una generalización de PUSH (amplía sus capacidades), porque puede trabajar con cualquier registro y no solo con SP. En otras palabras, STMFD puede usarse para almacenar un conjunto de registros en la dirección de memoria indicada.

La `ADR R0, aHelloWorld hello, world` instrucción suma o resta el valor de PC[16] al desplazamiento donde está la cadena. ?¿Cómo se usa aquí PC? Se trata del llamado «código independiente de la posición». [17] Este código puede ejecutarse en una dirección no fija de memoria. En otras palabras, es direccionamiento relativo al PC. .El opcode de ADR codifica la diferencia entre la dirección de esta instruc-

[13] STMFD[14]

[15] stack pointer. SP/ESP/RSP en x86/x64. SP en ARM.

[16] Program Counter. IP/EIP/RIP en x86/64. PC en ARM.

[17] Lee más sobre esto en la sección correspondiente (67.1 on page 1328)

ción y el lugar donde está la cadena. Esta diferencia (desplazamiento) siempre será la misma, independientemente de la dirección a la que el OS cargue nuestro código. Por eso, lo único necesario es sumar la dirección de la instrucción actual (desde PC) para obtener la dirección absoluta de nuestra cadena en C.

`BL __2printf`[18] . llama a la función `printf()`. Así es como funciona esta instrucción:

- guardar la dirección siguiente a BL (`0xC`) en LR;
- luego transferir el control a `printf()` escribiendo su dirección en el registro PC.

Cuando `printf()` termina debe saber adónde volver; por eso toda función devuelve el control a la dirección guardada en LR.

[19].

.Por cierto, una dirección absoluta de 32 bits (o un desplazamiento) no puede codificarse en la instrucción de 32 bits BL, porque solo hay 24 bits disponibles. Como recordamos, todas las instrucciones en modo ARM miden 4 bytes (32 bits) y solo pueden situarse en direcciones múltiplos de 4; por ello, los 2 bits bajos de la dirección (siempre a 0) no se codifican. $current\_PC \pm \approx 32M$.En resumen, tenemos 26 bits para codificar el desplazamiento; esto basta para $current\_PC \pm \approx 32M$.

[18]Branch with Link
[19]Lee más sobre esto en la siguiente sección (5 on page 63)

A continuación, la `MOV R0, #0`[20] simplemente escribe 0 en el registro R0. Nuestra función en C devuelve 0 y el valor de retorno debe ponerse en R0.

La última instrucción `LDMFD SP!, R4,PC`[21] es la inversa de STMFD. Carga desde la pila (o cualquier otra zona de memoria) los valores para guardarlos en R4 y PC, e incrementa el puntero de pila SP. Aquí actúa como un POP.
N.B. La primera instrucción STMFD guardó en la pila el par de registros R4 y LR, pero durante LDMFD se *restauran* R4 y PC.

Como ya sabemos, en LR suele guardarse la dirección a la que debe devolver el control cada función. La primera instrucción guarda ese valor en la pila, porque nuestra función `main()` usará ese registro al llamar a `printf()`. Al final de la función, ese valor puede escribirse directamente en PC, devolviendo el control al lugar desde el que fue llamada.

Como `main()` suele ser la función principal en C/C++, el control volverá al cargador del OS, o a algún punto del CRT, o similar.

Todo esto permite omitir la instrucción `BX LR` al final de la función.

DCB es una directiva del lenguaje ensamblador que define un array de bytes o cadenas ASCII, similar a la directiva DB en ensamblador x86.

[20]MOVe
[21]LDMFD[22]

3.4.2 Sin optimización Keil 6/2013 (Modo Thumb)

Compilamos el mismo ejemplo en Keil en modo Thumb:

```
armcc.exe --thumb --c90 -O0 1.c
```

Obtenemos (en IDA):

Listing 3.12: Sin optimización Keil 6/2013 (Modo Thumb) + IDA

```
.text:00000000                          main
.text:00000000 10 B5                       PUSH    {R4,LR↙
    ↘ }
.text:00000002 C0 A0                       ADR     R0, ↙
    ↘ aHelloWorld ; "hello, world"
.text:00000004 06 F0 2E F9                 BL      ↙
    ↘ __2printf
.text:00000008 00 20                       MOVS    R0, #0
.text:0000000A 10 BD                       POP     {R4,PC↙
    ↘ }

.text:00000304 68 65 6C 6C+aHelloWorld  DCB ↙
    ↘ "hello, world",0    ; DATA XREF: main↙
    ↘ +2
```

Saltan a la vista los opcodes de 2 bytes (16 bits); como ya se señaló, es Thumb. Excepto la instrucción BL. .En realidad consta de dos instrucciones de 16 bits. .Esto es porque en un solo opcode de 16 bits hay muy poco espacio para especificar el desplazamiento hasta la función `printf()`. .Así que la primera instrucción de 16 bits carga los 10 bits altos del desplazamiento y la segunda los 11 bits bajos. . Como se mencionó, todas las instrucciones en modo Thumb tienen longitud de 2 bytes (o 16 bits). Por tanto, es imposible que una instrucción Thumb empiece en una dirección impar. . Dado lo anterior, el último bit de la dirección puede omitirse al codificar instrucciones. $current\_PC \pm \approx 2M$.En resumen, en la instrucción Thumb BL se puede codificar la dirección $current\_PC \pm \approx 2M$.

. El resto de instrucciones de la función (PUSH y POP) funcionan aquí casi igual que las STMFD/LDMFD descritas; solo que el registro SP no se indica explícitamente. ADR funciona igual que en el ejemplo anterior. MOVS .escribe 0 en el registro R0 para devolver cero.

3.4.3 Con optimización Xcode 4.6.3 (LLVM) (Modo ARM)

Xcode 4.6.3 sin optimización genera demasiado código redundante, así que estudiaremos la salida optimizada, donde hay el menor número de instrucciones posible, activando el conmutador del compilador -O3.

Listing 3.13: Con optimización Xcode 4.6.3 (LLVM) (Modo ARM)

```
__text:000028C4                 _hello_world
__text:000028C4 80 40 2D E9   STMFD       ↙
    ↳   SP!, {R7,LR}
__text:000028C8 86 06 01 E3   MOV         ↙
    ↳   R0, #0x1686
__text:000028CC 0D 70 A0 E1   MOV         ↙
    ↳   R7, SP
```

```
__text:000028D0 00 00 40 E3   MOVT
    R0, #0
__text:000028D4 00 00 8F E0   ADD
    R0, PC, R0
__text:000028D8 C3 05 00 EB   BL
    _puts
__text:000028DC 00 00 A0 E3   MOV
    R0, #0
__text:000028E0 80 80 BD E8   LDMFD
    SP!, {R7,PC}

__cstring:00003F62 48 65 6C 6C+aHelloWorld_0
    DCB "Hello world!",0
```

Las instrucciones STMFD y LDMFD ya nos son familiares.

.La instrucción MOV simplemente escribe el número 0x1686 en R0: es el desplazamiento que apunta a la cadena «Hello world!».

El registro R7 (según el estándar de [App10]) es el frame pointer; lo veremos más abajo.

MOVT R0, #0 (MOVe Top) . escribe 0 en los 16 bits altos del registro. . El asunto es que la MOV genérica en modo ARM solo puede escribir en los 16 bits bajos del registro. Recuerda que en modo ARM todos los opcodes están limitados a 32 bits. Claro que esto no afecta a los movimientos de datos entre registros. . Por eso existe la instrucción adicional MOVT para escribir en los bits altos (del 16 al 31 inclusive). . Su uso aquí es redundante, porque MOV R0, #0x1686 ya limpió la parte alta del registro. . Probablemente sea una limitación del compilador.

`ADD R0, PC, R0` . suma PC a R0 para calcular la dirección absoluta de la cadena «Hello world!». Como ya sabemos, es «código independiente de la posición», así que esta corrección es esencial.

. La instrucción BL llama a `puts()` en lugar de `printf()`.

El compilador sustituyó `printf()` por `puts()`. En efecto, `printf()` con un único argumento es casi equivalente a `puts()`. *Casi*, porque ambas funciones producen el mismo resultado solo si la cadena no contiene especificadores de formato de `printf()` que empiezan por %. Si los contiene, el efecto será diferente[23].

¿Por qué el compilador sustituyó `printf()` por `puts()`? Probablemente porque `puts()` es más rápida[24]. Seguramente porque `puts()` envía los caracteres a stdout sin comparar cada uno con el símbolo %.

A continuación, vemos la conocida `MOV R0, #0` instrucción para establecer a 0 el valor devuelto en R0.

3.4.4 Con optimización Xcode 4.6.3 (LLVM) (Modo Thumb-2)

Xcode 4.6.3 :

[23]También hay que notar que `puts()` no requiere el símbolo de nueva línea '\n' al final de la cadena, por eso aquí no aparece.

[24]ciselant.de/projects/gcc_printf/gcc_printf.html

Listing 3.14: Con optimización Xcode 4.6.3 (LLVM) (Modo Thumb-2)

```
__text:00002B6C                                  ↙
    ↳ _hello_world
__text:00002B6C 80 B5               PUSH         ↙
    ↳    {R7,LR}
__text:00002B6E 41 F2 D8 30     MOVW         ↙
    ↳    R0, #0x13D8
__text:00002B72 6F 46               MOV          ↙
    ↳    R7, SP
__text:00002B74 C0 F2 00 00     MOVT.W       ↙
    ↳    R0, #0
__text:00002B78 78 44               ADD          ↙
    ↳    R0, PC
__text:00002B7A 01 F0 38 EA     BLX          ↙
    ↳    _puts
__text:00002B7E 00 20               MOVS         ↙
    ↳    R0, #0
__text:00002B80 80 BD               POP          ↙
    ↳    {R7,PC}

...

__cstring:00003E70 48 65 6C 6C 6F 20+↙
    ↳ aHelloWorld  DCB "Hello world!",0xA,0
```

...

- : 4-3-2-1;
- : 2-1;
- : 2-1-4-3.

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`0xFx`.

`MOVW R0, #0x13D8`

`MOVT.W R0, #0` `MOVT` Thumb-2.

`BLX` `BL`. :

```
__symbolstub1:00003FEC _puts              ; CODE XREF: _hello_world+E
__symbolstub1:00003FEC 44 F0 9F E5        LDR PC, =__imp__puts
```

.

. .

(Windows PE .exe, ELF Mach-O) .

.

__imp__puts . .

. .

.

> “A piece of coding which provides an address:”, according to P. Z. Ingerman, who invented thunks in 1961 as a way of binding actual parameters to their formal definitions in Algol-60 procedure calls. If a

 CAPÍTULO 3. HELLO, WORLD!

> procedure is called with an expression in the place of a formal parameter, the compiler generates a thunk which computes the expression and leaves the address of the result in some standard location.
>
> ...
>
> Microsoft and IBM have both defined, in their Intel-based systems, a “16-bit environment” (with bletcherous segment registers and 64K address limits) and a “32-bit environment” (with flat addressing and semi-real memory management). The two environments can both be running on the same computer and OS (thanks to what is called, in the Microsoft world, WOW which stands for Windows On Windows). MS and IBM have both decided that the process of getting from 16- to 32-bit and vice versa is called a “thunk”; for Windows 95, there is even a tool THUNK.EXE called a “thunk compiler”.

(The Jargon File)

3.4.5 ARM64

GCC

Compilamos el ejemplo con GCC 4.8.1 en ARM64:

Listing 3.15: Sin optimización GCC 4.8.1 + objdump

```
0000000000400590 <main>:
  400590:       a9bf7bfd        stp     x29, x30, [sp,#-16]!
  400594:       910003fd        mov     x29, sp
  400598:       90000000        adrp    x0, 400000 <_init-0x3b8>
  40059c:       91192000        add     x0, x0, #0x648
  4005a0:       97ffffa0        bl      400420 <puts@plt>
  4005a4:       52800000        mov     w0, #0x0                        // #0
  4005a8:       a8c17bfd        ldp     x29, x30, [sp],#16
  4005ac:       d65f03c0        ret

...

Contents of section .rodata:
 400640 01000200 00000000 48656c6c 6f210a00  ........Hello!..
```

En ARM64 no existen los modos Thumb ni Thumb-2, solo ARM, así que solo hay instrucciones de 32 bits. La cantidad de registros se duplica: B.4.1 on page 1905. Los registros de 64 bits tienen el prefijo X-W-, mientras que sus partes de 32 bits — W-.

STP (*Store Pair*) guarda dos registros en la pila simultáneamente: X29 en X30. Por supuesto, esta instrucción puede guardar ese par en cualquier lugar de la memoria, pero aquí se especifica el registro SP, así que el par se guarda en la pila. Los registros en ARM64 son de 64 bits y cada uno ocupa 8 bytes, por lo que se necesitan 16 bytes para guardar dos registros.

El signo de exclamación tras el operando significa que primero se resta 16 de SP y solo después los valores del par de registros se escriben en la pila. A esto también se le llama *pre-index*. Sobre la diferencia entre *post-index* y *pre-index* lee aquí: 28.2 on page 850.

Por lo tanto, en términos del más familiar x86, la primera instrucción es análoga a un par de PUSH X29 y PUSH X30. X29 , X30 LR, ; por eso se guardan en el prólogo de la función y se restauran en el epílogo.

La segunda instrucción copia SP en X29 (o FP[25]). Esto se hace para establecer el marco de pila de la función.

ADRP y ADD se usan para formar la dirección de la cadena «Hello!» en el registro X0, porque el primer argumento de la función se pasa por ese registro. (porque la longitud de la instrucción está limitada a 4 bytes; más sobre esto aquí: 28.3.1 on page 852). Así que deben usarse varias instrucciones. La primera instrucción (ADRP) escribe en X0 la dirección de la página de 4 KB donde está la cadena, y la segunda (ADD) simplemente suma el resto a esa dirección. Más sobre esto en: 28.4 on page 855.

0x400000 + 0x648 = 0x400648, y vemos que en la

[25]Frame Pointer

 CAPÍTULO 3. HELLO, WORLD!

sección de datos `.rodata` en esa dirección está nuestra cadena en C «Hello!».

Luego se llama a `puts()` usando la instrucción BL. Esto ya se comentó: 3.4.3 on page 39.

MOV La instrucción MOV escribe 0 en W0. W0 son los 32 bits bajos del registro de 64 bits X0:

Spanish text placeholder	Spanish text placeholder
X0	
	W0

El resultado de la función se devuelve por X0, y `main()` devuelve 0, así que así se prepara el valor de retorno. ¿Por qué usar la parte de 32 bits? Porque el tipo *int* en ARM64, igual que en x86-64, sigue siendo de 32 bits para una mejor compatibilidad. Por lo tanto, si una función devuelve un *int* de 32 bits, solo hay que llenar los 32 bits bajos del registro X0.

Para verificarlo, modifiquemos un poco este ejemplo y recompilémoslo. Ahora `main()` devuelve un valor de 64 bits:

Listing 3.16: `main()` que devuelve un valor de tipo `uint64_t`

```
#include <stdio.h>
#include <stdint.h>

uint64_t main()
{
        printf ("Hello!\n");
```

```
        return 0;
}
```

El resultado es el mismo, solo que así se ve ahora el MOV en esa línea:

Listing 3.17: Sin optimización GCC 4.8.1 + objdump

```
  4005a4:        d2800000        mov     x0, ↙
    ↘ #0x0                          // #0
```

Luego, con la instrucción LDP (*Load Pair*) se restauran los registros X29 y X30. No hay signo de exclamación tras la instrucción: esto implica que primero se cargan los valores de la pila y solo después SP se incrementa en 16. Esto se llama *post-index*.

En ARM64 apareció una instrucción nueva: RET. Funciona igual que BX LR, , solo que se añade un bit de *pista* especial que informa a la CPU de que es un retorno de función, no un salto cualquiera, para poder ejecutarla de forma más óptima.

Debido a la simplicidad de la función, el GCC con optimización genera exactamente el mismo código.

3.5 MIPS

3.5.1 Una palabra sobre el «global pointer»

Un concepto importante en MIPS es el «global pointer» (puntero global). Como tal vez ya sepamos, cada instrucción

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MIPS tiene un tamaño de 32 bits, por lo que es imposible incrustar una dirección de 32 bits en una sola instrucción: se debe usar un par de ellas para esto (como hizo GCC en nuestro ejemplo para cargar la dirección de la cadena de texto).

Sin embargo, es posible cargar datos desde la dirección en el rango de $register - 32768 ... register + 32767$ usando una sola instrucción (porque se pueden codificar 16 bits de desplazamiento con signo en una sola instrucción). Así que podemos asignar algún registro para este propósito y también asignar un área de 64 KiB para los datos más utilizados. Este registro asignado se llama «global pointer» y apunta al medio del área de 64 KiB.

Esta área usualmente contiene variables globales y direcciones de funciones importadas como `printf()`, porque los desarrolladores de GCC decidieron que obtener la dirección de alguna función debe ser tan rápido como la ejecución de una sola instrucción en lugar de dos. En un archivo ELF, esta área de 64 KiB se encuentra parcialmente en las secciones .sbss («BSS pequeño») para datos no inicializados y .sdata («datos pequeños») para datos inicializados.

Esto implica que el programador puede elegir qué datos desea acceder de forma rápida y colocarlos en .sdata/.sbss.

Algunos programadores de la «vieja escuela» pueden recordar el modelo de memoria de MS-DOS 94 on page 1776 o los gestores de memoria de MS-DOS como XMS/EMS, donde toda la memoria se dividía en bloques de 64 KiB.

Este concepto no es exclusivo de MIPS. Al menos PowerPC también utiliza esta técnica.

3.5.2 Con optimización GCC

Consideremos el siguiente ejemplo que ilustra el concepto del «global pointer».

Listing 3.18: Con optimización GCC 4.4.5 (salida de assembly)

```

$LC0:
; \000 :
        .ascii  "Hello, world!\012\000"
main:
;
;  GP:
        lui     $28,%hi(__gnu_local_gp)
        addiu   $sp,$sp,-32
        addiu   $28,$28,%lo(__gnu_local_gp)
; :
        sw      $31,28($sp)
;  $25:
        lw      $25,%call16(puts)($28)
;  $4 ($a0):
        lui     $4,%hi($LC0)
; :
        jalr    $25
        addiu   $4,$4,%lo($LC0) ; branch delay slot
;  RA:
        lw      $31,28($sp)
```

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```
;  $zero  $v0:
        move    $2,$0
; :
        j       $31
; :
        addiu   $sp,$sp,32 ; branch delay ↙
    ↳ slot
```

Como vemos, el registro $GP se configura en el prólogo de la función para apuntar al medio de esta área. El registro RA[26] también se guarda en la pila local. Aquí también se utiliza `puts()` en lugar de `printf()`. La dirección de la función `puts()` se carga en $25 utilizando la instrucción LW («Load Word»). Luego, la dirección de la cadena de texto se carga en $4 mediante el par de instrucciones LUI («Load Upper Immediate») y ADDIU («Add Immediate Unsigned Word»). LUI establece los 16 bits superiores del registro (de ahí la palabra «upper» en el nombre de la instrucción) y ADDIU suma los 16 bits inferiores de la dirección. ADDIU sigue a JALR (¿recuerdas los *branch delay slots*?). [27].El registro $4 también se llama $A0, que se utiliza para pasar el primer argumento de la función[28].

JALR («Jump and Link Register») salta a la dirección almacenada en el registro $25 (dirección de `puts()`) mientras guarda la dirección de la siguiente instrucción (LW) en RA. .Esto es muy similar a ARM. Ah, y una cosa importante es

[26]Dirección de retorno

[27]C.1 on page 1907

[28]La tabla de registros MIPS está disponible en el apéndice C.1 on page 1907

que la dirección guardada en RA no es la dirección de la siguiente instrucción (porque está en un *delay slot* y se ejecuta antes de la instrucción de salto), sino la dirección de la instrucción posterior a la siguiente (después del *delay slot*). Por lo tanto, se escribe $PC + 8$ en RA durante la ejecución de JALR; en nuestro caso, esta es la dirección de la instrucción LW que sigue a ADDIU.

LW («Load Word») en la línea 19 restaura RA de la pila local (esta instrucción es más bien parte del epílogo de la función).

MOVE en la línea 22 copia el valor del registro \$0 (\$ZERO) al \$2 (\$V0). MIPS tiene un registro *constante*, que siempre contiene cero. Aparentemente, a los desarrolladores de MIPS se les ocurrió la idea de que el cero es, de hecho, la constante más utilizada en la programación informática, así que simplemente usemos el registro \$0 cada vez que se necesite cero. Otro dato interesante es que MIPS carece de una instrucción que transfiera datos entre registros. De hecho, `MOVE DST, SRC` es `ADD DST, SRC, $ZERO` ($DST = SRC + 0$), .lo cual hace lo mismo. Al parecer, los desarrolladores de MIPS querían tener una tabla de opcodes compacta. Esto no significa que ocurra una suma real en cada instrucción MOVE. Lo más probable es que la CPU optimice estas pseudoinstrucciones y la ALU[29] nunca sea utilizada.

J en la línea 24 salta a la dirección en RA, lo que efectivamente realiza el retorno de la función. ADDIU después de J

[29] Unidad aritmético-lógica

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se ejecuta de hecho antes de J (¿recuerdas los *branch delay slots*?) y es parte del epílogo de la función.

Aquí también hay un listado generado por IDA. Cada registro aquí tiene su propio seudónimo:

Listing 3.19: Con optimización GCC 4.4.5 (IDA)

```

.text:00000000 main:
.text:00000000
.text:00000000 var_10          = -0x10
.text:00000000 var_4           = -4
.text:00000000
;
;  GP:
.text:00000000                 lui     $gp, (__gnu_local_gp >> 16)
.text:00000004                 addiu   $sp, -0x20
.text:00000008                 la      $gp, (__gnu_local_gp & 0xFFFF)
; :
.text:0000000C                 sw      $ra, 0x20+var_4($sp)
; :
; :
.text:00000010                 sw      $gp, 0x20+var_10($sp)
;  $t9:
.text:00000014                 lw      $t9, (puts & 0xFFFF)($gp)
;  $a0:
.text:00000018                 lui     $a0, 
```

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```
    ($LC0 >> 16)  # "Hello, world!"
21 ; :
22 .text:0000001C                    jalr    $t9
23 .text:00000020                    la      $a0, ($LC0 & 0xFFFF)  # "Hello, world!"
24 ;  RA:
25 .text:00000024                    lw      $ra, 0x20+var_4($sp)
26 ;  $zero  $v0:
27 .text:00000028                    move    $v0, $zero
28 ; :
29 .text:0000002C                    jr      $ra
30 ; :
31 .text:00000030                    addiu   $sp, 0x20
```

La instrucción en la línea 15 guarda el valor de GP en la pila local, y esta instrucción falta misteriosamente en el listado de salida de GCC, tal vez por un error de GCC[30]. El valor de GP debe guardarse de hecho, porque cada función puede usar su propia ventana de datos de 64 KiB.

El registro que contiene la dirección de `puts()` se llama $T9, porque los registros con prefijo T- se denominan «temporaries» (temporales) y no es necesario conservar su contenido.

[30] Aparentemente, las funciones que generan listados no son tan críticas para los usuarios de GCC, por lo que aún pueden existir algunos errores no corregidos.

3.5.3 Sin optimización GCC

Sin optimización GCC es más verboso.

Listing 3.20: Sin optimización GCC 4.4.5 (salida de assembly)

```

$LC0:
        .ascii  "Hello, world!\012\000"
main:
;
; :
        addiu   $sp,$sp,-32
        sw      $31,28($sp)
        sw      $fp,24($sp)
; :
        move    $fp,$sp
;  GP:
        lui     $28,%hi(__gnu_local_gp)
        addiu   $28,$28,%lo(__gnu_local_gp)
; :
        lui     $2,%hi($LC0)
        addiu   $4,$2,%lo($LC0)
; :
        lw      $2,%call16(puts)($28)
        nop
;  puts():
        move    $25,$2
        jalr    $25
        nop  ; branch delay slot

; :
        lw      $28,16($fp)
```

```
; $2 ($V0) :
        move    $2,$0
; .
; SP:
        move    $sp,$fp
; RA:
        lw      $31,28($sp)
; FP:
        lw      $fp,24($sp)
        addiu   $sp,$sp,32
; RA:
        j       $31
        nop  ; branch delay slot
```

Vemos aquí que el registro FP se utiliza como un puntero al marco de la pila. También vemos 3 NOP[31]s. El segundo y el tercero de los cuales siguen a las instrucciones de ramificación.

Tal vez, el compilador GCC siempre añade NOPs (debido a los *branch delay slots*) después de las instrucciones de ramificación y luego, si la optimización está activada, tal vez los elimine. .Así que en este caso se dejan aquí.

Aquí también está el listado de IDA:

Listing 3.21: Sin optimización GCC 4.4.5 (IDA)

```

.text:00000000 main:
.text:00000000
.text:00000000 var_10          = -0x10
```

[31] No OPeration

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```
.text:00000000 var_8           = -8
.text:00000000 var_4           = -4
.text:00000000
;
; :
.text:00000000                 addiu   $sp, -0x20
.text:00000004                 sw      $ra, 0x20+var_4($sp)
.text:00000008                 sw      $fp, 0x20+var_8($sp)
; :
.text:0000000C                 move    $fp, $sp
;  GP:
.text:00000010                 la      $gp, __gnu_local_gp
.text:00000018                 sw      $gp, 0x20+var_10($sp)
; :
.text:0000001C                 lui     $v0, (aHelloWorld >> 16)  # "Hello, world!"
.text:00000020                 addiu   $a0, $v0, (aHelloWorld & 0xFFFF)  # "Hello, world!"
; :
.text:00000024                 lw      $v0, (puts & 0xFFFF)($gp)
.text:00000028                 or      $at, $zero ; NOP
;  puts():
.text:0000002C                 move    $t9, $v0
```

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```
26 .text:00000030                 jalr    $t9
27 .text:00000034                 or      $at, $zero ; NOP
28 ; :
29 .text:00000038                 lw      $gp, 0x20+var_10($fp)
30 ;  $2 ($V0) :
31 .text:0000003C                 move    $v0, $zero
32 ; .
33 ;  SP:
34 .text:00000040                 move    $sp, $fp
35 ;  RA:
36 .text:00000044                 lw      $ra, 0x20+var_4($sp)
37 ;  FP:
38 .text:00000048                 lw      $fp, 0x20+var_8($sp)
39 .text:0000004C                 addiu   $sp, 0x20
40 ;  RA:
41 .text:00000050                 jr      $ra
42 .text:00000054                 or      $at, $zero ; NOP
```

Curiosamente, IDA reconoció el par de instrucciones LUI/ADDIU y las fusionó en una sola pseudoinstrucción LA («Load Address») en la línea 15. ¡También podemos ver que esta pseudoinstrucción tiene un tamaño de 8 bytes! Esta es una pseudoinstrucción (o *macro*) porque no es una instrucción MIPS real, sino más bien un nombre práctico para un par de instrucciones.

Otra cosa es que IDA no reconoce las instrucciones NOP, por lo que aquí están en las líneas 22, 26 y 41. Esto es `OR $AT, $ZERO`. Esencialmente, esta instrucción aplica la operación OR al contenido del registro $AT con cero, lo cual es, por supuesto, una instrucción ociosa (idle). MIPS, al igual que muchas otras ISA, no tiene una instrucción NOP separada.

3.5.4 El papel del marco de pila en este ejemplo

La dirección de la cadena de texto se pasa en un registro. Entonces, ¿por qué configurar una pila local de todos modos? La razón de esto radica en el hecho de que los valores de los registros RA y GP tienen que guardarse en algún lugar (porque se llama a `printf()`), y la pila local se utiliza para este propósito. 2.3 on page 12.Si esta fuera una leaf function, habría sido posible deshacerse del prólogo y epílogo de la función, por ejemplo: 2.3 on page 12.

3.5.5 Con optimización GCC: carguemos en GDB

Listing 3.22: sesión de ejemplo en GDB

```
root@debian-mips:~# gcc hw.c -O3 -o hw
root@debian-mips:~# gdb hw
GNU gdb (GDB) 7.0.1-debian
Copyright (C) 2009 Free Software Foundation,↙
    ↳ Inc.
License GPLv3+: GNU GPL version 3 or later <↙
    ↳ http://gnu.org/licenses/gpl.html>
```

 CAPÍTULO 3. HELLO, WORLD!

```
This is free software: you are free to ↙
    ↘ change and redistribute it.
There is NO WARRANTY, to the extent ↙
    ↘ permitted by law.  Type "show copying"
and "show warranty" for details.
This GDB was configured as "mips-linux-gnu".
For bug reporting instructions, please see:
<http://www.gnu.org/software/gdb/bugs/>...
Reading symbols from /root/hw...(no ↙
    ↘ debugging symbols found)...done.
(gdb) b main
Breakpoint 1 at 0x400654
(gdb) run
Starting program: /root/hw

Breakpoint 1, 0x00400654 in main ()
(gdb) set step-mode on
(gdb) disas
Dump of assembler code for function main:
0x00400640 <main+0>:    lui     gp,0x42
0x00400644 <main+4>:    addiu   sp,sp,-32
0x00400648 <main+8>:    addiu   gp,gp,-30624
0x0040064c <main+12>:   sw      ra,28(sp)
0x00400650 <main+16>:   sw      gp,16(sp)
0x00400654 <main+20>:   lw      t9,-32716(gp↙
    ↘ )
0x00400658 <main+24>:   lui     a0,0x40
0x0040065c <main+28>:   jalr    t9
0x00400660 <main+32>:   addiu   a0,a0,2080
0x00400664 <main+36>:   lw      ra,28(sp)
0x00400668 <main+40>:   move    v0,zero
0x0040066c <main+44>:   jr      ra
0x00400670 <main+48>:   addiu   sp,sp,32
```

```
End of assembler dump.
(gdb) s
0x00400658 in main ()
(gdb) s
0x0040065c in main ()
(gdb) s
0x2ab2de60 in printf () from /lib/libc.so.6
(gdb) x/s $a0
0x400820:        "hello, world"
(gdb)
```

3.6 Conclusión

La principal diferencia entre el código x86/ARM y x64/ARM64 es que el puntero a la cadena ahora es de 64 bits. En efecto, las CPU modernas son de 64 bits porque la memoria se abarató y ahora se puede instalar mucha más; para direccionarla, 32 bits ya no bastan. Por eso todos los punteros son ahora de 64 bits.

3.7 Ejercicios

3.7.1 Ejercicio #1

```
main:
    push 0xFFFFFFFF
    call MessageBeep
    xor  eax,eax
    retn
```

 CAPÍTULO 3. HELLO, WORLD!

?¿Qué hace esta función win32?

Responda: G.1.1 on page 1917.

3.7.2 Ejercicio #2

```
main:
        pushq   %rbp
        movq    %rsp, %rbp
        movl    $2, %edi
        call    sleep
        popq    %rbp
        ret
```

?¿Qué hace esta función Linux? Aquí se usa la sintaxis de ensamblador AT&T. Aquí se usa la sintaxis de ensamblador AT&T.

Responda: G.1.1 on page 1917.

CAPÍTULO 4. PRÓLOGO Y EPÍLOGO DE FUNCIONES

Capítulo 4

Prólogo y epílogo de funciones

El prólogo de una función es una secuencia de instrucciones al inicio de la misma. A menudo se ve como el siguiente fragmento de código:

```
    push    ebp
    mov     ebp, esp
    sub     esp, X
```

Lo que hacen estas instrucciones: guardan el valor del registro EBP para el futuro, establecen el valor del registro EBP al valor de ESP y luego asignan espacio en la pila para las variables locales.

CAPÍTULO 4. PRÓLOGO Y EPÍLOGO DE FUNCIONES

El valor en EBP permanece igual durante todo el período de ejecución de la función y se utiliza para el acceso a las variables locales y a los argumentos. Para el mismo propósito se podría usar ESP, pero dado que cambia con el tiempo, este enfoque no es muy conveniente.

El epílogo de la función libera el espacio asignado en la pila, restaura el valor del registro EBP a su estado inicial y devuelve el flujo de control a la caller:

```
    mov    esp, ebp
    pop    ebp
    ret    0
```

Los prólogos y epílogos de funciones usualmente son detectados por los desensambladores para delimitar las funciones.

4.1 Recursión

La presencia de prólogos y epílogos puede afectar negativamente el rendimiento de la recursión.

Más sobre la recursión en este libro: 36.3 on page 900.

Capítulo 5

Pila

1 .

STMFD/LDMFD, STMED[2]/LDMED[3] STMFA[4]/LDMFA[5], STMEA[6]/

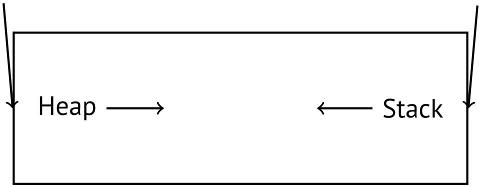
[1] wikipedia.org/wiki/Call_stack
[2] Store Multiple Empty Descending ()
[3] Load Multiple Empty Descending ()
[4] Store Multiple Full Ascending ()
[5] Load Multiple Full Ascending ()
[6] Store Multiple Empty Ascending ()
[7] Load Multiple Empty Ascending ()

 CAPÍTULO 5. PILA

5.1

.

. . .



[RT74] :

> The user-core part of an image is divided into three logical segments. The program text segment begins at location 0 in the virtual address space. During execution, this segment is write-protected and a single copy of it is shared among all processes executing the same program. At the first 8K byte boundary above the program text segment in the virtual address space begins a nonshared, writable data segment, the size of which may be extended by a system call. Starting at the highest address in the virtual address space is a stack segment, which automatically grows downward as the hardware's stack pointer

fluctuates.

5.2

5.2.1

x86

```
void f()
{
        f();
};
```

```
c:\tmp6>cl ss.cpp /Fass.asm
Microsoft (R) 32-bit C/C++ Optimizing ↙
    ↘ Compiler Version 15.00.21022.08 for 80↙
    ↘ x86
Copyright (C) Microsoft Corporation.  All ↙
    ↘ rights reserved.

ss.cpp
c:\tmp6\ss.cpp(4) : warning C4717: 'f' : ↙
    ↘ recursive on all control paths, ↙
    ↘ function will cause runtime stack ↙
    ↘ overflow
```

...:

```
?f@@YAXXZ PROC                              ↙
    ↘                          ; f
```

```
; File c:\tmp6\ss.cpp
; Line 2
        push    ebp
        mov     ebp, esp
; Line 3
        call    ?f@@YAXXZ                                    ↙
    ↘                       ; f
; Line 4
        pop     ebp
        ret     0
?f@@YAXXZ ENDP                                               ↙
    ↘                       ; f
```

...

```
?f@@YAXXZ PROC                                               ↙
    ↘                       ; f
; File c:\tmp6\ss.cpp
; Line 2
$LL3@f:
; Line 3
        jmp     SHORT $LL3@f
?f@@YAXXZ ENDP                                               ↙
    ↘                       ; f
```

ARM

. «Hello, world!» (3.4 on page 31), LR (link register). . PUSH R4-R7,LR POP R4-R7,PC LR.

leaf function[8]. . [9]. .

8.3.2 on page 199, 8.3.3 on page 199, 19.17 on page 595, 19.33 on page 623, 19.5.4 on page 624, 15.4 on page 388, 15.2 on page 386, 17.3 on page 421.

5.2.2

«cdecl»:

```
push arg3
push arg2
push arg1
call f
add esp, 12 ; 4*3=12
```

ESP	
ESP+4	argumento#1, Marcado en IDA como arg_0
ESP+8	argumento#2, Marcado en IDA como arg_4
ESP+0xC	argumento#3, Marcado en IDA como arg_8
…	…

(64 on page 1299).

[10].

[8] infocenter.arm.com/help/index.jsp?topic=/com.arm.doc.faqs/ka13785.html
[9] [Ray03, Chapter 4, Part II].
[10]

```
printf("%d %d %d", 1234);
```

`printf()`

: `main()`, `main(int argc, char *argv[])` `main(int argc, char *argv[], char *envp[])`.

```
push envp
push argv
push argc
call main
...
```

`main(int argc, char *argv[])`, `main(int argc)`,
.

5.2.3

5.2.4 x86:

[11].

```
#ifdef __GNUC__
#include <alloca.h> // GCC
#else
#include <malloc.h> // MSVC
#endif
#include <stdio.h>
```

[11] alloca16.asm y chkstk.asm en C:\Program Files (x86)\Microsoft Visual Studio 10.0\VC\crt\src\intel

```
void f()
{
    char *buf=(char*)alloca (600);
#ifdef __GNUC__
    snprintf (buf, 600, "hi! %d, %d, %d\n", ⤦
    ↪ 1, 2, 3); // GCC
#else
    _snprintf (buf, 600, "hi! %d, %d, %d\n",⤦
    ↪  1, 2, 3); // MSVC
#endif

    puts (buf);
};
```

MSVC

(MSVC 2010):

Listing 5.1: MSVC 2010

```
...

    mov     eax, 600            ; 00000258H
    call    __alloca_probe_16
    mov     esi, esp

    push    3
    push    2
    push    1
    push    OFFSET $SG2672
    push    600                 ; 00000258H
```

```
    push    esi
    call    __snprintf

    push    esi
    call    _puts
    add     esp, 28                 ; 0000001cH

...
```

12.

GCC + Sintaxis Intel

Listing 5.2: GCC 4.7.3

```

.LC0:
        .string "hi! %d, %d, %d\n"
f:
        push    ebp
        mov     ebp, esp
        push    ebx
        sub     esp, 660
        lea     ebx, [esp+39]
        and     ebx, -16                                       ↙
    ↪         ;
        mov     DWORD PTR [esp], ebx                           ↙
    ↪         ; s
        mov     DWORD PTR [esp+20], 3
        mov     DWORD PTR [esp+16], 2
```

12.

```
        mov     DWORD PTR [esp+12], 1
        mov     DWORD PTR [esp+8], OFFSET ↙
    ↘ FLAT:.LC0 ; "hi! %d, %d, %d\n"
        mov     DWORD PTR [esp+4], 600          ↙
    ↘           ; maxlen
        call    _snprintf
        mov     DWORD PTR [esp], ebx            ↙
    ↘           ; s
        call    puts
        mov     ebx, DWORD PTR [ebp-4]
        leave
        ret
```

GCC + Sintaxis AT&T

:

Listing 5.3: GCC 4.7.3

```
.LC0:
        .string "hi! %d, %d, %d\n"
f:
        pushl   %ebp
        movl    %esp, %ebp
        pushl   %ebx
        subl    $660, %esp
        leal    39(%esp), %ebx
        andl    $-16, %ebx
        movl    %ebx, (%esp)
        movl    $3, 20(%esp)
        movl    $2, 16(%esp)
        movl    $1, 12(%esp)
```

```
        movl    $.LC0, 8(%esp)
        movl    $600, 4(%esp)
        call    _snprintf
        movl    %ebx, (%esp)
        call    puts
        movl    -4(%ebp), %ebx
        leave
        ret
```

,`movl $3, 20(%esp)` `mov DWORD PTR [esp+20], 3`

5.2.5 (Windows) SEH

.

: (68.3 on page 1359).

5.2.6

(18.2 on page 496).

5.2.7

5.3

…	…
ESP-0xC	#2, Marcado en IDA como var_8
ESP-8	#1, Marcado en IDA como var_4
ESP-4	EBP
ESP	
ESP+4	argumento#1, Marcado en IDA como arg_0
ESP+8	argumento#2, Marcado en IDA como arg_4
ESP+0xC	argumento#3, Marcado en IDA como arg_8
…	…

5.4

? :

```
#include <stdio.h>

void f1()
{
        int a=1, b=2, c=3;
};

void f2()
{
        int a, b, c;
        printf ("%d, %d, %d\n", a, b, c);
};
```

```
int main()
{
        f1();
        f2();
};
```

...

Listing 5.4: Sin optimización MSVC 2010

```
$SG2752 DB      '%d, %d, %d', 0aH, 00H

_c$ = -12       ; size = 4
_b$ = -8        ; size = 4
_a$ = -4        ; size = 4
_f1     PROC
        push    ebp
        mov     ebp, esp
        sub     esp, 12
        mov     DWORD PTR _a$[ebp], 1
        mov     DWORD PTR _b$[ebp], 2
        mov     DWORD PTR _c$[ebp], 3
        mov     esp, ebp
        pop     ebp
        ret     0
_f1     ENDP

_c$ = -12       ; size = 4
_b$ = -8        ; size = 4
_a$ = -4        ; size = 4
_f2     PROC
        push    ebp
```

```
        mov     ebp, esp
        sub     esp, 12
        mov     eax, DWORD PTR _c$[ebp]
        push    eax
        mov     ecx, DWORD PTR _b$[ebp]
        push    ecx
        mov     edx, DWORD PTR _a$[ebp]
        push    edx
        push    OFFSET $SG2752 ; '%d, %d, %d↙
    ↘ '
        call    DWORD PTR __imp__printf
        add     esp, 16
        mov     esp, ebp
        pop     ebp
        ret     0
_f2     ENDP

_main   PROC
        push    ebp
        mov     ebp, esp
        call    _f1
        call    _f2
        xor     eax, eax
        pop     ebp
        ret     0
_main   ENDP
```

…

```
c:\Polygon\c>cl st.c /Fast.asm /MD
Microsoft (R) 32-bit C/C++ Optimizing ↙
    ↘ Compiler Version 16.00.40219.01 for 80↙
    ↘ x86
```

 CAPÍTULO 5. PILA

```
Copyright (C) Microsoft Corporation.  All rights reserved.

st.c
c:\polygon\c\st.c(11) : warning C4700: uninitialized local variable 'c' used
c:\polygon\c\st.c(11) : warning C4700: uninitialized local variable 'b' used
c:\polygon\c\st.c(11) : warning C4700: uninitialized local variable 'a' used
Microsoft (R) Incremental Linker Version 10.00.40219.01
Copyright (C) Microsoft Corporation.  All rights reserved.

/out:st.exe
st.obj
```

…

```
c:\Polygon\c>st
1, 2, 3
```

`f2()`.

OllyDbg:

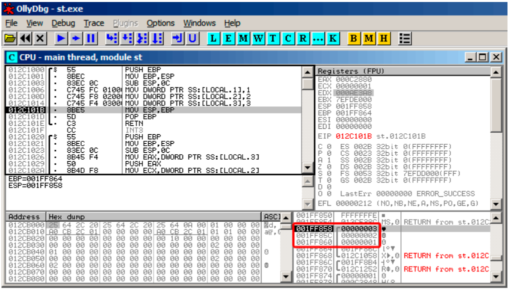


Figura 5.1: OllyDbg: f1()

f1() a, b y c 0x14F860

`f2()`:

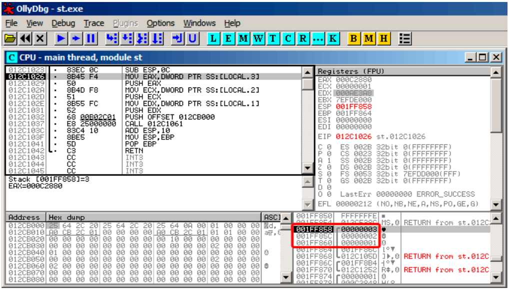


Figura 5.2: OllyDbg: `f2()`

… a, b y c `f2()`

?

5.5 Ejercicios

5.5.1 Ejercicio #1

?

```
#include <stdio.h>

int main()
{
        printf ("%d, %d, %d\n");
```

```
        return 0;
};
```

Responda: G.1.2 on page 1918.

5.5.2 Ejercicio #2

Lo que hace el código?

Listing 5.5: Con optimización MSVC 2010

```
$SG3103 DB      '%d', 0aH, 00H

_main   PROC
        push    0
        call    DWORD PTR __imp___time64
        push    edx
        push    eax
        push    OFFSET $SG3103 ; '%d'
        call    DWORD PTR __imp__printf
        add     esp, 16
        xor     eax, eax
        ret     0
_main   ENDP
```

Listing 5.6: Con optimización Keil 6/2013 (Modo ARM)

```
main PROC
        PUSH     {r4,lr}
        MOV      r0,#0
        BL       time
        MOV      r1,r0
        ADR      r0,|L0.32|
```

```
        BL      __2printf
        MOV     r0,#0
        POP     {r4,pc}
        ENDP

|L0.32|
        DCB     "%d\n",0
```

Listing 5.7: Con optimización Keil 6/2013 (Modo Thumb)

```
main PROC
        PUSH    {r4,lr}
        MOVS    r0,#0
        BL      time
        MOVS    r1,r0
        ADR     r0,|L0.20|
        BL      __2printf
        MOVS    r0,#0
        POP     {r4,pc}
        ENDP

|L0.20|
        DCB     "%d\n",0
```

Listing 5.8: Con optimización GCC 4.9 (ARM64)

```
main:
        stp     x29, x30, [sp, -16]!
        mov     x0, 0
        add     x29, sp, 0
        bl      time
        mov     x1, x0
        ldp     x29, x30, [sp], 16
```

```
        adrp    x0, .LC0
        add     x0, x0, :lo12:.LC0
        b       printf
.LC0:
        .string "%d\n"
```

Listing 5.9: Con optimización GCC 4.4.5 (MIPS) (IDA)

```
main:

var_10          = -0x10
var_4           = -4

                lui     $gp, (__gnu_local_gp ↙
    ↘ >> 16)
                addiu   $sp, -0x20
                la      $gp, (__gnu_local_gp ↙
    ↘ & 0xFFFF)
                sw      $ra, 0x20+var_4($sp)
                sw      $gp, 0x20+var_10($sp ↙
    ↘ )
                lw      $t9, (time & 0xFFFF) ↙
    ↘ ($gp)
                or      $at, $zero
                jalr    $t9
                move    $a0, $zero
                lw      $gp, 0x20+var_10($sp ↙
    ↘ )
                lui     $a0, ($LC0 >> 16)  # ↙
    ↘ "%d\n"
                lw      $t9, (printf & 0 ↙
    ↘ xFFFF)($gp)
                lw      $ra, 0x20+var_4($sp)
```

```
                la      $a0, ($LC0 & 0xFFFF)↙
    ↪   # "%d\n"
                move    $a1, $v0
                jr      $t9
                addiu   $sp, 0x20

$LC0:           .ascii "%d\n"<0>         # ↙
    ↪ DATA XREF: main+28
```

Responda: G.1.2 on page 1918.

CAPÍTULO 6. PRINTF() *CON VARIOS ARGUMENTOS*

Capítulo 6

`printf()` con varios argumentos

```
#include <stdio.h>

int main()
{
        printf("a=%d; b=%d; c=%d", 1, 2, 3);
        return 0;
};
```

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

6.1 x86

6.1.1 x86:

MSVC

```
$SG3830 DB      'a=%d; b=%d; c=%d', 00H

...

        push    3
        push    2
        push    1
        push    OFFSET $SG3830
        call    _printf
        add     esp, 16                          ↙
    ↘                      ; 00000010H
```

.

(64 on page 1299).

```
push a1
push a2
call ...
...
push a1
call ...
...
push a1
push a2
push a3
call ...
```

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

```
add esp, 24
```

Listing 6.1: x86

```
.text:100113E7                 push    3
.text:100113E9                 call    sub_100018B0 ;  (3)
.text:100113EE                 call    sub_100019D0 ;
.text:100113F3                 call    sub_10006A90 ;
.text:100113F8                 push    1
.text:100113FA                 call    sub_100018B0 ;  (1)
.text:100113FF                 add     esp, 8       ;
```

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

MSVC y OllyDbg

OllyDbg. . MSVC 2012 /MD MSVCR*.DLL,

OllyDbg. ntdll.dll, F9 (). CRT-. main().

:

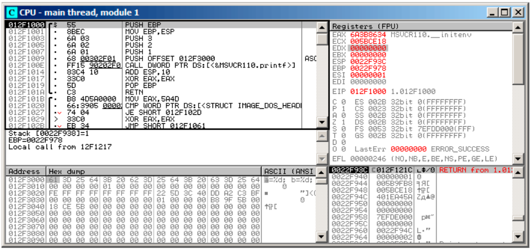


Figura 6.1: OllyDbg: main()

PUSH EBP F2 () F9 (). .

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

F8 (pasar por encima) 6 :

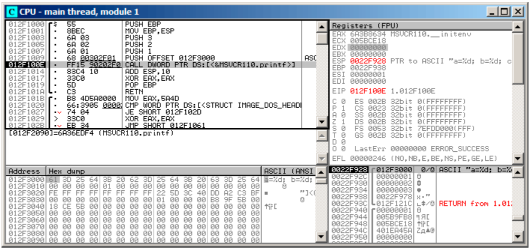


Figura 6.2: OllyDbg: `printf()`

PC `CALL printf`. OllyDbg, F8, EIP. ESP.

? :

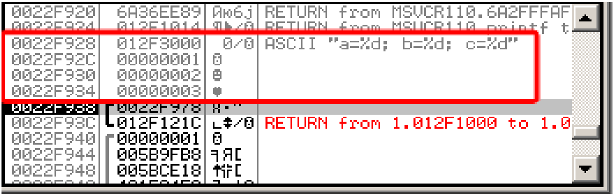


Figura 6.3: OllyDbg: ()

. OllyDbg `printf()`, .

. . . .

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

F8 (pasar por encima).

:

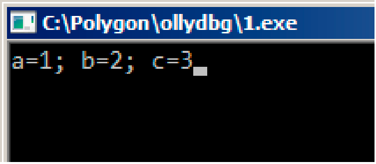


Figura 6.4: `printf()`

:

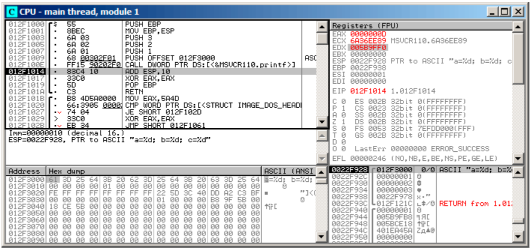


Figura 6.5: OllyDbg `printf()`

EAX 0xD (13). ECX y EDX. .

.

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

F8 ADD ESP, 10:

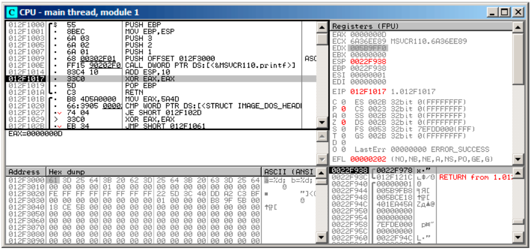


Figura 6.6: OllyDbg: ADD ESP, 10

ESP! (SP) o *Basura* . .

GCC

```
main            proc near

var_10          = dword ptr -10h
var_C           = dword ptr -0Ch
var_8           = dword ptr -8
var_4           = dword ptr -4

                push    ebp
                mov     ebp, esp
                and     esp, 0FFFFFFF0h
                sub     esp, 10h
                mov     eax, offset aADBDCD ↙
    ↳ ; "a=%d; b=%d; c=%d"
                mov     [esp+10h+var_4], 3
```

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

```
                mov     [esp+10h+var_8], 2
                mov     [esp+10h+var_C], 1
                mov     [esp+10h+var_10], ⤦
    ↳ eax
                call    _printf
                mov     eax, 0
                leave
                retn
main            endp
```

GCC y GDB

.

-g .

```
$ gcc 1.c -g -o 1
```

```
$ gdb 1
GNU gdb (GDB) 7.6.1-ubuntu
Copyright (C) 2013 Free Software Foundation,⤦
    ↳  Inc.
License GPLv3+: GNU GPL version 3 or later <⤦
    ↳ http://gnu.org/licenses/gpl.html>
This is free software: you are free to ⤦
    ↳ change and redistribute it.
There is NO WARRANTY, to the extent ⤦
    ↳ permitted by law.  Type "show copying"
and "show warranty" for details.
This GDB was configured as "i686-linux-gnu".
For bug reporting instructions, please see:
<http://www.gnu.org/software/gdb/bugs/>...
```

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

```
Reading symbols from /home/dennis/polygon⤦
    ↳ /1...done.
```

Listing 6.2: printf()

```
(gdb) b printf
Breakpoint 1 at 0x80482f0
```

. printf().

```
(gdb) run
Starting program: /home/dennis/polygon/1

Breakpoint 1, __printf (format=0x80484f0 "a⤦
    ↳ =%d; b=%d; c=%d") at printf.c:29
29      printf.c: No such file or directory.
```

```
(gdb) x/10w $esp
0xbffff11c:     0x0804844a      0x080484f0⤦
    ↳       0x00000001      0x00000002
0xbffff12c:     0x00000003      0x08048460⤦
    ↳       0x00000000      0x00000000
0xbffff13c:     0xb7e29905      0x00000001
```

RA (0x0804844a). :

```
(gdb) x/5i 0x0804844a
   0x804844a <main+45>: mov    $0x0,%eax
   0x804844f <main+50>: leave
   0x8048450 <main+51>: ret
   0x8048451:   xchg   %ax,%ax
   0x8048453:   xchg   %ax,%ax
```

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

```
(gdb) x/s 0x080484f0
0x80484f0:      "a=%d; b=%d; c=%d"
```

(1, 2, 3) `printf()`. .

«finish». . `printf()`.

```
(gdb) finish
Run till exit from #0  __printf (format=0x80484f0 "a=%d; b=%d; c=%d") at printf.c:29
main () at 1.c:6
6               return 0;
Value returned is $2 = 13
```

GDB[1] `printf()` EAX (13). .

«return 0;» . . ? . GDB .

. 13 en EAX:

```
(gdb) info registers
eax            0xd      13
ecx            0x0      0
edx            0x0      0
ebx            0xb7fc0000       -1208221696
esp            0xbffff120       0xbffff120
ebp            0xbffff138       0xbffff138
esi            0x0      0
edi            0x0      0
```

[1] GNU debugger

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

```
eip            0x804844a        0x804844a <
    ↳ main+45>
...
```

..

```
(gdb) disas
Dump of assembler code for function main:
   0x0804841d <+0>:     push   %ebp
   0x0804841e <+1>:     mov    %esp,%ebp
   0x08048420 <+3>:     and    $0xfffffff0,%
    ↳ esp
   0x08048423 <+6>:     sub    $0x10,%esp
   0x08048426 <+9>:     movl   $0x3,0xc(%esp
    ↳ )
   0x0804842e <+17>:    movl   $0x2,0x8(%esp
    ↳ )
   0x08048436 <+25>:    movl   $0x1,0x4(%esp
    ↳ )
   0x0804843e <+33>:    movl   $0x80484f0,(%
    ↳ esp)
   0x08048445 <+40>:    call   0x80482f0 <
    ↳ printf@plt>
=> 0x0804844a <+45>:    mov    $0x0,%eax
   0x0804844f <+50>:    leave
   0x08048450 <+51>:    ret
End of assembler dump.
```

GDB AT&T. :

```
(gdb) set disassembly-flavor intel
(gdb) disas
Dump of assembler code for function main:
```

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

```
   0x0804841d <+0>:     push   ebp
   0x0804841e <+1>:     mov    ebp,esp
   0x08048420 <+3>:     and    esp,0↙
    ↘ xfffffff0
   0x08048423 <+6>:     sub    esp,0x10
   0x08048426 <+9>:     mov    DWORD PTR [↙
    ↘ esp+0xc],0x3
   0x0804842e <+17>:    mov    DWORD PTR [↙
    ↘ esp+0x8],0x2
   0x08048436 <+25>:    mov    DWORD PTR [↙
    ↘ esp+0x4],0x1
   0x0804843e <+33>:    mov    DWORD PTR [↙
    ↘ esp],0x80484f0
   0x08048445 <+40>:    call   0x80482f0 <↙
    ↘ printf@plt>
=> 0x0804844a <+45>:    mov    eax,0x0
   0x0804844f <+50>:    leave
   0x08048450 <+51>:    ret
End of assembler dump.
```

. GDB

```
(gdb) step
7       };
```

MOV EAX, 0. EAX.

```
(gdb) info registers
eax            0x0      0
ecx            0x0      0
edx            0x0      0
ebx            0xb7fc0000       -1208221696
esp            0xbffff120       0xbffff120
```

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

```
ebp            0xbffff138       0xbffff138
esi            0x0      0
edi            0x0      0
eip            0x804844f        0x804844f <
    ↳ main+50>
...
```

6.1.2 x64:

:

```
#include <stdio.h>

int main()
{
        printf("a=%d; b=%d; c=%d; d=%d; e=%d
    ↳ ; f=%d; g=%d; h=%d\n", 1, 2, 3, 4, 5,
    ↳ 6, 7, 8);
        return 0;
};
```

MSVC

RCX, RDX, R8, R9 . . .

Listing 6.3: MSVC 2012 x64

```

$SG2923 DB      'a=%d; b=%d; c=%d; d=%d; e=%
    ↳ d; f=%d; g=%d; h=%d', 0aH, 00H

```

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

```
main    PROC
        sub     rsp, 88

        mov     DWORD PTR [rsp+64], 8
        mov     DWORD PTR [rsp+56], 7
        mov     DWORD PTR [rsp+48], 6
        mov     DWORD PTR [rsp+40], 5
        mov     DWORD PTR [rsp+32], 4
        mov     r9d, 3
        mov     r8d, 2
        mov     edx, 1
        lea     rcx, OFFSET FLAT:$SG2923
        call    printf

        ;  0
        xor     eax, eax

        add     rsp, 88
        ret     0
main    ENDP
_TEXT   ENDS
END
```

GCC

RDI, RSI, RDX, RCX, R8, R9. . : 3.2.2 on page 28.

: 3.2.2 on page 28.

Listing 6.4: Con optimización GCC 4.4.6 x64

```
.LC0:
```

 CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

```
        .string "a=%d; b=%d; c=%d; d=%d; e=%d; f=%d; g=%d; h=%d\n"

main:
        sub     rsp, 40

        mov     r9d, 5
        mov     r8d, 4
        mov     ecx, 3
        mov     edx, 2
        mov     esi, 1
        mov     edi, OFFSET FLAT:.LC0
        xor     eax, eax ;
        mov     DWORD PTR [rsp+16], 8
        mov     DWORD PTR [rsp+8], 7
        mov     DWORD PTR [rsp], 6
        call    printf

        ;  0

        xor     eax, eax
        add     rsp, 40
        ret
```

GCC + GDB

GDB.

```
$ gcc -g 2.c -o 2
```

```
$ gdb 2
GNU gdb (GDB) 7.6.1-ubuntu
```

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

```
Copyright (C) 2013 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.  Type "show copying"
and "show warranty" for details.
This GDB was configured as "x86_64-linux-gnu".
For bug reporting instructions, please see:
<http://www.gnu.org/software/gdb/bugs/>...
Reading symbols from /home/dennis/polygon/2...done.
```

Listing 6.5:

```
(gdb) b printf
Breakpoint 1 at 0x400410
(gdb) run
Starting program: /home/dennis/polygon/2

Breakpoint 1, __printf (format=0x400628 "a=%d; b=%d; c=%d; d=%d; e=%d; f=%d; g=%d; h=%d\n") at printf.c:29
29      printf.c: No such file or directory.
```

RSI/RDX/RCX/R8/R9 . RIP printf().

```
(gdb) info registers
rax            0x0      0
rbx            0x0      0
```

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

```
rcx            0x3      3
rdx            0x2      2
rsi            0x1      1
rdi            0x400628 4195880
rbp            0x7fffffffdf60   0↙
    ↘ x7fffffffdf60
rsp            0x7fffffffdf38   0↙
    ↘ x7fffffffdf38
r8             0x4      4
r9             0x5      5
r10            0x7fffffffdce0   ↙
    ↘ 140737488346336
r11            0x7ffff7a65f60   ↙
    ↘ 140737348263776
r12            0x400440 4195392
r13            0x7fffffffe040   ↙
    ↘ 140737488347200
r14            0x0      0
r15            0x0      0
rip            0x7ffff7a65f60   0↙
    ↘ x7ffff7a65f60 <__printf>
...
```

Listing 6.6:

```
(gdb) x/s $rdi
0x400628:       "a=%d; b=%d; c=%d; d=%d; e=%↙
    ↘ d; f=%d; g=%d; h=%d\n"
```

g *giant words*, .

```
(gdb) x/10g $rsp
```

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

```
0x7fffffffdf38: 0x0000000000400576      0↙
    ↳ x0000000000000006
0x7fffffffdf48: 0x0000000000000007      0↙
    ↳ x00007fff00000008
0x7fffffffdf58: 0x0000000000000000      0↙
    ↳ x0000000000000000
0x7fffffffdf68: 0x00007ffff7a33de5      0↙
    ↳ x0000000000000000
0x7fffffffdf78: 0x00007fffffffe048      0↙
    ↳ x0000000100000000
```

RA. : 6, 7, 8. : 0x00007fff00000008...

:

```
(gdb) set disassembly-flavor intel
(gdb) disas 0x0000000000400576
Dump of assembler code for function main:
   0x000000000040052d <+0>:     push   rbp
   0x000000000040052e <+1>:     mov    rbp,↙
    ↳ rsp
   0x0000000000400531 <+4>:     sub    rsp,0↙
    ↳ x20
   0x0000000000400535 <+8>:     mov    DWORD↙
    ↳  PTR [rsp+0x10],0x8
   0x000000000040053d <+16>:    mov    DWORD↙
    ↳  PTR [rsp+0x8],0x7
   0x0000000000400545 <+24>:    mov    DWORD↙
    ↳  PTR [rsp],0x6
   0x000000000040054c <+31>:    mov    r9d,0↙
    ↳ x5
   0x0000000000400552 <+37>:    mov    r8d,0↙
    ↳ x4
```

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

```
   0x0000000000400558 <+43>:    mov    ecx,0x3
   0x000000000040055d <+48>:    mov    edx,0x2
   0x0000000000400562 <+53>:    mov    esi,0x1
   0x0000000000400567 <+58>:    mov    edi,0x400628
   0x000000000040056c <+63>:    mov    eax,0x0
   0x0000000000400571 <+68>:    call   0x400410 <printf@plt>
   0x0000000000400576 <+73>:    mov    eax,0x0
   0x000000000040057b <+78>:    leave
   0x000000000040057c <+79>:    ret
End of assembler dump.
```

. RIP LEAVE .

```
(gdb) finish
Run till exit from #0  __printf (format=0x400628 "a=%d; b=%d; c=%d; d=%d; e=%d; f=%d; g=%d; h=%d\n") at printf.c:29
a=1; b=2; c=3; d=4; e=5; f=6; g=7; h=8
main () at 2.c:6
6               return 0;
Value returned is $1 = 39
(gdb) next
7       };
(gdb) info registers
rax            0x0      0
rbx            0x0      0
```

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

```
rcx            0x26     38
rdx            0x7ffff7dd59f0   ↙
    ↘ 140737351866864
rsi            0x7fffffd9       2147483609
rdi            0x0      0
rbp            0x7fffffffdf60   0↙
    ↘ x7fffffffdf60
rsp            0x7fffffffdf40   0↙
    ↘ x7fffffffdf40
r8             0x7ffff7dd26a0   ↙
    ↘ 140737351853728
r9             0x7ffff7a60134   ↙
    ↘ 140737348239668
r10            0x7fffffffd5b0   ↙
    ↘ 140737488344496
r11            0x7ffff7a95900   ↙
    ↘ 140737348458752
r12            0x400440 4195392
r13            0x7fffffffe040   ↙
    ↘ 140737488347200
r14            0x0      0
r15            0x0      0
rip            0x40057b 0x40057b <main+78>
...
```

6.2 ARM

6.2.1 ARM:

. fastcall (64.3 on page 1301) o win64 (64.5.1 on page 1303).

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

32- ARM

Sin optimización Keil 6/2013 (Modo ARM)

Listing 6.7: Sin optimización Keil 6/2013 (Modo ARM)

```
.text:00000000 main
.text:00000000 10 40 2D E9   STMFD   SP!, {R4,LR}
.text:00000004 03 30 A0 E3   MOV     R3, #3
.text:00000008 02 20 A0 E3   MOV     R2, #2
.text:0000000C 01 10 A0 E3   MOV     R1, #1
.text:00000010 08 00 8F E2   ADR     R0, aADBDCD     ; "a=%d; b=%d; c=%d"
.text:00000014 06 00 00 EB   BL      __2printf
.text:00000018 00 00 A0 E3   MOV     R0, #0          ; return 0
.text:0000001C 10 80 BD E8   LDMFD   SP!, {R4,PC}
```

0x18 0 R0 *return 0*.

.

Con optimización Keil 6/2013.

Con optimización Keil 6/2013 (Modo Thumb)

Listing 6.8: Con optimización Keil 6/2013 (Modo Thumb)

```
.text:00000000 main
```

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

```
.text:00000000 10 B5          PUSH    {R4,LR}
.text:00000002 03 23          MOVS    R3, #3
.text:00000004 02 22          MOVS    R2, #2
.text:00000006 01 21          MOVS    R1, #1
.text:00000008 02 A0          ADR     R0, ↙
    ↳ aADBDCD     ; "a=%d; b=%d; c=%d"
.text:0000000A 00 F0 0D F8  BL      ↙
    ↳ __2printf
.text:0000000E 00 20          MOVS    R0, #0
.text:00000010 10 BD          POP     {R4,PC}
```

Con optimización Keil 6/2013 (Modo ARM) + return

return 0:

```
#include <stdio.h>

void main()
{
        printf("a=%d; b=%d; c=%d", 1, 2, 3);
};
```

Listing 6.9: Con optimización Keil 6/2013 (Modo ARM)

```
.text:00000014 main
.text:00000014 03 30 A0 E3   MOV     R3, #3
.text:00000018 02 20 A0 E3   MOV     R2, #2
.text:0000001C 01 10 A0 E3   MOV     R1, #1
.text:00000020 1E 0E 8F E2   ADR     R0, ↙
    ↳ aADBDCD     ; "a=%d; b=%d; c=%d\n"
```

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

```
.text:00000024 CB 18 00 EA    B
    __2printf
```

... . .!.

: 13.1.1 on page 287.

ARM64

Sin optimización GCC (Linaro) 4.9

Listing 6.10: Sin optimización GCC (Linaro) 4.9

```
.LC1:
        .string "a=%d; b=%d; c=%d"
f2:
; :
        stp     x29, x30, [sp, -16]!
;  (FP=SP):
        add     x29, sp, 0
        adrp    x0, .LC1
        add     x0, x0, :lo12:.LC1
        mov     w1, 1
        mov     w2, 2
        mov     w3, 3
        bl      printf
        mov     w0, 0
;
        ldp     x29, x30, [sp], 16
        ret
```

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS
Con optimización GCC (Linaro) 4.9 .

6.2.2 ARM:

: 6.1.2 on page 95.

```
#include <stdio.h>

int main()
{
        printf("a=%d; b=%d; c=%d; d=%d; e=%d
    ; f=%d; g=%d; h=%d\n", 1, 2, 3, 4, 5, 
    6, 7, 8);
        return 0;
};
```

Con optimización Keil 6/2013: Modo ARM

```
.text:00000028             main
.text:00000028
.text:00000028             var_18 = -0x18
.text:00000028             var_14 = -0x14
.text:00000028             var_4  = -4
.text:00000028
.text:00000028 04 E0 2D E5  STR    LR, [SP,#
    var_4]!
.text:0000002C 14 D0 4D E2  SUB    SP, SP, 
    #0x14
.text:00000030 08 30 A0 E3  MOV    R3, #8
.text:00000034 07 20 A0 E3  MOV    R2, #7
.text:00000038 06 10 A0 E3  MOV    R1, #6
```

 CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

```
.text:0000003C 05 00 A0 E3  MOV     R0, #5
.text:00000040 04 C0 8D E2  ADD     R12, SP, #0x18+var_14
.text:00000044 0F 00 8C E8  STMIA   R12, {R0-R3}
.text:00000048 04 00 A0 E3  MOV     R0, #4
.text:0000004C 00 00 8D E5  STR     R0, [SP,#0x18+var_18]
.text:00000050 03 30 A0 E3  MOV     R3, #3
.text:00000054 02 20 A0 E3  MOV     R2, #2
.text:00000058 01 10 A0 E3  MOV     R1, #1
.text:0000005C 6E 0F 8F E2  ADR     R0, aADBDCDDDEDFDGD ; "a=%d; b=%d; c=%d; d=%d; e=%d; f=%d; g=%"...
.text:00000060 BC 18 00 EB  BL      __2printf
.text:00000064 14 D0 8D E2  ADD     SP, SP, #0x14
.text:00000068 04 F0 9D E4  LDR     PC, [SP+4+var_4],#4
```

:

- :

  `STR LR, [SP,#var_4]!` . *pre-index*. . : 28.2 on page 850.

  `SUB SP, SP, #0x14`

- :. `ADD R12, SP, #0x18+var_14` . *var_14*, -0x14 `STMIA R12, R0-R3` `STMIA` *Store Multiple Increment After*. *«Increment After»*

- : `STR R0, [SP,#0x18+var_18]` . *var_18* -0x18,

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

, .

- :

  , :

- .

- :

  ADD SP, SP, #0x14 .

  LDR PC, [SP+4+var_4],#4 $(4 + var\_4 = 4 + (-4) = 0$, LDR PC, [SP],#4), SP 4. *post-index*[2]. IDA? var_4 LR. POP PC en x86[3].

Con optimización Keil 6/2013: Modo Thumb

```
.text:0000001C                  printf_main2
.text:0000001C
.text:0000001C                  var_18 = -0x18
.text:0000001C                  var_14 = -0x14
.text:0000001C                  var_8  = -8
.text:0000001C
.text:0000001C 00 B5             PUSH    {LR}
.text:0000001E 08 23             MOVS    R3, #8
.text:00000020 85 B0             SUB     SP, SP, ↙
    ↘ #0x14
.text:00000022 04 93             STR     R3, [SP↙
    ↘ ,#0x18+var_8]
.text:00000024 07 22             MOVS    R2, #7
.text:00000026 06 21             MOVS    R1, #6
```

[2]: 28.2 on page 850.

[3].

 CAPÍTULO 6. PRINTF() *CON VARIOS ARGUMENTOS*

```
.text:00000028 05 20         MOVS    R0, #5
.text:0000002A 01 AB         ADD     R3, SP, ↙
    ↳ #0x18+var_14
.text:0000002C 07 C3         STMIA   R3!, {R0↙
    ↳ -R2}
.text:0000002E 04 20         MOVS    R0, #4
.text:00000030 00 90         STR     R0, [SP↙
    ↳ ,#0x18+var_18]
.text:00000032 03 23         MOVS    R3, #3
.text:00000034 02 22         MOVS    R2, #2
.text:00000036 01 21         MOVS    R1, #1
.text:00000038 A0 A0         ADR     R0, ↙
    ↳ aADBDCDDDEDFDGD ; "a=%d; b=%d; c=%d; d↙
    ↳ =%d; e=%d; f=%d; g=%"...
.text:0000003A 06 F0 D9 F8  BL      ↙
    ↳ __2printf
.text:0000003E
.text:0000003E              loc_3E   ; CODE ↙
    ↳ XREF: example13_f+16
.text:0000003E 05 B0         ADD     SP, SP, ↙
    ↳ #0x14
.text:00000040 00 BD         POP     {PC}
```

.

Con optimización Xcode 4.6.3 (LLVM): Modo ARM

```
__text:0000290C                  _printf_main2
__text:0000290C
__text:0000290C                  var_1C = -0x1C
__text:0000290C                  var_C  = -0xC
__text:0000290C
```

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

```
__text:0000290C 80 40 2D E9   STMFD  SP!, {⤦
    ↪ R7,LR}
__text:00002910 0D 70 A0 E1   MOV    R7, SP
__text:00002914 14 D0 4D E2   SUB    SP, SP,⤦
    ↪  #0x14
__text:00002918 70 05 01 E3   MOV    R0, #0⤦
    ↪ x1570
__text:0000291C 07 C0 A0 E3   MOV    R12, #7
__text:00002920 00 00 40 E3   MOVT   R0, #0
__text:00002924 04 20 A0 E3   MOV    R2, #4
__text:00002928 00 00 8F E0   ADD    R0, PC,⤦
    ↪  R0
__text:0000292C 06 30 A0 E3   MOV    R3, #6
__text:00002930 05 10 A0 E3   MOV    R1, #5
__text:00002934 00 20 8D E5   STR    R2, [SP⤦
    ↪ ,#0x1C+var_1C]
__text:00002938 0A 10 8D E9   STMFA  SP, {R1⤦
    ↪ ,R3,R12}
__text:0000293C 08 90 A0 E3   MOV    R9, #8
__text:00002940 01 10 A0 E3   MOV    R1, #1
__text:00002944 02 20 A0 E3   MOV    R2, #2
__text:00002948 03 30 A0 E3   MOV    R3, #3
__text:0000294C 10 90 8D E5   STR    R9, [SP⤦
    ↪ ,#0x1C+var_C]
__text:00002950 A4 05 00 EB   BL     _printf
__text:00002954 07 D0 A0 E1   MOV    SP, R7
__text:00002958 80 80 BD E8   LDMFD  SP!, {⤦
    ↪ R7,PC}
```

STMFA (Store Multiple Full Ascending)STMIB (Store Multiple Increment Before). .

 CAPÍTULO 6. PRINTF() *CON VARIOS ARGUMENTOS*

`. . . .` `MOVT R0, #0` y `ADD R0, PC, R0` . `MOVT R0, #0` y `MOV R2, #4` ..

Con optimización Xcode 4.6.3 (LLVM): Modo Thumb-2

```
__text:00002BA0             _printf_main2
__text:00002BA0
__text:00002BA0             var_1C = -0x1C
__text:00002BA0             var_18 = -0x18
__text:00002BA0             var_C  = -0xC
__text:00002BA0
__text:00002BA0 80 B5        PUSH     {R7, LR}
__text:00002BA2 6F 46        MOV      R7, SP
__text:00002BA4 85 B0        SUB      SP, SP, #0x14
__text:00002BA6 41 F2 D8 20  MOVW     R0, #0x12D8
__text:00002BAA 4F F0 07 0C  MOV.W    R12, #7
__text:00002BAE C0 F2 00 00  MOVT.W   R0, #0
__text:00002BB2 04 22        MOVS     R2, #4
__text:00002BB4 78 44        ADD      R0, PC  ; char *
__text:00002BB6 06 23        MOVS     R3, #6
__text:00002BB8 05 21        MOVS     R1, #5
```

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

```
__text:00002BBA 0D F1 04 0E     ADD.W    LR, ↙
    ↳ SP, #0x1C+var_18
__text:00002BBE 00 92           STR      R2, ↙
    ↳ [SP,#0x1C+var_1C]
__text:00002BC0 4F F0 08 09     MOV.W    R9, ↙
    ↳ #8
__text:00002BC4 8E E8 0A 10     STMIA.W  LR, ↙
    ↳ {R1,R3,R12}
__text:00002BC8 01 21           MOVS     R1, ↙
    ↳ #1
__text:00002BCA 02 22           MOVS     R2, ↙
    ↳ #2
__text:00002BCC 03 23           MOVS     R3, ↙
    ↳ #3
__text:00002BCE CD F8 10 90     STR.W    R9, ↙
    ↳ [SP,#0x1C+var_C]
__text:00002BD2 01 F0 0A EA     BLX      ↙
    ↳ _printf
__text:00002BD6 05 B0           ADD      SP, ↙
    ↳ SP, #0x14
__text:00002BD8 80 BD           POP      {R7,↙
    ↳ PC}
```

.

ARM64

Sin optimización GCC (Linaro) 4.9

Listing 6.11: Sin optimización GCC (Linaro) 4.9

 CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

```
.LC2:
        .string "a=%d; b=%d; c=%d; d=%d; e=%↙
    ↘ d; f=%d; g=%d; h=%d\n"
f3:
; :
        sub     sp, sp, #32
; :
        stp     x29, x30, [sp,16]
;  (FP=SP):
        add     x29, sp, 16
        adrp    x0, .LC2 ; "a=%d; b=%d; c=%d↙
    ↘ ; d=%d; e=%d; f=%d; g=%d; h=%d\n"
        add     x0, x0, :lo12:.LC2
        mov     w1, 8           ; 9th ↙
    ↘ argument
        str     w1, [sp]        ; store 9th ↙
    ↘ argument in the stack
        mov     w1, 1
        mov     w2, 2
        mov     w3, 3
        mov     w4, 4
        mov     w5, 5
        mov     w6, 6
        mov     w7, 7
        bl      printf
        sub     sp, x29, #16
;
        ldp     x29, x30, [sp,16]
        add     sp, sp, 32
        ret
```

: [ARM13c]. X0.

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS
Con optimización GCC (Linaro) 4.9 .

6.3 MIPS

6.3.1 3

Con optimización GCC 4.4.5

\$5…\$7 (o \$A0…\$A2).

Listing 6.12: Con optimización GCC 4.4.5 (salida de assembly)

```
$LC0:
        .ascii  "a=%d; b=%d; c=%d\000"
main:
; :
        lui     $28,%hi(__gnu_local_gp)
        addiu   $sp,$sp,-32
        addiu   $28,$28,%lo(__gnu_local_gp)
        sw      $31,28($sp)
;  printf():
        lw      $25,%call16(printf)($28)
;  printf():
        lui     $4,%hi($LC0)
        addiu   $4,$4,%lo($LC0)
;  printf():
        li      $5,1                     # 0
    ↳ x1
;  printf():
        li      $6,2                     # 0
    ↳ x2
```

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

```
;  printf():
        jalr    $25
;  printf() (branch delay slot):
        li      $7,3                    # 0
    ↳ x3

; :
        lw      $31,28($sp)
;  0:
        move    $2,$0
;
        j       $31
        addiu   $sp,$sp,32 ; branch delay 
    ↳ slot
```

Listing 6.13: Con optimización GCC 4.4.5 (IDA)

```

.text:00000000 main:
.text:00000000
.text:00000000 var_10          = -0x10
.text:00000000 var_4           = -4
.text:00000000
; :
.text:00000000                 lui      $gp, 
    ↳ (__gnu_local_gp >> 16)
.text:00000004                 addiu    $sp, 
    ↳ -0x20
.text:00000008                 la       $gp, 
    ↳ (__gnu_local_gp & 0xFFFF)
.text:0000000C                 sw       $ra, 
    ↳ 0x20+var_4($sp)
.text:00000010                 sw       $gp, 
```

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

```
    ↪ 0x20+var_10($sp)
; printf():
.text:00000014                     lw      $t9, ⤦
    ↪ (printf & 0xFFFF)($gp)
; printf():
.text:00000018                     la      $a0, ⤦
    ↪ $LC0        # "a=%d; b=%d; c=%d"
; printf():
.text:00000020                     li      $a1, ⤦
    ↪ 1
; printf():
.text:00000024                     li      $a2, ⤦
    ↪ 2
; printf():
.text:00000028                     jalr    $t9
; printf() (branch delay slot):
.text:0000002C                     li      $a3, ⤦
    ↪ 3
; :
.text:00000030                     lw      $ra, ⤦
    ↪ 0x20+var_4($sp)
; 0:
.text:00000034                     move    $v0, ⤦
    ↪ $zero
;
.text:00000038                     jr      $ra
.text:0000003C                     addiu   $sp, ⤦
    ↪ 0x20 ; branch delay slot
```

Sin optimización GCC 4.4.5

Sin optimización GCC :

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

Listing 6.14: Sin optimización GCC 4.4.5 (salida de assembly)

```
$LC0:
        .ascii  "a=%d; b=%d; c=%d\000"
main:
; :
        addiu   $sp,$sp,-32
        sw      $31,28($sp)
        sw      $fp,24($sp)
        move    $fp,$sp
        lui     $28,%hi(__gnu_local_gp)
        addiu   $28,$28,%lo(__gnu_local_gp)
; :
        lui     $2,%hi($LC0)
        addiu   $2,$2,%lo($LC0)
; printf():
        move    $4,$2
; printf():
        li      $5,1                    # 0↙
    ↘ x1
; printf():
        li      $6,2                    # 0↙
    ↘ x2
; printf():
        li      $7,3                    # 0↙
    ↘ x3
; printf():
        lw      $2,%call16(printf)($28)
        nop
; printf():
        move    $25,$2
```

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

```
        jalr    $25
        nop

; :
        lw      $28,16($fp)
;  0:
        move    $2,$0
        move    $sp,$fp
        lw      $31,28($sp)
        lw      $fp,24($sp)
        addiu   $sp,$sp,32
;
        j       $31
        nop
```

Listing 6.15: Sin optimización GCC 4.4.5 (IDA)

```

.text:00000000 main:
.text:00000000
.text:00000000 var_10          = -0x10
.text:00000000 var_8           = -8
.text:00000000 var_4           = -4
.text:00000000
; :
.text:00000000                 addiu   $sp, ⤦
    ↪ -0x20
.text:00000004                 sw      $ra, ⤦
    ↪ 0x20+var_4($sp)
.text:00000008                 sw      $fp, ⤦
    ↪ 0x20+var_8($sp)
.text:0000000C                 move    $fp, ⤦
    ↪ $sp
```

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

```
.text:00000010                 la      $gp, ⤦
    ↳ __gnu_local_gp
.text:00000018                 sw      $gp, ⤦
    ↳ 0x20+var_10($sp)
; :
.text:0000001C                 la      $v0, ⤦
    ↳ aADBDCD      # "a=%d; b=%d; c=%d"
; printf():
.text:00000024                 move    $a0, ⤦
    ↳ $v0
; printf():
.text:00000028                 li      $a1, ⤦
    ↳ 1
; printf():
.text:0000002C                 li      $a2, ⤦
    ↳ 2
; printf():
.text:00000030                 li      $a3, ⤦
    ↳ 3
; printf():
.text:00000034                 lw      $v0, ⤦
    ↳ (printf & 0xFFFF)($gp)
.text:00000038                 or      $at, ⤦
    ↳ $zero
; printf():
.text:0000003C                 move    $t9, ⤦
    ↳ $v0
.text:00000040                 jalr    $t9
.text:00000044                 or      $at, ⤦
    ↳ $zero ; NOP
; :
.text:00000048                 lw      $gp, ⤦
    ↳ 0x20+var_10($fp)
```

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

```
;  0:
.text:0000004C                          move    $v0, ↙
    ↘ $zero
.text:00000050                          move    $sp, ↙
    ↘ $fp
.text:00000054                          lw      $ra, ↙
    ↘ 0x20+var_4($sp)
.text:00000058                          lw      $fp, ↙
    ↘ 0x20+var_8($sp)
.text:0000005C                          addiu   $sp, ↙
    ↘ 0x20
;
.text:00000060                          jr      $ra
.text:00000064                          or      $at, ↙
    ↘ $zero ; NOP
```

6.3.2 8

: 6.1.2 on page 95.

```
#include <stdio.h>

int main()
{
        printf("a=%d; b=%d; c=%d; d=%d; e=%d↙
    ↘ ; f=%d; g=%d; h=%d\n", 1, 2, 3, 4, 5, ↙
    ↘ 6, 7, 8);
        return 0;
};
```

Con optimización GCC 4.4.5

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

Listing 6.16: Con optimización GCC 4.4.5 (salida de assembly)

```
$LC0:
        .ascii  "a=%d; b=%d; c=%d; d=%d; e=%
    ↳ d; f=%d; g=%d; h=%d\012\000"
main:
; :
        lui     $28,%hi(__gnu_local_gp)
        addiu   $sp,$sp,-56
        addiu   $28,$28,%lo(__gnu_local_gp)
        sw      $31,52($sp)
; :
        li      $2,4                    # 0
    ↳ x4
        sw      $2,16($sp)
; :
        li      $2,5                    # 0
    ↳ x5
        sw      $2,20($sp)
; :
        li      $2,6                    # 0
    ↳ x6
        sw      $2,24($sp)
; :
        li      $2,7                    # 0
    ↳ x7
        lw      $25,%call16(printf)($28)
        sw      $2,28($sp)
;  $a0:
        lui     $4,%hi($LC0)
; :
```

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

```
        li      $2,8                    # 0
    ↳ x8
        sw      $2,32($sp)
        addiu   $4,$4,%lo($LC0)
;  $a1:
        li      $5,1                    # 0
    ↳ x1
;  $a2:
        li      $6,2                    # 0
    ↳ x2
;  printf():
        jalr    $25
;  $a3 (branch delay slot):
        li      $7,3                    # 0
    ↳ x3

; :
        lw      $31,52($sp)
;  0:
        move    $2,$0
;
        j       $31
        addiu   $sp,$sp,56 ; branch delay
    ↳ slot
```

Listing 6.17: Con optimización GCC 4.4.5 (IDA)

```
.text:00000000 main:
.text:00000000
.text:00000000 var_28          = -0x28
.text:00000000 var_24          = -0x24
.text:00000000 var_20          = -0x20
```

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

```
.text:00000000 var_1C          = -0x1C
.text:00000000 var_18          = -0x18
.text:00000000 var_10          = -0x10
.text:00000000 var_4           = -4
.text:00000000
; :
.text:00000000                 lui     $gp, ⤦
    ↳ (__gnu_local_gp >> 16)
.text:00000004                 addiu   $sp, ⤦
    ↳ -0x38
.text:00000008                 la      $gp, ⤦
    ↳ (__gnu_local_gp & 0xFFFF)
.text:0000000C                 sw      $ra, ⤦
    ↳ 0x38+var_4($sp)
.text:00000010                 sw      $gp, ⤦
    ↳ 0x38+var_10($sp)
; :
.text:00000014                 li      $v0, ⤦
    ↳ 4
.text:00000018                 sw      $v0, ⤦
    ↳ 0x38+var_28($sp)
; :
.text:0000001C                 li      $v0, ⤦
    ↳ 5
.text:00000020                 sw      $v0, ⤦
    ↳ 0x38+var_24($sp)
; :
.text:00000024                 li      $v0, ⤦
    ↳ 6
.text:00000028                 sw      $v0, ⤦
    ↳ 0x38+var_20($sp)
; :
.text:0000002C                 li      $v0, ⤦
```

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

```
    ↳ 7
.text:00000030                  lw      $t9, ↙
    ↳ (printf & 0xFFFF)($gp)
.text:00000034                  sw      $v0, ↙
    ↳ 0x38+var_1C($sp)
;  $a0:
.text:00000038                  lui     $a0, ↙
    ↳ ($LC0 >> 16)  # "a=%d; b=%d; c=%d; d=%↙
    ↳ d; e=%d; f=%d; g=%"...
; :
.text:0000003C                  li      $v0, ↙
    ↳ 8
.text:00000040                  sw      $v0, ↙
    ↳ 0x38+var_18($sp)
;  $a1:
.text:00000044                  la      $a0, ↙
    ↳ ($LC0 & 0xFFFF)  # "a=%d; b=%d; c=%d; ↙
    ↳ d=%d; e=%d; f=%d; g=%"...
;  $a1:
.text:00000048                  li      $a1, ↙
    ↳ 1
;  $a2:
.text:0000004C                  li      $a2, ↙
    ↳ 2
;  printf():
.text:00000050                  jalr    $t9
;  $a3 (branch delay slot):
.text:00000054                  li      $a3, ↙
    ↳ 3
; :
.text:00000058                  lw      $ra, ↙
    ↳ 0x38+var_4($sp)
;  0:
```

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

```
.text:0000005C                 move    $v0, ⤦
    ↳ $zero
;
.text:00000060                 jr      $ra
.text:00000064                 addiu   $sp, ⤦
    ↳ 0x38 ; branch delay slot
```

Sin optimización GCC 4.4.5

Sin optimización GCC :

Listing 6.18: Sin optimización GCC 4.4.5 (salida de assembly)

```

$LC0:
        .ascii  "a=%d; b=%d; c=%d; d=%d; e=%⤦
    ↳ d; f=%d; g=%d; h=%d\012\000"
main:
; :
        addiu   $sp,$sp,-56
        sw      $31,52($sp)
        sw      $fp,48($sp)
        move    $fp,$sp
        lui     $28,%hi(__gnu_local_gp)
        addiu   $28,$28,%lo(__gnu_local_gp)
        lui     $2,%hi($LC0)
        addiu   $2,$2,%lo($LC0)
; :
        li      $3,4                    # 0⤦
    ↳ x4
        sw      $3,16($sp)
```

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

```
; :
        li      $3,5                    # 0↙
    ↘ x5
        sw      $3,20($sp)
; :
        li      $3,6                    # 0↙
    ↘ x6
        sw      $3,24($sp)
; :
        li      $3,7                    # 0↙
    ↘ x7
        sw      $3,28($sp)
; :
        li      $3,8                    # 0↙
    ↘ x8
        sw      $3,32($sp)
;  $a0:
        move    $4,$2
;  $a1:
        li      $5,1                    # 0↙
    ↘ x1
;  $a2:
        li      $6,2                    # 0↙
    ↘ x2
;  $a3:
        li      $7,3                    # 0↙
    ↘ x3
;  printf():
        lw      $2,%call16(printf)($28)
        nop
        move    $25,$2
        jalr    $25
        nop
```

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

```
; :
        lw      $28,40($fp)
;  0:
        move    $2,$0
        move    $sp,$fp
        lw      $31,52($sp)
        lw      $fp,48($sp)
        addiu   $sp,$sp,56
;
        j       $31
        nop
```

Listing 6.19: Sin optimización GCC 4.4.5 (IDA)

```
.text:00000000 main:
.text:00000000
.text:00000000 var_28          = -0x28
.text:00000000 var_24          = -0x24
.text:00000000 var_20          = -0x20
.text:00000000 var_1C          = -0x1C
.text:00000000 var_18          = -0x18
.text:00000000 var_10          = -0x10
.text:00000000 var_8           = -8
.text:00000000 var_4           = -4
.text:00000000
; :
.text:00000000                 addiu   $sp, ↙
    ↳ -0x38
.text:00000004                 sw      $ra, ↙
    ↳ 0x38+var_4($sp)
.text:00000008                 sw      $fp, ↙
    ↳ 0x38+var_8($sp)
```

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

```
.text:0000000C                 move    $fp, ↙
    ↘ $sp
.text:00000010                 la      $gp, ↙
    ↘ __gnu_local_gp
.text:00000018                 sw      $gp, ↙
    ↘ 0x38+var_10($sp)
.text:0000001C                 la      $v0, ↙
    ↘ aADBDCDDDEDFDGD  # "a=%d; b=%d; c=%d; ↙
    ↘ d=%d; e=%d; f=%d; g=%"...
; :
.text:00000024                 li      $v1, ↙
    ↘ 4
.text:00000028                 sw      $v1, ↙
    ↘ 0x38+var_28($sp)
; :
.text:0000002C                 li      $v1, ↙
    ↘ 5
.text:00000030                 sw      $v1, ↙
    ↘ 0x38+var_24($sp)
; :
.text:00000034                 li      $v1, ↙
    ↘ 6
.text:00000038                 sw      $v1, ↙
    ↘ 0x38+var_20($sp)
; :
.text:0000003C                 li      $v1, ↙
    ↘ 7
.text:00000040                 sw      $v1, ↙
    ↘ 0x38+var_1C($sp)
; :
.text:00000044                 li      $v1, ↙
    ↘ 8
.text:00000048                 sw      $v1, ↙
```

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

```
    ↳ 0x38+var_18($sp)
; $a0:
.text:0000004C                     move    $a0, ↙
    ↳ $v0
; $a1:
.text:00000050                     li      $a1, ↙
    ↳ 1
; $a2:
.text:00000054                     li      $a2, ↙
    ↳ 2
; $a3:
.text:00000058                     li      $a3, ↙
    ↳ 3
; printf():
.text:0000005C                     lw      $v0, ↙
    ↳ (printf & 0xFFFF)($gp)
.text:00000060                     or      $at, ↙
    ↳ $zero
.text:00000064                     move    $t9, ↙
    ↳ $v0
.text:00000068                     jalr    $t9
.text:0000006C                     or      $at, ↙
    ↳ $zero ; NOP
; :
.text:00000070                     lw      $gp, ↙
    ↳ 0x38+var_10($fp)
; 0:
.text:00000074                     move    $v0, ↙
    ↳ $zero
.text:00000078                     move    $sp, ↙
    ↳ $fp
.text:0000007C                     lw      $ra, ↙
    ↳ 0x38+var_4($sp)
```

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

```
.text:00000080                    lw      $fp, ⤦
    ↳ 0x38+var_8($sp)
.text:00000084                    addiu   $sp, ⤦
    ↳ 0x38
;
.text:00000088                    jr      $ra
.text:0000008C                    or      $at, ⤦
    ↳ $zero ; NOP
```

6.4 Conclusión

:

Listing 6.20: x86

```


...
PUSH
PUSH
PUSH
CALL
;
```

Listing 6.21: x64 (MSVC)

```


MOV RCX,
MOV RDX,
MOV R8,
MOV R9,
...
```

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

```
PUSH
CALL
;
```

Listing 6.22: x64 (GCC)

```
MOV RDI,
MOV RSI,
MOV RDX,
MOV RCX,
MOV R8,
MOV R9,
...
PUSH
CALL
;
```

Listing 6.23: ARM

```
MOV R0,
MOV R1,
MOV R2,
MOV R3,
;
BL
;
```

Listing 6.24: ARM64

```
MOV X0,
```

CAPÍTULO 6. PRINTF() CON VARIOS ARGUMENTOS

```
MOV X1,
MOV X2,
MOV X3,
MOV X4,
MOV X5,
MOV X6,
MOV X7,
;
BL CALL
;
```

Listing 6.25: MIPS ()

```
LI $4,  ; AKA $A0
LI $5,  ; AKA $A1
LI $6,  ; AKA $A2
LI $7,  ; AKA $A3
;
LW temp_reg,
JALR temp_reg
```

6.5

CPU .

Capítulo 7

scanf()

7.1

```
#include <stdio.h>

int main()
{
        int x;
        printf ("Enter X:\n");

        scanf ("%d", &x);

        printf ("You entered %d...\n", x);

        return 0;
```

```
};
```

7.1.1

C/C++

; memcpy(),, void*,

((10 on page 211)). *scanf()*.

C/C++

7.1.2 x86

MSVC

```
CONST    SEGMENT
$SG3831    DB    'Enter X:', 0aH, 00H
$SG3832    DB    '%d', 00H
$SG3833    DB    'You entered %d...', 0aH, ⤦
    ↳ 00H
CONST    ENDS
PUBLIC    _main
EXTRN    _scanf:PROC
EXTRN    _printf:PROC
; Function compile flags: /Odtp
_TEXT    SEGMENT
_x$ = -4                       ; size = 4
_main    PROC
    push   ebp
    mov    ebp, esp
```

```
    push   ecx
    push   OFFSET $SG3831 ; 'Enter X:'
    call   _printf
    add    esp, 4
    lea    eax, DWORD PTR _x$[ebp]
    push   eax
    push   OFFSET $SG3832 ; '%d'
    call   _scanf
    add    esp, 8
    mov    ecx, DWORD PTR _x$[ebp]
    push   ecx
    push   OFFSET $SG3833 ; 'You entered %d⤦
    ↪ ...'
    call   _printf
    add    esp, 8

    ;  0
    xor    eax, eax
    mov    esp, ebp
    pop    ebp
    ret    0
_main    ENDP
_TEXT    ENDS
```

:

…	…
EBP-8	#2, Marcado en IDA como `var_8`
EBP-4	#1, Marcado en IDA como `var_4`
EBP	EBP
EBP+4	
EBP+8	argumento#1, Marcado en IDA como `arg_0`
EBP+0xC	argumento#2, Marcado en IDA como `arg_4`
EBP+0x10	argumento#3, Marcado en IDA como `arg_8`
…	…

(A.6.2 on page 1880).

`lea eax, [ebp-4]`.

`You entered %d...\n`.

7.1.3 MSVC + OllyDbg

OllyDbg. `ntdll.dll`. .

:

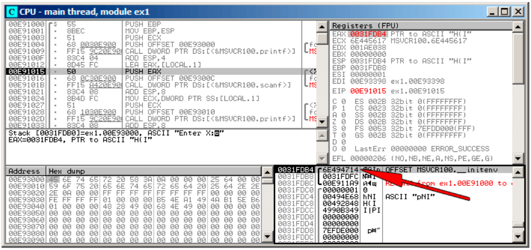


Figura 7.1: OllyDbg:

«Follow in stack». . (0x6E494714). . . :

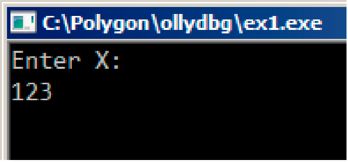


Figura 7.2:

scanf():

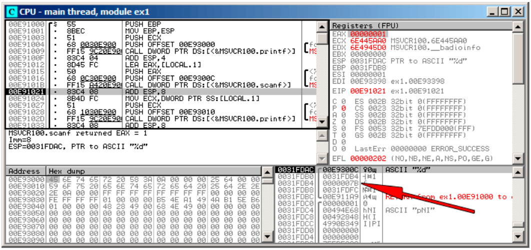


Figura 7.3: OllyDbg: scanf()

scanf() 1 en EAX, . 0x7B (123).

:

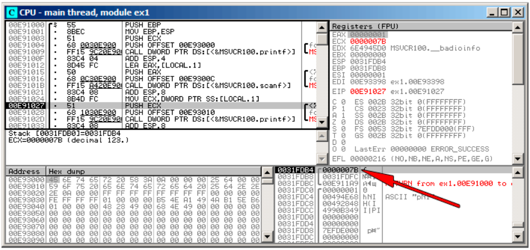


Figura 7.4: OllyDbg: `printf()`

GCC

```
main                proc near

var_20              = dword ptr -20h
var_1C              = dword ptr -1Ch
var_4               = dword ptr -4

                    push     ebp
                    mov      ebp, esp
                    and      esp, 0FFFFFFF0h
                    sub      esp, 20h
                    mov      [esp+20h+var_20], ↙
    ↳ offset aEnterX ; "Enter X:"
                    call     _puts
                    mov      eax, offset aD  ; "%↙
    ↳ d"
                    lea      edx, [esp+20h+var_4]
```

```
                mov     [esp+20h+var_1C], ↙
    ↘ edx
                mov     [esp+20h+var_20], ↙
    ↘ eax
                call    ___isoc99_scanf
                mov     edx, [esp+20h+var_4]
                mov     eax, offset ↙
    ↘ aYouEnteredD___ ; "You entered %d...\n↙
    ↘ "
                mov     [esp+20h+var_1C], ↙
    ↘ edx
                mov     [esp+20h+var_20], ↙
    ↘ eax
                call    _printf
                mov     eax, 0
                leave
                retn
main            endp
```

7.1.4 x64

.

MSVC

Listing 7.1: MSVC 2012 x64

```

_DATA   SEGMENT
$SG1289 DB      'Enter X:', 0aH, 00H
```

```
$SG1291 DB      '%d', 00H
$SG1292 DB      'You entered %d...', 0aH, 00
    ↳ H
_DATA   ENDS

_TEXT   SEGMENT
x$ = 32
main    PROC
$LN3:
        sub     rsp, 56
        lea     rcx, OFFSET FLAT:$SG1289 ; '
    ↳ Enter X:'
        call    printf
        lea     rdx, QWORD PTR x$[rsp]
        lea     rcx, OFFSET FLAT:$SG1291 ; 
    ↳ '%d'
        call    scanf
        mov     edx, DWORD PTR x$[rsp]
        lea     rcx, OFFSET FLAT:$SG1292 ; '
    ↳ You entered %d...'
        call    printf

        ;  0
        xor     eax, eax
        add     rsp, 56
        ret     0
main    ENDP
_TEXT   ENDS
```

GCC

Listing 7.2: Con optimización GCC 4.4.6 x64

```
.LC0:
        .string "Enter X:"
.LC1:
        .string "%d"
.LC2:
        .string "You entered %d...\n"

main:
        sub     rsp, 24
        mov     edi, OFFSET FLAT:.LC0 ; "
    Enter X:"
        call    puts
        lea     rsi, [rsp+12]
        mov     edi, OFFSET FLAT:.LC1 ; "%d"
        xor     eax, eax
        call    __isoc99_scanf
        mov     esi, DWORD PTR [rsp+12]
        mov     edi, OFFSET FLAT:.LC2 ; "You
     entered %d...\n"
        xor     eax, eax
        call    printf

        ;  0
        xor     eax, eax
        add     rsp, 24
        ret
```

7.1.5 ARM

Con optimización Keil 6/2013 (Modo Thumb)

```
.text:00000042                         scanf_main
.text:00000042
.text:00000042                         var_8                =↙
    ↘ -8
.text:00000042
.text:00000042 08 B5                                        ↙
    ↘ PUSH    {R3,LR}
.text:00000044 A9 A0                                        ↙
    ↘ ADR     R0, aEnterX      ; "Enter X:\n"
.text:00000046 06 F0 D3 F8                                  ↙
    ↘ BL      __2printf
.text:0000004A 69 46                                        ↙
    ↘ MOV     R1, SP
.text:0000004C AA A0                                        ↙
    ↘ ADR     R0, aD           ; "%d"
.text:0000004E 06 F0 CD F8                                  ↙
    ↘ BL      __0scanf
.text:00000052 00 99                                        ↙
    ↘ LDR     R1, [SP,#8+var_8]
.text:00000054 A9 A0                                        ↙
    ↘ ADR     R0, aYouEnteredD___ ; "You ↙
    ↘ entered %d...\n"
.text:00000056 06 F0 CB F8                                  ↙
    ↘ BL      __2printf
.text:0000005A 00 20                                        ↙
    ↘ MOVS    R0, #0
.text:0000005C 08 BD                                        ↙
    ↘ POP     {R3,PC}
```

int

ARM64

Listing 7.3: Sin optimización GCC 4.9.1 ARM64

```

.LC0:
        .string "Enter X:"
.LC1:
        .string "%d"
.LC2:
        .string "You entered %d...\n"
scanf_main:
; :
        stp     x29, x30, [sp, -32]!
;  (FP=SP)
        add     x29, sp, 0
;
        adrp    x0, .LC0
        add     x0, x0, :lo12:.LC0
; X0=
; :
        bl      puts
; :
        adrp    x0, .LC1
        add     x0, x0, :lo12:.LC1
;  (X1=FP+28):
        add     x1, x29, 28
; X1=
; :
        bl      __isoc99_scanf
; :
```

```
        ldr     w1, [x29,28]
; W1=x
;
; printf()
        adrp    x0, .LC2
        add     x0, x0, :lo12:.LC2
        bl      printf
;  0
        mov     w0, 0
; :
        ldp     x29, x30, [sp], 32
        ret
```

.

7.1.6 MIPS

Listing 7.4: Con optimización GCC 4.4.5 (salida de assembly)

```
$LC0:
        .ascii  "Enter X:\000"
$LC1:
        .ascii  "%d\000"
$LC2:
        .ascii  "You entered %d...\012\000"
main:
; :
        lui     $28,%hi(__gnu_local_gp)
        addiu   $sp,$sp,-40
        addiu   $28,$28,%lo(__gnu_local_gp)
        sw      $31,36($sp)
```

```
; puts():
        lw      $25,%call16(puts)($28)
        lui     $4,%hi($LC0)
        jalr    $25
        addiu   $4,$4,%lo($LC0) ; branch ↙
    ↘ delay slot
; scanf():
        lw      $28,16($sp)
        lui     $4,%hi($LC1)
        lw      $25,%call16(__isoc99_scanf)(↙
    ↘ $28)
; scanf(), $a1=$sp+24:
        addiu   $5,$sp,24
        jalr    $25
        addiu   $4,$4,%lo($LC1) ; branch ↙
    ↘ delay slot

; printf():
        lw      $28,16($sp)
; printf(),
; $sp+24:
        lw      $5,24($sp)
        lw      $25,%call16(printf)($28)
        lui     $4,%hi($LC2)
        jalr    $25
        addiu   $4,$4,%lo($LC2) ; branch ↙
    ↘ delay slot

; :
        lw      $31,36($sp)
; 0:
        move    $2,$0
; :
```

```
        j       $31
        addiu   $sp,$sp,40        ; branch ↙
    ↘ delay slot
```

Listing 7.5: Con optimización GCC 4.4.5 (IDA)

```
.text:00000000 main:
.text:00000000
.text:00000000 var_18          = -0x18
.text:00000000 var_10          = -0x10
.text:00000000 var_4           = -4
.text:00000000
; :
.text:00000000                 lui     $gp, ↙
    ↘ (__gnu_local_gp >> 16)
.text:00000004                 addiu   $sp, ↙
    ↘ -0x28
.text:00000008                 la      $gp, ↙
    ↘ (__gnu_local_gp & 0xFFFF)
.text:0000000C                 sw      $ra, ↙
    ↘ 0x28+var_4($sp)
.text:00000010                 sw      $gp, ↙
    ↘ 0x28+var_18($sp)
;  puts():
.text:00000014                 lw      $t9, ↙
    ↘ (puts & 0xFFFF)($gp)
.text:00000018                 lui     $a0, ↙
    ↘ ($LC0 >> 16)  # "Enter X:"
.text:0000001C                 jalr    $t9
.text:00000020                 la      $a0, ↙
    ↘ ($LC0 & 0xFFFF)  # "Enter X:" ; branch↙
    ↘  delay slot
```

```
; scanf():
.text:00000024                 lw      $gp, ↙
    ↘ 0x28+var_18($sp)
.text:00000028                 lui     $a0, ↙
    ↘ ($LC1 >> 16)  # "%d"
.text:0000002C                 lw      $t9, ↙
    ↘ (__isoc99_scanf & 0xFFFF)($gp)
; scanf(), $a1=$sp+24:
.text:00000030                 addiu   $a1, ↙
    ↘ $sp, 0x28+var_10
.text:00000034                 jalr    $t9  ↙
    ↘ ; branch delay slot
.text:00000038                 la      $a0, ↙
    ↘ ($LC1 & 0xFFFF)  # "%d"
; printf():
.text:0000003C                 lw      $gp, ↙
    ↘ 0x28+var_18($sp)
; printf(),
; $sp+24:
.text:00000040                 lw      $a1, ↙
    ↘ 0x28+var_10($sp)
.text:00000044                 lw      $t9, ↙
    ↘ (printf & 0xFFFF)($gp)
.text:00000048                 lui     $a0, ↙
    ↘ ($LC2 >> 16)  # "You entered %d...\n"
.text:0000004C                 jalr    $t9
.text:00000050                 la      $a0, ↙
    ↘ ($LC2 & 0xFFFF)  # "You entered %d...\↙
    ↘ n" ; branch delay slot
; :
.text:00000054                 lw      $ra, ↙
    ↘ 0x28+var_4($sp)
; 0:
```

```
.text:00000058                          move    $v0, ↙
    ↘ $zero
; :
.text:0000005C                          jr      $ra
.text:00000060                          addiu   $sp, ↙
    ↘ 0x28 ; branch delay slot
```

7.2

```

#include <stdio.h>

//
int x;

int main()
{
        printf ("Enter X:\n");

        scanf ("%d", &x);

        printf ("You entered %d...\n", x);

        return 0;
};
```

7.2.1 MSVC: x86

```
_DATA    SEGMENT
```

```
COMM    _x:DWORD
$SG2456    DB    'Enter X:', 0aH, 00H
$SG2457    DB    '%d', 00H
$SG2458    DB    'You entered %d...', 0aH, ↙
    ↘ 00H
_DATA    ENDS
PUBLIC    _main
EXTRN    _scanf:PROC
EXTRN    _printf:PROC
; Function compile flags: /Odtp
_TEXT    SEGMENT
_main    PROC
    push   ebp
    mov    ebp, esp
    push   OFFSET $SG2456
    call   _printf
    add    esp, 4
    push   OFFSET _x
    push   OFFSET $SG2457
    call   _scanf
    add    esp, 8
    mov    eax, DWORD PTR _x
    push   eax
    push   OFFSET $SG2458
    call   _printf
    add    esp, 8
    xor    eax, eax
    pop    ebp
    ret    0
_main    ENDP
_TEXT    ENDS
```

```
int x=10; //
```

```
_DATA   SEGMENT
_x      DD      0aH

...
```

```
.data:0040FA80 _x                    dd ?                   ↙
    ↘              ; DATA XREF: _main+10
.data:0040FA80                                              ↙
    ↘              ; _main+22
.data:0040FA84 dword_40FA84    dd ?                         ↙
    ↘              ; DATA XREF: _memset+1E
.data:0040FA84                                              ↙
    ↘              ; unknown_libname_1+28
.data:0040FA88 dword_40FA88    dd ?                         ↙
    ↘              ; DATA XREF: ↙
    ↘ ___sbh_find_block+5
.data:0040FA88                                              ↙
    ↘              ; ___sbh_free_block+2BC
.data:0040FA8C ; LPVOID lpMem
.data:0040FA8C lpMem           dd ?                         ↙
    ↘              ; DATA XREF: ↙
    ↘ ___sbh_find_block+B
.data:0040FA8C                                              ↙
    ↘              ; ___sbh_free_block+2CA
.data:0040FA90 dword_40FA90    dd ?                         ↙
    ↘              ; DATA XREF: _V6_HeapAlloc↙
    ↘ +13
.data:0040FA90                                              ↙
    ↘              ; __calloc_impl+72
```

```
.data:0040FA94 dword_40FA94    dd ?                      ↙
    ↘              ; DATA XREF: ↙
    ↘ ___sbh_free_block+2FE
```

7.2.2 MSVC: x86 + OllyDbg

:

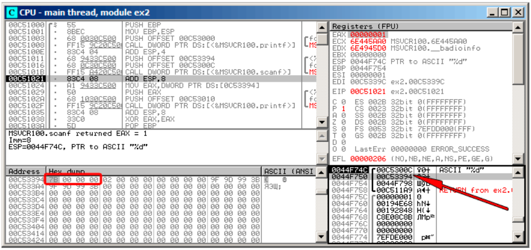


Figura 7.5: OllyDbg:

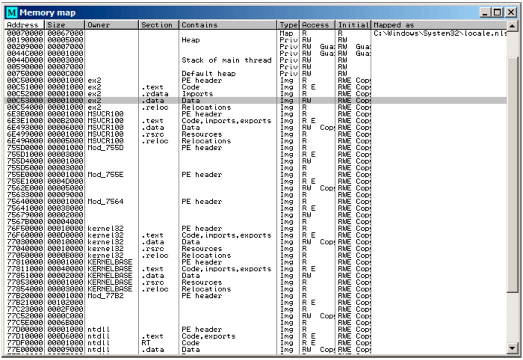
0x7B.

7B? 00 00 00 7B. endianness, *little-endian*. . : 31 on page 869.

printf().

0x00C53394.

:



Memory map

Address	Size	Owner	Section	Contains	Type	Access	Initial	Mapped as
00070000	00067000				Map	R	R	C:\Windows\System32\locale.nls
00190000	00005000			Heap	Priv	RW	RW	
00209000	00007000				Priv	RW Gua	RW Gua	
0044C000	00001000				Priv	RW Gua	RW Gua	
0044D000	00003000			Stack of main thread	Priv	RW	RW	
00590000	00007000				Priv	RW	RW	
00750000	0000C000			Default heap	Priv	RW	RW	
00C50000	00001000	ex2		PE header	Img	R	RWE Cop	
00C51000	00001000	ex2	.text	Code	Img	R E	RWE Cop	
00C52000	00001000	ex2	.rdata	Imports	Img	R	RWE Cop	
00C53000	00001000	ex2	.data	Data	Img	RW	RWE Cop	
00C54000	00001000	ex2	.reloc	Relocations	Img	R	RWE Cop	
6E3E0000	00001000	MSVCR100		PE header	Img	R	RWE Cop	
6E3E1000	000B2000	MSVCR100	.text	Code,imports,exports	Img	R E	RWE Cop	
6E493000	00006000	MSVCR100	.data	Data	Img	RW Cop	RWE Cop	
6E499000	00001000	MSVCR100	.rsrc	Resources	Img	R	RWE Cop	
6E49A000	00005000	MSVCR100	.reloc	Relocations	Img	R	RWE Cop	
755D0000	00001000	Mod_755D		PE header	Img	R	RWE Cop	
755D1000	00003000				Img	R E	RWE Cop	
755D4000	00001000				Img	RW	RWE Cop	
755D5000	00003000				Img	R	RWE Cop	
755E0000	00001000	Mod_755E		PE header	Img	R	RWE Cop	
755E1000	0004D000				Img	R E	RWE Cop	
7562E000	00005000				Img	RW Cop	RWE Cop	
75633000	00009000				Img	R	RWE Cop	
75640000	00001000	Mod_7564		PE header	Img	R	RWE Cop	
75641000	00038000				Img	R E	RWE Cop	
75679000	00002000				Img	RW	RWE Cop	
7567B000	00004000				Img	R	RWE Cop	
76F50000	00010000	kernel32		PE header	Img	R	RWE Cop	
76F60000	000D0000	kernel32	.text	Code,imports,exports	Img	R E	RWE Cop	
77030000	00010000	kernel32	.data	Data	Img	RW Cop	RWE Cop	
77040000	00010000	kernel32	.rsrc	Resources	Img	R	RWE Cop	
77050000	0000B000	kernel32	.reloc	Relocations	Img	R	RWE Cop	
77810000	00001000	KERNELBASE		PE header	Img	R	RWE Cop	
77811000	00040000	KERNELBASE	.text	Code,imports,exports	Img	R E	RWE Cop	
77851000	00002000	KERNELBASE	.data	Data	Img	RW	RWE Cop	
77853000	00001000	KERNELBASE	.rsrc	Resources	Img	R	RWE Cop	
77854000	00003000	KERNELBASE	.reloc	Relocations	Img	R	RWE Cop	
77B20000	00001000	Mod_77B2		PE header	Img	R	RWE Cop	
77B21000	00102000				Img	R E	RWE Cop	
77C23000	0002F000				Img	R	RWE Cop	
77C52000	0000C000				Img	RW Cop	RWE Cop	
77C5E000	0006B000				Img	R	RWE Cop	
77D00000	00001000	ntdll		PE header	Img	R	RWE Cop	
77D10000	000D6000	ntdll	.text	Code,exports	Img	R E	RWE Cop	
77DF0000	00001000	ntdll	RT	Code	Img	R E	RWE Cop	
77E00000	00009000	ntdll	.data	Data	Img	RW Cop	RWE Cop	

Figura 7.6: OllyDbg:

7.2.3 GCC: x86

```
; Segment type: Uninitialized
; Segment permissions: Read/Write
```

```
; Segment type: Pure data
; Segment permissions: Read/Write
```

7.2.4 MSVC: x64

Listing 7.6: MSVC 2012 x64

```
```

```
_DATA   SEGMENT
COMM    x:DWORD
$SG2924 DB      'Enter X:', 0aH, 00H
$SG2925 DB      '%d', 00H
$SG2926 DB      'You entered %d...', 0aH, 00⤦
    ↳ H
_DATA   ENDS

_TEXT   SEGMENT
main    PROC
$LN3:
        sub     rsp, 40

        lea     rcx, OFFSET FLAT:$SG2924 ; '⤦
    ↳ Enter X:'
        call    printf
        lea     rdx, OFFSET FLAT:x
        lea     rcx, OFFSET FLAT:$SG2925 ; ⤦
    ↳ '%d'
        call    scanf
        mov     edx, DWORD PTR x
        lea     rcx, OFFSET FLAT:$SG2926 ; '⤦
    ↳ You entered %d...'
        call    printf

        ; 0
        xor     eax, eax

        add     rsp, 40
        ret     0
main    ENDP
_TEXT   ENDS
```

x86. . DWORD PTR.

7.2.5 ARM: Con optimización Keil 6/2013 (Modo Thumb)

```
.text:00000000 ; Segment type: Pure code
.text:00000000                 AREA .text, ↙
    ↘ CODE
...
.text:00000000 main
.text:00000000                 PUSH    {R4,↙
    ↘ LR}
.text:00000002                 ADR     R0, ↙
    ↘ aEnterX     ; "Enter X:\n"
.text:00000004                 BL      ↙
    ↘ __2printf
.text:00000008                 LDR     R1, =↙
    ↘ x
.text:0000000A                 ADR     R0, ↙
    ↘ aD          ; "%d"
.text:0000000C                 BL      ↙
    ↘ __0scanf
.text:00000010                 LDR     R0, =↙
    ↘ x
.text:00000012                 LDR     R1, [↙
    ↘ R0]
.text:00000014                 ADR     R0, ↙
    ↘ aYouEnteredD___ ; "You entered %d...\n↙
    ↘ "
.text:00000016                 BL      ↙
    ↘ __2printf
```

```
.text:0000001A                  MOVS    R0, ↙
    ↘ #0
.text:0000001C                  POP     {R4,↙
    ↘ PC}
...
.text:00000020 aEnterX          DCB "Enter X↙
    ↘ :",0xA,0    ; DATA XREF: main+2
.text:0000002A                  DCB    0
.text:0000002B                  DCB    0
.text:0000002C off_2C           DCD x          ↙
    ↘               ; DATA XREF: main+8
.text:0000002C                                 ↙
    ↘               ; main+10
.text:00000030 aD               DCB "%d",0    ↙
    ↘               ; DATA XREF: main+A
.text:00000033                  DCB    0
.text:00000034 aYouEnteredD___ DCB "You ↙
    ↘ entered %d...",0xA,0 ; DATA XREF: main↙
    ↘ +14
.text:00000047                  DCB 0
.text:00000047 ; .text          ends
.text:00000047
...
.data:00000048 ; Segment type: Pure data
.data:00000048                  AREA .data, ↙
    ↘ DATA
.data:00000048                  ; ORG 0x48
.data:00000048                  EXPORT x
.data:00000048 x                DCD 0xA        ↙
    ↘               ; DATA XREF: main+8
.data:00000048                                 ↙
    ↘               ; main+10
.data:00000048 ; .data          ends
```

(.*data*).

7.2.6 ARM64

Listing 7.7: Sin optimización GCC 4.9.1 ARM64

```

        .comm   x,4,4
.LC0:
        .string "Enter X:"
.LC1:
        .string "%d"
.LC2:
        .string "You entered %d...\n"
f5:
; :
        stp     x29, x30, [sp, -16]!
;  (FP=SP)
        add     x29, sp, 0
; :
        adrp    x0, .LC0
        add     x0, x0, :lo12:.LC0
        bl      puts
; :
        adrp    x0, .LC1
        add     x0, x0, :lo12:.LC1
; :
        adrp    x1, x
        add     x1, x1, :lo12:x
        bl      __isoc99_scanf
; :
```

```
        adrp    x0, x
        add     x0, x0, :lo12:x
; :
        ldr     w1, [x0]
; :
        adrp    x0, .LC2
        add     x0, x0, :lo12:.LC2
        bl      printf
;  0
        mov     w0, 0
; :
        ldp     x29, x30, [sp], 16
        ret
```

7.2.7 MIPS

Listing 7.8: Con optimización GCC 4.4.5 (IDA)

```
.text:004006C0 main:
.text:004006C0
.text:004006C0 var_10          = -0x10
.text:004006C0 var_4           = -4
.text:004006C0
; :
.text:004006C0                 lui     $gp, 0x42
.text:004006C4                 addiu   $sp, -0x20
```

```
.text:004006C8                  li      $gp, ⤦
    ↳ 0x418940
.text:004006CC                  sw      $ra, ⤦
    ↳ 0x20+var_4($sp)
.text:004006D0                  sw      $gp, ⤦
    ↳ 0x20+var_10($sp)
; puts():
.text:004006D4                  la      $t9, ⤦
    ↳ puts
.text:004006D8                  lui     $a0, ⤦
    ↳ 0x40
.text:004006DC                  jalr    $t9 ;⤦
    ↳  puts
.text:004006E0                  la      $a0, ⤦
    ↳ aEnterX     # "Enter X:" ; branch ⤦
    ↳ delay slot
; scanf():
.text:004006E4                  lw      $gp, ⤦
    ↳ 0x20+var_10($sp)
.text:004006E8                  lui     $a0, ⤦
    ↳ 0x40
.text:004006EC                  la      $t9, ⤦
    ↳ __isoc99_scanf
; x:
.text:004006F0                  la      $a1, ⤦
    ↳ x
.text:004006F4                  jalr    $t9 ;⤦
    ↳  __isoc99_scanf
.text:004006F8                  la      $a0, ⤦
    ↳ aD          # "%d"       ; branch ⤦
    ↳ delay slot
; printf():
```

```
.text:004006FC                 lw      $gp, ↙
    ↘ 0x20+var_10($sp)
.text:00400700                 lui     $a0, ↙
    ↘ 0x40
;  x:
.text:00400704                 la      $v0, ↙
    ↘ x
.text:00400708                 la      $t9, ↙
    ↘ printf
;  printf()  $a1:
.text:0040070C                 lw      $a1, ↙
    ↘ (x - 0x41099C)($v0)
.text:00400710                 jalr    $t9 ;↙
    ↘  printf
.text:00400714                 la      $a0, ↙
    ↘ aYouEnteredD___  # "You entered %d...\↙
    ↘ n" ; branch delay slot
; :
.text:00400718                 lw      $ra, ↙
    ↘ 0x20+var_4($sp)
.text:0040071C                 move    $v0, ↙
    ↘ $zero
.text:00400720                 jr      $ra
.text:00400724                 addiu   $sp, ↙
    ↘ 0x20  ; branch delay slot


...

.sbss:0041099C  # Segment type: ↙
    ↘ Uninitialized
.sbss:0041099C                 .sbss
.sbss:0041099C                 .globl x
.sbss:0041099C x:              .space 4
```

```
.sbss:0041099C
```

Listing 7.9: Con optimización GCC 4.4.5 (objdump)

```

004006c0 <main>:
; :
  4006c0:       3c1c0042        lui     gp,0x42
  4006c4:       27bdffe0        addiu   sp,sp,-32
  4006c8:       279c8940        addiu   gp,gp,-30400
  4006cc:       afbf001c        sw      ra,28(sp)
  4006d0:       afbc0010        sw      gp,16(sp)
;  puts():
  4006d4:       8f998034        lw      t9,-32716(gp)
  4006d8:       3c040040        lui     a0,0x40
  4006dc:       0320f809        jalr    t9
  4006e0:       248408f0        addiu   a0,a0,2288 ; branch delay slot
;  scanf():
  4006e4:       8fbc0010        lw      gp,16(sp)
  4006e8:       3c040040        lui     a0,0x40
  4006ec:       8f998038        lw      t9,-32712(gp)
;  x:
```

```
  4006f0:       8f858044        lw      a1,-32700(gp)
  4006f4:       0320f809        jalr    t9
  4006f8:       248408fc        addiu   a0,a0,2300 ; branch delay slot
;  printf():
  4006fc:       8fbc0010        lw      gp,16(sp)
  400700:       3c040040        lui     a0,0x40
;  x:
  400704:       8f828044        lw      v0,-32700(gp)
  400708:       8f99803c        lw      t9,-32708(gp)
;  printf()  $a1:
  40070c:       8c450000        lw      a1,0(v0)
  400710:       0320f809        jalr    t9
  400714:       24840900        addiu   a0,a0,2304 ; branch delay slot
; :
  400718:       8fbf001c        lw      ra,28(sp)
  40071c:       00001021        move    v0,zero
  400720:       03e00008        jr      ra
  400724:       27bd0020        addiu   sp,sp,32   ; branch delay slot
; :
  400728:       00200825        move    at,at
  40072c:       00200825        move    at,
```

```
    ↳ at
```

.

```
int x=10; //
```

Listing 7.10: Con optimización GCC 4.4.5 (IDA)

```
.text:004006A0 main:
.text:004006A0
.text:004006A0 var_10          = -0x10
.text:004006A0 var_8           = -8
.text:004006A0 var_4           = -4
.text:004006A0
.text:004006A0                 lui     $gp, ↙
    ↳ 0x42
.text:004006A4                 addiu   $sp, ↙
    ↳ -0x20
.text:004006A8                 li      $gp, ↙
    ↳ 0x418930
.text:004006AC                 sw      $ra, ↙
    ↳ 0x20+var_4($sp)
.text:004006B0                 sw      $s0, ↙
    ↳ 0x20+var_8($sp)
.text:004006B4                 sw      $gp, ↙
    ↳ 0x20+var_10($sp)
.text:004006B8                 la      $t9, ↙
    ↳ puts
```

```
.text:004006BC                 lui     $a0, ↙
    ↘ 0x40
.text:004006C0                 jalr    $t9 ;↙
    ↘  puts
.text:004006C4                 la      $a0, ↙
    ↘ aEnterX      # "Enter X:"
.text:004006C8                 lw      $gp, ↙
    ↘ 0x20+var_10($sp)
; :
.text:004006CC                 lui     $s0, ↙
    ↘ 0x41
.text:004006D0                 la      $t9, ↙
    ↘ __isoc99_scanf
.text:004006D4                 lui     $a0, ↙
    ↘ 0x40
; :
.text:004006D8                 addiu   $a1, ↙
    ↘ $s0, (x - 0x410000)
;  $a1.
.text:004006DC                 jalr    $t9 ;↙
    ↘  __isoc99_scanf
.text:004006E0                 la      $a0, ↙
    ↘ aD           # "%d"
.text:004006E4                 lw      $gp, ↙
    ↘ 0x20+var_10($sp)
; :
.text:004006E8                 lw      $a1, ↙
    ↘ x
;  $a1.
.text:004006EC                 la      $t9, ↙
    ↘ printf
.text:004006F0                 lui     $a0, ↙
    ↘ 0x40
```

```
.text:004006F4                 jalr    $t9 ;
    ↳ printf
.text:004006F8                 la      $a0,
    ↳ aYouEnteredD___  # "You entered %d...\
    ↳ n"
.text:004006FC                 lw      $ra,
    ↳ 0x20+var_4($sp)
.text:00400700                 move    $v0,
    ↳ $zero
.text:00400704                 lw      $s0,
    ↳ 0x20+var_8($sp)
.text:00400708                 jr      $ra
.text:0040070C                 addiu   $sp,
    ↳ 0x20

...

.data:00410920                 .globl x
.data:00410920 x:              .word 0xA
```

Listing 7.11: Con optimización GCC 4.4.5 (objdump)

```
004006a0 <main>:
  4006a0:       3c1c0042        lui     gp,0
    ↳ x42
  4006a4:       27bdffe0        addiu   sp,
    ↳ sp,-32
  4006a8:       279c8930        addiu   gp,
    ↳ gp,-30416
  4006ac:       afbf001c        sw      ra
    ↳ ,28(sp)
  4006b0:       afb00018        sw      s0
```

```
  ↳ ,24(sp)
  4006b4:       afbc0010        sw      gp↙
  ↳ ,16(sp)
  4006b8:       8f998034        lw      t9↙
  ↳ ,-32716(gp)
  4006bc:       3c040040        lui     a0,0↙
  ↳ x40
  4006c0:       0320f809        jalr    t9
  4006c4:       248408d0        addiu   a0,↙
  ↳ a0,2256
  4006c8:       8fbc0010        lw      gp↙
  ↳ ,16(sp)
; :
  4006cc:       3c100041        lui     s0,0↙
  ↳ x41
  4006d0:       8f998038        lw      t9↙
  ↳ ,-32712(gp)
  4006d4:       3c040040        lui     a0,0↙
  ↳ x40
; :
  4006d8:       26050920        addiu   a1,↙
  ↳ s0,2336
;  $a1.
  4006dc:       0320f809        jalr    t9
  4006e0:       248408dc        addiu   a0,↙
  ↳ a0,2268
  4006e4:       8fbc0010        lw      gp↙
  ↳ ,16(sp)
;  $s0.
; :
  4006e8:       8e050920        lw      a1↙
  ↳ ,2336(s0)
;  $a1.
```

```
  4006ec:       8f99803c        lw      t9↙
    ↳ ,-32708(gp)
  4006f0:       3c040040        lui     a0,0↙
    ↳ x40
  4006f4:       0320f809        jalr    t9
  4006f8:       248408e0        addiu   a0,↙
    ↳ a0,2272
  4006fc:       8fbf001c        lw      ra↙
    ↳ ,28(sp)
  400700:       00001021        move    v0,↙
    ↳ zero
  400704:       8fb00018        lw      s0↙
    ↳ ,24(sp)
  400708:       03e00008        jr      ra
  40070c:       27bd0020        addiu   sp,↙
    ↳ sp,32
```

7.3

```
#include <stdio.h>

int main()
{
        int x;
        printf ("Enter X:\n");

        if (scanf ("%d", &x)==1)
                printf ("You entered %d...\n↙
    ↳ ", x);
        else
```

```
                printf ("What you entered? ↙
    ↘ Huh?\n");

        return 0;
};
```

scanf()[1]

:

```
C:\...>ex3.exe
Enter X:
123
You entered 123...

C:\...>ex3.exe
Enter X:
ouch
What you entered? Huh?
```

7.3.1 MSVC: x86

(MSVC 2010):

```
        lea     eax, DWORD PTR _x$[ebp]
        push    eax
        push    OFFSET $SG3833 ; '%d', 00H
        call    _scanf
        add     esp, 8
        cmp     eax, 1
```

[1]scanf, wscanf: MSDN

```
        jne     SHORT $LN2@main
        mov     ecx, DWORD PTR _x$[ebp]
        push    ecx
        push    OFFSET $SG3834 ; 'You ↙
    ↳ entered %d...', 0aH, 00H
        call    _printf
        add     esp, 8
        jmp     SHORT $LN1@main
$LN2@main:
        push    OFFSET $SG3836 ; 'What you ↙
    ↳ entered? Huh?', 0aH, 00H
        call    _printf
        add     esp, 4
$LN1@main:
        xor     eax, eax
```

7.3.2 MSVC: x86: IDA

.

. «exit». :

```
.text:00401000 _main proc near
.text:00401000
.text:00401000 var_4 = dword ptr -4
.text:00401000 argc  = dword ptr  8
.text:00401000 argv  = dword ptr  0Ch
.text:00401000 envp  = dword ptr  10h
.text:00401000
.text:00401000       push    ebp
.text:00401001       mov     ebp, esp
.text:00401003       push    ecx
```

```
.text:00401004          push    offset Format ;
    ↳  "Enter X:\n"
.text:00401009          call    ds:printf
.text:0040100F          add     esp, 4
.text:00401012          lea     eax, [ebp+var_4
    ↳ ]
.text:00401015          push    eax
.text:00401016          push    offset aD ; "%d
    ↳ "
.text:0040101B          call    ds:scanf
.text:00401021          add     esp, 8
.text:00401024          cmp     eax, 1
.text:00401027          jnz     short error
.text:00401029          mov     ecx, [ebp+var_4
    ↳ ]
.text:0040102C          push    ecx
.text:0040102D          push    offset aYou ; "
    ↳ You entered %d...\n"
.text:00401032          call    ds:printf
.text:00401038          add     esp, 8
.text:0040103B          jmp     short exit
.text:0040103D
.text:0040103D error: ; CODE XREF: _main+27
.text:0040103D          push    offset aWhat ;
    ↳ "What you entered? Huh?\n"
.text:00401042          call    ds:printf
.text:00401048          add     esp, 4
.text:0040104B
.text:0040104B exit: ; CODE XREF: _main+3B
.text:0040104B          xor     eax, eax
.text:0040104D          mov     esp, ebp
.text:0040104F          pop     ebp
.text:00401050          retn
```

```
.text:00401050 _main endp
```

.

.

:

```
.text:00401000 _text segment para public '
    CODE' use32
.text:00401000         assume cs:_text
.text:00401000         ;org 401000h
.text:00401000 ; ask for X
.text:00401012 ; get X
.text:00401024         cmp  eax, 1
.text:00401027         jnz  short error
.text:00401029 ; print result
.text:0040103B         jmp  short exit
.text:0040103D
.text:0040103D error: ; CODE XREF: _main+27
.text:0040103D         push offset aWhat ; "
    What you entered? Huh?\n"
.text:00401042         call ds:printf
.text:00401048         add  esp, 4
.text:0040104B
.text:0040104B exit: ; CODE XREF: _main+3B
.text:0040104B         xor  eax, eax
.text:0040104D         mov  esp, ebp
.text:0040104F         pop  ebp
.text:00401050         retn
.text:00401050 _main endp
```

.

-
-

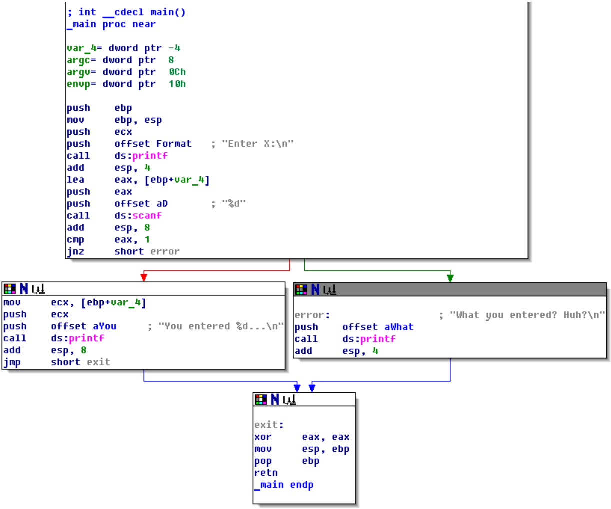

; int __cdecl main()
_main proc near
var_4= dword ptr -4
argc= dword ptr 8
argv= dword ptr 0Ch
envp= dword ptr 10h
push ebp
mov ebp, esp
push ecx
push offset Format ; "Enter X:\n"
call ds:printf
add esp, 4
lea eax, [ebp+var_4]
push eax
push offset aD ; "%d"
call ds:scanf
add esp, 8
cmp eax, 1
jnz short error
mov ecx, [ebp+var_4]
push ecx
push offset aYou ; "You entered %d...\n"
call ds:printf
add esp, 8
jmp short exit
error: ; "What you entered? Huh?\n"
push offset aWhat
call ds:printf
add esp, 4
exit:
xor eax, eax
mov esp, ebp
pop ebp
retn
_main endp

Figura 7.7:

-

 CAPÍTULO 7. SCANF()

(«group nodes»). :

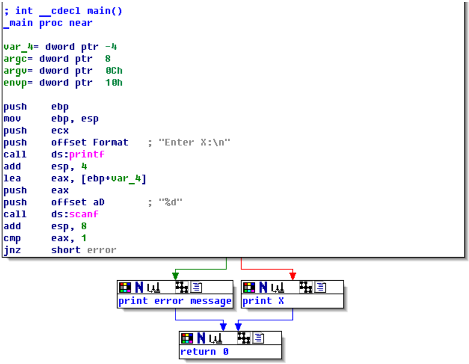


Figura 7.8:

.

7.3.3 MSVC: x86 + OllyDbg

0x6E494714:

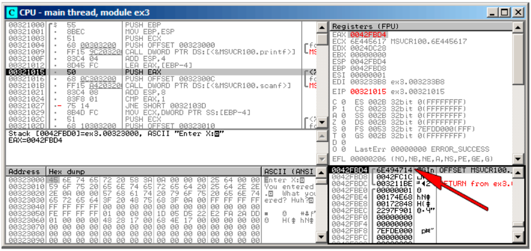


Figura 7.9: OllyDbg: `scanf()`

`scanf()` «asdasd». `scanf()` EAX, :

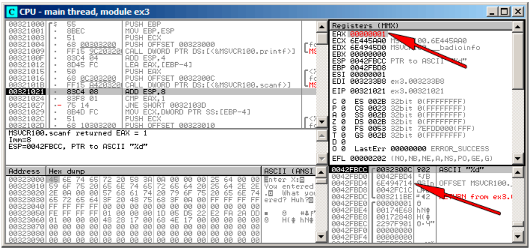


Figura 7.10: OllyDbg: `scanf()`

? .

. EAX, «Set to 1». .

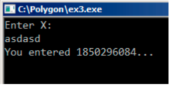


Figura 7.11:

, 1850296084 (0x6E494714)!

7.3.4 MSVC: x86 + Hiew

. .

MSVCR*.DLL (/MD)[2], main() .text. .text (Enter, F8, F6, Enter, Enter).

:

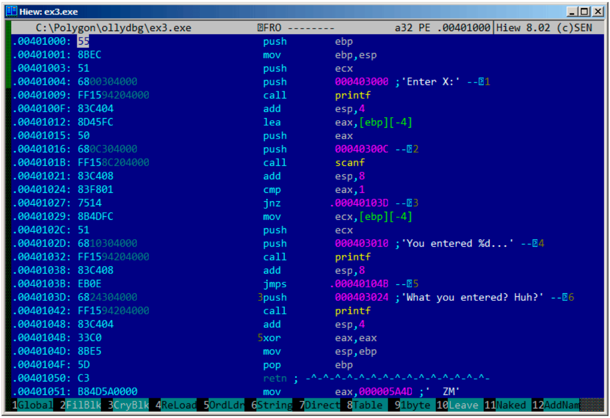


Figura 7.12: Hiew: main()

Hiew ASCIIZ[3].

[2] «dynamic linking»

[3] ASCII Zero ()

. 00401027 (JNZ), F3, «9090»(NOP):

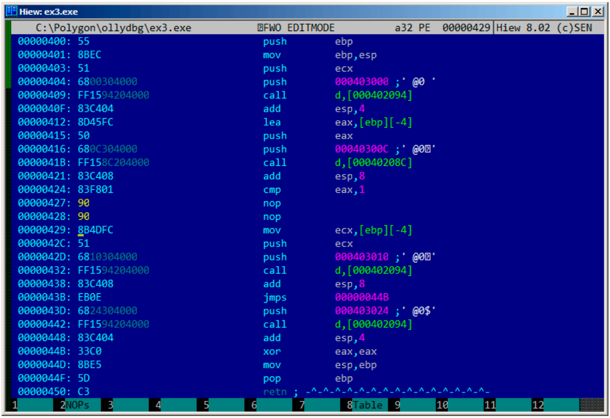


Figura 7.13: Hiew: JNZ NOP

F9 (update).

NOP . JNZ .

.

7.3.5 MSVC: x64

Listing 7.12: MSVC 2012 x64

```
_DATA   SEGMENT
$SG2924 DB      'Enter X:', 0aH, 00H
```

```
$SG2926 DB      '%d', 00H
$SG2927 DB      'You entered %d...', 0aH, 00⤦
    ↳ H
$SG2929 DB      'What you entered? Huh?', 0⤦
    ↳ aH, 00H
_DATA   ENDS

_TEXT   SEGMENT
x$ = 32
main    PROC
$LN5:
        sub     rsp, 56
        lea     rcx, OFFSET FLAT:$SG2924 ; '⤦
    ↳ Enter X:'
        call    printf
        lea     rdx, QWORD PTR x$[rsp]
        lea     rcx, OFFSET FLAT:$SG2926 ; ⤦
    ↳ '%d'
        call    scanf
        cmp     eax, 1
        jne     SHORT $LN2@main
        mov     edx, DWORD PTR x$[rsp]
        lea     rcx, OFFSET FLAT:$SG2927 ; '⤦
    ↳ You entered %d...'
        call    printf
        jmp     SHORT $LN1@main
$LN2@main:
        lea     rcx, OFFSET FLAT:$SG2929 ; '⤦
    ↳ What you entered? Huh?'
        call    printf
$LN1@main:
        ; 0
        xor     eax, eax
```

```
        add     rsp, 56
        ret     0
main    ENDP
_TEXT   ENDS
END
```

7.3.6 ARM

ARM: Con optimización Keil 6/2013 (Modo Thumb)

Listing 7.13: Con optimización Keil 6/2013 (Modo Thumb)

```
var_8           = -8

                PUSH    {R3,LR}
                ADR     R0, aEnterX      ; "↙
    ↳ Enter X:\n"
                BL      __2printf
                MOV     R1, SP
                ADR     R0, aD           ; "%↙
    ↳ d"
                BL      __0scanf
                CMP     R0, #1
                BEQ     loc_1E
                ADR     R0, aWhatYouEntered ↙
    ↳ ; "What you entered? Huh?\n"
                BL      __2printf

loc_1A                                   ; ↙
    ↳ CODE XREF: main+26
                MOVS    R0, #0
                POP     {R3,PC}
```

```
loc_1E                                  ; CODE XREF: main+12
                LDR     R1, [SP,#8+var_8]
                ADR     R0, aYouEnteredD___ ; "You entered %d...\n"
                BL      __2printf
                B       loc_1A
```

CMP

BEQ[4] x86.

ARM64

Listing 7.14: Sin optimización GCC 4.9.1 ARM64

```

.LC0:
        .string "Enter X:"
.LC1:
        .string "%d"
.LC2:
        .string "You entered %d...\n"
.LC3:
        .string "What you entered? Huh?"
f6:
; :
        stp     x29, x30, [sp, -32]!
; (FP=SP)
        add     x29, sp, 0
```

[4](PowerPC, ARM) Branch if Equal

```
;
        adrp    x0, .LC0
        add     x0, x0, :lo12:.LC0
        bl      puts
; :
        adrp    x0, .LC1
        add     x0, x0, :lo12:.LC1
;
        add     x1, x29, 28
        bl      __isoc99_scanf
; scanf()  W0.
; :
        cmp     w0, 1
; BNE  Branch if Not Equal
;  W0<>0,
        bne     .L2
;  W0=1,
;
        ldr     w1, [x29,28]
; :
        adrp    x0, .LC2
        add     x0, x0, :lo12:.LC2
        bl      printf
;  "What you entered? Huh?" :
        b       .L3
.L2:
;  "What you entered? Huh?" :
        adrp    x0, .LC3
        add     x0, x0, :lo12:.LC3
        bl      puts
.L3:
;  0
        mov     w0, 0
```

```
; :
        ldp     x29, x30, [sp], 32
        ret
```

7.3.7 MIPS

Listing 7.15: Con optimización GCC 4.4.5 (IDA)

```
.text:004006A0 main:
.text:004006A0
.text:004006A0 var_18          = -0x18
.text:004006A0 var_10          = -0x10
.text:004006A0 var_4           = -4
.text:004006A0
.text:004006A0                 lui     $gp, ⤦
    ↳ 0x42
.text:004006A4                 addiu   $sp, ⤦
    ↳ -0x28
.text:004006A8                 li      $gp, ⤦
    ↳ 0x418960
.text:004006AC                 sw      $ra, ⤦
    ↳ 0x28+var_4($sp)
.text:004006B0                 sw      $gp, ⤦
    ↳ 0x28+var_18($sp)
.text:004006B4                 la      $t9, ⤦
    ↳ puts
.text:004006B8                 lui     $a0, ⤦
    ↳ 0x40
.text:004006BC                 jalr    $t9 ;⤦
    ↳  puts
.text:004006C0                 la      $a0, ⤦
    ↳ aEnterX      # "Enter X:"
```

```
.text:004006C4                  lw      $gp, ↙
    ↳ 0x28+var_18($sp)
.text:004006C8                  lui     $a0, ↙
    ↳ 0x40
.text:004006CC                  la      $t9, ↙
    ↳ __isoc99_scanf
.text:004006D0                  la      $a0, ↙
    ↳ aD          # "%d"
.text:004006D4                  jalr    $t9 ;↙
    ↳  __isoc99_scanf
.text:004006D8                  addiu   $a1, ↙
    ↳ $sp, 0x28+var_10  # branch delay slot
.text:004006DC                  li      $v1, ↙
    ↳ 1
.text:004006E0                  lw      $gp, ↙
    ↳ 0x28+var_18($sp)
.text:004006E4                  beq     $v0, ↙
    ↳ $v1, loc_40070C
.text:004006E8                  or      $at, ↙
    ↳ $zero       # branch delay slot, NOP
.text:004006EC                  la      $t9, ↙
    ↳ puts
.text:004006F0                  lui     $a0, ↙
    ↳ 0x40
.text:004006F4                  jalr    $t9 ;↙
    ↳  puts
.text:004006F8                  la      $a0, ↙
    ↳ aWhatYouEntered  # "What you entered? ↙
    ↳ Huh?"
.text:004006FC                  lw      $ra, ↙
    ↳ 0x28+var_4($sp)
.text:00400700                  move    $v0, ↙
    ↳ $zero
```

```
.text:00400704                   jr      $ra
.text:00400708                   addiu   $sp, ⤦
    ↪ 0x28

.text:0040070C loc_40070C:
.text:0040070C                   la      $t9, ⤦
    ↪ printf
.text:00400710                   lw      $a1, ⤦
    ↪ 0x28+var_10($sp)
.text:00400714                   lui     $a0, ⤦
    ↪ 0x40
.text:00400718                   jalr    $t9 ;⤦
    ↪  printf
.text:0040071C                   la      $a0, ⤦
    ↪ aYouEnteredD___  # "You entered %d...\⤦
    ↪ n"
.text:00400720                   lw      $ra, ⤦
    ↪ 0x28+var_4($sp)
.text:00400724                   move    $v0, ⤦
    ↪ $zero
.text:00400728                   jr      $ra
.text:0040072C                   addiu   $sp, ⤦
    ↪ 0x28
```

BEQ «Branch Equal».

7.3.8 Ejercicio

7.4 Ejercicios

7.4.1 Ejercicio #1

CAPÍTULO 7. SCANF()

```
#include <string.h>
#include <stdio.h>

void alter_string(char *s)
{
        strcpy (s, "Goodbye!");
        printf ("Result: %s\n", s);
};

int main()
{
        alter_string ("Hello, world!\n");
};
```

Responda: G.1.3 on page 1919.

Capítulo 8

Listing 8.1:

```
#include <stdio.h>

int f (int a, int b, int c)
{
        return a*b+c;
};

int main()
{
        printf ("%d\n", f(1, 2, 3));
        return 0;
};
```

8.1 x86

8.1.1 MSVC

(MSVC 2010 Express):

Listing 8.2: MSVC 2010 Express

```
_TEXT   SEGMENT
_a$ = 8                                                    ↙
    ↘                 ; size = 4
_b$ = 12                                                   ↙
    ↘                 ; size = 4
_c$ = 16                                                   ↙
    ↘                 ; size = 4
_f      PROC
        push    ebp
        mov     ebp, esp
        mov     eax, DWORD PTR _a$[ebp]
        imul    eax, DWORD PTR _b$[ebp]
        add     eax, DWORD PTR _c$[ebp]
        pop     ebp
        ret     0
_f      ENDP

_main   PROC
        push    ebp
        mov     ebp, esp
        push    3 ;
        push    2 ;
        push    1 ;
        call    _f
```

```
        add     esp, 12
        push    eax
        push    OFFSET $SG2463 ; '%d', 0aH, ↙
    ↘ 00H
        call    _printf
        add     esp, 8
        ;  0
        xor     eax, eax
        pop     ebp
        ret     0
_main   ENDP
```

8.1.2 MSVC + OllyDbg

OllyDbg.

OllyDbg «RETURN from» y «Arg1 = …», Spanish text placeholder.

N.B.:

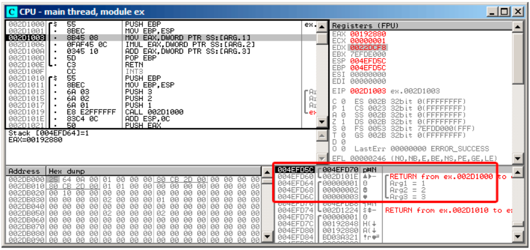


Figura 8.1: OllyDbg: `f()`

8.1.3 GCC

Listing 8.3: GCC 4.4.1

```
                public f
f               proc near

arg_0           = dword ptr  8
arg_4           = dword ptr  0Ch
arg_8           = dword ptr  10h

                push    ebp
                mov     ebp, esp
                mov     eax, [ebp+arg_0] ;
                imul    eax, [ebp+arg_4] ;
                add     eax, [ebp+arg_8] ;
                pop     ebp
                retn
f               endp

                public main
main            proc near

var_10          = dword ptr -10h
var_C           = dword ptr -0Ch
var_8           = dword ptr -8

                push    ebp
                mov     ebp, esp
                and     esp, 0FFFFFFF0h
                sub     esp, 10h
                mov     [esp+10h+var_8], 3 ;
```

```
                mov     [esp+10h+var_C], 2 ;
                mov     [esp+10h+var_10], 1 ↙
    ↘ ;
                call    f
                mov     edx, offset aD  ; "%↙
    ↘ d\n"
                mov     [esp+10h+var_C], eax
                mov     [esp+10h+var_10], ↙
    ↘ edx
                call    _printf
                mov     eax, 0
                leave
                retn
main            endp
```

LEAVE (A.6.2 on page 1880)

8.2 x64

8.2.1 MSVC

Con optimización MSVC:

Listing 8.4: Con optimización MSVC 2012 x64

```
$SG2997 DB      '%d', 0aH, 00H

main    PROC
        sub     rsp, 40
        mov     edx, 2
```

```
        lea     r8d, QWORD PTR [rdx+1] ; R8D
    ↪ =3
        lea     ecx, QWORD PTR [rdx-1] ; ECX
    ↪ =1
        call    f
        lea     rcx, OFFSET FLAT:$SG2997 ; 
    ↪ '%d'
        mov     edx, eax
        call    printf
        xor     eax, eax
        add     rsp, 40
        ret     0
main    ENDP

f       PROC
        ; ECX -
        ; EDX -
        ; R8D -
        imul    ecx, edx
        lea     eax, DWORD PTR [r8+rcx]
        ret     0
f       ENDP
```

:

Listing 8.5: MSVC 2012 x64

```

f               proc near

; shadow space:
arg_0           = dword ptr  8
arg_8           = dword ptr  10h
```

```
arg_10          = dword ptr  18h

                ; ECX -
                ; EDX -
                ; R8D -
                mov     [rsp+arg_10], r8d
                mov     [rsp+arg_8], edx
                mov     [rsp+arg_0], ecx
                mov     eax, [rsp+arg_0]
                imul    eax, [rsp+arg_8]
                add     eax, [rsp+arg_10]
                retn
f               endp

main            proc near
                sub     rsp, 28h
                mov     r8d, 3 ;
                mov     edx, 2 ;
                mov     ecx, 1 ;
                call    f
                mov     edx, eax
                lea     rcx, $SG2931    ; "%↙
    ↘ d\n"
                call    printf

                ;  0
                xor     eax, eax
                add     rsp, 28h
                retn
main            endp
```

«shadow space» [1]: : 1) 2) [2].

8.2.2 GCC

Con optimización GCC :

Listing 8.6: Con optimización GCC 4.4.6 x64

```
f:
        ; EDI -
        ; ESI -
        ; EDX -
        imul    esi, edi
        lea     eax, [rdx+rsi]
        ret

main:
        sub     rsp, 8
        mov     edx, 3
        mov     esi, 2
        mov     edi, 1
        call    f
        mov     edi, OFFSET FLAT:.LC0 ; "%d\
    ↳ n"
        mov     esi, eax
        xor     eax, eax  ;
        call    printf
        xor     eax, eax
        add     rsp, 8
```

[1] MSDN
[2] MSDN

```
        ret
```

Sin optimización GCC:

Listing 8.7: GCC 4.4.6 x64

```
f:
        ; EDI -
        ; ESI -
        ; EDX -
        push    rbp
        mov     rbp, rsp
        mov     DWORD PTR [rbp-4], edi
        mov     DWORD PTR [rbp-8], esi
        mov     DWORD PTR [rbp-12], edx
        mov     eax, DWORD PTR [rbp-4]
        imul    eax, DWORD PTR [rbp-8]
        add     eax, DWORD PTR [rbp-12]
        leave
        ret
main:
        push    rbp
        mov     rbp, rsp
        mov     edx, 3
        mov     esi, 2
        mov     edi, 1
        call    f
        mov     edx, eax
        mov     eax, OFFSET FLAT:.LC0 ; "%d\
    ↳ n"
        mov     esi, edx
        mov     rdi, rax
```

```
        mov     eax, 0  ;
        call    printf
        mov     eax, 0
        leave
        ret
```

8.2.3 GCC: uint64_t int

:

```
#include <stdio.h>
#include <stdint.h>

uint64_t f (uint64_t a, uint64_t b, uint64_t ⤦
    ↳  c)
{
        return a*b+c;
};

int main()
{
        printf ("%lld\n", f(0⤦
    ↳ x1122334455667788,
                                0⤦
    ↳ x1111111122222222,
                                0⤦
    ↳ x3333333344444444));
        return 0;
};
```

Listing 8.8: Con optimización GCC 4.4.6 x64

```
f                proc near
                 imul    rsi, rdi
                 lea     rax, [rdx+rsi]
                 retn
f                endp

main             proc near
                 sub     rsp, 8
                 mov     rdx, ↙
    ↘ 3333333344444444h ;
                 mov     rsi, ↙
    ↘ 1111111122222222h ;
                 mov     rdi, ↙
    ↘ 1122334455667788h ;
                 call    f
                 mov     edi, offset format ;↙
    ↘  "%lld\n"
                 mov     rsi, rax
                 xor     eax, eax ;
                 call    _printf
                 xor     eax, eax
                 add     rsp, 8
                 retn
main             endp
```

8.3 ARM

8.3.1 Sin optimización Keil 6/2013 (Modo ARM)

```
.text:000000A4 00 30 A0 E1                                    ↙
    ↘ MOV     R3, R0
.text:000000A8 93 21 20 E0                                    ↙
    ↘ MLA     R0, R3, R1, R2
.text:000000AC 1E FF 2F E1                                    ↙
    ↘ BX      LR
...
.text:000000B0                          main
.text:000000B0 10 40 2D E9                                    ↙
    ↘ STMFD   SP!, {R4,LR}
.text:000000B4 03 20 A0 E3                                    ↙
    ↘ MOV     R2, #3
.text:000000B8 02 10 A0 E3                                    ↙
    ↘ MOV     R1, #2
.text:000000BC 01 00 A0 E3                                    ↙
    ↘ MOV     R0, #1
.text:000000C0 F7 FF FF EB                                    ↙
    ↘ BL      f
.text:000000C4 00 40 A0 E1                                    ↙
    ↘ MOV     R4, R0
.text:000000C8 04 10 A0 E1                                    ↙
    ↘ MOV     R1, R4
.text:000000CC 5A 0F 8F E2                                    ↙
    ↘ ADR     R0, aD_0          ; "%d\n"
.text:000000D0 E3 18 00 EB                                    ↙
    ↘ BL      __2printf
.text:000000D4 00 00 A0 E3                                    ↙
    ↘ MOV     R0, #0
.text:000000D8 10 80 BD E8                                    ↙
    ↘ LDMFD   SP!, {R4,PC}
```

f()

`MLA` (*Multiply Accumulate*)

[3] (*Fused multiply–add*) [4].

`MOV R3, R0,`

8.3.2 Con optimización Keil 6/2013 (Modo ARM)

```
.text:00000098                  f
.text:00000098 91 20 20 E0
    MLA      R0, R1, R0, R2
.text:0000009C 1E FF 2F E1
    BX       LR
```

(-O3).

8.3.3 Con optimización Keil 6/2013 (Modo Thumb)

```
.text:0000005E 48 43
    MULS     R0, R1
.text:00000060 80 18
    ADDS     R0, R0, R2
.text:00000062 70 47
    BX       LR
```

[3] Spanish text placeholder
[4] wikipedia

8.3.4 ARM64

Con optimización GCC (Linaro) 4.9

.

Listing 8.9: Con optimización GCC (Linaro) 4.9

```
f:
        madd    w0, w0, w1, w2
        ret

main:
; :
        stp     x29, x30, [sp, -16]!
        mov     w2, 3
        mov     w1, 2
        add     x29, sp, 0
        mov     w0, 1
        bl      f
        mov     w1, w0
        adrp    x0, .LC7
        add     x0, x0, :lo12:.LC7
        bl      printf
;  0
        mov     w0, 0
;
        ldp     x29, x30, [sp], 16
        ret

.LC7:
        .string "%d\n"
```

```
#include <stdio.h>
#include <stdint.h>

uint64_t f (uint64_t a, uint64_t b, uint64_t
    c)
{
        return a*b+c;
};

int main()
{
        printf ("%lld\n", f(0
    x1122334455667788,
                                0
    x1111111122222222,
                                0
    x3333333344444444));
        return 0;
};
```

```
f:
        madd    x0, x0, x1, x2
        ret
main:
        mov     x1, 13396
        adrp    x0, .LC8
        stp     x29, x30, [sp, -16]!
        movk    x1, 0x27d0, lsl 16
        add     x0, x0, :lo12:.LC8
        movk    x1, 0x122, lsl 32
        add     x29, sp, 0
        movk    x1, 0x58be, lsl 48
```

```
        bl      printf
        mov     w0, 0
        ldp     x29, x30, [sp], 16
        ret

.LC8:
        .string "%lld\n"
```

: 28.3.1 on page 852.

Sin optimización GCC (Linaro) 4.9

```
f:
        sub     sp, sp, #16
        str     w0, [sp,12]
        str     w1, [sp,8]
        str     w2, [sp,4]
        ldr     w1, [sp,12]
        ldr     w0, [sp,8]
        mul     w1, w1, w0
        ldr     w0, [sp,4]
        add     w0, w1, w0
        add     sp, sp, 16
        ret
```

Register Save Area. [ARM13c] «Shadow Space»: 8.2.1 on page 193.

8.4 MIPS

Listing 8.10: Con optimización GCC 4.4.5

```
.text:00000000 f:
; $a0=a
; $a1=b
; $a2=c
.text:00000000                 mult    $a1, ⤦
    ↳ $a0
.text:00000004                 mflo    $v0
.text:00000008                 jr      $ra
.text:0000000C                 addu    $v0, ⤦
    ↳ $a2, $v0    ; branch delay slot
;

.text:00000010 main:
.text:00000010
.text:00000010 var_10          = -0x10
.text:00000010 var_4           = -4
.text:00000010
.text:00000010                 lui     $gp, ⤦
    ↳ (__gnu_local_gp >> 16)
.text:00000014                 addiu   $sp, ⤦
    ↳ -0x20
.text:00000018                 la      $gp, ⤦
    ↳ (__gnu_local_gp & 0xFFFF)
.text:0000001C                 sw      $ra, ⤦
    ↳ 0x20+var_4($sp)
.text:00000020                 sw      $gp, ⤦
    ↳ 0x20+var_10($sp)
;  c:
.text:00000024                 li      $a2, ⤦
    ↳ 3
```

```
;  a:
.text:00000028                  li      $a0, ↙
    ↘ 1
.text:0000002C                  jal     f
;  b:
.text:00000030                  li      $a1, ↙
    ↘ 2         ; branch delay slot
;
.text:00000034                  lw      $gp, ↙
    ↘ 0x20+var_10($sp)
.text:00000038                  lui     $a0, ↙
    ↘ ($LC0 >> 16)
.text:0000003C                  lw      $t9, ↙
    ↘ (printf & 0xFFFF)($gp)
.text:00000040                  la      $a0, ↙
    ↘ ($LC0 & 0xFFFF)
.text:00000044                  jalr    $t9
;
.text:00000048                  move    $a1, ↙
    ↘ $v0       ; branch delay slot
.text:0000004C                  lw      $ra, ↙
    ↘ 0x20+var_4($sp)
.text:00000050                  move    $v0, ↙
    ↘ $zero
.text:00000054                  jr      $ra
.text:00000058                  addiu   $sp, ↙
    ↘ 0x20      ; branch delay slot
```

ADD y ADDU. ADDU .

$V0.

JAL («Jump and Link»).

Capítulo 9

1

9.1

```
push envp
push argv
push argc
call main
push eax
call exit
```

```
exit(main(argc,argv,envp));
```

[1] Spanish text placeholder: MSDN: Return Values (C++): MSDN

Lo más probable es que quede un valor aleatorio de la ejecución de tu función, así que el código de salida será seudorandom.

. `main()` *void*:

```
#include <stdio.h>

void main()
{
        printf ("Hello, world!\n");
};
```

Linux.

GCC 4.8.1 `printf()` `puts()` (: 3.4.3 on page 39), `puts()` `printf()`. EAX `main()`.

Listing 9.1: GCC 4.8.1

```
.LC0:
        .string "Hello, world!"
main:
        push    ebp
        mov     ebp, esp
        and     esp, -16
        sub     esp, 16
        mov     DWORD PTR [esp], OFFSET FLAT↙
    ↳ :.LC0
        call    puts
        leave
        ret
```

:

Listing 9.2: tst.sh

```
#!/bin/sh
./hello_world
echo $?
```

:

```
$ tst.sh
Hello, world!
14
```

14 .

9.2

```
int f()
{
    // skip first 3 random values
    rand();
    rand();
    rand();
    // and use 4th
    return rand();
};
```

9.3

```
struct s
{
    int a;
    int b;
    int c;
};

struct s get_some_values (int a)
{
    struct s rt;

    rt.a=a+1;
    rt.b=a+2;
    rt.c=a+3;

    return rt;
};
```

… (MSVC 2010 /Ox):

```
$T3853 = 8                                ; size = 4
_a$ = 12                                  ; size = 4
?get_some_values@@YA?AUs@@H@Z PROC
    ; get_some_values
    mov     ecx, DWORD PTR _a$[esp-4]
    mov     eax, DWORD PTR $T3853[esp-4]
    lea     edx, DWORD PTR [ecx+1]
    mov     DWORD PTR [eax], edx
    lea     edx, DWORD PTR [ecx+2]
    add     ecx, 3
    mov     DWORD PTR [eax+4], edx
    mov     DWORD PTR [eax+8], ecx
```

```
    ret     0
?get_some_values@@YA?AUs@@H@Z ENDP                   ↙
    ↘   ; get_some_values
```

:

```
struct s
{
    int a;
    int b;
    int c;
};

struct s get_some_values (int a)
{
    return (struct s){.a=a+1, .b=a+2, .c=a↙
    ↘ +3};
};
```

Listing 9.3: GCC 4.8.1

```
_get_some_values proc near

ptr_to_struct   = dword ptr  4
a               = dword ptr  8

                mov     edx, [esp+a]
                mov     eax, [esp+↙
    ↘ ptr_to_struct]
                lea     ecx, [edx+1]
                mov     [eax], ecx
                lea     ecx, [edx+2]
                add     edx, 3
```

CAPÍTULO 9.

```
                mov     [eax+4], ecx
                mov     [eax+8], edx
                retn
_get_some_values endp
```

Capítulo 10

10.1

```
#include <stdio.h>

void f1 (int x, int y, int *sum, int *
    product)
{
        *sum=x+y;
        *product=x*y;
};

int sum, product;

void main()
{
        f1(123, 456, &sum, &product);
        printf ("sum=%d, product=%d\n", sum,
     product);
```

```
};
```

:

Listing 10.1: Con optimización MSVC 2010 (/Ob0)

```
COMM    _product:DWORD
COMM    _sum:DWORD
$SG2803 DB      'sum=%d, product=%d', 0aH, 00H

_x$ = 8                                 ; size = 4
_y$ = 12                                ; size = 4
_sum$ = 16                              ; size = 4
_product$ = 20                          ; size = 4
_f1     PROC
        mov     ecx, DWORD PTR _y$[esp-4]
        mov     eax, DWORD PTR _x$[esp-4]
        lea     edx, DWORD PTR [eax+ecx]
        imul    eax, ecx
        mov     ecx, DWORD PTR _product$[esp-4]
        push    esi
        mov     esi, DWORD PTR _sum$[esp]
        mov     DWORD PTR [esi], edx
        mov     DWORD PTR [ecx], eax
        pop     esi
        ret     0
_f1     ENDP
```

```
_main   PROC
        push    OFFSET _product
        push    OFFSET _sum
        push    456                                    ↙
    ↘                 ; 000001c8H
        push    123                                    ↙
    ↘                 ; 0000007bH
        call    _f1
        mov     eax, DWORD PTR _product
        mov     ecx, DWORD PTR _sum
        push    eax
        push    ecx
        push    OFFSET $SG2803
        call    DWORD PTR __imp__printf
        add     esp, 28                                ↙
    ↘                 ; 0000001cH
        xor     eax, eax
        ret     0
_main   ENDP
```

OllyDbg:

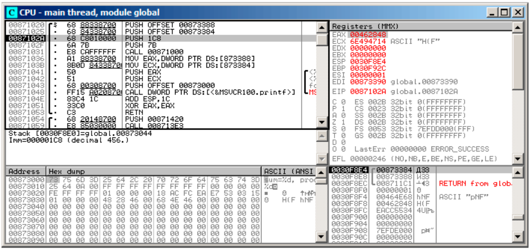
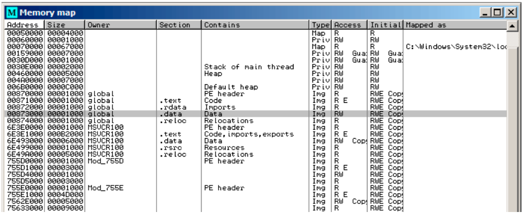


Figura 10.1: OllyDbg: `f1()`

`f1()`. «Follow in dump»

Alt-M :



Memory map

Address	Size	Owner	Section	Contains	Type	Access	Initial	Mapped as
00050000	00004000				Map	R	R	
00060000	00001000				Priv	RW	RW	
00070000	00067000				Map	R	R	C:\Windows\System32\loc
00159000	00007000				Priv	RW Gua	RW Gua	
0030D000	00001000				Priv	RW Gua	RW Gua	
0030E000	00002000			Stack of main thread	Priv	RW	RW	
00460000	00005000			Heap	Priv	RW	RW	
004A0000	00007000				Priv	RW	RW	
006B0000	0000C000			Default heap	Priv	RW	RW	
00870000	00001000	global		PE header	Img	R	RWE Cop	
00871000	00001000	global	.text	Code	Img	R E	RWE Cop	
00872000	00001000	global	.rdata	Imports	Img	R	RWE Cop	
00873000	00001000	global	.data	Data	Img	RW	RWE Cop	
00874000	00001000	global	.reloc	Relocations	Img	R	RWE Cop	
6E3E0000	00001000	MSVCR100		PE header	Img	R	RWE Cop	
6E3E1000	000B2000	MSVCR100	.text	Code,imports,exports	Img	R E	RWE Cop	
6E493000	00006000	MSVCR100	.data	Data	Img	RW Cop	RWE Cop	
6E499000	00001000	MSVCR100	.rsrc	Resources	Img	R	RWE Cop	
6E49A000	00005000	MSVCR100	.reloc	Relocations	Img	R	RWE Cop	
755D0000	00001000	Mod_755D		PE header	Img	R	RWE Cop	
755D1000	00003000				Img	R E	RWE Cop	
755D4000	00001000				Img	RW	RWE Cop	
755D5000	00003000				Img	R	RWE Cop	
755E0000	00001000	Mod_755E		PE header	Img	R	RWE Cop	
755E1000	0004D000				Img	R E	RWE Cop	
7562E000	00005000				Img	RW Cop	RWE Cop	
75633000	00009000				Img	R	RWE Cop	

Figura 10.2: OllyDbg:

(F7) `f1()`:

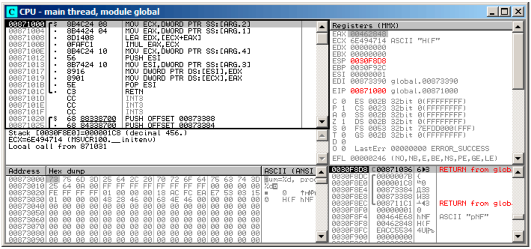


Figura 10.3: OllyDbg:

456 (0x1C8) y 123 (0x7B), .

f1().:

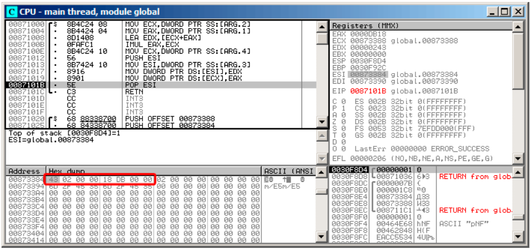


Figura 10.4: OllyDbg: f1()

`printf()`:

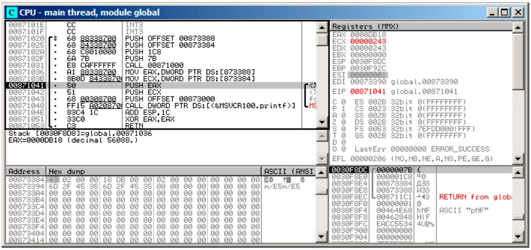


Figura 10.5: OllyDbg: `printf()`

10.2

:

Listing 10.2:

```
void main()
{
        int sum, product; //

        f1(123, 456, &sum, &product);
        printf ("sum=%d, product=%d\n", sum, ↙
    ↳ product);
};
```

`f1()`. :

Listing 10.3: Con optimización MSVC 2010 (/Ob0)

```
_product$ = -8                                          ↙
    ↘                    ; size = 4
_sum$ = -4                                              ↙
    ↘                    ; size = 4
_main   PROC
; Line 10
        sub     esp, 8
; Line 13
        lea     eax, DWORD PTR _product$[esp↙
    ↘ +8]
        push    eax
        lea     ecx, DWORD PTR _sum$[esp+12]
        push    ecx
        push    456                                     ↙
    ↘                    ; 000001c8H
        push    123                                     ↙
    ↘                    ; 0000007bH
        call    _f1
; Line 14
        mov     edx, DWORD PTR _product$[esp↙
    ↘ +24]
        mov     eax, DWORD PTR _sum$[esp+24]
        push    edx
        push    eax
        push    OFFSET $SG2803
        call    DWORD PTR __imp__printf
; Line 15
        xor     eax, eax
        add     esp, 36                                 ↙
    ↘                    ; 00000024H
        ret     0
```

OllyDbg. 0x2EF854 y 0x2EF858. :

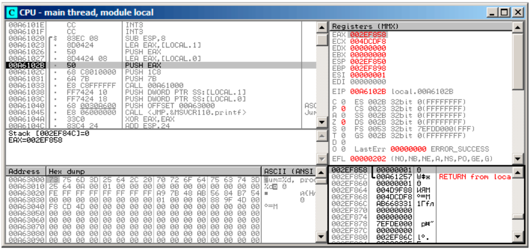


Figura 10.6: OllyDbg:

. 0x2EF854 y 0x2EF858:

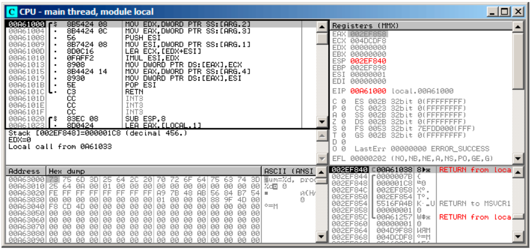


Figura 10.7: OllyDbg: f1()

:

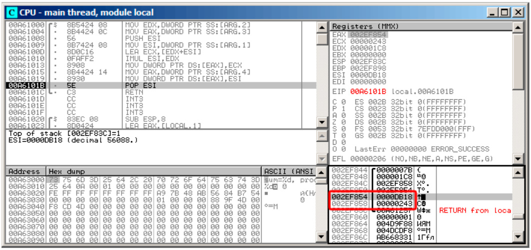


Figura 10.8: OllyDbg: f1()

10.3 Conclusión

references . : (51.3 on page 1072).

Capítulo 11

[Dij68], [Knu74], [Yur13, pág. 1.3.2].

:

```
#include <stdio.h>

int main()
{
        printf ("begin\n");
        goto exit;
        printf ("skip me!\n");
exit:
        printf ("end\n");
};
```

MSVC 2012:

Listing 11.1: MSVC 2012

```
$SG2934 DB      'begin', 0aH, 00H
$SG2936 DB      'skip me!', 0aH, 00H
$SG2937 DB      'end', 0aH, 00H

_main   PROC
        push    ebp
        mov     ebp, esp
        push    OFFSET $SG2934 ; 'begin'
        call    _printf
        add     esp, 4
        jmp     SHORT $exit$3
        push    OFFSET $SG2936 ; 'skip me!'
        call    _printf
        add     esp, 4
$exit$3:
        push    OFFSET $SG2937 ; 'end'
        call    _printf
        add     esp, 4
        xor     eax, eax
        pop     ebp
        ret     0
_main   ENDP
```

La segunda llamada a `printf()` solo podría ejecutarse con intervención humana, usando un depurador o parcheando el código.

Hiew:

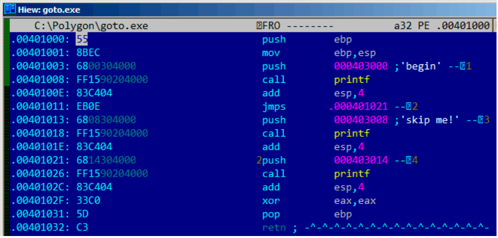


Figura 11.1: Hiew

JMP (0x410), F3 (), EB 00:

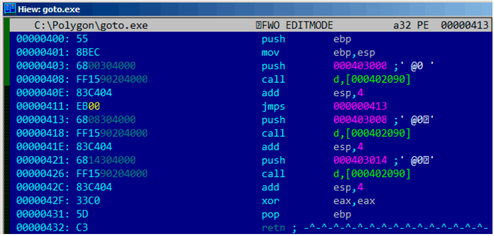


Figura 11.2: Hiew

:

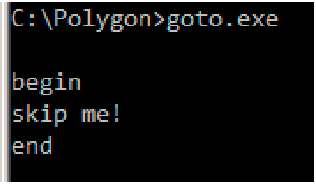


Figura 11.3:

11.1

Listing 11.2: Con optimización MSVC 2012

```
$SG2981 DB      'begin', 0aH, 00H
$SG2983 DB      'skip me!', 0aH, 00H
```

```
$SG2984 DB      'end', 0aH, 00H

_main   PROC
        push    OFFSET $SG2981 ; 'begin'
        call    _printf
        push    OFFSET $SG2984 ; 'end'
$exit$4:
        call    _printf
        add     esp, 8
        xor     eax, eax
        ret     0
_main   ENDP
```

«skip me!» .

11.2 Ejercicio

Capítulo 12

12.1

```
#include <stdio.h>

void f_signed (int a, int b)
{
    if (a>b)
        printf ("a>b\n");
    if (a==b)
        printf ("a==b\n");
    if (a<b)
        printf ("a<b\n");
};

void f_unsigned (unsigned int a, unsigned ↙
    ↘ int b)
{
    if (a>b)
```

```
        printf ("a>b\n");
    if (a==b)
        printf ("a==b\n");
    if (a<b)
        printf ("a<b\n");
};

int main()
{
    f_signed(1, 2);
    f_unsigned(1, 2);
    return 0;
};
```

12.1.1 x86

x86 + MSVC

Listing 12.1: Sin optimización MSVC 2010

```
_a$ = 8
_b$ = 12
_f_signed PROC
    push   ebp
    mov    ebp, esp
    mov    eax, DWORD PTR _a$[ebp]
    cmp    eax, DWORD PTR _b$[ebp]
    jle    SHORT $LN3@f_signed
    push   OFFSET $SG737         ; 'a>b'
    call   _printf
    add    esp, 4
$LN3@f_signed:
```

```
    mov     ecx, DWORD PTR _a$[ebp]
    cmp     ecx, DWORD PTR _b$[ebp]
    jne     SHORT $LN2@f_signed
    push    OFFSET $SG739           ; 'a==b'
    call    _printf
    add     esp, 4
$LN2@f_signed:
    mov     edx, DWORD PTR _a$[ebp]
    cmp     edx, DWORD PTR _b$[ebp]
    jge     SHORT $LN4@f_signed
    push    OFFSET $SG741           ; 'a<b'
    call    _printf
    add     esp, 4
$LN4@f_signed:
    pop     ebp
    ret     0
_f_signed ENDP
```

Jump if Less or Equal. JNE: *Jump if Not Equal.* .

JGE: *Jump if Greater or Equal.*

f_unsigned()

Listing 12.2: GCC

```
_a$ = 8   ; size = 4
_b$ = 12  ; size = 4
_f_unsigned PROC
    push    ebp
    mov     ebp, esp
    mov     eax, DWORD PTR _a$[ebp]
    cmp     eax, DWORD PTR _b$[ebp]
    jbe     SHORT $LN3@f_unsigned
```

```
    push   OFFSET $SG2761    ; 'a>b'
    call   _printf
    add    esp, 4
$LN3@f_unsigned:
    mov    ecx, DWORD PTR _a$[ebp]
    cmp    ecx, DWORD PTR _b$[ebp]
    jne    SHORT $LN2@f_unsigned
    push   OFFSET $SG2763    ; 'a==b'
    call   _printf
    add    esp, 4
$LN2@f_unsigned:
    mov    edx, DWORD PTR _a$[ebp]
    cmp    edx, DWORD PTR _b$[ebp]
    jae    SHORT $LN4@f_unsigned
    push   OFFSET $SG2765    ; 'a<b'
    call   _printf
    add    esp, 4
$LN4@f_unsigned:
    pop    ebp
    ret    0
_f_unsigned ENDP
```

JBE*Jump if Below or Equal* y JAE*Jump if Above or Equal*. (JA/JAE/ JG/JGE/JL/JLE

(30 on page 866).

Listing 12.3: main()

```
_main   PROC
        push    ebp
        mov     ebp, esp
        push    2
```

```
        push    1
        call    _f_signed
        add     esp, 8
        push    2
        push    1
        call    _f_unsigned
        add     esp, 8
        xor     eax, eax
        pop     ebp
        ret     0
_main   ENDP
```

x86 + MSVC + OllyDbg

. f_unsigned()

:

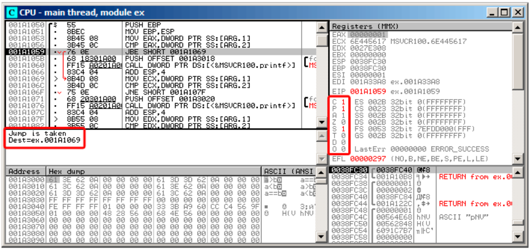


Figura 12.1: OllyDbg: f_unsigned():

: C=1, P=1, A=1, Z=0, S=1, T=0, D=0, O=0.

OllyDbg (JBE) . [Int13], JBE CF=1 o ZF=1. .

:

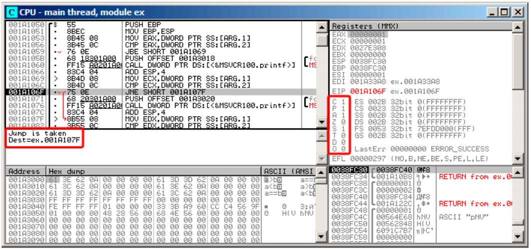


Figura 12.2: OllyDbg: `f_unsigned()`:

OllyDbg JNZ. , JNZ ZF=0 (zero flag).

JNB:

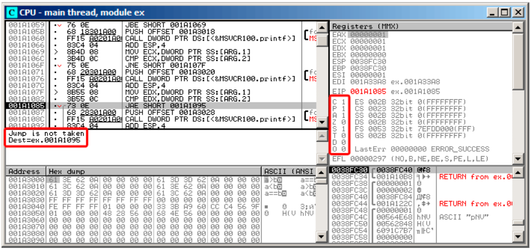


Figura 12.3: OllyDbg: `f_unsigned()`:

[Int13] JNB CF=0 (carry flag). `printf()`.

`f_signed()`.

: C=1, P=1, A=1, Z=0, S=1, T=0, D=0, O=0.

JLE:

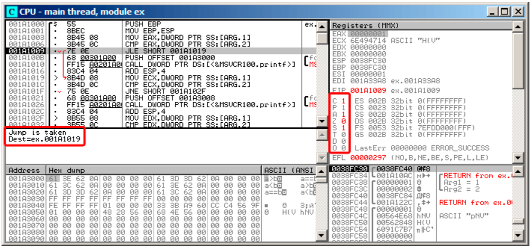


Figura 12.4: OllyDbg: `f_signed()`:

[Int13] ZF=1 o SF≠OF. SF≠OF.

JNZ: ZF=0 (zero flag):

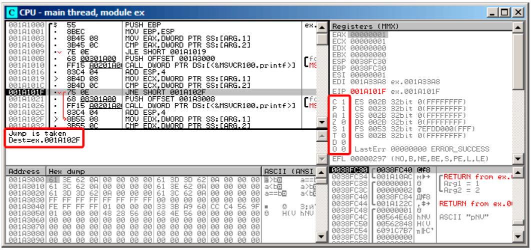


Figura 12.5: OllyDbg: `f_signed()`:

JGE SF=OF, :

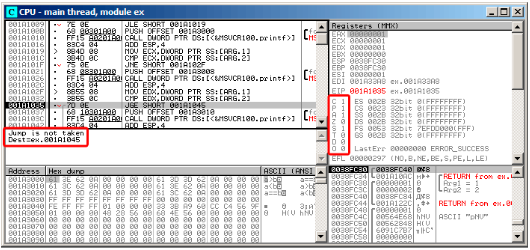


Figura 12.6: OllyDbg: `f_signed()`:

x86 + MSVC + Hiew

`f_unsigned()` «a==b», . Hiew:

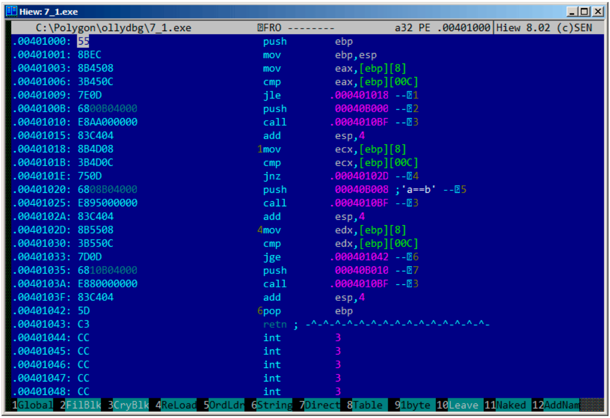


Figura 12.7: Hiew: `f_unsigned()`

:

- ;
- ;
- .

`printf()`, «a==b».

:

- JMP, .

 CAPÍTULO 12.

-
-

:

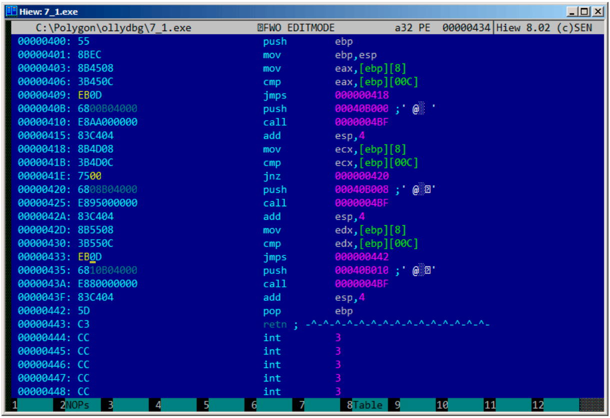


Figura 12.8: Hiew: `f_unsigned()`

Sin optimización GCC

Sin optimización GCC 4.4.1 `puts()` (3.4.3 on page 39) `printf()`.

Con optimización GCC

?

Listing 12.4: GCC 4.8.1 f_signed()

```
f_signed:
        mov     eax, DWORD PTR [esp+8]
```

```
        cmp     DWORD PTR [esp+4], eax
        jg      .L6
        je      .L7
        jge     .L1
        mov     DWORD PTR [esp+4], OFFSET ↙
    ↘ FLAT:.LC2 ; "a<b"
        jmp     puts
.L6:
        mov     DWORD PTR [esp+4], OFFSET ↙
    ↘ FLAT:.LC0 ; "a>b"
        jmp     puts
.L1:
        rep ret
.L7:
        mov     DWORD PTR [esp+4], OFFSET ↙
    ↘ FLAT:.LC1 ; "a==b"
        jmp     puts
```

`JMP puts` `CALL puts / RETN`. : 13.1.1 on page 286.

. MSVC 2012. `Jcc`

`f_unsigned()` :

Listing 12.5: GCC 4.8.1 f_unsigned()

```
f_unsigned:
        push    esi
        push    ebx
        sub     esp, 20
        mov     esi, DWORD PTR [esp+32]
        mov     ebx, DWORD PTR [esp+36]
        cmp     esi, ebx
```

```
        ja      .L13
        cmp     esi, ebx        ;
        je      .L14
.L10:
        jb      .L15
        add     esp, 20
        pop     ebx
        pop     esi
        ret
.L15:
        mov     DWORD PTR [esp+32], OFFSET ↙
    ↘ FLAT:.LC2 ; "a<b"
        add     esp, 20
        pop     ebx
        pop     esi
        jmp     puts
.L13:
        mov     DWORD PTR [esp], OFFSET FLAT↙
    ↘ :.LC0 ; "a>b"
        call    puts
        cmp     esi, ebx
        jne     .L10
.L14:
        mov     DWORD PTR [esp+32], OFFSET ↙
    ↘ FLAT:.LC1 ; "a==b"
        add     esp, 20
        pop     ebx
        pop     esi
        jmp     puts
```

12.1.2 ARM

32- ARM

Con optimización Keil 6/2013 (Modo ARM)

Listing 12.6: Con optimización Keil 6/2013 (Modo ARM)

```
.text:000000B8                                  EXPORT ↙
    ↘ f_signed
.text:000000B8                          f_signed          ↙
    ↘       ; CODE XREF: main+C
.text:000000B8 70 40 2D E9              STMFD   ↙
    ↘ SP!, {R4-R6,LR}
.text:000000BC 01 40 A0 E1              MOV     ↙
    ↘ R4, R1
.text:000000C0 04 00 50 E1              CMP     ↙
    ↘ R0, R4
.text:000000C4 00 50 A0 E1              MOV     ↙
    ↘ R5, R0
.text:000000C8 1A 0E 8F C2              ADRGT   ↙
    ↘ R0, aAB         ; "a>b\n"
.text:000000CC A1 18 00 CB              BLGT    ↙
    ↘ __2printf
.text:000000D0 04 00 55 E1              CMP     ↙
    ↘ R5, R4
.text:000000D4 67 0F 8F 02              ADREQ   ↙
    ↘ R0, aAB_0       ; "a==b\n"
.text:000000D8 9E 18 00 0B              BLEQ    ↙
    ↘ __2printf
.text:000000DC 04 00 55 E1              CMP     ↙
    ↘ R5, R4
.text:000000E0 70 80 BD A8              LDMGEFD ↙
```

```
    ↳ SP!, {R4-R6,PC}
.text:000000E4 70 40 BD E8          LDMFD   ↙
    ↳ SP!, {R4-R6,LR}
.text:000000E8 19 0E 8F E2          ADR     ↙
    ↳ R0, aAB_1       ; "a<b\n"
.text:000000EC 99 18 00 EA          B       ↙
    ↳ __2printf
.text:000000EC                 ; End of function↙
    ↳  f_signed
```

(*condition field*).

(*Greater Than*). `ADRGT a>b\n printf()`.

(*Greater or Equal*).

`LDMGEFD SP!, {R4-R6,PC}`

`printf()` «a<b\n» . «`printf()` con varios argumentos» (6.2.1 on page 104).

`f_unsigned` `ADRHI`, `BLHI`, y `LDMCSFD` (*HI* = *Unsigned higher, CS* = *Carry Set (greater than or equal)*)

Listing 12.7: `main()`

```
.text:00000128                    EXPORT main
.text:00000128               main
.text:00000128 10 40 2D E9        STMFD   SP!,↙
    ↳  {R4,LR}
.text:0000012C 02 10 A0 E3        MOV     R1, ↙
    ↳ #2
.text:00000130 01 00 A0 E3        MOV     R0, ↙
    ↳ #1
```

```
.text:00000134 DF FF FF EB       BL       ⤦
    ⤷ f_signed
.text:00000138 02 10 A0 E3       MOV      R1, ⤦
    ⤷ #2
.text:0000013C 01 00 A0 E3       MOV      R0, ⤦
    ⤷ #1
.text:00000140 EA FF FF EB       BL       ⤦
    ⤷ f_unsigned
.text:00000144 00 00 A0 E3       MOV      R0, ⤦
    ⤷ #0
.text:00000148 10 80 BD E8       LDMFD    SP!,⤦
    ⤷  {R4,PC}
.text:00000148                 ; End of function⤦
    ⤷  main
```

: 33.1 on page 873.

Con optimización Keil 6/2013 (Modo Thumb)

Listing 12.8: Con optimización Keil 6/2013 (Modo Thumb)

```
.text:00000072                 f_signed ; CODE ⤦
    ⤷ XREF: main+6
.text:00000072 70 B5            PUSH     {R4-R6,⤦
    ⤷ LR}
.text:00000074 0C 00            MOVS     R4, R1
.text:00000076 05 00            MOVS     R5, R0
.text:00000078 A0 42            CMP      R0, R4
.text:0000007A 02 DD            BLE      loc_82
.text:0000007C A4 A0            ADR      R0, aAB⤦
    ⤷          ; "a>b\n"
```

```
.text:0000007E 06 F0 B7 F8   BL       ⤦
    ↳ __2printf
.text:00000082
.text:00000082              loc_82 ; CODE ⤦
    ↳ XREF: f_signed+8
.text:00000082 A5 42         CMP      R5, R4
.text:00000084 02 D1         BNE      loc_8C
.text:00000086 A4 A0         ADR      R0, ⤦
    ↳ aAB_0        ; "a==b\n"
.text:00000088 06 F0 B2 F8   BL       ⤦
    ↳ __2printf
.text:0000008C
.text:0000008C              loc_8C ; CODE ⤦
    ↳ XREF: f_signed+12
.text:0000008C A5 42         CMP      R5, R4
.text:0000008E 02 DA         BGE      ⤦
    ↳ locret_96
.text:00000090 A3 A0         ADR      R0, ⤦
    ↳ aAB_1        ; "a<b\n"
.text:00000092 06 F0 AD F8   BL       ⤦
    ↳ __2printf
.text:00000096
.text:00000096              locret_96 ; CODE ⤦
    ↳ XREF: f_signed+1C
.text:00000096 70 BD         POP      {R4-R6,⤦
    ↳ PC}
.text:00000096              ; End of function⤦
    ↳  f_signed
```

BLE *Less than or Equal*, BNE*Not Equal*, BGE*Greater than or Equal*.

`f_unsigned` BLS (*Unsigned lower or same*) y BCS (*Carry Set (Greater than or equal)*).

ARM64: Con optimización GCC (Linaro) 4.9

Listing 12.9: f_signed()

```
f_signed:
; W0=a, W1=b
        cmp     w0, w1
        bgt     .L19    ; Branch if Greater ↙
    ↘ Than (a>b)
        beq     .L20    ; Branch if Equal (a↙
    ↘ ==b)
        bge     .L15    ; Branch if Greater ↙
    ↘ than or Equal (a>=b) ()
        ; a<b
        adrp    x0, .LC11       ; "a<b"
        add     x0, x0, :lo12:.LC11
        b       puts
.L19:
        adrp    x0, .LC9        ; "a>b"
        add     x0, x0, :lo12:.LC9
        b       puts
.L15:   ;
        ret
.L20:
        adrp    x0, .LC10       ; "a==b"
        add     x0, x0, :lo12:.LC10
        b       puts
```

Listing 12.10: f_unsigned()

```
f_unsigned:
        stp     x29, x30, [sp, -48]!
; W0=a, W1=b
        cmp     w0, w1
        add     x29, sp, 0
        str     x19, [sp,16]
        mov     w19, w0
        bhi     .L25    ; Branch if HIgher (a>b)
        cmp     w19, w1
        beq     .L26    ; Branch if Equal (a==b)
.L23:
        bcc     .L27    ; Branch if Carry Clear () (a<b)
;
        ldr     x19, [sp,16]
        ldp     x29, x30, [sp], 48
        ret
.L27:
        ldr     x19, [sp,16]
        adrp    x0, .LC11       ; "a<b"
        ldp     x29, x30, [sp], 48
        add     x0, x0, :lo12:.LC11
        b       puts
.L25:
        adrp    x0, .LC9        ; "a>b"
        str     x1, [x29,40]
        add     x0, x0, :lo12:.LC9
        bl      puts
```

```
        ldr     x1, [x29,40]
        cmp     w19, w1
        bne     .L23    ; Branch if Not ↙
    ↘ Equal
.L26:
        ldr     x19, [sp,16]
        adrp    x0, .LC10       ; "a==b"
        ldp     x29, x30, [sp], 48
        add     x0, x0, :lo12:.LC10
        b       puts
```

.

Ejercicio

12.1.3 MIPS

Listing 12.11: Sin optimización GCC 4.4.5 (IDA)

```

.text:00000000 f_signed:                      ↙
    ↘             # CODE XREF: main+18
.text:00000000
.text:00000000 var_10          = -0x10
.text:00000000 var_8           = -8
.text:00000000 var_4           = -4
.text:00000000 arg_0           =  0
.text:00000000 arg_4           =  4
.text:00000000
.text:00000000                 addiu   $sp, ↙
    ↘ -0x20
```

```
.text:00000004                  sw      $ra, ⤦
    ↳ 0x20+var_4($sp)
.text:00000008                  sw      $fp, ⤦
    ↳ 0x20+var_8($sp)
.text:0000000C                  move    $fp, ⤦
    ↳ $sp
.text:00000010                  la      $gp, ⤦
    ↳ __gnu_local_gp
.text:00000018                  sw      $gp, ⤦
    ↳ 0x20+var_10($sp)
; :
.text:0000001C                  sw      $a0, ⤦
    ↳ 0x20+arg_0($fp)
.text:00000020                  sw      $a1, ⤦
    ↳ 0x20+arg_4($fp)
; reload them.
.text:00000024                  lw      $v1, ⤦
    ↳ 0x20+arg_0($fp)
.text:00000028                  lw      $v0, ⤦
    ↳ 0x20+arg_4($fp)
; $v0=b
; $v1=a
.text:0000002C                  or      $at, ⤦
    ↳ $zero ; NOP
;  "slt $v0,$v0,$v1" .
;  $v0  $v0<$v1 (b<a) :
.text:00000030                  slt     $v0, ⤦
    ↳ $v1
; .
;  "beq $v0,$zero,loc_5c" :
.text:00000034                  beqz    $v0, ⤦
    ↳ loc_5C
;
```

```
.text:00000038                     or      $at, ↙
    ↘ $zero ; branch delay slot, NOP
.text:0000003C                     lui     $v0, ↙
    ↘ (unk_230 >> 16) # "a>b"
.text:00000040                     addiu   $a0, ↙
    ↘ $v0, (unk_230 & 0xFFFF) # "a>b"
.text:00000044                     lw      $v0, ↙
    ↘ (puts & 0xFFFF)($gp)
.text:00000048                     or      $at, ↙
    ↘ $zero ; NOP
.text:0000004C                     move    $t9, ↙
    ↘ $v0
.text:00000050                     jalr    $t9
.text:00000054                     or      $at, ↙
    ↘ $zero ; branch delay slot, NOP
.text:00000058                     lw      $gp, ↙
    ↘ 0x20+var_10($fp)
.text:0000005C
.text:0000005C loc_5C:                          ↙
    ↘             # CODE XREF: f_signed+34
.text:0000005C                     lw      $v1, ↙
    ↘ 0x20+arg_0($fp)
.text:00000060                     lw      $v0, ↙
    ↘ 0x20+arg_4($fp)
.text:00000064                     or      $at, ↙
    ↘ $zero ; NOP
; :
.text:00000068                     bne     $v1, ↙
    ↘ $v0, loc_90
.text:0000006C                     or      $at, ↙
    ↘ $zero ; branch delay slot, NOP
; :
```

```
.text:00000070                 lui     $v0, ⤦
    ↳ (aAB >> 16)  # "a==b"
.text:00000074                 addiu   $a0, ⤦
    ↳ $v0, (aAB & 0xFFFF)  # "a==b"
.text:00000078                 lw      $v0, ⤦
    ↳ (puts & 0xFFFF)($gp)
.text:0000007C                 or      $at, ⤦
    ↳ $zero ; NOP
.text:00000080                 move    $t9, ⤦
    ↳ $v0
.text:00000084                 jalr    $t9
.text:00000088                 or      $at, ⤦
    ↳ $zero ; branch delay slot, NOP
.text:0000008C                 lw      $gp, ⤦
    ↳ 0x20+var_10($fp)
.text:00000090
.text:00000090 loc_90:                                  ⤦
    ↳            # CODE XREF: f_signed+68
.text:00000090                 lw      $v1, ⤦
    ↳ 0x20+arg_0($fp)
.text:00000094                 lw      $v0, ⤦
    ↳ 0x20+arg_4($fp)
.text:00000098                 or      $at, ⤦
    ↳ $zero ; NOP
;  $v1<$v0 (a<b),  $v0 :
.text:0000009C                 slt     $v0, ⤦
    ↳ $v1, $v0
; :
.text:000000A0                 beqz    $v0, ⤦
    ↳ loc_C8
.text:000000A4                 or      $at, ⤦
    ↳ $zero ; branch delay slot, NOP
;
```

```
.text:000000A8                  lui     $v0, ↙
    ↳ (aAB_0 >> 16)  # "a<b"
.text:000000AC                  addiu   $a0, ↙
    ↳ $v0, (aAB_0 & 0xFFFF)  # "a<b"
.text:000000B0                  lw      $v0, ↙
    ↳ (puts & 0xFFFF)($gp)
.text:000000B4                  or      $at, ↙
    ↳ $zero ; NOP
.text:000000B8                  move    $t9, ↙
    ↳ $v0
.text:000000BC                  jalr    $t9
.text:000000C0                  or      $at, ↙
    ↳ $zero ; branch delay slot, NOP
.text:000000C4                  lw      $gp, ↙
    ↳ 0x20+var_10($fp)
.text:000000C8
; :
.text:000000C8 loc_C8:                         ↙
    ↳            # CODE XREF: f_signed+A0
.text:000000C8                  move    $sp, ↙
    ↳ $fp
.text:000000CC                  lw      $ra, ↙
    ↳ 0x20+var_4($sp)
.text:000000D0                  lw      $fp, ↙
    ↳ 0x20+var_8($sp)
.text:000000D4                  addiu   $sp, ↙
    ↳ 0x20
.text:000000D8                  jr      $ra
.text:000000DC                  or      $at, ↙
    ↳ $zero ; branch delay slot, NOP
.text:000000DC  # End of function f_signed
```

«SLT REG0, REG0, REG1» «SLT REG0, REG1».

Listing 12.12: Sin optimización GCC 4.4.5 (IDA)

```
.text:000000E0 f_unsigned:                                    ↙
    ↘               # CODE XREF: main+28
.text:000000E0
.text:000000E0 var_10          = -0x10
.text:000000E0 var_8           = -8
.text:000000E0 var_4           = -4
.text:000000E0 arg_0           =  0
.text:000000E0 arg_4           =  4
.text:000000E0
.text:000000E0                 addiu   $sp, ↙
    ↘ -0x20
.text:000000E4                 sw      $ra, ↙
    ↘ 0x20+var_4($sp)
.text:000000E8                 sw      $fp, ↙
    ↘ 0x20+var_8($sp)
.text:000000EC                 move    $fp, ↙
    ↘ $sp
.text:000000F0                 la      $gp, ↙
    ↘ __gnu_local_gp
.text:000000F8                 sw      $gp, ↙
    ↘ 0x20+var_10($sp)
.text:000000FC                 sw      $a0, ↙
    ↘ 0x20+arg_0($fp)
.text:00000100                 sw      $a1, ↙
    ↘ 0x20+arg_4($fp)
.text:00000104                 lw      $v1, ↙
    ↘ 0x20+arg_0($fp)
.text:00000108                 lw      $v0, ↙
    ↘ 0x20+arg_4($fp)
.text:0000010C                 or      $at, ↙
```

```
    ↳ $zero
.text:00000110                  sltu    $v0, ↙
    ↳ $v1
.text:00000114                  beqz    $v0, ↙
    ↳ loc_13C
.text:00000118                  or      $at, ↙
    ↳ $zero
.text:0000011C                  lui     $v0, ↙
    ↳ (unk_230 >> 16)
.text:00000120                  addiu   $a0, ↙
    ↳ $v0, (unk_230 & 0xFFFF)
.text:00000124                  lw      $v0, ↙
    ↳ (puts & 0xFFFF)($gp)
.text:00000128                  or      $at, ↙
    ↳ $zero
.text:0000012C                  move    $t9, ↙
    ↳ $v0
.text:00000130                  jalr    $t9
.text:00000134                  or      $at, ↙
    ↳ $zero
.text:00000138                  lw      $gp, ↙
    ↳ 0x20+var_10($fp)
.text:0000013C
.text:0000013C loc_13C:                          ↙
    ↳            # CODE XREF: f_unsigned+34
.text:0000013C                  lw      $v1, ↙
    ↳ 0x20+arg_0($fp)
.text:00000140                  lw      $v0, ↙
    ↳ 0x20+arg_4($fp)
.text:00000144                  or      $at, ↙
    ↳ $zero
.text:00000148                  bne     $v1, ↙
    ↳ $v0, loc_170
```

```
.text:0000014C                 or      $at, ⤦
    ↳ $zero
.text:00000150                 lui     $v0, ⤦
    ↳ (aAB >> 16)  # "a==b"
.text:00000154                 addiu   $a0, ⤦
    ↳ $v0, (aAB & 0xFFFF)  # "a==b"
.text:00000158                 lw      $v0, ⤦
    ↳ (puts & 0xFFFF)($gp)
.text:0000015C                 or      $at, ⤦
    ↳ $zero
.text:00000160                 move    $t9, ⤦
    ↳ $v0
.text:00000164                 jalr    $t9
.text:00000168                 or      $at, ⤦
    ↳ $zero
.text:0000016C                 lw      $gp, ⤦
    ↳ 0x20+var_10($fp)
.text:00000170
.text:00000170 loc_170:                                ⤦
    ↳                   # CODE XREF: f_unsigned+68
.text:00000170                 lw      $v1, ⤦
    ↳ 0x20+arg_0($fp)
.text:00000174                 lw      $v0, ⤦
    ↳ 0x20+arg_4($fp)
.text:00000178                 or      $at, ⤦
    ↳ $zero
.text:0000017C                 sltu    $v0, ⤦
    ↳ $v1, $v0
.text:00000180                 beqz    $v0, ⤦
    ↳ loc_1A8
.text:00000184                 or      $at, ⤦
    ↳ $zero
.text:00000188                 lui     $v0, ⤦
```

```
    ↳ (aAB_0 >> 16)  # "a<b"
.text:0000018C                   addiu   $a0, ↙
    ↳ $v0, (aAB_0 & 0xFFFF)  # "a<b"
.text:00000190                   lw      $v0, ↙
    ↳ (puts & 0xFFFF)($gp)
.text:00000194                   or      $at, ↙
    ↳ $zero
.text:00000198                   move    $t9, ↙
    ↳ $v0
.text:0000019C                   jalr    $t9
.text:000001A0                   or      $at, ↙
    ↳ $zero
.text:000001A4                   lw      $gp, ↙
    ↳ 0x20+var_10($fp)
.text:000001A8
.text:000001A8 loc_1A8:                       ↙
    ↳            # CODE XREF: f_unsigned+A0
.text:000001A8                   move    $sp, ↙
    ↳ $fp
.text:000001AC                   lw      $ra, ↙
    ↳ 0x20+var_4($sp)
.text:000001B0                   lw      $fp, ↙
    ↳ 0x20+var_8($sp)
.text:000001B4                   addiu   $sp, ↙
    ↳ 0x20
.text:000001B8                   jr      $ra
.text:000001BC                   or      $at, ↙
    ↳ $zero
.text:000001BC  # End of function f_unsigned
```

12.2

:

```
int my_abs (int i)
{
        if (i<0)
                return -i;
        else
                return i;
};
```

12.2.1 Con optimización MSVC

Listing 12.13: Con optimización MSVC 2012 x64

```


i$ = 8
my_abs  PROC
; ECX = input
        test    ecx, ecx
;
;
        jns     SHORT $LN2@my_abs
;
        neg     ecx
$LN2@my_abs:
;  EAX:
        mov     eax, ecx
        ret     0
my_abs  ENDP
```

12.2.2 Con optimización Keil 6/2013: Modo Thumb

Listing 12.14: Con optimización Keil 6/2013: Modo Thumb

```
my_abs PROC
        CMP      r0,#0
; ?
;
        BGE      |L0.6|
;  0:
        RSBS     r0,r0,#0
|L0.6|
        BX       lr
        ENDP
```

12.2.3 Con optimización Keil 6/2013: Modo ARM

Listing 12.15: Con optimización Keil 6/2013: Modo ARM

```
my_abs PROC
        CMP      r0,#0
;  "Reverse Subtract" :
        RSBLT    r0,r0,#0
        BX       lr
        ENDP
```

33.1 on page 873.

12.2.4 Sin optimización GCC 4.9 (ARM64)

Listing 12.16: Con optimización GCC 4.9 (ARM64)

```
my_abs:
        sub     sp, sp, #16
        str     w0, [sp,12]
        ldr     w0, [sp,12]
;
;
        cmp     w0, wzr
        bge     .L2
        ldr     w0, [sp,12]
        neg     w0, w0
        b       .L3
.L2:
        ldr     w0, [sp,12]
.L3:
        add     sp, sp, 16
        ret
```

12.2.5 MIPS

Listing 12.17: Con optimización GCC 4.4.5 (IDA)

```
my_abs:
;  $a0<0:
                bltz    $a0, locret_10
;  ($a0)  $v0:
                move    $v0, $a0
                jr      $ra
                or      $at, $zero ; branch ↙
    ↘ delay slot, NOP
```

```
locret_10:
;  $v0:
                jr      $ra
;  "subu $v0,$zero,$a0" ($v0=0-$a0)
                negu    $v0, $a0
```

: BLTZ («Branch if Less Than Zero»).

12.2.6 ?

45 on page 985.

12.3

C/C++:

```
expression ? expression : expression
```

:

```
const char* f (int a)
{
        return a==10 ? "it is ten" : "it is not ten";
};
```

12.3.1 x86

Listing 12.18: Sin optimización MSVC 2008

```
$SG746  DB      'it is ten', 00H
$SG747  DB      'it is not ten', 00H

tv65 = -4 ;
_a$ = 8
_f      PROC
        push    ebp
        mov     ebp, esp
        push    ecx
;  10
        cmp     DWORD PTR _a$[ebp], 10
;  $LN3@f
        jne     SHORT $LN3@f
; :
        mov     DWORD PTR tv65[ebp], OFFSET ⤦
    ↳ $SG746 ; 'it is ten'
;
        jmp     SHORT $LN4@f
$LN3@f:
; :
        mov     DWORD PTR tv65[ebp], OFFSET ⤦
    ↳ $SG747 ; 'it is not ten'
$LN4@f:
;
        mov     eax, DWORD PTR tv65[ebp]
        mov     esp, ebp
        pop     ebp
        ret     0
_f      ENDP
```

Listing 12.19: Con optimización MSVC 2008

```
$SG792  DB      'it is ten', 00H
$SG793  DB      'it is not ten', 00H

_a$ = 8 ; size = 4
_f      PROC
;  10
        cmp     DWORD PTR _a$[esp-4], 10
        mov     eax, OFFSET $SG792 ; 'it is ↙
    ↳ ten'
;  $LN4@f
        je      SHORT $LN4@f
        mov     eax, OFFSET $SG793 ; 'it is ↙
    ↳ not ten'
$LN4@f:
        ret     0
_f      ENDP
```

:

Listing 12.20: Con optimización MSVC 2012 x64

```
$SG1355 DB      'it is ten', 00H
$SG1356 DB      'it is not ten', 00H

a$ = 8
f       PROC
;
        lea     rdx, OFFSET FLAT:$SG1355 ; '↙
    ↳ it is ten'
```

```
        lea     rax, OFFSET FLAT:$SG1356 ; '↙
    ↘ it is not ten'
;  10
        cmp     ecx, 10
;  ("it is ten")
;
        cmove   rax, rdx
        ret     0
f       ENDP
```

Con optimización GCC 4.8 para x86 CMOVcc.

12.3.2 ARM

Con optimización Keil para:

Listing 12.21: Con optimización Keil 6/2013 (Modo ARM)

```

f PROC
;  10
        CMP      r0,#0xa
;
        ADREQ    r0,|L0.16| ; "it is ten"
;
        ADRNE    r0,|L0.28| ; "it is not ten↙
    ↘ "
        BX       lr
        ENDP

|L0.16|
        DCB      "it is ten",0
|L0.28|
```

```
        DCB       "it is not ten",0
```

Con optimización Keil para:

Listing 12.22: Con optimización Keil 6/2013 (Modo Thumb)

```
f PROC
;  10
        CMP       r0,#0xa
;  |L0.8|  EQual
        BEQ       |L0.8|
        ADR       r0,|L0.12| ; "it is not ten"
        BX        lr
|L0.8|
        ADR       r0,|L0.28| ; "it is ten"
        BX        lr
        ENDP

|L0.12|
        DCB       "it is not ten",0
|L0.28|
        DCB       "it is ten",0
```

12.3.3 ARM64

Con optimización GCC (Linaro) 4.9 para ARM64 :

Listing 12.23: Con optimización GCC (Linaro) 4.9

```
f:
```

```
        cmp     x0, 10
        beq     .L3             ; branch if ↙
    ↘ equal
        adrp    x0, .LC1        ; "it is ten↙
    ↘ "
        add     x0, x0, :lo12:.LC1
        ret
.L3:
        adrp    x0, .LC0        ; "it is not↙
    ↘  ten"
        add     x0, x0, :lo12:.LC0
        ret
.LC0:
        .string "it is ten"
.LC1:
        .string "it is not ten"
```

[ARM13a, p390, C5.5].

12.3.4 MIPS

:

Listing 12.24: Con optimización GCC 4.4.5 (salida de assembly)

```

$LC0:
        .ascii  "it is not ten\000"
$LC1:
        .ascii  "it is ten\000"
f:
```

```
        li      $2,10                          # 0↙
    ↘ xa
;  $a0  10, :
        beq     $4,$2,$L2
        nop ; branch delay slot

; :
        lui     $2,%hi($LC0)
        j       $31
        addiu   $2,$2,%lo($LC0)

$L2:
; :
        lui     $2,%hi($LC1)
        j       $31
        addiu   $2,$2,%lo($LC1)
```

12.3.5

```
const char* f (int a)
{
        if (a==10)
                return "it is ten";
        else
                return "it is not ten";
};
```

Listing 12.25: Con optimización GCC 4.8

```

.LC0:
```

```
        .string "it is ten"
.LC1:
        .string "it is not ten"
f:
.LFB0:
;  10
        cmp     DWORD PTR [esp+4], 10
        mov     edx, OFFSET FLAT:.LC1 ; "it is not ten"
        mov     eax, OFFSET FLAT:.LC0 ; "it is ten"
;
;
        cmovne  eax, edx
        ret
```

Con optimización Keil en Spanish text placeholder.12.21.

MSVC 2012 .

12.3.6 Conclusión

33.1 on page 873.

12.4

12.4.1 32-bit

```
int my_max(int a, int b)
{
        if (a>b)
```

```
                return a;
        else
                return b;
};

int my_min(int a, int b)
{
        if (a<b)
                return a;
        else
                return b;
};
```

Listing 12.26: Sin optimización MSVC 2013

```

_a$ = 8
_b$ = 12
_my_min PROC
        push    ebp
        mov     ebp, esp
        mov     eax, DWORD PTR _a$[ebp]
; :
        cmp     eax, DWORD PTR _b$[ebp]
; :
        jge     SHORT $LN2@my_min
;
        mov     eax, DWORD PTR _a$[ebp]
        jmp     SHORT $LN3@my_min
        jmp     SHORT $LN3@my_min ;  JMP
$LN2@my_min:
;  B
        mov     eax, DWORD PTR _b$[ebp]
```

```
$LN3@my_min:
        pop     ebp
        ret     0
_my_min ENDP

_a$ = 8
_b$ = 12
_my_max PROC
        push    ebp
        mov     ebp, esp
        mov     eax, DWORD PTR _a$[ebp]
; :
        cmp     eax, DWORD PTR _b$[ebp]
; :
        jle     SHORT $LN2@my_max
;
        mov     eax, DWORD PTR _a$[ebp]
        jmp     SHORT $LN3@my_max
        jmp     SHORT $LN3@my_max ;  JMP
$LN2@my_max:
;  B
        mov     eax, DWORD PTR _b$[ebp]
$LN3@my_max:
        pop     ebp
        ret     0
_my_max ENDP
```

Listing 12.27: Con optimización Keil 6/2013 (Modo Thumb)

```
my_max PROC
; R0=A
; R1=B
; :
        CMP      r0,r1
; :
        BGT      |L0.6|
; (A<=B) R1 (B):
        MOVS     r0,r1
|L0.6|
;
        BX       lr
        ENDP

my_min PROC
; R0=A
; R1=B
; :
        CMP      r0,r1
; :
        BLT      |L0.14|
; (A>=B) R1 (B):
        MOVS     r0,r1
|L0.14|
;
        BX       lr
        ENDP
```

Listing 12.28: Con optimización Keil 6/2013 (Modo ARM)

```
my_max PROC
; R0=A
```

```
; R1=B
; :
        CMP       r0,r1
;
;
;
        MOVLE     r0,r1
        BX        lr
        ENDP

my_min PROC
; R0=A
; R1=B
; :
        CMP       r0,r1
;
;
;
        MOVGE     r0,r1
        BX        lr
        ENDP
```

Con optimización GCC 4.8.1 MSVC 2013

Listing 12.29: Con optimización MSVC 2013

```

my_max:
        mov      edx, DWORD PTR [esp+4]
        mov      eax, DWORD PTR [esp+8]
; EDX=A
; EAX=B
; :
```

```
        cmp     edx, eax
;  A>=B,
;  ( A<B)
        cmovge  eax, edx
        ret

my_min:
        mov     edx, DWORD PTR [esp+4]
        mov     eax, DWORD PTR [esp+8]
; EDX=A
; EAX=B
; :
        cmp     edx, eax
;  A<=B,
;  ( A>B)
        cmovle  eax, edx
        ret
```

12.4.2 64-bit

```
#include <stdint.h>

int64_t my_max(int64_t a, int64_t b)
{
        if (a>b)
                return a;
        else
                return b;
};

int64_t my_min(int64_t a, int64_t b)
{
```

```
        if (a<b)
                return a;
        else
                return b;
};
```

Listing 12.30: Sin optimización GCC 4.9.1 ARM64

```
my_max:
        sub     sp, sp, #16
        str     x0, [sp,8]
        str     x1, [sp]
        ldr     x1, [sp,8]
        ldr     x0, [sp]
        cmp     x1, x0
        ble     .L2
        ldr     x0, [sp,8]
        b       .L3
.L2:
        ldr     x0, [sp]
.L3:
        add     sp, sp, 16
        ret

my_min:
        sub     sp, sp, #16
        str     x0, [sp,8]
        str     x1, [sp]
        ldr     x1, [sp,8]
        ldr     x0, [sp]
        cmp     x1, x0
        bge     .L5
        ldr     x0, [sp,8]
```

```
        b       .L6
.L5:
        ldr     x0, [sp]
.L6:
        add     sp, sp, 16
        ret
```

Listing 12.31: Con optimización GCC 4.9.1 x64

```
my_max:
; RDI=A
; RSI=B
; :
        cmp     rdi, rsi
; :
        mov     rax, rsi
;  A>=B,  A (RDI)  RAX .
;  ( A<B)
        cmovge  rax, rdi
        ret

my_min:
; RDI=A
; RSI=B
; :
        cmp     rdi, rsi
; :
        mov     rax, rsi
;  A<=B,  A (RDI)  RAX .
```

```
; ( A>B)
        cmovle  rax, rdi
        ret
```

MSVC 2013 .

«Conditional SELect».

Listing 12.32: Con optimización GCC 4.9.1 ARM64

```

my_max:
; X0=A
; X1=B
; :
        cmp     x0, x1
;
;   A<B
        csel    x0, x0, x1, ge
        ret

my_min:
; X0=A
; X1=B
; :
        cmp     x0, x1
;
;   A>B
        csel    x0, x0, x1, le
        ret
```

12.4.3 MIPS

:

Listing 12.33: Con optimización GCC 4.4.5 (IDA)

```
my_max:
;  $v1  $a1<$a0:
                slt     $v1, $a1, $a0
;  $a1<$a0:
                beqz    $v1, locret_10
;  branch delay slot
;  $a1  $v0 :
                move    $v0, $a1
;  $a0  $v0:
                move    $v0, $a0

locret_10:
                jr      $ra
                or      $at, $zero ; branch ↙
    ↘ delay slot, NOP

; :
my_min:
                slt     $v1, $a0, $a1
                beqz    $v1, locret_28
                move    $v0, $a1
                move    $v0, $a0

locret_28:
                jr      $ra
                or      $at, $zero ; branch ↙
    ↘ delay slot, NOP
```

12.5 Conclusión

12.5.1 x86

:

Listing 12.34: x86

```
CMP register, register/value
Jcc true ; cc=
false:
...  ...
JMP exit
true:
...  ...
exit:
```

12.5.2 ARM

Listing 12.35: ARM

```
CMP register, register/value
Bcc true ; cc=
false:
...  ...
JMP exit
true:
```

```
...  ...
exit:
```

12.5.3 MIPS

Listing 12.36:

```
BEQZ REG, label
...
```

Listing 12.37:

```
BLTZ REG, label
...
```

Listing 12.38:

```
BEQ REG1, REG2, label
...
```

Listing 12.39:

```
BNE REG1, REG2, label
...
```

Listing 12.40:

```
SLT REG1, REG2, REG3
BEQ REG1, label
...
```

Listing 12.41:

```
SLTU REG1, REG2, REG3
BEQ REG1, label
...
```

12.5.4

ARM

Listing 12.42: ARM (Modo ARM)

```

CMP register, register/value
instr1_cc ;
instr2_cc ;
... ...
```

: 17.7.2 on page 471.

Listing 12.43: ARM (Modo Thumb)

```

CMP register, register/value
ITEEE EQ ; : if-then-else-else-else
instr1   ;
instr2   ;
instr3   ;
instr4   ;
```

12.6 Ejercicio

(ARM64) Spanish text placeholder.12.23 .

 CAPÍTULO 13. SWITCH()/CASE/DEFAULT

Capítulo 13

switch()/case/default

13.1

```
#include <stdio.h>

void f (int a)
{
    switch (a)
    {
    case 0: printf ("zero\n"); break;
    case 1: printf ("one\n"); break;
    case 2: printf ("two\n"); break;
    default: printf ("something unknown\n");
    ↳ break;
    };
```

```
};

int main()
{
    f (2); // test
};
```

13.1.1 x86

Sin optimización MSVC

(MSVC 2010):

Listing 13.1: MSVC 2010

```
tv64 = -4 ; size = 4
_a$ = 8   ; size = 4
_f    PROC
    push   ebp
    mov    ebp, esp
    push   ecx
    mov    eax, DWORD PTR _a$[ebp]
    mov    DWORD PTR tv64[ebp], eax
    cmp    DWORD PTR tv64[ebp], 0
    je     SHORT $LN4@f
    cmp    DWORD PTR tv64[ebp], 1
    je     SHORT $LN3@f
    cmp    DWORD PTR tv64[ebp], 2
    je     SHORT $LN2@f
    jmp    SHORT $LN1@f
$LN4@f:
    push   OFFSET $SG739 ; 'zero', 0aH, 00H
```

```
    call   _printf
    add    esp, 4
    jmp    SHORT $LN7@f
$LN3@f:
    push   OFFSET $SG741 ; 'one', 0aH, 00H
    call   _printf
    add    esp, 4
    jmp    SHORT $LN7@f
$LN2@f:
    push   OFFSET $SG743 ; 'two', 0aH, 00H
    call   _printf
    add    esp, 4
    jmp    SHORT $LN7@f
$LN1@f:
    push   OFFSET $SG745 ; 'something ↙
    ↘ unknown', 0aH, 00H
    call   _printf
    add    esp, 4
$LN7@f:
    mov    esp, ebp
    pop    ebp
    ret    0
_f    ENDP
```

```
void f (int a)
{
    if (a==0)
        printf ("zero\n");
    else if (a==1)
        printf ("one\n");
    else if (a==2)
        printf ("two\n");
```

```
    else
        printf ("something unknown\n");
};
```

Con optimización MSVC

MSVC (/Ox): `cl 1.c /Fa1.asm /Ox`

Listing 13.2: MSVC

```
_a$ = 8 ; size = 4
_f    PROC
    mov    eax, DWORD PTR _a$[esp-4]
    sub    eax, 0
    je     SHORT $LN4@f
    sub    eax, 1
    je     SHORT $LN3@f
    sub    eax, 1
    je     SHORT $LN2@f
    mov    DWORD PTR _a$[esp-4], OFFSET ⤦
    ↳ $SG791 ; 'something unknown', 0aH, 00H
    jmp    _printf
$LN2@f:
    mov    DWORD PTR _a$[esp-4], OFFSET ⤦
    ↳ $SG789 ; 'two', 0aH, 00H
    jmp    _printf
$LN3@f:
    mov    DWORD PTR _a$[esp-4], OFFSET ⤦
    ↳ $SG787 ; 'one', 0aH, 00H
    jmp    _printf
$LN4@f:
    mov    DWORD PTR _a$[esp-4], OFFSET ⤦
    ↳ $SG785 ; 'zero', 0aH, 00H
```

```
    jmp    _printf
_f     ENDP
```

- ESP RA
- ESP+4

«printf() con varios argumentos» (6.2.1 on page 104).

 CAPÍTULO 13. SWITCH()/CASE/DEFAULT

OllyDbg

OllyDbg.

OllyDbg. EAX 2, :

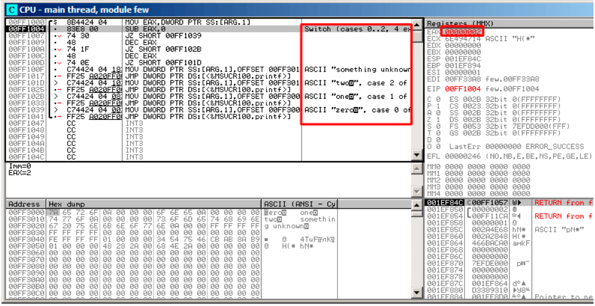


Figura 13.1: OllyDbg: EAX

0 2 en EAX. , EAX 2. ZF 0, :

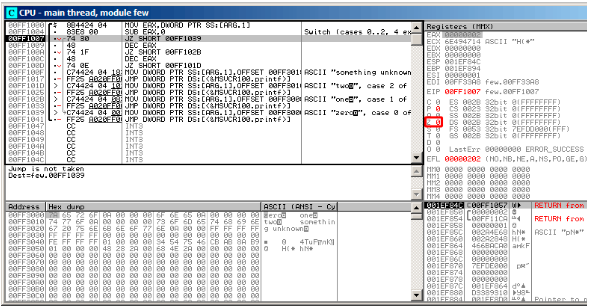


Figura 13.2: OllyDbg: SUB

DEC EAX 1. 1 ZF 0:

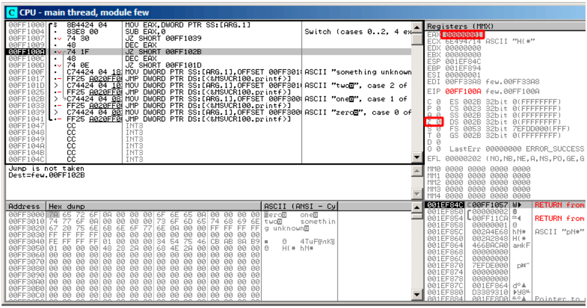


Figura 13.3: OllyDbg: DEC

DEC. EAX 0 ZF:

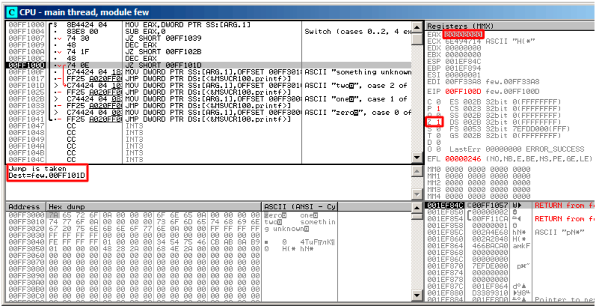


Figura 13.4: OllyDbg: DEC

OllyDbg

 CAPÍTULO 13. SWITCH()/CASE/DEFAULT

«two» :

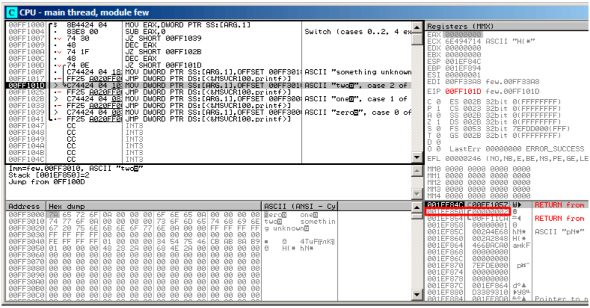


Figura 13.5: OllyDbg:

0x001EF850.

 CAPÍTULO 13. SWITCH()/CASE/DEFAULT

MOV 0x001EF850 (). . `printf()` MSVCR100.DLL ():

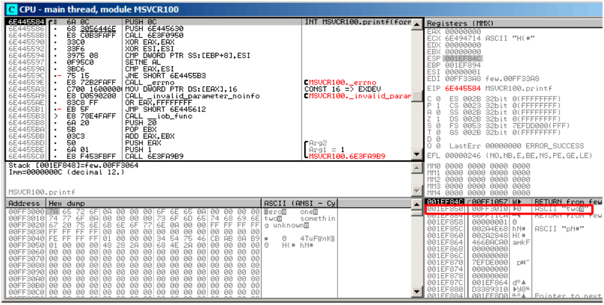


Figura 13.6: OllyDbg: `printf()` en MSVCR100.DLL

`printf()` 0x00FF3010 .

`printf()`:

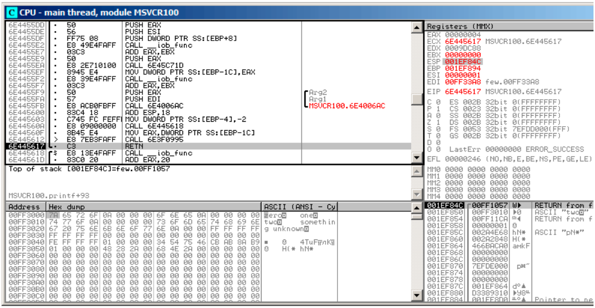


Figura 13.7: OllyDbg: `printf()` en MSVCR100.DLL

«two» .

 CAPÍTULO 13. SWITCH()/CASE/DEFAULT

F7 o F8 (pasar por encima) ... `f()` `main()`:

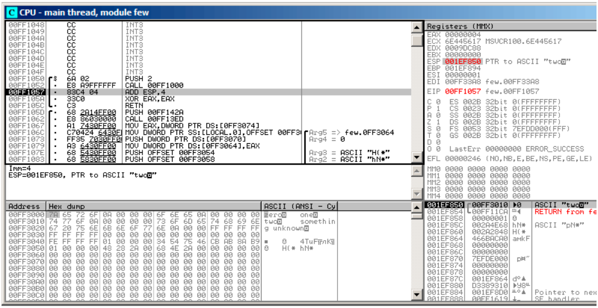


Figura 13.8: OllyDbg: `main()`

`printf()` `main()`. RA `f()` `main()`. CALL 0x00FF1000 `f()`.

13.1.2 ARM: Con optimización Keil 6/2013 (Modo ARM)

```
.text:0000014C                  f1:
.text:0000014C 00 00 50 E3        CMP      R0, #0
.text:00000150 13 0E 8F 02        ADREQ    R0, ↙
    ↳ aZero ; "zero\n"
.text:00000154 05 00 00 0A        BEQ      loc_170
.text:00000158 01 00 50 E3        CMP      R0, #1
.text:0000015C 4B 0F 8F 02        ADREQ    R0, ↙
    ↳ aOne  ; "one\n"
.text:00000160 02 00 00 0A        BEQ      loc_170
```

```
.text:00000164 02 00 50 E3    CMP      R0, #2
.text:00000168 4A 0F 8F 12    ADRNE    R0, ↙
    ↳ aSomethingUnkno ; "something unknown\n↙
    ↳ "
.text:0000016C 4E 0F 8F 02    ADREQ    R0, ↙
    ↳ aTwo   ; "two\n"
.text:00000170
.text:00000170               loc_170: ; CODE ↙
    ↳ XREF: f1+8
.text:00000170                        ; f1+14
.text:00000170 78 18 00 EA    B          ↙
    ↳ __2printf
```

BEQ loc_170, $R0 = 0$.

printf() (6.2.1 on page 104). printf().

CMP R0, #2 $R0 = 2$, *«two\n»* .

13.1.3 ARM: Con optimización Keil 6/2013 (Modo Thumb)

```
.text:000000D4               f1:
.text:000000D4 10 B5         PUSH     {R4,LR}
.text:000000D6 00 28         CMP      R0, #0
.text:000000D8 05 D0         BEQ      ↙
    ↳ zero_case
.text:000000DA 01 28         CMP      R0, #1
.text:000000DC 05 D0         BEQ      one_case
.text:000000DE 02 28         CMP      R0, #2
.text:000000E0 05 D0         BEQ      two_case
```

```
.text:000000E2 91 A0           ADR     R0, ⤦
    ↳ aSomethingUnkno ; "something unknown\n⤦
    ↳ "
.text:000000E4 04 E0           B       ⤦
    ↳ default_case

.text:000000E6                 zero_case: ; ⤦
    ↳ CODE XREF: f1+4
.text:000000E6 95 A0           ADR     R0, ⤦
    ↳ aZero ; "zero\n"
.text:000000E8 02 E0           B       ⤦
    ↳ default_case

.text:000000EA                 one_case: ; CODE⤦
    ↳  XREF: f1+8
.text:000000EA 96 A0           ADR     R0, aOne⤦
    ↳  ; "one\n"
.text:000000EC 00 E0           B       ⤦
    ↳ default_case

.text:000000EE                 two_case: ; CODE⤦
    ↳  XREF: f1+C
.text:000000EE 97 A0           ADR     R0, aTwo⤦
    ↳  ; "two\n"
.text:000000F0                 default_case ; ⤦
    ↳ CODE XREF: f1+10
.text:000000F0                              ⤦
    ↳        ; f1+14
.text:000000F0 06 F0 7E F8  BL      ⤦
    ↳ __2printf
.text:000000F4 10 BD           POP     {R4,PC}
```

13.1.4 ARM64: Sin optimización GCC (Linaro) 4.9

```
.LC12:
        .string "zero"
.LC13:
        .string "one"
.LC14:
        .string "two"
.LC15:
        .string "something unknown"
f12:
        stp     x29, x30, [sp, -32]!
        add     x29, sp, 0
        str     w0, [x29,28]
        ldr     w0, [x29,28]
        cmp     w0, 1
        beq     .L34
        cmp     w0, 2
        beq     .L35
        cmp     w0, wzr
        bne     .L38            ;
        adrp    x0, .LC12       ; "zero"
        add     x0, x0, :lo12:.LC12
        bl      puts
        b       .L32
.L34:
        adrp    x0, .LC13       ; "one"
        add     x0, x0, :lo12:.LC13
        bl      puts
        b       .L32
.L35:
        adrp    x0, .LC14       ; "two"
```

```
        add     x0, x0, :lo12:.LC14
        bl      puts
        b       .L32
.L38:
        adrp    x0, .LC15          ; "something ↙
    ↘  unknown"
        add     x0, x0, :lo12:.LC15
        bl      puts
        nop
.L32:
        ldp     x29, x30, [sp], 32
        ret
```

«Hello, world!»: 3.4.5 on page 44.

13.1.5 ARM64: Con optimización GCC (Linaro) 4.9

```

f12:
        cmp     w0, 1
        beq     .L31
        cmp     w0, 2
        beq     .L32
        cbz     w0, .L35
;
        adrp    x0, .LC15          ; "something ↙
    ↘  unknown"
        add     x0, x0, :lo12:.LC15
        b       puts
.L35:
        adrp    x0, .LC12          ; "zero"
```

```
        add     x0, x0, :lo12:.LC12
        b       puts
.L32:
        adrp    x0, .LC14        ; "two"
        add     x0, x0, :lo12:.LC14
        b       puts
.L31:
        adrp    x0, .LC13        ; "one"
        add     x0, x0, :lo12:.LC13
        b       puts
```

. CBZ (*Compare and Branch on Zero*) `W0` . `puts()` 13.1.1 on page 286.

13.1.6 MIPS

Listing 13.3: Con optimización GCC 4.4.5 (IDA)

```

f:
                lui     $gp, (__gnu_local_gp ↙
    ↘  >> 16)
;   1?
                li      $v0, 1
                beq     $a0, $v0, loc_60
                la      $gp, (__gnu_local_gp ↙
    ↘  & 0xFFFF) ; branch delay slot
;   2?
                li      $v0, 2
                beq     $a0, $v0, loc_4C
                or      $at, $zero ; branch ↙
    ↘ delay slot, NOP
```

 CAPÍTULO 13. SWITCH()/CASE/DEFAULT

```
; 0:
                bnez    $a0, loc_38
                or      $at, $zero ; branch ⤦
    ↳ delay slot, NOP
; :
                lui     $a0, ($LC0 >> 16)  #⤦
    ↳  "zero"
                lw      $t9, (puts & 0xFFFF)⤦
    ↳ ($gp)
                or      $at, $zero ; load ⤦
    ↳ delay slot, NOP
                jr      $t9 ; branch delay ⤦
    ↳ slot, NOP
                la      $a0, ($LC0 & 0xFFFF)⤦
    ↳   # "zero" ; branch delay slot
 # ⤦
    ↳ ---------------------------------------
    ↳

loc_38:                                 # ⤦
    ↳ CODE XREF: f+1C
                lui     $a0, ($LC3 >> 16)  #⤦
    ↳  "something unknown"
                lw      $t9, (puts & 0xFFFF)⤦
    ↳ ($gp)
                or      $at, $zero ; load ⤦
    ↳ delay slot, NOP
                jr      $t9
                la      $a0, ($LC3 & 0xFFFF)⤦
    ↳   # "something unknown" ; branch delay⤦
    ↳  slot
 # ⤦
    ↳ ---------------------------------------
```

 CAPÍTULO 13. SWITCH()/CASE/DEFAULT

```
    ↪

loc_4C:                                        # ↙
    ↪ CODE XREF: f+14
                lui     $a0, ($LC2 >> 16)  #↙
    ↪  "two"
                lw      $t9, (puts & 0xFFFF)↙
    ↪ ($gp)
                or      $at, $zero ; load ↙
    ↪ delay slot, NOP
                jr      $t9
                la      $a0, ($LC2 & 0xFFFF)↙
    ↪   # "two" ; branch delay slot
 # ↙
    ↪ ----------------------------------------
    ↪

loc_60:                                        # ↙
    ↪ CODE XREF: f+8
                lui     $a0, ($LC1 >> 16)  #↙
    ↪  "one"
                lw      $t9, (puts & 0xFFFF)↙
    ↪ ($gp)
                or      $at, $zero ; load ↙
    ↪ delay slot, NOP
                jr      $t9
                la      $a0, ($LC1 & 0xFFFF)↙
    ↪   # "one" ; branch delay slot
```

: 13.1.1 on page 286.

«load delay slot»: . .

CAPÍTULO 13. SWITCH()/CASE/DEFAULT

13.1.7 Conclusión

Spanish text placeholder.13.1.1.

13.2

```
#include <stdio.h>

void f (int a)
{
    switch (a)
    {
    case 0: printf ("zero\n"); break;
    case 1: printf ("one\n"); break;
    case 2: printf ("two\n"); break;
    case 3: printf ("three\n"); break;
    case 4: printf ("four\n"); break;
    default: printf ("something unknown\n");
    ↪ break;
    };
};

int main()
{
    f (2); // test
};
```

13.2.1 x86

Sin optimización MSVC

(MSVC 2010):

Listing 13.4: MSVC 2010

```
tv64 = -4  ; size = 4
_a$ = 8    ; size = 4
_f    PROC
    push   ebp
    mov    ebp, esp
    push   ecx
    mov    eax, DWORD PTR _a$[ebp]
    mov    DWORD PTR tv64[ebp], eax
    cmp    DWORD PTR tv64[ebp], 4
    ja     SHORT $LN1@f
    mov    ecx, DWORD PTR tv64[ebp]
    jmp    DWORD PTR $LN11@f[ecx*4]
$LN6@f:
    push   OFFSET $SG739 ; 'zero', 0aH, 00H
    call   _printf
    add    esp, 4
    jmp    SHORT $LN9@f
$LN5@f:
    push   OFFSET $SG741 ; 'one', 0aH, 00H
    call   _printf
    add    esp, 4
    jmp    SHORT $LN9@f
$LN4@f:
    push   OFFSET $SG743 ; 'two', 0aH, 00H
    call   _printf
```

```
    add     esp, 4
    jmp     SHORT $LN9@f
$LN3@f:
    push    OFFSET $SG745 ; 'three', 0aH, 00H
    call    _printf
    add     esp, 4
    jmp     SHORT $LN9@f
$LN2@f:
    push    OFFSET $SG747 ; 'four', 0aH, 00H
    call    _printf
    add     esp, 4
    jmp     SHORT $LN9@f
$LN1@f:
    push    OFFSET $SG749 ; 'something ↙
    ↘ unknown', 0aH, 00H
    call    _printf
    add     esp, 4
$LN9@f:
    mov     esp, ebp
    pop     ebp
    ret     0
    npad     2 ;
$LN11@f:
    DD     $LN6@f ; 0
    DD     $LN5@f ; 1
    DD     $LN4@f ; 2
    DD     $LN3@f ; 3
    DD     $LN2@f ; 4
_f     ENDP
```

jumptable o *branch table*[1].

[1] *computed GOTO* : wikipedia. !

 CAPÍTULO 13. SWITCH()/CASE/DEFAULT

npad (88 on page 1747)

OllyDbg

OllyDbg. (2) EAX:

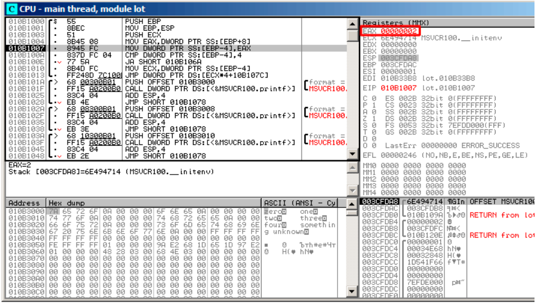


Figura 13.9: OllyDbg: EAX

 CAPÍTULO 13. SWITCH()/CASE/DEFAULT

4? :

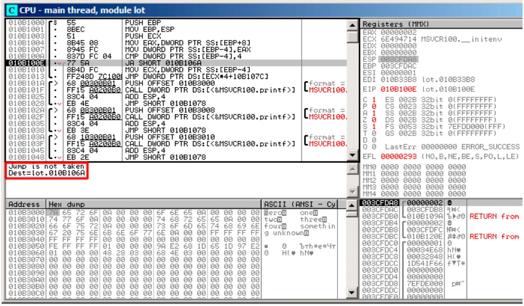


Figura 13.10: OllyDbg: 2 4:

jumptable:

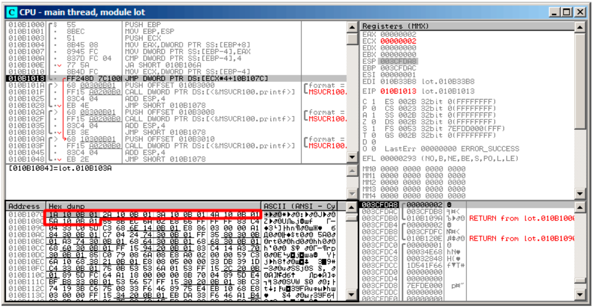


Figura 13.11: OllyDbg: jumptable

[2]. ECX 2. «Follow in Dump» → «Memory address» y OllyDbg. 0x010B103A.

[2]: 68.2.6 on page 1354,

 CAPÍTULO 13. SWITCH()/CASE/DEFAULT

0x010B103A: «two»:

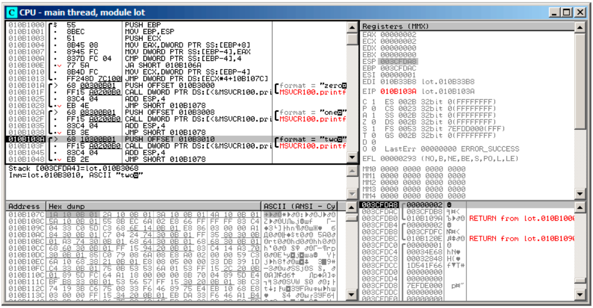


Figura 13.12: OllyDbg: *case:*

Sin optimización GCC

:

Listing 13.5: GCC 4.4.1

```
        public f
f       proc near ; CODE XREF: main+10

var_18 = dword ptr -18h
arg_0  = dword ptr  8

        push    ebp
        mov     ebp, esp
        sub     esp, 18h
        cmp     [ebp+arg_0], 4
        ja      short loc_8048444
```

```
        mov     eax, [ebp+arg_0]
        shl     eax, 2
        mov     eax, ds:off_804855C[eax]
        jmp     eax

loc_80483FE: ; DATA XREF: .rodata:⤦
    ↳ off_804855C
        mov     [esp+18h+var_18], offset ⤦
    ↳ aZero ; "zero"
        call    _puts
        jmp     short locret_8048450

loc_804840C: ; DATA XREF: .rodata:08048560
        mov     [esp+18h+var_18], offset aOne⤦
    ↳  ; "one"
        call    _puts
        jmp     short locret_8048450

loc_804841A: ; DATA XREF: .rodata:08048564
        mov     [esp+18h+var_18], offset aTwo⤦
    ↳  ; "two"
        call    _puts
        jmp     short locret_8048450

loc_8048428: ; DATA XREF: .rodata:08048568
        mov     [esp+18h+var_18], offset ⤦
    ↳ aThree ; "three"
        call    _puts
        jmp     short locret_8048450

loc_8048436: ; DATA XREF: .rodata:0804856C
        mov     [esp+18h+var_18], offset ⤦
    ↳ aFour ; "four"
```

```
        call    _puts
        jmp     short locret_8048450

loc_8048444: ; CODE XREF: f+A
        mov     [esp+18h+var_18], offset ↙
    ↘ aSomethingUnkno ; "something unknown"
        call    _puts

locret_8048450: ; CODE XREF: f+26
                ; f+34...
        leave
        retn
f       endp

off_804855C dd offset loc_80483FE   ; DATA ↙
    ↘ XREF: f+12
            dd offset loc_804840C
            dd offset loc_804841A
            dd offset loc_8048428
            dd offset loc_8048436
```

13.2.2 ARM: Con optimización Keil 6/2013 (Modo ARM)

Listing 13.6: Con optimización Keil 6/2013 (Modo ARM)

```
00000174                f2
00000174 05 00 50 E3     CMP     R0, #5       ↙
    ↘       ; switch 5 cases
00000178 00 F1 8F 30     ADDCC   PC, PC, R0,↙
    ↘ LSL#2 ; switch jump
```

```
0000017C 0E 00 00 EA     B          ↙
    ↳ default_case     ; jumptable 00000178 ↙
    ↳ default case

00000180
00000180                 loc_180 ; CODE XREF: f2↙
    ↳ +4
00000180 03 00 00 EA     B          zero_case  ↙
    ↳      ; jumptable 00000178 case 0

00000184
00000184                 loc_184 ; CODE XREF: f2↙
    ↳ +4
00000184 04 00 00 EA     B          one_case   ↙
    ↳      ; jumptable 00000178 case 1

00000188
00000188                 loc_188 ; CODE XREF: f2↙
    ↳ +4
00000188 05 00 00 EA     B          two_case   ↙
    ↳      ; jumptable 00000178 case 2

0000018C
0000018C                 loc_18C ; CODE XREF: f2↙
    ↳ +4
0000018C 06 00 00 EA     B          three_case ↙
    ↳      ; jumptable 00000178 case 3

00000190
00000190                 loc_190 ; CODE XREF: f2↙
    ↳ +4
00000190 07 00 00 EA     B          four_case  ↙
    ↳      ; jumptable 00000178 case 4
```

```
00000194
00000194                zero_case ; CODE XREF: ↙
    ↘ f2+4
00000194                             ; f2:loc_180
00000194 EC 00 8F E2     ADR     R0, aZero  ↙
    ↘       ; jumptable 00000178 case 0
00000198 06 00 00 EA     B       loc_1B8

0000019C
0000019C                one_case ; CODE XREF: ↙
    ↘ f2+4
0000019C                            ; f2:loc_184
0000019C EC 00 8F E2     ADR     R0, aOne   ↙
    ↘       ; jumptable 00000178 case 1
000001A0 04 00 00 EA     B       loc_1B8

000001A4
000001A4                two_case ; CODE XREF: ↙
    ↘ f2+4
000001A4                            ; f2:loc_188
000001A4 01 0C 8F E2     ADR     R0, aTwo   ↙
    ↘       ; jumptable 00000178 case 2
000001A8 02 00 00 EA     B       loc_1B8

000001AC
000001AC                three_case ; CODE XREF:↙
    ↘  f2+4
000001AC                             ; f2:loc_18C
000001AC 01 0C 8F E2     ADR     R0, aThree ↙
    ↘       ; jumptable 00000178 case 3
000001B0 00 00 00 EA     B       loc_1B8
```

```
000001B4
000001B4             four_case ; CODE XREF: ↙
    ↘ f2+4
000001B4                       ; f2:loc_190
000001B4 01 0C 8F E2     ADR     R0, aFour   ↙
    ↘     ; jumptable 00000178 case 4
000001B8
000001B8             loc_1B8   ; CODE XREF: ↙
    ↘ f2+24
000001B8                       ; f2+2C
000001B8 66 18 00 EA     B       __2printf

000001BC
000001BC             default_case ; CODE ↙
    ↘ XREF: f2+4
000001BC                          ; f2+8
000001BC D4 00 8F E2     ADR     R0, ↙
    ↘ aSomethingUnkno ; jumptable 00000178 ↙
    ↘ default case
000001C0 FC FF FF EA     B       loc_1B8
```

CMP R0, #5

ADDCC PC, PC, R0,LSL#2 [3] $R0 < 5$ (*CC=Carry clear / Less than*). ADDCC ($R0 \geq 5$), *default_case*.

$R0 < 5$ y ADDCC

. , LSL#2 (16.2.1 on page 407) «Shifts»,

$R0 * 4$,

[3]ADD

ADDCC, (0x180) ADDCC (0x178),

$a = 1,\ PC + 8 + a * 4 = PC + 8 + 1 * 4 = PC + 12 = 0x184.$

switch(). Spanish text placeholder.

13.2.3 ARM: Con optimización Keil 6/2013 (Modo Thumb)

Listing 13.7: Con optimización Keil 6/2013 (Modo Thumb)

```
000000F6                              EXPORT f2
000000F6                          f2
000000F6 10 B5                        PUSH    {R4,↙
    ↘ LR}
000000F8 03 00                        MOVS    R3, ↙
    ↘ R0
000000FA 06 F0 69 F8                  BL      ↙
    ↘ __ARM_common_switch8_thumb ; switch 6 ↙
    ↘ cases

000000FE 05                           DCB 5
000000FF 04 06 08 0A 0C 10            DCB 4, 6, 8, ↙
    ↘ 0xA, 0xC, 0x10 ; jump table for switch↙
    ↘  statement
00000105 00                           ALIGN 2
00000106
00000106                          zero_case ; CODE ↙
    ↘ XREF: f2+4
00000106 8D A0                        ADR     R0, ↙
    ↘ aZero ; jumptable 000000FA case 0
00000108 06 E0                        B       ↙
    ↘ loc_118
```

```
0000010A
0000010A                      one_case ; CODE ⤦
    ↳ XREF: f2+4
0000010A 8E A0                    ADR     R0, ⤦
    ↳ aOne ; jumptable 000000FA case 1
0000010C 04 E0                    B       ⤦
    ↳ loc_118

0000010E
0000010E                      two_case ; CODE ⤦
    ↳ XREF: f2+4
0000010E 8F A0                    ADR     R0, ⤦
    ↳ aTwo ; jumptable 000000FA case 2
00000110 02 E0                    B       ⤦
    ↳ loc_118

00000112
00000112                      three_case ; CODE⤦
    ↳  XREF: f2+4
00000112 90 A0                    ADR     R0, ⤦
    ↳ aThree ; jumptable 000000FA case 3
00000114 00 E0                    B       ⤦
    ↳ loc_118

00000116
00000116                      four_case ; CODE ⤦
    ↳ XREF: f2+4
00000116 91 A0                    ADR     R0, ⤦
    ↳ aFour ; jumptable 000000FA case 4
00000118
00000118                      loc_118 ; CODE ⤦
    ↳ XREF: f2+12
```

```
00000118                                ; f2+16
00000118 06 F0 6A F8            BL       ↙
    ↘ __2printf
0000011C 10 BD                  POP      {R4,↙
    ↘ PC}

0000011E
0000011E                    default_case ; ↙
    ↘ CODE XREF: f2+4
0000011E 82 A0                  ADR      R0, ↙
    ↘ aSomethingUnkno ; jumptable 000000FA ↙
    ↘ default case
00000120 FA E7                  B        ↙
    ↘ loc_118


000061D0                        EXPORT ↙
    ↘ __ARM_common_switch8_thumb
000061D0                    ↙
    ↘ __ARM_common_switch8_thumb ; CODE XREF↙
    ↘ : example6_f2+4
000061D0 78 47                  BX       PC

000061D2 00 00                  ALIGN 4
000061D2                    ; End of function↙
    ↘  __ARM_common_switch8_thumb
000061D2
000061D4                    ↙
    ↘ __32__ARM_common_switch8_thumb ; CODE ↙
    ↘ XREF: __ARM_common_switch8_thumb
000061D4 01 C0 5E E5            LDRB     R12, ↙
    ↘ [LR,#-1]
```

```
000061D8 0C 00 53 E1                 CMP      R3, ↙
    ↘ R12
000061DC 0C 30 DE 27                 LDRCSB   R3, [↙
    ↘ LR,R12]
000061E0 03 30 DE 37                 LDRCCB   R3, [↙
    ↘ LR,R3]
000061E4 83 C0 8E E0                 ADD      R12, ↙
    ↘ LR, R3,LSL#1
000061E8 1C FF 2F E1                 BX       R12
000061E8                         ; End of function↙
    ↘  __32__ARM_common_switch8_thumb
```

__ARM_common_switch8_thumb. `BX PC`

`BL __ARM_common_switch8_thumb`

IDA
`jumptable 000000FA case 0`.

13.2.4 MIPS

Listing 13.8: Con optimización GCC 4.4.5 (IDA)

```
f:
                lui     $gp, (__gnu_local_gp↙
    ↘  >> 16)
;
                sltiu   $v0, $a0, 5
                bnez    $v0, loc_24
                la      $gp, (__gnu_local_gp↙
    ↘  & 0xFFFF) ; branch delay slot
;
```

```
; "something unknown" :
                lui     $a0, ($LC5 >> 16)  #↙
    ↘ "something unknown"
                lw      $t9, (puts & 0xFFFF)↙
    ↘ ($gp)
                or      $at, $zero ; NOP
                jr      $t9
                la      $a0, ($LC5 & 0xFFFF)↙
    ↘  # "something unknown" ; branch delay↙
    ↘ slot

loc_24:                                     # ↙
    ↘ CODE XREF: f+8
;
;
                la      $v0, off_120
; 4:
                sll     $a0, 2
; :
                addu    $a0, $v0, $a0
; :
                lw      $v0, 0($a0)
                or      $at, $zero ; NOP
; :
                jr      $v0
                or      $at, $zero ; branch ↙
    ↘ delay slot, NOP

sub_44:                                     # ↙
    ↘ DATA XREF: .rodata:0000012C
; "three"
                lui     $a0, ($LC3 >> 16)  #↙
    ↘ "three"
```

```
                lw      $t9, (puts & 0xFFFF)↙
    ↳ ($gp)
                or      $at, $zero ; NOP
                jr      $t9
                la      $a0, ($LC3 & 0xFFFF)↙
    ↳   # "three" ; branch delay slot

sub_58:                                    # ↙
    ↳ DATA XREF: .rodata:00000130
;  "four"
                lui     $a0, ($LC4 >> 16)  #↙
    ↳  "four"
                lw      $t9, (puts & 0xFFFF)↙
    ↳ ($gp)
                or      $at, $zero ; NOP
                jr      $t9
                la      $a0, ($LC4 & 0xFFFF)↙
    ↳   # "four" ; branch delay slot

sub_6C:                                    # ↙
    ↳ DATA XREF: .rodata:off_120
;  "zero"
                lui     $a0, ($LC0 >> 16)  #↙
    ↳  "zero"
                lw      $t9, (puts & 0xFFFF)↙
    ↳ ($gp)
                or      $at, $zero ; NOP
                jr      $t9
                la      $a0, ($LC0 & 0xFFFF)↙
    ↳   # "zero" ; branch delay slot

sub_80:                                    # ↙
    ↳ DATA XREF: .rodata:00000124
```

 CAPÍTULO 13. SWITCH()/CASE/DEFAULT

```
;  "one"
                lui     $a0, ($LC1 >> 16)  #↙
    ↳  "one"
                lw      $t9, (puts & 0xFFFF)↙
    ↳ ($gp)
                or      $at, $zero ; NOP
                jr      $t9
                la      $a0, ($LC1 & 0xFFFF)↙
    ↳   # "one" ; branch delay slot

sub_94:                                     # ↙
    ↳ DATA XREF: .rodata:00000128
;  "two"
                lui     $a0, ($LC2 >> 16)  #↙
    ↳  "two"
                lw      $t9, (puts & 0xFFFF)↙
    ↳ ($gp)
                or      $at, $zero ; NOP
                jr      $t9
                la      $a0, ($LC2 & 0xFFFF)↙
    ↳   # "two" ; branch delay slot

; :
off_120:        .word sub_6C
                .word sub_80
                .word sub_94
                .word sub_44
                .word sub_58
```

BNEZ «Branch if Not Equal to Zero».

SLL («Shift Word Left Logical») .

13.2.5 Conclusión

switch():

Listing 13.9: x86

```
MOV REG, input
CMP REG, 4 ;
JA default
SHL REG, 2 ;
MOV REG, jump_table[REG]
JMP REG

case1:
    ;
    JMP exit
case2:
    ;
    JMP exit
case3:
    ;
    JMP exit
case4:
    ;
    JMP exit
case5:
    ;
    JMP exit

default:

    ...
```

```
exit:

    ....

jump_table dd case1
           dd case2
           dd case3
           dd case4
           dd case5
```

JMP jump_table[REG*4]. JMP jump_table[REG*8] en x64.

18.5 on page 514.

13.3

```
#include <stdio.h>

void f(int a)
{
        switch (a)
        {
        case 1:
        case 2:
        case 7:
        case 10:
                printf ("1, 2, 7, 10\n");
                break;
        case 3:
        case 4:
```

 CAPÍTULO 13. SWITCH()/CASE/DEFAULT

```
        case 5:
        case 6:
                printf ("3, 4, 5\n");
                break;
        case 8:
        case 9:
        case 20:
        case 21:
                printf ("8, 9, 21\n");
                break;
        case 22:
                printf ("22\n");
                break;
        default:
                printf ("default\n");
                break;
        };
};

int main()
{
        f(4);
};
```

13.3.1 MSVC

Listing 13.10: Con optimización MSVC 2010

```

$SG2798 DB      '1, 2, 7, 10', 0aH, 00H
$SG2800 DB      '3, 4, 5', 0aH, 00H
$SG2802 DB      '8, 9, 21', 0aH, 00H
```

```
$SG2804 DB      '22', 0aH, 00H
$SG2806 DB      'default', 0aH, 00H

_a$ = 8
_f      PROC
        mov     eax, DWORD PTR _a$[esp-4]
        dec     eax
        cmp     eax, 21
        ja      SHORT $LN1@f
        movzx   eax, BYTE PTR $LN10@f[eax]
        jmp     DWORD PTR $LN11@f[eax*4]
$LN5@f:
        mov     DWORD PTR _a$[esp-4], OFFSET $SG2798 ; '1, 2, 7, 10'
        jmp     DWORD PTR __imp__printf
$LN4@f:
        mov     DWORD PTR _a$[esp-4], OFFSET $SG2800 ; '3, 4, 5'
        jmp     DWORD PTR __imp__printf
$LN3@f:
        mov     DWORD PTR _a$[esp-4], OFFSET $SG2802 ; '8, 9, 21'
        jmp     DWORD PTR __imp__printf
$LN2@f:
        mov     DWORD PTR _a$[esp-4], OFFSET $SG2804 ; '22'
        jmp     DWORD PTR __imp__printf
$LN1@f:
        mov     DWORD PTR _a$[esp-4], OFFSET $SG2806 ; 'default'
        jmp     DWORD PTR __imp__printf
        npad    2 ;  $LN11@f
$LN11@f:
```

 CAPÍTULO 13. SWITCH()/CASE/DEFAULT

```
33          DD      $LN5@f ;  '1, 2, 7, 10'
34          DD      $LN4@f ;  '3, 4, 5'
35          DD      $LN3@f ;  '8, 9, 21'
36          DD      $LN2@f ;  '22'
37          DD      $LN1@f ;  'default'
38  $LN10@f:
39          DB      0 ; a=1
40          DB      0 ; a=2
41          DB      1 ; a=3
42          DB      1 ; a=4
43          DB      1 ; a=5
44          DB      1 ; a=6
45          DB      0 ; a=7
46          DB      2 ; a=8
47          DB      2 ; a=9
48          DB      0 ; a=10
49          DB      4 ; a=11
50          DB      4 ; a=12
51          DB      4 ; a=13
52          DB      4 ; a=14
53          DB      4 ; a=15
54          DB      4 ; a=16
55          DB      4 ; a=17
56          DB      4 ; a=18
57          DB      4 ; a=19
58          DB      2 ; a=20
59          DB      2 ; a=21
60          DB      3 ; a=22
61  _f      ENDP
```

: ($LN10@f) , ($LN11@f) .

(línea 13).

: 0 (1, 2, 7, 10), 1 (3, 4, 5), 2 (8, 9, 21), 3 (22), 4 .

(línea 14).

.

? (13.2.1 on page 310), ? .

13.3.2 GCC

GCC (13.2.1 on page 310), .

13.3.3 ARM64: Con optimización GCC 4.9.1

GCC 4.9.1 para ARM64 . .

Listing 13.11: Con optimización GCC 4.9.1 ARM64

```
f14:
;  W0
        sub     w0, w0, #1
        cmp     w0, 21
;
        bls     .L9
.L2:
;  "default":
        adrp    x0, .LC4
        add     x0, x0, :lo12:.LC4
        b       puts
.L9:
;  X1:
        adrp    x1, .L4
        add     x1, x1, :lo12:.L4
```

```
; W0=input_value-1
;
        ldrb    w0, [x1,w0,uxtw]
;
        adr     x1, .Lrtx4
;
        add     x0, x1, w0, sxtb #2
; :
        br      x0
; :
.Lrtx4:
        .section        .rodata
;
.L4:
        .byte   (.L3 - .Lrtx4) / 4      ; ↙
    ↘ case 1
        .byte   (.L3 - .Lrtx4) / 4      ; ↙
    ↘ case 2
        .byte   (.L5 - .Lrtx4) / 4      ; ↙
    ↘ case 3
        .byte   (.L5 - .Lrtx4) / 4      ; ↙
    ↘ case 4
        .byte   (.L5 - .Lrtx4) / 4      ; ↙
    ↘ case 5
        .byte   (.L5 - .Lrtx4) / 4      ; ↙
    ↘ case 6
        .byte   (.L3 - .Lrtx4) / 4      ; ↙
    ↘ case 7
        .byte   (.L6 - .Lrtx4) / 4      ; ↙
    ↘ case 8
        .byte   (.L6 - .Lrtx4) / 4      ; ↙
    ↘ case 9
        .byte   (.L3 - .Lrtx4) / 4      ; ↙
```

 CAPÍTULO 13. SWITCH()/CASE/DEFAULT

```
case 10
        .byte   (.L2 - .Lrtx4) / 4      ; case 11
        .byte   (.L2 - .Lrtx4) / 4      ; case 12
        .byte   (.L2 - .Lrtx4) / 4      ; case 13
        .byte   (.L2 - .Lrtx4) / 4      ; case 14
        .byte   (.L2 - .Lrtx4) / 4      ; case 15
        .byte   (.L2 - .Lrtx4) / 4      ; case 16
        .byte   (.L2 - .Lrtx4) / 4      ; case 17
        .byte   (.L2 - .Lrtx4) / 4      ; case 18
        .byte   (.L2 - .Lrtx4) / 4      ; case 19
        .byte   (.L6 - .Lrtx4) / 4      ; case 20
        .byte   (.L6 - .Lrtx4) / 4      ; case 21
        .byte   (.L7 - .Lrtx4) / 4      ; case 22
        .text
;
.L7:
; "22"
        adrp    x0, .LC3
        add     x0, x0, :lo12:.LC3
        b       puts
.L6:
```

```
; "8, 9, 21"
        adrp    x0, .LC2
        add     x0, x0, :lo12:.LC2
        b       puts
.L5:
; "3, 4, 5"
        adrp    x0, .LC1
        add     x0, x0, :lo12:.LC1
        b       puts
.L3:
; "1, 2, 7, 10"
        adrp    x0, .LC0
        add     x0, x0, :lo12:.LC0
        b       puts
.LC0:
        .string "1, 2, 7, 10"
.LC1:
        .string "3, 4, 5"
.LC2:
        .string "8, 9, 21"
.LC3:
        .string "22"
.LC4:
        .string "default"
```

Listing 13.12: jumptable in IDA

```
.rodata:0000000000000064                                        ↙
    ↘ AREA .rodata, DATA, READONLY
.rodata:0000000000000064                                      ; ↙
    ↘ ORG 0x64
.rodata:0000000000000064 $d                                  DCB↙
    ↘     9    ; case 1
```

CAPÍTULO 13. SWITCH()/CASE/DEFAULT

```
.rodata:0000000000000065                                     DCB↙
    ↘      9    ; case 2
.rodata:0000000000000066                                     DCB↙
    ↘      6    ; case 3
.rodata:0000000000000067                                     DCB↙
    ↘      6    ; case 4
.rodata:0000000000000068                                     DCB↙
    ↘      6    ; case 5
.rodata:0000000000000069                                     DCB↙
    ↘      6    ; case 6
.rodata:000000000000006A                                     DCB↙
    ↘      9    ; case 7
.rodata:000000000000006B                                     DCB↙
    ↘      3    ; case 8
.rodata:000000000000006C                                     DCB↙
    ↘      3    ; case 9
.rodata:000000000000006D                                     DCB↙
    ↘      9    ; case 10
.rodata:000000000000006E                                     DCB↙
    ↘   0xF7    ; case 11
.rodata:000000000000006F                                     DCB↙
    ↘   0xF7    ; case 12
.rodata:0000000000000070                                     DCB↙
    ↘   0xF7    ; case 13
.rodata:0000000000000071                                     DCB↙
    ↘   0xF7    ; case 14
.rodata:0000000000000072                                     DCB↙
    ↘   0xF7    ; case 15
.rodata:0000000000000073                                     DCB↙
    ↘   0xF7    ; case 16
.rodata:0000000000000074                                     DCB↙
    ↘   0xF7    ; case 17
.rodata:0000000000000075                                     DCB↙
```

 CAPÍTULO 13. SWITCH()/CASE/DEFAULT

```
    ↳  0xF7    ; case 18
.rodata:0000000000000076                         DCB↙
    ↳  0xF7    ; case 19
.rodata:0000000000000077                         DCB↙
    ↳     3    ; case 20
.rodata:0000000000000078                         DCB↙
    ↳     3    ; case 21
.rodata:0000000000000079                         DCB↙
    ↳     0    ; case 22
.rodata:000000000000007B ; .rodata            ↙
    ↳ ends
```

13.4 Fall-through

switch().:

```
#define R 1
#define W 2
#define RW 3

void f(int type)
{
        int read=0, write=0;

        switch (type)
        {
        case RW:
                read=1;
        case W:
                write=1;
                break;
```

```
        case R:
                read=1;
                break;
        default:
                break;
        };
        printf ("read=%d, write=%d\n", read, write);
};
```

$type = 1$ (R), $read$ 1, $type = 2$ (W), $write$ 2. $type = 3$ (RW), $read$ y $write$ 1.

$type = RW$ $type = W$..

13.4.1 MSVC x86

Listing 13.13: MSVC 2012

```
$SG1305 DB      'read=%d, write=%d', 0aH, 00H

_write$ = -12   ; size = 4
_read$ = -8     ; size = 4
tv64 = -4       ; size = 4
_type$ = 8      ; size = 4
_f      PROC
        push    ebp
        mov     ebp, esp
        sub     esp, 12
        mov     DWORD PTR _read$[ebp], 0
        mov     DWORD PTR _write$[ebp], 0
```

```
        mov     eax, DWORD PTR _type$[ebp]
        mov     DWORD PTR tv64[ebp], eax
        cmp     DWORD PTR tv64[ebp], 1 ; R
        je      SHORT $LN2@f
        cmp     DWORD PTR tv64[ebp], 2 ; W
        je      SHORT $LN3@f
        cmp     DWORD PTR tv64[ebp], 3 ; RW
        je      SHORT $LN4@f
        jmp     SHORT $LN5@f
$LN4@f: ; case RW:
        mov     DWORD PTR _read$[ebp], 1
$LN3@f: ; case W:
        mov     DWORD PTR _write$[ebp], 1
        jmp     SHORT $LN5@f
$LN2@f: ; case R:
        mov     DWORD PTR _read$[ebp], 1
$LN5@f: ; default
        mov     ecx, DWORD PTR _write$[ebp]
        push    ecx
        mov     edx, DWORD PTR _read$[ebp]
        push    edx
        push    OFFSET $SG1305 ; 'read=%d, ↙
    ↘ write=%d'
        call    _printf
        add     esp, 12
        mov     esp, ebp
        pop     ebp
        ret     0
_f      ENDP
```

`$LN4@f` y `$LN3@f`: `$LN4@f`, $read$ $write$. $type = W$, `$LN3@f`, .

13.4.2 ARM64

Listing 13.14: GCC (Linaro) 4.9

```
.LC0:
        .string "read=%d, write=%d\n"
f:
        stp     x29, x30, [sp, -48]!
        add     x29, sp, 0
        str     w0, [x29,28]
        str     wzr, [x29,44] ;
        str     wzr, [x29,40]
        ldr     w0, [x29,28]  ;
        cmp     w0, 2         ; type=W?
        beq     .L3
        cmp     w0, 3         ; type=RW?
        beq     .L4
        cmp     w0, 1         ; type=R?
        beq     .L5
        b       .L6           ; ...
.L4: ; case RW
        mov     w0, 1
        str     w0, [x29,44] ; read=1
.L3: ; case W
        mov     w0, 1
        str     w0, [x29,40] ; write=1
        b       .L6
.L5: ; case R
        mov     w0, 1
        str     w0, [x29,44] ; read=1
        nop
.L6: ; default
```

```
        adrp    x0, .LC0 ; "read=%d, write=%d\n"
        add     x0, x0, :lo12:.LC0
        ldr     w1, [x29,44] ;  "read"
        ldr     w2, [x29,40] ;  "write"
        bl      printf
        ldp     x29, x30, [sp], 48
        ret
```

. .L4 y .L3.

13.5 Ejercicios

13.5.1 Ejercicio #1

.

: G.1.5 on page 1920.

Capítulo 14

Lazos

14.1

14.1.1 x86

Hay una instrucción especial LOOP en el conjunto de instrucciones x86 para verificar el valor en el registro ECX y si no es 0, decrement ECX y pasar el flujo de control a la etiqueta en el operando LOOP. Probablemente esta instrucción no es muy conveniente, y no hay compiladores modernos que la emitan automáticamente. Así que, si ves esta instrucción en algún lugar del código, es muy probable que sea un fragmento de código ensamblador escrito manualmente.

 CAPÍTULO 14. LAZOS

for().

```


{
    ;
}
```

:

```
#include <stdio.h>

void printing_function(int i)
{
        printf ("f(%d)\n", i);
};

int main()
{
        int i;

        for (i=2; i<10; i++)
                printing_function(i);

        return 0;
};
```

(MSVC 2010):

Listing 14.1: MSVC 2010

```
_i$ = -4
_main    PROC
    push   ebp
    mov    ebp, esp
    push   ecx
    mov    DWORD PTR _i$[ebp], 2   ;
    jmp    SHORT $LN3@main
$LN2@main:
    mov    eax, DWORD PTR _i$[ebp] ; :
    add    eax, 1                  ;
    mov    DWORD PTR _i$[ebp], eax
$LN3@main:
    cmp    DWORD PTR _i$[ebp], 10  ;
    jge    SHORT $LN1@main         ;
    mov    ecx, DWORD PTR _i$[ebp] ;  ↙
    ↘ printing_function(i)
    push   ecx
    call   _printing_function
    add    esp, 4
    jmp    SHORT $LN2@main         ;
$LN1@main:                         ;
    xor    eax, eax
    mov    esp, ebp
    pop    ebp
    ret    0
_main    ENDP
```

Listing 14.2: GCC 4.4.1

```
main            proc near
```

```
var_20          = dword ptr -20h
var_4           = dword ptr -4

                push    ebp
                mov     ebp, esp
                and     esp, 0FFFFFFF0h
                sub     esp, 20h
                mov     [esp+20h+var_4], 2  ⤦
    ↳ ;
                jmp     short loc_8048476

loc_8048465:
                mov     eax, [esp+20h+var_4]
                mov     [esp+20h+var_20], ⤦
    ↳ eax
                call    printing_function
                add     [esp+20h+var_4], 1  ⤦
    ↳ ;

loc_8048476:
                cmp     [esp+20h+var_4], 9
                jle     short loc_8048465   ⤦
    ↳ ;
                mov     eax, 0
                leave
                retn
main            endp
```

(/Ox):

Listing 14.3: Con optimización MSVC

```
_main    PROC
```

```
    push   esi
    mov    esi, 2
$LL3@main:
    push   esi
    call   _printing_function
    inc    esi
    add    esp, 4
    cmp    esi, 10      ; 0000000aH
    jl     SHORT $LL3@main
    xor    eax, eax
    pop    esi
    ret    0
_main    ENDP
```

Listing 14.4: Con optimización GCC 4.4.1

```
main                proc near

var_10              = dword ptr -10h

                    push    ebp
                    mov     ebp, esp
                    and     esp, 0FFFFFFF0h
                    sub     esp, 10h
                    mov     [esp+10h+var_10], 2
                    call    printing_function
                    mov     [esp+10h+var_10], 3
                    call    printing_function
                    mov     [esp+10h+var_10], 4
                    call    printing_function
                    mov     [esp+10h+var_10], 5
                    call    printing_function
                    mov     [esp+10h+var_10], 6
```

```
                call    printing_function
                mov     [esp+10h+var_10], 7
                call    printing_function
                mov     [esp+10h+var_10], 8
                call    printing_function
                mov     [esp+10h+var_10], 9
                call    printing_function
                xor     eax, eax
                leave
                retn
main            endp
```

[1].

Listing 14.5: GCC

```

                public main
main            proc near

var_20          = dword ptr -20h

                push    ebp
                mov     ebp, esp
                and     esp, 0FFFFFFF0h
                push    ebx
                mov     ebx, 2      ; i=2
                sub     esp, 1Ch

```

[1]: [Dre07]. : [Int14, pág. 3.4.1.7].

```
; :
                nop

loc_80484D0:
; :
                mov     [esp+20h+var_20], ↙
    ↘ ebx
                add     ebx, 1    ; i++
                call    printing_function
                cmp     ebx, 64h  ; i==100?
                jnz     short loc_80484D0 ;
                add     esp, 1Ch
                xor     eax, eax  ;  0
                pop     ebx
                mov     esp, ebp
                pop     ebp
                retn
main            endp
```

14.1.2 x86: OllyDbg

MSVC 2010 /Ox y /Ob0 OllyDbg.

OllyDbg:

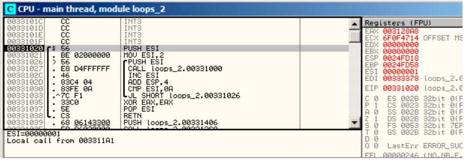


Figura 14.1: OllyDbg:

(F8 — pasar por encima) ESI $ESI = i = 6$:

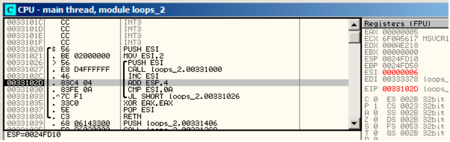


Figura 14.2: OllyDbg: $i = 6$

9 . JL

 CAPÍTULO 14. LAZOS

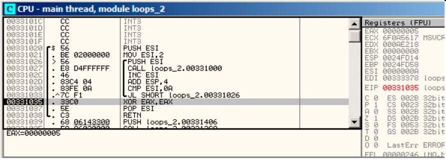


Figura 14.3: OllyDbg: $ESI = 10$,

14.1.3 x86: tracer

```
tracer.exe -l:loops_2.exe bpx=loops_2.exe!0↙
    ↳ x00401026
```

BPX

`tracer.log`:

```
PID=12884|New process loops_2.exe
(0) loops_2.exe!0x401026
EAX=0x00a328c8 EBX=0x00000000 ECX=0x6f0f4714↙
    ↳  EDX=0x00000000
ESI=0x00000002 EDI=0x00333378 EBP=0x0024fbfc↙
    ↳  ESP=0x0024fbb8
EIP=0x00331026
FLAGS=PF ZF IF
(0) loops_2.exe!0x401026
EAX=0x00000005 EBX=0x00000000 ECX=0x6f0a5617↙
    ↳  EDX=0x000ee188
ESI=0x00000003 EDI=0x00333378 EBP=0x0024fbfc↙
    ↳  ESP=0x0024fbb8
EIP=0x00331026
FLAGS=CF PF AF SF IF
```

```
(0) loops_2.exe!0x401026
EAX=0x00000005 EBX=0x00000000 ECX=0x6f0a5617
    ↳ EDX=0x000ee188
ESI=0x00000004 EDI=0x00333378 EBP=0x0024fbfc
    ↳ ESP=0x0024fbb8
EIP=0x00331026
FLAGS=CF PF AF SF IF
(0) loops_2.exe!0x401026
EAX=0x00000005 EBX=0x00000000 ECX=0x6f0a5617
    ↳ EDX=0x000ee188
ESI=0x00000005 EDI=0x00333378 EBP=0x0024fbfc
    ↳ ESP=0x0024fbb8
EIP=0x00331026
FLAGS=CF AF SF IF
(0) loops_2.exe!0x401026
EAX=0x00000005 EBX=0x00000000 ECX=0x6f0a5617
    ↳ EDX=0x000ee188
ESI=0x00000006 EDI=0x00333378 EBP=0x0024fbfc
    ↳ ESP=0x0024fbb8
EIP=0x00331026
FLAGS=CF PF AF SF IF
(0) loops_2.exe!0x401026
EAX=0x00000005 EBX=0x00000000 ECX=0x6f0a5617
    ↳ EDX=0x000ee188
ESI=0x00000007 EDI=0x00333378 EBP=0x0024fbfc
    ↳ ESP=0x0024fbb8
EIP=0x00331026
FLAGS=CF AF SF IF
(0) loops_2.exe!0x401026
EAX=0x00000005 EBX=0x00000000 ECX=0x6f0a5617
    ↳ EDX=0x000ee188
ESI=0x00000008 EDI=0x00333378 EBP=0x0024fbfc
    ↳ ESP=0x0024fbb8
```

 CAPÍTULO 14. LAZOS

```
EIP=0x00331026
FLAGS=CF AF SF IF
(0) loops_2.exe!0x401026
EAX=0x00000005 EBX=0x00000000 ECX=0x6f0a5617 ↙
    ↳  EDX=0x000ee188
ESI=0x00000009 EDI=0x00333378 EBP=0x0024fbfc ↙
    ↳  ESP=0x0024fbb8
EIP=0x00331026
FLAGS=CF PF AF SF IF
PID=12884|Process loops_2.exe exited. ↙
    ↳ ExitCode=0 (0x0)
```

ESI

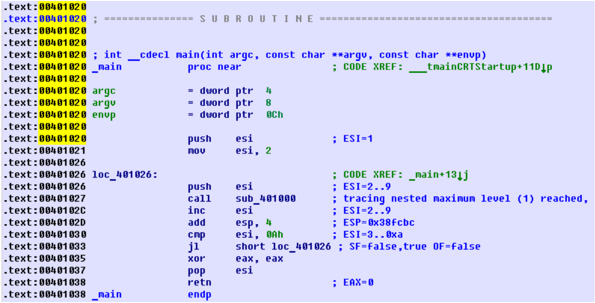
trace. . IDA `main()` `0x00401020` :

```
tracer.exe -l:loops_2.exe bpf=loops_2.exe!0 ↙
    ↳ x00401020,trace:cc
```

BPF .

`loops_2.exe.idc` y `loops_2.exe_clear.idc`.

`loops_2.exe.idc` IDA:



```
.text:00401020
.text:00401020 ; =============== S U B R O U T I N E =======================================
.text:00401020
.text:00401020
.text:00401020 ; int __cdecl main(int argc, const char **argv, const char **envp)
.text:00401020 _main           proc near               ; CODE XREF: ___tmainCRTStartup+11D↓p
.text:00401020
.text:00401020 argc            = dword ptr  4
.text:00401020 argv            = dword ptr  8
.text:00401020 envp            = dword ptr  0Ch
.text:00401020
.text:00401020                 push    esi             ; ESI=1
.text:00401021                 mov     esi, 2
.text:00401026
.text:00401026 loc_401026:                             ; CODE XREF: _main+13↓j
.text:00401026                 push    esi             ; ESI=2..9
.text:00401027                 call    sub_401000      ; tracing nested maximum level (1) reached,
.text:0040102C                 inc     esi             ; ESI=2..9
.text:0040102D                 add     esp, 4          ; ESP=0x38fcbc
.text:00401030                 cmp     esi, 0Ah        ; ESI=3..0xa
.text:00401033                 jl      short loc_401026 ; SF=false,true OF=false
.text:00401035                 xor     eax, eax
.text:00401037                 pop     esi
.text:00401038                 retn                    ; EAX=0
.text:00401038 _main           endp
```

Figura 14.4: IDA

ESI. `main()` EAX.

tracer `loops_2.exe.txt`, :

Listing 14.6: loops_2.exe.txt

```
0x401020 (.text+0x20), e=       1 [PUSH ESI] ESI=1
0x401021 (.text+0x21), e=       1 [MOV ESI, 2]
0x401026 (.text+0x26), e=       8 [PUSH ESI] ESI=2..9
0x401027 (.text+0x27), e=       8 [CALL 8D1000h] tracing nested maximum level (1) reached, skipping this CALL 8D1000h=0x8d1000
0x40102c (.text+0x2c), e=       8 [INC ESI] ESI=2..9
```

```
0x40102d (.text+0x2d), e=        8 [ADD ESP, ↙
    ↘ 4] ESP=0x38fcbc
0x401030 (.text+0x30), e=        8 [CMP ESI, ↙
    ↘ 0Ah] ESI=3..0xa
0x401033 (.text+0x33), e=        8 [JL 8↙
    ↘ D1026h] SF=false,true OF=false
0x401035 (.text+0x35), e=        1 [XOR EAX, ↙
    ↘ EAX]
0x401037 (.text+0x37), e=        1 [POP ESI]
0x401038 (.text+0x38), e=        1 [RETN] EAX↙
    ↘ =0
```

.

14.1.4 ARM

Sin optimización Keil 6/2013 (Modo ARM)

```
main
        STMFD   SP!, {R4,LR}
        MOV     R4, #2
        B       loc_368
loc_35C  ; CODE XREF: main+1C
        MOV     R0, R4
        BL      printing_function
        ADD     R4, R4, #1

loc_368  ; CODE XREF: main+8
        CMP     R4, #0xA
        BLT     loc_35C
        MOV     R0, #0
        LDMFD   SP!, {R4,PC}
```

«MOV R4, #2» i.

«MOV R0, R4» y «BL printing_function» ,

«ADD R4, R4, #1»

«CMP R4, #0xA» i 0xA (10).

Con optimización Keil 6/2013 (Modo Thumb)

```
_main
                PUSH    {R4,LR}
                MOVS    R4, #2

loc_132                                         ; ↙
    ↘ CODE XREF: _main+E
                MOVS    R0, R4
                BL      printing_function
                ADDS    R4, R4, #1
                CMP     R4, #0xA
                BLT     loc_132
                MOVS    R0, #0
                POP     {R4,PC}
```

Con optimización Xcode 4.6.3 (LLVM) (Modo Thumb-2)

```
_main
        PUSH            {R4,R7,LR}
        MOVW            R4, #0x1124 ; "%d\n"
        MOVS            R1, #2
        MOVT.W          R4, #0
        ADD             R7, SP, #4
```

```
        ADD             R4, PC
        MOV             R0, R4
        BLX             _printf
        MOV             R0, R4
        MOVS            R1, #3
        BLX             _printf
        MOV             R0, R4
        MOVS            R1, #4
        BLX             _printf
        MOV             R0, R4
        MOVS            R1, #5
        BLX             _printf
        MOV             R0, R4
        MOVS            R1, #6
        BLX             _printf
        MOV             R0, R4
        MOVS            R1, #7
        BLX             _printf
        MOV             R0, R4
        MOVS            R1, #8
        BLX             _printf
        MOV             R0, R4
        MOVS            R1, #9
        BLX             _printf
        MOVS            R0, #0
        POP             {R4,R7,PC}
```

```
void printing_function(int i)
{
    printf ("%d\n", i);
};
```

LLVM , , .

ARM64: Con optimización GCC 4.9.1

Listing 14.7: Con optimización GCC 4.9.1

```
printing_function:
;  printf():
        mov     w1, w0
;
        adrp    x0, .LC0
        add     x0, x0, :lo12:.LC0
; :
        b       printf
main:
; :
        stp     x29, x30, [sp, -32]!
; :
        add     x29, sp, 0
; :
        str     x19, [sp,16]
; .
; :
        mov     w19, 2
.L3:
;  printing_function():
        mov     w0, w19
; .
        add     w19, w19, 1
; .
        bl      printing_function
; ?
```

```
        cmp     w19, 10
; :
        bne     .L3
;   0
        mov     w0, 0
; :
        ldr     x19, [sp,16]
; :
        ldp     x29, x30, [sp], 32
        ret
.LC0:
        .string "f(%d)\n"
```

ARM64: Sin optimización GCC 4.9.1

Listing 14.8: Sin optimización GCC 4.9.1 -fno-inline

```
printing_function:
;  printf():
        mov     w1, w0
;
        adrp    x0, .LC0
        add     x0, x0, :lo12:.LC0
; :
        b       printf
main:
; :
        stp     x29, x30, [sp, -32]!
; :
        add     x29, sp, 0
; :
```

```
        str     x19, [sp,16]
; .
; :
        mov     w19, 2
.L3:
;  printing_function():
        mov     w0, w19
; .
        add     w19, w19, 1
; .
        bl      printing_function
; ?
        cmp     w19, 10
; :
        bne     .L3
;  0
        mov     w0, 0
; :
        ldr     x19, [sp,16]
; :
        ldp     x29, x30, [sp], 32
        ret
.LC0:
        .string "f(%d)\n"
```

14.1.5 MIPS

Listing 14.9: Sin optimización GCC 4.4.5 (IDA)

```
main:
```

```
;
; :
i               = -0x10
saved_FP        = -8
saved_RA        = -4

; :
                addiu   $sp, -0x28
                sw      $ra, 0x28+saved_RA(↙
    ↘ $sp)
                sw      $fp, 0x28+saved_FP(↙
    ↘ $sp)
                move    $fp, $sp
;
                li      $v0, 2
                sw      $v0, 0x28+i($fp)
; . "BEQ $ZERO, $ZERO, loc_9C":
                b       loc_9C
                or      $at, $zero ; branch ↙
    ↘ delay slot, NOP
 # ↙
    ↘ -------------------------------------------
    ↘

loc_80:                                     # ↙
    ↘ CODE XREF: main+48
;  printing_function():
                lw      $a0, 0x28+i($fp)
                jal     printing_function
                or      $at, $zero ; branch ↙
    ↘ delay slot, NOP
; :
                lw      $v0, 0x28+i($fp)
```

```
                or      $at, $zero ; NOP
                addiu   $v0, 1
                sw      $v0, 0x28+i($fp)

loc_9C:                                     # ↙
    ↳ CODE XREF: main+18
;
                lw      $v0, 0x28+i($fp)
                or      $at, $zero ; NOP
                slti    $v0, 0xA
; :
                bnez    $v0, loc_80
                or      $at, $zero ; branch ↙
    ↳ delay slot, NOP
; :
                move    $v0, $zero
; :
                move    $sp, $fp
                lw      $ra, 0x28+saved_RA(↙
    ↳ $sp)
                lw      $fp, 0x28+saved_FP(↙
    ↳ $sp)
                addiu   $sp, 0x28
                jr      $ra
                or      $at, $zero ; branch ↙
    ↳ delay slot, NOP
```

«B». (BEQ).

14.1.6

: i

```

for (i=0; i<total_entries_to_process; i++)
    ;
```

total_entries_to_process 0, .

14.2

```
#include <stdio.h>

void my_memcpy (unsigned char* dst, unsigned
    ↳  char* src, size_t cnt)
{
        size_t i;
        for (i=0; i<cnt; i++)
                dst[i]=src[i];
};
```

14.2.1

Listing 14.10: GCC 4.9 x64 (-Os)

```

my_memcpy:
; RDI =
; RSI =
; RDX =

;
```

```
        xor     eax, eax
.L2:
;
        cmp     rax, rdx
        je      .L5
;  RSI+i:
        mov     cl, BYTE PTR [rsi+rax]
;  RDI+i:
        mov     BYTE PTR [rdi+rax], cl
        inc     rax ; i++
        jmp     .L2
.L5:
        ret
```

Listing 14.11: GCC 4.9 ARM64 (-Os)

```


my_memcpy:
; X0 =
; X1 =
; X2 =

;
        mov     x3, 0
.L2:
;
        cmp     x3, x2
        beq     .L5
;  X1+i:
        ldrb    w4, [x1,x3]
;  X1+i:
        strb    w4, [x0,x3]
        add     x3, x3, 1 ; i++
```

```
        b       .L2
.L5:
        ret
```

Listing 14.12: Con optimización Keil 6/2013 (Modo Thumb)

```

my_memcpy PROC
; R0 =
; R1 =
; R2 =

        PUSH     {r4,lr}
;
        MOVS     r3,#0
;
        B        |L0.12|
|L0.6|
;  R1+i:
        LDRB     r4,[r1,r3]
;  R1+i:
        STRB     r4,[r0,r3]
; i++
        ADDS     r3,r3,#1
|L0.12|
; i<size?
        CMP      r3,r2
;
        BCC      |L0.6|
        POP      {r4,pc}
        ENDP
```

14.2.2 ARM

Listing 14.13: Con optimización Keil 6/2013 (Modo ARM)

```
my_memcpy PROC
; R0 =
; R1 =
; R2 =

;
        MOV      r3,#0
|L0.4|
;
        CMP      r3,r2
;
;  R2<R3  i<size.
;  R1+i:
        LDRBCC   r12,[r1,r3]
;  R1+i:
        STRBCC   r12,[r0,r3]
; i++
        ADDCC    r3,r3,#1
;
;  i<size
;  ( i>=size)
        BCC      |L0.4|
;
        BX       lr
        ENDP
```

14.2.3 MIPS

Listing 14.14: GCC 4.4.5 (-Os) (IDA)

```
my_memcpy:
;
                b       loc_14
;
;  \$v0:
                move    $v0, $zero ; branch ↙
    ↳ delay slot

loc_8:                                      # ↙
    ↳ CODE XREF: my_memcpy+1C
;
                lbu     $v1, 0($t0)
;  (i):
                addiu   $v0, 1
;  $a3
                sb      $v1, 0($a3)

loc_14:                                     # ↙
    ↳ CODE XREF: my_memcpy
;
                sltu    $v1, $v0, $a2
; :
                addu    $t0, $a1, $v0
; $t0 = $a1+$v0 = src+i
;  "cnt":
                bnez    $v1, loc_8
;  (\$a3 = \$a0+\$v0 = dst+i):
                addu    $a3, $a0, $v0 ; ↙
    ↳ branch delay slot
;  BNEZ
```

```
                jr      $ra
                or      $at, $zero ; branch ↙
    ↳ delay slot, NOP
```

LBU («Load Byte Unsigned») y SB («Store Byte»).

14.2.4

Con optimización GCC : 25.1.2 on page 802.

14.3 Conclusión

:

Listing 14.15: x86

```

    mov [counter], 2 ;
    jmp check
body:
    ;
    ;
    ;
    add [counter], 1 ;
check:
    cmp [counter], 9
    jle body
```

Listing 14.16: x86

```

```

```
    MOV [counter], 2 ;
    JMP check
body:
    ;
    ;
    ;
    MOV REG, [counter] ;
    INC REG
    MOV [counter], REG
check:
    CMP [counter], 9
    JLE body
```

Listing 14.17: x86

```
    MOV EBX, 2 ;
    JMP check
body:
    ;
    ;
    ;
    INC EBX ;
check:
    CMP EBX, 9
    JLE body
```

Listing 14.18: x86

```
    MOV [counter], 2 ;
    JMP label_check
label_increment:
```

```
    ADD [counter], 1 ;
label_check:
    CMP [counter], 10
    JGE exit
    ;
    ;
    ;
    JMP label_increment
exit:
```

Listing 14.19: x86

```

    MOV REG, 2 ;
body:
    ;
    ;
    ;
    INC REG ;
    CMP REG, 10
    JL body
```

Listing 14.20: x86

```

    ;
    MOV ECX, 10
body:
    ;
    ;
    ;
    LOOP body
```

ARM.

Listing 14.21: ARM

```
    MOV R4, 2 ;
    B check
body:
    ;
    ;
    ;
    ADD R4,R4, #1 ;
check:
    CMP R4, #10
    BLT body
```

14.4 Ejercicios

14.4.1 Ejercicio #1

LOOP?

14.4.2 Ejercicio #2

(14.1.1 on page 339), OS [6..20].

14.4.3 Ejercicio #3

Lo que hace el código?

Listing 14.22: Con optimización MSVC 2010

```
$SG2795 DB      '%d', 0aH, 00H

_main   PROC
        push    esi
        push    edi
        mov     edi, DWORD PTR __imp__printf
        mov     esi, 100
        npad    3 ; align next label
$LL3@main:
        push    esi
        push    OFFSET $SG2795 ; '%d'
        call    edi
        dec     esi
        add     esp, 8
        test    esi, esi
        jg      SHORT $LL3@main
        pop     edi
        xor     eax, eax
        pop     esi
        ret     0
_main   ENDP
```

Listing 14.23: Sin optimización Keil 6/2013 (Modo ARM)

```
main PROC
        PUSH     {r4,lr}
        MOV      r4,#0x64
|L0.8|
        MOV      r1,r4
        ADR      r0,|L0.40|
        BL       __2printf
        SUB      r4,r4,#1
```

```
        CMP     r4,#0
        MOVLE   r0,#0
        BGT     |L0.8|
        POP     {r4,pc}
        ENDP

|L0.40|
        DCB     "%d\n",0
```

Listing 14.24: Sin optimización Keil 6/2013 (Modo Thumb)

```
main PROC
        PUSH    {r4,lr}
        MOVS    r4,#0x64
|L0.4|
        MOVS    r1,r4
        ADR     r0,|L0.24|
        BL      __2printf
        SUBS    r4,r4,#1
        CMP     r4,#0
        BGT     |L0.4|
        MOVS    r0,#0
        POP     {r4,pc}
        ENDP

        DCW     0x0000
|L0.24|
        DCB     "%d\n",0
```

Listing 14.25: Con optimización GCC 4.9 (ARM64)

```
main:
        stp    x29, x30, [sp, -32]!
```

```
        add     x29, sp, 0
        stp     x19, x20, [sp,16]
        adrp    x20, .LC0
        mov     w19, 100
        add     x20, x20, :lo12:.LC0
.L2:
        mov     w1, w19
        mov     x0, x20
        bl      printf
        subs    w19, w19, #1
        bne     .L2
        ldp     x19, x20, [sp,16]
        ldp     x29, x30, [sp], 32
        ret
.LC0:
        .string "%d\n"
```

Listing 14.26: Con optimización GCC 4.4.5 (MIPS) (IDA)

```
main:

var_18          = -0x18
var_C           = -0xC
var_8           = -8
var_4           = -4

                lui     $gp, (__gnu_local_gp ↙
    ↳ >> 16)
                addiu   $sp, -0x28
                la      $gp, (__gnu_local_gp ↙
    ↳ & 0xFFFF)
                sw      $ra, 0x28+var_4($sp)
                sw      $s1, 0x28+var_8($sp)
```

 CAPÍTULO 14. LAZOS

```
                 sw      $s0, 0x28+var_C($sp)
                 sw      $gp, 0x28+var_18($sp↙
    ↳ )
                 la      $s1, $LC0         # ↙
    ↳ "%d\n"
                 li      $s0, 0x64   # 'd'

loc_28:                                    # ↙
    ↳ CODE XREF: main+40
                 lw      $t9, (printf & 0↙
    ↳ xFFFF)($gp)
                 move    $a1, $s0
                 move    $a0, $s1
                 jalr    $t9
                 addiu   $s0, -1
                 lw      $gp, 0x28+var_18($sp↙
    ↳ )
                 bnez    $s0, loc_28
                 or      $at, $zero
                 lw      $ra, 0x28+var_4($sp)
                 lw      $s1, 0x28+var_8($sp)
                 lw      $s0, 0x28+var_C($sp)
                 jr      $ra
                 addiu   $sp, 0x28

$LC0:            .ascii "%d\n"<0>          # ↙
    ↳ DATA XREF: main+1C
```

: G.1.7 on page 1921.

14.4.4 Ejercicio #4

Lo que hace el código?

Listing 14.27: Con optimización MSVC 2010

```
$SG2795 DB      '%d', 0aH, 00H

_main   PROC
        push    esi
        push    edi
        mov     edi, DWORD PTR __imp__printf
        mov     esi, 1
        npad    3 ; align next label
$LL3@main:
        push    esi
        push    OFFSET $SG2795 ; '%d'
        call    edi
        add     esi, 3
        add     esp, 8
        cmp     esi, 100
        jl      SHORT $LL3@main
        pop     edi
        xor     eax, eax
        pop     esi
        ret     0
_main   ENDP
```

Listing 14.28: Sin optimización Keil 6/2013 (Modo ARM)

```
main PROC
        PUSH     {r4,lr}
        MOV      r4,#1
|L0.8|
```

```
        MOV      r1,r4
        ADR      r0,|L0.40|
        BL       __2printf
        ADD      r4,r4,#3
        CMP      r4,#0x64
        MOVGE    r0,#0
        BLT      |L0.8|
        POP      {r4,pc}
        ENDP

|L0.40|
        DCB      "%d\n",0
```

Listing 14.29: Sin optimización Keil 6/2013 (Modo Thumb)

```
main PROC
        PUSH     {r4,lr}
        MOVS     r4,#1
|L0.4|
        MOVS     r1,r4
        ADR      r0,|L0.24|
        BL       __2printf
        ADDS     r4,r4,#3
        CMP      r4,#0x64
        BLT      |L0.4|
        MOVS     r0,#0
        POP      {r4,pc}
        ENDP

        DCW      0x0000
|L0.24|
        DCB      "%d\n",0
```

Listing 14.30: Con optimización GCC 4.9 (ARM64)

```
main:
        stp     x29, x30, [sp, -32]!
        add     x29, sp, 0
        stp     x19, x20, [sp,16]
        adrp    x20, .LC0
        mov     w19, 1
        add     x20, x20, :lo12:.LC0
.L2:
        mov     w1, w19
        mov     x0, x20
        add     w19, w19, 3
        bl      printf
        cmp     w19, 100
        bne     .L2
        ldp     x19, x20, [sp,16]
        ldp     x29, x30, [sp], 32
        ret
.LC0:
        .string "%d\n"
```

Listing 14.31: Con optimización GCC 4.4.5 (MIPS) (IDA)

```
main:

var_18          = -0x18
var_10          = -0x10
var_C           = -0xC
var_8           = -8
var_4           = -4

                lui     $gp, (__gnu_local_gp
    ↳ >> 16)
```

```
                addiu   $sp, -0x28
                la      $gp, (__gnu_local_gp ↙
    ↳  & 0xFFFF)
                sw      $ra, 0x28+var_4($sp)
                sw      $s2, 0x28+var_8($sp)
                sw      $s1, 0x28+var_C($sp)
                sw      $s0, 0x28+var_10($sp↙
    ↳ )
                sw      $gp, 0x28+var_18($sp↙
    ↳ )
                la      $s2, $LC0        # ↙
    ↳ "%d\n"
                li      $s0, 1
                li      $s1, 0x64  # 'd'

loc_30:                                  # ↙
    ↳ CODE XREF: main+48
                lw      $t9, (printf & 0↙
    ↳ xFFFF)($gp)
                move    $a1, $s0
                move    $a0, $s2
                jalr    $t9
                addiu   $s0, 3
                lw      $gp, 0x28+var_18($sp↙
    ↳ )
                bne     $s0, $s1, loc_30
                or      $at, $zero
                lw      $ra, 0x28+var_4($sp)
                lw      $s2, 0x28+var_8($sp)
                lw      $s1, 0x28+var_C($sp)
                lw      $s0, 0x28+var_10($sp↙
    ↳ )
                jr      $ra
```

 CAPÍTULO 14. LAZOS

```
                addiu   $sp, 0x28

$LC0:           .ascii "%d\n"<0>          #
    DATA XREF: main+20
```

: G.1.8 on page 1921.

CAPÍTULO 15. PROCESAMIENTO DE STRINGS C SIMPLES

Capítulo 15

Procesamiento de strings C simples

15.1 strlen()

```
int my_strlen (const char * str)
{
        const char *eos = str;

        while( *eos++ ) ;

        return( eos - str - 1 );
}

int main()
```

CAPÍTULO 15. PROCESAMIENTO DE STRINGS C SIMPLES

```
{
        // test
        return my_strlen("hello!");
};
```

15.1.1 x86

Sin optimización MSVC

```
_eos$ = -4                              ; size = 4
_str$ = 8                               ; size = 4
_strlen PROC
    push    ebp
    mov     ebp, esp
    push    ecx
    mov     eax, DWORD PTR _str$[ebp]  ;  "⤦
    ↳ str"
    mov     DWORD PTR _eos$[ebp], eax  ;  "⤦
    ↳ eos"
$LN2@strlen_:
    mov     ecx, DWORD PTR _eos$[ebp]  ; ECX⤦
    ↳ =eos

    ;

    movsx   edx, BYTE PTR [ecx]
    mov     eax, DWORD PTR _eos$[ebp]  ; EAX⤦
    ↳ =eos
    add     eax, 1                     ;  ⤦
    ↳ EAX
```

CAPÍTULO 15. PROCESAMIENTO DE STRINGS C SIMPLES

```
    mov     DWORD PTR _eos$[ebp], eax   ;  "↙
    ↳ eos"
    test    edx, edx                    ; EDX↙
    ↳  ?
    je      SHORT $LN1@strlen_          ;
    jmp     SHORT $LN2@strlen_          ;
$LN1@strlen_:

    ;

    mov    eax, DWORD PTR _eos$[ebp]
    sub    eax, DWORD PTR _str$[ebp]
    sub    eax, 1                       ;
    mov    esp, ebp
    pop    ebp
    ret    0
_strlen_ ENDP
```

«» (30 on page 866).

Sin optimización GCC

GCC 4.4.1:

```
                public strlen
strlen          proc near

eos             = dword ptr -4
arg_0           = dword ptr  8

                push    ebp
                mov     ebp, esp
```

CAPÍTULO 15. PROCESAMIENTO DE STRINGS C SIMPLES

```
                sub     esp, 10h
                mov     eax, [ebp+arg_0]
                mov     [ebp+eos], eax

loc_80483F0:
                mov     eax, [ebp+eos]
                movzx   eax, byte ptr [eax]
                test    al, al
                setnz   al
                add     [ebp+eos], 1
                test    al, al
                jnz     short loc_80483F0
                mov     edx, [ebp+eos]
                mov     eax, [ebp+arg_0]
                mov     ecx, edx
                sub     ecx, eax
                mov     eax, ecx
                sub     eax, 1
                leave
                retn
strlen          endp
```

Con optimización MSVC

:

Listing 15.1: Con optimización MSVC 2012 /Ob0

```
_str$ = 8                          ; size = 4
_strlen PROC
        mov     edx, DWORD PTR _str$[esp-4] ↙
    ↘ ; EDX ->
```

CAPÍTULO 15. PROCESAMIENTO DE STRINGS C SIMPLES

```
        mov     eax, edx
    ↳ ;  EAX
$LL2@strlen:
        mov     cl, BYTE PTR [eax]
    ↳ ; CL = *EAX
        inc     eax
    ↳ ; EAX++
        test    cl, cl
    ↳ ; CL==0?
        jne     SHORT $LL2@strlen
    ↳ ;
        sub     eax, edx
    ↳ ;
        dec     eax
    ↳ ;  EAX
        ret     0
_strlen ENDP
```

INC/DEC

 CAPÍTULO 15. PROCESAMIENTO DE STRINGS C SIMPLES

Con optimización MSVC + OllyDbg

OllyDbg. :

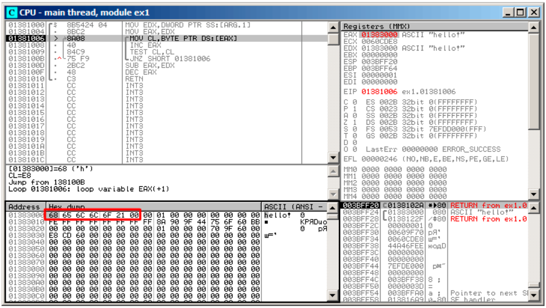


Figura 15.1: OllyDbg:

EAX, «Follow in Dump» «hello!». .

CAPÍTULO 15. PROCESAMIENTO DE STRINGS C SIMPLES

F8 (pasar por encima) :

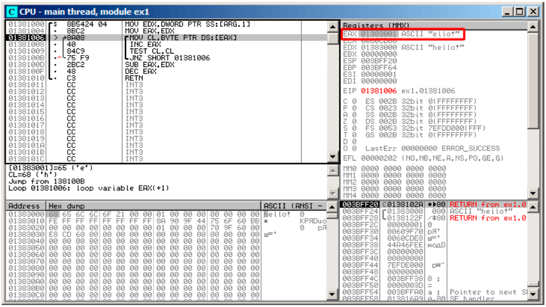


Figura 15.2: OllyDbg:

EAX

CAPÍTULO 15. PROCESAMIENTO DE STRINGS C SIMPLES

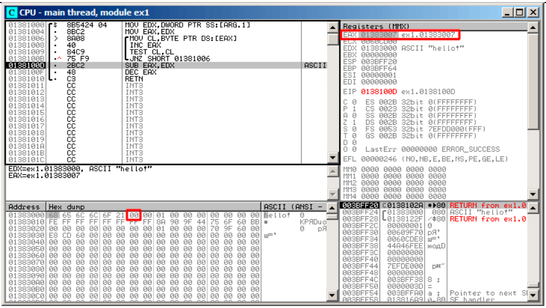


Figura 15.3: OllyDbg:

.

CAPÍTULO 15. PROCESAMIENTO DE STRINGS C SIMPLES

:

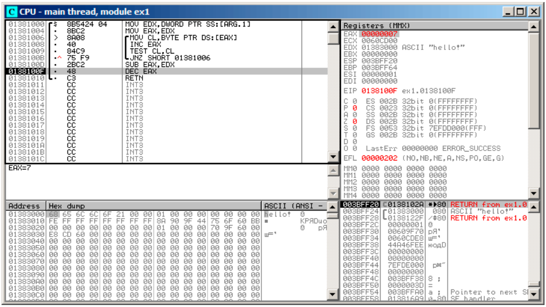


Figura 15.4: OllyDbg:

«hello!», 7. `strlen()`

Con optimización GCC

```
                public strlen
strlen          proc near

arg_0           = dword ptr  8

                push    ebp
                mov     ebp, esp
                mov     ecx, [ebp+arg_0]
                mov     eax, ecx

loc_8048418:
                movzx   edx, byte ptr [eax]
```

CAPÍTULO 15. PROCESAMIENTO DE STRINGS C SIMPLES

```
                add     eax, 1
                test    dl, dl
                jnz     short loc_8048418
                not     ecx
                add     eax, ecx
                pop     ebp
                retn
strlen          endp
```

`mov dl, byte ptr [eax]`.

«» (30 on page 866).

```
ecx=str;
eax=eos;
ecx=(-ecx)-1;
eax=eax+ecx
return eax
```

... :

```
ecx=str;
eax=eos;
eax=eax-ecx;
eax=eax-1;
return eax
```

CAPÍTULO 15. PROCESAMIENTO DE STRINGS C SIMPLES

15.1.2 ARM

32- ARM

Sin optimización Xcode 4.6.3 (LLVM) (Modo ARM)

Listing 15.2: Sin optimización Xcode 4.6.3 (LLVM) (Modo ARM)

```
_strlen

eos  = -8
str  = -4

     SUB    SP, SP, #8 ;
     STR    R0, [SP,#8+str]
     LDR    R0, [SP,#8+str]
     STR    R0, [SP,#8+eos]

loc_2CB8 ; CODE XREF: _strlen+28
     LDR    R0, [SP,#8+eos]
     ADD    R1, R0, #1
     STR    R1, [SP,#8+eos]
     LDRSB  R0, [R0]
     CMP    R0, #0
     BEQ    loc_2CD4
     B      loc_2CB8
loc_2CD4 ; CODE XREF: _strlen+24
     LDR    R0, [SP,#8+eos]
     LDR    R1, [SP,#8+str]
     SUB    R0, R0, R1 ; R0=eos-str
     SUB    R0, R0, #1 ; R0=R0-1
```

CAPÍTULO 15. PROCESAMIENTO DE STRINGS C SIMPLES

```
    ADD     SP, SP, #8 ;
    BX      LR
```

eos y *str*.

str y *eos*.

loc_2CB8.

(LDR, ADD, STR) *eos* R0.

LDRSB R0, [R0] («Load Register Signed Byte») [1]. MOVSX x86. (15.1.1 on page 378) .

, LDRSB CMP y BEQ

eos y *str*

N.B. .

Con optimización Xcode 4.6.3 (LLVM) (Modo Thumb)

Listing 15.3: Con optimización Xcode 4.6.3 (LLVM) (Modo Thumb)

```
_strlen
                MOV         R1, R0

loc_2DF6
                LDRB.W      R2, [R1],#1
                CMP         R2, #0
                BNE         loc_2DF6
```

[1] .

CAPÍTULO 15. PROCESAMIENTO DE STRINGS C SIMPLES

```
            MVNS        R0, R0
            ADD         R0, R1
            BX          LR
```

`LDRB.W R2, [R1],#1`

: 28.2 on page 850.

MVNS[2] $eos - str - 1$. $R0 = str + eos$, (15.1.1 on page 385).

Con optimización Keil 6/2013 (Modo ARM)

Listing 15.4: Con optimización Keil 6/2013 (Modo ARM)

```
_strlen
                MOV     R1, R0

loc_2C8
                LDRB    R2, [R1],#1
                CMP     R2, #0
                SUBEQ   R0, R1, R0
                SUBEQ   R0, R0, #1
                BNE     loc_2C8
                BX      LR
```

$str - eos - 1$ -EQ , , SUBEQ

[2] MoVe Not

CAPÍTULO 15. PROCESAMIENTO DE STRINGS C SIMPLES

ARM64

Con optimización GCC (Linaro) 4.9

```
my_strlen:
        mov     x1, x0
        ; X1
.L58:
        ;
        ldrb    w2, [x1],1
        ; Compare and Branch if NonZero:
        cbnz    w2, .L58
        ;
        sub     x0, x1, x0
        ;
        sub     w0, w0, #1
        ret
```

15.1.1 on page 379:..

Sin optimización GCC (Linaro) 4.9

```
my_strlen:
;
        sub     sp, sp, #32
;  [sp,8]
        str     x0, [sp,8]
        ldr     x0, [sp,8]
;
```

CAPÍTULO 15. PROCESAMIENTO DE STRINGS C SIMPLES

```
        str     x0, [sp,24]
        nop
.L62:
; eos++
        ldr     x0, [sp,24]  ;  "eos"  X0
        add     x1, x0, 1    ;  X0
        str     x1, [sp,24]  ;  X0  "eos"
;
        ldrb    w0, [x0]
;
        cmp     w0, wzr
;  (Branch Not Equal)
        bne     .L62
;
;  "eos"  X1
        ldr     x1, [sp,24]
;  "str"  X0
        ldr     x0, [sp,8]
;
        sub     x0, x1, x0
;
        sub     w0, w0, #1
;
        add     sp, sp, 32
        ret
```

...

15.1.3 MIPS

Listing 15.5: Con optimización GCC 4.4.5 (IDA)

CAPÍTULO 15. PROCESAMIENTO DE STRINGS C SIMPLES

```
my_strlen:
;  $v1:
                move    $v1, $a0

loc_4:
;  $a1:
                lb      $a1, 0($v1)
                or      $at, $zero ; load ↙
    ↘ delay slot, NOP
;  loc_4:
                bnez    $a1, loc_4
; :
                addiu   $v1, 1 ; branch ↙
    ↘ delay slot
; :
                nor     $v0, $zero, $a0
; $v0=-str-1
                jr      $ra
;  = $v1 + $v0 = eos + ( -str-1 ) = eos - ↙
    ↘ str - 1
                addu    $v0, $v1, $v0 ; ↙
    ↘ branch delay slot
```

NOR DST, $ZERO, SRC.

.

CAPÍTULO 15. PROCESAMIENTO DE STRINGS C SIMPLES

15.2 Ejercicios

15.2.1 Ejercicio #1

Lo que hace el código?

Listing 15.6: Con optimización MSVC 2010

```
_s$ = 8
_f      PROC
        mov     edx, DWORD PTR _s$[esp-4]
        mov     cl, BYTE PTR [edx]
        xor     eax, eax
        test    cl, cl
        je      SHORT $LN2@f
        npad    4 ; align next label
$LL4@f:
        cmp     cl, 32
        jne     SHORT $LN3@f
        inc     eax
$LN3@f:
        mov     cl, BYTE PTR [edx+1]
        inc     edx
        test    cl, cl
        jne     SHORT $LL4@f
$LN2@f:
        ret     0
_f      ENDP
```

Listing 15.7: GCC 4.8.1 -O3

```
f:
.LFB24:
```

CAPÍTULO 15. PROCESAMIENTO DE STRINGS C SIMPLES

```
        push    ebx
        mov     ecx, DWORD PTR [esp+8]
        xor     eax, eax
        movzx   edx, BYTE PTR [ecx]
        test    dl, dl
        je      .L2
.L3:
        cmp     dl, 32
        lea     ebx, [eax+1]
        cmove   eax, ebx
        add     ecx, 1
        movzx   edx, BYTE PTR [ecx]
        test    dl, dl
        jne     .L3
.L2:
        pop     ebx
        ret
```

Listing 15.8: Con optimización Keil 6/2013 (Modo ARM)

```
f PROC
        MOV      r1,#0
|L0.4|
        LDRB     r2,[r0,#0]
        CMP      r2,#0
        MOVEQ    r0,r1
        BXEQ     lr
        CMP      r2,#0x20
        ADDEQ    r1,r1,#1
        ADD      r0,r0,#1
        B        |L0.4|
        ENDP
```

CAPÍTULO 15. PROCESAMIENTO DE STRINGS C SIMPLES

Listing 15.9: Con optimización Keil 6/2013 (Modo Thumb)

```
f PROC
        MOVS     r1,#0
        B        |L0.12|
|L0.4|
        CMP      r2,#0x20
        BNE      |L0.10|
        ADDS     r1,r1,#1
|L0.10|
        ADDS     r0,r0,#1
|L0.12|
        LDRB     r2,[r0,#0]
        CMP      r2,#0
        BNE      |L0.4|
        MOVS     r0,r1
        BX       lr
        ENDP
```

Listing 15.10: Con optimización GCC 4.9 (ARM64)

```
f:
        ldrb    w1, [x0]
        cbz     w1, .L4
        mov     w2, 0
.L3:
        cmp     w1, 32
        ldrb    w1, [x0,1]!
        csinc   w2, w2, w2, ne
        cbnz    w1, .L3
.L2:
        mov     w0, w2
        ret
.L4:
```

CAPÍTULO 15. PROCESAMIENTO DE STRINGS C SIMPLES

```
        mov     w2, w1
        b       .L2
```

Listing 15.11: Con optimización GCC 4.4.5 (MIPS) (IDA)

```
f:
                lb      $v1, 0($a0)
                or      $at, $zero
                beqz    $v1, locret_48
                li      $a1, 0x20  # ' '
                b       loc_28
                move    $v0, $zero
loc_18:                                         # ⤦
    ↳ CODE XREF: f:loc_28
                lb      $v1, 0($a0)
                or      $at, $zero
                beqz    $v1, locret_40
                or      $at, $zero

loc_28:                                         # ⤦
    ↳ CODE XREF: f+10
                                                # f⤦
    ↳ +38
                bne     $v1, $a1, loc_18
                addiu   $a0, 1
                lb      $v1, 0($a0)
                or      $at, $zero
                bnez    $v1, loc_28
                addiu   $v0, 1

locret_40:                                      # ⤦
    ↳ CODE XREF: f+20
                jr      $ra
```

CAPÍTULO 15. PROCESAMIENTO DE STRINGS C SIMPLES

```
                or      $at, $zero
locret_48:                                      # CODE XREF: f+8
                jr      $ra
                move    $v0, $zero
```

Responda G.1.9 on page 1922.

Capítulo 16

Substituición de instrucciones aritméticas por otros

, ADD y SUB: línea 18 enSpanish text placeholder.52.1.

A.6.2 on page 1880.

16.1

16.1.1

:

Listing 16.1: Con optimización MSVC 2010

```
unsigned int f(unsigned int a)
{
        return a*8;
};
```

```
_TEXT   SEGMENT
_a$ = 8
                             ; size = 4
_f      PROC
; File c:\polygon\c\2.c
        mov     eax, DWORD PTR _a$[esp-4]
        add     eax, eax
        add     eax, eax
        add     eax, eax
        ret     0
_f      ENDP
_TEXT   ENDS
END
```

16.1.2

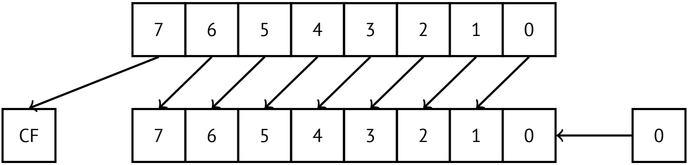
```
unsigned int f(unsigned int a)
{
        return a*4;
```

```
};
```

Listing 16.2: Sin optimización MSVC 2010

```
_a$ = 8         ; size = 4
_f      PROC
        push    ebp
        mov     ebp, esp
        mov     eax, DWORD PTR _a$[ebp]
        shl     eax, 2
        pop     ebp
        ret     0
_f      ENDP
```

:



.

ARM:

Listing 16.3: Sin optimización Keil 6/2013 (Modo ARM)

```
f PROC
        LSL     r0,r0,#2
        BX      lr
        ENDP
```

MIPS:

Listing 16.4: Con optimización GCC 4.4.5 (IDA)

```
                jr      $ra
                sll     $v0, $a0, 2 ; branch delay slot
```

SLL «Shift Left Logical».

16.1.3

.

32-

```
#include <stdint.h>

int f1(int a)
{
        return a*7;
};

int f2(int a)
{
        return a*28;
};

int f3(int a)
{
        return a*17;
};
```

x86

Listing 16.5: Con optimización MSVC 2012

```
; a*7
_a$ = 8
_f1     PROC
        mov     ecx, DWORD PTR _a$[esp-4]
; ECX=a
        lea     eax, DWORD PTR [ecx*8]
; EAX=ECX*8
        sub     eax, ecx
; EAX=EAX-ECX=ECX*8-ECX=ECX*7=a*7
        ret     0
_f1     ENDP

; a*28
_a$ = 8
_f2     PROC
        mov     ecx, DWORD PTR _a$[esp-4]
; ECX=a
        lea     eax, DWORD PTR [ecx*8]
; EAX=ECX*8
        sub     eax, ecx
; EAX=EAX-ECX=ECX*8-ECX=ECX*7=a*7
        shl     eax, 2
; EAX=EAX<<2=(a*7)*4=a*28
        ret     0
_f2     ENDP

; a*17
_a$ = 8
_f3     PROC
```

```
        mov     eax, DWORD PTR _a$[esp-4]
; EAX=a
        shl     eax, 4
; EAX=EAX<<4=EAX*16=a*16
        add     eax, DWORD PTR _a$[esp-4]
; EAX=EAX+a=a*16+a=a*17
        ret     0
_f3     ENDP
```

ARM

Listing 16.6: Con optimización Keil 6/2013 (Modo ARM)

```
; a*7
||f1|| PROC
        RSB      r0,r0,r0,LSL #3
; R0=R0<<3-R0=R0*8-R0=a*8-a=a*7
        BX       lr
        ENDP

; a*28
||f2|| PROC
        RSB      r0,r0,r0,LSL #3
; R0=R0<<3-R0=R0*8-R0=a*8-a=a*7
        LSL      r0,r0,#2
; R0=R0<<2=R0*4=a*7*4=a*28
        BX       lr
        ENDP

; a*17
||f3|| PROC
```

```
        ADD         r0,r0,r0,LSL #4
; R0=R0+R0<<4=R0+R0*16=R0*17=a*17
        BX          lr
        ENDP
```

:

Listing 16.7: Con optimización Keil 6/2013 (Modo Thumb)

```
; a*7
||f1|| PROC
        LSLS        r1,r0,#3
; R1=R0<<3=a<<3=a*8
        SUBS        r0,r1,r0
; R0=R1-R0=a*8-a=a*7
        BX          lr
        ENDP

; a*28
||f2|| PROC
        MOVS        r1,#0x1c ; 28
; R1=28
        MULS        r0,r1,r0
; R0=R1*R0=28*a
        BX          lr
        ENDP

; a*17
||f3|| PROC
        LSLS        r1,r0,#4
; R1=R0<<4=R0*16=a*16
        ADDS        r0,r0,r1
; R0=R0+R1=a+a*16=a*17
```

```
        BX      lr
        ENDP
```

MIPS

Listing 16.8: Con optimización GCC 4.4.5 (IDA)

```
_f1:
                sll     $v0, $a0, 3
; $v0 = $a0<<3 = $a0*8
                jr      $ra
                subu    $v0, $a0 ; branch delay slot
; $v0 = $v0-$a0 = $a0*8-$a0 = $a0*7

_f2:
                sll     $v0, $a0, 5
; $v0 = $a0<<5 = $a0*32
                sll     $a0, 2
; $a0 = $a0<<2 = $a0*4
                jr      $ra
                subu    $v0, $a0 ; branch delay slot
; $v0 = $a0*32-$a0*4 = $a0*28

_f3:
                sll     $v0, $a0, 4
; $v0 = $a0<<4 = $a0*16
                jr      $ra
                addu    $v0, $a0 ; branch delay slot
```

```
; $v0 = $a0*16+$a0 = $a0*17
```

64-

```
#include <stdint.h>

int64_t f1(int64_t a)
{
        return a*7;
};

int64_t f2(int64_t a)
{
        return a*28;
};

int64_t f3(int64_t a)
{
        return a*17;
};
```

x64

Listing 16.9: Con optimización MSVC 2012

```
; a*7
f1:
        lea     rax, [0+rdi*8]
; RAX=RDI*8=a*8
        sub     rax, rdi
```

```
; RAX=RAX-RDI=a*8-a=a*7
        ret

; a*28
f2:
        lea     rax, [0+rdi*4]
; RAX=RDI*4=a*4
        sal     rdi, 5
; RDI=RDI<<5=RDI*32=a*32
        sub     rdi, rax
; RDI=RDI-RAX=a*32-a*4=a*28
        mov     rax, rdi
        ret

; a*17
f3:
        mov     rax, rdi
        sal     rax, 4
; RAX=RAX<<4=a*16
        add     rax, rdi
; RAX=a*16+a=a*17
        ret
```

ARM64

Listing 16.10: Con optimización GCC (Linaro) 4.9 ARM64

```
; a*7
f1:
        lsl     x1, x0, 3
; X1=X0<<3=X0*8=a*8
```

```
        sub     x0, x1, x0
; X0=X1-X0=a*8-a=a*7
        ret

; a*28
f2:
        lsl     x1, x0, 5
; X1=X0<<5=a*32
        sub     x0, x1, x0, lsl 2
; X0=X1-X0<<2=a*32-a<<2=a*32-a*4=a*28
        ret

; a*17
f3:
        add     x0, x0, x0, lsl 4
; X0=X0+X0<<4=a+a*16=a*17
        ret
```

16.2

16.2.1

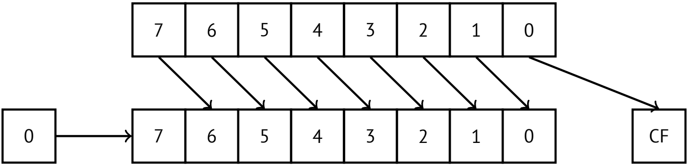
:

```
unsigned int f(unsigned int a)
{
        return a/4;
};
```

(MSVC 2010):

Listing 16.11: MSVC 2010

```
_a$ = 8                                 ; size = 4
_f      PROC
        mov     eax, DWORD PTR _a$[esp-4]
        shr     eax, 2
        ret     0
_f      ENDP
```



Es fácil de entender si imaginas el número 23 en el sistema numérico decimal. 23 se puede dividir fácilmente por 10 simplemente eliminando el último dígito (3—resto de la división). 2 queda después de la operación como quotient.

ARM:

Listing 16.12: Sin optimización Keil 6/2013 (Modo ARM)

```
f PROC
        LSR     r0,r0,#2
        BX      lr
        ENDP
```

MIPS:

Listing 16.13: Con optimización GCC 4.4.5 (IDA)

```
                jr      $ra
                srl     $v0, $a0, 2 ; branch delay slot
```

«Shift Right Logical».

16.3 Ejercicios

16.3.1 Ejercicio #2

Lo que hace el código?

Listing 16.14: Con optimización MSVC 2010

```
_a$ = 8
_f      PROC
        mov     ecx, DWORD PTR _a$[esp-4]
        lea     eax, DWORD PTR [ecx*8]
        sub     eax, ecx
        ret     0
_f      ENDP
```

Listing 16.15: Sin optimización Keil 6/2013 (Modo ARM)

```
f PROC
        RSB     r0,r0,r0,LSL #3
        BX      lr
        ENDP
```

Listing 16.16: Sin optimización Keil 6/2013 (Modo Thumb)

```
f PROC
        LSLS     r1,r0,#3
        SUBS     r0,r1,r0
        BX       lr
        ENDP
```

Listing 16.17: Con optimización GCC 4.9 (ARM64)

```
f:
        lsl     w1, w0, 3
        sub     w0, w1, w0
        ret
```

Listing 16.18: Con optimización GCC 4.4.5 (MIPS) (IDA)

```
f:
                sll     $v0, $a0, 3
                jr      $ra
                subu    $v0, $a0
```

Responda G.1.10 on page 1922.

 CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

Capítulo 17

Unidad de Punto flotante

17.1 IEEE 754

17.2 x86

[1].

.

1

 CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

17.3 ARM, MIPS, x86/x64 SIMD

17.4 C/C++

17.5

:

```
#include <stdio.h>

double f (double a, double b)
{
        return a/3.14 + b*4.1;
};

int main()
{
        printf ("%f\n", f(1.2, 3.4));
};
```

17.5.1 x86

MSVC

MSVC 2010:

Listing 17.1: MSVC 2010: f()

```
CONST    SEGMENT
__real@4010666666666666 DQ 04010666666666666↙
    ↘ r     ; 4.1
```

```
CONST    ENDS
CONST    SEGMENT
__real@40091eb851eb851f DQ 040091↙
    ↘ eb851eb851fr    ; 3.14
CONST    ENDS
_TEXT    SEGMENT
_a$ = 8            ; size = 8
_b$ = 16           ; size = 8
_f  PROC
    push   ebp
    mov    ebp, esp
    fld    QWORD PTR _a$[ebp]

; : ST(0) = _a

    fdiv   QWORD PTR __real@40091eb851eb851f

; : ST(0) =

    fld    QWORD PTR _b$[ebp]

; : ST(0) = _b; ST(1) =

    fmul   QWORD PTR __real@4010666666666666

; :
; ST(0) = ;
; ST(1) =

    faddp  ST(1), ST(0)

; : ST(0) =

```

 CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

```
    pop     ebp
    ret     0
_f  ENDP
```

CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

MSVC + OllyDbg

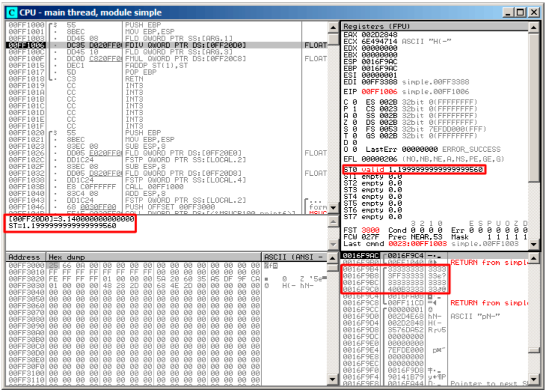


Figura 17.1: OllyDbg:

, OllyDbg

 CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

. FDIV, ST(0) (quotient):

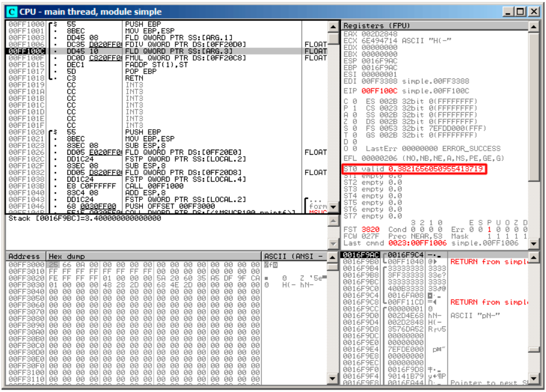


Figura 17.2: OllyDbg: FDIV

 CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

: FLD :

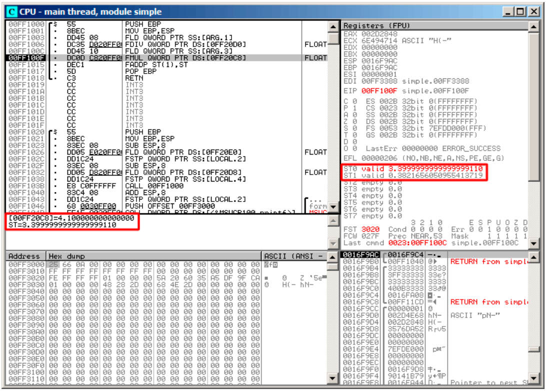


Figura 17.3: OllyDbg:

quotient ST(1). : FMUL. .

: FMUL:

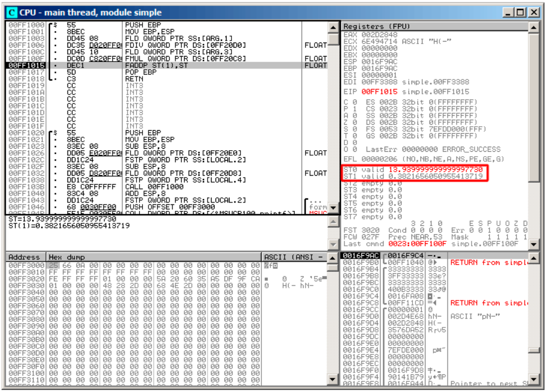


Figura 17.4: OllyDbg: FMUL

 CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

: FADDP:

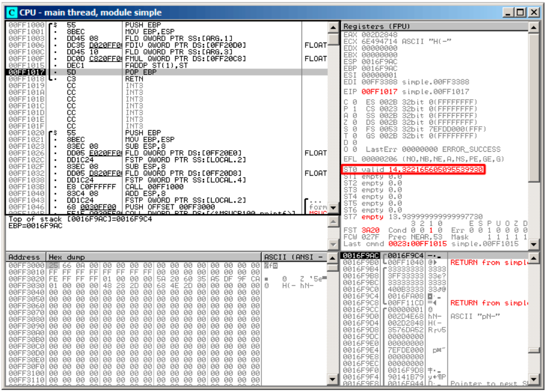


Figura 17.5: OllyDbg: FADDP

?

: 17.2 on page 411. .   (). .

GCC

Listing 17.2: Con optimización GCC 4.4.1

```
                public f
f               proc near

arg_0           = qword ptr  8
arg_8           = qword ptr  10h
```

```
                 push    ebp
                 fld     ds:dbl_8048608 ; ↙
    ↘ 3.14

; : ST(0) = 3.14

                 mov     ebp, esp
                 fdivr   [ebp+arg_0]

; : ST(0) =

                 fld     ds:dbl_8048610 ; 4.1

; : ST(0) = 4.1, ST(1) =

                 fmul    [ebp+arg_8]

; : ST(0) = , ST(1) =

                 pop     ebp
                 faddp   st(1), st

; : ST(0) =

                 retn
f                endp
```

 CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

17.5.2 ARM: Con optimización Xcode 4.6.3 (LLVM) (Modo ARM)

, . VFP (*Vector Floating Point*).

Listing 17.3: Con optimización Xcode 4.6.3 (LLVM) (Modo ARM)

```
f
                VLDR            D16, =3.14
                VMOV            D17, R0, R1 ↙
    ↘ ;  "a"
                VMOV            D18, R2, R3 ↙
    ↘ ;  "b"
                VDIV.F64        D16, D17, ↙
    ↘ D16 ; a/3.14
                VLDR            D17, =4.1
                VMUL.F64        D17, D18, ↙
    ↘ D17 ; b*4.1
                VADD.F64        D16, D17, ↙
    ↘ D16 ; +
                VMOV            R0, R1, D16
                BX              LR

dbl_2C98        DCFD 3.14               ; ↙
    ↘ DATA XREF: f
dbl_2CA0        DCFD 4.1                ; ↙
    ↘ DATA XREF: f+10
```

: B.3.3 on page 1904.

VLDR y VMOV

 CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

VMOV D17, R0, R1 D17.

VMOV R0, R1, D16 : D16 R0 y R1, ,

VDIV, VMUL y VADD,

17.5.3 ARM: Con optimización Keil 6/2013 (Modo Thumb)

```
f
                PUSH    {R3-R7,LR}
                MOVS    R7, R2
                MOVS    R4, R3
                MOVS    R5, R0
                MOVS    R6, R1
                LDR     R2, =0x66666666 ; ⤦
    ↪ 4.1
                LDR     R3, =0x40106666
                MOVS    R0, R7
                MOVS    R1, R4
                BL      __aeabi_dmul
                MOVS    R7, R0
                MOVS    R4, R1
                LDR     R2, =0x51EB851F ; ⤦
    ↪ 3.14
                LDR     R3, =0x40091EB8
                MOVS    R0, R5
                MOVS    R1, R6
                BL      __aeabi_ddiv
                MOVS    R2, R7
                MOVS    R3, R4
                BL      __aeabi_dadd
```

```
                POP     {R3-R7,PC}

; 4.1 :
dword_364       DCD 0x66666666            ; ↙
    ↘ DATA XREF: f+A
dword_368       DCD 0x40106666            ; ↙
    ↘ DATA XREF: f+C
; 3.14 :
dword_36C       DCD 0x51EB851F            ; ↙
    ↘ DATA XREF: f+1A
dword_370       DCD 0x40091EB8            ; ↙
    ↘ DATA XREF: f+1C
```

`__aeabi_dmul`, `__aeabi_ddiv`, `__aeabi_dadd`

17.5.4 ARM64: Con optimización GCC (Linaro) 4.9

:

Listing 17.4: Con optimización GCC (Linaro) 4.9

```
f:
; D0 = a, D1 = b
        ldr     d2, .LC25       ; 3.14
; D2 = 3.14
        fdiv    d0, d0, d2
; D0 = D0/D2 = a/3.14
        ldr     d2, .LC26       ; 4.1
; D2 = 4.1
        fmadd   d0, d1, d2, d0
```

```
; D0 = D1*D2+D0 = b*4.1+a/3.14
        ret

; :
.LC25:
        .word   1374389535      ; 3.14
        .word   1074339512
.LC26:
        .word   1717986918      ; 4.1
        .word   1074816614
```

17.5.5 ARM64: Sin optimización GCC (Linaro) 4.9

Listing 17.5: Sin optimización GCC (Linaro) 4.9

```
f:
        sub     sp, sp, #16
        str     d0, [sp,8]          ;   "a" ↙
    ↘ Register Save Area
        str     d1, [sp]            ;   "b" ↙
    ↘ Register Save Area
        ldr     x1, [sp,8]
; X1 = a
        ldr     x0, .LC25
; X0 = 3.14
        fmov    d0, x1
        fmov    d1, x0
; D0 = a, D1 = 3.14
        fdiv    d0, d0, d1
; D0 = D0/D1 = a/3.14
```

```
        fmov    x1, d0
; X1 = a/3.14
        ldr     x2, [sp]
; X2 = b
        ldr     x0, .LC26
; X0 = 4.1
        fmov    d0, x2
; D0 = b
        fmov    d1, x0
; D1 = 4.1
        fmul    d0, d0, d1
; D0 = D0*D1 = b*4.1

        fmov    x0, d0
; X0 = D0 = b*4.1
        fmov    d0, x1
; D0 = a/3.14
        fmov    d1, x0
; D1 = X0 = b*4.1
        fadd    d0, d0, d1
; D0 = D0+D1 = a/3.14 + b*4.1

        fmov    x0, d0 ; \
        fmov    d0, x0 ; /
        add     sp, sp, 16
        ret
.LC25:
        .word   1374389535      ; 3.14
        .word   1074339512
.LC26:
        .word   1717986918      ; 4.1
        .word   1074816614
```

 CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

Sin optimización GCC . , GCC 4.9 . .

17.5.6 MIPS

C.1.2 on page 1908.

Listing 17.6: Con optimización GCC 4.4.5 (IDA)

```
f:
; $f12-$f13=A
; $f14-$f15=B
                lui     $v0, (dword_C4 >> ↙
    ↘ 16) ; ?
;  $f0:
                lwc1    $f0, dword_BC
                or      $at, $zero                ↙
    ↘   ; load delay slot, NOP
;  $f1:
                lwc1    $f1, $LC0
                lui     $v0, ($LC1 >> 16)   ↙
    ↘   ; ?
; A  $f12-$f13,  $f0-$f1, :
                div.d   $f0, $f12, $f0
; $f0-$f1=A/3.14
;  $f2:
                lwc1    $f2, dword_C4
                or      $at, $zero                ↙
    ↘   ; load delay slot, NOP
;  $f3:
                lwc1    $f3, $LC1
                or      $at, $zero                ↙
    ↘   ; load delay slot, NOP
```

 CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

```
; B  $f14-$f15,  $f2-$f3, :
                mul.d   $f2, $f14, $f2
; $f2-$f3=B*4.1
                jr      $ra
;  $f0-$f1:
                add.d   $f0, $f2                   ↙
    ↘   ; branch delay slot, NOP


.rodata.cst8:000000B8 $LC0:           .word ↙
    ↘ 0x40091EB8          # DATA XREF: f+C
.rodata.cst8:000000BC dword_BC:       .word ↙
    ↘ 0x51EB851F          # DATA XREF: f+4
.rodata.cst8:000000C0 $LC1:           .word ↙
    ↘ 0x40106666          # DATA XREF: f+10
.rodata.cst8:000000C4 dword_C4:       .word ↙
    ↘ 0x66666666          # DATA XREF: f
```

:

- LWC1
- DIV.D, MUL.D, ADD.D («.D» , «.S»)

. [2] .

17.6

```
#include <math.h>
#include <stdio.h>
```

[2] dennis(a)yurichev.com

```
int main ()
{
        printf ("32.01 ^ 1.54 = %lf\n", pow ⤦
    ↳ (32.01,1.54));

        return 0;
}
```

17.6.1 x86

(MSVC 2010):

Listing 17.7: MSVC 2010

```
CONST    SEGMENT
__real@40400147ae147ae1 DQ 040400147⤦
    ↳ ae147ae1r    ; 32.01
__real@3ff8a3d70a3d70a4 DQ 03⤦
    ↳ ff8a3d70a3d70a4r    ; 1.54
CONST    ENDS

_main    PROC
    push   ebp
    mov    ebp, esp
    sub    esp, 8  ;
    fld    QWORD PTR __real@3ff8a3d70a3d70a4
    fstp   QWORD PTR [esp]
    sub    esp, 8  ;
    fld    QWORD PTR __real@40400147ae147ae1
    fstp   QWORD PTR [esp]
```

 CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

```
    call    _pow
    add     esp, 8   ;

;
;

    fstp    QWORD PTR [esp] ;
    push    OFFSET $SG2651
    call    _printf
    add     esp, 12
    xor     eax, eax
    pop     ebp
    ret     0
_main    ENDP
```

: 17.6.2.

17.6.2 ARM + Sin optimización Xcode 4.6.3 (LLVM) (Modo Thumb-2)

```
_main

var_C               = -0xC

                    PUSH            {R7,LR}
                    MOV             R7, SP
                    SUB             SP, SP, #4
                    VLDR            D16, =32.01
                    VMOV            R0, R1, D16
                    VLDR            D16, =1.54
                    VMOV            R2, R3, D16
                    BLX             _pow
```

```
                VMOV            D16, R0, R1
                MOV             R0, 0xFC1 ; ⤦
    ↳ "32.01 ^ 1.54 = %lf\n"
                ADD             R0, PC
                VMOV            R1, R2, D16
                BLX             _printf
                MOVS            R1, 0
                STR             R0, [SP,#0xC⤦
    ↳ +var_C]
                MOV             R0, R1
                ADD             SP, SP, #4
                POP             {R7,PC}

dbl_2F90        DCFD 32.01              ; ⤦
    ↳ DATA XREF: _main+6
dbl_2F98        DCFD 1.54               ; ⤦
    ↳ DATA XREF: _main+E
```

_pow R0 y R1, R2 y R3. R0 y R1. _pow D16, R1 y R2, printf()

17.6.3 ARM + Sin optimización Keil 6/2013 (Modo ARM)

```
_main
                STMFD   SP!, {R4-R6,LR}
                LDR     R2, =0xA3D70A4  ; y
                LDR     R3, =0x3FF8A3D7
                LDR     R0, =0xAE147AE1 ; x
                LDR     R1, =0x40400147
                BL      pow
```

```
                MOV     R4, R0
                MOV     R2, R4
                MOV     R3, R1
                ADR     R0, a32_011_54Lf ; ↙
    ↘ "32.01 ^ 1.54 = %lf\n"
                BL      __2printf
                MOV     R0, #0
                LDMFD   SP!, {R4-R6,PC}

y               DCD 0xA3D70A4           ; ↙
    ↘ DATA XREF: _main+4
dword_520       DCD 0x3FF8A3D7          ; ↙
    ↘ DATA XREF: _main+8
; double x
x               DCD 0xAE147AE1          ; ↙
    ↘ DATA XREF: _main+C
dword_528       DCD 0x40400147          ; ↙
    ↘ DATA XREF: _main+10
a32_011_54Lf    DCB "32.01 ^ 1.54 = %lf",0xA↙
    ↘ ,0
                                        ; ↙
    ↘ DATA XREF: _main+24
```

17.6.4 ARM64 + Con optimización GCC (Linaro) 4.9

Listing 17.8: Con optimización GCC (Linaro) 4.9

```
f:
        stp     x29, x30, [sp, -16]!
        add     x29, sp, 0
```

```
        ldr     d1, .LC1 ;  1.54  D1
        ldr     d0, .LC0 ;  32.01  D0
        bl      pow
;  pow()  D0
        adrp    x0, .LC2
        add     x0, x0, :lo12:.LC2
        bl      printf
        mov     w0, 0
        ldp     x29, x30, [sp], 16
        ret
.LC0:
; 32.01
        .word   -1374389535
        .word   1077936455
.LC1:
; 1.54
        .word   171798692
        .word   1073259479
.LC2:
        .string "32.01 ^ 1.54 = %lf\n"
```

D0 y D1: pow() . D0 pow(). `printf()`, `printf()`.

17.6.5 MIPS

Listing 17.9: Con optimización GCC 4.4.5 (IDA)

```
main:

var_10          = -0x10
var_4           = -4
```

```
; :
                lui     $gp, (dword_9C >> ↙
    ↳ 16)
                addiu   $sp, -0x20
                la      $gp, (__gnu_local_gp↙
    ↳  & 0xFFFF)
                sw      $ra, 0x20+var_4($sp)
                sw      $gp, 0x20+var_10($sp↙
    ↳ )
                lui     $v0, (dword_A4 >> ↙
    ↳ 16) ; ?
;  32.01:
                lwc1    $f12, dword_9C
; :
                lw      $t9, (pow & 0xFFFF)(↙
    ↳ $gp)
;  32.01:
                lwc1    $f13, $LC0
                lui     $v0, ($LC1 >> 16) ; ↙
    ↳ ?
;  1.54:
                lwc1    $f14, dword_A4
                or      $at, $zero ; load ↙
    ↳ delay slot, NOP
;  1.54:
                lwc1    $f15, $LC1
;  pow():
                jalr    $t9
                or      $at, $zero ; branch ↙
    ↳ delay slot, NOP
                lw      $gp, 0x20+var_10($sp↙
    ↳ )
```

 CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

```
; :
                mfc1    $a3, $f0
                lw      $t9, (printf & 0↙
    ↳ xFFFF)($gp)
                mfc1    $a2, $f1
;  printf():
                lui     $a0, ($LC2 >> 16)  #↙
    ↳  "32.01 ^ 1.54 = %lf\n"
                jalr    $t9
                la      $a0, ($LC2 & 0xFFFF)↙
    ↳   # "32.01 ^ 1.54 = %lf\n"
; :
                lw      $ra, 0x20+var_4($sp)
;  0:
                move    $v0, $zero
                jr      $ra
                addiu   $sp, 0x20

.rodata.str1.4:00000084 $LC2:          .↙
    ↳ ascii "32.01 ^ 1.54 = %lf\n"<0>

; 32.01:
.rodata.cst8:00000098 $LC0:            .word ↙
    ↳ 0x40400147         # DATA XREF: main↙
    ↳ +20
.rodata.cst8:0000009C dword_9C:        .word ↙
    ↳ 0xAE147AE1         # DATA XREF: main
.rodata.cst8:0000009C                        ↙
    ↳                    # main+18
; 1.54:
.rodata.cst8:000000A0 $LC1:            .word ↙
    ↳ 0x3FF8A3D7         # DATA XREF: main↙
    ↳ +24
```

 CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

```
.rodata.cst8:000000A0
                            # main+30
.rodata.cst8:000000A4 dword_A4:        .word
    0xA3D70A4               # DATA XREF: main
    +14
```

.

MFC1 («Move From Coprocessor 1»).

17.7

```
#include <stdio.h>

double d_max (double a, double b)
{
        if (a>b)
                return a;

        return b;
};

int main()
{
        printf ("%f\n", d_max (1.2, 3.4));
        printf ("%f\n", d_max (5.6, -4));
};
```

17.7.1 x86

Sin optimización MSVC

:

Listing 17.10: Sin optimización MSVC 2010

```
PUBLIC    _d_max
_TEXT    SEGMENT
_a$ = 8                    ; size = 8
_b$ = 16                   ; size = 8
_d_max     PROC
    push   ebp
    mov    ebp, esp
    fld    QWORD PTR _b$[ebp]

; : ST(0) = _b
;

    fcomp  QWORD PTR _a$[ebp]

;

    fnstsw ax
    test   ah, 5
    jp     SHORT $LN1@d_max

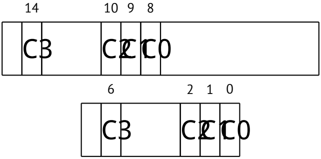
;  a>b

    fld    QWORD PTR _a$[ebp]
    jmp    SHORT $LN2@d_max
$LN1@d_max:
```

 CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

```
    fld     QWORD PTR _b$[ebp]
$LN2@d_max:
    pop     ebp
    ret     0
_d_max      ENDP
```

- 0, 0, 0.
- 0, 0, 1.
- 1, 0, 0.
- 1, 1, 1.



Wikipedia[3]:

> One common reason to test the parity flag actually has nothing to do with parity. The FPU has four condition flags (C0 to C3), but they can not be tested directly, and must instead be first copied to the flags register. When this happens, C0 is placed in the carry flag, C2 in the parity flag and C3 in the zero flag. The C2 flag is set when e.g.

[3]wikipedia.org/wiki/Parity_flag

 CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

incomparable floating point values (NaN or unsupported format) are compared with the FUCOM instructions.

?

OllyDbg:

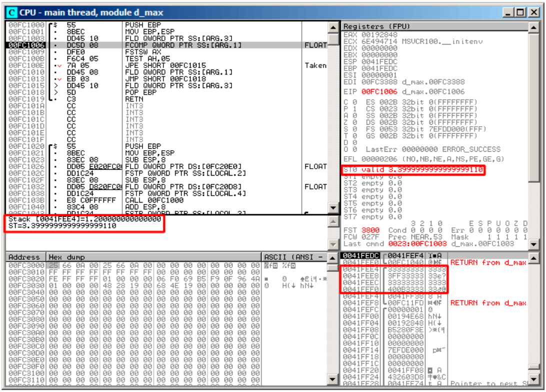


Figura 17.6: OllyDbg:

(). ST(0). . OllyDbg.

CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

FCOMP:

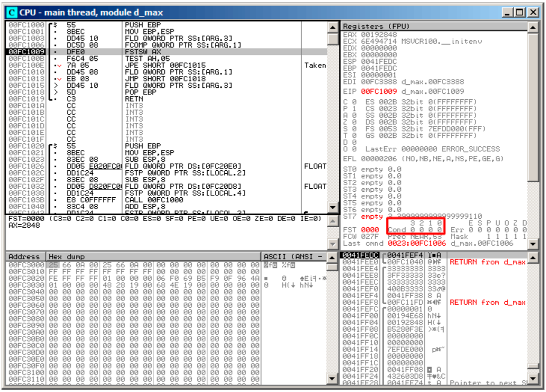


Figura 17.7: OllyDbg: FCOMP

: . : 17.5.1 on page 419.

 CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

FNSTSW:

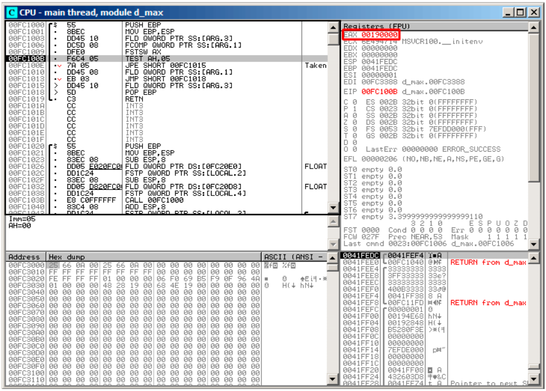


Figura 17.8: OllyDbg: FNSTSW

(OllyDbg FNSTSW FSTSW—).

TEST:

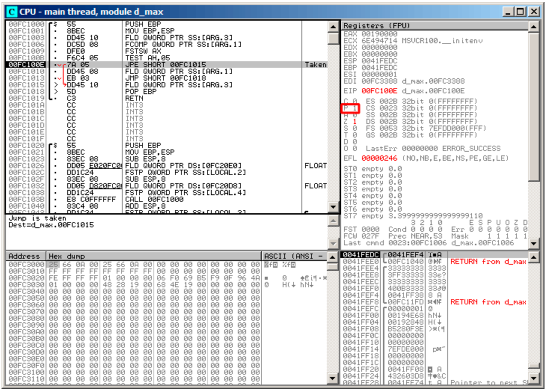


Figura 17.9: OllyDbg: TEST

OllyDbg JP JPE[4]—. .

[4]Jump Parity Even ()

 CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

JPE , FLD:

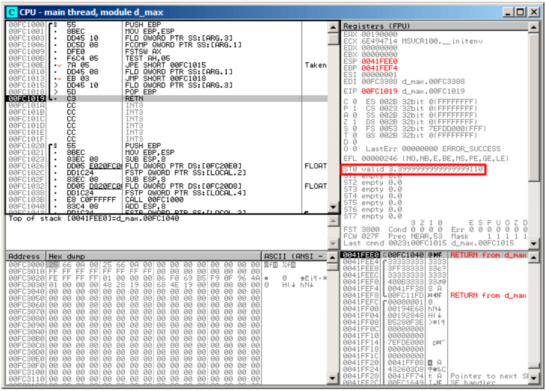


Figura 17.10: OllyDbg:

.

CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

OllyDbg:

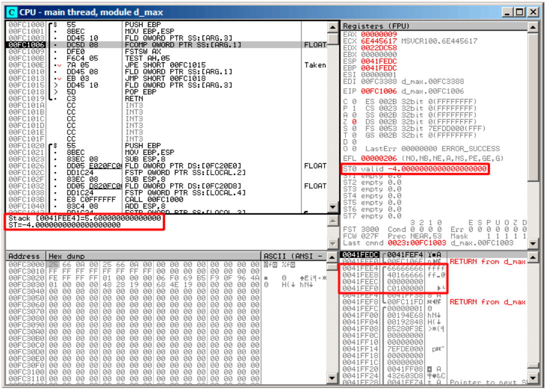


Figura 17.11: OllyDbg:

. b (-4) ST(0). . OllyDbg

 CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

FCOMP:

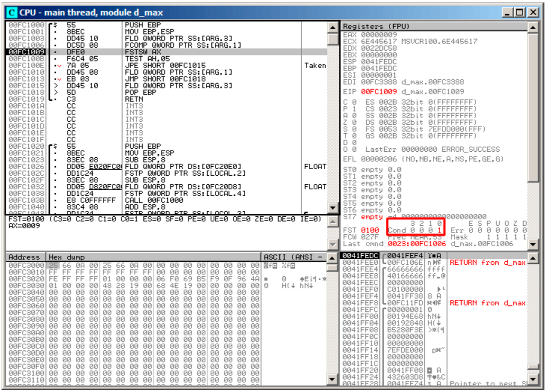


Figura 17.12: OllyDbg: FCOMP

FNSTSW:

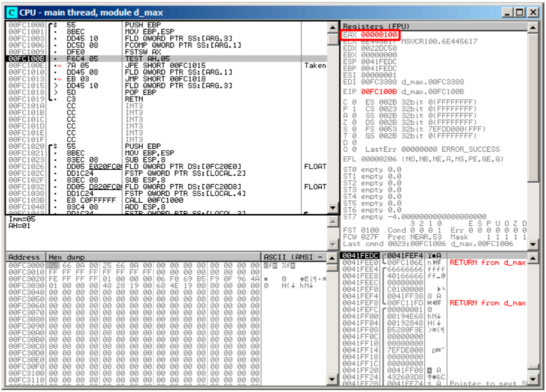


Figura 17.13: OllyDbg: FNSTSW

 CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

TEST:

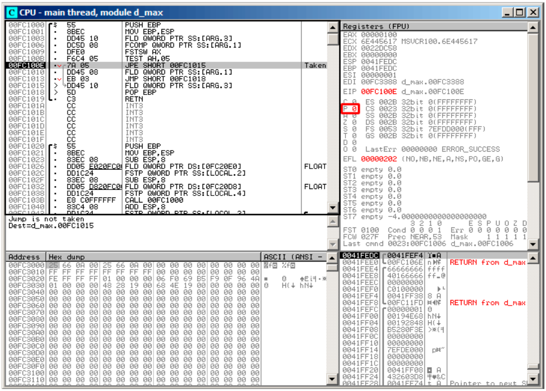


Figura 17.14: OllyDbg: TEST

PF . : JPE .

CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

JPE FLD:

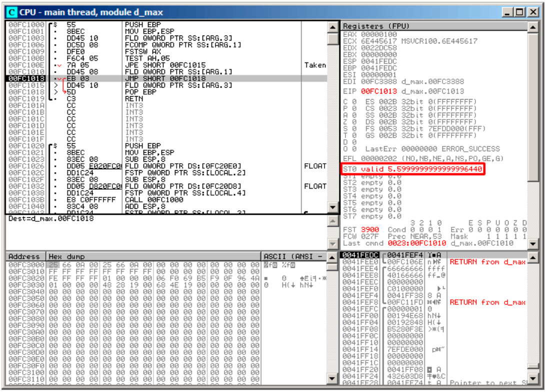


Figura 17.15: OllyDbg:

.

Con optimización MSVC 2010

Listing 17.11: Con optimización MSVC 2010

```
_a$ = 8                          ; size = 8
_b$ = 16                         ; size = 8
_d_max    PROC
    fld     QWORD PTR _b$[esp-4]
    fld     QWORD PTR _a$[esp-4]

; : ST(0) = _a, ST(1) = _b
```

```
    fcom    ST(1) ;  _a  ST(1) = (_b)
    fnstsw  ax
    test    ah, 65 ; 00000041H
    jne     SHORT $LN5@d_max
;
;
    fstp    ST(1)

; : ST(0) = _a

    ret    0
$LN5@d_max:
;
;
    fstp    ST(0)

; : ST(0) = _b

    ret    0
_d_max    ENDP
```

- 0, 0, 0.
- 0, 0, 1.
- 1, 0, 0.

 CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

FLD:

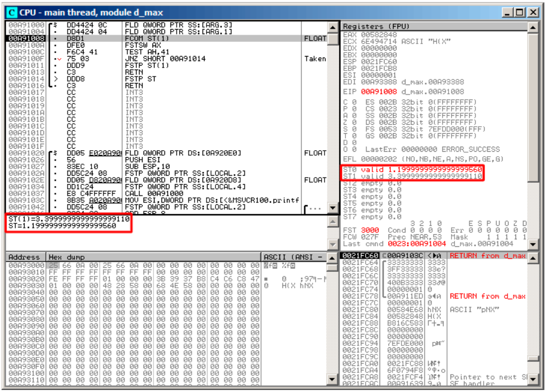


Figura 17.16: OllyDbg:

FCOM: OllyDbg ST(0) y ST(1) .

 CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

FCOM:

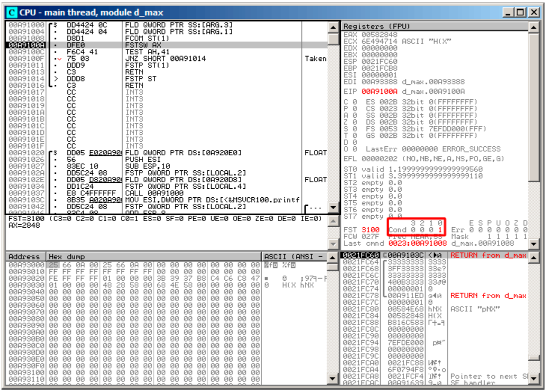


Figura 17.17: OllyDbg: FCOM

C0.

FNSTSW, AX=0x3100:

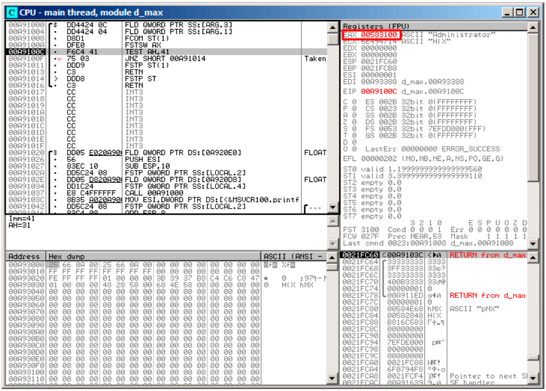


Figura 17.18: OllyDbg: FNSTSW

 CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

TEST:

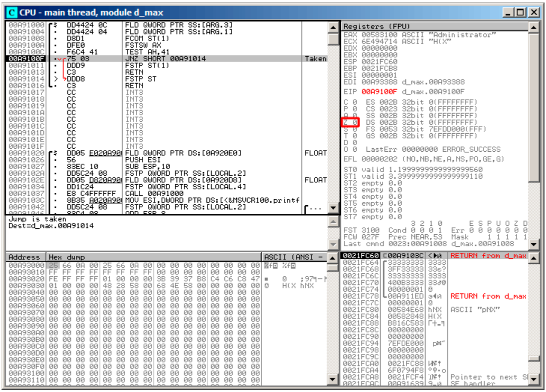


Figura 17.19: OllyDbg: TEST

ZF=0, .

CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

FSTP ST (o FSTP ST(0)) —:

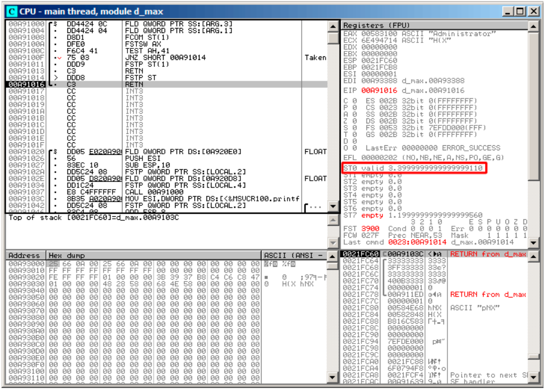


Figura 17.20: OllyDbg: FSTP

FSTP ST

CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

FLD:

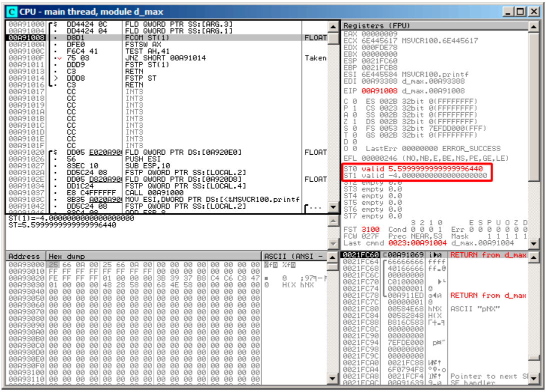


Figura 17.21: OllyDbg:

FCOM:

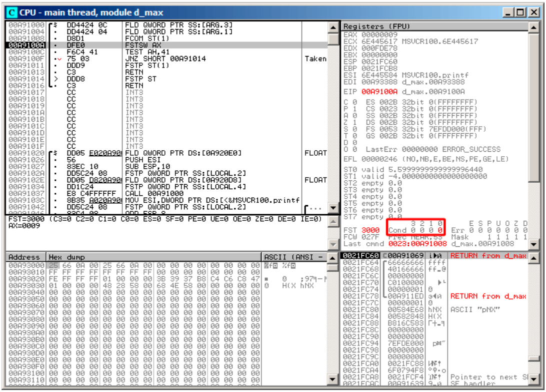


Figura 17.22: OllyDbg: FCOM

•

FNSTSW, AX=0x3000:

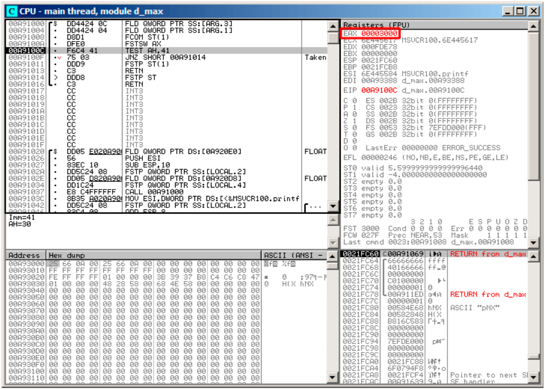


Figura 17.23: OllyDbg: FNSTSW

 CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

TEST:

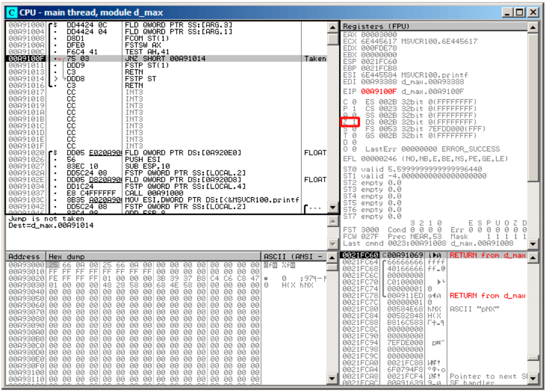


Figura 17.24: OllyDbg: TEST

ZF=1, .

 CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

FSTP ST(1) .

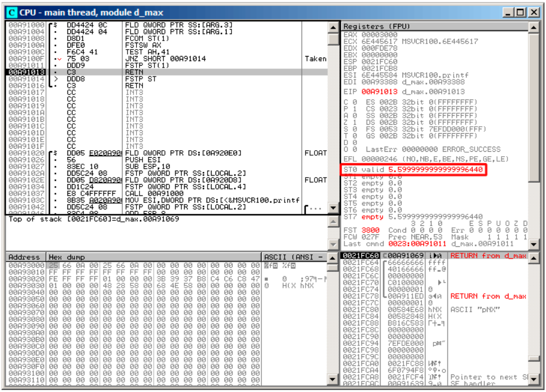


Figura 17.25: OllyDbg: FSTP

FSTP ST(1)

GCC 4.4.1

Listing 17.12: GCC 4.4.1

```
d_max proc near

b               = qword ptr -10h
a               = qword ptr -8
a_first_half    = dword ptr  8
a_second_half   = dword ptr  0Ch
```

```
b_first_half    = dword ptr  10h
b_second_half   = dword ptr  14h

    push    ebp
    mov     ebp, esp
    sub     esp, 10h

; :

    mov     eax, [ebp+a_first_half]
    mov     dword ptr [ebp+a], eax
    mov     eax, [ebp+a_second_half]
    mov     dword ptr [ebp+a+4], eax
    mov     eax, [ebp+b_first_half]
    mov     dword ptr [ebp+b], eax
    mov     eax, [ebp+b_second_half]
    mov     dword ptr [ebp+b+4], eax

; :

    fld     [ebp+a]
    fld     [ebp+b]

; : ST(0) - b; ST(1) - a

    fxch    st(1) ;

; : ST(0) - a; ST(1) - b

    fucompp    ;
    fnstsw  ax ;
    sahf       ;
    setnbe  al ;
```

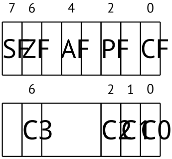
 CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

```
    test    al, al              ; AL==0 ?
    jz      short loc_8048453 ;
    fld     [ebp+a]
    jmp     short locret_8048456

loc_8048453:
    fld     [ebp+b]

locret_8048456:
    leave
    retn
d_max endp
```

.



- 0, 0, 0.
- 0, 0, 1.
- 1, 0, 0.
- ZF=0, PF=0, CF=0.
- ZF=0, PF=0, CF=1.
- ZF=1, PF=0, CF=0.

Con optimización GCC 4.4.1

Listing 17.13: Con optimización GCC 4.4.1

```
                public d_max
d_max           proc near

arg_0           = qword ptr  8
arg_8           = qword ptr  10h

                push    ebp
                mov     ebp, esp
                fld     [ebp+arg_0] ; _a
                fld     [ebp+arg_8] ; _b

; : ST(0) = _b, ST(1) = _a
                fxch    st(1)

; : ST(0) = _a, ST(1) = _b
                fucom   st(1) ;
                fnstsw  ax
                sahf
                ja      short loc_8048448

;
;
                fstp    st
                jmp     short loc_804844A

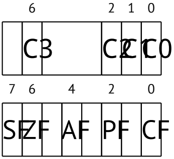
loc_8048448:
;
                fstp    st(1)

loc_804844A:
```

 CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

```
                    pop      ebp
                    retn
d_max               endp
```

Es casi lo mismo excepto que se usa JA después de SAHF. De hecho, las instrucciones de salto condicional que verifican «mayor», «menor» o «igual» para comparación de números sin signo (éstas son JA, JAE, JB, JBE, JEJZ, JNA, JNAE, JNB, JNBE, JNEJNZ) verifican solo las banderas CF y ZF.



GCC 4.8.1

[5]. FUCOMI () y FCMOVcc (CMOVcc,). .

:

Listing 17.14: Con optimización GCC 4.8.1

```
        fld     QWORD PTR [esp+4]       ;  "
    a"
```

[5] Pentium Pro, Pentium-II, Spanish text placeholder.

 CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

```
        fld     QWORD PTR [esp+12]      ;  "↙
    ↘ b"
        ; ST0=b, ST1=a
        fxch    st(1)
        ; ST0=a, ST1=b
        ;  "a"  "b"
        fucomi  st, st(1)
        ;  ST1 ()  ST0  a<=b
        ;
        fcmovbe st, st(1)
        ;  ST1
        fstp    st(1)
        ret
```

.

FUCOMI ST(0) (a) y ST(1) (b) . FCMOVBE ST(1) () ST(0) () $ST0(a) <= ST1(b)$. ($a > b$), a en ST(0).

FSTP ST(0) ST(1).

GDB:

Listing 17.15: Con optimización GCC 4.8.1 and GDB

```
dennis@ubuntuvm:~/polygon$ gcc -O3 d_max.c -↙
    ↘ o d_max -fno-inline
dennis@ubuntuvm:~/polygon$ gdb d_max
GNU gdb (GDB) 7.6.1-ubuntu
Copyright (C) 2013 Free Software Foundation,↙
    ↘  Inc.
License GPLv3+: GNU GPL version 3 or later <↙
    ↘ http://gnu.org/licenses/gpl.html>
```

 CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

```
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.  Type "show copying"
and "show warranty" for details.
This GDB was configured as "i686-linux-gnu".
For bug reporting instructions, please see:
<http://www.gnu.org/software/gdb/bugs/>...
Reading symbols from /home/dennis/polygon/d_max...(no debugging symbols found)...done.
(gdb) b d_max
Breakpoint 1 at 0x80484a0
(gdb) run
Starting program: /home/dennis/polygon/d_max

Breakpoint 1, 0x080484a0 in d_max ()
(gdb) ni
0x080484a4 in d_max ()
(gdb) disas $eip
Dump of assembler code for function d_max:
   0x080484a0 <+0>:     fldl   0x4(%esp)
=> 0x080484a4 <+4>:     fldl   0xc(%esp)
   0x080484a8 <+8>:     fxch   %st(1)
   0x080484aa <+10>:    fucomi %st(1),%st
   0x080484ac <+12>:    fcmovbe %st(1),%st
   0x080484ae <+14>:    fstp   %st(1)
   0x080484b0 <+16>:    ret
End of assembler dump.
(gdb) ni
0x080484a8 in d_max ()
(gdb) info float
  R7: Valid   0x3fff9999999999999800
```

 CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

```
      +1.19999999999999956
=>R6: Valid   0x4000d999999999999800 +3.39999999999999911
  R5: Empty   0x00000000000000000000
  R4: Empty   0x00000000000000000000
  R3: Empty   0x00000000000000000000
  R2: Empty   0x00000000000000000000
  R1: Empty   0x00000000000000000000
  R0: Empty   0x00000000000000000000

Status Word:         0x3000
                       TOP: 6
Control Word:        0x037f   IM DM ZM OM UM PM
                       PC: Extended Precision (64-bits)
                       RC: Round to nearest
Tag Word:            0x0fff
Instruction Pointer: 0x73:0x080484a4
Operand Pointer:     0x7b:0xbffff118
Opcode:              0x0000
(gdb) ni
0x080484aa in d_max ()
(gdb) info float
  R7: Valid   0x4000d999999999999800 +3.39999999999999911
=>R6: Valid   0x3fff9999999999999800 +1.19999999999999956
  R5: Empty   0x00000000000000000000
  R4: Empty   0x00000000000000000000
  R3: Empty   0x00000000000000000000
  R2: Empty   0x00000000000000000000
  R1: Empty   0x00000000000000000000
```

 CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

```
  R0: Empty   0x00000000000000000000

Status Word:         0x3000
                       TOP: 6
Control Word:        0x037f   IM DM ZM OM UM PM
                       PC: Extended Precision (64-bits)
                       RC: Round to nearest
Tag Word:            0x0fff
Instruction Pointer: 0x73:0x080484a8
Operand Pointer:     0x7b:0xbffff118
Opcode:              0x0000
(gdb) disas $eip
Dump of assembler code for function d_max:
   0x080484a0 <+0>:     fldl   0x4(%esp)
   0x080484a4 <+4>:     fldl   0xc(%esp)
   0x080484a8 <+8>:     fxch   %st(1)
=> 0x080484aa <+10>:    fucomi %st(1),%st
   0x080484ac <+12>:    fcmovbe %st(1),%st
   0x080484ae <+14>:    fstp   %st(1)
   0x080484b0 <+16>:    ret
End of assembler dump.
(gdb) ni
0x080484ac in d_max ()
(gdb) info registers
eax            0x1      1
ecx            0xbffff1c4       -1073745468
edx            0x8048340        134513472
ebx            0xb7fbf000       -1208225792
esp            0xbffff10c       0xbffff10c
ebp            0xbffff128       0xbffff128
esi            0x0      0
```

 CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

```
edi            0x0      0
eip            0x80484ac        0x80484ac <d_max+12>
eflags         0x203    [ CF IF ]
cs             0x73     115
ss             0x7b     123
ds             0x7b     123
es             0x7b     123
fs             0x0      0
gs             0x33     51
(gdb) ni
0x080484ae in d_max ()
(gdb) info float
  R7: Valid   0x4000d999999999999800 +3.399999999999999911
=>R6: Valid   0x4000d999999999999800 +3.399999999999999911
  R5: Empty   0x00000000000000000000
  R4: Empty   0x00000000000000000000
  R3: Empty   0x00000000000000000000
  R2: Empty   0x00000000000000000000
  R1: Empty   0x00000000000000000000
  R0: Empty   0x00000000000000000000

Status Word:         0x3000
                       TOP: 6
Control Word:        0x037f   IM DM ZM OM UM PM
                       PC: Extended Precision (64-bits)
                       RC: Round to nearest
Tag Word:            0x0fff
Instruction Pointer: 0x73:0x080484ac
```

 CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

```
Operand Pointer:      0x7b:0xbffff118
Opcode:               0x0000
(gdb) disas $eip
Dump of assembler code for function d_max:
   0x080484a0 <+0>:     fldl   0x4(%esp)
   0x080484a4 <+4>:     fldl   0xc(%esp)
   0x080484a8 <+8>:     fxch   %st(1)
   0x080484aa <+10>:    fucomi %st(1),%st
   0x080484ac <+12>:    fcmovbe %st(1),%st
=> 0x080484ae <+14>:    fstp   %st(1)
   0x080484b0 <+16>:    ret
End of assembler dump.
(gdb) ni
0x080484b0 in d_max ()
(gdb) info float
=>R7: Valid   0x4000d999999999999800 +3.399999999999999911
  R6: Empty   0x4000d999999999999800
  R5: Empty   0x00000000000000000000
  R4: Empty   0x00000000000000000000
  R3: Empty   0x00000000000000000000
  R2: Empty   0x00000000000000000000
  R1: Empty   0x00000000000000000000
  R0: Empty   0x00000000000000000000

Status Word:          0x3800
                        TOP: 7
Control Word:         0x037f   IM DM ZM OM UM PM
                        PC: Extended Precision (64-bits)
                        RC: Round to nearest
Tag Word:             0x3fff
```

 CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

```
51 Instruction Pointer: 0x73:0x080484ae
52 Operand Pointer:     0x7b:0xbffff118
53 Opcode:              0x0000
54 (gdb) quit
55 A debugging session is active.
56
57         Inferior 1 [process 30194] will be killed.
58
59 Quit anyway? (y or n) y
60 dennis@ubuntuvm:~/polygon$
```

«ni»,

(línea 33).

(17.5.1 on page 419). `GDB STx` (Rx). (35)

(línea 54).

`FUCOMI` (línea 83). : CF (línea 95).

`FCMOVBE`

`FSTP` (línea 136). .

17.7.2 ARM

Con optimización Xcode 4.6.3 (LLVM) (Modo ARM)

Listing 17.16: Con optimización Xcode 4.6.3 (LLVM) (Modo ARM)

 CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

```
VMOV            D16, R2, R3 ; b
VMOV            D17, R0, R1 ; a
VCMPE.F64       D17, D16
VMRS            APSR_nzcv, FPSCR
VMOVGT.F64      D16, D17 ;  "b"  D16
VMOV            R0, R1, D16
BX              LR
```

D17 y D16 VCMPE. (FPSCR[6]), , , . VMRS (N, Z, C, V)

VMOVGT MOVGT, (*GTGreater Than*).

, D17.

Con optimización Xcode 4.6.3 (LLVM) (Modo Thumb-2)

Listing 17.17: Con optimización Xcode 4.6.3 (LLVM) (Modo Thumb-2)

```
VMOV            D16, R2, R3 ; b
VMOV            D17, R0, R1 ; a
VCMPE.F64       D17, D16
VMRS            APSR_nzcv, FPSCR
IT GT
VMOVGT.F64      D16, D17
VMOV            R0, R1, D16
BX              LR
```

, ,

IT *if-then block*. IT GT *GT* (*Greater Than*) .

[6](ARM) Floating-Point Status and Control Register

Angry Birds (iOS):

Listing 17.18: Angry Birds Classic

```
...
ITE NE
VMOVNE          R2, R3, D16
VMOVEQ          R2, R3, D17
BLX             _objc_msgSend ; not prefixed
...
```

`ITE` *if-then-else* `ITE` (*NE, not equal*) , . (`NE` `EQ` (*equal*)).

, Angry Birds:

Listing 17.19: Angry Birds Classic

```
...
ITTTT EQ
MOVEQ           R0, R4
ADDEQ           SP, SP, #0x20
POPEQ.W         {R8,R10}
POPEQ           {R4-R7,PC}
BLX             ___stack_chk_fail ; not ↙
    ↘ prefixed
...
```

-EQ.

`ITEEE EQ` (*if-then-else-else-else*),

```
-EQ
-NE
-NE
-NE
```

Angry Birds:

Listing 17.20: Angry Birds Classic

```
...
CMP.W           R0, #0xFFFFFFFF
ITTE LE
SUBLE.W         R10, R0, #1
NEGLE           R0, R0
MOVGT           R10, R0
MOVS            R6, #0         ; not ↙
    ↘ prefixed
CBZ             R0, loc_1E7E32 ; not ↙
    ↘ prefixed
...
```

ITTE (*if-then-then-else*)

: IT, ITE, ITT, ITTE, ITTT, ITTTT.

```
cat AngryBirdsClassic.lst | grep " BF" | ↙
    ↘ grep "IT" > results.lst
```

Sin optimización Xcode 4.6.3 (LLVM) (Modo ARM)

Listing 17.21: Sin optimización Xcode 4.6.3 (LLVM) (Modo ARM)

```
b               = -0x20
a               = -0x18
val_to_return   = -0x10
saved_R7        = -4
```

```
                STR             R7, [SP,#↙
    ↘ saved_R7]!
                MOV             R7, SP
                SUB             SP, SP, #0↙
    ↘ x1C
                BIC             SP, SP, #7
                VMOV            D16, R2, R3
                VMOV            D17, R0, R1
                VSTR            D17, [SP,#0↙
    ↘ x20+a]
                VSTR            D16, [SP,#0↙
    ↘ x20+b]
                VLDR            D16, [SP,#0↙
    ↘ x20+a]
                VLDR            D17, [SP,#0↙
    ↘ x20+b]
                VCMPE.F64       D16, D17
                VMRS            APSR_nzcv, ↙
    ↘ FPSCR
                BLE             loc_2E08
                VLDR            D16, [SP,#0↙
    ↘ x20+a]
                VSTR            D16, [SP,#0↙
    ↘ x20+val_to_return]
                B               loc_2E10

loc_2E08
                VLDR            D16, [SP,#0↙
    ↘ x20+b]
                VSTR            D16, [SP,#0↙
    ↘ x20+val_to_return]

loc_2E10
```

```
                VLDR              D16, [SP,#0↙
    ↳ x20+val_to_return]
                VMOV              R0, R1, D16
                MOV               SP, R7
                LDR               R7, [SP+0x20↙
    ↳ +b],#4
                BX                LR
```

,

Con optimización Keil 6/2013 (Modo Thumb)

Listing 17.22: Con optimización Keil 6/2013 (Modo Thumb)

```
                PUSH    {R3-R7,LR}
                MOVS    R4, R2
                MOVS    R5, R3
                MOVS    R6, R0
                MOVS    R7, R1
                BL      __aeabi_cdrcmple
                BCS     loc_1C0
                MOVS    R0, R6
                MOVS    R1, R7
                POP     {R3-R7,PC}

loc_1C0
                MOVS    R0, R4
                MOVS    R1, R5
                POP     {R3-R7,PC}
```

`__aeabi_cdrcmple`.

 CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

N.B. BCS (*Carry set—Greater than or equal*)

17.7.3 ARM64

Con optimización GCC (Linaro) 4.9

```
d_max:
; D0 - a, D1 - b
        fcmpe   d0, d1
        fcsel   d0, d0, d1, gt
;  D0
        ret
```

ARM64 ISA APSR[7] FPSCR . FPU[8] . FCMPE. D0 y D1 () APSR (N, Z, C, V).

FCSEL (*Floating Conditional Select*) D0 o D1 D0 (GTGreater Than), APSR FPSCR.

(GT), D0 D0 (). D1 D0.

Sin optimización GCC (Linaro) 4.9

```
d_max:
;  "Register Save Area"
        sub     sp, sp, #16
        str     d0, [sp,8]
```

[7](ARM) Application Program Status Register

[8]Floating-point unit

 CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

```
        str     d1, [sp]
;
        ldr     x1, [sp,8]
        ldr     x0, [sp]
        fmov    d0, x1
        fmov    d1, x0
; D0 - a, D1 - b
        fcmpe   d0, d1
        ble     .L76
; a>b;  D0 (a)  X0
        ldr     x0, [sp,8]
        b       .L74
.L76:
; a<=b;  D1 (b)  X0
        ldr     x0, [sp]
.L74:
;  X0
        fmov    d0, x0
;  D0
        add     sp, sp, 16
        ret
```

. (*Register Save Area*). X0/X1 D0/D1 FCMPE. . FCMPE APSR. FCSEL: BLE (*Branch if Less than or Equal*). ($a > b$) a X0. ($a <= b$) b X0. X0 D0, .

Ejercicio

.

Con optimización GCC (Linaro) 4.9float

```
float f_max (float a, float b)
{
        if (a>b)
                return a;

        return b;
};
```

```
f_max:
; S0 - a, S1 - b
        fcmpe   s0, s1
        fcsel   s0, s0, s1, gt
;  S0
        ret
```

17.7.4 MIPS

Listing 17.23: Con optimización GCC 4.4.5 (IDA)

```
d_max:
;  $f14<$f12 (b<a):
                c.lt.d  $f14, $f12
                or      $at, $zero ; NOP
;  locret_14
                bc1t    locret_14
;  ( "a"):
                mov.d   $f0, $f12  ; branch ↙
    ↘ delay slot
```

 CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

```
;
;  "b":
                mov.d   $f0, $f14

locret_14:
                jr      $ra
                or      $at, $zero ; branch ↙
    ↳ delay slot, NOP
```

C.LT.D. LT «Less Than». D *double*.

BC1T. T («True»). «BC1F» («False»).

17.8

17.9 x64

17.10 Ejercicios

17.10.1 Ejercicio #1

FXCH 17.7.1 on page 463.

17.10.2 Ejercicio #2

Lo que hace el código?

Listing 17.24: Con optimización MSVC 2010

```
__real@4014000000000000 DQ 04014000000000000↙
    ↘ r   ; 5

_a1$ = 8        ; size = 8
_a2$ = 16       ; size = 8
_a3$ = 24       ; size = 8
_a4$ = 32       ; size = 8
_a5$ = 40       ; size = 8
_f      PROC
        fld     QWORD PTR _a1$[esp-4]
        fadd    QWORD PTR _a2$[esp-4]
        fadd    QWORD PTR _a3$[esp-4]
        fadd    QWORD PTR _a4$[esp-4]
        fadd    QWORD PTR _a5$[esp-4]
        fdiv    QWORD PTR ↙
    ↘ __real@4014000000000000
        ret     0
_f      ENDP
```

Listing 17.25: Sin optimización Keil 6/2013 (Modo Thumb / Cortex-R4F CPU)

```
f PROC
        VADD.F64 d0,d0,d1
        VMOV.F64 d1,#5.00000000
        VADD.F64 d0,d0,d2
        VADD.F64 d0,d0,d3
        VADD.F64 d2,d0,d4
        VDIV.F64 d0,d2,d1
        BX       lr
        ENDP
```

Listing 17.26: Con optimización GCC 4.9 (ARM64)

```
f:
        fadd    d0, d0, d1
        fmov    d1, 5.0e+0
        fadd    d2, d0, d2
        fadd    d3, d2, d3
        fadd    d0, d3, d4
        fdiv    d0, d0, d1
        ret
```

Listing 17.27: Con optimización GCC 4.4.5 (MIPS) (IDA)

```
f:

arg_10          =  0x10
arg_14          =  0x14
arg_18          =  0x18
arg_1C          =  0x1C
arg_20          =  0x20
arg_24          =  0x24

                lwc1    $f0, arg_14($sp)
                add.d   $f2, $f12, $f14
                lwc1    $f1, arg_10($sp)
                lui     $v0, ($LC0 >> 16)
                add.d   $f0, $f2, $f0
                lwc1    $f2, arg_1C($sp)
                or      $at, $zero
                lwc1    $f3, arg_18($sp)
                or      $at, $zero
                add.d   $f0, $f2
                lwc1    $f2, arg_24($sp)
                or      $at, $zero
```

CAPÍTULO 17. UNIDAD DE PUNTO FLOTANTE

```
                lwc1    $f3, arg_20($sp)
                or      $at, $zero
                add.d   $f0, $f2
                lwc1    $f2, dword_6C
                or      $at, $zero
                lwc1    $f3, $LC0
                jr      $ra
                div.d   $f0, $f2

$LC0:           .word 0x40140000        # ↙
    ↳ DATA XREF: f+C
                                        # f↙
    ↳ +44
dword_6C:       .word 0                 # ↙
    ↳ DATA XREF: f+3C
```

Responda G.1.11 on page 1923.

Capítulo 18

Matriz

18.1

```
#include <stdio.h>

int main()
{
    int a[20];
    int i;

    for (i=0; i<20; i++)
        a[i]=i*2;

    for (i=0; i<20; i++)
        printf ("a[%d]=%d\n", i, a[i]);
```

```
    return 0;
};
```

18.1.1 x86

MSVC

:

Listing 18.1: MSVC 2008

```
_TEXT    SEGMENT
_i$ = -84                              ; size = 4
_a$ = -80                              ; size = 80
_main      PROC
    push   ebp
    mov    ebp, esp
    sub    esp, 84          ; 00000054H
    mov    DWORD PTR _i$[ebp], 0
    jmp    SHORT $LN6@main
$LN5@main:
    mov    eax, DWORD PTR _i$[ebp]
    add    eax, 1
    mov    DWORD PTR _i$[ebp], eax
$LN6@main:
    cmp    DWORD PTR _i$[ebp], 20   ; ↙
    ↘ 00000014H
    jge    SHORT $LN4@main
    mov    ecx, DWORD PTR _i$[ebp]
    shl    ecx, 1
    mov    edx, DWORD PTR _i$[ebp]
```

```
    mov     DWORD PTR _a$[ebp+edx*4], ecx
    jmp     SHORT $LN5@main
$LN4@main:
    mov     DWORD PTR _i$[ebp], 0
    jmp     SHORT $LN3@main
$LN2@main:
    mov     eax, DWORD PTR _i$[ebp]
    add     eax, 1
    mov     DWORD PTR _i$[ebp], eax
$LN3@main:
    cmp     DWORD PTR _i$[ebp], 20    ; ↙
    ↘ 00000014H
    jge     SHORT $LN1@main
    mov     ecx, DWORD PTR _i$[ebp]
    mov     edx, DWORD PTR _a$[ebp+ecx*4]
    push    edx
    mov     eax, DWORD PTR _i$[ebp]
    push    eax
    push    OFFSET $SG2463
    call    _printf
    add     esp, 12         ; 0000000cH
    jmp     SHORT $LN2@main
$LN1@main:
    xor     eax, eax
    mov     esp, ebp
    pop     ebp
    ret     0
_main       ENDP
```

OllyDbg.

:

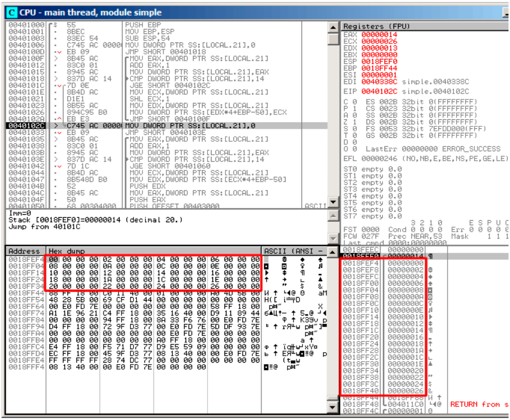


Figura 18.1: OllyDbg:

GCC

Listing 18.2: GCC 4.4.1

```
                public main
main            proc near               ; DATA XREF: _start+17

var_70          = dword ptr -70h
var_6C          = dword ptr -6Ch
var_68          = dword ptr -68h
i_2             = dword ptr -54h
```

```
i               = dword ptr -4

                push    ebp
                mov     ebp, esp
                and     esp, 0FFFFFFF0h
                sub     esp, 70h
                mov     [esp+70h+i], 0        ↙
    ↘      ; i=0
                jmp     short loc_804840A

loc_80483F7:
                mov     eax, [esp+70h+i]
                mov     edx, [esp+70h+i]
                add     edx, edx              ↙
    ↘      ; edx=i*2
                mov     [esp+eax*4+70h+i_2],↙
    ↘ edx
                add     [esp+70h+i], 1        ↙
    ↘      ; i++

loc_804840A:
                cmp     [esp+70h+i], 13h
                jle     short loc_80483F7
                mov     [esp+70h+i], 0
                jmp     short loc_8048441

loc_804841B:
                mov     eax, [esp+70h+i]
                mov     edx, [esp+eax*4+70h+↙
    ↘ i_2]
                mov     eax, offset aADD ; "↙
    ↘ a[%d]=%d\n"
```

```
                mov     [esp+70h+var_68], ↙
    ↘ edx
                mov     edx, [esp+70h+i]
                mov     [esp+70h+var_6C], ↙
    ↘ edx
                mov     [esp+70h+var_70], ↙
    ↘ eax
                call    _printf
                add     [esp+70h+i], 1

loc_8048441:
                cmp     [esp+70h+i], 13h
                jle     short loc_804841B
                mov     eax, 0
                leave
                retn
main            endp
```

18.1.2 ARM

Sin optimización Keil 6/2013 (Modo ARM)

```

            EXPORT _main
_main
            STMFD   SP!, {R4,LR}
            SUB     SP, SP, #0x50       ;

;

            MOV     R4, #0              ; i
            B       loc_4A0
```

```
loc_494
            MOV     R0, R4,LSL#1          ; R0=↙
    ↘ R4*2
            STR     R0, [SP,R4,LSL#2]    ;
            ADD     R4, R4, #1            ; i=i↙
    ↘ +1

loc_4A0
            CMP     R4, #20               ; i↙
    ↘ <20?
            BLT     loc_494               ;

;

            MOV     R4, #0                ; i
            B       loc_4C4
loc_4B0
            LDR     R2, [SP,R4,LSL#2]    ;
            MOV     R1, R4                ; () ↙
    ↘ R1=i
            ADR     R0, aADD              ; "a↙
    ↘ [%d]=%d\n"
            BL      __2printf
            ADD     R4, R4, #1            ; i=i↙
    ↘ +1

loc_4C4
            CMP     R4, #20               ; i↙
    ↘ <20?
            BLT     loc_4B0               ;
            MOV     R0, #0                ;
            ADD     SP, SP, #0x50         ;
            LDMFD   SP!, {R4,PC}
```

SUB SP, SP, #0x50

STR R0, [SP,R4,LSL#2].

Con optimización Keil 6/2013 (Modo Thumb)

```
_main
          PUSH    {R4,R5,LR}
;
          SUB     SP, SP, #0x54

;

          MOVS    R0, #0            ; i
          MOV     R5, SP            ;

loc_1CE
          LSLS    R1, R0, #1        ; R1=i<<1 ↙
    ↘ ( i*2)
          LSLS    R2, R0, #2        ; R2=i<<2 ↙
    ↘ ( i*4)
          ADDS    R0, R0, #1        ; i=i+1
          CMP     R0, #20           ; i<20?
          STR     R1, [R5,R2]       ;
          BLT     loc_1CE           ;

;

          MOVS    R4, #0            ; i=0
loc_1DC
```

```
        LSLS     R0, R4, #2        ; R0=i<<2 ⤦
    ↳ ( i*4)
        LDR      R2, [R5,R0]       ;  *(R5+R0⤦
    ↳ ) ( R5+i*4)
        MOVS     R1, R4
        ADR      R0, aADD          ; "a[%d]=%⤦
    ↳ d\n"
        BL       __2printf
        ADDS     R4, R4, #1        ; i=i+1
        CMP      R4, #20           ; i<20?
        BLT      loc_1DC           ;
        MOVS     R0, #0            ;
;
        ADD      SP, SP, #0x54
        POP      {R4,R5,PC}
```

Sin optimización GCC 4.9.1 (ARM64)

Listing 18.3: Sin optimización GCC 4.9.1 (ARM64)

```
.LC0:
        .string "a[%d]=%d\n"
main:
; :
        stp     x29, x30, [sp, -112]!
;  (FP=SP)
        add     x29, sp, 0
; :
        str     wzr, [x29,108]
; :
        b       .L2
```

```
.L3:
; :
        ldr     w0, [x29,108]
;  2:
        lsl     w2, w0, 1
; :
        add     x0, x29, 24
; :
        ldrsw   x1, [x29,108]
;  (X0+X1<<2=array address+i*4)  W2 (i*2) :
        str     w2, [x0,x1,lsl 2]
;  (i):
        ldr     w0, [x29,108]
        add     w0, w0, 1
        str     w0, [x29,108]
.L2:
; :
        ldr     w0, [x29,108]
        cmp     w0, 19
; :
        ble     .L3
; .
;  0.
;
; .
        str     wzr, [x29,108]
        b       .L4
.L5:
; :
        add     x0, x29, 24
; :
        ldrsw   x1, [x29,108]
;  (X0+X1<<2 =  + i*4)
```

```
        ldr     w2, [x0,x1,lsl 2]
;  "a[%d]=%d\n" :
        adrp    x0, .LC0
        add     x0, x0, :lo12:.LC0
; :
        ldr     w1, [x29,108]
; W2 .
;  printf():
        bl      printf
; :
        ldr     w0, [x29,108]
        add     w0, w0, 1
        str     w0, [x29,108]
.L4:
; ?
        ldr     w0, [x29,108]
        cmp     w0, 19
; :
        ble     .L5
;  0
        mov     w0, 0
; :
        ldp     x29, x30, [sp], 112
        ret
```

18.1.3 MIPS

Listing 18.4: Con optimización GCC 4.4.5 (IDA)

```
main:
```

```
var_70          = -0x70
var_68          = -0x68
var_14          = -0x14
var_10          = -0x10
var_C           = -0xC
var_8           = -8
var_4           = -4
; :
                lui     $gp, (__gnu_local_gp↙
    ↳  >> 16)
                addiu   $sp, -0x80
                la      $gp, (__gnu_local_gp↙
    ↳  & 0xFFFF)
                sw      $ra, 0x80+var_4($sp)
                sw      $s3, 0x80+var_8($sp)
                sw      $s2, 0x80+var_C($sp)
                sw      $s1, 0x80+var_10($sp↙
    ↳ )
                sw      $s0, 0x80+var_14($sp↙
    ↳ )
                sw      $gp, 0x80+var_70($sp↙
    ↳ )
                addiu   $s1, $sp, 0x80+↙
    ↳ var_68
                move    $v1, $s1
                move    $v0, $zero
; .
; :
                li      $a0, 0x28  # '('

loc_34:                                  # ↙
    ↳ CODE XREF: main+3C
; :
```

```
                sw      $v0, 0($v1)
; :
                addiu   $v0, 2
; ?
                bne     $v0, $a0, loc_34
; :
                addiu   $v1, 4
;
;
                la      $s3, $LC0        # "↙
    ↘ a[%d]=%d\n"
;  $s0:
                move    $s0, $zero
                li      $s2, 0x14

loc_54:                                  # ↙
    ↘ CODE XREF: main+70
;  printf():
                lw      $t9, (printf & 0↙
    ↘ xFFFF)($gp)
                lw      $a2, 0($s1)
                move    $a1, $s0
                move    $a0, $s3
                jalr    $t9
;  "i":
                addiu   $s0, 1
                lw      $gp, 0x80+var_70($sp↙
    ↘ )
; :
                bne     $s0, $s2, loc_54
; :
                addiu   $s1, 4
;
```

```
                lw      $ra, 0x80+var_4($sp)
                move    $v0, $zero
                lw      $s3, 0x80+var_8($sp)
                lw      $s2, 0x80+var_C($sp)
                lw      $s1, 0x80+var_10($sp↙
    ↳ )
                lw      $s0, 0x80+var_14($sp↙
    ↳ )
                jr      $ra
                addiu   $sp, 0x80

$LC0:           .ascii "a[%d]=%d\n"<0>    # ↙
    ↳ DATA XREF: main+44
```

39 on page 922.

18.2

18.2.1

```
#include <stdio.h>

int main()
{
        int a[20];
        int i;

        for (i=0; i<20; i++)
                a[i]=i*2;

        printf ("a[20]=%d\n", a[20]);
```

```
        return 0;
};
```

(MSVC 2008):

Listing 18.5: Sin optimización MSVC 2008

```
$SG2474 DB    'a[20]=%d', 0aH, 00H

_i$ = -84 ; size = 4
_a$ = -80 ; size = 80
_main    PROC
    push   ebp
    mov    ebp, esp
    sub    esp, 84
    mov    DWORD PTR _i$[ebp], 0
    jmp    SHORT $LN3@main
$LN2@main:
    mov    eax, DWORD PTR _i$[ebp]
    add    eax, 1
    mov    DWORD PTR _i$[ebp], eax
$LN3@main:
    cmp    DWORD PTR _i$[ebp], 20
    jge    SHORT $LN1@main
    mov    ecx, DWORD PTR _i$[ebp]
    shl    ecx, 1
    mov    edx, DWORD PTR _i$[ebp]
    mov    DWORD PTR _a$[ebp+edx*4], ecx
    jmp    SHORT $LN2@main
$LN1@main:
    mov    eax, DWORD PTR _a$[ebp+80]
    push   eax
```

```
    push   OFFSET $SG2474 ; 'a[20]=%d'
    call   DWORD PTR __imp__printf
    add    esp, 8
    xor    eax, eax
    mov    esp, ebp
    pop    ebp
    ret    0
_main    ENDP
_TEXT    ENDS
END
```

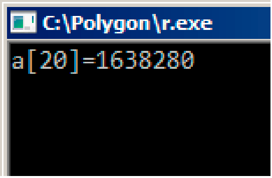


Figura 18.2: OllyDbg:

.

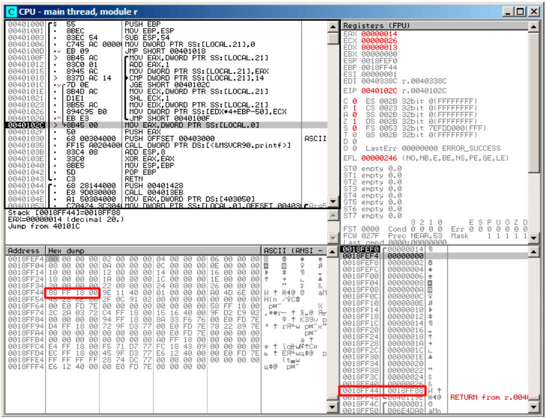


Figura 18.3: OllyDbg:

? .

CAPÍTULO 18. MATRIZ

:

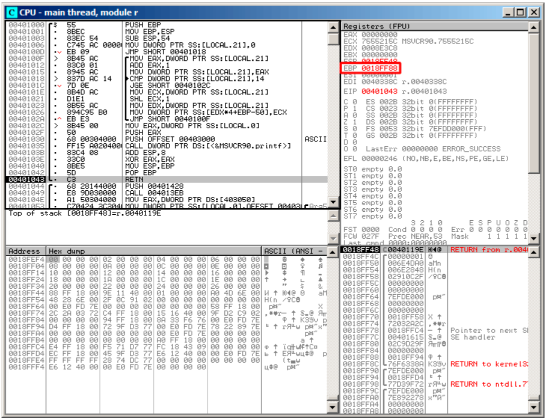


Figura 18.4: OllyDbg:

18.2.2

```
#include <stdio.h>

int main()
{
        int a[20];
        int i;

        for (i=0; i<30; i++)
                a[i]=i;

        return 0;
```

```
};
```

MSVC

Listing 18.6: Sin optimización MSVC 2008

```
_TEXT    SEGMENT
_i$ = -84 ; size = 4
_a$ = -80 ; size = 80
_main   PROC
 push   ebp
 mov    ebp, esp
 sub    esp, 84
 mov    DWORD PTR _i$[ebp], 0
 jmp    SHORT $LN3@main
$LN2@main:
 mov    eax, DWORD PTR _i$[ebp]
 add    eax, 1
 mov    DWORD PTR _i$[ebp], eax
$LN3@main:
 cmp    DWORD PTR _i$[ebp], 30 ; 0000001eH
 jge    SHORT $LN1@main
 mov    ecx, DWORD PTR _i$[ebp]
 mov    edx, DWORD PTR _i$[ebp]      ;
 mov    DWORD PTR _a$[ebp+ecx*4], edx ;
 jmp    SHORT $LN2@main
$LN1@main:
 xor    eax, eax
 mov    esp, ebp
 pop    ebp
 ret    0
```

 CAPÍTULO 18. MATRIZ

```
_main   ENDP
```

OllyDbg,

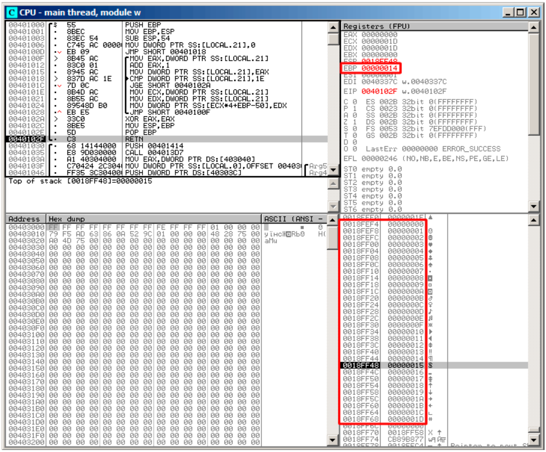


Figura 18.5: OllyDbg:

:

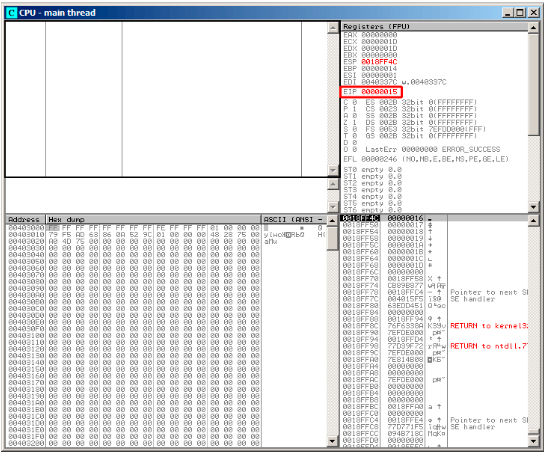


Figura 18.6: OllyDbg:

main():

ESP	
ESP+4	
ESP+84	
ESP+88	

buffer overflow[1].

[1] wikipedia

GCC

```
                public main
main            proc near

a               = dword ptr -54h
i               = dword ptr -4

                push    ebp
                mov     ebp, esp
                sub     esp, 60h ; 96
                mov     [ebp+i], 0
                jmp     short loc_80483D1
loc_80483C3:
                mov     eax, [ebp+i]
                mov     edx, [ebp+i]
                mov     [ebp+eax*4+a], edx
                add     [ebp+i], 1
loc_80483D1:
                cmp     [ebp+i], 1Dh
                jle     short loc_80483C3
                mov     eax, 0
                leave
                retn
main            endp
```

Segmentation fault.

```
(gdb) r
Starting program: /home/dennis/RE/1

Program received signal SIGSEGV, ↙
    ↘ Segmentation fault.
```

```
0x00000016 in ?? ()
(gdb) info registers
eax            0x0      0
ecx            0xd2f96388       -755407992
edx            0x1d     29
ebx            0x26eff4 2551796
esp            0xbffff4b0       0xbffff4b0
ebp            0x15     0x15
esi            0x0      0
edi            0x0      0
eip            0x16     0x16
eflags         0x10202  [ IF RF ]
cs             0x73     115
ss             0x7b     123
ds             0x7b     123
es             0x7b     123
fs             0x0      0
gs             0x33     51
(gdb)
```

18.3

[2]:

```
/RTCs Stack Frame runtime checking
/GZ Enable stack checks (/RTCs)
```

(18.1 on page 483) en MSVC[3] , `@_RTC_CheckStackVars@8`

[2]: wikipedia.org/wiki/Buffer_overflow_protection
[3]Microsoft Visual C++

. `alloca()` (5.2.4 on page 68):

```
#ifdef __GNUC__
#include <alloca.h> // GCC
#else
#include <malloc.h> // MSVC
#endif
#include <stdio.h>

void f()
{
    char *buf=(char*)alloca (600);
#ifdef __GNUC__
    snprintf (buf, 600, "hi! %d, %d, %d\n", ↙
    ↘ 1, 2, 3); // GCC
#else
    _snprintf (buf, 600, "hi! %d, %d, %d\n",↙
    ↘  1, 2, 3); // MSVC
#endif

    puts (buf);
};
```

Listing 18.7: GCC 4.7.3

```

.LC0:
        .string "hi! %d, %d, %d\n"
f:
        push    ebp
        mov     ebp, esp
        push    ebx
```

```
        sub     esp, 676
        lea     ebx, [esp+39]
        and     ebx, -16
        mov     DWORD PTR [esp+20], 3
        mov     DWORD PTR [esp+16], 2
        mov     DWORD PTR [esp+12], 1
        mov     DWORD PTR [esp+8], OFFSET ↙
    ↳ FLAT:.LC0  ; "hi! %d, %d, %d\n"
        mov     DWORD PTR [esp+4], 600
        mov     DWORD PTR [esp], ebx
        mov     eax, DWORD PTR gs:20                  ↙
    ↳           ;
        mov     DWORD PTR [ebp-12], eax
        xor     eax, eax
        call    _snprintf
        mov     DWORD PTR [esp], ebx
        call    puts
        mov     eax, DWORD PTR [ebp-12]
        xor     eax, DWORD PTR gs:20                  ↙
    ↳           ;
        jne     .L5
        mov     ebx, DWORD PTR [ebp-4]
        leave
        ret
.L5:
        call    __stack_chk_fail
```

`gs:20`. `gs:20`. `__stack_chk_fail` (Ubuntu 13.04 x86):

```
*** buffer overflow detected ***: ./2_1 ↙
    ↳ terminated
======= Backtrace: =========
```

```
/lib/i386-linux-gnu/libc.so.6(__fortify_fail↙
    ↳ +0x63)[0xb7699bc3]
/lib/i386-linux-gnu/libc.so.6(+0x10593a)[0↙
    ↳ xb769893a]
/lib/i386-linux-gnu/libc.so.6(+0x105008)[0↙
    ↳ xb7698008]
/lib/i386-linux-gnu/libc.so.6(↙
    ↳ _IO_default_xsputn+0x8c)[0xb7606e5c]
/lib/i386-linux-gnu/libc.so.6(_IO_vfprintf+0↙
    ↳ x165)[0xb75d7a45]
/lib/i386-linux-gnu/libc.so.6(__vsprintf_chk↙
    ↳ +0xc9)[0xb76980d9]
/lib/i386-linux-gnu/libc.so.6(__sprintf_chk↙
    ↳ +0x2f)[0xb7697fef]
./2_1[0x8048404]
/lib/i386-linux-gnu/libc.so.6(↙
    ↳ __libc_start_main+0xf5)[0xb75ac935]
======= Memory map: ========
08048000-08049000 r-xp 00000000 08:01 ↙
    ↳ 2097586    /home/dennis/2_1
08049000-0804a000 r--p 00000000 08:01 ↙
    ↳ 2097586    /home/dennis/2_1
0804a000-0804b000 rw-p 00001000 08:01 ↙
    ↳ 2097586    /home/dennis/2_1
094d1000-094f2000 rw-p 00000000 00:00 0     ↙
    ↳      [heap]
b7560000-b757b000 r-xp 00000000 08:01 ↙
    ↳ 1048602    /lib/i386-linux-gnu/↙
    ↳ libgcc_s.so.1
b757b000-b757c000 r--p 0001a000 08:01 ↙
    ↳ 1048602    /lib/i386-linux-gnu/↙
    ↳ libgcc_s.so.1
b757c000-b757d000 rw-p 0001b000 08:01 ↙
```

```
    ↳ 1048602    /lib/i386-linux-gnu/↙
    ↳ libgcc_s.so.1
b7592000-b7593000 rw-p 00000000 00:00 0
b7593000-b7740000 r-xp 00000000 08:01 ↙
    ↳ 1050781    /lib/i386-linux-gnu/libc↙
    ↳ -2.17.so
b7740000-b7742000 r--p 001ad000 08:01 ↙
    ↳ 1050781    /lib/i386-linux-gnu/libc↙
    ↳ -2.17.so
b7742000-b7743000 rw-p 001af000 08:01 ↙
    ↳ 1050781    /lib/i386-linux-gnu/libc↙
    ↳ -2.17.so
b7743000-b7746000 rw-p 00000000 00:00 0
b775a000-b775d000 rw-p 00000000 00:00 0
b775d000-b775e000 r-xp 00000000 00:00 0      ↙
    ↳      [vdso]
b775e000-b777e000 r-xp 00000000 08:01 ↙
    ↳ 1050794    /lib/i386-linux-gnu/ld↙
    ↳ -2.17.so
b777e000-b777f000 r--p 0001f000 08:01 ↙
    ↳ 1050794    /lib/i386-linux-gnu/ld↙
    ↳ -2.17.so
b777f000-b7780000 rw-p 00020000 08:01 ↙
    ↳ 1050794    /lib/i386-linux-gnu/ld↙
    ↳ -2.17.so
bff35000-bff56000 rw-p 00000000 00:00 0      ↙
    ↳      [stack]
Aborted (core dumped)
```

gs TIB[4] [5].

[4] Thread Information Block

[5] wikipedia.org/wiki/Win32_Thread_Information_Block

arch/x86/include/asm/stackprotector.h

18.3.1 Con optimización Xcode 4.6.3 (LLVM) (Modo Thumb-2)

(18.1 on page 483),

```
_main

var_64          = -0x64
var_60          = -0x60
var_5C          = -0x5C
var_58          = -0x58
var_54          = -0x54
var_50          = -0x50
var_4C          = -0x4C
var_48          = -0x48
var_44          = -0x44
var_40          = -0x40
var_3C          = -0x3C
var_38          = -0x38
var_34          = -0x34
var_30          = -0x30
var_2C          = -0x2C
var_28          = -0x28
var_24          = -0x24
var_20          = -0x20
var_1C          = -0x1C
var_18          = -0x18
canary          = -0x14
var_10          = -0x10
```

```
    PUSH     {R4-R7,LR}
    ADD      R7, SP, #0xC
    STR.W    R8, [SP,#0xC+var_10]!
    SUB      SP, SP, #0x54
    MOVW     R0, #aObjc_methtype ; "⤦
    ↳ objc_methtype"
    MOVS     R2, #0
    MOVT.W   R0, #0
    MOVS     R5, #0
    ADD      R0, PC
    LDR.W    R8, [R0]
    LDR.W    R0, [R8]
    STR      R0, [SP,#0x64+canary]
    MOVS     R0, #2
    STR      R2, [SP,#0x64+var_64]
    STR      R0, [SP,#0x64+var_60]
    MOVS     R0, #4
    STR      R0, [SP,#0x64+var_5C]
    MOVS     R0, #6
    STR      R0, [SP,#0x64+var_58]
    MOVS     R0, #8
    STR      R0, [SP,#0x64+var_54]
    MOVS     R0, #0xA
    STR      R0, [SP,#0x64+var_50]
    MOVS     R0, #0xC
    STR      R0, [SP,#0x64+var_4C]
    MOVS     R0, #0xE
    STR      R0, [SP,#0x64+var_48]
    MOVS     R0, #0x10
    STR      R0, [SP,#0x64+var_44]
    MOVS     R0, #0x12
    STR      R0, [SP,#0x64+var_40]
```

```
    MOVS      R0, #0x14
    STR       R0, [SP,#0x64+var_3C]
    MOVS      R0, #0x16
    STR       R0, [SP,#0x64+var_38]
    MOVS      R0, #0x18
    STR       R0, [SP,#0x64+var_34]
    MOVS      R0, #0x1A
    STR       R0, [SP,#0x64+var_30]
    MOVS      R0, #0x1C
    STR       R0, [SP,#0x64+var_2C]
    MOVS      R0, #0x1E
    STR       R0, [SP,#0x64+var_28]
    MOVS      R0, #0x20
    STR       R0, [SP,#0x64+var_24]
    MOVS      R0, #0x22
    STR       R0, [SP,#0x64+var_20]
    MOVS      R0, #0x24
    STR       R0, [SP,#0x64+var_1C]
    MOVS      R0, #0x26
    STR       R0, [SP,#0x64+var_18]
    MOV       R4, 0xFDA ; "a[%d]=%d\n"
    MOV       R0, SP
    ADDS      R6, R0, #4
    ADD       R4, PC
    B         loc_2F1C

;

loc_2F14
    ADDS      R0, R5, #1
    LDR.W     R2, [R6,R5,LSL#2]
    MOV       R5, R0
```

```
loc_2F1C
    MOV      R0, R4
    MOV      R1, R5
    BLX      _printf
    CMP      R5, #0x13
    BNE      loc_2F14
    LDR.W    R0, [R8]
    LDR      R1, [SP,#0x64+canary]
    CMP      R0, R1
    ITTTT EQ            ; ?
    MOVEQ    R0, #0
    ADDEQ    SP, SP, #0x54
    LDREQ.W  R8, [SP+0x64+var_64],#4
    POPEQ    {R4-R7,PC}
    BLX      ___stack_chk_fail
```

___stack_chk_fail

18.4

```
void f(int size)
{
    int a[size];
...
};
```

18.5

Listing 18.8:

```
#include <stdio.h>

const char* month1[]=
{
        "January",
        "February",
        "March",
        "April",
        "May",
        "June",
        "July",
        "August",
        "September",
        "October",
        "November",
        "December"
};

//
const char* get_month1 (int month)
{
        return month1[month];
};
```

18.5.1 x64

Listing 18.9: Con optimización MSVC 2013 x64

```
_DATA   SEGMENT
month1  DQ      FLAT:$SG3122
```

```
        DQ      FLAT:$SG3123
        DQ      FLAT:$SG3124
        DQ      FLAT:$SG3125
        DQ      FLAT:$SG3126
        DQ      FLAT:$SG3127
        DQ      FLAT:$SG3128
        DQ      FLAT:$SG3129
        DQ      FLAT:$SG3130
        DQ      FLAT:$SG3131
        DQ      FLAT:$SG3132
        DQ      FLAT:$SG3133
$SG3122 DB      'January', 00H
$SG3123 DB      'February', 00H
$SG3124 DB      'March', 00H
$SG3125 DB      'April', 00H
$SG3126 DB      'May', 00H
$SG3127 DB      'June', 00H
$SG3128 DB      'July', 00H
$SG3129 DB      'August', 00H
$SG3130 DB      'September', 00H
$SG3156 DB      '%s', 0aH, 00H
$SG3131 DB      'October', 00H
$SG3132 DB      'November', 00H
$SG3133 DB      'December', 00H
_DATA   ENDS

month$ = 8
get_month1 PROC
        movsxd  rax, ecx
        lea     rcx, OFFSET FLAT:month1
        mov     rax, QWORD PTR [rcx+rax*8]
        ret     0
get_month1 ENDP
```

:

- [6].
-
- .

Con optimización GCC 4.9 [7]:

Listing 18.10: Con optimización GCC 4.9 x64

```
        movsx   rdi, edi
        mov     rax, QWORD PTR month1[0+rdi*8]
        ret
```

32-bit MSVC

Listing 18.11: Con optimización MSVC 2013 x86

```
_month$ = 8
_get_month1 PROC
        mov     eax, DWORD PTR _month$[esp-4]
        mov     eax, DWORD PTR _month1[eax*4]
        ret     0
_get_month1 ENDP
```

6

7

18.5.2 32- ARM

ARM

Listing 18.12: Con optimización Keil 6/2013 (Modo ARM)

```
get_month1 PROC
        LDR      r1,|L0.100|
        LDR      r0,[r1,r0,LSL #2]
        BX       lr
        ENDP

|L0.100|
        DCD      ||.data||

        DCB      "January",0
        DCB      "February",0
        DCB      "March",0
        DCB      "April",0
        DCB      "May",0
        DCB      "June",0
        DCB      "July",0
        DCB      "August",0
        DCB      "September",0
        DCB      "October",0
        DCB      "November",0
        DCB      "December",0

        AREA ||.data||, DATA, ALIGN=2
month1
        DCD      ||.conststring||
        DCD      ||.conststring||+0x8
        DCD      ||.conststring||+0x11
        DCD      ||.conststring||+0x17
```

```
        DCD      ||.conststring||+0x1d
        DCD      ||.conststring||+0x21
        DCD      ||.conststring||+0x26
        DCD      ||.conststring||+0x2b
        DCD      ||.conststring||+0x32
        DCD      ||.conststring||+0x3c
        DCD      ||.conststring||+0x44
        DCD      ||.conststring||+0x4d
```

ARM

```
get_month1 PROC
        LSLS     r0,r0,#2
        LDR      r1,|L0.64|
        LDR      r0,[r1,r0]
        BX       lr
        ENDP
```

18.5.3 ARM64

Listing 18.13: Con optimización GCC 4.9 ARM64

```
get_month1:
        adrp    x1, .LANCHOR0
        add     x1, x1, :lo12:.LANCHOR0
        ldr     x0, [x1,w0,sxtw 3]
        ret

.LANCHOR0 = . + 0
        .type   month1, %object
        .size   month1, 96
```

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```
month1:
        .xword  .LC2
        .xword  .LC3
        .xword  .LC4
        .xword  .LC5
        .xword  .LC6
        .xword  .LC7
        .xword  .LC8
        .xword  .LC9
        .xword  .LC10
        .xword  .LC11
        .xword  .LC12
        .xword  .LC13
.LC2:
        .string "January"
.LC3:
        .string "February"
.LC4:
        .string "March"
.LC5:
        .string "April"
.LC6:
        .string "May"
.LC7:
        .string "June"
.LC8:
        .string "July"
.LC9:
        .string "August"
.LC10:
        .string "September"
.LC11:
        .string "October"
```

```
.LC12:
        .string "November"
.LC13:
        .string "December"
```

18.5.4 MIPS

Listing 18.14: Con optimización GCC 4.4.5 (IDA)

```

get_month1:
;  $v0:
                la      $v0, month1
;  4:
                sll     $a0, 2
; :
                addu    $a0, $v0
;  $v0:
                lw      $v0, 0($a0)
;
                jr      $ra
                or      $at, $zero ; branch ↙
    ↘ delay slot, NOP

                .data # .data.rel.local
                .globl month1
month1:         .word aJanuary              # "↙
    ↘ January"
                .word aFebruary             # "↙
    ↘ February"
                .word aMarch                # "↙
    ↘ March"
```

CAPÍTULO 18. MATRIZ

```
                .word aApril                # "↙
    ↳ April"
                .word aMay                  # "↙
    ↳ May"
                .word aJune                 # "↙
    ↳ June"
                .word aJuly                 # "↙
    ↳ July"
                .word aAugust               # "↙
    ↳ August"
                .word aSeptember            # "↙
    ↳ September"
                .word aOctober              # "↙
    ↳ October"
                .word aNovember             # "↙
    ↳ November"
                .word aDecember             # "↙
    ↳ December"

                .data # .rodata.str1.4
aJanuary:       .ascii "January"<0>
aFebruary:      .ascii "February"<0>
aMarch:         .ascii "March"<0>
aApril:         .ascii "April"<0>
aMay:           .ascii "May"<0>
aJune:          .ascii "June"<0>
aJuly:          .ascii "July"<0>
aAugust:        .ascii "August"<0>
aSeptember:     .ascii "September"<0>
aOctober:       .ascii "October"<0>
aNovember:      .ascii "November"<0>
aDecember:      .ascii "December"<0>
```

18.5.5

Listing 18.15: IDA

```
off_140011000   dq offset aJanuary_1      ; ↙
    ↳ DATA XREF: .text:0000000140001003
                                          ; "↙
    ↳ January"
                dq offset aFebruary_1     ; "↙
    ↳ February"
                dq offset aMarch_1        ; "↙
    ↳ March"
                dq offset aApril_1        ; "↙
    ↳ April"
                dq offset aMay_1          ; "↙
    ↳ May"
                dq offset aJune_1         ; "↙
    ↳ June"
                dq offset aJuly_1         ; "↙
    ↳ July"
                dq offset aAugust_1       ; "↙
    ↳ August"
                dq offset aSeptember_1    ; "↙
    ↳ September"
                dq offset aOctober_1      ; "↙
    ↳ October"
                dq offset aNovember_1     ; "↙
    ↳ November"
                dq offset aDecember_1     ; "↙
    ↳ December"
aJanuary_1      db 'January',0            ; ↙
    ↳ DATA XREF: sub_140001020+4
```

 CAPÍTULO 18. MATRIZ

```
                                                ; .↙
    ↘ data:off_140011000
aFebruary_1     db 'February',0                 ; ↙
    ↘ DATA XREF: .data:0000000140011008
                align 4
aMarch_1        db 'March',0                    ; ↙
    ↘ DATA XREF: .data:0000000140011010
                align 4
aApril_1        db 'April',0                    ; ↙
    ↘ DATA XREF: .data:0000000140011018
```

.

Listing 18.16: IDA

```
off_140011000   dq offset qword_140011060
                                                ; ↙
    ↘ DATA XREF: .text:0000000140001003
                dq offset aFebruary_1   ; "↙
    ↘ February"
                dq offset aMarch_1      ; "↙
    ↘ March"
                dq offset aApril_1      ; "↙
    ↘ April"
                dq offset aMay_1        ; "↙
    ↘ May"
                dq offset aJune_1       ; "↙
    ↘ June"
                dq offset aJuly_1       ; "↙
    ↘ July"
                dq offset aAugust_1     ; "↙
    ↘ August"
                dq offset aSeptember_1  ; "↙
```

 CAPÍTULO 18. MATRIZ

```
    ↘ September"
                dq offset aOctober_1     ; "↙
    ↘ October"
                dq offset aNovember_1    ; "↙
    ↘ November"
                dq offset aDecember_1    ; "↙
    ↘ December"
qword_140011060 dq 797261756E614Ah       ; ↙
    ↘ DATA XREF: sub_140001020+4
                                         ; .↙
    ↘ data:off_140011000
aFebruary_1     db 'February',0          ; ↙
    ↘ DATA XREF: .data:0000000140011008
                align 4
aMarch_1        db 'March',0             ; ↙
    ↘ DATA XREF: .data:0000000140011010
```

0x797261756E614A.

Si algo puede salir mal, saldrá mal

Ley de Murphy

Listing 18.17: assert()

```
const char* get_month1_checked (int month)
{
        assert (month<12);
        return month1[month];
};
```

Listing 18.18: Con optimización MSVC 2013 x64

```
$SG3143 DB      'm', 00H, 'o', 00H, 'n', 00H↙
    ↘ , 't', 00H, 'h', 00H, '.', 00H
        DB      'c', 00H, 00H, 00H
$SG3144 DB      'm', 00H, 'o', 00H, 'n', 00H↙
    ↘ , 't', 00H, 'h', 00H, '<', 00H
        DB      '1', 00H, '2', 00H, 00H, 00H

month$ = 48
get_month1_checked PROC
$LN5:
        push    rbx
        sub     rsp, 32
        movsxd  rbx, ecx
        cmp     ebx, 12
        jl      SHORT $LN3@get_month1
        lea     rdx, OFFSET FLAT:$SG3143
        lea     rcx, OFFSET FLAT:$SG3144
        mov     r8d, 29
        call    _wassert
$LN3@get_month1:
        lea     rcx, OFFSET FLAT:month1
        mov     rax, QWORD PTR [rcx+rbx*8]
        add     rsp, 32
        pop     rbx
        ret     0
get_month1_checked ENDP
```

.

:

Listing 18.19: Con optimización GCC 4.9 x64

```
.LC1:
        .string "month.c"
.LC2:
        .string "month<12"

get_month1_checked:
        cmp     edi, 11
        jg      .L6
        movsx   rdi, edi
        mov     rax, QWORD PTR month1[0+rdi*8]
        ret
.L6:
        push    rax
        mov     ecx, OFFSET FLAT:__PRETTY_FUNCTION__.2423
        mov     edx, 29
        mov     esi, OFFSET FLAT:.LC1
        mov     edi, OFFSET FLAT:.LC2
        call    __assert_fail

__PRETTY_FUNCTION__.2423:
        .string "get_month1_checked"
```

[8].

[8] msdn.microsoft.com/en-us/library/windows/hardware/ff543450(v=vs.85).a

18.6

0	[0][0]
1	[0][1]
2	[0][2]
3	[0][3]
4	[1][0]
5	[1][1]
6	[1][2]
7	[1][3]
8	[2][0]
9	[2][1]
10	[2][2]
11	[2][3]

Cuadro 18.1:

0	1	2	3
4	5	6	7
8	9	10	11

Cuadro 18.2:

row-major order, C/C++ y Python. *row-major order*

column-major order () FORTRAN, MATLAB y R. *column-major order*

?

CAPÍTULO 18. MATRIZ

18.6.1

0..3:

Listing 18.20:

```
#include <stdio.h>

char a[3][4];

int main()
{
        int x, y;

        //
        for (x=0; x<3; x++)
                for (y=0; y<4; y++)
                        a[x][y]=0;

        //  0..3:
        for (y=0; y<4; y++)
                a[1][y]=y;
};
```

. 0, 1, 2 y 3:

 CAPÍTULO 18. MATRIZ

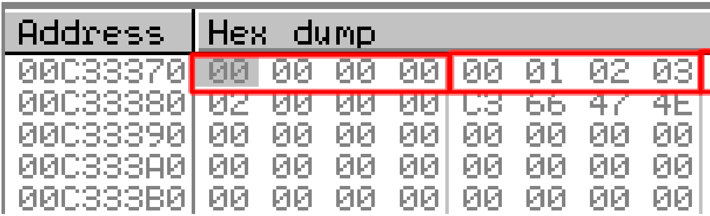

Figura 18.7: OllyDbg:

0..2:

Listing 18.21:

```
#include <stdio.h>

char a[3][4];

int main()
{
        int x, y;

        //
        for (x=0; x<3; x++)
                for (y=0; y<4; y++)
                        a[x][y]=0;

        //  0..2:
        for (x=0; x<3; x++)
                a[x][2]=x;
```

 CAPÍTULO 18. MATRIZ

```
};
```

. 0, 1 y 2.

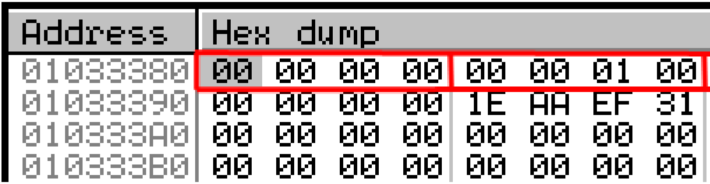

Figura 18.8: OllyDbg:

18.6.2

```
#include <stdio.h>

char a[3][4];

char get_by_coordinates1 (char array[3][4], int a, int b)
{
        return array[a][b];
};

char get_by_coordinates2 (char *array, int a, int b)
{
        //
        // 4
```

```
        return array[a*4+b];
};

char get_by_coordinates3 (char *array, int a
    ↳ , int b)
{
        // ,
        //
        // 4
        return *(array+a*4+b);
};

int main()
{
        a[2][3]=123;
        printf ("%d\n", get_by_coordinates1(
    ↳ a, 2, 3));
        printf ("%d\n", get_by_coordinates2(
    ↳ a, 2, 3));
        printf ("%d\n", get_by_coordinates3(
    ↳ a, 2, 3));
};
```

Listing 18.22: Con optimización MSVC 2013 x64

```

array$ = 8
a$ = 16
b$ = 24
get_by_coordinates3 PROC
; RCX=
; RDX=a
; R8=b
```

```
        movsxd  rax, r8d
; EAX=b
        movsxd  r9, edx
; R9=a
        add     rax, rcx
; RAX=b+
        movzx   eax, BYTE PTR [rax+r9*4]
; AL= RAX+R9*4=b++a*4=+a*4+b
        ret     0
get_by_coordinates3 ENDP

array$ = 8
a$ = 16
b$ = 24
get_by_coordinates2 PROC
        movsxd  rax, r8d
        movsxd  r9, edx
        add     rax, rcx
        movzx   eax, BYTE PTR [rax+r9*4]
        ret     0
get_by_coordinates2 ENDP

array$ = 8
a$ = 16
b$ = 24
get_by_coordinates1 PROC
        movsxd  rax, r8d
        movsxd  r9, edx
        add     rax, rcx
        movzx   eax, BYTE PTR [rax+r9*4]
        ret     0
get_by_coordinates1 ENDP
```

Listing 18.23: Con optimización GCC 4.9 x64

```
; RDI=
; RSI=a
; RDX=b

get_by_coordinates1:
;
        movsx   rsi, esi
        movsx   rdx, edx
        lea     rax, [rdi+rsi*4]
; RAX=RDI+RSI*4=+a*4
        movzx   eax, BYTE PTR [rax+rdx]
; AL= RAX+RDX=+a*4+b
        ret

get_by_coordinates2:
        lea     eax, [rdx+rsi*4]
; RAX=RDX+RSI*4=b+a*4
        cdqe
        movzx   eax, BYTE PTR [rdi+rax]
; AL= RDI+RAX=+b+a*4
        ret

get_by_coordinates3:
        sal     esi, 2
; ESI=a<<2=a*4
;
        movsx   rdx, edx
        movsx   rsi, esi
        add     rdi, rsi
; RDI=RDI+RSI=+a*4
```

```
        movzx   eax, BYTE PTR [rdi+rdx]
; AL= RDI+RDX=+a*4+b
        ret
```

18.6.3

:

Listing 18.24:

```
#include <stdio.h>

int a[10][20][30];

void insert(int x, int y, int z, int value)
{
        a[x][y][z]=value;
};
```

x86

(MSVC 2010):

Listing 18.25: MSVC 2010

```

_DATA    SEGMENT
COMM     _a:DWORD:01770H
_DATA    ENDS
PUBLIC   _insert
_TEXT    SEGMENT
_x$ = 8                    ; size = 4
```

```
_y$ = 12                    ; size = 4
_z$ = 16                    ; size = 4
_value$ = 20                ; size = 4
_insert    PROC
    push   ebp
    mov    ebp, esp
    mov    eax, DWORD PTR _x$[ebp]
    imul   eax, 2400                        ; ⤦
    ↳ eax=600*4*x
    mov    ecx, DWORD PTR _y$[ebp]
    imul   ecx, 120                         ; ⤦
    ↳ ecx=30*4*y
    lea    edx, DWORD PTR _a[eax+ecx]   ; ⤦
    ↳ edx=a + 600*4*x + 30*4*y
    mov    eax, DWORD PTR _z$[ebp]
    mov    ecx, DWORD PTR _value$[ebp]
    mov    DWORD PTR [edx+eax*4], ecx   ; *(⤦
    ↳ edx+z*4)=
    pop    ebp
    ret    0
_insert    ENDP
_TEXT      ENDS
```

Listing 18.26: GCC 4.4.1

```

                public insert
insert          proc near

x               = dword ptr  8
y               = dword ptr  0Ch
z               = dword ptr  10h
value           = dword ptr  14h
```

 CAPÍTULO 18. MATRIZ

```
                push    ebp
                mov     ebp, esp
                push    ebx
                mov     ebx, [ebp+x]
                mov     eax, [ebp+y]
                mov     ecx, [ebp+z]
                lea     edx, [eax+eax]          ; edx=y*2
                mov     eax, edx                ; eax=y*2
                shl     eax, 4                  ; eax=(y*2)<<4 = y*2*16 = y*32
                sub     eax, edx                ; eax=y*32 - y*2=y*30
                imul    edx, ebx, 600           ; edx=x*600
                add     eax, edx                ; eax=eax+edx=y*30 + x*600
                lea     edx, [eax+ecx]          ; edx=y*30 + x*600 + z
                mov     eax, [ebp+value]
                mov     dword ptr ds:a[edx*4], eax  ; *(a+edx*4)=
                pop     ebx
                pop     ebp
                retn
insert          endp
```

. : $(y+y) \ll 4 - (y+y) = (2y) \ll 4 - 2y = 2 \cdot 16 \cdot y - 2y = 32y - 2y = 30y$. .

ARM + Sin optimización Xcode 4.6.3 (LLVM) (Modo Thumb)

Listing 18.27: Sin optimización Xcode 4.6.3 (LLVM) (Modo Thumb)

```
_insert

value   = -0x10
z       = -0xC
y       = -8
x       = -4

;
SUB     SP, SP, #0x10
MOV     R9, 0xFC2 ; a
ADD     R9, PC
LDR.W   R9, [R9]
STR     R0, [SP,#0x10+x]
STR     R1, [SP,#0x10+y]
STR     R2, [SP,#0x10+z]
STR     R3, [SP,#0x10+value]
LDR     R0, [SP,#0x10+value]
LDR     R1, [SP,#0x10+z]
LDR     R2, [SP,#0x10+y]
LDR     R3, [SP,#0x10+x]
MOV     R12, 2400
MUL.W   R3, R3, R12
ADD     R3, R9
MOV     R9, 120
MUL.W   R2, R2, R9
ADD     R2, R3
LSLS    R1, R1, #2 ; R1=R1<<2
```

```
ADD      R1, R2
STR      R0, [R1]    ; R1 -
;
ADD      SP, SP, #0x10
BX       LR
```

Sin optimización LLVM

ARM + Con optimización Xcode 4.6.3 (LLVM) (Modo Thumb)

Listing 18.28: Con optimización Xcode 4.6.3 (LLVM) (Modo Thumb)

```

_insert
MOVW     R9, #0x10FC
MOV.W    R12, #2400
MOVT.W   R9, #0
RSB.W    R1, R1, R1,LSL#4  ; R1 - y. R1=y<<4 ⤦
    ↳ - y = y*16 - y = y*15
ADD      R9, PC            ; R9 =
LDR.W    R9, [R9]
MLA.W    R0, R0, R12, R9   ; R0 - x, R12 - ⤦
    ↳ 2400, R9 - . R0=x*2400 +
ADD.W    R0, R0, R1,LSL#3  ; R0 = R0+R1<<3 = ⤦
    ↳ R0+R1*8 = x*2400 +  + y*15*8 =
                           ;  + y*30*4 + x⤦
    ↳ *600*4
STR.W    R3, [R0,R2,LSL#2] ; R2 - z, R3 - . =⤦
    ↳ R0+z*4 =
                           ;  + y*30*4 + x⤦
    ↳ *600*4 + z*4
```

```
BX          LR
```

RSB (*Reverse Subtract*). SUB y RSB: (LSL#4).

`LDR.W R9, [R9]` LEA (A.6.2 on page 1880)

MIPS

Listing 18.29: Con optimización GCC 4.4.5 (IDA)

```
insert:
; $a0=x
; $a1=y
; $a2=z
; $a3=
                sll     $v0, $a0, 5
; $v0 = $a0<<5 = x*32
                sll     $a0, 3
; $a0 = $a0<<3 = x*8
                addu    $a0, $v0
; $a0 = $a0+$v0 = x*8+x*32 = x*40
                sll     $v1, $a1, 5
; $v1 = $a1<<5 = y*32
                sll     $v0, $a0, 4
; $v0 = $a0<<4 = x*40*16 = x*640
                sll     $a1, 1
; $a1 = $a1<<1 = y*2
                subu    $a1, $v1, $a1
; $a1 = $v1-$a1 = y*32-y*2 = y*30
                subu    $a0, $v0, $a0
; $a0 = $v0-$a0 = x*640-x*40 = x*600
                la      $gp, __gnu_local_gp
```

```
                addu    $a0, $a1, $a0
; $a0 = $a1+$a0 = y*30+x*600
                addu    $a0, $a2
; $a0 = $a0+$a2 = y*30+x*600+z
; :
                lw      $v0, (a & 0xFFFF)(
    $gp)
; :
                sll     $a0, 2
; :
                addu    $a0, $v0, $a0
; :
                jr      $ra
                sw      $a3, 0($a0)

                .comm a:0x1770
```

18.6.4

. : 83.2 on page 1689.

18.7

Spanish text placeholder.18.8.

```

#include <stdio.h>
#include <assert.h>

const char month2[12][10]=
```

 CAPÍTULO 18. MATRIZ

```
{
        { 'J','a','n','u','a','r','y', 0, 0, 0 },
        { 'F','e','b','r','u','a','r','y', 0, 0 },
        { 'M','a','r','c','h', 0, 0, 0, 0, 0 },
        { 'A','p','r','i','l', 0, 0, 0, 0, 0 },
        { 'M','a','y', 0, 0, 0, 0, 0, 0, 0 },
        { 'J','u','n','e', 0, 0, 0, 0, 0, 0 },
        { 'J','u','l','y', 0, 0, 0, 0, 0, 0 },
        { 'A','u','g','u','s','t', 0, 0, 0, 0 },
        { 'S','e','p','t','e','m','b','e','r', 0 },
        { 'O','c','t','o','b','e','r', 0, 0, 0 },
        { 'N','o','v','e','m','b','e','r', 0, 0 },
        { 'D','e','c','e','m','b','e','r', 0, 0 }
};

//
const char* get_month2 (int month)
{
        return &month2[month][0];
};
```

Listing 18.30: Con optimización MSVC 2013 x64

```
month2  DB      04aH
        DB      061H
        DB      06eH
        DB      075H
        DB      061H
        DB      072H
        DB      079H
        DB      00H
        DB      00H
        DB      00H
...

get_month2 PROC
;
        movsxd  rax, ecx
        lea     rcx, QWORD PTR [rax+rax*4]
; RCX=+*4=*5
        lea     rax, OFFSET FLAT:month2
; RAX=
        lea     rax, QWORD PTR [rax+rcx*2]
; RAX= + RCX*2= + *5*2= + *10
        ret     0
get_month2 ENDP
```

. $pointer\_to\_the\_table + month * 10$.

.

Con optimización GCC 4.9 :

Listing 18.31: Con optimización GCC 4.9 x64

```
        movsx   rdi, edi
        lea     rax, [rdi+rdi*4]
        lea     rax, month2[rax+rax]
        ret
```

Listing 18.32: Sin optimización GCC 4.9 x64

```

get_month2:
        push    rbp
        mov     rbp, rsp
        mov     DWORD PTR [rbp-4], edi
        mov     eax, DWORD PTR [rbp-4]
        movsx   rdx, eax
; RDX =
        mov     rax, rdx
; RAX =
        sal     rax, 2
; RAX = <<2 = *4
        add     rax, rdx
; RAX = RAX+RDX = *4+ = *5
        add     rax, rax
; RAX = RAX*2 = *5*2 = *10
        add     rax, OFFSET FLAT:month2
; RAX = *10 +
        pop     rbp
        ret
```

Sin optimización MSVC :

Listing 18.33: Sin optimización MSVC 2013 x64

```
month$ = 8
get_month2 PROC
        mov     DWORD PTR [rsp+8], ecx
        movsxd  rax, DWORD PTR month$[rsp]
; RAX =
        imul    rax, rax, 10
; RAX = RAX*10
        lea     rcx, OFFSET FLAT:month2
; RCX =
        add     rcx, rax
; RCX = RCX+RAX = +month*10
        mov     rax, rcx
; RAX = +*10
        mov     ecx, 1
; RCX = 1
        imul    rcx, rcx, 0
; RCX = 1*0 = 0
        add     rax, rcx
; RAX = +*10 + 0 = +*10
        ret     0
get_month2 ENDP
```

18.7.1 32-bit ARM

Con optimización Keil MULS:

Listing 18.34: Con optimización Keil 6/2013 (Modo Thumb)

```
; R0 =
        MOVS     r1,#0xa
```

```
; R1 = 10
        MULS     r0,r1,r0
; R0 = R1*R0 = 10*
        LDR      r1,|L0.68|
; R1 =
        ADDS     r0,r0,r1
; R0 = R0+R1 = 10* +
        BX       lr
```

Con optimización Keil :

Listing 18.35: Con optimización Keil 6/2013 (Modo ARM)

```

; R0 =
        LDR      r1,|L0.104|
; R1 =
        ADD      r0,r0,r0,LSL #2
; R0 = R0+R0<<2 = R0+R0*4 = *5
        ADD      r0,r1,r0,LSL #1
; R0 = R1+R0<<2 =  + *5*2 =  + *10
        BX       lr
```

18.7.2 ARM64

Listing 18.36: Con optimización GCC 4.9 ARM64

```

; W0 =
        sxtw    x0, w0
; X0 =
        adrp    x1, .LANCHOR1
```

```
        add     x1, x1, :lo12:.LANCHOR1
; X1 =
        add     x0, x0, x0, lsl 2
; X0 = X0+X0<<2 = X0+X0*4 = X0*5
        add     x0, x1, x0, lsl 1
; X0 = X1+X0<<1 = X1+X0*2 =  + X0*10
        ret
```

18.7.3 MIPS

Listing 18.37: Con optimización GCC 4.4.5 (IDA)

```

                .globl get_month2
get_month2:
; $a0=
                sll     $v0, $a0, 3
; $v0 = $a0<<3 = *8
                sll     $a0, 1
; $a0 = $a0<<1 = *2
                addu    $a0, $v0
; $a0 = *2+*8 = *10
; :
                la      $v0, month2
; :
                jr      $ra
                addu    $v0, $a0

month2:         .ascii "January"<0>
                .byte 0, 0
aFebruary:      .ascii "February"<0>
                .byte    0
```

```
aMarch:          .ascii "March"<0>
                 .byte 0, 0, 0, 0
aApril:          .ascii "April"<0>
                 .byte 0, 0, 0, 0
aMay:            .ascii "May"<0>
                 .byte 0, 0, 0, 0, 0, 0
aJune:           .ascii "June"<0>
                 .byte 0, 0, 0, 0, 0
aJuly:           .ascii "July"<0>
                 .byte 0, 0, 0, 0, 0
aAugust:         .ascii "August"<0>
                 .byte 0, 0, 0
aSeptember:      .ascii "September"<0>
aOctober:        .ascii "October"<0>
                 .byte 0, 0
aNovember:       .ascii "November"<0>
                 .byte    0
aDecember:       .ascii "December"<0>
                 .byte 0, 0, 0, 0, 0, 0, 0, ↙
    ↘ 0, 0
```

18.7.4 Conclusión

18.8 Conclusión

18.9 Ejercicios

18.9.1 Ejercicio #1

Lo que hace el código?

Listing 18.38: MSVC 2010 + /O1

```
_a$ = 8         ; size = 4
_b$ = 12        ; size = 4
_c$ = 16        ; size = 4
?s@@YAXPAN00@Z PROC     ; s, COMDAT
    mov    eax, DWORD PTR _b$[esp-4]
    mov    ecx, DWORD PTR _a$[esp-4]
    mov    edx, DWORD PTR _c$[esp-4]
    push   esi
    push   edi
    sub    ecx, eax
    sub    edx, eax
    mov    edi, 200     ; 000000c8H
$LL6@s:
    push   100          ; 00000064H
    pop    esi
$LL3@s:
    fld    QWORD PTR [ecx+eax]
    fadd   QWORD PTR [eax]
    fstp   QWORD PTR [edx+eax]
    add    eax, 8
    dec    esi
    jne    SHORT $LL3@s
    dec    edi
    jne    SHORT $LL6@s
    pop    edi
    pop    esi
    ret    0
?s@@YAXPAN00@Z ENDP     ; s
```

(/O1:).

Listing 18.39: Con optimización Keil 6/2013 (Modo ARM)

```
        PUSH     {r4-r12,lr}
        MOV      r9,r2
        MOV      r10,r1
        MOV      r11,r0
        MOV      r5,#0
|L0.20|
        ADD      r0,r5,r5,LSL #3
        ADD      r0,r0,r5,LSL #4
        MOV      r4,#0
        ADD      r8,r10,r0,LSL #5
        ADD      r7,r11,r0,LSL #5
        ADD      r6,r9,r0,LSL #5
|L0.44|
        ADD      r0,r8,r4,LSL #3
        LDM      r0,{r2,r3}
        ADD      r1,r7,r4,LSL #3
        LDM      r1,{r0,r1}
        BL       __aeabi_dadd
        ADD      r2,r6,r4,LSL #3
        ADD      r4,r4,#1
        STM      r2,{r0,r1}
        CMP      r4,#0x64
        BLT      |L0.44|
        ADD      r5,r5,#1
        CMP      r5,#0xc8
        BLT      |L0.20|
        POP      {r4-r12,pc}
```

Listing 18.40: Con optimización Keil 6/2013 (Modo Thumb)

```
        PUSH     {r0-r2,r4-r7,lr}
        MOVS     r4,#0
```

```
        SUB      sp,sp,#8
|L0.6|
        MOVS     r1,#0x19
        MOVS     r0,r4
        LSLS     r1,r1,#5
        MULS     r0,r1,r0
        LDR      r2,[sp,#8]
        LDR      r1,[sp,#0xc]
        ADDS     r2,r0,r2
        STR      r2,[sp,#0]
        LDR      r2,[sp,#0x10]
        MOVS     r5,#0
        ADDS     r7,r0,r2
        ADDS     r0,r0,r1
        STR      r0,[sp,#4]
|L0.32|
        LSLS     r6,r5,#3
        ADDS     r0,r0,r6
        LDM      r0!,{r2,r3}
        LDR      r0,[sp,#0]
        ADDS     r1,r0,r6
        LDM      r1,{r0,r1}
        BL       __aeabi_dadd
        ADDS     r2,r7,r6
        ADDS     r5,r5,#1
        STM      r2!,{r0,r1}
        CMP      r5,#0x64
        BGE      |L0.62|
        LDR      r0,[sp,#4]
        B        |L0.32|
|L0.62|
        ADDS     r4,r4,#1
        CMP      r4,#0xc8
```

```
        BLT      |L0.6|
        ADD      sp,sp,#0x14
        POP      {r4-r7,pc}
```

Listing 18.41: Sin optimización GCC 4.9 (ARM64)

```
s:
        sub     sp, sp, #48
        str     x0, [sp,24]
        str     x1, [sp,16]
        str     x2, [sp,8]
        str     wzr, [sp,44]
        b       .L2
.L5:
        str     wzr, [sp,40]
        b       .L3
.L4:
        ldr     w1, [sp,44]
        mov     w0, 100
        mul     w0, w1, w0
        sxtw    x1, w0
        ldrsw   x0, [sp,40]
        add     x0, x1, x0
        lsl     x0, x0, 3
        ldr     x1, [sp,8]
        add     x0, x1, x0
        ldr     w2, [sp,44]
        mov     w1, 100
        mul     w1, w2, w1
        sxtw    x2, w1
        ldrsw   x1, [sp,40]
        add     x1, x2, x1
        lsl     x1, x1, 3
```

```
        ldr     x2, [sp,24]
        add     x1, x2, x1
        ldr     x2, [x1]
        ldr     w3, [sp,44]
        mov     w1, 100
        mul     w1, w3, w1
        sxtw    x3, w1
        ldrsw   x1, [sp,40]
        add     x1, x3, x1
        lsl     x1, x1, 3
        ldr     x3, [sp,16]
        add     x1, x3, x1
        ldr     x1, [x1]
        fmov    d0, x2
        fmov    d1, x1
        fadd    d0, d0, d1
        fmov    x1, d0
        str     x1, [x0]
        ldr     w0, [sp,40]
        add     w0, w0, 1
        str     w0, [sp,40]
.L3:
        ldr     w0, [sp,40]
        cmp     w0, 99
        ble     .L4
        ldr     w0, [sp,44]
        add     w0, w0, 1
        str     w0, [sp,44]
.L2:
        ldr     w0, [sp,44]
        cmp     w0, 199
        ble     .L5
        add     sp, sp, 48
```

```
        ret
```

Listing 18.42: Con optimización GCC 4.4.5 (MIPS) (IDA)

```
sub_0:
                li      $t3, 0x27100
                move    $t2, $zero
                li      $t1, 0x64  # 'd'

loc_10:                                          # ⤦
    ↳ CODE XREF: sub_0+54
                addu    $t0, $a1, $t2
                addu    $a3, $a2, $t2
                move    $v1, $a0
                move    $v0, $zero

loc_20:                                          # ⤦
    ↳ CODE XREF: sub_0+48
                lwc1    $f2, 4($v1)
                lwc1    $f0, 4($t0)
                lwc1    $f3, 0($v1)
                lwc1    $f1, 0($t0)
                addiu   $v0, 1
                add.d   $f0, $f2, $f0
                addiu   $v1, 8
                swc1    $f0, 4($a3)
                swc1    $f1, 0($a3)
                addiu   $t0, 8
                bne     $v0, $t1, loc_20
                addiu   $a3, 8
                addiu   $t2, 0x320
                bne     $t2, $t3, loc_10
                addiu   $a0, 0x320
```

```
                jr      $ra
                or      $at, $zero
```

Responda G.1.12 on page 1923.

18.9.2 Ejercicio #2

Lo que hace el código?

Listing 18.43: MSVC 2010 + /O1

```
tv315 = -8                      ; size = 4
tv291 = -4                      ; size = 4
_a$ = 8                         ; size = 4
_b$ = 12                        ; size = 4
_c$ = 16                        ; size = 4
?m@@YAXPAN00@Z PROC             ; m, COMDAT
    push    ebp
    mov     ebp, esp
    push    ecx
    push    ecx
    mov     edx, DWORD PTR _a$[ebp]
    push    ebx
    mov     ebx, DWORD PTR _c$[ebp]
    push    esi
    mov     esi, DWORD PTR _b$[ebp]
    sub     edx, esi
    push    edi
    sub     esi, ebx
    mov     DWORD PTR tv315[ebp], 100   ; ↙
    ↪ 00000064H
$LL9@m:
    mov     eax, ebx
```

```
    mov     DWORD PTR tv291[ebp], 300  ; ↙
    ↘ 0000012cH
$LL6@m:
    fldz
    lea     ecx, DWORD PTR [esi+eax]
    fstp    QWORD PTR [eax]
    mov     edi, 200                  ; ↙
    ↘ 000000c8H
$LL3@m:
    dec     edi
    fld     QWORD PTR [ecx+edx]
    fmul    QWORD PTR [ecx]
    fadd    QWORD PTR [eax]
    fstp    QWORD PTR [eax]
    jne     HORT $LL3@m
    add     eax, 8
    dec     DWORD PTR tv291[ebp]
    jne     SHORT $LL6@m
    add     ebx, 800                  ; ↙
    ↘ 00000320H
    dec     DWORD PTR tv315[ebp]
    jne     SHORT $LL9@m
    pop     edi
    pop     esi
    pop     ebx
    leave
    ret     0
?m@@YAXPAN00@Z ENDP                   ; m
```

(/O1:).

Listing 18.44: Con optimización Keil 6/2013 (Modo ARM)

```
        PUSH      {r0-r2,r4-r11,lr}
        SUB       sp,sp,#8
        MOV       r5,#0
|L0.12|
        LDR       r1,[sp,#0xc]
        ADD       r0,r5,r5,LSL #3
        ADD       r0,r0,r5,LSL #4
        ADD       r1,r1,r0,LSL #5
        STR       r1,[sp,#0]
        LDR       r1,[sp,#8]
        MOV       r4,#0
        ADD       r11,r1,r0,LSL #5
        LDR       r1,[sp,#0x10]
        ADD       r10,r1,r0,LSL #5
|L0.52|
        MOV       r0,#0
        MOV       r1,r0
        ADD       r7,r10,r4,LSL #3
        STM       r7,{r0,r1}
        MOV       r6,r0
        LDR       r0,[sp,#0]
        ADD       r8,r11,r4,LSL #3
        ADD       r9,r0,r4,LSL #3
|L0.84|
        LDM       r9,{r2,r3}
        LDM       r8,{r0,r1}
        BL        __aeabi_dmul
        LDM       r7,{r2,r3}
        BL        __aeabi_dadd
        ADD       r6,r6,#1
        STM       r7,{r0,r1}
        CMP       r6,#0xc8
        BLT       |L0.84|
```

```
        ADD     r4,r4,#1
        CMP     r4,#0x12c
        BLT     |L0.52|
        ADD     r5,r5,#1
        CMP     r5,#0x64
        BLT     |L0.12|
        ADD     sp,sp,#0x14
        POP     {r4-r11,pc}
```

Listing 18.45: Con optimización Keil 6/2013 (Modo Thumb)

```
        PUSH    {r0-r2,r4-r7,lr}
        MOVS    r0,#0
        SUB     sp,sp,#0x10
        STR     r0,[sp,#0]
|L0.8|
        MOVS    r1,#0x19
        LSLS    r1,r1,#5
        MULS    r0,r1,r0
        LDR     r2,[sp,#0x10]
        LDR     r1,[sp,#0x14]
        ADDS    r2,r0,r2
        STR     r2,[sp,#4]
        LDR     r2,[sp,#0x18]
        MOVS    r5,#0
        ADDS    r7,r0,r2
        ADDS    r0,r0,r1
        STR     r0,[sp,#8]
|L0.32|
        LSLS    r4,r5,#3
        MOVS    r0,#0
        ADDS    r2,r7,r4
        STR     r0,[r2,#0]
```

```
        MOVS     r6,r0
        STR      r0,[r2,#4]
|L0.44|
        LDR      r0,[sp,#8]
        ADDS     r0,r0,r4
        LDM      r0!,{r2,r3}
        LDR      r0,[sp,#4]
        ADDS     r1,r0,r4
        LDM      r1,{r0,r1}
        BL       __aeabi_dmul
        ADDS     r3,r7,r4
        LDM      r3,{r2,r3}
        BL       __aeabi_dadd
        ADDS     r2,r7,r4
        ADDS     r6,r6,#1
        STM      r2!,{r0,r1}
        CMP      r6,#0xc8
        BLT      |L0.44|
        MOVS     r0,#0xff
        ADDS     r5,r5,#1
        ADDS     r0,r0,#0x2d
        CMP      r5,r0
        BLT      |L0.32|
        LDR      r0,[sp,#0]
        ADDS     r0,r0,#1
        CMP      r0,#0x64
        STR      r0,[sp,#0]
        BLT      |L0.8|
        ADD      sp,sp,#0x1c
        POP      {r4-r7,pc}
```

Listing 18.46: Sin optimización GCC 4.9 (ARM64)

```
m:
        sub     sp, sp, #48
        str     x0, [sp,24]
        str     x1, [sp,16]
        str     x2, [sp,8]
        str     wzr, [sp,44]
        b       .L2
.L7:
        str     wzr, [sp,40]
        b       .L3
.L6:
        ldr     w1, [sp,44]
        mov     w0, 100
        mul     w0, w1, w0
        sxtw    x1, w0
        ldrsw   x0, [sp,40]
        add     x0, x1, x0
        lsl     x0, x0, 3
        ldr     x1, [sp,8]
        add     x0, x1, x0
        ldr     x1, .LC0
        str     x1, [x0]
        str     wzr, [sp,36]
        b       .L4
.L5:
        ldr     w1, [sp,44]
        mov     w0, 100
        mul     w0, w1, w0
        sxtw    x1, w0
        ldrsw   x0, [sp,40]
        add     x0, x1, x0
        lsl     x0, x0, 3
        ldr     x1, [sp,8]
```

```
        add     x0, x1, x0
        ldr     w2, [sp,44]
        mov     w1, 100
        mul     w1, w2, w1
        sxtw    x2, w1
        ldrsw   x1, [sp,40]
        add     x1, x2, x1
        lsl     x1, x1, 3
        ldr     x2, [sp,8]
        add     x1, x2, x1
        ldr     x2, [x1]
        ldr     w3, [sp,44]
        mov     w1, 100
        mul     w1, w3, w1
        sxtw    x3, w1
        ldrsw   x1, [sp,40]
        add     x1, x3, x1
        lsl     x1, x1, 3
        ldr     x3, [sp,24]
        add     x1, x3, x1
        ldr     x3, [x1]
        ldr     w4, [sp,44]
        mov     w1, 100
        mul     w1, w4, w1
        sxtw    x4, w1
        ldrsw   x1, [sp,40]
        add     x1, x4, x1
        lsl     x1, x1, 3
        ldr     x4, [sp,16]
        add     x1, x4, x1
        ldr     x1, [x1]
        fmov    d0, x3
        fmov    d1, x1
```

```
        fmul    d0, d0, d1
        fmov    x1, d0
        fmov    d0, x2
        fmov    d1, x1
        fadd    d0, d0, d1
        fmov    x1, d0
        str     x1, [x0]
        ldr     w0, [sp,36]
        add     w0, w0, 1
        str     w0, [sp,36]
.L4:
        ldr     w0, [sp,36]
        cmp     w0, 199
        ble     .L5
        ldr     w0, [sp,40]
        add     w0, w0, 1
        str     w0, [sp,40]
.L3:
        ldr     w0, [sp,40]
        cmp     w0, 299
        ble     .L6
        ldr     w0, [sp,44]
        add     w0, w0, 1
        str     w0, [sp,44]
.L2:
        ldr     w0, [sp,44]
        cmp     w0, 99
        ble     .L7
        add     sp, sp, 48
        ret
```

Listing 18.47: Con optimización GCC 4.4.5 (MIPS) (IDA)

```
m:
                li      $t5, 0x13880
                move    $t4, $zero
                li      $t1, 0xC8
                li      $t3, 0x12C

loc_14:                                       # ⤦
    ↳ CODE XREF: m+7C
                addu    $t0, $a0, $t4
                addu    $a3, $a1, $t4
                move    $v1, $a2
                move    $t2, $zero

loc_24:                                       # ⤦
    ↳ CODE XREF: m+70
                mtc1    $zero, $f0
                move    $v0, $zero
                mtc1    $zero, $f1
                or      $at, $zero
                swc1    $f0, 4($v1)
                swc1    $f1, 0($v1)

loc_3C:                                       # ⤦
    ↳ CODE XREF: m+5C
                lwc1    $f4, 4($t0)
                lwc1    $f2, 4($a3)
                lwc1    $f5, 0($t0)
                lwc1    $f3, 0($a3)
                addiu   $v0, 1
                mul.d   $f2, $f4, $f2
                add.d   $f0, $f2
                swc1    $f0, 4($v1)
                bne     $v0, $t1, loc_3C
```

```
                swc1    $f1, 0($v1)
                addiu   $t2, 1
                addiu   $v1, 8
                addiu   $t0, 8
                bne     $t2, $t3, loc_24
                addiu   $a3, 8
                addiu   $t4, 0x320
                bne     $t4, $t5, loc_14
                addiu   $a2, 0x320
                jr      $ra
                or      $at, $zero
```

Responda G.1.12 on page 1923.

18.9.3 Ejercicio #3

Lo que hace el código?

Listing 18.48: Con optimización MSVC 2010

```
_array$ = 8
_x$ = 12
_y$ = 16
_f      PROC
        mov     eax, DWORD PTR _x$[esp-4]
        mov     edx, DWORD PTR _y$[esp-4]
        mov     ecx, eax
        shl     ecx, 4
        sub     ecx, eax
        lea     eax, DWORD PTR [edx+ecx*8]
        mov     ecx, DWORD PTR _array$[esp
    -4]
        fld     QWORD PTR [ecx+eax*8]
```

```
        ret     0
_f      ENDP
```

Listing 18.49: Sin optimización Keil 6/2013 (Modo ARM)

```
f PROC
        RSB      r1,r1,r1,LSL #4
        ADD      r0,r0,r1,LSL #6
        ADD      r1,r0,r2,LSL #3
        LDM      r1,{r0,r1}
        BX       lr
        ENDP
```

Listing 18.50: Sin optimización Keil 6/2013 (Modo Thumb)

```
f PROC
        MOVS     r3,#0xf
        LSLS     r3,r3,#6
        MULS     r1,r3,r1
        ADDS     r0,r1,r0
        LSLS     r1,r2,#3
        ADDS     r1,r0,r1
        LDM      r1,{r0,r1}
        BX       lr
        ENDP
```

Listing 18.51: Con optimización GCC 4.9 (ARM64)

```
f:
        sxtw    x1, w1
        add     x0, x0, x2, sxtw 3
        lsl     x2, x1, 10
        sub     x1, x2, x1, lsl 6
```

```
        ldr     d0, [x0,x1]
        ret
```

Listing 18.52: Con optimización GCC 4.4.5 (MIPS) (IDA)

```
f:
                sll     $v0, $a1, 10
                sll     $a1, 6
                subu    $a1, $v0, $a1
                addu    $a1, $a0, $a1
                sll     $a2, 3
                addu    $a1, $a2
                lwc1    $f0, 4($a1)
                or      $at, $zero
                lwc1    $f1, 0($a1)
                jr      $ra
                or      $at, $zero
```

Responda G.1.12 on page 1924

18.9.4 Ejercicio #4

Lo que hace el código?

Listing 18.53: Con optimización MSVC 2010

```
_array$ = 8
_x$ = 12
_y$ = 16
_z$ = 20
_f      PROC
        mov     eax, DWORD PTR _x$[esp-4]
        mov     edx, DWORD PTR _y$[esp-4]
```

```
        mov     ecx, eax
        shl     ecx, 4
        sub     ecx, eax
        lea     eax, DWORD PTR [edx+ecx*4]
        mov     ecx, DWORD PTR _array$[esp
    ↪ -4]
        lea     eax, DWORD PTR [eax+eax*4]
        shl     eax, 4
        add     eax, DWORD PTR _z$[esp-4]
        mov     eax, DWORD PTR [ecx+eax*4]
        ret     0
_f      ENDP
```

Listing 18.54: Sin optimización Keil 6/2013 (Modo ARM)

```
f PROC
        RSB      r1,r1,r1,LSL #4
        ADD      r1,r1,r1,LSL #2
        ADD      r0,r0,r1,LSL #8
        ADD      r1,r2,r2,LSL #2
        ADD      r0,r0,r1,LSL #6
        LDR      r0,[r0,r3,LSL #2]
        BX       lr
        ENDP
```

Listing 18.55: Sin optimización Keil 6/2013 (Modo Thumb)

```
f PROC
        PUSH     {r4,lr}
        MOVS     r4,#0x4b
        LSLS     r4,r4,#8
        MULS     r1,r4,r1
        ADDS     r0,r1,r0
```

```
        MOVS     r1,#0xff
        ADDS     r1,r1,#0x41
        MULS     r2,r1,r2
        ADDS     r0,r0,r2
        LSLS     r1,r3,#2
        LDR      r0,[r0,r1]
        POP      {r4,pc}
        ENDP
```

Listing 18.56: Con optimización GCC 4.9 (ARM64)

```
f:
        sxtw    x2, w2
        mov     w4, 19200
        add     x2, x2, x2, lsl 2
        smull   x1, w1, w4
        lsl     x2, x2, 4
        add     x3, x2, x3, sxtw
        add     x0, x0, x3, lsl 2
        ldr     w0, [x0,x1]
        ret
```

Listing 18.57: Con optimización GCC 4.4.5 (MIPS) (IDA)

```
f:
                sll     $v0, $a1, 10
                sll     $a1, 8
                addu    $a1, $v0
                sll     $v0, $a2, 6
                sll     $a2, 4
                addu    $a2, $v0
                sll     $v0, $a1, 4
                subu    $a1, $v0, $a1
```

```
                addu    $a2, $a3
                addu    $a1, $a0, $a1
                sll     $a2, 2
                addu    $a1, $a2
                lw      $v0, 0($a1)
                jr      $ra
                or      $at, $zero
```

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18.9.5 Ejercicio #5

Lo que hace el código?

Listing 18.58: Con optimización MSVC 2012 /GS-

```
COMM    _tbl:DWORD:064H

tv759 = -4      ; size = 4
_main   PROC
        push    ecx
        push    ebx
        push    ebp
        push    esi
        xor     edx, edx
        push    edi
        xor     esi, esi
        xor     edi, edi
        xor     ebx, ebx
        xor     ebp, ebp
        mov     DWORD PTR tv759[esp+20], edx
        mov     eax, OFFSET _tbl+4
        npad    8 ; align next label
```

```
$LL6@main:
        lea     ecx, DWORD PTR [edx+edx]
        mov     DWORD PTR [eax+4], ecx
        mov     ecx, DWORD PTR tv759[esp+20]
        add     DWORD PTR tv759[esp+20], 3
        mov     DWORD PTR [eax+8], ecx
        lea     ecx, DWORD PTR [edx*4]
        mov     DWORD PTR [eax+12], ecx
        lea     ecx, DWORD PTR [edx*8]
        mov     DWORD PTR [eax], edx
        mov     DWORD PTR [eax+16], ebp
        mov     DWORD PTR [eax+20], ebx
        mov     DWORD PTR [eax+24], edi
        mov     DWORD PTR [eax+32], esi
        mov     DWORD PTR [eax-4], 0
        mov     DWORD PTR [eax+28], ecx
        add     eax, 40
        inc     edx
        add     ebp, 5
        add     ebx, 6
        add     edi, 7
        add     esi, 9
        cmp     eax, OFFSET _tbl+404
        jl      SHORT $LL6@main
        pop     edi
        pop     esi
        pop     ebp
        xor     eax, eax
        pop     ebx
        pop     ecx
        ret     0
_main   ENDP
```

Listing 18.59: Sin optimización Keil 6/2013 (Modo ARM)

```
main PROC
        LDR      r12,|L0.60|
        MOV      r1,#0
|L0.8|
        ADD      r2,r1,r1,LSL #2
        MOV      r0,#0
        ADD      r2,r12,r2,LSL #3
|L0.20|
        MUL      r3,r1,r0
        STR      r3,[r2,r0,LSL #2]
        ADD      r0,r0,#1
        CMP      r0,#0xa
        BLT      |L0.20|
        ADD      r1,r1,#1
        CMP      r1,#0xa
        MOVGE    r0,#0
        BLT      |L0.8|
        BX       lr
        ENDP

|L0.60|
        DCD      ||.bss||

        AREA ||.bss||, DATA, NOINIT, ALIGN=2

tbl
        %        400
```

Listing 18.60: Sin optimización Keil 6/2013 (Modo Thumb)

```
main PROC
        PUSH     {r4,r5,lr}
```

```
        LDR      r4,|L0.40|
        MOVS     r1,#0
|L0.6|
        MOVS     r2,#0x28
        MULS     r2,r1,r2
        MOVS     r0,#0
        ADDS     r3,r2,r4
|L0.14|
        MOVS     r2,r1
        MULS     r2,r0,r2
        LSLS     r5,r0,#2
        ADDS     r0,r0,#1
        CMP      r0,#0xa
        STR      r2,[r3,r5]
        BLT      |L0.14|
        ADDS     r1,r1,#1
        CMP      r1,#0xa
        BLT      |L0.6|
        MOVS     r0,#0
        POP      {r4,r5,pc}
        ENDP

        DCW      0x0000
|L0.40|
        DCD      ||.bss||

        AREA ||.bss||, DATA, NOINIT, ALIGN=2

tbl
        %        400
```

Listing 18.61: Sin optimización GCC 4.9 (ARM64)

```
        .comm   tbl,400,8
main:
        sub     sp, sp, #16
        str     wzr, [sp,12]
        b       .L2
.L5:
        str     wzr, [sp,8]
        b       .L3
.L4:
        ldr     w1, [sp,12]
        ldr     w0, [sp,8]
        mul     w3, w1, w0
        adrp    x0, tbl
        add     x2, x0, :lo12:tbl
        ldrsw   x4, [sp,8]
        ldrsw   x1, [sp,12]
        mov     x0, x1
        lsl     x0, x0, 2
        add     x0, x0, x1
        lsl     x0, x0, 1
        add     x0, x0, x4
        str     w3, [x2,x0,lsl 2]
        ldr     w0, [sp,8]
        add     w0, w0, 1
        str     w0, [sp,8]
.L3:
        ldr     w0, [sp,8]
        cmp     w0, 9
        ble     .L4
        ldr     w0, [sp,12]
        add     w0, w0, 1
        str     w0, [sp,12]
.L2:
```

```
        ldr     w0, [sp,12]
        cmp     w0, 9
        ble     .L5
        mov     w0, 0
        add     sp, sp, 16
        ret
```

Listing 18.62: Sin optimización GCC 4.4.5 (MIPS) (IDA)

```
main:

var_18          = -0x18
var_10          = -0x10
var_C           = -0xC
var_4           = -4

                addiu   $sp, -0x18
                sw      $fp, 0x18+var_4($sp)
                move    $fp, $sp
                la      $gp, __gnu_local_gp
                sw      $gp, 0x18+var_18($sp↙
    ↘ )
                sw      $zero, 0x18+var_C(↙
    ↘ $fp)
                b       loc_A0
                or      $at, $zero

loc_24:                                     # ↙
    ↘ CODE XREF: main+AC
                sw      $zero, 0x18+var_10(↙
    ↘ $fp)
                b       loc_7C
                or      $at, $zero
```

```
loc_30:                                      # ↙
    ↘ CODE XREF: main+88
                lw      $v0, 0x18+var_C($fp)
                lw      $a0, 0x18+var_10($fp↙
    ↘ )
                lw      $a1, 0x18+var_C($fp)
                lw      $v1, 0x18+var_10($fp↙
    ↘ )
                or      $at, $zero
                mult    $a1, $v1
                mflo    $a2
                lw      $v1, (tbl & 0xFFFF)(↙
    ↘ $gp)
                sll     $v0, 1
                sll     $a1, $v0, 2
                addu    $v0, $a1
                addu    $v0, $a0
                sll     $v0, 2
                addu    $v0, $v1, $v0
                sw      $a2, 0($v0)
                lw      $v0, 0x18+var_10($fp↙
    ↘ )
                or      $at, $zero
                addiu   $v0, 1
                sw      $v0, 0x18+var_10($fp↙
    ↘ )

loc_7C:                                      # ↙
    ↘ CODE XREF: main+28
                lw      $v0, 0x18+var_10($fp↙
    ↘ )
                or      $at, $zero
```

```
                slti    $v0, 0xA
                bnez    $v0, loc_30
                or      $at, $zero
                lw      $v0, 0x18+var_C($fp)
                or      $at, $zero
                addiu   $v0, 1
                sw      $v0, 0x18+var_C($fp)

loc_A0:                                         # ↙
    ↳ CODE XREF: main+1C
                lw      $v0, 0x18+var_C($fp)
                or      $at, $zero
                slti    $v0, 0xA
                bnez    $v0, loc_24
                or      $at, $zero
                move    $v0, $zero
                move    $sp, $fp
                lw      $fp, 0x18+var_4($sp)
                addiu   $sp, 0x18
                jr      $ra
                or      $at, $zero

                .comm tbl:0x64                  # ↙
    ↳ DATA XREF: main+4C
```

Responda G.1.12 on page 1925

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

Capítulo 19

Manipulando bit(s) específicas

19.1

19.1.1 x86

```
        HANDLE fh;

        fh=CreateFile ("file", GENERIC_WRITE ↙
    ↪  | GENERIC_READ, FILE_SHARE_READ, NULL↙
    ↪ , OPEN_ALWAYS, FILE_ATTRIBUTE_NORMAL, ↙
    ↪ NULL);
```

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS
(MSVC 2010):

Listing 19.1: MSVC 2010

```
        push    0
        push    128                                  ↙
    ↘                       ; 00000080H
        push    4
        push    0
        push    1
        push    -1073741824                          ↙
    ↘                       ; c0000000H
        push    OFFSET $SG78813
        call    DWORD PTR ↙
    ↘ __imp__CreateFileA@28
        mov     DWORD PTR _fh$[ebp], eax
```

WinNT.h:

Listing 19.2: WinNT.h

```
#define GENERIC_READ                          (0↙
    ↘ x80000000L)
#define GENERIC_WRITE                         (0↙
    ↘ x40000000L)
#define GENERIC_EXECUTE                       (0↙
    ↘ x20000000L)
#define GENERIC_ALL                           (0↙
    ↘ x10000000L)
```

,`GENERIC_READ | GENERIC_WRITE = 0x80000000 | 0x40000000 = 0xC0000000`, `CreateFile()`[1].

[1] msdn.microsoft.com/en-us/library/aa363858(VS.85).aspx

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

Listing 19.3: KERNEL32.DLL (Windows XP SP3 x86)

```
.text:7C83D429                 test    byte ptr [ebp+dwDesiredAccess+3], 40h
.text:7C83D42D                 mov     [ebp+var_8], 1
.text:7C83D434                 jz      short loc_7C83D417
.text:7C83D436                 jmp     loc_7C810817
```

(7.3.1 on page 170)).

```
if ((dwDesiredAccess&0x40000000) == 0) goto loc_7C83D417
```

```
#include <stdio.h>
#include <fcntl.h>

void main()
{
        int handle;

        handle=open ("file", O_RDWR | O_CREAT);
};
```

:

Listing 19.4: GCC 4.4.1

```
                public main
main            proc near
```

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

```
var_20          = dword ptr -20h
var_1C          = dword ptr -1Ch
var_4           = dword ptr -4

                push    ebp
                mov     ebp, esp
                and     esp, 0FFFFFFF0h
                sub     esp, 20h
                mov     [esp+20h+var_1C], 42↙
    ↘ h
                mov     [esp+20h+var_20], ↙
    ↘ offset aFile ; "file"
                call    _open
                mov     [esp+20h+var_4], eax
                leave
                retn
main            endp
```

Listing 19.5: open() (libc.so.6)

```
.text:000BE69B                mov     edx, ↙
    ↘ [esp+4+mode] ; mode
.text:000BE69F                mov     ecx, ↙
    ↘ [esp+4+flags] ; flags
.text:000BE6A3                mov     ebx, ↙
    ↘ [esp+4+filename] ; filename
.text:000BE6A7                mov     eax, ↙
    ↘ 5
.text:000BE6AC                int     80h  ↙
    ↘           ; LINUX - sys_open
```

N.B.

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

Listing 19.6: do_filp_open() (linux kernel 2.6.31)

```
do_filp_open    proc near
...
                push    ebp
                mov     ebp, esp
                push    edi
                push    esi
                push    ebx
                mov     ebx, ecx
                add     ebx, 1
                sub     esp, 98h
                mov     esi, [ebp+arg_4] ; ↙
    ↘ acc_mode ()
                test    bl, 3
                mov     [ebp+var_80], eax ; ↙
    ↘ dfd ()
                mov     [ebp+var_7C], edx ; ↙
    ↘ pathname ()
                mov     [ebp+var_78], ecx ; ↙
    ↘ open_flag ()
                jnz     short loc_C01EF684
                mov     ebx, ecx        ; ↙
    ↘ ebx <- open_flag
```

:

Listing 19.7: do_filp_open() (linux kernel 2.6.31)

```
loc_C01EF6B4:                           ; ↙
    ↘ CODE XREF: do_filp_open+4F
                test    bl, 40h         ; ↙
    ↘ O_CREAT
```

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

```
                jnz     loc_C01EF810
                mov     edi, ebx
                shr     edi, 11h
                xor     edi, 1
                and     edi, 1
                test    ebx, 10000h
                jz      short loc_C01EF6D3
                or      edi, 2
```

19.1.2 ARM

Listing 19.8: linux kernel 3.8.0

```
struct file *do_filp_open(int dfd, struct ⤦
    ↳ filename *pathname,
                const struct open_flags *op)
{
...
        filp = path_openat(dfd, pathname, &⤦
    ↳ nd, op, flags | LOOKUP_RCU);
...
}

static struct file *path_openat(int dfd, ⤦
    ↳ struct filename *pathname,
                struct nameidata *nd, const ⤦
    ↳ struct open_flags *op, int flags)
{
...
        error = do_last(nd, &path, file, op,⤦
    ↳  &opened, pathname);
...
```

```
}


static int do_last(struct nameidata *nd, 
    ↳ struct path *path,
                  struct file *file, const 
    ↳ struct open_flags *op,
                  int *opened, struct 
    ↳ filename *name)
{
...
        if (!(open_flag & O_CREAT)) {
    ...
                error = lookup_fast(nd, path
    ↳ , &inode);
    ...
        } else {
    ...
                error = complete_walk(nd);
        }
...
}
```

Listing 19.9: do_last() (vmlinux)

```
...
.text:C0169EA8                 MOV             
    ↳    R9, R3  ; R3 - (4th argument) 
    ↳ open_flag
...
.text:C0169ED4                 LDR             
    ↳    R6, [R9] ; R6 - open_flag
...
```

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

```
.text:C0169F68                      TST                    ↙
    ↘    R6, #0x40 ; jumptable C0169F00 ↙
    ↘ default case
.text:C0169F6C                      BNE                    ↙
    ↘    loc_C016A128
.text:C0169F70                      LDR                    ↙
    ↘    R2, [R4,#0x10]
.text:C0169F74                      ADD                    ↙
    ↘    R12, R4, #8
.text:C0169F78                      LDR                    ↙
    ↘    R3, [R4,#0xC]
.text:C0169F7C                      MOV                    ↙
    ↘    R0, R4
.text:C0169F80                      STR                    ↙
    ↘    R12, [R11,#var_50]
.text:C0169F84                      LDRB                   ↙
    ↘    R3, [R2,R3]
.text:C0169F88                      MOV                    ↙
    ↘    R2, R8
.text:C0169F8C                      CMP                    ↙
    ↘    R3, #0
.text:C0169F90                      ORRNE                  ↙
    ↘    R1, R1, #3
.text:C0169F94                      STRNE                  ↙
    ↘    R1, [R4,#0x24]
.text:C0169F98                      ANDS                   ↙
    ↘    R3, R6, #0x200000
.text:C0169F9C                      MOV                    ↙
    ↘    R1, R12
.text:C0169FA0                      LDRNE                  ↙
    ↘    R3, [R4,#0x24]
.text:C0169FA4                      ANDNE                  ↙
    ↘    R3, R3, #1
```

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

```
.text:C0169FA8                  EORNE
        R3, R3, #1
.text:C0169FAC                  STR
        R3, [R11,#var_54]
.text:C0169FB0                  SUB
        R3, R11, #-var_38
.text:C0169FB4                  BL
        lookup_fast
...
.text:C016A128 loc_C016A128
            ; CODE XREF: do_last.isra
    .14+DC
.text:C016A128                  MOV
        R0, R4
.text:C016A12C                  BL
        complete_walk
...
```

TST

O_CREAT

19.2

:

```
#include <stdio.h>

#define IS_SET(flag, bit)       ((flag) & (
    bit))
#define SET_BIT(var, bit)       ((var) |= (
    bit))
```

```
#define REMOVE_BIT(var, bit)    ((var) &= ~(
    bit))

int f(int a)
{
    int rt=a;

    SET_BIT (rt, 0x4000);
    REMOVE_BIT (rt, 0x200);

    return rt;
};

int main()
{
    f(0x12340678);
};
```

19.2.1 x86

Sin optimización MSVC

(MSVC 2010):

Listing 19.10: MSVC 2010

```
_rt$ = -4          ; size = 4
_a$ = 8            ; size = 4
_f  PROC
    push   ebp
    mov    ebp, esp
    push   ecx
```

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

```
    mov     eax, DWORD PTR _a$[ebp]
    mov     DWORD PTR _rt$[ebp], eax
    mov     ecx, DWORD PTR _rt$[ebp]
    or      ecx, 16384                        ; ⤦
    ↳ 00004000H
    mov     DWORD PTR _rt$[ebp], ecx
    mov     edx, DWORD PTR _rt$[ebp]
    and     edx, -513                        ; ⤦
    ↳ fffffdffH
    mov     DWORD PTR _rt$[ebp], edx
    mov     eax, DWORD PTR _rt$[ebp]
    mov     esp, ebp
    pop     ebp
    ret     0
_f  ENDP
```

OllyDbg

OllyDbg. :

0x200 (00000000000000000001000000000) ().

0x200 0xFFFFFDFF (11111111111111111110111111111).

0x4000 (00000000000000100000000000000) ().

: 0x12340678 (10010001101000000011001111000). :

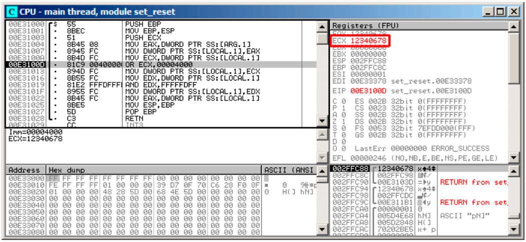

Figura 19.1: OllyDbg: ECX

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

OR:

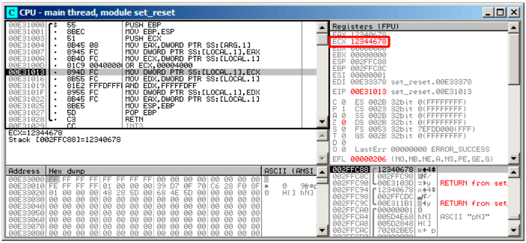

Figura 19.2: OllyDbg: OR

: 0x12344678 (10010001101000100011001111000).

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

:

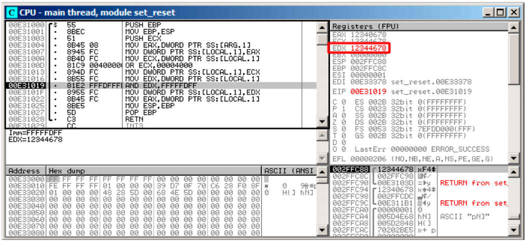

Figura 19.3: OllyDbg: EDX

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

AND:

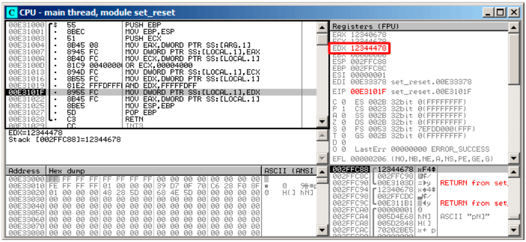

Figura 19.4: OllyDbg: AND

El 10º bit fue limpiado (o, en otras palabras, todos los bits fueron dejados excepto el 10º) y el valor final ahora es 0x12344478 (10010001101000100010001111000).

Con optimización MSVC

Listing 19.11: Con optimización MSVC

```
_a$ = 8                                 ; size = 4
_f      PROC
    mov     eax, DWORD PTR _a$[esp-4]
    and     eax, -513                   ; fffffdffH
    or      eax, 16384                  ; 00004000H
    ret     0
_f      ENDP
```

Sin optimización GCC

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

Listing 19.12: Sin optimización GCC

```
                public f
f               proc near

var_4           = dword ptr -4
arg_0           = dword ptr  8

                push    ebp
                mov     ebp, esp
                sub     esp, 10h
                mov     eax, [ebp+arg_0]
                mov     [ebp+var_4], eax
                or      [ebp+var_4], 4000h
                and     [ebp+var_4], 0
    FFFFFDFFh
                mov     eax, [ebp+var_4]
                leave
                retn
f               endp
```

-O3:

Con optimización GCC

Listing 19.13: Con optimización GCC

```
                public f
f               proc near

arg_0           = dword ptr  8

                push    ebp
```

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

```
                mov     ebp, esp
                mov     eax, [ebp+arg_0]
                pop     ebp
                or      ah, 40h
                and     ah, 0FDh
                retn
f               endp
```

<table>
<tr><td colspan="3">Spanish text placeholder</td></tr>
<tr><td colspan="3">RAX^{x64}</td></tr>
<tr><td></td><td colspan="2">EAX</td></tr>
<tr><td></td><td colspan="2">AX</td></tr>
<tr><td></td><td>AH</td><td>AL</td></tr>
</table>

N.B.

Con optimización GCC y regparm

Listing 19.14: Con optimización GCC

```
                public f
f               proc near
                push    ebp
                or      ah, 40h
                mov     ebp, esp
                and     ah, 0FDh
                pop     ebp
                retn
f               endp
```

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

19.2.2 ARM + Con optimización Keil 6/2013 (Modo ARM)

Listing 19.15: Con optimización Keil 6/2013 (Modo ARM)

```
02 0C C0 E3             BIC     R0, R0, #0x200
01 09 80 E3             ORR     R0, R0, #0x4000
1E FF 2F E1             BX      LR
```

BIC (*BItwise bit Clear*)

ORR OR en x86.

.

19.2.3 ARM + Con optimización Keil 6/2013 (Modo Thumb)

Listing 19.16: Con optimización Keil 6/2013 (Modo Thumb)

```
01 21 89 03             MOVS    R1, 0x4000
08 43                   ORRS    R0, R1
49 11                   ASRS    R1, R1, #5   ; ↙
    ↳ generate 0x200 and place to R1
88 43                   BICS    R0, R1
70 47                   BX      LR
```

0x200 0x4000, 0x200 .

ASRS (Spanish text placeholder), 0x4000 $\gg 5$.

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

19.2.4 ARM + Con optimización Xcode 4.6.3 (LLVM) (Modo ARM)

Listing 19.17: Con optimización Xcode 4.6.3 (LLVM) (Modo ARM)

```
42 0C C0 E3          BIC          R0, R0, #0x4200
01 09 80 E3          ORR          R0, R0, #0x4000
1E FF 2F E1          BX           LR
```

```
    REMOVE_BIT (rt, 0x4200);
    SET_BIT (rt, 0x4000);
```

. 0x4200? [2].

(91 on page 1754).

Con optimización Xcode 4.6.3 (LLVM)

19.2.5 ARM:

:

```
int f(int a)
{
    int rt=a;

    REMOVE_BIT (rt, 0x1234);

```

[2]LLVM build 2410.2.00 Apple Xcode 4.6.3

```
    return rt;
};
```

Keil 5.03 :

```
f PROC
        BIC      r0,r0,#0x1000
        BIC      r0,r0,#0x234
        BX       lr
        ENDP
```

19.2.6 ARM64: Con optimización GCC (Linaro) 4.9

Con optimización GCC BIC:

Listing 19.18: Con optimización GCC (Linaro) 4.9

```
f:
        and     w0, w0, -513    ; 0↙
    ↘ xFFFFFFFFFFFFFDFF
        orr     w0, w0, 16384   ; 0x4000
        ret
```

19.2.7 ARM64: Sin optimización GCC (Linaro) 4.9

Sin optimización GCC :

Listing 19.19: Sin optimización GCC (Linaro) 4.9

```
f:
```

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

```
        sub     sp, sp, #32
        str     w0, [sp,12]
        ldr     w0, [sp,12]
        str     w0, [sp,28]
        ldr     w0, [sp,28]
        orr     w0, w0, 16384   ; 0x4000
        str     w0, [sp,28]
        ldr     w0, [sp,28]
        and     w0, w0, -513    ; 0↙
    ↘ xFFFFFFFFFFFFFDFF
        str     w0, [sp,28]
        ldr     w0, [sp,28]
        add     sp, sp, 32
        ret
```

19.2.8 MIPS

Listing 19.20: Con optimización GCC 4.4.5 (IDA)

```
f:
; $a0=a
                ori     $a0, 0x4000
; $a0=a|0x4000
                li      $v0, 0xFFFFFDFF
                jr      $ra
                and     $v0, $a0, $v0
; : $v0 = $a0&$v0 = a|0x4000 & 0xFFFFFDFF
```

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

19.3 Shifts

SHL (SHift Left) y SHR (SHift Right) .

16.1.2 on page 398, 16.2.1 on page 407.

19.4

31	30 … 23	22 … 0
S	Spanish text placeholder	Spanish text placeholder

(SSpanish text placeholder)

MSB[3].

```
#include <stdio.h>

float my_abs (float i)
{
        unsigned int tmp=(*(unsigned int*)&i) & 0x7FFFFFFF;
        return *(float*)&tmp;
};

float set_sign (float i)
{
        unsigned int tmp=(*(unsigned int*)&i) | 0x80000000;
        return *(float*)&tmp;
};
```

[3] Most significant bit/byte

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

```
float negate (float i)
{
        unsigned int tmp=(*(unsigned int*)&i) ^ 0x80000000;
        return *(float*)&tmp;
};

int main()
{
        printf ("my_abs():\n");
        printf ("%f\n", my_abs (123.456));
        printf ("%f\n", my_abs (-456.123));
        printf ("set_sign():\n");
        printf ("%f\n", set_sign (123.456));
        printf ("%f\n", set_sign (-456.123));
        printf ("negate():\n");
        printf ("%f\n", negate (123.456));
        printf ("%f\n", negate (-456.123));
};
```

19.4.1

0	0	0
0	1	1
1	0	1
1	1	0

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

19.4.2 x86

:

Listing 19.21: Con optimización MSVC 2012

```
_tmp$ = 8
_i$ = 8
_my_abs PROC
        and     DWORD PTR _i$[esp-4], ↙
    ↘ 2147483647          ; 7fffffffH
        fld     DWORD PTR _tmp$[esp-4]
        ret     0
_my_abs ENDP

_tmp$ = 8
_i$ = 8
_set_sign PROC
        or      DWORD PTR _i$[esp-4], ↙
    ↘ -2147483648         ; 80000000H
        fld     DWORD PTR _tmp$[esp-4]
        ret     0
_set_sign ENDP

_tmp$ = 8
_i$ = 8
_negate PROC
        xor     DWORD PTR _i$[esp-4], ↙
    ↘ -2147483648         ; 80000000H
        fld     DWORD PTR _tmp$[esp-4]
        ret     0
_negate ENDP
```

AND y OR. XOR.

:

Listing 19.22: Con optimización MSVC 2012 x64

```
tmp$ = 8
i$ = 8
my_abs  PROC
        movss   DWORD PTR [rsp+8], xmm0
        mov     eax, DWORD PTR i$[rsp]
        btr     eax, 31
        mov     DWORD PTR tmp$[rsp], eax
        movss   xmm0, DWORD PTR tmp$[rsp]
        ret     0
my_abs  ENDP
_TEXT   ENDS

tmp$ = 8
i$ = 8
set_sign PROC
        movss   DWORD PTR [rsp+8], xmm0
        mov     eax, DWORD PTR i$[rsp]
        bts     eax, 31
        mov     DWORD PTR tmp$[rsp], eax
        movss   xmm0, DWORD PTR tmp$[rsp]
        ret     0
set_sign ENDP

tmp$ = 8
i$ = 8
negate  PROC
        movss   DWORD PTR [rsp+8], xmm0
        mov     eax, DWORD PTR i$[rsp]
        btc     eax, 31
        mov     DWORD PTR tmp$[rsp], eax
```

```
        movss   xmm0, DWORD PTR tmp$[rsp]
        ret     0
negate  ENDP
```

.

19.4.3 MIPS

GCC 4.4.5 para MIPS :

Listing 19.23: Con optimización GCC 4.4.5 (IDA)

```

my_abs:
;  1:
                mfc1    $v1, $f12
                li      $v0, 0x7FFFFFFF
; $v0=0x7FFFFFFF
; :
                and     $v0, $v1
;  1:
                mtc1    $v0, $f0
;
                jr      $ra
                or      $at, $zero ; branch ↙
    ↘ delay slot

set_sign:
;  1:
                mfc1    $v0, $f12
                lui     $v1, 0x8000
; $v1=0x80000000
; :
```

```
                or      $v0, $v1, $v0
;  1:
                mtc1    $v0, $f0
;
                jr      $ra
                or      $at, $zero ; branch ⤦
    ↳ delay slot

negate:
;  1:
                mfc1    $v0, $f12
                lui     $v1, 0x8000
; $v1=0x80000000
; :
                xor     $v0, $v1, $v0
;  1:
                mtc1    $v0, $f0
;
                jr      $ra
                or      $at, $zero ; branch ⤦
    ↳ delay slot
```

19.4.4 ARM

Con optimización Keil 6/2013 (Modo ARM)

Listing 19.24: Con optimización Keil 6/2013 (Modo ARM)

```
my_abs PROC
; :
        BIC      r0,r0,#0x80000000
```

```
        BX      lr
        ENDP

set_sign PROC
; :
        ORR     r0,r0,#0x80000000
        BX      lr
        ENDP

negate PROC
; :
        EOR     r0,r0,#0x80000000
        BX      lr
        ENDP
```

. («Exclusive OR»).

Con optimización Keil 6/2013 (Modo Thumb)

Listing 19.25: Con optimización Keil 6/2013 (Modo Thumb)

```
my_abs PROC
        LSLS    r0,r0,#1
; r0=i<<1
        LSRS    r0,r0,#1
; r0=(i<<1)>>1
        BX      lr
        ENDP

set_sign PROC
        MOVS    r1,#1
; r1=1
```

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

```
        LSLS     r1,r1,#31
; r1=1<<31=0x80000000
        ORRS     r0,r0,r1
; r0=r0 | 0x80000000
        BX       lr
        ENDP

negate PROC
        MOVS     r1,#1
; r1=1
        LSLS     r1,r1,#31
; r1=1<<31=0x80000000
        EORS     r0,r0,r1
; r0=r0 ^ 0x80000000
        BX       lr
        ENDP
```

: $1 << 31 = 0x80000000$.

: $(i << 1) >> 1$. .

Con optimización GCC 4.6.3 (Raspberry Pi, Modo ARM)

Listing 19.26: Con optimización GCC 4.6.3 para Raspberry Pi (Modo ARM)

```

my_abs
; :
                FMRS    R2, S0
; :
                BIC     R3, R2, #0x80000000
; :
```

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

```
                FMSR    S0, R3
                BX      LR

set_sign
; :
                FMRS    R2, S0
; :
                ORR     R3, R2, #0x80000000
; :
                FMSR    S0, R3
                BX      LR

negate
; :
                FMRS    R2, S0
; :
                ADD     R3, R2, #0x80000000
; :
                FMSR    S0, R3
                BX      LR
```

my_abs() y set_sign() negate()? ?

`ADD register, 0x80000000` `XOR register, 0x8000`
? : $1234567 + 10000 = 1244567$ ( ). . .

19.5

«population count»[4].

4

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

```
#include <stdio.h>

#define IS_SET(flag, bit)       ((flag) & (bit))

int f(unsigned int a)
{
    int i;
    int rt=0;

    for (i=0; i<32; i++)
        if (IS_SET (a, 1<<i))
            rt++;

    return rt;
};

int main()
{
    f(0x12345678); // test
};
```

$1 \ll i$ $i = 0 \ldots 31$:

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

C/C++			
$1 \ll 0$	1	1	1
$1 \ll 1$	2^1	2	2
$1 \ll 2$	2^2	4	4
$1 \ll 3$	2^3	8	8
$1 \ll 4$	2^4	16	0x10
$1 \ll 5$	2^5	32	0x20
$1 \ll 6$	2^6	64	0x40
$1 \ll 7$	2^7	128	0x80
$1 \ll 8$	2^8	256	0x100
$1 \ll 9$	2^9	512	0x200
$1 \ll 10$	2^{10}	1024	0x400
$1 \ll 11$	2^{11}	2048	0x800
$1 \ll 12$	2^{12}	4096	0x1000
$1 \ll 13$	2^{13}	8192	0x2000
$1 \ll 14$	2^{14}	16384	0x4000
$1 \ll 15$	2^{15}	32768	0x8000
$1 \ll 16$	2^{16}	65536	0x10000
$1 \ll 17$	2^{17}	131072	0x20000
$1 \ll 18$	2^{18}	262144	0x40000
$1 \ll 19$	2^{19}	524288	0x80000
$1 \ll 20$	2^{20}	1048576	0x100000
$1 \ll 21$	2^{21}	2097152	0x200000
$1 \ll 22$	2^{22}	4194304	0x400000
$1 \ll 23$	2^{23}	8388608	0x800000
$1 \ll 24$	2^{24}	16777216	0x1000000
$1 \ll 25$	2^{25}	33554432	0x2000000
$1 \ll 26$	2^{26}	67108864	0x4000000
$1 \ll 27$	2^{27}	134217728	0x8000000
$1 \ll 28$	2^{28}	268435456	0x10000000
$1 \ll 29$	2^{29}	536870912	0x20000000

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

ssl_private.h Apache 2.4.6:

```
/**
 * Define the SSL options
 */
#define SSL_OPT_NONE           (0)
#define SSL_OPT_RELSET         (1<<0)
#define SSL_OPT_STDENVVARS     (1<<1)
#define SSL_OPT_EXPORTCERTDATA (1<<3)
#define SSL_OPT_FAKEBASICAUTH  (1<<4)
#define SSL_OPT_STRICTREQUIRE  (1<<5)
#define SSL_OPT_OPTRENEGOTIATE (1<<6)
#define SSL_OPT_LEGACYDNFORMAT (1<<7)
```

.

19.5.1 x86

MSVC

(MSVC 2010):

Listing 19.27: MSVC 2010

```

_rt$ = -8             ; size = 4
_i$ = -4              ; size = 4
_a$ = 8               ; size = 4
_f  PROC
    push   ebp
    mov    ebp, esp
    sub    esp, 8
    mov    DWORD PTR _rt$[ebp], 0
```

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

```
    mov     DWORD PTR _i$[ebp], 0
    jmp     SHORT $LN4@f
$LN3@f:
    mov     eax, DWORD PTR _i$[ebp]   ;  i
    add     eax, 1
    mov     DWORD PTR _i$[ebp], eax
$LN4@f:
    cmp     DWORD PTR _i$[ebp], 32    ; ↙
    ↘ 00000020H
    jge     SHORT $LN2@f              ; ?
    mov     edx, 1
    mov     ecx, DWORD PTR _i$[ebp]
    shl     edx, cl                   ; EDX=↙
    ↘ EDX<<CL
    and     edx, DWORD PTR _a$[ebp]
    je      SHORT $LN1@f              ; ?
                                      ;
    mov     eax, DWORD PTR _rt$[ebp]  ;
    add     eax, 1                    ;  rt
    mov     DWORD PTR _rt$[ebp], eax
$LN1@f:
    jmp     SHORT $LN3@f
$LN2@f:
    mov     eax, DWORD PTR _rt$[ebp]
    mov     esp, ebp
    pop     ebp
    ret     0
_f    ENDP
```

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

OllyDbg

OllyDbg. 0x12345678.

$i = 1$, i ECX:

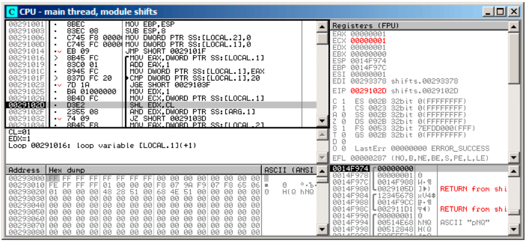

Figura 19.5: OllyDbg: $i = 1$, i ECX

EDX 1. SHL.

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

SHL:

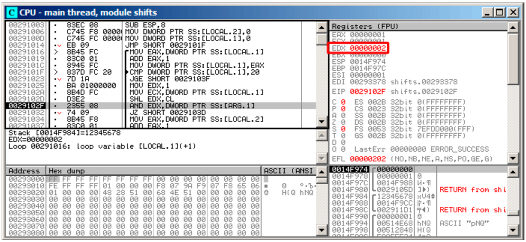

Figura 19.6: OllyDbg: $i = 1$, EDX $= 1 \ll 1 = 2$

EDX $1 \ll 1$ (o 2). .

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

AND ZF 1, (0x12345678) 2 0:

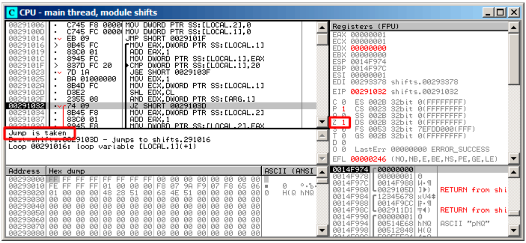

Figura 19.7: OllyDbg: $i = 1$, (ZF =1)

. .

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

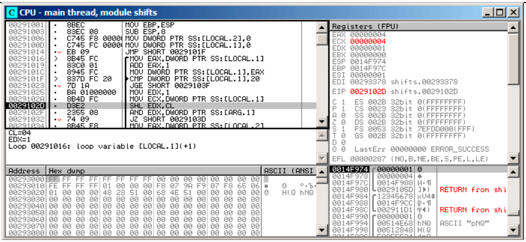

Figura 19.8: OllyDbg: $i = 4$, i ECX

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

EDX =$1 \ll 4$ (o 0x10 o 16):

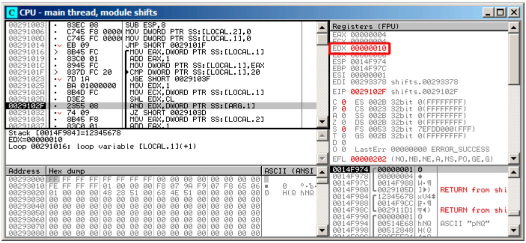

Figura 19.9: OllyDbg: $i = 4$, EDX =$1 \ll 4 = 0x10$

.

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

AND:

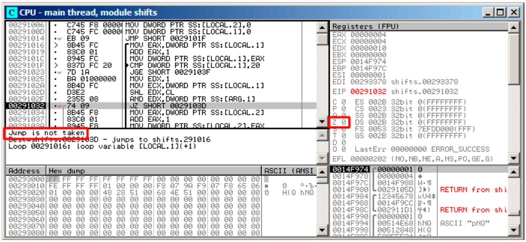

Figura 19.10: OllyDbg: $i = 4$, (ZF =0)

ZF 0 ., 0x12345678 & 0x10 = 0x10.

13. 0x12345678.

GCC

GCC 4.4.1:

Listing 19.28: GCC 4.4.1

```
                public f
f               proc near

rt              = dword ptr -0Ch
i               = dword ptr -8
arg_0           = dword ptr  8

                push    ebp
                mov     ebp, esp
```

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

```
                push    ebx
                sub     esp, 10h
                mov     [ebp+rt], 0
                mov     [ebp+i], 0
                jmp     short loc_80483EF
loc_80483D0:
                mov     eax, [ebp+i]
                mov     edx, 1
                mov     ebx, edx
                mov     ecx, eax
                shl     ebx, cl
                mov     eax, ebx
                and     eax, [ebp+arg_0]
                test    eax, eax
                jz      short loc_80483EB
                add     [ebp+rt], 1
loc_80483EB:
                add     [ebp+i], 1
loc_80483EF:
                cmp     [ebp+i], 1Fh
                jle     short loc_80483D0
                mov     eax, [ebp+rt]
                add     esp, 10h
                pop     ebx
                pop     ebp
                retn
f               endp
```

19.5.2 x64

:

```
#include <stdio.h>
#include <stdint.h>

#define IS_SET(flag, bit)       ((flag) & (⤦
    ↘ bit))

int f(uint64_t a)
{
    uint64_t i;
    int rt=0;

    for (i=0; i<64; i++)
        if (IS_SET (a, 1ULL<<i))
            rt++;

    return rt;
};
```

Sin optimización GCC 4.8.2

.

Listing 19.29: Sin optimización GCC 4.8.2

```
f:
        push    rbp
        mov     rbp, rsp
        mov     QWORD PTR [rbp-24], rdi ; a
        mov     DWORD PTR [rbp-12], 0   ; rt⤦
    ↘ =0
```

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

```
        mov     QWORD PTR [rbp-8], 0    ; i
    ↪ =0
        jmp     .L2
.L4:
        mov     rax, QWORD PTR [rbp-8]
        mov     rdx, QWORD PTR [rbp-24]
; RAX = i, RDX = a
        mov     ecx, eax
; ECX = i
        shr     rdx, cl
; RDX = RDX>>CL = a>>i
        mov     rax, rdx
; RAX = RDX = a>>i
        and     eax, 1
; EAX = EAX&1 = (a>>i)&1
        test    rax, rax
; ?
; .
        je      .L3
        add     DWORD PTR [rbp-12], 1   ; rt
    ↪ ++
.L3:
        add     QWORD PTR [rbp-8], 1    ; i
    ↪ ++
.L2:
        cmp     QWORD PTR [rbp-8], 63   ; i
    ↪ <63?
        jbe     .L4                     ;
        mov     eax, DWORD PTR [rbp-12] ;
    ↪ rt
        pop     rbp
        ret
```

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

Con optimización GCC 4.8.2

Listing 19.30: Con optimización GCC 4.8.2

```

f:
        xor     eax, eax        ;
        xor     ecx, ecx        ;
.L3:
        mov     rsi, rdi        ;
        lea     edx, [rax+1]    ; EDX=EAX+1
;
        shr     rsi, cl         ; RSI=RSI>>CL
        and     esi, 1          ; ESI=ESI&1
;
        cmovne  eax, edx
        add     rcx, 1          ; RCX++
        cmp     rcx, 64
        jne     .L3
        rep ret                 ; AKA fatret
```

. CMOVNE[5] (CMOVNZ[6]) «rt» EDX («») EAX (« rt»).

: 33.1 on page 873.

REP RET (F3 C3) FATRET MSVC. : [AMD13b, p.15] [7].

Con optimización MSVC 2010

[5] Conditional MOVe if Not Equal

[6] Conditional MOVe if Not Zero

[7] : http://go.yurichev.com/17328

Listing 19.31: MSVC 2010

```
a$ = 8
f       PROC
; RCX =
        xor     eax, eax
        mov     edx, 1
        lea     r8d, QWORD PTR [rax+64]
; R8D=64
        npad    5
$LL4@f:
        test    rdx, rcx
; ?
; .
        je      SHORT $LN3@f
        inc     eax     ; rt++
$LN3@f:
        rol     rdx, 1  ; RDX=RDX<<1
        dec     r8      ; R8--
        jne     SHORT $LL4@f
        fatret  0
f       ENDP
```

ROL SHL, «rotate left» «shift left», SHL.

: A.6.3 on page 1895.

R8 . .

:

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

RDX	R8
0x0000000000000001	64
0x0000000000000002	63
0x0000000000000004	62
0x0000000000000008	61
...	...
0x4000000000000000	2
0x8000000000000000	1

FATRET: 19.5.2 on page 620.

Con optimización MSVC 2012

Listing 19.32: MSVC 2012

```
a$ = 8
f       PROC
; RCX =
        xor     eax, eax
        mov     edx, 1
        lea     r8d, QWORD PTR [rax+32]
; EDX = 1, R8D = 32
        npad    5
$LL4@f:
;
        test    rdx, rcx
        je      SHORT $LN3@f
        inc     eax         ; rt++
$LN3@f:
        rol     rdx, 1   ; RDX=RDX<<1
; ↙
    ↳ ----------------------------------------
```

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

```
    ↳
;
        test    rdx, rcx
        je      SHORT $LN11@f
        inc     eax     ; rt++
$LN11@f:
        rol     rdx, 1  ; RDX=RDX<<1
; ↙
    ↳ ---------------------------------------
    ↳
        dec     r8      ; R8--
        jne     SHORT $LL4@f
        fatret  0
f       ENDP
```

Con optimización MSVC 2012 MSVC 2010, .

19.5.3 ARM + Con optimización Xcode 4.6.3 (LLVM) (Modo ARM)

Listing 19.33: Con optimización Xcode 4.6.3 (LLVM) (Modo ARM)

```

                MOV             R1, R0
                MOV             R0, #0
                MOV             R2, #1
                MOV             R3, R0
loc_2E54
                TST             R1, R2,LSL ↙
    ↳ R3 ;  R1 & (R2<<R3)
```

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

```
                ADD             R3, R3, #1   ↙
    ↳   ; R3++
                ADDNE           R0, R0, #1   ↙
    ↳   ;  R0++
                CMP             R3, #32
                BNE             loc_2E54
                BX              LR
```

TST TEST en x86.

 ([41.2.1 on page 942](#)), LSL (*Logical Shift Left*), LSR (*Logical Shift Right*), ASR (*Arithmetic Shift Right*), ROR (*Rotate Right*) y RRX (*Rotate Right with Extend*) MOV, TST, CMP, ADD, SUB, RSB[8].

«TST R1, R2,LSL R3» $R1 \wedge (R2 \ll R3)$.

19.5.4 ARM + Con optimización Xcode 4.6.3 (LLVM) (Modo Thumb-2)

, LSL.W/TST TST, LSL TST.

```
                MOV             R1, R0
                MOVS            R0, #0
                MOV.W           R9, #1
                MOVS            R3, #0
loc_2F7A
                LSL.W           R2, R9, R3
                TST             R2, R1
                ADD.W           R3, R3, #1
```

[8] Estas instrucciones también se llaman «data processing instructions»

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

```
                IT NE
                ADDNE           R0, #1
                CMP             R3, #32
                BNE             loc_2F7A
                BX              LR
```

19.5.5 ARM64 + Con optimización GCC 4.9

: 19.5.2 on page 617.

Listing 19.34: Con optimización GCC (Linaro) 4.8

```
f:
        mov     w2, 0           ; rt=0
        mov     x5, 1
        mov     w1, w2
.L2:
        lsl     x4, x5, x1      ; w4 = w5<<↙
    ↳ w1 = 1<<i
        add     w3, w2, 1       ; new_rt=rt↙
    ↳ +1
        tst     x4, x0          ; (1<<i) & a
        add     w1, w1, 1       ; i++
; ?
;  w2=w3  rt=new_rt.
; : w2=w2  rt=rt ()
        csel    w2, w3, w2, ne
        cmp     w1, 64          ; i<64?
        bne     .L2             ;
        mov     w0, w2          ;  rt
        ret
```

: 19.30 on page 620.

CSEL «Conditional SELect». TST W2.

19.5.6 ARM64 + Sin optimización GCC 4.9

: 19.5.2 on page 617.

.

Listing 19.35: Sin optimización GCC (Linaro) 4.8

```
f:
        sub     sp, sp, #32
        str     x0, [sp,8]        ;  Register ⤦
    ↳ Save Area
        str     wzr, [sp,24]      ; rt=0
        str     wzr, [sp,28]      ; i=0
        b       .L2
.L4:
        ldr     w0, [sp,28]
        mov     x1, 1
        lsl     x0, x1, x0        ; X0 = X1<<⤦
    ↳ X0 = 1<<i
        mov     x1, x0
; X1 = 1<<1
        ldr     x0, [sp,8]
; X0 = a
        and     x0, x1, x0
; X0 = X1&X0 = (1<<i) & a
; X0 ?
        cmp     x0, xzr
        beq     .L3
```

```
; rt++
        ldr     w0, [sp,24]
        add     w0, w0, 1
        str     w0, [sp,24]
.L3:
; i++
        ldr     w0, [sp,28]
        add     w0, w0, 1
        str     w0, [sp,28]
.L2:
; i<=63?  .L4
        ldr     w0, [sp,28]
        cmp     w0, 63
        ble     .L4
;  rt
        ldr     w0, [sp,24]
        add     sp, sp, 32
        ret
```

19.5.7 MIPS

Sin optimización GCC

Listing 19.36: Sin optimización GCC 4.4.5 (IDA)

```
f:
; :
rt              = -0x10
i               = -0xC
var_4           = -4
a               =  0
```

```
                addiu   $sp, -0x18
                sw      $fp, 0x18+var_4($sp)
                move    $fp, $sp
                sw      $a0, 0x18+a($fp)
; :
                sw      $zero, 0x18+rt($fp)
                sw      $zero, 0x18+i($fp)
; :
                b       loc_68
                or      $at, $zero ; branch ↙
    ↳ delay slot, NOP

loc_20:
                li      $v1, 1
                lw      $v0, 0x18+i($fp)
                or      $at, $zero ; load ↙
    ↳ delay slot, NOP
                sllv    $v0, $v1, $v0
; $v0 = 1<<i
                move    $v1, $v0
                lw      $v0, 0x18+a($fp)
                or      $at, $zero ; load ↙
    ↳ delay slot, NOP
                and     $v0, $v1, $v0
; $v0 = a&(1<<i)
; a&(1<<i) ? :
                beqz    $v0, loc_58
                or      $at, $zero
;  a&(1<<i)!=0, :
                lw      $v0, 0x18+rt($fp)
                or      $at, $zero ; load ↙
    ↳ delay slot, NOP
```

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

```
                addiu   $v0, 1
                sw      $v0, 0x18+rt($fp)

loc_58:
;  i:
                lw      $v0, 0x18+i($fp)
                or      $at, $zero ; load ⤦
    ↳ delay slot, NOP
                addiu   $v0, 1
                sw      $v0, 0x18+i($fp)

loc_68:
;  0x20 (32).
;  0x20 (32):
                lw      $v0, 0x18+i($fp)
                or      $at, $zero ; load ⤦
    ↳ delay slot, NOP
                slti    $v0, 0x20  # ' '
                bnez    $v0, loc_20
                or      $at, $zero ; branch ⤦
    ↳ delay slot, NOP
; .  rt:
                lw      $v0, 0x18+rt($fp)
                move    $sp, $fp   ; load ⤦
    ↳ delay slot
                lw      $fp, 0x18+var_4($sp)
                addiu   $sp, 0x18  ; load ⤦
    ↳ delay slot
                jr      $ra
                or      $at, $zero ; branch ⤦
    ↳ delay slot, NOP
```

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

Con optimización GCC

. ? : 33.1 on page 873.

Listing 19.37: Con optimización GCC 4.4.5 (IDA)

```
f:
; $a0=a
;  $v0:
                move    $v0, $zero
;  $v1:
                move    $v1, $zero
                li      $t0, 1
                li      $a3, 32
                sllv    $a1, $t0, $v1
; $a1 = $t0<<$v1 = 1<<i

loc_14:
                and     $a1, $a0
; $a1 = a&(1<<i)
;  i:
                addiu   $v1, 1
;  loc\_28  a&(1<<i)==0   rt:
                beqz    $a1, loc_28
                addiu   $a2, $v0, 1
;  $v0:
                move    $v0, $a2

loc_28:
;  i!=32,  loc_14 :
                bne     $v1, $a3, loc_14
                sllv    $a1, $t0, $v1
;
```

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

```
                jr      $ra
                or      $at, $zero ; branch ↙
    ↳ delay slot, NOP
```

19.6 Conclusión

19.6.1

Listing 19.38: C/C++

```
if (input&0x40)
    ...
```

Listing 19.39: x86

```
TEST REG, 40h
JNZ is_set
;
```

Listing 19.40: x86

```
TEST REG, 40h
JZ is_cleared
;
```

Listing 19.41: ARM (Modo ARM)

```
TST REG, #0x40
```

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

```
BNE is_set
;
```

19.6.2

Listing 19.42: C/C++

```
if ((value>>n)&1)
    ....
```

Listing 19.43: x86

```
; REG=input_value
; CL=n
SHR REG, CL
AND REG, 1
```

Listing 19.44: C/C++

```
if (value & (1<<n))
    ....
```

Listing 19.45: x86

```
; CL=n
MOV REG, 1
SHL REG, CL
AND input_value, REG
```

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

19.6.3

Listing 19.46: C/C++

```
value=value|0x40;
```

Listing 19.47: x86

```
OR REG, 40h
```

Listing 19.48: ARM (Modo ARM) y ARM64

```
ORR R0, R0, #0x40
```

19.6.4

Listing 19.49: C/C++

```
value=value|(1<<n);
```

Listing 19.50: x86

```
; CL=n
MOV REG, 1
SHL REG, CL
OR input_value, REG
```

19.6.5

Listing 19.51: C/C++

```
value=value&(~0x40);
```

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

Listing 19.52: x86

```
AND REG, 0FFFFFFBFh
```

Listing 19.53: x64

```
AND REG, 0FFFFFFFFFFFFFFBFh
```

Listing 19.54: ARM (Modo ARM)

```
BIC R0, R0, #0x40
```

19.6.6

Listing 19.55: C/C++

```
value=value&(~(1<<n));
```

Listing 19.56: x86

```
; CL=n
MOV REG, 1
SHL REG, CL
NOT REG
AND input_value, REG
```

19.7 Ejercicios

19.7.1 Ejercicio #1

Lo que hace el código?

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

Listing 19.57: Con optimización MSVC 2010

```
_a$ = 8
_f      PROC
        mov     ecx, DWORD PTR _a$[esp-4]
        mov     eax, ecx
        mov     edx, ecx
        shl     edx, 16             ; 00000010H
        and     eax, 65280          ; 0000ff00H
        or      eax, edx
        mov     edx, ecx
        and     edx, 16711680       ; 00ff0000H
        shr     ecx, 16             ; 00000010H
        or      edx, ecx
        shl     eax, 8
        shr     edx, 8
        or      eax, edx
        ret     0
_f      ENDP
```

Listing 19.58: Con optimización Keil 6/2013 (Modo ARM)

```
f PROC
        MOV      r1,#0xff0000
        AND      r1,r1,r0,LSL #8
        MOV      r2,#0xff00
        ORR      r1,r1,r0,LSR #24
        AND      r2,r2,r0,LSR #8
        ORR      r1,r1,r2
        ORR      r0,r1,r0,LSL #24
        BX       lr
        ENDP
```

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

Listing 19.59: Con optimización Keil 6/2013 (Modo Thumb)

```
f PROC
        MOVS     r3,#0xff
        LSLS     r2,r0,#8
        LSLS     r3,r3,#16
        ANDS     r2,r2,r3
        LSRS     r1,r0,#24
        ORRS     r1,r1,r2
        LSRS     r2,r0,#8
        ASRS     r3,r3,#8
        ANDS     r2,r2,r3
        ORRS     r1,r1,r2
        LSLS     r0,r0,#24
        ORRS     r0,r0,r1
        BX       lr
        ENDP
```

Listing 19.60: Con optimización GCC 4.9 (ARM64)

```
f:
        rev     w0, w0
        ret
```

Listing 19.61: Con optimización GCC 4.4.5 (MIPS) (IDA)

```
f:
                srl     $v0, $a0, 24
                sll     $v1, $a0, 24
                sll     $a1, $a0, 8
                or      $v1, $v0
                lui     $v0, 0xFF
                and     $v0, $a1, $v0
```

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

```
                srl     $a0, 8
                or      $v0, $v1, $v0
                andi    $a0, 0xFF00
                jr      $ra
                or      $v0, $a0
```

Responda: G.1.13 on page 1925.

19.7.2 Ejercicio #2

Lo que hace el código?

Listing 19.62: Con optimización MSVC 2010

```
_a$ = 8                                                         ↙
    ↘                        ; size = 4
_f      PROC
        push    esi
        mov     esi, DWORD PTR _a$[esp]
        xor     ecx, ecx
        push    edi
        lea     edx, DWORD PTR [ecx+1]
        xor     eax, eax
        npad    3 ; align next label
$LL3@f:
        mov     edi, esi
        shr     edi, cl
        add     ecx, 4
        and     edi, 15
        imul    edi, edx
        lea     edx, DWORD PTR [edx+edx*4]
        add     eax, edi
        add     edx, edx
```

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

```
        cmp     ecx, 28
        jle     SHORT $LL3@f
        pop     edi
        pop     esi
        ret     0
_f      ENDP
```

Listing 19.63: Con optimización Keil 6/2013 (Modo ARM)

```
f PROC
        MOV      r3,r0
        MOV      r1,#0
        MOV      r2,#1
        MOV      r0,r1
|L0.16|
        LSR      r12,r3,r1
        AND      r12,r12,#0xf
        MLA      r0,r12,r2,r0
        ADD      r1,r1,#4
        ADD      r2,r2,r2,LSL #2
        CMP      r1,#0x1c
        LSL      r2,r2,#1
        BLE      |L0.16|
        BX       lr
        ENDP
```

Listing 19.64: Con optimización Keil 6/2013 (Modo Thumb)

```
f PROC
        PUSH     {r4,lr}
        MOVS     r3,r0
        MOVS     r1,#0
        MOVS     r2,#1
```

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

```
        MOVS     r0,r1
|L0.10|
        MOVS     r4,r3
        LSRS     r4,r4,r1
        LSLS     r4,r4,#28
        LSRS     r4,r4,#28
        MULS     r4,r2,r4
        ADDS     r0,r4,r0
        MOVS     r4,#0xa
        MULS     r2,r4,r2
        ADDS     r1,r1,#4
        CMP      r1,#0x1c
        BLE      |L0.10|
        POP      {r4,pc}
        ENDP
```

Listing 19.65: Sin optimización GCC 4.9 (ARM64)

```
f:
        sub     sp, sp, #32
        str     w0, [sp,12]
        str     wzr, [sp,28]
        mov     w0, 1
        str     w0, [sp,24]
        str     wzr, [sp,20]
        b       .L2
.L3:
        ldr     w0, [sp,28]
        ldr     w1, [sp,12]
        lsr     w0, w1, w0
        and     w1, w0, 15
        ldr     w0, [sp,24]
        mul     w0, w1, w0
```

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

```
        ldr     w1, [sp,20]
        add     w0, w1, w0
        str     w0, [sp,20]
        ldr     w0, [sp,28]
        add     w0, w0, 4
        str     w0, [sp,28]
        ldr     w1, [sp,24]
        mov     w0, w1
        lsl     w0, w0, 2
        add     w0, w0, w1
        lsl     w0, w0, 1
        str     w0, [sp,24]
.L2:
        ldr     w0, [sp,28]
        cmp     w0, 28
        ble     .L3
        ldr     w0, [sp,20]
        add     sp, sp, 32
        ret
```

Listing 19.66: Con optimización GCC 4.4.5 (MIPS) (IDA)

```
f:
                srl     $v0, $a0, 8
                srl     $a3, $a0, 20
                andi    $a3, 0xF
                andi    $v0, 0xF
                srl     $a1, $a0, 12
                srl     $a2, $a0, 16
                andi    $a1, 0xF
                andi    $a2, 0xF
                sll     $t2, $v0, 4
                sll     $v1, $a3, 2
```

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

```
                sll     $t0, $v0, 2
                srl     $t1, $a0, 4
                sll     $t5, $a3, 7
                addu    $t0, $t2
                subu    $t5, $v1
                andi    $t1, 0xF
                srl     $v1, $a0, 28
                sll     $t4, $a1, 7
                sll     $t2, $a2, 2
                sll     $t3, $a1, 2
                sll     $t7, $a2, 7
                srl     $v0, $a0, 24
                addu    $a3, $t5, $a3
                subu    $t3, $t4, $t3
                subu    $t7, $t2
                andi    $v0, 0xF
                sll     $t5, $t1, 3
                sll     $t6, $v1, 8
                sll     $t2, $t0, 2
                sll     $t4, $t1, 1
                sll     $t1, $v1, 3
                addu    $a2, $t7, $a2
                subu    $t1, $t6, $t1
                addu    $t2, $t0, $t2
                addu    $t4, $t5
                addu    $a1, $t3, $a1
                sll     $t5, $a3, 2
                sll     $t3, $v0, 8
                sll     $t0, $v0, 3
                addu    $a3, $t5
                subu    $t0, $t3, $t0
                addu    $t4, $t2, $t4
                sll     $t3, $a2, 2
```

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

```
                sll     $t2, $t1, 6
                sll     $a1, 3
                addu    $a1, $t4, $a1
                subu    $t1, $t2, $t1
                addu    $a2, $t3
                sll     $t2, $t0, 6
                sll     $t3, $a3, 2
                andi    $a0, 0xF
                addu    $v1, $t1, $v1
                addu    $a0, $a1
                addu    $a3, $t3
                subu    $t0, $t2, $t0
                sll     $a2, 4
                addu    $a2, $a0, $a2
                addu    $v0, $t0, $v0
                sll     $a1, $v1, 2
                sll     $a3, 5
                addu    $a3, $a2, $a3
                addu    $v1, $a1
                sll     $v0, 6
                addu    $v0, $a3, $v0
                sll     $v1, 7
                jr      $ra
                addu    $v0, $v1, $v0
```

Responda: G.1.13 on page 1926.

19.7.3 Ejercicio #3

Listing 19.67: Con optimización MSVC 2010

```
_main   PROC
```

```
        push    278595              ; 00044043H
        push    OFFSET $SG79792 ; 'caption'
        push    OFFSET $SG79793 ; 'hello, world!'
        push    0
        call    DWORD PTR __imp__MessageBoxA@16
        xor     eax, eax
        ret     0
_main   ENDP
```

Responda: G.1.13 on page 1927.

19.7.4 Ejercicio #4

Lo que hace el código?

Listing 19.68: Con optimización MSVC 2010

```
_m$ = 8         ; size = 4
_n$ = 12        ; size = 4
_f      PROC
        mov     ecx, DWORD PTR _n$[esp-4]
        xor     eax, eax
        xor     edx, edx
        test    ecx, ecx
        je      SHORT $LN2@f
        push    esi
        mov     esi, DWORD PTR _m$[esp]
$LL3@f:
        test    cl, 1
        je      SHORT $LN1@f
        add     eax, esi
```

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

```
        adc     edx, 0
$LN1@f:
        add     esi, esi
        shr     ecx, 1
        jne     SHORT $LL3@f
        pop     esi
$LN2@f:
        ret     0
_f      ENDP
```

Listing 19.69: Con optimización Keil 6/2013 (Modo ARM)

```
f PROC
        PUSH     {r4,lr}
        MOV      r3,r0
        MOV      r0,#0
        MOV      r2,r0
        MOV      r12,r0
        B        |L0.48|
|L0.24|
        TST      r1,#1
        BEQ      |L0.40|
        ADDS     r0,r0,r3
        ADC      r2,r2,r12
|L0.40|
        LSL      r3,r3,#1
        LSR      r1,r1,#1
|L0.48|
        CMP      r1,#0
        MOVEQ    r1,r2
        BNE      |L0.24|
        POP      {r4,pc}
        ENDP
```

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

Listing 19.70: Con optimización Keil 6/2013 (Modo Thumb)

```
f PROC
        PUSH     {r4,r5,lr}
        MOVS     r3,r0
        MOVS     r0,#0
        MOVS     r2,r0
        MOVS     r4,r0
        B        |L0.24|
|L0.12|
        LSLS     r5,r1,#31
        BEQ      |L0.20|
        ADDS     r0,r0,r3
        ADCS     r2,r2,r4
|L0.20|
        LSLS     r3,r3,#1
        LSRS     r1,r1,#1
|L0.24|
        CMP      r1,#0
        BNE      |L0.12|
        MOVS     r1,r2
        POP      {r4,r5,pc}
        ENDP
```

Listing 19.71: Con optimización GCC 4.9 (ARM64)

```
f:
        mov     w2, w0
        mov     x0, 0
        cbz     w1, .L2
.L3:
        and     w3, w1, 1
        lsr     w1, w1, 1
        cmp     w3, wzr
```

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

```
        add     x3, x0, x2, uxtw
        lsl     w2, w2, 1
        csel    x0, x3, x0, ne
        cbnz    w1, .L3
.L2:
        ret
```

Listing 19.72: Con optimización GCC 4.4.5 (MIPS) (IDA)

```
mult:
                beqz    $a1, loc_40
                move    $a3, $zero
                move    $a2, $zero
                addu    $t0, $a3, $a0

loc_10:                                      # ↙
    ↘ CODE XREF: mult+38
                sltu    $t1, $t0, $a3
                move    $v1, $t0
                andi    $t0, $a1, 1
                addu    $t1, $a2
                beqz    $t0, loc_30
                srl     $a1, 1
                move    $a3, $v1
                move    $a2, $t1

loc_30:                                      # ↙
    ↘ CODE XREF: mult+20
                beqz    $a1, loc_44
                sll     $a0, 1
                b       loc_10
                addu    $t0, $a3, $a0
loc_40:                                      # ↙
```

CAPÍTULO 19. MANIPULANDO BIT(S) ESPECÍFICAS

```
    ↳ CODE XREF: mult
                move    $a2, $zero

loc_44:                                         # ↙
    ↳ CODE XREF: mult:loc_30
                move    $v0, $a2
                jr      $ra
                move    $v1, $a3
```

Responda: G.1.13 on page 1927.

Capítulo 20

```
#include <stdint.h>

//
#define RNG_a 1664525
#define RNG_c 1013904223

static uint32_t rand_state;

void my_srand (uint32_t init)
{
        rand_state=init;
}

int my_rand ()
{
        rand_state=rand_state*RNG_a;
        rand_state=rand_state+RNG_c;
```

```
        return rand_state & 0x7fff;
}
```

. [Pre+07].

20.1 x86

Listing 20.1: Con optimización MSVC 2013

```
_BSS    SEGMENT
_rand_state DD  01H DUP (?)
_BSS    ENDS

_init$ = 8
_srand  PROC
        mov     eax, DWORD PTR _init$[esp-4]
        mov     DWORD PTR _rand_state, eax
        ret     0
_srand  ENDP

_TEXT   SEGMENT
_rand   PROC
        imul    eax, DWORD PTR _rand_state, ↙
    ↘ 1664525
        add     eax, 1013904223         ; 3↙
    ↘ c6ef35fH
        mov     DWORD PTR _rand_state, eax
        and     eax, 32767              ; ↙
    ↘ 00007fffH
        ret     0
_rand   ENDP
```

```
_TEXT   ENDS
```

:

Listing 20.2: Sin optimización MSVC 2013

```
_BSS    SEGMENT
_rand_state DD  01H DUP (?)
_BSS    ENDS

_init$ = 8
_srand  PROC
        push    ebp
        mov     ebp, esp
        mov     eax, DWORD PTR _init$[ebp]
        mov     DWORD PTR _rand_state, eax
        pop     ebp
        ret     0
_srand  ENDP

_TEXT   SEGMENT
_rand   PROC
        push    ebp
        mov     ebp, esp
        imul    eax, DWORD PTR _rand_state, ↙
    ↘ 1664525
        mov     DWORD PTR _rand_state, eax
        mov     ecx, DWORD PTR _rand_state
        add     ecx, 1013904223         ; 3↙
    ↘ c6ef35fH
        mov     DWORD PTR _rand_state, ecx
        mov     eax, DWORD PTR _rand_state
```

```
        and     eax, 32767                      ; ⤦
    ↪ 00007fffH
        pop     ebp
        ret     0
_rand   ENDP

_TEXT   ENDS
```

20.2 x64

Listing 20.3: Con optimización MSVC 2013 x64

```

_BSS    SEGMENT
rand_state DD   01H DUP (?)
_BSS    ENDS

init$ = 8
my_srand PROC
; ECX =
        mov     DWORD PTR rand_state, ecx
        ret     0
my_srand ENDP

_TEXT   SEGMENT
my_rand PROC
        imul    eax, DWORD PTR rand_state, ⤦
    ↪ 1664525      ; 0019660dH
        add     eax, 1013904223           ⤦
    ↪                    ; 3c6ef35fH
        mov     DWORD PTR rand_state, eax
```

```
        and     eax, 32767                    ↙
    ↘                   ; 00007fffH
        ret     0
my_rand ENDP

_TEXT   ENDS
```

.

20.3 32-bit ARM

Listing 20.4: Con optimización Keil 6/2013 (Modo ARM)

```

my_srand PROC
        LDR      r1,|L0.52|  ;  rand_state
        STR      r0,[r1,#0]  ;  rand_state
        BX       lr
        ENDP

my_rand PROC
        LDR      r0,|L0.52|  ;  rand_state
        LDR      r2,|L0.56|  ;  RNG_a
        LDR      r1,[r0,#0]  ;  rand_state
        MUL      r1,r2,r1
        LDR      r2,|L0.60|  ;  RNG_c
        ADD      r1,r1,r2
        STR      r1,[r0,#0]  ;  rand_state
; AND  0x7FFF:
        LSL      r0,r1,#17
        LSR      r0,r0,#17
```

```
        BX       lr
        ENDP

|L0.52|
        DCD      ||.data||
|L0.56|
        DCD      0x0019660d
|L0.60|
        DCD      0x3c6ef35f

        AREA ||.data||, DATA, ALIGN=2

rand_state
        DCD      0x00000000
```

Parece una operación inútil, pero lo que hace es limpiar los 17 bits altos, dejando intactos los 15 bits bajos, y ese es nuestro objetivo.

Con optimización Keil .

20.4 MIPS

Listing 20.5: Con optimización GCC 4.4.5 (IDA)

```

my_srand:
; rand_state:
                lui     $v0, (rand_state >> ↙
    ↘ 16)
                jr      $ra
```

```
                sw      $a0, rand_state
my_rand:
;
                lui     $v1, (rand_state >> ↙
    ↘ 16)
                lw      $v0, rand_state
                or      $at, $zero  ; load ↙
    ↘ delay slot
;  1664525 (RNG_a):
                sll     $a1, $v0, 2
                sll     $a0, $v0, 4
                addu    $a0, $a1, $a0
                sll     $a1, $a0, 6
                subu    $a0, $a1, $a0
                addu    $a0, $v0
                sll     $a1, $a0, 5
                addu    $a0, $a1
                sll     $a0, 3
                addu    $v0, $a0, $v0
                sll     $a0, $v0, 2
                addu    $v0, $a0
;  1013904223 (RNG_c)
;
                li      $a0, 0x3C6EF35F
                addu    $v0, $a0
;  rand_state:
                sw      $v0, (rand_state & 0↙
    ↘ xFFFF)($v1)
                jr      $ra
                andi    $v0, 0x7FFF ; branch↙
    ↘  delay slot
```

(0x3C6EF35F o 1013904223). (1664525)?

:

```
#define RNG_a 1664525

int f (int a)
{
        return a*RNG_a;
}
```

Listing 20.6: Con optimización GCC 4.4.5 (IDA)

```
f:
                sll     $v1, $a0, 2
                sll     $v0, $a0, 4
                addu    $v0, $v1, $v0
                sll     $v1, $v0, 6
                subu    $v0, $v1, $v0
                addu    $v0, $a0
                sll     $v1, $v0, 5
                addu    $v0, $v1
                sll     $v0, 3
                addu    $a0, $v0, $a0
                sll     $v0, $a0, 2
                jr      $ra
                addu    $v0, $a0, $v0 ; ↙
    ↘ branch delay slot
```

!

20.4.1

Listing 20.7: Con optimización GCC 4.4.5 (objdump)

```
# objdump -D rand_O3.o

...

00000000 <my_srand>:
   0:   3c020000        lui     v0,0x0
   4:   03e00008        jr      ra
   8:   ac440000        sw      a0,0(v0)

0000000c <my_rand>:
   c:   3c030000        lui     v1,0x0
  10:   8c620000        lw      v0,0(v1)
  14:   00200825        move    at,at
  18:   00022880        sll     a1,v0,0x2
  1c:   00022100        sll     a0,v0,0x4
  20:   00a42021        addu    a0,a1,a0
  24:   00042980        sll     a1,a0,0x6
  28:   00a42023        subu    a0,a1,a0
  2c:   00822021        addu    a0,a0,v0
  30:   00042940        sll     a1,a0,0x5
  34:   00852021        addu    a0,a0,a1
  38:   000420c0        sll     a0,a0,0x3
  3c:   00821021        addu    v0,a0,v0
  40:   00022080        sll     a0,v0,0x2
  44:   00441021        addu    v0,v0,a0
  48:   3c043c6e        lui     a0,0x3c6e
  4c:   3484f35f        ori     a0,a0,0xf35f
  50:   00441021        addu    v0,v0,a0
  54:   ac620000        sw      v0,0(v1)
  58:   03e00008        jr      ra
  5c:   30427fff        andi    v0,v0,0x7fff
```

```
...

# objdump -r rand_O3.o

...

RELOCATION RECORDS FOR [.text]:
OFFSET   TYPE              VALUE
00000000 R_MIPS_HI16       .bss
00000008 R_MIPS_LO16       .bss
0000000c R_MIPS_HI16       .bss
00000010 R_MIPS_LO16       .bss
00000054 R_MIPS_LO16       .bss

...
```

20.5

: 65.1 on page 1315.

CAPÍTULO 21. ESTRUCTURAS

Capítulo 21

Estructuras

21.1 MSVC:

Listing 21.1: WinBase.h

```
typedef struct _SYSTEMTIME {
  WORD wYear;
  WORD wMonth;
  WORD wDayOfWeek;
  WORD wDay;
  WORD wHour;
  WORD wMinute;
  WORD wSecond;
  WORD wMilliseconds;
} SYSTEMTIME, *PSYSTEMTIME;
```

 CAPÍTULO 21. ESTRUCTURAS

```
#include <windows.h>
#include <stdio.h>

void main()
{
    SYSTEMTIME t;
    GetSystemTime (&t);

    printf ("%04d-%02d-%02d %02d:%02d:%02d\n↙
    ↳ ",
        t.wYear, t.wMonth, t.wDay,
        t.wHour, t.wMinute, t.wSecond);

    return;
};
```

(MSVC 2010):

Listing 21.2: MSVC 2010 /GS-

```
_t$ = -16 ; size = 16
_main      PROC
    push   ebp
    mov    ebp, esp
    sub    esp, 16
    lea    eax, DWORD PTR _t$[ebp]
    push   eax
    call   DWORD PTR __imp__GetSystemTime@4
    movzx  ecx, WORD PTR _t$[ebp+12] ; ↙
    ↳ wSecond
    push   ecx
    movzx  edx, WORD PTR _t$[ebp+10] ; ↙
    ↳ wMinute
```

```
    push   edx
    movzx  eax, WORD PTR _t$[ebp+8] ; wHour
    push   eax
    movzx  ecx, WORD PTR _t$[ebp+6] ; wDay
    push   ecx
    movzx  edx, WORD PTR _t$[ebp+2] ; wMonth
    push   edx
    movzx  eax, WORD PTR _t$[ebp] ; wYear
    push   eax
    push   OFFSET $SG78811 ; '%04d-%02d-%02d↙
    ↘ %02d:%02d:%02d', 0aH, 00H
    call   _printf
    add    esp, 28
    xor    eax, eax
    mov    esp, ebp
    pop    ebp
    ret    0
_main      ENDP
```

21.1.1 OllyDbg

MSVC 2010 /GS- /MD OllyDbg. GetSystemTime(),:

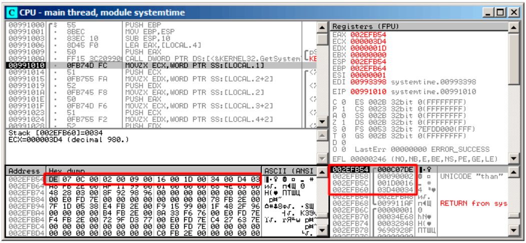

Figura 21.1: OllyDbg: `GetSystemTime()`

9 2014, 22:29:52:

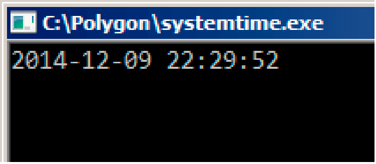

Figura 21.2: OllyDbg:

:

```
DE 07 0C 00 02 00 09 00 16 00 1D 00 34 00 D4 03
```

. . :

0x07DE	2014	wYear
0x000C	12	wMonth
0x0002	2	wDayOfWeek
0x0009	9	wDay
0x0016	22	wHour
0x001D	29	wMinute
0x0034	52	wSecond
0x03D4	980	wMilliseconds

.

`printf()` .

`printf()` (`wDayOfWeek` y `wMilliseconds`),

21.1.2

```
#include <windows.h>
#include <stdio.h>

void main()
{
    WORD array[8];
    GetSystemTime (array);

    printf ("%04d-%02d-%02d %02d:%02d:%02d\n
    ↪ ",
        array[0] /* wYear */, array[1] /*
    ↪ wMonth */, array[3] /* wDay */,
```

```
        array[4] /* wHour */, array[5] /* ⤦
    ↳ wMinute */, array[6] /* wSecond */);

    return;
};
```

```
systemtime2.c(7) : warning C4133: 'function'⤦
    ↳  : incompatible types - from 'WORD ⤦
    ↳ [8]' to 'LPSYSTEMTIME'
```

:

Listing 21.3: Sin optimización MSVC 2010

```
$SG78573 DB     '%04d-%02d-%02d %02d:%02d⤦
    ↳ :%02d', 0aH, 00H

_array$ = -16   ; size = 16
_main   PROC
        push    ebp
        mov     ebp, esp
        sub     esp, 16
        lea     eax, DWORD PTR _array$[ebp]
        push    eax
        call    DWORD PTR ⤦
    ↳ __imp__GetSystemTime@4
        movzx   ecx, WORD PTR _array$[ebp⤦
    ↳ +12] ; wSecond
        push    ecx
        movzx   edx, WORD PTR _array$[ebp⤦
    ↳ +10] ; wMinute
        push    edx
```

```
        movzx   eax, WORD PTR _array$[ebp+8]
    ↳  ; wHoure
        push    eax
        movzx   ecx, WORD PTR _array$[ebp+6]
    ↳  ; wDay
        push    ecx
        movzx   edx, WORD PTR _array$[ebp+2]
    ↳  ; wMonth
        push    edx
        movzx   eax, WORD PTR _array$[ebp] ;
    ↳  wYear
        push    eax
        push    OFFSET $SG78573
        call    _printf
        add     esp, 28
        xor     eax, eax
        mov     esp, ebp
        pop     ebp
        ret     0
_main   ENDP
```

!

.

.

21.2

```
#include <windows.h>
#include <stdio.h>
```

```
void main()
{
    SYSTEMTIME *t;

    t=(SYSTEMTIME *)malloc (sizeof (
    ↳ SYSTEMTIME));

    GetSystemTime (t);

    printf ("%04d-%02d-%02d %02d:%02d:%02d\n
    ↳ ",
        t->wYear, t->wMonth, t->wDay,
        t->wHour, t->wMinute, t->wSecond);

    free (t);

    return;
};
```

Listing 21.4: Con optimización MSVC

```
_main       PROC
    push   esi
    push   16
    call   _malloc
    add    esp, 4
    mov    esi, eax
    push   esi
    call   DWORD PTR __imp__GetSystemTime@4
    movzx  eax, WORD PTR [esi+12] ; wSecond
    movzx  ecx, WORD PTR [esi+10] ; wMinute
    movzx  edx, WORD PTR [esi+8] ; wHour
    push   eax
```

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```
    movzx  eax, WORD PTR [esi+6] ; wDay
    push   ecx
    movzx  ecx, WORD PTR [esi+2] ; wMonth
    push   edx
    movzx  edx, WORD PTR [esi] ; wYear
    push   eax
    push   ecx
    push   edx
    push   OFFSET $SG78833
    call   _printf
    push   esi
    call   _free
    add    esp, 32
    xor    eax, eax
    pop    esi
    ret    0
_main      ENDP
```

```
#include <windows.h>
#include <stdio.h>

void main()
{
    WORD *t;

    t=(WORD *)malloc (16);

    GetSystemTime (t);

    printf ("%04d-%02d-%02d %02d:%02d:%02d\n⤦
    ↪ ",
        t[0] /* wYear */, t[1] /* wMonth */,⤦
```

```
    ↳  t[3] /* wDay */,
         t[4] /* wHour */, t[5] /* wMinute ↙
    ↳ */, t[6] /* wSecond */);

    free (t);

    return;
};
```

:

Listing 21.5: Con optimización MSVC

```
$SG78594 DB     '%04d-%02d-%02d %02d:%02d↙
    ↳ :%02d', 0aH, 00H

_main   PROC
        push    esi
        push    16
        call    _malloc
        add     esp, 4
        mov     esi, eax
        push    esi
        call    DWORD PTR ↙
    ↳ __imp__GetSystemTime@4
        movzx   eax, WORD PTR [esi+12]
        movzx   ecx, WORD PTR [esi+10]
        movzx   edx, WORD PTR [esi+8]
        push    eax
        movzx   eax, WORD PTR [esi+6]
        push    ecx
        movzx   ecx, WORD PTR [esi+2]
        push    edx
```

```
        movzx   edx, WORD PTR [esi]
        push    eax
        push    ecx
        push    edx
        push    OFFSET $SG78594
        call    _printf
        push    esi
        call    _free
        add     esp, 32
        xor     eax, eax
        pop     esi
        ret     0
_main   ENDP
```

21.3 UNIX: struct tm

21.3.1 Linux

```
#include <stdio.h>
#include <time.h>

void main()
{
    struct tm t;
    time_t unix_time;

    unix_time=time(NULL);

    localtime_r (&unix_time, &t);

    printf ("Year: %d\n", t.tm_year+1900);
```

```
    printf ("Month: %d\n", t.tm_mon);
    printf ("Day: %d\n", t.tm_mday);
    printf ("Hour: %d\n", t.tm_hour);
    printf ("Minutes: %d\n", t.tm_min);
    printf ("Seconds: %d\n", t.tm_sec);
};
```

GCC 4.4.1:

Listing 21.6: GCC 4.4.1

```
main proc near
     push    ebp
     mov     ebp, esp
     and     esp, 0FFFFFFF0h
     sub     esp, 40h
     mov     dword ptr [esp], 0 ;  time()
     call    time
     mov     [esp+3Ch], eax
     lea     eax, [esp+3Ch]  ;
     lea     edx, [esp+10h]  ;
     mov     [esp+4], edx    ;
     mov     [esp], eax      ;  time()
     call    localtime_r
     mov     eax, [esp+24h]  ; tm_year
     lea     edx, [eax+76Ch] ; edx=eax+1900
     mov     eax, offset format ; "Year: %d\
    n"
     mov     [esp+4], edx
     mov     [esp], eax
     call    printf
     mov     edx, [esp+20h]       ; tm_mon
```

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```
    mov     eax, offset aMonthD ; "Month: %d\n"
    mov     [esp+4], edx
    mov     [esp], eax
    call    printf
    mov     edx, [esp+1Ch]      ; tm_mday
    mov     eax, offset aDayD  ; "Day: %d\n"
    mov     [esp+4], edx
    mov     [esp], eax
    call    printf
    mov     edx, [esp+18h]      ; tm_hour
    mov     eax, offset aHourD ; "Hour: %d\n"
    mov     [esp+4], edx
    mov     [esp], eax
    call    printf
    mov     edx, [esp+14h]          ; tm_min
    mov     eax, offset aMinutesD ; "Minutes: %d\n"
    mov     [esp+4], edx
    mov     [esp], eax
    call    printf
    mov     edx, [esp+10h]
    mov     eax, offset aSecondsD ; "Seconds: %d\n"
    mov     [esp+4], edx            ; tm_sec
    mov     [esp], eax
    call    printf
    leave
    retn
main endp
```

GDB

1.
:

Listing 21.7: GDB

```
dennis@ubuntuvm:~/polygon$ date
Mon Jun  2 18:10:37 EEST 2014
dennis@ubuntuvm:~/polygon$ gcc GCC_tm.c -o ↙
    ↳ GCC_tm
dennis@ubuntuvm:~/polygon$ gdb GCC_tm
GNU gdb (GDB) 7.6.1-ubuntu
Copyright (C) 2013 Free Software Foundation,↙
    ↳  Inc.
License GPLv3+: GNU GPL version 3 or later <↙
    ↳ http://gnu.org/licenses/gpl.html>
This is free software: you are free to ↙
    ↳ change and redistribute it.
There is NO WARRANTY, to the extent ↙
    ↳ permitted by law.  Type "show copying"
and "show warranty" for details.
This GDB was configured as "i686-linux-gnu".
For bug reporting instructions, please see:
<http://www.gnu.org/software/gdb/bugs/>...
Reading symbols from /home/dennis/polygon/↙
    ↳ GCC_tm...(no debugging symbols found)↙
    ↳ ...done.
(gdb) b printf
Breakpoint 1 at 0x8048330
(gdb) run
Starting program: /home/dennis/polygon/↙
    ↳ GCC_tm
```

1

 CAPÍTULO 21. ESTRUCTURAS

```
Breakpoint 1, __printf (format=0x80485c0 "
    ↳ Year: %d\n") at printf.c:29
29      printf.c: No such file or directory.
(gdb) x/20x $esp
0xbffff0dc:     0x080484c3      0x080485c0
    ↳       0x000007de      0x00000000
0xbffff0ec:     0x08048301      0x538c93ed
    ↳       0x00000025      0x0000000a
0xbffff0fc:     0x00000012      0x00000002
    ↳       0x00000005      0x00000072
0xbffff10c:     0x00000001      0x00000098
    ↳       0x00000001      0x00002a30
0xbffff11c:     0x0804b090      0x08048530
    ↳       0x00000000      0x00000000
(gdb)
```

. *time.h*:

Listing 21.8: time.h

```
struct tm
{
  int   tm_sec;
  int   tm_min;
  int   tm_hour;
  int   tm_mday;
  int   tm_mon;
  int   tm_year;
  int   tm_wday;
  int   tm_yday;
  int   tm_isdst;
};
```

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. .

:

```
0xbffff0dc:     0x080484c3      0x080485c0
        0x000007de      0x00000000
0xbffff0ec:     0x08048301      0x538c93ed
        0x00000025 sec  0x0000000a min
0xbffff0fc:     0x00000012 hour 0x00000002
    mday 0x00000005 mon  0x00000072 year
0xbffff10c:     0x00000001 wday 0x00000098
    yday 0x00000001 isdst0x00002a30
0xbffff11c:     0x0804b090      0x08048530
        0x00000000      0x00000000
```

:

0x00000025	37	tm_sec
0x0000000a	10	tm_min
0x00000012	18	tm_hour
0x00000002	2	tm_mday
0x00000005	5	tm_mon
0x00000072	114	tm_year
0x00000001	1	tm_wday
0x00000098	152	tm_yday
0x00000001	1	tm_isdst

SYSTEMTIME (21.1 on page 658), tm_wday, tm_yday, tm_isdst.

21.3.2 ARM

Con optimización Keil 6/2013 (Modo Thumb)

:

Listing 21.9: Con optimización Keil 6/2013 (Modo Thumb)

```
var_38 = -0x38
var_34 = -0x34
var_30 = -0x30
var_2C = -0x2C
var_28 = -0x28
var_24 = -0x24
timer  = -0xC

       PUSH    {LR}
       MOVS    R0, #0          ; timer
       SUB     SP, SP, #0x34
       BL      time
       STR     R0, [SP,#0x38+timer]
       MOV     R1, SP          ; tp
       ADD     R0, SP, #0x38+timer ; timer
       BL      localtime_r
       LDR     R1, =0x76C
       LDR     R0, [SP,#0x38+var_24]
       ADDS    R1, R0, R1
       ADR     R0, aYearD      ; "Year: %d\n↙
    ↘ "
       BL      __2printf
       LDR     R1, [SP,#0x38+var_28]
       ADR     R0, aMonthD     ; "Month: %d\↙
    ↘ n"
       BL      __2printf
```

```
        LDR     R1, [SP,#0x38+var_2C]
        ADR     R0, aDayD       ; "Day: %d\n"
        BL      __2printf
        LDR     R1, [SP,#0x38+var_30]
        ADR     R0, aHourD      ; "Hour: %d\n
    ↳ "
        BL      __2printf
        LDR     R1, [SP,#0x38+var_34]
        ADR     R0, aMinutesD   ; "Minutes: %
    ↳ d\n"
        BL      __2printf
        LDR     R1, [SP,#0x38+var_38]
        ADR     R0, aSecondsD   ; "Seconds: %
    ↳ d\n"
        BL      __2printf
        ADD     SP, SP, #0x34
        POP     {PC}
```

Con optimización Xcode 4.6.3 (LLVM) (Modo Thumb-2)

IDA «» `tm` (IDA «» `localtime_r()`), .

Listing 21.10: Con optimización Xcode 4.6.3 (LLVM) (Modo Thumb-2)

```
var_38 = -0x38
var_34 = -0x34

        PUSH {R7,LR}
        MOV  R7, SP
        SUB  SP, SP, #0x30
        MOVS R0, #0   ; time_t *
```

```
        BLX  _time
        ADD  R1, SP, #0x38+var_34 ; struct tm
    ↪   *
        STR  R0, [SP,#0x38+var_38]
        MOV  R0, SP  ; time_t *
        BLX  _localtime_r
        LDR  R1, [SP,#0x38+var_34.tm_year]
        MOV  R0, 0xF44 ; "Year: %d\n"
        ADD  R0, PC  ; char *
        ADDW R1, R1, #0x76C
        BLX  _printf
        LDR  R1, [SP,#0x38+var_34.tm_mon]
        MOV  R0, 0xF3A ; "Month: %d\n"
        ADD  R0, PC  ; char *
        BLX  _printf
        LDR  R1, [SP,#0x38+var_34.tm_mday]
        MOV  R0, 0xF35 ; "Day: %d\n"
        ADD  R0, PC  ; char *
        BLX  _printf
        LDR  R1, [SP,#0x38+var_34.tm_hour]
        MOV  R0, 0xF2E ; "Hour: %d\n"
        ADD  R0, PC  ; char *
        BLX  _printf
        LDR  R1, [SP,#0x38+var_34.tm_min]
        MOV  R0, 0xF28 ; "Minutes: %d\n"
        ADD  R0, PC  ; char *
        BLX  _printf
        LDR  R1, [SP,#0x38+var_34]
        MOV  R0, 0xF25 ; "Seconds: %d\n"
        ADD  R0, PC  ; char *
        BLX  _printf
        ADD  SP, SP, #0x30
        POP  {R7,PC}
```

```
...

00000000 tm         struc ; (sizeof=0x2C, ↙
    ↘ standard type)
00000000 tm_sec     DCD ?
00000004 tm_min     DCD ?
00000008 tm_hour    DCD ?
0000000C tm_mday    DCD ?
00000010 tm_mon     DCD ?
00000014 tm_year    DCD ?
00000018 tm_wday    DCD ?
0000001C tm_yday    DCD ?
00000020 tm_isdst   DCD ?
00000024 tm_gmtoff DCD ?
00000028 tm_zone    DCD ? ; offset
0000002C tm         ends
```

21.3.3 MIPS

Listing 21.11: Con optimización GCC 4.4.5 (IDA)

```

main:

; :

var_40          = -0x40
var_38          = -0x38
seconds         = -0x34
minutes         = -0x30
hour            = -0x2C
```

 CAPÍTULO 21. ESTRUCTURAS

```
day             = -0x28
month           = -0x24
year            = -0x20
var_4           = -4

                lui     $gp, (__gnu_local_gp >> 16)
                addiu   $sp, -0x50
                la      $gp, (__gnu_local_gp & 0xFFFF)
                sw      $ra, 0x50+var_4($sp)
                sw      $gp, 0x50+var_40($sp)
                lw      $t9, (time & 0xFFFF)($gp)
                or      $at, $zero ; load delay slot, NOP
                jalr    $t9
                move    $a0, $zero ; branch delay slot, NOP
                lw      $gp, 0x50+var_40($sp)
                addiu   $a0, $sp, 0x50+var_38
                lw      $t9, (localtime_r & 0xFFFF)($gp)
                addiu   $a1, $sp, 0x50+seconds
                jalr    $t9
                sw      $v0, 0x50+var_38($sp) ; branch delay slot
                lw      $gp, 0x50+var_40($sp)
```

 CAPÍTULO 21. ESTRUCTURAS

```
                lw      $a1, 0x50+year($sp)
                lw      $t9, (printf & 0xFFFF)($gp)
                la      $a0, $LC0          # "Year: %d\n"
                jalr    $t9
                addiu   $a1, 1900 ; branch delay slot
                lw      $gp, 0x50+var_40($sp)
                lw      $a1, 0x50+month($sp)
                lw      $t9, (printf & 0xFFFF)($gp)
                lui     $a0, ($LC1 >> 16)  # "Month: %d\n"
                jalr    $t9
                la      $a0, ($LC1 & 0xFFFF)  # "Month: %d\n" ; branch delay slot
                lw      $gp, 0x50+var_40($sp)
                lw      $a1, 0x50+day($sp)
                lw      $t9, (printf & 0xFFFF)($gp)
                lui     $a0, ($LC2 >> 16)  # "Day: %d\n"
                jalr    $t9
                la      $a0, ($LC2 & 0xFFFF)  # "Day: %d\n" ; branch delay slot
                lw      $gp, 0x50+var_40($sp)
                lw      $a1, 0x50+hour($sp)
                lw      $t9, (printf & 0xFFFF)($gp)
```

 CAPÍTULO 21. ESTRUCTURAS

```
                lui     $a0, ($LC3 >> 16)  # "Hour: %d\n"
                jalr    $t9
                la      $a0, ($LC3 & 0xFFFF)  # "Hour: %d\n" ; branch delay slot
                lw      $gp, 0x50+var_40($sp)
                lw      $a1, 0x50+minutes($sp)
                lw      $t9, (printf & 0xFFFF)($gp)
                lui     $a0, ($LC4 >> 16)  # "Minutes: %d\n"
                jalr    $t9
                la      $a0, ($LC4 & 0xFFFF)  # "Minutes: %d\n" ; branch delay slot
                lw      $gp, 0x50+var_40($sp)
                lw      $a1, 0x50+seconds($sp)
                lw      $t9, (printf & 0xFFFF)($gp)
                lui     $a0, ($LC5 >> 16)  # "Seconds: %d\n"
                jalr    $t9
                la      $a0, ($LC5 & 0xFFFF)  # "Seconds: %d\n" ; branch delay slot
                lw      $ra, 0x50+var_4($sp)
                or      $at, $zero ; load delay slot, NOP
                jr      $ra
```

 CAPÍTULO 21. ESTRUCTURAS

```
                addiu   $sp, 0x50

$LC0:           .ascii "Year: %d\n"<0>
$LC1:           .ascii "Month: %d\n"<0>
$LC2:           .ascii "Day: %d\n"<0>
$LC3:           .ascii "Hour: %d\n"<0>
$LC4:           .ascii "Minutes: %d\n"<0>
$LC5:           .ascii "Seconds: %d\n"<0>
```

21.3.4

: Spanish text placeholder.21.8.

```
#include <stdio.h>
#include <time.h>

void main()
{
    int tm_sec, tm_min, tm_hour, tm_mday, tm_mon, tm_year, tm_wday, tm_yday, tm_isdst;
    time_t unix_time;

    unix_time=time(NULL);

    localtime_r (&unix_time, &tm_sec);

    printf ("Year: %d\n", tm_year+1900);
    printf ("Month: %d\n", tm_mon);
    printf ("Day: %d\n", tm_mday);
    printf ("Hour: %d\n", tm_hour);
    printf ("Minutes: %d\n", tm_min);
```

 CAPÍTULO 21. ESTRUCTURAS

```
    printf ("Seconds: %d\n", tm_sec);
};
```

N.B.

:

Listing 21.12: GCC 4.7.3

```
GCC_tm2.c: In function 'main':
GCC_tm2.c:11:5: warning: passing argument 2 ↙
    ↘ of 'localtime_r' from incompatible ↙
    ↘ pointer type [enabled by default]
In file included from GCC_tm2.c:2:0:
/usr/include/time.h:59:12: note: expected '↙
    ↘ struct tm *' but argument is of type '↙
    ↘ int *'
```

:

Listing 21.13: GCC 4.7.3

```
main       proc near

var_30     = dword ptr -30h
var_2C     = dword ptr -2Ch
unix_time  = dword ptr -1Ch
tm_sec     = dword ptr -18h
tm_min     = dword ptr -14h
tm_hour    = dword ptr -10h
tm_mday    = dword ptr -0Ch
tm_mon     = dword ptr -8
tm_year    = dword ptr -4

```

```
        push    ebp
        mov     ebp, esp
        and     esp, 0FFFFFFF0h
        sub     esp, 30h
        call    __main
        mov     [esp+30h+var_30], 0 ; arg ⤦
    ↳ 0
        call    time
        mov     [esp+30h+unix_time], eax
        lea     eax, [esp+30h+tm_sec]
        mov     [esp+30h+var_2C], eax
        lea     eax, [esp+30h+unix_time]
        mov     [esp+30h+var_30], eax
        call    localtime_r
        mov     eax, [esp+30h+tm_year]
        add     eax, 1900
        mov     [esp+30h+var_2C], eax
        mov     [esp+30h+var_30], offset ⤦
    ↳ aYearD ; "Year: %d\n"
        call    printf
        mov     eax, [esp+30h+tm_mon]
        mov     [esp+30h+var_2C], eax
        mov     [esp+30h+var_30], offset ⤦
    ↳ aMonthD ; "Month: %d\n"
        call    printf
        mov     eax, [esp+30h+tm_mday]
        mov     [esp+30h+var_2C], eax
        mov     [esp+30h+var_30], offset ⤦
    ↳ aDayD ; "Day: %d\n"
        call    printf
        mov     eax, [esp+30h+tm_hour]
        mov     [esp+30h+var_2C], eax
        mov     [esp+30h+var_30], offset ⤦
```

```
    ↳ aHourD ; "Hour: %d\n"
            call    printf
            mov     eax, [esp+30h+tm_min]
            mov     [esp+30h+var_2C], eax
            mov     [esp+30h+var_30], offset ↙
    ↳ aMinutesD ; "Minutes: %d\n"
            call    printf
            mov     eax, [esp+30h+tm_sec]
            mov     [esp+30h+var_2C], eax
            mov     [esp+30h+var_30], offset ↙
    ↳ aSecondsD ; "Seconds: %d\n"
            call    printf
            leave
            retn
main        endp
```

. .

tm_year, tm_mon, tm_mday, tm_hour, tm_min tm_sec . localtime_r().

GetSystemTime() ( ). : «Organización de campos en la estructura» (21.4 on page 690).

[Rit93].

21.3.5

```
#include <stdio.h>
#include <time.h>

void main()
{
```

 CAPÍTULO 21. ESTRUCTURAS

```
    struct tm t;
    time_t unix_time;
    int i;

    unix_time=time(NULL);

    localtime_r (&unix_time, &t);

    for (i=0; i<9; i++)
    {
        int tmp=((int*)&t)[i];
        printf ("0x%08X (%d)\n", tmp, tmp);
    };
};
```

. ! 23:51:45 26-July-2014.

```
0x0000002D (45)
0x00000033 (51)
0x00000017 (23)
0x0000001A (26)
0x00000006 (6)
0x00000072 (114)
0x00000006 (6)
0x000000CE (206)
0x00000001 (1)
```

: 21.8 on page 672.

:

Listing 21.14: Con optimización GCC 4.8.1

```
main                proc near
                    push    ebp
                    mov     ebp, esp
                    push    esi
                    push    ebx
                    and     esp, 0FFFFFFF0h
                    sub     esp, 40h
                    mov     dword ptr [esp], 0 ;↙
    ↘  timer
                    lea     ebx, [esp+14h]
                    call    _time
                    lea     esi, [esp+38h]
                    mov     [esp+4], ebx     ; tp
                    mov     [esp+10h], eax
                    lea     eax, [esp+10h]
                    mov     [esp], eax       ; ↙
    ↘ timer
                    call    _localtime_r
                    nop
                    lea     esi, [esi+0]     ; ↙
    ↘ NOP
loc_80483D8:
; .
                    mov     eax, [ebx]       ;
                    add     ebx, 4           ;
                    mov     dword ptr [esp+4], ↙
    ↘ offset a0x08xD ; "0x%08X (%d)\n"
                    mov     dword ptr [esp], 1
                    mov     [esp+0Ch], eax  ;  ↙
    ↘ printf()
                    mov     [esp+8], eax     ;  ↙
    ↘ printf()
```

 CAPÍTULO 21. ESTRUCTURAS

```
                call    ___printf_chk
                cmp     ebx, esi        ; ?
                jnz     short loc_80483D8↙
    ↪           ;
                lea     esp, [ebp-8]
                pop     ebx
                pop     esi
                pop     ebp
                retn
main            endp
```

Ejercicio

21.3.6

```
#include <stdio.h>
#include <time.h>

void main()
{
    struct tm t;
    time_t unix_time;
    int i, j;

    unix_time=time(NULL);

    localtime_r (&unix_time, &t);

    for (i=0; i<9; i++)
    {
```

```
        for (j=0; j<4; j++)
            printf ("0x%02X ", ((unsigned ↙
    ↘ char*)&t)[i*4+j]);
        printf ("\n");
    };
};
```

```
0x2D 0x00 0x00 0x00
0x33 0x00 0x00 0x00
0x17 0x00 0x00 0x00
0x1A 0x00 0x00 0x00
0x06 0x00 0x00 0x00
0x72 0x00 0x00 0x00
0x06 0x00 0x00 0x00
0xCE 0x00 0x00 0x00
0x01 0x00 0x00 0x00
```

23:51:45 26-July-2014 [2]. (21.3.5 on page 685), (31 on page 869).

Listing 21.15: Con optimización GCC 4.8.1

```

main                proc near
                    push    ebp
                    mov     ebp, esp
                    push    edi
                    push    esi
                    push    ebx
                    and     esp, 0FFFFFFF0h
                    sub     esp, 40h
```

2

```
                mov     dword ptr [esp], 0 ;↙
    ↘  timer
                lea     esi, [esp+14h]
                call    _time
                lea     edi, [esp+38h]  ; ↙
    ↘ struct end
                mov     [esp+4], esi    ; tp
                mov     [esp+10h], eax
                lea     eax, [esp+10h]
                mov     [esp], eax      ; ↙
    ↘ timer
                call    _localtime_r
                lea     esi, [esi+0]    ; ↙
    ↘ NOP
;
loc_8048408:
                xor     ebx, ebx        ; j↙
    ↘ =0

loc_804840A:
                movzx   eax, byte ptr [esi+↙
    ↘ ebx] ;
                add     ebx, 1          ; j=↙
    ↘ j+1
                mov     dword ptr [esp+4], ↙
    ↘ offset a0x02x ; "0x%02X "
                mov     dword ptr [esp], 1
                mov     [esp+8], eax    ;  ↙
    ↘ printf()
                call    ___printf_chk
                cmp     ebx, 4
                jnz     short loc_804840A
;  (CR)
```

```
                mov     dword ptr [esp], 0Ah ↙
    ↘ ; c
                add     esi, 4
                call    _putchar
                cmp     esi, edi        ; ?
                jnz     short loc_8048408 ; ↙
    ↘ j=0
                lea     esp, [ebp-0Ch]
                pop     ebx
                pop     esi
                pop     edi
                pop     ebp
                retn
main            endp
```

21.4 Organización de campos en la estructura

```
#include <stdio.h>

struct s
{
    char a;
    int b;
    char c;
    int d;
};

void f(struct s s)
{
```

```
    printf ("a=%d; b=%d; c=%d; d=%d\n", s.a,
    ↳ s.b, s.c, s.d);
};

int main()
{
    struct s tmp;
    tmp.a=1;
    tmp.b=2;
    tmp.c=3;
    tmp.d=4;
    f(tmp);
};
```

21.4.1 x86

Listing 21.16: MSVC 2012 /GS- /Ob0

```

_tmp$ = -16
_main    PROC
    push   ebp
    mov    ebp, esp
    sub    esp, 16
    mov    BYTE PTR _tmp$[ebp], 1      ;  a
    mov    DWORD PTR _tmp$[ebp+4], 2  ;  b
    mov    BYTE PTR _tmp$[ebp+8], 3    ;  c
    mov    DWORD PTR _tmp$[ebp+12], 4 ;  d
    sub    esp, 16                     ;
    mov    eax, esp
    mov    ecx, DWORD PTR _tmp$[ebp]  ;
    mov    DWORD PTR [eax], ecx
```

```
    mov     edx, DWORD PTR _tmp$[ebp+4]
    mov     DWORD PTR [eax+4], edx
    mov     ecx, DWORD PTR _tmp$[ebp+8]
    mov     DWORD PTR [eax+8], ecx
    mov     edx, DWORD PTR _tmp$[ebp+12]
    mov     DWORD PTR [eax+12], edx
    call     _f
    add     esp, 16
    xor     eax, eax
    mov     esp, ebp
    pop     ebp
    ret     0
_main    ENDP

_s$ = 8 ; size = 16
?f@@YAXUs@@@Z PROC ; f
    push    ebp
    mov     ebp, esp
    mov     eax, DWORD PTR _s$[ebp+12]
    push    eax
    movsx   ecx, BYTE PTR _s$[ebp+8]
    push    ecx
    mov     edx, DWORD PTR _s$[ebp+4]
    push    edx
    movsx   eax, BYTE PTR _s$[ebp]
    push    eax
    push    OFFSET $SG3842
    call    _printf
    add     esp, 20
    pop     ebp
    ret     0
?f@@YAXUs@@@Z ENDP ; f
_TEXT    ENDS
```

(/Zp1) (*/Zp[n] pack structures on n-byte boundary*).

Listing 21.17: MSVC 2012 /GS- /Zp1

```

_main    PROC
    push   ebp
    mov    ebp, esp
    sub    esp, 12
    mov    BYTE PTR _tmp$[ebp], 1      ;  a
    mov    DWORD PTR _tmp$[ebp+1], 2  ;  b
    mov    BYTE PTR _tmp$[ebp+5], 3   ;  c
    mov    DWORD PTR _tmp$[ebp+6], 4  ;  d
    sub    esp, 12                    ;
    mov    eax, esp
    mov    ecx, DWORD PTR _tmp$[ebp]  ;
    mov    DWORD PTR [eax], ecx
    mov    edx, DWORD PTR _tmp$[ebp+4]
    mov    DWORD PTR [eax+4], edx
    mov    cx, WORD PTR _tmp$[ebp+8]
    mov    WORD PTR [eax+8], cx
    call    _f
    add    esp, 12
    xor    eax, eax
    mov    esp, ebp
    pop    ebp
    ret    0
_main    ENDP

_TEXT    SEGMENT
_s$ = 8 ; size = 10
```

 CAPÍTULO 21. ESTRUCTURAS

```
?f@@YAXUs@@@Z PROC    ; f
    push   ebp
    mov    ebp, esp
    mov    eax, DWORD PTR _s$[ebp+6]
    push   eax
    movsx  ecx, BYTE PTR _s$[ebp+5]
    push   ecx
    mov    edx, DWORD PTR _s$[ebp+1]
    push   edx
    movsx  eax, BYTE PTR _s$[ebp]
    push   eax
    push   OFFSET $SG3842
    call   _printf
    add    esp, 20
    pop    ebp
    ret    0
?f@@YAXUs@@@Z ENDP    ; f
```

Listing 21.18: WinNT.h

```
#include "pshpack1.h"
```

Listing 21.19: WinNT.h

```
#include "pshpack4.h"                    // 4 byte packing is the default
```

Listing 21.20: PshPack1.h

```
#if ! (defined(lint) || defined(RC_INVOKED))
#if ( _MSC_VER >= 800 && !defined(_M_I86)) || defined(_PUSHPOP_SUPPORTED)
#pragma warning(disable:4103)
```

```
#if !(defined( MIDL_PASS )) || defined( ↙
    ↪ __midl )
#pragma pack(push,1)
#else
#pragma pack(1)
#endif
#else
#pragma pack(1)
#endif
#endif /* ! (defined(lint) || defined(↙
    ↪ RC_INVOKED)) */
```

 CAPÍTULO 21. ESTRUCTURAS

OllyDbg +

:

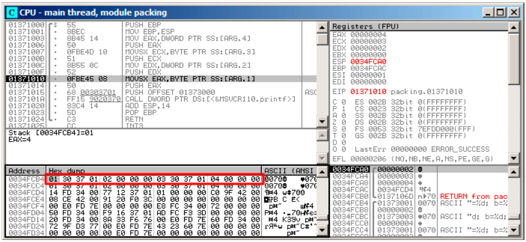


Figura 21.3: OllyDbg:

. : Spanish text placeholder.21.16 (34 y 38).

`unsigned char` o `uint8_t`, .

OllyDbg +

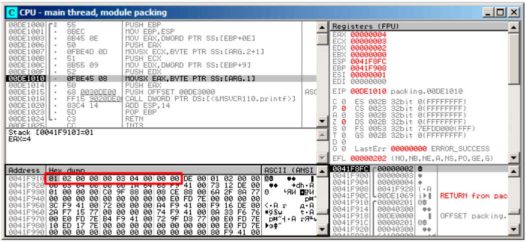


Figura 21.4: OllyDbg:

21.4.2 ARM

Con optimización Keil 6/2013 (Modo Thumb)

Listing 21.21: Con optimización Keil 6/2013 (Modo Thumb)

```
.text:0000003E                exit ; CODE XREF: ↙
    ↳  f+16
.text:0000003E 05 B0               ADD     SP↙
    ↳ , SP, #0x14
.text:00000040 00 BD               POP     {↙
    ↳ PC}


.text:00000280                f
.text:00000280
.text:00000280                var_18 = -0x18
.text:00000280                a      = -0x14
.text:00000280                b      = -0x10
```

```
.text:00000280                c        = -0xC
.text:00000280                d        = -8
.text:00000280
.text:00000280 0F B5                   PUSH     {↙
    ↘ R0-R3,LR}
.text:00000282 81 B0                   SUB      SP↙
    ↘ , SP, #4
.text:00000284 04 98                   LDR      R0↙
    ↘ , [SP,#16]    ; d
.text:00000286 02 9A                   LDR      R2↙
    ↘ , [SP,#8]     ; b
.text:00000288 00 90                   STR      R0↙
    ↘ , [SP]
.text:0000028A 68 46                   MOV      R0↙
    ↘ , SP
.text:0000028C 03 7B                   LDRB     R3↙
    ↘ , [R0,#12]    ; c
.text:0000028E 01 79                   LDRB     R1↙
    ↘ , [R0,#4]     ; a
.text:00000290 59 A0                   ADR      R0↙
    ↘ , aADBDCDDD   ; "a=%d; b=%d; c=%d; d=%↙
    ↘ d\n"
.text:00000292 05 F0 AD FF             BL       ↙
    ↘ __2printf
.text:00000296 D2 E6                   B        ↙
    ↘ exit
```

R0-R3.

LDRB MOVSX x86. a y c .

($(5 * 4 = 0x14)$).

 CAPÍTULO 21. ESTRUCTURAS

ARM + Con optimización Xcode 4.6.3 (LLVM) (Modo Thumb-2)

Listing 21.22: Con optimización Xcode 4.6.3 (LLVM) (Modo Thumb-2)

```
var_C = -0xC

      PUSH   {R7,LR}
      MOV    R7, SP
      SUB    SP, SP, #4
      MOV    R9, R1 ; b
      MOV    R1, R0 ; a
      MOVW   R0, #0xF10 ; "a=%d; b=%d; c=%d; d=%d\n"
      SXTB   R1, R1  ; prepare a
      MOVT.W R0, #0
      STR    R3, [SP,#0xC+var_C] ; place d to stack for printf()
      ADD    R0, PC  ; format-string
      SXTB   R3, R2  ; prepare c
      MOV    R2, R9  ; b
      BLX    _printf
      ADD    SP, SP, #4
      POP    {R7,PC}
```

SXTB (*Signed Extend Byte*) MOVSX en x86.

21.4.3 MIPS

Listing 21.23: Con optimización GCC 4.4.5 (IDA)

```
f:
```

```

var_18          = -0x18
var_10          = -0x10
var_4           = -4
arg_0           =  0
arg_4           =  4
arg_8           =  8
arg_C           =  0xC

; $a0=s.a
; $a1=s.b
; $a2=s.c
; $a3=s.d
                lui     $gp, (__gnu_local_gp >> 16)
                addiu   $sp, -0x28
                la      $gp, (__gnu_local_gp & 0xFFFF)
                sw      $ra, 0x28+var_4($sp)
                sw      $gp, 0x28+var_10($sp)
; prepare byte from 32-bit big-endian integer:
                sra     $t0, $a0, 24
                move    $v1, $a1
; prepare byte from 32-bit big-endian integer:
                sra     $v0, $a2, 24
                lw      $t9, (printf & 0xFFFF)($gp)
                sw      $a0, 0x28+arg_0($sp)
                lui     $a0, ($LC0 >> 16)  # "a=%d; b=%d; c=%d; d=%d\n"
```

 CAPÍTULO 21. ESTRUCTURAS

```
                sw      $a3, 0x28+var_18($sp)
                sw      $a1, 0x28+arg_4($sp)
                sw      $a2, 0x28+arg_8($sp)
                sw      $a3, 0x28+arg_C($sp)
                la      $a0, ($LC0 & 0xFFFF)  # "a=%d; b=%d; c=%d; d=%d\n"
                move    $a1, $t0
                move    $a2, $v1
                jalr    $t9
                move    $a3, $v0 ; branch delay slot
                lw      $ra, 0x28+var_4($sp)
                or      $at, $zero ; load delay slot, NOP
                jr      $ra
                addiu   $sp, 0x28 ; branch delay slot

$LC0:           .ascii "a=%d; b=%d; c=%d; d=%d\n"<0>
```

?

21.4.4

```
void f(char a, int b, char c, int d)
{
    printf ("a=%d; b=%d; c=%d; d=%d\n", a, b, c, d);
};
```

 CAPÍTULO 21. ESTRUCTURAS

.

21.5

```
#include <stdio.h>

struct inner_struct
{
    int a;
    int b;
};

struct outer_struct
{
    char a;
    int b;
    struct inner_struct c;
    char d;
    int e;
};

void f(struct outer_struct s)
{
    printf ("a=%d; b=%d; c.a=%d; c.b=%d; d=%d; e=%d\n",
        s.a, s.b, s.c.a, s.c.b, s.d, s.e);
};

int main()
{
    struct outer_struct s;
```

```
    s.a=1;
    s.b=2;
    s.c.a=100;
    s.c.b=101;
    s.d=3;
    s.e=4;
    f(s);
};
```

...

(MSVC 2010):

Listing 21.24: Con optimización MSVC 2010 /Ob0

```
$SG2802 DB    'a=%d; b=%d; c.a=%d; c.b=%d; d⤦
    ⤷ =%d; e=%d', 0aH, 00H

_TEXT    SEGMENT
_s$ = 8
_f    PROC
    mov    eax, DWORD PTR _s$[esp+16]
    movsx  ecx, BYTE PTR _s$[esp+12]
    mov    edx, DWORD PTR _s$[esp+8]
    push   eax
    mov    eax, DWORD PTR _s$[esp+8]
    push   ecx
    mov    ecx, DWORD PTR _s$[esp+8]
    push   edx
    movsx  edx, BYTE PTR _s$[esp+8]
    push   eax
    push   ecx
    push   edx
```

```
    push   OFFSET $SG2802 ; 'a=%d; b=%d; c.a↙
    ↘ =%d; c.b=%d; d=%d; e=%d'
    call   _printf
    add    esp, 28
    ret    0
_f    ENDP

_s$ = -24
_main    PROC
    sub    esp, 24
    push   ebx
    push   esi
    push   edi
    mov    ecx, 2
    sub    esp, 24
    mov    eax, esp
    mov    BYTE PTR _s$[esp+60], 1
    mov    ebx, DWORD PTR _s$[esp+60]
    mov    DWORD PTR [eax], ebx
    mov    DWORD PTR [eax+4], ecx
    lea    edx, DWORD PTR [ecx+98]
    lea    esi, DWORD PTR [ecx+99]
    lea    edi, DWORD PTR [ecx+2]
    mov    DWORD PTR [eax+8], edx
    mov    BYTE PTR _s$[esp+76], 3
    mov    ecx, DWORD PTR _s$[esp+76]
    mov    DWORD PTR [eax+12], esi
    mov    DWORD PTR [eax+16], ecx
    mov    DWORD PTR [eax+20], edi
    call   _f
    add    esp, 24
    pop    edi
    pop    esi
```

 CAPÍTULO 21. ESTRUCTURAS

```
    xor     eax, eax
    pop     ebx
    add     esp, 24
    ret     0
_main    ENDP
```

21.5.1 OllyDbg

OllyDbg `outer_struct` :

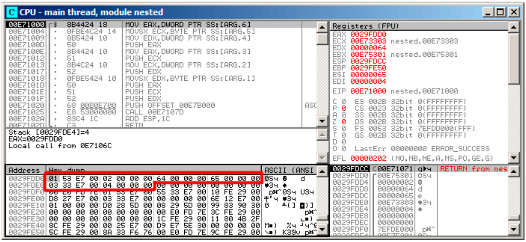


Figura 21.5: OllyDbg:

:

- *(outer_struct.a)* ;
- *(outer_struct.b)* () 2;
- *(inner_struct.a)* () 0x64 (100);
- *(inner_struct.b)* () 0x65 (101);
- *(outer_struct.d)* ;
- *(outer_struct.e)* () 4.

21.6

21.6.1

3:0 (4 Spanish text placeholder)	Stepping
7:4 (4 Spanish text placeholder)	Model
11:8 (4 Spanish text placeholder)	Family
13:12 (2 Spanish text placeholder)	Processor Type
19:16 (4 Spanish text placeholder)	Extended Model
27:20 (8 Spanish text placeholder)	Extended Family

```
#include <stdio.h>

#ifdef __GNUC__
static inline void cpuid(int code, int *a, ↙
    ↘ int *b, int *c, int *d) {
  asm volatile("cpuid":"=a"(*a),"=b"(*b),"=c↙
    ↘ "(*c),"=d"(*d):"a"(code));
}
#endif

#ifdef _MSC_VER
#include <intrin.h>
#endif

struct CPUID_1_EAX
{
    unsigned int stepping:4;
    unsigned int model:4;
    unsigned int family_id:4;
    unsigned int processor_type:2;
    unsigned int reserved1:2;
    unsigned int extended_model_id:4;
```

```
    unsigned int extended_family_id:8;
    unsigned int reserved2:4;
};

int main()
{
    struct CPUID_1_EAX *tmp;
    int b[4];

#ifdef _MSC_VER
    __cpuid(b,1);
#endif

#ifdef __GNUC__
    cpuid (1, &b[0], &b[1], &b[2], &b[3]);
#endif

    tmp=(struct CPUID_1_EAX *)&b[0];

    printf ("stepping=%d\n", tmp->stepping);
    printf ("model=%d\n", tmp->model);
    printf ("family_id=%d\n", tmp->family_id
    ↳ );
    printf ("processor_type=%d\n", tmp->
    ↳ processor_type);
    printf ("extended_model_id=%d\n", tmp->
    ↳ extended_model_id);
    printf ("extended_family_id=%d\n", tmp->
    ↳ extended_family_id);

    return 0;
};
```

MSVC

:

Listing 21.25: Con optimización MSVC 2008

```
_b$ = -16  ; size = 16
_main    PROC
    sub    esp, 16
    push   ebx

    xor    ecx, ecx
    mov    eax, 1
    cpuid
    push   esi
    lea    esi, DWORD PTR _b$[esp+24]
    mov    DWORD PTR [esi], eax
    mov    DWORD PTR [esi+4], ebx
    mov    DWORD PTR [esi+8], ecx
    mov    DWORD PTR [esi+12], edx

    mov    esi, DWORD PTR _b$[esp+24]
    mov    eax, esi
    and    eax, 15
    push   eax
    push   OFFSET $SG15435 ; 'stepping=%d', ↙
    ↘ 0aH, 00H
    call   _printf

    mov    ecx, esi
    shr    ecx, 4
    and    ecx, 15
    push   ecx
    push   OFFSET $SG15436 ; 'model=%d', 0aH↙
```

```
↳ , 00H
    call    _printf

    mov     edx, esi
    shr     edx, 8
    and     edx, 15
    push    edx
    push    OFFSET $SG15437 ; 'family_id=%d',↙
↳  0aH, 00H
    call    _printf

    mov     eax, esi
    shr     eax, 12
    and     eax, 3
    push    eax
    push    OFFSET $SG15438 ; 'processor_type↙
↳ =%d', 0aH, 00H
    call    _printf

    mov     ecx, esi
    shr     ecx, 16
    and     ecx, 15
    push    ecx
    push    OFFSET $SG15439 ; '↙
↳ extended_model_id=%d', 0aH, 00H
    call    _printf

    shr     esi, 20
    and     esi, 255
    push    esi
    push    OFFSET $SG15440 ; '↙
↳ extended_family_id=%d', 0aH, 00H
    call    _printf
```

```
    add     esp, 48
    pop     esi

    xor     eax, eax
    pop     ebx

    add     esp, 16
    ret     0
_main   ENDP
```

 CAPÍTULO 21. ESTRUCTURAS

MSVC + OllyDbg

OllyDbg CPUID:

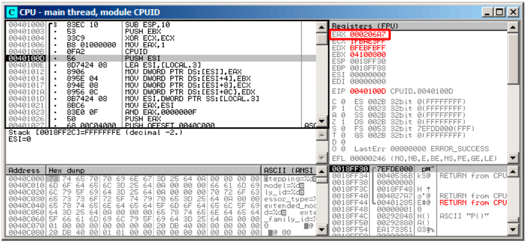


Figura 21.6: OllyDbg:

0x000206A7 (CPU Intel Xeon E3-1220).
00000000000000100000011010100111.

:

reserved2	0000	0
extended_family_id	00000000	0
extended_model_id	0010	2
reserved1	00	0
processor_id	00	0
family_id	0110	6
model	1010	10
stepping	0111	7

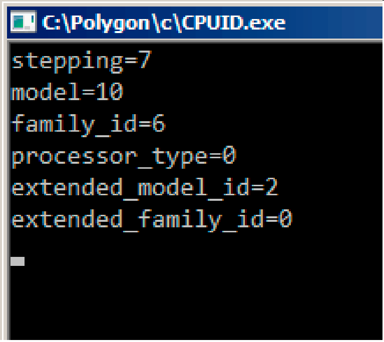


Figura 21.7: OllyDbg:

GCC

Listing 21.26: Con optimización GCC 4.4.1

```
main                proc near ; DATA XREF: ↙
    ↳ _start+17
    push    ebp
    mov     ebp, esp
    and     esp, 0FFFFFFF0h
    push    esi
    mov     esi, 1
    push    ebx
    mov     eax, esi
    sub     esp, 18h
    cpuid
    mov     esi, eax
    and     eax, 0Fh
    mov     [esp+8], eax
```

```
    mov     dword ptr [esp+4], offset ↙
    ↘ aSteppingD ; "stepping=%d\n"
    mov     dword ptr [esp], 1
    call    ___printf_chk
    mov     eax, esi
    shr     eax, 4
    and     eax, 0Fh
    mov     [esp+8], eax
    mov     dword ptr [esp+4], offset ↙
    ↘ aModelD ; "model=%d\n"
    mov     dword ptr [esp], 1
    call    ___printf_chk
    mov     eax, esi
    shr     eax, 8
    and     eax, 0Fh
    mov     [esp+8], eax
    mov     dword ptr [esp+4], offset ↙
    ↘ aFamily_idD ; "family_id=%d\n"
    mov     dword ptr [esp], 1
    call    ___printf_chk
    mov     eax, esi
    shr     eax, 0Ch
    and     eax, 3
    mov     [esp+8], eax
    mov     dword ptr [esp+4], offset ↙
    ↘ aProcessor_type ; "processor_type=%d\n↙
    ↘ "
    mov     dword ptr [esp], 1
    call    ___printf_chk
    mov     eax, esi
    shr     eax, 10h
    shr     esi, 14h
    and     eax, 0Fh
```

```
    and     esi, 0FFh
    mov     [esp+8], eax
    mov     dword ptr [esp+4], offset ↙
    ↳ aExtended_model ; "extended_model_id=%↙
    ↳ d\n"
    mov     dword ptr [esp], 1
    call    ___printf_chk
    mov     [esp+8], esi
    mov     dword ptr [esp+4], offset ↙
    ↳ unk_80486D0
    mov     dword ptr [esp], 1
    call    ___printf_chk
    add     esp, 18h
    xor     eax, eax
    pop     ebx
    pop     esi
    mov     esp, ebp
    pop     ebp
    retn
main            endp
```

21.6.2 Trabajando con el tipo float como una estructura

(SSpanish text placeholder)

```
#include <stdio.h>
```

 CAPÍTULO 21. ESTRUCTURAS

```
#include <assert.h>
#include <stdlib.h>
#include <memory.h>

struct float_as_struct
{
    unsigned int fraction : 23; //
    unsigned int exponent : 8;  //  + 0x3FF
    unsigned int sign : 1;      //
};

float f(float )
{
    float f=;
    struct float_as_struct t;

    assert (sizeof (struct float_as_struct) ↙
    ↘ == sizeof (float));

    memcpy (&t, &f, sizeof (float));

    t.sign=1; //
    t.exponent=t.exponent+2; //

    memcpy (&f, &t, sizeof (float));

    return f;
};

int main()
{
    printf ("%f\n", f(1.234));
};
```

Listing 21.27: Sin optimización MSVC 2008

```
_t$ = -8   ; size = 4
_f$ = -4   ; size = 4
__in$ = 8  ; size = 4
?f@@YAMM@Z PROC ; f
    push   ebp
    mov    ebp, esp
    sub    esp, 8

    fld    DWORD PTR __in$[ebp]
    fstp   DWORD PTR _f$[ebp]

    push   4
    lea    eax, DWORD PTR _f$[ebp]
    push   eax
    lea    ecx, DWORD PTR _t$[ebp]
    push   ecx
    call   _memcpy
    add    esp, 12

    mov    edx, DWORD PTR _t$[ebp]
    or     edx, -2147483648 ; 80000000H -
    mov    DWORD PTR _t$[ebp], edx

    mov    eax, DWORD PTR _t$[ebp]
    shr    eax, 23             ; 00000017H -
    and    eax, 255            ; 000000ffH -
    add    eax, 2              ;
    and    eax, 255            ; 000000ffH
```

```
    shl     eax, 23             ; 00000017H -
    mov     ecx, DWORD PTR _t$[ebp]
    and     ecx, -2139095041 ; 807fffffH -

;
    or      ecx, eax
    mov     DWORD PTR _t$[ebp], ecx

    push    4
    lea     edx, DWORD PTR _t$[ebp]
    push    edx
    lea     eax, DWORD PTR _f$[ebp]
    push    eax
    call    _memcpy
    add     esp, 12

    fld     DWORD PTR _f$[ebp]

    mov     esp, ebp
    pop     ebp
    ret     0
?f@@YAMM@Z ENDP    ; f
```

Listing 21.28: Con optimización GCC 4.4.1

```

; f(float)
       public _Z1ff
_Z1ff  proc near

var_4  = dword ptr -4
arg_0  = dword ptr  8

```

```
        push    ebp
        mov     ebp, esp
        sub     esp, 4
        mov     eax, [ebp+arg_0]
        or      eax, 80000000h  ;
        mov     edx, eax
        and     eax, 807FFFFFh  ;
        shr     edx, 23         ;
        add     edx, 2          ;
        movzx   edx, dl         ;
        shl     edx, 23         ;
        or      eax, edx        ;
        mov     [ebp+var_4], eax
        fld     [ebp+var_4]
        leave
        retn
_Z1ff   endp

        public main
main    proc near
        push    ebp
        mov     ebp, esp
        and     esp, 0FFFFFFF0h
        sub     esp, 10h
        fld     ds:dword_8048614 ; -4.936
        fstp    qword ptr [esp+8]
        mov     dword ptr [esp+4], offset ⤦
    ↳ asc_8048610 ; "%f\n"
        mov     dword ptr [esp], 1
        call    ___printf_chk
        xor     eax, eax
        leave
        retn
```

```
main    endp
```

21.7 Ejercicios

21.7.1 Ejercicio #1

Linux x86 (beginners.re)[3]
Linux MIPS (beginners.re)[4]

Responda: G.1.14 on page 1928.

21.7.2 Ejercicio #2

Listing 21.29: Con optimización MSVC 2010

```
$SG2802 DB      '%f', 0aH, 00H
$SG2803 DB      '%c, %d', 0aH, 00H
$SG2805 DB      'error #2', 0aH, 00H
$SG2807 DB      'error #1', 0aH, 00H

__real@405ec00000000000 DQ 0405↙
    ↳ ec00000000000r    ; 123
__real@407bc00000000000 DQ 0407↙
    ↳ bc00000000000r    ; 444

_s$ = 8
_f      PROC
```

[3] GCC 4.8.1 -O3
[4] GCC 4.4.5 -O3

```
        push    esi
        mov     esi, DWORD PTR _s$[esp]
        cmp     DWORD PTR [esi], 1000
        jle     SHORT $LN4@f
        cmp     DWORD PTR [esi+4], 10
        jbe     SHORT $LN3@f
        fld     DWORD PTR [esi+8]
        sub     esp, 8
        fmul    QWORD PTR ⤦
    ↳ __real@407bc00000000000
        fld     QWORD PTR [esi+16]
        fmul    QWORD PTR ⤦
    ↳ __real@405ec00000000000
        faddp   ST(1), ST(0)
        fstp    QWORD PTR [esp]
        push    OFFSET $SG2802 ; '%f'
        call    _printf
        movzx   eax, BYTE PTR [esi+25]
        movsx   ecx, BYTE PTR [esi+24]
        push    eax
        push    ecx
        push    OFFSET $SG2803 ; '%c, %d'
        call    _printf
        add     esp, 24
        pop     esi
        ret     0
$LN3@f:
        pop     esi
        mov     DWORD PTR _s$[esp-4], OFFSET⤦
    ↳  $SG2805 ; 'error #2'
        jmp     _printf
$LN4@f:
        pop     esi
```

```
        mov     DWORD PTR _s$[esp-4], OFFSET
    ↳ $SG2807 ; 'error #1'
        jmp     _printf
_f      ENDP
```

Listing 21.30: Sin optimización Keil 6/2013 (Modo ARM)

```
f PROC
        PUSH     {r4-r6,lr}
        MOV      r4,r0
        LDR      r0,[r0,#0]
        CMP      r0,#0x3e8
        ADRLE    r0,|L0.140|
        BLE      |L0.132|
        LDR      r0,[r4,#4]
        CMP      r0,#0xa
        ADRLS    r0,|L0.152|
        BLS      |L0.132|
        ADD      r0,r4,#0x10
        LDM      r0,{r0,r1}
        LDR      r3,|L0.164|
        MOV      r2,#0
        BL       __aeabi_dmul
        MOV      r5,r0
        MOV      r6,r1
        LDR      r0,[r4,#8]
        LDR      r1,|L0.168|
        BL       __aeabi_fmul
        BL       __aeabi_f2d
        MOV      r2,r5
        MOV      r3,r6
        BL       __aeabi_dadd
        MOV      r2,r0
```

```
        MOV      r3,r1
        ADR      r0,|L0.172|
        BL       __2printf
        LDRB     r2,[r4,#0x19]
        LDRB     r1,[r4,#0x18]
        POP      {r4-r6,lr}
        ADR      r0,|L0.176|
        B        __2printf
|L0.132|
        POP      {r4-r6,lr}
        B        __2printf
        ENDP

|L0.140|
        DCB      "error #1\n",0
        DCB      0
        DCB      0
|L0.152|
        DCB      "error #2\n",0
        DCB      0
        DCB      0
|L0.164|
        DCD      0x405ec000
|L0.168|
        DCD      0x43de0000
|L0.172|
        DCB      "%f\n",0
|L0.176|
        DCB      "%c, %d\n",0
```

Listing 21.31: Sin optimización Keil 6/2013 (Modo Thumb)

```
f PROC
```

```
        PUSH     {r4-r6,lr}
        MOV      r4,r0
        LDR      r0,[r0,#0]
        CMP      r0,#0x3e8
        ADRLE    r0,|L0.140|
        BLE      |L0.132|
        LDR      r0,[r4,#4]
        CMP      r0,#0xa
        ADRLS    r0,|L0.152|
        BLS      |L0.132|
        ADD      r0,r4,#0x10
        LDM      r0,{r0,r1}
        LDR      r3,|L0.164|
        MOV      r2,#0
        BL       __aeabi_dmul
        MOV      r5,r0
        MOV      r6,r1
        LDR      r0,[r4,#8]
        LDR      r1,|L0.168|
        BL       __aeabi_fmul
        BL       __aeabi_f2d
        MOV      r2,r5
        MOV      r3,r6
        BL       __aeabi_dadd
        MOV      r2,r0
        MOV      r3,r1
        ADR      r0,|L0.172|
        BL       __2printf
        LDRB     r2,[r4,#0x19]
        LDRB     r1,[r4,#0x18]
        POP      {r4-r6,lr}
        ADR      r0,|L0.176|
        B        __2printf
```

```
|L0.132|
        POP      {r4-r6,lr}
        B        __2printf
        ENDP

|L0.140|
        DCB      "error #1\n",0
        DCB      0
        DCB      0
|L0.152|
        DCB      "error #2\n",0
        DCB      0
        DCB      0
|L0.164|
        DCD      0x405ec000
|L0.168|
        DCD      0x43de0000
|L0.172|
        DCB      "%f\n",0
|L0.176|
        DCB      "%c, %d\n",0
```

Listing 21.32: Con optimización GCC 4.9 (ARM64)

```
f:
        stp     x29, x30, [sp, -32]!
        add     x29, sp, 0
        ldr     w1, [x0]
        str     x19, [sp,16]
        cmp     w1, 1000
        ble     .L2
        ldr     w1, [x0,4]
        cmp     w1, 10
```

```
        bls     .L3
        ldr     s1, [x0,8]
        mov     x19, x0
        ldr     s0, .LC1
        adrp    x0, .LC0
        ldr     d2, [x19,16]
        add     x0, x0, :lo12:.LC0
        fmul    s1, s1, s0
        ldr     d0, .LC2
        fmul    d0, d2, d0
        fcvt    d1, s1
        fadd    d0, d1, d0
        bl      printf
        ldrb    w1, [x19,24]
        adrp    x0, .LC3
        ldrb    w2, [x19,25]
        add     x0, x0, :lo12:.LC3
        ldr     x19, [sp,16]
        ldp     x29, x30, [sp], 32
        b       printf
.L3:
        ldr     x19, [sp,16]
        adrp    x0, .LC4
        ldp     x29, x30, [sp], 32
        add     x0, x0, :lo12:.LC4
        b       puts
.L2:
        ldr     x19, [sp,16]
        adrp    x0, .LC5
        ldp     x29, x30, [sp], 32
        add     x0, x0, :lo12:.LC5
        b       puts
        .size   f, .-f
```

```
.LC1:
        .word   1138622464
.LC2:
        .word   0
        .word   1079951360
.LC0:
        .string "%f\n"
.LC3:
        .string "%c, %d\n"
.LC4:
        .string "error #2"
.LC5:
        .string "error #1"
```

Listing 21.33: Con optimización GCC 4.4.5 (MIPS) (IDA)

```
f:

var_10          = -0x10
var_8           = -8
var_4           = -4

                lui     $gp, (__gnu_local_gp ↙
    ↘ >> 16)
                addiu   $sp, -0x20
                la      $gp, (__gnu_local_gp ↙
    ↘ & 0xFFFF)
                sw      $ra, 0x20+var_4($sp)
                sw      $s0, 0x20+var_8($sp)
                sw      $gp, 0x20+var_10($sp ↙
    ↘ )
                lw      $v0, 0($a0)
                or      $at, $zero
```

 CAPÍTULO 21. ESTRUCTURAS

```
                slti    $v0, 0x3E9
                bnez    $v0, loc_C8
                move    $s0, $a0
                lw      $v0, 4($a0)
                or      $at, $zero
                sltiu   $v0, 0xB
                bnez    $v0, loc_AC
                lui     $v0, (dword_134 >> ↙
    ↘ 16)
                lwc1    $f4, $LC1
                lwc1    $f2, 8($a0)
                lui     $v0, ($LC2 >> 16)
                lwc1    $f0, 0x14($a0)
                mul.s   $f2, $f4, $f2
                lwc1    $f4, dword_134
                lwc1    $f1, 0x10($a0)
                lwc1    $f5, $LC2
                cvt.d.s $f2, $f2
                mul.d   $f0, $f4, $f0
                lw      $t9, (printf & 0↙
    ↘ xFFFF)($gp)
                lui     $a0, ($LC0 >> 16)  #↙
    ↘  "%f\n"
                add.d   $f4, $f2, $f0
                mfc1    $a2, $f5
                mfc1    $a3, $f4
                jalr    $t9
                la      $a0, ($LC0 & 0xFFFF)↙
    ↘   # "%f\n"
                lw      $gp, 0x20+var_10($sp↙
    ↘ )
                lbu     $a2, 0x19($s0)
                lb      $a1, 0x18($s0)
```

 CAPÍTULO 21. ESTRUCTURAS

```
                lui     $a0, ($LC3 >> 16)  #↙
    ↳  "%c, %d\n"
                lw      $t9, (printf & 0↙
    ↳ xFFFF)($gp)
                lw      $ra, 0x20+var_4($sp)
                lw      $s0, 0x20+var_8($sp)
                la      $a0, ($LC3 & 0xFFFF)↙
    ↳   # "%c, %d\n"
                jr      $t9
                addiu   $sp, 0x20
loc_AC:                                     # ↙
    ↳ CODE XREF: f+38
                lui     $a0, ($LC4 >> 16)  #↙
    ↳  "error #2"
                lw      $t9, (puts & 0xFFFF)↙
    ↳ ($gp)
                lw      $ra, 0x20+var_4($sp)
                lw      $s0, 0x20+var_8($sp)
                la      $a0, ($LC4 & 0xFFFF)↙
    ↳   # "error #2"
                jr      $t9
                addiu   $sp, 0x20
loc_C8:                                     # ↙
    ↳ CODE XREF: f+24
                lui     $a0, ($LC5 >> 16)  #↙
    ↳  "error #1"
                lw      $t9, (puts & 0xFFFF)↙
    ↳ ($gp)
                lw      $ra, 0x20+var_4($sp)
                lw      $s0, 0x20+var_8($sp)
                la      $a0, ($LC5 & 0xFFFF)↙
    ↳   # "error #1"
                jr      $t9
```

```
                 addiu   $sp, 0x20

$LC0:            .ascii "%f\n"<0>
$LC3:            .ascii "%c, %d\n"<0>
$LC4:            .ascii "error #2"<0>
$LC5:            .ascii "error #1"<0>

                 .data # .rodata.cst4
$LC1:            .word 0x43DE0000

                 .data # .rodata.cst8
$LC2:            .word 0x405EC000
dword_134:       .word 0
```

Responda: G.1.14 on page 1929.

Capítulo 22

22.1

20 on page 648.

```
#include <stdio.h>
#include <stdint.h>
#include <time.h>

//

//
const uint32_t RNG_a=1664525;
const uint32_t RNG_c=1013904223;
uint32_t RNG_state; //

void my_srand(uint32_t i)
{
```

```
        RNG_state=i;
};

uint32_t my_rand()
{
        RNG_state=RNG_state*RNG_a+RNG_c;
        return RNG_state;
};

//

union uint32_t_float
{
        uint32_t i;
        float f;
};

float float_rand()
{
        union uint32_t_float tmp;
        tmp.i=my_rand() & 0x007fffff | 0↙
    ↘ x3F800000;
        return tmp.f-1;
};

//

int main()
{
        my_srand(time(NULL)); //

        for (int i=0; i<100; i++)
                printf ("%f\n", float_rand()↙
```

```
    ↳ );

        return 0;
};
```

22.1.1 x86

Listing 22.1: Con optimización MSVC 2010

```

$SG4238 DB      '%f', 0aH, 00H

__real@3ff0000000000000 DQ 03↙
    ↳ ff0000000000000r   ; 1

tv130 = -4
_tmp$ = -4
?float_rand@@YAMXZ PROC
        push    ecx
        call    ?my_rand@@YAIXZ
; EAX=
        and     eax, 8388607                ↙
    ↳                   ; 007fffffH
        or      eax, 1065353216             ↙
    ↳                   ; 3f800000H
; EAX= & 0x007fffff | 0x3f800000
;
        mov     DWORD PTR _tmp$[esp+4], eax
;
        fld     DWORD PTR _tmp$[esp+4]
;  1.0:
```

```
        fsub    QWORD PTR ⤦
    ↪ __real@3ff0000000000000
;
        fstp    DWORD PTR tv130[esp+4] ; \
        fld     DWORD PTR tv130[esp+4] ; /
        pop     ecx
        ret     0
?float_rand@@YAMXZ ENDP

_main   PROC
        push    esi
        xor     eax, eax
        call    _time
        push    eax
        call    ?my_srand@@YAXI@Z
        add     esp, 4
        mov     esi, 100
$LL3@main:
        call    ?float_rand@@YAMXZ
        sub     esp, 8
        fstp    QWORD PTR [esp]
        push    OFFSET $SG4238
        call    _printf
        add     esp, 12
        dec     esi
        jne     SHORT $LL3@main
        xor     eax, eax
        pop     esi
        ret     0
_main   ENDP
```

51.1.1 on page 1040.

27.5 on page 847.

22.1.2 MIPS

Listing 22.2: Con optimización GCC 4.4.5

```
float_rand:

var_10          = -0x10
var_4           = -4

                lui     $gp, (__gnu_local_gp↙
    ↳  >> 16)
                addiu   $sp, -0x20
                la      $gp, (__gnu_local_gp↙
    ↳  & 0xFFFF)
                sw      $ra, 0x20+var_4($sp)
                sw      $gp, 0x20+var_10($sp↙
    ↳ )
;  my_rand():
                jal     my_rand
                or      $at, $zero ; branch ↙
    ↳ delay slot, NOP
; $v0=
                li      $v1, 0x7FFFFF
; $v1=0x7FFFFF
                and     $v1, $v0, $v1
; $v1= & 0x7FFFFF
                lui     $a0, 0x3F80
; $a0=0x3F800000
                or      $v1, $a0
; $v1= & 0x7FFFFF | 0x3F800000
```

```
;
                lui     $v0, ($LC0 >> 16)
;  $f0:
                lwc1    $f0, $LC0
;
;
                mtc1    $v1, $f2
                lw      $ra, 0x20+var_4($sp)
;
                sub.s   $f0, $f2, $f0
                jr      $ra
                addiu   $sp, 0x20  ; branch ↙
    ↳ delay slot

main:

var_18          = -0x18
var_10          = -0x10
var_C           = -0xC
var_8           = -8
var_4           = -4

                lui     $gp, (__gnu_local_gp↙
    ↳  >> 16)
                addiu   $sp, -0x28
                la      $gp, (__gnu_local_gp↙
    ↳  & 0xFFFF)
                sw      $ra, 0x28+var_4($sp)
                sw      $s2, 0x28+var_8($sp)
                sw      $s1, 0x28+var_C($sp)
                sw      $s0, 0x28+var_10($sp↙
    ↳ )
```

```
                sw      $gp, 0x28+var_18($sp⤦
    ↳ )
                lw      $t9, (time & 0xFFFF)⤦
    ↳ ($gp)
                or      $at, $zero ; load ⤦
    ↳ delay slot, NOP
                jalr    $t9
                move    $a0, $zero ; branch ⤦
    ↳ delay slot
                lui     $s2, ($LC1 >> 16)  #⤦
    ↳  "%f\n"
                move    $a0, $v0
                la      $s2, ($LC1 & 0xFFFF)⤦
    ↳   # "%f\n"
                move    $s0, $zero
                jal     my_srand
                li      $s1, 0x64  # 'd' ; ⤦
    ↳ branch delay slot

loc_104:
                jal     float_rand
                addiu   $s0, 1
                lw      $gp, 0x28+var_18($sp⤦
    ↳ )
;
                cvt.d.s $f2, $f0
                lw      $t9, (printf & 0⤦
    ↳ xFFFF)($gp)
                mfc1    $a3, $f2
                mfc1    $a2, $f3
                jalr    $t9
                move    $a0, $s2
                bne     $s0, $s1, loc_104
```

```
                move    $v0, $zero
                lw      $ra, 0x28+var_4($sp)
                lw      $s2, 0x28+var_8($sp)
                lw      $s1, 0x28+var_C($sp)
                lw      $s0, 0x28+var_10($sp⤦
    ↳ )
                jr      $ra
                addiu   $sp, 0x28 ; branch ⤦
    ↳ delay slot

$LC1:           .ascii "%f\n"<0>
$LC0:           .float 1.0
```

17.5.6 on page 427.

22.1.3 ARM (Modo ARM)

Listing 22.3: Con optimización GCC 4.6.3 (IDA)

```

float_rand
                STMFD   SP!, {R3,LR}
                BL      my_rand
; R0=
                FLDS    S0, =1.0
; S0=1.0
                BIC     R3, R0, #0xFF000000
                BIC     R3, R3, #0x800000
                ORR     R3, R3, #0x3F800000
; R3= & 0x007fffff | 0x3f800000
;
;
```

```
                FMSR    S15, R3
;
                FSUBS   S0, S15, S0
                LDMFD   SP!, {R3,PC}

flt_5C          DCFS 1.0

main
                STMFD   SP!, {R4,LR}
                MOV     R0, #0
                BL      time
                BL      my_srand
                MOV     R4, #0x64 ; 'd'

loc_78
                BL      float_rand
; S0=
                LDR     R0, =aF          ; "%
    f"
;
                FCVTDS  D7, S0
;
                FMRRD   R2, R3, D7
                BL      printf
                SUBS    R4, R4, #1
                BNE     loc_78
                MOV     R0, R4
                LDMFD   SP!, {R4,PC}

aF              DCB "%f",0xA,0
```

Listing 22.4: Con optimización GCC 4.6.3 (objdump)

```
00000038 <float_rand>:
  38:   e92d4008        push    {r3, lr}
  3c:   ebfffffe        bl      10 <my_rand>
  40:   ed9f0a05        vldr    s0, [pc, ↙
    ↘ #20]   ; 5c <float_rand+0x24>
  44:   e3c034ff        bic     r3, r0, ↙
    ↘ #-16777216       ; 0xff000000
  48:   e3c33502        bic     r3, r3, ↙
    ↘ #8388608         ; 0x800000
  4c:   e38335fe        orr     r3, r3, ↙
    ↘ #1065353216      ; 0x3f800000
  50:   ee073a90        vmov    s15, r3
  54:   ee370ac0        vsub.f32        s0, ↙
    ↘ s15, s0
  58:   e8bd8008        pop     {r3, pc}
  5c:   3f800000        svccc   0x00800000

00000000 <main>:
   0:   e92d4010        push    {r4, lr}
   4:   e3a00000        mov     r0, #0
   8:   ebfffffe        bl      0 <time>
   c:   ebfffffe        bl      0 <main>
  10:   e3a04064        mov     r4, #100    ↙
    ↘     ; 0x64
  14:   ebfffffe        bl      38 <main+0↙
    ↘ x38>
  18:   e59f0018        ldr     r0, [pc, ↙
    ↘ #24]   ; 38 <main+0x38>
  1c:   eeb77ac0        vcvt.f64.f32    d7, ↙
    ↘ s0
  20:   ec532b17        vmov    r2, r3, d7
  24:   ebfffffe        bl      0 <printf>
  28:   e2544001        subs    r4, r4, #1
```

```
  2c:   1affff f8         bne      14 <main+0
    ↳ x14>
  30:   e1a00004          mov      r0, r4
  34:   e8bd8010          pop      {r4, pc}
  38:   00000000          andeq    r0, r0, r0
```

22.2

$2^{-23} = 1.19e - 07$ para *float* y $2^{-52} = 2.22e - 16$ para *double*.

```
#include <stdio.h>
#include <stdint.h>

union uint_float
{
        uint32_t i;
        float f;
};

float calculate_machine_epsilon(float start)
{
        union uint_float v;
        v.f=start;
        v.i++;
        return v.f-start;
}

void main()
{
        printf ("%g\n", 
    ↳ calculate_machine_epsilon(1.0));
```

```
};
```

22.2.1 x86

Listing 22.5: Con optimización MSVC 2010

```
tv130 = 8
_v$ = 8
_start$ = 8
_calculate_machine_epsilon PROC
        fld     DWORD PTR _start$[esp-4]
        fst     DWORD PTR _v$[esp-4]     ;
        inc     DWORD PTR _v$[esp-4]
        fsubr   DWORD PTR _v$[esp-4]
        fstp    DWORD PTR tv130[esp-4]   ; \
        fld     DWORD PTR tv130[esp-4]   ; /
        ret     0
_calculate_machine_epsilon ENDP
```

22.2.2 ARM64

```
#include <stdio.h>
#include <stdint.h>

typedef union
{
        uint64_t i;
        double d;
} uint_double;
```

```
double calculate_machine_epsilon(double ↙
    ↘ start)
{
        uint_double v;
        v.d=start;
        v.i++;
        return v.d-start;
}

void main()
{
        printf ("%g\n", ↙
    ↘ calculate_machine_epsilon(1.0));
};
```

Listing 22.6: Con optimización GCC 4.9 ARM64

```

calculate_machine_epsilon:
        fmov    x0, d0          ;
        add     x0, x0, 1       ; X0++
        fmov    d1, x0          ;
        fsub    d0, d1, d0      ;
        ret
```

: 27.4 on page 846.

22.2.3 MIPS

Listing 22.7: Con optimización GCC 4.4.5 (IDA)

```
calculate_machine_epsilon:
```

```
                mfc1    $v0, $f12
                or      $at, $zero ; NOP
                addiu   $v1, $v0, 1
                mtc1    $v1, $f2
                jr      $ra
                sub.s   $f0, $f2, $f12 ; ⤦
    ↪ branch delay slot
```

22.2.4 Conclusión

22.3

Listing 22.8: : http://go.yurichev.com/17364

```
/* Assumes that float is in the IEEE 754 ⤦
    ↪ single precision floating point format
 * and that int is 32 bits. */
float sqrt_approx(float z)
{
    int val_int = *(int*)&z; /* Same bits, ⤦
    ↪ but as an int */
    /*
     * To justify the following code, prove ⤦
    ↪ that
     *
     * ((((val_int / 2^m) - b) / 2) + b) * ⤦
    ↪ 2^m = ((val_int - 2^m) / 2) + ((b + 1)⤦
    ↪  / 2) * 2^m)
     *
     * where
     *
```

```
     * b = exponent bias
     * m = number of mantissa bits
     *
     * .
     */

    val_int -= 1 << 23; /* Subtract 2^m. */
    val_int >>= 1; /* Divide by 2. */
    val_int += 1 << 29; /* Add ((b + 1) / 2)↙
    ↳  * 2^m. */

    return *(float*)&val_int; /* Interpret ↙
    ↳ again as float */
}
```

Como ejercicio, puedes compilar esta función y entender cómo funciona.

$\frac{1}{\sqrt{x}}$.

Wikipedia: .

Capítulo 23

[1].

- `qsort()`[2], `atexit()`[3] ;
- [4];
- : `CreateThread()` (win32), `pthread_create()` (POSIX);
- `EnumChildWindows()`[5].
- : http://go.yurichev.com/17076
- : http://go.yurichev.com/17077

[1] wikipedia
[2] wikipedia
[3] http://go.yurichev.com/17073
[4] wikipedia
[5] MSDN

 CAPÍTULO 23.

- GitHub.

```
int (*compare)(const void *, const void *)
```

:

```
/* ex3 Sorting ints with qsort */

#include <stdio.h>
#include <stdlib.h>

int comp(const void * _a, const void * _b)
{
  const int *a=(const int *)_a;
  const int *b=(const int *)_b;

  if (*a==*b)
    return 0;
  else
    if (*a < *b)
        return -1;
     else
      return 1;
}

int main(int argc, char* argv[])
{
   int numbers[10]={1892,45,200,-98,4087,5,-12345,1087,88
   int i;

  /* Sort the array */
```

```
  qsort(numbers,10,sizeof(int),comp) ;
  for (i=0;i<9;i++)
    printf("Number = %d\n",numbers[ i ]) ;
  return 0;
}
```

23.1 MSVC

:

Listing 23.1: Con optimización MSVC 2010: /GS- /MD

```
__a$ = 8                        ; size = 4
__b$ = 12                       ; size = 4
_comp   PROC
        mov     eax, DWORD PTR __a$[esp-4]
        mov     ecx, DWORD PTR __b$[esp-4]
        mov     eax, DWORD PTR [eax]
        mov     ecx, DWORD PTR [ecx]
        cmp     eax, ecx
        jne     SHORT $LN4@comp
        xor     eax, eax
        ret     0
$LN4@comp:
        xor     edx, edx
        cmp     eax, ecx
        setge   dl
        lea     eax, DWORD PTR [edx+edx-1]
        ret     0
```

```
_comp   ENDP


_numbers$ = -40                                    ⤦
    ↳                   ; size = 40
_argc$ = 8                                         ⤦
    ↳                   ; size = 4
_argv$ = 12                                        ⤦
    ↳                   ; size = 4
_main   PROC
        sub     esp, 40                            ⤦
    ↳                   ; 00000028H
        push    esi
        push    OFFSET _comp
        push    4
        lea     eax, DWORD PTR _numbers$[esp⤦
    ↳ +52]
        push    10                                 ⤦
    ↳                   ; 0000000aH
        push    eax
        mov     DWORD PTR _numbers$[esp+60],⤦
    ↳ 1892          ; 00000764H
        mov     DWORD PTR _numbers$[esp+64],⤦
    ↳ 45            ; 0000002dH
        mov     DWORD PTR _numbers$[esp+68],⤦
    ↳ 200           ; 000000c8H
        mov     DWORD PTR _numbers$[esp+72],⤦
    ↳ -98           ; ffffff9eH
        mov     DWORD PTR _numbers$[esp+76],⤦
    ↳ 4087          ; 00000ff7H
        mov     DWORD PTR _numbers$[esp+80],⤦
    ↳ 5
        mov     DWORD PTR _numbers$[esp+84],⤦
```

```
    ↳ -12345     ; ffffcfc7H
     mov     DWORD PTR _numbers$[esp+88],↙
    ↳ 1087       ; 0000043fH
     mov     DWORD PTR _numbers$[esp+92],↙
    ↳ 88         ; 00000058H
     mov     DWORD PTR _numbers$[esp+96],↙
    ↳ -100000    ; fffe7960H
     call    _qsort
     add     esp, 16                    ↙
    ↳                ; 00000010H

...
```

Listing 23.2: MSVCR80.DLL

```
.text:7816CBF0 ; void __cdecl qsort(void *, ↙
    ↳ unsigned int, unsigned int, int (↙
    ↳ __cdecl *)(const void *, const void *)↙
    ↳ )
.text:7816CBF0                  public _qsort
.text:7816CBF0 _qsort           proc near
.text:7816CBF0
.text:7816CBF0 lo               = dword ptr ↙
    ↳ -104h
.text:7816CBF0 hi               = dword ptr ↙
    ↳ -100h
.text:7816CBF0 var_FC           = dword ptr ↙
    ↳ -0FCh
.text:7816CBF0 stkptr           = dword ptr ↙
    ↳ -0F8h
.text:7816CBF0 lostk            = dword ptr ↙
    ↳ -0F4h
.text:7816CBF0 histk            = dword ptr ↙
```

```
    ↳ -7Ch
.text:7816CBF0 base                    = dword ptr  ↙
    ↳ 4
.text:7816CBF0 num                     = dword ptr  ↙
    ↳ 8
.text:7816CBF0 width                   = dword ptr  ↙
    ↳ 0Ch
.text:7816CBF0 comp                    = dword ptr  ↙
    ↳ 10h
.text:7816CBF0
.text:7816CBF0                         sub     esp, ↙
    ↳ 100h

....

.text:7816CCE0 loc_7816CCE0:                        ↙
    ↳             ; CODE XREF: _qsort+B1
.text:7816CCE0                         shr     eax, ↙
    ↳ 1
.text:7816CCE2                         imul    eax, ↙
    ↳ ebp
.text:7816CCE5                         add     eax, ↙
    ↳ ebx
.text:7816CCE7                         mov     edi, ↙
    ↳ eax
.text:7816CCE9                         push    edi
.text:7816CCEA                         push    ebx
.text:7816CCEB                         call    [esp↙
    ↳ +118h+comp]
.text:7816CCF2                         add     esp, ↙
    ↳ 8
.text:7816CCF5                         test    eax, ↙
    ↳ eax
```

```
.text:7816CCF7                    jle      short ↙
    ↳ loc_7816CD04
```

comp

23.1.1 MSVC + OllyDbg

.

:

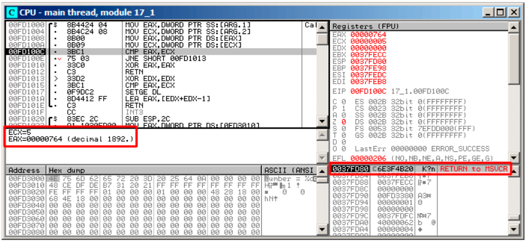


Figura 23.1: OllyDbg: `comp()`

OllyDbg . SP RA `qsort()` (`MSVCR100.DLL`).

(F8) `RETN qsort()`:

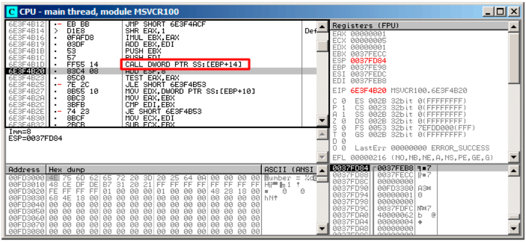


Figura 23.2: OllyDbg: `qsort()` `comp()`

.

comp() :

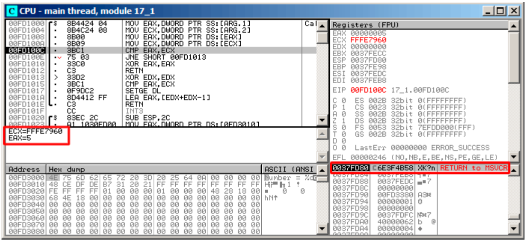


Figura 23.3: OllyDbg: comp()

23.1.2 MSVC + tracer

. : 1892, 45, 200, -98, 4087, 5, -12345, 1087, 88, -100000.

```
tracer.exe -l:17_1.exe bpx=17_1.exe!0x0040100C
```

:

```
PID=4336|New process 17_1.exe
(0) 17_1.exe!0x40100c
EAX=0x00000764 EBX=0x0051f7c8 ECX=0x00000005 EDX=0x00000000
ESI=0x0051f7d8 EDI=0x0051f7b4 EBP=0x0051f794 ESP=0x0051f67c
EIP=0x0028100c
FLAGS=IF
(0) 17_1.exe!0x40100c
```

```
EAX=0x00000005 EBX=0x0051f7c8 ECX=0xfffe7960 ↙
    ↪ EDX=0x00000000
ESI=0x0051f7d8 EDI=0x0051f7b4 EBP=0x0051f794 ↙
    ↪ ESP=0x0051f67c
EIP=0x0028100c
FLAGS=PF ZF IF
(0) 17_1.exe!0x40100c
EAX=0x00000764 EBX=0x0051f7c8 ECX=0x00000005 ↙
    ↪ EDX=0x00000000
ESI=0x0051f7d8 EDI=0x0051f7b4 EBP=0x0051f794 ↙
    ↪ ESP=0x0051f67c
EIP=0x0028100c
FLAGS=CF PF ZF IF
...
```

```
EAX=0x00000764 ECX=0x00000005
EAX=0x00000005 ECX=0xfffe7960
EAX=0x00000764 ECX=0x00000005
EAX=0x0000002d ECX=0x00000005
EAX=0x00000058 ECX=0x00000005
EAX=0x0000043f ECX=0x00000005
EAX=0xffffcfc7 ECX=0x00000005
EAX=0x000000c8 ECX=0x00000005
EAX=0xffffff9e ECX=0x00000005
EAX=0x00000ff7 ECX=0x00000005
EAX=0x00000ff7 ECX=0x00000005
EAX=0xffffff9e ECX=0x00000005
EAX=0xffffff9e ECX=0x00000005
EAX=0xffffcfc7 ECX=0xfffe7960
EAX=0x00000005 ECX=0xffffcfc7
EAX=0xffffff9e ECX=0x00000005
EAX=0xffffcfc7 ECX=0xfffe7960
```

```
EAX=0xffffff9e ECX=0xffffcfc7
EAX=0xffffcfc7 ECX=0xfffe7960
EAX=0x000000c8 ECX=0x00000ff7
EAX=0x0000002d ECX=0x00000ff7
EAX=0x0000043f ECX=0x00000ff7
EAX=0x00000058 ECX=0x00000ff7
EAX=0x00000764 ECX=0x00000ff7
EAX=0x000000c8 ECX=0x00000764
EAX=0x0000002d ECX=0x00000764
EAX=0x0000043f ECX=0x00000764
EAX=0x00000058 ECX=0x00000764
EAX=0x000000c8 ECX=0x00000058
EAX=0x0000002d ECX=0x000000c8
EAX=0x0000043f ECX=0x000000c8
EAX=0x000000c8 ECX=0x00000058
EAX=0x0000002d ECX=0x000000c8
EAX=0x0000002d ECX=0x00000058
```

34 . .

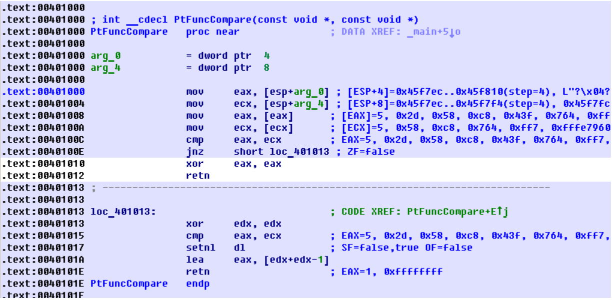
23.1.3 MSVC + tracer (code coverage)

.

:

```
tracer.exe -l:17_1.exe bpf=17_1.exe!0
    x00401000,trace:cc
```

:



```
.text:00401000
.text:00401000 ; int __cdecl PtFuncCompare(const void *, const void *)
.text:00401000 PtFuncCompare   proc near               ; DATA XREF: _main+5↓o
.text:00401000
.text:00401000 arg_0           = dword ptr  4
.text:00401000 arg_4           = dword ptr  8
.text:00401000
.text:00401000                 mov     eax, [esp+arg_0] ; [ESP+4]=0x45f7ec..0x45f810(step=4), L"?\x04?
.text:00401004                 mov     ecx, [esp+arg_4] ; [ESP+8]=0x45f7ec..0x45f7f4(step=4), 0x45f7fc
.text:00401008                 mov     eax, [eax]       ; [EAX]=5, 0x2d, 0x58, 0xc8, 0x43f, 0x764, 0xff
.text:0040100A                 mov     ecx, [ecx]       ; [ECX]=5, 0x58, 0xc8, 0x764, 0xff7, 0xfffe7960
.text:0040100C                 cmp     eax, ecx         ; EAX=5, 0x2d, 0x58, 0xc8, 0x43f, 0x764, 0xff7,
.text:0040100E                 jnz     short loc_401013 ; ZF=false
.text:00401010                 xor     eax, eax
.text:00401012                 retn
.text:00401013 ; ---------------------------------------------------------------------------
.text:00401013
.text:00401013 loc_401013:                             ; CODE XREF: PtFuncCompare+E↑j
.text:00401013                 xor     edx, edx
.text:00401015                 cmp     eax, ecx         ; EAX=5, 0x2d, 0x58, 0xc8, 0x43f, 0x764, 0xff7,
.text:00401017                 setnl   dl               ; SF=false,true OF=false
.text:0040101A                 lea     eax, [edx+edx-1]
.text:0040101E                 retn                     ; EAX=1, 0xffffffff
.text:0040101E PtFuncCompare   endp
.text:0040101E
```

Figura 23.4: tracer y IDA. N.B.:

.

.

0x401010 y 0x401012 (): comp().

23.2 GCC

:

Listing 23.3: GCC

```
                lea     eax, [esp+40h+var_28↙
    ↘ ]
                mov     [esp+40h+var_40], ↙
    ↘ eax
                mov     [esp+40h+var_28], ↙
    ↘ 764h
                mov     [esp+40h+var_24], 2↙
    ↘ Dh
                mov     [esp+40h+var_20], 0↙
    ↘ C8h
                mov     [esp+40h+var_1C], 0↙
    ↘ FFFFFF9Eh
                mov     [esp+40h+var_18], 0↙
    ↘ FF7h
                mov     [esp+40h+var_14], 5
                mov     [esp+40h+var_10], 0↙
    ↘ FFFFCFC7h
                mov     [esp+40h+var_C], 43↙
    ↘ Fh
                mov     [esp+40h+var_8], 58h
                mov     [esp+40h+var_4], 0↙
    ↘ FFFE7960h
                mov     [esp+40h+var_34], ↙
    ↘ offset comp
                mov     [esp+40h+var_38], 4
                mov     [esp+40h+var_3C], 0↙
    ↘ Ah
```

```
                call    _qsort
```

:

```
                public comp
comp            proc near

arg_0           = dword ptr  8
arg_4           = dword ptr  0Ch

                push    ebp
                mov     ebp, esp
                mov     eax, [ebp+arg_4]
                mov     ecx, [ebp+arg_0]
                mov     edx, [eax]
                xor     eax, eax
                cmp     [ecx], edx
                jnz     short loc_8048458
                pop     ebp
                retn
loc_8048458:
                setnl   al
                movzx   eax, al
                lea     eax, [eax+eax-1]
                pop     ebp
                retn
comp            endp
```

Listing 23.4: (2.10.1)

```

.text:0002DDF6                 mov     edx, ↙
    ↘ [ebp+arg_10]
```

```
.text:0002DDF9                 mov     [esp↙
    ↘ +4], esi
.text:0002DDFD                 mov     [esp↙
    ↘ ], edi
.text:0002DE00                 mov     [esp↙
    ↘ +8], edx
.text:0002DE04                 call    [ebp+↙
    ↘ arg_C]
...
```

23.2.1 GCC + GDB ()

(). . (p): .

(bt) `msort_with_tmp()`.

Listing 23.5: GDB

```
dennis@ubuntuvm:~/polygon$ gcc 17_1.c -g
dennis@ubuntuvm:~/polygon$ gdb ./a.out
GNU gdb (GDB) 7.6.1-ubuntu
Copyright (C) 2013 Free Software Foundation,↙
    ↘  Inc.
License GPLv3+: GNU GPL version 3 or later <↙
    ↘ http://gnu.org/licenses/gpl.html>
This is free software: you are free to ↙
    ↘ change and redistribute it.
There is NO WARRANTY, to the extent ↙
    ↘ permitted by law.  Type "show copying"
and "show warranty" for details.
This GDB was configured as "i686-linux-gnu".
For bug reporting instructions, please see:
<http://www.gnu.org/software/gdb/bugs/>...
```

```
Reading symbols from /home/dennis/polygon/a.
    out...done.
(gdb) b 17_1.c:11
Breakpoint 1 at 0x804845f: file 17_1.c, line
     11.
(gdb) run
Starting program: /home/dennis/polygon/./a.
    out

Breakpoint 1, comp (_a=0xbffff0f8, _b=
    _b@entry=0xbffff0fc) at 17_1.c:11
11          if (*a==*b)
(gdb) p *a
$1 = 1892
(gdb) p *b
$2 = 45
(gdb) c
Continuing.

Breakpoint 1, comp (_a=0xbffff104, _b=
    _b@entry=0xbffff108) at 17_1.c:11
11          if (*a==*b)
(gdb) p *a
$3 = -98
(gdb) p *b
$4 = 4087
(gdb) bt
#0  comp (_a=0xbffff0f8, _b=_b@entry=0
    xbffff0fc) at 17_1.c:11
#1  0xb7e42872 in msort_with_tmp (p=p@entry
    =0xbffff07c, b=b@entry=0xbffff0f8, n=
    n@entry=2)
    at msort.c:65
```

 CAPÍTULO 23.

```
#2  0xb7e4273e in msort_with_tmp (n=2, b=0xbffff0f8, p=0xbffff07c) at msort.c:45
#3  msort_with_tmp (p=p@entry=0xbffff07c, b=b@entry=0xbffff0f8, n=n@entry=5) at msort.c:53
#4  0xb7e4273e in msort_with_tmp (n=5, b=0xbffff0f8, p=0xbffff07c) at msort.c:45
#5  msort_with_tmp (p=p@entry=0xbffff07c, b=b@entry=0xbffff0f8, n=n@entry=10) at msort.c:53
#6  0xb7e42cef in msort_with_tmp (n=10, b=0xbffff0f8, p=0xbffff07c) at msort.c:45
#7  __GI_qsort_r (b=b@entry=0xbffff0f8, n=n@entry=10, s=s@entry=4, cmp=cmp@entry=0x804844d <comp>,
    arg=arg@entry=0x0) at msort.c:297
#8  0xb7e42dcf in __GI_qsort (b=0xbffff0f8, n=10, s=4, cmp=0x804844d <comp>) at msort.c:307
#9  0x0804850d in main (argc=1, argv=0xbffff1c4) at 17_1.c:26
(gdb)
```

23.2.2 GCC + GDB ()

(`disas`), (*b*) . (`info registers`). (`bt`), .

Listing 23.6: GDB

```
dennis@ubuntuvm:~/polygon$ gcc 17_1.c
dennis@ubuntuvm:~/polygon$ gdb ./a.out
GNU gdb (GDB) 7.6.1-ubuntu
```

```
Copyright (C) 2013 Free Software Foundation,↙
    ↳ Inc.
License GPLv3+: GNU GPL version 3 or later <↙
    ↳ http://gnu.org/licenses/gpl.html>
This is free software: you are free to ↙
    ↳ change and redistribute it.
There is NO WARRANTY, to the extent ↙
    ↳ permitted by law.  Type "show copying"
and "show warranty" for details.
This GDB was configured as "i686-linux-gnu".
For bug reporting instructions, please see:
<http://www.gnu.org/software/gdb/bugs/>...
Reading symbols from /home/dennis/polygon/a.↙
    ↳ out...(no debugging symbols found)...↙
    ↳ done.
(gdb) set disassembly-flavor intel
(gdb) disas comp
Dump of assembler code for function comp:
   0x0804844d <+0>:     push   ebp
   0x0804844e <+1>:     mov    ebp,esp
   0x08048450 <+3>:     sub    esp,0x10
   0x08048453 <+6>:     mov    eax,DWORD PTR↙
    ↳  [ebp+0x8]
   0x08048456 <+9>:     mov    DWORD PTR [↙
    ↳ ebp-0x8],eax
   0x08048459 <+12>:    mov    eax,DWORD PTR↙
    ↳  [ebp+0xc]
   0x0804845c <+15>:    mov    DWORD PTR [↙
    ↳ ebp-0x4],eax
   0x0804845f <+18>:    mov    eax,DWORD PTR↙
    ↳  [ebp-0x8]
   0x08048462 <+21>:    mov    edx,DWORD PTR↙
    ↳  [eax]
```

```
   0x08048464 <+23>:     mov     eax,DWORD PTR↙
    ↳ [ebp-0x4]
   0x08048467 <+26>:     mov     eax,DWORD PTR↙
    ↳ [eax]
   0x08048469 <+28>:     cmp     edx,eax
   0x0804846b <+30>:     jne     0x8048474 <↙
    ↳ comp+39>
   0x0804846d <+32>:     mov     eax,0x0
   0x08048472 <+37>:     jmp     0x804848e <↙
    ↳ comp+65>
   0x08048474 <+39>:     mov     eax,DWORD PTR↙
    ↳ [ebp-0x8]
   0x08048477 <+42>:     mov     edx,DWORD PTR↙
    ↳ [eax]
   0x08048479 <+44>:     mov     eax,DWORD PTR↙
    ↳ [ebp-0x4]
   0x0804847c <+47>:     mov     eax,DWORD PTR↙
    ↳ [eax]
   0x0804847e <+49>:     cmp     edx,eax
   0x08048480 <+51>:     jge     0x8048489 <↙
    ↳ comp+60>
   0x08048482 <+53>:     mov     eax,0↙
    ↳ xffffffff
   0x08048487 <+58>:     jmp     0x804848e <↙
    ↳ comp+65>
   0x08048489 <+60>:     mov     eax,0x1
   0x0804848e <+65>:     leave
   0x0804848f <+66>:     ret
End of assembler dump.
(gdb) b *0x08048469
Breakpoint 1 at 0x8048469
(gdb) run
Starting program: /home/dennis/polygon/./a.↙
```

```
    ↳ out

Breakpoint 1, 0x08048469 in comp ()
(gdb) info registers
eax            0x2d     45
ecx            0xbffff0f8       -1073745672
edx            0x764    1892
ebx            0xb7fc0000       -1208221696
esp            0xbfffeeb8       0xbfffeeb8
ebp            0xbfffeec8       0xbfffeec8
esi            0xbffff0fc       -1073745668
edi            0xbffff010       -1073745904
eip            0x8048469        0x8048469 <↙
    ↳ comp+28>
eflags         0x286    [ PF SF IF ]
cs             0x73     115
ss             0x7b     123
ds             0x7b     123
es             0x7b     123
fs             0x0      0
gs             0x33     51
(gdb) c
Continuing.

Breakpoint 1, 0x08048469 in comp ()
(gdb) info registers
eax            0xff7    4087
ecx            0xbffff104       -1073745660
edx            0xffffff9e       -98
ebx            0xb7fc0000       -1208221696
esp            0xbfffee58       0xbfffee58
ebp            0xbfffee68       0xbfffee68
esi            0xbffff108       -1073745656
```

```
edi            0xbffff010       -1073745904
eip            0x8048469        0x8048469 <
    comp+28>
eflags         0x282    [ SF IF ]
cs             0x73     115
ss             0x7b     123
ds             0x7b     123
es             0x7b     123
fs             0x0      0
gs             0x33     51
(gdb) c
Continuing.

Breakpoint 1, 0x08048469 in comp ()
(gdb) info registers
eax            0xffffff9e       -98
ecx            0xbffff100       -1073745664
edx            0xc8     200
ebx            0xb7fc0000       -1208221696
esp            0xbfffeeb8       0xbfffeeb8
ebp            0xbfffeec8       0xbfffeec8
esi            0xbffff104       -1073745660
edi            0xbffff010       -1073745904
eip            0x8048469        0x8048469 <
    comp+28>
eflags         0x286    [ PF SF IF ]
cs             0x73     115
ss             0x7b     123
ds             0x7b     123
es             0x7b     123
fs             0x0      0
gs             0x33     51
(gdb) bt
```

```
#0  0x08048469 in comp ()
#1  0xb7e42872 in msort_with_tmp (p=p@entry
    =0xbffff07c, b=b@entry=0xbffff0f8, n=
    n@entry=2)
    at msort.c:65
#2  0xb7e4273e in msort_with_tmp (n=2, b=0
    xbffff0f8, p=0xbffff07c) at msort.c:45
#3  msort_with_tmp (p=p@entry=0xbffff07c, b=
    b@entry=0xbffff0f8, n=n@entry=5) at
    msort.c:53
#4  0xb7e4273e in msort_with_tmp (n=5, b=0
    xbffff0f8, p=0xbffff07c) at msort.c:45
#5  msort_with_tmp (p=p@entry=0xbffff07c, b=
    b@entry=0xbffff0f8, n=n@entry=10) at
    msort.c:53
#6  0xb7e42cef in msort_with_tmp (n=10, b=0
    xbffff0f8, p=0xbffff07c) at msort.c:45
#7  __GI_qsort_r (b=b@entry=0xbffff0f8, n=
    n@entry=10, s=s@entry=4, cmp=cmp@entry
    =0x804844d <comp>,
    arg=arg@entry=0x0) at msort.c:297
#8  0xb7e42dcf in __GI_qsort (b=0xbffff0f8,
    n=10, s=4, cmp=0x804844d <comp>) at
    msort.c:307
#9  0x0804850d in main ()
```

Capítulo 24

[1].

24.1

```
#include <stdint.h>

uint64_t f ()
{
        return 0x1234567890ABCDEF;
};
```

24.1.1 x86

[1]: 53.4 on page 1164

Listing 24.1: Con optimización MSVC 2010

```
_f      PROC
        mov     eax, -1867788817                ; 90abcdefH
        mov     edx, 305419896                  ; 12345678H
        ret     0
_f      ENDP
```

24.1.2 ARM

Listing 24.2: Con optimización Keil 6/2013 (Modo ARM)

```
||f|| PROC
        LDR      r0,|L0.12|
        LDR      r1,|L0.16|
        BX       lr
        ENDP

|L0.12|
        DCD      0x90abcdef
|L0.16|
        DCD      0x12345678
```

24.1.3 MIPS

Listing 24.3: Con optimización GCC 4.4.5 (assembly listing)

```
        li      $3,-1867841536                  # 0xffffffff90ab0000
```

```
        li      $2,305397760                          ↙
    ↘     # 0x12340000
        ori     $3,$3,0xcdef
        j       $31
        ori     $2,$2,0x5678
```

Listing 24.4: Con optimización GCC 4.4.5 (IDA)

```
                 lui      $v1, 0x90AB
                 lui      $v0, 0x1234
                 li       $v1, 0x90ABCDEF
                 jr       $ra
                 li       $v0, 0x12345678
```

24.2

```
#include <stdint.h>

uint64_t f_add (uint64_t a, uint64_t b)
{
        return a+b;
};

void f_add_test ()
{
#ifdef __GNUC__
        printf ("%lld\n", f_add↙
    ↘ (12345678901234, 23456789012345));
#else
        printf ("%I64d\n", f_add↙
    ↘ (12345678901234, 23456789012345));
```

```
#endif
};

uint64_t f_sub (uint64_t a, uint64_t b)
{
        return a-b;
};
```

24.2.1 x86

Listing 24.5: Con optimización MSVC 2012 /Ob1

```
_a$ = 8         ; size = 8
_b$ = 16        ; size = 8
_f_add  PROC
        mov     eax, DWORD PTR _a$[esp-4]
        add     eax, DWORD PTR _b$[esp-4]
        mov     edx, DWORD PTR _a$[esp]
        adc     edx, DWORD PTR _b$[esp]
        ret     0
_f_add  ENDP

_f_add_test PROC
        push    5461                ; 00001555H
        push    1972608889          ; 75939f79H
        push    2874                ; 00000b3aH
        push    1942892530          ; 73↙
    ↘ ce2ff_subH
        call    _f_add
        push    edx
        push    eax
```

```
        push    OFFSET $SG1436 ; '%I64d', 0⤦
    ↳ aH, 00H
        call    _printf
        add     esp, 28
        ret     0
_f_add_test ENDP

_f_sub  PROC
        mov     eax, DWORD PTR _a$[esp-4]
        sub     eax, DWORD PTR _b$[esp-4]
        mov     edx, DWORD PTR _a$[esp]
        sbb     edx, DWORD PTR _b$[esp]
        ret     0
_f_sub  ENDP
```

`f_add_test()` .

.

...

. SUB : .

.

Listing 24.6: GCC 4.8.1 -O1 -fno-inline

```
_f_add:
        mov     eax, DWORD PTR [esp+12]
        mov     edx, DWORD PTR [esp+16]
        add     eax, DWORD PTR [esp+4]
        adc     edx, DWORD PTR [esp+8]
```

```
        ret

_f_add_test:
        sub     esp, 28
        mov     DWORD PTR [esp+8], ⤦
    ↳ 1972608889   ; 75939f79H
        mov     DWORD PTR [esp+12], 5461⤦
    ↳          ; 00001555H
        mov     DWORD PTR [esp], 1942892530⤦
    ↳       ; 73ce2ff_subH
        mov     DWORD PTR [esp+4], 2874 ⤦
    ↳          ; 00000b3aH
        call    _f_add
        mov     DWORD PTR [esp+4], eax
        mov     DWORD PTR [esp+8], edx
        mov     DWORD PTR [esp], OFFSET FLAT⤦
    ↳ :LC0 ; "%lld\12\0"
        call    _printf
        add     esp, 28
        ret

_f_sub:
        mov     eax, DWORD PTR [esp+4]
        mov     edx, DWORD PTR [esp+8]
        sub     eax, DWORD PTR [esp+12]
        sbb     edx, DWORD PTR [esp+16]
        ret
```

.

24.2.2 ARM

Listing 24.7: Con optimización Keil 6/2013 (Modo ARM)

```
f_add PROC
        ADDS     r0,r0,r2
        ADC      r1,r1,r3
        BX       lr
        ENDP

f_sub PROC
        SUBS     r0,r0,r2
        SBC      r1,r1,r3
        BX       lr
        ENDP

f_add_test PROC
        PUSH     {r4,lr}
        LDR      r2,|L0.68| ; 0x75939f79
        LDR      r3,|L0.72| ; 0x00001555
        LDR      r0,|L0.76| ; 0x73ce2ff2
        LDR      r1,|L0.80| ; 0x00000b3a
        BL       f_add
        POP      {r4,lr}
        MOV      r2,r0
        MOV      r3,r1
        ADR      r0,|L0.84| ; "%I64d\n"
        B        __2printf
        ENDP

|L0.68|
        DCD      0x75939f79
|L0.72|
        DCD      0x00001555
|L0.76|
```

```
        DCD       0x73ce2ff2
|L0.80|
        DCD       0x00000b3a
|L0.84|
        DCB       "%I64d\n",0
```

24.2.3 MIPS

Listing 24.8: Con optimización GCC 4.4.5 (IDA)

```
f_add:
; $a0 -  a
; $a1 -  a
; $a2 -  b
; $a3 -  b
                addu    $v1, $a3, $a1 ;
                addu    $a0, $a2, $a0 ;
;
;
                sltu    $v0, $v1, $a3
                jr      $ra
;
                addu    $v0, $a0 ; branch ↙
    ↘ delay slot
; $v0 -
; $v1 -

f_sub:
; $a0 -  a
; $a1 -  a
; $a2 -  b
```

```
; $a3 -  b
                subu    $v1, $a1, $a3 ;
                subu    $v0, $a0, $a2 ;
;
;
                sltu    $a1, $v1
                jr      $ra
;
                subu    $v0, $a1 ; branch ⤦
    ↪ delay slot
; $v0 -
; $v1 -

f_add_test:

var_10          = -0x10
var_4           = -4

                lui     $gp, (__gnu_local_gp⤦
    ↪  >> 16)
                addiu   $sp, -0x20
                la      $gp, (__gnu_local_gp⤦
    ↪  & 0xFFFF)
                sw      $ra, 0x20+var_4($sp)
                sw      $gp, 0x20+var_10($sp⤦
    ↪ )
                lui     $a1, 0x73CE
                lui     $a3, 0x7593
                li      $a0, 0xB3A
                li      $a3, 0x75939F79
                li      $a2, 0x1555
                jal     f_add
                li      $a1, 0x73CE2FF2
```

```
                lw      $gp, 0x20+var_10($sp
    ↪ )
                lui     $a0, ($LC0 >> 16)  #
    ↪  "%lld\n"
                lw      $t9, (printf & 0
    ↪ xFFFF)($gp)
                lw      $ra, 0x20+var_4($sp)
                la      $a0, ($LC0 & 0xFFFF)
    ↪   # "%lld\n"
                move    $a3, $v1
                move    $a2, $v0
                jr      $t9
                addiu   $sp, 0x20

$LC0:           .ascii "%lld\n"<0>
```

24.3

```
#include <stdint.h>

uint64_t f_mul (uint64_t a, uint64_t b)
{
        return a*b;
};

uint64_t f_div (uint64_t a, uint64_t b)
{
        return a/b;
};

uint64_t f_rem (uint64_t a, uint64_t b)
```

```
{
        return a % b;
};
```

24.3.1 x86

Listing 24.9: Con optimización MSVC 2013 /Ob1

```
_a$ = 8 ; size = 8
_b$ = 16 ; size = 8
_f_mul  PROC
        push    ebp
        mov     ebp, esp
        mov     eax, DWORD PTR _b$[ebp+4]
        push    eax
        mov     ecx, DWORD PTR _b$[ebp]
        push    ecx
        mov     edx, DWORD PTR _a$[ebp+4]
        push    edx
        mov     eax, DWORD PTR _a$[ebp]
        push    eax
        call    __allmul ; long long ↙
    ↘ multiplication
        pop     ebp
        ret     0
_f_mul  ENDP

_a$ = 8 ; size = 8
_b$ = 16 ; size = 8
_f_div  PROC
        push    ebp
```

```
        mov     ebp, esp
        mov     eax, DWORD PTR _b$[ebp+4]
        push    eax
        mov     ecx, DWORD PTR _b$[ebp]
        push    ecx
        mov     edx, DWORD PTR _a$[ebp+4]
        push    edx
        mov     eax, DWORD PTR _a$[ebp]
        push    eax
        call    __aulldiv ; unsigned long ↙
    ↘ long division
        pop     ebp
        ret     0
_f_div  ENDP

_a$ = 8 ; size = 8
_b$ = 16 ; size = 8
_f_rem  PROC
        push    ebp
        mov     ebp, esp
        mov     eax, DWORD PTR _b$[ebp+4]
        push    eax
        mov     ecx, DWORD PTR _b$[ebp]
        push    ecx
        mov     edx, DWORD PTR _a$[ebp+4]
        push    edx
        mov     eax, DWORD PTR _a$[ebp]
        push    eax
        call    __aullrem ; unsigned long ↙
    ↘ long remainder
        pop     ebp
        ret     0
_f_rem  ENDP
```

.

: E on page 1911.

Listing 24.10: Con optimización GCC 4.8.1 -fno-inline

```
_f_mul:
        push    ebx
        mov     edx, DWORD PTR [esp+8]
        mov     eax, DWORD PTR [esp+16]
        mov     ebx, DWORD PTR [esp+12]
        mov     ecx, DWORD PTR [esp+20]
        imul    ebx, eax
        imul    ecx, edx
        mul     edx
        add     ecx, ebx
        add     edx, ecx
        pop     ebx
        ret

_f_div:
        sub     esp, 28
        mov     eax, DWORD PTR [esp+40]
        mov     edx, DWORD PTR [esp+44]
        mov     DWORD PTR [esp+8], eax
        mov     eax, DWORD PTR [esp+32]
        mov     DWORD PTR [esp+12], edx
        mov     edx, DWORD PTR [esp+36]
        mov     DWORD PTR [esp], eax
        mov     DWORD PTR [esp+4], edx
```

```
        call    ___udivdi3 ; unsigned ⤦
    ↳ division
        add     esp, 28
        ret

_f_rem:
        sub     esp, 28
        mov     eax, DWORD PTR [esp+40]
        mov     edx, DWORD PTR [esp+44]
        mov     DWORD PTR [esp+8], eax
        mov     eax, DWORD PTR [esp+32]
        mov     DWORD PTR [esp+12], edx
        mov     edx, DWORD PTR [esp+36]
        mov     DWORD PTR [esp], eax
        mov     DWORD PTR [esp+4], edx
        call    ___umoddi3 ; unsigned modulo
        add     esp, 28
        ret
```

. : D on page 1910.

24.3.2 ARM

Listing 24.11: Con optimización Keil 6/2013 (Modo Thumb)

```
||f_mul|| PROC
        PUSH     {r4,lr}
        BL       __aeabi_lmul
        POP      {r4,pc}
        ENDP

||f_div|| PROC
```

```
        PUSH     {r4,lr}
        BL       __aeabi_uldivmod
        POP      {r4,pc}
        ENDP

||f_rem|| PROC
        PUSH     {r4,lr}
        BL       __aeabi_uldivmod
        MOVS     r0,r2
        MOVS     r1,r3
        POP      {r4,pc}
        ENDP
```

Listing 24.12: Con optimización Keil 6/2013 (Modo ARM)

```
||f_mul|| PROC
        PUSH     {r4,lr}
        UMULL    r12,r4,r0,r2
        MLA      r1,r2,r1,r4
        MLA      r1,r0,r3,r1
        MOV      r0,r12
        POP      {r4,pc}
        ENDP

||f_div|| PROC
        PUSH     {r4,lr}
        BL       __aeabi_uldivmod
        POP      {r4,pc}
        ENDP

||f_rem|| PROC
        PUSH     {r4,lr}
        BL       __aeabi_uldivmod
```

```
        MOV     r0,r2
        MOV     r1,r3
        POP     {r4,pc}
        ENDP
```

24.3.3 MIPS

Con optimización GCC para MIPS

Listing 24.13: Con optimización GCC 4.4.5 (IDA)

```
f_mul:
                mult    $a2, $a1
                mflo    $v0
                or      $at, $zero  ; NOP
                or      $at, $zero  ; NOP
                mult    $a0, $a3
                mflo    $a0
                addu    $v0, $a0
                or      $at, $zero  ; NOP
                multu   $a3, $a1
                mfhi    $a2
                mflo    $v1
                jr      $ra
                addu    $v0, $a2

f_div:

var_10          = -0x10
var_4           = -4

                lui     $gp, (__gnu_local_gp >> 16)
```

```
                addiu   $sp, -0x20
                la      $gp, (__gnu_local_gp↙
    ↳ & 0xFFFF)
                sw      $ra, 0x20+var_4($sp)
                sw      $gp, 0x20+var_10($sp↙
    ↳ )
                lw      $t9, (__udivdi3 & 0↙
    ↳ xFFFF)($gp)
                or      $at, $zero
                jalr    $t9
                or      $at, $zero
                lw      $ra, 0x20+var_4($sp)
                or      $at, $zero
                jr      $ra
                addiu   $sp, 0x20

f_rem:

var_10          = -0x10
var_4           = -4

                lui     $gp, (__gnu_local_gp↙
    ↳ >> 16)
                addiu   $sp, -0x20
                la      $gp, (__gnu_local_gp↙
    ↳ & 0xFFFF)
                sw      $ra, 0x20+var_4($sp)
                sw      $gp, 0x20+var_10($sp↙
    ↳ )
                lw      $t9, (__umoddi3 & 0↙
    ↳ xFFFF)($gp)
                or      $at, $zero
                jalr    $t9
```

```
                or      $at, $zero
                lw      $ra, 0x20+var_4($sp)
                or      $at, $zero
                jr      $ra
                addiu   $sp, 0x20
```

24.4

```
#include <stdint.h>

uint64_t f (uint64_t a)
{
        return a>>7;
};
```

24.4.1 x86

Listing 24.14: Con optimización MSVC 2012 /Ob1

```
_a$ = 8         ; size = 8
_f      PROC
        mov     eax, DWORD PTR _a$[esp-4]
        mov     edx, DWORD PTR _a$[esp]
        shrd    eax, edx, 7
        shr     edx, 7
        ret     0
_f      ENDP
```

Listing 24.15: Con optimización GCC 4.8.1 -fno-inline

```
_f:
        mov     edx, DWORD PTR [esp+8]
        mov     eax, DWORD PTR [esp+4]
        shrd    eax, edx, 7
        shr     edx, 7
        ret
```

. .

24.4.2 ARM

Listing 24.16: Con optimización Keil 6/2013 (Modo ARM)

```
||f|| PROC
        LSR      r0,r0,#7
        ORR      r0,r0,r1,LSL #25
        LSR      r1,r1,#7
        BX       lr
        ENDP
```

Listing 24.17: Con optimización Keil 6/2013 (Modo Thumb)

```
||f|| PROC
        LSLS     r2,r1,#25
        LSRS     r0,r0,#7
        ORRS     r0,r0,r2
        LSRS     r1,r1,#7
        BX       lr
        ENDP
```

24.4.3 MIPS

Listing 24.18: Con optimización GCC 4.4.5 (IDA)

```
f:
                sll     $v0, $a0, 25
                srl     $v1, $a1, 7
                or      $v1, $v0, $v1
                jr      $ra
                srl     $v0, $a0, 7
```

24.5

```
#include <stdint.h>

int64_t f (int32_t a)
{
        return a;
};
```

24.5.1 x86

Listing 24.19: Con optimización MSVC 2012

```
_a$ = 8
_f      PROC
        mov     eax, DWORD PTR _a$[esp-4]
        cdq
        ret     0
_f      ENDP
```

. . . . .

24.5.2 ARM

Listing 24.20: Con optimización Keil 6/2013 (Modo ARM)

```
||f|| PROC
        ASR      r1,r0,#31
        BX       lr
        ENDP
```

24.5.3 MIPS

Listing 24.21: Con optimización GCC 4.4.5 (IDA)

```
f:
                sra     $v0, $a0, 31
                jr      $ra
                move    $v1, $a0
```

Capítulo 25

SIMD

SIMD[1] : *Single Instruction, Multiple Data.*

SSE

AVX

http://go.yurichev.com/17313

25.1

.

```
for (i = 0; i < 1024; i++)
```

[1] Single instruction, multiple data

```
{
    C[i] = A[i]*B[i];
}
```

25.1.1

```
int f (int sz, int *ar1, int *ar2, int *ar3)
{
        for (int i=0; i<sz; i++)
                ar3[i]=ar1[i]+ar2[i];

        return 0;
};
```

Intel C++

Intel C++ 11.1.051 win32:

`icl intel.cpp /QaxSSE2 /Faintel.asm /Ox`

```
; int __cdecl f(int, int *, int *, int *)
                public ?f@@YAHHPAH00@Z
?f@@YAHHPAH00@Z proc near

var_10 = dword ptr -10h
sz     = dword ptr  4
ar1    = dword ptr  8
ar2    = dword ptr  0Ch
ar3    = dword ptr  10h
```

```
        push    edi
        push    esi
        push    ebx
        push    esi
        mov     edx, [esp+10h+sz]
        test    edx, edx
        jle     loc_15B
        mov     eax, [esp+10h+ar3]
        cmp     edx, 6
        jle     loc_143
        cmp     eax, [esp+10h+ar2]
        jbe     short loc_36
        mov     esi, [esp+10h+ar2]
        sub     esi, eax
        lea     ecx, ds:0[edx*4]
        neg     esi
        cmp     ecx, esi
        jbe     short loc_55

loc_36: ; CODE XREF: f(int,int *,int *,int ↙
    ↘ *)+21
        cmp     eax, [esp+10h+ar2]
        jnb     loc_143
        mov     esi, [esp+10h+ar2]
        sub     esi, eax
        lea     ecx, ds:0[edx*4]
        cmp     esi, ecx
        jb      loc_143

loc_55: ; CODE XREF: f(int,int *,int *,int ↙
    ↘ *)+34
        cmp     eax, [esp+10h+ar1]
        jbe     short loc_67
```

```
        mov     esi, [esp+10h+ar1]
        sub     esi, eax
        neg     esi
        cmp     ecx, esi
        jbe     short loc_7F

loc_67: ; CODE XREF: f(int,int *,int *,int ↙
    ↘ *)+59
        cmp     eax, [esp+10h+ar1]
        jnb     loc_143
        mov     esi, [esp+10h+ar1]
        sub     esi, eax
        cmp     esi, ecx
        jb      loc_143

loc_7F: ; CODE XREF: f(int,int *,int *,int ↙
    ↘ *)+65
        mov     edi, eax        ; edi = ar1
        and     edi, 0Fh        ; ?
        jz      short loc_9A    ;
        test    edi, 3
        jnz     loc_162
        neg     edi
        add     edi, 10h
        shr     edi, 2

loc_9A: ; CODE XREF: f(int,int *,int *,int ↙
    ↘ *)+84
        lea     ecx, [edi+4]
        cmp     edx, ecx
        jl      loc_162
        mov     ecx, edx
        sub     ecx, edi
```

```
        and     ecx, 3
        neg     ecx
        add     ecx, edx
        test    edi, edi
        jbe     short loc_D6
        mov     ebx, [esp+10h+ar2]
        mov     [esp+10h+var_10], ecx
        mov     ecx, [esp+10h+ar1]
        xor     esi, esi

loc_C1: ; CODE XREF: f(int,int *,int *,int ↙
    ↪ *)+CD
        mov     edx, [ecx+esi*4]
        add     edx, [ebx+esi*4]
        mov     [eax+esi*4], edx
        inc     esi
        cmp     esi, edi
        jb      short loc_C1
        mov     ecx, [esp+10h+var_10]
        mov     edx, [esp+10h+sz]

loc_D6: ; CODE XREF: f(int,int *,int *,int ↙
    ↪ *)+B2
        mov     esi, [esp+10h+ar2]
        lea     esi, [esi+edi*4] ; ar2+i*4 ?
        test    esi, 0Fh
        jz      short loc_109   ; !
        mov     ebx, [esp+10h+ar1]
        mov     esi, [esp+10h+ar2]

loc_ED: ; CODE XREF: f(int,int *,int *,int ↙
    ↪ *)+105
        movdqu  xmm1, xmmword ptr [ebx+edi*4]↙
```

```
    ↘ ; ar1+i*4
        movdqu  xmm0, xmmword ptr [esi+edi*4]↙
    ↘ ; ar2+i*4  XMM0
        paddd   xmm1, xmm0
        movdqa  xmmword ptr [eax+edi*4], xmm1↙
    ↘ ; ar3+i*4
        add     edi, 4
        cmp     edi, ecx
        jb      short loc_ED
        jmp     short loc_127

loc_109: ; CODE XREF: f(int,int *,int *,int ↙
    ↘ *)+E3
        mov     ebx, [esp+10h+ar1]
        mov     esi, [esp+10h+ar2]

loc_111: ; CODE XREF: f(int,int *,int *,int ↙
    ↘ *)+125
        movdqu  xmm0, xmmword ptr [ebx+edi*4]
        paddd   xmm0, xmmword ptr [esi+edi*4]
        movdqa  xmmword ptr [eax+edi*4], xmm0
        add     edi, 4
        cmp     edi, ecx
        jb      short loc_111

loc_127: ; CODE XREF: f(int,int *,int *,int ↙
    ↘ *)+107
          ; f(int,int *,int *,int *)+164
        cmp     ecx, edx
        jnb     short loc_15B
        mov     esi, [esp+10h+ar1]
        mov     edi, [esp+10h+ar2]
```

```
loc_133: ; CODE XREF: f(int,int *,int *,int ⤦
    ↳ *)+13F
        mov     ebx, [esi+ecx*4]
        add     ebx, [edi+ecx*4]
        mov     [eax+ecx*4], ebx
        inc     ecx
        cmp     ecx, edx
        jb      short loc_133
        jmp     short loc_15B

loc_143: ; CODE XREF: f(int,int *,int *,int ⤦
    ↳ *)+17
          ; f(int,int *,int *,int *)+3A ...
        mov     esi, [esp+10h+ar1]
        mov     edi, [esp+10h+ar2]
        xor     ecx, ecx

loc_14D: ; CODE XREF: f(int,int *,int *,int ⤦
    ↳ *)+159
        mov     ebx, [esi+ecx*4]
        add     ebx, [edi+ecx*4]
        mov     [eax+ecx*4], ebx
        inc     ecx
        cmp     ecx, edx
        jb      short loc_14D

loc_15B: ; CODE XREF: f(int,int *,int *,int ⤦
    ↳ *)+A
          ; f(int,int *,int *,int *)+129 ...
        xor     eax, eax
        pop     ecx
        pop     ebx
        pop     esi
```

```
        pop     edi
        retn

loc_162: ; CODE XREF: f(int,int *,int *,int ⤦
    ↪ *)+8C
          ; f(int,int *,int *,int *)+9F
        xor     ecx, ecx
        jmp     short loc_127
?f@@YAHHPAH00@Z endp
```

- MOVDQU (*Move Unaligned Double Quadword*) .
- PADDD (*Add Packed Integers*)
- MOVDQA (*Move Aligned Double Quadword*)

```
movdqu  xmm0, xmmword ptr [ebx+edi*4] ; ar1+⤦
    ↪ i*4
paddd   xmm0, xmmword ptr [esi+edi*4] ; ar2+⤦
    ↪ i*4
movdqa  xmmword ptr [eax+edi*4], xmm0 ; ar3+⤦
    ↪ i*4
```

```
movdqu  xmm1, xmmword ptr [ebx+edi*4] ; ar1+⤦
    ↪ i*4
movdqu  xmm0, xmmword ptr [esi+edi*4] ; ar2+⤦
    ↪ i*4  XMM0
paddd   xmm1, xmm0
movdqa  xmmword ptr [eax+edi*4], xmm1 ; ar3+⤦
    ↪ i*4
```

GCC

(GCC 4.4.1):

```
; f(int, int *, int *, int *)
            public _Z1fiPiS_S_
_Z1fiPiS_S_ proc near

var_18      = dword ptr -18h
var_14      = dword ptr -14h
var_10      = dword ptr -10h
arg_0       = dword ptr  8
arg_4       = dword ptr  0Ch
arg_8       = dword ptr  10h
arg_C       = dword ptr  14h

            push    ebp
            mov     ebp, esp
            push    edi
            push    esi
            push    ebx
            sub     esp, 0Ch
            mov     ecx, [ebp+arg_0]
            mov     esi, [ebp+arg_4]
            mov     edi, [ebp+arg_8]
            mov     ebx, [ebp+arg_C]
            test    ecx, ecx
            jle     short loc_80484D8
            cmp     ecx, 6
            lea     eax, [ebx+10h]
            ja      short loc_80484E8

loc_80484C1: ; CODE XREF: f(int,int *,int *,↙
```

```
    ↳ int *)+4B
             ; f(int,int *,int *,int *)+61 ↙
    ↳ ...
            xor     eax, eax
            nop
            lea     esi, [esi+0]

loc_80484C8: ; CODE XREF: f(int,int *,int *,↙
    ↳ int *)+36
            mov     edx, [edi+eax*4]
            add     edx, [esi+eax*4]
            mov     [ebx+eax*4], edx
            add     eax, 1
            cmp     eax, ecx
            jnz     short loc_80484C8

loc_80484D8: ; CODE XREF: f(int,int *,int *,↙
    ↳ int *)+17
             ; f(int,int *,int *,int *)+A5
            add     esp, 0Ch
            xor     eax, eax
            pop     ebx
            pop     esi
            pop     edi
            pop     ebp
            retn

            align 8

loc_80484E8: ; CODE XREF: f(int,int *,int *,↙
    ↳ int *)+1F
            test    bl, 0Fh
            jnz     short loc_80484C1
```

```
            lea     edx, [esi+10h]
            cmp     ebx, edx
            jbe     loc_8048578

loc_80484F8: ; CODE XREF: f(int,int *,int *,↙
    ↘ int *)+E0
            lea     edx, [edi+10h]
            cmp     ebx, edx
            ja      short loc_8048503
            cmp     edi, eax
            jbe     short loc_80484C1

loc_8048503: ; CODE XREF: f(int,int *,int *,↙
    ↘ int *)+5D
            mov     eax, ecx
            shr     eax, 2
            mov     [ebp+var_14], eax
            shl     eax, 2
            test    eax, eax
            mov     [ebp+var_10], eax
            jz      short loc_8048547
            mov     [ebp+var_18], ecx
            mov     ecx, [ebp+var_14]
            xor     eax, eax
            xor     edx, edx
            nop

loc_8048520: ; CODE XREF: f(int,int *,int *,↙
    ↘ int *)+9B
            movdqu  xmm1, xmmword ptr [edi+↙
    ↘ eax]
            movdqu  xmm0, xmmword ptr [esi+↙
    ↘ eax]
```

```
            add     edx, 1
            paddd   xmm0, xmm1
            movdqa  xmmword ptr [ebx+eax], ⤦
    ⤷ xmm0
            add     eax, 10h
            cmp     edx, ecx
            jb      short loc_8048520
            mov     ecx, [ebp+var_18]
            mov     eax, [ebp+var_10]
            cmp     ecx, eax
            jz      short loc_80484D8

loc_8048547: ; CODE XREF: f(int,int *,int *,⤦
    ⤷ int *)+73
            lea     edx, ds:0[eax*4]
            add     esi, edx
            add     edi, edx
            add     ebx, edx
            lea     esi, [esi+0]

loc_8048558: ; CODE XREF: f(int,int *,int *,⤦
    ⤷ int *)+CC
            mov     edx, [edi]
            add     eax, 1
            add     edi, 4
            add     edx, [esi]
            add     esi, 4
            mov     [ebx], edx
            add     ebx, 4
            cmp     ecx, eax
            jg      short loc_8048558
            add     esp, 0Ch
            xor     eax, eax
```

```
            pop     ebx
            pop     esi
            pop     edi
            pop     ebp
            retn

loc_8048578: ; CODE XREF: f(int,int *,int *,↙
    ↘ int *)+52
            cmp     eax, esi
            jnb     loc_80484C1
            jmp     loc_80484F8
_Z1fiPiS_S_ endp
```

25.1.2

(14.2 on page 358):

```
#include <stdio.h>

void my_memcpy (unsigned char* dst, unsigned↙
    ↘  char* src, size_t cnt)
{
        size_t i;
        for (i=0; i<cnt; i++)
                dst[i]=src[i];
};
```

Listing 25.1: Con optimización GCC 4.9.1 x64

```


my_memcpy:
; RDI =
```

```
; RSI =
; RDX =
        test    rdx, rdx
        je      .L41
        lea     rax, [rdi+16]
        cmp     rsi, rax
        lea     rax, [rsi+16]
        setae   cl
        cmp     rdi, rax
        setae   al
        or      cl, al
        je      .L13
        cmp     rdx, 22
        jbe     .L13
        mov     rcx, rsi
        push    rbp
        push    rbx
        neg     rcx
        and     ecx, 15
        cmp     rcx, rdx
        cmova   rcx, rdx
        xor     eax, eax
        test    rcx, rcx
        je      .L4
        movzx   eax, BYTE PTR [rsi]
        cmp     rcx, 1
        mov     BYTE PTR [rdi], al
        je      .L15
        movzx   eax, BYTE PTR [rsi+1]
        cmp     rcx, 2
        mov     BYTE PTR [rdi+1], al
        je      .L16
        movzx   eax, BYTE PTR [rsi+2]
```

```
        cmp     rcx, 3
        mov     BYTE PTR [rdi+2], al
        je      .L17
        movzx   eax, BYTE PTR [rsi+3]
        cmp     rcx, 4
        mov     BYTE PTR [rdi+3], al
        je      .L18
        movzx   eax, BYTE PTR [rsi+4]
        cmp     rcx, 5
        mov     BYTE PTR [rdi+4], al
        je      .L19
        movzx   eax, BYTE PTR [rsi+5]
        cmp     rcx, 6
        mov     BYTE PTR [rdi+5], al
        je      .L20
        movzx   eax, BYTE PTR [rsi+6]
        cmp     rcx, 7
        mov     BYTE PTR [rdi+6], al
        je      .L21
        movzx   eax, BYTE PTR [rsi+7]
        cmp     rcx, 8
        mov     BYTE PTR [rdi+7], al
        je      .L22
        movzx   eax, BYTE PTR [rsi+8]
        cmp     rcx, 9
        mov     BYTE PTR [rdi+8], al
        je      .L23
        movzx   eax, BYTE PTR [rsi+9]
        cmp     rcx, 10
        mov     BYTE PTR [rdi+9], al
        je      .L24
        movzx   eax, BYTE PTR [rsi+10]
        cmp     rcx, 11
```

 CAPÍTULO 25. SIMD

```
        mov     BYTE PTR [rdi+10], al
        je      .L25
        movzx   eax, BYTE PTR [rsi+11]
        cmp     rcx, 12
        mov     BYTE PTR [rdi+11], al
        je      .L26
        movzx   eax, BYTE PTR [rsi+12]
        cmp     rcx, 13
        mov     BYTE PTR [rdi+12], al
        je      .L27
        movzx   eax, BYTE PTR [rsi+13]
        cmp     rcx, 15
        mov     BYTE PTR [rdi+13], al
        jne     .L28
        movzx   eax, BYTE PTR [rsi+14]
        mov     BYTE PTR [rdi+14], al
        mov     eax, 15
.L4:
        mov     r10, rdx
        lea     r9, [rdx-1]
        sub     r10, rcx
        lea     r8, [r10-16]
        sub     r9, rcx
        shr     r8, 4
        add     r8, 1
        mov     r11, r8
        sal     r11, 4
        cmp     r9, 14
        jbe     .L6
        lea     rbp, [rsi+rcx]
        xor     r9d, r9d
        add     rcx, rdi
        xor     ebx, ebx
```

```
.L7:
        movdqa  xmm0, XMMWORD PTR [rbp+0+r9]
        add     rbx, 1
        movups  XMMWORD PTR [rcx+r9], xmm0
        add     r9, 16
        cmp     rbx, r8
        jb      .L7
        add     rax, r11
        cmp     r10, r11
        je      .L1
.L6:
        movzx   ecx, BYTE PTR [rsi+rax]
        mov     BYTE PTR [rdi+rax], cl
        lea     rcx, [rax+1]
        cmp     rdx, rcx
        jbe     .L1
        movzx   ecx, BYTE PTR [rsi+1+rax]
        mov     BYTE PTR [rdi+1+rax], cl
        lea     rcx, [rax+2]
        cmp     rdx, rcx
        jbe     .L1
        movzx   ecx, BYTE PTR [rsi+2+rax]
        mov     BYTE PTR [rdi+2+rax], cl
        lea     rcx, [rax+3]
        cmp     rdx, rcx
        jbe     .L1
        movzx   ecx, BYTE PTR [rsi+3+rax]
        mov     BYTE PTR [rdi+3+rax], cl
        lea     rcx, [rax+4]
        cmp     rdx, rcx
        jbe     .L1
        movzx   ecx, BYTE PTR [rsi+4+rax]
        mov     BYTE PTR [rdi+4+rax], cl
```

```
        lea     rcx, [rax+5]
        cmp     rdx, rcx
        jbe     .L1
        movzx   ecx, BYTE PTR [rsi+5+rax]
        mov     BYTE PTR [rdi+5+rax], cl
        lea     rcx, [rax+6]
        cmp     rdx, rcx
        jbe     .L1
        movzx   ecx, BYTE PTR [rsi+6+rax]
        mov     BYTE PTR [rdi+6+rax], cl
        lea     rcx, [rax+7]
        cmp     rdx, rcx
        jbe     .L1
        movzx   ecx, BYTE PTR [rsi+7+rax]
        mov     BYTE PTR [rdi+7+rax], cl
        lea     rcx, [rax+8]
        cmp     rdx, rcx
        jbe     .L1
        movzx   ecx, BYTE PTR [rsi+8+rax]
        mov     BYTE PTR [rdi+8+rax], cl
        lea     rcx, [rax+9]
        cmp     rdx, rcx
        jbe     .L1
        movzx   ecx, BYTE PTR [rsi+9+rax]
        mov     BYTE PTR [rdi+9+rax], cl
        lea     rcx, [rax+10]
        cmp     rdx, rcx
        jbe     .L1
        movzx   ecx, BYTE PTR [rsi+10+rax]
        mov     BYTE PTR [rdi+10+rax], cl
        lea     rcx, [rax+11]
        cmp     rdx, rcx
        jbe     .L1
```

```
        movzx   ecx, BYTE PTR [rsi+11+rax]
        mov     BYTE PTR [rdi+11+rax], cl
        lea     rcx, [rax+12]
        cmp     rdx, rcx
        jbe     .L1
        movzx   ecx, BYTE PTR [rsi+12+rax]
        mov     BYTE PTR [rdi+12+rax], cl
        lea     rcx, [rax+13]
        cmp     rdx, rcx
        jbe     .L1
        movzx   ecx, BYTE PTR [rsi+13+rax]
        mov     BYTE PTR [rdi+13+rax], cl
        lea     rcx, [rax+14]
        cmp     rdx, rcx
        jbe     .L1
        movzx   edx, BYTE PTR [rsi+14+rax]
        mov     BYTE PTR [rdi+14+rax], dl
.L1:
        pop     rbx
        pop     rbp
.L41:
        rep ret
.L13:
        xor     eax, eax
.L3:
        movzx   ecx, BYTE PTR [rsi+rax]
        mov     BYTE PTR [rdi+rax], cl
        add     rax, 1
        cmp     rax, rdx
        jne     .L3
        rep ret
.L28:
        mov     eax, 14
```

```
        jmp     .L4
.L15:
        mov     eax, 1
        jmp     .L4
.L16:
        mov     eax, 2
        jmp     .L4
.L17:
        mov     eax, 3
        jmp     .L4
.L18:
        mov     eax, 4
        jmp     .L4
.L19:
        mov     eax, 5
        jmp     .L4
.L20:
        mov     eax, 6
        jmp     .L4
.L21:
        mov     eax, 7
        jmp     .L4
.L22:
        mov     eax, 8
        jmp     .L4
.L23:
        mov     eax, 9
        jmp     .L4
.L24:
        mov     eax, 10
        jmp     .L4
.L25:
        mov     eax, 11
```

```
        jmp     .L4
.L26:
        mov     eax, 12
        jmp     .L4
.L27:
        mov     eax, 13
        jmp     .L4
```

25.2

[2].

```
size_t strlen_sse2(const char *str)
{
    register size_t len = 0;
    const char *s=str;
    bool str_is_aligned=(((unsigned int)str)↙
    ↪ &0xFFFFFFF0) == (unsigned int)str;

    if (str_is_aligned==false)
        return strlen (str);

    __m128i xmm0 = _mm_setzero_si128();
    __m128i xmm1;
    int mask = 0;

    for (;;)
    {
        xmm1 = _mm_load_si128((__m128i *)s);
```

2

```
        xmm1 = _mm_cmpeq_epi8(xmm1, xmm0);
        if ((mask = _mm_movemask_epi8(xmm1)) ↙
    ↘ != 0)
        {
            unsigned long pos;
            _BitScanForward(&pos, mask);
            len += (size_t)pos;

            break;
        }
        s += sizeof(__m128i);
        len += sizeof(__m128i);
    };

    return len;
}
```

Listing 25.2: Con optimización MSVC 2010

```

_pos$75552 = -4          ; size = 4
_str$ = 8                ; size = 4
?strlen_sse2@@YAIPBD@Z PROC ; strlen_sse2

    push     ebp
    mov      ebp, esp
    and      esp, -16              ; fffffff0H
    mov      eax, DWORD PTR _str$[ebp]
    sub      esp, 12               ; 0000000cH
    push     esi
    mov      esi, eax
    and      esi, -16              ; fffffff0H
    xor      edx, edx
```

```
    mov      ecx, eax
    cmp      esi, eax
    je       SHORT $LN4@strlen_sse
    lea      edx, DWORD PTR [eax+1]
    npad     3 ;
$LL11@strlen_sse:
    mov      cl, BYTE PTR [eax]
    inc      eax
    test     cl, cl
    jne      SHORT $LL11@strlen_sse
    sub      eax, edx
    pop      esi
    mov      esp, ebp
    pop      ebp
    ret      0
$LN4@strlen_sse:
    movdqa   xmm1, XMMWORD PTR [eax]
    pxor     xmm0, xmm0
    pcmpeqb  xmm1, xmm0
    pmovmskb eax, xmm1
    test     eax, eax
    jne      SHORT $LN9@strlen_sse
$LL3@strlen_sse:
    movdqa   xmm1, XMMWORD PTR [ecx+16]
    add      ecx, 16                          ; ↙
    ↘ 00000010H
    pcmpeqb  xmm1, xmm0
    add      edx, 16                          ; ↙
    ↘ 00000010H
    pmovmskb eax, xmm1
    test     eax, eax
    je       SHORT $LL3@strlen_sse
$LN9@strlen_sse:
```

```
    bsf       eax, eax
    mov       ecx, eax
    mov       DWORD PTR _pos$75552[esp+16], ↙
    ↳ eax
    lea       eax, DWORD PTR [ecx+edx]
    pop       esi
    mov       esp, ebp
    pop       ebp
    ret       0
?strlen_sse2@@YAIPBD@Z ENDP                    ;↙
    ↳  strlen_sse2
```

0x008c1ff8	'h'
0x008c1ff9	'e'
0x008c1ffa	'l'
0x008c1ffb	'l'
0x008c1ffc	'o'
0x008c1ffd	'\x00'
0x008c1ffe	
0x008c1fff	

.

`_mm_setzero_si128()`.

`_mm_load_si128()`

`_mm_cmpeq_epi8()`

```
XMM1: 11223344556677880000000000000000
XMM0: 11ab3444007877881111111111111111
```

```
XMM1: ff0000ff0000ffff0000000000000000
```

 CAPÍTULO 25. SIMD

`pmovmskb eax, xmm1`

:

15	14	13	12	11	10	9	3	2	1	0
«h»	«e»	«l»	«l»	«o»	0	Basura		0	Basura	

`XMM1: 0000ff0000000000000ff0000000000`

0010000000100000b.

BSF (*Bit Scan Forward*).

EAX=0010000000100000b

`_BitScanForward`.

http://go.yurichev.com/17331

Capítulo 26

26.1 x86-64

- RAX, RBX, RCX, RDX, RBP, RSP, RSI, RDI, R8, R9, R10, R11, R12, R13, R14, R15.

Spanish text placeholder		
RAX^{x64}		
	EAX	
	AX	
	AH	AL

Spanish text placeholder	
R8	
	R8D
	R8W
	R8L

XMM0-XMM15.

- System V AMD64 ABI (Linux, *BSD, Mac OS X)[Mit13] fastcall, RDI, RSI, RDX, RCX, R8, R9 . .

  (64 on page 1299).

-

- .

```
/*
 * Generated S-box files.
 *
 * This software may be modified, ↙
    ↳ redistributed, and used for any ↙
    ↳ purpose,
 * so long as its origin is acknowledged.
 *
 * Produced by Matthew Kwan - March 1998
 */

#ifdef _WIN64
#define DES_type unsigned __int64
#else
#define DES_type unsigned int
#endif

void
s1 (
    DES_type    a1,
    DES_type    a2,
    DES_type    a3,
    DES_type    a4,
```

```
    DES_type    a5,
    DES_type    a6,
    DES_type    *out1,
    DES_type    *out2,
    DES_type    *out3,
    DES_type    *out4
) {
    DES_type    x1, x2, x3, x4, x5, x6, x7, ↙
    ↳ x8;
    DES_type    x9, x10, x11, x12, x13, x14,↙
    ↳  x15, x16;
    DES_type    x17, x18, x19, x20, x21, x22↙
    ↳ , x23, x24;
    DES_type    x25, x26, x27, x28, x29, x30↙
    ↳ , x31, x32;
    DES_type    x33, x34, x35, x36, x37, x38↙
    ↳ , x39, x40;
    DES_type    x41, x42, x43, x44, x45, x46↙
    ↳ , x47, x48;
    DES_type    x49, x50, x51, x52, x53, x54↙
    ↳ , x55, x56;

    x1 = a3 & ~a5;
    x2 = x1 ^ a4;
    x3 = a3 & ~a4;
    x4 = x3 | a5;
    x5 = a6 & x4;
    x6 = x2 ^ x5;
    x7 = a4 & ~a5;
    x8 = a3 ^ a4;
    x9 = a6 & ~x8;
    x10 = x7 ^ x9;
    x11 = a2 | x10;
```

```
    x12 = x6 ^ x11;
    x13 = a5 ^ x5;
    x14 = x13 & x8;
    x15 = a5 & ~a4;
    x16 = x3 ^ x14;
    x17 = a6 | x16;
    x18 = x15 ^ x17;
    x19 = a2 | x18;
    x20 = x14 ^ x19;
    x21 = a1 & x20;
    x22 = x12 ^ ~x21;
    *out2 ^= x22;
    x23 = x1 | x5;
    x24 = x23 ^ x8;
    x25 = x18 & ~x2;
    x26 = a2 & ~x25;
    x27 = x24 ^ x26;
    x28 = x6 | x7;
    x29 = x28 ^ x25;
    x30 = x9 ^ x24;
    x31 = x18 & ~x30;
    x32 = a2 & x31;
    x33 = x29 ^ x32;
    x34 = a1 & x33;
    x35 = x27 ^ x34;
    *out4 ^= x35;
    x36 = a3 & x28;
    x37 = x18 & ~x36;
    x38 = a2 | x3;
    x39 = x37 ^ x38;
    x40 = a3 | x31;
    x41 = x24 & ~x37;
    x42 = x41 | x3;
```

```
    x43 = x42 & ~a2;
    x44 = x40 ^ x43;
    x45 = a1 & ~x44;
    x46 = x39 ^ ~x45;
    *out1 ^= x46;
    x47 = x33 & ~x9;
    x48 = x47 ^ x39;
    x49 = x4 ^ x36;
    x50 = x49 & ~x5;
    x51 = x42 | x18;
    x52 = x51 ^ a5;
    x53 = a2 & ~x52;
    x54 = x50 ^ x53;
    x55 = a1 | x54;
    x56 = x48 ^ ~x55;
    *out3 ^= x56;
}
```

Listing 26.1: Con optimización MSVC 2008

```
PUBLIC    _s1
; Function compile flags: /Ogtpy
_TEXT    SEGMENT
_x6$ = -20          ; size = 4
_x3$ = -16          ; size = 4
_x1$ = -12          ; size = 4
_x8$ = -8           ; size = 4
_x4$ = -4           ; size = 4
_a1$ = 8            ; size = 4
_a2$ = 12           ; size = 4
_a3$ = 16           ; size = 4
_x33$ = 20          ; size = 4
_x7$ = 20           ; size = 4
```

```
_a4$ = 20           ; size = 4
_a5$ = 24           ; size = 4
tv326 = 28          ; size = 4
_x36$ = 28          ; size = 4
_x28$ = 28          ; size = 4
_a6$ = 28           ; size = 4
_out1$ = 32         ; size = 4
_x24$ = 36          ; size = 4
_out2$ = 36         ; size = 4
_out3$ = 40         ; size = 4
_out4$ = 44         ; size = 4
_s1    PROC
    sub    esp, 20                          ; ↙
    ↘ 00000014H
    mov    edx, DWORD PTR _a5$[esp+16]
    push   ebx
    mov    ebx, DWORD PTR _a4$[esp+20]
    push   ebp
    push   esi
    mov    esi, DWORD PTR _a3$[esp+28]
    push   edi
    mov    edi, ebx
    not    edi
    mov    ebp, edi
    and    edi, DWORD PTR _a5$[esp+32]
    mov    ecx, edx
    not    ecx
    and    ebp, esi
    mov    eax, ecx
    and    eax, esi
    and    ecx, ebx
    mov    DWORD PTR _x1$[esp+36], eax
    xor    eax, ebx
```

```
    mov     esi, ebp
    or      esi, edx
    mov     DWORD PTR _x4$[esp+36], esi
    and     esi, DWORD PTR _a6$[esp+32]
    mov     DWORD PTR _x7$[esp+32], ecx
    mov     edx, esi
    xor     edx, eax
    mov     DWORD PTR _x6$[esp+36], edx
    mov     edx, DWORD PTR _a3$[esp+32]
    xor     edx, ebx
    mov     ebx, esi
    xor     ebx, DWORD PTR _a5$[esp+32]
    mov     DWORD PTR _x8$[esp+36], edx
    and     ebx, edx
    mov     ecx, edx
    mov     edx, ebx
    xor     edx, ebp
    or      edx, DWORD PTR _a6$[esp+32]
    not     ecx
    and     ecx, DWORD PTR _a6$[esp+32]
    xor     edx, edi
    mov     edi, edx
    or      edi, DWORD PTR _a2$[esp+32]
    mov     DWORD PTR _x3$[esp+36], ebp
    mov     ebp, DWORD PTR _a2$[esp+32]
    xor     edi, ebx
    and     edi, DWORD PTR _a1$[esp+32]
    mov     ebx, ecx
    xor     ebx, DWORD PTR _x7$[esp+32]
    not     edi
    or      ebx, ebp
    xor     edi, ebx
    mov     ebx, edi
```

```
    mov     edi, DWORD PTR _out2$[esp+32]
    xor     ebx, DWORD PTR [edi]
    not     eax
    xor     ebx, DWORD PTR _x6$[esp+36]
    and     eax, edx
    mov     DWORD PTR [edi], ebx
    mov     ebx, DWORD PTR _x7$[esp+32]
    or      ebx, DWORD PTR _x6$[esp+36]
    mov     edi, esi
    or      edi, DWORD PTR _x1$[esp+36]
    mov     DWORD PTR _x28$[esp+32], ebx
    xor     edi, DWORD PTR _x8$[esp+36]
    mov     DWORD PTR _x24$[esp+32], edi
    xor     edi, ecx
    not     edi
    and     edi, edx
    mov     ebx, edi
    and     ebx, ebp
    xor     ebx, DWORD PTR _x28$[esp+32]
    xor     ebx, eax
    not     eax
    mov     DWORD PTR _x33$[esp+32], ebx
    and     ebx, DWORD PTR _a1$[esp+32]
    and     eax, ebp
    xor     eax, ebx
    mov     ebx, DWORD PTR _out4$[esp+32]
    xor     eax, DWORD PTR [ebx]
    xor     eax, DWORD PTR _x24$[esp+32]
    mov     DWORD PTR [ebx], eax
    mov     eax, DWORD PTR _x28$[esp+32]
    and     eax, DWORD PTR _a3$[esp+32]
    mov     ebx, DWORD PTR _x3$[esp+36]
    or      edi, DWORD PTR _a3$[esp+32]
```

```
    mov     DWORD PTR _x36$[esp+32], eax
    not     eax
    and     eax, edx
    or      ebx, ebp
    xor     ebx, eax
    not     eax
    and     eax, DWORD PTR _x24$[esp+32]
    not     ebp
    or      eax, DWORD PTR _x3$[esp+36]
    not     esi
    and     ebp, eax
    or      eax, edx
    xor     eax, DWORD PTR _a5$[esp+32]
    mov     edx, DWORD PTR _x36$[esp+32]
    xor     edx, DWORD PTR _x4$[esp+36]
    xor     ebp, edi
    mov     edi, DWORD PTR _out1$[esp+32]
    not     eax
    and     eax, DWORD PTR _a2$[esp+32]
    not     ebp
    and     ebp, DWORD PTR _a1$[esp+32]
    and     edx, esi
    xor     eax, edx
    or      eax, DWORD PTR _a1$[esp+32]
    not     ebp
    xor     ebp, DWORD PTR [edi]
    not     ecx
    and     ecx, DWORD PTR _x33$[esp+32]
    xor     ebp, ebx
    not     eax
    mov     DWORD PTR [edi], ebp
    xor     eax, ecx
    mov     ecx, DWORD PTR _out3$[esp+32]
```

```
    xor     eax, DWORD PTR [ecx]
    pop     edi
    pop     esi
    xor     eax, ebx
    pop     ebp
    mov     DWORD PTR [ecx], eax
    pop     ebx
    add     esp, 20                          ; ⤦
    ↳ 00000014H
    ret     0
_s1    ENDP
```

Listing 26.2: Con optimización MSVC 2008

```
a1$ = 56
a2$ = 64
a3$ = 72
a4$ = 80
x36$1$ = 88
a5$ = 88
a6$ = 96
out1$ = 104
out2$ = 112
out3$ = 120
out4$ = 128
s1    PROC
$LN3:
    mov     QWORD PTR [rsp+24], rbx
    mov     QWORD PTR [rsp+32], rbp
    mov     QWORD PTR [rsp+16], rdx
    mov     QWORD PTR [rsp+8], rcx
    push    rsi
    push    rdi
```

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```
    push    r12
    push    r13
    push    r14
    push    r15
    mov     r15, QWORD PTR a5$[rsp]
    mov     rcx, QWORD PTR a6$[rsp]
    mov     rbp, r8
    mov     r10, r9
    mov     rax, r15
    mov     rdx, rbp
    not     rax
    xor     rdx, r9
    not     r10
    mov     r11, rax
    and     rax, r9
    mov     rsi, r10
    mov     QWORD PTR x36$1$[rsp], rax
    and     r11, r8
    and     rsi, r8
    and     r10, r15
    mov     r13, rdx
    mov     rbx, r11
    xor     rbx, r9
    mov     r9, QWORD PTR a2$[rsp]
    mov     r12, rsi
    or      r12, r15
    not     r13
    and     r13, rcx
    mov     r14, r12
    and     r14, rcx
    mov     rax, r14
    mov     r8, r14
    xor     r8, rbx
```

```
    xor     rax, r15
    not     rbx
    and     rax, rdx
    mov     rdi, rax
    xor     rdi, rsi
    or      rdi, rcx
    xor     rdi, r10
    and     rbx, rdi
    mov     rcx, rdi
    or      rcx, r9
    xor     rcx, rax
    mov     rax, r13
    xor     rax, QWORD PTR x36$1$[rsp]
    and     rcx, QWORD PTR a1$[rsp]
    or      rax, r9
    not     rcx
    xor     rcx, rax
    mov     rax, QWORD PTR out2$[rsp]
    xor     rcx, QWORD PTR [rax]
    xor     rcx, r8
    mov     QWORD PTR [rax], rcx
    mov     rax, QWORD PTR x36$1$[rsp]
    mov     rcx, r14
    or      rax, r8
    or      rcx, r11
    mov     r11, r9
    xor     rcx, rdx
    mov     QWORD PTR x36$1$[rsp], rax
    mov     r8, rsi
    mov     rdx, rcx
    xor     rdx, r13
    not     rdx
    and     rdx, rdi
```

```
    mov     r10, rdx
    and     r10, r9
    xor     r10, rax
    xor     r10, rbx
    not     rbx
    and     rbx, r9
    mov     rax, r10
    and     rax, QWORD PTR a1$[rsp]
    xor     rbx, rax
    mov     rax, QWORD PTR out4$[rsp]
    xor     rbx, QWORD PTR [rax]
    xor     rbx, rcx
    mov     QWORD PTR [rax], rbx
    mov     rbx, QWORD PTR x36$1$[rsp]
    and     rbx, rbp
    mov     r9, rbx
    not     r9
    and     r9, rdi
    or      r8, r11
    mov     rax, QWORD PTR out1$[rsp]
    xor     r8, r9
    not     r9
    and     r9, rcx
    or      rdx, rbp
    mov     rbp, QWORD PTR [rsp+80]
    or      r9, rsi
    xor     rbx, r12
    mov     rcx, r11
    not     rcx
    not     r14
    not     r13
    and     rcx, r9
    or      r9, rdi
```

```
    and     rbx, r14
    xor     r9, r15
    xor     rcx, rdx
    mov     rdx, QWORD PTR a1$[rsp]
    not     r9
    not     rcx
    and     r13, r10
    and     r9, r11
    and     rcx, rdx
    xor     r9, rbx
    mov     rbx, QWORD PTR [rsp+72]
    not     rcx
    xor     rcx, QWORD PTR [rax]
    or      r9, rdx
    not     r9
    xor     rcx, r8
    mov     QWORD PTR [rax], rcx
    mov     rax, QWORD PTR out3$[rsp]
    xor     r9, r13
    xor     r9, QWORD PTR [rax]
    xor     r9, r8
    mov     QWORD PTR [rax], r9
    pop     r15
    pop     r14
    pop     r13
    pop     r12
    pop     rdi
    pop     rsi
    ret     0
s1    ENDP
```

CAPÍTULO 26.

26.2 ARM

ARMv8.

26.3

Capítulo 27

.

SIMD (SSE2) .

(IEEE 754).

.

: 17 on page 411.

27.1

```
#include <stdio.h>

double f (double a, double b)
{
        return a/3.14 + b*4.1;
};
```

```
int main()
{
        printf ("%f\n", f(1.2, 3.4));
};
```

27.1.1 x64

Listing 27.1: Con optimización MSVC 2012 x64

```
__real@4010666666666666 DQ 04010666666666666↙
    ↘ r   ; 4.1
__real@40091eb851eb851f DQ 040091↙
    ↘ eb851eb851fr    ; 3.14

a$ = 8
b$ = 16
f       PROC
        divsd   xmm0, QWORD PTR ↙
    ↘ __real@40091eb851eb851f
        mulsd   xmm1, QWORD PTR ↙
    ↘ __real@4010666666666666
        addsd   xmm0, xmm1
        ret     0
f       ENDP
```

[1].

a XMM0, b XMM1. .

[1] MSDN: Parameter Passing

DIVSD «Divide Scalar Double-Precision Floating-Point Values», .

.

MULSD y ADDSD .

.

:

Listing 27.2: MSVC 2012 x64

```
__real@4010666666666666 DQ 04010666666666666↙
    ↘ r   ; 4.1
__real@40091eb851eb851f DQ 040091↙
    ↘ eb851eb851fr   ; 3.14

a$ = 8
b$ = 16
f       PROC
        movsdx  QWORD PTR [rsp+16], xmm1
        movsdx  QWORD PTR [rsp+8], xmm0
        movsdx  xmm0, QWORD PTR a$[rsp]
        divsd   xmm0, QWORD PTR ↙
    ↘ __real@40091eb851eb851f
        movsdx  xmm1, QWORD PTR b$[rsp]
        mulsd   xmm1, QWORD PTR ↙
    ↘ __real@4010666666666666
        addsd   xmm0, xmm1
        ret     0
f       ENDP
```

. «shadow space» (8.2.1 on page 193), . .

27.1.2 x86

Listing 27.3: Sin optimización MSVC 2012 x86

```
tv70 = -8           ; size = 8
_a$ = 8             ; size = 8
_b$ = 16            ; size = 8
_f       PROC
         push    ebp
         mov     ebp, esp
         sub     esp, 8
         movsd   xmm0, QWORD PTR _a$[ebp]
         divsd   xmm0, QWORD PTR ⤦
    ↳ __real@40091eb851eb851f
         movsd   xmm1, QWORD PTR _b$[ebp]
         mulsd   xmm1, QWORD PTR ⤦
    ↳ __real@4010666666666666
         addsd   xmm0, xmm1
         movsd   QWORD PTR tv70[ebp], xmm0
         fld     QWORD PTR tv70[ebp]
         mov     esp, ebp
         pop     ebp
         ret     0
_f       ENDP
```

Listing 27.4: Con optimización MSVC 2012 x86

```
tv67 = 8            ; size = 8
_a$ = 8             ; size = 8
_b$ = 16            ; size = 8
_f       PROC
         movsd   xmm1, QWORD PTR _a$[esp-4]
         divsd   xmm1, QWORD PTR ⤦
    ↳ __real@40091eb851eb851f
```

```
        movsd   xmm0, QWORD PTR _b$[esp-4]
        mulsd   xmm0, QWORD PTR ↙
    ↪ __real@4010666666666666
        addsd   xmm1, xmm0
        movsd   QWORD PTR tv67[esp-4], xmm1
        fld     QWORD PTR tv67[esp-4]
        ret     0
_f      ENDP
```

1) 2)

OllyDbg:

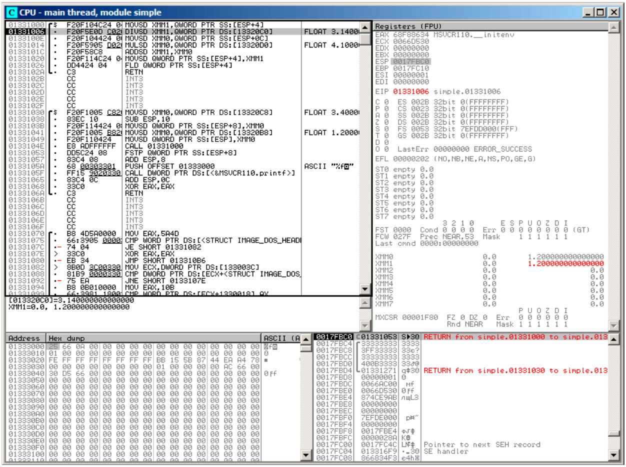

Figura 27.1: OllyDbg: MOVSD a XMM1

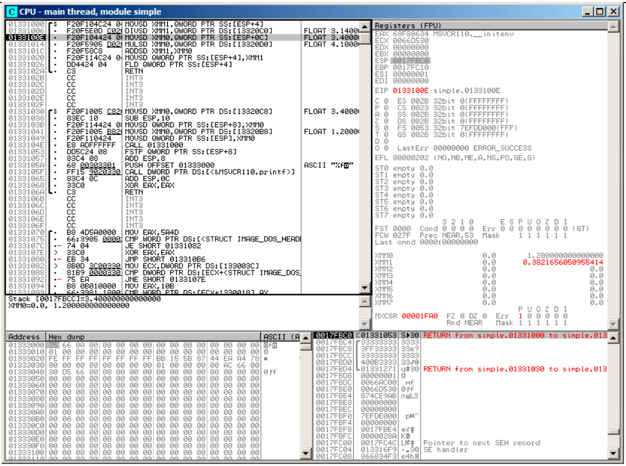

Figura 27.2: OllyDbg: DIVSD quotient XMM1

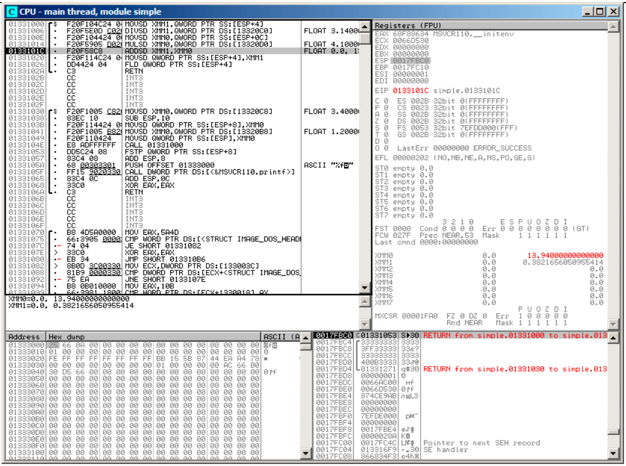

Figura 27.3: OllyDbg: MULSD product XMM0

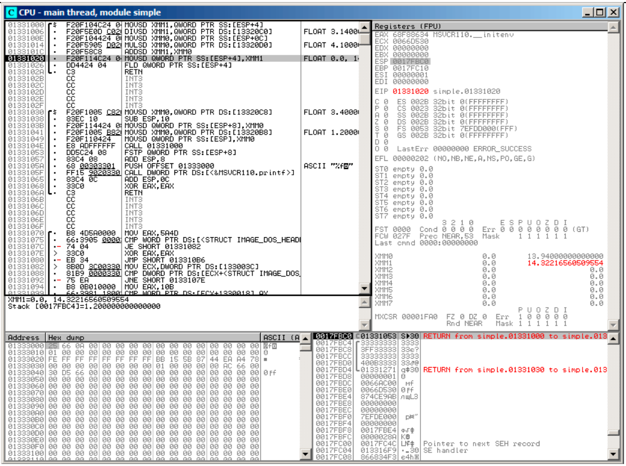

Figura 27.4: OllyDbg: ADDSD XMM0 XMM1

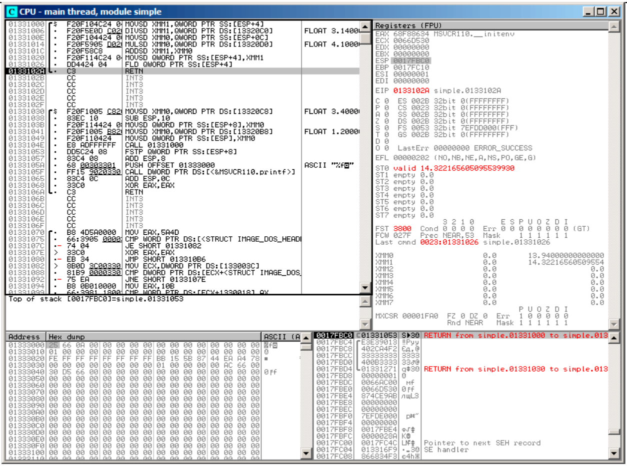

Figura 27.5: OllyDbg: FLD ST(0)

27.2

```
#include <math.h>
#include <stdio.h>

int main ()
{
        printf ("32.01 ^ 1.54 = %lf\n", pow ⤦
    ↳ (32.01,1.54));

        return 0;
}
```

XMM0-XMM3.

Listing 27.5: Con optimización MSVC 2012 x64

```
$SG1354 DB      '32.01 ^ 1.54 = %lf', 0aH, ⤦
    ↳ 00H

__real@40400147ae147ae1 DQ 040400147⤦
    ↳ ae147ae1r   ; 32.01
__real@3ff8a3d70a3d70a4 DQ 03⤦
    ↳ ff8a3d70a3d70a4r   ; 1.54

main    PROC
        sub     rsp, 40                       ⤦
    ↳                    ; 00000028H
        movsdx  xmm1, QWORD PTR ⤦
    ↳ __real@3ff8a3d70a3d70a4
        movsdx  xmm0, QWORD PTR ⤦
    ↳ __real@40400147ae147ae1
        call    pow
```

```
        lea     rcx, OFFSET FLAT:$SG1354
        movaps  xmm1, xmm0
        movd    rdx, xmm1
        call    printf
        xor     eax, eax
        add     rsp, 40                                    ↙
    ↘                 ; 00000028H
        ret     0
main    ENDP
```

MOVSDX Intel [Int13] y AMD [AMD13a], MOVSD. (: A.6.2 on page 1882). MOVSDX. .

pow() XMM0 y XMM1, XMM0. RDX para printf(). ? printf()?

Listing 27.6: Con optimización GCC 4.4.6 x64

```

.LC2:
        .string "32.01 ^ 1.54 = %lf\n"
main:
        sub     rsp, 8
        movsd   xmm1, QWORD PTR .LC0[rip]
        movsd   xmm0, QWORD PTR .LC1[rip]
        call    pow
        ;  XMM0
        mov     edi, OFFSET FLAT:.LC2
        mov     eax, 1 ;
        call    printf
        xor     eax, eax
        add     rsp, 8
        ret
```

```
.LC0:
        .long   171798692
        .long   1073259479
.LC1:
        .long   2920577761
        .long   1077936455
```

GCC. `printf()` XMM0. EAX para `printf()` [Mit13].

27.3

```
#include <stdio.h>

double d_max (double a, double b)
{
        if (a>b)
                return a;

        return b;
};

int main()
{
        printf ("%f\n", d_max (1.2, 3.4));
        printf ("%f\n", d_max (5.6, -4));
};
```

27.3.1 x64

Listing 27.7: Con optimización MSVC 2012 x64

```
a$ = 8
b$ = 16
d_max   PROC
        comisd  xmm0, xmm1
        ja      SHORT $LN2@d_max
        movaps  xmm0, xmm1
$LN2@d_max:
        fatret  0
d_max   ENDP
```

Con optimización MSVC .

`COMISD` «Compare Scalar Ordered Double-Precision Floating-Point Values and Set EFLAGS». .

Sin optimización MSVC :

Listing 27.8: MSVC 2012 x64

```
a$ = 8
b$ = 16
d_max   PROC
        movsdx  QWORD PTR [rsp+16], xmm1
        movsdx  QWORD PTR [rsp+8], xmm0
        movsdx  xmm0, QWORD PTR a$[rsp]
        comisd  xmm0, QWORD PTR b$[rsp]
        jbe     SHORT $LN1@d_max
        movsdx  xmm0, QWORD PTR a$[rsp]
        jmp     SHORT $LN2@d_max
$LN1@d_max:
        movsdx  xmm0, QWORD PTR b$[rsp]
$LN2@d_max:
```

```
        fatret  0
d_max   ENDP
```

GCC 4.4.6 MAXSD («Return Maximum Scalar Double-Precision Floating-Point Value»)!

Listing 27.9: Con optimización GCC 4.4.6 x64

```
d_max:
        maxsd   xmm0, xmm1
        ret
```

27.3.2 x86

Listing 27.10: Con optimización MSVC 2012 x86

```
_a$ = 8         ; size = 8
_b$ = 16        ; size = 8
_d_max  PROC
        movsd   xmm0, QWORD PTR _a$[esp-4]
        comisd  xmm0, QWORD PTR _b$[esp-4]
        jbe     SHORT $LN1@d_max
        fld     QWORD PTR _a$[esp-4]
        ret     0
$LN1@d_max:
        fld     QWORD PTR _b$[esp-4]
        ret     0
_d_max  ENDP
```

a y b `ST(0)`.

OllyDbg, `COMISD` `CF` y `PF`:

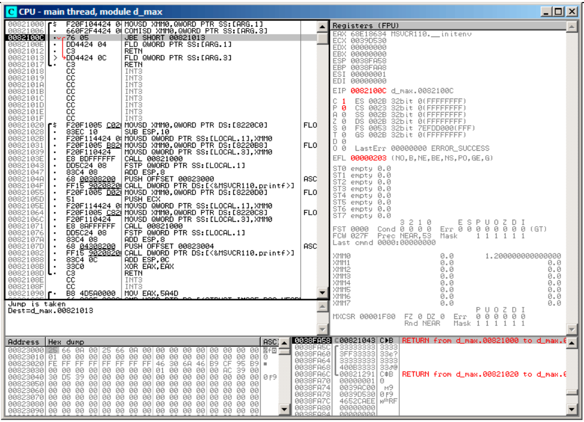

Figura 27.6: OllyDbg: COMISD CF y PF

27.4 : x64 y SIMD

:

Listing 27.11: Con optimización MSVC 2012 x64

```
v$ = 8
calculate_machine_epsilon PROC
        movsdx   QWORD PTR v$[rsp], xmm0
        movaps   xmm1, xmm0
        inc      QWORD PTR v$[rsp]
        movsdx   xmm0, QWORD PTR v$[rsp]
        subsd    xmm0, xmm1
        ret      0
calculate_machine_epsilon ENDP
```

2.

SUBSD «Subtract Scalar Double-Precision Floating-Point Values», . .

27.5

Listing 27.12: Con optimización MSVC 2012

```
__real@3f800000 DD 03f800000r                  ↙
    ↘           ; 1

tv128 = -4
_tmp$ = -4
?float_rand@@YAMXZ PROC
        push    ecx
        call    ?my_rand@@YAIXZ
; EAX=
        and     eax, 8388607                   ↙
    ↘                    ; 007fffffH
        or      eax, 1065353216                ↙
    ↘                    ; 3f800000H
; EAX= & 0x007fffff | 0x3f800000
;
        mov     DWORD PTR _tmp$[esp+4], eax
;
        movss   xmm0, DWORD PTR _tmp$[esp+4]
```

2.

```
;  1.0:
        subss   xmm0, DWORD PTR ⤦
    ↪ __real@3f800000
;
        movss   DWORD PTR tv128[esp+4], xmm0
; ...
        fld     DWORD PTR tv128[esp+4]
        pop     ecx
        ret     0
?float_rand@@YAMXZ ENDP
```

27.6

.

-SD («Scalar Double-Precision»).

.

double float -SS («Scalar Single-Precision»), , MOVSS, COMISS, ADDSS, Spanish text placeholder.

«Scalar» .

CAPÍTULO 28. DETALLES ESPECÍFICOS DE ARM

Capítulo 28

Detalles específicos de ARM

28.1 (#) Signo numeral (#) antes del número

El compilador Keil, IDA y objdump anteponen a todos los números el signo «#», por ejemplo: Spanish text placeholder.14.1.4. Pero cuando GCC 4.9 genera salida en lenguaje ensamblador, no lo hace, por ejemplo: Spanish text placeholder.39.3.

Así que los listados para ARM en este libro están, en cierto sentido, mezclados.

 CAPÍTULO 28. DETALLES ESPECÍFICOS DE ARM
Es difícil decir cuál es lo correcto. Probablemente conviene ceñirse a las convenciones aceptadas en el entorno en el que se trabaja.

28.2 Modos de direccionamiento

Esta instrucción es posible en ARM64:

```
ldr     x0, [x29,24]
```

Esto significa sumar 24 al valor en X29 y cargar el valor desde esa dirección. Ten en cuenta que 24 está dentro de los corchetes. El significado cambia si el número está fuera de los corchetes:

```
ldr     w4, [x1],28
```

Esto significa cargar el valor en la dirección contenida en X1 y luego sumar 28 a X1.

ARM permite sumar o restar una constante a/de la dirección usada para la carga. Y es posible hacerlo tanto antes como después de la carga.

Este modo de direccionamiento no existe en x86, pero sí en otros procesadores, incluso en el PDP-11. , *ptr++, *++ptr, *ptr--, *--ptr. Por cierto, esta es una característica de C difícil de recordar. Así es como funciona:

 CAPÍTULO 28. DETALLES ESPECÍFICOS DE ARM

Término en C	Término en ARM	Expresión en
Post-incremento	post-indexed addressing	`*ptr++`
Post-decremento	post-indexed addressing	`*ptr--`
Pre-incremento	pre-indexed addressing	`*++ptr`
Pre-decremento	pre-indexed addressing	`*--ptr`

Pre-indexing se marca con un signo de exclamación en el lenguaje ensamblador de ARM. Por ejemplo, ver la línea 2 en Spanish text placeholder.3.15.

Dennis Ritchie (uno de los creadores del lenguaje C) mencionó que probablemente fue idea de Ken Thompson (otro creador de C), porque esta característica del procesador ya estaba presente en el PDP-7 [Rit86][Rit93]. Por lo tanto, los compiladores de C pueden usarlo si está presente en el procesador de destino.

Todo esto es muy conveniente para trabajar con arreglos.

28.3 Cargar una constante en un registro

28.3.1 32 bits ARM

Como ya sabemos, todas las instrucciones tienen una longitud de 4 bytes en modo ARM y de 2 bytes en modo Thumb. Entonces, ¿cómo podemos cargar un valor de 32 bits en un registro si no es posible codificarlo en una sola instrucción?

Probemos:

```
unsigned int f()
{
        return 0x12345678;
};
```

Listing 28.1: GCC 4.6.3 -O3 Modo ARM

```
f:
        ldr     r0, .L2
        bx      lr
.L2:
        .word   305419896 ; 0x12345678
```

Es decir, el valor `0x12345678` se almacena aparte en memoria y se carga cuando es necesario. Pero también es posible evitar el acceso adicional a memoria.

Listing 28.2: GCC 4.6.3 -O3 -march=armv7-a (Modo ARM)

```
movw    r0, #22136       ; 0x5678
movt    r0, #4660        ; 0x1234
```

 CAPÍTULO 28. DETALLES ESPECÍFICOS DE ARM

```
bx      lr
```

Se ve que el valor se carga en el registro por partes: primero la parte baja (con MOVW) y luego la alta (con MOVT).

Por lo tanto, se necesitan 2 instrucciones en modo ARM para cargar un valor de 32 bits en un registro. No es un gran problema, porque en el código real no hay tantas constantes (excepto 0 y 1). ¿Significa eso que la versión de dos instrucciones es más lenta que la de una sola instrucción? Probablemente no. Lo más seguro es que los procesadores ARM modernos detecten estas secuencias y las ejecuten rápidamente.

Por otro lado, IDA reconoce fácilmente estos patrones en el código y desensambla esta función como:

```
MOV    R0, 0x12345678
BX     LR
```

28.3.2 ARM64

```
uint64_t f()
{
        return 0x12345678ABCDEF01;
};
```

Listing 28.3: GCC 4.9.1 -O3

```
mov     x0, 61185   ; 0xef01
movk    x0, 0xabcd, lsl 16
```

 CAPÍTULO 28. DETALLES ESPECÍFICOS DE ARM

```
movk    x0, 0x5678, lsl 32
movk    x0, 0x1234, lsl 48
ret
```

MOVK «MOV Keep», i.e., escribe un valor de 16 bits en el registro sin tocar el resto de los bits. LSL Por lo tanto, se necesitan 4 instrucciones para cargar un valor de 64 bits en un registro.

Almacenar un número en coma flotante en un registro

Algunos números se pueden escribir en un registro D usando solo una instrucción.

Por ejemplo:

```
double a()
{
        return 1.5;
};
```

Listing 28.4: GCC 4.9.1 -O3 + objdump

```
0000000000000000 <a>:
   0:   1e6f1000        fmov    d0, ↙
    ↘ #1.5000000000000000000e+000
   4:   d65f03c0        ret
```

El número 1.5 efectivamente fue codificado en una instrucción de 32 bits. ?¿Pero cómo? En ARM64, la instrucción FMOV dispone de 8 bits para codificar ciertos números en coma flotante. El algoritmo se denomina VFPExpandImm()

en [ARM13a]. Esto también se llama *minifloat*[1]. Podemos probar con distintos valores: el compilador puede codificar 30.0 y 31.0, pero no pudo codificar 32.0, por lo que se deben reservar 8 bytes en memoria y escribirlo allí en formato IEEE 754:

```
double a()
{
        return 32;
};
```

Listing 28.5: GCC 4.9.1 -O3

```
a:
        ldr     d0, .LC0
        ret
.LC0:
        .word   0
        .word   1077936128
```

28.4 en ARM64Relocaciones en ARM64

Como sabemos, en ARM64 las instrucciones tienen 4 bytes, por lo que es imposible escribir un número grande en un registro con una sola instrucción. Sin embargo, una imagen ejecutable puede cargarse en cualquier dirección aleatoria de memoria; para eso existen las relocaciones. Más sobre ellas (en relación con Win32 PE): 68.2.6 on page 1354.

[1]wikipedia

 CAPÍTULO 28. DETALLES ESPECÍFICOS DE ARM

En ARM64 se adopta el siguiente método: la dirección se forma mediante el par de instrucciones ADRP y ADD. La primera carga en un registro la dirección de una página de 4 KB y la segunda suma el resto. Compilaremos el ejemplo de «Hello, world!» (Spanish text placeholder.6) en GCC (Linaro) 4.9 en win32:

Listing 28.6: GCC (Linaro) 4.9 y objdump

```
...>aarch64-linux-gnu-gcc.exe hw.c -c

...>aarch64-linux-gnu-objdump.exe -d hw.o

...

0000000000000000 <main>:
   0:   a9bf7bfd        stp     x29, x30, [⤦
    ↳ sp,#-16]!
   4:   910003fd        mov     x29, sp
   8:   90000000        adrp    x0, 0 <main>
   c:   91000000        add     x0, x0, #0x0
  10:   94000000        bl      0 <printf>
  14:   52800000        mov     w0, #0x0    ⤦
    ↳                     // #0
  18:   a8c17bfd        ldp     x29, x30, [⤦
    ↳ sp],#16
  1c:   d65f03c0        ret

...>aarch64-linux-gnu-objdump.exe -r hw.o

...

RELOCATION RECORDS FOR [.text]:
```

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```
OFFSET           TYPE                  VALUE
0000000000000008 R_AARCH64_ADR_PREL_PG_HI21 ⤦
    ↳  .rodata
000000000000000c R_AARCH64_ADD_ABS_LO12_NC  ⤦
    ↳ .rodata
0000000000000010 R_AARCH64_CALL26  printf
```

Así que en este archivo objeto hay 3 relocaciones.

- La primera toma la dirección de la página, recorta los 12 bits inferiores y escribe los 21 bits altos restantes en los campos de bits de la instrucción ADRP. Esto se debe a que no es necesario codificar los 12 bits bajos y la instrucción ADRP solo tiene espacio para 21 bits.
- La segunda coloca los 12 bits de la dirección relativa al inicio de la página en los campos de bits de la instrucción ADD.
- La última, de 26 bits, se aplica a la instrucción en la dirección `0x10` donde está el salto a la función `printf()`. Todas las direcciones de instrucciones en ARM64 (y en ARM en modo ARM) tienen ceros en los dos bits menos significativos (porque todas las instrucciones tienen un tamaño de 4 bytes), así que solo es necesario codificar los 26 bits superiores del espacio de direcciones de 28 bits (± 128 MB).

En el archivo ejecutable enlazado no hay tales relocaciones en estos lugares, porque ya se conoce dónde estará la cadena «Hello!», en qué página, y también se conoce la dirección de la función `puts()`. Por lo tanto, los valores ya

 CAPÍTULO 28. DETALLES ESPECÍFICOS DE ARM

están establecidos en las instrucciones ADRP, ADD y BL (el enlazador los escribió durante el enlace):

Listing 28.7: objdump

```
0000000000400590 <main>:
  400590:       a9bf7bfd        stp     x29,↙
    ↳  x30, [sp,#-16]!
  400594:       910003fd        mov     x29,↙
    ↳  sp
  400598:       90000000        adrp    x0, ↙
    ↳ 400000 <_init-0x3b8>
  40059c:       91192000        add     x0, ↙
    ↳ x0, #0x648
  4005a0:       97ffffa0        bl      ↙
    ↳ 400420 <puts@plt>
  4005a4:       52800000        mov     w0, ↙
    ↳ #0x0                                // #0
  4005a8:       a8c17bfd        ldp     x29,↙
    ↳  x30, [sp],#16
  4005ac:       d65f03c0        ret

...

Contents of section .rodata:
 400640 01000200 00000000 48656c6c 6f210000 ↙
    ↳  ........Hello!..
```

Como ejemplo, intentemos desensamblar manualmente la instrucción BL.
`0x97ffffa0` $10010111111111111111111110100000b$. De acuerdo con [ARM13a, pág. C5.6.26], imm26 son los últimos 26 bits: $imm26 = 11111111111111111110100000$. Es

 CAPÍTULO 28. DETALLES ESPECÍFICOS DE ARM
0x3FFFFA0, pero el MSB es 1, por lo que el número es negativo y, en términos de aritmética modular módulo 2^{32}, es 0xFFFFFFA0. De nuevo, módulo 2^{32}, 0xFFFFFFA0 * 4 = 0xFFFFFE80, y 0x4005a0 + FFFFFE80 = 0x400420 (ten en cuenta: consideramos la dirección de la instrucción BL, no el valor actual de PC, que puede ser distinto). Así que la dirección de destino es 0x400420.

Más sobre relocaciones relacionadas con ARM64: [ARM13b].

 CAPÍTULO 29. DETALLES ESPECÍFICOS DE MIPS

Capítulo 29

Detalles específicos de MIPS

29.1 Carga de constantes en un registro

```
unsigned int f()
{
        return 0x12345678;
};
```

En MIPS, al igual que en ARM, todas las instrucciones tienen 32 bits, por lo que no es posible incrustar una constan-

 CAPÍTULO 29. DETALLES ESPECÍFICOS DE MIPS

te de 32 bits en una sola instrucción. Por lo tanto, esto se traduce en al menos dos instrucciones: la primera carga la parte alta del número de 32 bits y la segunda aplica una operación OR, que en la práctica establece los 16 bits bajos del registro de destino.

Listing 29.1: GCC 4.4.5 -O3 (salida de assembly)

```
        li      $2,305397760                      ↙
    ↳     # 0x12340000
        j       $31
        ori     $2,$2,0x5678 ; branch delay ↙
    ↳ slot
```

IDA IDA conoce estos patrones de código frecuentes y, para mayor comodidad, muestra la última instrucción ORI como la pseudoinstrucción LI, que supuestamente carga un número completo de 32 bits en el registro $V0.

Listing 29.2: GCC 4.4.5 -O3 (IDA)

```
         lui     $v0, 0x1234
         jr      $ra
         li      $v0, 0x12345678 ; branch ↙
    ↳ delay slot
```

En la salida de ensamblador de GCC aparece la pseudoinstrucción LI, pero en realidad se trata de LUI («Load Upper Immediate»), que escribe un valor de 16 bits en la parte alta del registro.

CAPÍTULO 29. DETALLES ESPECÍFICOS DE MIPS

29.2 Lecturas adicionales sobre MIPS

[Swe10].

Parte II

CAPÍTULO 30.

Capítulo 30

			()
01111111	0x7f	127	127
01111110	0x7e	126	126
...			
00000110	0x6	6	6
00000101	0x5	5	5
00000100	0x4	4	4
00000011	0x3	3	3
00000010	0x2	2	2
00000001	0x1	1	1
00000000	0x0	0	0
11111111	0xff	255	-1
11111110	0xfe	254	-2
11111101	0xfd	253	-3
11111100	0xfc	252	-4
11111011	0xfb	251	-5
11111010	0xfa	250	-6

La diferencia entre números con y sin signo es que si representamos 0xFFFFFFFE y 0x0000002 como sin signo, entonces el primer número (4294967294) es mayor que el segundo (2). Si los representamos ambos como con signo, el primero es -2, que es menor que el segundo (2). Esa es la razón por la que los saltos condicionales (12 on page 228) están presentes tanto para comparaciones con signo (ej. JG, JL) como sin signo (JA, JB).

:

- .
- :
  - `int64_t` (-9,223,372,036,854,775,808..9,223,372,0 (- 9.2.. 9.2) o `0x8000000000000000..0x7FFFFFFFFFFFFF`
  - *int* (-2,147,483,648..2,147,483,647 (- 2.15.. 2.15Gb) o `0x80000000..0x7FFFFFFF`),
  - *char* (-128..127 o `0x80..0x7F`),
  - `ssize_t`.

  :

  - `uint64_t` (0..18,446,744,073,709,551,615 (18) o `0..0xFFFFFFFFFFFFFFFF`),
  - `unsigned int` (0..4,294,967,295 (4.3Gb) o `0..0x`
  - `unsigned char` (0..255 o `0..0xFF`),

  - size_t.

- .
- 24.5 on page 788.
-
- . : IDIV/IMUL y DIV/MUL .
- : CBW/CWD/CWDE/CDQ/CDQE (A.6.3 on page 1886), MOVSX (15.1.1 on page 378), SAR (A.6.3 on page 1895).

 CAPÍTULO 31. ENDIANNESS

Capítulo 31

Endianness

.

31.1 Big-endian

0x12345678 :

+0	0x12
+1	0x34
+2	0x56
+3	0x78

Motorola 68k, IBM POWER.

31.2 Little-endian

0x12345678 :

+0	0x78
+1	0x56
+2	0x34
+3	0x12

Intel x86.

31.3 Ejemplo

[1].

:

```
#include <stdio.h>

int main()
{
        int v, i;

        v=123;

        printf ("%02X %02X %02X %02X\n",
                *(char*)&v,
                *(((char*)&v)+1),
                *(((char*)&v)+2),
                *(((char*)&v)+3));
```

[1]: http://go.yurichev.com/17008

```
};
```

:

```
root@debian-mips:~# ./a.out
00 00 00 7B
```

. 0x7B 123 .

(«mips» (big-endian) y «mipsel» (little-endian)).

21.4.3 on page 699.

31.4 Bi-endian

ARM, PowerPC, SPARC, MIPS, IA64[2], Spanish text placeholder.

31.5

The BSWAP .

`htonl()` y `htons()`.

«network byte order», «host byte order».

[2] Intel Architecture 64 (Itanium): 93 on page 1769

Capítulo 32

- . AKA[1] «static memory allocation». : 7.2 on page 149.
- . AKA «allocate on stack». . : 18.2 on page 496.
- . AKA «dynamic memory allocation». `malloc()/free()` o `new/delete` en C++. . . : 21.2 on page 664.

[1] Also Known As (También conocido como)

Capítulo 33

CPU

33.1

. : 12.1.2 on page 244, 12.3 on page 262, 19.5.2 on page 617.

33.2

Capítulo 34

. . .

. . . .

34.1 ?

```
4 6 0 1 3 5 7 8 9 2
```

:

- ;
- ;
- .

:

 CAPÍTULO 34.

```
4 6 0 1 3 5 7 8 9 2
        ^   ^
```

:

```
4 6 0 1 7 5 3 8 9 2
```

Parte III

Capítulo 35

.

$$C = \frac{5 \cdot (F - 32)}{9}$$

: 1) ; 2) .

`exit()` .

35.1

```
#include <stdio.h>
#include <stdlib.h>

int main()
{
        int celsius, fahr;
```

```
        printf ("Enter temperature in ↙
    ↘ Fahrenheit:\n");
        if (scanf ("%d", &fahr)!=1)
        {
                printf ("Error while parsing↙
    ↘  your input\n");
                exit(0);
        };

        celsius = 5 * (fahr-32) / 9;

        if (celsius<-273)
        {
                printf ("Error: incorrect ↙
    ↘ temperature!\n");
                exit(0);
        };
        printf ("Celsius: %d\n", celsius);
};
```

35.1.1 Con optimización MSVC 2012 x86

Listing 35.1: Con optimización MSVC 2012 x86

```
$SG4228 DB      'Enter temperature in ↙
    ↘ Fahrenheit:', 0aH, 00H
$SG4230 DB      '%d', 00H
$SG4231 DB      'Error while parsing your ↙
    ↘ input', 0aH, 00H
$SG4233 DB      'Error: incorrect ↙
    ↘ temperature!', 0aH, 00H
```

```
$SG4234 DB      'Celsius: %d', 0aH, 00H

_fahr$ = -4                                         ↙
    ↘                       ; size = 4
_main   PROC
        push    ecx
        push    esi
        mov     esi, DWORD PTR __imp__printf
        push    OFFSET $SG4228              ; '↙
    ↘ Enter temperature in Fahrenheit:'
        call    esi                         ;   ↙
    ↘ printf()
        lea     eax, DWORD PTR _fahr$[esp↙
    ↘ +12]
        push    eax
        push    OFFSET $SG4230              ; '%↙
    ↘ d'
        call    DWORD PTR __imp__scanf
        add     esp, 12                     ↙
    ↘                       ; 0000000cH
        cmp     eax, 1
        je      SHORT $LN2@main
        push    OFFSET $SG4231              ; '↙
    ↘ Error while parsing your input'
        call    esi                         ;   ↙
    ↘ printf()
        add     esp, 4
        push    0
        call    DWORD PTR __imp__exit
$LN9@main:
$LN2@main:
        mov     eax, DWORD PTR _fahr$[esp+8]
```

```
        add     eax, -32                        ⤦
    ↳                   ; ffffffe0H
        lea     ecx, DWORD PTR [eax+eax*4]
        mov     eax, 954437177                  ⤦
    ↳                   ; 38e38e39H
        imul    ecx
        sar     edx, 1
        mov     eax, edx
        shr     eax, 31                         ⤦
    ↳                   ; 0000001fH
        add     eax, edx
        cmp     eax, -273                       ⤦
    ↳                   ; fffffeefH
        jge     SHORT $LN1@main
        push    OFFSET $SG4233                  ; '⤦
    ↳ Error: incorrect temperature!'
        call    esi                             ;   ⤦
    ↳ printf()
        add     esp, 4
        push    0
        call    DWORD PTR __imp__exit
$LN10@main:
$LN1@main:
        push    eax
        push    OFFSET $SG4234                  ; '⤦
    ↳ Celsius: %d'
        call    esi                             ;   ⤦
    ↳ printf()
        add     esp, 8
        ;
        xor     eax, eax
        pop     esi
        pop     ecx
```

```
        ret     0
$LN8@main:
_main   ENDP
```

:

- `printf()` ESI `printf()` `CALL ESI`. .
- `ADD EAX, -32` . $EAX = EAX + (-32)$ $EAX = EAX - 32$ `ADD` `SUB`.
- LEA 5: `lea ecx, DWORD PTR [eax+eax*4]`., $i + i * 4$ $i * 5$ y LEA IMUL. `SHL EAX, 2 / ADD EAX, EAX` .
- (41 on page 938) .
- `main()` 0 `return 0` . [ISO07, pág. 5.1.2.2.3] `main()` `0 return`. .

35.1.2 Con optimización MSVC 2012 x64

```
        xor     ecx, ecx
        call    QWORD PTR __imp_exit
        int     3
```

`INT 3` .

`exit()` [1], .

[1] `longjmp()`

35.2

```
#include <stdio.h>
#include <stdlib.h>

int main()
{
        double celsius, fahr;
        printf ("Enter temperature in ↙
    ↳ Fahrenheit:\n");
        if (scanf ("%lf", &fahr)!=1)
        {
                printf ("Error while parsing↙
    ↳  your input\n");
                exit(0);
        };

        celsius = 5 * (fahr-32) / 9;

        if (celsius<-273)
        {
                printf ("Error: incorrect ↙
    ↳ temperature!\n");
                exit(0);
        };
        printf ("Celsius: %lf\n", celsius);
};
```

MSVC 2010 x86 FPU…

Listing 35.2: Con optimización MSVC 2010 x86

```
$SG4038 DB      'Enter temperature in ⤦
    ↪ Fahrenheit:', 0aH, 00H
$SG4040 DB      '%lf', 00H
$SG4041 DB      'Error while parsing your ⤦
    ↪ input', 0aH, 00H
$SG4043 DB      'Error: incorrect ⤦
    ↪ temperature!', 0aH, 00H
$SG4044 DB      'Celsius: %lf', 0aH, 00H

__real@c071100000000000 DQ 0⤦
    ↪ c071100000000000r   ; -273
__real@4022000000000000 DQ 04022000000000000⤦
    ↪ r   ; 9
__real@4014000000000000 DQ 04014000000000000⤦
    ↪ r   ; 5
__real@4040000000000000 DQ 04040000000000000⤦
    ↪ r   ; 32

_fahr$ = -8                                 ⤦
    ↪                   ; size = 8
_main   PROC
        sub     esp, 8
        push    esi
        mov     esi, DWORD PTR __imp__printf
        push    OFFSET $SG4038          ; '⤦
    ↪ Enter temperature in Fahrenheit:'
        call    esi                     ;  ⤦
    ↪ printf()
        lea     eax, DWORD PTR _fahr$[esp⤦
    ↪ +16]
        push    eax
        push    OFFSET $SG4040          ; '%⤦
    ↪ lf'
```

```
	call	DWORD PTR __imp__scanf
	add	esp, 12                     ↙
    ↘                   ; 0000000cH
	cmp	eax, 1
	je	SHORT $LN2@main
	push	OFFSET $SG4041              ; '↙
    ↘ Error while parsing your input'
	call	esi                         ;  ↙
    ↘ printf()
	add	esp, 4
	push	0
	call	DWORD PTR __imp__exit
$LN2@main:
	fld	QWORD PTR _fahr$[esp+12]
	fsub	QWORD PTR ↙
    ↘ __real@4040000000000000 ; 32
	fmul	QWORD PTR ↙
    ↘ __real@4014000000000000 ; 5
	fdiv	QWORD PTR ↙
    ↘ __real@4022000000000000 ; 9
	fld	QWORD PTR ↙
    ↘ __real@c071100000000000 ; -273
	fcomp	ST(1)
	fnstsw	ax
	test	ah, 65                      ↙
    ↘                   ; 00000041H
	jne	SHORT $LN1@main
	push	OFFSET $SG4043              ; '↙
    ↘ Error: incorrect temperature!'
	fstp	ST(0)
	call	esi                         ;  ↙
    ↘ printf()
	add	esp, 4
```

```
        push    0
        call    DWORD PTR __imp__exit
$LN1@main:
        sub     esp, 8
        fstp    QWORD PTR [esp]
        push    OFFSET $SG4044                  ; '
    ↳ Celsius: %lf'
        call    esi
        add     esp, 12                        
    ↳                       ; 0000000cH
        ;
        xor     eax, eax
        pop     esi
        add     esp, 8
        ret     0
$LN10@main:
_main   ENDP
```

... MSVC 2012 SIMD :

Listing 35.3: Con optimización MSVC 2010 x86

```
$SG4228 DB      'Enter temperature in 
    ↳ Fahrenheit:', 0aH, 00H
$SG4230 DB      '%lf', 00H
$SG4231 DB      'Error while parsing your 
    ↳ input', 0aH, 00H
$SG4233 DB      'Error: incorrect 
    ↳ temperature!', 0aH, 00H
$SG4234 DB      'Celsius: %lf', 0aH, 00H
__real@c071100000000000 DQ 0
    ↳ c071100000000000r   ; -273
```

```
__real@4040000000000000 DQ 04040000000000000↙
    ↘ r   ; 32
__real@4022000000000000 DQ 04022000000000000↙
    ↘ r   ; 9
__real@4014000000000000 DQ 04014000000000000↙
    ↘ r   ; 5

_fahr$ = -8                                 ↙
    ↘                   ; size = 8
_main   PROC
        sub     esp, 8
        push    esi
        mov     esi, DWORD PTR __imp__printf
        push    OFFSET $SG4228              ; '↙
    ↘ Enter temperature in Fahrenheit:'
        call    esi                         ;  ↙
    ↘ printf()
        lea     eax, DWORD PTR _fahr$[esp↙
    ↘ +16]
        push    eax
        push    OFFSET $SG4230              ; '%↙
    ↘ lf'
        call    DWORD PTR __imp__scanf
        add     esp, 12                     ↙
    ↘                   ; 0000000cH
        cmp     eax, 1
        je      SHORT $LN2@main
        push    OFFSET $SG4231              ; '↙
    ↘ Error while parsing your input'
        call    esi                         ;  ↙
    ↘ printf()
        add     esp, 4
        push    0
```

```
        call    DWORD PTR __imp__exit
$LN9@main:
$LN2@main:
        movsd   xmm1, QWORD PTR _fahr$[esp
    ↳ +12]
        subsd   xmm1, QWORD PTR 
    ↳ __real@4040000000000000 ; 32
        movsd   xmm0, QWORD PTR 
    ↳ __real@c071100000000000 ; -273
        mulsd   xmm1, QWORD PTR 
    ↳ __real@4014000000000000 ; 5
        divsd   xmm1, QWORD PTR 
    ↳ __real@4022000000000000 ; 9
        comisd  xmm0, xmm1
        jbe     SHORT $LN1@main
        push    OFFSET $SG4233          ; '
    ↳ Error: incorrect temperature!'
        call    esi                     ; 
    ↳ printf()
        add     esp, 4
        push    0
        call    DWORD PTR __imp__exit
$LN10@main:
$LN1@main:
        sub     esp, 8
        movsd   QWORD PTR [esp], xmm1
        push    OFFSET $SG4234          ; '
    ↳ Celsius: %lf'
        call    esi                     ; 
    ↳ printf()
        add     esp, 12                 
    ↳                   ; 0000000cH
        ;
```

```
        xor     eax, eax
        pop     esi
        add     esp, 8
        ret     0
$LN8@main:
_main   ENDP
```

, SIMD. .

-273 XMM0 . .

Capítulo 36

[1]...

:

$$0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, 377, 610, 987, 1597, 25$$

36.1 #1

21.

```
#include <stdio.h>

void fib (int a, int b, int limit)
{
        printf ("%d\n", a+b);
```

[1] http://go.yurichev.com/17332

```
        if (a+b > limit)
                return;
        fib (b, a+b, limit);
};

int main()
{
        printf ("0\n1\n1\n");
        fib (1, 1, 20);
};
```

Listing 36.1: MSVC 2010 x86

```
_TEXT   SEGMENT
_a$ = 8         ; size = 4
_b$ = 12        ; size = 4
_limit$ = 16    ; size = 4
_fib    PROC
        push    ebp
        mov     ebp, esp
        mov     eax, DWORD PTR _a$[ebp]
        add     eax, DWORD PTR _b$[ebp]
        push    eax
        push    OFFSET $SG2750 ; "%d"
        call    DWORD PTR __imp__printf
        add     esp, 8
        mov     ecx, DWORD PTR _limit$[ebp]
        push    ecx
        mov     edx, DWORD PTR _a$[ebp]
        add     edx, DWORD PTR _b$[ebp]
        push    edx
        mov     eax, DWORD PTR _b$[ebp]
        push    eax
```

```
        call    _fib
        add     esp, 12
        pop     ebp
        ret     0
_fib    ENDP

_main   PROC
        push    ebp
        mov     ebp, esp
        push    OFFSET $SG2753 ; "0\n1\n1\n"
        call    DWORD PTR __imp__printf
        add     esp, 4
        push    20
        push    1
        push    1
        call    _fib
        add     esp, 12
        xor     eax, eax
        pop     ebp
        ret     0
_main   ENDP
```

.

:

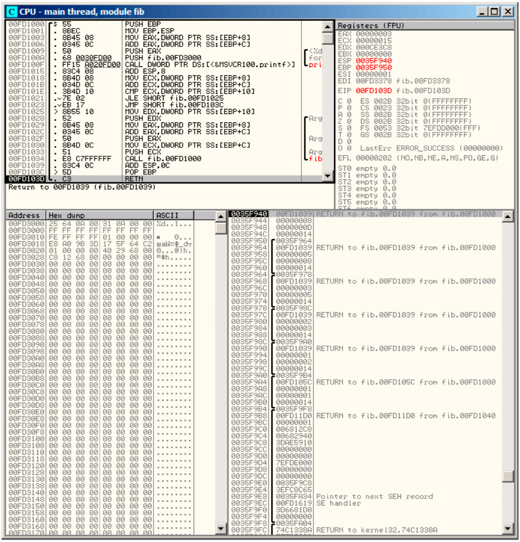


Figura 36.1: OllyDbg:

. [2]:

```
0035F940   00FD1039  RETURN to fib.00FD1039 ⤦
    ↳ from fib.00FD1000
0035F944   00000008  : a
0035F948   0000000D  : b
0035F94C   00000014  : limit
0035F950  /0035F964
0035F954  |00FD1039  RETURN to fib.00FD1039 ⤦
    ↳ from fib.00FD1000
0035F958  |00000005  : a
0035F95C  |00000008  : b
0035F960  |00000014  : limit
0035F964  ]0035F978
0035F968  |00FD1039  RETURN to fib.00FD1039 ⤦
    ↳ from fib.00FD1000
0035F96C  |00000003  : a
0035F970  |00000005  : b
0035F974  |00000014  : limit
0035F978  ]0035F98C
0035F97C  |00FD1039  RETURN to fib.00FD1039 ⤦
    ↳ from fib.00FD1000
0035F980  |00000002  : a
0035F984  |00000003  : b
0035F988  |00000014  : limit
0035F98C  ]0035F9A0
0035F990  |00FD1039  RETURN to fib.00FD1039 ⤦
    ↳ from fib.00FD1000
0035F994  |00000001  : a
0035F998  |00000002  : b
```

2

```
0035F99C  |00000014  : limit
0035F9A0  ]0035F9B4
0035F9A4  |00FD105C  RETURN to fib.00FD105C ↙
    ↘ from fib.00FD1000
0035F9A8  |00000001  : a                \
0035F9AC  |00000001  : b                |  f1()
0035F9B0  |00000014  : limit            /
0035F9B4  ]0035F9F8
0035F9B8  |00FD11D0  RETURN to fib.00FD11D0 ↙
    ↘ from fib.00FD1040
0035F9BC  |00000001  main() : argc   \
0035F9C0  |006812C8  main() : argv   |  main↙
    ↘ ()
0035F9C4  |00682940  main() : envp   /
```

[3], . *limit* (0x14 o 20), a y b . . OllyDbg . . . .

.

main(). argc 1 ().

0xC00000FD (.)

36.2 #2

:

```
#include <stdio.h>

void fib (int a, int b, int limit)
{
```

3

```
        int next=a+b;
        printf ("%d\n", next);
        if (next > limit)
                return;
        fib (b, next, limit);
};

int main()
{
        printf ("0\n1\n1\n");
        fib (1, 1, 20);
};
```

:

Listing 36.2: MSVC 2010 x86

```
_next$ = -4     ; size = 4
_a$ = 8         ; size = 4
_b$ = 12        ; size = 4
_limit$ = 16    ; size = 4
_fib    PROC
        push    ebp
        mov     ebp, esp
        push    ecx
        mov     eax, DWORD PTR _a$[ebp]
        add     eax, DWORD PTR _b$[ebp]
        mov     DWORD PTR _next$[ebp], eax
        mov     ecx, DWORD PTR _next$[ebp]
        push    ecx
        push    OFFSET $SG2751 ; '%d'
        call    DWORD PTR __imp__printf
        add     esp, 8
```

```
        mov     edx, DWORD PTR _next$[ebp]
        cmp     edx, DWORD PTR _limit$[ebp]
        jle     SHORT $LN1@fib
        jmp     SHORT $LN2@fib
$LN1@fib:
        mov     eax, DWORD PTR _limit$[ebp]
        push    eax
        mov     ecx, DWORD PTR _next$[ebp]
        push    ecx
        mov     edx, DWORD PTR _b$[ebp]
        push    edx
        call    _fib
        add     esp, 12
$LN2@fib:
        mov     esp, ebp
        pop     ebp
        ret     0
_fib    ENDP

_main   PROC
        push    ebp
        mov     ebp, esp
        push    OFFSET $SG2753 ; "0\n1\n1\n"
        call    DWORD PTR __imp__printf
        add     esp, 4
        push    20
        push    1
        push    1
        call    _fib
        add     esp, 12
        xor     eax, eax
        pop     ebp
        ret     0
```

```
_main   ENDP
```

OllyDbg:

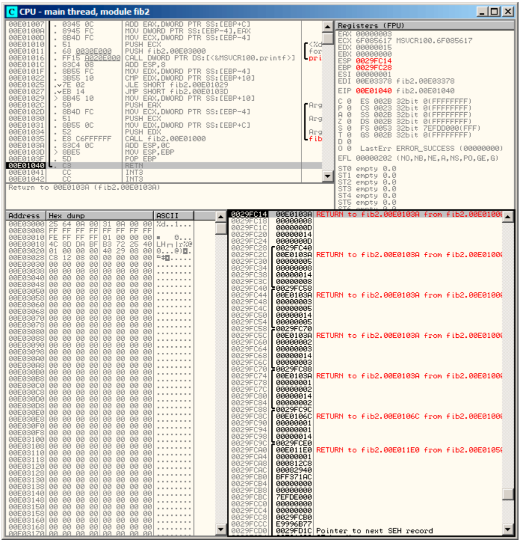


Figura 36.2: OllyDbg:

:

```
0029FC14   00E0103A  RETURN to fib2.00E0103A↙
    ↘ from fib2.00E01000
0029FC18   00000008  : a
0029FC1C   0000000D  : b
0029FC20   00000014  : limit
0029FC24   0000000D
0029FC28  /0029FC40
0029FC2C  |00E0103A  RETURN to fib2.00E0103A↙
    ↘ from fib2.00E01000
0029FC30  |00000005  : a
0029FC34  |00000008  : b
0029FC38  |00000014  : limit
0029FC3C  |00000008
0029FC40  ]0029FC58
0029FC44  |00E0103A  RETURN to fib2.00E0103A↙
    ↘ from fib2.00E01000
0029FC48  |00000003  : a
0029FC4C  |00000005  : b
0029FC50  |00000014  : limit
0029FC54  |00000005
0029FC58  ]0029FC70
0029FC5C  |00E0103A  RETURN to fib2.00E0103A↙
    ↘ from fib2.00E01000
0029FC60  |00000002  : a
0029FC64  |00000003  : b
0029FC68  |00000014  : limit
0029FC6C  |00000003
0029FC70  ]0029FC88
0029FC74  |00E0103A  RETURN to fib2.00E0103A↙
    ↘ from fib2.00E01000
```

```
0029FC78  |00000001  : a              \
0029FC7C  |00000002  : b              | ↙
    ↳ prepared in f1() for next f1()
0029FC80  |00000014  : limit          /
0029FC84  |00000002
0029FC88  ]0029FC9C
0029FC8C  |00E0106C  RETURN to fib2.00E0106C↙
    ↳  from fib2.00E01000
0029FC90  |00000001  : a              \
0029FC94  |00000001  : b              | ↙
    ↳ prepared in main() for f1()
0029FC98  |00000014  : limit          /
0029FC9C  ]0029FCE0
0029FCA0  |00E011E0  RETURN to fib2.00E011E0↙
    ↳  from fib2.00E01050
0029FCA4  |00000001  main() : argc  \
0029FCA8  |000812C8  main() : argv  |  main↙
    ↳ ()
0029FCAC  |00082940  main() : envp  /
```

.

36.3

. .

Capítulo 37

```
/* By Bob Jenkins, (c) 2006, Public Domain ⤦
    ↳ */

#include <stdio.h>
#include <stddef.h>
#include <string.h>

typedef  unsigned long  ub4;
typedef  unsigned char  ub1;

static const ub4 crctab[256] = {
  0x00000000, 0x77073096, 0xee0e612c, 0⤦
    ↳ x990951ba, 0x076dc419, 0x706af48f,
  0xe963a535, 0x9e6495a3, 0x0edb8832, 0⤦
    ↳ x79dcb8a4, 0xe0d5e91e, 0x97d2d988,
  0x09b64c2b, 0x7eb17cbd, 0xe7b82d07, 0⤦
    ↳ x90bf1d91, 0x1db71064, 0x6ab020f2,
```

```
0xf3b97148, 0x84be41de, 0x1adad47d, 0↙
  ↪ x6ddde4eb, 0xf4d4b551, 0x83d385c7,
0x136c9856, 0x646ba8c0, 0xfd62f97a, 0↙
  ↪ x8a65c9ec, 0x14015c4f, 0x63066cd9,
0xfa0f3d63, 0x8d080df5, 0x3b6e20c8, 0↙
  ↪ x4c69105e, 0xd56041e4, 0xa2677172,
0x3c03e4d1, 0x4b04d447, 0xd20d85fd, 0↙
  ↪ xa50ab56b, 0x35b5a8fa, 0x42b2986c,
0xdbbbc9d6, 0xacbcf940, 0x32d86ce3, 0↙
  ↪ x45df5c75, 0xdcd60dcf, 0xabd13d59,
0x26d930ac, 0x51de003a, 0xc8d75180, 0↙
  ↪ xbfd06116, 0x21b4f4b5, 0x56b3c423,
0xcfba9599, 0xb8bda50f, 0x2802b89e, 0↙
  ↪ x5f058808, 0xc60cd9b2, 0xb10be924,
0x2f6f7c87, 0x58684c11, 0xc1611dab, 0↙
  ↪ xb6662d3d, 0x76dc4190, 0x01db7106,
0x98d220bc, 0xefd5102a, 0x71b18589, 0↙
  ↪ x06b6b51f, 0x9fbfe4a5, 0xe8b8d433,
0x7807c9a2, 0x0f00f934, 0x9609a88e, 0↙
  ↪ xe10e9818, 0x7f6a0dbb, 0x086d3d2d,
0x91646c97, 0xe6635c01, 0x6b6b51f4, 0↙
  ↪ x1c6c6162, 0x856530d8, 0xf262004e,
0x6c0695ed, 0x1b01a57b, 0x8208f4c1, 0↙
  ↪ xf50fc457, 0x65b0d9c6, 0x12b7e950,
0x8bbeb8ea, 0xfcb9887c, 0x62dd1ddf, 0↙
  ↪ x15da2d49, 0x8cd37cf3, 0xfbd44c65,
0x4db26158, 0x3ab551ce, 0xa3bc0074, 0↙
  ↪ xd4bb30e2, 0x4adfa541, 0x3dd895d7,
0xa4d1c46d, 0xd3d6f4fb, 0x4369e96a, 0↙
  ↪ x346ed9fc, 0xad678846, 0xda60b8d0,
0x44042d73, 0x33031de5, 0xaa0a4c5f, 0↙
  ↪ xdd0d7cc9, 0x5005713c, 0x270241aa,
```

```
0xbe0b1010, 0xc90c2086, 0x5768b525, 0↙
   ↳ x206f85b3, 0xb966d409, 0xce61e49f,
0x5edef90e, 0x29d9c998, 0xb0d09822, 0↙
   ↳ xc7d7a8b4, 0x59b33d17, 0x2eb40d81,
0xb7bd5c3b, 0xc0ba6cad, 0xedb88320, 0↙
   ↳ x9abfb3b6, 0x03b6e20c, 0x74b1d29a,
0xead54739, 0x9dd277af, 0x04db2615, 0↙
   ↳ x73dc1683, 0xe3630b12, 0x94643b84,
0x0d6d6a3e, 0x7a6a5aa8, 0xe40ecf0b, 0↙
   ↳ x9309ff9d, 0x0a00ae27, 0x7d079eb1,
0xf00f9344, 0x8708a3d2, 0x1e01f268, 0↙
   ↳ x6906c2fe, 0xf762575d, 0x806567cb,
0x196c3671, 0x6e6b06e7, 0xfed41b76, 0↙
   ↳ x89d32be0, 0x10da7a5a, 0x67dd4acc,
0xf9b9df6f, 0x8ebeeff9, 0x17b7be43, 0↙
   ↳ x60b08ed5, 0xd6d6a3e8, 0xa1d1937e,
0x38d8c2c4, 0x4fdff252, 0xd1bb67f1, 0↙
   ↳ xa6bc5767, 0x3fb506dd, 0x48b2364b,
0xd80d2bda, 0xaf0a1b4c, 0x36034af6, 0↙
   ↳ x41047a60, 0xdf60efc3, 0xa867df55,
0x316e8eef, 0x4669be79, 0xcb61b38c, 0↙
   ↳ xbc66831a, 0x256fd2a0, 0x5268e236,
0xcc0c7795, 0xbb0b4703, 0x220216b9, 0↙
   ↳ x5505262f, 0xc5ba3bbe, 0xb2bd0b28,
0x2bb45a92, 0x5cb36a04, 0xc2d7ffa7, 0↙
   ↳ xb5d0cf31, 0x2cd99e8b, 0x5bdeae1d,
0x9b64c2b0, 0xec63f226, 0x756aa39c, 0↙
   ↳ x026d930a, 0x9c0906a9, 0xeb0e363f,
0x72076785, 0x05005713, 0x95bf4a82, 0↙
   ↳ xe2b87a14, 0x7bb12bae, 0x0cb61b38,
0x92d28e9b, 0xe5d5be0d, 0x7cdcefb7, 0↙
   ↳ x0bdbdf21, 0x86d3d2d4, 0xf1d4e242,
```

```
  0x68ddb3f8, 0x1fda836e, 0x81be16cd, 0↙
    ↳ xf6b9265b, 0x6fb077e1, 0x18b74777,
  0x88085ae6, 0xff0f6a70, 0x66063bca, 0↙
    ↳ x11010b5c, 0x8f659eff, 0xf862ae69,
  0x616bffd3, 0x166ccf45, 0xa00ae278, 0↙
    ↳ xd70dd2ee, 0x4e048354, 0x3903b3c2,
  0xa7672661, 0xd06016f7, 0x4969474d, 0↙
    ↳ x3e6e77db, 0xaed16a4a, 0xd9d65adc,
  0x40df0b66, 0x37d83bf0, 0xa9bcae53, 0↙
    ↳ xdebb9ec5, 0x47b2cf7f, 0x30b5ffe9,
  0xbdbdf21c, 0xcabac28a, 0x53b39330, 0↙
    ↳ x24b4a3a6, 0xbad03605, 0xcdd70693,
  0x54de5729, 0x23d967bf, 0xb3667a2e, 0↙
    ↳ xc4614ab8, 0x5d681b02, 0x2a6f2b94,
  0xb40bbe37, 0xc30c8ea1, 0x5a05df1b, 0↙
    ↳ x2d02ef8d,
};

/* how to derive the values in crctab[] from↙
    ↳  polynomial 0xedb88320 */
void build_table()
{
  ub4 i, j;
  for (i=0; i<256; ++i) {
    j = i;
    j = (j>>1) ^ ((j&1) ? 0xedb88320 : 0);
    j = (j>>1) ^ ((j&1) ? 0xedb88320 : 0);
    j = (j>>1) ^ ((j&1) ? 0xedb88320 : 0);
    j = (j>>1) ^ ((j&1) ? 0xedb88320 : 0);
    j = (j>>1) ^ ((j&1) ? 0xedb88320 : 0);
    j = (j>>1) ^ ((j&1) ? 0xedb88320 : 0);
    j = (j>>1) ^ ((j&1) ? 0xedb88320 : 0);
    j = (j>>1) ^ ((j&1) ? 0xedb88320 : 0);
```

```
    printf("0x%.8lx, ", j);
    if (i%6 == 5) printf("\n");
  }
}


/* the hash function */
ub4 crc(const void *key, ub4 len, ub4 hash)
{
  ub4  i;
  const ub1 *k = key;
  for (hash=len, i=0; i<len; ++i)
    hash = (hash >> 8) ^ crctab[(hash & 0xff
    ↪ ) ^ k[i]];
  return hash;
}

/* To use, try "gcc -O crc.c -o crc; crc < 
    ↪ crc.c" */
int main()
{
  char s[1000];
  while (gets(s)) printf("%.8lx\n", crc(s, 
    ↪ strlen(s), 0));
  return 0;
}
```

```
_key$ = 8                         ; size = 4
_len$ = 12                        ; size = 4
_hash$ = 16                       ; size = 4
_crc    PROC
```

```
    mov     edx, DWORD PTR _len$[esp-4]
    xor     ecx, ecx ;
    mov     eax, edx
    test    edx, edx
    jbe     SHORT $LN1@crc
    push    ebx
    push    esi
    mov     esi, DWORD PTR _key$[esp+4] ; ESI⤦
    ↳  = key
    push    edi
$LL3@crc:

;

    movzx   edi, BYTE PTR [ecx+esi]
    mov     ebx, eax ; EBX = (hash = len)
    and     ebx, 255 ; EBX = hash & 0xff

; XOR EDI, EBX (EDI=EDI^EBX) -
;
;
;

    xor     edi, ebx

; EAX=EAX>>8;
    shr     eax, 8

; EAX=EAX^crctab[EDI*4] -
    xor     eax, DWORD PTR _crctab[edi*4]
    inc     ecx                 ; i++
    cmp     ecx, edx            ; i<len ?
    jb      SHORT $LL3@crc ;
```

```
    pop    edi
    pop    esi
    pop    ebx
$LN1@crc:
    ret    0
_crc    ENDP
```

```

                public crc
crc             proc near

key             = dword ptr  8
hash            = dword ptr  0Ch

                push    ebp
                xor     edx, edx
                mov     ebp, esp
                push    esi
                mov     esi, [ebp+key]
                push    ebx
                mov     ebx, [ebp+hash]
                test    ebx, ebx
                mov     eax, ebx
                jz      short loc_80484D3
                nop                          ;
                lea     esi, [esi+0]         ;

loc_80484B8:
                mov     ecx, eax             ;
                xor     al, [esi+edx]        ; AL⤦
    ↳ =*(key+i)
                add     edx, 1               ; i⤦
```

```
    ↳ ++
                shr     ecx, 8          ; ↙
    ↳ ECX=hash>>8
                movzx   eax, al         ; ↙
    ↳ EAX=*(key+i)
                mov     eax, dword ptr ds:↙
    ↳ crctab[eax*4] ; EAX=crctab[EAX]
                xor     eax, ecx        ; ↙
    ↳ hash=EAX^ECX
                cmp     ebx, edx
                ja      short loc_80484B8

loc_80484D3:
                pop     ebx
                pop     esi
                pop     ebp
                retn
crc             endp
\
```

CAPÍTULO 38.

Capítulo 38

CAPÍTULO 38.

/30	4	2	255.255.255.252	ffffff
/29	8	6	255.255.255.248	ffffff
/28	16	14	255.255.255.240	ffffff
/27	32	30	255.255.255.224	ffffff
/26	64	62	255.255.255.192	ffffff
/24	256	254	255.255.255.0	ffffff
/23	512	510	255.255.254.0	fffffe
/22	1024	1022	255.255.252.0	fffffc
/21	2048	2046	255.255.248.0	fffff8
/20	4096	4094	255.255.240.0	fffff0
/19	8192	8190	255.255.224.0	ffffe0
/18	16384	16382	255.255.192.0	ffffc0
/17	32768	32766	255.255.128.0	ffff80
/16	65536	65534	255.255.0.0	ffff00
/8	16777216	16777214	255.0.0.0	ff0000

```
#include <stdio.h>
#include <stdint.h>

uint32_t form_IP (uint8_t ip1, uint8_t ip2, uint8_t ip3, uint8_t ip4)
{
        return (ip1<<24) | (ip2<<16) | (ip3<<8) | ip4;
};

void print_as_IP (uint32_t a)
{
        printf ("%d.%d.%d.%d\n",
                (a>>24)&0xFF,
```

```
                (a>>16)&0xFF,
                (a>>8)&0xFF,
                (a)&0xFF);
};

// bit=31..0
uint32_t set_bit (uint32_t input, int bit)
{
        return input=input|(1<<bit);
};

uint32_t form_netmask (uint8_t netmask_bits)
{
        uint32_t netmask=0;
        uint8_t i;

        for (i=0; i<netmask_bits; i++)
                netmask=set_bit(netmask, 31-↙
    ↘ i);

        return netmask;
};

void calc_network_address (uint8_t ip1, ↙
    ↘ uint8_t ip2, uint8_t ip3, uint8_t ip4,↙
    ↘  uint8_t netmask_bits)
{
        uint32_t netmask=form_netmask(↙
    ↘ netmask_bits);
        uint32_t ip=form_IP(ip1, ip2, ip3, ↙
    ↘ ip4);
        uint32_t netw_adr;
```

```
        printf ("netmask=");
        print_as_IP (netmask);

        netw_adr=ip&netmask;

        printf ("network address=");
        print_as_IP (netw_adr);
};

int main()
{
        calc_network_address (10, 1, 2, 4, 24);    // 10.1.2.4, /24
        calc_network_address (10, 1, 2, 4, 8);     // 10.1.2.4, /8
        calc_network_address (10, 1, 2, 4, 25);    // 10.1.2.4, /25
        calc_network_address (10, 1, 2, 64, 26);  // 10.1.2.4, /26
};
```

38.1 calc_network_address()

`calc_network_address()` :

Listing 38.1: Con optimización MSVC 2012 /Ob0

```
_ip1$ = 8                       ; size = 1
_ip2$ = 12                      ; size = 1
_ip3$ = 16                      ; size = 1
_ip4$ = 20                      ; size = 1
```

```
_netmask_bits$ = 24     ; size = 1
_calc_network_address PROC
        push    edi
        push    DWORD PTR _netmask_bits$[esp]
        call    _form_netmask
        push    OFFSET $SG3045 ; 'netmask='
        mov     edi, eax
        call    DWORD PTR __imp__printf
        push    edi
        call    _print_as_IP
        push    OFFSET $SG3046 ; 'network address='
        call    DWORD PTR __imp__printf
        push    DWORD PTR _ip4$[esp+16]
        push    DWORD PTR _ip3$[esp+20]
        push    DWORD PTR _ip2$[esp+24]
        push    DWORD PTR _ip1$[esp+28]
        call    _form_IP
        and     eax, edi        ; network address = host address & netmask
        push    eax
        call    _print_as_IP
        add     esp, 36
        pop     edi
        ret     0
_calc_network_address ENDP
```

38.2 form_IP()

`form_IP()`.

:

-
-
- .
-
-

Listing 38.2: Sin optimización MSVC 2012

```
;  ip1  "dd", ip2  "cc", ip3  "bb", ip4  "aa".
_ip1$ = 8       ; size = 1
_ip2$ = 12      ; size = 1
_ip3$ = 16      ; size = 1
_ip4$ = 20      ; size = 1
_form_IP PROC
        push    ebp
        mov     ebp, esp
        movzx   eax, BYTE PTR _ip1$[ebp]
        ; EAX=000000dd
        shl     eax, 24
        ; EAX=dd000000
        movzx   ecx, BYTE PTR _ip2$[ebp]
        ; ECX=000000cc
        shl     ecx, 16
        ; ECX=00cc0000
        or      eax, ecx
        ; EAX=ddcc0000
        movzx   edx, BYTE PTR _ip3$[ebp]
```

```
        ; EDX=000000bb
        shl     edx, 8
        ; EDX=0000bb00
        or      eax, edx
        ; EAX=ddccbb00
        movzx   ecx, BYTE PTR _ip4$[ebp]
        ; ECX=000000aa
        or      eax, ecx
        ; EAX=ddccbbaa
        pop     ebp
        ret     0
_form_IP ENDP
```

Con optimización MSVC 2012 :

Listing 38.3: Con optimización MSVC 2012 /Ob0

```

;  ip1  "dd", ip2  "cc", ip3  "bb", ip4  "aa".
_ip1$ = 8       ; size = 1
_ip2$ = 12      ; size = 1
_ip3$ = 16      ; size = 1
_ip4$ = 20      ; size = 1
_form_IP PROC
        movzx   eax, BYTE PTR _ip1$[esp-4]
        ; EAX=000000dd
        movzx   ecx, BYTE PTR _ip2$[esp-4]
        ; ECX=000000cc
        shl     eax, 8
        ; EAX=0000dd00
        or      eax, ecx
        ; EAX=0000ddcc
```

```
        movzx   ecx, BYTE PTR _ip3$[esp-4]
        ; ECX=000000bb
        shl     eax, 8
        ; EAX=00ddcc00
        or      eax, ecx
        ; EAX=00ddccbb
        movzx   ecx, BYTE PTR _ip4$[esp-4]
        ; ECX=000000aa
        shl     eax, 8
        ; EAX=ddccbb00
        or      eax, ecx
        ; EAX=ddccbbaa
        ret     0
_form_IP ENDP
```

Repetir 4 veces, para cada byte de entrada.

!

38.3 print_as_IP()

`print_as_IP()`

Listing 38.4: Sin optimización MSVC 2012

```
_a$ = 8                         ; size = 4
_print_as_IP PROC
        push    ebp
        mov     ebp, esp
        mov     eax, DWORD PTR _a$[ebp]
        ; EAX=ddccbbaa
```

```
        and     eax, 255
        ; EAX=000000aa
        push    eax
        mov     ecx, DWORD PTR _a$[ebp]
        ; ECX=ddccbbaa
        shr     ecx, 8
        ; ECX=00ddccbb
        and     ecx, 255
        ; ECX=000000bb
        push    ecx
        mov     edx, DWORD PTR _a$[ebp]
        ; EDX=ddccbbaa
        shr     edx, 16
        ; EDX=0000ddcc
        and     edx, 255
        ; EDX=000000cc
        push    edx
        mov     eax, DWORD PTR _a$[ebp]
        ; EAX=ddccbbaa
        shr     eax, 24
        ; EAX=000000dd
        and     eax, 255 ;
        ; EAX=000000dd
        push    eax
        push    OFFSET $SG2973 ; '%d.%d.%d.%
    ↘ d'
        call    DWORD PTR __imp__printf
        add     esp, 20
        pop     ebp
        ret     0
_print_as_IP ENDP
```

Con optimización MSVC 2012

Listing 38.5: Con optimización MSVC 2012 /Ob0

```
_a$ = 8         ; size = 4
_print_as_IP PROC
        mov     ecx, DWORD PTR _a$[esp-4]
        ; ECX=ddccbbaa
        movzx   eax, cl
        ; EAX=000000aa
        push    eax
        mov     eax, ecx
        ; EAX=ddccbbaa
        shr     eax, 8
        ; EAX=00ddccbb
        and     eax, 255
        ; EAX=000000bb
        push    eax
        mov     eax, ecx
        ; EAX=ddccbbaa
        shr     eax, 16
        ; EAX=0000ddcc
        and     eax, 255
        ; EAX=000000cc
        push    eax
        ; ECX=ddccbbaa
        shr     ecx, 24
        ; ECX=000000dd
        push    ecx
        push    OFFSET $SG3020 ; '%d.%d.%d.%↙
    ↘ d'
        call    DWORD PTR __imp__printf
        add     esp, 20
        ret     0
```

```
_print_as_IP ENDP
```

38.4 form_netmask() y set_bit()

form_netmask(). set_bit().

Listing 38.6: Con optimización MSVC 2012 /Ob0

```
_input$ = 8                 ; size = 4
_bit$ = 12                  ; size = 4
_set_bit PROC
        mov     ecx, DWORD PTR _bit$[esp-4]
        mov     eax, 1
        shl     eax, cl
        or      eax, DWORD PTR _input$[esp
    -4]
        ret     0
_set_bit ENDP

_netmask_bits$ = 8          ; size = 1
_form_netmask PROC
        push    ebx
        push    esi
        movzx   esi, BYTE PTR _netmask_bits$
    [esp+4]
        xor     ecx, ecx
        xor     bl, bl
        test    esi, esi
        jle     SHORT $LN9@form_netma
        xor     edx, edx
$LL3@form_netma:
```

```
        mov     eax, 31
        sub     eax, edx
        push    eax
        push    ecx
        call    _set_bit
        inc     bl
        movzx   edx, bl
        add     esp, 8
        mov     ecx, eax
        cmp     edx, esi
        jl      SHORT $LL3@form_netma
$LN9@form_netma:
        pop     esi
        mov     eax, ecx
        pop     ebx
        ret     0
_form_netmask ENDP
```

set_bit() . form_netmask()

38.5

! :

```
netmask=255.255.255.0
network address=10.1.2.0
netmask=255.0.0.0
network address=10.0.0.0
netmask=255.255.255.128
network address=10.1.2.0
netmask=255.255.255.192
```

CAPÍTULO 38.

```
network address=10.1.2.64
```

Capítulo 39

:

```
#include <stdio.h>

void f(int *a1, int *a2, size_t cnt)
{
        size_t i;

        //
        for (i=0; i<cnt; i++)
                a1[i*3]=a2[i*7];
};
```

?

39.1

Listing 39.1: Con optimización MSVC 2013 x64

```
f       PROC
; RDX=a1
; RCX=a2
; R8=cnt
        test    r8, r8          ; cnt==0?
        je      SHORT $LN1@f
        npad    11
$LL3@f:
        mov     eax, DWORD PTR [rdx]
        lea     rcx, QWORD PTR [rcx+12]
        lea     rdx, QWORD PTR [rdx+28]
        mov     DWORD PTR [rcx-12], eax
        dec     r8
        jne     SHORT $LL3@f
$LN1@f:
        ret     0
f       ENDP
```

C/C++:

```
#include <stdio.h>

void f(int *a1, int *a2, size_t cnt)
{
        size_t i;
        size_t idx1=0; idx2=0;

        //
        for (i=0; i<cnt; i++)
```

```
        {
                a1[idx1]=a2[idx2];
                idx1+=3;
                idx2+=7;
        };
};
```

39.2

Listing 39.2: Con optimización GCC 4.9 x64

```

; RDI=a1
; RSI=a2
; RDX=cnt
f:
        test    rdx, rdx  ; cnt==0?
        je      .L1
;
        lea     rax, [0+rdx*4]
; RAX=RDX*4=cnt*4
        sal     rdx, 5
; RDX=RDX<<5=cnt*32
        sub     rdx, rax
; RDX=RDX-RAX=cnt*32-cnt*4=cnt*28
        add     rdx, rsi
; RDX=RDX+RSI=a2+cnt*28
.L3:
        mov     eax, DWORD PTR [rsi]
        add     rsi, 28
        add     rdi, 12
```

```
        mov     DWORD PTR [rdi-12], eax
        cmp     rsi, rdx
        jne     .L3
.L1:
        rep ret
```

16.1.3 on page 400.

```

#include <stdio.h>

void f(int *a1, int *a2, size_t cnt)
{
        size_t i;
        size_t idx1=0; idx2=0;
        size_t last_idx2=cnt*7;

        //
        for (;;)
        {
                a1[idx1]=a2[idx2];
                idx1+=3;
                idx2+=7;
                if (idx2==last_idx2)
                        break;
        };
};
```

GCC (Linaro) 4.9 para ARM64 :

Listing 39.3: Con optimización GCC (Linaro) 4.9 ARM64

```
; X0=a1
; X1=a2
; X2=cnt
f:
        cbz     x2, .L1                 ; cnt==0?
;
        add     x2, x2, x2, lsl 1
; X2=X2+X2<<1=X2+X2*2=X2*3
        mov     x3, 0
        lsl     x2, x2, 2
; X2=X2<<2=X2*4=X2*3*4=X2*12
.L3:
        ldr     w4, [x1],28             ;
        str     w4, [x0,x3]             ;  X0+X3=⤦
    ⤥ a1+X3
        add     x3, x3, 12              ;  X3
        cmp     x3, x2                  ; ?
        bne     .L3
.L1:
        ret
```

GCC 4.4.5 para MIPS :

Listing 39.4: Con optimización GCC 4.4.5 para MIPS (IDA)

```

; $a0=a1
; $a1=a2
; $a2=cnt
f:
; :
                beqz    $a2, locret_24
; :
```

```
                move    $v0, $zero ; branch ↙
    ↘ delay slot, NOP

loc_8:
;  $a1
                lw      $a3, 0($a1)
;  (i):
                addiu   $v0, 1
; :
                sltu    $v1, $v0, $a2
;  $a0:
                sw      $a3, 0($a0)
;  0x1C (28)  \$a1 :
                addiu   $a1, 0x1C
;  i<cnt:
                bnez    $v1, loc_8
;  0xC (12)  \$a0 :
                addiu   $a0, 0xC ; branch ↙
    ↘ delay slot

locret_24:
                jr      $ra
                or      $at, $zero ; branch ↙
    ↘ delay slot, NOP
```

39.3 Intel C++ 2011

:

Listing 39.5: Con optimización Intel C++ 2011 x64

```
f       PROC
; parameter 1: rcx = a1
; parameter 2: rdx = a2
; parameter 3: r8  = cnt
.B1.1::                         ; Preds .B1⤦
    ↪ .0
        test        r8, r8                          ⤦
    ↪                          ;8.14
        jbe         exit            ; Prob 50%    ⤦
    ↪                        ;8.14
                                ; LOE rdx ⤦
    ↪ rcx rbx rbp rsi rdi r8 r12 r13 r14 r15⤦
    ↪  xmm6 xmm7 xmm8 xmm9 xmm10 xmm11 xmm12⤦
    ↪  xmm13 xmm14 xmm15
.B1.2::                         ; Preds .B1⤦
    ↪ .1
        cmp         r8, 6                           ⤦
    ↪                          ;8.2
        jbe         just_copy       ; Prob ⤦
    ↪ 50%                           ;8.2
                                ; LOE rdx ⤦
    ↪ rcx rbx rbp rsi rdi r8 r12 r13 r14 r15⤦
    ↪  xmm6 xmm7 xmm8 xmm9 xmm10 xmm11 xmm12⤦
    ↪  xmm13 xmm14 xmm15
.B1.3::                         ; Preds .B1⤦
    ↪ .2
        cmp         rcx, rdx                        ⤦
    ↪                          ;9.11
        jbe         .B1.5           ; Prob 50%  ⤦
    ↪                          ;9.11
                                ; LOE rdx ⤦
    ↪ rcx rbx rbp rsi rdi r8 r12 r13 r14 r15⤦
    ↪  xmm6 xmm7 xmm8 xmm9 xmm10 xmm11 xmm12⤦
```

```
    ↳  xmm13 xmm14 xmm15
.B1.4::                              ; Preds .B1↙
    ↳ .3
        mov         r10, r8                      ↙
    ↳                       ;9.11
        mov         r9, rcx                      ↙
    ↳                       ;9.11
        shl         r10, 5                       ↙
    ↳                       ;9.11
        lea         rax, QWORD PTR [r8*4]        ↙
    ↳                       ;9.11
        sub         r9, rdx                      ↙
    ↳                       ;9.11
        sub         r10, rax                     ↙
    ↳                       ;9.11
        cmp         r9, r10                      ↙
    ↳                       ;9.11
        jge         just_copy2          ; Prob ↙
    ↳ 50%                        ;9.11
                                     ; LOE rdx ↙
    ↳ rcx rbx rbp rsi rdi r8 r12 r13 r14 r15↙
    ↳  xmm6 xmm7 xmm8 xmm9 xmm10 xmm11 xmm12↙
    ↳  xmm13 xmm14 xmm15
.B1.5::                              ; Preds .B1↙
    ↳ .3 .B1.4
        cmp         rdx, rcx                     ↙
    ↳                       ;9.11
        jbe         just_copy         ; Prob ↙
    ↳ 50%                        ;9.11
                                     ; LOE rdx ↙
    ↳ rcx rbx rbp rsi rdi r8 r12 r13 r14 r15↙
    ↳  xmm6 xmm7 xmm8 xmm9 xmm10 xmm11 xmm12↙
    ↳  xmm13 xmm14 xmm15
```

```
.B1.6::                               ; Preds .B1↙
    ↪ .5
        mov         r9, rdx                          ↙
    ↪                       ;9.11
        lea         rax, QWORD PTR [r8*8]            ↙
    ↪                       ;9.11
        sub         r9, rcx                          ↙
    ↪                       ;9.11
        lea         r10, QWORD PTR [rax+r8*4]        ↙
    ↪                       ;9.11
        cmp         r9, r10                          ↙
    ↪                       ;9.11
        jl          just_copy           ; Prob ↙
    ↪ 50%                               ;9.11
                                      ; LOE rdx ↙
    ↪ rcx rbx rbp rsi rdi r8 r12 r13 r14 r15↙
    ↪  xmm6 xmm7 xmm8 xmm9 xmm10 xmm11 xmm12↙
    ↪  xmm13 xmm14 xmm15
just_copy2::                                   ; Preds↙
    ↪  .B1.4 .B1.6
; R8 = cnt
; RDX = a2
; RCX = a1
        xor         r10d, r10d                       ↙
    ↪                       ;8.2
        xor         r9d, r9d                         ↙
    ↪                       ;
        xor         eax, eax                         ↙
    ↪                       ;
                                      ; LOE rax ↙
    ↪ rdx rcx rbx rbp rsi rdi r8 r9 r10 r12 ↙
    ↪ r13 r14 r15 xmm6 xmm7 xmm8 xmm9 xmm10 ↙
    ↪ xmm11 xmm12 xmm13 xmm14 xmm15
```

```
.B1.8::                         ; Preds .B1↙
    ↳ .8 just_copy2
        mov         r11d, DWORD PTR [rax+rdx] ↙
    ↳                       ;3.6
        inc         r10                        ↙
    ↳                       ;8.2
        mov         DWORD PTR [r9+rcx], r11d   ↙
    ↳                       ;3.6
        add         r9, 12                     ↙
    ↳                       ;8.2
        add         rax, 28                    ↙
    ↳                       ;8.2
        cmp         r10, r8                    ↙
    ↳                       ;8.2
        jb          .B1.8           ; Prob 82% ↙
    ↳                       ;8.2
        jmp         exit          ; Prob 100%  ↙
    ↳                     ;8.2
                                ; LOE rax ↙
    ↳ rdx rcx rbx rbp rsi rdi r8 r9 r10 r12 ↙
    ↳ r13 r14 r15 xmm6 xmm7 xmm8 xmm9 xmm10 ↙
    ↳ xmm11 xmm12 xmm13 xmm14 xmm15
just_copy::                        ; Preds .↙
    ↳ B1.2 .B1.5 .B1.6
; R8 = cnt
; RDX = a2
; RCX = a1
        xor         r10d, r10d                 ↙
    ↳                       ;8.2
        xor         r9d, r9d                   ↙
    ↳                       ;
        xor         eax, eax                   ↙
    ↳                       ;
```

```
                                   ; LOE rax ⤦
    ↳ rdx rcx rbx rbp rsi rdi r8 r9 r10 r12 ⤦
    ↳ r13 r14 r15 xmm6 xmm7 xmm8 xmm9 xmm10 ⤦
    ↳ xmm11 xmm12 xmm13 xmm14 xmm15
.B1.11::                           ; Preds .B1⤦
    ↳ .11 just_copy
      mov       r11d, DWORD PTR [rax+rdx] ⤦
    ↳                        ;3.6
      inc       r10                       ⤦
    ↳                        ;8.2
      mov       DWORD PTR [r9+rcx], r11d  ⤦
    ↳                        ;3.6
      add       r9, 12                    ⤦
    ↳                        ;8.2
      add       rax, 28                   ⤦
    ↳                        ;8.2
      cmp       r10, r8                   ⤦
    ↳                        ;8.2
      jb        .B1.11         ; Prob 82% ⤦
    ↳                        ;8.2
                                   ; LOE rax ⤦
    ↳ rdx rcx rbx rbp rsi rdi r8 r9 r10 r12 ⤦
    ↳ r13 r14 r15 xmm6 xmm7 xmm8 xmm9 xmm10 ⤦
    ↳ xmm11 xmm12 xmm13 xmm14 xmm15
exit::                           ; Preds .B1.11⤦
    ↳  .B1.8 .B1.1
      ret                                 ⤦
    ↳                        ;10.1
```

!

Capítulo 40

Duff's device

Duff's device

. ?

:

-
-

. :

```
#include <stdint.h>
#include <stdio.h>

void bzero(uint8_t* dst, size_t count)
```

 CAPÍTULO 40. DUFF'S DEVICE

```
{
        int i;

        if (count&(~7))
                //
                for (i=0; i<count>>3; i++)
                {
                        *(uint64_t*)dst=0;
                        dst=dst+8;
                };

        //
        switch(count & 7)
        {
        case 7: *dst++ = 0;
        case 6: *dst++ = 0;
        case 5: *dst++ = 0;
        case 4: *dst++ = 0;
        case 3: *dst++ = 0;
        case 2: *dst++ = 0;
        case 1: *dst++ = 0;
        case 0: //
                break;
        }
}
```

	7	6	5	4	3	2	1	0
...	B	B	B	B	B	S	S	S

.

Spanish text placeholder.

.

:

```
dst$ = 8
count$ = 16
bzero   PROC
        test    rdx, -8
        je      SHORT $LN11@bzero
;
        xor     r10d, r10d
        mov     r9, rdx
        shr     r9, 3
        mov     r8d, r10d
        test    r9, r9
        je      SHORT $LN11@bzero
        npad    5
$LL19@bzero:
        inc     r8d
        mov     QWORD PTR [rcx], r10
        add     rcx, 8
        movsxd  rax, r8d
        cmp     rax, r9
        jb      SHORT $LL19@bzero
$LN11@bzero:
;
        and     edx, 7
        dec     rdx
        cmp     rdx, 6
        ja      SHORT $LN9@bzero
        lea     r8, OFFSET FLAT:__ImageBase
        mov     eax, DWORD PTR $LN22@bzero[
```

```
    ↳ r8+rdx*4]
        add     rax, r8
        jmp     rax
$LN8@bzero:
        mov     BYTE PTR [rcx], 0
        inc     rcx
$LN7@bzero:
        mov     BYTE PTR [rcx], 0
        inc     rcx
$LN6@bzero:
        mov     BYTE PTR [rcx], 0
        inc     rcx
$LN5@bzero:
        mov     BYTE PTR [rcx], 0
        inc     rcx
$LN4@bzero:
        mov     BYTE PTR [rcx], 0
        inc     rcx
$LN3@bzero:
        mov     BYTE PTR [rcx], 0
        inc     rcx
$LN2@bzero:
        mov     BYTE PTR [rcx], 0
$LN9@bzero:
        fatret  0
        npad    1
$LN22@bzero:
        DD      $LN2@bzero
        DD      $LN3@bzero
        DD      $LN4@bzero
        DD      $LN5@bzero
        DD      $LN6@bzero
        DD      $LN7@bzero
```

```
        DD      $LN8@bzero
bzero   ENDP
```

.

Capítulo 41

División por 9

```
int f(int a)
{
        return a/9;
};
```

41.1 x86

…

Listing 41.1: MSVC

```
_a$ = 8                 ; size = 4
_f     PROC
```

```
    push   ebp
    mov    ebp, esp
    mov    eax, DWORD PTR _a$[ebp]
    cdq             ;
    mov    ecx, 9
    idiv   ecx
    pop    ebp
    ret    0
_f  ENDP
```

Listing 41.2: Con optimización MSVC

```
_a$ = 8                         ; size = 4
_f    PROC

    mov    ecx, DWORD PTR _a$[esp-4]
    mov    eax, 954437177   ; 38e38e39H
    imul   ecx
    sar    edx, 1
    mov    eax, edx
    shr    eax, 31          ; 0000001fH
    add    eax, edx
    ret    0
_f    ENDP
```

«strength reduction».

Listing 41.3: Sin optimización GCC 4.4.1

```
      public f
f     proc near

arg_0  = dword ptr  8
```

```
        push    ebp
        mov     ebp, esp
        mov     ecx, [ebp+arg_0]
        mov     edx, 954437177 ; 38E38E39h
        mov     eax, ecx
        imul    edx
        sar     edx, 1
        mov     eax, ecx
        sar     eax, 1Fh
        mov     ecx, edx
        sub     ecx, eax
        mov     eax, ecx
        pop     ebp
        retn
f       endp
```

41.2 ARM

(19 on page 577).

[Ltd94, 3.3 Division by a Constant]. .

```
; takes argument in a1
; returns quotient in a1, remainder in a2
; cycles could be saved if only divide or ↙
    ↘ remainder is required
    SUB     a2, a1, #10             ; keep (x↙
    ↘ -10) for later
    SUB     a1, a1, a1, lsr #2
    ADD     a1, a1, a1, lsr #4
```

```
    ADD     a1, a1, a1, lsr #8
    ADD     a1, a1, a1, lsr #16
    MOV     a1, a1, lsr #3
    ADD     a3, a1, a1, asl #2
    SUBS    a2, a2, a3, asl #1          ; calc (x↙
    ↪ -10) - (x/10)*10
    ADDPL   a1, a1, #1                  ; fix-up ↙
    ↪ quotient
    ADDMI   a2, a2, #10                 ; fix-up ↙
    ↪ remainder
    MOV     pc, lr
```

41.2.1 Con optimización Xcode 4.6.3 (LLVM) (Modo ARM)

```
__text:00002C58 39 1E 08 E3 E3 18 43 E3   MOV↙
    ↪     R1, 0x38E38E39
__text:00002C60 10 F1 50 E7                    ↙
    ↪ SMMUL  R0, R0, R1
__text:00002C64 C0 10 A0 E1               MOV↙
    ↪     R1, R0,ASR#1
__text:00002C68 A0 0F 81 E0               ADD↙
    ↪     R0, R1, R0,LSR#31
__text:00002C6C 1E FF 2F E1               BX ↙
    ↪     LR
```

IDA

SMMUL (*Signed Most Significant Word Multiply*)

«MOV R1, R0,ASR#1»

«ADD R0, R1, R0,LSR#31» $R0 = R1 + R0 >> 31$

(MOV, ADD, SUB, RSB)[1] ASR *Arithmetic Shift Right*, LSR*Logical Shift Right*.

41.2.2 Con optimización Xcode 4.6.3 (LLVM) (Modo Thumb-2)

```
MOV             R1, 0x38E38E39
SMMUL.W         R0, R0, R1
ASRS            R1, R0, #1
ADD.W           R0, R1, R0,LSR#31
BX              LR
```

, ASRS ().

41.2.3 Sin optimización Xcode 4.6.3 (LLVM) y Keil 6/2013

Sin optimización LLVM .

__aeabi_idivmod.

41.3 MIPS

Listing 41.4: Con optimización GCC 4.4.5 (IDA)

[1]Estas instrucciones también se llaman «data processing instructions»

```
f:
                li      $v0, 9
                bnez    $v0, loc_10
                div     $a0, $v0 ; branch ↙
    ↳ delay slot
                break   0x1C00   ; "break 7"

loc_10:
                mflo    $v0
                jr      $ra
                or      $at, $zero ; branch ↙
    ↳ delay slot, NOP
```

.

«a % 9»,

41.4

:

$$result = \frac{input}{divisor} = \frac{input \cdot \frac{2^n}{divisor}}{2^n} = \frac{input \cdot M}{2^n}$$

M *magic.*

:

$$M = \frac{2^n}{divisor}$$

:

$$result = \frac{input \cdot M}{2^n}$$

$n < 32$, (en EAX o RAX). $n \geq 32$, (en EDX o RDX).

n .

.

:

```
int f3_32_signed(int a)
{
        return a/3;
};

unsigned int f3_32_unsigned(unsigned int a)
{
        return a/3;
};
```

magic 0xAAAAAAAB 2^{33}.

magic 0x55555556 2^{32}. EDX.

1 .

Listing 41.5: Con optimización MSVC 2012

```
_f3_32_unsigned PROC
        mov     eax, -1431655765        ; ↙
    ↘ aaaaaaabH
```

```
        mul     DWORD PTR _a$[esp-4] ;
; EDX=(input*0xaaaaaaab)/2^32
        shr     edx, 1
; EDX=(input*0xaaaaaaab)/2^33
        mov     eax, edx
        ret     0
_f3_32_unsigned ENDP

_f3_32_signed PROC
        mov     eax, 1431655766         ; ↙
    ↳ 55555556H
        imul    DWORD PTR _a$[esp-4] ;
;
;  2^32
        mov     eax, edx        ; EAX=EDX=(↙
    ↳ input*0x55555556)/2^32
        shr     eax, 31         ; 0000001fH
        add     eax, edx        ;
        ret     0
_f3_32_signed ENDP
```

41.4.1

$$\frac{x}{c} = x\frac{1}{c}$$

[2].

[War02, págs. 10-3].

[2] Wikipedia

41.5

41.5.1 #1

:

```
        mov     eax, MAGICAL CONSTANT
        imul
        sar     edx, SHIFTING COEFFICIENT ;
        mov     eax, edx
        shr     eax, 31
        add     eax, edx
```

M, C D.

:

$$D = \frac{2^{32+C}}{M}$$

:

Listing 41.6: Con optimización MSVC 2012

```
        mov     eax, 2021161081                 ↙
    ↘                   ; 78787879H
        imul    DWORD PTR _a$[esp-4]
        sar     edx, 3
        mov     eax, edx
        shr     eax, 31                         ↙
    ↘                   ; 0000001fH
        add     eax, edx
```

:

$$D = \frac{2^{32+3}}{2021161081}$$

Wolfram Mathematica :

Listing 41.7: Wolfram Mathematica

```
In[1]:=N[2^(32+3)/2021161081]
Out[1]:=17.
```

17.

2^{64} 2^{32}:

```
uint64_t f1234(uint64_t a)
{
        return a/1234;
};
```

Listing 41.8: Con optimización MSVC 2012 x64

```
f1234   PROC
        mov     rax, 7653754429286296943 ; 6a37991a23aead6fH
        mul     rcx
        shr     rdx, 9
        mov     rax, rdx
        ret     0
f1234   ENDP
```

Listing 41.9: Wolfram Mathematica

```
In[1]:=N[2^(64+9)/16^^6a37991a23aead6f]
Out[1]:=1234.
```

41.5.2 #2

:

```
                mov     eax, 55555556h ; 
    1431655766
                imul    ecx
                mov     eax, edx
                shr     eax, 1Fh
```

:

$$D = \frac{2^{32}}{M}$$

:

$$D = \frac{2^{32}}{1431655766}$$

Wolfram Mathematica:

Listing 41.10: Wolfram Mathematica

```
In[1]:=N[2^32/16^^55555556]
Out[1]:=3.
```

3.

41.6 Ejercicio #1

Lo que hace el código?

Listing 41.11: Con optimización MSVC 2010

```
_a$ = 8
_f      PROC
        mov     ecx, DWORD PTR _a$[esp-4]
        mov     eax, -968154503 ; c64b2279H
        imul    ecx
        add     edx, ecx
        sar     edx, 9
        mov     eax, edx
        shr     eax, 31             ; 0000001fH
        add     eax, edx
        ret     0
_f      ENDP
```

Listing 41.12: Con optimización GCC 4.9 (ARM64)

```
f:
        mov     w1, 8825
        movk    w1, 0xc64b, lsl 16
        smull   x1, w0, w1
        lsr     x1, x1, 32
        add     w1, w0, w1
        asr     w1, w1, 9
        sub     w0, w1, w0, asr 31
        ret
```

Responda G.1.16 on page 1931.

Capítulo 42

(atoi())

42.1

```
#include <stdio.h>

int my_atoi (char *s)
{
        int rt=0;

        while (*s)
        {
                rt=rt*10 + (*s-'0');
                s++;
        };

```

```
        return rt;
};

int main()
{
        printf ("%d\n", my_atoi ("1234"));
        printf ("%d\n", my_atoi ⤦
    ↳ ("1234567890"));
};
```

42.1.1 Con optimización MSVC 2013 x64

Listing 42.1: Con optimización MSVC 2013 x64

```
s$ = 8
my_atoi PROC
;
        movzx   r8d, BYTE PTR [rcx]
;
;
        xor     eax, eax
;
; .
        test    r8b, r8b
        je      SHORT $LN9@my_atoi
$LL2@my_atoi:
        lea     edx, DWORD PTR [rax+rax*4]
; EDX=RAX+RAX*4=rt+rt*4=rt*5
        movsx   eax, r8b
; EAX=
;  R8D
```

```
        movzx   r8d, BYTE PTR [rcx+1]
; :
        lea     rcx, QWORD PTR [rcx+1]
        lea     eax, DWORD PTR [rax+rdx*2]
; EAX=RAX+RDX*2= + rt*5*2= + rt*10
;  48 (0x30  '0')
        add     eax, -48                                 ↙
    ↳                   ; ffffffffffffffd0H
; ?
        test    r8b, r8b
;
        jne     SHORT $LL2@my_atoi
$LN9@my_atoi:
        ret     0
my_atoi ENDP
```

. .

42.1.2 Con optimización GCC 4.9.1 x64

Con optimización GCC 4.9.1 .

Listing 42.2: Con optimización GCC 4.9.1 x64

```
my_atoi:
;  EDX
        movsx   edx, BYTE PTR [rdi]
; EAX
        xor     eax, eax
;
        test    dl, dl
        je      .L4
```

```
.L3:
        lea     eax, [rax+rax*4]
; EAX=RAX*5=rt*5
; :
        add     rdi, 1
        lea     eax, [rdx-48+rax*2]
; EAX= - 48 + RAX*2 =  - '0' + rt*10
; :
        movsx   edx, BYTE PTR [rdi]
;
        test    dl, dl
        jne     .L3
        rep ret
.L4:
        rep ret
```

42.1.3 Con optimización Keil 6/2013 (Modo ARM)

Listing 42.3: Con optimización Keil 6/2013 (Modo ARM)

```
my_atoi PROC
; R1
        MOV      r1,r0
; R0
        MOV      r0,#0
        B        |L0.28|
|L0.12|
        ADD      r0,r0,r0,LSL #2
; R0=R0+R0<<2=rt*5
        ADD      r0,r2,r0,LSL #1
; R0= + rt*5<<1 =  + rt*10
```

```
;  '0'  rt:
        SUB       r0,r0,#0x30
; :
        ADD       r1,r1,#1
|L0.28|
;  R2
        LDRB      r2,[r1,#0]
;
        CMP       r2,#0
        BNE       |L0.12|
; .
;
        BX        lr
        ENDP
```

42.1.4 Con optimización Keil 6/2013 (Modo Thumb)

Listing 42.4: Con optimización Keil 6/2013 (Modo Thumb)

```
my_atoi PROC
; R1
        MOVS      r1,r0
; R0
        MOVS      r0,#0
        B         |L0.16|
|L0.6|
        MOVS      r3,#0xa
; R3=10
        MULS      r0,r3,r0
; R0=R3*R0=rt*10
; :
```

```
        ADDS      r1,r1,#1
; :
        SUBS      r0,r0,#0x30
        ADDS      r0,r2,r0
; rt=R2+R0= + (rt*10 - '0')
|L0.16|
;  R2
        LDRB      r2,[r1,#0]
; ?
        CMP       r2,#0
;
        BNE       |L0.6|
;
        BX        lr
        ENDP
```

42.1.5 Con optimización GCC 4.9.1 ARM64

Listing 42.5: Con optimización GCC 4.9.1 ARM64

```
my_atoi:
;  W1
        ldrb    w1, [x0]
        mov     x2, x0
; X2=
; ?
;
; W1 .
; .
        cbz     w1, .L4
; W0
```

```
; :
        mov     w0, 0
.L3:
;  48  '0'  W3:
        sub     w3, w1, #48
;  X2+1  W1 :
        ldrb    w1, [x2,1]!
        add     w0, w0, w0, lsl 2
; W0=W0+W0<<2=W0+W0*4=rt*5
        add     w0, w3, w0, lsl 1
; W0= + W0<<1 =  + rt*5*2 =  + rt*10
;
        cbnz    w1, .L3
;
        ret
.L4:
        mov     w0, w1
        ret
```

42.2

```
#include <stdio.h>

int my_atoi (char *s)
{
        int negative=0;
        int rt=0;

        if (*s=='-')
        {
                negative=1;
```

```
                s++;
        };

        while (*s)
        {
                if (*s<'0' || *s>'9')
                {
                        printf ("Error! ↙
    ↳ Unexpected char: '%c'\n", *s);
                        exit(0);
                };
                rt=rt*10 + (*s-'0');
                s++;
        };

        if (negative)
                return -rt;
        return rt;
};

int main()
{
        printf ("%d\n", my_atoi ("1234"));
        printf ("%d\n", my_atoi ↙
    ↳ ("1234567890"));
        printf ("%d\n", my_atoi ("-1234"));
        printf ("%d\n", my_atoi ↙
    ↳ ("-1234567890"));
        printf ("%d\n", my_atoi ("-↙
    ↳ a1234567890")); // error
};
```

42.2.1 Con optimización GCC 4.9.1 x64

Listing 42.6: Con optimización GCC 4.9.1 x64

```
.LC0:
        .string "Error! Unexpected char: '%c↙
    ↘ '\n"

my_atoi:
        sub     rsp, 8
        movsx   edx, BYTE PTR [rdi]
;
        cmp     dl, 45 ; '-'
        je      .L22
        xor     esi, esi
        test    dl, dl
        je      .L20
.L10:
; ESI=0
        lea     eax, [rdx-48]
;
;
        cmp     al, 9
        ja      .L4
        xor     eax, eax
        jmp     .L6
.L7:
        lea     ecx, [rdx-48]
        cmp     cl, 9
        ja      .L4
.L6:
        lea     eax, [rax+rax*4]
```

```
        add     rdi, 1
        lea     eax, [rdx-48+rax*2]
        movsx   edx, BYTE PTR [rdi]
        test    dl, dl
        jne     .L7
;
; .
        test    esi, esi
        je      .L18
        neg     eax
.L18:
        add     rsp, 8
        ret
.L22:
        movsx   edx, BYTE PTR [rdi+1]
        lea     rax, [rdi+1]
        test    dl, dl
        je      .L20
        mov     rdi, rax
        mov     esi, 1
        jmp     .L10
.L20:
        xor     eax, eax
        jmp     .L18
.L4:
;
        mov     edi, 1
        mov     esi, OFFSET FLAT:.LC0 ; "
    ↳ Error! Unexpected char: '%c'\n"
        xor     eax, eax
        call    __printf_chk
        xor     edi, edi
        call    exit
```

.

. :

```
if (*s<'0' || *s>'9')
    ...
```

. ,

$46 - 48 = -2$. 0xFFFFFFFE (4294967294)

: 48.1.2 on page 1017.

Con optimización MSVC 2013 x64 .

42.2.2 Con optimización Keil 6/2013 (Modo ARM)

Listing 42.7: Con optimización Keil 6/2013 (Modo ARM)

```

my_atoi PROC
        PUSH     {r4-r6,lr}
        MOV      r4,r0
        LDRB     r0,[r0,#0]
        MOV      r6,#0
        MOV      r5,r6
        CMP      r0,#0x2d '-'
; R6
        MOVEQ    r6,#1
        ADDEQ    r4,r4,#1
        B        |L0.80|
|L0.36|
```

```
        SUB      r0,r1,#0x30
        CMP      r0,#0xa
        BCC      |L0.64|
        ADR      r0,|L0.220|
        BL       __2printf
        MOV      r0,#0
        BL       exit
|L0.64|
        LDRB     r0,[r4],#1
        ADD      r1,r5,r5,LSL #2
        ADD      r0,r0,r1,LSL #1
        SUB      r5,r0,#0x30
|L0.80|
        LDRB     r1,[r4,#0]
        CMP      r1,#0
        BNE      |L0.36|
        CMP      r6,#0
;
        RSBNE    r0,r5,#0
        MOVEQ    r0,r5
        POP      {r4-r6,pc}
        ENDP

|L0.220|
        DCB      "Error! Unexpected char: '%c'\n",0
```

: $0 - x = -x$.

GCC 4.9 para ARM64 ARM64.

CAPÍTULO 42. (ATOI())

42.3 Ejercicio

Capítulo 43

Listing 43.1:

```
#include <stdio.h>

int celsius_to_fahrenheit (int celsius)
{
        return celsius * 9 / 5 + 32;
};

int main(int argc, char *argv[])
{
        int celsius=atol(argv[1]);
        printf ("%d\n", ↙
    ↘ celsius_to_fahrenheit (celsius));
};
```

…

Listing 43.2: Con optimización GCC 4.8.1

```
_main:
        push    ebp
        mov     ebp, esp
        and     esp, -16
        sub     esp, 16
        call    ___main
        mov     eax, DWORD PTR [ebp+12]
        mov     eax, DWORD PTR [eax+4]
        mov     DWORD PTR [esp], eax
        call    _atol
        mov     edx, 1717986919
        mov     DWORD PTR [esp], OFFSET FLAT↙
    ↳ :LC2 ; "%d\12\0"
        lea     ecx, [eax+eax*8]
        mov     eax, ecx
        imul    edx
        sar     ecx, 31
        sar     edx
        sub     edx, ecx
        add     edx, 32
        mov     DWORD PTR [esp+4], edx
        call    _printf
        leave
        ret
```

((41 on page 938).)

43.1

strcpy(), *strcmp()*, *strlen()*, *memset()*, *memcmp()*, *memcpy()*, Spanish text placeholder.

43.1.1 strcmp()

Listing 43.3:

```
bool is_bool (char *s)
{
        if (strcmp (s, "true")==0)
                return true;
        if (strcmp (s, "false")==0)
                return false;

        assert(0);
};
```

Listing 43.4: Con optimización GCC 4.8.1

```
.LC0:
        .string "true"
.LC1:
        .string "false"
is_bool:
.LFB0:
        push    edi
        mov     ecx, 5
        push    esi
        mov     edi, OFFSET FLAT:.LC0
        sub     esp, 20
```

```
        mov     esi, DWORD PTR [esp+32]
        repz cmpsb
        je      .L3
        mov     esi, DWORD PTR [esp+32]
        mov     ecx, 6
        mov     edi, OFFSET FLAT:.LC1
        repz cmpsb
        seta    cl
        setb    dl
        xor     eax, eax
        cmp     cl, dl
        jne     .L8
        add     esp, 20
        pop     esi
        pop     edi
        ret
.L8:
        mov     DWORD PTR [esp], 0
        call    assert
        add     esp, 20
        pop     esi
        pop     edi
        ret
.L3:
        add     esp, 20
        mov     eax, 1
        pop     esi
        pop     edi
        ret
```

Listing 43.5: Con optimización MSVC 2010

```
$SG3454 DB      'true', 00H
$SG3456 DB      'false', 00H

_s$ = 8         ; size = 4
?is_bool@@YA_NPAD@Z PROC ; is_bool
        push    esi
        mov     esi, DWORD PTR _s$[esp]
        mov     ecx, OFFSET $SG3454 ; 'true'
        mov     eax, esi
        npad    4 ;
$LL6@is_bool:
        mov     dl, BYTE PTR [eax]
        cmp     dl, BYTE PTR [ecx]
        jne     SHORT $LN7@is_bool
        test    dl, dl
        je      SHORT $LN8@is_bool
        mov     dl, BYTE PTR [eax+1]
        cmp     dl, BYTE PTR [ecx+1]
        jne     SHORT $LN7@is_bool
        add     eax, 2
        add     ecx, 2
        test    dl, dl
        jne     SHORT $LL6@is_bool
$LN8@is_bool:
        xor     eax, eax
        jmp     SHORT $LN9@is_bool
$LN7@is_bool:
        sbb     eax, eax
        sbb     eax, -1
$LN9@is_bool:
        test    eax, eax
        jne     SHORT $LN2@is_bool
```

```
        mov     al, 1
        pop     esi

        ret     0
$LN2@is_bool:

        mov     ecx, OFFSET $SG3456 ; 'false↙
    ↘ '
        mov     eax, esi
$LL10@is_bool:
        mov     dl, BYTE PTR [eax]
        cmp     dl, BYTE PTR [ecx]
        jne     SHORT $LN11@is_bool
        test    dl, dl
        je      SHORT $LN12@is_bool
        mov     dl, BYTE PTR [eax+1]
        cmp     dl, BYTE PTR [ecx+1]
        jne     SHORT $LN11@is_bool
        add     eax, 2
        add     ecx, 2
        test    dl, dl
        jne     SHORT $LL10@is_bool
$LN12@is_bool:
        xor     eax, eax
        jmp     SHORT $LN13@is_bool
$LN11@is_bool:
        sbb     eax, eax
        sbb     eax, -1
$LN13@is_bool:
        test    eax, eax
        jne     SHORT $LN1@is_bool

        xor     al, al
```

```
        pop     esi

        ret     0
$LN1@is_bool:

        push    11
        push    OFFSET $SG3458
        push    OFFSET $SG3459
        call    DWORD PTR __imp___wassert
        add     esp, 12
        pop     esi

        ret     0
?is_bool@@YA_NPAD@Z ENDP ; is_bool
```

43.1.2 strlen()

Listing 43.6:

```
int strlen_test(char *s1)
{
        return strlen(s1);
};
```

Listing 43.7: Con optimización MSVC 2010

```
_s1$ = 8 ; size = 4
_strlen_test PROC
        mov     eax, DWORD PTR _s1$[esp-4]
        lea     edx, DWORD PTR [eax+1]
$LL3@strlen_tes:
        mov     cl, BYTE PTR [eax]
```

```
        inc     eax
        test    cl, cl
        jne     SHORT $LL3@strlen_tes
        sub     eax, edx
        ret     0
_strlen_test ENDP
```

43.1.3 strcpy()

Listing 43.8:

```
void strcpy_test(char *s1, char *outbuf)
{
        strcpy(outbuf, s1);
};
```

Listing 43.9: Con optimización MSVC 2010

```

_s1$ = 8        ; size = 4
_outbuf$ = 12   ; size = 4
_strcpy_test PROC
        mov     eax, DWORD PTR _s1$[esp-4]
        mov     edx, DWORD PTR _outbuf$[esp
    ↘ -4]
        sub     edx, eax
        npad    6 ;
$LL3@strcpy_tes:
        mov     cl, BYTE PTR [eax]
        mov     BYTE PTR [edx+eax], cl
        inc     eax
        test    cl, cl
```

```
        jne     SHORT $LL3@strcpy_tes
        ret     0
_strcpy_test ENDP
```

43.1.4 memset()

Ejemplo#1

Listing 43.10: 32

```
#include <stdio.h>

void f(char *out)
{
        memset(out, 0, 32);
};
```

Listing 43.11: Con optimización GCC 4.9.1 x64

```
f:
        mov     QWORD PTR [rdi], 0
        mov     QWORD PTR [rdi+8], 0
        mov     QWORD PTR [rdi+16], 0
        mov     QWORD PTR [rdi+24], 0
        ret
```

: 14.1.4 on page 351.

Ejemplo#2

Listing 43.12: 67

```
#include <stdio.h>

void f(char *out)
{
        memset(out, 0, 67);
};
```

Listing 43.13: Con optimización MSVC 2012 x64

```
out$ = 8
f       PROC
        xor     eax, eax
        mov     QWORD PTR [rcx], rax
        mov     QWORD PTR [rcx+8], rax
        mov     QWORD PTR [rcx+16], rax
        mov     QWORD PTR [rcx+24], rax
        mov     QWORD PTR [rcx+32], rax
        mov     QWORD PTR [rcx+40], rax
        mov     QWORD PTR [rcx+48], rax
        mov     QWORD PTR [rcx+56], rax
        mov     WORD PTR [rcx+64], ax
        mov     BYTE PTR [rcx+66], al
        ret     0
f       ENDP
```

Listing 43.14: Con optimización GCC 4.9.1 x64

```
f:
        mov     QWORD PTR [rdi], 0
        mov     QWORD PTR [rdi+59], 0
        mov     rcx, rdi
```

```
        lea     rdi, [rdi+8]
        xor     eax, eax
        and     rdi, -8
        sub     rcx, rdi
        add     ecx, 67
        shr     ecx, 3
        rep stosq
        ret
```

43.1.5 memcpy()

Listing 43.15:

```
void memcpy_7(char *inbuf, char *outbuf)
{
        memcpy(outbuf+10, inbuf, 7);
};
```

Listing 43.16: Con optimización MSVC 2010

```
_inbuf$ = 8     ; size = 4
_outbuf$ = 12   ; size = 4
_memcpy_7 PROC
        mov     ecx, DWORD PTR _inbuf$[esp⤦
    ↪ -4]
        mov     edx, DWORD PTR [ecx]
        mov     eax, DWORD PTR _outbuf$[esp⤦
    ↪ -4]
        mov     DWORD PTR [eax+10], edx
        mov     dx, WORD PTR [ecx+4]
```

```
        mov     WORD PTR [eax+14], dx
        mov     cl, BYTE PTR [ecx+6]
        mov     BYTE PTR [eax+16], cl
        ret     0
_memcpy_7 ENDP
```

Listing 43.17: Con optimización GCC 4.8.1

```
memcpy_7:
        push    ebx
        mov     eax, DWORD PTR [esp+8]
        mov     ecx, DWORD PTR [esp+12]
        mov     ebx, DWORD PTR [eax]
        lea     edx, [ecx+10]
        mov     DWORD PTR [ecx+10], ebx
        movzx   ecx, WORD PTR [eax+4]
        mov     WORD PTR [edx+4], cx
        movzx   eax, BYTE PTR [eax+6]
        mov     BYTE PTR [edx+6], al
        pop     ebx
        ret
```

: 21.4.1 on page 694.

Listing 43.18:

```
void memcpy_128(char *inbuf, char *outbuf)
{
        memcpy(outbuf+10, inbuf, 128);
};
```

```
void memcpy_123(char *inbuf, char *outbuf)
{
        memcpy(outbuf+10, inbuf, 123);
};
```

Listing 43.19: Con optimización MSVC 2010

```
_inbuf$ = 8                 ; size = 4
_outbuf$ = 12               ; size = 4
_memcpy_128 PROC
        push    esi
        mov     esi, DWORD PTR _inbuf$[esp]
        push    edi
        mov     edi, DWORD PTR _outbuf$[esp
    ↘ +4]
        add     edi, 10
        mov     ecx, 32
        rep movsd
        pop     edi
        pop     esi
        ret     0
_memcpy_128 ENDP
```

Listing 43.20: Con optimización MSVC 2010

```
_inbuf$ = 8                 ; size = 4
_outbuf$ = 12               ; size = 4
_memcpy_123 PROC
        push    esi
        mov     esi, DWORD PTR _inbuf$[esp]
        push    edi
```

```
        mov     edi, DWORD PTR _outbuf$[esp
    +4]
        add     edi, 10
        mov     ecx, 30
        rep movsd
        movsw
        movsb
        pop     edi
        pop     esi
        ret     0
_memcpy_123 ENDP
```

Listing 43.21: Con optimización GCC 4.8.1

```
memcpy_123:
.LFB3:
        push    edi
        mov     eax, 123
        push    esi
        mov     edx, DWORD PTR [esp+16]
        mov     esi, DWORD PTR [esp+12]
        lea     edi, [edx+10]
        test    edi, 1
        jne     .L24
        test    edi, 2
        jne     .L25
.L7:
        mov     ecx, eax
        xor     edx, edx
        shr     ecx, 2
        test    al, 2
        rep movsd
        je      .L8
```

```
        movzx   edx, WORD PTR [esi]
        mov     WORD PTR [edi], dx
        mov     edx, 2
.L8:
        test    al, 1
        je      .L5
        movzx   eax, BYTE PTR [esi+edx]
        mov     BYTE PTR [edi+edx], al
.L5:
        pop     esi
        pop     edi
        ret
.L24:
        movzx   eax, BYTE PTR [esi]
        lea     edi, [edx+11]
        add     esi, 1
        test    edi, 2
        mov     BYTE PTR [edx+10], al
        mov     eax, 122
        je      .L7
.L25:
        movzx   edx, WORD PTR [esi]
        add     edi, 2
        add     esi, 2
        sub     eax, 2
        mov     WORD PTR [edi-2], dx
        jmp     .L7
.LFE3:
```

: 25.2 on page 810.

43.1.6 memcmp()

Listing 43.22:

```
void memcpy_1235(char *inbuf, char *outbuf)
{
        memcpy(outbuf+10, inbuf, 1235);
};
```

Listing 43.23: Con optimización MSVC 2010

```
_buf1$ = 8       ; size = 4
_buf2$ = 12      ; size = 4
_memcmp_1235 PROC
        mov     edx, DWORD PTR _buf2$[esp-4]
        mov     ecx, DWORD PTR _buf1$[esp-4]
        push    esi
        push    edi
        mov     esi, 1235
        add     edx, 10
$LL4@memcmp_123:
        mov     eax, DWORD PTR [edx]
        cmp     eax, DWORD PTR [ecx]
        jne     SHORT $LN10@memcmp_123
        sub     esi, 4
        add     ecx, 4
        add     edx, 4
        cmp     esi, 4
        jae     SHORT $LL4@memcmp_123
$LN10@memcmp_123:
        movzx   edi, BYTE PTR [ecx]
        movzx   eax, BYTE PTR [edx]
        sub     eax, edi
        jne     SHORT $LN7@memcmp_123
```

```
        movzx   eax, BYTE PTR [edx+1]
        movzx   edi, BYTE PTR [ecx+1]
        sub     eax, edi
        jne     SHORT $LN7@memcmp_123
        movzx   eax, BYTE PTR [edx+2]
        movzx   edi, BYTE PTR [ecx+2]
        sub     eax, edi
        jne     SHORT $LN7@memcmp_123
        cmp     esi, 3
        jbe     SHORT $LN6@memcmp_123
        movzx   eax, BYTE PTR [edx+3]
        movzx   ecx, BYTE PTR [ecx+3]
        sub     eax, ecx
$LN7@memcmp_123:
        sar     eax, 31
        pop     edi
        or      eax, 1
        pop     esi
        ret     0
$LN6@memcmp_123:
        pop     edi
        xor     eax, eax
        pop     esi
        ret     0
_memcmp_1235 ENDP
```

43.1.7

También hay un pequeño script para IDA que busca y contrae estos fragmentos de código inline tan frecuentes. GitHub.

CAPÍTULO 44. C99 RESTRICT

Capítulo 44

C99 restrict

```
void f1 (int* x, int* y, int* sum, int* ⤦
    ↳ product, int* sum_product, int* ⤦
    ↳ update_me, size_t s)
{
        for (int i=0; i<s; i++)
        {
                sum[i]=x[i]+y[i];
                product[i]=x[i]*y[i];
                update_me[i]=i*123; // some ⤦
    ↳ dummy value
                sum_product[i]=sum[i]+⤦
    ↳ product[i];
        };
};
```

 CAPÍTULO 44. C99 RESTRICT

: `update_me` `sum`, `product`, `sum_product`

- `sum[i]`
- `product[i]`
- `update_me[i]`
- `sum_product[i]` `sum[i]` y `product[i]`

`sum[i]` y `product[i]` «pointer aliasing»,

restrict [ISO07, págs. 6.7.3/1]

restrict.

:

```
void f2 (int* restrict x, int* restrict y, ↙
    ↳ int* restrict sum, int* restrict ↙
    ↳ product, int* restrict sum_product,
        int* restrict update_me, size_t s)
{
        for (int i=0; i<s; i++)
        {
                sum[i]=x[i]+y[i];
                product[i]=x[i]*y[i];
                update_me[i]=i*123; // some ↙
    ↳ dummy value
                sum_product[i]=sum[i]+↙
    ↳ product[i];
        };
};
```

:

Listing 44.1: GCC x64: f1()

```
f1:
        push    r15 r14 r13 r12 rbp rdi rsi ↙
    ↘ rbx
        mov     r13, QWORD PTR 120[rsp]
        mov     rbp, QWORD PTR 104[rsp]
        mov     r12, QWORD PTR 112[rsp]
        test    r13, r13
        je      .L1
        add     r13, 1
        xor     ebx, ebx
        mov     edi, 1
        xor     r11d, r11d
        jmp     .L4
.L6:
        mov     r11, rdi
        mov     rdi, rax
.L4:
        lea     rax, 0[0+r11*4]
        lea     r10, [rcx+rax]
        lea     r14, [rdx+rax]
        lea     rsi, [r8+rax]
        add     rax, r9
        mov     r15d, DWORD PTR [r10]
        add     r15d, DWORD PTR [r14]
        mov     DWORD PTR [rsi], r15d         ↙
    ↘   ;  sum[]
        mov     r10d, DWORD PTR [r10]
        imul    r10d, DWORD PTR [r14]
        mov     DWORD PTR [rax], r10d         ↙
    ↘   ;  product[]
```

```
        mov     DWORD PTR [r12+r11*4], ebx
    ↪   ; update_me[]
        add     ebx, 123
        mov     r10d, DWORD PTR [rsi]
    ↪   ; sum[i]
        add     r10d, DWORD PTR [rax]
    ↪   ; product[i]
        lea     rax, 1[rdi]
        cmp     rax, r13
        mov     DWORD PTR 0[rbp+r11*4], r10d
    ↪   ; sum_product[]
        jne     .L6
.L1:
        pop     rbx rsi rdi rbp r12 r13 r14
    ↪ r15
        ret
```

Listing 44.2: GCC x64: f2()

```
f2:
        push    r13 r12 rbp rdi rsi rbx
        mov     r13, QWORD PTR 104[rsp]
        mov     rbp, QWORD PTR 88[rsp]
        mov     r12, QWORD PTR 96[rsp]
        test    r13, r13
        je      .L7
        add     r13, 1
        xor     r10d, r10d
        mov     edi, 1
        xor     eax, eax
        jmp     .L10
.L11:
```

```
        mov     rax, rdi
        mov     rdi, r11
.L10:
        mov     esi, DWORD PTR [rcx+rax*4]
        mov     r11d, DWORD PTR [rdx+rax*4]
        mov     DWORD PTR [r12+rax*4], r10d ⤦
    ↳  ;  update_me[]
        add     r10d, 123
        lea     ebx, [rsi+r11]
        imul    r11d, esi
        mov     DWORD PTR [r8+rax*4], ebx   ⤦
    ↳  ;  sum[]
        mov     DWORD PTR [r9+rax*4], r11d  ⤦
    ↳  ;  product[]
        add     r11d, ebx
        mov     DWORD PTR 0[rbp+rax*4], r11d⤦
    ↳  ;  sum_product[]
        lea     r11, 1[rdi]
        cmp     r11, r13
        jne     .L11
.L7:
        pop     rbx rsi rdi rbp r12 r13
        ret
```

: en `f1()`, `sum[i]` y `product[i]` , `f2()` , `sum[i]` y `product[i]` , .

FORTRAN. , *restrict* en, FORTRAN .

? [Loh10].

Capítulo 45

```
int my_abs (int i)
{
        if (i<0)
                return -i;
        else
                return i;
};
```

.

45.1 Con optimización GCC 4.9.1 x64

Listing 45.1: Con optimización GCC 4.9 x64

```
my_abs:
        mov     edx, edi
```

```
        mov     eax, edi
        sar     edx, 31
;
;
;
;
        xor     eax, edx
        sub     eax, edx
        ret
```

:

[War02, págs. 2-4].

45.2 Con optimización GCC 4.9 ARM64

Listing 45.2: Con optimización GCC 4.9 ARM64

```

my_abs:
;
        sxtw    x0, w0
        eor     x1, x0, x0, asr 63
; X1=X0^(X0>>63)
        sub     x0, x1, x0, asr 63
; X0=X1-(X0>>63)=X0^(X0>>63)-(X0>>63)
        ret
```

Capítulo 46

46.1

```
#include <stdio.h>
#include <stdarg.h>

int arith_mean(int v, ...)
{
        va_list args;
        int sum=v, count=1, i;
        va_start(args, v);

        while(1)
        {
                i=va_arg(args, int);
                if (i==-1) //
                        break;
                sum=sum+i;
```

```
                count++;
        }

        va_end(args);
        return sum/count;
};

int main()
{
        printf ("%d\n", arith_mean (1, 2, 7,↙
    ↘ 10, 15, -1 /*  */));
};
```

?

46.1.1

Listing 46.1: Con optimización MSVC 6.0

```
_v$ = 8
_arith_mean PROC NEAR
        mov     eax, DWORD PTR _v$[esp-4] ; ↙
    ↘ sum
        push    esi
        mov     esi, 1                    ; ↙
    ↘ count=1
        lea     edx, DWORD PTR _v$[esp]   ;
$L838:
        mov     ecx, DWORD PTR [edx+4]    ;
        add     edx, 4                    ;
        cmp     ecx, -1                   ; ↙
    ↘ -1?
```

```
        je      SHORT $L856                     ;
        add     eax, ecx                        ; ↙
    ↘ sum = sum +
        inc     esi                             ; ↙
    ↘ count++
        jmp     SHORT $L838
$L856:
;

        cdq
        idiv    esi
        pop     esi
        ret     0
_arith_mean ENDP

$SG851  DB      '%d', 0aH, 00H

_main   PROC NEAR
        push    -1
        push    15
        push    10
        push    7
        push    2
        push    1
        call    _arith_mean
        push    eax
        push    OFFSET FLAT:$SG851 ; '%d'
        call    _printf
        add     esp, 32
        ret     0
_main   ENDP
```

46.1.2

:

Listing 46.2: Con optimización MSVC 2012 x64

```
$SG3013 DB      '%d', 0aH, 00H

v$ = 8
arith_mean PROC
        mov     DWORD PTR [rsp+8], ecx   ;
        mov     QWORD PTR [rsp+16], rdx  ;
        mov     QWORD PTR [rsp+24], r8   ;
        mov     eax, ecx                 ; ↙
    ↳ sum =
        lea     rcx, QWORD PTR v$[rsp+8] ;
        mov     QWORD PTR [rsp+32], r9   ;
        mov     edx, DWORD PTR [rcx]     ;
        mov     r8d, 1                   ; ↙
    ↳ count=1
        cmp     edx, -1                  ;  ↙
    ↳ -1?
        je      SHORT $LN8@arith_mean    ;
$LL3@arith_mean:
        add     eax, edx                 ; ↙
    ↳ sum = sum +
        mov     edx, DWORD PTR [rcx+8]   ;
        lea     rcx, QWORD PTR [rcx+8]   ;
        inc     r8d                      ; ↙
    ↳ count++
        cmp     edx, -1                  ;  ↙
    ↳ -1?
        jne     SHORT $LL3@arith_mean    ;
```

```
$LN8@arith_mean:
;
        cdq
        idiv    r8d
        ret     0
arith_mean ENDP

main    PROC
        sub     rsp, 56
        mov     edx, 2
        mov     DWORD PTR [rsp+40], -1
        mov     DWORD PTR [rsp+32], 15
        lea     r9d, QWORD PTR [rdx+8]
        lea     r8d, QWORD PTR [rdx+5]
        lea     ecx, QWORD PTR [rdx-1]
        call    arith_mean
        lea     rcx, OFFSET FLAT:$SG3013
        mov     edx, eax
        call    printf
        xor     eax, eax
        add     rsp, 56
        ret     0
main    ENDP
```

Listing 46.3: Con optimización GCC 4.9.1 x64

```

arith_mean:
        lea     rax, [rsp+8]
        ;
        mov     QWORD PTR [rsp-40], rsi
        mov     QWORD PTR [rsp-32], rdx
        mov     QWORD PTR [rsp-16], r8
```

```
        mov     QWORD PTR [rsp-24], rcx
        mov     esi, 8
        mov     QWORD PTR [rsp-64], rax
        lea     rax, [rsp-48]
        mov     QWORD PTR [rsp-8], r9
        mov     DWORD PTR [rsp-72], 8
        lea     rdx, [rsp+8]
        mov     r8d, 1
        mov     QWORD PTR [rsp-56], rax
        jmp     .L5
.L7:
        ;
        lea     rax, [rsp-48]
        mov     ecx, esi
        add     esi, 8
        add     rcx, rax
        mov     ecx, DWORD PTR [rcx]
        cmp     ecx, -1
        je      .L4
.L8:
        add     edi, ecx
        add     r8d, 1
.L5:
        ;
        ;
        cmp     esi, 47
        jbe     .L7             ;
        ;
        mov     rcx, rdx
        add     rdx, 8
        mov     ecx, DWORD PTR [rcx]
        cmp     ecx, -1
        jne     .L8
```

```
.L4:
        mov     eax, edi
        cdq
        idiv    r8d
        ret

.LC1:
        .string "%d\n"
main:
        sub     rsp, 8
        mov     edx, 7
        mov     esi, 2
        mov     edi, 1
        mov     r9d, -1
        mov     r8d, 15
        mov     ecx, 10
        xor     eax, eax
        call    arith_mean
        mov     esi, OFFSET FLAT:.LC1
        mov     edx, eax
        mov     edi, 1
        xor     eax, eax
        add     rsp, 8
        jmp     __printf_chk
```

: 64.8 on page 1310.

46.2

?

```
#include <stdlib.h>
#include <stdarg.h>

void die (const char * fmt, ...)
{
        va_list va;
        va_start (va, fmt);

        vprintf (fmt, va);
        exit(0);
};
```

:

Listing 46.4: Con optimización MSVC 2010

```

_fmt$ = 8
_die    PROC
        ;
        mov     ecx, DWORD PTR _fmt$[esp-4]
        ;
        lea     eax, DWORD PTR _fmt$[esp]
        push    eax                 ;
        push    ecx
        call    _vprintf
        add     esp, 8
        push    0
        call    _exit
$LN3@die:
        int     3
_die    ENDP
```

Listing 46.5: Con optimización MSVC 2012 x64

```
fmt$ = 48
die     PROC
        ;  Shadow Space
        mov     QWORD PTR [rsp+8], rcx
        mov     QWORD PTR [rsp+16], rdx
        mov     QWORD PTR [rsp+24], r8
        mov     QWORD PTR [rsp+32], r9
        sub     rsp, 40
        lea     rdx, QWORD PTR fmt$[rsp+8] ;
        ; RCX  die()
        ;  vprintf()  RCX
        call    vprintf
        xor     ecx, ecx
        call    exit
        int     3
die     ENDP
```

Capítulo 47

```
#include <stdio.h>
#include <string.h>

char* str_trim (char *s)
{
        char c;
        size_t str_len;

        //  \r  \n
        //
        // ()
        for (str_len=strlen(s); str_len>0 &&
    ↳  (c=s[str_len-1]); str_len--)
        {
                if (c=='\r' || c=='\n')
                        s[str_len-1]=0;
                else
```

```
                                break;
                };
                return s;
};

int main()
{
                //

                //
                //
                //
                // .

                printf ("[%s]\n", str_trim (strdup↙
        ↘ ("")));
                printf ("[%s]\n", str_trim (strdup↙
        ↘ ("\n")));
                printf ("[%s]\n", str_trim (strdup↙
        ↘ ("\r")));
                printf ("[%s]\n", str_trim (strdup↙
        ↘ ("\n\r")));
                printf ("[%s]\n", str_trim (strdup↙
        ↘ ("\r\n")));
                printf ("[%s]\n", str_trim (strdup("↙
        ↘ test1\r\n")));
                printf ("[%s]\n", str_trim (strdup("↙
        ↘ test2\n\r")));
                printf ("[%s]\n", str_trim (strdup("↙
        ↘ test3\n\r\n\r")));
                printf ("[%s]\n", str_trim (strdup("↙
        ↘ test4\n")));
```

```
        printf ("[%s]\n", str_trim (strdup("↙
    ↘ test5\r")));
        printf ("[%s]\n", str_trim (strdup("↙
    ↘ test6\r\r\r")));
};
```

47.1 x64: Con optimización MSVC 2013

Listing 47.1: Con optimización MSVC 2013 x64

```
s$ = 8
str_trim PROC

; RCX

; :
;  RAX  0xFFFFFFFFFFFFFFFF (-1)
        or      rax, -1
$LL14@str_trim:
        inc     rax
        cmp     BYTE PTR [rcx+rax], 0
        jne     SHORT $LL14@str_trim
;
        test    eax, eax
$LN18@str_trim:
        je      SHORT $LN15@str_trim
; RAX
;
;
        lea     edx, DWORD PTR [rax-1]
```

```
;
        mov     eax, edx
;  s[str_len-1]
        movzx   eax, BYTE PTR [rdx+rcx]
;  R8
        lea     r8, QWORD PTR [rdx+rcx]
        cmp     al, 13  ; is it '\r'?
        je      SHORT $LN2@str_trim
        cmp     al, 10  ; is it '\n'?
        jne     SHORT $LN15@str_trim
$LN2@str_trim:
;
        mov     BYTE PTR [r8], 0
        mov     eax, edx
;
        test    edx, edx
        jmp     SHORT $LN18@str_trim
$LN15@str_trim:
;  "s"
        mov     rax, rcx
        ret     0
str_trim ENDP
```

: 43 on page 963.

OR RAX, 0xFFFFFFFFFFFFFFFF.

. :

Listing 47.2: MSVC 2013 x64

```


; RCX =
; RAX =
```

```
        or      rax, -1
label:
        inc     rax
        cmp     BYTE PTR [rcx+rax], 0
        jne     SHORT label
; RAX =
```

:

Listing 47.3: `strlen()`

```
; RCX =
; RAX =
        xor     rax, rax
label:
        cmp     byte ptr [rcx+rax], 0
        jz      exit
        inc     rax
        jmp     label
exit:
; RAX =
```

. «$LN18@str_trim» JE !

33.1 on page 873.

47.2 x64: Sin optimización GCC 4.9.1

```
str_trim:
        push    rbp
```

```
        mov     rbp, rsp
        sub     rsp, 32
        mov     QWORD PTR [rbp-24], rdi
;
        mov     rax, QWORD PTR [rbp-24]
        mov     rdi, rax
        call    strlen
        mov     QWORD PTR [rbp-8], rax   ; ↙
    ↘ str_len
;
        jmp     .L2
;
.L5:
        cmp     BYTE PTR [rbp-9], 13     ; c↙
    ↘ =='\r'?
        je      .L3
        cmp     BYTE PTR [rbp-9], 10     ; c↙
    ↘ =='\n'?
        jne     .L4
.L3:
        mov     rax, QWORD PTR [rbp-8]   ; ↙
    ↘ str_len
        lea     rdx, [rax-1]             ; ↙
    ↘ EDX=str_len-1
        mov     rax, QWORD PTR [rbp-24]  ; s
        add     rax, rdx                 ; ↙
    ↘ RAX=s+str_len-1
        mov     BYTE PTR [rax], 0        ; s↙
    ↘ [str_len-1]=0
;
;
        sub     QWORD PTR [rbp-8], 1     ; ↙
    ↘ str_len--
```

```
;
.L2:
;
        cmp     QWORD PTR [rbp-8], 0     ; ↙
    ↘ str_len==0?
        je      .L4                      ;
;
        mov     rax, QWORD PTR [rbp-8]   ; ↙
    ↘ RAX=str_len
        lea     rdx, [rax-1]             ; ↙
    ↘ RDX=str_len-1
        mov     rax, QWORD PTR [rbp-24]  ; ↙
    ↘ RAX=s
        add     rax, rdx                 ; ↙
    ↘ RAX=s+str_len-1
        movzx   eax, BYTE PTR [rax]      ; ↙
    ↘ AL=s[str_len-1]
        mov     BYTE PTR [rbp-9], al     ;  ↙
    ↘ "c"
        cmp     BYTE PTR [rbp-9], 0      ; ?
        jne     .L5                      ;
;
.L4:
;  "s"
        mov     rax, QWORD PTR [rbp-24]
        leave
        ret
```

.

-

- goto L2

- L5:
-
- L2:
- L4: //
- return s

47.3 x64: Con optimización GCC 4.9.1

```
str_trim:
        push    rbx
        mov     rbx, rdi
; RBX  "s"
        call    strlen
;  str_len==0
        test    rax, rax
        je      .L9
        lea     rdx, [rax-1]
; RDX  str_len-1  str_len
;
        lea     rsi, [rbx+rdx]      ; RSI=s+
    ↳ str_len-1
        movzx   ecx, BYTE PTR [rsi] ;
        test    cl, cl
        je      .L9                 ;
        cmp     cl, 10
        je      .L4
        cmp     cl, 13              ;  '\n' 
    ↳  '\r'
```

```
        jne     .L9
.L4:
;
;  MOV RSI, EBX / DEC RSI
;
        sub     rsi, rax
; RSI = s+str_len-1-str_len = s-1
;
.L12:
        test    rdx, rdx
;  s-1+str_len-1+1 = s-1+str_len = s+str_len⤦
    ↳ -1
        mov     BYTE PTR [rsi+1+rdx], 0
;  str_len-1==0. .
        je      .L9
        sub     rdx, 1                       ;  ⤦
    ↳ str_len--
;  s+str_len-1
        movzx   ecx, BYTE PTR [rbx+rdx]
        test    cl, cl                       ;
        je      .L9
        cmp     cl, 10                       ;  ⤦
    ↳ '\n'?
        je      .L12
        cmp     cl, 13                       ;  ⤦
    ↳ '\r'?
        je      .L12
.L9:
;  "s"
        mov     rax, rbx
        pop     rbx
        ret
```

. ?

- () CR[1]/LF[2]
-

.

-
- ()
-

.

47.4 ARM64: Sin optimización GCC (Linaro) 4.9

:

Listing 47.4: Sin optimización GCC (Linaro) 4.9

```
str_trim:
        stp     x29, x30, [sp, -48]!
        add     x29, sp, 0
        str     x0, [x29,24] ;
        ldr     x0, [x29,24] ; s
        bl      strlen
        str     x0, [x29,40] ; str_len
```

[1] Carriage return (13 o'\r' en C/C++)

[2] Line feed (10 o'\n' en C/C++)

```
        b       .L2
;
.L5:
        ldrb    w0, [x29,39]
; W0=c
        cmp     w0, 13          ;   '\r'?
        beq     .L3
        ldrb    w0, [x29,39]
; W0=c
        cmp     w0, 10          ;   '\n'?
        bne     .L4             ;
.L3:
        ldr     x0, [x29,40]
; X0=str_len
        sub     x0, x0, #1
; X0=str_len-1
        ldr     x1, [x29,24]
; X1=s
        add     x0, x1, x0
; X0=s+str_len-1
        strb    wzr, [x0]       ;   s+str_len-1
;  str_len:
        ldr     x0, [x29,40]
; X0=str_len
        sub     x0, x0, #1
; X0=str_len-1
        str     x0, [x29,40]
;
.L2:
        ldr     x0, [x29,40]
; str_len==0?
        cmp     x0, xzr
;
```

```
        beq     .L4
        ldr     x0, [x29,40]
; X0=str_len
        sub     x0, x0, #1
; X0=str_len-1
        ldr     x1, [x29,24]
; X1=s
        add     x0, x1, x0
; X0=s+str_len-1
;  s+str_len-1  W0
        ldrb    w0, [x0]
        strb    w0, [x29,39] ;  "c"
        ldrb    w0, [x29,39] ;
; ?
        cmp     w0, wzr
;
        bne     .L5
.L4:
;  s
        ldr     x0, [x29,24]
        ldp     x29, x30, [sp], 48
        ret
```

47.5 ARM64: Con optimización GCC (Linaro) 4.9

.

Listing 47.5: Con optimización GCC (Linaro) 4.9

```
str_trim:
        stp     x29, x30, [sp, -32]!
        add     x29, sp, 0
        str     x19, [sp,16]
        mov     x19, x0
; X19  "s"
        bl      strlen
; X0=str_len
        cbz     x0, .L9          ;  str_len==0
        sub     x1, x0, #1
; X1=X0-1=str_len-1
        add     x3, x19, x1
; X3=X19+X1=s+str_len-1
        ldrb    w2, [x19,x1]     ;  X19+X1=s+⤦
    ↳ str_len-1
; W2=
        cbz     w2, .L9          ;
        cmp     w2, 10           ;  '\n'?
        bne     .L15
.L12:
;
        sub     x2, x1, x0
; X2=X1-X0=str_len-1-str_len=-1
        add     x2, x3, x2
; X2=X3+X2=s+str_len-1+(-1)=s+str_len-2
        strb    wzr, [x2,1]      ;  s+str_len⤦
    ↳ -2+1=s+str_len-1
        cbz     x1, .L9          ; str_len⤦
    ↳ -1==0?
        sub     x1, x1, #1       ; str_len--
        ldrb    w2, [x19,x1]     ;  X19+X1=s+⤦
    ↳ str_len-1
        cmp     w2, 10           ;  '\n'?
```

```
        cbz     w2, .L9          ;
        beq     .L12             ;  '\n'
.L15:
        cmp     w2, 13           ;  '\r'?
        beq     .L12             ;
.L9:
;  "s"
        mov     x0, x19
        ldr     x19, [sp,16]
        ldp     x29, x30, [sp], 32
        ret
```

47.6 ARM: Con optimización Keil 6/2013 (Modo ARM)

Listing 47.6: Con optimización Keil 6/2013 (Modo ARM)

```

str_trim PROC
        PUSH     {r4,lr}
; R0=s
        MOV      r4,r0
; R4=s
        BL       strlen         ; strlen()
; R0=str_len
        MOV      r3,#0
; R3  0
|L0.16|
        CMP      r0,#0          ; str_len==0?
        ADDNE    r2,r4,r0       ; ( str_len⤦
    ↳ !=0) R2=R4+R0=s+str_len
```

```
        LDRBNE   r1,[r2,#-1]  ; ( str_len⤦
    ⤥ !=0) R1= R2-1=s+str_len-1
        CMPNE    r1,#0        ; ( str_len⤦
    ⤥ !=0)  0
        BEQ      |L0.56|      ;  str_len==0 ⤦
    ⤥  0
        CMP      r1,#0xd      ;  '\r'?
        CMPNE    r1,#0xa      ; ( '\r')  '\r⤦
    ⤥ '?
        SUBEQ    r0,r0,#1     ; ( '\r'  '\n⤦
    ⤥ ') R0--  str_len--
        STRBEQ   r3,[r2,#-1]  ; ( '\r'  '\n⤦
    ⤥ ')  R2-1=s+str_len-1
        BEQ      |L0.16|      ;  '\r'  '\n'
|L0.56|
;  "s"
        MOV      r0,r4
        POP      {r4,pc}
        ENDP
```

47.7 ARM: Con optimización Keil 6/2013 (Modo Thumb)

? .

Listing 47.7: Con optimización Keil 6/2013 (Modo Thumb)

```
str_trim PROC
        PUSH     {r4,lr}
        MOVS     r4,r0
```

```
; R4=s
        BL          strlen          ; strlen()
; R0=str_len
        MOVS        r3,#0
; R3  0
        B           |L0.24|
|L0.12|
        CMP         r1,#0xd         ;  '\r'?
        BEQ         |L0.20|
        CMP         r1,#0xa         ;  '\n'?
        BNE         |L0.38|         ;
|L0.20|
        SUBS        r0,r0,#1        ; R0--   ↙
    ↳ str_len--
        STRB        r3,[r2,#0x1f] ;  0  R2+0x1F↙
    ↳ =s+str_len-0x20+0x1F=s+str_len-1
|L0.24|
        CMP         r0,#0           ; str_len==0?
        BEQ         |L0.38|         ;
        ADDS        r2,r4,r0        ; R2=R4+R0=s+↙
    ↳ str_len
        SUBS        r2,r2,#0x20     ; R2=R2-0x20=↙
    ↳ s+str_len-0x20
        LDRB        r1,[r2,#0x1f] ;  R2+0x1F=s+↙
    ↳ str_len-0x20+0x1F=s+str_len-1  R1
        CMP         r1,#0           ;  0?
        BNE         |L0.12|         ;
|L0.38|
;  "s"
        MOVS        r0,r4
        POP         {r4,pc}
        ENDP
```

47.8 MIPS

Listing 47.8: Con optimización GCC 4.4.5 (IDA)

```
str_trim:
; :
saved_GP            = -0x10
saved_S0             = -8
saved_RA             = -4

                lui     $gp, (__gnu_local_gp ↙
    ↳  >> 16)
                addiu   $sp, -0x20
                la      $gp, (__gnu_local_gp ↙
    ↳  & 0xFFFF)
                sw      $ra, 0x20+saved_RA( ↙
    ↳ $sp)
                sw      $s0, 0x20+saved_S0( ↙
    ↳ $sp)
                sw      $gp, 0x20+saved_GP( ↙
    ↳ $sp)
;  strlen().  $a0, strlen() :
                lw      $t9, (strlen & 0 ↙
    ↳ xFFFF)($gp)
                or      $at, $zero ; load ↙
    ↳ delay slot, NOP
                jalr    $t9
;  $a0,  $s0:
                move    $s0, $a0   ; branch ↙
    ↳ delay slot
;  strlen() ()
;  $v0==0 ( 0):
```

```
                beqz    $v0, exit
                or      $at, $zero ; branch ↙
    ↘ delay slot, NOP
                addiu   $a1, $v0, -1
; $a1 = $v0-1 = str_len-1
                addu    $a1, $s0, $a1
; $a1 =  + $a1 = s+strlen-1
;  $a1:
                lb      $a0, 0($a1)
                or      $at, $zero ; load ↙
    ↘ delay slot, NOP
; :
                beqz    $a0, exit
                or      $at, $zero ; branch ↙
    ↘ delay slot, NOP
                addiu   $v1, $v0, -2
; $v1 = str_len-2
                addu    $v1, $s0, $v1
; $v1 = $s0+$v1 = s+str_len-2
                li      $a2, 0xD
; :
                b       loc_6C
                li      $a3, 0xA   ; branch ↙
    ↘ delay slot
loc_5C:
;  $a0:
                lb      $a0, 0($v1)
                move    $a1, $v1
; $a1=s+str_len-2
; :
                beqz    $a0, exit
;  str_len:
```

```
                addiu   $v1, -1    ; branch ↙
    ↘ delay slot
loc_6C:
; , $a0=, $a2=0xD ()  $a3=0xA ()
; :
                beq     $a0, $a2, loc_7C
                addiu   $v0, -1    ; branch ↙
    ↘ delay slot
; :
                bne     $a0, $a3, exit
                or      $at, $zero ; branch ↙
    ↘ delay slot, NOP
loc_7C:
;
; :
                bnez    $v0, loc_5C
; :
                sb      $zero, 0($a1) ; ↙
    ↘ branch delay slot
; :
exit:
                lw      $ra, 0x20+saved_RA(↙
    ↘ $sp)
                move    $v0, $s0
                lw      $s0, 0x20+saved_S0(↙
    ↘ $sp)
                jr      $ra
                addiu   $sp, 0x20     ; ↙
    ↘ branch delay slot
```

Capítulo 48

```
char toupper (char c)
{
        if(c>='a' && c<='z')
                return c-'a'+'A';
        else
                return c;
}
```

[1].

1

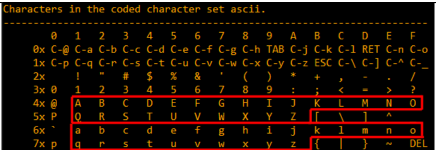


Figura 48.1:

48.1 x64

48.1.1

Listing 48.1: Sin optimización MSVC 2013 (x64)

```
c$ = 8
toupper PROC
        mov     BYTE PTR [rsp+8], cl
        movsx   eax, BYTE PTR c$[rsp]
        cmp     eax, 97
        jl      SHORT $LN2@toupper
        movsx   eax, BYTE PTR c$[rsp]
        cmp     eax, 122
        jg      SHORT $LN2@toupper
        movsx   eax, BYTE PTR c$[rsp]
        sub     eax, 32
        jmp     SHORT $LN3@toupper
        jmp     SHORT $LN1@toupper ; compiler artefact
$LN2@toupper:
        movzx   eax, BYTE PTR c$[rsp] ; unnecessary casting
$LN1@toupper:
```

```
$LN3@toupper: ; compiler artefact
        ret     0
toupper ENDP
```

Sin optimización GCC :

Listing 48.2: Sin optimización GCC 4.9 (x64)

```
toupper:
        push    rbp
        mov     rbp, rsp
        mov     eax, edi
        mov     BYTE PTR [rbp-4], al
        cmp     BYTE PTR [rbp-4], 96
        jle     .L2
        cmp     BYTE PTR [rbp-4], 122
        jg      .L2
        movzx   eax, BYTE PTR [rbp-4]
        sub     eax, 32
        jmp     .L3
.L2:
        movzx   eax, BYTE PTR [rbp-4]
.L3:
        pop     rbp
        ret
```

48.1.2

Con optimización MSVC :

Listing 48.3: Con optimización MSVC 2013 (x64)

```
toupper PROC
        lea     eax, DWORD PTR [rcx-97]
        cmp     al, 25
        ja      SHORT $LN2@toupper
        movsx   eax, cl
        sub     eax, 32
        ret     0
$LN2@toupper:
        movzx   eax, cl
        ret     0
toupper ENDP
```

: 42.2.1 on page 960.

```
int tmp=c-97;

if (tmp>25)
        return c;
else
        return c-32;
```

Con optimización GCC

Listing 48.4: Con optimización GCC 4.9 (x64)

```
toupper:
        lea     edx, [rdi-97] ; 0x61
        lea     eax, [rdi-32] ; 0x20
        cmp     dl, 25
        cmova   eax, edi
        ret
```

48.2 ARM

Listing 48.5: Con optimización Keil 6/2013 (Modo ARM)

```
toupper PROC
        SUB      r1,r0,#0x61
        CMP      r1,#0x19
        SUBLS    r0,r0,#0x20
        ANDLS    r0,r0,#0xff
        BX       lr
        ENDP
```

Listing 48.6: Con optimización Keil 6/2013 (Modo Thumb)

```
toupper PROC
        MOVS     r1,r0
        SUBS     r1,r1,#0x61
        CMP      r1,#0x19
        BHI      |L0.14|
        SUBS     r0,r0,#0x20
        LSLS     r0,r0,#24
        LSRS     r0,r0,#24
|L0.14|
        BX       lr
        ENDP
```

48.2.1 GCC para ARM64

Listing 48.7: Sin optimización GCC 4.9 (ARM64)

```
toupper:
        sub     sp, sp, #16
```

```
        strb    w0, [sp,15]
        ldrb    w0, [sp,15]
        cmp     w0, 96
        bls     .L2
        ldrb    w0, [sp,15]
        cmp     w0, 122
        bhi     .L2
        ldrb    w0, [sp,15]
        sub     w0, w0, #32
        uxtb    w0, w0
        b       .L3
.L2:
        ldrb    w0, [sp,15]
.L3:
        add     sp, sp, 16
        ret
```

Listing 48.8: Con optimización GCC 4.9 (ARM64)

```
toupper:
        uxtb    w0, w0
        sub     w1, w0, #97
        uxtb    w1, w1
        cmp     w1, 25
        bhi     .L2
        sub     w0, w0, #32
        uxtb    w0, w0
.L2:
        ret
```

48.3

Capítulo 49

.

49.1 (x86)

:

```
add     [ebp-31F7Bh], cl
dec     dword ptr [ecx-3277Bh]
dec     dword ptr [ebp-2CF7Bh]
inc     dword ptr [ebx-7A76F33Ch]
fdiv    st(4), st
db 0FFh
dec     dword ptr [ecx-21F7Bh]
dec     dword ptr [ecx-22373h]
dec     dword ptr [ecx-2276Bh]
dec     dword ptr [ecx-22B63h]
```

```
dec      dword ptr [ecx-22F4Bh]
dec      dword ptr [ecx-23343h]
jmp      dword ptr [esi-74h]
xchg     eax, ebp
clc
std
db 0FFh
db 0FFh
mov      word ptr [ebp-214h], cs ; <-
mov      word ptr [ebp-238h], ds
mov      word ptr [ebp-23Ch], es
mov      word ptr [ebp-240h], fs
mov      word ptr [ebp-244h], gs
pushf
pop      dword ptr [ebp-210h]
mov      eax, [ebp+4]
mov      [ebp-218h], eax
lea      eax, [ebp+4]
mov      [ebp-20Ch], eax
mov      dword ptr [ebp-2D0h], 10001h
mov      eax, [eax-4]
mov      [ebp-21Ch], eax
mov      eax, [ebp+0Ch]
mov      [ebp-320h], eax
mov      eax, [ebp+10h]
mov      [ebp-31Ch], eax
mov      eax, [ebp+4]
mov      [ebp-314h], eax
call     ds:IsDebuggerPresent
mov      edi, eax
lea      eax, [ebp-328h]
push     eax
call     sub_407663
```

```
pop     ecx
test    eax, eax
jnz     short loc_402D7B
```

.

49.2 ?

:

- . PUSH, MOV, CALL, .
- immediates.
- .

Listing 49.1: Ruido aleatorio (x86)

```
    mov     bl, 0Ch
    mov     ecx, 0D38558Dh
    mov     eax, ds:2C869A86h
    db      67h
    mov     dl, 0CCh
    insb
    movsb
    push    eax
    xor     [edx-53h], ah
    fcom    qword ptr [edi-45A0EF72h]
    pop     esp
    pop     ss
    in      eax, dx
    dec     ebx
    push    esp
```

```
    lds     esp, [esi-41h]
    retf
    rcl     dword ptr [eax], cl
    mov     cl, 9Ch
    mov     ch, 0DFh
    push    cs
    insb
    mov     esi, 0D9C65E4Dh
    imul    ebp, [ecx], 66h
    pushf
    sal     dword ptr [ebp-64h], cl
    sub     eax, 0AC433D64h
    out     8Ch, eax
    pop     ss
    sbb     [eax], ebx
    aas
    xchg    cl, [ebx+ebx*4+14B31Eh]
    jecxz   short near ptr loc_58+1
    xor     al, 0C6h
    inc     edx
    db      36h
    pusha
    stosb
    test    [ebx], ebx
    sub     al, 0D3h ; 'L'
    pop     eax
    stosb

loc_58: ; CODE XREF: seg000:0000004A
    test    [esi], eax
    inc     ebp
    das
    db      64h
```

```
    pop     ecx
    das
    hlt

    pop     edx
    out     0B0h, al
    lodsb
    push    ebx
    cdq
    out     dx, al
    sub     al, 0Ah
    sti
    outsd
    add     dword ptr [edx], 96FCBE4Bh
    and     eax, 0E537EE4Fh
    inc     esp
    stosd
    cdq
    push    ecx
    in      al, 0CBh
    mov     ds:0D114C45Ch, al
    mov     esi, 659D1985h
```

Listing 49.2: Ruido aleatorio (x86-64)

```
    lea     esi, [rax+rdx*4+43558D29h]

loc_AF3: ; CODE XREF: seg000:0000000000000⤦
    ↳ B46
    rcl     byte ptr [rsi+rax*8+29BB423Ah], ⤦
    ↳ 1
    lea     ecx, cs:0FFFFFFFFB2A6780Fh
    mov     al, 96h
```

```
    mov     ah, 0CEh
    push    rsp
    lods    byte ptr [esi]

    db  2Fh ; /

    pop     rsp
    db      64h
    retf    0E993h

    cmp     ah, [rax+4Ah]
    movzx   rsi, dword ptr [rbp-25h]
    push    4Ah
    movzx   rdi, dword ptr [rdi+rdx*8]

    db  9Ah

    rcr     byte ptr [rax+1Dh], cl
    lodsd
    xor     [rbp+6CF20173h], edx
    xor     [rbp+66F8B593h], edx
    push    rbx
    sbb     ch, [rbx-0Fh]
    stosd
    int     87h
    db      46h, 4Ch
    out     33h, rax
    xchg    eax, ebp
    test    ecx, ebp
    movsd
    leave
    push    rsp
```

```
    db  16h

    xchg    eax, esi
    pop     rdi

loc_B3D: ; CODE XREF: seg000:0000000000000⤦
    ↳ B5F
    mov     ds:93CA685DF98A90F9h, eax
    jnz     short near ptr loc_AF3+6
    out     dx, eax
    cwde
    mov     bh, 5Dh ; ']'
    movsb
    pop     rbp
```

Listing 49.3: Ruido aleatorio (ARM (Modo ARM))

```
BLNE    0xFE16A9D8
BGE     0x1634D0C
SVCCS   0x450685
STRNVT  R5, [PC],#-0x964
LDCGE   p6, c14, [R0],#0x168
STCCSL  p9, c9, [LR],#0x14C
CMNHIP  PC, R10,LSL#22
FLDMIADNV LR!, {D4}
MCR     p5, 2, R2,c15,c6, 4
BLGE    0x1139558
BLGT    0xFF9146E4
STRNEB  R5, [R4],#0xCA2
STMNEIB R5, {R0,R4,R6,R7,R9-SP,PC}
STMIA   R8, {R0,R2-R4,R7,R8,R10,SP,LR}^
STRB    SP, [R8],PC,ROR#18
LDCCS   p9, c13, [R6,#0x1BC]
```

```
LDRGE   R8, [R9,#0x66E]
STRNEB  R5, [R8],#-0x8C3
STCCSL  p15, c9, [R7,#-0x84]
RSBLS   LR, R2, R11,ASR LR
SVCGT   0x9B0362
SVCGT   0xA73173
STMNEDB R11!, {R0,R1,R4-R6,R8,R10,R11,SP}
STR     R0, [R3],#-0xCE4
LDCGT   p15, c8, [R1,#0x2CC]
LDRCCB  R1, [R11],-R7,ROR#30
BLLT    0xFED9D58C
BL      0x13E60F4
LDMVSIB R3!, {R1,R4-R7}^
USATNE  R10, #7, SP,LSL#11
LDRGEB  LR, [R1],#0xE56
STRPLT  R9, [LR],#0x567
LDRLT   R11, [R1],#-0x29B
SVCNV   0x12DB29
MVNNVS  R5, SP,LSL#25
LDCL    p8, c14, [R12,#-0x288]
STCNEL  p2, c6, [R6,#-0xBC]!
SVCNV   0x2E5A2F
BLX     0x1A8C97E
TEQGE   R3, #0x1100000
STMLSIA R6, {R3,R6,R10,R11,SP}
BICPLS  R12, R2, #0x5800
BNE     0x7CC408
TEQGE   R2, R4,LSL#20
SUBS    R1, R11, #0x28C
BICVS   R3, R12, R7,ASR R0
LDRMI   R7, [LR],R3,LSL#21
BLMI    0x1A79234
STMVCDB R6, {R0-R3,R6,R7,R10,R11}
```

```
EORMI   R12, R6, #0xC5
MCRRCS  p1, 0xF, R1,R3,c2
```

Listing 49.4: Ruido aleatorio (ARM (Modo Thumb))

```
    LSRS     R3, R6, #0x12
    LDRH     R1, [R7,#0x2C]
    SUBS     R0, #0x55 ; 'U'
    ADR      R1, loc_3C
    LDR      R2, [SP,#0x218]
    CMP      R4, #0x86
    SXTB     R7, R4
    LDR      R4, [R1,#0x4C]
    STR      R4, [R4,R2]
    STR      R0, [R6,#0x20]
    BGT      0xFFFFFF72
    LDRH     R7, [R2,#0x34]
    LDRSH    R0, [R2,R4]
    LDRB     R2, [R7,R2]

    DCB 0x17
    DCB 0xED

    STRB     R3, [R1,R1]
    STR      R5, [R0,#0x6C]
    LDMIA    R3, {R0-R5,R7}
    ASRS     R3, R2, #3
    LDR      R4, [SP,#0x2C4]
    SVC      0xB5
    LDR      R6, [R1,#0x40]
    LDR      R5, =0xB2C5CA32
    STMIA    R6, {R1-R4,R6}
    LDR      R1, [R3,#0x3C]
```

```
    STR     R1, [R5,#0x60]
    BCC     0xFFFFFF70
    LDR     R4, [SP,#0x1D4]
    STR     R5, [R5,#0x40]
    ORRS    R5, R7

loc_3C ; DATA XREF: ROM:00000006
    B       0xFFFFFF98
```

Listing 49.5: Ruido aleatorio (MIPS little endian)

```
    lw      $t9, 0xCB3($t5)
    sb      $t5, 0x3855($t0)
    sltiu   $a2, $a0, -0x657A
    ldr     $t4, -0x4D99($a2)
    daddi   $s0, $s1, 0x50A4
    lw      $s7, -0x2353($s4)
    bgtzl   $a1, 0x17C5C

    .byte 0x17
    .byte 0xED
    .byte 0x4B  # K
    .byte 0x54  # T

    lwc2    $31, 0x66C5($sp)
    lwu     $s1, 0x10D3($a1)
    ldr     $t6, -0x204B($zero)
    lwc1    $f30, 0x4DBE($s2)
    daddiu  $t1, $s1, 0x6BD9
    lwu     $s5, -0x2C64($v1)
    cop0    0x13D642D
    bne     $gp, $t4, 0xFFFF9EF0
    lh      $ra, 0x1819($s1)
```

```
    sdl     $fp, -0x6474($t8)
    jal     0x78C0050
    ori     $v0, $s2, 0xC634
    blez    $gp, 0xFFFEA9D4
    swl     $t8, -0x2CD4($s2)
    sltiu   $a1, $k0, 0x685
    sdc1    $f15, 0x5964($at)
    sw      $s0, -0x19A6($a1)
    sltiu   $t6, $a3, -0x66AD
    lb      $t7, -0x4F6($t3)
    sd      $fp, 0x4B02($a1)
```

Capítulo 50

.

50.1

(57 on page 1273) . .

.

:

```
mov     byte ptr [ebx], 'h'
mov     byte ptr [ebx+1], 'e'
mov     byte ptr [ebx+2], 'l'
mov     byte ptr [ebx+3], 'l'
mov     byte ptr [ebx+4], 'o'
mov     byte ptr [ebx+5], ' '
mov     byte ptr [ebx+6], 'w'
mov     byte ptr [ebx+7], 'o'
```

```
mov     byte ptr [ebx+8], 'r'
mov     byte ptr [ebx+9], 'l'
mov     byte ptr [ebx+10], 'd'
```

:

```
mov     ebx, offset username
cmp     byte ptr [ebx], 'j'
jnz     fail
cmp     byte ptr [ebx+1], 'o'
jnz     fail
cmp     byte ptr [ebx+2], 'h'
jnz     fail
cmp     byte ptr [ebx+3], 'n'
jnz     fail
jz      it_is_john
```

`sprintf()`:

```
sprintf(buf, "%s%c%s%c%s", "hel",'l',"o w",'↙
    ↳ o',"rld");
```

: 78.2 on page 1486.

50.2

50.2.1

.

:

Listing 50.1:

```
add      eax, ebx
mul      ecx
```

Listing 50.2: obfuscated code

```


xor      esi, 011223344h ;
add      esi, eax        ;
add      eax, ebx
mov      edx, eax        ;
shl      edx, 4          ;
mul      ecx
xor      esi, ecx        ;
```

(ESI y EDX). ?

50.2.2

- `MOV op1, op2` `PUSH op2 / POP op1`.
- `JMP label` `PUSH label / RET`. IDA .
- `CALL label`:`PUSH label_after_CALL_instruct` `/ PUSH label / RET`.
- `PUSH op` : `SUB ESP, 4 (o 8) / MOV [ESP], op`.

50.2.3

ESI :

```
mov     esi, 1
...     ;
dec     esi
...     ;
cmp     esi, 0
jz      real_code
;
real_code:
```

everse engineer.

opaque predicate.

(ESI):

```
add     eax, ebx        ;
mul     ecx             ;
add     eax, esi        ; opaque predicate.
```

50.2.4

```
instruction 1
instruction 2
instruction 3
```

:

```
begin:          jmp     ins1_label
```

```
ins2_label:     instruction 2
                jmp     ins3_label

ins3_label:     instruction 3
                jmp     exit:

ins1_label:     instruction 1
                jmp     ins2_label
exit:
```

50.2.5

```
dummy_data1     db      100h dup (0)
message1        db      'hello world',0

dummy_data2     db      200h dup (0)
message2        db      'another message',0

func            proc
                ...
                mov     eax, offset ↙
    ↘ dummy_data1 ; PE or ELF reloc here
                add     eax, 100h
                push    eax
                call    dump_string
                ...
                mov     eax, offset ↙
    ↘ dummy_data2 ; PE or ELF reloc here
                add     eax, 200h
                push    eax
                call    dump_string
                ...
```

```
func                    endp
```

IDA `dummy_data1` y `dummy_data2`, .

.

50.3

PL o ISA . (, .NET or Java machines). .

50.4

: `http://go.yurichev.com/17220`.

MOV : [Dol13].

50.5 Ejercicios

50.5.1 Ejercicio #1

[1]. .

beginners.re.

: G.1.15 on page 1931.

[1] blog.yurichev.com

Capítulo 51

C++

51.1

51.1.1

```
#include <stdio.h>

class c
{
private:
    int v1;
    int v2;
public:
    c() //
```

```
    {
        v1=667;
        v2=999;
    };

    c(int a, int b) //
    {
        v1=a;
        v2=b;
    };

    void dump()
    {
        printf ("%d; %d\n", v1, v2);
    };
};

int main()
{
    class c c1;
    class c c2(5,6);

    c1.dump();
    c2.dump();

    return 0;
};
```

MSVCx86

Listing 51.1: MSVC

```
_c2$ = -16 ; size = 8
_c1$ = -8  ; size = 8
_main PROC
    push ebp
    mov  ebp, esp
    sub  esp, 16
    lea  ecx, DWORD PTR _c1$[ebp]
    call ??0c@@QAE@XZ ; c::c
    push 6
    push 5
    lea  ecx, DWORD PTR _c2$[ebp]
    call ??0c@@QAE@HH@Z ; c::c
    lea  ecx, DWORD PTR _c1$[ebp]
    call ?dump@c@@QAEXXZ ; c::dump
    lea  ecx, DWORD PTR _c2$[ebp]
    call ?dump@c@@QAEXXZ ; c::dump
    xor  eax, eax
    mov  esp, ebp
    pop  ebp
    ret  0
_main ENDP
```

Listing 51.2: MSVC

```
_this$ = -4       ; size = 4
??0c@@QAE@XZ PROC ; c::c, COMDAT
; _this$ = ecx
    push ebp
    mov  ebp, esp
    push ecx
    mov  DWORD PTR _this$[ebp], ecx
    mov  eax, DWORD PTR _this$[ebp]
    mov  DWORD PTR [eax], 667
```

```
    mov  ecx, DWORD PTR _this$[ebp]
    mov  DWORD PTR [ecx+4], 999
    mov  eax, DWORD PTR _this$[ebp]
    mov  esp, ebp
    pop  ebp
    ret  0
??0c@@QAE@XZ ENDP ; c::c

_this$ = -4 ; size = 4
_a$ = 8     ; size = 4
_b$ = 12    ; size = 4
??0c@@QAE@HH@Z PROC ; c::c, COMDAT
; _this$ = ecx
    push ebp
    mov  ebp, esp
    push ecx
    mov  DWORD PTR _this$[ebp], ecx
    mov  eax, DWORD PTR _this$[ebp]
    mov  ecx, DWORD PTR _a$[ebp]
    mov  DWORD PTR [eax], ecx
    mov  edx, DWORD PTR _this$[ebp]
    mov  eax, DWORD PTR _b$[ebp]
    mov  DWORD PTR [edx+4], eax
    mov  eax, DWORD PTR _this$[ebp]
    mov  esp, ebp
    pop  ebp
    ret  8
??0c@@QAE@HH@Z ENDP ; c::c
```

[ISO13, pág. 12.1] . , *this*.

Listing 51.3: MSVC

```
_this$ = -4          ; size = 4
?dump@c@@QAEXXZ PROC ; c::dump, COMDAT
; _this$ = ecx
    push ebp
    mov  ebp, esp
    push ecx
    mov  DWORD PTR _this$[ebp], ecx
    mov  eax, DWORD PTR _this$[ebp]
    mov  ecx, DWORD PTR [eax+4]
    push ecx
    mov  edx, DWORD PTR _this$[ebp]
    mov  eax, DWORD PTR [edx]
    push eax
    push OFFSET ??_C@_07NJBDCIEC@?$CFd?$DL↙
    ↘ ?5?$CFd?6?$AA@
    call _printf
    add  esp, 12
    mov  esp, ebp
    pop  ebp
    ret  0
?dump@c@@QAEXXZ ENDP ; c::dump
```

Listing 51.4: MSVC

```
??0c@@QAE@XZ PROC ; c::c, COMDAT
; _this$ = ecx
    mov  eax, ecx
    mov  DWORD PTR [eax], 667
    mov  DWORD PTR [eax+4], 999
    ret  0
??0c@@QAE@XZ ENDP ; c::c

_a$ = 8  ; size = 4
```

```
_b$ = 12 ; size = 4
??0c@@QAE@HH@Z PROC ; c::c, COMDAT
; _this$ = ecx
    mov  edx, DWORD PTR _b$[esp-4]
    mov  eax, ecx
    mov  ecx, DWORD PTR _a$[esp-4]
    mov  DWORD PTR [eax], ecx
    mov  DWORD PTR [eax+4], edx
    ret  8
??0c@@QAE@HH@Z ENDP ; c::c

?dump@c@@QAEXXZ PROC ; c::dump, COMDAT
; _this$ = ecx
    mov  eax, DWORD PTR [ecx+4]
    mov  ecx, DWORD PTR [ecx]
    push eax
    push ecx
    push OFFSET ??_C@_07NJBDCIEC@?$CFd?$DL
    ?5?$CFd?6?$AA@
    call _printf
    add  esp, 12
    ret  0
?dump@c@@QAEXXZ ENDP ; c::dump
```

(64 on page 1299).

MSVCx86-64

. Spanish text placeholder. `c(int a, int b)`:

Listing 51.5: Con optimización MSVC 2012 x64

```
; void dump()

?dump@c@@QEAAXXZ PROC ; c::dump
    mov    r8d, DWORD PTR [rcx+4]
    mov    edx, DWORD PTR [rcx]
    lea    rcx, OFFSET FLAT:??⤦
    ↳ _C@_07NJBDCIEC@?$CFd?$DL?5?$CFd?6?$AA@⤦
    ↳  ; '%d; %d'
    jmp    printf
?dump@c@@QEAAXXZ ENDP ; c::dump

; c(int a, int b)

??0c@@QEAA@HH@Z PROC ; c::c
    mov    DWORD PTR [rcx], edx    ; : a
    mov    DWORD PTR [rcx+4], r8d ; : b
    mov    rax, rcx
    ret    0
??0c@@QEAA@HH@Z ENDP ; c::c

;

??0c@@QEAA@XZ PROC ; c::c
    mov    DWORD PTR [rcx], 667
    mov    DWORD PTR [rcx+4], 999
    mov    rax, rcx
    ret    0
??0c@@QEAA@XZ ENDP ; c::c
```

.

: 13.1.1 on page 286.

GCCx86

Listing 51.6: GCC 4.4.1

```
     public main
main proc near

var_20 = dword ptr -20h
var_1C = dword ptr -1Ch
var_18 = dword ptr -18h
var_10 = dword ptr -10h
var_8  = dword ptr -8

    push ebp
    mov  ebp, esp
    and  esp, 0FFFFFFF0h
    sub  esp, 20h
    lea  eax, [esp+20h+var_8]
    mov  [esp+20h+var_20], eax
    call _ZN1cC1Ev
    mov  [esp+20h+var_18], 6
    mov  [esp+20h+var_1C], 5
    lea  eax, [esp+20h+var_10]
    mov  [esp+20h+var_20], eax
    call _ZN1cC1Eii
    lea  eax, [esp+20h+var_8]
    mov  [esp+20h+var_20], eax
    call _ZN1c4dumpEv
    lea  eax, [esp+20h+var_10]
    mov  [esp+20h+var_20], eax
    call _ZN1c4dumpEv
    mov  eax, 0
    leave
```

```
    retn
main endp
```

```
                public _ZN1cC1Ev ; weak
_ZN1cC1Ev       proc near               ; ⤦
    ↳ CODE XREF: main+10

arg_0           = dword ptr  8

                push    ebp
                mov     ebp, esp
                mov     eax, [ebp+arg_0]
                mov     dword ptr [eax], 667
                mov     eax, [ebp+arg_0]
                mov     dword ptr [eax+4], ⤦
    ↳ 999
                pop     ebp
                retn
_ZN1cC1Ev       endp
```

```
                public _ZN1cC1Eii
_ZN1cC1Eii      proc near

arg_0           = dword ptr  8
arg_4           = dword ptr  0Ch
arg_8           = dword ptr  10h

                push    ebp
                mov     ebp, esp
                mov     eax, [ebp+arg_0]
                mov     edx, [ebp+arg_4]
                mov     [eax], edx
```

```
                mov     eax, [ebp+arg_0]
                mov     edx, [ebp+arg_8]
                mov     [eax+4], edx
                pop     ebp
                retn
_ZN1cC1Eii      endp
```

```
void ZN1cC1Eii (int *obj, int a, int b)
{
  *obj=a;
  *(obj+1)=b;
};
```

....

```
                public _ZN1c4dumpEv
_ZN1c4dumpEv    proc near

var_18          = dword ptr -18h
var_14          = dword ptr -14h
var_10          = dword ptr -10h
arg_0           = dword ptr  8

                push    ebp
                mov     ebp, esp
                sub     esp, 18h
                mov     eax, [ebp+arg_0]
                mov     edx, [eax+4]
                mov     eax, [ebp+arg_0]
                mov     eax, [eax]
                mov     [esp+18h+var_10], ↙
    ↘ edx
```

```
                mov     [esp+18h+var_14], ↙
    ↘ eax
                mov     [esp+18h+var_18], ↙
    ↘ offset aDD ; "%d; %d\n"
                call    _printf
                leave
                retn
_ZN1c4dumpEv    endp
```

:

```
void ZN1c4dumpEv (int *obj)
{
  printf ("%d; %d\n", *obj, *(obj+1));
};
```

GCCx86-64

RDI, RSI, RDX, RCX, R8 and R9 [Mit13]. . .

Listing 51.7: GCC 4.4.6 x64

```

;

_ZN1cC2Ev:
    mov  DWORD PTR [rdi], 667
    mov  DWORD PTR [rdi+4], 999
    ret

; c(int a, int b)

```

```
_ZN1cC2Eii:
    mov  DWORD PTR [rdi], esi
    mov  DWORD PTR [rdi+4], edx
    ret

; dump()

_ZN1c4dumpEv:
    mov  edx, DWORD PTR [rdi+4]
    mov  esi, DWORD PTR [rdi]
    xor  eax, eax
    mov  edi, OFFSET FLAT:.LC0 ; "%d; %d\n"
    jmp  printf
```

51.1.2

:

```
#include <stdio.h>

class object
{
    public:
        int color;
        object() { };
        object (int color) { this->color=
    ↳ color; };
        void print_color() { printf ("color
    ↳ =%d\n", color); };
};

class box : public object
```

```
{
    private:
        int width, height, depth;
    public:
        box(int color, int width, int height⤦
    ↳ , int depth)
        {
            this->color=color;
            this->width=width;
            this->height=height;
            this->depth=depth;
        };
        void dump()
        {
            printf ("this is box. color=%d, ⤦
    ↳ width=%d, height=%d, depth=%d\n", ⤦
    ↳ color, width, height, depth);
        };
};

class sphere : public object
{
private:
    int radius;
public:
    sphere(int color, int radius)
    {
        this->color=color;
        this->radius=radius;
    };
    void dump()
    {
        printf ("this is sphere. color=%d, ⤦
```

```
    ↳ radius=%d\n", color, radius);
    };
};

int main()
{
    box b(1, 10, 20, 30);
    sphere s(2, 40);

    b.print_color();
    s.print_color();

    b.dump();
    s.dump();

    return 0;
};
```

1

Listing 51.8: Con optimización MSVC 2008 /Ob0

```
??_C@_09GCEDOLPA@color?$DN?$CFd?6?$AA@ DB '↙
    ↳ color=%d', 0aH, 00H ; `string'
?print_color@object@@QAEXXZ PROC ; object::↙
    ↳ print_color, COMDAT
; _this$ = ecx
    mov  eax, DWORD PTR [ecx]
    push eax

; 'color=%d', 0aH, 00H
```

1

```
    push OFFSET ??_C@_09GCEDOLPA@color?$DN?⤦
    ↳ $CFd?6?$AA@
    call _printf
    add  esp, 8
    ret  0
?print_color@object@@QAEXXZ ENDP ; object::⤦
    ↳ print_color
```

Listing 51.9: Con optimización MSVC 2008 /Ob0

```
?dump@box@@QAEXXZ PROC ; box::dump, COMDAT
; _this$ = ecx
    mov  eax, DWORD PTR [ecx+12]
    mov  edx, DWORD PTR [ecx+8]
    push eax
    mov  eax, DWORD PTR [ecx+4]
    mov  ecx, DWORD PTR [ecx]
    push edx
    push eax
    push ecx

; 'this is box. color=%d, width=%d, height=%⤦
    ↳ d, depth=%d', 0aH, 00H ; `string'
    push OFFSET ??_C@_0DG@NCNGAADL@this?5is⤦
    ↳ ?5box?4?5color?$DN?$CFd?0?5width?$DN?⤦
    ↳ $CFd?0@
    call _printf
    add  esp, 20
    ret  0
?dump@box@@QAEXXZ ENDP ; box::dump
```

Listing 51.10: Con optimización MSVC 2008 /Ob0

```
?dump@sphere@@QAEXXZ PROC ; sphere::dump, ⤦
    ↳ COMDAT
; _this$ = ecx
    mov  eax, DWORD PTR [ecx+4]
    mov  ecx, DWORD PTR [ecx]
    push eax
    push ecx

; 'this is sphere. color=%d, radius=%d', 0aH⤦
    ↳ , 00H
    push OFFSET ??_C@_0CF@EFEDJLDC@this?5is⤦
    ↳ ?5sphere?4?5color?$DN?$CFd?0?5radius@
    call _printf
    add  esp, 12
    ret  0
?dump@sphere@@QAEXXZ ENDP ; sphere::dump
```

:

Spanish text placeholder	Spanish text placeholder
+0x0	int color

box:

Spanish text placeholder	Spanish text placeholder
+0x0	int color
+0x4	int width
+0x8	int height
+0xC	int depth

sphere:

Spanish text placeholder	Spanish text placeholder
+0x0	int color
+0x4	int radius

:

Listing 51.11: Con optimización MSVC 2008 /Ob0

```
PUBLIC _main
_TEXT SEGMENT
_s$ = -24 ; size = 8
_b$ = -16 ; size = 16
_main PROC
    sub  esp, 24
    push 30
    push 20
    push 10
    push 1
    lea  ecx, DWORD PTR _b$[esp+40]
    call ??0box@@QAE@HHHH@Z ; box::box
    push 40
    push 2
    lea  ecx, DWORD PTR _s$[esp+32]
    call ??0sphere@@QAE@HH@Z ; sphere::⤦
    ↳ sphere
    lea  ecx, DWORD PTR _b$[esp+24]
    call ?print_color@object@@QAEXXZ ; ⤦
    ↳ object::print_color
    lea  ecx, DWORD PTR _s$[esp+24]
    call ?print_color@object@@QAEXXZ ; ⤦
    ↳ object::print_color
    lea  ecx, DWORD PTR _b$[esp+24]
    call ?dump@box@@QAEXXZ ; box::dump
    lea  ecx, DWORD PTR _s$[esp+24]
```

```
    call ?dump@sphere@@QAEXXZ ; sphere::dump
    xor  eax, eax
    add  esp, 24
    ret  0
_main ENDP
```

51.1.3

```
#include <stdio.h>

class box
{
    private:
        int color, width, height, depth;
    public:
        box(int color, int width, int height
    ↳ , int depth)
        {
            this->color=color;
            this->width=width;
            this->height=height;
            this->depth=depth;
        };
        void dump()
        {
            printf ("this is box. color=%d, 
    ↳ width=%d, height=%d, depth=%d\n", 
    ↳ color, width, height, depth);
        };
};
```

```
?dump@box@@QAEXXZ PROC ; box::dump, COMDAT
; _this$ = ecx
    mov  eax, DWORD PTR [ecx+12]
    mov  edx, DWORD PTR [ecx+8]
    push eax
    mov  eax, DWORD PTR [ecx+4]
    mov  ecx, DWORD PTR [ecx]
    push edx
    push eax
    push ecx
; 'this is box. color=%d, width=%d, height=%↙
    ↘ d, depth=%d', 0aH, 00H
    push OFFSET ??_C@_0DG@NCNGAADL@this?5is↙
    ↘ ?5box?4?5color?$DN?$CFd?0?5width?$DN?↙
    ↘ $CFd?0@
    call _printf
    add  esp, 20
    ret  0
?dump@box@@QAEXXZ ENDP ; box::dump
```

Spanish text placeholder	Spanish text placeholder
+0x0	int color
+0x4	int width
+0x8	int height
+0xC	int depth

```
void hack_oop_encapsulation(class box * o)
{
    o->width=1; // :
                // "error C2248: 'box::width↙
    ↘ ' : cannot access private member ↙
```

```
    ↳ declared in class 'box'"
};
```

```
void hack_oop_encapsulation(class box * o)
{
    unsigned int *ptr_to_object=↙
    ↳ reinterpret_cast<unsigned int*>(o);
    ptr_to_object[1]=123;
};
```

```
?hack_oop_encapsulation@@YAXPAVbox@@@Z PROC ↙
    ↳ ; hack_oop_encapsulation
    mov eax, DWORD PTR _o$[esp-4]
    mov DWORD PTR [eax+4], 123
    ret 0
?hack_oop_encapsulation@@YAXPAVbox@@@Z ENDP ↙
    ↳ ; hack_oop_encapsulation
```

```
int main()
{
    box b(1, 10, 20, 30);

    b.dump();

    hack_oop_encapsulation(&b);

    b.dump();

    return 0;
};
```

```
this is box. color=1, width=10, height=20, ↙
    ↘ depth=30
this is box. color=1, width=123, height=20, ↙
    ↘ depth=30
```

51.1.4

```
#include <stdio.h>

class box
{
    public:
        int width, height, depth;
        box() { };
        box(int width, int height, int depth↙
    ↘ )
        {
            this->width=width;
            this->height=height;
            this->depth=depth;
        };
        void dump()
        {
            printf ("this is box. width=%d, ↙
    ↘ height=%d, depth=%d\n", width, height,↙
    ↘  depth);
        };
        int get_volume()
        {
            return width * height * depth;
        };
```

```
};

class solid_object
{
    public:
        int density;
        solid_object() { };
        solid_object(int density)
        {
            this->density=density;
        };
        int get_density()
        {
            return density;
        };
        void dump()
        {
            printf ("this is solid_object. ↙
    ↘ density=%d\n", density);
        };
};

class solid_box: box, solid_object
{
    public:
        solid_box (int width, int height, ↙
    ↘ int depth, int density)
        {
            this->width=width;
            this->height=height;
            this->depth=depth;
            this->density=density;
        };
```

```
        void dump()
        {
            printf ("this is solid_box. ↙
    ↘ width=%d, height=%d, depth=%d, density↙
    ↘ =%d\n", width, height, depth, density)↙
    ↘ ;
        };
        int get_weight() { return get_volume↙
    ↘ () * get_density(); };
};

int main()
{
    box b(10, 20, 30);
    solid_object so(100);
    solid_box sb(10, 20, 30, 3);

    b.dump();
    so.dump();
    sb.dump();
    printf ("%d\n", sb.get_weight());

    return 0;
};
```

Listing 51.12: Con optimización MSVC 2008 /Ob0

```
?dump@box@@QAEXXZ PROC ; box::dump, COMDAT
; _this$ = ecx
    mov  eax, DWORD PTR [ecx+8]
    mov  edx, DWORD PTR [ecx+4]
    push eax
    mov  eax, DWORD PTR [ecx]
```

```
    push edx
    push eax
; 'this is box. width=%d, height=%d, depth=%↙
    ↘ d', 0aH, 00H
    push OFFSET ??_C@_0CM@DIKPHDFI@this?5is↙
    ↘ ?5box?4?5width?$DN?$CFd?0?5height?$DN?↙
    ↘ $CFd@
    call _printf
    add  esp, 16
    ret  0
?dump@box@@QAEXXZ ENDP ; box::dump
```

Listing 51.13: Con optimización MSVC 2008 /Ob0

```
?dump@solid_object@@QAEXXZ PROC ; ↙
    ↘ solid_object::dump, COMDAT
; _this$ = ecx
    mov  eax, DWORD PTR [ecx]
    push eax
; 'this is solid_object. density=%d', 0aH
    push OFFSET ??_C@_0CC@KICFJINL@this?5is↙
    ↘ ?5solid_object?4?5density?$DN?$CFd@
    call _printf
    add  esp, 8
    ret  0
?dump@solid_object@@QAEXXZ ENDP ; ↙
    ↘ solid_object::dump
```

Listing 51.14: Con optimización MSVC 2008 /Ob0

```
?dump@solid_box@@QAEXXZ PROC ; solid_box::↙
    ↘ dump, COMDAT
; _this$ = ecx
```

```
    mov  eax, DWORD PTR [ecx+12]
    mov  edx, DWORD PTR [ecx+8]
    push eax
    mov  eax, DWORD PTR [ecx+4]
    mov  ecx, DWORD PTR [ecx]
    push edx
    push eax
    push ecx
; 'this is solid_box. width=%d, height=%d, ↙
    ↘ depth=%d, density=%d', 0aH
    push OFFSET ??_C@_0DO@HNCNIHNN@this?5is↙
    ↘ ?5solid_box?4?5width?$DN?$CFd?0?5hei@
    call _printf
    add  esp, 20
    ret  0
?dump@solid_box@@QAEXXZ ENDP ; solid_box::↙
    ↘ dump
```

:

Spanish text placeholder	Spanish text placeholder
+0x0	width
+0x4	height
+0x8	depth

:

Spanish text placeholder	Spanish text placeholder
+0x0	density

:

Spanish text placeholder	Spanish text placeholder
+0x0	width
+0x4	height
+0x8	depth
+0xC	density

Listing 51.15: Con optimización MSVC 2008 /Ob0

```
?get_volume@box@@QAEHXZ PROC ; box::⤦
    ↳ get_volume, COMDAT
; _this$ = ecx
    mov  eax, DWORD PTR [ecx+8]
    imul eax, DWORD PTR [ecx+4]
    imul eax, DWORD PTR [ecx]
    ret  0
?get_volume@box@@QAEHXZ ENDP ; box::⤦
    ↳ get_volume
```

Listing 51.16: Con optimización MSVC 2008 /Ob0

```
?get_density@solid_object@@QAEHXZ PROC ; ⤦
    ↳ solid_object::get_density, COMDAT
; _this$ = ecx
    mov eax, DWORD PTR [ecx]
    ret 0
?get_density@solid_object@@QAEHXZ ENDP ; ⤦
    ↳ solid_object::get_density
```

Listing 51.17: Con optimización MSVC 2008 /Ob0

```
?get_weight@solid_box@@QAEHXZ PROC ; ⤦
    ↳ solid_box::get_weight, COMDAT
; _this$ = ecx
```

```
    push esi
    mov  esi, ecx
    push edi
    lea  ecx, DWORD PTR [esi+12]
    call ?get_density@solid_object@@QAEHXZ ;↙
    ↳  solid_object::get_density
    mov  ecx, esi
    mov  edi, eax
    call ?get_volume@box@@QAEHXZ ; box::↙
    ↳ get_volume
    imul eax, edi
    pop  edi
    pop  esi
    ret  0
?get_weight@solid_box@@QAEHXZ ENDP ; ↙
    ↳ solid_box::get_weight
```

51.1.5

```
#include <stdio.h>

class object
{
    public:
        int color;
        object() { };
        object (int color) { this->color=↙
    ↳ color; };
        virtual void dump()
        {
            printf ("color=%d\n", color);
        };
```

```
};

class box : public object
{
    private:
        int width, height, depth;
    public:
        box(int color, int width, int height
    ↳ , int depth)
        {
            this->color=color;
            this->width=width;
            this->height=height;
            this->depth=depth;
        };
        void dump()
        {
            printf ("this is box. color=%d, 
    ↳ width=%d, height=%d, depth=%d\n", 
    ↳ color, width, height, depth);
        };
};

class sphere : public object
{
    private:
        int radius;
    public:
        sphere(int color, int radius)
        {
            this->color=color;
            this->radius=radius;
        };
```

```
        void dump()
        {
            printf ("this is sphere. color=%
    ↘ d, radius=%d\n", color, radius);
        };
};

int main()
{
    box b(1, 10, 20, 30);
    sphere s(2, 40);

    object *o1=&b;
    object *o2=&s;

    o1->dump();
    o2->dump();
    return 0;
};
```

```
_s$ = -32 ; size = 12
_b$ = -20 ; size = 20
_main PROC
    sub  esp, 32
    push 30
    push 20
    push 10
    push 1
    lea  ecx, DWORD PTR _b$[esp+48]
    call ??0box@@QAE@HHHH@Z ; box::box
    push 40
    push 2
```

```
    lea  ecx, DWORD PTR _s$[esp+40]
    call ??0sphere@@QAE@HH@Z ; sphere::↙
    ↘ sphere
    mov  eax, DWORD PTR _b$[esp+32]
    mov  edx, DWORD PTR [eax]
    lea  ecx, DWORD PTR _b$[esp+32]
    call edx
    mov  eax, DWORD PTR _s$[esp+32]
    mov  edx, DWORD PTR [eax]
    lea  ecx, DWORD PTR _s$[esp+32]
    call edx
    xor  eax, eax
    add  esp, 32
    ret  0
_main ENDP
```

2

```
??_R0?AVbox@@@8 DD FLAT:??_7type_info@@6B@ ;↙
    ↘  box `RTTI Type Descriptor'
    DD    00H
    DB    '.?AVbox@@', 00H

??_R1A@?0A@EA@box@@8 DD FLAT:??_R0?AVbox@@@8↙
    ↘  ; box::`RTTI Base Class Descriptor at↙
    ↘  (0,-1,0,64)'
    DD    01H
    DD    00H
    DD    0ffffffffH
    DD    00H
    DD    040H
```

[2]:(23 on page 746)

```
    DD      FLAT:??_R3box@@8

??_R2box@@8 DD     FLAT:??_R1A@?0A@EA@box@@8 ⤦
    ↳ ; box::`RTTI Base Class Array'
    DD      FLAT:??_R1A@?0A@EA@object@@8

??_R3box@@8 DD     00H ; box::`RTTI Class ⤦
    ↳ Hierarchy Descriptor'
    DD      00H
    DD      02H
    DD      FLAT:??_R2box@@8

??_R4box@@6B@ DD 00H ; box::`RTTI Complete ⤦
    ↳ Object Locator'
    DD      00H
    DD      00H
    DD      FLAT:??_R0?AVbox@@@8
    DD      FLAT:??_R3box@@8

??_7box@@6B@ DD     FLAT:??_R4box@@6B@ ; box⤦
    ↳ ::`vftable'
    DD      FLAT:?dump@box@@UAEXXZ

_color$ = 8   ; size = 4
_width$ = 12  ; size = 4
_height$ = 16 ; size = 4
_depth$ = 20  ; size = 4
??0box@@QAE@HHHH@Z PROC ; box::box, COMDAT
; _this$ = ecx
    push esi
    mov  esi, ecx
    call ??0object@@QAE@XZ ; object::object
    mov  eax, DWORD PTR _color$[esp]
```

```
    mov  ecx, DWORD PTR _width$[esp]
    mov  edx, DWORD PTR _height$[esp]
    mov  DWORD PTR [esi+4], eax
    mov  eax, DWORD PTR _depth$[esp]
    mov  DWORD PTR [esi+16], eax
    mov  DWORD PTR [esi], OFFSET ??↙
    ↘ _7box@@6B@
    mov  DWORD PTR [esi+8], ecx
    mov  DWORD PTR [esi+12], edx
    mov  eax, esi
    pop  esi
    ret  16
??0box@@QAE@HHHH@Z ENDP ; box::box
```

51.2 ostream

```
#include <iostream>

int main()
{
    std::cout << "Hello, world!\n";
}
```

ostream. :

Listing 51.18: MSVC 2012 (reduced listing)

```
$SG37112 DB 'Hello, world!', 0aH, 00H

_main PROC
    push OFFSET $SG37112
```

```
    push OFFSET ?cout@std@@3V?↙
    ↪ $basic_ostream@DU?↙
    ↪ $char_traits@D@std@@@1@A ; std::cout
    call ??$?6U?↙
    ↪ $char_traits@D@std@@@std@@YAAAV?↙
    ↪ $basic_ostream@DU?↙
    ↪ $char_traits@D@std@@@0@AAV10@PBD@Z ; ↙
    ↪ std::operator<<<std::char_traits<char>↙
    ↪  >
    add  esp, 8
    xor  eax, eax
    ret  0
_main ENDP
```

:

```
#include <iostream>

int main()
{
    std::cout << "Hello, " << "world!\n";
}
```

:

Listing 51.19: MSVC 2012

```

$SG37112 DB 'world!', 0aH, 00H
$SG37113 DB 'Hello, ', 00H

_main PROC
    push OFFSET $SG37113 ; 'Hello, '
```

```
    push OFFSET ?cout@std@@3V?↙
    ↳ $basic_ostream@DU?↙
    ↳ $char_traits@D@std@@@1@A ; std::cout
    call ??$?6U?↙
    ↳ $char_traits@D@std@@@std@@YAAAV?↙
    ↳ $basic_ostream@DU?↙
    ↳ $char_traits@D@std@@@0@AAV10@PBD@Z ; ↙
    ↳ std::operator<<<std::char_traits<char>↙
    ↳  >
    add  esp, 8

    push OFFSET $SG37112 ; 'world!'
    push eax             ;
    call ??$?6U?↙
    ↳ $char_traits@D@std@@@std@@YAAAV?↙
    ↳ $basic_ostream@DU?↙
    ↳ $char_traits@D@std@@@0@AAV10@PBD@Z ; ↙
    ↳ std::operator<<<std::char_traits<char>↙
    ↳  >
    add  esp, 8

    xor  eax, eax
    ret  0
_main ENDP
```

```
f(f(std::cout, "Hello, "), "world!");
```

GCC MSVC.

51.3 References

[ISO13, pág. 8.3.2]. [Cli, pág. 8.6]. [Cli, pág. 8.5].

```
void f2 (int x, int y, int & sum, int & ↙
    ↘ product)
{
        sum=x+y;
        product=x*y;
};
```

Listing 51.20: Con optimización MSVC 2010

```
_x$ = 8                                          ↙
    ↘                     ; size = 4
_y$ = 12                                         ↙
    ↘                     ; size = 4
_sum$ = 16                                       ↙
    ↘                     ; size = 4
_product$ = 20                                   ↙
    ↘                     ; size = 4
?f2@@YAXHHAAH0@Z PROC                            ↙
    ↘                     ; f2
        mov     ecx, DWORD PTR _y$[esp-4]
        mov     eax, DWORD PTR _x$[esp-4]
        lea     edx, DWORD PTR [eax+ecx]
        imul eax, ecx
        mov ecx, DWORD PTR _product$[esp-4]
        push esi
        mov     esi, DWORD PTR _sum$[esp]
        mov     DWORD PTR [esi], edx
        mov     DWORD PTR [ecx], eax
        pop     esi
```

```
        ret     0
?f2@@YAXHHAAH0@Z ENDP                           ↙
     ↘                      ; f2
```

(: 51.1.1 on page 1040.)

51.4 STL

N.B.: . .

51.4.1 std::string

.

MSVC

$16+4+4=24$ $16+8+8=32$.

Listing 51.21: MSVC

```
#include <string>
#include <stdio.h>

struct std_string
{
    union
    {
```

```
        char buf[16];
        char* ptr;
    } u;
    size_t size;     // AKA 'Mysize'  MSVC
    size_t capacity; // AKA 'Myres'  MSVC
};

void dump_std_string(std::string s)
{
    struct std_string *p=(struct std_string ↙
    ↳ *)&s;
    printf ("[%s] size:%d capacity:%d\n", p↙
    ↳ ->size>16 ? p->u.ptr : p->u.buf, p->↙
    ↳ size, p->capacity);
};

int main()
{
    std::string s1="short string";
    std::string s2="string longer that 16 ↙
    ↳ bytes";

    dump_std_string(s1);
    dump_std_string(s2);

    //  c_str()
    printf ("%s\n", &s1);
    printf ("%s\n", s2);
};
```

:

. .

.

. printf() .

. .

GCC

.

. libstdc++-v3\include\bits\basic_string.h :

```
   *  The reason you want _M_data pointing ↙
    ↪ to the character %array and
   *  not the _Rep is so that the debugger ↙
    ↪ can see the string
   *  contents. (Probably we should add a ↙
    ↪ non-inline member to get
   *  the _Rep for the debugger to use, so ↙
    ↪ users can check the actual
   *  string length.)
```

basic_string.h

:

Listing 51.22: GCC

```

#include <string>
#include <stdio.h>

struct std_string
{
```

```
    size_t length;
    size_t capacity;
    size_t refcount;
};

void dump_std_string(std::string s)
{
    char *p1=*(char**)&s; //
    struct std_string *p2=(struct std_string
    *)(p1-sizeof(struct std_string));
    printf ("[%s] size:%d capacity:%d\n", p1
    , p2->length, p2->capacity);
};

int main()
{
    std::string s1="short string";
    std::string s2="string longer that 16 
    bytes";

    dump_std_string(s1);
    dump_std_string(s2);

    // :
    printf ("%s\n", *(char**)&s1);
    printf ("%s\n", *(char**)&s2);
};
```

.

```
#include <string>
#include <stdio.h>

int main()
{
    std::string s1="Hello, ";
    std::string s2="world!\n";
    std::string s3=s1+s2;

    printf ("%s\n", s3.c_str());
}
```

Listing 51.23: MSVC 2012

```


$SG39512 DB 'Hello, ', 00H
$SG39514 DB 'world!', 0aH, 00H
$SG39581 DB '%s', 0aH, 00H

_s2$ = -72 ; size = 24
_s3$ = -48 ; size = 24
_s1$ = -24 ; size = 24
_main PROC
    sub  esp, 72

    push 7
    push OFFSET $SG39512
    lea  ecx, DWORD PTR _s1$[esp+80]
    mov  DWORD PTR _s1$[esp+100], 15
    mov  DWORD PTR _s1$[esp+96], 0
    mov  BYTE PTR _s1$[esp+80], 0
    call ?assign@?$basic_string@DU?↙
    ↘ $char_traits@D@std@@V?↙
```

```
    ↳ $allocator@D@2@@std@@QAEAAV12@PBDI@Z ;↙
    ↳  std::basic_string<char,std::↙
    ↳ char_traits<char>,std::allocator<char>↙
    ↳  >::assign

    push 7
    push OFFSET $SG39514
    lea  ecx, DWORD PTR _s2$[esp+80]
    mov  DWORD PTR _s2$[esp+100], 15
    mov  DWORD PTR _s2$[esp+96], 0
    mov  BYTE PTR _s2$[esp+80], 0
    call ?assign@?$basic_string@DU?↙
    ↳ $char_traits@D@std@@V?↙
    ↳ $allocator@D@2@@std@@QAEAAV12@PBDI@Z ;↙
    ↳  std::basic_string<char,std::↙
    ↳ char_traits<char>,std::allocator<char>↙
    ↳  >::assign

    lea  eax, DWORD PTR _s2$[esp+72]
    push eax
    lea  eax, DWORD PTR _s1$[esp+76]
    push eax
    lea  eax, DWORD PTR _s3$[esp+80]
    push eax
    call ??$?HDU?$char_traits@D@std@@V?↙
    ↳ $allocator@D@1@@std@@YA?AV?↙
    ↳ $basic_string@DU?$char_traits@D@std@@V↙
    ↳ ?$allocator@D@2@@0@ABV10@0@Z ; std::↙
    ↳ operator+<char,std::char_traits<char>,↙
    ↳ std::allocator<char> >

    ; :
    cmp  DWORD PTR _s3$[esp+104], 16
```

```
    lea  eax, DWORD PTR _s3$[esp+84]
    cmovae eax, DWORD PTR _s3$[esp+84]

    push eax
    push OFFSET $SG39581
    call _printf
    add  esp, 20

    cmp  DWORD PTR _s3$[esp+92], 16
    jb   SHORT $LN119@main
    push DWORD PTR _s3$[esp+72]
    call ??3@YAXPAX@Z                  ; ↙
    ↘ operator delete
    add  esp, 4
$LN119@main:
    cmp  DWORD PTR _s2$[esp+92], 16
    mov  DWORD PTR _s3$[esp+92], 15
    mov  DWORD PTR _s3$[esp+88], 0
    mov  BYTE PTR _s3$[esp+72], 0
    jb   SHORT $LN151@main
    push DWORD PTR _s2$[esp+72]
    call ??3@YAXPAX@Z                  ; ↙
    ↘ operator delete
    add  esp, 4
$LN151@main:
    cmp  DWORD PTR _s1$[esp+92], 16
    mov  DWORD PTR _s2$[esp+92], 15
    mov  DWORD PTR _s2$[esp+88], 0
    mov  BYTE PTR _s2$[esp+72], 0
    jb   SHORT $LN195@main
    push DWORD PTR _s1$[esp+72]
    call ??3@YAXPAX@Z                  ; ↙
    ↘ operator delete
```

```
    add  esp, 4
$LN195@main:
    xor  eax, eax
    add  esp, 72
    ret  0
_main ENDP
```

..

.

..

.

:

Listing 51.24: GCC 4.8.1

```

.LC0:
    .string "Hello, "
.LC1:
    .string "world!\n"
main:
    push ebp
    mov  ebp, esp
    push edi
    push esi
    push ebx
    and  esp, -16
    sub  esp, 32
    lea  ebx, [esp+28]
    lea  edi, [esp+20]
```

```
    mov  DWORD PTR [esp+8], ebx
    lea  esi, [esp+24]
    mov  DWORD PTR [esp+4], OFFSET FLAT:.LC0
    mov  DWORD PTR [esp], edi

    call _ZNSsC1EPKcRKSaIcE

    mov  DWORD PTR [esp+8], ebx
    mov  DWORD PTR [esp+4], OFFSET FLAT:.LC1
    mov  DWORD PTR [esp], esi

    call _ZNSsC1EPKcRKSaIcE

    mov  DWORD PTR [esp+4], edi
    mov  DWORD PTR [esp], ebx

    call _ZNSsC1ERKSs

    mov  DWORD PTR [esp+4], esi
    mov  DWORD PTR [esp], ebx

    call _ZNSs6appendERKSs

    ;  c_str():
    mov  eax, DWORD PTR [esp+28]
    mov  DWORD PTR [esp], eax

    call puts

    mov  eax, DWORD PTR [esp+28]
    lea  ebx, [esp+19]
    mov  DWORD PTR [esp+4], ebx
    sub  eax, 12
```

```
    mov  DWORD PTR [esp], eax
    call _ZNSs4_Rep10_M_disposeERKSaIcE
    mov  eax, DWORD PTR [esp+24]
    mov  DWORD PTR [esp+4], ebx
    sub  eax, 12
    mov  DWORD PTR [esp], eax
    call _ZNSs4_Rep10_M_disposeERKSaIcE
    mov  eax, DWORD PTR [esp+20]
    mov  DWORD PTR [esp+4], ebx
    sub  eax, 12
    mov  DWORD PTR [esp], eax
    call _ZNSs4_Rep10_M_disposeERKSaIcE
    lea  esp, [ebp-12]
    xor  eax, eax
    pop  ebx
    pop  esi
    pop  edi
    pop  ebp
    ret
```

.

std::string

:

```
#include <stdio.h>
#include <string>

std::string s="a string";

int main()
```

```
{
    printf ("%s\n", s.c_str());
};
```

Listing 51.25: MSVC 2012:

```
??__Es@@YAXXZ PROC
    push 8
    push OFFSET $SG39512 ; 'a string'
    mov  ecx, OFFSET ?s@@3V?$basic_string@DU⤦
    ↳ ?$char_traits@D@std@@V?⤦
    ↳ $allocator@D@2@@std@@A ; s
    call ?assign@?$basic_string@DU?⤦
    ↳ $char_traits@D@std@@V?⤦
    ↳ $allocator@D@2@@std@@QAEAAV12@PBDI@Z ;⤦
    ↳  std::basic_string<char,std::⤦
    ↳ char_traits<char>,std::allocator<char>⤦
    ↳  >::assign
    push OFFSET ??__Fs@@YAXXZ ; `dynamic ⤦
    ↳ atexit destructor for 's''
    call _atexit
    pop  ecx
    ret  0
??__Es@@YAXXZ ENDP
```

Listing 51.26: MSVC 2012: main()

```
$SG39512 DB 'a string', 00H
$SG39519 DB '%s', 0aH, 00H

_main PROC
    cmp  DWORD PTR ?s@@3V?$basic_string@DU?⤦
    ↳ $char_traits@D@std@@V?⤦
```

```
    ↳ $allocator@D@2@@std@@A+20, 16
    mov  eax, OFFSET ?s@@3V?$basic_string@DU↙
    ↳ ?$char_traits@D@std@@V?↙
    ↳ $allocator@D@2@@std@@A ; s
    cmovae eax, DWORD PTR ?s@@3V?↙
    ↳ $basic_string@DU?$char_traits@D@std@@V↙
    ↳ ?$allocator@D@2@@std@@A
    push eax
    push OFFSET $SG39519 ; '%s'
    call _printf
    add  esp, 8
    xor  eax, eax
    ret  0
_main ENDP
```

Listing 51.27: MSVC 2012:

```
??__Fs@@YAXXZ PROC
    push ecx
    cmp  DWORD PTR ?s@@3V?$basic_string@DU?↙
    ↳ $char_traits@D@std@@V?↙
    ↳ $allocator@D@2@@std@@A+20, 16
    jb   SHORT $LN23@dynamic
    push esi
    mov  esi, DWORD PTR ?s@@3V?↙
    ↳ $basic_string@DU?$char_traits@D@std@@V↙
    ↳ ?$allocator@D@2@@std@@A
    lea  ecx, DWORD PTR $T2[esp+8]
    call ??0?$_Wrap_alloc@V?↙
    ↳ $allocator@D@std@@@std@@QAE@XZ
    push OFFSET ?s@@3V?$basic_string@DU?↙
    ↳ $char_traits@D@std@@V?↙
    ↳ $allocator@D@2@@std@@A ; s
```

```
    lea  ecx, DWORD PTR $T2[esp+12]
    call ??$destroy@PAD@?$_Wrap_alloc@V?↙
    ↘ $allocator@D@std@@@std@@QAEXPAPAD@Z
    lea  ecx, DWORD PTR $T1[esp+8]
    call ??0?$_Wrap_alloc@V?↙
    ↘ $allocator@D@std@@@std@@QAE@XZ
    push esi
    call ??3@YAXPAX@Z ; operator delete
    add  esp, 4
    pop  esi
$LN23@dynamic:
    mov  DWORD PTR ?s@@3V?$basic_string@DU?↙
    ↘ $char_traits@D@std@@V?↙
    ↘ $allocator@D@2@@std@@A+20, 15
    mov  DWORD PTR ?s@@3V?$basic_string@DU?↙
    ↘ $char_traits@D@std@@V?↙
    ↘ $allocator@D@2@@std@@A+16, 0
    mov  BYTE PTR ?s@@3V?$basic_string@DU?↙
    ↘ $char_traits@D@std@@V?↙
    ↘ $allocator@D@2@@std@@A, 0
    pop  ecx
    ret  0
??__Fs@@YAXXZ ENDP
```

.

GCC :

Listing 51.28: GCC 4.8.1

```
main:
    push ebp
    mov  ebp, esp
```

```
    and  esp, -16
    sub  esp, 16
    mov  eax, DWORD PTR s
    mov  DWORD PTR [esp], eax
    call puts
    xor  eax, eax
    leave
    ret
.LC0:
    .string "a string"
_GLOBAL__sub_I_s:
    sub  esp, 44
    lea  eax, [esp+31]
    mov  DWORD PTR [esp+8], eax
    mov  DWORD PTR [esp+4], OFFSET FLAT:.LC0
    mov  DWORD PTR [esp], OFFSET FLAT:s
    call _ZNSsC1EPKcRKSaIcE
    mov  DWORD PTR [esp+8], OFFSET FLAT:↙
    ↘ __dso_handle
    mov  DWORD PTR [esp+4], OFFSET FLAT:s
    mov  DWORD PTR [esp], OFFSET FLAT:↙
    ↘ _ZNSsD1Ev
    call __cxa_atexit
    add  esp, 44
    ret
.LFE645:
    .size  _GLOBAL__sub_I_s, .-↙
    ↘ _GLOBAL__sub_I_s
    .section .init_array,"aw"
    .align 4
    .long  _GLOBAL__sub_I_s
    .globl s
    .bss
```

```
    .align 4
    .type  s, @object
    .size  s, 4
s:
    .zero  4
    .hidden __dso_handle
```

51.4.2 std::list

.

.

.

.

```

#include <stdio.h>
#include <list>
#include <iostream>

struct a
{
    int x;
    int y;
};

struct List_node
{
    struct List_node* _Next;
    struct List_node* _Prev;
    int x;
```

```
    int y;
};

void dump_List_node (struct List_node *n)
{
    printf ("ptr=0x%p _Next=0x%p _Prev=0x%p ↙
    ↪ x=%d y=%d\n",
        n, n->_Next, n->_Prev, n->x, n->y);
};

void dump_List_vals (struct List_node* n)
{
    struct List_node* current=n;

    for (;;)
    {
        dump_List_node (current);
        current=current->_Next;
        if (current==n) // end
            break;
    };
};

void dump_List_val (unsigned int *a)
{
#ifdef _MSC_VER
    //
    printf ("_Myhead=0x%p, _Mysize=%d\n", a↙
    ↪ [0], a[1]);
#endif
    dump_List_vals ((struct List_node*)a[0])↙
    ↪ ;
};
```

```
int main()
{
    std::list<struct a> l;

    printf ("* empty list:\n");
    dump_List_val((unsigned int*)(void*)&l);

    struct a t1;
    t1.x=1;
    t1.y=2;
    l.push_front (t1);
    t1.x=3;
    t1.y=4;
    l.push_front (t1);
    t1.x=5;
    t1.y=6;
    l.push_back (t1);

    printf ("* 3-elements list:\n");
    dump_List_val((unsigned int*)(void*)&l);

    std::list<struct a>::iterator tmp;
    printf ("node at .begin:\n");
    tmp=l.begin();
    dump_List_node ((struct List_node *)*(↙
    ↘ void**)&tmp);
    printf ("node at .end:\n");
    tmp=l.end();
    dump_List_node ((struct List_node *)*(↙
    ↘ void**)&tmp);

    printf ("* let's count from the begin:\n↙
```

```
↳ ");
    std::list<struct a>::iterator it=l.begin↙
    ↳ ();
    printf ("1st element: %d %d\n", (*it).x,↙
    ↳  (*it).y);
    it++;
    printf ("2nd element: %d %d\n", (*it).x,↙
    ↳  (*it).y);
    it++;
    printf ("3rd element: %d %d\n", (*it).x,↙
    ↳  (*it).y);
    it++;
    printf ("element at .end(): %d %d\n", (*↙
    ↳ it).x, (*it).y);

    printf ("* let's count from the end:\n")↙
    ↳ ;
    std::list<struct a>::iterator it2=l.end↙
    ↳ ();
    printf ("element at .end(): %d %d\n", (*↙
    ↳ it2).x, (*it2).y);
    it2--;
    printf ("3rd element: %d %d\n", (*it2).x↙
    ↳ , (*it2).y);
    it2--;
    printf ("2nd element: %d %d\n", (*it2).x↙
    ↳ , (*it2).y);
    it2--;
    printf ("1st element: %d %d\n", (*it2).x↙
    ↳ , (*it2).y);

    printf ("removing last element...\n");
    l.pop_back();
```

```
    dump_List_val((unsigned int*)(void*)&l);
};
```

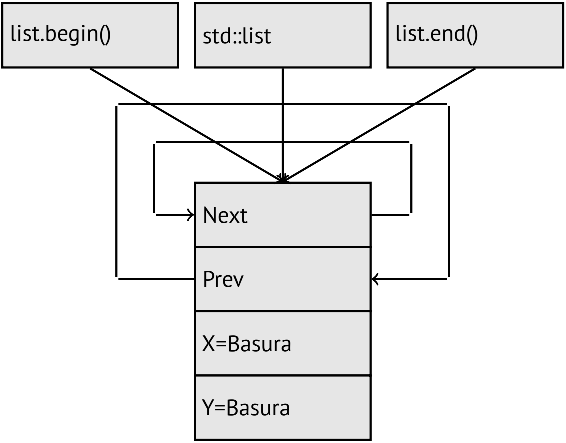
GCC

GCC.

.

```
* empty list:
ptr=0x0028fe90 _Next=0x0028fe90 _Prev=0
    x0028fe90 x=3 y=0
```

. «next» y «prev» :



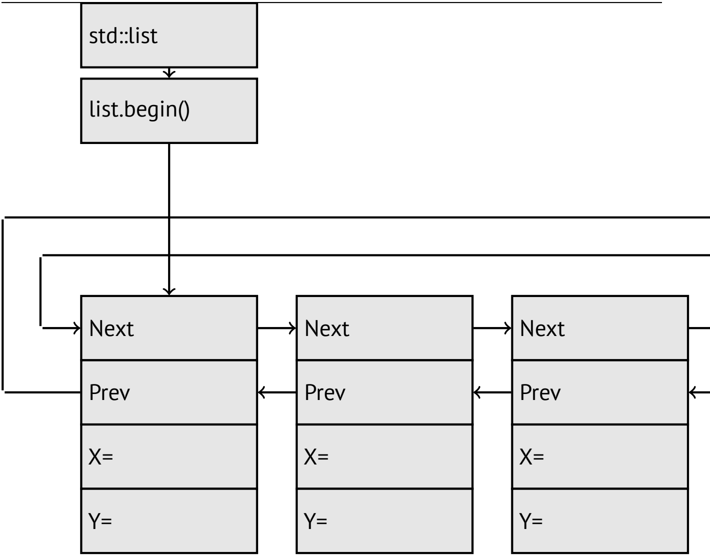
.

```
* 3-elements list:
ptr=0x000349a0 _Next=0x00034988 _Prev=0x0028fe90 x=3 y=4
ptr=0x00034988 _Next=0x00034b40 _Prev=0x000349a0 x=1 y=2
ptr=0x00034b40 _Next=0x0028fe90 _Prev=0x00034988 x=5 y=6
ptr=0x0028fe90 _Next=0x000349a0 _Prev=0x00034b40 x=5 y=6
```

..

:

 CAPÍTULO 51. C++



l .

.

. list.begin() y list.end() .

```
node at .begin:
ptr=0x000349a0 _Next=0x00034988 _Prev=0x0028fe90 x=3 y=4
node at .end:
ptr=0x0028fe90 _Next=0x000349a0 _Prev=0x00034b40 x=5 y=6
```

. .

operator-- y operator++ current_node->prev o
current_node->next. (.rbegin, .rend) .

operator* (x).

.

dangling pointer.

. .

Listing 51.29: Con optimización GCC 4.8.1 -fno-inline-small-functions

```
main proc near
    push ebp
    mov  ebp, esp
    push esi
    push ebx
    and  esp, 0FFFFFFF0h
    sub  esp, 20h
    lea  ebx, [esp+10h]
    mov  dword ptr [esp], offset s ; "* ⤦
    ↳ empty list:"
    mov  [esp+10h], ebx
    mov  [esp+14h], ebx
    call puts
    mov  [esp], ebx
    call _Z13dump_List_valPj ; dump_List_val⤦
    ↳ (uint *)
    lea  esi, [esp+18h]
    mov  [esp+4], esi
    mov  [esp], ebx
```

```
    mov  dword ptr [esp+18h], 1 ; X
    mov  dword ptr [esp+1Ch], 2 ; Y
    call ↙
    ↘ _ZNSt4listI1aSaIS0_EE10push_frontERKS0_↙
    ↘  ; std::list<a,std::allocator<a>>::↙
    ↘ push_front(a const&)
    mov  [esp+4], esi
    mov  [esp], ebx
    mov  dword ptr [esp+18h], 3 ; X
    mov  dword ptr [esp+1Ch], 4 ; Y
    call ↙
    ↘ _ZNSt4listI1aSaIS0_EE10push_frontERKS0_↙
    ↘  ; std::list<a,std::allocator<a>>::↙
    ↘ push_front(a const&)
    mov  dword ptr [esp], 10h
    mov  dword ptr [esp+18h], 5 ; X
    mov  dword ptr [esp+1Ch], 6 ; Y
    call _Znwj             ; operator new(uint↙
    ↘ )
    cmp  eax, 0FFFFFFF8h
    jz   short loc_80002A6
    mov  ecx, [esp+1Ch]
    mov  edx, [esp+18h]
    mov  [eax+0Ch], ecx
    mov  [eax+8], edx

loc_80002A6: ; CODE XREF: main+86
    mov  [esp+4], ebx
    mov  [esp], eax
    call ↙
    ↘ _ZNSt8__detail15_List_node_base7_M_hookEPS0_
    ↘  ; std::__detail::_List_node_base::↙
    ↘ _M_hook(std::__detail::_List_node_base↙
```

```
↳ *)
mov  dword ptr [esp], offset ↙
↳ a3ElementsList ; "* 3-elements list:"
call puts
mov  [esp], ebx
call _Z13dump_List_valPj ; dump_List_val↙
↳ (uint *)
mov  dword ptr [esp], offset ↙
↳ aNodeAt_begin ; "node at .begin:"
call puts
mov  eax, [esp+10h]
mov  [esp], eax
call _Z14dump_List_nodeP9List_node ; ↙
↳ dump_List_node(List_node *)
mov  dword ptr [esp], offset aNodeAt_end↙
↳  ; "node at .end:"
call puts
mov  [esp], ebx
call _Z14dump_List_nodeP9List_node ; ↙
↳ dump_List_node(List_node *)
mov  dword ptr [esp], offset ↙
↳ aLetSCountFromT ; "* let's count from ↙
↳ the begin:"
call puts
mov  esi, [esp+10h]
mov  eax, [esi+0Ch]
mov  [esp+0Ch], eax
mov  eax, [esi+8]
mov  dword ptr [esp+4], offset ↙
↳ a1stElementDD ; "1st element: %d %d\n"
mov  dword ptr [esp], 1
mov  [esp+8], eax
call __printf_chk
```

```
    mov  esi, [esi] ; operator++: get ->next
    ↳  pointer
    mov  eax, [esi+0Ch]
    mov  [esp+0Ch], eax
    mov  eax, [esi+8]
    mov  dword ptr [esp+4], offset 
    ↳ a2ndElementDD ; "2nd element: %d %d\n"
    mov  dword ptr [esp], 1
    mov  [esp+8], eax
    call __printf_chk
    mov  esi, [esi] ; operator++: get ->next
    ↳  pointer
    mov  eax, [esi+0Ch]
    mov  [esp+0Ch], eax
    mov  eax, [esi+8]
    mov  dword ptr [esp+4], offset 
    ↳ a3rdElementDD ; "3rd element: %d %d\n"
    mov  dword ptr [esp], 1
    mov  [esp+8], eax
    call __printf_chk
    mov  eax, [esi] ; operator++: get ->next
    ↳  pointer
    mov  edx, [eax+0Ch]
    mov  [esp+0Ch], edx
    mov  eax, [eax+8]
    mov  dword ptr [esp+4], offset 
    ↳ aElementAt_endD ; "element at .end(): 
    ↳ %d %d\n"
    mov  dword ptr [esp], 1
    mov  [esp+8], eax
    call __printf_chk
    mov  dword ptr [esp], offset 
    ↳ aLetSCountFro_0 ; "* let's count from 
```

```
    ↳ the end:"
    call puts
    mov  eax, [esp+1Ch]
    mov  dword ptr [esp+4], offset ↙
    ↳ aElementAt_endD ; "element at .end(): ↙
    ↳ %d %d\n"
    mov  dword ptr [esp], 1
    mov  [esp+0Ch], eax
    mov  eax, [esp+18h]
    mov  [esp+8], eax
    call __printf_chk
    mov  esi, [esp+14h]
    mov  eax, [esi+0Ch]
    mov  [esp+0Ch], eax
    mov  eax, [esi+8]
    mov  dword ptr [esp+4], offset ↙
    ↳ a3rdElementDD ; "3rd element: %d %d\n"
    mov  dword ptr [esp], 1
    mov  [esp+8], eax
    call __printf_chk
    mov  esi, [esi+4] ; operator--: get ->↙
    ↳ prev pointer
    mov  eax, [esi+0Ch]
    mov  [esp+0Ch], eax
    mov  eax, [esi+8]
    mov  dword ptr [esp+4], offset ↙
    ↳ a2ndElementDD ; "2nd element: %d %d\n"
    mov  dword ptr [esp], 1
    mov  [esp+8], eax
    call __printf_chk
    mov  eax, [esi+4] ; operator--: get ->↙
    ↳ prev pointer
    mov  edx, [eax+0Ch]
```

```
    mov  [esp+0Ch], edx
    mov  eax, [eax+8]
    mov  dword ptr [esp+4], offset ↙
    ↘ a1stElementDD ; "1st element: %d %d\n"
    mov  dword ptr [esp], 1
    mov  [esp+8], eax
    call __printf_chk
    mov  dword ptr [esp], offset ↙
    ↘ aRemovingLastEl ; "removing last ↙
    ↘ element..."
    call puts
    mov  esi, [esp+14h]
    mov  [esp], esi
    call ↙
    ↘ _ZNSt8__detail15_List_node_base9_M_unhookEv↙
    ↘  ; std::__detail::_List_node_base::↙
    ↘ _M_unhook(void)
    mov  [esp], esi ; void *
    call _ZdlPv ; operator delete(void *)
    mov  [esp], ebx
    call _Z13dump_List_valPj ; dump_List_val↙
    ↘ (uint *)
    mov  [esp], ebx
    call ↙
    ↘ _ZNSt10_List_baseI1aSaIS0_EE8_M_clearEv↙
    ↘  ; std::_List_base<a,std::allocator<a↙
    ↘ >>::_M_clear(void)
    lea  esp, [ebp-8]
    xor  eax, eax
    pop  ebx
    pop  esi
    pop  ebp
    retn
```

```
main endp
```

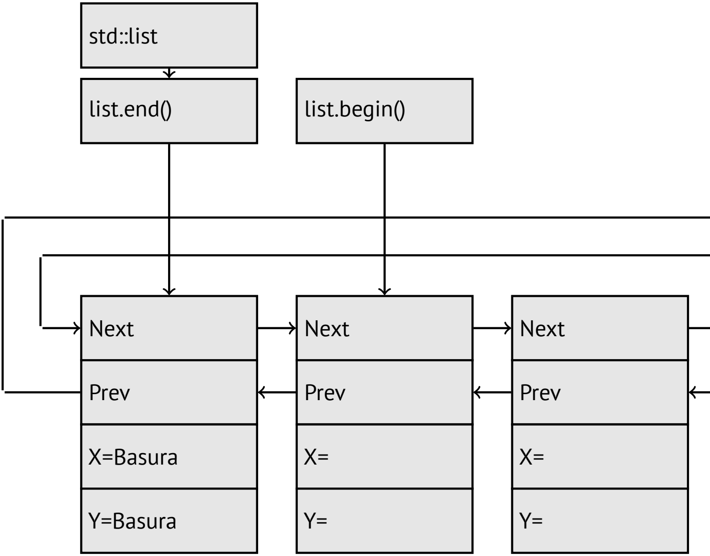
Listing 51.30:

```
* empty list:
ptr=0x0028fe90 _Next=0x0028fe90 _Prev=0
    ↳ x0028fe90 x=3 y=0
* 3-elements list:
ptr=0x000349a0 _Next=0x00034988 _Prev=0
    ↳ x0028fe90 x=3 y=4
ptr=0x00034988 _Next=0x00034b40 _Prev=0
    ↳ x000349a0 x=1 y=2
ptr=0x00034b40 _Next=0x0028fe90 _Prev=0
    ↳ x00034988 x=5 y=6
ptr=0x0028fe90 _Next=0x000349a0 _Prev=0
    ↳ x00034b40 x=5 y=6
node at .begin:
ptr=0x000349a0 _Next=0x00034988 _Prev=0
    ↳ x0028fe90 x=3 y=4
node at .end:
ptr=0x0028fe90 _Next=0x000349a0 _Prev=0
    ↳ x00034b40 x=5 y=6
* let's count from the begin:
1st element: 3 4
2nd element: 1 2
3rd element: 5 6
element at .end(): 5 6
* let's count from the end:
element at .end(): 5 6
3rd element: 5 6
2nd element: 1 2
1st element: 3 4
removing last element...
```

```
ptr=0x000349a0 _Next=0x00034988 _Prev=0x0028fe90 x=3 y=4
ptr=0x00034988 _Next=0x0028fe90 _Prev=0x000349a0 x=1 y=2
ptr=0x0028fe90 _Next=0x000349a0 _Prev=0x00034988 x=5 y=6
```

MSVC

. . .



Listing 51.31: Con optimización MSVC 2012 /Fa2.asm /GS- /Ob1

```
_l$ = -16 ; size = 8
_t1$ = -8 ; size = 8
_main    PROC
    sub  esp, 16
    push ebx
    push esi
    push edi
    push 0
    push 0
    lea  ecx, DWORD PTR _l$[esp+36]
    mov  DWORD PTR _l$[esp+40], 0
    ;
    call ?_Buynode0@?$_List_alloc@$0A@U?↙
    ↘ $_List_base_types@Ua@@V?↙
    ↘ $allocator@Ua@@@std@@@std@@@std@@QAEPAU↙
    ↘ ?$_List_node@Ua@@PAX@2@PAU32@0@Z ; std↙
    ↘ ::_List_alloc<0,std::_List_base_types<↙
    ↘ a,std::allocator<a> > >::_Buynode0
    mov  edi, DWORD PTR __imp__printf
    mov  ebx, eax
    push OFFSET $SG40685 ; '* empty list:'
    mov  DWORD PTR _l$[esp+32], ebx
    call edi  ; printf
    lea  eax, DWORD PTR _l$[esp+32]
    push eax
    call ?dump_List_val@@YAXPAI@Z ; ↙
    ↘ dump_List_val
    mov  esi, DWORD PTR [ebx]
    add  esp, 8
```

```
    lea  eax, DWORD PTR _t1$[esp+28]
    push eax
    push DWORD PTR [esi+4]
    lea  ecx, DWORD PTR _l$[esp+36]
    push esi
    mov  DWORD PTR _t1$[esp+40], 1 ;
    mov  DWORD PTR _t1$[esp+44], 2 ;
    ; allocate new node
    call ??$_Buynode@ABUa@@@?↙
    ↳ $_List_buy@Ua@@V?↙
    ↳ $allocator@Ua@@@std@@@std@@QAEPAU?↙
    ↳ $_List_node@Ua@@PAX@1@PAU21@0ABUa@@@Z ↙
    ↳ ; std::_List_buy<a,std::allocator<a> ↙
    ↳ >::_Buynode<a const &>
    mov  DWORD PTR [esi+4], eax
    mov  ecx, DWORD PTR [eax+4]
    mov  DWORD PTR _t1$[esp+28], 3 ;
    mov  DWORD PTR [ecx], eax
    mov  esi, DWORD PTR [ebx]
    lea  eax, DWORD PTR _t1$[esp+28]
    push eax
    push DWORD PTR [esi+4]
    lea  ecx, DWORD PTR _l$[esp+36]
    push esi
    mov  DWORD PTR _t1$[esp+44], 4 ;
    ; allocate new node
    call ??$_Buynode@ABUa@@@?↙
    ↳ $_List_buy@Ua@@V?↙
    ↳ $allocator@Ua@@@std@@@std@@QAEPAU?↙
    ↳ $_List_node@Ua@@PAX@1@PAU21@0ABUa@@@Z ↙
    ↳ ; std::_List_buy<a,std::allocator<a> ↙
    ↳ >::_Buynode<a const &>
    mov  DWORD PTR [esi+4], eax
```

```
    mov  ecx, DWORD PTR [eax+4]
    mov  DWORD PTR _t1$[esp+28], 5 ;
    mov  DWORD PTR [ecx], eax
    lea  eax, DWORD PTR _t1$[esp+28]
    push eax
    push DWORD PTR [ebx+4]
    lea  ecx, DWORD PTR _l$[esp+36]
    push ebx
    mov  DWORD PTR _t1$[esp+44], 6 ;
    ; allocate new node
    call ??$_Buynode@ABUa@@@?↙
    ↘ $_List_buy@Ua@@V?↙
    ↘ $allocator@Ua@@@std@@@std@@QAEPAU?↙
    ↘ $_List_node@Ua@@PAX@1@PAU21@0ABUa@@@Z ↙
    ↘ ; std::_List_buy<a,std::allocator<a> ↙
    ↘ >::_Buynode<a const &>
    mov  DWORD PTR [ebx+4], eax
    mov  ecx, DWORD PTR [eax+4]
    push OFFSET $SG40689 ; '* 3-elements ↙
    ↘ list:'
    mov  DWORD PTR _l$[esp+36], 3
    mov  DWORD PTR [ecx], eax
    call edi ; printf
    lea  eax, DWORD PTR _l$[esp+32]
    push eax
    call ?dump_List_val@@YAXPAI@Z ; ↙
    ↘ dump_List_val
    push OFFSET $SG40831 ; 'node at .begin:'
    call edi ; printf
    push DWORD PTR [ebx] ;
    call ?↙
    ↘ dump_List_node@@YAXPAUList_node@@@Z ; ↙
    ↘ dump_List_node
```

```
push OFFSET $SG40835 ; 'node at .end:'
call edi ; printf
push ebx ; pointer to the node $l$ ⤦
↳ variable points to!
call ?⤦
↳ dump_List_node@@YAXPAUList_node@@@Z ; ⤦
↳ dump_List_node
push OFFSET $SG40839 ; '* let''s count ⤦
↳ from the begin:'
call edi ; printf
mov  esi, DWORD PTR [ebx] ; operator++: ⤦
↳ get ->next pointer
push DWORD PTR [esi+12]
push DWORD PTR [esi+8]
push OFFSET $SG40846 ; '1st element: %d ⤦
↳ %d'
call edi ; printf
mov  esi, DWORD PTR [esi] ; operator++: ⤦
↳ get ->next pointer
push DWORD PTR [esi+12]
push DWORD PTR [esi+8]
push OFFSET $SG40848 ; '2nd element: %d ⤦
↳ %d'
call edi ; printf
mov  esi, DWORD PTR [esi] ; operator++: ⤦
↳ get ->next pointer
push DWORD PTR [esi+12]
push DWORD PTR [esi+8]
push OFFSET $SG40850 ; '3rd element: %d ⤦
↳ %d'
call edi ; printf
mov  eax, DWORD PTR [esi] ; operator++: ⤦
↳ get ->next pointer
```

```
    add  esp, 64
    push DWORD PTR [eax+12]
    push DWORD PTR [eax+8]
    push OFFSET $SG40852 ; 'element at .end⤦
    ↳ (): %d %d'
    call edi ; printf
    push OFFSET $SG40853 ; '* let''s count ⤦
    ↳ from the end:'
    call edi ; printf
    push DWORD PTR [ebx+12] ;
    push DWORD PTR [ebx+8]
    push OFFSET $SG40860 ; 'element at .end⤦
    ↳ (): %d %d'
    call edi ; printf
    mov  esi, DWORD PTR [ebx+4] ; operator⤦
    ↳ --: get ->prev pointer
    push DWORD PTR [esi+12]
    push DWORD PTR [esi+8]
    push OFFSET $SG40862 ; '3rd element: %d ⤦
    ↳ %d'
    call edi ; printf
    mov  esi, DWORD PTR [esi+4] ; operator⤦
    ↳ --: get ->prev pointer
    push DWORD PTR [esi+12]
    push DWORD PTR [esi+8]
    push OFFSET $SG40864 ; '2nd element: %d ⤦
    ↳ %d'
    call edi ; printf
    mov  eax, DWORD PTR [esi+4] ; operator⤦
    ↳ --: get ->prev pointer
    push DWORD PTR [eax+12]
    push DWORD PTR [eax+8]
    push OFFSET $SG40866 ; '1st element: %d ⤦
```

```
	↳ %d'
	call edi ; printf
	add  esp, 64
	push OFFSET $SG40867 ; 'removing last ↙
	↳ element...'
	call edi ; printf
	mov  edx, DWORD PTR [ebx+4]
	add  esp, 4

	; prev=next?
	; ?
	; !
	cmp  edx, ebx
	je   SHORT $LN349@main
	mov  ecx, DWORD PTR [edx+4]
	mov  eax, DWORD PTR [edx]
	mov  DWORD PTR [ecx], eax
	mov  ecx, DWORD PTR [edx]
	mov  eax, DWORD PTR [edx+4]
	push edx
	mov  DWORD PTR [ecx+4], eax
	call ??3@YAXPAX@Z ; operator delete
	add  esp, 4
	mov  DWORD PTR _l$[esp+32], 2
$LN349@main:
	lea  eax, DWORD PTR _l$[esp+28]
	push eax
	call ?dump_List_val@@YAXPAI@Z ; ↙
	↳ dump_List_val
	mov  eax, DWORD PTR [ebx]
	add  esp, 4
	mov  DWORD PTR [ebx], ebx
	mov  DWORD PTR [ebx+4], ebx
```

```
    cmp  eax, ebx
    je   SHORT $LN412@main
$LL414@main:
    mov  esi, DWORD PTR [eax]
    push eax
    call ??3@YAXPAX@Z ; operator delete
    add  esp, 4
    mov  eax, esi
    cmp  esi, ebx
    jne  SHORT $LL414@main
$LN412@main:
    push  ebx
    call  ??3@YAXPAX@Z ; operator delete
    add   esp, 4
    xor   eax, eax
    pop   edi
    pop   esi
    pop   ebx
    add   esp, 16
    ret   0
_main ENDP
```

( ).

Listing 51.32:

```
* empty list:
_Myhead=0x003CC258, _Mysize=0
ptr=0x003CC258 _Next=0x003CC258 _Prev=0↙
    ↘ x003CC258 x=6226002 y=4522072
* 3-elements list:
_Myhead=0x003CC258, _Mysize=3
ptr=0x003CC258 _Next=0x003CC288 _Prev=0↙
```

```
    ↳ x003CC2A0 x=6226002 y=4522072
ptr=0x003CC288 _Next=0x003CC270 _Prev=0↙
    ↳ x003CC258 x=3 y=4
ptr=0x003CC270 _Next=0x003CC2A0 _Prev=0↙
    ↳ x003CC288 x=1 y=2
ptr=0x003CC2A0 _Next=0x003CC258 _Prev=0↙
    ↳ x003CC270 x=5 y=6
node at .begin:
ptr=0x003CC288 _Next=0x003CC270 _Prev=0↙
    ↳ x003CC258 x=3 y=4
node at .end:
ptr=0x003CC258 _Next=0x003CC288 _Prev=0↙
    ↳ x003CC2A0 x=6226002 y=4522072
* let's count from the begin:
1st element: 3 4
2nd element: 1 2
3rd element: 5 6
element at .end(): 6226002 4522072
* let's count from the end:
element at .end(): 6226002 4522072
3rd element: 5 6
2nd element: 1 2
1st element: 3 4
removing last element...
_Myhead=0x003CC258, _Mysize=2
ptr=0x003CC258 _Next=0x003CC288 _Prev=0↙
    ↳ x003CC270 x=6226002 y=4522072
ptr=0x003CC288 _Next=0x003CC270 _Prev=0↙
    ↳ x003CC258 x=3 y=4
ptr=0x003CC270 _Next=0x003CC258 _Prev=0↙
    ↳ x003CC288 x=1 y=2
```

C++11 std::forward_list

. .

51.4.3 std::vector

std::string (51.4.1 on page 1073):

(18 on page 483). C++11 .data() .c_str() en std::stri

[3], . std::vector:

```
#include <stdio.h>
#include <vector>
#include <algorithm>
#include <functional>

struct vector_of_ints
{
    // MSVC names:
    int *Myfirst;
    int *Mylast;
    int *Myend;

    // : _M_start, _M_finish, ⤦
    ↳ _M_end_of_storage
};

void dump(struct vector_of_ints *in)
{
```

[3]: http://go.yurichev.com/17086

```
    printf ("_Myfirst=%p, _Mylast=%p, _Myend⤦
    ↪ =%p\n", in->Myfirst, in->Mylast, in->⤦
    ↪ Myend);
    size_t size=(in->Mylast-in->Myfirst);
    size_t capacity=(in->Myend-in->Myfirst);
    printf ("size=%d, capacity=%d\n", size, ⤦
    ↪ capacity);
    for (size_t i=0; i<size; i++)
        printf ("element %d: %d\n", i, in->⤦
    ↪ Myfirst[i]);
};

int main()
{
    std::vector<int> c;
    dump ((struct vector_of_ints*)(void*)&c)⤦
    ↪ ;
    c.push_back(1);
    dump ((struct vector_of_ints*)(void*)&c)⤦
    ↪ ;
    c.push_back(2);
    dump ((struct vector_of_ints*)(void*)&c)⤦
    ↪ ;
    c.push_back(3);
    dump ((struct vector_of_ints*)(void*)&c)⤦
    ↪ ;
    c.push_back(4);
    dump ((struct vector_of_ints*)(void*)&c)⤦
    ↪ ;
    c.reserve (6);
    dump ((struct vector_of_ints*)(void*)&c)⤦
    ↪ ;
    c.push_back(5);
```

```
    dump ((struct vector_of_ints*)(void*)&c)↙
    ↘ ;
    c.push_back(6);
    dump ((struct vector_of_ints*)(void*)&c)↙
    ↘ ;
    printf ("%d\n", c.at(5)); //
    printf ("%d\n", c[8]); // operator[],
};
```

MSVC:

```
_Myfirst=00000000, _Mylast=00000000, _Myend↙
    ↘ =00000000
size=0, capacity=0
_Myfirst=0051CF48, _Mylast=0051CF4C, _Myend↙
    ↘ =0051CF4C
size=1, capacity=1
element 0: 1
_Myfirst=0051CF58, _Mylast=0051CF60, _Myend↙
    ↘ =0051CF60
size=2, capacity=2
element 0: 1
element 1: 2
_Myfirst=0051C278, _Mylast=0051C284, _Myend↙
    ↘ =0051C284
size=3, capacity=3
element 0: 1
element 1: 2
element 2: 3
_Myfirst=0051C290, _Mylast=0051C2A0, _Myend↙
    ↘ =0051C2A0
size=4, capacity=4
element 0: 1
```

```
element 1: 2
element 2: 3
element 3: 4
_Myfirst=0051B180, _Mylast=0051B190, _Myend⤦
    ⤷ =0051B198
size=4, capacity=6
element 0: 1
element 1: 2
element 2: 3
element 3: 4
_Myfirst=0051B180, _Mylast=0051B194, _Myend⤦
    ⤷ =0051B198
size=5, capacity=6
element 0: 1
element 1: 2
element 2: 3
element 3: 4
element 4: 5
_Myfirst=0051B180, _Mylast=0051B198, _Myend⤦
    ⤷ =0051B198
size=6, capacity=6
element 0: 1
element 1: 2
element 2: 3
element 3: 4
element 4: 5
element 5: 6
6
6619158
```

. push_back() . push_back() . push_back() . .reserve(). . operator[] std::vector . .at()

std::out_of_range .

:

Listing 51.33: MSVC 2012 /GS- /Ob1

```
$SG52650 DB '%d', 0aH, 00H
$SG52651 DB '%d', 0aH, 00H

_this$ = -4 ; size = 4
__Pos$ = 8  ; size = 4
?at@?$vector@HV?⤦
    ↘ $allocator@H@std@@@std@@QAEAAHI@Z PROC⤦
    ↘  ; std::vector<int,std::allocator<int>⤦
    ↘  >::at, COMDAT
; _this$ = ecx
    push ebp
    mov  ebp, esp
    push ecx
    mov  DWORD PTR _this$[ebp], ecx
    mov  eax, DWORD PTR _this$[ebp]
    mov  ecx, DWORD PTR _this$[ebp]
    mov  edx, DWORD PTR [eax+4]
    sub  edx, DWORD PTR [ecx]
    sar  edx, 2
    cmp  edx, DWORD PTR __Pos$[ebp]
    ja   SHORT $LN1@at
    push OFFSET ??_C@_0BM@NMJKDPPO@invalid?5⤦
    ↘ vector?$DMT?$DO?5subscript?$AA@
    call DWORD PTR __imp_?⤦
    ↘ _Xout_of_range@std@@YAXPBD@Z
$LN1@at:
    mov  eax, DWORD PTR _this$[ebp]
    mov  ecx, DWORD PTR [eax]
```

```
    mov  edx, DWORD PTR __Pos$[ebp]
    lea  eax, DWORD PTR [ecx+edx*4]
$LN3@at:
    mov  esp, ebp
    pop  ebp
    ret  4
?at@?$vector@HV?⤦
    ↳ $allocator@H@std@@@std@@QAEAAHI@Z ENDP⤦
    ↳  ; std::vector<int,std::allocator<int>⤦
    ↳  >::at

_c$ = -36 ; size = 12
$T1 = -24 ; size = 4
$T2 = -20 ; size = 4
$T3 = -16 ; size = 4
$T4 = -12 ; size = 4
$T5 = -8  ; size = 4
$T6 = -4  ; size = 4
_main PROC
    push ebp
    mov  ebp, esp
    sub  esp, 36
    mov  DWORD PTR _c$[ebp], 0     ; Myfirst
    mov  DWORD PTR _c$[ebp+4], 0   ; Mylast
    mov  DWORD PTR _c$[ebp+8], 0   ; Myend
    lea  eax, DWORD PTR _c$[ebp]
    push eax
    call ?dump@@YAXPAUvector_of_ints@@@Z ; ⤦
    ↳ dump
    add  esp, 4
    mov  DWORD PTR $T6[ebp], 1
    lea  ecx, DWORD PTR $T6[ebp]
    push ecx
```

```
    lea  ecx, DWORD PTR _c$[ebp]
    call ?push_back@?$vector@HV?⤦
    ↳ $allocator@H@std@@@std@@QAEX$$QAH@Z ; ⤦
    ↳ std::vector<int,std::allocator<int> ⤦
    ↳ >::push_back
    lea  edx, DWORD PTR _c$[ebp]
    push edx
    call ?dump@@YAXPAUvector_of_ints@@@Z ; ⤦
    ↳ dump
    add  esp, 4
    mov  DWORD PTR $T5[ebp], 2
    lea  eax, DWORD PTR $T5[ebp]
    push eax
    lea  ecx, DWORD PTR _c$[ebp]
    call ?push_back@?$vector@HV?⤦
    ↳ $allocator@H@std@@@std@@QAEX$$QAH@Z ; ⤦
    ↳ std::vector<int,std::allocator<int> ⤦
    ↳ >::push_back
    lea  ecx, DWORD PTR _c$[ebp]
    push ecx
    call ?dump@@YAXPAUvector_of_ints@@@Z ; ⤦
    ↳ dump
    add  esp, 4
    mov  DWORD PTR $T4[ebp], 3
    lea  edx, DWORD PTR $T4[ebp]
    push edx
    lea  ecx, DWORD PTR _c$[ebp]
    call ?push_back@?$vector@HV?⤦
    ↳ $allocator@H@std@@@std@@QAEX$$QAH@Z ; ⤦
    ↳ std::vector<int,std::allocator<int> ⤦
    ↳ >::push_back
    lea  eax, DWORD PTR _c$[ebp]
    push eax
```

```
    call ?dump@@YAXPAUvector_of_ints@@@Z ; ↙
    ↘ dump
    add  esp, 4
    mov  DWORD PTR $T3[ebp], 4
    lea  ecx, DWORD PTR $T3[ebp]
    push ecx
    lea  ecx, DWORD PTR _c$[ebp]
    call ?push_back@?$vector@HV?↙
    ↘ $allocator@H@std@@@std@@QAEX$$QAH@Z ; ↙
    ↘ std::vector<int,std::allocator<int> ↙
    ↘ >::push_back
    lea  edx, DWORD PTR _c$[ebp]
    push edx
    call ?dump@@YAXPAUvector_of_ints@@@Z ; ↙
    ↘ dump
    add  esp, 4
    push 6
    lea  ecx, DWORD PTR _c$[ebp]
    call ?reserve@?$vector@HV?↙
    ↘ $allocator@H@std@@@std@@QAEXI@Z ; std↙
    ↘ ::vector<int,std::allocator<int> >::↙
    ↘ reserve
    lea  eax, DWORD PTR _c$[ebp]
    push eax
    call ?dump@@YAXPAUvector_of_ints@@@Z ; ↙
    ↘ dump
    add  esp, 4
    mov  DWORD PTR $T2[ebp], 5
    lea  ecx, DWORD PTR $T2[ebp]
    push ecx
    lea  ecx, DWORD PTR _c$[ebp]
    call ?push_back@?$vector@HV?↙
    ↘ $allocator@H@std@@@std@@QAEX$$QAH@Z ; ↙
```

```
    ↳ std::vector<int,std::allocator<int> ↙
    ↳ >::push_back
    lea  edx, DWORD PTR _c$[ebp]
    push edx
    call ?dump@@YAXPAUvector_of_ints@@@Z ; ↙
    ↳ dump
    add  esp, 4
    mov  DWORD PTR $T1[ebp], 6
    lea  eax, DWORD PTR $T1[ebp]
    push eax
    lea  ecx, DWORD PTR _c$[ebp]
    call ?push_back@?$vector@HV?↙
    ↳ $allocator@H@std@@@std@@QAEX$$QAH@Z ; ↙
    ↳ std::vector<int,std::allocator<int> ↙
    ↳ >::push_back
    lea  ecx, DWORD PTR _c$[ebp]
    push ecx
    call ?dump@@YAXPAUvector_of_ints@@@Z ; ↙
    ↳ dump
    add  esp, 4
    push 5
    lea  ecx, DWORD PTR _c$[ebp]
    call ?at@?$vector@HV?↙
    ↳ $allocator@H@std@@@std@@QAEAAHI@Z ; ↙
    ↳ std::vector<int,std::allocator<int> ↙
    ↳ >::at
    mov  edx, DWORD PTR [eax]
    push edx
    push OFFSET $SG52650 ; '%d'
    call DWORD PTR __imp__printf
    add  esp, 8
    mov  eax, 8
    shl  eax, 2
```

```
    mov  ecx, DWORD PTR _c$[ebp]
    mov  edx, DWORD PTR [ecx+eax]
    push edx
    push OFFSET $SG52651 ; '%d'
    call DWORD PTR __imp__printf
    add  esp, 8
    lea  ecx, DWORD PTR _c$[ebp]
    call ?_Tidy@?$vector@HV?↙
    ↳ $allocator@H@std@@@std@@IAEXXZ ; std::↙
    ↳ vector<int,std::allocator<int> >::↙
    ↳ _Tidy
    xor  eax, eax
    mov  esp, ebp
    pop  ebp
    ret  0
_main ENDP
```

`.at()` . `printf()`.

«size» y «capacity», `std::string`. .

`.at()` :

Listing 51.34: GCC 4.8.1 -fno-inline-small-functions -O1

```
main proc near
     push ebp
     mov  ebp, esp
     push edi
     push esi
     push ebx
     and  esp, 0FFFFFFF0h
     sub  esp, 20h
     mov  dword ptr [esp+14h], 0
```

```
 mov  dword ptr [esp+18h], 0
 mov  dword ptr [esp+1Ch], 0
 lea  eax, [esp+14h]
 mov  [esp], eax
 call _Z4dumpP14vector_of_ints ; dump(↙
↘ vector_of_ints *)
 mov  dword ptr [esp+10h], 1
 lea  eax, [esp+10h]
 mov  [esp+4], eax
 lea  eax, [esp+14h]
 mov  [esp], eax
 call _ZNSt6vectorIiSaIiEE9push_backERKi↙
↘  ; std::vector<int,std::allocator<int↙
↘ >>::push_back(int  const&)
 lea  eax, [esp+14h]
 mov  [esp], eax
 call _Z4dumpP14vector_of_ints ; dump(↙
↘ vector_of_ints *)
 mov  dword ptr [esp+10h], 2
 lea  eax, [esp+10h]
 mov  [esp+4], eax
 lea  eax, [esp+14h]
 mov  [esp], eax
 call _ZNSt6vectorIiSaIiEE9push_backERKi↙
↘  ; std::vector<int,std::allocator<int↙
↘ >>::push_back(int  const&)
 lea  eax, [esp+14h]
 mov  [esp], eax
 call _Z4dumpP14vector_of_ints ; dump(↙
↘ vector_of_ints *)
 mov  dword ptr [esp+10h], 3
 lea  eax, [esp+10h]
 mov  [esp+4], eax
```

```
    lea  eax, [esp+14h]
    mov  [esp], eax
    call _ZNSt6vectorIiSaIiEE9push_backERKi ↙
↳  ; std::vector<int,std::allocator<int↙
↳ >>::push_back(int  const&)
    lea  eax, [esp+14h]
    mov  [esp], eax
    call _Z4dumpP14vector_of_ints ; dump(↙
↳ vector_of_ints *)
    mov  dword ptr [esp+10h], 4
    lea  eax, [esp+10h]
    mov  [esp+4], eax
    lea  eax, [esp+14h]
    mov  [esp], eax
    call _ZNSt6vectorIiSaIiEE9push_backERKi ↙
↳  ; std::vector<int,std::allocator<int↙
↳ >>::push_back(int  const&)
    lea  eax, [esp+14h]
    mov  [esp], eax
    call _Z4dumpP14vector_of_ints ; dump(↙
↳ vector_of_ints *)
    mov  ebx, [esp+14h]
    mov  eax, [esp+1Ch]
    sub  eax, ebx
    cmp  eax, 17h
    ja   short loc_80001CF
    mov  edi, [esp+18h]
    sub  edi, ebx
    sar  edi, 2
    mov  dword ptr [esp], 18h
    call _Znwj                ; operator new(↙
↳ uint)
    mov  esi, eax
```

```
     test edi, edi
     jz   short loc_80001AD
     lea  eax, ds:0[edi*4]
     mov  [esp+8], eax    ; n
     mov  [esp+4], ebx    ; src
     mov  [esp], esi      ; dest
     call memmove

loc_80001AD: ; CODE XREF: main+F8
     mov  eax, [esp+14h]
     test eax, eax
     jz   short loc_80001BD
     mov  [esp], eax      ; void *
     call _ZdlPv          ; operator delete(↙
    ↘ void *)

loc_80001BD: ; CODE XREF: main+117
     mov  [esp+14h], esi
     lea  eax, [esi+edi*4]
     mov  [esp+18h], eax
     add  esi, 18h
     mov  [esp+1Ch], esi

loc_80001CF: ; CODE XREF: main+DD
     lea  eax, [esp+14h]
     mov  [esp], eax
     call _Z4dumpP14vector_of_ints ; dump(↙
    ↘ vector_of_ints *)
     mov  dword ptr [esp+10h], 5
     lea  eax, [esp+10h]
     mov  [esp+4], eax
     lea  eax, [esp+14h]
     mov  [esp], eax
```

```
     call _ZNSt6vectorIiSaIiEE9push_backERKi↙
    ↘ ; std::vector<int,std::allocator<int↙
    ↘ >>::push_back(int  const&)
     lea  eax, [esp+14h]
     mov  [esp], eax
     call _Z4dumpP14vector_of_ints ; dump(↙
    ↘ vector_of_ints *)
     mov  dword ptr [esp+10h], 6
     lea  eax, [esp+10h]
     mov  [esp+4], eax
     lea  eax, [esp+14h]
     mov  [esp], eax
     call _ZNSt6vectorIiSaIiEE9push_backERKi↙
    ↘ ; std::vector<int,std::allocator<int↙
    ↘ >>::push_back(int  const&)
     lea  eax, [esp+14h]
     mov  [esp], eax
     call _Z4dumpP14vector_of_ints ; dump(↙
    ↘ vector_of_ints *)
     mov  eax, [esp+14h]
     mov  edx, [esp+18h]
     sub  edx, eax
     cmp  edx, 17h
     ja   short loc_8000246
     mov  dword ptr [esp], offset ↙
    ↘ aVector_m_range ; "vector::↙
    ↘ _M_range_check"
     call _ZSt20__throw_out_of_rangePKc ; ↙
    ↘ std::__throw_out_of_range(char  const↙
    ↘ *)

loc_8000246:                               ; ↙
    ↘ CODE XREF: main+19C
```

```
     mov  eax, [eax+14h]
     mov  [esp+8], eax
     mov  dword ptr [esp+4], offset aD ; "%d
    ↳ \n"
     mov  dword ptr [esp], 1
     call __printf_chk
     mov  eax, [esp+14h]
     mov  eax, [eax+20h]
     mov  [esp+8], eax
     mov  dword ptr [esp+4], offset aD ; "%d
    ↳ \n"
     mov  dword ptr [esp], 1
     call __printf_chk
     mov  eax, [esp+14h]
     test eax, eax
     jz   short loc_80002AC
     mov  [esp], eax       ; void *
     call _ZdlPv           ; operator delete(
    ↳ void *)
     jmp  short loc_80002AC

     mov  ebx, eax
     mov  edx, [esp+14h]
     test edx, edx
     jz   short loc_80002A4
     mov  [esp], edx       ; void *
     call _ZdlPv           ; operator delete(
    ↳ void *)

loc_80002A4: ; CODE XREF: main+1FE
     mov  [esp], ebx
     call _Unwind_Resume
```

```
loc_80002AC: ; CODE XREF: main+1EA
             ; main+1F4
     mov  eax, 0
     lea  esp, [ebp-0Ch]
     pop  ebx
     pop  esi
     pop  edi
     pop  ebp

locret_80002B8: ; DATA XREF: .eh_frame↙
    ↘ :08000510
                ; .eh_frame:080005BC
     retn
main endp
```

.reserve() . new() , memmove() , delete() .

:

```
_Myfirst=0x(nil), _Mylast=0x(nil), _Myend=0x↙
    ↘ (nil)
size=0, capacity=0
_Myfirst=0x8257008, _Mylast=0x825700c, ↙
    ↘ _Myend=0x825700c
size=1, capacity=1
element 0: 1
_Myfirst=0x8257018, _Mylast=0x8257020, ↙
    ↘ _Myend=0x8257020
size=2, capacity=2
element 0: 1
element 1: 2
_Myfirst=0x8257028, _Mylast=0x8257034, ↙
```

```
    ↳ _Myend=0x8257038
size=3, capacity=4
element 0: 1
element 1: 2
element 2: 3
_Myfirst=0x8257028, _Mylast=0x8257038, ↙
    ↳ _Myend=0x8257038
size=4, capacity=4
element 0: 1
element 1: 2
element 2: 3
element 3: 4
_Myfirst=0x8257040, _Mylast=0x8257050, ↙
    ↳ _Myend=0x8257058
size=4, capacity=6
element 0: 1
element 1: 2
element 2: 3
element 3: 4
_Myfirst=0x8257040, _Mylast=0x8257054, ↙
    ↳ _Myend=0x8257058
size=5, capacity=6
element 0: 1
element 1: 2
element 2: 3
element 3: 4
element 4: 5
_Myfirst=0x8257040, _Mylast=0x8257058, ↙
    ↳ _Myend=0x8257058
size=6, capacity=6
element 0: 1
element 1: 2
element 2: 3
```

```
element 3: 4
element 4: 5
element 5: 6
6
0
```

MSVC.

~50% , 100% .

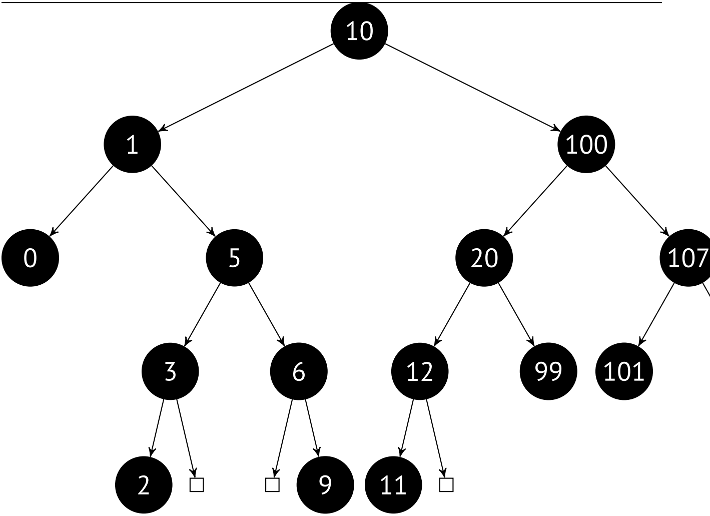
51.4.4 std::map y std::set

. . .

.

- .
- . .
- .

: 0, 1, 2, 3, 5, 6, 9, 10, 11, 12, 20, 99, 100, 101, 107, 1001, 1010.



. .

.

.

$\approx \log_2 n$. ≈ 10 ≈ 1000 ≈ 13 ≈ 10000 . . .

. .

.

`std::set` . `std::map` .

MSVC

```
#include <map>
#include <set>
#include <string>
#include <iostream>

// !
struct tree_node
{
    struct tree_node *Left;
    struct tree_node *Parent;
    struct tree_node *Right;
    char Color; // 0 - Red, 1 - Black
    char Isnil;
    //std::pair Myval;
    unsigned int first; //  Myval  std::set
    const char *second; //  std::set
};

struct tree_struct
{
    struct tree_node *Myhead;
    size_t Mysize;
};

void dump_tree_node (struct tree_node *n, ↙
    ↘ bool is_set, bool traverse)
{
    printf ("ptr=0x%p Left=0x%p Parent=0x%p ↙
    ↘ Right=0x%p Color=%d Isnil=%d\n",
            n, n->Left, n->Parent, n->Right,↙
    ↘  n->Color, n->Isnil);
```

```
    if (n->Isnil==0)
    {
        if (is_set)
            printf ("first=%d\n", n->first);
        else
            printf ("first=%d second=[%s]\n↙
    ↘ ", n->first, n->second);
    }

    if (traverse)
    {
        if (n->Isnil==1)
            dump_tree_node (n->Parent, ↙
    ↘ is_set, true);
        else
        {
            if (n->Left->Isnil==0)
                dump_tree_node (n->Left, ↙
    ↘ is_set, true);
            if (n->Right->Isnil==0)
                dump_tree_node (n->Right, ↙
    ↘ is_set, true);
        };
    };
};

const char* ALOT_OF_TABS="\t\t\t\t\t\t\t\t\t↙
    ↘ \t\t";

void dump_as_tree (int tabs, struct ↙
    ↘ tree_node *n, bool is_set)
{
    if (is_set)
```

```
        printf ("%d\n", n->first);
    else
        printf ("%d [%s]\n", n->first, n->⤦
    ↳ second);
    if (n->Left->Isnil==0)
    {
        printf ("%.*sL-------", tabs, ⤦
    ↳ ALOT_OF_TABS);
        dump_as_tree (tabs+1, n->Left, ⤦
    ↳ is_set);
    };
    if (n->Right->Isnil==0)
    {
        printf ("%.*sR-------", tabs, ⤦
    ↳ ALOT_OF_TABS);
        dump_as_tree (tabs+1, n->Right, ⤦
    ↳ is_set);
    };
};

void dump_map_and_set(struct tree_struct *m,⤦
    ↳  bool is_set)
{
    printf ("ptr=0x%p, Myhead=0x%p, Mysize=%⤦
    ↳ d\n", m, m->Myhead, m->Mysize);
    dump_tree_node (m->Myhead, is_set, true)⤦
    ↳ ;
    printf ("As a tree:\n");
    printf ("root----");
    dump_as_tree (1, m->Myhead->Parent, ⤦
    ↳ is_set);
};
```

```
int main()
{
    // map

    std::map<int, const char*> m;

    m[10]="ten";
    m[20]="twenty";
    m[3]="three";
    m[101]="one hundred one";
    m[100]="one hundred";
    m[12]="twelve";
    m[107]="one hundred seven";
    m[0]="zero";
    m[1]="one";
    m[6]="six";
    m[99]="ninety-nine";
    m[5]="five";
    m[11]="eleven";
    m[1001]="one thousand one";
    m[1010]="one thousand ten";
    m[2]="two";
    m[9]="nine";
    printf ("dumping m as map:\n");
    dump_map_and_set ((struct tree_struct *)↙
    ↘ (void*)&m, false);

    std::map<int, const char*>::iterator it1↙
    ↘ =m.begin();
    printf ("m.begin():\n");
    dump_tree_node ((struct tree_node *)*(↙
    ↘ void**)&it1, false, false);
    it1=m.end();
```

```
    printf ("m.end():\n");
    dump_tree_node ((struct tree_node *)*(↙
    ↳ void**)&it1, false, false);

    // set

    std::set<int> s;
    s.insert(123);
    s.insert(456);
    s.insert(11);
    s.insert(12);
    s.insert(100);
    s.insert(1001);
    printf ("dumping s as set:\n");
    dump_map_and_set ((struct tree_struct *)↙
    ↳ (void*)&s, true);
    std::set<int>::iterator it2=s.begin();
    printf ("s.begin():\n");
    dump_tree_node ((struct tree_node *)*(↙
    ↳ void**)&it2, true, false);
    it2=s.end();
    printf ("s.end():\n");
    dump_tree_node ((struct tree_node *)*(↙
    ↳ void**)&it2, true, false);
};
```

Listing 51.35: MSVC 2012

```
dumping m as map:
ptr=0x0020FE04, Myhead=0x005BB3A0, Mysize=17
ptr=0x005BB3A0 Left=0x005BB4A0 Parent=0↙
    ↳ x005BB3C0 Right=0x005BB580 Color=1 ↙
    ↳ Isnil=1
```

```
ptr=0x005BB3C0 Left=0x005BB4C0 Parent=0↙
    ↳ x005BB3A0 Right=0x005BB440 Color=1 ↙
    ↳ Isnil=0
first=10 second=[ten]
ptr=0x005BB4C0 Left=0x005BB4A0 Parent=0↙
    ↳ x005BB3C0 Right=0x005BB520 Color=1 ↙
    ↳ Isnil=0
first=1 second=[one]
ptr=0x005BB4A0 Left=0x005BB3A0 Parent=0↙
    ↳ x005BB4C0 Right=0x005BB3A0 Color=1 ↙
    ↳ Isnil=0
first=0 second=[zero]
ptr=0x005BB520 Left=0x005BB400 Parent=0↙
    ↳ x005BB4C0 Right=0x005BB4E0 Color=0 ↙
    ↳ Isnil=0
first=5 second=[five]
ptr=0x005BB400 Left=0x005BB5A0 Parent=0↙
    ↳ x005BB520 Right=0x005BB3A0 Color=1 ↙
    ↳ Isnil=0
first=3 second=[three]
ptr=0x005BB5A0 Left=0x005BB3A0 Parent=0↙
    ↳ x005BB400 Right=0x005BB3A0 Color=0 ↙
    ↳ Isnil=0
first=2 second=[two]
ptr=0x005BB4E0 Left=0x005BB3A0 Parent=0↙
    ↳ x005BB520 Right=0x005BB5C0 Color=1 ↙
    ↳ Isnil=0
first=6 second=[six]
ptr=0x005BB5C0 Left=0x005BB3A0 Parent=0↙
    ↳ x005BB4E0 Right=0x005BB3A0 Color=0 ↙
    ↳ Isnil=0
first=9 second=[nine]
ptr=0x005BB440 Left=0x005BB3E0 Parent=0↙
```

```
    ↳ x005BB3C0 Right=0x005BB480 Color=1 ↙
    ↳ Isnil=0
first=100 second=[one hundred]
ptr=0x005BB3E0 Left=0x005BB460 Parent=0↙
    ↳ x005BB440 Right=0x005BB500 Color=0 ↙
    ↳ Isnil=0
first=20 second=[twenty]
ptr=0x005BB460 Left=0x005BB540 Parent=0↙
    ↳ x005BB3E0 Right=0x005BB3A0 Color=1 ↙
    ↳ Isnil=0
first=12 second=[twelve]
ptr=0x005BB540 Left=0x005BB3A0 Parent=0↙
    ↳ x005BB460 Right=0x005BB3A0 Color=0 ↙
    ↳ Isnil=0
first=11 second=[eleven]
ptr=0x005BB500 Left=0x005BB3A0 Parent=0↙
    ↳ x005BB3E0 Right=0x005BB3A0 Color=1 ↙
    ↳ Isnil=0
first=99 second=[ninety-nine]
ptr=0x005BB480 Left=0x005BB420 Parent=0↙
    ↳ x005BB440 Right=0x005BB560 Color=0 ↙
    ↳ Isnil=0
first=107 second=[one hundred seven]
ptr=0x005BB420 Left=0x005BB3A0 Parent=0↙
    ↳ x005BB480 Right=0x005BB3A0 Color=1 ↙
    ↳ Isnil=0
first=101 second=[one hundred one]
ptr=0x005BB560 Left=0x005BB3A0 Parent=0↙
    ↳ x005BB480 Right=0x005BB580 Color=1 ↙
    ↳ Isnil=0
first=1001 second=[one thousand one]
ptr=0x005BB580 Left=0x005BB3A0 Parent=0↙
    ↳ x005BB560 Right=0x005BB3A0 Color=0 ↙
```

```
    ↳ Isnil=0
first=1010 second=[one thousand ten]
As a tree:
root----10 [ten]
       L-------1 [one]
               L-------0 [zero]
               R-------5 [five]
                       L-------3 [three]
                               L-------2 [↙
    ↳ two]
                       R-------6 [six]
                               R-------9 [↙
    ↳ nine]
       R-------100 [one hundred]
               L-------20 [twenty]
                       L-------12 [twelve]
                               L-------11 [↙
    ↳ eleven]
                       R-------99 [ninety-↙
    ↳ nine]
               R-------107 [one hundred ↙
    ↳ seven]
                       L-------101 [one ↙
    ↳ hundred one]
                       R-------1001 [one ↙
    ↳ thousand one]
                               R-------1010↙
    ↳  [one thousand ten]
m.begin():
ptr=0x005BB4A0 Left=0x005BB3A0 Parent=0↙
    ↳ x005BB4C0 Right=0x005BB3A0 Color=1 ↙
    ↳ Isnil=0
first=0 second=[zero]
```

```
m.end():
ptr=0x005BB3A0 Left=0x005BB4A0 Parent=0↙
    ↳ x005BB3C0 Right=0x005BB580 Color=1 ↙
    ↳ Isnil=1

dumping s as set:
ptr=0x0020FDFC, Myhead=0x005BB5E0, Mysize=6
ptr=0x005BB5E0 Left=0x005BB640 Parent=0↙
    ↳ x005BB600 Right=0x005BB6A0 Color=1 ↙
    ↳ Isnil=1
ptr=0x005BB600 Left=0x005BB660 Parent=0↙
    ↳ x005BB5E0 Right=0x005BB620 Color=1 ↙
    ↳ Isnil=0
first=123
ptr=0x005BB660 Left=0x005BB640 Parent=0↙
    ↳ x005BB600 Right=0x005BB680 Color=1 ↙
    ↳ Isnil=0
first=12
ptr=0x005BB640 Left=0x005BB5E0 Parent=0↙
    ↳ x005BB660 Right=0x005BB5E0 Color=0 ↙
    ↳ Isnil=0
first=11
ptr=0x005BB680 Left=0x005BB5E0 Parent=0↙
    ↳ x005BB660 Right=0x005BB5E0 Color=0 ↙
    ↳ Isnil=0
first=100
ptr=0x005BB620 Left=0x005BB5E0 Parent=0↙
    ↳ x005BB600 Right=0x005BB6A0 Color=1 ↙
    ↳ Isnil=0
first=456
ptr=0x005BB6A0 Left=0x005BB5E0 Parent=0↙
    ↳ x005BB620 Right=0x005BB5E0 Color=0 ↙
    ↳ Isnil=0
```

```
first=1001
As a tree:
root----123
        L-------12
                L-------11
                R-------100
        R-------456
                R-------1001
s.begin():
ptr=0x005BB640 Left=0x005BB5E0 Parent=0⤦
    ⤥ x005BB660 Right=0x005BB5E0 Color=0 ⤦
    ⤥ Isnil=0
first=11
s.end():
ptr=0x005BB5E0 Left=0x005BB640 Parent=0⤦
    ⤥ x005BB600 Right=0x005BB6A0 Color=1 ⤦
    ⤥ Isnil=1
```

.

`std::map`, `first` y `second` `std::pair`. `std::set` .

(51.4.2 on page 1101).

`std::list`, . `.begin()`. . `operator--` y `operator++` . [Cor+09].

`.end()` «landing zone» en HDD[4].

GCC

[4]Hard disk drive

```
#include <stdio.h>
#include <map>
#include <set>
#include <string>
#include <iostream>

struct map_pair
{
    int key;
    const char *value;
};

struct tree_node
{
    int M_color; // 0 - Red, 1 - Black
    struct tree_node *M_parent;
    struct tree_node *M_left;
    struct tree_node *M_right;
};

struct tree_struct
{
    int M_key_compare;
    struct tree_node M_header;
    size_t M_node_count;
};

void dump_tree_node (struct tree_node *n, ⤦
    ↳ bool is_set, bool traverse, bool ⤦
    ↳ dump_keys_and_values)
{
```

```
    printf ("ptr=0x%p M_left=0x%p M_parent=0↙
    ↳ x%p M_right=0x%p M_color=%d\n",
            n, n->M_left, n->M_parent, n->↙
    ↳ M_right, n->M_color);

    void *point_after_struct=((char*)n)+↙
    ↳ sizeof(struct tree_node);

    if (dump_keys_and_values)
    {
        if (is_set)
            printf ("key=%d\n", *(int*)↙
    ↳ point_after_struct);
        else
        {
            struct map_pair *p=(struct ↙
    ↳ map_pair *)point_after_struct;
            printf ("key=%d value=[%s]\n", p↙
    ↳ ->key, p->value);
        };
    };

    if (traverse==false)
        return;

    if (n->M_left)
        dump_tree_node (n->M_left, is_set, ↙
    ↳ traverse, dump_keys_and_values);
    if (n->M_right)
        dump_tree_node (n->M_right, is_set, ↙
    ↳ traverse, dump_keys_and_values);
};
```

```
const char* ALOT_OF_TABS="\t\t\t\t\t\t\t\t\t↙
    ↘ \t\t";

void dump_as_tree (int tabs, struct ↙
    ↘ tree_node *n, bool is_set)
{
    void *point_after_struct=((char*)n)+↙
    ↘ sizeof(struct tree_node);

    if (is_set)
        printf ("%d\n", *(int*)↙
    ↘ point_after_struct);
    else
    {
        struct map_pair *p=(struct map_pair ↙
    ↘ *)point_after_struct;
        printf ("%d [%s]\n", p->key, p->↙
    ↘ value);
    }

    if (n->M_left)
    {
        printf ("%.*sL-------", tabs, ↙
    ↘ ALOT_OF_TABS);
        dump_as_tree (tabs+1, n->M_left, ↙
    ↘ is_set);
    };
    if (n->M_right)
    {
        printf ("%.*sR-------", tabs, ↙
    ↘ ALOT_OF_TABS);
        dump_as_tree (tabs+1, n->M_right, ↙
    ↘ is_set);
```

```
    };
};

void dump_map_and_set(struct tree_struct *m, ↙
    ↘ bool is_set)
{
    printf ("ptr=0x%p, M_key_compare=0x%x, ↙
    ↘ M_header=0x%p, M_node_count=%d\n",
        m, m->M_key_compare, &m->M_header, m↙
    ↘ ->M_node_count);
    dump_tree_node (m->M_header.M_parent, ↙
    ↘ is_set, true, true);
    printf ("As a tree:\n");
    printf ("root----");
    dump_as_tree (1, m->M_header.M_parent, ↙
    ↘ is_set);
};

int main()
{
    // map

    std::map<int, const char*> m;

    m[10]="ten";
    m[20]="twenty";
    m[3]="three";
    m[101]="one hundred one";
    m[100]="one hundred";
    m[12]="twelve";
    m[107]="one hundred seven";
    m[0]="zero";
    m[1]="one";
```

```
    m[6]="six";
    m[99]="ninety-nine";
    m[5]="five";
    m[11]="eleven";
    m[1001]="one thousand one";
    m[1010]="one thousand ten";
    m[2]="two";
    m[9]="nine";

    printf ("dumping m as map:\n");
    dump_map_and_set ((struct tree_struct *)↙
    ↘ (void*)&m, false);

    std::map<int, const char*>::iterator it1↙
    ↘ =m.begin();
    printf ("m.begin():\n");
    dump_tree_node ((struct tree_node *)*(↙
    ↘ void**)&it1, false, false, true);
    it1=m.end();
    printf ("m.end():\n");
    dump_tree_node ((struct tree_node *)*(↙
    ↘ void**)&it1, false, false, false);

    // set

    std::set<int> s;
    s.insert(123);
    s.insert(456);
    s.insert(11);
    s.insert(12);
    s.insert(100);
    s.insert(1001);
    printf ("dumping s as set:\n");
```

```
    dump_map_and_set ((struct tree_struct *)↙
    ↳ (void*)&s, true);
    std::set<int>::iterator it2=s.begin();
    printf ("s.begin():\n");
    dump_tree_node ((struct tree_node *)*(↙
    ↳ void**)&it2, true, false, true);
    it2=s.end();
    printf ("s.end():\n");
    dump_tree_node ((struct tree_node *)*(↙
    ↳ void**)&it2, true, false, false);
};
```

Listing 51.36: GCC 4.8.1

```
dumping m as map:
ptr=0x0028FE3C, M_key_compare=0x402b70, ↙
    ↳ M_header=0x0028FE40, M_node_count=17
ptr=0x007A4988 M_left=0x007A4C00 M_parent=0↙
    ↳ x0028FE40 M_right=0x007A4B80 M_color=1
key=10 value=[ten]
ptr=0x007A4C00 M_left=0x007A4BE0 M_parent=0↙
    ↳ x007A4988 M_right=0x007A4C60 M_color=1
key=1 value=[one]
ptr=0x007A4BE0 M_left=0x00000000 M_parent=0↙
    ↳ x007A4C00 M_right=0x00000000 M_color=1
key=0 value=[zero]
ptr=0x007A4C60 M_left=0x007A4B40 M_parent=0↙
    ↳ x007A4C00 M_right=0x007A4C20 M_color=0
key=5 value=[five]
ptr=0x007A4B40 M_left=0x007A4CE0 M_parent=0↙
    ↳ x007A4C60 M_right=0x00000000 M_color=1
key=3 value=[three]
ptr=0x007A4CE0 M_left=0x00000000 M_parent=0↙
```

```
    ↳ x007A4B40 M_right=0x00000000 M_color=0
key=2 value=[two]
ptr=0x007A4C20 M_left=0x00000000 M_parent=0↙
    ↳ x007A4C60 M_right=0x007A4D00 M_color=1
key=6 value=[six]
ptr=0x007A4D00 M_left=0x00000000 M_parent=0↙
    ↳ x007A4C20 M_right=0x00000000 M_color=0
key=9 value=[nine]
ptr=0x007A4B80 M_left=0x007A49A8 M_parent=0↙
    ↳ x007A4988 M_right=0x007A4BC0 M_color=1
key=100 value=[one hundred]
ptr=0x007A49A8 M_left=0x007A4BA0 M_parent=0↙
    ↳ x007A4B80 M_right=0x007A4C40 M_color=0
key=20 value=[twenty]
ptr=0x007A4BA0 M_left=0x007A4C80 M_parent=0↙
    ↳ x007A49A8 M_right=0x00000000 M_color=1
key=12 value=[twelve]
ptr=0x007A4C80 M_left=0x00000000 M_parent=0↙
    ↳ x007A4BA0 M_right=0x00000000 M_color=0
key=11 value=[eleven]
ptr=0x007A4C40 M_left=0x00000000 M_parent=0↙
    ↳ x007A49A8 M_right=0x00000000 M_color=1
key=99 value=[ninety-nine]
ptr=0x007A4BC0 M_left=0x007A4B60 M_parent=0↙
    ↳ x007A4B80 M_right=0x007A4CA0 M_color=0
key=107 value=[one hundred seven]
ptr=0x007A4B60 M_left=0x00000000 M_parent=0↙
    ↳ x007A4BC0 M_right=0x00000000 M_color=1
key=101 value=[one hundred one]
ptr=0x007A4CA0 M_left=0x00000000 M_parent=0↙
    ↳ x007A4BC0 M_right=0x007A4CC0 M_color=1
key=1001 value=[one thousand one]
ptr=0x007A4CC0 M_left=0x00000000 M_parent=0↙
```

```
    ↳ x007A4CA0 M_right=0x00000000 M_color=0
key=1010 value=[one thousand ten]
As a tree:
root----10 [ten]
        L-------1 [one]
                L-------0 [zero]
                R-------5 [five]
                        L-------3 [three]
                                L-------2 [↙
    ↳ two]
                        R-------6 [six]
                                R-------9 [↙
    ↳ nine]
        R-------100 [one hundred]
                L-------20 [twenty]
                        L-------12 [twelve]
                                L-------11 [↙
    ↳ eleven]
                        R-------99 [ninety-↙
    ↳ nine]
                R-------107 [one hundred ↙
    ↳ seven]
                        L-------101 [one ↙
    ↳ hundred one]
                        R-------1001 [one ↙
    ↳ thousand one]
                                R-------1010↙
    ↳  [one thousand ten]
m.begin():
ptr=0x007A4BE0 M_left=0x00000000 M_parent=0↙
    ↳ x007A4C00 M_right=0x00000000 M_color=1
key=0 value=[zero]
m.end():
```

```
ptr=0x0028FE40 M_left=0x007A4BE0 M_parent=0
    ↳ x007A4988 M_right=0x007A4CC0 M_color=0

dumping s as set:
ptr=0x0028FE20, M_key_compare=0x8, M_header
    ↳ =0x0028FE24, M_node_count=6
ptr=0x007A1E80 M_left=0x01D5D890 M_parent=0
    ↳ x0028FE24 M_right=0x01D5D850 M_color=1
key=123
ptr=0x01D5D890 M_left=0x01D5D870 M_parent=0
    ↳ x007A1E80 M_right=0x01D5D8B0 M_color=1
key=12
ptr=0x01D5D870 M_left=0x00000000 M_parent=0
    ↳ x01D5D890 M_right=0x00000000 M_color=0
key=11
ptr=0x01D5D8B0 M_left=0x00000000 M_parent=0
    ↳ x01D5D890 M_right=0x00000000 M_color=0
key=100
ptr=0x01D5D850 M_left=0x00000000 M_parent=0
    ↳ x007A1E80 M_right=0x01D5D8D0 M_color=1
key=456
ptr=0x01D5D8D0 M_left=0x00000000 M_parent=0
    ↳ x01D5D850 M_right=0x00000000 M_color=0
key=1001
As a tree:
root----123
        L-------12
                L-------11
                R-------100
        R-------456
                R-------1001
s.begin():
ptr=0x01D5D870 M_left=0x00000000 M_parent=0
```

```
    ↳ x01D5D890 M_right=0x00000000 M_color=0
key=11
s.end():
ptr=0x0028FE24 M_left=0x01D5D870 M_parent=0↙
    ↳ x007A1E80 M_right=0x01D5D8D0 M_color=0
```

[5].,

(GCC)

Listing 51.37: GCC

```
#include <stdio.h>
#include <map>
#include <set>
#include <string>
#include <iostream>

struct map_pair
{
    int key;
    const char *value;
};

struct tree_node
{
    int M_color; // 0 - Red, 1 - Black
    struct tree_node *M_parent;
    struct tree_node *M_left;
    struct tree_node *M_right;
};
```

[5]http://go.yurichev.com/17084

```
struct tree_struct
{
    int M_key_compare;
    struct tree_node M_header;
    size_t M_node_count;
};

const char* ALOT_OF_TABS="\t\t\t\t\t\t\t\t\t↙
    ↘ \t\t";

void dump_as_tree (int tabs, struct ↙
    ↘ tree_node *n)
{
    void *point_after_struct=((char*)n)+↙
    ↘ sizeof(struct tree_node);

    printf ("%d\n", *(int*)↙
    ↘ point_after_struct);

    if (n->M_left)
    {
        printf ("%.*sL-------", tabs, ↙
    ↘ ALOT_OF_TABS);
        dump_as_tree (tabs+1, n->M_left);
    };
    if (n->M_right)
    {
        printf ("%.*sR-------", tabs, ↙
    ↘ ALOT_OF_TABS);
        dump_as_tree (tabs+1, n->M_right);
    };
};
```

```
void dump_map_and_set(struct tree_struct *m)
{
    printf ("root----");
    dump_as_tree (1, m->M_header.M_parent);
};

int main()
{
    std::set<int> s;
    s.insert(123);
    s.insert(456);
    printf ("123, 456 are inserted\n");
    dump_map_and_set ((struct tree_struct *)
    ↳ (void*)&s);
    s.insert(11);
    s.insert(12);
    printf ("\n");
    printf ("11, 12 are inserted\n");
    dump_map_and_set ((struct tree_struct *)
    ↳ (void*)&s);
    s.insert(100);
    s.insert(1001);
    printf ("\n");
    printf ("100, 1001 are inserted\n");
    dump_map_and_set ((struct tree_struct *)
    ↳ (void*)&s);
    s.insert(667);
    s.insert(1);
    s.insert(4);
    s.insert(7);
    printf ("\n");
    printf ("667, 1, 4, 7 are inserted\n");
```

```
    dump_map_and_set ((struct tree_struct *)
    ↳ (void*)&s);
    printf ("\n");
};
```

Listing 51.38: GCC 4.8.1

```
123, 456 are inserted
root----123
        R-------456

11, 12 are inserted
root----123
        L-------11
                R-------12
        R-------456

100, 1001 are inserted
root----123
        L-------12
                L-------11
                R-------100
        R-------456
                R-------1001

667, 1, 4, 7 are inserted
root----12
        L-------4
                L-------1
                R-------11
                        L-------7
        R-------123
                L-------100
```

CAPÍTULO 51. C++

```
                    R-------667
                              L-------456
                              R-------1001
```

Capítulo 52

, $array[-1]$.

. :

```
#include <stdio.h>

int main()
{
        int random_value=0x11223344;
        unsigned char array[10];
        int i;
        unsigned char *fakearray=&array[-1];

        for (i=0; i<10; i++)
                array[i]=i;

        printf ("first element %d\n", ↙
    ↘ fakearray[1]);
        printf ("second element %d\n", ↙
```

```
    fakearray[2]);
        printf ("last element %d\n", fakearray[10]);

        printf ("array[-1]=%02X, array[-2]=%02X, array[-3]=%02X, array[-4]=%02X\n",
                array[-1],
                array[-2],
                array[-3],
                array[-4]);
};
```

Listing 52.1: Sin optimización MSVC 2010

```

$SG2751 DB      'first element %d', 0aH, 00H
$SG2752 DB      'second element %d', 0aH, 00H
$SG2753 DB      'last element %d', 0aH, 00H
$SG2754 DB      'array[-1]=%02X, array[-2]=%02X, array[-3]=%02X, array[-4'
        DB      ']=%02X', 0aH, 00H

_fakearray$ = -24               ; size = 4
_random_value$ = -20    ; size = 4
_array$ = -16           ; size = 10
_i$ = -4                ; size = 4
_main   PROC
        push    ebp
        mov     ebp, esp
        sub     esp, 24
        mov     DWORD PTR _random_value$[ebp
```

```
    ↳ ], 287454020 ; 11223344H
        ;
        lea     eax, DWORD PTR _array$[ebp]
        add     eax, -1 ; eax=eax-1
        mov     DWORD PTR _fakearray$[ebp], ↙
    ↳ eax
        mov     DWORD PTR _i$[ebp], 0
        jmp     SHORT $LN3@main
        ;
$LN2@main:
        mov     ecx, DWORD PTR _i$[ebp]
        add     ecx, 1
        mov     DWORD PTR _i$[ebp], ecx
$LN3@main:
        cmp     DWORD PTR _i$[ebp], 10
        jge     SHORT $LN1@main
        mov     edx, DWORD PTR _i$[ebp]
        mov     al, BYTE PTR _i$[ebp]
        mov     BYTE PTR _array$[ebp+edx], ↙
    ↳ al
        jmp     SHORT $LN2@main
$LN1@main:
        mov     ecx, DWORD PTR _fakearray$[↙
    ↳ ebp]
        ; ecx= fakearray[0], ecx+1 ↙
    ↳ fakearray[1]  array[0]
        movzx   edx, BYTE PTR [ecx+1]
        push    edx
        push    OFFSET $SG2751 ; 'first ↙
    ↳ element %d'
        call    _printf
        add     esp, 8
        mov     eax, DWORD PTR _fakearray$[↙
```

```
ebp]
    ; eax= fakearray[0], eax+2 fakearray[2]  array[1]
    movzx   ecx, BYTE PTR [eax+2]
    push    ecx
    push    OFFSET $SG2752 ; 'second element %d'
    call    _printf
    add     esp, 8
    mov     edx, DWORD PTR _fakearray$[ebp]
    ; edx= fakearray[0], edx+10 fakearray[10]  array[9]
    movzx   eax, BYTE PTR [edx+10]
    push    eax
    push    OFFSET $SG2753 ; 'last element %d'
    call    _printf
    add     esp, 8
    ;  4, 3, 2  1  array[0]  array[]
    lea     ecx, DWORD PTR _array$[ebp]
    movzx   edx, BYTE PTR [ecx-4]
    push    edx
    lea     eax, DWORD PTR _array$[ebp]
    movzx   ecx, BYTE PTR [eax-3]
    push    ecx
    lea     edx, DWORD PTR _array$[ebp]
    movzx   eax, BYTE PTR [edx-2]
    push    eax
    lea     ecx, DWORD PTR _array$[ebp]
    movzx   edx, BYTE PTR [ecx-1]
    push    edx
    push    OFFSET $SG2754 ; 'array
```

```
    ↳ [-1]=%02X, array[-2]=%02X, array↙
    ↳ [-3]=%02X, array[-4]=%02X'
        call    _printf
        add     esp, 20
        xor     eax, eax
        mov     esp, ebp
        pop     ebp
        ret     0
_main   ENDP
```

fakearray[1] array[0]. array[]? random_value array[] 0x11223344. .

:

```
first element 0
second element 1
last element 9
array[-1]=11, array[-2]=22, array[-3]=33, ↙
    ↳ array[-4]=44
```

Listing 52.2: Sin optimización MSVC 2010

```


CPU Stack
Address   Value
001DFBCC  /001DFBD3 ;
001DFBD0  |11223344 ; random_value
001DFBD4  |03020100 ; 4  array[]
001DFBD8  |07060504 ; 4  array[]
001DFBDC  |00CB0908 ;  array[]
001DFBE0  |0000000A ;
001DFBE4  |001DFC2C ;
```

```
001DFBE8   \00CB129D ;
```

`fakearray[]` (0x001DFBD3) `array[]` (0x001DFBD4), .

Capítulo 53

Windows 16-bit

3.11. 96/98/ME Windows NT. Windows NT .

.

OpenWatcom 1.9 :
`wcl.exe -i=C:/WATCOM/h/win/ -s -os -bt=window`
`-bcl=windows example.c`

53.1 Ejemplo#1

```
#include <windows.h>

int PASCAL WinMain( HINSTANCE hInstance,
                    HINSTANCE hPrevInstance,
```

```
                    LPSTR lpCmdLine,
                    int nCmdShow )
{
        MessageBeep(MB_ICONEXCLAMATION);
        return 0;
};
```

```
WinMain         proc near
                push    bp
                mov     bp, sp
                mov     ax, 30h ; '0'    ; ↙
    ↘ MB_ICONEXCLAMATION constant
                push    ax
                call    MESSAGEBEEP
                xor     ax, ax           ; ↙
    ↘ return 0
                pop     bp
                retn    0Ah
WinMain         endp
```

.

53.2 Ejemplo #2

```
#include <windows.h>

int PASCAL WinMain( HINSTANCE hInstance,
                    HINSTANCE hPrevInstance,
                    LPSTR lpCmdLine,
                    int nCmdShow )
{
```

```
        MessageBox (NULL, "hello, world", "
    ↳ caption", MB_YESNOCANCEL);
        return 0;
};
```

```
WinMain         proc near
                push    bp
                mov     bp, sp
                xor     ax, ax          ; 
    ↳ NULL
                push    ax
                push    ds
                mov     ax, offset 
    ↳ aHelloWorld ; 0x18. "hello, world"
                push    ax
                push    ds
                mov     ax, offset aCaption 
    ↳ ; 0x10. "caption"
                push    ax
                mov     ax, 3           ; 
    ↳ MB_YESNOCANCEL
                push    ax
                call    MESSAGEBOX
                xor     ax, ax          ; 
    ↳ return 0
                pop     bp
                retn    0Ah
WinMain         endp

dseg02:0010 aCaption        db 'caption',0
dseg02:0018 aHelloWorld     db 'hello, world
    ↳ ',0
```

(MB_YESNOCANCEL), (NULL). stack pointer: RETN 0Ah . stdcall (64.2 on page 1299), .

. .

53.3 Ejemplo #3

```
#include <windows.h>

int PASCAL WinMain( HINSTANCE hInstance,
                    HINSTANCE hPrevInstance,
                    LPSTR lpCmdLine,
                    int nCmdShow )
{
    int result=MessageBox (NULL, "hello, ↙
    ↘ world", "caption", MB_YESNOCANCEL);

    if (result==IDCANCEL)
        MessageBox (NULL, "you pressed ↙
    ↘ cancel", "caption", MB_OK);
    else if (result==IDYES)
        MessageBox (NULL, "you pressed yes",↙
    ↘  "caption", MB_OK);
    else if (result==IDNO)
        MessageBox (NULL, "you pressed no", ↙
    ↘ "caption", MB_OK);

    return 0;
};
```

```
WinMain               proc near
```

```
                push    bp
                mov     bp, sp
                xor     ax, ax          ; NULL
                push    ax
                push    ds
                mov     ax, offset ↙
    ↘ aHelloWorld ; "hello, world"
                push    ax
                push    ds
                mov     ax, offset aCaption ↙
    ↘ ; "caption"
                push    ax
                mov     ax, 3              ; ↙
    ↘ MB_YESNOCANCEL
                push    ax
                call    MESSAGEBOX
                cmp     ax, 2              ; ↙
    ↘ IDCANCEL
                jnz     short loc_2F
                xor     ax, ax
                push    ax
                push    ds
                mov     ax, offset ↙
    ↘ aYouPressedCanc ; "you pressed cancel"
                jmp     short loc_49
loc_2F:
                cmp     ax, 6              ; ↙
    ↘ IDYES
                jnz     short loc_3D
                xor     ax, ax
                push    ax
                push    ds
                mov     ax, offset ↙
```

```
    ↳ aYouPressedYes ; "you pressed yes"
                jmp     short loc_49
loc_3D:
                cmp     ax, 7           ; ↙
    ↳ IDNO
                jnz     short loc_57
                xor     ax, ax
                push    ax
                push    ds
                mov     ax, offset ↙
    ↳ aYouPressedNo ; "you pressed no"
loc_49:
                push    ax
                push    ds
                mov     ax, offset aCaption ↙
    ↳ ; "caption"
                push    ax
                xor     ax, ax
                push    ax
                call    MESSAGEBOX
loc_57:
                xor     ax, ax
                pop     bp
                retn    0Ah
WinMain         endp
```

.

53.4 Ejemplo #4

```
#include <windows.h>
```

```
int PASCAL func1 (int a, int b, int c)
{
        return a*b+c;
};

long PASCAL func2 (long a, long b, long c)
{
        return a*b+c;
};

long PASCAL func3 (long a, long b, long c, ↙
    ↘ int d)
{
        return a*b+c-d;
};

int PASCAL WinMain( HINSTANCE hInstance,
                    HINSTANCE hPrevInstance,
                    LPSTR lpCmdLine,
                    int nCmdShow )
{
        func1 (123, 456, 789);
        func2 (600000, 700000, 800000);
        func3 (600000, 700000, 800000, 123);
        return 0;
};
```

```
func1           proc near

c               = word ptr  4
b               = word ptr  6
```

```
a               = word ptr  8

                push    bp
                mov     bp, sp
                mov     ax, [bp+a]
                imul    [bp+b]
                add     ax, [bp+c]
                pop     bp
                retn    6
func1           endp

func2           proc near

arg_0           = word ptr  4
arg_2           = word ptr  6
arg_4           = word ptr  8
arg_6           = word ptr  0Ah
arg_8           = word ptr  0Ch
arg_A           = word ptr  0Eh

                push    bp
                mov     bp, sp
                mov     ax, [bp+arg_8]
                mov     dx, [bp+arg_A]
                mov     bx, [bp+arg_4]
                mov     cx, [bp+arg_6]
                call    sub_B2 ; long 32-bit multiplication
                add     ax, [bp+arg_0]
                adc     dx, [bp+arg_2]
                pop     bp
                retn    12
func2           endp
```

```
func3           proc near

arg_0           = word ptr  4
arg_2           = word ptr  6
arg_4           = word ptr  8
arg_6           = word ptr  0Ah
arg_8           = word ptr  0Ch
arg_A           = word ptr  0Eh
arg_C           = word ptr  10h

                push    bp
                mov     bp, sp
                mov     ax, [bp+arg_A]
                mov     dx, [bp+arg_C]
                mov     bx, [bp+arg_6]
                mov     cx, [bp+arg_8]
                call    sub_B2 ; long 32-bit↙
    ↳  multiplication
                mov     cx, [bp+arg_2]
                add     cx, ax
                mov     bx, [bp+arg_4]
                adc     bx, dx              ; BX↙
    ↳ =high part, CX=low part
                mov     ax, [bp+arg_0]
                cwd                         ; AX↙
    ↳ =low part d, DX=high part d
                sub     cx, ax
                mov     ax, cx
                sbb     bx, dx
                mov     dx, bx
                pop     bp
                retn    14
```

```
func3           endp

WinMain         proc near
                push    bp
                mov     bp, sp
                mov     ax, 123
                push    ax
                mov     ax, 456
                push    ax
                mov     ax, 789
                push    ax
                call    func1
                mov     ax, 9       ; high ↙
    ↘ part of 600000
                push    ax
                mov     ax, 27C0h   ; low ↙
    ↘ part of 600000
                push    ax
                mov     ax, 0Ah     ; high ↙
    ↘ part of 700000
                push    ax
                mov     ax, 0AE60h ; low ↙
    ↘ part of 700000
                push    ax
                mov     ax, 0Ch     ; high ↙
    ↘ part of 800000
                push    ax
                mov     ax, 3500h   ; low ↙
    ↘ part of 800000
                push    ax
                call    func2
                mov     ax, 9       ; high ↙
    ↘ part of 600000
```

```
                push    ax
                mov     ax, 27C0h   ; low ↙
    ↳ part of 600000
                push    ax
                mov     ax, 0Ah     ; high ↙
    ↳ part of 700000
                push    ax
                mov     ax, 0AE60h ; low ↙
    ↳ part of 700000
                push    ax
                mov     ax, 0Ch     ; high ↙
    ↳ part of 800000
                push    ax
                mov     ax, 3500h   ; low ↙
    ↳ part of 800000
                push    ax
                mov     ax, 7Bh     ; 123
                push    ax
                call    func3
                xor     ax, ax      ; return ↙
    ↳ 0
                pop     bp
                retn    0Ah
WinMain         endp
```

. (24 on page 769).

sub_B2 here «long multiplication», . : E on page 1911, D on page 1910.

ADD/ADC :

SUB/SBB :

.

WinMain().

.

53.5 Ejemplo #5

```
#include <windows.h>

int PASCAL string_compare (char *s1, char *
    s2)
{
        while (1)
        {
                if (*s1!=*s2)
                        return 0;
                if (*s1==0 || *s2==0)
                        return 1; // end of 
    string
                s1++;
                s2++;
        };

};

int PASCAL string_compare_far (char far *s1,
     char far *s2)
{
        while (1)
        {
                if (*s1!=*s2)
```

```
                                return 0;
                if (*s1==0 || *s2==0)
                                return 1; // end of ↙
    ↘ string
                s1++;
                s2++;
        };

};

void PASCAL remove_digits (char *s)
{
        while (*s)
        {
                if (*s>='0' && *s<='9')
                        *s='-';
                s++;
        };
};

char str[]="hello 1234 world";

int PASCAL WinMain( HINSTANCE hInstance,
                    HINSTANCE hPrevInstance,
                    LPSTR lpCmdLine,
                    int nCmdShow )
{
        string_compare ("asd", "def");
        string_compare_far ("asd", "def");
        remove_digits (str);
        MessageBox (NULL, str, "caption", ↙
    ↘ MB_YESNOCANCEL);
        return 0;
```

```
};
```

```
string_compare  proc near

arg_0 = word ptr  4
arg_2 = word ptr  6

      push    bp
      mov     bp, sp
      push    si
      mov     si, [bp+arg_0]
      mov     bx, [bp+arg_2]

loc_12: ; CODE XREF: string_compare+21j
      mov     al, [bx]
      cmp     al, [si]
      jz      short loc_1C
      xor     ax, ax
      jmp     short loc_2B


loc_1C: ; CODE XREF: string_compare+Ej
      test    al, al
      jz      short loc_22
      jnz     short loc_27

loc_22: ; CODE XREF: string_compare+16j
      mov     ax, 1
      jmp     short loc_2B


loc_27: ; CODE XREF: string_compare+18j
```

```
      inc     bx
      inc     si
      jmp     short loc_12


loc_2B: ; CODE XREF: string_compare+12j
        ; string_compare+1Dj
      pop     si
      pop     bp
      retn    4
string_compare  endp

string_compare_far proc near ; CODE XREF: ↙
    ↘ WinMain+18p

arg_0 = word ptr  4
arg_2 = word ptr  6
arg_4 = word ptr  8
arg_6 = word ptr  0Ah

      push    bp
      mov     bp, sp
      push    si
      mov     si, [bp+arg_0]
      mov     bx, [bp+arg_4]

loc_3A: ; CODE XREF: string_compare_far+35j
      mov     es, [bp+arg_6]
      mov     al, es:[bx]
      mov     es, [bp+arg_2]
      cmp     al, es:[si]
      jz      short loc_4C
      xor     ax, ax
```

```
      jmp     short loc_67


loc_4C: ; CODE XREF: string_compare_far+16j
      mov     es, [bp+arg_6]
      cmp     byte ptr es:[bx], 0
      jz      short loc_5E
      mov     es, [bp+arg_2]
      cmp     byte ptr es:[si], 0
      jnz     short loc_63

loc_5E: ; CODE XREF: string_compare_far+23j
      mov     ax, 1
      jmp     short loc_67


loc_63: ; CODE XREF: string_compare_far+2Cj
      inc     bx
      inc     si
      jmp     short loc_3A


loc_67: ; CODE XREF: string_compare_far+1Aj
        ; string_compare_far+31j
      pop     si
      pop     bp
      retn    8
string_compare_far endp

remove_digits   proc near ; CODE XREF: ↙
    ↘ WinMain+1Fp

arg_0 = word ptr  4
```

```
      push    bp
      mov     bp, sp
      mov     bx, [bp+arg_0]

loc_72: ; CODE XREF: remove_digits+18j
      mov     al, [bx]
      test    al, al
      jz      short loc_86
      cmp     al, 30h ; '0'
      jb      short loc_83
      cmp     al, 39h ; '9'
      ja      short loc_83
      mov     byte ptr [bx], 2Dh ; '-'

loc_83: ; CODE XREF: remove_digits+Ej
        ; remove_digits+12j
      inc     bx
      jmp     short loc_72


loc_86: ; CODE XREF: remove_digits+Aj
      pop     bp
      retn    2
remove_digits   endp

WinMain proc near ; CODE XREF: start+EDp
      push    bp
      mov     bp, sp
      mov     ax, offset aAsd ; "asd"
      push    ax
      mov     ax, offset aDef ; "def"
      push    ax
```

```
        call    string_compare
        push    ds
        mov     ax, offset aAsd ; "asd"
        push    ax
        push    ds
        mov     ax, offset aDef ; "def"
        push    ax
        call    string_compare_far
        mov     ax, offset aHello1234World ; "↙
    ↘ hello 1234 world"
        push    ax
        call    remove_digits
        xor     ax, ax
        push    ax
        push    ds
        mov     ax, offset aHello1234World ; "↙
    ↘ hello 1234 world"
        push    ax
        push    ds
        mov     ax, offset aCaption ; "caption↙
    ↘ "
        push    ax
        mov     ax, 3 ; MB_YESNOCANCEL
        push    ax
        call    MESSAGEBOX
        xor     ax, ax
        pop     bp
        retn    0Ah
WinMain endp
```

«near» «far» .

: 94 on page 1776.

«near» . `string_compare()` (`mov al, [bx]` `mov al, ds:[bx]DS`).

«far» . `string_compare_far()` (`mov al, es:[bx]`).
«far» `MessageBox()`: 53.2 on page 1160. .

. .

.

«memory model».

.

53.6 Ejemplo #6

```
#include <windows.h>
#include <time.h>
#include <stdio.h>

char strbuf[256];

int PASCAL WinMain( HINSTANCE hInstance,
                    HINSTANCE hPrevInstance,
                    LPSTR lpCmdLine,
                    int nCmdShow )
{

        struct tm *t;
        time_t unix_time;

        unix_time=time(NULL);

```

```
        t=localtime (&unix_time);

        sprintf (strbuf, "%04d-%02d-%02d %02↙
    ↳ d:%02d:%02d", t->tm_year+1900, t->↙
    ↳ tm_mon, t->tm_mday,
                t->tm_hour, t->tm_min, t->↙
    ↳ tm_sec);

        MessageBox (NULL, strbuf, "caption",↙
    ↳  MB_OK);
        return 0;
};
```

```
WinMain         proc near

var_4           = word ptr -4
var_2           = word ptr -2

                push    bp
                mov     bp, sp
                push    ax
                push    ax
                xor     ax, ax
                call    time_
                mov     [bp+var_4], ax   ; ↙
    ↳ low part of UNIX time
                mov     [bp+var_2], dx   ; ↙
    ↳ high part of UNIX time
                lea     ax, [bp+var_4]   ; ↙
    ↳ take a pointer of high part
                call    localtime_
                mov     bx, ax           ; t
```

```
                push    word ptr [bx]    ; ↙
    ↘ second
                push    word ptr [bx+2]  ; ↙
    ↘ minute
                push    word ptr [bx+4]  ; ↙
    ↘ hour
                push    word ptr [bx+6]  ; ↙
    ↘ day
                push    word ptr [bx+8]  ; ↙
    ↘ month
                mov     ax, [bx+0Ah]     ; ↙
    ↘ year
                add     ax, 1900
                push    ax
                mov     ax, offset ↙
    ↘ a04d02d02d02d02 ; "%04d-%02d-%02d %02d↙
    ↘ :%02d:%02d"
                push    ax
                mov     ax, offset strbuf
                push    ax
                call    sprintf_
                add     sp, 10h
                xor     ax, ax            ; ↙
    ↘ NULL
                push    ax
                push    ds
                mov     ax, offset strbuf
                push    ax
                push    ds
                mov     ax, offset aCaption ↙
    ↘ ; "caption"
                push    ax
                xor     ax, ax          ; ↙
```

```
    ↳ MB_OK
                push    ax
                call    MESSAGEBOX
                xor     ax, ax
                mov     sp, bp
                pop     bp
                retn    0Ah
WinMain         endp
```

. localtime(). localtime() struct tm . .

time() y localtime() AX, DX, BX y CX, . .

sprintf() .

53.6.1

:

```
#include <windows.h>
#include <time.h>
#include <stdio.h>

char strbuf[256];
struct tm *t;
time_t unix_time;

int PASCAL WinMain( HINSTANCE hInstance,
                    HINSTANCE hPrevInstance,
                    LPSTR lpCmdLine,
                    int nCmdShow )
{
```

```
        unix_time=time(NULL);

        t=localtime (&unix_time);

        sprintf (strbuf, "%04d-%02d-%02d %02↙
    ↳ d:%02d:%02d", t->tm_year+1900, t->↙
    ↳ tm_mon, t->tm_mday,
                t->tm_hour, t->tm_min, t->↙
    ↳ tm_sec);

        MessageBox (NULL, strbuf, "caption",↙
    ↳  MB_OK);
        return 0;
};
```

```
unix_time_low   dw 0
unix_time_high  dw 0
t               dw 0

WinMain         proc near
                push    bp
                mov     bp, sp
                xor     ax, ax
                call    time_
                mov     unix_time_low, ax
                mov     unix_time_high, dx
                mov     ax, offset ↙
    ↳ unix_time_low
                call    localtime_
                mov     bx, ax
                mov     t, ax                   ↙
    ↳ ; will not be used in future...
```

```
                push    word ptr [bx]        ↙
    ↘ ; seconds
                push    word ptr [bx+2]      ↙
    ↘ ; minutes
                push    word ptr [bx+4]      ↙
    ↘ ; hour
                push    word ptr [bx+6]      ↙
    ↘ ; day
                push    word ptr [bx+8]      ↙
    ↘ ; month
                mov     ax, [bx+0Ah]         ↙
    ↘ ; year
                add     ax, 1900
                push    ax
                mov     ax, offset ↙
    ↘ a04d02d02d02d02 ; "%04d-%02d-%02d %02d↙
    ↘ :%02d:%02d"
                push    ax
                mov     ax, offset strbuf
                push    ax
                call    sprintf_
                add     sp, 10h
                xor     ax, ax           ; ↙
    ↘ NULL
                push    ax
                push    ds
                mov     ax, offset strbuf
                push    ax
                push    ds
                mov     ax, offset aCaption ↙
    ↘ ; "caption"
                push    ax
                xor     ax, ax           ; ↙
```

```
    ↳ MB_OK
                push    ax
                call    MESSAGEBOX
                xor     ax, ax              ; ↙
    ↳ return 0
                pop     bp
                retn    0Ah
WinMain         endp
```

Parte IV

Java

Capítulo 54

Java

54.1

-
-
-
-
- .
- .
-

.

: `javap -c -verbose`

: [Jav13].

54.2

```
public class ret
{
        public static int main(String[] args
    ↳ )
        {
                return 0;
        }
}
```

:

```
javac ret.java
```

...:

```
javap -c -verbose ret.class
```

:

Listing 54.1: JDK 1.7 (excerpt)

```
  public static int main(java.lang.String[])
    ↳ ;
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=1, locals=1, args_size=1
```

```
          0: iconst_0
          1: ireturn
```

[1]. `iconst_m1` -1.

Stack is used in JVM for data passing into functions to be called and also returning values. So `iconst_0` pushed 0 into stack. `ireturn` returns integer value (*i* in name mean *integer*) from the TOS[2].

1234:

```
public class ret
{
        public static int main(String[] args↙
    ↘ )
        {
                return 1234;
        }
}
```

...:

Listing 54.2: JDK 1.7 (excerpt)

```
  public static int main(java.lang.String[])↙
    ↘ ;
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=1, locals=1, args_size=1
         0: sipush        1234
```

[1]: 3.5.2 on page 50.

[2]Top Of Stack

```
        3: ireturn
```

`sipush` (*short integer*) 1234 . *short* .

```
public class ret
{
        public static int main(String[] args
    ↳ )
        {
                return 12345678;
        }
}
```

Listing 54.3: Constant pool

```
...
   #2 = Integer            12345678
...
```

```
  public static int main(java.lang.String[])
    ↳ ;
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=1, locals=1, args_size=1
         0: ldc           #2
    ↳    // int 12345678
         2: ireturn
```

28.3 on page 852, 29.1 on page 860.

!

Boolean:

```
public class ret
{
        public static boolean main(String[] ↙
    ↘ args)
        {
                return true;
        }
}
```

```
  public static boolean main(java.lang.↙
    ↘ String[]);
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=1, locals=1, args_size=1
         0: iconst_1
         1: ireturn
```

short:

```
public class ret
{
        public static short main(String[] ↙
    ↘ args)
        {
                return 1234;
        }
}
```

```
  public static short main(java.lang.String↙
    ↘ []);
    flags: ACC_PUBLIC, ACC_STATIC
```

```
    Code:
      stack=1, locals=1, args_size=1
         0: sipush        1234
         3: ireturn
```

...y *char*!

```
public class ret
{
        public static char main(String[] ↙
    ↘ args)
        {
                return 'A';
        }
}
```

```
  public static char main(java.lang.String↙
    ↘ []);
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=1, locals=1, args_size=1
         0: bipush        65
         2: ireturn
```

`bipush` «push byte».

byte:

```
public class retc
{
        public static byte main(String[] ↙
    ↘ args)
        {
```

```
                return 123;
        }
}
```

```
  public static byte main(java.lang.String
    [];
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=1, locals=1, args_size=1
         0: bipush        123
         2: ireturn
```

```
public class ret3
{
        public static long main(String[]
    args)
        {
                return 1234567890123456789L;
        }
}
```

Listing 54.4: Constant pool

```
...
   #2 = Long
    123456789012345678 9l
...
```

```
  public static long main(java.lang.String
    []);
    flags: ACC_PUBLIC, ACC_STATIC
```

```
    Code:
      stack=2, locals=1, args_size=1
         0: ldc2_w        #2                  ↙
    ↘   // long 1234567890123456789l
         3: lreturn
```

```
public class ret
{
        public static double main(String[] ↙
    ↘ args)
        {
                return 123.456d;
        }
}
```

Listing 54.5: Constant pool

```
...
   #2 = Double             123.456d
...
```

```
  public static double main(java.lang.String↙
    ↘ []);
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=2, locals=1, args_size=1
         0: ldc2_w        #2                  ↙
    ↘   // double 123.456d
         3: dreturn
```

```
public class ret
```

```
{
        public static float main(String[] ↙
    ↘ args)
        {
                return 123.456f;
        }
}
```

Listing 54.6: Constant pool

```
...
   #2 = Float              123.456f
...
```

```
  public static float main(java.lang.String↙
    ↘ []);
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=1, locals=1, args_size=1
         0: ldc           #2                  ↙
    ↘   // float 123.456f
         2: freturn
```

```
public class ret
{
        public static void main(String[] ↙
    ↘ args)
        {
                return;
        }
}
```

```
  public static void main(java.lang.String
    ↳ []);
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=0, locals=1, args_size=1
         0: return
```

54.3

.

```
public class calc
{
        public static int half(int a)
        {
                return a/2;
        }
}
```

```
  public static int half(int);
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=2, locals=1, args_size=1
         0: iload_0
         1: iconst_2
         2: idiv
         3: ireturn
```

`iload_0` `iconst_2`.

```
      +---+
TOS ->| 2 |
      +---+
      | a |
      +---+
```

```
      +--------+
TOS ->| result |
      +--------+
```

ireturn.

```
public class calc
{
        public static double half_double(
    double a)
        {
                return a/2.0;
        }
}
```

Listing 54.7: Constant pool

```
...
   #2 = Double             2.0d
...
```

```
  public static double half_double(double);
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=4, locals=2, args_size=1
```

```
        0: dload_0
        1: ldc2_w        #2                  ↙
    ↳   // double 2.0d
        4: ddiv
        5: dreturn
```

```
public class calc
{
        public static int sum(int a, int b)
        {
                return a+b;
        }
}
```

```
  public static int sum(int, int);
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=2, locals=2, args_size=2
         0: iload_0
         1: iload_1
         2: iadd
         3: ireturn
```

```
      +---+
TOS ->| b |
      +---+
      | a |
      +---+
```

```
      +--------+
TOS ->| result |
```

```
    +--------+
```

```
        public static long lsum(long a, long
    ↳  b)
        {
                return a+b;
        }
```

...:

```
  public static long lsum(long, long);
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=4, locals=4, args_size=2
         0: lload_0
         1: lload_2
         2: ladd
         3: lreturn
```

```
public class calc
{
        public static int mult_add(int a,
    ↳ int b, int c)
        {
                return a*b+c;
        }
}
```

```
  public static int mult_add(int, int, int);
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
```

```
      stack=2, locals=3, args_size=3
         0: iload_0
         1: iload_1
         2: imul
         3: iload_2
         4: iadd
         5: ireturn
```

```
     +---------+
TOS ->| product |
     +---------+
```

`iload_2`:

```
     +---------+
TOS ->|    c    |
     +---------+
     | product |
     +---------+
```

54.4

JVM[3] .

:

- (LVA[4]). `istore`.
- .

[3]Java virtual machine

[4](Java) Local Variable Array

•

54.5

```
public class HalfRandom
{
        public static double f()
        {
                return Math.random()/2;
        }
}
```

Listing 54.8: Constant pool

```
...
   #2 = Methodref          #18.#19        //⤦
    ↳   java/lang/Math.random:()D
   #3 = Double             2.0d
...
  #12 = Utf8               ()D
...
  #18 = Class              #22            //⤦
    ↳   java/lang/Math
  #19 = NameAndType        #23:#12        //⤦
    ↳   random:()D
  #22 = Utf8               java/lang/Math
  #23 = Utf8               random
```

```
  public static double f();
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
```

```
      stack=4, locals=0, args_size=0
         0: invokestatic  #2                  ⤦
    ↪   // Method java/lang/Math.random:()D
         3: ldc2_w        #3                  ⤦
    ↪   // double 2.0d
         6: ddiv
         7: dreturn
```

```
public class HelloWorld
{
        public static void main(String[] ⤦
    ↪ args)
        {
                System.out.println("Hello, ⤦
    ↪ World");
        }
}
```

Listing 54.9: Constant pool

```
...
   #2 = Fieldref           #16.#17        //⤦
    ↪   java/lang/System.out:Ljava/io/⤦
    ↪ PrintStream;
   #3 = String             #18            //⤦
    ↪   Hello, World
   #4 = Methodref          #19.#20        //⤦
    ↪   java/io/PrintStream.println:(Ljava/⤦
    ↪ lang/String;)V
...
  #16 = Class              #23            //⤦
    ↪   java/lang/System
```

```
  #17 = NameAndType        #24:#25          //↙
    ↳  out:Ljava/io/PrintStream;
  #18 = Utf8               Hello, World
  #19 = Class              #26              //↙
    ↳  java/io/PrintStream
  #20 = NameAndType        #27:#28          //↙
    ↳  println:(Ljava/lang/String;)V
...
  #23 = Utf8               java/lang/System
  #24 = Utf8               out
  #25 = Utf8               Ljava/io/↙
    ↳ PrintStream;
  #26 = Utf8               java/io/↙
    ↳ PrintStream
  #27 = Utf8               println
  #28 = Utf8               (Ljava/lang/↙
    ↳ String;)V
...
```

```
  public static void main(java.lang.String↙
    ↳ []);
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=2, locals=1, args_size=1
         0: getstatic     #2                  ↙
    ↳  // Field java/lang/System.out:Ljava/↙
    ↳ io/PrintStream;
         3: ldc           #3                  ↙
    ↳  // String Hello, World
         5: invokevirtual #4                  ↙
    ↳  // Method java/io/PrintStream.↙
    ↳ println:(Ljava/lang/String;)V
```

```
        8: return
```

5
.

54.6 beep()

```
        public static void main(String[] ↙
    ↳ args)
        {
                java.awt.Toolkit.↙
    ↳ getDefaultToolkit().beep();
        };
```

```
  public static void main(java.lang.String↙
    ↳ []);
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=1, locals=1, args_size=1
         0: invokestatic  #2                  ↙
    ↳   // Method java/awt/Toolkit.↙
    ↳ getDefaultToolkit:()Ljava/awt/Toolkit;
         3: invokevirtual #3                  ↙
    ↳   // Method java/awt/Toolkit.beep:()V
         6: return
```

54.7

[5]: 51.3 on page 1072.

```
public class LCG
{
        public static int rand_state;

        public void my_srand (int init)
        {
                rand_state=init;
        }

        public static int RNG_a=1664525;
        public static int RNG_c=1013904223;

        public int my_rand ()
        {
                rand_state=rand_state*RNG_a;
                rand_state=rand_state+RNG_c;
                return rand_state & 0x7fff;
        }
}
```

?

```
  static {};
    flags: ACC_STATIC
    Code:
      stack=1, locals=0, args_size=0
         0: ldc           #5                  
    ↳   // int 1664525
         2: putstatic     #3                  
    ↳   // Field RNG_a:I
         5: ldc           #6                  
    ↳   // int 1013904223
```

```
        7: putstatic     #4                          ⤦
    ↳   // Field RNG_c:I
       10: return
```

```
  public void my_srand(int);
    flags: ACC_PUBLIC
    Code:
      stack=1, locals=2, args_size=2
         0: iload_1
         1: putstatic     #2                         ⤦
    ↳   // Field rand_state:I
         4: return
```

my_rand():

```
  public int my_rand();
    flags: ACC_PUBLIC
    Code:
      stack=2, locals=1, args_size=1
         0: getstatic     #2                         ⤦
    ↳   // Field rand_state:I
         3: getstatic     #3                         ⤦
    ↳   // Field RNG_a:I
         6: imul
         7: putstatic     #2                         ⤦
    ↳   // Field rand_state:I
        10: getstatic     #2                         ⤦
    ↳   // Field rand_state:I
        13: getstatic     #4                         ⤦
    ↳   // Field RNG_c:I
        16: iadd
        17: putstatic     #2                         ⤦
    ↳   // Field rand_state:I
```

```
      20: getstatic     #2                  
   // Field rand_state:I
      23: sipush        32767
      26: iand
      27: ireturn
```

54.8

.

```
public class abs
{
        public static int abs(int a)
        {
                if (a<0)
                        return -a;
                return a;
        }
}
```

```
  public static int abs(int);
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=1, locals=1, args_size=1
         0: iload_0
         1: ifge          7
         4: iload_0
         5: ineg
         6: ireturn
         7: iload_0
         8: ireturn
```

:

```
        public static int min (int a, int b)
        {
                if (a>b)
                        return b;
                return a;
        }
```

:

```
  public static int min(int, int);
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=2, locals=2, args_size=2
         0: iload_0
         1: iload_1
         2: if_icmple     7
         5: iload_1
         6: ireturn
         7: iload_0
         8: ireturn
```

`if_icmple`.

```
        public static int max (int a, int b)
        {
                if (a>b)
                        return a;
                return b;
        }
```

```
  public static int max(int, int);
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=2, locals=2, args_size=2
         0: iload_0
         1: iload_1
         2: if_icmple     7
         5: iload_0
         6: ireturn
         7: iload_1
         8: ireturn
```

:

```
public class cond
{
        public static void f(int i)
        {
                if (i<100)
                        System.out.print↙
    ↳ ("<100");
                if (i==100)
                        System.out.print↙
    ↳ ("==100");
                if (i>100)
                        System.out.print↙
    ↳ (">100");
                if (i==0)
                        System.out.print↙
    ↳ ("==0");
        }
}
```

```
  public static void f(int);
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=2, locals=1, args_size=1
         0: iload_0
         1: bipush        100
         3: if_icmpge     14
         6: getstatic     #2                  ↙
    ↘    // Field java/lang/System.out:Ljava/↙
    ↘ io/PrintStream;
         9: ldc           #3                  ↙
    ↘    // String <100
        11: invokevirtual #4                  ↙
    ↘    // Method java/io/PrintStream.print↙
    ↘ :(Ljava/lang/String;)V
        14: iload_0
        15: bipush        100
        17: if_icmpne     28
        20: getstatic     #2                  ↙
    ↘    // Field java/lang/System.out:Ljava/↙
    ↘ io/PrintStream;
        23: ldc           #5                  ↙
    ↘    // String ==100
        25: invokevirtual #4                  ↙
    ↘    // Method java/io/PrintStream.print↙
    ↘ :(Ljava/lang/String;)V
        28: iload_0
        29: bipush        100
        31: if_icmple     42
        34: getstatic     #2                  ↙
    ↘    // Field java/lang/System.out:Ljava/↙
    ↘ io/PrintStream;
```

```
      37: ldc             #6                   ⤦
  ↪   // String >100
      39: invokevirtual #4                     ⤦
  ↪   // Method java/io/PrintStream.print⤦
  ↪ :(Ljava/lang/String;)V
      42: iload_0
      43: ifne            54
      46: getstatic       #2                   ⤦
  ↪   // Field java/lang/System.out:Ljava/⤦
  ↪ io/PrintStream;
      49: ldc             #7                   ⤦
  ↪   // String ==0
      51: invokevirtual #4                     ⤦
  ↪   // Method java/io/PrintStream.print⤦
  ↪ :(Ljava/lang/String;)V
      54: return
```

if_icmpge.

54.9

```
public class minmax
{
        public static int min (int a, int b)
        {
                if (a>b)
                        return b;
                return a;
        }

        public static int max (int a, int b)
```

```
        {
                if (a>b)
                        return a;
                return b;
        }

        public static void main(String[] ↙
    ↳ args)
        {
                int a=123, b=456;
                int max_value=max(a, b);
                int min_value=min(a, b);
                System.out.println(min_value↙
    ↳ );
                System.out.println(max_value↙
    ↳ );
        }
}
```

```
  public static void main(java.lang.String↙
    ↳ []);
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=2, locals=5, args_size=1
         0: bipush        123
         2: istore_1
         3: sipush        456
         6: istore_2
         7: iload_1
         8: iload_2
         9: invokestatic  #2                 ↙
    ↳   // Method max:(II)I
```

```
      12: istore_3
      13: iload_1
      14: iload_2
      15: invokestatic  #3                  ↙
  ↪   // Method min:(II)I
      18: istore        4
      20: getstatic     #4                  ↙
  ↪   // Field java/lang/System.out:Ljava/↙
  ↪ io/PrintStream;
      23: iload         4
      25: invokevirtual #5                  ↙
  ↪   // Method java/io/PrintStream.↙
  ↪ println:(I)V
      28: getstatic     #4                  ↙
  ↪   // Field java/lang/System.out:Ljava/↙
  ↪ io/PrintStream;
      31: iload_3
      32: invokevirtual #5                  ↙
  ↪   // Method java/io/PrintStream.↙
  ↪ println:(I)V
      35: return
```

54.10

```
      public static int set (int a, int b)
      {
              return a | 1<<b;
      }

      public static int clear (int a, int ↙
  ↪ b)
```

```
        {
                return a & (~(1<<b));
        }
```

```
  public static int set(int, int);
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=3, locals=2, args_size=2
         0: iload_0
         1: iconst_1
         2: iload_1
         3: ishl
         4: ior
         5: ireturn

  public static int clear(int, int);
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=3, locals=2, args_size=2
         0: iload_0
         1: iconst_1
         2: iload_1
         3: ishl
         4: iconst_m1
         5: ixor
         6: iand
         7: ireturn
```

(A.6.2 on page 1886).

```
        public static long lset (long a, int↙
    ↘ b)
```

```
        {
                return a | 1<<b;
        }

        public static long lclear (long a, ↙
    ↘ int b)
        {
                return a & (~(1<<b));
        }
```

```
  public static long lset(long, int);
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=4, locals=3, args_size=2
         0: lload_0
         1: iconst_1
         2: iload_2
         3: ishl
         4: i2l
         5: lor
         6: lreturn

  public static long lclear(long, int);
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=4, locals=3, args_size=2
         0: lload_0
         1: iconst_1
         2: iload_2
         3: ishl
         4: iconst_m1
         5: ixor
```

```
        6: i2l
        7: land
        8: lreturn
```

54.11

```
public class Loop
{
        public static void main(String[] ↙
    ↘ args)
        {
                for (int i = 1; i <= 10; i↙
    ↘ ++)
                {
                        System.out.println(i↙
    ↘ );
                }
        }
}
```

```
  public static void main(java.lang.String↙
    ↘ []);
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=2, locals=2, args_size=1
         0: iconst_1
         1: istore_1
         2: iload_1
         3: bipush        10
         5: if_icmpgt     21
```

```
         8: getstatic     #2                  
    ↳   // Field java/lang/System.out:Ljava/
    ↳ io/PrintStream;
        11: iload_1
        12: invokevirtual #3                  
    ↳   // Method java/io/PrintStream.
    ↳ println:(I)V
        15: iinc          1, 1
        18: goto          2
        21: return
```

?

```
public class Fibonacci
{
        public static void main(String[] 
    ↳ args)
        {
                int limit = 20, f = 0, g = 
    ↳ 1;

                for (int i = 1; i <= limit; 
    ↳ i++)
                {
                        f = f + g;
                        g = f - g;
                        System.out.println(f
    ↳ );
                }
        }
}
```

```
  public static void main(java.lang.String⤦
    ↳ []);
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=2, locals=5, args_size=1
         0: bipush        20
         2: istore_1
         3: iconst_0
         4: istore_2
         5: iconst_1
         6: istore_3
         7: iconst_1
         8: istore        4
        10: iload         4
        12: iload_1
        13: if_icmpgt     37
        16: iload_2
        17: iload_3
        18: iadd
        19: istore_2
        20: iload_2
        21: iload_3
        22: isub
        23: istore_3
        24: getstatic     #2                  ⤦
    ↳   // Field java/lang/System.out:Ljava/⤦
    ↳ io/PrintStream;
        27: iload_2
        28: invokevirtual #3                  ⤦
    ↳   // Method java/io/PrintStream.⤦
    ↳ println:(I)V
        31: iinc          4, 1
        34: goto          10
```

```
        37: return
```

- 0 —
- 1 — *limit*, 20
- 2 — *f*
- 3 — *g*
- 4 — *i*

54.12 switch()

```
        public static void f(int a)
        {
                switch (a)
                {
                case 0: System.out.println("↙
    ↳ zero"); break;
                case 1: System.out.println("↙
    ↳ one\n"); break;
                case 2: System.out.println("↙
    ↳ two\n"); break;
                case 3: System.out.println("↙
    ↳ three\n"); break;
                case 4: System.out.println("↙
    ↳ four\n"); break;
                default: System.out.println↙
    ↳ ("something unknown\n"); break;
                };
        }
```

:

```
  public static void f(int);
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=2, locals=1, args_size=1
         0: iload_0
         1: tableswitch   { // 0 to 4
                       0: 36
                       1: 47
                       2: 58
                       3: 69
                       4: 80
                 default: 91
            }
        36: getstatic     #2                  ↙
    ↳   // Field java/lang/System.out:Ljava/↙
    ↳ io/PrintStream;
        39: ldc           #3                  ↙
    ↳   // String zero
        41: invokevirtual #4                  ↙
    ↳   // Method java/io/PrintStream.↙
    ↳ println:(Ljava/lang/String;)V
        44: goto          99
        47: getstatic     #2                  ↙
    ↳   // Field java/lang/System.out:Ljava/↙
    ↳ io/PrintStream;
        50: ldc           #5                  ↙
    ↳   // String one\n
        52: invokevirtual #4                  ↙
    ↳   // Method java/io/PrintStream.↙
    ↳ println:(Ljava/lang/String;)V
        55: goto          99
```

```
    58: getstatic     #2                  ↙
↘   // Field java/lang/System.out:Ljava/↙
↘ io/PrintStream;
    61: ldc           #6                  ↙
↘   // String two\n
    63: invokevirtual #4                  ↙
↘   // Method java/io/PrintStream.↙
↘ println:(Ljava/lang/String;)V
    66: goto          99
    69: getstatic     #2                  ↙
↘   // Field java/lang/System.out:Ljava/↙
↘ io/PrintStream;
    72: ldc           #7                  ↙
↘   // String three\n
    74: invokevirtual #4                  ↙
↘   // Method java/io/PrintStream.↙
↘ println:(Ljava/lang/String;)V
    77: goto          99
    80: getstatic     #2                  ↙
↘   // Field java/lang/System.out:Ljava/↙
↘ io/PrintStream;
    83: ldc           #8                  ↙
↘   // String four\n
    85: invokevirtual #4                  ↙
↘   // Method java/io/PrintStream.↙
↘ println:(Ljava/lang/String;)V
    88: goto          99
    91: getstatic     #2                  ↙
↘   // Field java/lang/System.out:Ljava/↙
↘ io/PrintStream;
    94: ldc           #9                  ↙
↘   // String something unknown\n
    96: invokevirtual #4                  ↙
```

```
↳   // Method java/io/PrintStream.↙
↳ println:(Ljava/lang/String;)V
    99: return
```

54.13

54.13.1

```
        public static void main(String[] ↙
    ↳ args)
        {
                int a[]=new int[10];
                for (int i=0; i<10; i++)
                        a[i]=i;
                dump (a);
        }
```

```
  public static void main(java.lang.String↙
    ↳ []);
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=3, locals=3, args_size=1
         0: bipush        10
         2: newarray       int
         4: astore_1
         5: iconst_0
         6: istore_2
         7: iload_2
         8: bipush        10
        10: if_icmpge     23
```

```
      13: aload_1
      14: iload_2
      15: iload_2
      16: iastore
      17: iinc          2, 1
      20: goto          7
      23: aload_1
      24: invokestatic  #4
  ↳   // Method dump:([I)V
      27: return
```

```
      public static void dump(int a[])
      {
              for (int i=0; i<a.length; i
  ↳ ++)
                      System.out.println(a
  ↳ [i]);
      }
```

```
  public static void dump(int[]);
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=3, locals=2, args_size=1
         0: iconst_0
         1: istore_1
         2: iload_1
         3: aload_0
         4: arraylength
         5: if_icmpge     23
         8: getstatic     #2
  ↳    // Field java/lang/System.out:Ljava/
  ↳ io/PrintStream;
```

```
        11: aload_0
        12: iload_1
        13: iaload
        14: invokevirtual #3                  ↙
    ↘   // Method java/io/PrintStream.↙
    ↘ println:(I)V
        17: iinc          1, 1
        20: goto          2
        23: return
```

.

54.13.2

:

```
public class ArraySum
{
        public static int f (int[] a)
        {
                int sum=0;
                for (int i=0; i<a.length; i↙
    ↘ ++)
                        sum=sum+a[i];
                return sum;
        }
}
```

```
  public static int f(int[]);
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=3, locals=3, args_size=1
```

```
         0: iconst_0
         1: istore_1
         2: iconst_0
         3: istore_2
         4: iload_2
         5: aload_0
         6: arraylength
         7: if_icmpge     22
        10: iload_1
        11: aload_0
        12: iload_2
        13: iaload
        14: iadd
        15: istore_1
        16: iinc          2, 1
        19: goto          4
        22: iload_1
        23: ireturn
```

54.13.3

```
public class UseArgument
{
        public static void main(String[] 
    ↳ args)
        {
                System.out.print("Hi, ");
                System.out.print(args[1]);
                System.out.println(". How 
    ↳ are you?");
        }
}
```

```
public static void main(java.lang.String
  ↘ []);
  flags: ACC_PUBLIC, ACC_STATIC
  Code:
    stack=3, locals=1, args_size=1
       0: getstatic     #2
  ↘   // Field java/lang/System.out:Ljava/
  ↘ io/PrintStream;
       3: ldc           #3
  ↘   // String Hi,
       5: invokevirtual #4
  ↘   // Method java/io/PrintStream.print
  ↘ :(Ljava/lang/String;)V
       8: getstatic     #2
  ↘   // Field java/lang/System.out:Ljava/
  ↘ io/PrintStream;
      11: aload_0
      12: iconst_1
      13: aaload
      14: invokevirtual #4
  ↘   // Method java/io/PrintStream.print
  ↘ :(Ljava/lang/String;)V
      17: getstatic     #2
  ↘   // Field java/lang/System.out:Ljava/
  ↘ io/PrintStream;
      20: ldc           #5
  ↘   // String . How are you?
      22: invokevirtual #6
  ↘   // Method java/io/PrintStream.
  ↘ println:(Ljava/lang/String;)V
      25: return
```

CAPÍTULO 54. JAVA

54.13.4

```
class Month
{
        public static String[] months =
        {
                "January",
                "February",
                "March",
                "April",
                "May",
                "June",
                "July",
                "August",
                "September",
                "October",
                "November",
                "December"
        };

        public String get_month (int i)
        {
                return months[i];
        };
}
```

```
  public java.lang.String get_month(int);
    flags: ACC_PUBLIC
    Code:
```

```
      stack=2, locals=2, args_size=2
         0: getstatic     #2                  
    // Field months:[Ljava/lang/String;
         3: iload_1
         4: aaload
         5: areturn
```

```
  static {};
    flags: ACC_STATIC
    Code:
      stack=4, locals=0, args_size=0
         0: bipush        12
         2: anewarray     #3                  
    // class java/lang/String
         5: dup
         6: iconst_0
         7: ldc           #4                  
    // String January
         9: aastore
        10: dup
        11: iconst_1
        12: ldc           #5                  
    // String February
        14: aastore
        15: dup
        16: iconst_2
        17: ldc           #6                  
    // String March
        19: aastore
        20: dup
        21: iconst_3
        22: ldc           #7                  
```

```
↳   // String April
    24: aastore
    25: dup
    26: iconst_4
    27: ldc            #8                         ↙
↳   // String May
    29: aastore
    30: dup
    31: iconst_5
    32: ldc            #9                         ↙
↳   // String June
    34: aastore
    35: dup
    36: bipush         6
    38: ldc            #10                        ↙
↳   // String July
    40: aastore
    41: dup
    42: bipush         7
    44: ldc            #11                        ↙
↳   // String August
    46: aastore
    47: dup
    48: bipush         8
    50: ldc            #12                        ↙
↳   // String September
    52: aastore
    53: dup
    54: bipush         9
    56: ldc            #13                        ↙
↳   // String October
    58: aastore
    59: dup
```

```
     60: bipush        10
     62: ldc           #14                    ↙
↘    // String November
     64: aastore
     65: dup
     66: bipush        11
     68: ldc           #15                    ↙
↘    // String December
     70: aastore
     71: putstatic     #2                     ↙
↘    // Field months:[Ljava/lang/String;
     74: return
```

54.13.5

```
        public static void f(int... values)
        {
                for (int i=0; i<values.↙
    ↘ length; i++)
                        System.out.println(↙
    ↘ values[i]);
        }

        public static void main(String[] ↙
    ↘ args)
        {
                f (1,2,3,4,5);
        }
```

```
  public static void f(int...);
```

```
    flags: ACC_PUBLIC, ACC_STATIC, ↙
    ↳ ACC_VARARGS
    Code:
      stack=3, locals=2, args_size=1
         0: iconst_0
         1: istore_1
         2: iload_1
         3: aload_0
         4: arraylength
         5: if_icmpge     23
         8: getstatic     #2                  ↙
    ↳   // Field java/lang/System.out:Ljava/↙
    ↳ io/PrintStream;
        11: aload_0
        12: iload_1
        13: iaload
        14: invokevirtual #3                  ↙
    ↳   // Method java/io/PrintStream.↙
    ↳ println:(I)V
        17: iinc          1, 1
        20: goto          2
        23: return
```

```
  public static void main(java.lang.String↙
    ↳ []);
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=4, locals=1, args_size=1
         0: iconst_5
         1: newarray       int
         3: dup
         4: iconst_0
```

```
     5: iconst_1
     6: iastore
     7: dup
     8: iconst_1
     9: iconst_2
    10: iastore
    11: dup
    12: iconst_2
    13: iconst_3
    14: iastore
    15: dup
    16: iconst_3
    17: iconst_4
    18: iastore
    19: dup
    20: iconst_4
    21: iconst_5
    22: iastore
    23: invokestatic  #4                  ↙
↳   // Method f:([I)V
    26: return
```

```
    public PrintStream format(String ↙
↳ format, Object... args)
```

(http://docs.oracle.com/javase/tutorial/java/data/numberformat.html)

:

```
    public static void main(String[] ↙
↳ args)
    {
```

```
                int i=123;
                double d=123.456;
                System.out.format("int: %d ⤦
    ↳ double: %f.%n", i, d);
        }
```

```
  public static void main(java.lang.String⤦
    ↳ []);
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=7, locals=4, args_size=1
         0: bipush        123
         2: istore_1
         3: ldc2_w        #2                  ⤦
    ↳   // double 123.456d
         6: dstore_2
         7: getstatic     #4                  ⤦
    ↳   // Field java/lang/System.out:Ljava/⤦
    ↳ io/PrintStream;
        10: ldc           #5                  ⤦
    ↳   // String int: %d double: %f.%n
        12: iconst_2
        13: anewarray     #6                  ⤦
    ↳   // class java/lang/Object
        16: dup
        17: iconst_0
        18: iload_1
        19: invokestatic  #7                  ⤦
    ↳   // Method java/lang/Integer.valueOf⤦
    ↳ :(I)Ljava/lang/Integer;
        22: aastore
        23: dup
```

```
      24: iconst_1
      25: dload_2
      26: invokestatic  #8                  ↙
   ↳   // Method java/lang/Double.valueOf:(↙
   ↳ D)Ljava/lang/Double;
      29: aastore
      30: invokevirtual #9                  ↙
   ↳   // Method java/io/PrintStream.format↙
   ↳ :(Ljava/lang/String;[Ljava/lang/Object↙
   ↳ ;)Ljava/io/PrintStream;
      33: pop
      34: return
```

54.13.6

:

```
      public static void main(String[] ↙
   ↳ args)
      {
              int[][] a = new int[5][10];
              a[1][2]=3;
      }
```

```
  public static void main(java.lang.String↙
    ↳ []);
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=3, locals=2, args_size=1
         0: iconst_5
         1: bipush        10
```

```
         3: multianewarray #2,  2
   // class "[[I"
         7: astore_1
         8: aload_1
         9: iconst_1
        10: aaload
        11: iconst_2
        12: iconst_3
        13: iastore
        14: return
```

?

```
        public static int get12 (int[][] in)
        {
                return in[1][2];
        }
```

```
  public static int get12(int[][]);
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=2, locals=1, args_size=1
         0: aload_0
         1: iconst_1
         2: aaload
         3: iconst_2
         4: iaload
         5: ireturn
```

54.13.7

```
    public static void main(String[] ↙
  ↘ args)
    {
            int[][][] a = new int↙
  ↘ [5][10][15];

            a[1][2][3]=4;

            get_elem(a);
    }
```

```
  public static void main(java.lang.String↙
    ↘ []);
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=3, locals=2, args_size=1
         0: iconst_5
         1: bipush        10
         3: bipush        15
         5: multianewarray #2,  3              ↙
    ↘   // class "[[[I"
         9: astore_1
        10: aload_1
        11: iconst_1
        12: aaload
        13: iconst_2
        14: aaload
        15: iconst_3
        16: iconst_4
        17: iastore
        18: aload_1
        19: invokestatic  #3                   ↙
```

```
    ↳   // Method get_elem:([[[I)I
        22: pop
        23: return
```

```
        public static int get_elem (int↙
    ↳ [][][] a)
        {
                return a[1][2][3];
        }
```

```
  public static int get_elem(int[][][]);
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=2, locals=1, args_size=1
         0: aload_0
         1: iconst_1
         2: aaload
         3: iconst_2
         4: aaload
         5: iconst_3
         6: iaload
         7: ireturn
```

54.13.8

54.14

54.14.1

```
    public static void main(String[] ↙
↪ args)
    {
            System.out.println("What is ↙
↪ your name?");
            String input = System.↙
↪ console().readLine();
            System.out.println("Hello, ↙
↪ "+input);
    }
```

```
  public static void main(java.lang.String↙
  ↪ []);
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=3, locals=2, args_size=1
         0: getstatic     #2                  ↙
  ↪   // Field java/lang/System.out:Ljava/↙
  ↪ io/PrintStream;
         3: ldc           #3                  ↙
  ↪   // String What is your name?
         5: invokevirtual #4                  ↙
  ↪   // Method java/io/PrintStream.↙
  ↪ println:(Ljava/lang/String;)V
         8: invokestatic  #5                  ↙
  ↪   // Method java/lang/System.console↙
  ↪ :()Ljava/io/Console;
        11: invokevirtual #6                  ↙
  ↪   // Method java/io/Console.readLine↙
  ↪ :()Ljava/lang/String;
        14: astore_1
        15: getstatic     #2                  ↙
```

```
    ↳   // Field java/lang/System.out:Ljava/↙
    ↳ io/PrintStream;
       18: new           #7                  ↙
    ↳   // class java/lang/StringBuilder
       21: dup
       22: invokespecial #8                  ↙
    ↳   // Method java/lang/StringBuilder."<↙
    ↳ init>":()V
       25: ldc           #9                  ↙
    ↳   // String Hello,
       27: invokevirtual #10                 ↙
    ↳   // Method java/lang/StringBuilder.↙
    ↳ append:(Ljava/lang/String;)Ljava/lang/↙
    ↳ StringBuilder;
       30: aload_1
       31: invokevirtual #10                 ↙
    ↳   // Method java/lang/StringBuilder.↙
    ↳ append:(Ljava/lang/String;)Ljava/lang/↙
    ↳ StringBuilder;
       34: invokevirtual #11                 ↙
    ↳   // Method java/lang/StringBuilder.↙
    ↳ toString:()Ljava/lang/String;
       37: invokevirtual #4                  ↙
    ↳   // Method java/io/PrintStream.↙
    ↳ println:(Ljava/lang/String;)V
       40: return
```

54.14.2

:

```
public class strings
```

```
{
        public static char test (String a)
        {
                return a.charAt(3);
        };

        public static String concat (String ↙
    ↘ a, String b)
        {
                return a+b;
        }
}
```

```
  public static char test(java.lang.String);
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=2, locals=1, args_size=1
         0: aload_0
         1: iconst_3
         2: invokevirtual #2                  ↙
    ↘   // Method java/lang/String.charAt:(I↙
    ↘ )C
         5: ireturn
```

```
  public static java.lang.String concat(java↙
    ↘ .lang.String, java.lang.String);
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=2, locals=2, args_size=2
         0: new           #3                  ↙
    ↘   // class java/lang/StringBuilder
         3: dup
```

```
     4: invokespecial #4
   ↳   // Method java/lang/StringBuilder."<
   ↳ init>":()V
     7: aload_0
     8: invokevirtual #5
   ↳   // Method java/lang/StringBuilder.
   ↳ append:(Ljava/lang/String;)Ljava/lang/
   ↳ StringBuilder;
    11: aload_1
    12: invokevirtual #5
   ↳   // Method java/lang/StringBuilder.
   ↳ append:(Ljava/lang/String;)Ljava/lang/
   ↳ StringBuilder;
    15: invokevirtual #6
   ↳   // Method java/lang/StringBuilder.
   ↳ toString:()Ljava/lang/String;
    18: areturn
```

:

```
        public static void main(String[]
   ↳ args)
        {
                String s="Hello!";
                int n=123;
                System.out.println("s=" + s
   ↳ + " n=" + n);
        }
```

```
  public static void main(java.lang.String
   ↳ []);
    flags: ACC_PUBLIC, ACC_STATIC
```

```
    Code:
      stack=3, locals=3, args_size=1
         0: ldc           #2                  ↙
    ↘   // String Hello!
         2: astore_1
         3: bipush        123
         5: istore_2
         6: getstatic     #3                  ↙
    ↘   // Field java/lang/System.out:Ljava/↙
    ↘ io/PrintStream;
         9: new           #4                  ↙
    ↘   // class java/lang/StringBuilder
        12: dup
        13: invokespecial #5                  ↙
    ↘   // Method java/lang/StringBuilder."<↙
    ↘ init>":()V
        16: ldc           #6                  ↙
    ↘   // String s=
        18: invokevirtual #7                  ↙
    ↘   // Method java/lang/StringBuilder.↙
    ↘ append:(Ljava/lang/String;)Ljava/lang/↙
    ↘ StringBuilder;
        21: aload_1
        22: invokevirtual #7                  ↙
    ↘   // Method java/lang/StringBuilder.↙
    ↘ append:(Ljava/lang/String;)Ljava/lang/↙
    ↘ StringBuilder;
        25: ldc           #8                  ↙
    ↘   // String  n=
        27: invokevirtual #7                  ↙
    ↘   // Method java/lang/StringBuilder.↙
    ↘ append:(Ljava/lang/String;)Ljava/lang/↙
    ↘ StringBuilder;
```

```
        30: iload_2
        31: invokevirtual #9                     ↙
    ↘   // Method java/lang/StringBuilder.↙
    ↘ append:(I)Ljava/lang/StringBuilder;
        34: invokevirtual #10                    ↙
    ↘   // Method java/lang/StringBuilder.↙
    ↘ toString:()Ljava/lang/String;
        37: invokevirtual #11                    ↙
    ↘   // Method java/io/PrintStream.↙
    ↘ println:(Ljava/lang/String;)V
        40: return
```

54.15

Listing 54.10: IncorrectMonthException.java

```
public class IncorrectMonthException extends↙
    ↘  Exception
{
        private int index;

        public IncorrectMonthException(int ↙
    ↘ index)
        {
                this.index = index;
        }
        public int getIndex()
        {
                return index;
        }
}
```

Listing 54.11: Month2.java

```
class Month2
{
        public static String[] months =
        {
                "January",
                "February",
                "March",
                "April",
                "May",
                "June",
                "July",
                "August",
                "September",
                "October",
                "November",
                "December"
        };

        public static String get_month (int ⤦
    ↳ i) throws IncorrectMonthException
        {
                if (i<0 || i>11)
                        throw new ⤦
    ↳ IncorrectMonthException(i);
                return months[i];
        };

        public static void main (String[] ⤦
    ↳ args)
        {
                try
```

```
                {
                        System.out.println(⤦
    ↪ get_month(100));
                }
                catch(⤦
    ↪ IncorrectMonthException e)
                {
                        System.out.println("⤦
    ↪ incorrect month index: "+ e.getIndex()⤦
    ↪ );
                        e.printStackTrace();
                }
        };
}
```

```
  public IncorrectMonthException(int);
    flags: ACC_PUBLIC
    Code:
      stack=2, locals=2, args_size=2
         0: aload_0
         1: invokespecial #1                    ⤦
    ↪   // Method java/lang/Exception."<init⤦
    ↪ >":()V
         4: aload_0
         5: iload_1
         6: putfield      #2                    ⤦
    ↪   // Field index:I
         9: return
```

getIndex() .

```
  public int getIndex();
```

```
    flags: ACC_PUBLIC
    Code:
      stack=1, locals=1, args_size=1
         0: aload_0
         1: getfield      #2                  ↙
    ↘   // Field index:I
         4: ireturn
```

Listing 54.12: Month2.class

```
  public static java.lang.String get_month(↙
    ↘ int) throws IncorrectMonthException;
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=3, locals=1, args_size=1
         0: iload_0
         1: iflt          10
         4: iload_0
         5: bipush        11
         7: if_icmple     19
        10: new           #2                  ↙
    ↘   // class IncorrectMonthException
        13: dup
        14: iload_0
        15: invokespecial #3                  ↙
    ↘   // Method IncorrectMonthException."<↙
    ↘ init>":(I)V
        18: athrow
        19: getstatic     #4                  ↙
    ↘   // Field months:[Ljava/lang/String;
        22: iload_0
        23: aaload
        24: areturn
```

? `main()` en `Month2.class`:

Listing 54.13: Month2.class

```
public static void main(java.lang.String
  ↳ []);
  flags: ACC_PUBLIC, ACC_STATIC
  Code:
    stack=3, locals=2, args_size=1
       0: getstatic     #5
  ↳   // Field java/lang/System.out:Ljava/
  ↳ io/PrintStream;
       3: bipush        100
       5: invokestatic  #6
  ↳   // Method get_month:(I)Ljava/lang/
  ↳ String;
       8: invokevirtual #7
  ↳   // Method java/io/PrintStream.
  ↳ println:(Ljava/lang/String;)V
      11: goto          47
      14: astore_1
      15: getstatic     #5
  ↳   // Field java/lang/System.out:Ljava/
  ↳ io/PrintStream;
      18: new           #8
  ↳   // class java/lang/StringBuilder
      21: dup
      22: invokespecial #9
  ↳   // Method java/lang/StringBuilder."<
  ↳ init>":()V
      25: ldc           #10
  ↳   // String incorrect month index:
      27: invokevirtual #11
  ↳   // Method java/lang/StringBuilder.
```

```
    ↳ append:(Ljava/lang/String;)Ljava/lang/↙
    ↳ StringBuilder;
      30: aload_1
      31: invokevirtual #12                        ↙
    ↳   // Method IncorrectMonthException.↙
    ↳ getIndex:()I
      34: invokevirtual #13                        ↙
    ↳   // Method java/lang/StringBuilder.↙
    ↳ append:(I)Ljava/lang/StringBuilder;
      37: invokevirtual #14                        ↙
    ↳   // Method java/lang/StringBuilder.↙
    ↳ toString:()Ljava/lang/String;
      40: invokevirtual #7                         ↙
    ↳   // Method java/io/PrintStream.↙
    ↳ println:(Ljava/lang/String;)V
      43: aload_1
      44: invokevirtual #15                        ↙
    ↳   // Method IncorrectMonthException.↙
    ↳ printStackTrace:()V
      47: return
    Exception table:
       from    to  target type
           0    11    14   Class ↙
    ↳ IncorrectMonthException
```

Listing 54.14:

```
   .catch java/io/FileNotFoundException ↙
   ↳ from met001_335 to met001_360\
using met001_360
   .catch java/io/FileNotFoundException ↙
   ↳ from met001_185 to met001_214\
using met001_214
```

```
    .catch java/io/FileNotFoundException ⤦
    ↳ from met001_181 to met001_192\
 using met001_195
    .catch java/io/FileNotFoundException ⤦
    ↳ from met001_155 to met001_176\
 using met001_176
    .catch java/io/FileNotFoundException ⤦
    ↳ from met001_83 to met001_129 using \
met001_129
    .catch java/io/FileNotFoundException ⤦
    ↳ from met001_42 to met001_66 using \
met001_69
    .catch java/io/FileNotFoundException ⤦
    ↳ from met001_begin to met001_37\
 using met001_37
```

54.16

:

Listing 54.15: test.java

```
public class test
{
        public static int a;
        private static int b;

        public test()
        {
            a=0;
            b=0;
```

```
        }
        public static void set_a (int input)
        {
                a=input;
        }
        public static int get_a ()
        {
                return a;
        }
        public static void set_b (int input)
        {
                b=input;
        }
        public static int get_b ()
        {
                return b;
        }
}
```

```
  public test();
    flags: ACC_PUBLIC
    Code:
      stack=1, locals=1, args_size=1
         0: aload_0
         1: invokespecial #1                  ↙
    ↪   // Method java/lang/Object."<init↙
    ↪ >":()V
         4: iconst_0
         5: putstatic     #2                  ↙
    ↪   // Field a:I
         8: iconst_0
         9: putstatic     #3                  ↙
```

```
↳    // Field b:I
     12: return
```

a:

```
public static void set_a(int);
  flags: ACC_PUBLIC, ACC_STATIC
  Code:
    stack=1, locals=1, args_size=1
       0: iload_0
       1: putstatic     #2                    ↙
↳    // Field a:I
       4: return
```

a:

```
public static int get_a();
  flags: ACC_PUBLIC, ACC_STATIC
  Code:
    stack=1, locals=0, args_size=0
       0: getstatic     #2                    ↙
↳    // Field a:I
       3: ireturn
```

b:

```
public static void set_b(int);
  flags: ACC_PUBLIC, ACC_STATIC
  Code:
    stack=1, locals=1, args_size=1
       0: iload_0
       1: putstatic     #3                    ↙
↳    // Field b:I
```

```
        4: return
```

b:

```
  public static int get_b();
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=1, locals=0, args_size=0
         0: getstatic     #3                  ↙
    ↘   // Field b:I
         3: ireturn
```

Listing 54.16: ex1.java

```
public class ex1
{
        public static void main(String[] ↙
    ↘ args)
        {
                test obj=new test();
                obj.set_a (1234);
                System.out.println(obj.a);
        }
}
```

```
  public static void main(java.lang.String↙
    ↘ []);
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=2, locals=2, args_size=1
         0: new           #2                  ↙
    ↘   // class test
```

```
         3: dup
         4: invokespecial #3                  ↙
    ↘    // Method test."<init>":()V
         7: astore_1
         8: aload_1
         9: pop
        10: sipush        1234
        13: invokestatic  #4                  ↙
    ↘    // Method test.set_a:(I)V
        16: getstatic     #5                  ↙
    ↘    // Field java/lang/System.out:Ljava/↙
    ↘ io/PrintStream;
        19: aload_1
        20: pop
        21: getstatic     #6                  ↙
    ↘    // Field test.a:I
        24: invokevirtual #7                  ↙
    ↘    // Method java/io/PrintStream.↙
    ↘ println:(I)V
        27: return
```

54.17

54.17.1

```
public class nag
{
        public static void nag_screen()
        {
                System.out.println("This ↙
    ↘ program is not registered");
```

```
        };
        public static void main(String[] args)
        {
                System.out.println("Greetings from the mega-software");
                nag_screen();
        }
}
```

```
              ; Segment type: Pure code
                .method public static nag_screen()V
                .limit stack 2
                .line 4
178 000 002 |    getstatic java/lang/System.out Ljava/io/PrintStream;  ; CODE XREF: main+8↓P
018 003          ldc "This program is not registered"
182 000 004      invokevirtual java/io/PrintStream.println(Ljava/lang/String;)V
                .line 5
177              return
??? ??? ???+    .end method
??? ??? ???+
???           ; ---------------------------------------------------------------------------


              ; Segment type: Pure code
                .method public static main([Ljava/lang/String;)V
                .limit stack 2
                .limit locals 1
                .line 8
178 000 002      getstatic java/lang/System.out Ljava/io/PrintStream;
018 005          ldc "Greetings from the mega-software"
182 000 004      invokevirtual java/io/PrintStream.println(Ljava/lang/String;)V
                .line 9
184 000 006      invokestatic nag.nag_screen()V
                .line 10
177              return
```

Figura 54.1: IDA

```
                 ; Segment type: Pure code
                   .method public static nag_screen()V
                   .limit stack 2
                   .line 4

                 nag_screen:                              ; CODE XREF: main+8↓P
177                  return
000                  0 ; 0x00
002                  2 ; 0x02
018 003              ldc "This program is not registered"
182 000 004          invokevirtual java/io/PrintStream.println(Ljava/lang/String;)V
                   .line 5
177                  return
??? ??? ???+       .end method
??? ??? ???+
???              ; ---------------------------------------------------------------------------
```

Figura 54.2: IDA

(JRE 1.7):

```
Exception in thread "main" java.lang.VerifyError: Expecting a stack map frame
Exception Details:
  Location:
    nag.nag_screen()V @1: nop
  Reason:
    Error exists in the bytecode
  Bytecode:
    0000000: b100 0212 03b6 0004 b1

        at java.lang.Class.getDeclaredMethods0(Native Method)
        at java.lang.Class.privateGetDeclaredMethods(Class.java:2615)
        at java.lang.Class.getMethod0(Class.java:2856)
        at java.lang.Class.getMethod(Class.java:1668)
```

 CAPÍTULO 54. JAVA

```
        at sun.launcher.LauncherHelper.getMainMethod(LauncherHelper.java:494)
        at sun.launcher.LauncherHelper.checkAndLoadMain(LauncherHelper.java:486)
```

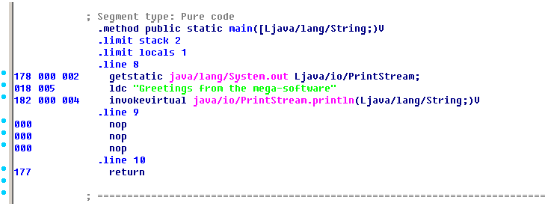


Figura 54.3: IDA

!

54.17.2

:

```
public class password
{
        public static void main(String[] args)
        {
                System.out.println("Please enter the password");
                String input = System.console().readLine();
```

```
                if (input.equals("secret"))
                        System.out.println("password is correct");
                else
                        System.out.println("password is not correct");
        }
}
```

IDA:

Figura 54.4: IDA

 CAPÍTULO 54. JAVA

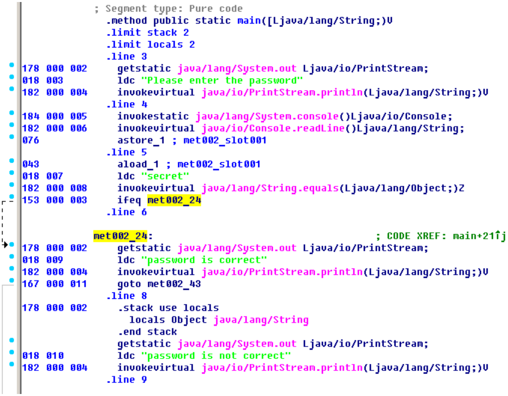


Figura 54.5: IDA

(JRE 1.7):

```
Exception in thread "main" java.lang.↙
    ↳ VerifyError: Expecting a stackmap ↙
    ↳ frame at branch target 24
Exception Details:
  Location:
    password.main([Ljava/lang/String;)V @21:↙
    ↳  ifeq
  Reason:
    Expected stackmap frame at this location↙
    ↳ .
  Bytecode:
    0000000: b200 0212 03b6 0004 b800 05b6 ↙
    ↳ 0006 4c2b
    0000010: 1207 b600 0899 0003 b200 0212 ↙
```

```
    ↳ 09b6 0004
    0000020: a700 0bb2 0002 120a b600 04b1
  Stackmap Table:
    append_frame(@35,Object[#20])
    same_frame(@43)

      at java.lang.Class.↙
    ↳ getDeclaredMethods0(Native Method)
      at java.lang.Class.↙
    ↳ privateGetDeclaredMethods(Class.java↙
    ↳ :2615)
      at java.lang.Class.getMethod0(Class.↙
    ↳ java:2856)
      at java.lang.Class.getMethod(Class.↙
    ↳ java:1668)
      at sun.launcher.LauncherHelper.↙
    ↳ getMainMethod(LauncherHelper.java:494)
      at sun.launcher.LauncherHelper.↙
    ↳ checkAndLoadMain(LauncherHelper.java↙
    ↳ :486)
```

JRE 1.6.

 CAPÍTULO 54. JAVA

```
                ; Segment type: Pure code
                .method public static main([Ljava/lang/String;)V
                .limit stack 2
                .limit locals 2
                .line 3
178 000 002       getstatic java/lang/System.out Ljava/io/PrintStream;
018 003           ldc "Please enter the password"
182 000 004       invokevirtual java/io/PrintStream.println(Ljava/lang/String;)V
                .line 4
184 000 005       invokestatic java/lang/System.console()Ljava/io/Console;
182 000 006       invokevirtual java/io/Console.readLine()Ljava/lang/String;
076               astore_1 ; met002_slot001
                .line 5
004               iconst_1
000               nop
000               nop
000               nop
000               nop
000               nop
153 000 014       ifeq met002_35
                .line 6
178 000 002       getstatic java/lang/System.out Ljava/io/PrintStream;
018 009           ldc "password is correct"
182 000 004       invokevirtual java/io/PrintStream.println(Ljava/lang/String;)V
167 000 011       goto met002_43
                .line 8

              met002_35:                              ; CODE XREF: main+21↑j
178 000 002       .stack use locals
                    locals Object java/lang/String
                  .end stack
```

Figura 54.6: IDA

.

54.18

- .
- .
- .
- .

Parte V

http://www.google.com/search?q=CsDecomprLZC

Capítulo 55

55.1 Microsoft Visual C++

:

		CL.EXE	
6	6.0	12.00	msvcrt.dll, msvcp60.dll
.NET (2002)	7.0	13.00	msvcr70.dll, msvcp70.dll
.NET 2003	7.1	13.10	msvcr71.dll, msvcp71.dll
2005	8.0	14.00	msvcr80.dll, msvcp80.dll
2008	9.0	15.00	msvcr90.dll, msvcp90.dll
2010	10.0	16.00	msvcr100.dll, msvcp100.dll
2012	11.0	17.00	msvcr110.dll, msvcp110.dll
2013	12.0	18.00	msvcr120.dll, msvcp120.dll

msvcp*.dll .

55.1.1 Name mangling

?.

name mangling : 51.1.1 on page 1040.

55.2 GCC

Cygwin y MinGW.

55.2.1 Name mangling

_Z.

name mangling : 51.1.1 on page 1040.

55.2.2 Cygwin

cygwin1.dll .

55.2.3 MinGW

msvcrt.dll .

55.3 Intel FORTRAN

libifcoremd.dll, libifportmd.dll y libiomp5md.dll (OpenMP) .

libifcoremd.dll `for_`, FORTRAN.

55.4 Watcom, OpenWatcom

55.4.1 Name mangling

W.

:

```
W?method$_class$n__v
```

55.5 Borland

:

```
@TApplication@IdleAction$qv
@TApplication@ProcessMDIAccels$qp6tagMSG
@TModule@$bctr$qpcpvt1
@TModule@$bdtr$qv
@TModule@ValidWindow$qp14TWindowsObject
@TrueColorTo8BitN$qpviiiiiit1iiiiii
@TrueColorTo16BitN$qpviiiiiit1iiiiii
@DIB24BitTo8BitBitmap$qpviiiiiit1iiiii
@TrueBitmap@$bctr$qpcl
@TrueBitmap@$bctr$qpvl
@TrueBitmap@$bctr$qiilll
```

@ .

Spanish text placeholder.

Borland Visual Component Libraries (VCL) , vcl50.dll, rtl60.dll.

: BORLNDMM.DLL.

55.5.1 Delphi

«Boolean» .

:

```
00000400  04 10 40 00 03 07 42 6f  6f 6c 65 ⤦
    ↘ 61 6e 01 00 00  |..@...Boolean...|
00000410  00 00 01 00 00 00 00 10  40 00 05 ⤦
    ↘ 46 61 6c 73 65  |........@..False|
00000420  04 54 72 75 65 8d 40 00  2c 10 40 ⤦
    ↘ 00 09 08 57 69  |.True.@.,.@...Wi|
00000430  64 65 43 68 61 72 03 00  00 00 00 ⤦
    ↘ ff ff 00 00 90  |deChar..........|
00000440  44 10 40 00 02 04 43 68  61 72 01 ⤦
    ↘ 00 00 00 00 ff  |D.@...Char......|
00000450  00 00 00 90 58 10 40 00  01 08 53 ⤦
    ↘ 6d 61 6c 6c 69  |....X.@...Smalli|
00000460  6e 74 02 00 80 ff ff ff  7f 00 00 ⤦
    ↘ 90 70 10 40 00  |nt..........p.@.|
00000470  01 07 49 6e 74 65 67 65  72 04 00 ⤦
    ↘ 00 00 80 ff ff  |..Integer.......|
00000480  ff 7f 8b c0 88 10 40 00  01 04 42 ⤦
    ↘ 79 74 65 01 00  |......@...Byte..|
00000490  00 00 00 ff 00 00 00 90  9c 10 40 ⤦
    ↘ 00 01 04 57 6f  |..........@...Wo|
000004a0  72 64 03 00 00 00 00 ff  ff 00 00 ⤦
    ↘ 90 b0 10 40 00  |rd............@.|
000004b0  01 08 43 61 72 64 69 6e  61 6c 05 ⤦
    ↘ 00 00 00 00 ff  |..Cardinal......|
000004c0  ff ff ff 90 c8 10 40 00  10 05 49 ⤦
    ↘ 6e 74 36 34 00  |......@...Int64.|
000004d0  00 00 00 00 00 00 80 ff  ff ff ff ⤦
    ↘ ff ff ff 7f 90  |................|
```

```
000004e0  e4 10 40 00 04 08 45 78  74 65 6e ⤦
    ↳ 64 65 64 02 90  |..@...Extended..|
000004f0  f4 10 40 00 04 06 44 6f  75 62 6c ⤦
    ↳ 65 01 8d 40 00  |..@...Double..@.|
00000500  04 11 40 00 04 08 43 75  72 72 65 ⤦
    ↳ 6e 63 79 04 90  |..@...Currency..|
00000510  14 11 40 00 0a 06 73 74  72 69 6e ⤦
    ↳ 67 20 11 40 00  |..@...string .@.|
00000520  0b 0a 57 69 64 65 53 74  72 69 6e ⤦
    ↳ 67 30 11 40 00  |..WideString0.@.|
00000530  0c 07 56 61 72 69 61 6e  74 8d 40 ⤦
    ↳ 00 40 11 40 00  |..Variant.@.@.@.|
00000540  0c 0a 4f 6c 65 56 61 72  69 61 6e ⤦
    ↳ 74 98 11 40 00  |..OleVariant..@.|
00000550  00 00 00 00 00 00 00 00  00 00 00 ⤦
    ↳ 00 00 00 00 00  |................|
00000560  00 00 00 00 00 00 00 00  00 00 00 ⤦
    ↳ 00 98 11 40 00  |.............@.|
00000570  04 00 00 00 00 00 00 00  18 4d 40 ⤦
    ↳ 00 24 4d 40 00  |.........M@.$M@.|
00000580  28 4d 40 00 2c 4d 40 00  20 4d 40 ⤦
    ↳ 00 68 4a 40 00  |(M@.,M@. M@.hJ@.|
00000590  84 4a 40 00 c0 4a 40 00  07 54 4f ⤦
    ↳ 62 6a 65 63 74  |.J@..J@..TObject|
000005a0  a4 11 40 00 07 07 54 4f  62 6a 65 ⤦
    ↳ 63 74 98 11 40  |..@...TObject..@|
000005b0  00 00 00 00 00 00 00 06  53 79 73 ⤦
    ↳ 74 65 6d 00 00  |........System..|
000005c0  c4 11 40 00 0f 0a 49 49  6e 74 65 ⤦
    ↳ 72 66 61 63 65  |..@...IInterface|
000005d0  00 00 00 00 01 00 00 00  00 00 00 ⤦
    ↳ 00 00 c0 00 00  |................|
000005e0  00 00 00 00 46 06 53 79  73 74 65 ⤦
```

```
    ↳ 6d 03 00 ff ff  |....F.System....|
000005f0  f4 11 40 00 0f 09 49 44  69 73 70 ⤦
    ↳ 61 74 63 68 c0  |..@...IDispatch.|
00000600  11 40 00 01 00 04 02 00  00 00 00 ⤦
    ↳ 00 c0 00 00 00  |.@..............|
00000610  00 00 00 46 06 53 79 73  74 65 6d ⤦
    ↳ 04 00 ff ff 90  |...F.System.....|
00000620  cc 83 44 24 04 f8 e9 51  6c 00 00 ⤦
    ↳ 83 44 24 04 f8  |..D$...Ql...D$..|
00000630  e9 6f 6c 00 00 83 44 24  04 f8 e9 ⤦
    ↳ 79 6c 00 00 cc  |.ol...D$...yl...|
00000640  cc 21 12 40 00 2b 12 40  00 35 12 ⤦
    ↳ 40 00 01 00 00  |.!.@.+.@.5.@....|
00000650  00 00 00 00 00 00 00 00  00 c0 00 ⤦
    ↳ 00 00 00 00 00  |................|
00000660  46 41 12 40 00 08 00 00  00 00 00 ⤦
    ↳ 00 00 8d 40 00  |FA.@..........@.|
00000670  bc 12 40 00 4d 12 40 00  00 00 00 ⤦
    ↳ 00 00 00 00 00  |..@.M.@.........|
00000680  00 00 00 00 00 00 00 00  00 00 00 ⤦
    ↳ 00 00 00 00 00  |................|
00000690  bc 12 40 00 0c 00 00 00  4c 11 40 ⤦
    ↳ 00 18 4d 40 00  |..@.....L.@..M@.|
000006a0  50 7e 40 00 5c 7e 40 00  2c 4d 40 ⤦
    ↳ 00 20 4d 40 00  |P~@.\~@.,M@. M@.|
000006b0  6c 7e 40 00 84 4a 40 00  c0 4a 40 ⤦
    ↳ 00 11 54 49 6e  |l~@..J@..J@..TIn|
000006c0  74 65 72 66 61 63 65 64  4f 62 6a ⤦
    ↳ 65 63 74 8b c0  |terfacedObject..|
000006d0  d4 12 40 00 07 11 54 49  6e 74 65 ⤦
    ↳ 72 66 61 63 65  |..@...TInterface|
000006e0  64 4f 62 6a 65 63 74 bc  12 40 00 ⤦
    ↳ a0 11 40 00 00  |dObject..@...@..|
```

```
000006f0  00 06 53 79 73 74 65 6d  00 00 8b ⤦
    ↳ c0 00 13 40 00  |..System......@.|
00000700  11 0b 54 42 6f 75 6e 64  41 72 72 ⤦
    ↳ 61 79 04 00 00  |..TBoundArray...|
00000710  00 00 00 00 00 03 00 00  00 6c 10 ⤦
    ↳ 40 00 06 53 79  |.........l.@..Sy|
00000720  73 74 65 6d 28 13 40 00  04 09 54 ⤦
    ↳ 44 61 74 65 54  |stem(.@...TDateT|
00000730  69 6d 65 01 ff 25 48 e0  c4 00 8b ⤦
    ↳ c0 ff 25 44 e0  |ime..%H......%D.|
```

00 00 00 00, 32 13 8B C0 o FF FF FF FF.

55.6

- vcomp*.dll.

Capítulo 56

(win32)

.

: Process Monitor[1] SysInternals.

Wireshark[2].

.

: 75.

blog.yurichev.com.

[1] http://go.yurichev.com/17301
[2] http://go.yurichev.com/17303

56.1 Windows API

. . CRT.

- (advapi32.dll): RegEnumKeyEx[3] [4], RegEnumValue[5] [4], RegGetValue[6] [4], RegOpenKeyEx[7] [4], RegQueryValueEx[8] [4].
- (kernel32.dll): GetPrivateProfileString [9] [4].
- (user32.dll): MessageBox [10] [4], MessageBoxEx [11] [4], SetDlgItemText [12] [4], GetDlgItemText [13] [4].
- (68.2.8 on page 1358) : (user32.dll): LoadMenu [14] [4].
- (ws2_32.dll): WSARecv [15], WSASend [16].
- (kernel32.dll): CreateFile [17] [4], ReadFile [18], ReadFileEx

[3] MSDN
[4]
[5] MSDN
[6] MSDN
[7] MSDN
[8] MSDN
[9] MSDN
[10] MSDN
[11] MSDN
[12] MSDN
[13] MSDN
[14] MSDN
[15] MSDN
[16] MSDN
[17] MSDN
[18] MSDN

[19], WriteFile [20], WriteFileEx [21].

- Internet (wininet.dll): WinHttpOpen [22].
- (wintrust.dll): WinVerifyTrust [23].
- (msvcr*.dll): assert, itoa, ltoa, open, printf, read, strcmp, atol, atoi, fopen, fread, fwrite, memcmp, rand, strlen, strstr, strchr.

56.2 tracer:

```
--one-time-INT3-bp:somedll.dll!.*
```

```
--one-time-INT3-bp:somedll.dll!xml.*
```

Y hay muchas funciones.

```
tracer -l:uptime.exe --one-time-INT3-bp:↙
    ↘ cygwin1.dll!.*
```

:

[19]MSDN
[20]MSDN
[21]MSDN
[22]MSDN
[23]MSDN

```
One-time INT3 breakpoint: cygwin1.dll!__main↙
    ↳ (called from uptime.exe!OEP+0x6d (0↙
    ↳ x40106d))
One-time INT3 breakpoint: cygwin1.dll!↙
    ↳ _geteuid32 (called from uptime.exe!OEP↙
    ↳ +0xba3 (0x401ba3))
One-time INT3 breakpoint: cygwin1.dll!↙
    ↳ _getuid32 (called from uptime.exe!OEP↙
    ↳ +0xbaa (0x401baa))
One-time INT3 breakpoint: cygwin1.dll!↙
    ↳ _getegid32 (called from uptime.exe!OEP↙
    ↳ +0xcb7 (0x401cb7))
One-time INT3 breakpoint: cygwin1.dll!↙
    ↳ _getgid32 (called from uptime.exe!OEP↙
    ↳ +0xcbe (0x401cbe))
One-time INT3 breakpoint: cygwin1.dll!↙
    ↳ sysconf (called from uptime.exe!OEP+0↙
    ↳ x735 (0x401735))
One-time INT3 breakpoint: cygwin1.dll!↙
    ↳ setlocale (called from uptime.exe!OEP↙
    ↳ +0x7b2 (0x4017b2))
One-time INT3 breakpoint: cygwin1.dll!↙
    ↳ _open64 (called from uptime.exe!OEP+0↙
    ↳ x994 (0x401994))
One-time INT3 breakpoint: cygwin1.dll!↙
    ↳ _lseek64 (called from uptime.exe!OEP+0↙
    ↳ x7ea (0x4017ea))
One-time INT3 breakpoint: cygwin1.dll!read (↙
    ↳ called from uptime.exe!OEP+0x809 (0↙
    ↳ x401809))
One-time INT3 breakpoint: cygwin1.dll!sscanf↙
    ↳ (called from uptime.exe!OEP+0x839 (0↙
```

```
    ↳ x401839))
One-time INT3 breakpoint: cygwin1.dll!uname ↙
    ↳ (called from uptime.exe!OEP+0x139 (0↙
    ↳ x401139))
One-time INT3 breakpoint: cygwin1.dll!time (↙
    ↳ called from uptime.exe!OEP+0x22e (0↙
    ↳ x40122e))
One-time INT3 breakpoint: cygwin1.dll!↙
    ↳ localtime (called from uptime.exe!OEP↙
    ↳ +0x236 (0x401236))
One-time INT3 breakpoint: cygwin1.dll!↙
    ↳ sprintf (called from uptime.exe!OEP+0↙
    ↳ x25a (0x40125a))
One-time INT3 breakpoint: cygwin1.dll!↙
    ↳ setutent (called from uptime.exe!OEP+0↙
    ↳ x3b1 (0x4013b1))
One-time INT3 breakpoint: cygwin1.dll!↙
    ↳ getutent (called from uptime.exe!OEP+0↙
    ↳ x3c5 (0x4013c5))
One-time INT3 breakpoint: cygwin1.dll!↙
    ↳ endutent (called from uptime.exe!OEP+0↙
    ↳ x3e6 (0x4013e6))
One-time INT3 breakpoint: cygwin1.dll!puts (↙
    ↳ called from uptime.exe!OEP+0x4c3 (0↙
    ↳ x4014c3))
```

Capítulo 57

57.1

57.1.1 C/C++

(ASCIIZ-).

. [Rit79] :

> A minor difference was that the unit of I/O was the word, not the byte, because the PDP-7 was a word-addressed machine. In practice this meant merely that all programs dealing with character streams ignored null characters, because null was used to pad a file to an even number of characters.

:

```
int main()
{
        printf ("Hello, world!\n");
};
```

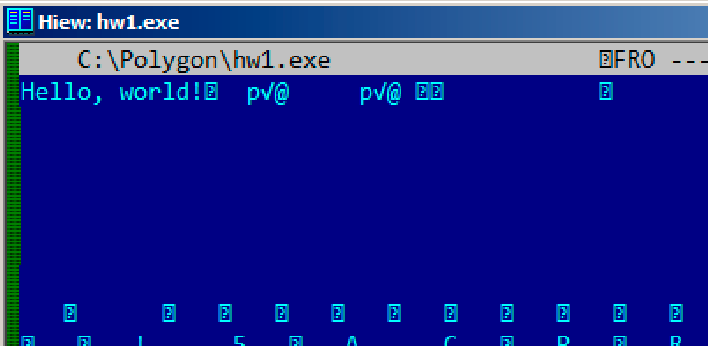


Figura 57.1: Hiew

57.1.2 Borland Delphi

:

Listing 57.1: Delphi

```
CODE:00518AC8                dd 19h
CODE:00518ACC aLoading___Plea db 'Loading...↙
    ↘ , please wait.',0
```

```
...

CODE:00518AFC                  dd 10h
CODE:00518B00 aPreparingRun__ db 'Preparing ⤦
    ↳ run...',0
```

57.1.3 Unicode

...

: UTF-8 () y UTF-16LE (Windows).

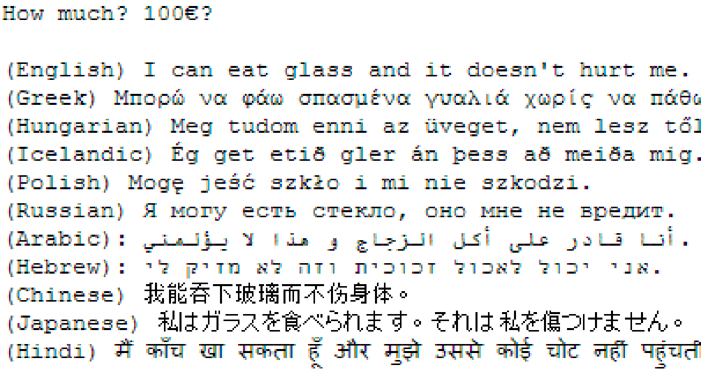
UTF-8

UTF-8 ...

1 :

[1]: http://go.yurichev.com/17304

CAPÍTULO 57.



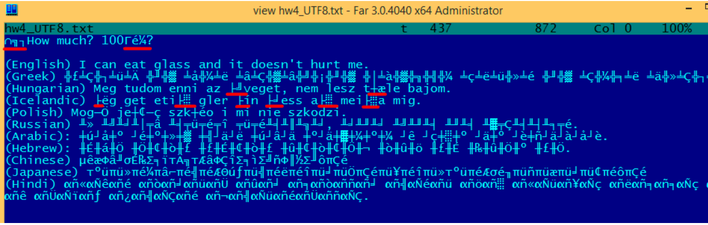


Figura 57.2: FAR: UTF-8

. . . .

. BOM[2] .

[2] Byte order mark

UTF-16LE

-A y -W. (*wide*). .

:

```
int wmain()
{
        wprintf (L"Hello, world!\n");
};
```

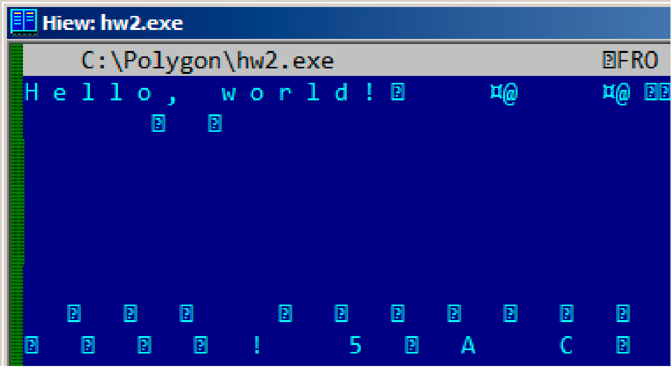


Figura 57.3: Hiew

:

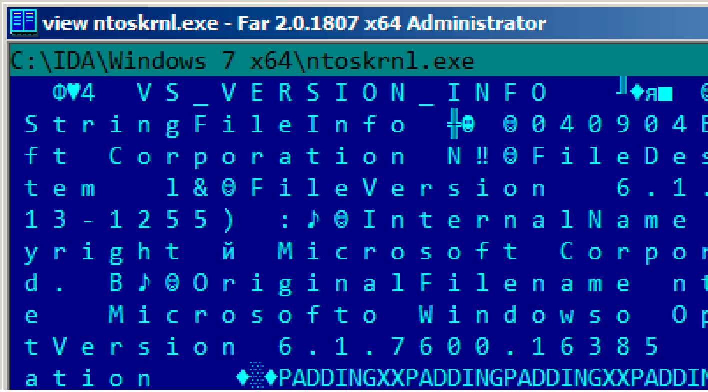


Figura 57.4: Hiew

:

```
.data:0040E000 aHelloWorld:
.data:0040E000                 unicode 0, <Hello, world!>
.data:0040E000                 dw 0Ah, 0
```

:

 CAPÍTULO 57.

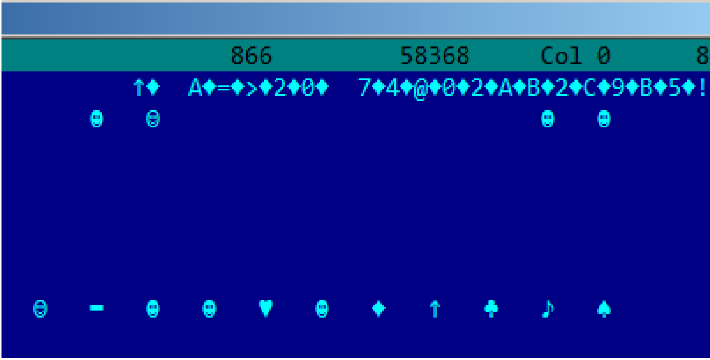


Figura 57.5: Hiew: UTF-16LE

. [3]. 0x400-0x4FF.

. .

[3]wikipedia

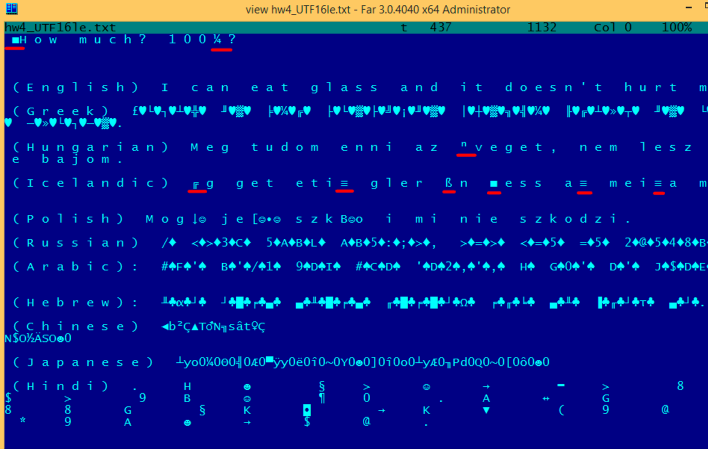


Figura 57.6: FAR: UTF-16LE

...

57.1.4 Base64

.


```
AVjbbVSVfcUMu1xvjaMgjNtueRwBbxnyJw8dpGnLW8ZW8aKG3
    ↳ +qEJAp9lAOuWs=
```

```
WVjbbVSVfcUMu1xvjaMgjNtueRwBbxnyJw8dpGnLW8ZW8aKG3
    ↳ +qEJAp9lAOuQ==
```

57.2

[4].

: blog.yurichev.com.

: 78.2 on page 1486.

57.3

[5].

http://192.168.0.1/userRpmNatDebugRpm26525557
start_art.html.

base64.

[4] blog.yurichev.com

[5] http://sekurak.pl/tp-link-httptftp-backdoor/

Capítulo 58

Listing 58.1:

```
.text:107D4B29 mov  dx, [ecx+42h]
.text:107D4B2D cmp  edx, 1
.text:107D4B30 jz   short loc_107D4B4A
.text:107D4B32 push 1ECh
.text:107D4B37 push offset aWrite_c ; "write.c"
.text:107D4B3C push offset aTdTd_planarcon ; "td->td_planarconfig == PLANARCONFIG_CON"...
.text:107D4B41 call ds:_assert

...

.text:107D52CA mov  edx, [ebp-4]
.text:107D52CD and  edx, 3
.text:107D52D0 test edx, edx
.text:107D52D2 jz   short loc_107D52E9
```

```
.text:107D52D4 push 58h
.text:107D52D6 push offset aDumpmode_c ; "
    dumpmode.c"
.text:107D52DB push offset aN30      ; "(n & 
    3) == 0"
.text:107D52E0 call ds:_assert

...

.text:107D6759 mov  cx, [eax+6]
.text:107D675D cmp  ecx, 0Ch
.text:107D6760 jle  short loc_107D677A
.text:107D6762 push 2D8h
.text:107D6767 push offset aLzw_c    ; "lzw.c
    "
.text:107D676C push offset aSpLzw_nbitsBit ;
     "sp->lzw_nbits <= BITS_MAX"
.text:107D6771 call ds:_assert
```

Capítulo 59

10, 100, 1000, .

: 10=0xA, 100=0x64, 1000=0x3E8, 10000=0x2710.

0xAAAAAAAA (10101010101010101010101010101010) y 0x55555555 (01010101010101010101010101010101) . 0x55AA , MBR[1], y en ROM[2] .

```
var int h0 := 0x67452301
var int h1 := 0xEFCDAB89
var int h2 := 0x98BADCFE
var int h3 := 0x10325476
```

Si encuentras estas cuatro constantes usadas en el código en fila, es muy altamente probable que esta función esté relacionada con MD5.

[1] Master Boot Record
[2] Memoria de solo lectura

:

Listing 59.1: linux/lib/crc16.c

```
/** CRC table for the CRC-16. The poly is 0
    ↳ x8005 (x^16 + x^15 + x^2 + 1) */
u16 const crc16_table[256] = {
        0x0000, 0xC0C1, 0xC181, 0x0140, 0
    ↳ xC301, 0x03C0, 0x0280, 0xC241,
        0xC601, 0x06C0, 0x0780, 0xC741, 0
    ↳ x0500, 0xC5C1, 0xC481, 0x0440,
        0xCC01, 0x0CC0, 0x0D80, 0xCD41, 0
    ↳ x0F00, 0xCFC1, 0xCE81, 0x0E40,
        ...
```

: 37 on page 901.

59.1 Magic numbers

«MZ»[3].

```
cmp [buf], 0x6468544D ; "MThd"
jnz _error_not_a_MIDI_file
```

59.1.1 DHCP

DhcpExtractOptionsForValidation() y DhcpExtrac

[3] wikipedia

Listing 59.2: dhcpcore.dll (Windows 7 x64)

```
.rdata:000007FF6483CBE8 dword_7FF6483CBE8 dd↙
    ↘ 63538263h           ; DATA XREF: ↙
    ↘ DhcpExtractOptionsForValidation+79
.rdata:000007FF6483CBEC dword_7FF6483CBEC dd↙
    ↘ 63538263h           ; DATA XREF: ↙
    ↘ DhcpExtractFullOptions+97
```

Listing 59.3: dhcpcore.dll (Windows 7 x64)

```
.text:000007FF6480875F  mov     eax, [rsi]
.text:000007FF64808761  cmp     eax, cs:↙
    ↘ dword_7FF6483CBE8
.text:000007FF64808767  jnz     ↙
    ↘ loc_7FF64817179
```

Listing 59.4: dhcpcore.dll (Windows 7 x64)

```
.text:000007FF648082C7  mov     eax, [r12]
.text:000007FF648082CB  cmp     eax, cs:↙
    ↘ dword_7FF6483CBEC
.text:000007FF648082D1  jnz     ↙
    ↘ loc_7FF648173AF
```

59.2

binary grep[4].

[4] GitHub

Capítulo 60

se puede intentar verificar cada una manualmente con un depurador.

operandoobviamente, no nos sirven):

```
cat EXCEL.lst | grep fdiv | grep -v dbl_ > ↙
    ↳ EXCEL.fdiv
```

...entonces vemos que hay 144 de ellas.

Spanish text placeholder

```
.text:3011E919 DC 33                                   ↙
    ↳           fdiv    qword ptr [ebx]
```

```
PID=13944|TID=28744|(0) 0x2f64e919 (Excel.↙
    ↳ exe!BASE+0x11e919)
EAX=0x02088006 EBX=0x02088018 ECX=0x00000001↙
    ↳  EDX=0x00000001
ESI=0x02088000 EDI=0x00544804 EBP=0x0274FA3C↙
    ↳  ESP=0x0274F9F8
EIP=0x2F64E919
FLAGS=PF IF
FPU ControlWord=IC RC=NEAR PC=64bits PM UM ↙
    ↳ OM ZM DM IM
FPU StatusWord=
FPU ST(0): 1.000000
```

[EBX].

```
.text:3011E91B DD 1E                                   ↙
    ↳           fstp    qword ptr [esi]
```

```
PID=32852|TID=36488|(0) 0x2f40e91b (Excel.↙
    ↳ exe!BASE+0x11e91b)
EAX=0x00598006 EBX=0x00598018 ECX=0x00000001↙
    ↳  EDX=0x00000001
ESI=0x00598000 EDI=0x00294804 EBP=0x026CF93C↙
    ↳  ESP=0x026CF8F8
EIP=0x2F40E91B
FLAGS=PF IF
```

```
FPU ControlWord=IC RC=NEAR PC=64bits PM UM ↙
    ↘ OM ZM DM IM
FPU StatusWord=C1 P
FPU ST(0): 0.333333
```

También como broma práctica, podemos modificarlo sobre la marcha:

```
tracer -l:excel.exe bpx=excel.exe!BASE+0↙
    ↘ x11E91B,set(st0,666)
```

```
PID=36540|TID=24056|(0) 0x2f40e91b (Excel.↙
    ↘ exe!BASE+0x11e91b)
EAX=0x00680006 EBX=0x00680018 ECX=0x00000001↙
    ↘  EDX=0x00000001
ESI=0x00680000 EDI=0x00395404 EBP=0x0290FD9C↙
    ↘  ESP=0x0290FD58
EIP=0x2F40E91B
FLAGS=PF IF
FPU ControlWord=IC RC=NEAR PC=64bits PM UM ↙
    ↘ OM ZM DM IM
FPU StatusWord=C1 P
FPU ST(0): 0.333333
Set ST0 register to 666.000000
```

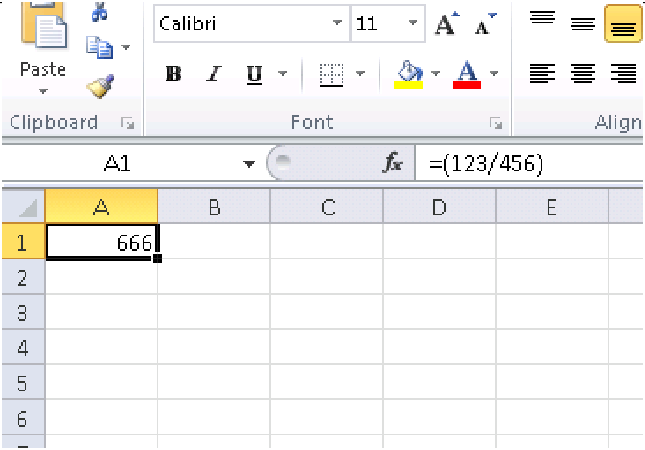


Figura 60.1:

```
tracer.exe -l:excel.exe bpx=excel.exe!BASE+0x1B7FCC,set(st0,666)
```

Capítulo 61

61.1

XOR op, op (, XOR EAX, EAX) «». . Spanish text placeholder.

:

```
gawk -e '$2=="xor" { tmp=substr($3, 0, 
    ↳ length($3)-1); if (tmp!=$4) if($4!="
    ↳ esp") if ($4!="ebp") { print $1, $2, 
    ↳ tmp, ",", $4 } }' filename.lst
```

(49 on page 1021).

61.2

Los compiladores modernos no emiten las instrucciones LOOP y RCL. Por otro lado, estas instrucciones son bien conocidas por los codificadores que les gusta programar directamente en lenguaje ensamblador. Si las ves, se puede decir que hay una alta probabilidad de que este fragmento de código haya sido escrito a mano. Tales instrucciones están marcadas como (M) en la lista de instrucciones en este apéndice: A.6 on page 1877.

.

Windows 2003 (ntoskrnl.exe):

```
MultiplyTest proc near                        ; CODE ↙
    ↳ XREF: Get386Stepping
             xor     cx, cx
loc_620555:                                   ; CODE ↙
    ↳ XREF: MultiplyTest+E
             push    cx
             call    Multiply
             pop     cx
             jb      short locret_620563
             loop    loc_620555
             clc
locret_620563:                                ; CODE ↙
    ↳ XREF: MultiplyTest+C
             retn
MultiplyTest endp
```

```
Multiply     proc near                    ; CODE ↙
    ↳ XREF: MultiplyTest+5
             mov     ecx, 81h
             mov     eax, 417A000h
             mul     ecx
             cmp     edx, 2
             stc
             jnz     short locret_62057F
             cmp     eax, 0FE7A000h
             stc
             jnz     short locret_62057F
             clc
locret_62057F:                            ; CODE ↙
    ↳ XREF: Multiply+10
                                          ; ↙
    ↳ Multiply+18
             retn
Multiply     endp
```

WRK[1] v1.2 *WRK-v1.2\base\ntos\ke\i386\cpu.asm*.

[1]Windows Research Kernel

Capítulo 62

:

```
0x150bf66 (_kziaia+0x14), e=        1 [MOV ↙
    ↘ EBX, [EBP+8]] [EBP+8]=0xf59c934
0x150bf69 (_kziaia+0x17), e=        1 [MOV ↙
    ↘ EDX, [69AEB08h]] [69AEB08h]=0
0x150bf6f (_kziaia+0x1d), e=        1 [FS: ↙
    ↘ MOV EAX, [2Ch]]
0x150bf75 (_kziaia+0x23), e=        1 [MOV ↙
    ↘ ECX, [EAX+EDX*4]] [EAX+EDX*4]=0↙
    ↘ xf1ac360
0x150bf78 (_kziaia+0x26), e=        1 [MOV [↙
    ↘ EBP-4], ECX] ECX=0xf1ac360
```

Capítulo 63

63.1

63.2 C++

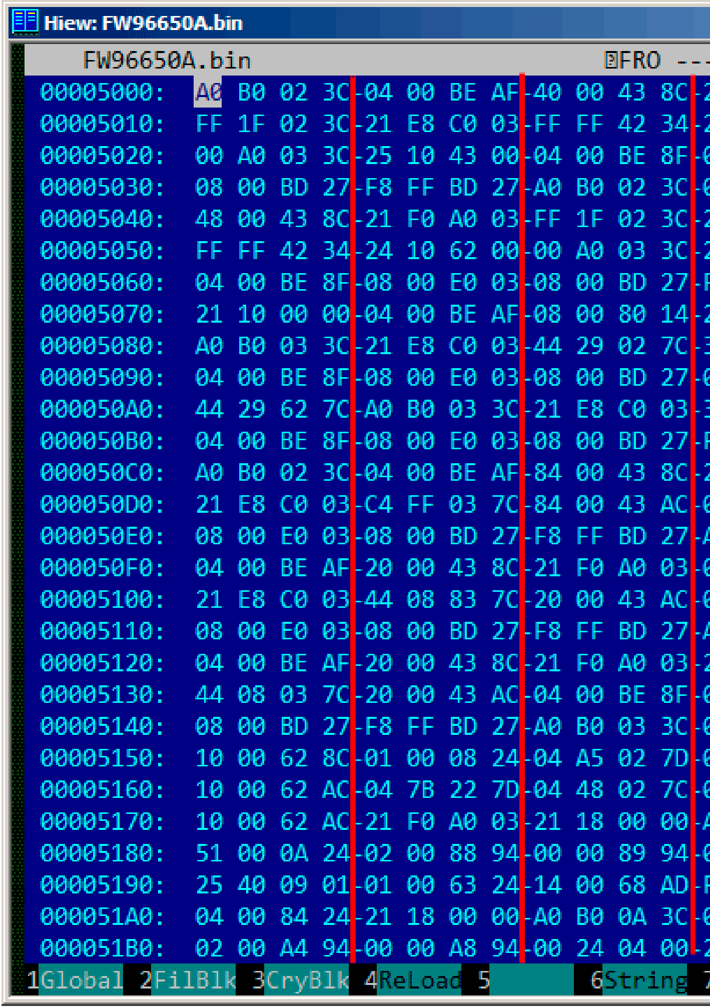
RTTI[1] (51.1.5 on page 1069)-.

63.3

:

[1] Run-time type information

 CAPÍTULO 63.


Hiew: FW96650A.bin
FW96650A.bin
FRO
00005000: A0 B0 02 3C-04 00 BE AF-40 00 43 8C
00005010: FF 1F 02 3C-21 E8 C0 03-FF FF 42 34
00005020: 00 A0 03 3C-25 10 43 00-04 00 BE 8F
00005030: 08 00 BD 27-F8 FF BD 27-A0 B0 02 3C
00005040: 48 00 43 8C-21 F0 A0 03-FF 1F 02 3C
00005050: FF FF 42 34-24 10 62 00-00 A0 03 3C
00005060: 04 00 BE 8F-08 00 E0 03-08 00 BD 27
00005070: 21 10 00 00-04 00 BE AF-08 00 80 14
00005080: A0 B0 03 3C-21 E8 C0 03-44 29 02 7C
00005090: 04 00 BE 8F-08 00 E0 03-08 00 BD 27
000050A0: 44 29 62 7C-A0 B0 03 3C-21 E8 C0 03
000050B0: 04 00 BE 8F-08 00 E0 03-08 00 BD 27
000050C0: A0 B0 02 3C-04 00 BE AF-84 00 43 8C
000050D0: 21 E8 C0 03-C4 FF 03 7C-84 00 43 AC
000050E0: 08 00 E0 03-08 00 BD 27-F8 FF BD 27
000050F0: 04 00 BE AF-20 00 43 8C-21 F0 A0 03
00005100: 21 E8 C0 03-44 08 83 7C-20 00 43 AC
00005110: 08 00 E0 03-08 00 BD 27-F8 FF BD 27
00005120: 04 00 BE AF-20 00 43 8C-21 F0 A0 03
00005130: 44 08 03 7C-20 00 43 AC-04 00 BE 8F
00005140: 08 00 BD 27-F8 FF BD 27-A0 B0 03 3C
00005150: 10 00 62 8C-01 00 08 24-04 A5 02 7D
00005160: 10 00 62 AC-04 7B 22 7D-04 48 02 7C
00005170: 10 00 62 AC-21 F0 A0 03-21 18 00 00
00005180: 51 00 0A 24-02 00 88 94-00 00 89 94
00005190: 25 40 09 01-01 00 63 24-14 00 68 AD
000051A0: 04 00 84 24-21 18 00 00-A0 B0 0A 3C
000051B0: 02 00 A4 94-00 00 A8 94-00 24 04 00
1Global 2FilBlk 3CryBlk 4ReLoad 5 6String

: 86 on page 1722.

63.4

wikipedia.

: 85 on page 1714.

63.4.1

.

63.4.2

Parte VI

Capítulo 64

64.1 cdecl

Listing 64.1: cdecl

```
push arg3
push arg2
push arg1
call function
add esp, 12 ; returns ESP
```

64.2 stdcall

Listing 64.2: stdcall

```
push arg3
push arg2
```

```
push arg1
call function

function:
... do something ...
ret 12
```

Este método es ubicuo en las bibliotecas estándar de win32, pero no en win64 (ver más abajo sobre win64).

8.1 on page 187 `__stdcall`:

```
int __stdcall f2 (int a, int b, int c)
{
        return a*b+c;
};
```

8.2 on page 188, `RET 12` RET. SP .

`RETN n`.

Listing 64.3: MSVC 2010

```
_a$ = 8                                                  ↙
    ↘                  ; size = 4
_b$ = 12                                                 ↙
    ↘                  ; size = 4
_c$ = 16                                                 ↙
    ↘                  ; size = 4
_f2@12  PROC
        push    ebp
        mov     ebp, esp
        mov     eax, DWORD PTR _a$[ebp]
```

```
        imul    eax, DWORD PTR _b$[ebp]
        add     eax, DWORD PTR _c$[ebp]
        pop     ebp
        ret     12                                  ↙
    ↘                   ; 0000000cH
_f2@12  ENDP

; ...
        push    3
        push    2
        push    1
        call    _f2@12
        push    eax
        push    OFFSET $SG81369
        call    _printf
        add     esp, 8
```

64.2.1

64.3 fastcall

Listing 64.4: fastcall

```
push arg3
mov edx, arg2
mov ecx, arg1
call function

function:
.. do something ..
ret 4
```

8.1 on page 187 __fastcall:

```
int __fastcall f3 (int a, int b, int c)
{
        return a*b+c;
};
```

:

Listing 64.5: Con optimización MSVC 2010 /Ob0

```
_c$ = 8                                                    ↙
    ↘                      ; size = 4
@f3@12  PROC
; _a$ = ecx
; _b$ = edx
        mov     eax, ecx
        imul    eax, edx
        add     eax, DWORD PTR _c$[esp-4]
        ret     4
@f3@12  ENDP

; ...

        mov     edx, 2
        push    3
        lea     ecx, DWORD PTR [edx-1]
        call    @f3@12
        push    eax
        push    OFFSET $SG81390
        call    _printf
        add     esp, 8
```

SP RETN.

64.3.1 GCC regparm

(19.1.1 on page 580).

64.3.2 Watcom/OpenWatcom

«register calling convention». EAX, EDX, EBX y ECX. . .

64.4 thiscall

(51.1.1 on page 1040).

64.5 x86-64

64.5.1 Windows x64

.

:

```
#include <stdio.h>

void f1(int a, int b, int c, int d, int e, ↙
    ↘ int f, int g)
{
        printf ("%d %d %d %d %d %d %d\n", a,↙
    ↘  b, c, d, e, f, g);
};

int main()
```

```
{
        f1(1,2,3,4,5,6,7);
};
```

Listing 64.6: MSVC 2012 /0b

```
$SG2937 DB      '%d %d %d %d %d %d %d', 0aH, ↙
    ↘ 00H

main    PROC
        sub     rsp, 72                         ↙
    ↘                   ; 00000048H

        mov     DWORD PTR [rsp+48], 7
        mov     DWORD PTR [rsp+40], 6
        mov     DWORD PTR [rsp+32], 5
        mov     r9d, 4
        mov     r8d, 3
        mov     edx, 2
        mov     ecx, 1
        call    f1

        xor     eax, eax
        add     rsp, 72                         ↙
    ↘                   ; 00000048H
        ret     0
main    ENDP

a$ = 80
b$ = 88
c$ = 96
d$ = 104
e$ = 112
```

```
f$ = 120
g$ = 128
f1      PROC
$LN3:
        mov     DWORD PTR [rsp+32], r9d
        mov     DWORD PTR [rsp+24], r8d
        mov     DWORD PTR [rsp+16], edx
        mov     DWORD PTR [rsp+8], ecx
        sub     rsp, 72                         ⤦
    ↪                 ; 00000048H

        mov     eax, DWORD PTR g$[rsp]
        mov     DWORD PTR [rsp+56], eax
        mov     eax, DWORD PTR f$[rsp]
        mov     DWORD PTR [rsp+48], eax
        mov     eax, DWORD PTR e$[rsp]
        mov     DWORD PTR [rsp+40], eax
        mov     eax, DWORD PTR d$[rsp]
        mov     DWORD PTR [rsp+32], eax
        mov     r9d, DWORD PTR c$[rsp]
        mov     r8d, DWORD PTR b$[rsp]
        mov     edx, DWORD PTR a$[rsp]
        lea     rcx, OFFSET FLAT:$SG2937
        call    printf

        add     rsp, 72                         ⤦
    ↪                 ; 00000048H
        ret     0
f1      ENDP
```

. . .

Listing 64.7: Con optimización MSVC 2012 /0b

```
$SG2777 DB      '%d %d %d %d %d %d %d', 0aH, ↙
    ↘ 00H

a$ = 80
b$ = 88
c$ = 96
d$ = 104
e$ = 112
f$ = 120
g$ = 128
f1      PROC
$LN3:
        sub     rsp, 72                         ↙
    ↘                   ; 00000048H

        mov     eax, DWORD PTR g$[rsp]
        mov     DWORD PTR [rsp+56], eax
        mov     eax, DWORD PTR f$[rsp]
        mov     DWORD PTR [rsp+48], eax
        mov     eax, DWORD PTR e$[rsp]
        mov     DWORD PTR [rsp+40], eax
        mov     DWORD PTR [rsp+32], r9d
        mov     r9d, r8d
        mov     r8d, edx
        mov     edx, ecx
        lea     rcx, OFFSET FLAT:$SG2777
        call    printf

        add     rsp, 72                         ↙
    ↘                   ; 00000048H
        ret     0
```

```
f1      ENDP

main    PROC
        sub     rsp, 72                                  ↙
    ↘                       ; 00000048H

        mov     edx, 2
        mov     DWORD PTR [rsp+48], 7
        mov     DWORD PTR [rsp+40], 6
        lea     r9d, QWORD PTR [rdx+2]
        lea     r8d, QWORD PTR [rdx+1]
        lea     ecx, QWORD PTR [rdx-1]
        mov     DWORD PTR [rsp+32], 5
        call    f1

        xor     eax, eax
        add     rsp, 72                                  ↙
    ↘                       ; 00000048H
        ret     0
main    ENDP
```

. .

: 74.1 on page 1427.

Windows x64: (C/C++)

this RCX, RDX, Spanish text placeholder. : 51.1.1 on page 1043.

64.5.2 Linux x64

Listing 64.8: Con optimización GCC 4.7.3

```
.LC0:
        .string "%d %d %d %d %d %d %d\n"
f1:
        sub     rsp, 40
        mov     eax, DWORD PTR [rsp+48]
        mov     DWORD PTR [rsp+8], r9d
        mov     r9d, ecx
        mov     DWORD PTR [rsp], r8d
        mov     ecx, esi
        mov     r8d, edx
        mov     esi, OFFSET FLAT:.LC0
        mov     edx, edi
        mov     edi, 1
        mov     DWORD PTR [rsp+16], eax
        xor     eax, eax
        call    __printf_chk
        add     rsp, 40
        ret
main:
        sub     rsp, 24
        mov     r9d, 6
        mov     r8d, 5
        mov     DWORD PTR [rsp], 7
        mov     ecx, 4
        mov     edx, 3
        mov     esi, 2
        mov     edi, 1
        call    f1
        add     rsp, 24
        ret
```

64.6

64.7

```
#include <stdio.h>

void f(int a, int b)
{
        a=a+b;
        printf ("%d\n", a);
};
```

Listing 64.9: MSVC 2012

```
_a$ = 8                                                  ↙
    ↳                       ; size = 4
_b$ = 12                                                 ↙
    ↳                       ; size = 4
_f      PROC
        push    ebp
        mov     ebp, esp
        mov     eax, DWORD PTR _a$[ebp]
        add     eax, DWORD PTR _b$[ebp]
        mov     DWORD PTR _a$[ebp], eax
        mov     ecx, DWORD PTR _a$[ebp]
        push    ecx
        push    OFFSET $SG2938 ; '%d', 0aH
        call    _printf
        add     esp, 8
        pop     ebp
        ret     0
_f      ENDP
```

references en C++ (51.3 on page 1072), , .

64.8

...

```
#include <stdio.h>

// located in some other file
void modify_a (int *a);

void f (int a)
{
        modify_a (&a);
        printf ("%d\n", a);
};
```

Listing 64.10: Con optimización MSVC 2010

```
$SG2796 DB      '%d', 0aH, 00H

_a$ = 8
_f      PROC
        lea     eax, DWORD PTR _a$[esp-4] ; just get the address of value in local stack
        push    eax                       ; and pass it to modify_a()
        call    _modify_a
        mov     ecx, DWORD PTR _a$[esp]   ; reload it from the local stack
```

```
        push    ecx                             ; ↙
    ↘ and pass it to printf()
        push    OFFSET $SG2796 ; '%d'
        call    _printf
        add     esp, 12
        ret     0
_f      ENDP
```

Modifica el valor al que apunta el puntero y luego `printf()` imprime el valor modificado.

Listing 64.11: Con optimización MSVC 2012 x64

```
$SG2994 DB      '%d', 0aH, 00H

a$ = 48
f       PROC
        mov     DWORD PTR [rsp+8], ecx    ; ↙
    ↘ save input value in Shadow Space
        sub     rsp, 40
        lea     rcx, QWORD PTR a$[rsp]    ; ↙
    ↘ get address of value and pass it to ↙
    ↘ modify_a()
        call    modify_a
        mov     edx, DWORD PTR a$[rsp]    ; ↙
    ↘ reload value from Shadow Space and ↙
    ↘ pass it to printf()
        lea     rcx, OFFSET FLAT:$SG2994 ; ↙
    ↘ '%d'
        call    printf
        add     rsp, 40
```

```
        ret     0
f       ENDP
```

Listing 64.12: Con optimización GCC 4.9.1 x64

```
.LC0:
        .string "%d\n"
f:
        sub     rsp, 24
        mov     DWORD PTR [rsp+12], edi  ; ↙
    ↘ store input value to the local stack
        lea     rdi, [rsp+12]            ; ↙
    ↘ take an address of the value and pass ↙
    ↘ it to modify_a()
        call    modify_a
        mov     edx, DWORD PTR [rsp+12]  ; ↙
    ↘ reload value from the local stack and ↙
    ↘ pass it to printf()
        mov     esi, OFFSET FLAT:.LC0    ; ↙
    ↘ '%d'
        mov     edi, 1
        xor     eax, eax
        call    __printf_chk
        add     rsp, 24
        ret
```

Listing 64.13: Con optimización GCC 4.9.1 ARM64

```
f:
        stp     x29, x30, [sp, -32]!
        add     x29, sp, 0             ; setup ↙
    ↘ FP
```

```
        add     x1, x29, 32          ; ↙
    ↘ calculate address of variable in ↙
    ↘ Register Save Area
        str     w0, [x1,-4]!         ; store ↙
    ↘ input value there
        mov     x0, x1               ; pass ↙
    ↘ address of variable to the modify_a()
        bl      modify_a
        ldr     w1, [x29,28]         ; load ↙
    ↘ value from the variable and pass it to↙
    ↘  printf()
        adrp    x0, .LC0 ; '%d'
        add     x0, x0, :lo12:.LC0
        bl      printf               ; call ↙
    ↘ printf()
        ldp     x29, x30, [sp], 32
        ret
.LC0:
        .string "%d\n"
```

: 46.1.2 on page 990.

CAPÍTULO 65. THREAD LOCAL STORAGE

Capítulo 65

Thread Local Storage

. *errno*.

C++11 *thread_local* , TLS [1]:

Listing 65.1: C++11

```
#include <iostream>
#include <thread>

thread_local int tmp=3;

int main()
{
	std::cout << tmp << std::endl;
```

1

CAPÍTULO 65. THREAD LOCAL STORAGE

```
};
```

MinGW GCC 4.8.1, MSVC 2012.

tmp TLS.

65.1

65.1.1 Win32

```
#include <stdint.h>
#include <windows.h>
#include <winnt.h>

// from the Numerical Recipes book:
#define RNG_a 1664525
#define RNG_c 1013904223

__declspec( thread ) uint32_t rand_state;

void my_srand (uint32_t init)
{
        rand_state=init;
}

int my_rand ()
{
        rand_state=rand_state*RNG_a;
        rand_state=rand_state+RNG_c;
        return rand_state & 0x7fff;
```

```
}

int main()
{
        my_srand(0x12345678);
        printf ("%d\n", my_rand());
};
```

.tls.

Listing 65.2: Con optimización MSVC 2013 x86

```
_TLS    SEGMENT
_rand_state DD  01H DUP (?)
_TLS    ENDS

_DATA   SEGMENT
$SG84851 DB     '%d', 0aH, 00H
_DATA   ENDS
_TEXT   SEGMENT

_init$ = 8                                              ; size = 4
_my_srand PROC
; FS:0=address of TIB
        mov     eax, DWORD PTR fs:__tls_array  ; displayed in IDA as FS:2Ch
; EAX=address of TLS of process
        mov     ecx, DWORD PTR __tls_index
        mov     ecx, DWORD PTR [eax+ecx*4]
; ECX=current TLS segment
        mov     eax, DWORD PTR _init$[esp-4]
```

```
        mov     DWORD PTR _rand_state[ecx], ⤦
    ↳ eax
        ret     0
_my_srand ENDP

_my_rand PROC
; FS:0=address of TIB
        mov     eax, DWORD PTR fs:⤦
    ↳ __tls_array  ; displayed in IDA as FS⤦
    ↳ :2Ch
; EAX=address of TLS of process
        mov     ecx, DWORD PTR __tls_index
        mov     ecx, DWORD PTR [eax+ecx*4]
; ECX=current TLS segment
        imul    eax, DWORD PTR _rand_state[⤦
    ↳ ecx], 1664525
        add     eax, 1013904223            ⤦
    ↳                  ; 3c6ef35fH
        mov     DWORD PTR _rand_state[ecx], ⤦
    ↳ eax
        and     eax, 32767                 ⤦
    ↳                  ; 00007fffH
        ret     0
_my_rand ENDP

_TEXT   ENDS
```

Listing 65.3: Con optimización MSVC 2013 x64

```
_TLS    SEGMENT
rand_state DD   01H DUP (?)
_TLS    ENDS

```

```
_DATA   SEGMENT
$SG85451 DB     '%d', 0aH, 00H
_DATA   ENDS

_TEXT   SEGMENT

init$ = 8
my_srand PROC
        mov     edx, DWORD PTR _tls_index
        mov     rax, QWORD PTR gs:88 ; 58h
        mov     r8d, OFFSET FLAT:rand_state
        mov     rax, QWORD PTR [rax+rdx*8]
        mov     DWORD PTR [r8+rax], ecx
        ret     0
my_srand ENDP

my_rand PROC
        mov     rax, QWORD PTR gs:88 ; 58h
        mov     ecx, DWORD PTR _tls_index
        mov     edx, OFFSET FLAT:rand_state
        mov     rcx, QWORD PTR [rax+rcx*8]
        imul    eax, DWORD PTR [rcx+rdx], ↙
    ↘ 1664525       ; 0019660dH
        add     eax, 1013904223         ↙
    ↘                 ; 3c6ef35fH
        mov     DWORD PTR [rcx+rdx], eax
        and     eax, 32767              ↙
    ↘                 ; 00007fffH
        ret     0
my_rand ENDP

_TEXT   ENDS
```

 CAPÍTULO 65. THREAD LOCAL STORAGE

```
#include <stdint.h>
#include <windows.h>
#include <winnt.h>

// from the Numerical Recipes book:
#define RNG_a 1664525
#define RNG_c 1013904223

__declspec( thread ) uint32_t rand_state=1234;

void my_srand (uint32_t init)
{
        rand_state=init;
}

int my_rand ()
{
        rand_state=rand_state*RNG_a;
        rand_state=rand_state+RNG_c;
        return rand_state & 0x7fff;
}

int main()
{
        printf ("%d\n", my_rand());
};
```

```
.tls:00404000 ; Segment type: Pure data
.tls:00404000 ; Segment permissions: Read/Write
```

```
.tls:00404000 _tls                segment para ↙
    ↘ public 'DATA' use32
.tls:00404000                     assume cs:_tls
.tls:00404000                     ;org 404000h
.tls:00404000 TlsStart            db    0         ↙
    ↘           ; DATA XREF: .rdata:↙
    ↘ TlsDirectory
.tls:00404001                     db    0
.tls:00404002                     db    0
.tls:00404003                     db    0
.tls:00404004                     dd 1234
.tls:00404008 TlsEnd              db    0         ↙
    ↘           ; DATA XREF: .rdata:↙
    ↘ TlsEnd_ptr
...
```

:

-
-
-

.

```
#include <stdint.h>
#include <windows.h>
#include <winnt.h>

// from the Numerical Recipes book:
```

```
#define RNG_a 1664525
#define RNG_c 1013904223

__declspec( thread ) uint32_t rand_state;

void my_srand (uint32_t init)
{
        rand_state=init;
}

void NTAPI tls_callback(PVOID a, DWORD ↙
    ↘ dwReason, PVOID b)
{
        my_srand (GetTickCount());
}

#pragma data_seg(".CRT$XLB")
PIMAGE_TLS_CALLBACK p_thread_callback = ↙
    ↘ tls_callback;
#pragma data_seg()

int my_rand ()
{
        rand_state=rand_state*RNG_a;
        rand_state=rand_state+RNG_c;
        return rand_state & 0x7fff;
}

int main()
{
        // rand_state is already initialized↙
    ↘  at the moment (using GetTickCount())
        printf ("%d\n", my_rand());
```

```
};
```

IDA:

Listing 65.4: Con optimización MSVC 2013

```
.text:00401020 TlsCallback_0   proc near      ⤦
    ↳              ; DATA XREF: .rdata:⤦
    ↳ TlsCallbacks
.text:00401020                 call    ds:⤦
    ↳ GetTickCount
.text:00401026                 push    eax
.text:00401027                 call    ⤦
    ↳ my_srand
.text:0040102C                 pop     ecx
.text:0040102D                 retn    0Ch
.text:0040102D TlsCallback_0   endp

...

.rdata:004020C0 TlsCallbacks    dd offset ⤦
    ↳ TlsCallback_0 ; DATA XREF: .rdata:⤦
    ↳ TlsCallbacks_ptr

...

.rdata:00402118 TlsDirectory    dd offset ⤦
    ↳ TlsStart
.rdata:0040211C TlsEnd_ptr      dd offset ⤦
    ↳ TlsEnd
.rdata:00402120 TlsIndex_ptr    dd offset ⤦
    ↳ TlsIndex
.rdata:00402124 TlsCallbacks_ptr dd offset ⤦
```

```
    ↳ TlsCallbacks
.rdata:00402128 TlsSizeOfZeroFill dd 0
.rdata:0040212C TlsCharacteristics dd 300000↙
    ↳ h
```

65.1.2 Linux

```
__thread uint32_t rand_state=1234;
```

[2].

Listing 65.5: Con optimización GCC 4.8.1 x86

```
.text:08048460 my_srand        proc near
.text:08048460
.text:08048460 arg_0           = dword ptr  ↙
    ↳ 4
.text:08048460
.text:08048460                 mov     eax, ↙
    ↳ [esp+arg_0]
.text:08048464                 mov     gs:0↙
    ↳ FFFFFFFCh, eax
.text:0804846A                 retn
.text:0804846A my_srand        endp

.text:08048470 my_rand         proc near
.text:08048470                 imul    eax, ↙
    ↳ gs:0FFFFFFFCh, 19660Dh
.text:0804847B                 add     eax, ↙
    ↳ 3C6EF35Fh
```

[2] http://go.yurichev.com/17062

```
.text:08048480                 mov     gs:0⤦
    ↳ FFFFFFFCh, eax
.text:08048486                 and     eax, ⤦
    ↳ 7FFFh
.text:0804848B                 retn
.text:0804848B my_rand         endp
```

: [Dre13].

Capítulo 66

: («kernel space») («user space»).

.

.

. kernel panic o BSOD[1].

: ring0 («kernel space») y ring3 («user space»).

(syscall-) .

.

.

[1]Black Screen of Death

66.1 Linux

int 0x80. EAX.

Listing 66.1:

```
section .text
global  _start

_start:
        mov     edx,len ; buffer len
        mov     ecx,msg ; buffer
        mov     ebx,1   ; file descriptor. 1
    ↳  is for stdout
        mov     eax,4   ; syscall number. 4
    ↳ is for sys_write
        int     0x80

        mov     eax,1   ; syscall number. 4
    ↳ is for sys_exit
        int     0x80

section .data

msg     db  'Hello, world!',0xa
len     equ $ - msg
```

:

```
nasm -f elf32 1.s
ld 1.o
```

Linux: http://go.yurichev.com/17319.

strace(71 on page 1417).

66.2 Windows

`int 0x2e` `SYSENTER`.

Windows: http://go.yurichev.com/17320.

:

«Windows Syscall Shellcode» by Piotr Bania:
http://go.yurichev.com/17321.

Capítulo 67

Linux

67.1 Código independiente de lá posición

:

Listing 67.1: libc-2.17.so x86

```
.text:0012D5E3 __x86_get_pc_thunk_bx proc ↙
    ↳ near          ; CODE XREF: sub_17350+3
.text:0012D5E3                                              ↙
    ↳               ; sub_173CC+4 ...
.text:0012D5E3                          mov      ebx, ↙
    ↳ [esp+0]
.text:0012D5E6                          retn
```

```
.text:0012D5E6 __x86_get_pc_thunk_bx endp

...

.text:000576C0 sub_576C0       proc near     ↙
    ↘              ; CODE XREF: tmpfile+73

...

.text:000576C0                 push    ebp
.text:000576C1                 mov     ecx, ↙
    ↘ large gs:0
.text:000576C8                 push    edi
.text:000576C9                 push    esi
.text:000576CA                 push    ebx
.text:000576CB                 call    ↙
    ↘ __x86_get_pc_thunk_bx
.text:000576D0                 add     ebx, ↙
    ↘ 157930h
.text:000576D6                 sub     esp, ↙
    ↘ 9Ch

...

.text:000579F0                 lea     eax, ↙
    ↘ (a__gen_tempname - 1AF000h)[ebx] ; "↙
    ↘ __gen_tempname"
.text:000579F6                 mov     [esp↙
    ↘ +0ACh+var_A0], eax
.text:000579FA                 lea     eax, ↙
    ↘ (a__SysdepsPosix - 1AF000h)[ebx] ; ↙
    ↘ "../sysdeps/posix/tempname.c"
.text:00057A00                 mov     [esp↙
```

```
    ↳ +0ACh+var_A8], eax
.text:00057A04                 lea     eax, ↙
    ↳ (aInvalidKindIn_ - 1AF000h)[ebx] ; "! ↙
    ↳ \"invalid KIND in __gen_tempname\""
.text:00057A0A                 mov     [esp↙
    ↳ +0ACh+var_A4], 14Ah
.text:00057A12                 mov     [esp↙
    ↳ +0ACh+var_AC], eax
.text:00057A15                 call    ↙
    ↳ __assert_fail
```

.

. ..

:

```
#include <stdio.h>

int global_variable=123;

int f1(int var)
{
    int rt=global_variable+var;
    printf ("returning %d\n", rt);
    return rt;
};
```

IDA:

```
gcc -fPIC -shared -O3 -o 1.so 1.c
```

Listing 67.2: GCC 4.7.3

```
.text:00000440                         public ⤦
    ↳ __x86_get_pc_thunk_bx
.text:00000440 __x86_get_pc_thunk_bx proc ⤦
    ↳ near          ; CODE XREF: _init_proc+4
.text:00000440                                 ⤦
    ↳             ; deregister_tm_clones+4 ⤦
    ↳ ...
.text:00000440                         mov     ebx, ⤦
    ↳ [esp+0]
.text:00000443                         retn
.text:00000443 __x86_get_pc_thunk_bx endp

.text:00000570                         public f1
.text:00000570 f1                      proc near
.text:00000570
.text:00000570 var_1C                  = dword ptr ⤦
    ↳ -1Ch
.text:00000570 var_18                  = dword ptr ⤦
    ↳ -18h
.text:00000570 var_14                  = dword ptr ⤦
    ↳ -14h
.text:00000570 var_8                   = dword ptr ⤦
    ↳ -8
.text:00000570 var_4                   = dword ptr ⤦
    ↳ -4
.text:00000570 arg_0                   = dword ptr  ⤦
    ↳ 4
.text:00000570
.text:00000570                         sub     esp, ⤦
    ↳ 1Ch
.text:00000573                         mov     [esp⤦
```

```
	↘ +1Ch+var_8], ebx
.text:00000577                 call    ↙
	↘ __x86_get_pc_thunk_bx
.text:0000057C                 add     ebx, ↙
	↘ 1A84h
.text:00000582                 mov     [esp↙
	↘ +1Ch+var_4], esi
.text:00000586                 mov     eax, ↙
	↘ ds:(global_variable_ptr - 2000h)[ebx]
.text:0000058C                 mov     esi, ↙
	↘ [eax]
.text:0000058E                 lea     eax, ↙
	↘ (aReturningD - 2000h)[ebx] ; "↙
	↘ returning %d\n"
.text:00000594                 add     esi, ↙
	↘ [esp+1Ch+arg_0]
.text:00000598                 mov     [esp↙
	↘ +1Ch+var_18], eax
.text:0000059C                 mov     [esp↙
	↘ +1Ch+var_1C], 1
.text:000005A3                 mov     [esp↙
	↘ +1Ch+var_14], esi
.text:000005A7                 call    ↙
	↘ ___printf_chk
.text:000005AC                 mov     eax, ↙
	↘ esi
.text:000005AE                 mov     ebx, ↙
	↘ [esp+1Ch+var_8]
.text:000005B2                 mov     esi, ↙
	↘ [esp+1Ch+var_4]
.text:000005B6                 add     esp, ↙
	↘ 1Ch
.text:000005B9                 retn
```

```
.text:000005B9 f1                         endp
```

Spanish text placeholder
`__x86_get_pc_thunk_bx()` . . `0x1A84` *Global Offset Table Procedure Linkage Table* (GOT PLT), *Global Offset Table* (GOT), . IDA:

```
.text:00000577                  call     ↙
    ↘ __x86_get_pc_thunk_bx
.text:0000057C                  add      ebx, ↙
    ↘ 1A84h
.text:00000582                  mov      [esp↙
    ↘ +1Ch+var_4], esi
.text:00000586                  mov      eax, ↙
    ↘ [ebx-0Ch]
.text:0000058C                  mov      esi, ↙
    ↘ [eax]
.text:0000058E                  lea      eax, ↙
    ↘ [ebx-1A30h]
```

. .

.

.

:

```
0000000000000720 <f1>:
 720:   48 8b 05 b9 08 20 00    mov    rax,↙
    ↘ QWORD PTR [rip+0x2008b9]        # 200↙
    ↘ fe0 <_DYNAMIC+0x1d0>
 727:   53                      push   rbx
```

```
 728:   89 fb                   mov    ebx,↙
    ↪ edi
 72a:   48 8d 35 20 00 00 00    lea    rsi,[↙
    ↪ rip+0x20]        # 751 <_fini+0x9>
 731:   bf 01 00 00 00          mov    edi,0↙
    ↪ x1
 736:   03 18                   add    ebx,↙
    ↪ DWORD PTR [rax]
 738:   31 c0                   xor    eax,↙
    ↪ eax
 73a:   89 da                   mov    edx,↙
    ↪ ebx
 73c:   e8 df fe ff ff          call   620 <↙
    ↪ __printf_chk@plt>
 741:   89 d8                   mov    eax,↙
    ↪ ebx
 743:   5b                      pop    rbx
 744:   c3                      ret
```

`0x2008b9` .

. [Fog13a].

67.1.1 Windows

67.2 *LD_PRELOAD* en Linux

libc.so.6.

time(), read(), write(), Spanish text placeholder.

. . strace(71 on page 1417), `/proc/uptime` :

```
$ strace uptime
...
open("/proc/uptime", O_RDONLY)              = 3
lseek(3, 0, SEEK_SET)                       = 0
read(3, "416166.86 414629.38\n", 2047)   = 20
...
```

:

```
$ cat /proc/uptime
416690.91 415152.03
```

[1]:

> The first number is the total number of seconds the system has been up. The second number is how much of that time the machine has spent idle, in seconds.

open(), read(), close() .

read() libc.so.6. close(),

[2].

```
#include <stdio.h>
#include <stdarg.h>
#include <stdlib.h>
```

[1] wikipedia

[2] en [3] Yong Huang

```
#include <stdbool.h>
#include <unistd.h>
#include <dlfcn.h>
#include <string.h>

void *libc_handle = NULL;
int (*open_ptr)(const char *, int) = NULL;
int (*close_ptr)(int) = NULL;
ssize_t (*read_ptr)(int, void*, size_t) = ↙
    ↳ NULL;

bool inited = false;

_Noreturn void die (const char * fmt, ...)
{
        va_list va;
        va_start (va, fmt);

        vprintf (fmt, va);
        exit(0);
};

static void find_original_functions ()
{
        if (inited)
                return;

        libc_handle = dlopen ("libc.so.6", ↙
    ↳ RTLD_LAZY);
        if (libc_handle==NULL)
                die ("can't open libc.so.6\n↙
    ↳ ");
```

```
        open_ptr = dlsym (libc_handle, "open
    ↪ ");
        if (open_ptr==NULL)
                die ("can't find open()\n");

        close_ptr = dlsym (libc_handle, "
    ↪ close");
        if (close_ptr==NULL)
                die ("can't find close()\n")
    ↪ ;

        read_ptr = dlsym (libc_handle, "read
    ↪ ");
        if (read_ptr==NULL)
                die ("can't find read()\n");

        inited = true;
}

static int opened_fd=0;

int open(const char *pathname, int flags)
{
        find_original_functions();

        int fd=(*open_ptr)(pathname, flags);
        if (strcmp(pathname, "/proc/uptime")
    ↪ ==0)
                opened_fd=fd; // that's our
    ↪ file! record its file descriptor
        else
                opened_fd=0;
        return fd;
```

 CAPÍTULO 67. LINUX

```
};

int close(int fd)
{
        find_original_functions();

        if (fd==opened_fd)
                opened_fd=0; // the file is not opened anymore
        return (*close_ptr)(fd);
};

ssize_t read(int fd, void *buf, size_t count)
{
        find_original_functions();

        if (opened_fd!=0 && fd==opened_fd)
        {
                // that's our file!
                return snprintf (buf, count, "%d %d", 0x7fffffff, 0x7fffffff)+1;
        };
        // not our file, go to real read() function
        return (*read_ptr)(fd, buf, count);
};
```

(GitHub)

:

```
gcc -fpic -shared -Wall -o fool_uptime.so ↙
    ↳ fool_uptime.c -ldl
```

:

```
LD_PRELOAD=`pwd`/fool_uptime.so uptime
```

:

```
 01:23:02 up 24855 days,  3:14,  3 users,  ↙
    ↳ load average: 0.00, 0.01, 0.05
```

LD_PRELOAD

:

- strcmp() (Yong Huang) http://go.yurichev.com/17043
- Kevin PuloFun with LD_PRELOAD. . yurichev.com
- (zlibc). http://go.yurichev.com/17146

Capítulo 68

Windows NT

68.1 CRT (win32)

?. ..

... :

```
int CALLBACK WinMain(
  _In_  HINSTANCE hInstance,
  _In_  HINSTANCE hPrevInstance,
  _In_  LPSTR lpCmdLine,
  _In_  int nCmdShow
);
```

.

. .

.

.

```
___tmainCRTStartup proc near

var_24 = dword ptr -24h
var_20 = dword ptr -20h
var_1C = dword ptr -1Ch
ms_exc = CPPEH_RECORD ptr -18h

        push    14h
        push    offset stru_4092D0
        call    __SEH_prolog4
        mov     eax, 5A4Dh
        cmp     ds:400000h, ax
        jnz     short loc_401096
        mov     eax, ds:40003Ch
        cmp     dword ptr [eax+400000h], 4550h
        jnz     short loc_401096
        mov     ecx, 10Bh
        cmp     [eax+400018h], cx
        jnz     short loc_401096
        cmp     dword ptr [eax+400074h], 0Eh
        jbe     short loc_401096
        xor     ecx, ecx
        cmp     [eax+4000E8h], ecx
        setnz   cl
        mov     [ebp+var_1C], ecx
        jmp     short loc_40109A
```

```


loc_401096: ; CODE XREF: ___tmainCRTStartup+18
            ; ___tmainCRTStartup+29 ...
        and     [ebp+var_1C], 0

loc_40109A: ; CODE XREF: ___tmainCRTStartup+50
        push    1
        call    __heap_init
        pop     ecx
        test    eax, eax
        jnz     short loc_4010AE
        push    1Ch
        call    _fast_error_exit
        pop     ecx

loc_4010AE: ; CODE XREF: ___tmainCRTStartup+60
        call    __mtinit
        test    eax, eax
        jnz     short loc_4010BF
        push    10h
        call    _fast_error_exit
        pop     ecx

loc_4010BF: ; CODE XREF: ___tmainCRTStartup+71
        call    sub_401F2B
        and     [ebp+ms_exc.disabled], 0
        call    __ioinit
        test    eax, eax
```

```
        jge     short loc_4010D9
        push    1Bh
        call    __amsg_exit
        pop     ecx

loc_4010D9: ; CODE XREF: ___tmainCRTStartup+8B
        call    ds:GetCommandLineA
        mov     dword_40B7F8, eax
        call    ___crtGetEnvironmentStringsA
        mov     dword_40AC60, eax
        call    __setargv
        test    eax, eax
        jge     short loc_4010FF
        push    8
        call    __amsg_exit
        pop     ecx

loc_4010FF: ; CODE XREF: ___tmainCRTStartup+B1
        call    __setenvp
        test    eax, eax
        jge     short loc_401110
        push    9
        call    __amsg_exit
        pop     ecx

loc_401110: ; CODE XREF: ___tmainCRTStartup+C2
        push    1
        call    __cinit
        pop     ecx
        test    eax, eax
```

```
        jz      short loc_401123
        push    eax
        call    __amsg_exit
        pop     ecx

loc_401123: ; CODE XREF: ___tmainCRTStartup+D6
        mov     eax, envp
        mov     dword_40AC80, eax
        push    eax             ; envp
        push    argv            ; argv
        push    argc            ; argc
        call    _main
        add     esp, 0Ch
        mov     [ebp+var_20], eax
        cmp     [ebp+var_1C], 0
        jnz     short $LN28
        push    eax             ; uExitCode
        call    $LN32

$LN28:      ; CODE XREF: ___tmainCRTStartup+105
        call    __cexit
        jmp     short loc_401186


$LN27:      ; DATA XREF: .rdata:stru_4092D0
        mov     eax, [ebp+ms_exc.exc_ptr] ; Exception filter 0 for function 401044
        mov     ecx, [eax]
        mov     ecx, [ecx]
        mov     [ebp+var_24], ecx
        push    eax
```

```
        push    ecx
        call    __XcptFilter
        pop     ecx
        pop     ecx

$LN24:
        retn


$LN14:       ; DATA XREF: .rdata:stru_4092D0
        mov     esp, [ebp+ms_exc.old_esp] ; Exception handler 0 for function 401044
        mov     eax, [ebp+var_24]
        mov     [ebp+var_20], eax
        cmp     [ebp+var_1C], 0
        jnz     short $LN29
        push    eax             ; int
        call    __exit


$LN29:       ; CODE XREF: ___tmainCRTStartup+135
        call    __c_exit

loc_401186: ; CODE XREF: ___tmainCRTStartup+112
        mov     [ebp+ms_exc.disabled], 0FFFFFFFEh
        mov     eax, [ebp+var_20]
        call    __SEH_epilog4
        retn
```

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`GetCommandLineA()` (línea 62), `setargv()` (línea 66) y `setenvp()` (línea 74), `argc`, `argv`, `envp`.

(línea 97).

`heap_init()` (línea 35), `ioinit()` (línea 54).

. `malloc()` CRT, :

```
runtime error R6030
- CRT not initialized
```

: 51.4.1 on page 1082.

`main()` `cexit()`, `$LN32`, `doexit()`.

.

```
#include <windows.h>

int main()
{
        MessageBox (NULL, "hello, world", "
    caption", MB_OK);
};
```

MSVC 2008.

```
cl no_crt.c user32.lib /link /entry:main
```

.

```
cl no_crt.c user32.lib /link /entry:main /
    align:16
```

:

```
LINK : warning LNK4108: /ALIGN specified ↙
    ↘ without /DRIVER; image may not run
```

. Windows 7 x86, x64 ().

68.2 Win32 PE

PE es un formato de archivo ejecutable usado en Windows.

La diferencia entre .exe, .dll y .sys es que .exe y .sys normalmente no tienen exportaciones, solo importaciones.

Una DLL[1], igual que cualquier otro archivo PE, tiene un punto de entrada (OEP[2]) (allí está la función DllMain()), pero normalmente esa función no hace nada.

.sys normalmente es un controlador de dispositivo.

Para los controladores, Windows exige que la suma de comprobación esté presente en el archivo PE y sea correcta [3].

Windows Vista, Desde Windows Vista, los archivos de los controladores también deben ir firmados digitalmente; de lo contrario no se cargarán.

«This program cannot be run in DOS mode.» Al comienzo de cualquier archivo PE hay un pequeño programa DOS que

[1]Dynamic-link library
[2]Original Entry Point
[3]Por ejemplo, Hiew(73 on page 1420) puede calcularla

muestra un mensaje como «This program cannot be run in DOS mode.»si se ejecuta en DOS o Windows 3.1 (OS que no conocen el formato PE), se mostrará ese mensaje.

68.2.1

- Móduloes un archivo independiente, .exe o .dll.
- Procesoes un programa cargado en memoria y ejecutándose. Normalmente consta de un archivo .exe y un montón de archivos .dll.
- Memoria del procesola memoria con la que trabaja el proceso. Cada proceso tiene la suya. Suele haber módulos cargados, memoria de la pila, heap(s),Spanish text placeholder.
- VA[4] es una dirección que se usa en el propio programa en tiempo de ejecución.
- es la dirección en la memoria del proceso donde debe cargarse el módulo. El cargador de la OS puede cambiarla si ese destino ya está ocupado por otro módulo cargado antes.
- RVA[5] VA-.es la dirección VA menos la dirección base. RVA-Muchas direcciones en las tablas de un archivo PE usan direcciones RVA.

[4]Virtual Address
[5]Relative Virtual Address

- IAT[6]un arreglo de direcciones de símbolos importados [7]. IMAGE_DIRECTORY_ENTRY_IAT IATA veces, la entrada de directorio de datos IMAGE_DIRECTORY apunta a la IAT. IDA (6.1) .idata para IAT, IAT !Es importante notar que IDA (al menos hasta la 6.1) puede crear una seudosección llamada .idata para la IAT, incluso si la IAT forma parte de otra sección!
- INT[8] un arreglo con los nombres de los símbolos a importar[9].

68.2.2

El problema es que varios autores de módulos pueden preparar DLL para que otros las usen y no es posible acordar qué direcciones se asignan a quién.

Por eso, si dos DLL necesarias para un proceso tienen la misma dirección base, una se cargará en esa base y la otra en otro espacio libre de la memoria del proceso, y todas las direcciones virtuales de la segunda DLL se ajustarán.

MSVC 0x400000 [10], 0x401000. RVA 0x1000. 0x1000000
[11]Con frecuencia, el enlazador de MSVC genera archivos .exe con dirección base 0x400000 y la sección de códi-

[6]Import Address Table
[7][Pie02]
[8]Import Name Table
[9][Pie02]
[10]: MSDN
[11]

go arrancando en 0x401000; esto significa que el RVA del inicio de la sección de código es 0x1000. Las DLL suelen generarse con base 0x10000000, y se puede cambiar con la opción /BASE del enlazador.

Además hay otra razón para cargar los módulos en direcciones distintas, concretamente en direcciones aleatorias.

ASLR[12][13]Esto es ASLR.

Un shellcode que intenta ejecutarse en un sistema comprometido debe llamar funciones del sistema y, por lo tanto, conocer sus direcciones.

OS (Windows Vista), DLL (kernel32.dll, user32.dll) , , las DLL del sistema (como kernel32.dll o user32.dll) se cargaban siempre en direcciones conocidas y, si recordamos que sus versiones cambiaban rara vez, las direcciones de las funciones eran prácticamente fijas y el shellcode podía llamarlas directamente.

ASLR carga tanto tu programa como todos los módulos que necesita en direcciones base aleatorias, diferentes en cada ejecución.

El soporte de ASLR en un archivo PE se marca activando el indicador
`IMAGE_DLL_CHARACTERISTICS_DYNAMIC_BASE` [RA09].

[12] Address Space Layout Randomization
[13]

68.2.3 Subsystem

También hay un campo *subsystem*, que suele ser:

- native[14] (.sys-controlador),
- console (aplicación de consola) o
- GUI[15] (no es de consola).

68.2.4

El archivo PE también especifica el número mínimo de versión de Windows necesario para cargar el módulo. aquí[16].

, MSVC 2005 Windows NT4 (4.00), MSVC 2008 (5.00, los archivos generados tienen versión 5.00 y requieren al menos Windows 2000 para ejecutarse).

Por defecto, MSVC 2012 genera archivos .exe de versión 6.00, que necesitan al menos Windows Vista; sin embargo, cambiando las opciones del compilador[17] es posible forzar la generación para Windows XP.

68.2.5

- *IMAGE_SCN_CNT_CODE* o *IMAGE_SCN_MEM_EXECUTE*.

[14] Significa que el módulo usa Native API en lugar de Win32
[15] Graphical user interface
[16] wikipedia
[17] MSDN

- *IMAGE_SCN_CNT_INITIALIZED_DATA*, *IMAGE_SCN_MEM_REA* y *IMAGE_SCN_MEM_WRITE*.
- *IMAGE_SCN_CNT_UNINITIALIZED_DATA*, *IMAGE_SCN_MEM_* y *IMAGE_SCN_MEM_WRITE*.
- , En una sección de datos constantes (protegida contra escritura) pueden estar los indicadores *IMAGE_SCN_CNT_INITIALIZED_DATA* y *IMAGE_SCN_MEM_RE* *IMAGE_SCN_MEM_WRITE*. Si se intenta escribir algo en esa sección, el proceso fallará.

En un archivo PE se puede asignar un nombre a cada sección, aunque no es algo crucial. `.text`la sección de código se llama `.text`, `.data`, — `.rdata` *(readable data)*(datos legibles). Otros nombres comunes de secciones son:

- `.idata`sección de importaciones. IDA puede crear una seudosección con el mismo nombre: 68.2.1 on page 1349.
- `.edata`sección de exportaciones (poco común)
- `.pdata` sección que contiene toda la información sobre excepciones en Windows NT para MIPS, IA64 y x64: 68.3.3 on page 1400
- `.reloc`sección de relocs
- `.bss`datos no inicializados (BSS[18])

[18]Block Started by Symbol

- `.tls`thread local storage (TLS)almacenamiento local de hilo (TLS)
- `.rsrc`recursos
- `.CRT` puede aparecer en binarios compilados con versiones muy antiguas de MSVC

Los empaquetadores o cifradores de archivos PE suelen borrar los nombres de las secciones o cambiarlos por los suyos propios.

En MSVC se pueden declarar datos en una sección con nombre arbitrario [19].

Algunos compiladores y enlazadores pueden añadir también una sección con símbolos de depuración y, en general, información de depuración (por ejemplo, MinGW). MSVC ( )allí se suele guardar la información de depuración en archivos PDB separados.

Así se describe una sección PE en el archivo:

```
typedef struct _IMAGE_SECTION_HEADER {
  BYTE  Name[IMAGE_SIZEOF_SHORT_NAME];
  union {
    DWORD PhysicalAddress;
    DWORD VirtualSize;
  } Misc;
  DWORD VirtualAddress;
  DWORD SizeOfRawData;
  DWORD PointerToRawData;
```

[19] MSDN

```
  DWORD PointerToRelocations;
  DWORD PointerToLinenumbers;
  WORD  NumberOfRelocations;
  WORD  NumberOfLinenumbers;
  DWORD Characteristics;
} IMAGE_SECTION_HEADER, *↙
    ↘ PIMAGE_SECTION_HEADER;
```

[20]

Algo más de terminología: *PointerToRawData* «Offset» en Hiew y *VirtualAddress* «RVA» allí.

68.2.6

(Hiew).

[21].

. código independiente de la posición (67.1 on page 1328).
.

0x410000 :

```
A1 00 00 41 00          mov          eax↙
    ↘ ,[000410000]
```

0x400000, RVA 0x10000.

0x500000, 0x510000.

[20] MSDN
[21] MS-DOS

 CAPÍTULO 68. WINDOWS NT

MOV, 0xA1.

0xA1, .

, , , (RVA), .

.

.

,
IMAGE_REL_BASED_HIGHLOW.

: Spanish text placeholder.7.12.

OllyDbg: Spanish text placeholder.13.11.

68.2.7

, .

.

.

.

.

.

. .

, MFC[22], .

. *mfc80_123*.

[22]Microsoft Foundation Classes

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.

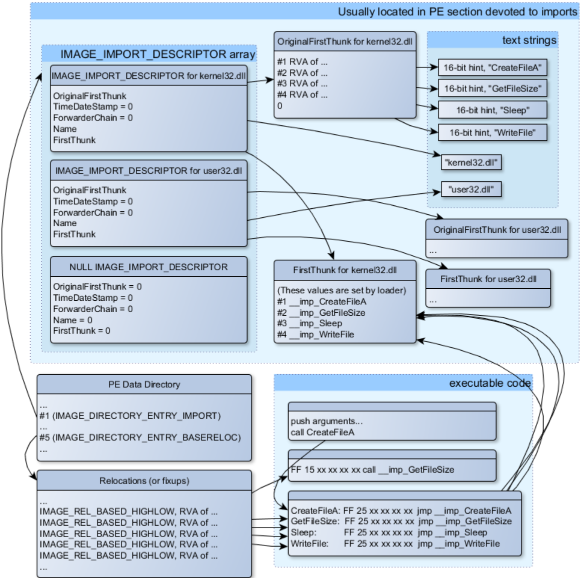

Figura 68.1:

IMAGE_IMPORT_DESCRIPTOR. .

RVA (*Name*).

OriginalFirstThink RVA . RVA, . («hint»).

FirstThunk, RVA .

: *__imp_CreateFileA*, Spanish text placeholder.

.

- *call __imp_CreateFileA*, , , .

  . . .

- .. ..

.

. PE_add_import[23] .

```
MOV EAX, [yourdll.dll!function]
JMP EAX
```

,
IMAGE_SCN_MEM_WRITE 5 (access denied).

.
IMAGE_IMPORT_DESCRIPTOR . . «old-style binding» [24]. . , Matt Pietrek en [Pie02], .

. . *LoadLibrary()* y *GetProcAddress()*.

.

[23] yurichev.com
[24] MSDN. «new-style binding».

68.2.8

. .

(68.2.11).

68.2.9 .NET

. :

```
jmp          mscoree.dll!_CorExeMain
```

[25].

68.2.10 TLS

TLS(65 on page 1314) (). TLS .

TLS TLS callbacks. (OEP).

68.2.11

- objdump (cygwin) .
- Hiew(73 on page 1420) .
- pefilePython- [26].
- ResHack AKA Resource Hacker [27].

[25] MSDN
[26] http://go.yurichev.com/17052
[27] http://go.yurichev.com/17052

- PE_add_import[28] .
- PE_patcher[29] .
- PE_search_str_refs[30] .

68.2.12 Further reading

- Daniel PistelliThe .NET File Format [31]

68.3 Windows SEH

68.3.1

. .

. . .

.

[28] http://go.yurichev.com/17049
[29] yurichev.com
[30] yurichev.com
[31] http://go.yurichev.com/17056

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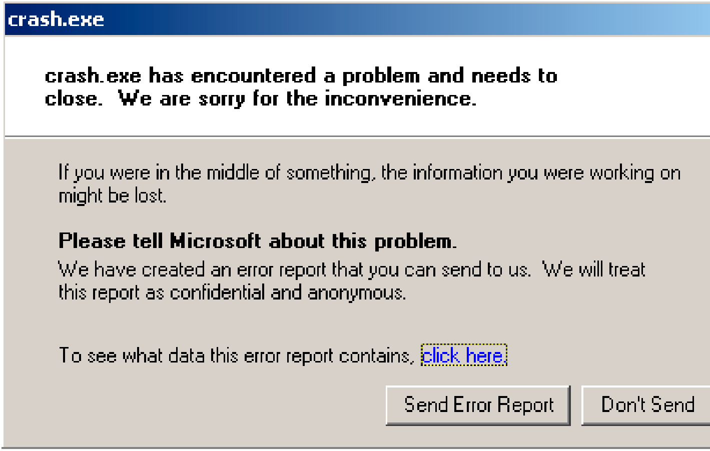

Figura 68.2: Windows XP

 CAPÍTULO 68. WINDOWS NT

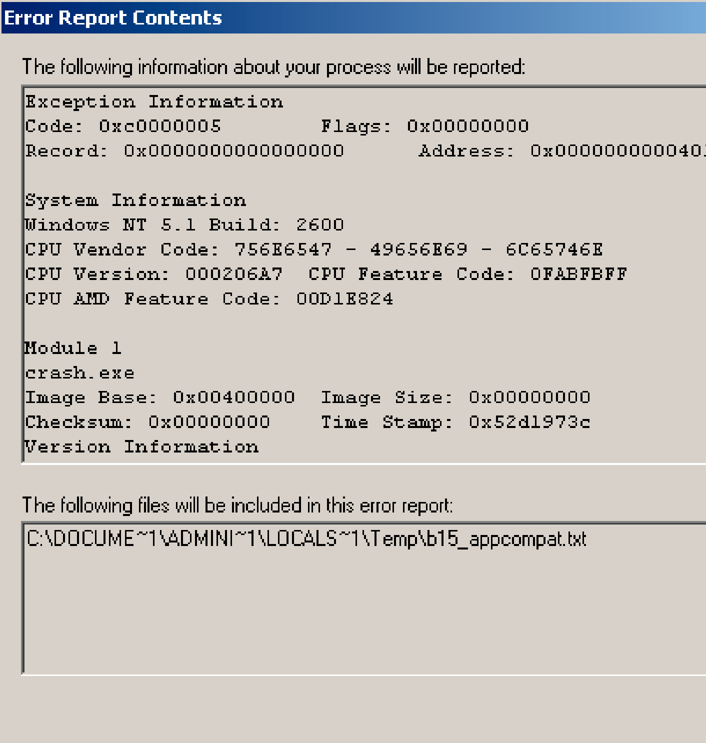

Figura 68.3: Windows XP

CAPÍTULO 68. WINDOWS NT

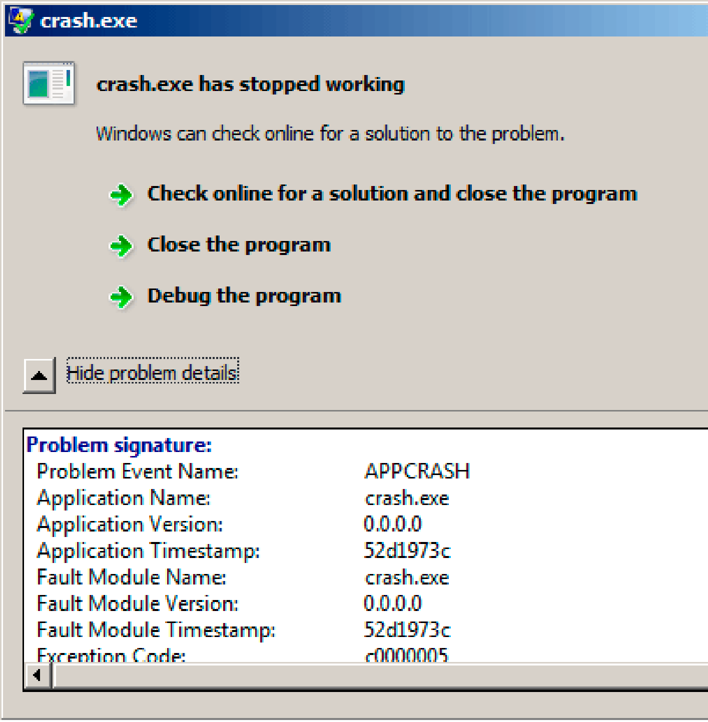

Figura 68.4: Windows 7

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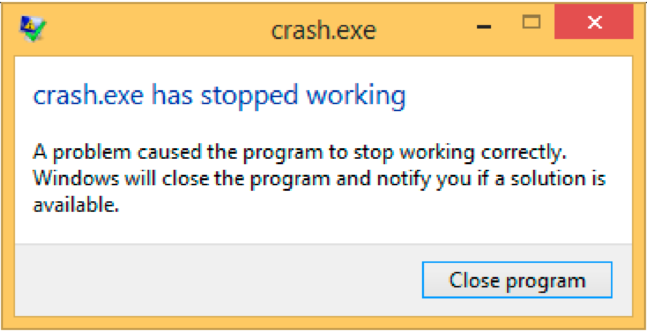

Figura 68.5: Windows 8.1

32 .

. `SetUnhandledExceptionFilter()` Oracle RDBMS

.

33 :

```
#include <windows.h>
#include <stdio.h>

DWORD new_value=1234;

EXCEPTION_DISPOSITION __cdecl except_handler↙
    ↳ (
```

32 wikipedia

33 [Pie]

SAFESEH: `cl seh1.cpp /link /safeseh:no`

SAFESEH :

MSDN

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```
                struct _EXCEPTION_RECORD *
    ↳ ExceptionRecord,
                void * EstablisherFrame,
                struct _CONTEXT *
    ↳ ContextRecord,
                void * DispatcherContext )
{
        unsigned i;

        printf ("%s\n", __FUNCTION__);
        printf ("ExceptionRecord->
    ↳ ExceptionCode=0x%p\n", ExceptionRecord
    ↳ ->ExceptionCode);
        printf ("ExceptionRecord->
    ↳ ExceptionFlags=0x%p\n", 
    ↳ ExceptionRecord->ExceptionFlags);
        printf ("ExceptionRecord->
    ↳ ExceptionAddress=0x%p\n", 
    ↳ ExceptionRecord->ExceptionAddress);

        if (ExceptionRecord->ExceptionCode
    ↳ ==0xE1223344)
        {
                printf ("That's for us\n");
                // yes, we "handled" the 
    ↳ exception
                return 
    ↳ ExceptionContinueExecution;
        }
        else if (ExceptionRecord->
    ↳ ExceptionCode==
    ↳ EXCEPTION_ACCESS_VIOLATION)
        {
```

```
                printf ("ContextRecord->Eax⤦
    ↳ =0x%08X\n", ContextRecord->Eax);
                // will it be possible to '⤦
    ↳ fix' it?
                printf ("Trying to fix wrong⤦
    ↳  pointer address\n");
                ContextRecord->Eax=(DWORD)&⤦
    ↳ new_value;
                // yes, we "handled" the ⤦
    ↳ exception
                return ⤦
    ↳ ExceptionContinueExecution;
        }
        else
        {
                printf ("We do not handle ⤦
    ↳ this\n");
                // someone else's problem
                return ⤦
    ↳ ExceptionContinueSearch;
        };
}

int main()
{
        DWORD handler = (DWORD)⤦
    ↳ except_handler; // take a pointer to ⤦
    ↳ our handler

        // install exception handler
        __asm
        {                               // ⤦
    ↳ make EXCEPTION_REGISTRATION record:
```

```
                push    handler             // ↙
    ↳ address of handler function
                push    FS:[0]              // ↙
    ↳ address of previous handler
                mov     FS:[0],ESP          // ↙
    ↳ add new EXECEPTION_REGISTRATION
        }

        RaiseException (0xE1223344, 0, 0, ↙
    ↳ NULL);

        // now do something very bad
        int* ptr=NULL;
        int val=0;
        val=*ptr;
        printf ("val=%d\n", val);

        // deinstall exception handler
        __asm
        {                                   // ↙
    ↳ remove our EXECEPTION_REGISTRATION ↙
    ↳ record
                mov     eax,[ESP]           // ↙
    ↳ get pointer to previous record
                mov     FS:[0], EAX         // ↙
    ↳ install previous record
                add     esp, 8              // ↙
    ↳ clean our EXECEPTION_REGISTRATION off ↙
    ↳ stack
        }

        return 0;
}
```

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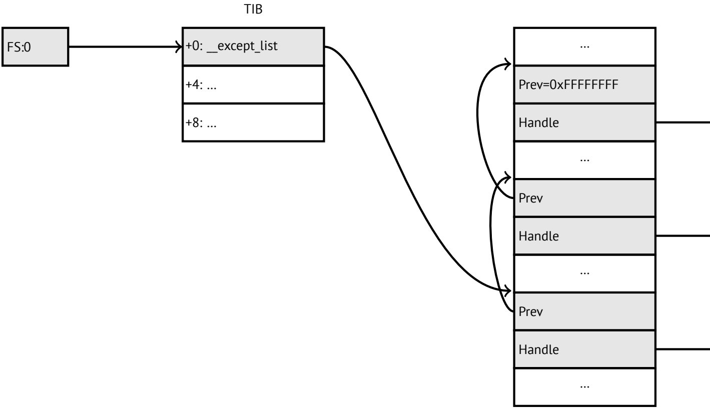
. .. _EXCEPTION_REGISTRATION, .

Listing 68.1: MSVC/VC/crt/src/exsup.inc

```
\_EXCEPTION\_REGISTRATION struc
     prev    dd      ?
     handler dd      ?
\_EXCEPTION\_REGISTRATION ends
```

«handler» «prev» . 0xFFFFFFFF (-1) «prev».



RaiseException()[34]... 0xE1223344, ExceptionConti
. ExceptionContinueSearch, . .

? :

[34] MSDN

CAPÍTULO 68. WINDOWS NT

WinBase.h	ntstatus.h
EXCEPTION_ACCESS_VIOLATION	STATUS_ACCE
EXCEPTION_DATATYPE_MISALIGNMENT	STATUS_DATA
EXCEPTION_BREAKPOINT	STATUS_BREA
EXCEPTION_SINGLE_STEP	STATUS_SING
EXCEPTION_ARRAY_BOUNDS_EXCEEDED	STATUS_ARRA
EXCEPTION_FLT_DENORMAL_OPERAND	STATUS_FLOA
EXCEPTION_FLT_DIVIDE_BY_ZERO	STATUS_FLOA
EXCEPTION_FLT_INEXACT_RESULT	STATUS_FLOA
EXCEPTION_FLT_INVALID_OPERATION	STATUS_FLOA
EXCEPTION_FLT_OVERFLOW	STATUS_FLOA
EXCEPTION_FLT_STACK_CHECK	STATUS_FLOA
EXCEPTION_FLT_UNDERFLOW	STATUS_FLOA
EXCEPTION_INT_DIVIDE_BY_ZERO	STATUS_INTE
EXCEPTION_INT_OVERFLOW	STATUS_INTE
EXCEPTION_PRIV_INSTRUCTION	STATUS_PRIV
EXCEPTION_IN_PAGE_ERROR	STATUS_IN_P
EXCEPTION_ILLEGAL_INSTRUCTION	STATUS_ILLE
EXCEPTION_NONCONTINUABLE_EXCEPTION	STATUS_NON
EXCEPTION_STACK_OVERFLOW	STATUS_STAC
EXCEPTION_INVALID_DISPOSITION	STATUS_INVA
EXCEPTION_GUARD_PAGE	STATUS_GUA
EXCEPTION_INVALID_HANDLE	STATUS_INVA
EXCEPTION_POSSIBLE_DEADLOCK	STATUS_POSS
CONTROL_C_EXIT	STATUS_CON

:

31	29	28 27	16 15	0
S	U	0	Facility code	Error code

S : 11; 10; 01; 00. U.

0xE1223344 0xE (1110b) . .

. . .

:

Listing 68.2: MSVC 2010

```
        ...
        xor     eax, eax
        mov     eax, DWORD PTR [eax] ; ↙
    ↳ exception will occur here
        push    eax
        push    OFFSET msg
        call    _printf
        add     esp, 8
        ...
```

? . . printf() 1234, EAX 0, new_value. .

:, , , .

? alloca(): (5.2.4 on page 68).

68.3.2

[35] . .

[35] MSDN

 CAPÍTULO 68. WINDOWS NT

```
__try
{
    ...
}
__except(filter code)
{
    handler code
}
```

:

```
__try
{
    ...
}
__finally
{
    ...
}
```

. .

. (WRK):

Listing 68.3: WRK-v1.2/base/ntos/ob/obwait.c

```
try {

    KeReleaseMutant( (PKMUTANT)SignalObject,
                     MUTANT_INCREMENT,
                     FALSE,
                     TRUE );
```

```
} except((GetExceptionCode () == ↙
    ↳ STATUS_ABANDONED ||
          GetExceptionCode () == ↙
    ↳ STATUS_MUTANT_NOT_OWNED)?
             EXCEPTION_EXECUTE_HANDLER :
             EXCEPTION_CONTINUE_SEARCH) {
    Status = GetExceptionCode();

    goto WaitExit;
}
```

Listing 68.4: WRK-v1.2/base/ntos/cache/cachesub.c

```
try {

    RtlCopyBytes( (PVOID)((PCHAR)CacheBuffer↙
    ↳  + PageOffset),
                  UserBuffer,
                  MorePages ?
                    (PAGE_SIZE - PageOffset)↙
    ↳  :
                    (ReceivedLength - ↙
    ↳ PageOffset) );

} except( CcCopyReadExceptionFilter( ↙
    ↳ GetExceptionInformation(),
                                         ↙
    ↳                  &Status ) ) {
```

:

Listing 68.5: WRK-v1.2/base/ntos/cache/copysup.c

```
LONG
CcCopyReadExceptionFilter(
    IN PEXCEPTION_POINTERS ExceptionPointer,
    IN PNTSTATUS ExceptionCode
    )

/*++

Routine Description:

    This routine serves as a exception ↙
    ↳ filter and has the special job of
    extracting the "real" I/O error when Mm ↙
    ↳ raises STATUS_IN_PAGE_ERROR
    beneath us.

Arguments:

    ExceptionPointer - A pointer to the ↙
    ↳ exception record that contains
                       the real Io Status.

    ExceptionCode - A pointer to an NTSTATUS↙
    ↳  that is to receive the real
                    status.

Return Value:

    EXCEPTION_EXECUTE_HANDLER

--*/
```

```
{
    *ExceptionCode = ExceptionPointer->⤦
    ↳ ExceptionRecord->ExceptionCode;

    if ( (*ExceptionCode == ⤦
    ↳ STATUS_IN_PAGE_ERROR) &&
         (ExceptionPointer->ExceptionRecord⤦
    ↳ ->NumberParameters >= 3) ) {

        *ExceptionCode = (NTSTATUS) ⤦
    ↳ ExceptionPointer->ExceptionRecord->⤦
    ↳ ExceptionInformation[2];
    }

    ASSERT( !NT_SUCCESS(*ExceptionCode) );

    return EXCEPTION_EXECUTE_HANDLER;
}
```

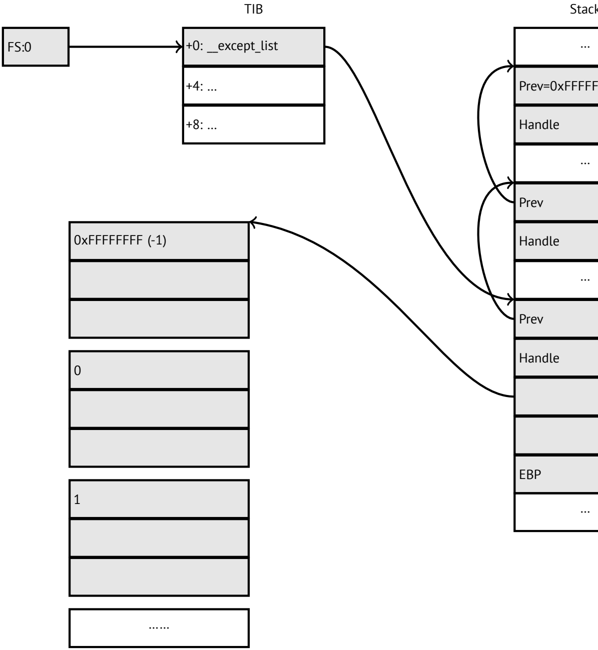
. `_except_handler3` (para SEH3) o `_except_handler4` (para SEH4). msvcr*.dll. .

SEH3

SEH3 `_except_handler3` `_EXCEPTION_REGISTRATION` . SEH4 *scope table* .

scope table .

 CAPÍTULO 68. WINDOWS NT



prev/handle . `_except_handler3` .

`_except_handler3` . [36]. Wine[37] y ReactOS[38].

filter handler .

[36] http://go.yurichev.com/17058
[37] GitHub
[38] http://go.yurichev.com/17060

SEH3:

```
#include <stdio.h>
#include <windows.h>
#include <excpt.h>

int main()
{
    int* p = NULL;
    __try
    {
        printf("hello #1!\n");
        *p = 13;    // causes an access ⤦
    ↳ violation exception;
        printf("hello #2!\n");
    }
    __except(GetExceptionCode()==⤦
    ↳ EXCEPTION_ACCESS_VIOLATION ?
            EXCEPTION_EXECUTE_HANDLER : ⤦
    ↳ EXCEPTION_CONTINUE_SEARCH)
    {
        printf("access violation, can't ⤦
    ↳ recover\n");
    }
}
```

Listing 68.6: MSVC 2003

```
$SG74605 DB    'hello #1!', 0aH, 00H
```

```
$SG74606 DB    'hello #2!', 0aH, 00H
$SG74608 DB    'access violation, can''t ↙
    ↘ recover', 0aH, 00H
_DATA    ENDS

; scope table:
CONST     SEGMENT
$T74622   DD    0ffffffffH    ; previous try↙
    ↘ level
          DD    FLAT:$L74617  ; filter
          DD    FLAT:$L74618  ; handler

CONST    ENDS
_TEXT    SEGMENT
$T74621 = -32 ; size = 4
_p$ = -28     ; size = 4
__$SEHRec$ = -24 ; size = 24
_main    PROC NEAR
    push   ebp
    mov    ebp, esp
    push   -1                                ;↙
    ↘ previous try level
    push   OFFSET FLAT:$T74622               ;↙
    ↘ scope table
    push   OFFSET FLAT:__except_handler3     ;↙
    ↘ handler
    mov    eax, DWORD PTR fs:__except_list
    push   eax                               ;↙
    ↘ prev
    mov    DWORD PTR fs:__except_list, esp
    add    esp, -16
; 3 registers to be saved:
    push   ebx
```

```
    push   esi
    push   edi
    mov    DWORD PTR __$SEHRec$[ebp], esp
    mov    DWORD PTR _p$[ebp], 0
    mov    DWORD PTR __$SEHRec$[ebp+20], 0  ↙
    ↳ ; previous try level
    push   OFFSET FLAT:$SG74605 ; 'hello ↙
    ↳ #1!'
    call   _printf
    add    esp, 4
    mov    eax, DWORD PTR _p$[ebp]
    mov    DWORD PTR [eax], 13
    push   OFFSET FLAT:$SG74606 ; 'hello ↙
    ↳ #2!'
    call   _printf
    add    esp, 4
    mov    DWORD PTR __$SEHRec$[ebp+20], -1 ↙
    ↳  ; previous try level
    jmp    SHORT $L74616

    ; filter code:
$L74617:
$L74627:
    mov    ecx, DWORD PTR __$SEHRec$[ebp+4]
    mov    edx, DWORD PTR [ecx]
    mov    eax, DWORD PTR [edx]
    mov    DWORD PTR $T74621[ebp], eax
    mov    eax, DWORD PTR $T74621[ebp]
    sub    eax, -1073741819; c0000005H
    neg    eax
    sbb    eax, eax
    inc    eax
$L74619:
```

```
$L74626:
    ret    0

    ; handler code:
$L74618:
    mov    esp, DWORD PTR __$SEHRec$[ebp]
    push   OFFSET FLAT:$SG74608 ; 'access ⤦
    ↳ violation, can''t recover'
    call   _printf
    add    esp, 4
    mov    DWORD PTR __$SEHRec$[ebp+20], -1 ⤦
    ↳ ; setting previous try level back to ⤦
    ↳ -1
$L74616:
    xor    eax, eax
    mov    ecx, DWORD PTR __$SEHRec$[ebp+8]
    mov    DWORD PTR fs:__except_list, ecx
    pop    edi
    pop    esi
    pop    ebx
    mov    esp, ebp
    pop    ebp
    ret    0
_main    ENDP
_TEXT    ENDS
END
```

. CONST . *previous try level*. 0xFFFFFFFF (-1). try 0 . try , -1 . . .

SEH_prolog() . .

tracer:

```
tracer.exe -l:2.exe --dump-seh
```

Listing 68.7: tracer.exe output

```
EXCEPTION_ACCESS_VIOLATION at 2.exe!main+0x44 (0x401054) ExceptionInformation[0]=1
EAX=0x00000000 EBX=0x7efde000 ECX=0x0040cbc8 EDX=0x0008e3c8
ESI=0x00001db1 EDI=0x00000000 EBP=0x0018feac ESP=0x0018fe80
EIP=0x00401054
FLAGS=AF IF RF
* SEH frame at 0x18fe9c prev=0x18ff78 handler=0x401204 (2.exe!_except_handler3)
SEH3 frame. previous trylevel=0
scopetable entry[0]. previous try level=-1, filter=0x401070 (2.exe!main+0x60) handler=0x401088 (2.exe!main+0x78)
* SEH frame at 0x18ff78 prev=0x18ffc4 handler=0x401204 (2.exe!_except_handler3)
SEH3 frame. previous trylevel=0
scopetable entry[0]. previous try level=-1, filter=0x401531 (2.exe!mainCRTStartup+0x18d) handler=0x401545 (2.exe!mainCRTStartup+0x1a1)
* SEH frame at 0x18ffc4 prev=0x18ffe4 handler=0x771f71f5 (ntdll.dll!__except_handler4)
```

```
SEH4 frame. previous trylevel=0
SEH4 header:    GSCookieOffset=0xfffffffe ⤦
    ↳ GSCookieXOROffset=0x0
                EHCookieOffset=0xffffffcc ⤦
    ↳ EHCookieXOROffset=0x0
scopetable entry[0]. previous try level=-2, ⤦
    ↳ filter=0x771f74d0 (ntdll.dll!⤦
    ↳ ___safe_se_handler_table+0x20) handler⤦
    ↳ =0x771f90eb (ntdll.dll!⤦
    ↳ _TppTerminateProcess@4+0x43)
* SEH frame at 0x18ffe4 prev=0xffffffff ⤦
    ↳ handler=0x77247428 (ntdll.dll!⤦
    ↳ _FinalExceptionHandler@16)
```

.

.?? `_mainCRTStartup()`, .:`crt/src/winxfltr.c`.

ntdll.dll, ntdll.dll, .

: : SEH3 y SEH4.

SEH3:

```
#include <stdio.h>
#include <windows.h>
#include <excpt.h>

int filter_user_exceptions (unsigned int ⤦
    ↳ code, struct _EXCEPTION_POINTERS *ep)
{
```

```
    printf("in filter. code=0x%08X\n", code)↙
    ↳ ;
    if (code == 0x112233)
    {
        printf("yes, that is our exception\n↙
    ↳ ");
        return EXCEPTION_EXECUTE_HANDLER;
    }
    else
    {
        printf("not our exception\n");
        return EXCEPTION_CONTINUE_SEARCH;
    };
}
int main()
{
    int* p = NULL;
    __try
    {
        __try
        {
            printf ("hello!\n");
            RaiseException (0x112233, 0, 0, ↙
    ↳ NULL);
            printf ("0x112233 raised. now ↙
    ↳ let's crash\n");
            *p = 13;    // causes an access ↙
    ↳ violation exception;
        }
        __except(GetExceptionCode()==↙
    ↳ EXCEPTION_ACCESS_VIOLATION ?
                EXCEPTION_EXECUTE_HANDLER : ↙
    ↳ EXCEPTION_CONTINUE_SEARCH)
```

```
        {
            printf("access violation, can't ↙
    ↘ recover\n");
        }
    }
    __except(filter_user_exceptions(↙
    ↘ GetExceptionCode(), ↙
    ↘ GetExceptionInformation()))
    {
        // the filter_user_exceptions() ↙
    ↘ function answering to the question
        // "is this exception belongs to ↙
    ↘ this block?"
        // if yes, do the follow:
        printf("user exception caught\n");
    }
}
```

. *scope table* . *Previous try level* .

Listing 68.8: MSVC 2003

```
$SG74606 DB    'in filter. code=0x%08X', 0aH↙
    ↘ , 00H
$SG74608 DB    'yes, that is our exception',↙
    ↘  0aH, 00H
$SG74610 DB    'not our exception', 0aH, 00H
$SG74617 DB    'hello!', 0aH, 00H
$SG74619 DB    '0x112233 raised. now let''s ↙
    ↘ crash', 0aH, 00H
$SG74621 DB    'access violation, can''t ↙
    ↘ recover', 0aH, 00H
$SG74623 DB    'user exception caught', 0aH,↙
```

```
    ↳  00H

_code$ = 8   ; size = 4
_ep$ = 12    ; size = 4
_filter_user_exceptions PROC NEAR
    push   ebp
    mov    ebp, esp
    mov    eax, DWORD PTR _code$[ebp]
    push   eax
    push   OFFSET FLAT:$SG74606 ; 'in filter↙
    ↳ . code=0x%08X'
    call   _printf
    add    esp, 8
    cmp    DWORD PTR _code$[ebp], 1122867; ↙
    ↳ 00112233H
    jne    SHORT $L74607
    push   OFFSET FLAT:$SG74608 ; 'yes, that↙
    ↳  is our exception'
    call   _printf
    add    esp, 4
    mov    eax, 1
    jmp    SHORT $L74605
$L74607:
    push   OFFSET FLAT:$SG74610 ; 'not our ↙
    ↳ exception'
    call   _printf
    add    esp, 4
    xor    eax, eax
$L74605:
    pop    ebp
    ret    0
_filter_user_exceptions ENDP
```

```
; scope table:
CONST    SEGMENT
$T74644   DD    0ffffffffH   ; previous try ↙
    ↘ level for outer block
          DD    FLAT:$L74634 ; outer block ↙
    ↘ filter
          DD    FLAT:$L74635 ; outer block ↙
    ↘ handler
          DD    00H          ; previous try ↙
    ↘ level for inner block
          DD    FLAT:$L74638 ; inner block ↙
    ↘ filter
          DD    FLAT:$L74639 ; inner block ↙
    ↘ handler
CONST    ENDS

$T74643 = -36     ; size = 4
$T74642 = -32     ; size = 4
_p$ = -28         ; size = 4
__$SEHRec$ = -24  ; size = 24
_main    PROC NEAR
    push   ebp
    mov    ebp, esp
    push   -1 ; previous try level
    push   OFFSET FLAT:$T74644
    push   OFFSET FLAT:__except_handler3
    mov    eax, DWORD PTR fs:__except_list
    push   eax
    mov    DWORD PTR fs:__except_list, esp
    add    esp, -20
    push   ebx
    push   esi
    push   edi
```

```
    mov     DWORD PTR __$SEHRec$[ebp], esp
    mov     DWORD PTR _p$[ebp], 0
    mov     DWORD PTR __$SEHRec$[ebp+20], 0 ;↙
    ↘ outer try block entered. set previous↙
    ↘ try level to 0
    mov     DWORD PTR __$SEHRec$[ebp+20], 1 ;↙
    ↘ inner try block entered. set previous↙
    ↘ try level to 1
    push    OFFSET FLAT:$SG74617 ; 'hello!'
    call    _printf
    add     esp, 4
    push    0
    push    0
    push    0
    push    1122867     ; 00112233H
    call    DWORD PTR ↙
    ↘ __imp__RaiseException@16
    push    OFFSET FLAT:$SG74619 ; '0x112233 ↙
    ↘ raised. now let''s crash'
    call    _printf
    add     esp, 4
    mov     eax, DWORD PTR _p$[ebp]
    mov     DWORD PTR [eax], 13
    mov     DWORD PTR __$SEHRec$[ebp+20], 0 ;↙
    ↘ inner try block exited. set previous ↙
    ↘ try level back to 0
    jmp     SHORT $L74615

; inner block filter:
$L74638:
$L74650:
    mov     ecx, DWORD PTR __$SEHRec$[ebp+4]
    mov     edx, DWORD PTR [ecx]
```

```
    mov     eax, DWORD PTR [edx]
    mov     DWORD PTR $T74643[ebp], eax
    mov     eax, DWORD PTR $T74643[ebp]
    sub     eax, -1073741819; c0000005H
    neg     eax
    sbb     eax, eax
    inc     eax
$L74640:
$L74648:
    ret     0

; inner block handler:
$L74639:
    mov     esp, DWORD PTR __$SEHRec$[ebp]
    push    OFFSET FLAT:$SG74621 ; 'access ↙
    ↘ violation, can''t recover'
    call    _printf
    add     esp, 4
    mov     DWORD PTR __$SEHRec$[ebp+20], 0 ;↙
    ↘  inner try block exited. set previous ↙
    ↘ try level back to 0

$L74615:
    mov     DWORD PTR __$SEHRec$[ebp+20], -1 ↙
    ↘ ; outer try block exited, set previous↙
    ↘  try level back to -1
    jmp     SHORT $L74633

; outer block filter:
$L74634:
$L74651:
    mov     ecx, DWORD PTR __$SEHRec$[ebp+4]
    mov     edx, DWORD PTR [ecx]
```

```
    mov     eax, DWORD PTR [edx]
    mov     DWORD PTR $T74642[ebp], eax
    mov     ecx, DWORD PTR __$SEHRec$[ebp+4]
    push    ecx
    mov     edx, DWORD PTR $T74642[ebp]
    push    edx
    call    _filter_user_exceptions
    add     esp, 8
$L74636:
$L74649:
    ret     0

; outer block handler:
$L74635:
    mov     esp, DWORD PTR __$SEHRec$[ebp]
    push    OFFSET FLAT:$SG74623 ; 'user ⤦
    ↪ exception caught'
    call    _printf
    add     esp, 4
    mov     DWORD PTR __$SEHRec$[ebp+20], -1 ⤦
    ↪ ; both try blocks exited. set previous⤦
    ↪  try level back to -1
$L74633:
    xor     eax, eax
    mov     ecx, DWORD PTR __$SEHRec$[ebp+8]
    mov     DWORD PTR fs:__except_list, ecx
    pop     edi
    pop     esi
    pop     ebx
    mov     esp, ebp
    pop     ebp
    ret     0
_main    ENDP
```

printf() . . *scope table* .

```
tracer.exe -l:3.exe bpx=3.exe!printf --dump-
    seh
```

Listing 68.9: tracer.exe output

```
(0) 3.exe!printf
EAX=0x0000001b EBX=0x00000000 ECX=0x0040cc58
     EDX=0x0008e3c8
ESI=0x00000000 EDI=0x00000000 EBP=0x0018f840
     ESP=0x0018f838
EIP=0x004011b6
FLAGS=PF ZF IF
* SEH frame at 0x18f88c prev=0x18fe9c
    handler=0x771db4ad (ntdll.dll!
    ExecuteHandler2@20+0x3a)
* SEH frame at 0x18fe9c prev=0x18ff78
    handler=0x4012e0 (3.exe!
    _except_handler3)
SEH3 frame. previous trylevel=1
scopetable entry[0]. previous try level=-1,
    filter=0x401120 (3.exe!main+0xb0)
    handler=0x40113b (3.exe!main+0xcb)
scopetable entry[1]. previous try level=0,
    filter=0x4010e8 (3.exe!main+0x78)
    handler=0x401100 (3.exe!main+0x90)
* SEH frame at 0x18ff78 prev=0x18ffc4
    handler=0x4012e0 (3.exe!
    _except_handler3)
SEH3 frame. previous trylevel=0
```

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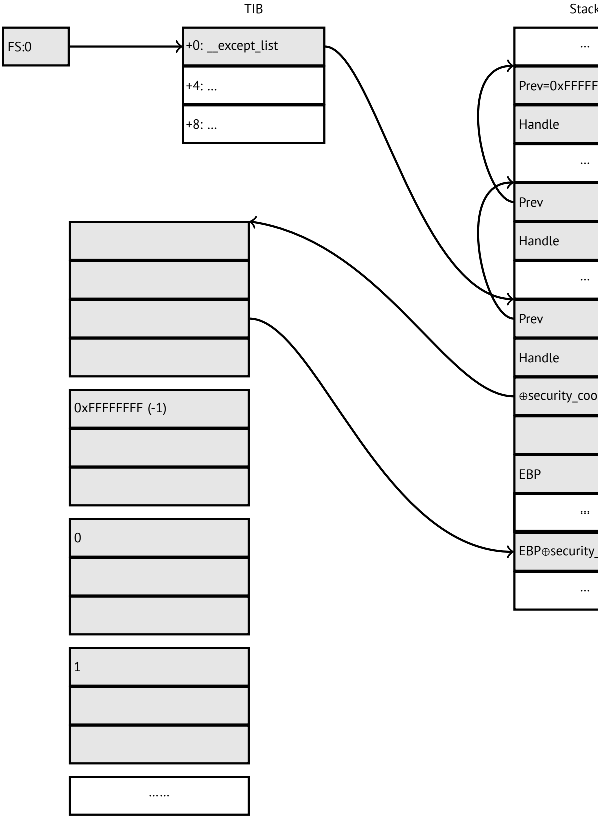
```
scopetable entry[0]. previous try level=-1, ⤦
    ↳ filter=0x40160d (3.exe!mainCRTStartup⤦
    ↳ +0x18d) handler=0x401621 (3.exe!⤦
    ↳ mainCRTStartup+0x1a1)
* SEH frame at 0x18ffc4 prev=0x18ffe4 ⤦
    ↳ handler=0x771f71f5 (ntdll.dll!⤦
    ↳ __except_handler4)
SEH4 frame. previous trylevel=0
SEH4 header:    GSCookieOffset=0xfffffffe ⤦
    ↳ GSCookieXOROffset=0x0
                EHCookieOffset=0xffffffcc ⤦
    ↳ EHCookieXOROffset=0x0
scopetable entry[0]. previous try level=-2, ⤦
    ↳ filter=0x771f74d0 (ntdll.dll!⤦
    ↳ ___safe_se_handler_table+0x20) handler⤦
    ↳ =0x771f90eb (ntdll.dll!⤦
    ↳ _TppTerminateProcess@4+0x43)
* SEH frame at 0x18ffe4 prev=0xffffffff ⤦
    ↳ handler=0x77247428 (ntdll.dll!⤦
    ↳ _FinalExceptionHandler@16)
```

SEH4

(18.2 on page 496) *scope table* MSVC 2005, SEH3 . *scope table* security cookie. *security cookies*. `security_cookie` .

previous try level -2 en SEH4 -1.



MSVC 2012 SEH4:

Listing 68.10: MSVC 2012: one try block example

```
$SG85485 DB    'hello #1!', 0aH, 00H
$SG85486 DB    'hello #2!', 0aH, 00H
$SG85488 DB    'access violation, can''t ⤦
    ↳ recover', 0aH, 00H

; scope table:
xdata$x          SEGMENT
__sehtable$_main DD 0fffffffeH   ; GS Cookie⤦
    ↳  Offset
    DD           00H             ; GS Cookie⤦
    ↳  XOR Offset
    DD           0ffffffccH      ; EH Cookie⤦
    ↳  Offset
    DD           00H             ; EH Cookie⤦
    ↳  XOR Offset
    DD           0fffffffeH      ; previous ⤦
    ↳ try level
    DD           FLAT:$LN12@main ; filter
    DD           FLAT:$LN8@main  ; handler
xdata$x          ENDS

$T2 = -36         ; size = 4
_p$ = -32         ; size = 4
tv68 = -28        ; size = 4
__$SEHRec$ = -24 ; size = 24
_main    PROC
    push   ebp
    mov    ebp, esp
    push   -2
    push   OFFSET __sehtable$_main
    push   OFFSET __except_handler4
    mov    eax, DWORD PTR fs:0
    push   eax
```

```
    add     esp, -20
    push    ebx
    push    esi
    push    edi
    mov     eax, DWORD PTR ___security_cookie
    xor     DWORD PTR __$SEHRec$[ebp+16], eax
    ↳  ; xored pointer to scope table
    xor     eax, ebp
    push    eax
    ↳ ; ebp ^ security_cookie
    lea     eax, DWORD PTR __$SEHRec$[ebp+8]
    ↳ ; pointer to
    ↳ VC_EXCEPTION_REGISTRATION_RECORD
    mov     DWORD PTR fs:0, eax
    mov     DWORD PTR __$SEHRec$[ebp], esp
    mov     DWORD PTR _p$[ebp], 0
    mov     DWORD PTR __$SEHRec$[ebp+20], 0 ;
    ↳  previous try level
    push    OFFSET $SG85485 ; 'hello #1!'
    call    _printf
    add     esp, 4
    mov     eax, DWORD PTR _p$[ebp]
    mov     DWORD PTR [eax], 13
    push    OFFSET $SG85486 ; 'hello #2!'
    call    _printf
    add     esp, 4
    mov     DWORD PTR __$SEHRec$[ebp+20], -2
    ↳ ; previous try level
    jmp     SHORT $LN6@main

; filter:
$LN7@main:
$LN12@main:
```

```
    mov     ecx, DWORD PTR __$SEHRec$[ebp+4]
    mov     edx, DWORD PTR [ecx]
    mov     eax, DWORD PTR [edx]
    mov     DWORD PTR $T2[ebp], eax
    cmp     DWORD PTR $T2[ebp], -1073741819 ; ↙
    ↘ c0000005H
    jne     SHORT $LN4@main
    mov     DWORD PTR tv68[ebp], 1
    jmp     SHORT $LN5@main
$LN4@main:
    mov     DWORD PTR tv68[ebp], 0
$LN5@main:
    mov     eax, DWORD PTR tv68[ebp]
$LN9@main:
$LN11@main:
    ret     0

; handler:
$LN8@main:
    mov     esp, DWORD PTR __$SEHRec$[ebp]
    push    OFFSET $SG85488 ; 'access ↙
    ↘ violation, can''t recover'
    call    _printf
    add     esp, 4
    mov     DWORD PTR __$SEHRec$[ebp+20], -2 ↙
    ↘ ; previous try level
$LN6@main:
    xor     eax, eax
    mov     ecx, DWORD PTR __$SEHRec$[ebp+8]
    mov     DWORD PTR fs:0, ecx
    pop     ecx
    pop     edi
    pop     esi
```

```
    pop    ebx
    mov    esp, ebp
    pop    ebp
    ret    0
_main    ENDP
```

Listing 68.11: MSVC 2012: two try blocks example

```
$SG85486 DB    'in filter. code=0x%08X', 0aH↙
    ↘ , 00H
$SG85488 DB    'yes, that is our exception',↙
    ↘  0aH, 00H
$SG85490 DB    'not our exception', 0aH, 00H
$SG85497 DB    'hello!', 0aH, 00H
$SG85499 DB    '0x112233 raised. now let''s ↙
    ↘ crash', 0aH, 00H
$SG85501 DB    'access violation, can''t ↙
    ↘ recover', 0aH, 00H
$SG85503 DB    'user exception caught', 0aH,↙
    ↘  00H

xdata$x    SEGMENT
__sehtable$_main DD 0fffffffeH        ; GS ↙
    ↘ Cookie Offset
                 DD    00H            ; GS ↙
    ↘ Cookie XOR Offset
                 DD    0ffffffc8H     ; EH ↙
    ↘ Cookie Offset
                 DD    00H            ; EH ↙
    ↘ Cookie Offset
                 DD    0fffffffeH     ; ↙
    ↘ previous try level for outer block
                 DD    FLAT:$LN19@main ; ↙
```

```
    ↘ outer block filter
                DD    FLAT:$LN9@main  ; ↙
    ↘ outer block handler
                DD    00H             ; ↙
    ↘ previous try level for inner block
                DD    FLAT:$LN18@main ; ↙
    ↘ inner block filter
                DD    FLAT:$LN13@main ; ↙
    ↘ inner block handler
xdata$x    ENDS

$T2 = -40        ; size = 4
$T3 = -36        ; size = 4
_p$ = -32        ; size = 4
tv72 = -28       ; size = 4
__$SEHRec$ = -24 ; size = 24
_main    PROC
    push   ebp
    mov    ebp, esp
    push   -2  ; initial previous try level
    push   OFFSET __sehtable$_main
    push   OFFSET __except_handler4
    mov    eax, DWORD PTR fs:0
    push   eax ; prev
    add    esp, -24
    push   ebx
    push   esi
    push   edi
    mov    eax, DWORD PTR ___security_cookie
    xor    DWORD PTR __$SEHRec$[ebp+16], eax↙
    ↘        ; xored pointer to scope table
    xor    eax, ebp                        ↙
    ↘        ; ebp ^ security_cookie
```

```
    push   eax
    lea    eax, DWORD PTR __$SEHRec$[ebp+8] ↙
    ↘        ; pointer to ↙
    ↘ VC_EXCEPTION_REGISTRATION_RECORD
    mov    DWORD PTR fs:0, eax
    mov    DWORD PTR __$SEHRec$[ebp], esp
    mov    DWORD PTR _p$[ebp], 0
    mov    DWORD PTR __$SEHRec$[ebp+20], 0 ;↙
    ↘  entering outer try block, setting ↙
    ↘ previous try level=0
    mov    DWORD PTR __$SEHRec$[ebp+20], 1 ;↙
    ↘  entering inner try block, setting ↙
    ↘ previous try level=1
    push   OFFSET $SG85497 ; 'hello!'
    call   _printf
    add    esp, 4
    push   0
    push   0
    push   0
    push   1122867 ; 00112233H
    call   DWORD PTR ↙
    ↘ __imp__RaiseException@16
    push   OFFSET $SG85499 ; '0x112233 ↙
    ↘ raised. now let''s crash'
    call   _printf
    add    esp, 4
    mov    eax, DWORD PTR _p$[ebp]
    mov    DWORD PTR [eax], 13
    mov    DWORD PTR __$SEHRec$[ebp+20], 0 ;↙
    ↘  exiting inner try block, set previous↙
    ↘  try level back to 0
    jmp    SHORT $LN2@main
```

```
; inner block filter:
$LN12@main:
$LN18@main:
    mov     ecx, DWORD PTR __$SEHRec$[ebp+4]
    mov     edx, DWORD PTR [ecx]
    mov     eax, DWORD PTR [edx]
    mov     DWORD PTR $T3[ebp], eax
    cmp     DWORD PTR $T3[ebp], -1073741819 ;↙
    ↘ c0000005H
    jne     SHORT $LN5@main
    mov     DWORD PTR tv72[ebp], 1
    jmp     SHORT $LN6@main
$LN5@main:
    mov     DWORD PTR tv72[ebp], 0
$LN6@main:
    mov     eax, DWORD PTR tv72[ebp]
$LN14@main:
$LN16@main:
    ret     0

; inner block handler:
$LN13@main:
    mov     esp, DWORD PTR __$SEHRec$[ebp]
    push    OFFSET $SG85501 ; 'access ↙
    ↘ violation, can''t recover'
    call    _printf
    add     esp, 4
    mov     DWORD PTR __$SEHRec$[ebp+20], 0 ;↙
    ↘  exiting inner try block, setting ↙
    ↘ previous try level back to 0
$LN2@main:
    mov     DWORD PTR __$SEHRec$[ebp+20], -2 ↙
    ↘ ; exiting both blocks, setting ↙
```

```
    ↳ previous try level back to -2
    jmp     SHORT $LN7@main

; outer block filter:
$LN8@main:
$LN19@main:
    mov     ecx, DWORD PTR __$SEHRec$[ebp+4]
    mov     edx, DWORD PTR [ecx]
    mov     eax, DWORD PTR [edx]
    mov     DWORD PTR $T2[ebp], eax
    mov     ecx, DWORD PTR __$SEHRec$[ebp+4]
    push    ecx
    mov     edx, DWORD PTR $T2[ebp]
    push    edx
    call    _filter_user_exceptions
    add     esp, 8
$LN10@main:
$LN17@main:
    ret     0

; outer block handler:
$LN9@main:
    mov     esp, DWORD PTR __$SEHRec$[ebp]
    push    OFFSET $SG85503 ; 'user exception↙
    ↳  caught'
    call    _printf
    add     esp, 4
    mov     DWORD PTR __$SEHRec$[ebp+20], -2 ↙
    ↳ ; exiting both blocks, setting ↙
    ↳ previous try level back to -2
$LN7@main:
    xor     eax, eax
    mov     ecx, DWORD PTR __$SEHRec$[ebp+8]
```

```
    mov     DWORD PTR fs:0, ecx
    pop     ecx
    pop     edi
    pop     esi
    pop     ebx
    mov     esp, ebp
    pop     ebp
    ret     0
_main    ENDP

_code$ = 8  ; size = 4
_ep$ = 12   ; size = 4
_filter_user_exceptions PROC
    push    ebp
    mov     ebp, esp
    mov     eax, DWORD PTR _code$[ebp]
    push    eax
    push    OFFSET $SG85486 ; 'in filter. ↙
    ↳ code=0x%08X'
    call    _printf
    add     esp, 8
    cmp     DWORD PTR _code$[ebp], 1122867 ; ↙
    ↳ 00112233H
    jne     SHORT $LN2@filter_use
    push    OFFSET $SG85488 ; 'yes, that is ↙
    ↳ our exception'
    call    _printf
    add     esp, 4
    mov     eax, 1
    jmp     SHORT $LN3@filter_use
    jmp     SHORT $LN3@filter_use
$LN2@filter_use:
    push    OFFSET $SG85490 ; 'not our ↙
```

```
    ↘ exception'
    call    _printf
    add     esp, 4
    xor     eax, eax
$LN3@filter_use:
    pop     ebp
    ret     0
_filter_user_exceptions ENDP
```

cookies:`Cookie Offset` $EBP \oplus security\_cookie$.`Cookie XOR Offset` $EBP \oplus security\_cookie$. :

$$security\_cookie \oplus (CookieXOROffset + address\_of\_saved\_EBP) == stack[address\_of\_saved\_EBP + CookieOffset]$$

`Cookie Offset` -2, .

tracer, GitHub .

68.3.3 Windows x64

. *previous try level* . `.pdata`, .

x64:

Listing 68.12: MSVC 2012

```
$SG86276 DB     'hello #1!', 0aH, 00H
$SG86277 DB     'hello #2!', 0aH, 00H
$SG86279 DB     'access violation, can''t ↙
    ↘ recover', 0aH, 00H
```

```
pdata   SEGMENT
$pdata$main DD  imagerel $LN9
        DD      imagerel $LN9+61
        DD      imagerel $unwind$main
pdata   ENDS
pdata   SEGMENT
$pdata$main$filt$0 DD imagerel main$filt$0
        DD      imagerel main$filt$0+32
        DD      imagerel $unwind$main$filt$0
pdata   ENDS
xdata   SEGMENT
$unwind$main DD 020609H
        DD      030023206H
        DD      imagerel ⤦
    ↳ __C_specific_handler
        DD      01H
        DD      imagerel $LN9+8
        DD      imagerel $LN9+40
        DD      imagerel main$filt$0
        DD      imagerel $LN9+40
$unwind$main$filt$0 DD 020601H
        DD      050023206H
xdata   ENDS

_TEXT   SEGMENT
main    PROC
$LN9:
        push    rbx
        sub     rsp, 32
        xor     ebx, ebx
        lea     rcx, OFFSET FLAT:$SG86276 ; ⤦
    ↳ 'hello #1!'
```

```
        call    printf
        mov     DWORD PTR [rbx], 13
        lea     rcx, OFFSET FLAT:$SG86277 ; ⤦
    ↳ 'hello #2!'
        call    printf
        jmp     SHORT $LN8@main
$LN6@main:
        lea     rcx, OFFSET FLAT:$SG86279 ; ⤦
    ↳ 'access violation, can''t recover'
        call    printf
        npad    1 ; align next label
$LN8@main:
        xor     eax, eax
        add     rsp, 32
        pop     rbx
        ret     0
main    ENDP
_TEXT   ENDS

text$x  SEGMENT
main$filt$0 PROC
        push    rbp
        sub     rsp, 32
        mov     rbp, rdx
$LN5@main$filt$:
        mov     rax, QWORD PTR [rcx]
        xor     ecx, ecx
        cmp     DWORD PTR [rax], ⤦
    ↳ -1073741819; c0000005H
        sete    cl
        mov     eax, ecx
$LN7@main$filt$:
        add     rsp, 32
```

```
        pop     rbp
        ret     0
        int     3
main$filt$0 ENDP
text$x  ENDS
```

Listing 68.13: MSVC 2012

```
$SG86277 DB     'in filter. code=0x%08X', 0↙
    ↘ aH, 00H
$SG86279 DB     'yes, that is our exception↙
    ↘ ', 0aH, 00H
$SG86281 DB     'not our exception', 0aH, 00↙
    ↘ H
$SG86288 DB     'hello!', 0aH, 00H
$SG86290 DB     '0x112233 raised. now let''s↙
    ↘  crash', 0aH, 00H
$SG86292 DB     'access violation, can''t ↙
    ↘ recover', 0aH, 00H
$SG86294 DB     'user exception caught', 0aH↙
    ↘ , 00H

pdata   SEGMENT
$pdata$filter_user_exceptions DD imagerel ↙
    ↘ $LN6
        DD      imagerel $LN6+73
        DD      imagerel ↙
    ↘ $unwind$filter_user_exceptions
$pdata$main DD  imagerel $LN14
        DD      imagerel $LN14+95
        DD      imagerel $unwind$main
pdata   ENDS
pdata   SEGMENT
```

```
$pdata$main$filt$0 DD imagerel main$filt$0
        DD      imagerel main$filt$0+32
        DD      imagerel $unwind$main$filt$0
$pdata$main$filt$1 DD imagerel main$filt$1
        DD      imagerel main$filt$1+30
        DD      imagerel $unwind$main$filt$1
pdata   ENDS

xdata   SEGMENT
$unwind$filter_user_exceptions DD 020601H
        DD      030023206H
$unwind$main DD 020609H
        DD      030023206H
        DD      imagerel ↙
    ↳ __C_specific_handler
        DD      02H
        DD      imagerel $LN14+8
        DD      imagerel $LN14+59
        DD      imagerel main$filt$0
        DD      imagerel $LN14+59
        DD      imagerel $LN14+8
        DD      imagerel $LN14+74
        DD      imagerel main$filt$1
        DD      imagerel $LN14+74
$unwind$main$filt$0 DD 020601H
        DD      050023206H
$unwind$main$filt$1 DD 020601H
        DD      050023206H
xdata   ENDS

_TEXT   SEGMENT
main    PROC
$LN14:
```

```
        push    rbx
        sub     rsp, 32
        xor     ebx, ebx
        lea     rcx, OFFSET FLAT:$SG86288 ; ↙
    ↘ 'hello!'
        call    printf
        xor     r9d, r9d
        xor     r8d, r8d
        xor     edx, edx
        mov     ecx, 1122867 ; 00112233H
        call    QWORD PTR ↙
    ↘ __imp_RaiseException
        lea     rcx, OFFSET FLAT:$SG86290 ; ↙
    ↘ '0x112233 raised. now let''s crash'
        call    printf
        mov     DWORD PTR [rbx], 13
        jmp     SHORT $LN13@main
$LN11@main:
        lea     rcx, OFFSET FLAT:$SG86292 ; ↙
    ↘ 'access violation, can''t recover'
        call    printf
        npad    1 ; align next label
$LN13@main:
        jmp     SHORT $LN9@main
$LN7@main:
        lea     rcx, OFFSET FLAT:$SG86294 ; ↙
    ↘ 'user exception caught'
        call    printf
        npad    1 ; align next label
$LN9@main:
        xor     eax, eax
        add     rsp, 32
        pop     rbx
```

```
        ret     0
main    ENDP

text$x  SEGMENT
main$filt$0 PROC
        push    rbp
        sub     rsp, 32
        mov     rbp, rdx
$LN10@main$filt$:
        mov     rax, QWORD PTR [rcx]
        xor     ecx, ecx
        cmp     DWORD PTR [rax], ↙
    ↘ -1073741819; c0000005H
        sete    cl
        mov     eax, ecx
$LN12@main$filt$:
        add     rsp, 32
        pop     rbp
        ret     0
        int     3
main$filt$0 ENDP

main$filt$1 PROC
        push    rbp
        sub     rsp, 32
        mov     rbp, rdx
$LN6@main$filt$:
        mov     rax, QWORD PTR [rcx]
        mov     rdx, rcx
        mov     ecx, DWORD PTR [rax]
        call    filter_user_exceptions
        npad    1 ; align next label
$LN8@main$filt$:
```

```
        add     rsp, 32
        pop     rbp
        ret     0
        int     3
main$filt$1 ENDP
text$x  ENDS

_TEXT   SEGMENT
code$ = 48
ep$ = 56
filter_user_exceptions PROC
$LN6:
        push    rbx
        sub     rsp, 32
        mov     ebx, ecx
        mov     edx, ecx
        lea     rcx, OFFSET FLAT:$SG86277 ; ⤦
    ↳ 'in filter. code=0x%08X'
        call    printf
        cmp     ebx, 1122867; 00112233H
        jne     SHORT $LN2@filter_use
        lea     rcx, OFFSET FLAT:$SG86279 ; ⤦
    ↳ 'yes, that is our exception'
        call    printf
        mov     eax, 1
        add     rsp, 32
        pop     rbx
        ret     0
$LN2@filter_use:
        lea     rcx, OFFSET FLAT:$SG86281 ; ⤦
    ↳ 'not our exception'
        call    printf
        xor     eax, eax
```

```
        add     rsp, 32
        pop     rbx
        ret     0
filter_user_exceptions ENDP
_TEXT   ENDS
```

[Sko12] .

.pdata .

68.3.4 SEH

[Pie], [Sko12].

68.4 Windows NT:

Las secciones críticas en cualquier OS son muy importantes en un entorno multihilo, principalmente para garantizar que solo un hilo pueda acceder a algunos datos en un momento dado, bloqueando otros hilos e interrupciones.

CRITICAL_SECTION Windows NT:

Listing 68.14: (Windows Research Kernel v1.2) public/sdk/inc/nturtl.h

```
typedef struct _RTL_CRITICAL_SECTION {
    PRTL_CRITICAL_SECTION_DEBUG DebugInfo;

    //
```

```
    //  The following three fields control ⤦
    ↪ entering and exiting the critical
    //  section for the resource
    //

    LONG LockCount;
    LONG RecursionCount;
    HANDLE OwningThread;        // from the ⤦
    ↪ thread's ClientId->UniqueThread
    HANDLE LockSemaphore;
    ULONG_PTR SpinCount;        // force ⤦
    ↪ size on 64-bit systems when packed
} RTL_CRITICAL_SECTION, *⤦
    ↪ PRTL_CRITICAL_SECTION;
```

EnterCriticalSection():

Listing 68.15: Windows 2008/ntdll.dll/x86 (begin)

```
_RtlEnterCriticalSection@4

var_C           = dword ptr -0Ch
var_8           = dword ptr -8
var_4           = dword ptr -4
arg_0           = dword ptr  8

                mov     edi, edi
                push    ebp
                mov     ebp, esp
                sub     esp, 0Ch
                push    esi
                push    edi
                mov     edi, [ebp+arg_0]
```

```
                lea     esi, [edi+4] ; ↙
    ↘ LockCount
                mov     eax, esi
                lock btr dword ptr [eax], 0
                jnb     wait ; jump if CF=0

loc_7DE922DD:
                mov     eax, large fs:18h
                mov     ecx, [eax+24h]
                mov     [edi+0Ch], ecx
                mov     dword ptr [edi+8], 1
                pop     edi
                xor     eax, eax
                pop     esi
                mov     esp, ebp
                pop     ebp
                retn    4

... skipped
```

BTR (LOCK): . , LockCount 1, . . WaitForSingleObject().

LeaveCriticalSection():

Listing 68.16: Windows 2008/ntdll.dll/x86 (begin)

```
_RtlLeaveCriticalSection@4 proc near

arg_0           = dword ptr  8

                mov     edi, edi
                push    ebp
```

```
                mov     ebp, esp
                push    esi
                mov     esi, [ebp+arg_0]
                add     dword ptr [esi+8], 0↙
    ↘ FFFFFFFFh ; RecursionCount
                jnz     short loc_7DE922B2
                push    ebx
                push    edi
                lea     edi, [esi+4]    ; ↙
    ↘ LockCount
                mov     dword ptr [esi+0Ch],↙
    ↘  0
                mov     ebx, 1
                mov     eax, edi
                lock xadd [eax], ebx
                inc     ebx
                cmp     ebx, 0FFFFFFFFh
                jnz     loc_7DEA8EB7

loc_7DE922B0:
                pop     edi
                pop     ebx

loc_7DE922B2:
                xor     eax, eax
                pop     esi
                pop     ebp
                retn    4

... skipped
```

XADD . . .

 CAPÍTULO 68. WINDOWS NT

LOCK :

Parte VII

Herramientas

Capítulo 69

Desensamblador

69.1 IDA

Una versión gratuita más antigua está disponible para descarga [1].

Cheatsheet de teclas de acceso rápido: F.1 on page 1912

[1] hex-rays.com/products/ida/support/download_freeware.shtml

Capítulo 70

Depurador

70.1 OllyDbg

Depurador de modo usuario win32 muy popular: ollydbg.de.

Cheatsheet de teclas de acceso rápido: F.2 on page 1914

70.2 GDB

Depurador no muy popular entre ingenieros inversos, pero muy cómodo de todos modos. Algunos comandos: F.5 on page 1915.

70.3 tracer

El autor a menudo usa *tracer*[1] en lugar de un depurador.

El autor de estas líneas dejó de usar un depurador eventualmente, ya que todo lo que necesita de él es detectar argumentos de función mientras se ejecuta, o el estado de los registros en algún punto. Cargar un depurador cada vez es demasiado, así que nació una pequeña utilidad llamada *tracer*. Funciona desde la línea de comandos, permite interceptar la ejecución de funciones, establecer puntos de interrupción en lugares arbitrarios, leer y cambiar el estado de los registros, etc.

Sin embargo, para fines de aprendizaje es altamente recomendable trazar código en un depurador manualmente, observar cómo cambia el estado de los registros (ej. SoftICE clásico, OllyDbg, WinDbg resaltan registros cambiados), flags, datos, cambiarlos manualmente, observar la reacción, etc.

1

CAPÍTULO 71. RASTREO DE LLAMADAS AL SISTEMA

Capítulo 71

Rastreo de llamadas al sistema

71.0.1 strace / dtruss

Muestra qué llamadas al sistema (syscalls(66 on page 1325)) están siendo llamadas por un proceso en este momento. Por ejemplo:

```
# strace df -h

...

access("/etc/ld.so.nohwcap", F_OK)      = -1
    ENOENT (No such file or directory)
```

CAPÍTULO 71. RASTREO DE LLAMADAS AL SISTEMA

```
open("/lib/i386-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\1\1\1\0\0\0\0\0\0\0\0\0\3\0\3\0\1\0\0\0\220
 512) = 512
fstat64(3, {st_mode=S_IFREG|0755, st_size=1770984, ...}) = 0
mmap2(NULL, 1780508, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0xb75b3000
```

En Mac OS X hay dtruss para hacer lo mismo.

Cygwin también tiene strace, pero hasta donde se sabe, funciona solo para archivos .exe compilados para el propio entorno cygwin.

 CAPÍTULO 72. DESCOMPILADORES

Capítulo 72

Descompiladores

Hay solo uno conocido, públicamente disponible, descompilador de alta calidad para código C: Hex-Rays:
hex-rays.com/products/decompiler/

CAPÍTULO 73. OTRAS HERRAMIENTAS

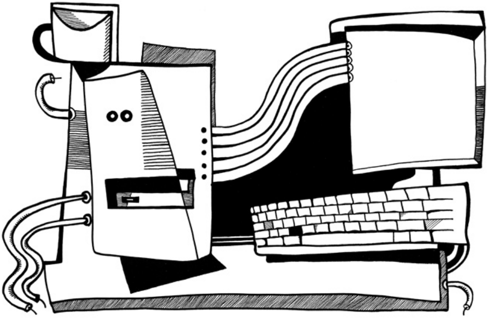
Capítulo 73

Otras herramientas

- Microsoft Visual Studio Express[1]: Versión gratuita reducida de Visual Studio, conveniente para experimentos simples. Algunas opciones útiles: F.3 on page 1914.
- Hiew[2] para pequeñas modificaciones de código en archivos binarios.
- binary grep: una pequeña utilidad para buscar cualquier secuencia de bytes en una gran pila de archivos, incluyendo no ejecutables: GitHub.

[1] visualstudio.com/en-US/products/visual-studio-express-vs
[2] hiew.ru

Parte VIII



Capítulo 74

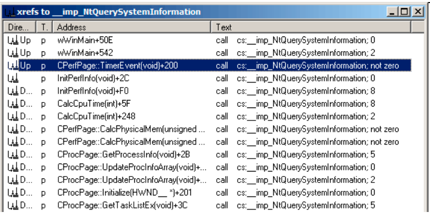
(Windows Vista)

`taskmgr.exe`. [1]).

`taskmgr.exe` en IDA. CAdapter, CNetPage, CPerfPage, CProcInfo, CProcPage, CSvcPage, CTaskPage, CUserPage.

[1] MSDN

 CAPÍTULO 74. (WINDOWS VISTA)



Dire...	T.	Address	Text
Up	p	wWinMain+50E	call cs:__imp_NtQuerySystemInformation; 0
Up	p	wWinMain+542	call cs:__imp_NtQuerySystemInformation; 2
Up	p	CPerfPage::TimerEvent(void)+200	call cs:__imp_NtQuerySystemInformation; not zero
	p	InitPerfInfo(void)+2C	call cs:__imp_NtQuerySystemInformation; 0
D...	p	InitPerfInfo(void)+F0	call cs:__imp_NtQuerySystemInformation; 8
D...	p	CalcCpuTime(int)+5F	call cs:__imp_NtQuerySystemInformation; 8
D...	p	CalcCpuTime(int)+248	call cs:__imp_NtQuerySystemInformation; 2
D...	p	CPerfPage::CalcPhysicalMem(unsigned ...	call cs:__imp_NtQuerySystemInformation; not zero
D...	p	CPerfPage::CalcPhysicalMem(unsigned ...	call cs:__imp_NtQuerySystemInformation; not zero
D...	p	CProcPage::GetProcessInfo(void)+2B	call cs:__imp_NtQuerySystemInformation; 5
D...	p	CProcPage::UpdateProcInfoArray(void)+...	call cs:__imp_NtQuerySystemInformation; 0
D...	p	CProcPage::UpdateProcInfoArray(void)+...	call cs:__imp_NtQuerySystemInformation; 2
D...	p	CProcPage::Initialize(HWND__ *)+201	call cs:__imp_NtQuerySystemInformation; 0
D...	p	CProcPage::GetTaskListEx(void)+3C	call cs:__imp_NtQuerySystemInformation; 5

Figura 74.1: IDA: NtQuerySystemInformation()

Listing 74.1: taskmgr.exe (Windows Vista)

```
.text:10000B4B3                 xor     r9d, r9d
.text:10000B4B6                 lea     rdx, [rsp+0C78h+var_C58] ; buffer
.text:10000B4BB                 xor     ecx, ecx
.text:10000B4BD                 lea     ebp, [r9+40h]
.text:10000B4C1                 mov     r8d, ebp
.text:10000B4C4                 call    cs:__imp_NtQuerySystemInformation ; 0
.text:10000B4CA                 xor     ebx, ebx
.text:10000B4CC                 cmp     eax, ebx
.text:10000B4CE                 jge     short loc_10000B4D7
.text:10000B4D0
.text:10000B4D0 loc_10000B4D0:                          ; CODE XREF: InitPerfInfo(void)+97
.text:10000B4D0                                         ; InitPerfInfo(void)+AF
.text:10000B4D0                 xor     al, al
```

```
.text:10000B4D2                 jmp     ⤦
    ↳ loc_10000B5EA
.text:10000B4D7 ; ⤦
    ↳ ---------------------------------------------------------------------------
    ↳
.text:10000B4D7
.text:10000B4D7 loc_10000B4D7:                          ⤦
    ↳             ; CODE XREF: InitPerfInfo(⤦
    ↳ void)+36
.text:10000B4D7                 mov     eax, [rsp+0⤦
    ↳ C78h+var_C50]
.text:10000B4DB                 mov     esi, ebx
.text:10000B4DD                 mov     r12d, 3E80h
.text:10000B4E3                 mov     cs:?⤦
    ↳ g_PageSize@@3KA, eax ; ulong ⤦
    ↳ g_PageSize
.text:10000B4E9                 shr     eax, 0Ah
.text:10000B4EC                 lea     r13, ⤦
    ↳ __ImageBase
.text:10000B4F3                 imul    eax, [rsp+0⤦
    ↳ C78h+var_C4C]
.text:10000B4F8                 cmp     [rsp+0C78h+⤦
    ↳ var_C20], bpl
.text:10000B4FD                 mov     cs:?⤦
    ↳ g_MEMMax@@3_JA, rax ; __int64 g_MEMMax
.text:10000B504                 movzx   eax, [rsp+0⤦
    ↳ C78h+var_C20] ; number of CPUs
.text:10000B509                 cmova   eax, ebp
.text:10000B50C                 cmp     al, bl
.text:10000B50E                 mov     cs:?⤦
    ↳ g_cProcessors@@3EA, al ; uchar ⤦
    ↳ g_cProcessors
```

 CAPÍTULO 74. (WINDOWS VISTA)

`g_cProcessors`.

`var_C20`. `var_C58` `NtQuerySystemInformation()`. 0xC20 y 0xC58 0x38 (56).

```
typedef struct _SYSTEM_BASIC_INFORMATION {
    BYTE Reserved1[24];
    PVOID Reserved2[4];
    CCHAR NumberOfProcessors;
} SYSTEM_BASIC_INFORMATION;
```

. `taskmgr.exe` `C:\Windows\System32` (*Windows Resource Protection* `taskmgr.exe`).

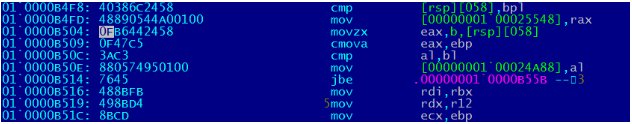


Figura 74.2: Hiew:

.

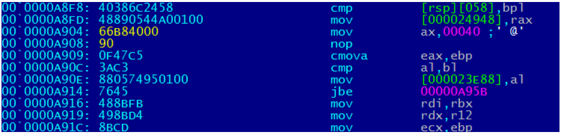


Figura 74.3: Hiew:

!.

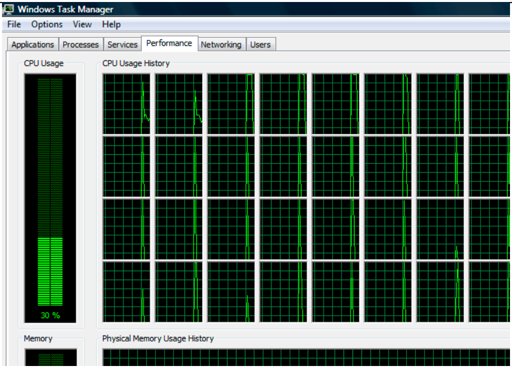


Figura 74.4: Windows Task Manager

74.1

Listing 74.2: taskmgr.exe (Windows Vista)

```
                xor     r9d, r9d
                div     dword ptr [rsp+4C8h↙
    ↳ +WndClass.lpfnWndProc]
                lea     rdx, [rsp+4C8h+↙
    ↳ VersionInformation]
                lea     ecx, [r9+2]      ; ↙
    ↳ put 2 to ECX
                mov     r8d, 138h
                mov     ebx, eax
; ECX=SystemPerformanceInformation
                call    cs:↙
    ↳ __imp_NtQuerySystemInformation ; 2
```

```
                ...

                mov     r8d, 30h
                lea     r9, [rsp+298h+↙
    ↳ var_268]
                lea     rdx, [rsp+298h+↙
    ↳ var_258]
                lea     ecx, [r8-2Dh]   ; ↙
    ↳ put 3 to ECX
; ECX=SystemTimeOfDayInformation
                call    cs:↙
    ↳ __imp_NtQuerySystemInformation ; not ↙
    ↳ zero

                ...

                mov     rbp, [rsi+8]
                mov     r8d, 20h
                lea     r9, [rsp+98h+arg_0]
                lea     rdx, [rsp+98h+↙
    ↳ var_78]
                lea     ecx, [r8+2Fh]   ; ↙
    ↳ put 0x4F to ECX
                mov     [rsp+98h+var_60], ↙
    ↳ ebx
                mov     [rsp+98h+var_68], ↙
    ↳ rbp
; ECX=SystemSuperfetchInformation
                call    cs:↙
    ↳ __imp_NtQuerySystemInformation ; not ↙
    ↳ zero
```

: 64.5.1 on page 1307.

Capítulo 75

. BallTriX, 1997, http://go.yurichev.com/17311.
:

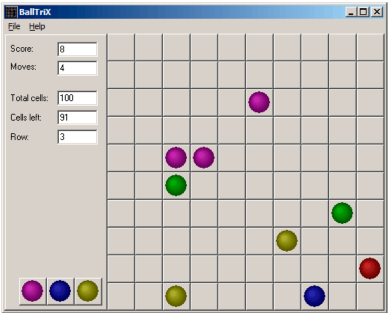


Figura 75.1:

IDA _rand balltrix.exe 0x00403DA0. IDA:

```
.text:00402C9C sub_402C9C      proc near      ↙
    ↘            ; CODE XREF: sub_402ACA+52
.text:00402C9C                                ↙
    ↘            ; sub_402ACA+64 ...
.text:00402C9C
.text:00402C9C arg_0           = dword ptr    ↙
    ↘ 8
.text:00402C9C
.text:00402C9C                 push    ebp
.text:00402C9D                 mov     ebp,   ↙
    ↘ esp
.text:00402C9F                 push    ebx
.text:00402CA0                 push    esi
.text:00402CA1                 push    edi
.text:00402CA2                 mov     eax,   ↙
    ↘ dword_40D430
.text:00402CA7                 imul    eax,   ↙
    ↘ dword_40D440
.text:00402CAE                 add     eax,   ↙
    ↘ dword_40D5C8
.text:00402CB4                 mov     ecx,   ↙
    ↘ 32000
.text:00402CB9                 cdq
.text:00402CBA                 idiv    ecx
.text:00402CBC                 mov     ↙
    ↘ dword_40D440, edx
.text:00402CC2                 call    _rand
.text:00402CC7                 cdq
.text:00402CC8                 idiv    [ebp+↙
    ↘ arg_0]
.text:00402CCB                 mov     ↙
```

```
    ↳ dword_40D430, edx
.text:00402CD1                mov     eax, ↙
    ↳ dword_40D430
.text:00402CD6                jmp     $+5
.text:00402CDB                pop     edi
.text:00402CDC                pop     esi
.text:00402CDD                pop     ebx
.text:00402CDE                leave
.text:00402CDF                retn
.text:00402CDF sub_402C9C     endp
```

«random». .

.

:

```
.text:00402B16                mov     eax, ↙
    ↳ dword_40C03C ; 10 here
.text:00402B1B                push    eax
.text:00402B1C                call    ↙
    ↳ random
.text:00402B21                add     esp, ↙
    ↳ 4
.text:00402B24                inc     eax
.text:00402B25                mov     [ebp+↙
    ↳ var_C], eax
.text:00402B28                mov     eax, ↙
    ↳ dword_40C040 ; 10 here
.text:00402B2D                push    eax
.text:00402B2E                call    ↙
    ↳ random
.text:00402B33                add     esp, ↙
```

```
    ↳ 4
```

:

```
.text:00402BBB                    mov      eax, ↙
    ↳ dword_40C058 ; 5 here
.text:00402BC0                    push     eax
.text:00402BC1                    call     ↙
    ↳ random
.text:00402BC6                    add      esp, ↙
    ↳ 4
.text:00402BC9                    inc      eax
```

. .! rand() 0..0x7FFF . $0..n-1$ y n . .

. (PUSH/CALL/ADD) NOPs. XOR EAX, EAX.

```
.00402BB8: 83C410          add             esp↙
    ↳ ,010
.00402BBB: A158C04000      mov             eax↙
    ↳ ,[00040C058]
.00402BC0: 31C0            xor             eax,↙
    ↳ eax
.00402BC2: 90              nop
.00402BC3: 90              nop
.00402BC4: 90              nop
.00402BC5: 90              nop
.00402BC6: 90              nop
.00402BC7: 90              nop
.00402BC8: 90              nop
.00402BC9: 40              inc             eax
.00402BCA: 8B4DF8          mov             ecx,[↙
    ↳ ebp][-8]
```

```
.00402BCD: 8D0C49           lea           ecx,[
    ecx][ecx]*2
.00402BD0: 8B15F4D54000     mov           edx
    ,[00040D5F4]
```

random().

:

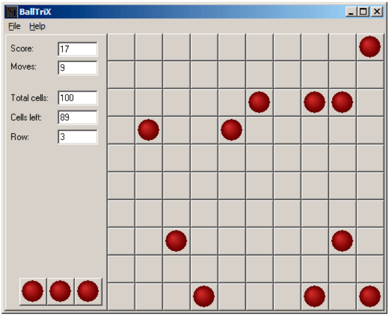


Figura 75.2:

1
.

random( ) ? . 10 y 5 .

1

CAPÍTULO 76. BUSCAMINAS (WINDOWS XP)

Capítulo 76

Buscaminas (Windows XP)

winmine.exe IDA, PDB .

rand() :

```
.text:01003940 ; __stdcall Rnd(x)
.text:01003940 _Rnd@4          proc near        ↙
    ↘            ; CODE XREF: StartGame()+53
.text:01003940                                  ↙
    ↘            ; StartGame()+61
.text:01003940
.text:01003940 arg_0           = dword ptr      ↙
    ↘ 4
.text:01003940
```

```
.text:01003940                 call    ds:⤦
    ↳ __imp__rand
.text:01003946                 cdq
.text:01003947                 idiv    [esp+⤦
    ↳ arg_0]
.text:0100394B                 mov     eax, ⤦
    ↳ edx
.text:0100394D                 retn    4
.text:0100394D _Rnd@4          endp
```

:

```
int Rnd(int limit)
{
    return rand() % limit;
};
```

.

Rnd() StartGame(), :

```
.text:010036C7                 push    ⤦
    ↳ _xBoxMac
.text:010036CD                 call    ⤦
    ↳ _Rnd@4          ; Rnd(x)
.text:010036D2                 push    ⤦
    ↳ _yBoxMac
.text:010036D8                 mov     esi, ⤦
    ↳ eax
.text:010036DA                 inc     esi
.text:010036DB                 call    ⤦
    ↳ _Rnd@4          ; Rnd(x)
.text:010036E0                 inc     eax
```

```
.text:010036E1                 mov     ecx, ⤦
    ↳ eax
.text:010036E3                 shl     ecx, ⤦
    ↳ 5          ; ECX=ECX*32
.text:010036E6                 test    ⤦
    ↳ _rgBlk[ecx+esi], 80h
.text:010036EE                 jnz     short⤦
    ↳  loc_10036C7
.text:010036F0                 shl     eax, ⤦
    ↳ 5          ; EAX=EAX*32
.text:010036F3                 lea     eax, ⤦
    ↳ _rgBlk[eax+esi]
.text:010036FA                 or      byte ⤦
    ↳ ptr [eax], 80h
.text:010036FD                 dec     ⤦
    ↳ _cBombStart
.text:01003703                 jnz     short⤦
    ↳  loc_10036C7
```

Rnd() . OR 0x010036FA. (Rnd()), TEST y JNZ 0x010036

.

cBombStart

.

rgBlk rgBlk . 0x360 (864):

```
.data:01005340 _rgBlk          db 360h dup⤦
    ↳ (?)           ; DATA XREF: MainWndProc(⤦
    ↳ x,x,x,x)+574
.data:01005340                             ⤦
    ↳               ; DisplayBlk(x,x)+23
```

```
.data:010056A0 _Preferences    dd ?                    ↙
    ↳              ; DATA XREF: FixMenus()+2
...
```

$864/32 = 27$.

$27 * 32$? . . .

OllyDbg. [1].

:

```
Address   Hex dump
01005340  10 10 10 10|10 10 10 10|10 10 10 0↙
    ↳ F|0F 0F 0F 0F|
01005350  0F 0F 0F 0F|0F 0F 0F 0F|0F 0F 0F 0↙
    ↳ F|0F 0F 0F 0F|
01005360  10 0F 0F 0F|0F 0F 0F 0F|0F 0F 10 0↙
    ↳ F|0F 0F 0F 0F|
01005370  0F 0F 0F 0F|0F 0F 0F 0F|0F 0F 0F 0↙
    ↳ F|0F 0F 0F 0F|
01005380  10 0F 0F 0F|0F 0F 0F 0F|0F 0F 10 0↙
    ↳ F|0F 0F 0F 0F|
01005390  0F 0F 0F 0F|0F 0F 0F 0F|0F 0F 0F 0↙
    ↳ F|0F 0F 0F 0F|
010053A0  10 0F 0F 0F|0F 0F 0F 0F|8F 0F 10 0↙
    ↳ F|0F 0F 0F 0F|
010053B0  0F 0F 0F 0F|0F 0F 0F 0F|0F 0F 0F 0↙
    ↳ F|0F 0F 0F 0F|
010053C0  10 0F 0F 0F|0F 0F 0F 0F|0F 0F 10 0↙
    ↳ F|0F 0F 0F 0F|
```

[1]Windows XP SP3 English. .

 CAPÍTULO 76. BUSCAMINAS (WINDOWS XP)

```
010053D0  0F 0F 0F 0F|0F 0F 0F 0F|0F 0F 0F 0↙
    ↳ F|0F 0F 0F 0F|
010053E0  10 0F 0F 0F|0F 0F 0F 0F|0F 0F 10 0↙
    ↳ F|0F 0F 0F 0F|
010053F0  0F 0F 0F 0F|0F 0F 0F 0F|0F 0F 0F 0↙
    ↳ F|0F 0F 0F 0F|
01005400  10 0F 0F 8F|0F 0F 8F 0F|0F 0F 10 0↙
    ↳ F|0F 0F 0F 0F|
01005410  0F 0F 0F 0F|0F 0F 0F 0F|0F 0F 0F 0↙
    ↳ F|0F 0F 0F 0F|
01005420  10 8F 0F 0F|8F 0F 0F 0F|0F 0F 10 0↙
    ↳ F|0F 0F 0F 0F|
01005430  0F 0F 0F 0F|0F 0F 0F 0F|0F 0F 0F 0↙
    ↳ F|0F 0F 0F 0F|
01005440  10 8F 0F 0F|0F 0F 8F 0F|0F 8F 10 0↙
    ↳ F|0F 0F 0F 0F|
01005450  0F 0F 0F 0F|0F 0F 0F 0F|0F 0F 0F 0↙
    ↳ F|0F 0F 0F 0F|
01005460  10 0F 0F 0F|0F 8F 0F 0F|0F 8F 10 0↙
    ↳ F|0F 0F 0F 0F|
01005470  0F 0F 0F 0F|0F 0F 0F 0F|0F 0F 0F 0↙
    ↳ F|0F 0F 0F 0F|
01005480  10 10 10 10|10 10 10 10|10 10 10 0↙
    ↳ F|0F 0F 0F 0F|
01005490  0F 0F 0F 0F|0F 0F 0F 0F|0F 0F 0F 0↙
    ↳ F|0F 0F 0F 0F|
010054A0  0F 0F 0F 0F|0F 0F 0F 0F|0F 0F 0F 0↙
    ↳ F|0F 0F 0F 0F|
010054B0  0F 0F 0F 0F|0F 0F 0F 0F|0F 0F 0F 0↙
    ↳ F|0F 0F 0F 0F|
010054C0  0F 0F 0F 0F|0F 0F 0F 0F|0F 0F 0F 0↙
    ↳ F|0F 0F 0F 0F|
```

 CAPÍTULO 76. BUSCAMINAS (WINDOWS XP)

OllyDbg, . .

.

.

:

Figura 76.1:

```
border:
01005340  10 10 10 10 10 10 10 10 10 10 10 0↙
    ↳ F 0F 0F 0F 0F
01005350  0F 0F 0F 0F 0F 0F 0F 0F 0F 0F 0F 0↙
    ↳ F 0F 0F 0F 0F
line #1:
01005360  10 0F 0F 0F 0F 0F 0F 0F 0F 0F 10 0↙
    ↳ F 0F 0F 0F 0F
01005370  0F 0F 0F 0F 0F 0F 0F 0F 0F 0F 0F 0↙
    ↳ F 0F 0F 0F 0F
line #2:
01005380  10 0F 0F 0F 0F 0F 0F 0F 0F 0F 10 0↙
    ↳ F 0F 0F 0F 0F
01005390  0F 0F 0F 0F 0F 0F 0F 0F 0F 0F 0F 0↙
    ↳ F 0F 0F 0F 0F
```

```
line #3:
010053A0  10 0F 0F 0F 0F 0F 0F 0F[8F]0F 10 0↙
    ↘ F 0F 0F 0F 0F
010053B0  0F 0F 0F 0F 0F 0F 0F 0F 0F 0F 0F 0↙
    ↘ F 0F 0F 0F 0F
line #4:
010053C0  10 0F 0F 0F 0F 0F 0F 0F 0F 0F 10 0↙
    ↘ F 0F 0F 0F 0F
010053D0  0F 0F 0F 0F 0F 0F 0F 0F 0F 0F 0F 0↙
    ↘ F 0F 0F 0F 0F
line #5:
010053E0  10 0F 0F 0F 0F 0F 0F 0F 0F 0F 10 0↙
    ↘ F 0F 0F 0F 0F
010053F0  0F 0F 0F 0F 0F 0F 0F 0F 0F 0F 0F 0↙
    ↘ F 0F 0F 0F 0F
line #6:
01005400  10 0F 0F[8F]0F 0F[8F]0F 0F 0F 10 0↙
    ↘ F 0F 0F 0F 0F
01005410  0F 0F 0F 0F 0F 0F 0F 0F 0F 0F 0F 0↙
    ↘ F 0F 0F 0F 0F
line #7:
01005420  10[8F]0F 0F[8F]0F 0F 0F 0F 0F 10 0↙
    ↘ F 0F 0F 0F 0F
01005430  0F 0F 0F 0F 0F 0F 0F 0F 0F 0F 0F 0↙
    ↘ F 0F 0F 0F 0F
line #8:
01005440  10[8F]0F 0F 0F 0F[8F]0F 0F[8F]10 0↙
    ↘ F 0F 0F 0F 0F
01005450  0F 0F 0F 0F 0F 0F 0F 0F 0F 0F 0F 0↙
    ↘ F 0F 0F 0F 0F
line #9:
01005460  10 0F 0F 0F 0F[8F]0F 0F 0F[8F]10 0↙
    ↘ F 0F 0F 0F 0F
```

 CAPÍTULO 76. BUSCAMINAS (WINDOWS XP)

```
01005470  0F 0F 0F 0F 0F 0F 0F 0F 0F 0F 0F 0↙
    ↘ F 0F 0F 0F 0F
border:
01005480  10 10 10 10 10 10 10 10 10 10 10 0↙
    ↘ F 0F 0F 0F 0F
01005490  0F 0F 0F 0F 0F 0F 0F 0F 0F 0F 0F 0↙
    ↘ F 0F 0F 0F 0F
```

```
 0F 0F 0F 0F 0F 0F 0F 0F 0F
 0F 0F 0F 0F 0F 0F 0F 0F 0F
 0F 0F 0F 0F 0F 0F 0F[8F]0F
 0F 0F 0F 0F 0F 0F 0F 0F 0F
 0F 0F 0F 0F 0F 0F 0F 0F 0F
 0F 0F[8F]0F 0F[8F]0F 0F 0F
[8F]0F 0F[8F]0F 0F 0F 0F 0F
[8F]0F 0F 0F 0F[8F]0F 0F[8F]
 0F 0F 0F 0F[8F]0F 0F 0F[8F]
```

 CAPÍTULO 76. BUSCAMINAS (WINDOWS XP)

:

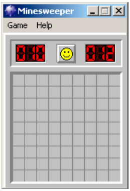


Figura 76.2:

:

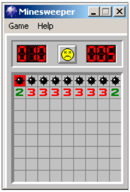


Figura 76.3:

```
// Windows XP MineSweeper cheater
// written by dennis(a)yurichev.com for http
    ↪ ://beginners.re/ book
#include <windows.h>
#include <assert.h>
#include <stdio.h>

int main (int argc, char * argv[])
```

```
{
        int i, j;
        HANDLE h;
        DWORD PID, address, rd;
        BYTE board[27][32];

        if (argc!=3)
        {
                printf ("Usage: %s <PID> <↙
    ↘ address>\n", argv[0]);
                return 0;
        };

        assert (argv[1]!=NULL);
        assert (argv[2]!=NULL);

        assert (sscanf (argv[1], "%d", &PID)↙
    ↘ ==1);
        assert (sscanf (argv[2], "%x", &↙
    ↘ address)==1);

        h=OpenProcess (PROCESS_VM_OPERATION ↙
    ↘ | PROCESS_VM_READ | PROCESS_VM_WRITE, ↙
    ↘ FALSE, PID);

        if (h==NULL)
        {
                DWORD e=GetLastError();
                printf ("OpenProcess error: ↙
    ↘ %08X\n", e);
                return 0;
        };

```

```
      if (ReadProcessMemory (h, (LPVOID)↙
    ↘ address, board, sizeof(board), &rd)!=↙
    ↘ TRUE)
      {
              printf ("ReadProcessMemory()↙
    ↘  failed\n");
              return 0;
      };

      for (i=1; i<26; i++)
      {
              if (board[i][0]==0x10 && ↙
    ↘ board[i][1]==0x10)
                      break; // end of ↙
    ↘ board
              for (j=1; j<31; j++)
              {
                      if (board[i][j]==0↙
    ↘ x10)
                              break; // ↙
    ↘ board border
                      if (board[i][j]==0↙
    ↘ x8F)
                              printf ("*")↙
    ↘ ;
                      else
                              printf (" ")↙
    ↘ ;

              };
              printf ("\n");
      };
```

```
        CloseHandle (h);
};
```

PID[2] [3] (0x01005340 Windows XP SP3 English) [4].

76.1 Ejercicios

- ?
- .
-
- : G.5.1 on page 1937.

[2] ID de programa/proceso
[3] PID Task Manager («View → Select Columns»)
[4] : beginners.re

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.

.

77.1

IDA:

```
sub_401510      proc near
                ; ECX = input
                mov     rdx, 5↙
    ↘ D7E0D1F2E0F1F84h
                mov     rax, rcx        ; ↙
    ↘ input
                imul    rax, rdx
                mov     rdx, 388↙
    ↘ D76AEE8CB1500h
                mov     ecx, eax
```

```
                and     ecx, 0Fh
                ror     rax, cl
                xor     rax, rdx
                mov     rdx, 0
    ↘ D2E9EE7E83C4285Bh
                mov     ecx, eax
                and     ecx, 0Fh
                rol     rax, cl
                lea     r8, [rax+rdx]
                mov     rdx, 
    ↘ 8888888888888889h
                mov     rax, r8
                mul     rdx
                shr     rdx, 5
                mov     rax, rdx
                lea     rcx, [r8+rdx*4]
                shl     rax, 6
                sub     rcx, rax
                mov     rax, r8
                rol     rax, cl
                ; EAX = output
                retn
sub_401510      endp
```

GCC, ECX.

Hex-Rays. :

```
uint64_t f(uint64_t input)
{
        uint64_t rax, rbx, rcx, rdx, r8;

        ecx=input;
```

```
        rdx=0x5D7E0D1F2E0F1F84;
        rax=rcx;
        rax*=rdx;
        rdx=0x388D76AEE8CB1500;
        rax=_lrotr(rax, rax&0xF); // rotate ↙
    ↳ right
        rax^=rdx;
        rdx=0xD2E9EE7E83C4285B;
        rax=_lrotl(rax, rax&0xF); // rotate ↙
    ↳ left
        r8=rax+rdx;
        rdx=0x8888888888888889;
        rax=r8;
        rax*=rdx;
        rdx=rdx>>5;
        rax=rdx;
        rcx=r8+rdx*4;
        rax=rax<<6;
        rcx=rcx-rax;
        rax=r8
        rax=_lrotl (rax, rcx&0xFF); // ↙
    ↳ rotate left
        return rax;
};
```

!

:

```
uint64_t f(uint64_t input)
{
        uint64_t rax, rbx, rcx, rdx, r8;
```

```
        ecx=input;

        rdx=0x5D7E0D1F2E0F1F84;
        rax=rcx;
        rax*=rdx;
        rdx=0x388D76AEE8CB1500;
        rax=_lrotr(rax, rax&0xF); // rotate ↙
    ↳ right
        rax^=rdx;
        rdx=0xD2E9EE7E83C4285B;
        rax=_lrotl(rax, rax&0xF); // rotate ↙
    ↳ left
        r8=rax+rdx;

        rdx=0x8888888888888889;
        rax=r8;
        rax*=rdx;
        // RDX here is a high part of ↙
    ↳ multiplication result
        rdx=rdx>>5;
        // RDX here is division result!
        rax=rdx;

        rcx=r8+rdx*4;
        rax=rax<<6;
        rcx=rcx-rax;
        rax=r8
        rax=_lrotl (rax, rcx&0xFF); // ↙
    ↳ rotate left
        return rax;
};
```

:

```
uint64_t f(uint64_t input)
{
        uint64_t rax, rbx, rcx, rdx, r8;

        ecx=input;

        rdx=0x5D7E0D1F2E0F1F84;
        rax=rcx;
        rax*=rdx;
        rdx=0x388D76AEE8CB1500;
        rax=_lrotr(rax, rax&0xF); // rotate ↙
    ↳ right
        rax^=rdx;
        rdx=0xD2E9EE7E83C4285B;
        rax=_lrotl(rax, rax&0xF); // rotate ↙
    ↳ left
        r8=rax+rdx;

        rdx=0x8888888888888889;
        rax=r8;
        rax*=rdx;
        // RDX here is a high part of ↙
    ↳ multiplication result
        rdx=rdx>>5;
        // RDX here is division result!
        rax=rdx;

        rcx=(r8+rdx*4)-(rax<<6);
        rax=r8
        rax=_lrotl (rax, rcx&0xFF); // ↙
    ↳ rotate left
```

```
        return rax;
};
```

(41 on page 938). Wolfram Mathematica:

Listing 77.1: Wolfram Mathematica

```
In[1]:=N[2^(64 + 5)/16^^8888888888888889]
Out[1]:=60.
```

:

```
uint64_t f(uint64_t input)
{
        uint64_t rax, rbx, rcx, rdx, r8;

        ecx=input;

        rdx=0x5D7E0D1F2E0F1F84;
        rax=rcx;
        rax*=rdx;
        rdx=0x388D76AEE8CB1500;
        rax=_lrotr(rax, rax&0xF); // rotate ↙
    ↳ right
        rax^=rdx;
        rdx=0xD2E9EE7E83C4285B;
        rax=_lrotl(rax, rax&0xF); // rotate ↙
    ↳ left
        r8=rax+rdx;

        rax=rdx=r8/60;

        rcx=(r8+rax*4)-(rax*64);
```

```
        rax=r8
        rax=_lrotl (rax, rcx&0xFF); // ↙
    ↘ rotate left
        return rax;
};
```

:

```
uint64_t f(uint64_t input)
{
        uint64_t rax, rbx, rcx, rdx, r8;

        rax=input;
        rax*=0x5D7E0D1F2E0F1F84;
        rax=_lrotr(rax, rax&0xF); // rotate ↙
    ↘ right
        rax^=0x388D76AEE8CB1500;
        rax=_lrotl(rax, rax&0xF); // rotate ↙
    ↘ left
        r8=rax+0xD2E9EE7E83C4285B;

        rcx=r8-(r8/60)*60;
        rax=r8
        rax=_lrotl (rax, rcx&0xFF); // ↙
    ↘ rotate left
        return rax;
};
```

:

```
uint64_t f(uint64_t input)
{
        uint64_t rax, rbx, rcx, rdx, r8;
```

```
        rax=input;
        rax*=0x5D7E0D1F2E0F1F84;
        rax=_lrotr(rax, rax&0xF); // rotate ↙
    ↘ right
        rax^=0x388D76AEE8CB1500;
        rax=_lrotl(rax, rax&0xF); // rotate ↙
    ↘ left
        r8=rax+0xD2E9EE7E83C4285B;

        return _lrotl (r8, r8 % 60); // ↙
    ↘ rotate left
};
```

:

```
#include <stdio.h>
#include <stdint.h>
#include <stdbool.h>
#include <string.h>
#include <intrin.h>

#define C1 0x5D7E0D1F2E0F1F84
#define C2 0x388D76AEE8CB1500
#define C3 0xD2E9EE7E83C4285B

uint64_t hash(uint64_t v)
{
        v*=C1;
        v=_lrotr(v, v&0xF); // rotate right
        v^=C2;
        v=_lrotl(v, v&0xF); // rotate left
        v+=C3;
```

```
        v=_lrotl(v, v % 60); // rotate left
        return v;
};

int main()
{
        printf ("%llu\n", hash(...));
};
```

. .

.

77.2

Microsoft Research Z3[1]. .

:

```
from z3 import *

C1=0x5D7E0D1F2E0F1F84
C2=0x388D76AEE8CB1500
C3=0xD2E9EE7E83C4285B

inp, i1, i2, i3, i4, i5, i6, outp = BitVecs('inp i1 i2 i3 i4 i5 i6 outp', 64)

s = Solver()
s.add(i1==inp*C1)
```

[1] http://go.yurichev.com/17314

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```
11 s.add(i2==RotateRight (i1, i1 & 0xF))
12 s.add(i3==i2 ^ C2)
13 s.add(i4==RotateLeft(i3, i3 & 0xF))
14 s.add(i5==i4 + C3)
15 s.add(outp==RotateLeft (i5, URem(i5, 60)))
16
17 s.add(outp==10816636949158156260)
18
19 print s.check()
20 m=s.model()
21 print m
22 print (" inp=0x%X" % m[inp].as_long())
23 print ("outp=0x%X" % m[outp].as_long())
```

.

.. i1..i6 .

10..15. 10816636949158156260.

.

RotateRight, RotateLeft, URem .

:

```
...>python.exe 1.py
sat
[i1 = 3959740824832824396,
 i3 = 8957124831728646493,
 i5 = 10816636949158156260,
 inp = 1364123924608584563,
 outp = 10816636949158156260,
 i4 = 14065440378185297801,
```

```
 i2 = 4954926323707358301]
 inp=0x12EE577B63E80B73
outp=0x961C69FF0AEFD7E4
```

«sat» «satisfiable», .. . 0x12EE577B63E80B73 .

. :

```
from z3 import *

C1=0x5D7E0D1F2E0F1F84
C2=0x388D76AEE8CB1500
C3=0xD2E9EE7E83C4285B

inp, i1, i2, i3, i4, i5, i6, outp = BitVecs('inp i1 i2 i3 i4 i5 i6 outp', 64)

s = Solver()
s.add(i1==inp*C1)
s.add(i2==RotateRight (i1, i1 & 0xF))
s.add(i3==i2 ^ C2)
s.add(i4==RotateLeft(i3, i3 & 0xF))
s.add(i5==i4 + C3)
s.add(outp==RotateLeft (i5, URem(i5, 60)))

s.add(outp==10816636949158156260)

s.add(inp!=0x12EE577B63E80B73)

print s.check()
m=s.model()
print m
print (" inp=0x%X" % m[inp].as_long())
```

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```
print ("outp=0x%X" % m[outp].as_long())
```

:

```
...>python.exe 2.py
sat
[i1 = 3959740824832824396,
 i3 = 8957124831728646493,
 i5 = 10816636949158156260,
 inp = 10587495961463360371,
 outp = 10816636949158156260,
 i4 = 14065440378185297801,
 i2 = 4954926323707358301]
 inp=0x92EE577B63E80B73
outp=0x961C69FF0AEFD7E4
```

. :

```
from z3 import *

C1=0x5D7E0D1F2E0F1F84
C2=0x388D76AEE8CB1500
C3=0xD2E9EE7E83C4285B

inp, i1, i2, i3, i4, i5, i6, outp = BitVecs('inp i1 i2 i3 i4 i5 i6 outp', 64)

s = Solver()
s.add(i1==inp*C1)
s.add(i2==RotateRight (i1, i1 & 0xF))
s.add(i3==i2 ^ C2)
s.add(i4==RotateLeft(i3, i3 & 0xF))
s.add(i5==i4 + C3)
```

```
s.add(outp==RotateLeft (i5, URem(i5, 60)))

s.add(outp==10816636949158156260)

# copypasted from http://stackoverflow.com/questions/11867611/z3py-checking-all-solutions-for-equation
result=[]
while True:
    if s.check() == sat:
        m = s.model()
        print m[inp]
        result.append(m)
        # Create a new constraint the blocks the current model
        block = []
        for d in m:
            # d is a declaration
            if d.arity() > 0:
                raise Z3Exception("uninterpreted functions are not supported")
            # create a constant from declaration
            c=d()
            if is_array(c) or c.sort().kind() == Z3_UNINTERPRETED_SORT:
                raise Z3Exception("arrays and uninterpreted sorts are not supported")
            block.append(c != m[d])
        s.add(Or(block))
    else:
```

```
                print "results total=",len(result)
                break
```

:

```
13641239246085843
1234567890
9223372038089343698
4611686019661955794
13835058056516731602
3096040143925676201
12319412180780452009
7707726162353064105
16931098199207839913
1906652839273745429
11130024876128521237
15741710894555909141
6518338857701133333
5975809943035972467
15199181979890748275
10587495961463360371
results total= 16
```

0x92EE577B63E80B73 .

1234567890.

. ?

outp :

```
from z3 import *

```

```
C1=0x5D7E0D1F2E0F1F84
C2=0x388D76AEE8CB1500
C3=0xD2E9EE7E83C4285B

inp, i1, i2, i3, i4, i5, i6, outp = BitVecs('inp i1 i2 i3 i4 i5 i6 outp', 64)

s = Solver()
s.add(i1==inp*C1)
s.add(i2==RotateRight (i1, i1 & 0xF))
s.add(i3==i2 ^ C2)
s.add(i4==RotateLeft(i3, i3 & 0xF))
s.add(i5==i4 + C3)
s.add(outp==RotateLeft (i5, URem(i5, 60)))

s.add(outp & 0xFFFFFFFF == inp & 0xFFFFFFFF)

print s.check()
m=s.model()
print m
print (" inp=0x%X" % m[inp].as_long())
print ("outp=0x%X" % m[outp].as_long())
```

:

```
sat
[i1 = 14869545517796235860,
 i3 = 8388171335828825253,
 i5 = 6918262285561543945,
 inp = 1370377541658871093,
 outp = 14543180351754208565,
 i4 = 10167065714588685486,
 i2 = 5541032613289652645]
```

```
 inp=0x13048F1D12C00535
outp=0xC9D3C17A12C00535
```

0x1234:

```
from z3 import *

C1=0x5D7E0D1F2E0F1F84
C2=0x388D76AEE8CB1500
C3=0xD2E9EE7E83C4285B

inp, i1, i2, i3, i4, i5, i6, outp = BitVecs('inp i1 i2 i3 i4 i5 i6 outp', 64)

s = Solver()
s.add(i1==inp*C1)
s.add(i2==RotateRight (i1, i1 & 0xF))
s.add(i3==i2 ^ C2)
s.add(i4==RotateLeft(i3, i3 & 0xF))
s.add(i5==i4 + C3)
s.add(outp==RotateLeft (i5, URem(i5, 60)))

s.add(outp & 0xFFFFFFFF == inp & 0xFFFFFFFF)
s.add(outp & 0xFFFF == 0x1234)

print s.check()
m=s.model()
print m
print (" inp=0x%X" % m[inp].as_long())
print ("outp=0x%X" % m[outp].as_long())
```

:

```
sat
[i1 = 2834222860503985872,
 i3 = 2294680776671411152,
 i5 = 17492621421353821227,
 inp = 461881484695179828,
 outp = 419247225543463476,
 i4 = 2294680776671411152,
 i2 = 2834222860503985872]
 inp=0x668EEC35F961234
outp=0x5D177215F961234
```

.

? AES, RSA, . .

[Yur12].

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: [Yur12].

78.1 #1: MacOS Classic y PowerPC

MacOS Classic [1], PowerPC. .

"Invalid Security Device". .

.

IDA "PEF (Mac OS or Be OS executable)" ( ).

:

```
...
```

1

```
seg000:000C87FC 38 60 00 01
    ↳ li       %r3, 1
seg000:000C8800 48 03 93 41
    ↳ bl       check1
seg000:000C8804 60 00 00 00
    ↳ nop
seg000:000C8808 54 60 06 3F
    ↳ clrlwi. %r0, %r3, 24
seg000:000C880C 40 82 00 40
    ↳ bne      OK
seg000:000C8810 80 62 9F D8
    ↳ lwz      %r3, TC_aInvalidSecurityDevice

...
```

. . .

check1() . BL *Branch Link* . .

[SK95] .

CLRLWI. [IBM00] . MOVZX en x86 (15.1.1 on page 379), BNE[2] .

check1():

```
seg000:00101B40                    check1: # CODE
    ↳ XREF: seg000:00063E7Cp
seg000:00101B40                             #
    ↳ sub_64070+160p ...
seg000:00101B40
seg000:00101B40                    .set arg_8,  8
```

[2](PowerPC, ARM) Branch if Not Equal

```
seg000:00101B40
seg000:00101B40 7C 08 02 A6              mflr    ⤦
    ↳ %r0
seg000:00101B44 90 01 00 08              stw     ⤦
    ↳ %r0, arg_8(%sp)
seg000:00101B48 94 21 FF C0              stwu    ⤦
    ↳ %sp, -0x40(%sp)
seg000:00101B4C 48 01 6B 39              bl      ⤦
    ↳ check2
seg000:00101B50 60 00 00 00              nop
seg000:00101B54 80 01 00 48              lwz     ⤦
    ↳ %r0, 0x40+arg_8(%sp)
seg000:00101B58 38 21 00 40              addi    ⤦
    ↳ %sp, %sp, 0x40
seg000:00101B5C 7C 08 03 A6              mtlr    ⤦
    ↳ %r0
seg000:00101B60 4E 80 00 20              blr
seg000:00101B60                 # End of ⤦
    ↳ function check1
```

. thunk function: `check1()` `check2()`.

BLR[3] . link register, ARM.

`check2()` :

```
seg000:00118684                 check2: # CODE ⤦
    ↳ XREF: check1+Cp
seg000:00118684
seg000:00118684                 .set var_18, -0⤦
    ↳ x18
seg000:00118684                 .set var_C, -0xC
```

[3](PowerPC) Branch to Link Register

```
seg000:00118684                 .set var_8, -8
seg000:00118684                 .set var_4, -4
seg000:00118684                 .set arg_8,  8
seg000:00118684
seg000:00118684 93 E1 FF FC   stw     %r31, ↙
    ↘ var_4(%sp)
seg000:00118688 7C 08 02 A6   mflr    %r0
seg000:0011868C 83 E2 95 A8   lwz     %r31, ↙
    ↘ off_1485E8 # dword_24B704
seg000:00118690                 .using ↙
    ↘ dword_24B704, %r31
seg000:00118690 93 C1 FF F8   stw     %r30, ↙
    ↘ var_8(%sp)
seg000:00118694 93 A1 FF F4   stw     %r29, ↙
    ↘ var_C(%sp)
seg000:00118698 7C 7D 1B 78   mr      %r29, ↙
    ↘ %r3
seg000:0011869C 90 01 00 08   stw     %r0, ↙
    ↘ arg_8(%sp)
seg000:001186A0 54 60 06 3E   clrlwi  %r0, %↙
    ↘ r3, 24
seg000:001186A4 28 00 00 01   cmplwi  %r0, 1
seg000:001186A8 94 21 FF B0   stwu    %sp, ↙
    ↘ -0x50(%sp)
seg000:001186AC 40 82 00 0C   bne     ↙
    ↘ loc_1186B8
seg000:001186B0 38 60 00 01   li      %r3, 1
seg000:001186B4 48 00 00 6C   b       exit
seg000:001186B8
seg000:001186B8               loc_1186B8: # ↙
    ↘ CODE XREF: check2+28j
seg000:001186B8 48 00 03 D5   bl      ↙
    ↘ sub_118A8C
```

```
seg000:001186BC 60 00 00 00   nop
seg000:001186C0 3B C0 00 00   li      %r30, ⤦
    ↳ 0
seg000:001186C4
seg000:001186C4              skip:       # ⤦
    ↳ CODE XREF: check2+94j
seg000:001186C4 57 C0 06 3F   clrlwi. %r0, %⤦
    ↳ r30, 24
seg000:001186C8 41 82 00 18   beq     ⤦
    ↳ loc_1186E0
seg000:001186CC 38 61 00 38   addi    %r3, %⤦
    ↳ sp, 0x50+var_18
seg000:001186D0 80 9F 00 00   lwz     %r4, ⤦
    ↳ dword_24B704
seg000:001186D4 48 00 C0 55   bl      .⤦
    ↳ RBEFINDNEXT
seg000:001186D8 60 00 00 00   nop
seg000:001186DC 48 00 00 1C   b       ⤦
    ↳ loc_1186F8
seg000:001186E0
seg000:001186E0              loc_1186E0: # ⤦
    ↳ CODE XREF: check2+44j
seg000:001186E0 80 BF 00 00   lwz     %r5, ⤦
    ↳ dword_24B704
seg000:001186E4 38 81 00 38   addi    %r4, %⤦
    ↳ sp, 0x50+var_18
seg000:001186E8 38 60 08 C2   li      %r3, 0⤦
    ↳ x1234
seg000:001186EC 48 00 BF 99   bl      .⤦
    ↳ RBEFINDFIRST
seg000:001186F0 60 00 00 00   nop
seg000:001186F4 3B C0 00 01   li      %r30, ⤦
    ↳ 1
```

```
seg000:001186F8
seg000:001186F8                loc_1186F8: # ↙
    ↘ CODE XREF: check2+58j
seg000:001186F8 54 60 04 3F   clrlwi. %r0, %↙
    ↘ r3, 16
seg000:001186FC 41 82 00 0C   beq      ↙
    ↘ must_jump
seg000:00118700 38 60 00 00   li      %r3, 0↙
    ↘           # error
seg000:00118704 48 00 00 1C   b       exit
seg000:00118708
seg000:00118708                must_jump:  # ↙
    ↘ CODE XREF: check2+78j
seg000:00118708 7F A3 EB 78   mr      %r3, %↙
    ↘ r29
seg000:0011870C 48 00 00 31   bl      check3
seg000:00118710 60 00 00 00   nop
seg000:00118714 54 60 06 3F   clrlwi. %r0, %↙
    ↘ r3, 24
seg000:00118718 41 82 FF AC   beq     skip
seg000:0011871C 38 60 00 01   li      %r3, 1
seg000:00118720
seg000:00118720                exit:       # ↙
    ↘ CODE XREF: check2+30j
seg000:00118720                            # ↙
    ↘ check2+80j
seg000:00118720 80 01 00 58   lwz     %r0, 0↙
    ↘ x50+arg_8(%sp)
seg000:00118724 38 21 00 50   addi    %sp, %↙
    ↘ sp, 0x50
seg000:00118728 83 E1 FF FC   lwz     %r31, ↙
    ↘ var_4(%sp)
seg000:0011872C 7C 08 03 A6   mtlr    %r0
```

```
seg000:00118730 83 C1 FF F8   lwz     %r30, ↙
    ↳ var_8(%sp)
seg000:00118734 83 A1 FF F4   lwz     %r29, ↙
    ↳ var_C(%sp)
seg000:00118738 4E 80 00 20   blr
seg000:00118738                # End of ↙
    ↳ function check2
```

`.GetNextDeviceViaUSB()`, `.USBSendPKT()`, .

`.GetNextEve3Device()` Sentinel Eve3 .

.

`li %r3, 1` `li %r3, 0` (*Load Immediate*,). `0x001186B0`.

: `.RBEFINDFIRST()` `.RBEFINDNEXT()` .

N.B.: `clrlwi. %r0, %r3, 16` , `.RBEFINDFIRST()` .

B (*branch*) .

BEQ BNE.

`check3()`:

```
seg000:0011873C                check3: # CODE ↙
    ↳ XREF: check2+88p
seg000:0011873C
seg000:0011873C                .set var_18, -0↙
    ↳ x18
seg000:0011873C                .set var_C, -0xC
seg000:0011873C                .set var_8, -8
seg000:0011873C                .set var_4, -4
seg000:0011873C                .set arg_8,  8
```

```
seg000:0011873C
seg000:0011873C 93 E1 FF FC   stw     %r31, ⤦
    ↳ var_4(%sp)
seg000:00118740 7C 08 02 A6   mflr    %r0
seg000:00118744 38 A0 00 00   li      %r5, 0
seg000:00118748 93 C1 FF F8   stw     %r30, ⤦
    ↳ var_8(%sp)
seg000:0011874C 83 C2 95 A8   lwz     %r30, ⤦
    ↳ off_1485E8 # dword_24B704
seg000:00118750                       .using ⤦
    ↳ dword_24B704, %r30
seg000:00118750 93 A1 FF F4   stw     %r29, ⤦
    ↳ var_C(%sp)
seg000:00118754 3B A3 00 00   addi    %r29, ⤦
    ↳ %r3, 0
seg000:00118758 38 60 00 00   li      %r3, 0
seg000:0011875C 90 01 00 08   stw     %r0, ⤦
    ↳ arg_8(%sp)
seg000:00118760 94 21 FF B0   stwu    %sp, ⤦
    ↳ -0x50(%sp)
seg000:00118764 80 DE 00 00   lwz     %r6, ⤦
    ↳ dword_24B704
seg000:00118768 38 81 00 38   addi    %r4, %⤦
    ↳ sp, 0x50+var_18
seg000:0011876C 48 00 C0 5D   bl      .⤦
    ↳ RBEREAD
seg000:00118770 60 00 00 00   nop
seg000:00118774 54 60 04 3F   clrlwi. %r0, %⤦
    ↳ r3, 16
seg000:00118778 41 82 00 0C   beq     ⤦
    ↳ loc_118784
seg000:0011877C 38 60 00 00   li      %r3, 0
seg000:00118780 48 00 02 F0   b       exit
```

```
seg000:00118784
seg000:00118784                   loc_118784: # ↙
    ↘ CODE XREF: check3+3Cj
seg000:00118784 A0 01 00 38   lhz     %r0, 0↙
    ↘ x50+var_18(%sp)
seg000:00118788 28 00 04 B2   cmplwi  %r0, 0↙
    ↘ x1100
seg000:0011878C 41 82 00 0C   beq     ↙
    ↘ loc_118798
seg000:00118790 38 60 00 00   li      %r3, 0
seg000:00118794 48 00 02 DC   b       exit
seg000:00118798
seg000:00118798                   loc_118798: # ↙
    ↘ CODE XREF: check3+50j
seg000:00118798 80 DE 00 00   lwz     %r6, ↙
    ↘ dword_24B704
seg000:0011879C 38 81 00 38   addi    %r4, %↙
    ↘ sp, 0x50+var_18
seg000:001187A0 38 60 00 01   li      %r3, 1
seg000:001187A4 38 A0 00 00   li      %r5, 0
seg000:001187A8 48 00 C0 21   bl      .↙
    ↘ RBEREAD
seg000:001187AC 60 00 00 00   nop
seg000:001187B0 54 60 04 3F   clrlwi. %r0, %↙
    ↘ r3, 16
seg000:001187B4 41 82 00 0C   beq     ↙
    ↘ loc_1187C0
seg000:001187B8 38 60 00 00   li      %r3, 0
seg000:001187BC 48 00 02 B4   b       exit
seg000:001187C0
seg000:001187C0                   loc_1187C0: # ↙
    ↘ CODE XREF: check3+78j
seg000:001187C0 A0 01 00 38   lhz     %r0, 0↙
```

```
    ↳ x50+var_18(%sp)
seg000:001187C4 28 00 06 4B   cmplwi  %r0, 0↙
    ↳ x09AB
seg000:001187C8 41 82 00 0C   beq     ↙
    ↳ loc_1187D4
seg000:001187CC 38 60 00 00   li      %r3, 0
seg000:001187D0 48 00 02 A0   b       exit
seg000:001187D4
seg000:001187D4                      loc_1187D4: # ↙
    ↳ CODE XREF: check3+8Cj
seg000:001187D4 4B F9 F3 D9   bl      ↙
    ↳ sub_B7BAC
seg000:001187D8 60 00 00 00   nop
seg000:001187DC 54 60 06 3E   clrlwi  %r0, %↙
    ↳ r3, 24
seg000:001187E0 2C 00 00 05   cmpwi   %r0, 5
seg000:001187E4 41 82 01 00   beq     ↙
    ↳ loc_1188E4
seg000:001187E8 40 80 00 10   bge     ↙
    ↳ loc_1187F8
seg000:001187EC 2C 00 00 04   cmpwi   %r0, 4
seg000:001187F0 40 80 00 58   bge     ↙
    ↳ loc_118848
seg000:001187F4 48 00 01 8C   b       ↙
    ↳ loc_118980
seg000:001187F8
seg000:001187F8                      loc_1187F8: # ↙
    ↳ CODE XREF: check3+ACj
seg000:001187F8 2C 00 00 0B   cmpwi   %r0, 0↙
    ↳ xB
seg000:001187FC 41 82 00 08   beq     ↙
    ↳ loc_118804
seg000:00118800 48 00 01 80   b       ↙
```

```
    ↳ loc_118980
seg000:00118804
seg000:00118804                  loc_118804: # ↙
    ↳ CODE XREF: check3+C0j
seg000:00118804 80 DE 00 00   lwz     %r6, ↙
    ↳ dword_24B704
seg000:00118808 38 81 00 38   addi    %r4, %↙
    ↳ sp, 0x50+var_18
seg000:0011880C 38 60 00 08   li      %r3, 8
seg000:00118810 38 A0 00 00   li      %r5, 0
seg000:00118814 48 00 BF B5   bl      .↙
    ↳ RBEREAD
seg000:00118818 60 00 00 00   nop
seg000:0011881C 54 60 04 3F   clrlwi. %r0, %↙
    ↳ r3, 16
seg000:00118820 41 82 00 0C   beq     ↙
    ↳ loc_11882C
seg000:00118824 38 60 00 00   li      %r3, 0
seg000:00118828 48 00 02 48   b       exit
seg000:0011882C
seg000:0011882C                  loc_11882C: # ↙
    ↳ CODE XREF: check3+E4j
seg000:0011882C A0 01 00 38   lhz     %r0, 0↙
    ↳ x50+var_18(%sp)
seg000:00118830 28 00 11 30   cmplwi  %r0, 0↙
    ↳ xFEA0
seg000:00118834 41 82 00 0C   beq     ↙
    ↳ loc_118840
seg000:00118838 38 60 00 00   li      %r3, 0
seg000:0011883C 48 00 02 34   b       exit
seg000:00118840
seg000:00118840                  loc_118840: # ↙
    ↳ CODE XREF: check3+F8j
```

```
seg000:00118840 38 60 00 01   li      %r3, 1
seg000:00118844 48 00 02 2C   b       exit
seg000:00118848
seg000:00118848                    loc_118848: # ⤦
    ↳ CODE XREF: check3+B4j
seg000:00118848 80 DE 00 00   lwz     %r6, ⤦
    ↳ dword_24B704
seg000:0011884C 38 81 00 38   addi    %r4, %⤦
    ↳ sp, 0x50+var_18
seg000:00118850 38 60 00 0A   li      %r3, 0⤦
    ↳ xA
seg000:00118854 38 A0 00 00   li      %r5, 0
seg000:00118858 48 00 BF 71   bl      .⤦
    ↳ RBEREAD
seg000:0011885C 60 00 00 00   nop
seg000:00118860 54 60 04 3F   clrlwi. %r0, %⤦
    ↳ r3, 16
seg000:00118864 41 82 00 0C   beq     ⤦
    ↳ loc_118870
seg000:00118868 38 60 00 00   li      %r3, 0
seg000:0011886C 48 00 02 04   b       exit
seg000:00118870
seg000:00118870                    loc_118870: # ⤦
    ↳ CODE XREF: check3+128j
seg000:00118870 A0 01 00 38   lhz     %r0, 0⤦
    ↳ x50+var_18(%sp)
seg000:00118874 28 00 03 F3   cmplwi  %r0, 0⤦
    ↳ xA6E1
seg000:00118878 41 82 00 0C   beq     ⤦
    ↳ loc_118884
seg000:0011887C 38 60 00 00   li      %r3, 0
seg000:00118880 48 00 01 F0   b       exit
seg000:00118884
```

```
seg000:00118884                 loc_118884: # ⤦
    ↳ CODE XREF: check3+13Cj
seg000:00118884 57 BF 06 3E   clrlwi  %r31, ⤦
    ↳ %r29, 24
seg000:00118888 28 1F 00 02   cmplwi  %r31, ⤦
    ↳ 2
seg000:0011888C 40 82 00 0C   bne     ⤦
    ↳ loc_118898
seg000:00118890 38 60 00 01   li      %r3, 1
seg000:00118894 48 00 01 DC   b       exit
seg000:00118898
seg000:00118898                 loc_118898: # ⤦
    ↳ CODE XREF: check3+150j
seg000:00118898 80 DE 00 00   lwz     %r6, ⤦
    ↳ dword_24B704
seg000:0011889C 38 81 00 38   addi    %r4, %⤦
    ↳ sp, 0x50+var_18
seg000:001188A0 38 60 00 0B   li      %r3, 0⤦
    ↳ xB
seg000:001188A4 38 A0 00 00   li      %r5, 0
seg000:001188A8 48 00 BF 21   bl      .⤦
    ↳ RBEREAD
seg000:001188AC 60 00 00 00   nop
seg000:001188B0 54 60 04 3F   clrlwi. %r0, %⤦
    ↳ r3, 16
seg000:001188B4 41 82 00 0C   beq     ⤦
    ↳ loc_1188C0
seg000:001188B8 38 60 00 00   li      %r3, 0
seg000:001188BC 48 00 01 B4   b       exit
seg000:001188C0
seg000:001188C0                 loc_1188C0: # ⤦
    ↳ CODE XREF: check3+178j
seg000:001188C0 A0 01 00 38   lhz     %r0, 0⤦
```

```
    ↳ x50+var_18(%sp)
seg000:001188C4 28 00 23 1C   cmplwi  %r0, 0↙
    ↳ x1C20
seg000:001188C8 41 82 00 0C   beq     ↙
    ↳ loc_1188D4
seg000:001188CC 38 60 00 00   li      %r3, 0
seg000:001188D0 48 00 01 A0   b       exit
seg000:001188D4
seg000:001188D4              loc_1188D4: # ↙
    ↳ CODE XREF: check3+18Cj
seg000:001188D4 28 1F 00 03   cmplwi  %r31, ↙
    ↳ 3
seg000:001188D8 40 82 01 94   bne     error
seg000:001188DC 38 60 00 01   li      %r3, 1
seg000:001188E0 48 00 01 90   b       exit
seg000:001188E4
seg000:001188E4              loc_1188E4: # ↙
    ↳ CODE XREF: check3+A8j
seg000:001188E4 80 DE 00 00   lwz     %r6, ↙
    ↳ dword_24B704
seg000:001188E8 38 81 00 38   addi    %r4, %↙
    ↳ sp, 0x50+var_18
seg000:001188EC 38 60 00 0C   li      %r3, 0↙
    ↳ xC
seg000:001188F0 38 A0 00 00   li      %r5, 0
seg000:001188F4 48 00 BE D5   bl      .↙
    ↳ RBEREAD
seg000:001188F8 60 00 00 00   nop
seg000:001188FC 54 60 04 3F   clrlwi. %r0, %↙
    ↳ r3, 16
seg000:00118900 41 82 00 0C   beq     ↙
    ↳ loc_11890C
seg000:00118904 38 60 00 00   li      %r3, 0
```

```
seg000:00118908 48 00 01 68   b       exit
seg000:0011890C
seg000:0011890C                       loc_11890C: # ⤦
    ↳ CODE XREF: check3+1C4j
seg000:0011890C A0 01 00 38   lhz     %r0, 0⤦
    ↳ x50+var_18(%sp)
seg000:00118910 28 00 1F 40   cmplwi  %r0, 0⤦
    ↳ x40FF
seg000:00118914 41 82 00 0C   beq     ⤦
    ↳ loc_118920
seg000:00118918 38 60 00 00   li      %r3, 0
seg000:0011891C 48 00 01 54   b       exit
seg000:00118920
seg000:00118920                       loc_118920: # ⤦
    ↳ CODE XREF: check3+1D8j
seg000:00118920 57 BF 06 3E   clrlwi  %r31, ⤦
    ↳ %r29, 24
seg000:00118924 28 1F 00 02   cmplwi  %r31, ⤦
    ↳ 2
seg000:00118928 40 82 00 0C   bne     ⤦
    ↳ loc_118934
seg000:0011892C 38 60 00 01   li      %r3, 1
seg000:00118930 48 00 01 40   b       exit
seg000:00118934
seg000:00118934                       loc_118934: # ⤦
    ↳ CODE XREF: check3+1ECj
seg000:00118934 80 DE 00 00   lwz     %r6, ⤦
    ↳ dword_24B704
seg000:00118938 38 81 00 38   addi    %r4, %⤦
    ↳ sp, 0x50+var_18
seg000:0011893C 38 60 00 0D   li      %r3, 0⤦
    ↳ xD
seg000:00118940 38 A0 00 00   li      %r5, 0
```

```
seg000:00118944 48 00 BE 85   bl        .↙
    ↘ RBEREAD
seg000:00118948 60 00 00 00   nop
seg000:0011894C 54 60 04 3F   clrlwi. %r0, %↙
    ↘ r3, 16
seg000:00118950 41 82 00 0C   beq       ↙
    ↘ loc_11895C
seg000:00118954 38 60 00 00   li        %r3, 0
seg000:00118958 48 00 01 18   b         exit
seg000:0011895C
seg000:0011895C                       loc_11895C: # ↙
    ↘ CODE XREF: check3+214j
seg000:0011895C A0 01 00 38   lhz       %r0, 0↙
    ↘ x50+var_18(%sp)
seg000:00118960 28 00 07 CF   cmplwi    %r0, 0↙
    ↘ xFC7
seg000:00118964 41 82 00 0C   beq       ↙
    ↘ loc_118970
seg000:00118968 38 60 00 00   li        %r3, 0
seg000:0011896C 48 00 01 04   b         exit
seg000:00118970
seg000:00118970                       loc_118970: # ↙
    ↘ CODE XREF: check3+228j
seg000:00118970 28 1F 00 03   cmplwi    %r31, ↙
    ↘ 3
seg000:00118974 40 82 00 F8   bne       error
seg000:00118978 38 60 00 01   li        %r3, 1
seg000:0011897C 48 00 00 F4   b         exit
seg000:00118980
seg000:00118980                       loc_118980: # ↙
    ↘ CODE XREF: check3+B8j
seg000:00118980                                   # ↙
    ↘ check3+C4j
```

```
seg000:00118980 80 DE 00 00   lwz      %r6, ↙
    ↳ dword_24B704
seg000:00118984 38 81 00 38   addi     %r4, %↙
    ↳ sp, 0x50+var_18
seg000:00118988 3B E0 00 00   li       %r31, ↙
    ↳ 0
seg000:0011898C 38 60 00 04   li       %r3, 4
seg000:00118990 38 A0 00 00   li       %r5, 0
seg000:00118994 48 00 BE 35   bl       .↙
    ↳ RBEREAD
seg000:00118998 60 00 00 00   nop
seg000:0011899C 54 60 04 3F   clrlwi. %r0, %↙
    ↳ r3, 16
seg000:001189A0 41 82 00 0C   beq      ↙
    ↳ loc_1189AC
seg000:001189A4 38 60 00 00   li       %r3, 0
seg000:001189A8 48 00 00 C8   b        exit
seg000:001189AC
seg000:001189AC                  loc_1189AC: # ↙
    ↳ CODE XREF: check3+264j
seg000:001189AC A0 01 00 38   lhz      %r0, 0↙
    ↳ x50+var_18(%sp)
seg000:001189B0 28 00 1D 6A   cmplwi   %r0, 0↙
    ↳ xAED0
seg000:001189B4 40 82 00 0C   bne      ↙
    ↳ loc_1189C0
seg000:001189B8 3B E0 00 01   li       %r31, ↙
    ↳ 1
seg000:001189BC 48 00 00 14   b        ↙
    ↳ loc_1189D0
seg000:001189C0
seg000:001189C0                  loc_1189C0: # ↙
    ↳ CODE XREF: check3+278j
```

```
seg000:001189C0 28 00 18 28   cmplwi  %r0, 0↙
    ↘ x2818
seg000:001189C4 41 82 00 0C   beq     ↙
    ↘ loc_1189D0
seg000:001189C8 38 60 00 00   li      %r3, 0
seg000:001189CC 48 00 00 A4   b       exit
seg000:001189D0
seg000:001189D0                  loc_1189D0: # ↙
    ↘ CODE XREF: check3+280j
seg000:001189D0                              # ↙
    ↘ check3+288j
seg000:001189D0 57 A0 06 3E   clrlwi  %r0, %↙
    ↘ r29, 24
seg000:001189D4 28 00 00 02   cmplwi  %r0, 2
seg000:001189D8 40 82 00 20   bne     ↙
    ↘ loc_1189F8
seg000:001189DC 57 E0 06 3F   clrlwi. %r0, %↙
    ↘ r31, 24
seg000:001189E0 41 82 00 10   beq     good2
seg000:001189E4 48 00 4C 69   bl      ↙
    ↘ sub_11D64C
seg000:001189E8 60 00 00 00   nop
seg000:001189EC 48 00 00 84   b       exit
seg000:001189F0
seg000:001189F0                  good2:      # ↙
    ↘ CODE XREF: check3+2A4j
seg000:001189F0 38 60 00 01   li      %r3, 1
seg000:001189F4 48 00 00 7C   b       exit
seg000:001189F8
seg000:001189F8                  loc_1189F8: # ↙
    ↘ CODE XREF: check3+29Cj
seg000:001189F8 80 DE 00 00   lwz     %r6, ↙
    ↘ dword_24B704
```

```
seg000:001189FC 38 81 00 38   addi    %r4, %↙
    ↘ sp, 0x50+var_18
seg000:00118A00 38 60 00 05   li      %r3, 5
seg000:00118A04 38 A0 00 00   li      %r5, 0
seg000:00118A08 48 00 BD C1   bl      .↙
    ↘ RBEREAD
seg000:00118A0C 60 00 00 00   nop
seg000:00118A10 54 60 04 3F   clrlwi. %r0, %↙
    ↘ r3, 16
seg000:00118A14 41 82 00 0C   beq     ↙
    ↘ loc_118A20
seg000:00118A18 38 60 00 00   li      %r3, 0
seg000:00118A1C 48 00 00 54   b       exit
seg000:00118A20
seg000:00118A20                      loc_118A20: # ↙
    ↘ CODE XREF: check3+2D8j
seg000:00118A20 A0 01 00 38   lhz     %r0, 0↙
    ↘ x50+var_18(%sp)
seg000:00118A24 28 00 11 D3   cmplwi  %r0, 0↙
    ↘ xD300
seg000:00118A28 40 82 00 0C   bne     ↙
    ↘ loc_118A34
seg000:00118A2C 3B E0 00 01   li      %r31, ↙
    ↘ 1
seg000:00118A30 48 00 00 14   b       good1
seg000:00118A34
seg000:00118A34                      loc_118A34: # ↙
    ↘ CODE XREF: check3+2ECj
seg000:00118A34 28 00 1A EB   cmplwi  %r0, 0↙
    ↘ xEBA1
seg000:00118A38 41 82 00 0C   beq     good1
seg000:00118A3C 38 60 00 00   li      %r3, 0
seg000:00118A40 48 00 00 30   b       exit
```

```
seg000:00118A44
seg000:00118A44                     good1:       # ↙
    ↘ CODE XREF: check3+2F4j
seg000:00118A44                                  # ↙
    ↘ check3+2FCj
seg000:00118A44 57 A0 06 3E   clrlwi  %r0, %↙
    ↘ r29, 24
seg000:00118A48 28 00 00 03   cmplwi  %r0, 3
seg000:00118A4C 40 82 00 20   bne     error
seg000:00118A50 57 E0 06 3F   clrlwi. %r0, %↙
    ↘ r31, 24
seg000:00118A54 41 82 00 10   beq     good
seg000:00118A58 48 00 4B F5   bl      ↙
    ↘ sub_11D64C
seg000:00118A5C 60 00 00 00   nop
seg000:00118A60 48 00 00 10   b       exit
seg000:00118A64
seg000:00118A64                     good:        # ↙
    ↘ CODE XREF: check3+318j
seg000:00118A64 38 60 00 01   li      %r3, 1
seg000:00118A68 48 00 00 08   b       exit
seg000:00118A6C
seg000:00118A6C                     error:       # ↙
    ↘ CODE XREF: check3+19Cj
seg000:00118A6C                                  # ↙
    ↘ check3+238j ...
seg000:00118A6C 38 60 00 00   li      %r3, 0
seg000:00118A70
seg000:00118A70                     exit:        # ↙
    ↘ CODE XREF: check3+44j
seg000:00118A70                                  # ↙
    ↘ check3+58j ...
seg000:00118A70 80 01 00 58   lwz     %r0, 0↙
```

```
    ↳ x50+arg_8(%sp)
seg000:00118A74 38 21 00 50   addi    %sp, %↙
    ↳ sp, 0x50
seg000:00118A78 83 E1 FF FC   lwz     %r31, ↙
    ↳ var_4(%sp)
seg000:00118A7C 7C 08 03 A6   mtlr    %r0
seg000:00118A80 83 C1 FF F8   lwz     %r30, ↙
    ↳ var_8(%sp)
seg000:00118A84 83 A1 FF F4   lwz     %r29, ↙
    ↳ var_C(%sp)
seg000:00118A88 4E 80 00 20   blr
seg000:00118A88                    # End of ↙
    ↳ function check3
```

.RBEREAD(). CMPLWI.

.RBEREAD() : 0, 1, 8, 0xA, 0xB, 0xC, 0xD, 4, 5. ?

Sentinel Eve3!

.RBEREAD(), .

.

0x001186FC 0x48 y 0 BEQ B (): [IBM00].

0x00118718 0x60 y NOP: .

.

.

78.2 #2: SCO OpenServer

SCO OpenServer 1997 .

: «Copyright 1989, Rainbow Technologies, Inc., Irvine, CA» y «Sentinel Integrated Driver Ver. 3.0 ».

:

```
/dev/rbsl8
/dev/rbsl9
/dev/rbsl10
```

.

SCO OpenServer.

:

```
.text:00022AB8         public SSQC
.text:00022AB8 SSQC    proc near ; CODE XREF:↙
    ↳  SSQ+7p
.text:00022AB8
.text:00022AB8 var_44 = byte ptr -44h
.text:00022AB8 var_29 = byte ptr -29h
.text:00022AB8 arg_0  = dword ptr  8
.text:00022AB8
.text:00022AB8         push    ebp
.text:00022AB9         mov     ebp, esp
.text:00022ABB         sub     esp, 44h
.text:00022ABE         push    edi
.text:00022ABF         mov     edi, offset ↙
    ↳ unk_4035D0
.text:00022AC4         push    esi
```

```
.text:00022AC5             mov     esi, [ebp+↙
    ↘ arg_0]
.text:00022AC8             push    ebx
.text:00022AC9             push    esi
.text:00022ACA             call    strlen
.text:00022ACF             add     esp, 4
.text:00022AD2             cmp     eax, 2
.text:00022AD7             jnz     loc_22BA4
.text:00022ADD             inc     esi
.text:00022ADE             mov     al, [esi-1]
.text:00022AE1             movsx   eax, al
.text:00022AE4             cmp     eax, '3'
.text:00022AE9             jz      loc_22B84
.text:00022AEF             cmp     eax, '4'
.text:00022AF4             jz      loc_22B94
.text:00022AFA             cmp     eax, '5'
.text:00022AFF             jnz     short ↙
    ↘ loc_22B6B
.text:00022B01             movsx   ebx, byte ptr ↙
    ↘ [esi]
.text:00022B04             sub     ebx, '0'
.text:00022B07             mov     eax, 7
.text:00022B0C             add     eax, ebx
.text:00022B0E             push    eax
.text:00022B0F             lea     eax, [ebp+↙
    ↘ var_44]
.text:00022B12             push    offset aDevSlD↙
    ↘   ; "/dev/sl%d"
.text:00022B17             push    eax
.text:00022B18             call    nl_sprintf
.text:00022B1D             push    0              ↙
    ↘   ; int
.text:00022B1F             push    offset ↙
```

```
    ↘ aDevRbsl8 ; char *
.text:00022B24         call    _access
.text:00022B29         add     esp, 14h
.text:00022B2C         cmp     eax, 0↙
    ↘ FFFFFFFFh
.text:00022B31         jz      short ↙
    ↘ loc_22B48
.text:00022B33         lea     eax, [ebx+7]
.text:00022B36         push    eax
.text:00022B37         lea     eax, [ebp+↙
    ↘ var_44]
.text:00022B3A         push    offset ↙
    ↘ aDevRbslD ; "/dev/rbsl%d"
.text:00022B3F         push    eax
.text:00022B40         call    nl_sprintf
.text:00022B45         add     esp, 0Ch
.text:00022B48
.text:00022B48 loc_22B48: ; CODE XREF: SSQC↙
    ↘ +79j
.text:00022B48         mov     edx, [edi]
.text:00022B4A         test    edx, edx
.text:00022B4C         jle     short ↙
    ↘ loc_22B57
.text:00022B4E         push    edx             ↙
    ↘   ; int
.text:00022B4F         call    _close
.text:00022B54         add     esp, 4
.text:00022B57
.text:00022B57 loc_22B57: ; CODE XREF: SSQC↙
    ↘ +94j
.text:00022B57         push    2               ↙
    ↘   ; int
.text:00022B59         lea     eax, [ebp+↙
```

```
    ↳ var_44]
.text:00022B5C         push    eax            ↙
    ↳   ; char *
.text:00022B5D         call    _open
.text:00022B62         add     esp, 8
.text:00022B65         test    eax, eax
.text:00022B67         mov     [edi], eax
.text:00022B69         jge     short ↙
    ↳ loc_22B78
.text:00022B6B
.text:00022B6B loc_22B6B: ; CODE XREF: SSQC↙
    ↳ +47j
.text:00022B6B         mov     eax, 0↙
    ↳ FFFFFFFFh
.text:00022B70         pop     ebx
.text:00022B71         pop     esi
.text:00022B72         pop     edi
.text:00022B73         mov     esp, ebp
.text:00022B75         pop     ebp
.text:00022B76         retn
.text:00022B78
.text:00022B78 loc_22B78: ; CODE XREF: SSQC+↙
    ↳ B1j
.text:00022B78         pop     ebx
.text:00022B79         pop     esi
.text:00022B7A         pop     edi
.text:00022B7B         xor     eax, eax
.text:00022B7D         mov     esp, ebp
.text:00022B7F         pop     ebp
.text:00022B80         retn
.text:00022B84
.text:00022B84 loc_22B84: ; CODE XREF: SSQC↙
    ↳ +31j
```

```
.text:00022B84          mov     al, [esi]
.text:00022B86          pop     ebx
.text:00022B87          pop     esi
.text:00022B88          pop     edi
.text:00022B89          mov     ds:byte_407224↙
    ↘ , al
.text:00022B8E          mov     esp, ebp
.text:00022B90          xor     eax, eax
.text:00022B92          pop     ebp
.text:00022B93          retn
.text:00022B94
.text:00022B94 loc_22B94: ; CODE XREF: SSQC↙
    ↘ +3Cj
.text:00022B94          mov     al, [esi]
.text:00022B96          pop     ebx
.text:00022B97          pop     esi
.text:00022B98          pop     edi
.text:00022B99          mov     ds:byte_407225↙
    ↘ , al
.text:00022B9E          mov     esp, ebp
.text:00022BA0          xor     eax, eax
.text:00022BA2          pop     ebp
.text:00022BA3          retn
.text:00022BA4
.text:00022BA4 loc_22BA4: ; CODE XREF: SSQC↙
    ↘ +1Fj
.text:00022BA4          movsx   eax, ds:↙
    ↘ byte_407225
.text:00022BAB          push    esi
.text:00022BAC          push    eax
.text:00022BAD          movsx   eax, ds:↙
    ↘ byte_407224
.text:00022BB4          push    eax
```

```
.text:00022BB5           lea     eax, [ebp+↙
    ↘ var_44]
.text:00022BB8           push    offset a46CCS ↙
    ↘   ; "46%c%c%s"
.text:00022BBD           push    eax
.text:00022BBE           call    nl_sprintf
.text:00022BC3           lea     eax, [ebp+↙
    ↘ var_44]
.text:00022BC6           push    eax
.text:00022BC7           call    strlen
.text:00022BCC           add     esp, 18h
.text:00022BCF           cmp     eax, 1Bh
.text:00022BD4           jle     short ↙
    ↘ loc_22BDA
.text:00022BD6           mov     [ebp+var_29], ↙
    ↘ 0
.text:00022BDA
.text:00022BDA loc_22BDA: ; CODE XREF: SSQC↙
    ↘ +11Cj
.text:00022BDA           lea     eax, [ebp+↙
    ↘ var_44]
.text:00022BDD           push    eax
.text:00022BDE           call    strlen
.text:00022BE3           push    eax             ↙
    ↘   ; unsigned int
.text:00022BE4           lea     eax, [ebp+↙
    ↘ var_44]
.text:00022BE7           push    eax             ↙
    ↘   ; void *
.text:00022BE8           mov     eax, [edi]
.text:00022BEA           push    eax             ↙
    ↘   ; int
.text:00022BEB           call    _write
```

```
.text:00022BF0          add     esp, 10h
.text:00022BF3          pop     ebx
.text:00022BF4          pop     esi
.text:00022BF5          pop     edi
.text:00022BF6          mov     esp, ebp
.text:00022BF8          pop     ebp
.text:00022BF9          retn
.text:00022BFA          db 0Eh dup(90h)
.text:00022BFA SSQC     endp
```

.

SSQC() thunk function:

```
.text:0000DBE8          public SSQ
.text:0000DBE8 SSQ      proc near ; CODE XREF:↙
    ↳  sys_info+A9p
.text:0000DBE8                    ; sys_info+↙
    ↳ CBp ...
.text:0000DBE8
.text:0000DBE8 arg_0    = dword ptr  8
.text:0000DBE8
.text:0000DBE8          push    ebp
.text:0000DBE9          mov     ebp, esp
.text:0000DBEB          mov     edx, [ebp+↙
    ↳ arg_0]
.text:0000DBEE          push    edx
.text:0000DBEF          call    SSQC
.text:0000DBF4          add     esp, 4
.text:0000DBF7          mov     esp, ebp
.text:0000DBF9          pop     ebp
.text:0000DBFA          retn
.text:0000DBFB SSQ      endp
```

SSQ() .

:

```
.data:0040169C _51_52_53       dd offset ⤦
    ↳ aPressAnyKeyT_0 ; DATA XREF: init_sys⤦
    ↳ +392r
.data:0040169C                                  ⤦
    ↳            ; sys_info+A1r
.data:0040169C                                  ⤦
    ↳            ; "PRESS ANY KEY TO ⤦
    ↳ CONTINUE: "
.data:004016A0                 dd offset a51⤦
    ↳            ; "51"
.data:004016A4                 dd offset a52⤦
    ↳            ; "52"
.data:004016A8                 dd offset a53⤦
    ↳            ; "53"

...

.data:004016B8 _3C_or_3E       dd offset a3c⤦
    ↳            ; DATA XREF: sys_info:⤦
    ↳ loc_D67Br
.data:004016B8                                  ⤦
    ↳            ; "3C"
.data:004016BC                 dd offset a3e⤦
    ↳            ; "3E"

; these names we gave to the labels:
.data:004016C0 answers1        dd 6B05h        ⤦
    ↳            ; DATA XREF: sys_info+E7r
.data:004016C4                 dd 3D87h
```

```
.data:004016C8 answers2         dd 3Ch       ⤦
    ↳            ; DATA XREF: sys_info+F2r
.data:004016CC                  dd 832h
.data:004016D0 _C_and_B         db 0Ch       ⤦
    ↳            ; DATA XREF: sys_info+BAr
.data:004016D0                               ⤦
    ↳            ; sys_info:OKr
.data:004016D1 byte_4016D1      db 0Bh       ⤦
    ↳            ; DATA XREF: sys_info+FDr
.data:004016D2                  db    0

...

.text:0000D652                  xor     eax, ⤦
    ↳ eax
.text:0000D654                  mov     al, ⤦
    ↳ ds:ctl_port
.text:0000D659                  mov     ecx, ⤦
    ↳ _51_52_53[eax*4]
.text:0000D660                  push    ecx
.text:0000D661                  call    SSQ
.text:0000D666                  add     esp, ⤦
    ↳ 4
.text:0000D669                  cmp     eax, ⤦
    ↳ 0FFFFFFFFh
.text:0000D66E                  jz      short⤦
    ↳  loc_D6D1
.text:0000D670                  xor     ebx, ⤦
    ↳ ebx
.text:0000D672                  mov     al, ⤦
    ↳ _C_and_B
.text:0000D677                  test    al, ⤦
    ↳ al
```

```
.text:0000D679                 jz      short ⤦
    ↳  loc_D6C0
.text:0000D67B
.text:0000D67B loc_D67B: ; CODE XREF: ⤦
    ↳ sys_info+106j
.text:0000D67B                 mov     eax, ⤦
    ↳ _3C_or_3E[ebx*4]
.text:0000D682                 push    eax
.text:0000D683                 call    SSQ
.text:0000D688                 push    ⤦
    ↳ offset a4g        ; "4G"
.text:0000D68D                 call    SSQ
.text:0000D692                 push    ⤦
    ↳ offset a0123456789 ; "0123456789"
.text:0000D697                 call    SSQ
.text:0000D69C                 add     esp, ⤦
    ↳ 0Ch
.text:0000D69F                 mov     edx, ⤦
    ↳ answers1[ebx*4]
.text:0000D6A6                 cmp     eax, ⤦
    ↳ edx
.text:0000D6A8                 jz      short ⤦
    ↳  OK
.text:0000D6AA                 mov     ecx, ⤦
    ↳ answers2[ebx*4]
.text:0000D6B1                 cmp     eax, ⤦
    ↳ ecx
.text:0000D6B3                 jz      short ⤦
    ↳  OK
.text:0000D6B5                 mov     al, ⤦
    ↳ byte_4016D1[ebx]
.text:0000D6BB                 inc     ebx
.text:0000D6BC                 test    al, ⤦
```

```
    ↳ al
.text:0000D6BE                  jnz     short↙
    ↳  loc_D67B
.text:0000D6C0
.text:0000D6C0 loc_D6C0: ; CODE XREF: ↙
    ↳ sys_info+C1j
.text:0000D6C0                  inc     ds:↙
    ↳ ctl_port
.text:0000D6C6                  xor     eax, ↙
    ↳ eax
.text:0000D6C8                  mov     al, ↙
    ↳ ds:ctl_port
.text:0000D6CD                  cmp     eax, ↙
    ↳ edi
.text:0000D6CF                  jle     short↙
    ↳  loc_D652
.text:0000D6D1
.text:0000D6D1 loc_D6D1: ; CODE XREF: ↙
    ↳ sys_info+98j
.text:0000D6D1           ; sys_info+B6j
.text:0000D6D1                  mov     edx, ↙
    ↳ [ebp+var_8]
.text:0000D6D4                  inc     edx
.text:0000D6D5                  mov     [ebp+↙
    ↳ var_8], edx
.text:0000D6D8                  cmp     edx, ↙
    ↳ 3
.text:0000D6DB                  jle     ↙
    ↳ loc_D641
.text:0000D6E1
.text:0000D6E1 loc_D6E1: ; CODE XREF: ↙
    ↳ sys_info+16j
.text:0000D6E1           ; sys_info+51j ...
```

```
.text:0000D6E1                 pop     ebx
.text:0000D6E2                 pop     edi
.text:0000D6E3                 mov     esp, ↙
    ↳ ebp
.text:0000D6E5                 pop     ebp
.text:0000D6E6                 retn
.text:0000D6E8 OK:           ; CODE XREF: ↙
    ↳ sys_info+F0j
.text:0000D6E8               ; sys_info+FBj
.text:0000D6E8                 mov     al, ↙
    ↳ _C_and_B[ebx]
.text:0000D6EE                 pop     ebx
.text:0000D6EF                 pop     edi
.text:0000D6F0                 mov     ds:↙
    ↳ ctl_model, al
.text:0000D6F5                 mov     esp, ↙
    ↳ ebp
.text:0000D6F7                 pop     ebp
.text:0000D6F8                 retn
.text:0000D6F8 sys_info        endp
```

«3C» y «3E» Sentinel Pro .

: 34 on page 874.

. . . (SSQC( )) . . . .

.

51/52/53 . 3x/4x «family» .

”0123456789”. . , 0xC o 0xB ctl_model.

”PRESS ANY KEY TO CONTINUE: ”, . [4].

ctl_mode.

:

```
.text:0000D708 prep_sys proc near ; CODE ↙
    ↳ XREF: init_sys+46Ap
.text:0000D708
.text:0000D708 var_14   = dword ptr -14h
.text:0000D708 var_10   = byte ptr -10h
.text:0000D708 var_8    = dword ptr -8
.text:0000D708 var_2    = word ptr -2
.text:0000D708
.text:0000D708          push    ebp
.text:0000D709          mov     eax, ds:↙
    ↳ net_env
.text:0000D70E          mov     ebp, esp
.text:0000D710          sub     esp, 1Ch
.text:0000D713          test    eax, eax
.text:0000D715          jnz     short ↙
    ↳ loc_D734
.text:0000D717          mov     al, ds:↙
    ↳ ctl_model
.text:0000D71C          test    al, al
.text:0000D71E          jnz     short ↙
    ↳ loc_D77E
.text:0000D720          mov     [ebp+var_8],↙
    ↳  offset aIeCvulnvvOkgT_ ; "Ie-cvulnvV↙
    ↳ \\\bOKG]T_"
.text:0000D727          mov     edx, 7
.text:0000D72C          jmp     loc_D7E7
```

4

```
...

.text:0000D7E7 loc_D7E7: ; CODE XREF: ↙
    ↘ prep_sys+24j
.text:0000D7E7              ; prep_sys+33j
.text:0000D7E7             push    edx
.text:0000D7E8             mov     edx, [ebp+↙
    ↘ var_8]
.text:0000D7EB             push    20h
.text:0000D7ED             push    edx
.text:0000D7EE             push    16h
.text:0000D7F0             call    err_warn
.text:0000D7F5             push    offset ↙
    ↘ station_sem
.text:0000D7FA             call    ClosSem
.text:0000D7FF             call    startup_err
```

.

:

```
.text:0000A43C err_warn          proc near     ↙
    ↘           ; CODE XREF: prep_sys+E8p
.text:0000A43C                                 ↙
    ↘           ; prep_sys2+2Fp ...
.text:0000A43C
.text:0000A43C var_55            = byte ptr ↙
    ↘ -55h
.text:0000A43C var_54            = byte ptr ↙
    ↘ -54h
.text:0000A43C arg_0             = dword ptr  ↙
    ↘ 8
```

```
.text:0000A43C arg_4                = dword ptr  ↙
    ↘ 0Ch
.text:0000A43C arg_8                = dword ptr  ↙
    ↘ 10h
.text:0000A43C arg_C                = dword ptr  ↙
    ↘ 14h
.text:0000A43C
.text:0000A43C                      push    ebp
.text:0000A43D                      mov     ebp, ↙
    ↘ esp
.text:0000A43F                      sub     esp, ↙
    ↘ 54h
.text:0000A442                      push    edi
.text:0000A443                      mov     ecx, ↙
    ↘ [ebp+arg_8]
.text:0000A446                      xor     edi, ↙
    ↘ edi
.text:0000A448                      test    ecx, ↙
    ↘ ecx
.text:0000A44A                      push    esi
.text:0000A44B                      jle     short↙
    ↘  loc_A466
.text:0000A44D                      mov     esi, ↙
    ↘ [ebp+arg_C] ; key
.text:0000A450                      mov     edx, ↙
    ↘ [ebp+arg_4] ; string
.text:0000A453
.text:0000A453 loc_A453:                         ↙
    ↘           ; CODE XREF: err_warn+28j
.text:0000A453                      xor     eax, ↙
    ↘ eax
.text:0000A455                      mov     al, [↙
    ↘ edx+edi]
```

```
.text:0000A458                 xor     eax, ↙
    ↘ esi
.text:0000A45A                 add     esi, ↙
    ↘ 3
.text:0000A45D                 inc     edi
.text:0000A45E                 cmp     edi, ↙
    ↘ ecx
.text:0000A460                 mov     [ebp+↙
    ↘ edi+var_55], al
.text:0000A464                 jl      short↙
    ↘  loc_A453
.text:0000A466
.text:0000A466 loc_A466:                               ↙
    ↘           ; CODE XREF: err_warn+Fj
.text:0000A466                 mov     [ebp+↙
    ↘ edi+var_54], 0
.text:0000A46B                 mov     eax, ↙
    ↘ [ebp+arg_0]
.text:0000A46E                 cmp     eax, ↙
    ↘ 18h
.text:0000A473                 jnz     short↙
    ↘  loc_A49C
.text:0000A475                 lea     eax, ↙
    ↘ [ebp+var_54]
.text:0000A478                 push    eax
.text:0000A479                 call    ↙
    ↘ status_line
.text:0000A47E                 add     esp, ↙
    ↘ 4
.text:0000A481
.text:0000A481 loc_A481:                               ↙
    ↘           ; CODE XREF: err_warn+72j
.text:0000A481                 push    50h
```

```
.text:0000A483                 push    0
.text:0000A485                 lea     eax, ⤦
    ↳ [ebp+var_54]
.text:0000A488                 push    eax
.text:0000A489                 call    ⤦
    ↳ memset
.text:0000A48E                 call    ⤦
    ↳ pcv_refresh
.text:0000A493                 add     esp, ⤦
    ↳ 0Ch
.text:0000A496                 pop     esi
.text:0000A497                 pop     edi
.text:0000A498                 mov     esp, ⤦
    ↳ ebp
.text:0000A49A                 pop     ebp
.text:0000A49B                 retn
.text:0000A49C
.text:0000A49C loc_A49C:                        ⤦
    ↳            ; CODE XREF: err_warn+37j
.text:0000A49C                 push    0
.text:0000A49E                 lea     eax, ⤦
    ↳ [ebp+var_54]
.text:0000A4A1                 mov     edx, ⤦
    ↳ [ebp+arg_0]
.text:0000A4A4                 push    edx
.text:0000A4A5                 push    eax
.text:0000A4A6                 call    ⤦
    ↳ pcv_lputs
.text:0000A4AB                 add     esp, ⤦
    ↳ 0Ch
.text:0000A4AE                 jmp     short⤦
    ↳  loc_A481
.text:0000A4AE err_warn        endp
```

.

«offln» 0xFE81 y 0x12A9. timer().

```
.text:0000DA55 loc_DA55:                                      ↙
    ↘               ; CODE XREF: sync_sys+24Cj
.text:0000DA55                       push    ↙
    ↘ offset aOffln    ; "offln"
.text:0000DA5A                       call    SSQ
.text:0000DA5F                       add     esp, ↙
    ↘ 4
.text:0000DA62                       mov     dl, [↙
    ↘ ebx]
.text:0000DA64                       mov     esi, ↙
    ↘ eax
.text:0000DA66                       cmp     dl, 0↙
    ↘ Bh
.text:0000DA69                       jnz     short↙
    ↘  loc_DA83
.text:0000DA6B                       cmp     esi, ↙
    ↘ 0FE81h
.text:0000DA71                       jz      OK
.text:0000DA77                       cmp     esi, ↙
    ↘ 0FFFFF8EFh
.text:0000DA7D                       jz      OK
.text:0000DA83
.text:0000DA83 loc_DA83:                                      ↙
    ↘               ; CODE XREF: sync_sys+201j
.text:0000DA83                       mov     cl, [↙
    ↘ ebx]
.text:0000DA85                       cmp     cl, 0↙
```

```
    ↳ Ch
.text:0000DA88                  jnz     short↙
    ↳  loc_DA9F
.text:0000DA8A                  cmp     esi, ↙
    ↳ 12A9h
.text:0000DA90                  jz      OK
.text:0000DA96                  cmp     esi, ↙
    ↳ 0FFFFFFF5h
.text:0000DA99                  jz      OK
.text:0000DA9F
.text:0000DA9F loc_DA9F:                        ↙
    ↳           ; CODE XREF: sync_sys+220j
.text:0000DA9F                  mov     eax, ↙
    ↳ [ebp+var_18]
.text:0000DAA2                  test    eax, ↙
    ↳ eax
.text:0000DAA4                  jz      short↙
    ↳  loc_DAB0
.text:0000DAA6                  push    24h
.text:0000DAA8                  call    timer
.text:0000DAAD                  add     esp, ↙
    ↳ 4
.text:0000DAB0
.text:0000DAB0 loc_DAB0:                        ↙
    ↳           ; CODE XREF: sync_sys+23Cj
.text:0000DAB0                  inc     edi
.text:0000DAB1                  cmp     edi, ↙
    ↳ 3
.text:0000DAB4                  jle     short↙
    ↳  loc_DA55
.text:0000DAB6                  mov     eax, ↙
    ↳ ds:net_env
.text:0000DABB                  test    eax, ↙
```

```
    ↳ eax
.text:0000DABD                 jz      short↙
    ↳  error

...

.text:0000DAF7 error:                                  ↙
    ↳             ; CODE XREF: sync_sys+255j
.text:0000DAF7                                         ↙
    ↳             ; sync_sys+274j ...
.text:0000DAF7                 mov     [ebp+↙
    ↳ var_8], offset ↙
    ↳ encrypted_error_message2
.text:0000DAFE                 mov     [ebp+↙
    ↳ var_C], 17h ; decrypting key
.text:0000DB05                 jmp     ↙
    ↳ decrypt_end_print_message

...

; this name we gave to label:
.text:0000D9B6 decrypt_end_print_message:   ↙
    ↳             ; CODE XREF: sync_sys+29Dj
.text:0000D9B6                                         ↙
    ↳             ; sync_sys+2ABj
.text:0000D9B6                 mov     eax, ↙
    ↳ [ebp+var_18]
.text:0000D9B9                 test    eax, ↙
    ↳ eax
.text:0000D9BB                 jnz     short↙
    ↳  loc_D9FB
.text:0000D9BD                 mov     edx, ↙
    ↳ [ebp+var_C] ; key
```

```
.text:0000D9C0                 mov     ecx, ↙
    ↘ [ebp+var_8] ; string
.text:0000D9C3                 push    edx
.text:0000D9C4                 push    20h
.text:0000D9C6                 push    ecx
.text:0000D9C7                 push    18h
.text:0000D9C9                 call    ↙
    ↘ err_warn
.text:0000D9CE                 push    0Fh
.text:0000D9D0                 push    190h
.text:0000D9D5                 call    sound
.text:0000D9DA                 mov     [ebp+↙
    ↘ var_18], 1
.text:0000D9E1                 add     esp, ↙
    ↘ 18h
.text:0000D9E4                 call    ↙
    ↘ pcv_kbhit
.text:0000D9E9                 test    eax, ↙
    ↘ eax
.text:0000D9EB                 jz      short↙
    ↘  loc_D9FB

...

; this name we gave to label:
.data:00401736 encrypted_error_message2 db ↙
    ↘ 74h, 72h, 78h, 43h, 48h, 6, 5Ah, 49h, ↙
    ↘ 4Ch, 2 dup(47h)
.data:00401736                 db 51h, 4Fh, ↙
    ↘ 47h, 61h, 20h, 22h, 3Ch, 24h, 33h, 36h↙
    ↘ , 76h
.data:00401736                 db 3Ah, 33h, ↙
    ↘ 31h, 0Ch, 0, 0Bh, 1Fh, 7, 1Eh, 1Ah
```

.

.

78.2.1

. err_warn() :

Listing 78.1:

```
.text:0000A44D                 mov     esi, [ebp+arg_C] ; key
.text:0000A450                 mov     edx, [ebp+arg_4] ; string
.text:0000A453 loc_A453:
.text:0000A453                 xor     eax, eax
.text:0000A455                 mov     al, [edx+edi] ;
.text:0000A458                 xor     eax, esi      ;
.text:0000A45A                 add     esi, 3        ;
.text:0000A45D                 inc     edi
.text:0000A45E                 cmp     edi, ecx
.text:0000A460                 mov     [ebp+edi+var_55], al
.text:0000A464                 jl      short loc_A453
```

:

```
.text:0000DAF7 error:                                    ↙
    ↘               ; CODE XREF: sync_sys+255j
.text:0000DAF7                                           ↙
    ↘               ; sync_sys+274j ...
.text:0000DAF7                 mov     [ebp+↙
    ↘ var_8], offset ↙
    ↘ encrypted_error_message2
.text:0000DAFE                 mov     [ebp+↙
    ↘ var_C], 17h ; decrypting key
.text:0000DB05                 jmp     ↙
    ↘ decrypt_end_print_message

...

; this name we gave to label manually:
.text:0000D9B6 decrypt_end_print_message:                ↙
    ↘               ; CODE XREF: sync_sys+29Dj
.text:0000D9B6                                           ↙
    ↘               ; sync_sys+2ABj
.text:0000D9B6                 mov     eax, ↙
    ↘ [ebp+var_18]
.text:0000D9B9                 test    eax, ↙
    ↘ eax
.text:0000D9BB                 jnz     short↙
    ↘  loc_D9FB
.text:0000D9BD                 mov     edx, ↙
    ↘ [ebp+var_C] ; key
.text:0000D9C0                 mov     ecx, ↙
    ↘ [ebp+var_8] ; string
.text:0000D9C3                 push    edx
.text:0000D9C4                 push    20h
```

```
.text:0000D9C6                 push    ecx
.text:0000D9C7                 push    18h
.text:0000D9C9                 call    ↙
    ↘ err_warn
```

xoring: .

:

Listing 78.2: Python 3.x

```
#!/usr/bin/python
import sys

msg=[0x74, 0x72, 0x78, 0x43, 0x48, 0x6, 0x5A↙
    ↘ , 0x49, 0x4C, 0x47, 0x47,
0x51, 0x4F, 0x47, 0x61, 0x20, 0x22, 0x3C, 0↙
    ↘ x24, 0x33, 0x36, 0x76,
0x3A, 0x33, 0x31, 0x0C, 0x0, 0x0B, 0x1F, 0x7↙
    ↘ , 0x1E, 0x1A]

key=0x17
tmp=key
for i in msg:
        sys.stdout.write ("%c" % (i^tmp))
        tmp=tmp+3
sys.stdout.flush()
```

: «check security device connection». .

. . . (XOR) . `ESI`. `255`, .

0..255. .

Listing 78.3: Python 3.x

```
#!/usr/bin/python
import sys, curses.ascii

msgs=[
[0x74, 0x72, 0x78, 0x43, 0x48, 0x6, 0x5A, 0↙
    ↘ x49, 0x4C, 0x47, 0x47,
0x51, 0x4F, 0x47, 0x61, 0x20, 0x22, 0x3C, 0↙
    ↘ x24, 0x33, 0x36, 0x76,
0x3A, 0x33, 0x31, 0x0C, 0x0, 0x0B, 0x1F, 0x7↙
    ↘ , 0x1E, 0x1A],

[0x49, 0x65, 0x2D, 0x63, 0x76, 0x75, 0x6C, 0↙
    ↘ x6E, 0x76, 0x56, 0x5C,
8, 0x4F, 0x4B, 0x47, 0x5D, 0x54, 0x5F, 0x1D,↙
    ↘  0x26, 0x2C, 0x33,
0x27, 0x28, 0x6F, 0x72, 0x75, 0x78, 0x7B, 0↙
    ↘ x7E, 0x41, 0x44],

[0x45, 0x61, 0x31, 0x67, 0x72, 0x79, 0x68, 0↙
    ↘ x52, 0x4A, 0x52, 0x50,
0x0C, 0x4B, 0x57, 0x43, 0x51, 0x58, 0x5B, 0↙
    ↘ x61, 0x37, 0x33, 0x2B,
0x39, 0x39, 0x3C, 0x38, 0x79, 0x3A, 0x30, 0↙
    ↘ x17, 0x0B, 0x0C],

[0x40, 0x64, 0x79, 0x75, 0x7F, 0x6F, 0x0, 0↙
    ↘ x4C, 0x40, 0x9, 0x4D, 0x5A,
0x46, 0x5D, 0x57, 0x49, 0x57, 0x3B, 0x21, 0↙
    ↘ x23, 0x6A, 0x38, 0x23,
0x36, 0x24, 0x2A, 0x7C, 0x3A, 0x1A, 0x6, 0↙
    ↘ x0D, 0x0E, 0x0A, 0x14,
```

```
0x10],

[0x72, 0x7C, 0x72, 0x79, 0x76, 0x0,
0x50, 0x43, 0x4A, 0x59, 0x5D, 0x5B, 0x41, 0
    x41, 0x1B, 0x5A,
0x24, 0x32, 0x2E, 0x29, 0x28, 0x70, 0x20, 0
    x22, 0x38, 0x28, 0x36,
0x0D, 0x0B, 0x48, 0x4B, 0x4E]]


def is_string_printable(s):
    return all(list(map(lambda x: curses.
    ascii.isprint(x), s)))

cnt=1
for msg in msgs:
        print ("message #%d" % cnt)
        for key in range(0,256):
                result=[]
                tmp=key
                for i in msg:
                        result.append (i^tmp
    )
                        tmp=tmp+3
                if is_string_printable (
    result):
                        print ("key=", key,
    "value=", "".join(list(map(chr, result
    ))))
        cnt=cnt+1
```

:

Listing 78.4: Results

```
message #1
key= 20 value= `eb^h%|``hudw|_af{n~f%ljmSbnwlpk
key= 21 value= ajc]i"}cawtgv{^bgto}g"millcmvkqh
key= 22 value= bkd\j#rbbvsfuz!cduh|d#bhomdlujni
key= 23 value= check security device connection
key= 24 value= lifbl!pd|tqhsx#ejwjbb!`nQofbshlo
message #2
key= 7 value= No security device found
key= 8 value= An#rbbvsVuz!cduhld#ghtme?!#!'!#!
message #3
key= 7 value= Bk<waoqNUpu$`yreoa\wpmpusj,bkIjh
key= 8 value= Mj?vfnrOjqv%gxqd``_vwlstlk/clHii
key= 9 value= Lm>ugasLkvw&fgpgag^uvcrwml.`mwhj
key= 10 value= Ol!td`tMhwx'efwfbf!tubuvnm!anvok
key= 11 value= No security device station found
key= 12 value= In#rjbvsnuz!{duhdd#r{`whho#gPtme
message #4
key= 14 value= Number of authorized users exceeded
```

```
key= 15 value= Ovlmdq!hg#`juknuhydk!vrbsp!Zy
    `dbefe
message #5
key= 17 value= check security device station
key= 18 value= `ijbh!td`tmhwx'efwfbf!tubuVnm
    !'!
```

!

..

78.3 #3: MS-DOS

MS-DOS 1995 .

. .

.

. :

```
seg030:0034             out_port proc far ;
    CODE XREF: sent_pro+22p
seg030:0034                                ;
    sent_pro+2Ap ...
seg030:0034
seg030:0034             arg_0    = byte ptr
     6
seg030:0034
seg030:0034 55                   push    bp
seg030:0035 8B EC                mov     bp,
     sp
```

```
seg030:0037 8B 16 7E E7                 mov     dx,↙
    ↘  _out_port ; 0x378
seg030:003B 8A 46 06                    mov     al,↙
    ↘  [bp+arg_0]
seg030:003E EE                          out     dx,↙
    ↘  al
seg030:003F 5D                          pop     bp
seg030:0040 CB                          retf
seg030:0040             out_port endp
```

().

out_port() :

```
seg030:0041             sent_pro proc far ; ↙
    ↘ CODE XREF: check_dongle+34p
seg030:0041
seg030:0041             var_3    = byte ptr ↙
    ↘ -3
seg030:0041             var_2    = word ptr ↙
    ↘ -2
seg030:0041             arg_0    = dword ptr↙
    ↘   6
seg030:0041
seg030:0041 C8 04 00 00                 enter   4, ↙
    ↘ 0
seg030:0045 56                          push    si
seg030:0046 57                          push    di
seg030:0047 8B 16 82 E7                 mov     dx,↙
    ↘  _in_port_1 ; 0x37A
seg030:004B EC                          in      al,↙
    ↘  dx
seg030:004C 8A D8                       mov     bl,↙
```

```
    ↳  al
seg030:004E 80 E3 FE                    and     bl,↙
    ↳  0FEh
seg030:0051 80 CB 04                    or      bl,↙
    ↳  4
seg030:0054 8A C3                       mov     al,↙
    ↳  bl
seg030:0056 88 46 FD                    mov     [bp↙
    ↳ +var_3], al
seg030:0059 80 E3 1F                    and     bl,↙
    ↳  1Fh
seg030:005C 8A C3                       mov     al,↙
    ↳  bl
seg030:005E EE                          out     dx,↙
    ↳  al
seg030:005F 68 FF 00                    push    0↙
    ↳ FFh
seg030:0062 0E                          push    cs
seg030:0063 E8 CE FF                    call    ↙
    ↳ near ptr out_port
seg030:0066 59                          pop     cx
seg030:0067 68 D3 00                    push    0↙
    ↳ D3h
seg030:006A 0E                          push    cs
seg030:006B E8 C6 FF                    call    ↙
    ↳ near ptr out_port
seg030:006E 59                          pop     cx
seg030:006F 33 F6                       xor     si,↙
    ↳  si
seg030:0071 EB 01                       jmp     ↙
    ↳ short loc_359D4
seg030:0073
seg030:0073                  loc_359D3: ; CODE ↙
```

```
    ↳ XREF: sent_pro+37j
seg030:0073 46                          inc     si
seg030:0074
seg030:0074                 loc_359D4: ; CODE ↙
    ↳ XREF: sent_pro+30j
seg030:0074 81 FE 96 00                 cmp     si,↙
    ↳  96h
seg030:0078 7C F9                       jl      ↙
    ↳ short loc_359D3
seg030:007A 68 C3 00                    push    0↙
    ↳ C3h
seg030:007D 0E                          push    cs
seg030:007E E8 B3 FF                    call    ↙
    ↳ near ptr out_port
seg030:0081 59                          pop     cx
seg030:0082 68 C7 00                    push    0↙
    ↳ C7h
seg030:0085 0E                          push    cs
seg030:0086 E8 AB FF                    call    ↙
    ↳ near ptr out_port
seg030:0089 59                          pop     cx
seg030:008A 68 D3 00                    push    0↙
    ↳ D3h
seg030:008D 0E                          push    cs
seg030:008E E8 A3 FF                    call    ↙
    ↳ near ptr out_port
seg030:0091 59                          pop     cx
seg030:0092 68 C3 00                    push    0↙
    ↳ C3h
seg030:0095 0E                          push    cs
seg030:0096 E8 9B FF                    call    ↙
    ↳ near ptr out_port
seg030:0099 59                          pop     cx
```

```
seg030:009A 68 C7 00                    push    0↙
    ↘ C7h
seg030:009D 0E                          push    cs
seg030:009E E8 93 FF                    call    ↙
    ↘ near ptr out_port
seg030:00A1 59                          pop     cx
seg030:00A2 68 D3 00                    push    0↙
    ↘ D3h
seg030:00A5 0E                          push    cs
seg030:00A6 E8 8B FF                    call    ↙
    ↘ near ptr out_port
seg030:00A9 59                          pop     cx
seg030:00AA BF FF FF                    mov     di,↙
    ↘  0FFFFh
seg030:00AD EB 40                       jmp     ↙
    ↘ short loc_35A4F
seg030:00AF
seg030:00AF                   loc_35A0F: ; CODE ↙
    ↘ XREF: sent_pro+BDj
seg030:00AF BE 04 00                    mov     si,↙
    ↘  4
seg030:00B2
seg030:00B2                   loc_35A12: ; CODE ↙
    ↘ XREF: sent_pro+ACj
seg030:00B2 D1 E7                       shl     di,↙
    ↘  1
seg030:00B4 8B 16 80 E7                 mov     dx,↙
    ↘  _in_port_2 ; 0x379
seg030:00B8 EC                          in      al,↙
    ↘  dx
seg030:00B9 A8 80                       test    al,↙
    ↘  80h
seg030:00BB 75 03                       jnz     ↙
```

```
    ↳ short loc_35A20
seg030:00BD 83 CF 01                    or      di,↙
    ↳  1
seg030:00C0
seg030:00C0               loc_35A20: ; CODE ↙
    ↳ XREF: sent_pro+7Aj
seg030:00C0 F7 46 FE 08+                test    [bp↙
    ↳ +var_2], 8
seg030:00C5 74 05                       jz      ↙
    ↳ short loc_35A2C
seg030:00C7 68 D7 00                    push    0↙
    ↳ D7h ; '+'
seg030:00CA EB 0B                       jmp     ↙
    ↳ short loc_35A37
seg030:00CC
seg030:00CC               loc_35A2C: ; CODE ↙
    ↳ XREF: sent_pro+84j
seg030:00CC 68 C3 00                    push    0↙
    ↳ C3h
seg030:00CF 0E                          push    cs
seg030:00D0 E8 61 FF                    call    ↙
    ↳ near ptr out_port
seg030:00D3 59                          pop     cx
seg030:00D4 68 C7 00                    push    0↙
    ↳ C7h
seg030:00D7
seg030:00D7               loc_35A37: ; CODE ↙
    ↳ XREF: sent_pro+89j
seg030:00D7 0E                          push    cs
seg030:00D8 E8 59 FF                    call    ↙
    ↳ near ptr out_port
seg030:00DB 59                          pop     cx
seg030:00DC 68 D3 00                    push    0↙
```

```
    ↳ D3h
seg030:00DF 0E                        push    cs
seg030:00E0 E8 51 FF                  call    ↙
    ↳ near ptr out_port
seg030:00E3 59                        pop     cx
seg030:00E4 8B 46 FE                  mov     ax,↙
    ↳  [bp+var_2]
seg030:00E7 D1 E0                     shl     ax,↙
    ↳  1
seg030:00E9 89 46 FE                  mov     [bp↙
    ↳ +var_2], ax
seg030:00EC 4E                        dec     si
seg030:00ED 75 C3                     jnz     ↙
    ↳ short loc_35A12
seg030:00EF
seg030:00EF              loc_35A4F: ; CODE ↙
    ↳ XREF: sent_pro+6Cj
seg030:00EF C4 5E 06                  les     bx,↙
    ↳  [bp+arg_0]
seg030:00F2 FF 46 06                  inc     ↙
    ↳ word ptr [bp+arg_0]
seg030:00F5 26 8A 07                  mov     al,↙
    ↳  es:[bx]
seg030:00F8 98                        cbw
seg030:00F9 89 46 FE                  mov     [bp↙
    ↳ +var_2], ax
seg030:00FC 0B C0                     or      ax,↙
    ↳  ax
seg030:00FE 75 AF                     jnz     ↙
    ↳ short loc_35A0F
seg030:0100 68 FF 00                  push    0↙
    ↳ FFh
seg030:0103 0E                        push    cs
```

```
seg030:0104 E8 2D FF                    call     ↙
    ↘ near ptr out_port
seg030:0107 59                          pop      cx
seg030:0108 8B 16 82 E7                 mov      dx,↙
    ↘ _in_port_1 ; 0x37A
seg030:010C EC                          in       al,↙
    ↘ dx
seg030:010D 8A C8                       mov      cl,↙
    ↘ al
seg030:010F 80 E1 5F                    and      cl,↙
    ↘ 5Fh
seg030:0112 8A C1                       mov      al,↙
    ↘ cl
seg030:0114 EE                          out      dx,↙
    ↘ al
seg030:0115 EC                          in       al,↙
    ↘ dx
seg030:0116 8A C8                       mov      cl,↙
    ↘ al
seg030:0118 F6 C1 20                    test     cl,↙
    ↘ 20h
seg030:011B 74 08                       jz       ↙
    ↘ short loc_35A85
seg030:011D 8A 5E FD                    mov      bl,↙
    ↘ [bp+var_3]
seg030:0120 80 E3 DF                    and      bl,↙
    ↘ 0DFh
seg030:0123 EB 03                       jmp      ↙
    ↘ short loc_35A88
seg030:0125
seg030:0125                  loc_35A85: ; CODE ↙
    ↘ XREF: sent_pro+DAj
seg030:0125 8A 5E FD                    mov      bl,↙
```

```
    ↳  [bp+var_3]
seg030:0128
seg030:0128                loc_35A88: ; CODE ↙
    ↳ XREF: sent_pro+E2j
seg030:0128 F6 C1 80               test    cl,↙
    ↳  80h
seg030:012B 74 03                  jz      ↙
    ↳ short loc_35A90
seg030:012D 80 E3 7F               and     bl,↙
    ↳  7Fh
seg030:0130
seg030:0130                loc_35A90: ; CODE ↙
    ↳ XREF: sent_pro+EAj
seg030:0130 8B 16 82 E7            mov     dx,↙
    ↳  _in_port_1 ; 0x37A
seg030:0134 8A C3                  mov     al,↙
    ↳  bl
seg030:0136 EE                     out     dx,↙
    ↳  al
seg030:0137 8B C7                  mov     ax,↙
    ↳  di
seg030:0139 5F                     pop     di
seg030:013A 5E                     pop     si
seg030:013B C9                     leave
seg030:013C CB                     retf
seg030:013C                sent_pro endp
```

. .

. . [5]. . .

5.

`_in_port_2` (0x379) y `_in_port_1` (0x37A).

`seg030:00B9`: .

? .

:

```
00000000 struct_0        struc ; (sizeof=0x1B)
00000000 field_0         db 25 dup(?)            ; string(C)
00000019 _A              dw ?
0000001B struct_0        ends

dseg:3CBC 61 63 72 75+_Q   struct_0 <'hello', 01122h>
dseg:3CBC 6E 00 00 00+       ; DATA XREF: check_dongle+2Eo

... skipped ...

dseg:3E00 63 6F 66 66+     struct_0 <'coffee', 7EB7h>
dseg:3E1B 64 6F 67 00+     struct_0 <'dog', 0FFADh>
dseg:3E36 63 61 74 00+     struct_0 <'cat', 0FF5Fh>
dseg:3E51 70 61 70 65+     struct_0 <'paper', 0FFDFh>
dseg:3E6C 63 6F 6B 65+     struct_0 <'coke', 0F568h>
dseg:3E87 63 6C 6F 63+     struct_0 <'clock', 55EAh>
```

```
dseg:3EA2 64 69 72 00+      struct_0 <'dir', ↙
    ↘ 0FFAEh>
dseg:3EBD 63 6F 70 79+      struct_0 <'copy',↙
    ↘  0F557h>



seg030:0145                check_dongle proc ↙
    ↘ far ; CODE XREF: sub_3771D+3EP
seg030:0145
seg030:0145                var_6 = dword ptr -6
seg030:0145                var_2 = word ptr -2
seg030:0145
seg030:0145 C8 06 00 00          enter   6, 0
seg030:0149 56                   push    si
seg030:014A 66 6A 00             push    large ↙
    ↘ 0          ; newtime
seg030:014D 6A 00                push    0     ↙
    ↘            ; cmd
seg030:014F 9A C1 18 00+         call    ↙
    ↘ _biostime
seg030:0154 52                   push    dx
seg030:0155 50                   push    ax
seg030:0156 66 58                pop     eax
seg030:0158 83 C4 06             add     sp, 6
seg030:015B 66 89 46 FA          mov     [bp+↙
    ↘ var_6], eax
seg030:015F 66 3B 06 D8+         cmp     eax, ↙
    ↘ _expiration
seg030:0164 7E 44                jle     short ↙
    ↘ loc_35B0A
seg030:0166 6A 14                push    14h
seg030:0168 90                   nop
seg030:0169 0E                   push    cs
```

```
seg030:016A E8 52 00                call    near ↙
    ↘ ptr get_rand
seg030:016D 59                      pop     cx
seg030:016E 8B F0                   mov     si, ax
seg030:0170 6B C0 1B                imul    ax, 1↙
    ↘ Bh
seg030:0173 05 BC 3C                add     ax, ↙
    ↘ offset _Q
seg030:0176 1E                      push    ds
seg030:0177 50                      push    ax
seg030:0178 0E                      push    cs
seg030:0179 E8 C5 FE                call    near ↙
    ↘ ptr sent_pro
seg030:017C 83 C4 04                add     sp, 4
seg030:017F 89 46 FE                mov     [bp+↙
    ↘ var_2], ax
seg030:0182 8B C6                   mov     ax, si
seg030:0184 6B C0 12                imul    ax, 18
seg030:0187 66 0F BF C0             movsx   eax, ↙
    ↘ ax
seg030:018B 66 8B 56 FA             mov     edx, [↙
    ↘ bp+var_6]
seg030:018F 66 03 D0                add     edx, ↙
    ↘ eax
seg030:0192 66 89 16 D8+            mov     ↙
    ↘ _expiration, edx
seg030:0197 8B DE                   mov     bx, si
seg030:0199 6B DB 1B                imul    bx, 27
seg030:019C 8B 87 D5 3C             mov     ax, _Q↙
    ↘ ._A[bx]
seg030:01A0 3B 46 FE                cmp     ax, [↙
    ↘ bp+var_2]
seg030:01A3 74 05                   jz      short ↙
```

```
    ↳ loc_35B0A
seg030:01A5 B8 01 00                  mov     ax, 1
seg030:01A8 EB 02                     jmp     short ↙
    ↳ loc_35B0C
seg030:01AA
seg030:01AA                loc_35B0A: ; CODE ↙
    ↳ XREF: check_dongle+1Fj
seg030:01AA                           ; ↙
    ↳ check_dongle+5Ej
seg030:01AA 33 C0                     xor     ax, ax
seg030:01AC
seg030:01AC                loc_35B0C: ; CODE ↙
    ↳ XREF: check_dongle+63j
seg030:01AC 5E                        pop     si
seg030:01AD C9                        leave
seg030:01AE CB                        retf
seg030:01AE                check_dongle endp
```

.

get_rand() :

```
seg030:01BF                get_rand proc far ; ↙
    ↳ CODE XREF: check_dongle+25p
seg030:01BF
seg030:01BF                arg_0    = word ptr ↙
    ↳  6
seg030:01BF
seg030:01BF 55                           push    bp
seg030:01C0 8B EC                        mov     bp,↙
    ↳  sp
seg030:01C2 9A 3D 21 00+                 call    ↙
    ↳ _rand
```

```
seg030:01C7 66 0F BF C0                 movsx   eax⤦
    ↳ , ax
seg030:01CB 66 0F BF 56+                movsx   edx⤦
    ↳ , [bp+arg_0]
seg030:01D0 66 0F AF C2                 imul    eax⤦
    ↳ , edx
seg030:01D4 66 BB 00 80+                mov     ebx⤦
    ↳ , 8000h
seg030:01DA 66 99                       cdq
seg030:01DC 66 F7 FB                    idiv    ebx
seg030:01DF 5D                          pop     bp
seg030:01E0 CB                          retf
seg030:01E0             get_rand endp
```

.

.

:

```
seg033:087B 9A 45 01 96+   call     ⤦
    ↳ check_dongle
seg033:0880 0B C0          or       ax, ax
seg033:0882 74 62          jz       short OK
seg033:0884 83 3E 60 42+   cmp      ⤦
    ↳ word_620E0, 0
seg033:0889 75 5B          jnz      short OK
seg033:088B FF 06 60 42    inc      ⤦
    ↳ word_620E0
seg033:088F 1E             push     ds
seg033:0890 68 22 44       push     offset ⤦
    ↳ aTrupcRequiresA ; "This Software ⤦
    ↳ Requires a Software Lock\n"
```

```
seg033:0893 1E               push    ds
seg033:0894 68 60 E9         push    offset ⤦
    ↳ byte_6C7E0 ; dest
seg033:0897 9A 79 65 00+     call    _strcpy
seg033:089C 83 C4 08         add     sp, 8
seg033:089F 1E               push    ds
seg033:08A0 68 42 44         push    offset ⤦
    ↳ aPleaseContactA ; "Please Contact ..."
seg033:08A3 1E               push    ds
seg033:08A4 68 60 E9         push    offset ⤦
    ↳ byte_6C7E0 ; dest
seg033:08A7 9A CD 64 00+     call    _strcat
```

.

:

```
mov ax,0
retf
```

:

```
seg033:088F 1E                                  push⤦
    ↳     ds
seg033:0890 68 22 44                            push⤦
    ↳     offset aTrupcRequiresA ; "This ⤦
    ↳ Software Requires a Software Lock\n"
seg033:0893 1E                                  push⤦
    ↳     ds
seg033:0894 68 60 E9                            push⤦
    ↳     offset byte_6C7E0 ; dest
seg033:0897 9A 79 65 00+                        call⤦
    ↳     _strcpy
```

```
seg033:089C 83 C4 08                            add ↙
    ↘      sp, 8
```

: 94 on page 1776.

.

. DS

.

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:

```
.text:00541000 set_bit              proc near    ↙
    ↘               ; CODE XREF: rotate1+42
.text:00541000                                   ↙
    ↘               ; rotate2+42 ...
.text:00541000
.text:00541000 arg_0                = dword ptr  ↙
    ↘ 4
.text:00541000 arg_4                = dword ptr  ↙
    ↘ 8
.text:00541000 arg_8                = dword ptr  ↙
    ↘ 0Ch
.text:00541000 arg_C                = byte ptr  ↙
    ↘ 10h
.text:00541000
.text:00541000                      mov     al, [↙
    ↘ esp+arg_C]
.text:00541004                      mov     ecx, ↙
```

```
    ↳ [esp+arg_8]
.text:00541008                        push    esi
.text:00541009                        mov     esi, ↙
    ↳ [esp+4+arg_0]
.text:0054100D                        test    al, ↙
    ↳ al
.text:0054100F                        mov     eax, ↙
    ↳ [esp+4+arg_4]
.text:00541013                        mov     dl, 1
.text:00541015                        jz      short↙
    ↳  loc_54102B
.text:00541017                        shl     dl, ↙
    ↳ cl
.text:00541019                        mov     cl, ↙
    ↳ cube64[eax+esi*8]
.text:00541020                        or      cl, ↙
    ↳ dl
.text:00541022                        mov     ↙
    ↳ cube64[eax+esi*8], cl
.text:00541029                        pop     esi
.text:0054102A                        retn
.text:0054102B
.text:0054102B loc_54102B:                            ↙
    ↳             ; CODE XREF: set_bit+15
.text:0054102B                        shl     dl, ↙
    ↳ cl
.text:0054102D                        mov     cl, ↙
    ↳ cube64[eax+esi*8]
.text:00541034                        not     dl
.text:00541036                        and     cl, ↙
    ↳ dl
.text:00541038                        mov     ↙
    ↳ cube64[eax+esi*8], cl
```

```
.text:0054103F                 pop     esi
.text:00541040                 retn
.text:00541040 set_bit         endp
.text:00541040
.text:00541041                 align 10h
.text:00541050
.text:00541050 ; =============== S U B R O U ↙
    ↘ T I N E ↙
    ↘ =======================================↙
    ↘
.text:00541050
.text:00541050
.text:00541050 get_bit         proc near     ↙
    ↘          ; CODE XREF: rotate1+16
.text:00541050                               ↙
    ↘          ; rotate2+16 ...
.text:00541050
.text:00541050 arg_0           = dword ptr   ↙
    ↘ 4
.text:00541050 arg_4           = dword ptr   ↙
    ↘ 8
.text:00541050 arg_8           = byte ptr  0↙
    ↘ Ch
.text:00541050
.text:00541050                 mov     eax, ↙
    ↘ [esp+arg_4]
.text:00541054                 mov     ecx, ↙
    ↘ [esp+arg_0]
.text:00541058                 mov     al, ↙
    ↘ cube64[eax+ecx*8]
.text:0054105F                 mov     cl, [↙
    ↘ esp+arg_8]
.text:00541063                 shr     al, ↙
```

```
    ↘ cl
.text:00541065                 and     al, 1
.text:00541067                 retn
.text:00541067 get_bit         endp
.text:00541067
.text:00541068                 align 10h
.text:00541070
.text:00541070 ; =============== S U B R O U↙
    ↘  T I N E ↙
    ↘ =======================================↙
    ↘
.text:00541070
.text:00541070
.text:00541070 rotate1         proc near     ↙
    ↘          ; CODE XREF: ↙
    ↘ rotate_all_with_password+8E
.text:00541070
.text:00541070 internal_array_64= byte ptr ↙
    ↘ -40h
.text:00541070 arg_0           = dword ptr  ↙
    ↘ 4
.text:00541070
.text:00541070                 sub     esp, ↙
    ↘ 40h
.text:00541073                 push    ebx
.text:00541074                 push    ebp
.text:00541075                 mov     ebp, ↙
    ↘ [esp+48h+arg_0]
.text:00541079                 push    esi
.text:0054107A                 push    edi
.text:0054107B                 xor     edi, ↙
    ↘ edi          ; EDI is loop1 counter
.text:0054107D                 lea     ebx, ↙
```

```
    ↳ [esp+50h+internal_array_64]
.text:00541081
.text:00541081 first_loop1_begin:                    ↙
    ↳           ; CODE XREF: rotate1+2E
.text:00541081                 xor     esi, ↙
    ↳ esi       ; ESI is loop2 counter
.text:00541083
.text:00541083 first_loop2_begin:                    ↙
    ↳           ; CODE XREF: rotate1+25
.text:00541083                 push    ebp  ↙
    ↳           ; arg_0
.text:00541084                 push    esi
.text:00541085                 push    edi
.text:00541086                 call    ↙
    ↳ get_bit
.text:0054108B                 add     esp, ↙
    ↳ 0Ch
.text:0054108E                 mov     [ebx+↙
    ↳ esi], al   ; store to internal array
.text:00541091                 inc     esi
.text:00541092                 cmp     esi, ↙
    ↳ 8
.text:00541095                 jl      short↙
    ↳  first_loop2_begin
.text:00541097                 inc     edi
.text:00541098                 add     ebx, ↙
    ↳ 8
.text:0054109B                 cmp     edi, ↙
    ↳ 8
.text:0054109E                 jl      short↙
    ↳  first_loop1_begin
.text:005410A0                 lea     ebx, ↙
    ↳ [esp+50h+internal_array_64]
```

```
.text:005410A4                 mov     edi, ⤦
    ↳ 7           ; EDI is loop1 counter, ⤦
    ↳ initial state is 7
.text:005410A9
.text:005410A9 second_loop1_begin:             ⤦
    ↳             ; CODE XREF: rotate1+57
.text:005410A9                 xor     esi, ⤦
    ↳ esi         ; ESI is loop2 counter
.text:005410AB
.text:005410AB second_loop2_begin:             ⤦
    ↳             ; CODE XREF: rotate1+4E
.text:005410AB                 mov     al, [⤦
    ↳ ebx+esi]    ; value from internal array
.text:005410AE                 push    eax
.text:005410AF                 push    ebp  ⤦
    ↳             ; arg_0
.text:005410B0                 push    edi
.text:005410B1                 push    esi
.text:005410B2                 call    ⤦
    ↳ set_bit
.text:005410B7                 add     esp, ⤦
    ↳ 10h
.text:005410BA                 inc     esi  ⤦
    ↳             ; increment loop2 counter
.text:005410BB                 cmp     esi, ⤦
    ↳ 8
.text:005410BE                 jl      short⤦
    ↳  second_loop2_begin
.text:005410C0                 dec     edi  ⤦
    ↳             ; decrement loop2 counter
.text:005410C1                 add     ebx, ⤦
    ↳ 8
.text:005410C4                 cmp     edi, ⤦
```

```
    ↳ 0FFFFFFFFh
.text:005410C7                 jg      short↙
    ↳  second_loop1_begin
.text:005410C9                 pop     edi
.text:005410CA                 pop     esi
.text:005410CB                 pop     ebp
.text:005410CC                 pop     ebx
.text:005410CD                 add     esp, ↙
    ↳ 40h
.text:005410D0                 retn
.text:005410D0 rotate1         endp
.text:005410D0
.text:005410D1                 align 10h
.text:005410E0
.text:005410E0 ; =============== S U B R O U↙
    ↳  T I N E ↙
    ↳ =======================================↙
    ↳
.text:005410E0
.text:005410E0
.text:005410E0 rotate2         proc near      ↙
    ↳          ; CODE XREF: ↙
    ↳ rotate_all_with_password+7A
.text:005410E0
.text:005410E0 internal_array_64= byte ptr ↙
    ↳ -40h
.text:005410E0 arg_0           = dword ptr  ↙
    ↳ 4
.text:005410E0
.text:005410E0                 sub     esp, ↙
    ↳ 40h
.text:005410E3                 push    ebx
.text:005410E4                 push    ebp
```

```
.text:005410E5                 mov     ebp, ↙
    ↘ [esp+48h+arg_0]
.text:005410E9                 push    esi
.text:005410EA                 push    edi
.text:005410EB                 xor     edi, ↙
    ↘ edi          ; loop1 counter
.text:005410ED                 lea     ebx, ↙
    ↘ [esp+50h+internal_array_64]
.text:005410F1
.text:005410F1 loc_5410F1:                          ↙
    ↘              ; CODE XREF: rotate2+2E
.text:005410F1                 xor     esi, ↙
    ↘ esi          ; loop2 counter
.text:005410F3
.text:005410F3 loc_5410F3:                          ↙
    ↘              ; CODE XREF: rotate2+25
.text:005410F3                 push    esi  ↙
    ↘              ; loop2
.text:005410F4                 push    edi  ↙
    ↘              ; loop1
.text:005410F5                 push    ebp  ↙
    ↘              ; arg_0
.text:005410F6                 call        ↙
    ↘ get_bit
.text:005410FB                 add     esp, ↙
    ↘ 0Ch
.text:005410FE                 mov     [ebx+↙
    ↘ esi], al    ; store to internal array
.text:00541101                 inc     esi  ↙
    ↘              ; increment loop1 counter
.text:00541102                 cmp     esi, ↙
    ↘ 8
.text:00541105                 jl      short↙
```

```
    ↳ loc_5410F3
.text:00541107                 inc     edi   ↙
    ↳          ; increment loop2 counter
.text:00541108                 add     ebx, ↙
    ↳ 8
.text:0054110B                 cmp     edi, ↙
    ↳ 8
.text:0054110E                 jl      short↙
    ↳ loc_5410F1
.text:00541110                 lea     ebx, ↙
    ↳ [esp+50h+internal_array_64]
.text:00541114                 mov     edi, ↙
    ↳ 7        ; loop1 counter is initial ↙
    ↳ state 7
.text:00541119
.text:00541119 loc_541119:                  ↙
    ↳          ; CODE XREF: rotate2+57
.text:00541119                 xor     esi, ↙
    ↳ esi      ; loop2 counter
.text:0054111B
.text:0054111B loc_54111B:                  ↙
    ↳          ; CODE XREF: rotate2+4E
.text:0054111B                 mov     al, [↙
    ↳ ebx+esi] ; get byte from internal ↙
    ↳ array
.text:0054111E                 push    eax
.text:0054111F                 push    edi   ↙
    ↳          ; loop1 counter
.text:00541120                 push    esi   ↙
    ↳          ; loop2 counter
.text:00541121                 push    ebp   ↙
    ↳          ; arg_0
.text:00541122                 call    ↙
```

```
    ↘ set_bit
.text:00541127                  add     esp, ↙
    ↘ 10h
.text:0054112A                  inc     esi  ↙
    ↘           ; increment loop2 counter
.text:0054112B                  cmp     esi, ↙
    ↘ 8
.text:0054112E                  jl      short↙
    ↘  loc_54111B
.text:00541130                  dec     edi  ↙
    ↘           ; decrement loop2 counter
.text:00541131                  add     ebx, ↙
    ↘ 8
.text:00541134                  cmp     edi, ↙
    ↘ 0FFFFFFFFh
.text:00541137                  jg      short↙
    ↘  loc_541119
.text:00541139                  pop     edi
.text:0054113A                  pop     esi
.text:0054113B                  pop     ebp
.text:0054113C                  pop     ebx
.text:0054113D                  add     esp, ↙
    ↘ 40h
.text:00541140                  retn
.text:00541140 rotate2          endp
.text:00541140
.text:00541141                  align 10h
.text:00541150
.text:00541150 ; =============== S U B R O U↙
    ↘  T I N E ↙
    ↘ =======================================↙
    ↘
.text:00541150
```

```
.text:00541150
.text:00541150 rotate3                 proc near
    ↳              ; CODE XREF: 
    ↳ rotate_all_with_password+66
.text:00541150
.text:00541150 var_40                  = byte ptr 
    ↳ -40h
.text:00541150 arg_0                   = dword ptr 
    ↳ 4
.text:00541150
.text:00541150                         sub     esp, 
    ↳ 40h
.text:00541153                         push    ebx
.text:00541154                         push    ebp
.text:00541155                         mov     ebp, 
    ↳ [esp+48h+arg_0]
.text:00541159                         push    esi
.text:0054115A                         push    edi
.text:0054115B                         xor     edi, 
    ↳ edi
.text:0054115D                         lea     ebx, 
    ↳ [esp+50h+var_40]
.text:00541161
.text:00541161 loc_541161:                             
    ↳              ; CODE XREF: rotate3+2E
.text:00541161                         xor     esi, 
    ↳ esi
.text:00541163
.text:00541163 loc_541163:                             
    ↳              ; CODE XREF: rotate3+25
.text:00541163                         push    esi
.text:00541164                         push    ebp
.text:00541165                         push    edi
```

```
.text:00541166                 call    ⤦
    ↳ get_bit
.text:0054116B                 add     esp, ⤦
    ↳ 0Ch
.text:0054116E                 mov     [ebx+⤦
    ↳ esi], al
.text:00541171                 inc     esi
.text:00541172                 cmp     esi, ⤦
    ↳ 8
.text:00541175                 jl      short⤦
    ↳  loc_541163
.text:00541177                 inc     edi
.text:00541178                 add     ebx, ⤦
    ↳ 8
.text:0054117B                 cmp     edi, ⤦
    ↳ 8
.text:0054117E                 jl      short⤦
    ↳  loc_541161
.text:00541180                 xor     ebx, ⤦
    ↳ ebx
.text:00541182                 lea     edi, ⤦
    ↳ [esp+50h+var_40]
.text:00541186
.text:00541186 loc_541186:                             ⤦
    ↳            ; CODE XREF: rotate3+54
.text:00541186                 mov     esi, ⤦
    ↳ 7
.text:0054118B
.text:0054118B loc_54118B:                             ⤦
    ↳            ; CODE XREF: rotate3+4E
.text:0054118B                 mov     al, [⤦
    ↳ edi]
.text:0054118D                 push    eax
```

```
.text:0054118E                 push    ebx
.text:0054118F                 push    ebp
.text:00541190                 push    esi
.text:00541191                 call    ⤦
    ↳ set_bit
.text:00541196                 add     esp, ⤦
    ↳ 10h
.text:00541199                 inc     edi
.text:0054119A                 dec     esi
.text:0054119B                 cmp     esi, ⤦
    ↳ 0FFFFFFFFh
.text:0054119E                 jg      short⤦
    ↳  loc_54118B
.text:005411A0                 inc     ebx
.text:005411A1                 cmp     ebx, ⤦
    ↳ 8
.text:005411A4                 jl      short⤦
    ↳  loc_541186
.text:005411A6                 pop     edi
.text:005411A7                 pop     esi
.text:005411A8                 pop     ebp
.text:005411A9                 pop     ebx
.text:005411AA                 add     esp, ⤦
    ↳ 40h
.text:005411AD                 retn
.text:005411AD rotate3         endp
.text:005411AD
.text:005411AE                 align 10h
.text:005411B0
.text:005411B0 ; =============== S U B R O U⤦
    ↳  T I N E ⤦
    ↳ =======================================⤦
    ↳
```

```
.text:005411B0
.text:005411B0
.text:005411B0 rotate_all_with_password proc
    ↳ near        ; CODE XREF: crypt+1F
.text:005411B0
    ↳             ; decrypt+36
.text:005411B0
.text:005411B0 arg_0              = dword ptr
    ↳ 4
.text:005411B0 arg_4              = dword ptr
    ↳ 8
.text:005411B0
.text:005411B0                    mov     eax,
    ↳ [esp+arg_0]
.text:005411B4                    push    ebp
.text:005411B5                    mov     ebp,
    ↳ eax
.text:005411B7                    cmp     byte
    ↳ ptr [eax], 0
.text:005411BA                    jz      exit
.text:005411C0                    push    ebx
.text:005411C1                    mov     ebx,
    ↳ [esp+8+arg_4]
.text:005411C5                    push    esi
.text:005411C6                    push    edi
.text:005411C7
.text:005411C7 loop_begin:
    ↳             ; CODE XREF:
    ↳ rotate_all_with_password+9F
.text:005411C7                    movsx   eax,
    ↳ byte ptr [ebp+0]
.text:005411CB                    push    eax
    ↳             ; C
```

```
.text:005411CC                 call    ⤦
    ↳ _tolower
.text:005411D1                 add     esp, ⤦
    ↳ 4
.text:005411D4                 cmp     al, '⤦
    ↳ a'
.text:005411D6                 jl      short⤦
    ↳  next_character_in_password
.text:005411D8                 cmp     al, '⤦
    ↳ z'
.text:005411DA                 jg      short⤦
    ↳  next_character_in_password
.text:005411DC                 movsx   ecx, ⤦
    ↳ al
.text:005411DF                 sub     ecx, ⤦
    ↳ 'a'
.text:005411E2                 cmp     ecx, ⤦
    ↳ 24
.text:005411E5                 jle     short⤦
    ↳  skip_subtracting
.text:005411E7                 sub     ecx, ⤦
    ↳ 24
.text:005411EA
.text:005411EA skip_subtracting:               ⤦
    ↳           ; CODE XREF: ⤦
    ↳ rotate_all_with_password+35
.text:005411EA                 mov     eax, ⤦
    ↳ 55555556h
.text:005411EF                 imul    ecx
.text:005411F1                 mov     eax, ⤦
    ↳ edx
.text:005411F3                 shr     eax, ⤦
    ↳ 1Fh
```

```
.text:005411F6                  add     edx, ↙
    ↘ eax
.text:005411F8                  mov     eax, ↙
    ↘ ecx
.text:005411FA                  mov     esi, ↙
    ↘ edx
.text:005411FC                  mov     ecx, ↙
    ↘ 3
.text:00541201                  cdq
.text:00541202                  idiv    ecx
.text:00541204                  sub     edx, ↙
    ↘ 0
.text:00541207                  jz      short↙
    ↘  call_rotate1
.text:00541209                  dec     edx
.text:0054120A                  jz      short↙
    ↘  call_rotate2
.text:0054120C                  dec     edx
.text:0054120D                  jnz     short↙
    ↘  next_character_in_password
.text:0054120F                  test    ebx, ↙
    ↘ ebx
.text:00541211                  jle     short↙
    ↘  next_character_in_password
.text:00541213                  mov     edi, ↙
    ↘ ebx
.text:00541215
.text:00541215 call_rotate3:                    ↙
    ↘            ; CODE XREF: ↙
    ↘ rotate_all_with_password+6F
.text:00541215                  push    esi
.text:00541216                  call    ↙
    ↘ rotate3
```

```
.text:0054121B                  add     esp, ↙
    ↳ 4
.text:0054121E                  dec     edi
.text:0054121F                  jnz     short↙
    ↳  call_rotate3
.text:00541221                  jmp     short↙
    ↳  next_character_in_password
.text:00541223
.text:00541223 call_rotate2:                 ↙
    ↳           ; CODE XREF: ↙
    ↳ rotate_all_with_password+5A
.text:00541223                  test    ebx, ↙
    ↳ ebx
.text:00541225                  jle     short↙
    ↳  next_character_in_password
.text:00541227                  mov     edi, ↙
    ↳ ebx
.text:00541229
.text:00541229 loc_541229:                   ↙
    ↳           ; CODE XREF: ↙
    ↳ rotate_all_with_password+83
.text:00541229                  push    esi
.text:0054122A                  call    ↙
    ↳ rotate2
.text:0054122F                  add     esp, ↙
    ↳ 4
.text:00541232                  dec     edi
.text:00541233                  jnz     short↙
    ↳  loc_541229
.text:00541235                  jmp     short↙
    ↳  next_character_in_password
.text:00541237
.text:00541237 call_rotate1:                 ↙
```

```
    ↘             ; CODE XREF: ↙
    ↘ rotate_all_with_password+57
.text:00541237                 test    ebx, ↙
    ↘ ebx
.text:00541239                 jle     short↙
    ↘  next_character_in_password
.text:0054123B                 mov     edi, ↙
    ↘ ebx
.text:0054123D
.text:0054123D loc_54123D:                  ↙
    ↘             ; CODE XREF: ↙
    ↘ rotate_all_with_password+97
.text:0054123D                 push    esi
.text:0054123E                 call    ↙
    ↘ rotate1
.text:00541243                 add     esp, ↙
    ↘ 4
.text:00541246                 dec     edi
.text:00541247                 jnz     short↙
    ↘  loc_54123D
.text:00541249
.text:00541249 next_character_in_password:  ↙
    ↘             ; CODE XREF: ↙
    ↘ rotate_all_with_password+26
.text:00541249                              ↙
    ↘             ; rotate_all_with_password↙
    ↘ +2A ...
.text:00541249                 mov     al, [↙
    ↘ ebp+1]
.text:0054124C                 inc     ebp
.text:0054124D                 test    al, ↙
    ↘ al
.text:0054124F                 jnz     ↙
```

```
    ↳ loop_begin
.text:00541255                 pop     edi
.text:00541256                 pop     esi
.text:00541257                 pop     ebx
.text:00541258
.text:00541258 exit:                                   ↙
    ↳           ; CODE XREF: ↙
    ↳ rotate_all_with_password+A
.text:00541258                 pop     ebp
.text:00541259                 retn
.text:00541259 rotate_all_with_password endp
.text:00541259
.text:0054125A                 align 10h
.text:00541260
.text:00541260 ; =============== S U B R O U↙
    ↳  T I N E ↙
    ↳ =======================================↙
    ↳
.text:00541260
.text:00541260
.text:00541260 crypt           proc near      ↙
    ↳           ; CODE XREF: crypt_file+8A
.text:00541260
.text:00541260 arg_0           = dword ptr  ↙
    ↳ 4
.text:00541260 arg_4           = dword ptr  ↙
    ↳ 8
.text:00541260 arg_8           = dword ptr  ↙
    ↳ 0Ch
.text:00541260
.text:00541260                 push    ebx
.text:00541261                 mov     ebx, ↙
    ↳ [esp+4+arg_0]
```

```
.text:00541265                 push    ebp
.text:00541266                 push    esi
.text:00541267                 push    edi
.text:00541268                 xor     ebp, ⤦
    ↘ ebp
.text:0054126A
.text:0054126A loc_54126A:                             ⤦
    ↘           ; CODE XREF: crypt+41
.text:0054126A                 mov     eax, ⤦
    ↘ [esp+10h+arg_8]
.text:0054126E                 mov     ecx, ⤦
    ↘ 10h
.text:00541273                 mov     esi, ⤦
    ↘ ebx
.text:00541275                 mov     edi, ⤦
    ↘ offset cube64
.text:0054127A                 push    1
.text:0054127C                 push    eax
.text:0054127D                 rep movsd
.text:0054127F                 call    ⤦
    ↘ rotate_all_with_password
.text:00541284                 mov     eax, ⤦
    ↘ [esp+18h+arg_4]
.text:00541288                 mov     edi, ⤦
    ↘ ebx
.text:0054128A                 add     ebp, ⤦
    ↘ 40h
.text:0054128D                 add     esp, ⤦
    ↘ 8
.text:00541290                 mov     ecx, ⤦
    ↘ 10h
.text:00541295                 mov     esi, ⤦
    ↘ offset cube64
```

```
.text:0054129A                 add     ebx, ↙
    ↘ 40h
.text:0054129D                 cmp     ebp, ↙
    ↘ eax
.text:0054129F                 rep movsd
.text:005412A1                 jl      short↙
    ↘  loc_54126A
.text:005412A3                 pop     edi
.text:005412A4                 pop     esi
.text:005412A5                 pop     ebp
.text:005412A6                 pop     ebx
.text:005412A7                 retn
.text:005412A7 crypt           endp
.text:005412A7
.text:005412A8                 align 10h
.text:005412B0
.text:005412B0 ; =============== S U B R O U↙
    ↘  T I N E ↙
    ↘ =======================================↙
    ↘
.text:005412B0
.text:005412B0
.text:005412B0 ; int __cdecl decrypt(int, ↙
    ↘ int, void *Src)
.text:005412B0 decrypt         proc near     ↙
    ↘           ; CODE XREF: decrypt_file↙
    ↘ +99
.text:005412B0
.text:005412B0 arg_0           = dword ptr  ↙
    ↘ 4
.text:005412B0 arg_4           = dword ptr  ↙
    ↘ 8
.text:005412B0 Src             = dword ptr  ↙
```

```
    ↳ 0Ch
.text:005412B0
.text:005412B0                     mov     eax, ⤦
    ↳ [esp+Src]
.text:005412B4                     push    ebx
.text:005412B5                     push    ebp
.text:005412B6                     push    esi
.text:005412B7                     push    edi
.text:005412B8                     push    eax  ⤦
    ↳             ; Src
.text:005412B9                     call    ⤦
    ↳ __strdup
.text:005412BE                     push    eax  ⤦
    ↳             ; Str
.text:005412BF                     mov     [esp⤦
    ↳ +18h+Src], eax
.text:005412C3                     call    ⤦
    ↳ __strrev
.text:005412C8                     mov     ebx, ⤦
    ↳ [esp+18h+arg_0]
.text:005412CC                     add     esp, ⤦
    ↳ 8
.text:005412CF                     xor     ebp, ⤦
    ↳ ebp
.text:005412D1
.text:005412D1 loc_5412D1:                      ⤦
    ↳             ; CODE XREF: decrypt+58
.text:005412D1                     mov     ecx, ⤦
    ↳ 10h
.text:005412D6                     mov     esi, ⤦
    ↳ ebx
.text:005412D8                     mov     edi, ⤦
    ↳ offset cube64
```

```
.text:005412DD                  push    3
.text:005412DF                  rep movsd
.text:005412E1                  mov     ecx, ↙
    ↘ [esp+14h+Src]
.text:005412E5                  push    ecx
.text:005412E6                  call    ↙
    ↘ rotate_all_with_password
.text:005412EB                  mov     eax, ↙
    ↘ [esp+18h+arg_4]
.text:005412EF                  mov     edi, ↙
    ↘ ebx
.text:005412F1                  add     ebp, ↙
    ↘ 40h
.text:005412F4                  add     esp, ↙
    ↘ 8
.text:005412F7                  mov     ecx, ↙
    ↘ 10h
.text:005412FC                  mov     esi, ↙
    ↘ offset cube64
.text:00541301                  add     ebx, ↙
    ↘ 40h
.text:00541304                  cmp     ebp, ↙
    ↘ eax
.text:00541306                  rep movsd
.text:00541308                  jl      short↙
    ↘  loc_5412D1
.text:0054130A                  mov     edx, ↙
    ↘ [esp+10h+Src]
.text:0054130E                  push    edx  ↙
    ↘           ; Memory
.text:0054130F                  call    _free
.text:00541314                  add     esp, ↙
    ↘ 4
```

```
.text:00541317                 pop     edi
.text:00541318                 pop     esi
.text:00541319                 pop     ebp
.text:0054131A                 pop     ebx
.text:0054131B                 retn
.text:0054131B decrypt         endp
.text:0054131B
.text:0054131C                 align 10h
.text:00541320
.text:00541320 ; =============== S U B R O U ↙
    ↘ T I N E ↙
    ↘ =======================================↙
    ↘
.text:00541320
.text:00541320
.text:00541320 ; int __cdecl crypt_file(int ↙
    ↘ Str, char *Filename, int password)
.text:00541320 crypt_file      proc near     ↙
    ↘          ; CODE XREF: _main+42
.text:00541320
.text:00541320 Str             = dword ptr  ↙
    ↘ 4
.text:00541320 Filename        = dword ptr  ↙
    ↘ 8
.text:00541320 password        = dword ptr  ↙
    ↘ 0Ch
.text:00541320
.text:00541320                 mov     eax, ↙
    ↘ [esp+Str]
.text:00541324                 push    ebp
.text:00541325                 push    ↙
    ↘ offset Mode     ; "rb"
.text:0054132A                 push    eax  ↙
```

```
    ↳               ; Filename
.text:0054132B                  call    ↙
    ↳ _fopen            ; open file
.text:00541330                  mov     ebp, ↙
    ↳ eax
.text:00541332                  add     esp, ↙
    ↳ 8
.text:00541335                  test    ebp, ↙
    ↳ ebp
.text:00541337                  jnz     short↙
    ↳  loc_541348
.text:00541339                  push    ↙
    ↳ offset Format   ; "Cannot open input ↙
    ↳ file!\n"
.text:0054133E                  call    ↙
    ↳ _printf
.text:00541343                  add     esp, ↙
    ↳ 4
.text:00541346                  pop     ebp
.text:00541347                  retn
.text:00541348
.text:00541348 loc_541348:                      ↙
    ↳               ; CODE XREF: crypt_file+17
.text:00541348                  push    ebx
.text:00541349                  push    esi
.text:0054134A                  push    edi
.text:0054134B                  push    2       ↙
    ↳               ; Origin
.text:0054134D                  push    0       ↙
    ↳               ; Offset
.text:0054134F                  push    ebp     ↙
    ↳               ; File
.text:00541350                  call    ↙
```

```
    ↘ _fseek
.text:00541355                  push    ebp  ↙
    ↘           ; File
.text:00541356                  call    ↙
    ↘ _ftell          ; get file size
.text:0054135B                  push    0    ↙
    ↘           ; Origin
.text:0054135D                  push    0    ↙
    ↘           ; Offset
.text:0054135F                  push    ebp  ↙
    ↘           ; File
.text:00541360                  mov     [esp↙
    ↘ +2Ch+Str], eax
.text:00541364                  call    ↙
    ↘ _fseek          ; rewind to start
.text:00541369                  mov     esi, ↙
    ↘ [esp+2Ch+Str]
.text:0054136D                  and     esi, ↙
    ↘ 0FFFFFFC0h ; reset all lowest 6 bits
.text:00541370                  add     esi, ↙
    ↘ 40h       ; align size to 64-byte ↙
    ↘ border
.text:00541373                  push    esi  ↙
    ↘           ; Size
.text:00541374                  call    ↙
    ↘ _malloc
.text:00541379                  mov     ecx, ↙
    ↘ esi
.text:0054137B                  mov     ebx, ↙
    ↘ eax       ; allocated buffer pointer ↙
    ↘ -> to EBX
.text:0054137D                  mov     edx, ↙
    ↘ ecx
```

```
.text:0054137F                 xor     eax, ↙
    ↘ eax
.text:00541381                 mov     edi, ↙
    ↘ ebx
.text:00541383                 push    ebp  ↙
    ↘          ; File
.text:00541384                 shr     ecx, ↙
    ↘ 2
.text:00541387                 rep stosd
.text:00541389                 mov     ecx, ↙
    ↘ edx
.text:0054138B                 push    1    ↙
    ↘          ; Count
.text:0054138D                 and     ecx, ↙
    ↘ 3
.text:00541390                 rep stosb    ↙
    ↘          ; memset (buffer, 0, ↙
    ↘ aligned_size)
.text:00541392                 mov     eax, ↙
    ↘ [esp+38h+Str]
.text:00541396                 push    eax  ↙
    ↘          ; ElementSize
.text:00541397                 push    ebx  ↙
    ↘          ; DstBuf
.text:00541398                 call    ↙
    ↘ _fread           ; read file
.text:0054139D                 push    ebp  ↙
    ↘          ; File
.text:0054139E                 call    ↙
    ↘ _fclose
.text:005413A3                 mov     ecx, ↙
    ↘ [esp+44h+password]
.text:005413A7                 push    ecx  ↙
```

```
    ↘               ; password
.text:005413A8                         push    esi  ↙
    ↘               ; aligned size
.text:005413A9                         push    ebx  ↙
    ↘               ; buffer
.text:005413AA                         call    crypt↙
    ↘               ; do crypt
.text:005413AF                         mov     edx, ↙
    ↘ [esp+50h+Filename]
.text:005413B3                         add     esp, ↙
    ↘ 40h
.text:005413B6                         push    ↙
    ↘ offset aWb        ; "wb"
.text:005413BB                         push    edx  ↙
    ↘               ; Filename
.text:005413BC                         call    ↙
    ↘ _fopen
.text:005413C1                         mov     edi, ↙
    ↘ eax
.text:005413C3                         push    edi  ↙
    ↘               ; File
.text:005413C4                         push    1    ↙
    ↘               ; Count
.text:005413C6                         push    3    ↙
    ↘               ; Size
.text:005413C8                         push    ↙
    ↘ offset aQr9       ; "QR9"
.text:005413CD                         call    ↙
    ↘ _fwrite           ; write file signature
.text:005413D2                         push    edi  ↙
    ↘               ; File
.text:005413D3                         push    1    ↙
    ↘               ; Count
```

```
.text:005413D5                 lea     eax, ⤦
    ↳ [esp+30h+Str]
.text:005413D9                 push    4       ⤦
    ↳            ; Size
.text:005413DB                 push    eax     ⤦
    ↳            ; Str
.text:005413DC                 call    ⤦
    ↳ _fwrite           ; write original file ⤦
    ↳ size
.text:005413E1                 push    edi     ⤦
    ↳            ; File
.text:005413E2                 push    1       ⤦
    ↳            ; Count
.text:005413E4                 push    esi     ⤦
    ↳            ; Size
.text:005413E5                 push    ebx     ⤦
    ↳            ; Str
.text:005413E6                 call    ⤦
    ↳ _fwrite           ; write encrypted file
.text:005413EB                 push    edi     ⤦
    ↳            ; File
.text:005413EC                 call    ⤦
    ↳ _fclose
.text:005413F1                 push    ebx     ⤦
    ↳            ; Memory
.text:005413F2                 call    _free
.text:005413F7                 add     esp, ⤦
    ↳ 40h
.text:005413FA                 pop     edi
.text:005413FB                 pop     esi
.text:005413FC                 pop     ebx
.text:005413FD                 pop     ebp
.text:005413FE                 retn
```

```
.text:005413FE crypt_file      endp
.text:005413FE
.text:005413FF                 align 10h
.text:00541400
.text:00541400 ; =============== S U B R O U ↙
    ↘ T I N E ↙
    ↘ =======================================↙
    ↘
.text:00541400
.text:00541400
.text:00541400 ; int __cdecl decrypt_file(↙
    ↘ char *Filename, int, void *Src)
.text:00541400 decrypt_file    proc near      ↙
    ↘             ; CODE XREF: _main+6E
.text:00541400
.text:00541400 Filename        = dword ptr  ↙
    ↘ 4
.text:00541400 arg_4           = dword ptr  ↙
    ↘ 8
.text:00541400 Src             = dword ptr  ↙
    ↘ 0Ch
.text:00541400
.text:00541400                 mov     eax, ↙
    ↘ [esp+Filename]
.text:00541404                 push    ebx
.text:00541405                 push    ebp
.text:00541406                 push    esi
.text:00541407                 push    edi
.text:00541408                 push    ↙
    ↘ offset aRb      ; "rb"
.text:0054140D                 push    eax  ↙
    ↘             ; Filename
.text:0054140E                 call    ↙
```

```
    ↳ _fopen
.text:00541413                  mov     esi, ↙
    ↳ eax
.text:00541415                  add     esp, ↙
    ↳ 8
.text:00541418                  test    esi, ↙
    ↳ esi
.text:0054141A                  jnz     short↙
    ↳  loc_54142E
.text:0054141C                  push    ↙
    ↳ offset aCannotOpenIn_0 ; "Cannot open ↙
    ↳ input file!\n"
.text:00541421                  call    ↙
    ↳ _printf
.text:00541426                  add     esp, ↙
    ↳ 4
.text:00541429                  pop     edi
.text:0054142A                  pop     esi
.text:0054142B                  pop     ebp
.text:0054142C                  pop     ebx
.text:0054142D                  retn
.text:0054142E
.text:0054142E loc_54142E:                  ↙
    ↳           ; CODE XREF: decrypt_file+1↙
    ↳ A
.text:0054142E                  push    2    ↙
    ↳           ; Origin
.text:00541430                  push    0    ↙
    ↳           ; Offset
.text:00541432                  push    esi  ↙
    ↳           ; File
.text:00541433                  call    ↙
    ↳ _fseek
```

```
.text:00541438                  push    esi    ⤦
    ↳         ; File
.text:00541439                  call   ⤦
    ↳ _ftell
.text:0054143E                  push    0      ⤦
    ↳         ; Origin
.text:00541440                  push    0      ⤦
    ↳         ; Offset
.text:00541442                  push    esi    ⤦
    ↳         ; File
.text:00541443                  mov     ebp,   ⤦
    ↳ eax
.text:00541445                  call   ⤦
    ↳ _fseek
.text:0054144A                  push    ebp    ⤦
    ↳         ; Size
.text:0054144B                  call   ⤦
    ↳ _malloc
.text:00541450                  push    esi    ⤦
    ↳         ; File
.text:00541451                  mov     ebx,   ⤦
    ↳ eax
.text:00541453                  push    1      ⤦
    ↳         ; Count
.text:00541455                  push    ebp    ⤦
    ↳         ; ElementSize
.text:00541456                  push    ebx    ⤦
    ↳         ; DstBuf
.text:00541457                  call   ⤦
    ↳ _fread
.text:0054145C                  push    esi    ⤦
    ↳         ; File
.text:0054145D                  call   ⤦
```

```
    ↳ _fclose
.text:00541462                    add     esp, ↙
    ↳ 34h
.text:00541465                    mov     ecx, ↙
    ↳ 3
.text:0054146A                    mov     edi, ↙
    ↳ offset aQr9_0 ; "QR9"
.text:0054146F                    mov     esi, ↙
    ↳ ebx
.text:00541471                    xor     edx, ↙
    ↳ edx
.text:00541473                    repe cmpsb
.text:00541475                    jz      short↙
    ↳  loc_541489
.text:00541477                    push    ↙
    ↳ offset aFileIsNotCrypt ; "File is not ↙
    ↳ encrypted!\n"
.text:0054147C                    call    ↙
    ↳ _printf
.text:00541481                    add     esp, ↙
    ↳ 4
.text:00541484                    pop     edi
.text:00541485                    pop     esi
.text:00541486                    pop     ebp
.text:00541487                    pop     ebx
.text:00541488                    retn
.text:00541489
.text:00541489 loc_541489:                    ↙
    ↳             ; CODE XREF: decrypt_file↙
    ↳ +75
.text:00541489                    mov     eax, ↙
    ↳ [esp+10h+Src]
.text:0054148D                    mov     edi, ↙
```

```
    ↳ [ebx+3]
.text:00541490                       add     ebp, ↙
    ↳ 0FFFFFFF9h
.text:00541493                       lea     esi, ↙
    ↳ [ebx+7]
.text:00541496                       push    eax  ↙
    ↳            ; Src
.text:00541497                       push    ebp  ↙
    ↳            ; int
.text:00541498                       push    esi  ↙
    ↳            ; int
.text:00541499                       call    ↙
    ↳ decrypt
.text:0054149E                       mov     ecx, ↙
    ↳ [esp+1Ch+arg_4]
.text:005414A2                       push    ↙
    ↳ offset aWb_0    ; "wb"
.text:005414A7                       push    ecx  ↙
    ↳            ; Filename
.text:005414A8                       call    ↙
    ↳ _fopen
.text:005414AD                       mov     ebp, ↙
    ↳ eax
.text:005414AF                       push    ebp  ↙
    ↳            ; File
.text:005414B0                       push    1    ↙
    ↳            ; Count
.text:005414B2                       push    edi  ↙
    ↳            ; Size
.text:005414B3                       push    esi  ↙
    ↳            ; Str
.text:005414B4                       call    ↙
    ↳ _fwrite
```

```
.text:005414B9                 push    ebp  ↙
    ↘            ; File
.text:005414BA                 call    ↙
    ↘ _fclose
.text:005414BF                 push    ebx  ↙
    ↘            ; Memory
.text:005414C0                 call    _free
.text:005414C5                 add     esp, ↙
    ↘ 2Ch
.text:005414C8                 pop     edi
.text:005414C9                 pop     esi
.text:005414CA                 pop     ebp
.text:005414CB                 pop     ebx
.text:005414CC                 retn
.text:005414CC decrypt_file    endp
```

```
.text:00541320 ; int __cdecl crypt_file(int ↙
    ↘ Str, char *Filename, int password)
.text:00541320 crypt_file      proc near
.text:00541320
.text:00541320 Str             = dword ptr  ↙
    ↘ 4
.text:00541320 Filename        = dword ptr  ↙
    ↘ 8
.text:00541320 password        = dword ptr  ↙
    ↘ 0Ch
.text:00541320
```

```
.text:00541320                 mov     eax, ↙
    ↘ [esp+Str]
.text:00541324                 push    ebp
```

```
.text:00541325                  push     ⤦
    ↳ offset Mode      ; "rb"
.text:0054132A                  push     eax  ⤦
    ↳            ; Filename
.text:0054132B                  call     ⤦
    ↳ _fopen           ; open file
.text:00541330                  mov      ebp, ⤦
    ↳ eax
.text:00541332                  add      esp, ⤦
    ↳ 8
.text:00541335                  test     ebp, ⤦
    ↳ ebp
.text:00541337                  jnz      short⤦
    ↳  loc_541348
.text:00541339                  push     ⤦
    ↳ offset Format   ; "Cannot open input ⤦
    ↳ file!\n"
.text:0054133E                  call     ⤦
    ↳ _printf
.text:00541343                  add      esp, ⤦
    ↳ 4
.text:00541346                  pop      ebp
.text:00541347                  retn
.text:00541348
.text:00541348 loc_541348:
```

fseek()/ftell():

```

.text:00541348 push    ebx
.text:00541349 push    esi
.text:0054134A push    edi
.text:0054134B push    2                  ; ⤦
```

```
    ↘ Origin
.text:0054134D push    0                    ; ↙
    ↘ Offset
.text:0054134F push    ebp                  ; ↙
    ↘ File

;
.text:00541350 call    _fseek
.text:00541355 push    ebp                  ; ↙
    ↘ File
.text:00541356 call    _ftell               ;
.text:0054135B push    0                    ; ↙
    ↘ Origin
.text:0054135D push    0                    ; ↙
    ↘ Offset
.text:0054135F push    ebp                  ; ↙
    ↘ File
.text:00541360 mov     [esp+2Ch+Str], eax

;
.text:00541364 call    _fseek
```

```

.text:00541369 mov     esi, [esp+2Ch+Str]
;
.text:0054136D and     esi, 0FFFFFFC0h
;
.text:00541370 add     esi, 40h
```

```
.text:00541373                      push    esi  ↙
    ↘           ; Size
```

```
.text:00541374                  call     ⤦
    ↳ _malloc
```

```


.text:00541379 mov      ecx, esi
.text:0054137B mov      ebx, eax          ;
.text:0054137D mov      edx, ecx
.text:0054137F xor      eax, eax
.text:00541381 mov      edi, ebx
.text:00541383 push     ebp               ; ⤦
    ↳ File
.text:00541384 shr      ecx, 2
.text:00541387 rep stosd
.text:00541389 mov      ecx, edx
.text:0054138B push     1                 ; ⤦
    ↳ Count
.text:0054138D and      ecx, 3
.text:00541390 rep stosb                  ; ⤦
    ↳ memset (buffer, 0, )
```

fread().

```
.text:00541392                  mov      eax, ⤦
    ↳ [esp+38h+Str]
.text:00541396                  push     eax  ⤦
    ↳           ; ElementSize
.text:00541397                  push     ebx  ⤦
    ↳           ; DstBuf
.text:00541398                  call     ⤦
    ↳ _fread            ; read file
.text:0054139D                  push     ebp  ⤦
    ↳           ; File
```

```
.text:0054139E                 call    ↙
    ↘ _fclose
```

```
.text:005413A3                 mov     ecx, ↙
    ↘ [esp+44h+password]
.text:005413A7                 push    ecx  ↙
    ↘            ; password
.text:005413A8                 push    esi  ↙
    ↘            ; aligned size
.text:005413A9                 push    ebx  ↙
    ↘            ; buffer
.text:005413AA                 call    crypt↙
    ↘            ; do crypt
```

```
.text:005413AF                 mov     edx, ↙
    ↘ [esp+50h+Filename]
.text:005413B3                 add     esp, ↙
    ↘ 40h
.text:005413B6                 push    ↙
    ↘ offset aWb      ; "wb"
.text:005413BB                 push    edx  ↙
    ↘            ; Filename
.text:005413BC                 call    ↙
    ↘ _fopen
.text:005413C1                 mov     edi, ↙
    ↘ eax
```

```
.text:005413C3                 push    edi  ↙
    ↘            ; File
.text:005413C4                 push    1    ↙
    ↘            ; Count
```

```
.text:005413C6                  push    3       ↙
    ↘               ; Size
.text:005413C8                  push    ↙
    ↘ offset aQr9       ; "QR9"
.text:005413CD                  call    ↙
    ↘ _fwrite           ; write file signature
```

:

```
.text:005413D2                  push    edi     ↙
    ↘               ; File
.text:005413D3                  push    1       ↙
    ↘               ; Count
.text:005413D5                  lea     eax,    ↙
    ↘ [esp+30h+Str]
.text:005413D9                  push    4       ↙
    ↘               ; Size
.text:005413DB                  push    eax     ↙
    ↘               ; Str
.text:005413DC                  call    ↙
    ↘ _fwrite           ; write original file ↙
    ↘ size
```

:

```
.text:005413E1                  push    edi     ↙
    ↘               ; File
.text:005413E2                  push    1       ↙
    ↘               ; Count
.text:005413E4                  push    esi     ↙
    ↘               ; Size
.text:005413E5                  push    ebx     ↙
    ↘               ; Str
```

```
.text:005413E6                 call    ⤦
    ↳ _fwrite           ; write encrypted file
```

:

```
.text:005413EB                 push    edi  ⤦
    ↳             ; File
.text:005413EC                 call    ⤦
    ↳ _fclose
.text:005413F1                 push    ebx  ⤦
    ↳             ; Memory
.text:005413F2                 call    _free
.text:005413F7                 add     esp, ⤦
    ↳ 40h
.text:005413FA                 pop     edi
.text:005413FB                 pop     esi
.text:005413FC                 pop     ebx
.text:005413FD                 pop     ebp
.text:005413FE                 retn
.text:005413FE crypt_file      endp
```

:

```
void crypt_file(char *fin, char* fout, char ⤦
    ↳ *pw)
{
        FILE *f;
        int flen, flen_aligned;
        BYTE *buf;

        f=fopen(fin, "rb");

        if (f==NULL)
```

```
        {
                printf ("Cannot open input ↙
    ↘ file!\n");
                return;
        };

        fseek (f, 0, SEEK_END);
        flen=ftell (f);
        fseek (f, 0, SEEK_SET);

        flen_aligned=(flen&0xFFFFFFC0)+0x40;

        buf=(BYTE*)malloc (flen_aligned);
        memset (buf, 0, flen_aligned);

        fread (buf, flen, 1, f);

        fclose (f);

        crypt (buf, flen_aligned, pw);

        f=fopen(fout, "wb");

        fwrite ("QR9", 3, 1, f);
        fwrite (&flen, 4, 1, f);
        fwrite (buf, flen_aligned, 1, f);

        fclose (f);

        free (buf);
};
```

:

```
.text:00541400 ; int __cdecl decrypt_file(⤦
    ⤷ char *Filename, int, void *Src)
.text:00541400 decrypt_file    proc near
.text:00541400
.text:00541400 Filename        = dword ptr  ⤦
    ⤷ 4
.text:00541400 arg_4           = dword ptr  ⤦
    ⤷ 8
.text:00541400 Src             = dword ptr  ⤦
    ⤷ 0Ch
.text:00541400
.text:00541400                 mov     eax, ⤦
    ⤷ [esp+Filename]
.text:00541404                 push    ebx
.text:00541405                 push    ebp
.text:00541406                 push    esi
.text:00541407                 push    edi
.text:00541408                 push    ⤦
    ⤷ offset aRb      ; "rb"
.text:0054140D                 push    eax  ⤦
    ⤷           ; Filename
.text:0054140E                 call    ⤦
    ⤷ _fopen
.text:00541413                 mov     esi, ⤦
    ⤷ eax
.text:00541415                 add     esp, ⤦
    ⤷ 8
.text:00541418                 test    esi, ⤦
    ⤷ esi
.text:0054141A                 jnz     short⤦
    ⤷  loc_54142E
```

```
.text:0054141C                 push    ⤦
    ↘ offset aCannotOpenIn_0 ; "Cannot open ⤦
    ↘ input file!\n"
.text:00541421                 call    ⤦
    ↘ _printf
.text:00541426                 add     esp, ⤦
    ↘ 4
.text:00541429                 pop     edi
.text:0054142A                 pop     esi
.text:0054142B                 pop     ebp
.text:0054142C                 pop     ebx
.text:0054142D                 retn
.text:0054142E
.text:0054142E loc_54142E:
.text:0054142E                 push    2     ⤦
    ↘           ; Origin
.text:00541430                 push    0     ⤦
    ↘           ; Offset
.text:00541432                 push    esi   ⤦
    ↘           ; File
.text:00541433                 call    ⤦
    ↘ _fseek
.text:00541438                 push    esi   ⤦
    ↘           ; File
.text:00541439                 call    ⤦
    ↘ _ftell
.text:0054143E                 push    0     ⤦
    ↘           ; Origin
.text:00541440                 push    0     ⤦
    ↘           ; Offset
.text:00541442                 push    esi   ⤦
    ↘           ; File
.text:00541443                 mov     ebp, ⤦
```

```
    ↳ eax
.text:00541445                  call    ↙
    ↳ _fseek
.text:0054144A                  push    ebp  ↙
    ↳           ; Size
.text:0054144B                  call    ↙
    ↳ _malloc
.text:00541450                  push    esi  ↙
    ↳           ; File
.text:00541451                  mov     ebx, ↙
    ↳ eax
.text:00541453                  push    1    ↙
    ↳           ; Count
.text:00541455                  push    ebp  ↙
    ↳           ; ElementSize
.text:00541456                  push    ebx  ↙
    ↳           ; DstBuf
.text:00541457                  call    ↙
    ↳ _fread
.text:0054145C                  push    esi  ↙
    ↳           ; File
.text:0054145D                  call    ↙
    ↳ _fclose
```

:

```
.text:00541462                  add     esp, ↙
    ↳ 34h
.text:00541465                  mov     ecx, ↙
    ↳ 3
.text:0054146A                  mov     edi, ↙
    ↳ offset aQr9_0 ; "QR9"
.text:0054146F                  mov     esi, ↙
```

```
    ↳ ebx
.text:00541471                  xor     edx, ↙
    ↳ edx
.text:00541473                  repe cmpsb
.text:00541475                  jz      short↙
    ↳  loc_541489
```

:

```
.text:00541477                  push    ↙
    ↳ offset aFileIsNotCrypt ; "File is not ↙
    ↳ encrypted!\n"
.text:0054147C                  call    ↙
    ↳ _printf
.text:00541481                  add     esp, ↙
    ↳ 4
.text:00541484                  pop     edi
.text:00541485                  pop     esi
.text:00541486                  pop     ebp
.text:00541487                  pop     ebx
.text:00541488                  retn
.text:00541489
.text:00541489 loc_541489:
```

decrypt().

```
.text:00541489                  mov     eax, ↙
    ↳ [esp+10h+Src]
.text:0054148D                  mov     edi, ↙
    ↳ [ebx+3]
.text:00541490                  add     ebp, ↙
    ↳ 0FFFFFFF9h
```

```
.text:00541493                 lea     esi, ⤦
    ↳ [ebx+7]
.text:00541496                 push    eax  ⤦
    ↳           ; Src
.text:00541497                 push    ebp  ⤦
    ↳           ; int
.text:00541498                 push    esi  ⤦
    ↳           ; int
.text:00541499                 call    ⤦
    ↳ decrypt
.text:0054149E                 mov     ecx, ⤦
    ↳ [esp+1Ch+arg_4]
.text:005414A2                 push    ⤦
    ↳ offset aWb_0    ; "wb"
.text:005414A7                 push    ecx  ⤦
    ↳           ; Filename
.text:005414A8                 call    ⤦
    ↳ _fopen
.text:005414AD                 mov     ebp, ⤦
    ↳ eax
.text:005414AF                 push    ebp  ⤦
    ↳           ; File
.text:005414B0                 push    1    ⤦
    ↳           ; Count
.text:005414B2                 push    edi  ⤦
    ↳           ; Size
.text:005414B3                 push    esi  ⤦
    ↳           ; Str
.text:005414B4                 call    ⤦
    ↳ _fwrite
.text:005414B9                 push    ebp  ⤦
    ↳           ; File
.text:005414BA                 call    ⤦
```

```
    ↳ _fclose
.text:005414BF                 push    ebx  ↙
    ↳            ; Memory
.text:005414C0                 call    _free
.text:005414C5                 add     esp, ↙
    ↳ 2Ch
.text:005414C8                 pop     edi
.text:005414C9                 pop     esi
.text:005414CA                 pop     ebp
.text:005414CB                 pop     ebx
.text:005414CC                 retn
.text:005414CC decrypt_file    endp
```

:

```
void decrypt_file(char *fin, char* fout, ↙
    ↳ char *pw)
{
        FILE *f;
        int real_flen, flen;
        BYTE *buf;

        f=fopen(fin, "rb");

        if (f==NULL)
        {
                printf ("Cannot open input ↙
    ↳ file!\n");
                return;
        };

        fseek (f, 0, SEEK_END);
        flen=ftell (f);
```

```
        fseek (f, 0, SEEK_SET);

        buf=(BYTE*)malloc (flen);

        fread (buf, flen, 1, f);

        fclose (f);

        if (memcmp (buf, "QR9", 3)!=0)
        {
                printf ("File is not ↙
    ↳ encrypted!\n");
                return;
        };

        memcpy (&real_flen, buf+3, 4);

        decrypt (buf+(3+4), flen-(3+4), pw);

        f=fopen(fout, "wb");

        fwrite (buf+(3+4), real_flen, 1, f);

        fclose (f);

        free (buf);
};
```

.

crypt():

```
.text:00541260 crypt           proc near
```

```
.text:00541260
.text:00541260 arg_0            = dword ptr ↙
    ↘ 4
.text:00541260 arg_4            = dword ptr ↙
    ↘ 8
.text:00541260 arg_8            = dword ptr ↙
    ↘ 0Ch
.text:00541260
.text:00541260                  push    ebx
.text:00541261                  mov     ebx, ↙
    ↘ [esp+4+arg_0]
.text:00541265                  push    ebp
.text:00541266                  push    esi
.text:00541267                  push    edi
.text:00541268                  xor     ebp, ↙
    ↘ ebp
.text:0054126A
.text:0054126A loc_54126A:
```

```
.text:0054126A                  mov     eax, ↙
    ↘ [esp+10h+arg_8]
.text:0054126E                  mov     ecx, ↙
    ↘ 10h
.text:00541273                  mov     esi, ↙
    ↘ ebx   ; EBX is pointer within input ↙
    ↘ buffer
.text:00541275                  mov     edi, ↙
    ↘ offset cube64
.text:0054127A                  push    1
.text:0054127C                  push    eax
.text:0054127D                  rep movsd
```

rotate_all_with_password():

```
.text:0054127F                 call    ⤦
    ↳ rotate_all_with_password
```

:

```
.text:00541284                 mov     eax, ⤦
    ↳ [esp+18h+arg_4]
.text:00541288                 mov     edi, ⤦
    ↳ ebx
.text:0054128A                 add     ebp, ⤦
    ↳ 40h
.text:0054128D                 add     esp, ⤦
    ↳ 8
.text:00541290                 mov     ecx, ⤦
    ↳ 10h
.text:00541295                 mov     esi, ⤦
    ↳ offset cube64
.text:0054129A                 add     ebx, ⤦
    ↳ 40h  ; add 64 to input buffer pointer
.text:0054129D                 cmp     ebp, ⤦
    ↳ eax  ; EBP contain amount of encrypted⤦
    ↳  data.
.text:0054129F                 rep movsd
```

```
.text:005412A1                 jl      short⤦
    ↳  loc_54126A
.text:005412A3                 pop     edi
.text:005412A4                 pop     esi
.text:005412A5                 pop     ebp
.text:005412A6                 pop     ebx
.text:005412A7                 retn
```

```
.text:005412A7 crypt           endp
```

:

```
void crypt (BYTE *buf, int sz, char *pw)
{
        int i=0;

        do
        {
                memcpy (cube, buf+i, 8*8);
                rotate_all (pw, 1);
                memcpy (buf+i, cube, 8*8);
                i+=64;
        }
        while (i<sz);
};
```

```
.text:005411B0 rotate_all_with_password proc ↙
    ↳ near
.text:005411B0
.text:005411B0 arg_0           = dword ptr  ↙
    ↳ 4
.text:005411B0 arg_4           = dword ptr  ↙
    ↳ 8
.text:005411B0
.text:005411B0                 mov     eax, ↙
    ↳ [esp+arg_0]
.text:005411B4                 push    ebp
.text:005411B5                 mov     ebp, ↙
    ↳ eax
```

```
.text:005411B7                    cmp      byte ⤦
    ↳ ptr [eax], 0
.text:005411BA                    jz       exit
.text:005411C0                    push     ebx
.text:005411C1                    mov      ebx, ⤦
    ↳ [esp+8+arg_4]
.text:005411C5                    push     esi
.text:005411C6                    push     edi
.text:005411C7
.text:005411C7 loop_begin:
```

```
.text:005411C7                    movsx    eax, ⤦
    ↳ byte ptr [ebp+0]
.text:005411CB                    push     eax  ⤦
    ↳           ; C
.text:005411CC                    call     ⤦
    ↳ _tolower
.text:005411D1                    add      esp, ⤦
    ↳ 4
```

```
.text:005411D4                    cmp      al, '⤦
    ↳ a'
.text:005411D6                    jl       short⤦
    ↳  next_character_in_password
.text:005411D8                    cmp      al, '⤦
    ↳ z'
.text:005411DA                    jg       short⤦
    ↳  next_character_in_password
.text:005411DC                    movsx    ecx, ⤦
    ↳ al
```

```
.text:005411DF                     sub     ecx, ↙
    ↘ 'a'  ; 97
```

```
.text:005411E2                     cmp     ecx, ↙
    ↘ 24
.text:005411E5                     jle     short↙
    ↘  skip_subtracting
.text:005411E7                     sub     ecx, ↙
    ↘ 24
```

```
.text:005411EA
.text:005411EA skip_subtracting:               ↙
    ↘             ; CODE XREF: ↙
    ↘ rotate_all_with_password+35
```

```
.text:005411EA                     mov     eax, ↙
    ↘ 55555556h
.text:005411EF                     imul    ecx
.text:005411F1                     mov     eax, ↙
    ↘ edx
.text:005411F3                     shr     eax, ↙
    ↘ 1Fh
.text:005411F6                     add     edx, ↙
    ↘ eax
.text:005411F8                     mov     eax, ↙
    ↘ ecx
.text:005411FA                     mov     esi, ↙
    ↘ edx
.text:005411FC                     mov     ecx, ↙
    ↘ 3
.text:00541201                     cdq
```

```
.text:00541202                  idiv    ecx
```

```
.text:00541204 sub     edx, 0
.text:00541207 jz      short call_rotate1 ;
.text:00541209 dec     edx
.text:0054120A jz      short call_rotate2 ; ↙
    ↳ ..
.text:0054120C dec     edx
.text:0054120D jnz     short ↙
    ↳ next_character_in_password
.text:0054120F test    ebx, ebx
.text:00541211 jle     short ↙
    ↳ next_character_in_password
.text:00541213 mov     edi, ebx
```

```
.text:00541215 call_rotate3:
.text:00541215                  push    esi
.text:00541216                  call    ↙
    ↳ rotate3
.text:0054121B                  add     esp, ↙
    ↳ 4
.text:0054121E                  dec     edi
.text:0054121F                  jnz     short↙
    ↳  call_rotate3
.text:00541221                  jmp     short↙
    ↳  next_character_in_password
.text:00541223
.text:00541223 call_rotate2:
.text:00541223                  test    ebx, ↙
    ↳ ebx
```

```
.text:00541225                  jle     short↙
    ↳  next_character_in_password
.text:00541227                  mov     edi, ↙
    ↳ ebx
.text:00541229
.text:00541229 loc_541229:
.text:00541229                  push    esi
.text:0054122A                  call    ↙
    ↳ rotate2
.text:0054122F                  add     esp, ↙
    ↳ 4
.text:00541232                  dec     edi
.text:00541233                  jnz     short↙
    ↳  loc_541229
.text:00541235                  jmp     short↙
    ↳  next_character_in_password
.text:00541237
.text:00541237 call_rotate1:
.text:00541237                  test    ebx, ↙
    ↳ ebx
.text:00541239                  jle     short↙
    ↳  next_character_in_password
.text:0054123B                  mov     edi, ↙
    ↳ ebx
.text:0054123D
.text:0054123D loc_54123D:
.text:0054123D                  push    esi
.text:0054123E                  call    ↙
    ↳ rotate1
.text:00541243                  add     esp, ↙
    ↳ 4
.text:00541246                  dec     edi
.text:00541247                  jnz     short↙
```

```
    ↳  loc_54123D
.text:00541249
```

```
.text:00541249 next_character_in_password:
.text:00541249                  mov     al, [↙
    ↳ ebp+1]
```

```
.text:0054124C                  inc     ebp
.text:0054124D                  test    al, ↙
    ↳ al
.text:0054124F                  jnz     ↙
    ↳ loop_begin
.text:00541255                  pop     edi
.text:00541256                  pop     esi
.text:00541257                  pop     ebx
.text:00541258
.text:00541258 exit:
.text:00541258                  pop     ebp
.text:00541259                  retn
.text:00541259 rotate_all_with_password endp
```

```
void rotate_all (char *pwd, int v)
{
        char *p=pwd;

        while (*p)
        {
                char c=*p;
                int q;

                c=tolower (c);

```

```
                if (c>='a' && c<='z')
                {
                        q=c-'a';
                        if (q>24)
                                q-=24;

                        int quotient=q/3;
                        int remainder=q % 3;

                        switch (remainder)
                        {
                        case 0: for (int i⤦
    ↳ =0; i<v; i++) rotate1 (quotient); ⤦
    ↳ break;
                        case 1: for (int i⤦
    ↳ =0; i<v; i++) rotate2 (quotient); ⤦
    ↳ break;
                        case 2: for (int i⤦
    ↳ =0; i<v; i++) rotate3 (quotient); ⤦
    ↳ break;
                        };
                };

                p++;
        };
};
```

```
.text:00541050 get_bit         proc near
.text:00541050
.text:00541050 arg_0           = dword ptr  ⤦
    ↳ 4
.text:00541050 arg_4           = dword ptr  ⤦
```

```
    ↳ 8
.text:00541050 arg_8              = byte ptr  0↙
    ↳ Ch
.text:00541050
.text:00541050                    mov     eax, ↙
    ↳ [esp+arg_4]
.text:00541054                    mov     ecx, ↙
    ↳ [esp+arg_0]
.text:00541058                    mov     al, ↙
    ↳ cube64[eax+ecx*8]
.text:0054105F                    mov     cl, [↙
    ↳ esp+arg_8]
.text:00541063                    shr     al, ↙
    ↳ cl
.text:00541065                    and     al, 1
.text:00541067                    retn
.text:00541067 get_bit            endp
```

: *arg_4 + arg_0 * 8*.

, `set_bit()`:

```
.text:00541000 set_bit            proc near
.text:00541000
.text:00541000 arg_0              = dword ptr  ↙
    ↳ 4
.text:00541000 arg_4              = dword ptr  ↙
    ↳ 8
.text:00541000 arg_8              = dword ptr  ↙
    ↳ 0Ch
.text:00541000 arg_C              = byte ptr  ↙
    ↳ 10h
.text:00541000
```

```
.text:00541000                 mov     al, [esp+arg_C]
.text:00541004                 mov     ecx, [esp+arg_8]
.text:00541008                 push    esi
.text:00541009                 mov     esi, [esp+4+arg_0]
.text:0054100D                 test    al, al
.text:0054100F                 mov     eax, [esp+4+arg_4]
.text:00541013                 mov     dl, 1
.text:00541015                 jz      short loc_54102B
```

```
.text:00541017                 shl     dl, cl
.text:00541019                 mov     cl, cube64[eax+esi*8]
```

```
.text:00541020                 or      cl, dl
```

```
.text:00541022                 mov     cube64[eax+esi*8], cl
.text:00541029                 pop     esi
.text:0054102A                 retn
.text:0054102B
.text:0054102B loc_54102B:
.text:0054102B                 shl     dl, cl
```

```
.text:0054102D                 mov     cl, ↙
    ↘ cube64[eax+esi*8]
```

```
.text:00541034                 not     dl
```

```
.text:00541036                 and     cl, ↙
    ↘ dl
```

```
.text:00541038                 mov     ↙
    ↘ cube64[eax+esi*8], cl
.text:0054103F                 pop     esi
.text:00541040                 retn
.text:00541040 set_bit         endp
```

```
#define IS_SET(flag, bit)       ((flag) & (↙
    ↘ bit))
#define SET_BIT(var, bit)       ((var) |= (↙
    ↘ bit))
#define REMOVE_BIT(var, bit)    ((var) &= ~(↙
    ↘ bit))

static BYTE cube[8][8];

void set_bit (int x, int y, int shift, int ↙
    ↘ bit)
{
        if (bit)
                SET_BIT (cube[x][y], 1<<↙
    ↘ shift);
        else
```

```
                REMOVE_BIT (cube[x][y], 1<<⤦
    ↳ shift);
};

bool get_bit (int x, int y, int shift)
{
        if ((cube[x][y]>>shift)&1==1)
                return 1;
        return 0;
};
```

```
.text:00541070 rotate1         proc near
.text:00541070
```

```
.text:00541070 internal_array_64= byte ptr ⤦
    ↳ -40h
.text:00541070 arg_0           = dword ptr  ⤦
    ↳ 4
.text:00541070
.text:00541070                 sub     esp, ⤦
    ↳ 40h
.text:00541073                 push    ebx
.text:00541074                 push    ebp
.text:00541075                 mov     ebp, ⤦
    ↳ [esp+48h+arg_0]
.text:00541079                 push    esi
.text:0054107A                 push    edi
.text:0054107B                 xor     edi, ⤦
    ↳ edi       ; EDI is loop1 counter
```

EBX

```
.text:0054107D                  lea     ebx, ⤦
    ↳ [esp+50h+internal_array_64]
.text:00541081
```

```
.text:00541081 first_loop1_begin:
.text:00541081    xor      esi, esi          ;
.text:00541083
.text:00541083 first_loop2_begin:
.text:00541083    push     ebp               ; ⤦
    ↳ arg_0
.text:00541084    push     esi               ;
.text:00541085    push     edi               ;
.text:00541086    call     get_bit
.text:0054108B    add      esp, 0Ch
.text:0054108E    mov      [ebx+esi], al     ;
.text:00541091    inc      esi               ;
.text:00541092    cmp      esi, 8
.text:00541095    jl       short ⤦
    ↳ first_loop2_begin
.text:00541097    inc      edi               ;

;
.text:00541098    add      ebx, 8
.text:0054109B    cmp      edi, 8
.text:0054109E    jl       short ⤦
    ↳ first_loop1_begin
```

```
.text:005410A0    lea      ebx, [esp+50h+⤦
    ↳ internal_array_64]
```

```
.text:005410A4    mov     edi, 7          ;
.text:005410A9
.text:005410A9 second_loop1_begin:
.text:005410A9    xor     esi, esi        ;
.text:005410AB
.text:005410AB second_loop2_begin:
.text:005410AB    mov     al, [ebx+esi]   ; ↙
    ↳ EN(value from internal array)
.text:005410AE    push    eax
.text:005410AF    push    ebp             ; ↙
    ↳ arg_0
.text:005410B0    push    edi             ;
.text:005410B1    push    esi             ;
.text:005410B2    call    set_bit
.text:005410B7    add     esp, 10h
.text:005410BA    inc     esi             ;
.text:005410BB    cmp     esi, 8
.text:005410BE    jl      short ↙
    ↳ second_loop2_begin
.text:005410C0    dec     edi             ;
.text:005410C1    add     ebx, 8          ;
.text:005410C4    cmp     edi, 0FFFFFFFFh
.text:005410C7    jg      short ↙
    ↳ second_loop1_begin
.text:005410C9    pop     edi
.text:005410CA    pop     esi
.text:005410CB    pop     ebp
.text:005410CC    pop     ebx
.text:005410CD    add     esp, 40h
.text:005410D0    retn
.text:005410D0 rotate1        endp
```

```
void rotate1 (int v)
{
        bool tmp[8][8]; // internal array
        int i, j;

        for (i=0; i<8; i++)
                for (j=0; j<8; j++)
                        tmp[i][j]=get_bit (i↙
    ↘ , j, v);

        for (i=0; i<8; i++)
                for (j=0; j<8; j++)
                        set_bit (j, 7-i, v, ↙
    ↘ tmp[x][y]);
};
```

```

.text:005410E0 rotate2 proc near
.text:005410E0
.text:005410E0 internal_array_64 = byte ptr ↙
    ↘ -40h
.text:005410E0 arg_0 = dword ptr  4
.text:005410E0
.text:005410E0     sub     esp, 40h
.text:005410E3     push    ebx
.text:005410E4     push    ebp
.text:005410E5     mov     ebp, [esp+48h+↙
    ↘ arg_0]
.text:005410E9     push    esi
.text:005410EA     push    edi
.text:005410EB     xor     edi, edi        ;
.text:005410ED     lea     ebx, [esp+50h+↙
```

```
    ↳ internal_array_64]
.text:005410F1
.text:005410F1 loc_5410F1:
.text:005410F1     xor     esi, esi          ;
.text:005410F3
.text:005410F3 loc_5410F3:
.text:005410F3     push    esi               ;
.text:005410F4     push    edi               ;
.text:005410F5     push    ebp               ;↙
    ↳  arg_0
.text:005410F6     call    get_bit
.text:005410FB     add     esp, 0Ch
.text:005410FE     mov     [ebx+esi], al     ;
.text:00541101     inc     esi               ;
.text:00541102     cmp     esi, 8
.text:00541105     jl      short loc_5410F3
.text:00541107     inc     edi               ;
.text:00541108     add     ebx, 8
.text:0054110B     cmp     edi, 8
.text:0054110E     jl      short loc_5410F1
.text:00541110     lea     ebx, [esp+50h+↙
    ↳ internal_array_64]
.text:00541114     mov     edi, 7            ;
.text:00541119
.text:00541119 loc_541119:
.text:00541119     xor     esi, esi          ;
.text:0054111B
.text:0054111B loc_54111B:
.text:0054111B     mov     al, [ebx+esi]     ;
.text:0054111E     push    eax
.text:0054111F     push    edi               ;
.text:00541120     push    esi               ;
.text:00541121     push    ebp               ;↙
```

```
    ↳ arg_0
.text:00541122      call    set_bit
.text:00541127      add     esp, 10h
.text:0054112A      inc     esi             ;
.text:0054112B      cmp     esi, 8
.text:0054112E      jl      short loc_54111B
.text:00541130      dec     edi             ;
.text:00541131      add     ebx, 8
.text:00541134      cmp     edi, 0FFFFFFFFh
.text:00541137      jg      short loc_541119
.text:00541139      pop     edi
.text:0054113A      pop     esi
.text:0054113B      pop     ebp
.text:0054113C      pop     ebx
.text:0054113D      add     esp, 40h
.text:00541140      retn
.text:00541140 rotate2 endp
```

```
void rotate2 (int v)
{
        bool tmp[8][8]; // internal array
        int i, j;

        for (i=0; i<8; i++)
                for (j=0; j<8; j++)
                        tmp[i][j]=get_bit (v↙
    ↳ , i, j);

        for (i=0; i<8; i++)
                for (j=0; j<8; j++)
                        set_bit (v, j, 7-i, ↙
    ↳ tmp[i][j]);
```

```
};
```

```
void rotate3 (int v)
{
        bool tmp[8][8];
        int i, j;

        for (i=0; i<8; i++)
                for (j=0; j<8; j++)
                        tmp[i][j]=get_bit (i, v, j);

        for (i=0; i<8; i++)
                for (j=0; j<8; j++)
                        set_bit (7-j, v, i, tmp[i][j]);
};
```

[1]?!

```
void decrypt (BYTE *buf, int sz, char *pw)
{
        char *p=strdup (pw);
        strrev (p);
        int i=0;

        do
        {
                memcpy (cube, buf+i, 8*8);
                rotate_all (p, 3);
```

[1] wikipedia

```
                memcpy (buf+i, cube, 8*8);
                i+=64;
        }
        while (i<sz);

        free (p);
};
```

strrev() [2]

```
q=c-'a';
if (q>24)
        q-=24;

int quotient=q/3; // in range 0..7
int remainder=q % 3;

switch (remainder)
{
    case 0: for (int i=0; i<v; i++) rotate1 ↙
    ↳ (quotient); break; // front
    case 1: for (int i=0; i<v; i++) rotate2 ↙
    ↳ (quotient); break; // top
    case 2: for (int i=0; i<v; i++) rotate3 ↙
    ↳ (quotient); break; // left
};
```

```
#include <windows.h>

#include <stdio.h>
```

[2] MSDN

```
#include <assert.h>

#define IS_SET(flag, bit)       ((flag) & (↙
    ↘ bit))
#define SET_BIT(var, bit)       ((var) |= (↙
    ↘ bit))
#define REMOVE_BIT(var, bit)    ((var) &= ~(↙
    ↘ bit))

static BYTE cube[8][8];

void set_bit (int x, int y, int z, bool bit)
{
        if (bit)
                SET_BIT (cube[x][y], 1<<z);
        else
                REMOVE_BIT (cube[x][y], 1<<z↙
    ↘ );
};

bool get_bit (int x, int y, int z)
{
        if ((cube[x][y]>>z)&1==1)
                return true;
        return false;
};

void rotate_f (int row)
{
        bool tmp[8][8];
        int x, y;

        for (x=0; x<8; x++)
```

```
                for (y=0; y<8; y++)
                        tmp[x][y]=get_bit (x
    ↳ , y, row);

        for (x=0; x<8; x++)
                for (y=0; y<8; y++)
                        set_bit (y, 7-x, row
    ↳ , tmp[x][y]);
};

void rotate_t (int row)
{
        bool tmp[8][8];
        int y, z;

        for (y=0; y<8; y++)
                for (z=0; z<8; z++)
                        tmp[y][z]=get_bit (
    ↳ row, y, z);

        for (y=0; y<8; y++)
                for (z=0; z<8; z++)
                        set_bit (row, z, 7-y
    ↳ , tmp[y][z]);
};

void rotate_l (int row)
{
        bool tmp[8][8];
        int x, z;

        for (x=0; x<8; x++)
                for (z=0; z<8; z++)
```

```
                                tmp[x][z]=get_bit (x↙
    ↘ , row, z);

        for (x=0; x<8; x++)
                for (z=0; z<8; z++)
                        set_bit (7-z, row, x↙
    ↘ , tmp[x][z]);
};

void rotate_all (char *pwd, int v)
{
        char *p=pwd;

        while (*p)
        {
                char c=*p;
                int q;

                c=tolower (c);

                if (c>='a' && c<='z')
                {
                        q=c-'a';
                        if (q>24)
                                q-=24;

                        int quotient=q/3;
                        int remainder=q % 3;

                        switch (remainder)
                        {
                        case 0: for (int i↙
    ↘ =0; i<v; i++) rotate_f (quotient); ↙
```

```
    ↳ break;
                                case 1: for (int i↙
    ↳ =0; i<v; i++) rotate_t (quotient); ↙
    ↳ break;
                                case 2: for (int i↙
    ↳ =0; i<v; i++) rotate_l (quotient); ↙
    ↳ break;
                                };
                        };

                        p++;
                };
};

void crypt (BYTE *buf, int sz, char *pw)
{
        int i=0;

        do
        {
                memcpy (cube, buf+i, 8*8);
                rotate_all (pw, 1);
                memcpy (buf+i, cube, 8*8);
                i+=64;
        }
        while (i<sz);
};

void decrypt (BYTE *buf, int sz, char *pw)
{
        char *p=strdup (pw);
        strrev (p);
        int i=0;
```

```
        do
        {
                memcpy (cube, buf+i, 8*8);
                rotate_all (p, 3);
                memcpy (buf+i, cube, 8*8);
                i+=64;
        }
        while (i<sz);

        free (p);
};

void crypt_file(char *fin, char* fout, char ⤦
    ↳ *pw)
{
        FILE *f;
        int flen, flen_aligned;
        BYTE *buf;

        f=fopen(fin, "rb");

        if (f==NULL)
        {
                printf ("Cannot open input ⤦
    ↳ file!\n");
                return;
        };

        fseek (f, 0, SEEK_END);
        flen=ftell (f);
        fseek (f, 0, SEEK_SET);
```

```
        flen_aligned=(flen&0xFFFFFFC0)+0x40;

        buf=(BYTE*)malloc (flen_aligned);
        memset (buf, 0, flen_aligned);

        fread (buf, flen, 1, f);

        fclose (f);

        crypt (buf, flen_aligned, pw);

        f=fopen(fout, "wb");

        fwrite ("QR9", 3, 1, f);
        fwrite (&flen, 4, 1, f);
        fwrite (buf, flen_aligned, 1, f);

        fclose (f);

        free (buf);

};

void decrypt_file(char *fin, char* fout, ↙
    ↘ char *pw)
{
        FILE *f;
        int real_flen, flen;
        BYTE *buf;

        f=fopen(fin, "rb");

        if (f==NULL)
```

```
	{
		printf ("Cannot open input ↙
↘ file!\n");
		return;
	};

	fseek (f, 0, SEEK_END);
	flen=ftell (f);
	fseek (f, 0, SEEK_SET);

	buf=(BYTE*)malloc (flen);

	fread (buf, flen, 1, f);

	fclose (f);

	if (memcmp (buf, "QR9", 3)!=0)
	{
		printf ("File is not ↙
↘ encrypted!\n");
		return;
	};

	memcpy (&real_flen, buf+3, 4);

	decrypt (buf+(3+4), flen-(3+4), pw);

	f=fopen(fout, "wb");

	fwrite (buf+(3+4), real_flen, 1, f);

	fclose (f);
```

```
        free (buf);
};

// run: input output 0/1 password
// 0 for encrypt, 1 for decrypt

int main(int argc, char *argv[])
{
        if (argc!=5)
        {
                printf ("Incorrect ↙
    ↘ parameters!\n");
                return 1;
        };

        if (strcmp (argv[3], "0")==0)
                crypt_file (argv[1], argv↙
    ↘ [2], argv[4]);
        else
                if (strcmp (argv[3], "1")↙
    ↘ ==0)
                        decrypt_file (argv↙
    ↘ [1], argv[2], argv[4]);
                else
                        printf ("Wrong param↙
    ↘  %s\n", argv[3]);

        return 0;
};
```

Capítulo 80

SAP

80.1

([1] y[2]).

:

[1] http://go.yurichev.com/17221
[2] blog.yurichev.com

 CAPÍTULO 80. SAP

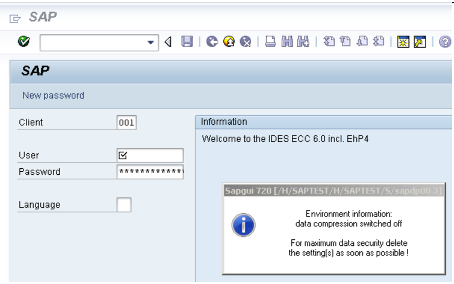


Figura 80.1:

.

.

:

```
.text:6440D51B                 lea     eax, [ebp+2108h+var_211C]
.text:6440D51E                 push    eax             ; int
.text:6440D51F                 push    offset aTdw_nocompress ; "TDW_NOCOMPRESS"
.text:6440D524                 mov     byte ptr [edi+15h], 0
.text:6440D528                 call    chk_env
.text:6440D52D                 pop     ecx
.text:6440D52E                 pop     ecx
.text:6440D52F                 push    
```

```
    ↳ offset byte_64443AF8
.text:6440D534                  lea     ecx, ↙
    ↳ [ebp+2108h+var_211C]

; demangled name: int ATL::CStringT::Compare↙
    ↳ (char const *)const
.text:6440D537                  call    ds:↙
    ↳ mfc90_1603
.text:6440D53D                  test    eax, ↙
    ↳ eax
.text:6440D53F                  jz      short↙
    ↳  loc_6440D55A
.text:6440D541                  lea     ecx, ↙
    ↳ [ebp+2108h+var_211C]

; demangled name: const char* ATL::↙
    ↳ CSimpleStringT::operator PCXSTR
.text:6440D544                  call    ds:↙
    ↳ mfc90_910
.text:6440D54A                  push    eax  ↙
    ↳             ; Str
.text:6440D54B                  call    ds:↙
    ↳ atoi
.text:6440D551                  test    eax, ↙
    ↳ eax
.text:6440D553                  setnz   al
.text:6440D556                  pop     ecx
.text:6440D557                  mov     [edi↙
    ↳ +15h], al
```

.

:

```
.text:64413F20 ; int __cdecl chk_env(char *↙
    ↘ VarName, int)
.text:64413F20 chk_env          proc near
.text:64413F20
.text:64413F20 DstSize          = dword ptr ↙
    ↘ -0Ch
.text:64413F20 var_8            = dword ptr ↙
    ↘ -8
.text:64413F20 DstBuf           = dword ptr ↙
    ↘ -4
.text:64413F20 VarName          = dword ptr  ↙
    ↘ 8
.text:64413F20 arg_4            = dword ptr  ↙
    ↘ 0Ch
.text:64413F20
.text:64413F20                  push    ebp
.text:64413F21                  mov     ebp, ↙
    ↘ esp
.text:64413F23                  sub     esp, ↙
    ↘ 0Ch
.text:64413F26                  mov     [ebp+↙
    ↘ DstSize], 0
.text:64413F2D                  mov     [ebp+↙
    ↘ DstBuf], 0
.text:64413F34                  push    ↙
    ↘ offset unk_6444C88C
.text:64413F39                  mov     ecx, ↙
    ↘ [ebp+arg_4]

; (demangled name) ATL::CStringT::operator=(↙
    ↘ char const *)
.text:64413F3C                  call    ds:↙
```

```
    ↘ mfc90_820
.text:64413F42                          mov     eax, ↙
    ↘ [ebp+VarName]
.text:64413F45                          push    eax  ↙
    ↘            ; VarName
.text:64413F46                          mov     ecx, ↙
    ↘ [ebp+DstSize]
.text:64413F49                          push    ecx  ↙
    ↘            ; DstSize
.text:64413F4A                          mov     edx, ↙
    ↘ [ebp+DstBuf]
.text:64413F4D                          push    edx  ↙
    ↘            ; DstBuf
.text:64413F4E                          lea     eax, ↙
    ↘ [ebp+DstSize]
.text:64413F51                          push    eax  ↙
    ↘            ; ReturnSize
.text:64413F52                          call    ds:↙
    ↘ getenv_s
.text:64413F58                          add     esp, ↙
    ↘ 10h
.text:64413F5B                          mov     [ebp+↙
    ↘ var_8], eax
.text:64413F5E                          cmp     [ebp+↙
    ↘ var_8], 0
.text:64413F62                          jz      short↙
    ↘  loc_64413F68
.text:64413F64                          xor     eax, ↙
    ↘ eax
.text:64413F66                          jmp     short↙
    ↘  loc_64413FBC
.text:64413F68
.text:64413F68 loc_64413F68:
```

```
.text:64413F68                 cmp     [ebp+⤦
    ↳ DstSize], 0
.text:64413F6C                 jnz     short⤦
    ↳  loc_64413F72
.text:64413F6E                 xor     eax, ⤦
    ↳ eax
.text:64413F70                 jmp     short⤦
    ↳  loc_64413FBC
.text:64413F72
.text:64413F72 loc_64413F72:
.text:64413F72                 mov     ecx, ⤦
    ↳ [ebp+DstSize]
.text:64413F75                 push    ecx
.text:64413F76                 mov     ecx, ⤦
    ↳ [ebp+arg_4]

; demangled name: ATL::CSimpleStringT<char, ⤦
    ↳ 1>::Preallocate(int)
.text:64413F79                 call    ds:⤦
    ↳ mfc90_2691
.text:64413F7F                 mov     [ebp+⤦
    ↳ DstBuf], eax
.text:64413F82                 mov     edx, ⤦
    ↳ [ebp+VarName]
.text:64413F85                 push    edx     ⤦
    ↳           ; VarName
.text:64413F86                 mov     eax, ⤦
    ↳ [ebp+DstSize]
.text:64413F89                 push    eax     ⤦
    ↳           ; DstSize
.text:64413F8A                 mov     ecx, ⤦
    ↳ [ebp+DstBuf]
.text:64413F8D                 push    ecx     ⤦
```

```
    ↳                ; DstBuf
.text:64413F8E                   lea     edx, ↙
    ↳ [ebp+DstSize]
.text:64413F91                   push    edx   ↙
    ↳                ; ReturnSize
.text:64413F92                   call    ds:↙
    ↳ getenv_s
.text:64413F98                   add     esp, ↙
    ↳ 10h
.text:64413F9B                   mov     [ebp+↙
    ↳ var_8], eax
.text:64413F9E                   push    0↙
    ↳ FFFFFFFFh
.text:64413FA0                   mov     ecx, ↙
    ↳ [ebp+arg_4]

; demangled name: ATL::CSimpleStringT::↙
    ↳ ReleaseBuffer(int)
.text:64413FA3                   call    ds:↙
    ↳ mfc90_5835
.text:64413FA9                   cmp     [ebp+↙
    ↳ var_8], 0
.text:64413FAD                   jz      short↙
    ↳  loc_64413FB3
.text:64413FAF                   xor     eax, ↙
    ↳ eax
.text:64413FB1                   jmp     short↙
    ↳  loc_64413FBC
.text:64413FB3
.text:64413FB3 loc_64413FB3:
.text:64413FB3                   mov     ecx, ↙
    ↳ [ebp+arg_4]
```

```
; demangled name: const char* ATL::↙
    ↘ CSimpleStringT::operator PCXSTR
.text:64413FB6                  call    ds:↙
    ↘ mfc90_910
.text:64413FBC
.text:64413FBC loc_64413FBC:
.text:64413FBC
.text:64413FBC                  mov     esp, ↙
    ↘ ebp
.text:64413FBE                  pop     ebp
.text:64413FBF                  retn
.text:64413FBF chk_env          endp
```

. getenv_s()[3] .

.

:

[3] MSDN

DPTRACE	"GUI-OPTION: Trace set to %d
TDW_HEXDUMP	"GUI-OPTION: Hexdump enab
TDW_WORKDIR	"GUI-OPTION: working directo
TDW_SPLASHSRCEENOFF	"GUI-OPTION: Splash Screen
TDW_REPLYTIMEOUT	"GUI-OPTION: reply timeout %
TDW_PLAYBACKTIMEOUT	"GUI-OPTION: PlaybackTimeo
TDW_NOCOMPRESS	"GUI-OPTION: no compression
TDW_EXPERT	"GUI-OPTION: expert mode"
TDW_PLAYBACKPROGRESS	"GUI-OPTION: PlaybackProgre
TDW_PLAYBACKNETTRAFFIC	"GUI-OPTION: PlaybackNetTra
TDW_PLAYLOG	"GUI-OPTION: /PlayLog is YES
TDW_PLAYTIME	"GUI-OPTION: /PlayTime set t
TDW_LOGFILE	"GUI-OPTION: TDW_LOGFILE
TDW_WAN	"GUI-OPTION: WAN - low spe
TDW_FULLMENU	"GUI-OPTION: FullMenu enab
SAP_CP / SAP_CODEPAGE	"GUI-OPTION: SAP_CODEPAG
UPDOWNLOAD_CP	"GUI-OPTION: UPDOWNLOAD
SNC_PARTNERNAME	"GUI-OPTION: SNC name '%s'
SNC_QOP	"GUI-OPTION: SNC_QOP '%s'"
SNC_LIB	"GUI-OPTION: SNC is set to: %
SAPGUI_INPLACE	"GUI-OPTION: environment va

:

```
.text:6440EE00                 lea     edi, ↙
    ↘ [ebp+2884h+var_2884] ; options here ↙
    ↘ like +0x15...
.text:6440EE03                 lea     ecx, ↙
    ↘ [esi+24h]
.text:6440EE06                 call    ↙
    ↘ load_command_line
.text:6440EE0B                 mov     edi, ↙
```

```
    ↳ eax
.text:6440EE0D                  xor     ebx, ↙
    ↳ ebx
.text:6440EE0F                  cmp     edi, ↙
    ↳ ebx
.text:6440EE11                  jz      short↙
    ↳  loc_6440EE42
.text:6440EE13                  push    edi
.text:6440EE14                  push    ↙
    ↳ offset aSapguiStoppedA ; "Sapgui ↙
    ↳ stopped after commandline interp"...
.text:6440EE19                  push    ↙
    ↳ dword_644F93E8
.text:6440EE1F                  call    ↙
    ↳ FEWTraceError
```

Sí, y la única referencia está en `CDwsGui::PrepareInfoWindow()`.:

```
.text:64405160                  push    dword↙
    ↳  ptr [esi+2854h]
.text:64405166                  push    ↙
    ↳ offset aCdwsguiPrepare ; "\nCDwsGui::↙
    ↳ PrepareInfoWindow: sapgui env"...
.text:6440516B                  push    dword↙
    ↳  ptr [esi+2848h]
.text:64405171                  call    dbg
.text:64405176                  add     esp, ↙
    ↳ 0Ch
```

...o:

```
.text:6440237A                  push    eax
.text:6440237B                  push    ⤦
    ↳ offset aCclientStart_6 ; "CClient::⤦
    ↳ Start: set shortcut user to '\%"...
.text:64402380                  push    dword⤦
    ↳  ptr [edi+4]
.text:64402383                  call    dbg
.text:64402388                  add     esp, ⤦
    ↳ 0Ch
```

.

:

```
.text:64404F4F CDwsGui__PrepareInfoWindow ⤦
    ↳ proc near
.text:64404F4F
.text:64404F4F pvParam          = byte ptr -3⤦
    ↳ Ch
.text:64404F4F var_38           = dword ptr ⤦
    ↳ -38h
.text:64404F4F var_34           = dword ptr ⤦
    ↳ -34h
.text:64404F4F rc               = tagRECT ptr⤦
    ↳  -2Ch
.text:64404F4F cy               = dword ptr ⤦
    ↳ -1Ch
.text:64404F4F h                = dword ptr ⤦
    ↳ -18h
.text:64404F4F var_14           = dword ptr ⤦
    ↳ -14h
.text:64404F4F var_10           = dword ptr ⤦
    ↳ -10h
```

```
.text:64404F4F var_4                = dword ptr ↙
    ↘ -4
.text:64404F4F
.text:64404F4F                        push    30h
.text:64404F51                        mov     eax, ↙
    ↘ offset loc_64438E00
.text:64404F56                        call    ↙
    ↘ __EH_prolog3
.text:64404F5B                        mov     esi, ↙
    ↘ ecx          ; ECX is pointer to object
.text:64404F5D                        xor     ebx, ↙
    ↘ ebx
.text:64404F5F                        lea     ecx, ↙
    ↘ [ebp+var_14]
.text:64404F62                        mov     [ebp+↙
    ↘ var_10], ebx

; demangled name: ATL::CStringT(void)
.text:64404F65                        call    ds:↙
    ↘ mfc90_316
.text:64404F6B                        mov     [ebp+↙
    ↘ var_4], ebx
.text:64404F6E                        lea     edi, ↙
    ↘ [esi+2854h]
.text:64404F74                        push    ↙
    ↘ offset aEnvironmentInf ; "Environment ↙
    ↘ information:\n"
.text:64404F79                        mov     ecx, ↙
    ↘ edi

; demangled name: ATL::CStringT::operator=(↙
    ↘ char const *)
.text:64404F7B                        call    ds:↙
```

```
    ↳ mfc90_820
.text:64404F81                 cmp     [esi↙
    ↳ +38h], ebx
.text:64404F84                 mov     ebx, ↙
    ↳ ds:mfc90_2539
.text:64404F8A                 jbe     short↙
    ↳  loc_64404FA9
.text:64404F8C                 push    dword↙
    ↳  ptr [esi+34h]
.text:64404F8F                 lea     eax, ↙
    ↳ [ebp+var_14]
.text:64404F92                 push    ↙
    ↳ offset aWorkingDirecto ; "working ↙
    ↳ directory: '\%s'\n"
.text:64404F97                 push    eax

; demangled name: ATL::CStringT::Format(char↙
    ↳  const *,...)
.text:64404F98                 call    ebx ;↙
    ↳  mfc90_2539
.text:64404F9A                 add     esp, ↙
    ↳ 0Ch
.text:64404F9D                 lea     eax, ↙
    ↳ [ebp+var_14]
.text:64404FA0                 push    eax
.text:64404FA1                 mov     ecx, ↙
    ↳ edi

; demangled name: ATL::CStringT::operator+=(↙
    ↳ class ATL::CSimpleStringT<char, 1> ↙
    ↳ const &)
.text:64404FA3                 call    ds:↙
    ↳ mfc90_941
```

```
.text:64404FA9
.text:64404FA9 loc_64404FA9:
.text:64404FA9                 mov     eax, ↙
    ↘ [esi+38h]
.text:64404FAC                 test    eax, ↙
    ↘ eax
.text:64404FAE                 jbe     short↙
    ↘  loc_64404FD3
.text:64404FB0                 push    eax
.text:64404FB1                 lea     eax, ↙
    ↘ [ebp+var_14]
.text:64404FB4                 push    ↙
    ↘ offset aTraceLevelDAct ; "trace level ↙
    ↘ \%d activated\n"
.text:64404FB9                 push    eax

; demangled name: ATL::CStringT::Format(char↙
    ↘  const *,...)
.text:64404FBA                 call    ebx ;↙
    ↘  mfc90_2539
.text:64404FBC                 add     esp, ↙
    ↘ 0Ch
.text:64404FBF                 lea     eax, ↙
    ↘ [ebp+var_14]
.text:64404FC2                 push    eax
.text:64404FC3                 mov     ecx, ↙
    ↘ edi

; demangled name: ATL::CStringT::operator+=(↙
    ↘ class ATL::CSimpleStringT<char, 1> ↙
    ↘ const &)
.text:64404FC5                 call    ds:↙
    ↘ mfc90_941
```

```
.text:64404FCB                   xor     ebx, ↙
    ↘ ebx
.text:64404FCD                   inc     ebx
.text:64404FCE                   mov     [ebp+↙
    ↘ var_10], ebx
.text:64404FD1                   jmp     short↙
    ↘ loc_64404FD6
.text:64404FD3
.text:64404FD3 loc_64404FD3:
.text:64404FD3                   xor     ebx, ↙
    ↘ ebx
.text:64404FD5                   inc     ebx
.text:64404FD6
.text:64404FD6 loc_64404FD6:
.text:64404FD6                   cmp     [esi↙
    ↘ +38h], ebx
.text:64404FD9                   jbe     short↙
    ↘ loc_64404FF1
.text:64404FDB                   cmp     dword↙
    ↘ ptr [esi+2978h], 0
.text:64404FE2                   jz      short↙
    ↘ loc_64404FF1
.text:64404FE4                   push    ↙
    ↘ offset aHexdumpInTrace ; "hexdump in ↙
    ↘ trace activated\n"
.text:64404FE9                   mov     ecx, ↙
    ↘ edi

; demangled name: ATL::CStringT::operator+=(↙
    ↘ char const *)
.text:64404FEB                   call    ds:↙
    ↘ mfc90_945
.text:64404FF1
```

```
.text:64404FF1 loc_64404FF1:
.text:64404FF1
.text:64404FF1                 cmp     byte ↙
    ↘ ptr [esi+78h], 0
.text:64404FF5                 jz      short↙
    ↘  loc_64405007
.text:64404FF7                 push    ↙
    ↘ offset aLoggingActivat ; "logging ↙
    ↘ activated\n"
.text:64404FFC                 mov     ecx, ↙
    ↘ edi

; demangled name: ATL::CStringT::operator+=(↙
    ↘ char const *)
.text:64404FFE                 call    ds:↙
    ↘ mfc90_945
.text:64405004                 mov     [ebp+↙
    ↘ var_10], ebx
.text:64405007
.text:64405007 loc_64405007:
.text:64405007                 cmp     byte ↙
    ↘ ptr [esi+3Dh], 0
.text:6440500B                 jz      short↙
    ↘  bypass
.text:6440500D                 push    ↙
    ↘ offset aDataCompressio ; "data ↙
    ↘ compression switched off\n"
.text:64405012                 mov     ecx, ↙
    ↘ edi

; demangled name: ATL::CStringT::operator+=(↙
    ↘ char const *)
.text:64405014                 call    ds:↙
```

```
    ↘ mfc90_945
.text:6440501A                     mov     [ebp+↙
    ↘ var_10], ebx
.text:6440501D
.text:6440501D bypass:
.text:6440501D                     mov     eax, ↙
    ↘ [esi+20h]
.text:64405020                     test    eax, ↙
    ↘ eax
.text:64405022                     jz      short↙
    ↘  loc_6440503A
.text:64405024                     cmp     dword↙
    ↘  ptr [eax+28h], 0
.text:64405028                     jz      short↙
    ↘  loc_6440503A
.text:6440502A                     push    ↙
    ↘ offset aDataRecordMode ; "data record ↙
    ↘ mode switched on\n"
.text:6440502F                     mov     ecx, ↙
    ↘ edi

; demangled name: ATL::CStringT::operator+=(↙
    ↘ char const *)
.text:64405031                     call    ds:↙
    ↘ mfc90_945
.text:64405037                     mov     [ebp+↙
    ↘ var_10], ebx
.text:6440503A
.text:6440503A loc_6440503A:
.text:6440503A
.text:6440503A                     mov     ecx, ↙
    ↘ edi
.text:6440503C                     cmp     [ebp+↙
```

```
    ↳ var_10], ebx
.text:6440503F                   jnz      ↙
    ↳ loc_64405142
.text:64405045                   push     ↙
    ↳ offset aForMaximumData ; "\nFor ↙
    ↳ maximum data security delete\nthe s↙
    ↳ "...

; demangled name: ATL::CStringT::operator+=(↙
    ↳ char const *)
.text:6440504A                   call    ds:↙
    ↳ mfc90_945
.text:64405050                   xor     edi, ↙
    ↳ edi
.text:64405052                   push    edi  ↙
    ↳           ; fWinIni
.text:64405053                   lea     eax, ↙
    ↳ [ebp+pvParam]
.text:64405056                   push    eax  ↙
    ↳           ; pvParam
.text:64405057                   push    edi  ↙
    ↳           ; uiParam
.text:64405058                   push    30h  ↙
    ↳           ; uiAction
.text:6440505A                   call    ds:↙
    ↳ SystemParametersInfoA
.text:64405060                   mov     eax, ↙
    ↳ [ebp+var_34]
.text:64405063                   cmp     eax, ↙
    ↳ 1600
.text:64405068                   jle     short↙
    ↳  loc_64405072
.text:6440506A                   cdq
```

```
.text:6440506B                 sub     eax, ↙
    ↘ edx
.text:6440506D                 sar     eax, ↙
    ↘ 1
.text:6440506F                 mov     [ebp+↙
    ↘ var_34], eax
.text:64405072
.text:64405072 loc_64405072:
.text:64405072                 push    edi   ↙
    ↘            ; hWnd
.text:64405073                 mov     [ebp+↙
    ↘ cy], 0A0h
.text:6440507A                 call    ds:↙
    ↘ GetDC
.text:64405080                 mov     [ebp+↙
    ↘ var_10], eax
.text:64405083                 mov     ebx, ↙
    ↘ 12Ch
.text:64405088                 cmp     eax, ↙
    ↘ edi
.text:6440508A                 jz      ↙
    ↘ loc_64405113
.text:64405090                 push    11h   ↙
    ↘            ; i
.text:64405092                 call    ds:↙
    ↘ GetStockObject
.text:64405098                 mov     edi, ↙
    ↘ ds:SelectObject
.text:6440509E                 push    eax   ↙
    ↘            ; h
.text:6440509F                 push    [ebp+↙
    ↘ var_10]    ; hdc
.text:644050A2                 call    edi ;↙
```

```
    ↳  SelectObject
.text:644050A4                  and     [ebp+↙
    ↳ rc.left], 0
.text:644050A8                  and     [ebp+↙
    ↳ rc.top], 0
.text:644050AC                  mov     [ebp+↙
    ↳ h], eax
.text:644050AF                  push    401h ↙
    ↳           ; format
.text:644050B4                  lea     eax, ↙
    ↳ [ebp+rc]
.text:644050B7                  push    eax  ↙
    ↳           ; lprc
.text:644050B8                  lea     ecx, ↙
    ↳ [esi+2854h]
.text:644050BE                  mov     [ebp+↙
    ↳ rc.right], ebx
.text:644050C1                  mov     [ebp+↙
    ↳ rc.bottom], 0B4h

; demangled name: ATL::CSimpleStringT::↙
    ↳ GetLength(void)
.text:644050C8                  call    ds:↙
    ↳ mfc90_3178
.text:644050CE                  push    eax  ↙
    ↳           ; cchText
.text:644050CF                  lea     ecx, ↙
    ↳ [esi+2854h]

; demangled name: const char* ATL::↙
    ↳ CSimpleStringT::operator PCXSTR
.text:644050D5                  call    ds:↙
    ↳ mfc90_910
```

```
.text:644050DB                 push    eax            ↙
    ↘            ; lpchText
.text:644050DC                 push    [ebp+↙
    ↘ var_10]    ; hdc
.text:644050DF                 call    ds:↙
    ↘ DrawTextA
.text:644050E5                 push    4              ↙
    ↘            ; nIndex
.text:644050E7                 call    ds:↙
    ↘ GetSystemMetrics
.text:644050ED                 mov     ecx, ↙
    ↘ [ebp+rc.bottom]
.text:644050F0                 sub     ecx, ↙
    ↘ [ebp+rc.top]
.text:644050F3                 cmp     [ebp+↙
    ↘ h], 0
.text:644050F7                 lea     eax, ↙
    ↘ [eax+ecx+28h]
.text:644050FB                 mov     [ebp+↙
    ↘ cy], eax
.text:644050FE                 jz      short↙
    ↘  loc_64405108
.text:64405100                 push    [ebp+↙
    ↘ h]         ; h
.text:64405103                 push    [ebp+↙
    ↘ var_10]    ; hdc
.text:64405106                 call    edi ;↙
    ↘  SelectObject
.text:64405108
.text:64405108 loc_64405108:
.text:64405108                 push    [ebp+↙
    ↘ var_10]    ; hDC
.text:6440510B                 push    0              ↙
```

```
    ↳              ; hWnd
.text:6440510D                 call    ds:↙
    ↳ ReleaseDC
.text:64405113
.text:64405113 loc_64405113:
.text:64405113                 mov     eax, ↙
    ↳ [ebp+var_38]
.text:64405116                 push    80h  ↙
    ↳              ; uFlags
.text:6440511B                 push    [ebp+↙
    ↳ cy]          ; cy
.text:6440511E                 inc     eax
.text:6440511F                 push    ebx  ↙
    ↳              ; cx
.text:64405120                 push    eax  ↙
    ↳              ; Y
.text:64405121                 mov     eax, ↙
    ↳ [ebp+var_34]
.text:64405124                 add     eax, ↙
    ↳ 0FFFFFED4h
.text:64405129                 cdq
.text:6440512A                 sub     eax, ↙
    ↳ edx
.text:6440512C                 sar     eax, ↙
    ↳ 1
.text:6440512E                 push    eax  ↙
    ↳              ; X
.text:6440512F                 push    0    ↙
    ↳              ; hWndInsertAfter
.text:64405131                 push    dword↙
    ↳  ptr [esi+285Ch] ; hWnd
.text:64405137                 call    ds:↙
    ↳ SetWindowPos
```

```
.text:6440513D                 xor     ebx, ⤦
    ↳ ebx
.text:6440513F                 inc     ebx
.text:64405140                 jmp     short⤦
    ↳  loc_6440514D
.text:64405142
.text:64405142 loc_64405142:
.text:64405142                 push    ⤦
    ↳ offset byte_64443AF8

; demangled name: ATL::CStringT::operator=(⤦
    ↳ char const *)
.text:64405147                 call    ds:⤦
    ↳ mfc90_820
.text:6440514D
.text:6440514D loc_6440514D:
.text:6440514D                 cmp     ⤦
    ↳ dword_6450B970, ebx
.text:64405153                 jl      short⤦
    ↳  loc_64405188
.text:64405155                 call    ⤦
    ↳ sub_6441C910
.text:6440515A                 mov     ⤦
    ↳ dword_644F858C, ebx
.text:64405160                 push    dword⤦
    ↳  ptr [esi+2854h]
.text:64405166                 push    ⤦
    ↳ offset aCdwsguiPrepare ; "\nCDwsGui::⤦
    ↳ PrepareInfoWindow: sapgui env"...
.text:6440516B                 push    dword⤦
    ↳  ptr [esi+2848h]
.text:64405171                 call    dbg
.text:64405176                 add     esp, ⤦
```

```
    ↳ 0Ch
.text:64405179                  mov     ↙
    ↳ dword_644F858C, 2
.text:64405183                  call    ↙
    ↳ sub_6441C920
.text:64405188
.text:64405188 loc_64405188:
.text:64405188                  or      [ebp+↙
    ↳ var_4], 0FFFFFFFFh
.text:6440518C                  lea     ecx, ↙
    ↳ [ebp+var_14]

; demangled name: ATL::CStringT::~CStringT()
.text:6440518F                  call    ds:↙
    ↳ mfc90_601
.text:64405195                  call    ↙
    ↳ __EH_epilog3
.text:6440519A                  retn
.text:6440519A CDwsGui__PrepareInfoWindow ↙
    ↳ endp
```

:

```
.text:64405007 loc_64405007:
.text:64405007                  cmp     byte ↙
    ↳ ptr [esi+3Dh], 0
.text:6440500B                  jz      short↙
    ↳  bypass
.text:6440500D                  push    ↙
    ↳ offset aDataCompressio ; "data ↙
    ↳ compression switched off\n"
.text:64405012                  mov     ecx, ↙
    ↳ edi
```

```
; demangled name: ATL::CStringT::operator+=(
    char const *)
.text:64405014                  call    ds:
    mfc90_945
.text:6440501A                  mov     [ebp+
    var_10], ebx
.text:6440501D
.text:6440501D bypass:
```

:

```
.text:6440503C                  cmp     [ebp+
    var_10], ebx
.text:6440503F                  jnz     exit 
    ;

;  "For maximum data security delete" / "the
     setting(s) as soon as possible !":

.text:64405045                  push    
    offset aForMaximumData ; "\nFor 
    maximum data security delete\nthe s
    "...
.text:6440504A                  call    ds:
    mfc90_945 ; ATL::CStringT::operator+=(
    char const *)
.text:64405050                  xor     edi, 
    edi
.text:64405052                  push    edi 
                  ; fWinIni
.text:64405053                  lea     eax, 
```

```
    ↳ [ebp+pvParam]
.text:64405056                  push    eax   ↙
    ↳           ; pvParam
.text:64405057                  push    edi   ↙
    ↳           ; uiParam
.text:64405058                  push    30h   ↙
    ↳           ; uiAction
.text:6440505A                  call    ds:↙
    ↳ SystemParametersInfoA
.text:64405060                  mov     eax, ↙
    ↳ [ebp+var_34]
.text:64405063                  cmp     eax, ↙
    ↳ 1600
.text:64405068                  jle     short↙
    ↳  loc_64405072
.text:6440506A                  cdq
.text:6440506B                  sub     eax, ↙
    ↳ edx
.text:6440506D                  sar     eax, ↙
    ↳ 1
.text:6440506F                  mov     [ebp+↙
    ↳ var_34], eax
.text:64405072
.text:64405072 loc_64405072:

:

.text:64405072                  push    edi   ↙
    ↳           ; hWnd
.text:64405073                  mov     [ebp+↙
    ↳ cy], 0A0h
.text:6440507A                  call    ds:↙
    ↳ GetDC
```

.

JNZ ...

```
.text:6440503F                    jnz     exit ;
```

...

```
.text:64404C19 sub_64404C19       proc near
.text:64404C19
.text:64404C19 arg_0              = dword ptr  4
.text:64404C19
.text:64404C19                    push    ebx
.text:64404C1A                    push    ebp
.text:64404C1B                    push    esi
.text:64404C1C                    push    edi
.text:64404C1D                    mov     edi, [esp+10h+arg_0]
.text:64404C21                    mov     eax, [edi]
.text:64404C23                    mov     esi, ecx ; ESI/ECX are pointers to some unknown object.
.text:64404C25                    mov     [esi], eax
.text:64404C27                    mov     eax, [edi+4]
.text:64404C2A                    mov     [esi+4], eax
```

```
.text:64404C2D                 mov     eax, ↙
    ↘ [edi+8]
.text:64404C30                 mov     [esi↙
    ↘ +8], eax
.text:64404C33                 lea     eax, ↙
    ↘ [edi+0Ch]
.text:64404C36                 push    eax
.text:64404C37                 lea     ecx, ↙
    ↘ [esi+0Ch]

; demangled name:  ATL::CStringT::operator=(↙
    ↘ class ATL::CStringT ... &)
.text:64404C3A                 call    ds:↙
    ↘ mfc90_817
.text:64404C40                 mov     eax, ↙
    ↘ [edi+10h]
.text:64404C43                 mov     [esi↙
    ↘ +10h], eax
.text:64404C46                 mov     al, [↙
    ↘ edi+14h]
.text:64404C49                 mov     [esi↙
    ↘ +14h], al
.text:64404C4C                 mov     al, [↙
    ↘ edi+15h] ; copy byte from 0x15 offset
.text:64404C4F                 mov     [esi↙
    ↘ +15h], al ; to 0x15 offset in CDwsGui ↙
    ↘ object
```

CDwsGui::Init():

```
.text:6440B0BF loc_6440B0BF:
.text:6440B0BF                 mov     eax, ↙
    ↘ [ebp+arg_0]
```

```
.text:6440B0C2                 push    [ebp+⤦
    ↳ arg_4]
.text:6440B0C5                 mov     [esi⤦
    ↳ +2844h], eax
.text:6440B0CB                 lea     eax, ⤦
    ↳ [esi+28h] ; ESI is pointer to CDwsGui ⤦
    ↳ object
.text:6440B0CE                 push    eax
.text:6440B0CF                 call    ⤦
    ↳ CDwsGui__CopyOptions
```

```
.text:64409D58                 cmp     [esi⤦
    ↳ +3Dh], bl   ; ESI is pointer to ⤦
    ↳ CDwsGui object
.text:64409D5B                 lea     ecx, ⤦
    ↳ [esi+2B8h]
.text:64409D61                 setz    al
.text:64409D64                 push    eax ⤦
    ↳           ; arg_10 of ⤦
    ↳ CConnectionContext::CreateNetwork
.text:64409D65                 push    dword⤦
    ↳  ptr [esi+64h]

; demangled name: const char* ATL::⤦
    ↳ CSimpleStringT::operator PCXSTR
.text:64409D68                 call    ds:⤦
    ↳ mfc90_910
.text:64409D68                         ⤦
    ↳           ; no arguments
.text:64409D6E                 push    eax
.text:64409D6F                 lea     ecx, ⤦
    ↳ [esi+2BCh]
```

```
; demangled name: const char* ATL::⤦
    ↳ CSimpleStringT::operator PCXSTR
.text:64409D75                  call    ds:⤦
    ↳ mfc90_910
.text:64409D75                                  ⤦
    ↳           ; no arguments
.text:64409D7B                  push    eax
.text:64409D7C                  push    esi
.text:64409D7D                  lea     ecx, ⤦
    ↳ [esi+8]
.text:64409D80                  call    ⤦
    ↳ CConnectionContext__CreateNetwork
```

```
...
.text:64403476                  push    [ebp+⤦
    ↳ compression]
.text:64403479                  push    [ebp+⤦
    ↳ arg_C]
.text:6440347C                  push    [ebp+⤦
    ↳ arg_8]
.text:6440347F                  push    [ebp+⤦
    ↳ arg_4]
.text:64403482                  push    [ebp+⤦
    ↳ arg_0]
.text:64403485                  call    ⤦
    ↳ CNetwork__CNetwork
```

```
.text:64411DF1                  cmp     [ebp+⤦
    ↳ compression], esi
.text:64411DF7                  jz      short⤦
    ↳  set_EAX_to_0
```

```
.text:64411DF9                 mov     al, [⤦
    ↳ ebx+78h]   ; another value may affect ⤦
    ↳ compression?
.text:64411DFC                 cmp     al, ⤦
    ↳ '3'
.text:64411DFE                 jz      short⤦
    ↳  set_EAX_to_1
.text:64411E00                 cmp     al, ⤦
    ↳ '4'
.text:64411E02                 jnz     short⤦
    ↳  set_EAX_to_0
.text:64411E04
.text:64411E04 set_EAX_to_1:
.text:64411E04                 xor     eax, ⤦
    ↳ eax
.text:64411E06                 inc     eax  ⤦
    ↳            ; EAX -> 1
.text:64411E07                 jmp     short⤦
    ↳  loc_64411E0B
.text:64411E09
.text:64411E09 set_EAX_to_0:
.text:64411E09
.text:64411E09                 xor     eax, ⤦
    ↳ eax        ; EAX -> 0
.text:64411E0B
.text:64411E0B loc_64411E0B:
.text:64411E0B                 mov     [ebx⤦
    ↳ +3A4h], eax ; EBX is pointer to ⤦
    ↳ CNetwork object
```

```
.text:64406F76 loc_64406F76:
.text:64406F76                 mov     ecx, ⤦
```

```
    ↘ [ebp+7728h+var_7794]
.text:64406F79                 cmp      dword↙
    ↘  ptr [ecx+3A4h], 1
.text:64406F80                 jnz      ↙
    ↘ compression_flag_is_zero
.text:64406F86                 mov      byte ↙
    ↘ ptr [ebx+7], 1
.text:64406F8A                 mov      eax, ↙
    ↘ [esi+18h]
.text:64406F8D                 mov      ecx, ↙
    ↘ eax
.text:64406F8F                 test     eax, ↙
    ↘ eax
.text:64406F91                 ja       short↙
    ↘  loc_64406FFF
.text:64406F93                 mov      ecx, ↙
    ↘ [esi+14h]
.text:64406F96                 mov      eax, ↙
    ↘ [esi+20h]
.text:64406F99
.text:64406F99 loc_64406F99:
.text:64406F99                 push     dword↙
    ↘  ptr [edi+2868h] ; int
.text:64406F9F                 lea      edx, ↙
    ↘ [ebp+7728h+var_77A4]
.text:64406FA2                 push     edx  ↙
    ↘           ; int
.text:64406FA3                 push     30000↙
    ↘           ; int
.text:64406FA8                 lea      edx, ↙
    ↘ [ebp+7728h+Dst]
.text:64406FAB                 push     edx  ↙
    ↘           ; Dst
```

```
.text:64406FAC                  push    ecx  ↙
    ↘           ; int
.text:64406FAD                  push    eax  ↙
    ↘           ; Src
.text:64406FAE                  push    dword↙
    ↘  ptr [edi+28C0h] ; int
.text:64406FB4                  call    ↙
    ↘ sub_644055C5       ; actual ↙
    ↘ compression routine
.text:64406FB9                  add     esp, ↙
    ↘ 1Ch
.text:64406FBC                  cmp     eax, ↙
    ↘ 0FFFFFFF6h
.text:64406FBF                  jz      short↙
    ↘  loc_64407004
.text:64406FC1                  cmp     eax, ↙
    ↘ 1
.text:64406FC4                  jz      ↙
    ↘ loc_6440708C
.text:64406FCA                  cmp     eax, ↙
    ↘ 2
.text:64406FCD                  jz      short↙
    ↘  loc_64407004
.text:64406FCF                  push    eax
.text:64406FD0                  push    ↙
    ↘ offset aCompressionErr ; "compression ↙
    ↘ error [rc = \%d]- program wi"...
.text:64406FD5                  push    ↙
    ↘ offset aGui_err_compre ; "↙
    ↘ GUI_ERR_COMPRESS"
.text:64406FDA                  push    dword↙
    ↘  ptr [edi+28D0h]
.text:64406FE0                  call    ↙
```

```
    ↳ SapPcTxtRead
```

```
.text:6441747C                  push    ↙
    ↳ offset aErrorCsrcompre ; "\nERROR: ↙
    ↳ CsRCompress: invalid handle"
.text:64417481                  call    eax ;↙
    ↳  dword_644F94C8
.text:64417483                  add     esp, ↙
    ↳ 4
```

Voilà! [4],

```
.text:64406F79                  cmp     dword↙
    ↳  ptr [ecx+3A4h], 1
.text:64406F80                  jnz     ↙
    ↳ compression_flag_is_zero
```

80.2

«Password logon no longer possible - too many failed attempts», .

TYPEINFODUMP[5] .

```
FUNCTION ThVmcSysEvent
  Address:        10143190  Size:       675 ↙
    ↳ bytes  Index:    60483  TypeIndex:    ↙
    ↳ 60484
```

[4] http://go.yurichev.com/17312
[5] http://go.yurichev.com/17038

```
  Type: int NEAR_C ThVmcSysEvent (unsigned ⤦
    ↳ int, unsigned char, unsigned short*)
Flags: 0
PARAMETER events
  Address: Reg335+288  Size:        4 bytes ⤦
    ↳  Index:    60488  TypeIndex:    60489
  Type: unsigned int
Flags: d0
PARAMETER opcode
  Address: Reg335+296  Size:        1 bytes ⤦
    ↳  Index:    60490  TypeIndex:    60491
  Type: unsigned char
Flags: d0
PARAMETER serverName
  Address: Reg335+304  Size:        8 bytes ⤦
    ↳  Index:    60492  TypeIndex:    60493
  Type: unsigned short*
Flags: d0
STATIC_LOCAL_VAR func
  Address:         12274af0  Size:        8 ⤦
    ↳ bytes  Index:    60495  TypeIndex:   ⤦
    ↳ 60496
  Type: wchar_t*
Flags: 80
LOCAL_VAR admhead
  Address: Reg335+304  Size:        8 bytes ⤦
    ↳  Index:    60498  TypeIndex:    60499
  Type: unsigned char*
Flags: 90
LOCAL_VAR record
  Address: Reg335+64  Size:      204 bytes  ⤦
    ↳ Index:    60501  TypeIndex:    60502
  Type: AD_RECORD
```

```
Flags: 90
LOCAL_VAR adlen
  Address: Reg335+296  Size:        4 bytes ↙
    ↳  Index:    60508  TypeIndex:    60509
  Type: int
Flags: 90
```

```
STRUCT DBSL_STMTID
Size: 120  Variables: 4  Functions: 0  Base ↙
    ↳ classes: 0
MEMBER moduletype
  Type:  DBSL_MODULETYPE
  Offset:        0  Index:        3  ↙
    ↳ TypeIndex:    38653
MEMBER module
  Type:  wchar_t module[40]
  Offset:        4  Index:        3  ↙
    ↳ TypeIndex:      831
MEMBER stmtnum
  Type:  long
  Offset:       84  Index:        3  ↙
    ↳ TypeIndex:      440
MEMBER timestamp
  Type:  wchar_t timestamp[15]
  Offset:       88  Index:        3  ↙
    ↳ TypeIndex:     6612
```

[6], .

:

[6]: http://go.yurichev.com/17039

```
cmp     cs:ct_level, 1
jl      short loc_1400375DA
call    DpLock
lea     rcx, aDpxxtool4_c ; "dpxxtool4.c"
mov     edx, 4Eh         ; line
call    CTrcSaveLocation
mov     r8, cs:func_48
mov     rcx, cs:hdl      ; hdl
lea     rdx, aSDpreadmemvalu ; "%s: ⤦
    ↳ DpReadMemValue (%d)"
mov     r9d, ebx
call    DpTrcErr
call    DpUnlock
```

```
cat "disp+work.pdb.d" | grep FUNCTION | grep⤦
    ↳  -i password
```

```
FUNCTION rcui::AgiPassword::DiagISelection
FUNCTION ssf_password_encrypt
FUNCTION ssf_password_decrypt
FUNCTION password_logon_disabled
FUNCTION dySignSkipUserPassword
FUNCTION migrate_password_history
FUNCTION password_is_initial
FUNCTION rcui::AgiPassword::IsVisible
FUNCTION password_distance_ok
FUNCTION ⤦
    ↳ get_password_downwards_compatibility
FUNCTION dySignUnSkipUserPassword
FUNCTION rcui::AgiPassword::GetTypeName
FUNCTION `rcui::AgiPassword::AgiPassword⤦
    ↳ '::`1'::dtor$2
```

```
FUNCTION `rcui::AgiPassword::AgiPassword↙
    ↳ '::`1'::dtor$0
FUNCTION `rcui::AgiPassword::AgiPassword↙
    ↳ '::`1'::dtor$1
FUNCTION usm_set_password
FUNCTION rcui::AgiPassword::TraceTo
FUNCTION days_since_last_password_change
FUNCTION rsecgrp_generate_random_password
FUNCTION rcui::AgiPassword::`scalar deleting↙
    ↳  destructor'
FUNCTION password_attempt_limit_exceeded
FUNCTION handle_incorrect_password
FUNCTION `rcui::AgiPassword::`scalar ↙
    ↳ deleting destructor''::`1'::dtor$1
FUNCTION calculate_new_password_hash
FUNCTION shift_password_to_history
FUNCTION rcui::AgiPassword::GetType
FUNCTION found_password_in_history
FUNCTION `rcui::AgiPassword::`scalar ↙
    ↳ deleting destructor''::`1'::dtor$0
FUNCTION rcui::AgiObj::IsaPassword
FUNCTION password_idle_check
FUNCTION SlicHwPasswordForDay
FUNCTION rcui::AgiPassword::IsaPassword
FUNCTION rcui::AgiPassword::AgiPassword
FUNCTION delete_user_password
FUNCTION usm_set_user_password
FUNCTION Password_API
FUNCTION get_password_change_for_SSO
FUNCTION password_in_USR40
FUNCTION ↙
    ↳ rsec_agrp_abap_generate_random_password↙
    ↳
```

«user was locked by subsequently failed password logon attempts»
password_attempt_limit_exceeded().

: *«password logon attempt will be rejected immediately (preventing dictionary attacks)»*, *«failed-logon lock: expired (but not removed due to 'read-only' operation)»*, *«failed-logon lock: expired => removed»*.

. .

:

tracer:

```
tracer64.exe -a:disp+work.exe bpf=disp+work.exe!chckpass,args:3,unicode
```

```
PID=2236|TID=2248|(0) disp+work.exe!chckpass (0x202c770, L"Brewered1                               ", 0x41) (called from 0x1402f1060 (disp+work.exe!usrexist+0x3c0))
PID=2236|TID=2248|(0) disp+work.exe!chckpass -> 0x35
```

: *syssigni() -> DylSigni() -> dychkusr() -> usrexist() -> chckpass()*.

0x35 *chckpass()* :

```
.text:00000001402ED567 loc_1402ED567:                  ↙
    ↘                           ; CODE XREF: ↙
    ↘ chckpass+B4
.text:00000001402ED567                          mov   ↙
    ↘     rcx, rbx          ; usr02
.text:00000001402ED56A                          call  ↙
    ↘     password_idle_check
.text:00000001402ED56F                          cmp   ↙
    ↘     eax, 33h
.text:00000001402ED572                          jz    ↙
    ↘     loc_1402EDB4E
.text:00000001402ED578                          cmp   ↙
    ↘     eax, 36h
.text:00000001402ED57B                          jz    ↙
    ↘     loc_1402EDB3D
.text:00000001402ED581                          xor   ↙
    ↘     edx, edx          ; usr02_readonly
.text:00000001402ED583                          mov   ↙
    ↘     rcx, rbx          ; usr02
.text:00000001402ED586                          call  ↙
    ↘     password_attempt_limit_exceeded
.text:00000001402ED58B                          test  ↙
    ↘     al, al
.text:00000001402ED58D                          jz    ↙
    ↘     short loc_1402ED5A0
.text:00000001402ED58F                          mov   ↙
    ↘     eax, 35h
.text:00000001402ED594                          add   ↙
    ↘     rsp, 60h
.text:00000001402ED598                          pop   ↙
    ↘     r14
.text:00000001402ED59A                          pop   ↙
```

```
    ↳    r12
.text:00000001402ED59C                        pop  ↙
    ↳    rdi
.text:00000001402ED59D                        pop  ↙
    ↳    rsi
.text:00000001402ED59E                        pop  ↙
    ↳    rbx
.text:00000001402ED59F                        retn
```

:

```
tracer64.exe -a:disp+work.exe bpf=disp+work.↙
    ↳ exe!password_attempt_limit_exceeded,↙
    ↳ args:4,unicode,rt:0
```

```
PID=2744|TID=360|(0) disp+work.exe!↙
    ↳ password_attempt_limit_exceeded (0↙
    ↳ x202c770, 0, 0x257758, 0) (called from↙
    ↳  0x1402ed58b (disp+work.exe!chckpass+0↙
    ↳ xeb))
PID=2744|TID=360|(0) disp+work.exe!↙
    ↳ password_attempt_limit_exceeded -> 1
PID=2744|TID=360|We modify return value (EAX↙
    ↳ /RAX) of this function to 0
PID=2744|TID=360|(0) disp+work.exe!↙
    ↳ password_attempt_limit_exceeded (0↙
    ↳ x202c770, 0, 0, 0) (called from 0↙
    ↳ x1402e9794 (disp+work.exe!chngpass+0↙
    ↳ xe4))
PID=2744|TID=360|(0) disp+work.exe!↙
    ↳ password_attempt_limit_exceeded -> 1
PID=2744|TID=360|We modify return value (EAX↙
    ↳ /RAX) of this function to 0
```

```
tracer64.exe -a:disp+work.exe bpf=disp+work.
    ↳ exe!chckpass,args:3,unicode,rt:0
```

```
PID=2744|TID=360|(0) disp+work.exe!chckpass
    ↳ (0x202c770, L"bogus
    ↳                    ", 0x41) (called from
    ↳ 0x1402f1060 (disp+work.exe!usrexist+0
    ↳ x3c0))
PID=2744|TID=360|(0) disp+work.exe!chckpass
    ↳ -> 0x35
PID=2744|TID=360|We modify return value (EAX
    ↳ /RAX) of this function to 0
```

```
lea     rcx, aLoginFailed_us ; "login/
    ↳ failed_user_auto_unlock"
call    sapgparam
test    rax, rax
jz      short loc_1402E19DE
movzx   eax, word ptr [rax]
cmp     ax, 'N'
jz      short loc_1402E19D4
cmp     ax, 'n'
jz      short loc_1402E19D4
cmp     ax, '0'
jnz     short loc_1402E19DE
```

Capítulo 81

Oracle RDBMS

81.1

```
SQL> select * from V$VERSION;
```

:

```
BANNER
--------------------------------------------------
    ↘

Oracle Database 11g Enterprise Edition ↙
    ↘ Release 11.2.0.1.0 - Production
PL/SQL Release 11.2.0.1.0 - Production
CORE    11.2.0.1.0      Production
```

```
TNS for 32-bit Windows: Version 11.2.0.1.0 -↙
    ↳ Production
NLSRTL Version 11.2.0.1.0 - Production
```

V$VERSION?

:

Listing 81.1: kqf.o

```
.rodata:0800C4A0 kqfviw          dd 0Bh      ↙
    ↳               ; DATA XREF: kqfchk:↙
    ↳ loc_8003A6D
.rodata:0800C4A0                             ↙
    ↳               ; kqfgbn+34
.rodata:0800C4A4                 dd offset ↙
    ↳ _2__STRING_10102_0 ; "GV$WAITSTAT"
.rodata:0800C4A8                 dd 4
.rodata:0800C4AC                 dd offset ↙
    ↳ _2__STRING_10103_0 ; "NULL"
.rodata:0800C4B0                 dd 3
.rodata:0800C4B4                 dd 0
.rodata:0800C4B8                 dd 195h
.rodata:0800C4BC                 dd 4
.rodata:0800C4C0                 dd 0
.rodata:0800C4C4                 dd 0↙
    ↳ FFFFC1CBh
.rodata:0800C4C8                 dd 3
.rodata:0800C4CC                 dd 0
.rodata:0800C4D0                 dd 0Ah
.rodata:0800C4D4                 dd offset ↙
    ↳ _2__STRING_10104_0 ; "V$WAITSTAT"
.rodata:0800C4D8                 dd 4
```

```
.rodata:0800C4DC                    dd offset ⤦
    ↳ _2__STRING_10103_0 ; "NULL"
.rodata:0800C4E0                    dd 3
.rodata:0800C4E4                    dd 0
.rodata:0800C4E8                    dd 4Eh
.rodata:0800C4EC                    dd 3
.rodata:0800C4F0                    dd 0
.rodata:0800C4F4                    dd 0⤦
    ↳ FFFFC003h
.rodata:0800C4F8                    dd 4
.rodata:0800C4FC                    dd 0
.rodata:0800C500                    dd 5
.rodata:0800C504                    dd offset ⤦
    ↳ _2__STRING_10105_0 ; "GV$BH"
.rodata:0800C508                    dd 4
.rodata:0800C50C                    dd offset ⤦
    ↳ _2__STRING_10103_0 ; "NULL"
.rodata:0800C510                    dd 3
.rodata:0800C514                    dd 0
.rodata:0800C518                    dd 269h
.rodata:0800C51C                    dd 15h
.rodata:0800C520                    dd 0
.rodata:0800C524                    dd 0⤦
    ↳ FFFFC1EDh
.rodata:0800C528                    dd 8
.rodata:0800C52C                    dd 0
.rodata:0800C530                    dd 4
.rodata:0800C534                    dd offset ⤦
    ↳ _2__STRING_10106_0 ; "V$BH"
.rodata:0800C538                    dd 4
.rodata:0800C53C                    dd offset ⤦
    ↳ _2__STRING_10103_0 ; "NULL"
.rodata:0800C540                    dd 3
```

```
.rodata:0800C544                dd 0
.rodata:0800C548                dd 0F5h
.rodata:0800C54C                dd 14h
.rodata:0800C550                dd 0
.rodata:0800C554                dd 0↙
    ↘ FFFFC1EEh
.rodata:0800C558                dd 5
.rodata:0800C55C                dd 0
```

. . : «6 significant initial characters in an external identifier»[1]

V$FIXED_VIEW_DEFINITION

```
SQL> select * from V$FIXED_VIEW_DEFINITION ↙
    ↘ where view_name='V$VERSION';

VIEW_NAME
------------------------------
VIEW_DEFINITION
------------------------------------------------------------
    ↘

V$VERSION
select  BANNER from GV$VERSION where inst_id↙
    ↘  = USERENV('Instance')
```

```
SQL> select * from V$FIXED_VIEW_DEFINITION ↙
    ↘ where view_name='GV$VERSION';

```

[1]Draft ANSI C Standard (ANSI X3J11/88-090) (May 13, 1988) (yurichev.com)

```
VIEW_NAME
------------------------------
VIEW_DEFINITION
-----------------------------------------------------
    ↪

GV$VERSION
select inst_id, banner from x$version
```

select BANNER from GV$VERSION where inst_id = USERENV('Instance') :

Listing 81.2: kqf.o

```
rodata:080185A0 kqfvip          dd offset ↙
    ↪ _2__STRING_11126_0 ; DATA XREF: ↙
    ↪ kqfgvcn+18
.rodata:080185A0                                ↙
    ↪             ; kqfgvt+F
.rodata:080185A0                                ↙
    ↪             ; "select inst_id,decode(↙
    ↪ indx,1,'data bloc"...
.rodata:080185A4                 dd offset ↙
    ↪ kqfv459_c_0
.rodata:080185A8                 dd 0
.rodata:080185AC                 dd 0

...

.rodata:08019570                 dd offset ↙
    ↪ _2__STRING_11378_0 ; "select  BANNER ↙
    ↪ from GV$VERSION where in"...
.rodata:08019574                 dd offset ↙
```

```
    ↳ kqfv133_c_0
.rodata:08019578                 dd 0
.rodata:0801957C                 dd 0
.rodata:08019580                 dd offset ↙
    ↳ _2__STRING_11379_0 ; "select inst_id,↙
    ↳ decode(bitand(cfflg,1),0"...
.rodata:08019584                 dd offset ↙
    ↳ kqfv403_c_0
.rodata:08019588                 dd 0
.rodata:0801958C                 dd 0
.rodata:08019590                 dd offset ↙
    ↳ _2__STRING_11380_0 ; "select  STATUS ,↙
    ↳  NAME, IS_RECOVERY_DEST"...
.rodata:08019594                 dd offset ↙
    ↳ kqfv199_c_0
```

Listing 81.3: kqf.o

```
.rodata:080BBAC4 kqfv133_c_0     dd 6            ↙
    ↳                ; DATA XREF: .rodata↙
    ↳ :08019574
.rodata:080BBAC8                 dd offset ↙
    ↳ _2__STRING_5017_0 ; "BANNER"
.rodata:080BBACC                 dd 0
.rodata:080BBAD0                 dd offset ↙
    ↳ _2__STRING_0_0
```

Listing 81.4: oracle tables

```
kqfviw_element.viewname: [V$VERSION] ?: 0x3 ↙
    ↳ 0x43 0x1 0xffffc085 0x4
kqfvip_element.statement: [select  BANNER ↙
    ↳ from GV$VERSION where inst_id = ↙
```

```
    ↳ USERENV('Instance')]
kqfvip_element.params:
[BANNER]
```

y:

Listing 81.5: oracle tables

```
kqfviw_element.viewname: [GV$VERSION] ?: 0x3↙
    ↳  0x26 0x2 0xffffc192 0x1
kqfvip_element.statement: [select inst_id, ↙
    ↳ banner from x$version]
kqfvip_element.params:
[INST_ID] [BANNER]
```

:

```
SQL> select * from x$version;

ADDR           INDX    INST_ID
-------- ---------- ----------
BANNER
------------------------------------------------------------------
    ↳

0DBAF574          0          1
Oracle Database 11g Enterprise Edition ↙
    ↳ Release 11.2.0.1.0 - Production

...
```

Listing 81.6: kqf.o

```
.rodata:0803CAC0                   dd 9               ↙
    ↘                ; element number 0x1f6
.rodata:0803CAC4                   dd offset ↙
    ↘ _2__STRING_13113_0 ; "X$VERSION"
.rodata:0803CAC8                   dd 4
.rodata:0803CACC                   dd offset ↙
    ↘ _2__STRING_13114_0 ; "kqvt"
.rodata:0803CAD0                   dd 4
.rodata:0803CAD4                   dd 4
.rodata:0803CAD8                   dd 0
.rodata:0803CADC                   dd 4
.rodata:0803CAE0                   dd 0Ch
.rodata:0803CAE4                   dd 0↙
    ↘ FFFFC075h
.rodata:0803CAE8                   dd 3
.rodata:0803CAEC                   dd 0
.rodata:0803CAF0                   dd 7
.rodata:0803CAF4                   dd offset ↙
    ↘ _2__STRING_13115_0 ; "X$KQFSZ"
.rodata:0803CAF8                   dd 5
.rodata:0803CAFC                   dd offset ↙
    ↘ _2__STRING_13116_0 ; "kqfsz"
.rodata:0803CB00                   dd 1
.rodata:0803CB04                   dd 38h
.rodata:0803CB08                   dd 0
.rodata:0803CB0C                   dd 7
.rodata:0803CB10                   dd 0
.rodata:0803CB14                   dd 0↙
    ↘ FFFFC09Dh
.rodata:0803CB18                   dd 2
.rodata:0803CB1C                   dd 0
```

`kqf.o`:

Listing 81.7: kqf.o

```
.rodata:0808C360 kqvt_c_0 kqftap_param <4, offset _2__STRING_19_0, 917h, 0, 0, 0, 4, 0, 0>
.rodata:0808C360                           ; DATA XREF: .rodata:08042680
.rodata:0808C360                           ; "ADDR"
.rodata:0808C384 kqftap_param <4, offset _2__STRING_20_0, 0B02h, 0, 0, 0, 4, 0, 0> ; "INDX"
.rodata:0808C3A8 kqftap_param <7, offset _2__STRING_21_0, 0B02h, 0, 0, 0, 4, 0, 0> ; "INST_ID"
.rodata:0808C3CC kqftap_param <6, offset _2__STRING_5017_0, 601h, 0, 0, 0, 50h, 0, 0> ; "BANNER"
.rodata:0808C3F0 kqftap_param <0, offset _2__STRING_0_0, 0, 0, 0, 0, 0, 0, 0>
```

Listing 81.8: kqf.o

```
.rodata:08042680 kqftap_element <0, offset kqvt_c_0, offset kqvrow, 0> ; element 0x1f6
```

Listing 81.9: oracle tables

```
kqftab_element.name: [X$VERSION] ?: [kqvt] 0x4 0x4 0x4 0xc 0xffffc075 0x3
kqftap_param.name=[ADDR] ?: 0x917 0x0 0x0 0x0 0x4 0x0 0x0
kqftap_param.name=[INDX] ?: 0xb02 0x0 0x0 0x0 0x4 0x0 0x0
kqftap_param.name=[INST_ID] ?: 0xb02 0x0 0x0 0x0 0x4 0x0 0x0
kqftap_param.name=[BANNER] ?: 0x601 0x0 0x0 0x0 0x50 0x0 0x0
kqftap_element.fn1=kqvrow
kqftap_element.fn2=NULL
```

```
tracer -a:oracle.exe bpf=oracle.exe!_kqvrow,trace:cc
```

```
_kqvrow_  proc near

var_7C    = byte ptr -7Ch
var_18    = dword ptr -18h
var_14    = dword ptr -14h
Dest      = dword ptr -10h
var_C     = dword ptr -0Ch
var_8     = dword ptr -8
var_4     = dword ptr -4
arg_8     = dword ptr  10h
arg_C     = dword ptr  14h
arg_14    = dword ptr  1Ch
arg_18    = dword ptr  20h

```

```
; FUNCTION CHUNK AT .text1:056C11A0 SIZE ↙
    ↘ 00000049 BYTES

        push    ebp
        mov     ebp, esp
        sub     esp, 7Ch
        mov     eax, [ebp+arg_14] ; [EBP+1↙
    ↘ Ch]=1
        mov     ecx, TlsIndex   ; [69↙
    ↘ AEB08h]=0
        mov     edx, large fs:2Ch
        mov     edx, [edx+ecx*4] ; [EDX+↙
    ↘ ECX*4]=0xc98c938
        cmp     eax, 2          ; EAX=1
        mov     eax, [ebp+arg_8] ; [EBP+10↙
    ↘ h]=0xcdfe554
        jz      loc_2CE1288
        mov     ecx, [eax]      ; [EAX↙
    ↘ ]=0..5
        mov     [ebp+var_4], edi ; EDI=0↙
    ↘ xc98c938

loc_2CE10F6: ; CODE XREF: _kqvrow_+10A
           ; _kqvrow_+1A9
        cmp     ecx, 5          ; ECX=0..5
        ja      loc_56C11C7
        mov     edi, [ebp+arg_18] ; [EBP↙
    ↘ +20h]=0
        mov     [ebp+var_14], edx ; EDX=0↙
    ↘ xc98c938
        mov     [ebp+var_8], ebx ; EBX=0
        mov     ebx, eax        ; EAX=0↙
    ↘ xcdfe554
```

```
        mov     [ebp+var_C], esi ; ESI=0↙
    ↘ xcdfe248

loc_2CE110D: ; CODE XREF: _kqvrow_+29E00E6
        mov     edx, ds:off_628B09C[ecx*4]↙
    ↘  ; [ECX*4+628B09Ch]=0x2ce1116, 0↙
    ↘ x2ce11ac, 0x2ce11db, 0x2ce11f6, 0↙
    ↘ x2ce1236, 0x2ce127a
        jmp     edx                 ; EDX=0↙
    ↘ x2ce1116, 0x2ce11ac, 0x2ce11db, 0↙
    ↘ x2ce11f6, 0x2ce1236, 0x2ce127a

loc_2CE1116: ; DATA XREF: .rdata:off_628B09C
        push    offset aXKqvvsnBuffer ; "↙
    ↘ x$kqvvsn buffer"
        mov     ecx, [ebp+arg_C] ; [EBP+14↙
    ↘ h]=0x8a172b4
        xor     edx, edx
        mov     esi, [ebp+var_14] ; [EBP↙
    ↘ -14h]=0xc98c938
        push    edx                 ; EDX=0
        push    edx                 ; EDX=0
        push    50h
        push    ecx                 ; ECX=0↙
    ↘ x8a172b4
        push    dword ptr [esi+10494h] ; [↙
    ↘ ESI+10494h]=0xc98cd58
        call    _kghalf             ; tracing ↙
    ↘ nested maximum level (1) reached, ↙
    ↘ skipping this CALL
        mov     esi, ds:__imp__vsnnum ; ↙
    ↘ [59771A8h]=0x61bc49e0
        mov     [ebp+Dest], eax ; EAX=0↙
```

```
↳ xce2ffb0
      mov     [ebx+8], eax     ; EAX=0↙
↳ xce2ffb0
      mov     [ebx+4], eax     ; EAX=0↙
↳ xce2ffb0
      mov     edi, [esi]       ; [ESI]=0↙
↳ xb200100
      mov     esi, ds:__imp__vsnstr ; ↙
↳ [597D6D4h]=0x65852148, "- Production"
      push    esi              ; ESI=0↙
↳ x65852148, "- Production"
      mov     ebx, edi         ; EDI=0↙
↳ xb200100
      shr     ebx, 18h         ; EBX=0↙
↳ xb200100
      mov     ecx, edi         ; EDI=0↙
↳ xb200100
      shr     ecx, 14h         ; ECX=0↙
↳ xb200100
      and     ecx, 0Fh         ; ECX=0xb2
      mov     edx, edi         ; EDI=0↙
↳ xb200100
      shr     edx, 0Ch         ; EDX=0↙
↳ xb200100
      movzx   edx, dl          ; DL=0
      mov     eax, edi         ; EDI=0↙
↳ xb200100
      shr     eax, 8           ; EAX=0↙
↳ xb200100
      and     eax, 0Fh         ; EAX=0↙
↳ xb2001
      and     edi, 0FFh        ; EDI=0↙
↳ xb200100
```

```
        push    edi               ; EDI=0
        mov     edi, [ebp+arg_18] ; [EBP↙
    ↘ +20h]=0
        push    eax               ; EAX=1
        mov     eax, ds:__imp__vsnban ; ↙
    ↘ [597D6D8h]=0x65852100, "Oracle ↙
    ↘ Database 11g Enterprise Edition ↙
    ↘ Release %d.%d.%d.%d.%d %s"
        push    edx               ; EDX=0
        push    ecx               ; ECX=2
        push    ebx               ; EBX=0xb
        mov     ebx, [ebp+arg_8] ; [EBP+10↙
    ↘ h]=0xcdfe554
        push    eax               ; EAX=0↙
    ↘ x65852100, "Oracle Database 11g ↙
    ↘ Enterprise Edition Release %d.%d.%d.%d↙
    ↘ .%d %s"
        mov     eax, [ebp+Dest] ; [EBP-10h↙
    ↘ ]=0xce2ffb0
        push    eax               ; EAX=0↙
    ↘ xce2ffb0
        call    ds:__imp__sprintf ; op1=↙
    ↘ MSVCR80.dll!sprintf tracing nested ↙
    ↘ maximum level (1) reached, skipping ↙
    ↘ this CALL
        add     esp, 38h
        mov     dword ptr [ebx], 1

loc_2CE1192: ; CODE XREF: _kqvrow_+FB
             ; _kqvrow_+128 ...
        test    edi, edi          ; EDI=0
        jnz     __VInfreq__kqvrow
        mov     esi, [ebp+var_C] ; [EBP-0↙
```

```
    ↳ Ch]=0xcdfe248
         mov     edi, [ebp+var_4] ; [EBP↙
    ↳ -4]=0xc98c938
         mov     eax, ebx         ; EBX=0↙
    ↳ xcdfe554
         mov     ebx, [ebp+var_8] ; [EBP↙
    ↳ -8]=0
         lea     eax, [eax+4]     ; [EAX↙
    ↳ +4]=0xce2ffb0, "NLSRTL Version ↙
    ↳ 11.2.0.1.0 - Production", "Oracle ↙
    ↳ Database 11g Enterprise Edition ↙
    ↳ Release 11.2.0.1.0 - Production", "PL/↙
    ↳ SQL Release 11.2.0.1.0 - Production", ↙
    ↳ "TNS for 32-bit Windows: Version ↙
    ↳ 11.2.0.1.0 - Production"

loc_2CE11A8: ; CODE XREF: _kqvrow_+29E00F6
         mov     esp, ebp
         pop     ebp
         retn                     ; EAX=0↙
    ↳ xcdfe558

loc_2CE11AC: ; DATA XREF: .rdata:0628B0A0
         mov     edx, [ebx+8]     ; [EBX↙
    ↳ +8]=0xce2ffb0, "Oracle Database 11g ↙
    ↳ Enterprise Edition Release 11.2.0.1.0 ↙
    ↳ - Production"
         mov     dword ptr [ebx], 2
         mov     [ebx+4], edx     ; EDX=0↙
    ↳ xce2ffb0, "Oracle Database 11g ↙
    ↳ Enterprise Edition Release 11.2.0.1.0 ↙
    ↳ - Production"
         push    edx              ; EDX=0↙
```

```
    ↳ xce2ffb0, "Oracle Database 11g ↙
    ↳ Enterprise Edition Release 11.2.0.1.0 ↙
    ↳ - Production"
        call    _kkxvsn             ; tracing ↙
    ↳ nested maximum level (1) reached, ↙
    ↳ skipping this CALL
        pop     ecx
        mov     edx, [ebx+4]    ; [EBX↙
    ↳ +4]=0xce2ffb0, "PL/SQL Release ↙
    ↳ 11.2.0.1.0 - Production"
        movzx   ecx, byte ptr [edx] ; [EDX↙
    ↳ ]=0x50
        test    ecx, ecx        ; ECX=0x50
        jnz     short loc_2CE1192
        mov     edx, [ebp+var_14]
        mov     esi, [ebp+var_C]
        mov     eax, ebx
        mov     ebx, [ebp+var_8]
        mov     ecx, [eax]
        jmp     loc_2CE10F6

loc_2CE11DB: ; DATA XREF: .rdata:0628B0A4
        push    0
        push    50h
        mov     edx, [ebx+8]    ; [EBX↙
    ↳ +8]=0xce2ffb0, "PL/SQL Release ↙
    ↳ 11.2.0.1.0 - Production"
        mov     [ebx+4], edx    ; EDX=0↙
    ↳ xce2ffb0, "PL/SQL Release 11.2.0.1.0 -↙
    ↳  Production"
        push    edx             ; EDX=0↙
    ↳ xce2ffb0, "PL/SQL Release 11.2.0.1.0 -↙
    ↳  Production"
```

```
        call    _lmxver             ; tracing ↙
    ↘ nested maximum level (1) reached, ↙
    ↘ skipping this CALL
        add     esp, 0Ch
        mov     dword ptr [ebx], 3
        jmp     short loc_2CE1192

loc_2CE11F6: ; DATA XREF: .rdata:0628B0A8
        mov     edx, [ebx+8]    ; [EBX↙
    ↘ +8]=0xce2ffb0
        mov     [ebp+var_18], 50h
        mov     [ebx+4], edx    ; EDX=0↙
    ↘ xce2ffb0
        push    0
        call    _npinli             ; tracing ↙
    ↘ nested maximum level (1) reached, ↙
    ↘ skipping this CALL
        pop     ecx
        test    eax, eax            ; EAX=0
        jnz     loc_56C11DA
        mov     ecx, [ebp+var_14] ; [EBP↙
    ↘ -14h]=0xc98c938
        lea     edx, [ebp+var_18] ; [EBP↙
    ↘ -18h]=0x50
        push    edx                 ; EDX=0↙
    ↘ xd76c93c
        push    dword ptr [ebx+8] ; [EBX↙
    ↘ +8]=0xce2ffb0
        push    dword ptr [ecx+13278h] ; [↙
    ↘ ECX+13278h]=0xacce190
        call    _nrtnsvrs           ; tracing ↙
    ↘ nested maximum level (1) reached, ↙
    ↘ skipping this CALL
```

```
        add     esp, 0Ch

loc_2CE122B: ; CODE XREF: _kqvrow_+29E0118
        mov     dword ptr [ebx], 4
        jmp     loc_2CE1192

loc_2CE1236: ; DATA XREF: .rdata:0628B0AC
        lea     edx, [ebp+var_7C] ; [EBP-7↙
    ↘ Ch]=1
        push    edx               ; EDX=0↙
    ↘ xd76c8d8
        push    0
        mov     esi, [ebx+8]      ; [EBX↙
    ↘ +8]=0xce2ffb0, "TNS for 32-bit Windows↙
    ↘ : Version 11.2.0.1.0 - Production"
        mov     [ebx+4], esi      ; ESI=0↙
    ↘ xce2ffb0, "TNS for 32-bit Windows: ↙
    ↘ Version 11.2.0.1.0 - Production"
        mov     ecx, 50h
        mov     [ebp+var_18], ecx ; ECX=0↙
    ↘ x50
        push    ecx               ; ECX=0x50
        push    esi               ; ESI=0↙
    ↘ xce2ffb0, "TNS for 32-bit Windows: ↙
    ↘ Version 11.2.0.1.0 - Production"
        call    _lxvers           ; tracing ↙
    ↘ nested maximum level (1) reached, ↙
    ↘ skipping this CALL
        add     esp, 10h
        mov     edx, [ebp+var_18] ; [EBP↙
    ↘ -18h]=0x50
        mov     dword ptr [ebx], 5
        test    edx, edx          ; EDX=0x50
```

```
        jnz     loc_2CE1192
        mov     edx, [ebp+var_14]
        mov     esi, [ebp+var_C]
        mov     eax, ebx
        mov     ebx, [ebp+var_8]
        mov     ecx, 5
        jmp     loc_2CE10F6

loc_2CE127A: ; DATA XREF: .rdata:0628B0B0
        mov     edx, [ebp+var_14] ; [EBP↙
    ↘ -14h]=0xc98c938
        mov     esi, [ebp+var_C] ; [EBP-0↙
    ↘ Ch]=0xcdfe248
        mov     edi, [ebp+var_4] ; [EBP↙
    ↘ -4]=0xc98c938
        mov     eax, ebx          ; EBX=0↙
    ↘ xcdfe554
        mov     ebx, [ebp+var_8] ; [EBP↙
    ↘ -8]=0

loc_2CE1288: ; CODE XREF: _kqvrow_+1F
        mov     eax, [eax+8]     ; [EAX↙
    ↘ +8]=0xce2ffb0, "NLSRTL Version ↙
    ↘ 11.2.0.1.0 - Production"
        test    eax, eax          ; EAX=0↙
    ↘ xce2ffb0, "NLSRTL Version 11.2.0.1.0 -↙
    ↘  Production"
        jz      short loc_2CE12A7
        push    offset aXKqvvsnBuffer ; "↙
    ↘ x$kqvvsn buffer"
        push    eax               ; EAX=0↙
    ↘ xce2ffb0, "NLSRTL Version 11.2.0.1.0 -↙
    ↘  Production"
```

```
        mov     eax, [ebp+arg_C] ; [EBP+14h]=0x8a172b4
        push    eax                  ; EAX=0x8a172b4
        push    dword ptr [edx+10494h] ; [EDX+10494h]=0xc98cd58
        call    _kghfrf              ; tracing nested maximum level (1) reached, skipping this CALL
        add     esp, 10h

loc_2CE12A7: ; CODE XREF: _kqvrow_+1C1
        xor     eax, eax
        mov     esp, ebp
        pop     ebp
        retn                         ; EAX=0
_kqvrow_  endp
```

1	
2	kkxvsn().
3	lmxver().
4	npinli(), nrtnsvrs().
5	lxvers().

81.2 Oracle RDBMS

Diagnosing and Resolving Error ORA-04031 on the Shared Pool or Other Memory Pools [Video] [ID 146599.1] :

There is a fixed table called X$KSMLRU that tracks allocations in the shared pool that cause other objects in the shared pool to be aged out. This fixed table can be used to identify what is causing the large allocation.

If many objects are being periodically flushed from the shared pool then this will cause response time problems and will likely cause library cache latch contention problems when the objects are reloaded into the shared pool.

One unusual thing about the X$KSMLRU fixed table is that the contents of the fixed table are erased whenever someone selects from the fixed table. This is done since the fixed table stores only the largest allocations that have occurred. The values are reset after being selected so that subsequent large allocations can be noted even if they were not quite as large as others that occurred previously. Because of this resetting, the output of selecting from this table should be carefully kept since it cannot be retrieved back after the query is issued.

Listing 81.10: oracle tables

```
kqftab_element.name: [X$KSMLRU] ?: [ksmlr] 0↙
    ↪ x4 0x64 0x11 0xc 0xffffc0bb 0x5
kqftap_param.name=[ADDR] ?: 0x917 0x0 0x0 0↙
    ↪ x0 0x4 0x0 0x0
kqftap_param.name=[INDX] ?: 0xb02 0x0 0x0 0↙
    ↪ x0 0x4 0x0 0x0
kqftap_param.name=[INST_ID] ?: 0xb02 0x0 0x0↙
    ↪  0x0 0x4 0x0 0x0
kqftap_param.name=[KSMLRIDX] ?: 0xb02 0x0 0↙
    ↪ x0 0x0 0x4 0x0 0x0
kqftap_param.name=[KSMLRDUR] ?: 0xb02 0x0 0↙
    ↪ x0 0x0 0x4 0x4 0x0
kqftap_param.name=[KSMLRSHRPOOL] ?: 0xb02 0↙
    ↪ x0 0x0 0x0 0x4 0x8 0x0
kqftap_param.name=[KSMLRCOM] ?: 0x501 0x0 0↙
    ↪ x0 0x0 0x14 0xc 0x0
kqftap_param.name=[KSMLRSIZ] ?: 0x2 0x0 0x0 ↙
    ↪ 0x0 0x4 0x20 0x0
kqftap_param.name=[KSMLRNUM] ?: 0x2 0x0 0x0 ↙
    ↪ 0x0 0x4 0x24 0x0
kqftap_param.name=[KSMLRHON] ?: 0x501 0x0 0↙
    ↪ x0 0x0 0x20 0x28 0x0
kqftap_param.name=[KSMLROHV] ?: 0xb02 0x0 0↙
    ↪ x0 0x0 0x4 0x48 0x0
kqftap_param.name=[KSMLRSES] ?: 0x17 0x0 0x0↙
    ↪  0x0 0x4 0x4c 0x0
kqftap_param.name=[KSMLRADU] ?: 0x2 0x0 0x0 ↙
    ↪ 0x0 0x4 0x50 0x0
kqftap_param.name=[KSMLRNID] ?: 0x2 0x0 0x0 ↙
    ↪ 0x0 0x4 0x54 0x0
kqftap_param.name=[KSMLRNSD] ?: 0x2 0x0 0x0 ↙
```

```
    ↳ 0x0 0x4 0x58 0x0
kqftap_param.name=[KSMLRNCD] ?: 0x2 0x0 0x0 ↙
    ↳ 0x0 0x4 0x5c 0x0
kqftap_param.name=[KSMLRNED] ?: 0x2 0x0 0x0 ↙
    ↳ 0x0 0x4 0x60 0x0
kqftap_element.fn1=ksmlrs
kqftap_element.fn2=NULL
```

Listing 81.11: ksm.o

```
...

.text:00434C50 loc_434C50:                  ↙
    ↳            ; DATA XREF: .rdata:↙
    ↳ off_5E50EA8
.text:00434C50                   mov     edx, ↙
    ↳ [ebp-4]
.text:00434C53                   mov     [eax↙
    ↳ ], esi
.text:00434C55                   mov     esi, ↙
    ↳ [edi]
.text:00434C57                   mov     [eax↙
    ↳ +4], esi
.text:00434C5A                   mov     [edi↙
    ↳ ], eax
.text:00434C5C                   add     edx, ↙
    ↳ 1
.text:00434C5F                   mov     [ebp↙
    ↳ -4], edx
.text:00434C62                   jnz     ↙
    ↳ loc_434B7D
.text:00434C68                   mov     ecx, ↙
    ↳ [ebp+14h]
```

```
.text:00434C6B                  mov     ebx, ↙
    ↘ [ebp-10h]
.text:00434C6E                  mov     esi, ↙
    ↘ [ebp-0Ch]
.text:00434C71                  mov     edi, ↙
    ↘ [ebp-8]
.text:00434C74                  lea     eax, ↙
    ↘ [ecx+8Ch]
.text:00434C7A                  push    370h ↙
    ↘           ; Size
.text:00434C7F                  push    0    ↙
    ↘           ; Val
.text:00434C81                  push    eax  ↙
    ↘           ; Dst
.text:00434C82                  call    ↙
    ↘ __intel_fast_memset
.text:00434C87                  add     esp, ↙
    ↘ 0Ch
.text:00434C8A                  mov     esp, ↙
    ↘ ebp
.text:00434C8C                  pop     ebp
.text:00434C8D                  retn
.text:00434C8D _ksmsplu         endp
```

0x434C7A 0x434C8A.

```
tracer -a:oracle.exe bpx=oracle.exe!0↙
    ↘ x00434C7A,set(eip,0x00434C8A)
```

81.3 Oracle RDBMS

V$TIMER

> V$TIMER displays the elapsed time in hundredths of a second. Time is measured since the beginning of the epoch, which is operating system specific, and wraps around to 0 again whenever the value overflows four bytes (roughly 497 days).

([2])

```
SQL> select * from V$FIXED_VIEW_DEFINITION ↙
    ↳ where view_name='V$TIMER';

VIEW_NAME
------------------------------
VIEW_DEFINITION
-----------------------------------------------
    ↳

V$TIMER
select  HSECS from GV$TIMER where inst_id = ↙
    ↳ USERENV('Instance')

SQL> select * from V$FIXED_VIEW_DEFINITION ↙
    ↳ where view_name='GV$TIMER';

```

[2] http://go.yurichev.com/17088

```
VIEW_NAME
------------------------------
VIEW_DEFINITION
--------------------------------------------------
    ↘

GV$TIMER
select inst_id,ksutmtim from x$ksutm
```

:

Listing 81.12: oracle tables

```
kqftab_element.name: [X$KSUTM] ?: [ksutm] 0↙
    ↘ x1 0x4 0x4 0x0 0xffffc09b 0x3
kqftap_param.name=[ADDR] ?: 0x10917 0x0 0x0 ↙
    ↘ 0x0 0x4 0x0 0x0
kqftap_param.name=[INDX] ?: 0x20b02 0x0 0x0 ↙
    ↘ 0x0 0x4 0x0 0x0
kqftap_param.name=[INST_ID] ?: 0xb02 0x0 0x0↙
    ↘  0x0 0x4 0x0 0x0
kqftap_param.name=[KSUTMTIM] ?: 0x1302 0x0 0↙
    ↘ x0 0x0 0x4 0x0 0x1e
kqftap_element.fn1=NULL
kqftap_element.fn2=NULL
```

```
kqfd_DRN_ksutm_c proc near               ; ↙
    ↘ DATA XREF: .rodata:0805B4E8

arg_0           = dword ptr  8
arg_8           = dword ptr  10h
arg_C           = dword ptr  14h

```

```
                push    ebp
                mov     ebp, esp
                push    [ebp+arg_C]
                push    offset ksugtm
                push    offset ⤦
    ⤷ _2__STRING_1263_0 ; "KSUTMTIM"
                push    [ebp+arg_8]
                push    [ebp+arg_0]
                call    kqfd_cfui_drain
                add     esp, 14h
                mov     esp, ebp
                pop     ebp
                retn
kqfd_DRN_ksutm_c endp
```

kqfd_DRN_ksutm_c() kqfd_tab_registry_0 :

```
dd offset _2__STRING_62_0 ; "X$KSUTM"
dd offset kqfd_OPN_ksutm_c
dd offset kqfd_tabl_fetch
dd 0
dd 0
dd offset kqfd_DRN_ksutm_c
```

. :

Listing 81.13: ksu.o

```
ksugtm          proc near

var_1C          = byte ptr -1Ch
arg_4           = dword ptr  0Ch

```

```
                push    ebp
                mov     ebp, esp
                sub     esp, 1Ch
                lea     eax, [ebp+var_1C]
                push    eax
                call    slgcs
                pop     ecx
                mov     edx, [ebp+arg_4]
                mov     [edx], eax
                mov     eax, 4
                mov     esp, ebp
                pop     ebp
                retn
ksugtm          endp
```

.

:

```
tracer -a:oracle.exe bpf=oracle.exe!_ksugtm,
    args:2,dump_args:0x4
```

:

```
SQL> select * from V$TIMER;

     HSECS
----------
  27294929

SQL> select * from V$TIMER;

     HSECS
```

```
----------
  27295006

SQL> select * from V$TIMER;

     HSECS
----------
  27295167
```

Listing 81.14:

```
TID=2428|(0) oracle.exe!_ksugtm (0x0, 0↙
    ↘ xd76c5f0) (called from oracle.exe!↙
    ↘ __VInfreq__qerfxFetch+0xfad (0x56bb6d5↙
    ↘ ))
Argument 2/2
0D76C5F0: 38 C9                                            ↙
    ↘                "8.                 "
TID=2428|(0) oracle.exe!_ksugtm () -> 0x4 (0↙
    ↘ x4)
Argument 2/2 difference
00000000: D1 7C A0 01                                      ↙
    ↘                ".|..               "
TID=2428|(0) oracle.exe!_ksugtm (0x0, 0↙
    ↘ xd76c5f0) (called from oracle.exe!↙
    ↘ __VInfreq__qerfxFetch+0xfad (0x56bb6d5↙
    ↘ ))
Argument 2/2
0D76C5F0: 38 C9                                            ↙
    ↘                "8.                 "
TID=2428|(0) oracle.exe!_ksugtm () -> 0x4 (0↙
    ↘ x4)
Argument 2/2 difference
```

```
00000000: 1E 7D A0 01                                       ↙
    ↘               ".}..               "
TID=2428|(0) oracle.exe!_ksugtm (0x0, 0↙
    ↘ xd76c5f0) (called from oracle.exe!↙
    ↘ __VInfreq__qerfxFetch+0xfad (0x56bb6d5↙
    ↘ ))
Argument 2/2
0D76C5F0: 38 C9                                             ↙
    ↘               "8.                 "
TID=2428|(0) oracle.exe!_ksugtm () -> 0x4 (0↙
    ↘ x4)
Argument 2/2 difference
00000000: BF 7D A0 01                                       ↙
    ↘               ".}..               "
```

.

`slgcs()` (Linux x86):

```
slgcs               proc near

var_4               = dword ptr -4
arg_0               = dword ptr  8

                    push    ebp
                    mov     ebp, esp
                    push    esi
                    mov     [ebp+var_4], ebx
                    mov     eax, [ebp+arg_0]
                    call    $+5
                    pop     ebx
                    nop                         ; ↙
    ↘ PIC mode
```

```
                mov     ebx, offset ↙
    ↘ _GLOBAL_OFFSET_TABLE_
                mov     dword ptr [eax], 0
                call    sltrgatime64    ; ↙
    ↘ PIC mode
                push    0
                push    0Ah
                push    edx
                push    eax
                call    __udivdi3       ; ↙
    ↘ PIC mode
                mov     ebx, [ebp+var_4]
                add     esp, 10h
                mov     esp, ebp
                pop     ebp
                retn
slgcs           endp
```

(sltrgatime64() 10 (41 on page 938))

:

```
_slgcs          proc near               ; ↙
    ↘ CODE XREF: _dbgefgHtElResetCount+15
                                        ; ↙
    ↘ _dbgerRunActions+1528
                db      66h
                nop
                push    ebp
                mov     ebp, esp
                mov     eax, [ebp+8]
                mov     dword ptr [eax], 0
                call    ds:↙
```

```
    ↳ __imp__GetTickCount@0 ; GetTickCount()
                 mov     edx, eax
                 mov     eax, 0CCCCCCCDh
                 mul     edx
                 shr     edx, 3
                 mov     eax, edx
                 mov     esp, ebp
                 pop     ebp
                 retn
_slgcs           endp
```

`GetTickCount()` [3] 10 (41 on page 938).

`kqfd_tab_registry_0` oracle tables[4], :

```
[X$KSUTM] [kqfd_OPN_ksutm_c] [↙
    ↳ kqfd_tabl_fetch] [NULL] [NULL] [↙
    ↳ kqfd_DRN_ksutm_c]
[X$KSUSGIF] [kqfd_OPN_ksusg_c] [↙
    ↳ kqfd_tabl_fetch] [NULL] [NULL] [↙
    ↳ kqfd_DRN_ksusg_c]
```

OPN, , *open*, *DRN*, *drain*.

[3] MSDN

[4] yurichev.com

Capítulo 82

82.1 EICAR

: «EICAR-STANDARD-ANTIVIRUS-TEST-FILE!» [1].

:

```
X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-
    ANTIVIRUS-TEST-FILE!$H+H*
```

:

```

; : SP=0FFFEh, SS:[SP]=0
0100 58            pop     ax
; AX=0, SP=0
0101 35 4F 21      xor     ax, 214Fh
; AX = 214Fh and SP = 0
```

[1] wikipedia

```
0104 50             push    ax
; AX = 214Fh, SP = FFFEh and SS:[FFFE] = 214↙
    ↳ Fh
0105 25 40 41       and     ax, 4140h
; AX = 140h, SP = FFFEh and SS:[FFFE] = 214↙
    ↳ Fh
0108 50             push    ax
; AX = 140h, SP = FFFCh, SS:[FFFC] = 140h ↙
    ↳ and SS:[FFFE] = 214Fh
0109 5B             pop     bx
; AX = 140h, BX = 140h, SP = FFFEh and SS:[↙
    ↳ FFFE] = 214Fh
010A 34 5C          xor     al, 5Ch
; AX = 11Ch, BX = 140h, SP = FFFEh and SS:[↙
    ↳ FFFE] = 214Fh
010C 50             push    ax
010D 5A             pop     dx
; AX = 11Ch, BX = 140h, DX = 11Ch, SP = ↙
    ↳ FFFEh and SS:[FFFE] = 214Fh
010E 58             pop     ax
; AX = 214Fh, BX = 140h, DX = 11Ch and SP = ↙
    ↳ 0
010F 35 34 28       xor     ax, 2834h
; AX = 97Bh, BX = 140h, DX = 11Ch and SP = 0
0112 50             push    ax
0113 5E             pop     si
; AX = 97Bh, BX = 140h, DX = 11Ch, SI = 97Bh↙
    ↳  and SP = 0
0114 29 37          sub     [bx], si
0116 43             inc     bx
0117 43             inc     bx
0118 29 37          sub     [bx], si
011A 7D 24          jge     short near ptr ↙
```

```
    ↳ word_10140
011C 45 49 43 ...  db 'EICAR-STANDARD-↙
    ↳ ANTIVIRUS-TEST-FILE!$'
0140 48 2B   word_10140  dw 2B48h  ; CD 21 (↙
    ↳ INT 21)
0142 48 2A               dw 2A48h  ; CD 20 (↙
    ↳ INT 20)
0144 0D                  db  0Dh
0145 0A                  db  0Ah
```

.

:

```
B4 09     MOV AH, 9
BA 1C 01  MOV DX, 11Ch
CD 21     INT 21h
CD 20     INT 20h
```

INT 21h DS:DX. . CP/M . INT 20h DOS.

. :

- ;
- ;
- INT 21 y INT 20.

.

: A.6.5 on page 1899.

Capítulo 83

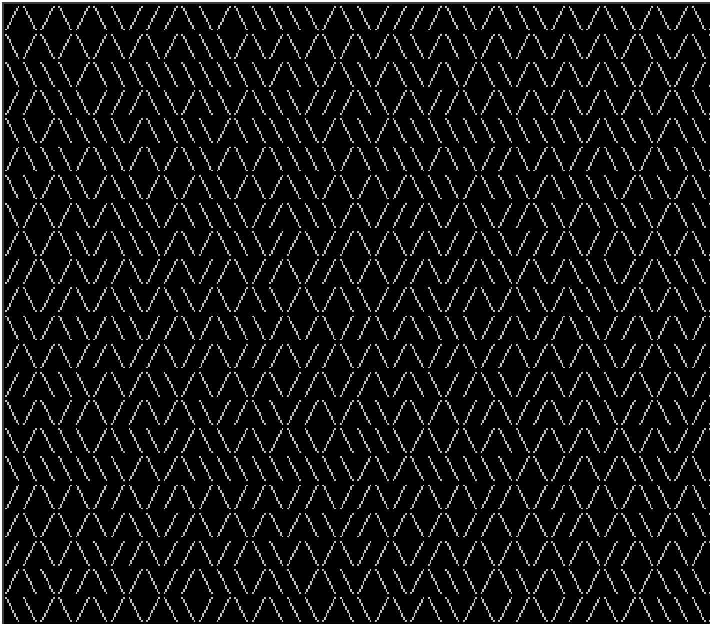
.

83.1 10 PRINT CHR$(205.5+RND(1)); : GOTO 10

.

[al12] . :

 CAPÍTULO 83.



.

83.1.1

[1], .

[1]http://go.yurichev.com/17305

```
00000000: B001              mov             al,1      ↙
    ↘     ;
00000002: CD10              int             010
00000004: 30FF              xor             bh,bh     ↙
    ↘     ;
00000006: B9D007            mov             cx,007D0 ↙
    ↘     ;
00000009: 31C0              xor             ax,ax
0000000B: 9C                pushf                     ↙
    ↘     ;
;
0000000C: FA                cli                       ↙
    ↘     ;
0000000D: E643              out             043,al    ↙
    ↘     ;
;
0000000F: E440              in              al,040
00000011: 88C4              mov             ah,al
00000013: E440              in              al,040
00000015: 9D                popf                      ↙
    ↘     ;
00000016: 86C4              xchg            ah,al
;
00000018: D1E8              shr             ax,1
0000001A: D1E8              shr             ax,1
;
0000001C: B05C              mov             al,05C ↙
    ↘ ;'\'
;
0000001E: 7202              jc              000000022
;
00000020: B02F              mov             al,02F ↙
    ↘ ;'/'
```

```
;
00000022: B40E          mov         ah,00E
00000024: CD10          int         010
00000026: E2E1          loop        000000009
    ;
00000028: CD20          int         020
    ;
```

.

43h, « 0», ”counter latch”, ”” ().

POPF.

AL, .

83.1.2

. .

:

```

00000000: B9D007   mov    cx,007D0 ;
00000003: 31C0     xor    ax,ax    ;
00000005: E643     out    043,al
00000007: E440     in     al,040   ;
00000009: D1E8     shr    ax,1     ;
0000000B: D1E8     shr    ax,1
0000000D: B05C     mov    al,05C   ;   '\'
0000000F: 7202     jc     000000013
00000011: B02F     mov    al,02F   ;   '/'
;
00000013: B40E     mov    ah,00E
```

```
00000015: CD10      int     010
00000017: E2EA      loop    000000003
;
00000019: CD20      int     020
```

83.1.3

MS-DOS, . LODSB DS:SI

[2] LODSB .

SCASB .

INT 29h AL.

Peter Ferrie y Andrey «herm1t» Baranovich (11 y 10) [3]:

Listing 83.1: Andrey «herm1t» Baranovich: 11

```

00000000: B05C          mov          al,05C    ↙
    ↘ ;'\'
;
00000002: AE            scasb
;
00000003: 7A02          jp           000000007
00000005: B02F          mov          al,02F    ↙
    ↘ ;'/'
00000007: CD29          int          029       ;
00000009: EBF5          jmp          000000000 ;
```

[2] http://go.yurichev.com/17305
[3] http://go.yurichev.com/17087

SCASB AL 5Ch AL. JP . .

SALC (AKA SETALC) («Set AL CF»). CPU AL 0xFF CF . 8086/8088.

Listing 83.2: Peter Ferrie: 10

```
;
00000000: AE          scasb
;
;
00000001: D6          setalc
;
00000002: 242D        and         al,02D ;'-'
;
00000004: 042F        add         al,02F ;'/'
;
00000006: CD29        int         029        ;
00000008: EBF6        jmps        000000000 ;
```

. ASCII[4] («\») 0x5C y 0x2F («/»). CF 0x5C o 0x2F.

AL () 0x2D 0 o 0x2D. 0x2F 0x5C o 0x2F. .

83.1.4 Conclusión

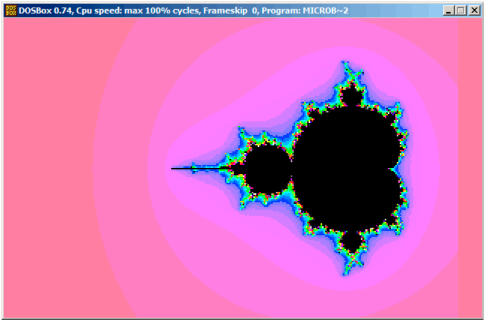
DOSBox, Windows NT MS-DOS, .

[4]American Standard Code for Information Interchange

83.2

[5] «Sir_Lagsalot» en 2009, . .

:



.

5 ,

83.2.1

:

- : $(a+bi)+(c+di)=(a+c)+(b+d)i$

  :

  $\text{Re}(sum) = \text{Re}(a) + \text{Re}(b)$

  $\text{Im}(sum) = \text{Im}(a) + \text{Im}(b)$

- : $(a+bi)(c+di)=(ac-bd)+(bc+ad)i$

  :

  $\text{Re}(product) = \text{Re}(a)\cdot\text{Re}(c) - \text{Re}(b)\cdot\text{Re}(d)$

  $\text{Im}(product) = \text{Im}(b)\cdot\text{Im}(c) + \text{Im}(a)\cdot\text{Im}(d)$

- : $(a+bi)^2=(a+bi)(a+bi)=(a^2-b^2)+(2ab)i$

  :

  $\text{Re}(square) = \text{Re}(a)^2 - \text{Im}(a)^2$

  $\text{Im}(square) = 2\cdot\text{Re}(a)\cdot\text{Im}(a)$

$z_{n+1} = {z_n}^2 + c$ (z y c c) .

:

- .

- .
- :
  - .
  - .
  - .
  - ?.
  - .
  - .
- ?.
- ?
  - .
  -

Listing 83.3:

```
def check_if_is_in_set(P):
    P_start=P
    iterations=0

    while True:
        if (P>bounds):
            break
        P=P^2+P_start
        if iterations > max_iterations:
            break
```

```
        iterations++

    return iterations

#
for each point on screen P:
    if check_if_is_in_set (P) < ↙
    ↳ max_iterations:


#
for each point on screen P:
    iterations = if check_if_is_in_set (P)
```

Listing 83.4:

```

def check_if_is_in_set(X, Y):
    X_start=X
    Y_start=Y
    iterations=0

    while True:
        if (X^2 + Y^2 > bounds):
            break
        new_X=X^2 - Y^2 + X_start
        new_Y=2*X*Y + Y_start
        if iterations > max_iterations:
            break
        iterations++

    return iterations

```

```
#
for X = min_X to max_X:
    for Y = min_Y to max_Y:
        if check_if_is_in_set (X,Y) < ↙
    ↳ max_iterations:
              X, Y

#
for X = min_X to max_X:
    for Y = min_Y to max_Y:
        iterations = if check_if_is_in_set (↙
    ↳ X,Y)

         X,Y
```

[6], [7]:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;

namespace Mnoj
{
    class Program
    {
        static void Main(string[] args)
        {
            double realCoord, imagCoord;
```

[6]wikipedia
[7]: beginners.re

```
        double realTemp, imagTemp, ↙
↳ realTemp2, arg;
        int iterations;
        for (imagCoord = 1.2; imagCoord ↙
↳ >= -1.2; imagCoord -= 0.05)
        {
            for (realCoord = -0.6; ↙
↳ realCoord <= 1.77; realCoord += 0.03)
            {
                iterations = 0;
                realTemp = realCoord;
                imagTemp = imagCoord;
                arg = (realCoord * ↙
↳ realCoord) + (imagCoord * imagCoord);
                while ((arg < 2*2) && (↙
↳ iterations < 40))
                {
                    realTemp2 = (↙
↳ realTemp * realTemp) - (imagTemp * ↙
↳ imagTemp) - realCoord;
                    imagTemp = (2 * ↙
↳ realTemp * imagTemp) - imagCoord;
                    realTemp = realTemp2↙
↳ ;
                    arg = (realTemp * ↙
↳ realTemp) + (imagTemp * imagTemp);
                    iterations += 1;
                }
                Console.Write("{0,2:D} ↙
↳ ", iterations);
            }
            Console.Write("\n");
        }
```

CAPÍTULO 83.

```
                Console.ReadKey();
            }
        }
}
```

:
beginners.re.

http://go.yurichev.com/17309, . .

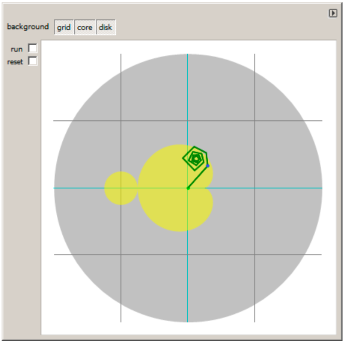


Figura 83.1:

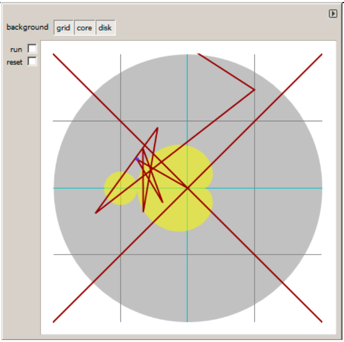


Figura 83.2:

.

: http://go.yurichev.com/17310.

83.2.2

:

Listing 83.5:

```
; X
; Y


; X=0, Y=0                  X=319, Y=0
; +------------------------------->
; |
; |
; |
; |
; |
; |
; v
; X=0, Y=199                X=319, Y=199


;
mov al,13h
int 10h
;
;
; DS:BX ( DS:0)  Program Segment Prefix
; ...  CD 20 FF 9F
les ax,[bx]
; ES:AX=9FFF:20CD

FillLoop:
```

```
; DX 0. CWD : DX:AX = sign_extend(AX).
;
; .
cwd
mov ax,di
; AX
; 320
mov cx,320
div cx
; DX (start_X) - (: 0..319); AX - (: 0..199)
sub ax,100
; AX=AX-100, AX (start_Y) -100..99
; DX 0..319 0x0000..0x013F
dec dh
; DX 0xFF00..0x003F (-256..63)

xor bx,bx
xor si,si
; BX (temp_X)=0; SI (temp_Y)=0

;
;
MandelLoop:
mov bp,si       ; BP = temp_Y
imul si,bx      ; SI = temp_X*temp_Y
add si,si       ; SI = SI*2 = (temp_X*temp_Y)*2
imul bx,bx      ; BX = BX^2 = temp_X^2
jo MandelBreak ; ?
imul bp,bp      ; BP = BP^2 = temp_Y^2
jo MandelBreak ; ?
add bx,bp       ; BX = BX+BP = temp_X^2 +
```

```
    temp_Y^2
jo MandelBreak ; ?
sub bx,bp       ; BX = BX-BP = temp_X^2 + temp_Y^2 - temp_Y^2 = temp_X^2
sub bx,bp       ; BX = BX-BP = temp_X^2 - temp_Y^2

; :
sar bx,6        ; BX=BX/64
add bx,dx       ; BX=BX+start_X
;  temp_X = temp_X^2 - temp_Y^2 + start_X
sar si,6        ; SI=SI/64
add si,ax       ; SI=SI+start_Y
;  temp_Y = (temp_X*temp_Y)*2 + start_Y

loop MandelLoop

MandelBreak:
; CX=
xchg ax,cx
; AX=.  AL  ES:[DI]
stosb
; stosb
;
jmp FillLoop
```

:

- 320*200 . $320 * 200 = 64000$ (0xFA00). .

  ES 0xA000 (`PUSH 0A000h` / `POP ES`). : 94 on page 1776.

MS-DOS, . . .

- . . $-100..99$ y Y $-256..63$. . $-160..159$? `SUB DX, 160` , `DEC DH2` (0x100 (256) DX). .
  - ..
  - . .
  -
  - . . . -32768..32767,
  -
- `MandelBreak`, : CX=0 ( ); . . !
- .

:

- 1- `CWD` `XOR DX, DX` `MOV DX, 0`.
- 1- `XCHG AX, CX` `MOV AX,CX`. .
- DI () [8]. .

83.2.3

Listing 83.6:

```

org 100h
mov al,13h
int 10h
```

[8]: http://go.yurichev.com/17004

```

;
mov dx, 3c8h
mov al, 0
out dx, al
mov cx, 100h
inc dx
l00:
mov al, cl
shl ax, 2
out dx, al ;
out dx, al ;
out dx, al ;
loop l00

push 0a000h
pop es

xor di, di

FillLoop:
cwd
mov ax,di
mov cx,320
div cx
sub ax,100
sub dx,160

xor bx,bx
xor si,si

MandelLoop:
mov bp,si
```

```
imul si,bx
add si,si
imul bx,bx
jo MandelBreak
imul bp,bp
jo MandelBreak
add bx,bp
jo MandelBreak
sub bx,bp
sub bx,bp

sar bx,6
add bx,dx
sar si,6
add si,ax

loop MandelLoop

MandelBreak:
xchg ax,cx
stosb
cmp di, 0FA00h
jb FillLoop

;
xor ax,ax
int 16h
;
mov ax, 3
int 10h
;
int 20h
```

[9]
.

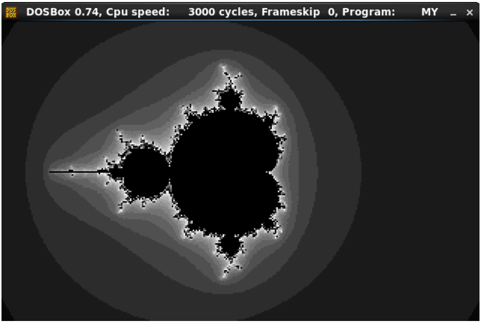


Figura 83.3:

[9] nasm fiole.asm -fbin -o file.com

Parte IX

Ejemplos de ingeniería inversa de formatos de archivo propietarios

Capítulo 84

84.1 Norton Guide:

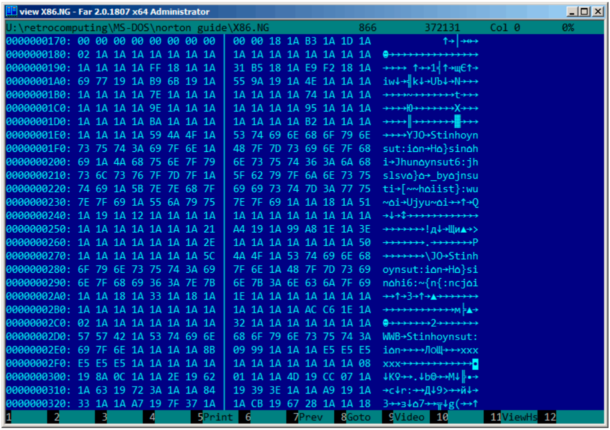


Figura 84.1:

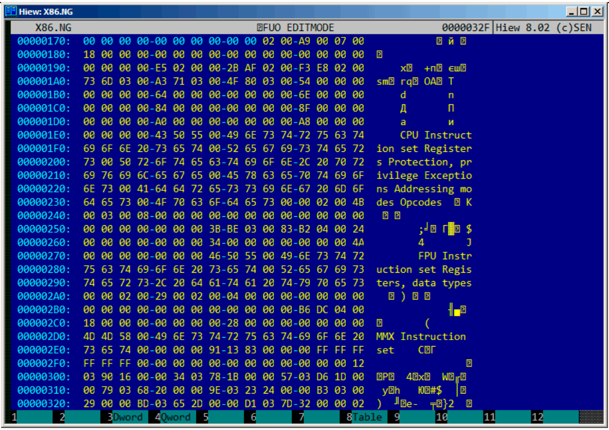


Figura 84.2: Hiew

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http://go.yurichev.com/17317.

84.1.1

Wolfram Mathematica 10.

Listing 84.1: Wolfram Mathematica 10

```
In[1]:= input = BinaryReadList["X86.NG"];

In[2]:= Entropy[2, input] // N
Out[2]= 5.62724
```

```
In[3]:= decrypted = Map[BitXor[#, 16^^1A] &,
    input];

In[4]:= Export["X86_decrypted.NG", decrypted
    , "Binary"];

In[5]:= Entropy[2, decrypted] // N
Out[5]= 5.62724

In[6]:= Entropy[2, ExampleData[{"Text", "
    ShakespearesSonnets"}]] // N
Out[6]= 4.42366
```

: http://go.yurichev.com/17350.

84.2

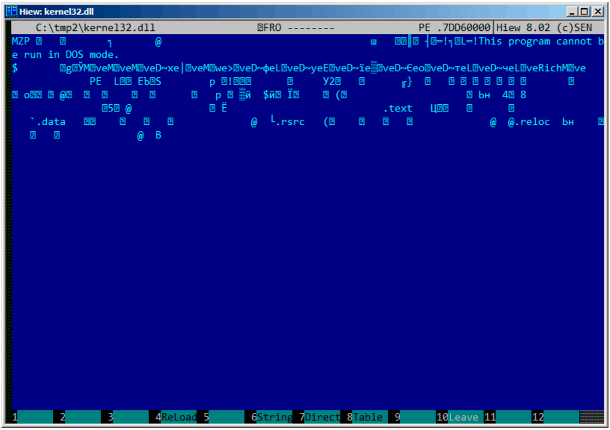

Figura 84.3:

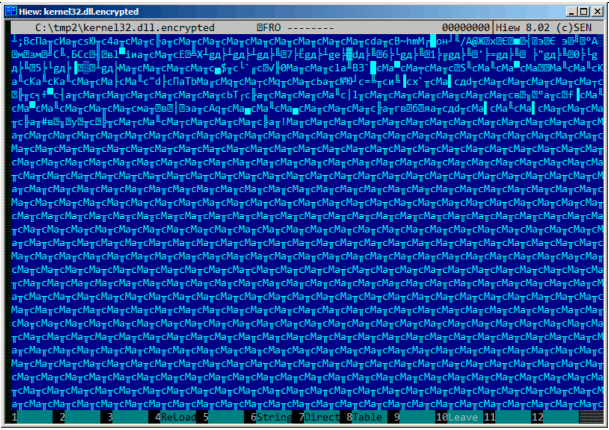
Hiew: kernel32.dll.encrypted
C:\tmp2\kernel32.dll.encrypted
FRO --------
00000000
Hiew 8.02 (c)SEN
1
2
3
4ReLoad
5
6String
7Direct
8Table
9
10Leave
11
12

Figura 84.4:

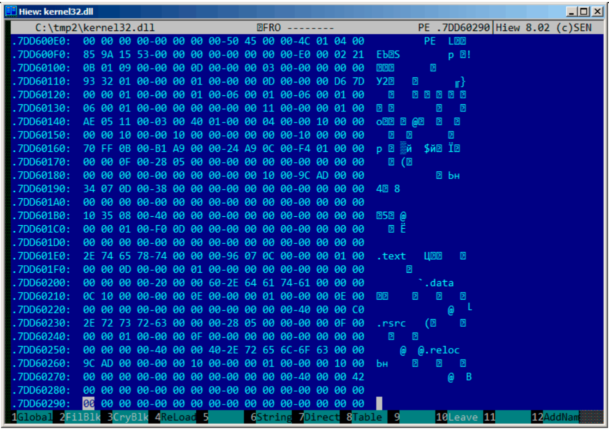

Figura 84.5: PE-

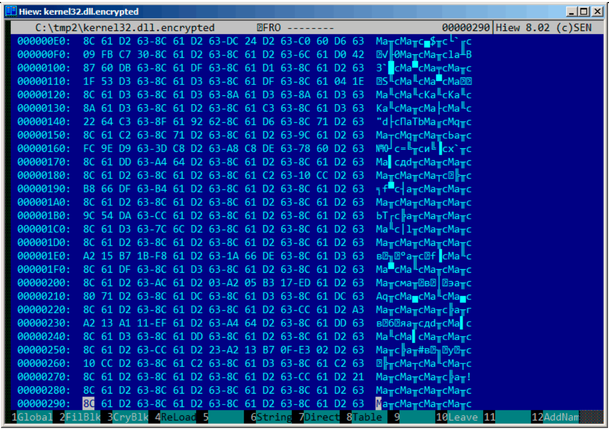

Figura 84.6:

: `8C 61 D2 63`.

http://go.yurichev.com/17352.

84.2.1 Ejercicio

http://go.yurichev.com/17353.

Capítulo 85

«Millenium Return to Earth» [1].

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1

 CAPÍTULO 85.

. . .

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Figura 85.1:

. 9538 .

:

Figura 85.2:

.

```
...> FC /b 2200save.i.v1 2200SAVE.I.V2

Comparing files 2200save.i.v1 and 2200SAVE.I.V2
00000016: 0D 04
00000017: 03 04
0000001C: 1F 1E
00000146: 27 3B
00000BDA: 0E 16
00000BDC: 66 9B
00000BDE: 0E 16
00000BE0: 0E 16
00000BE6: DB 4C
00000BE7: 00 01
00000BE8: 99 E8
00000BEC: A1 F3
```

```
00000BEE:  83  C7
00000BFB:  A8  28
00000BFD:  98  18
00000BFF:  A8  28
00000C01:  A8  28
00000C07:  D8  58
00000C09:  E4  A4
00000C0D:  38  B8
00000C0F:  E8  68
...
```

. . 0x0E (14) 0xBDA 0x16 (22) . . 0x66 (102) 0xBDC 0x9B (155) . .

: beginners.re.

:

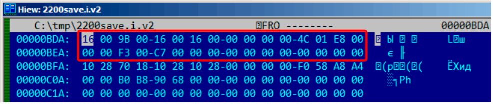

Figura 85.3: Hiew:

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. :

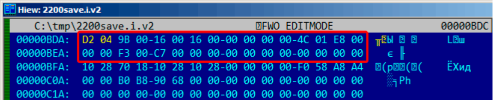

Figura 85.4: Hiew: (0x4D2)

:

Figura 85.5:

.

:

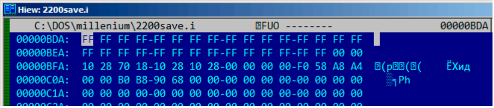

Figura 85.6: Hiew:

0xFFFF 65535, :

Figura 85.7: 65535 (0xFFFF)

! :

Figura 85.8:

. .

.

. .

: 63.4 on page 1297.

Capítulo 86

Oracle RDBMS:

```
----- Call Stack Trace -----
calling              call     entry                ↙
    ↘        argument values in hex
location             type     point                ↙
    ↘        (? means dubious value)
-------------------- -------- ↙
    ↘ -------------------- ↙
    ↘ ----------------------------
_kqvrow()                     00000000
_opifch2()+2729      CALLptr  00000000             ↙
    ↘        23D4B914 E47F264 1F19AE2
                                                   ↙
    ↘        EB1C8A8 1
_kpoal8()+2832       CALLrel  _opifch2()           ↙
    ↘        89 5 EB1CC74
```

```
_opiodr()+1248        CALLreg  00000000      ↙
    ↘        5E 1C EB1F0A0
_ttcpip()+1051        CALLreg  00000000      ↙
    ↘        5E 1C EB1F0A0 0
_opitsk()+1404        CALL???  00000000      ↙
    ↘        C96C040 5E EB1F0A0 0 EB1ED30
                                             ↙
    ↘        EB1F1CC 53E52E 0 EB1F1F8
_opiino()+980         CALLrel  _opitsk()     ↙
    ↘        0 0
_opiodr()+1248        CALLreg  00000000      ↙
    ↘        3C 4 EB1FBF4
_opidrv()+1201        CALLrel  _opiodr()     ↙
    ↘        3C 4 EB1FBF4 0
_sou2o()+55           CALLrel  _opidrv()     ↙
    ↘        3C 4 EB1FBF4
_opimai_real()+124    CALLrel  _sou2o()      ↙
    ↘        EB1FC04 3C 4 EB1FBF4
_opimai()+125         CALLrel  _opimai_real()↙
    ↘        2 EB1FC2C
_OracleThreadStart@   CALLrel  _opimai()     ↙
    ↘        2 EB1FF6C 7C88A7F4 EB1FC34 0
4()+830                                      ↙
    ↘        EB1FD04
77E6481C              CALLreg  00000000      ↙
    ↘        E41FF9C 0 0 E41FF9C 0 EB1FFC4
00000000              CALL???  00000000
```

. [1] .

1

Hiew:

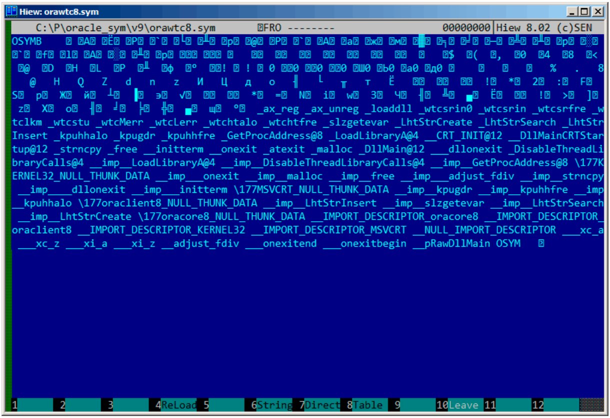

Figura 86.1:

•

 CAPÍTULO 86. ORACLE RDBMS:

:

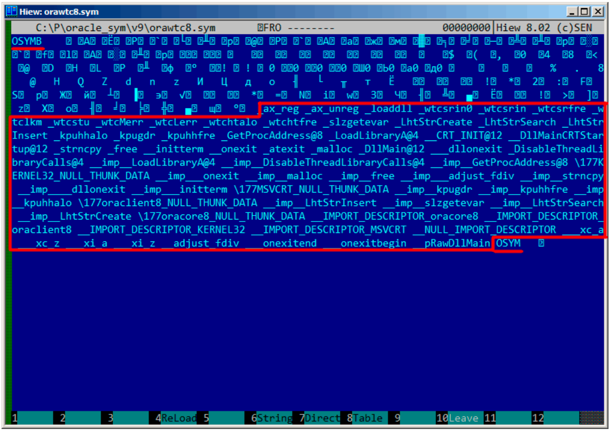

Figura 86.2:

.

```
strings strings_block | wc -l
66
```

. .

```
$ hexdump -C orawtc8.sym
00000000  4f 53 59 4d 42 00 00 00  00 10 00 10 80 10 00 10  |OSYMB...........|
00000010  f0 10 00 10 50 11 00 10  60 11 00 10 c0 11 00 10  |....P...`.......|
```

CAPÍTULO 86. ORACLE RDBMS:

```
00000020  d0 11 00 10 70 13 00 10  40 15 00 
    10 50 15 00 10  |....p...@...P...|
00000030  60 15 00 10 80 15 00 10  a0 15 00 
    10 a6 15 00 10  |`...............|
....
```

? . .

.

Hiew:

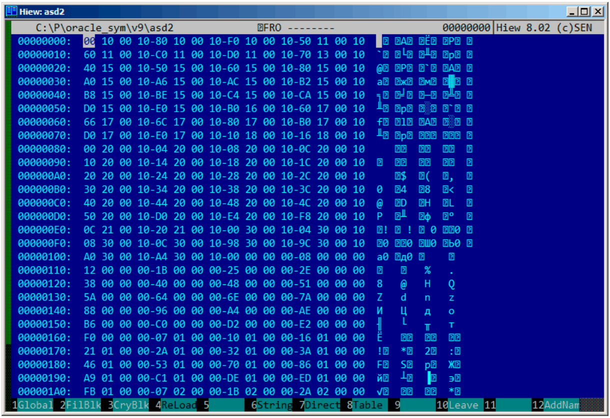

Figura 86.3:

•

:

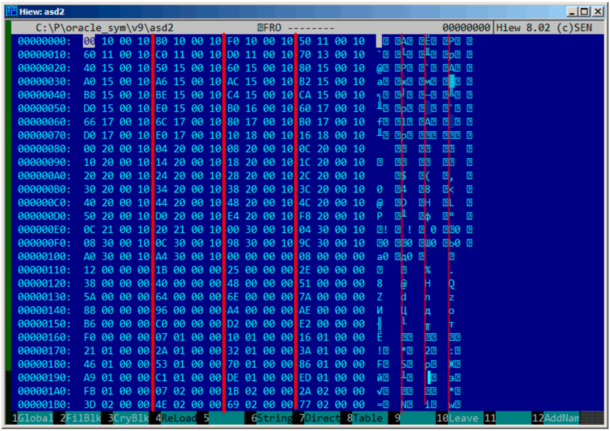

Figura 86.4:

.

:

Listing 86.1:

```
$ od -v -t x4 binary_block
0000000 10001000 10001080 100010f0 10001150
0000020 10001160 100011c0 100011d0 10001370
0000040 10001540 10001550 10001560 10001580
0000060 100015a0 100015a6 100015ac 100015b2
0000100 100015b8 100015be 100015c4 100015ca
0000120 100015d0 100015e0 100016b0 10001760
```

```
0000140 10001766 1000176c 10001780 100017b0
0000160 100017d0 100017e0 10001810 10001816
0000200 10002000 10002004 10002008 1000200c
0000220 10002010 10002014 10002018 1000201c
0000240 10002020 10002024 10002028 1000202c
0000260 10002030 10002034 10002038 1000203c
0000300 10002040 10002044 10002048 1000204c
0000320 10002050 100020d0 100020e4 100020f8
0000340 1000210c 10002120 10003000 10003004
0000360 10003008 1000300c 10003098 1000309c
0000400 100030a0 100030a4 00000000 00000008
0000420 00000012 0000001b 00000025 0000002e
0000440 00000038 00000040 00000048 00000051
0000460 0000005a 00000064 0000006e 0000007a
0000500 00000088 00000096 000000a4 000000ae
0000520 000000b6 000000c0 000000d2 000000e2
0000540 000000f0 00000107 00000110 00000116
0000560 00000121 0000012a 00000132 0000013a
0000600 00000146 00000153 00000170 00000186
0000620 000001a9 000001c1 000001de 000001ed
0000640 000001fb 00000207 0000021b 0000022a
0000660 0000023d 0000024e 00000269 00000277
0000700 00000287 00000297 000002b6 000002ca
0000720 000002dc 000002f0 00000304 00000321
0000740 0000033e 0000035d 0000037a 00000395
0000760 000003ae 000003b6 000003be 000003c6
0001000 000003ce 000003dc 000003e9 000003f8
0001020
```

. ? .

0x1000 .

```
.text:60351000 sub_60351000    proc near
.text:60351000
.text:60351000 arg_0           = dword ptr  ↙
    ↘ 8
.text:60351000 arg_4           = dword ptr  ↙
    ↘ 0Ch
.text:60351000 arg_8           = dword ptr  ↙
    ↘ 10h
.text:60351000
.text:60351000                 push    ebp
.text:60351001                 mov     ebp, ↙
    ↘ esp
.text:60351003                 mov     eax, ↙
    ↘ dword_60353014
.text:60351008                 cmp     eax, ↙
    ↘ 0FFFFFFFFh
.text:6035100B                 jnz     short↙
    ↘  loc_6035104F
.text:6035100D                 mov     ecx, ↙
    ↘ hModule
.text:60351013                 xor     eax, ↙
    ↘ eax
.text:60351015                 cmp     ecx, ↙
    ↘ 0FFFFFFFFh
.text:60351018                 mov     ↙
    ↘ dword_60353014, eax
.text:6035101D                 jnz     short↙
    ↘  loc_60351031
.text:6035101F                 call    ↙
    ↘ sub_603510F0
.text:60351024                 mov     ecx, ↙
    ↘ eax
.text:60351026                 mov     eax, ↙
```

```
    ↳ dword_60353014
.text:6035102B                 mov     ⤦
    ↳ hModule, ecx
.text:60351031
.text:60351031 loc_60351031:                           ⤦
    ↳             ; CODE XREF: sub_60351000+1⤦
    ↳ D
.text:60351031                 test    ecx, ⤦
    ↳ ecx
.text:60351033                 jbe     short⤦
    ↳  loc_6035104F
.text:60351035                 push    ⤦
    ↳ offset ProcName ; "ax_reg"
.text:6035103A                 push    ecx  ⤦
    ↳             ; hModule
.text:6035103B                 call    ds:⤦
    ↳ GetProcAddress
...
```

, «ax_reg» .   «ax_reg».

:

```
.text:60351080 sub_60351080    proc near
.text:60351080
.text:60351080 arg_0           = dword ptr  ⤦
    ↳ 8
.text:60351080 arg_4           = dword ptr  ⤦
    ↳ 0Ch
.text:60351080
.text:60351080                 push    ebp
.text:60351081                 mov     ebp, ⤦
    ↳ esp
```

```
.text:60351083                 mov     eax, ⤦
    ↳ dword_60353018
.text:60351088                 cmp     eax, ⤦
    ↳ 0FFFFFFFFh
.text:6035108B                 jnz     short⤦
    ↳  loc_603510CF
.text:6035108D                 mov     ecx, ⤦
    ↳ hModule
.text:60351093                 xor     eax, ⤦
    ↳ eax
.text:60351095                 cmp     ecx, ⤦
    ↳ 0FFFFFFFFh
.text:60351098                 mov     ⤦
    ↳ dword_60353018, eax
.text:6035109D                 jnz     short⤦
    ↳  loc_603510B1
.text:6035109F                 call    ⤦
    ↳ sub_603510F0
.text:603510A4                 mov     ecx, ⤦
    ↳ eax
.text:603510A6                 mov     eax, ⤦
    ↳ dword_60353018
.text:603510AB                 mov     ⤦
    ↳ hModule, ecx
.text:603510B1
.text:603510B1 loc_603510B1:                           ⤦
    ↳           ; CODE XREF: sub_60351080+1⤦
    ↳ D
.text:603510B1                 test    ecx, ⤦
    ↳ ecx
.text:603510B3                 jbe     short⤦
    ↳  loc_603510CF
.text:603510B5                 push    ⤦
```

```
    ↳ offset aAx_unreg ; "ax_unreg"
.text:603510BA                  push    ecx  ↙
    ↳           ; hModule
.text:603510BB                  call    ds:↙
    ↳ GetProcAddress
...
```

.

? 0x0000? [0...0x3F8]..

```
#include <stdio.h>
#include <stdint.h>
#include <io.h>
#include <assert.h>
#include <malloc.h>
#include <fcntl.h>
#include <string.h>

int main (int argc, char *argv[])
{
        uint32_t sig, cnt, offset;
        uint32_t *d1, *d2;
        int     h, i, remain, file_len;
        char    *d3;
        uint32_t array_size_in_bytes;

        assert (argv[1]); // file name
        assert (argv[2]); // additional ↙
    ↳ offset (if needed)

        // additional offset
        assert (sscanf (argv[2], "%X", &↙
```

```
↳ offset)==1);

    // get file length
    assert ((h=open (argv[1], _O_RDONLY ↙
↳ | _O_BINARY, 0))!=-1);
    assert ((file_len=lseek (h, 0, ↙
↳ SEEK_END))!=-1);
    assert (lseek (h, 0, SEEK_SET)!=-1);

    // read signature
    assert (read (h, &sig, 4)==4);
    // read count
    assert (read (h, &cnt, 4)==4);

    assert (sig==0x4D59534F); // OSYM

    // skip timedatestamp (for 11g)
    //_lseek (h, 4, 1);

    array_size_in_bytes=cnt*sizeof(↙
↳ uint32_t);

    // load symbol addresses array
    d1=(uint32_t*)malloc (↙
↳ array_size_in_bytes);
    assert (d1);
    assert (read (h, d1, ↙
↳ array_size_in_bytes)==↙
↳ array_size_in_bytes);

    // load string offsets array
    d2=(uint32_t*)malloc (↙
↳ array_size_in_bytes);
```

```
        assert (d2);
        assert (read (h, d2, ↙
    ↘ array_size_in_bytes)==↙
    ↘ array_size_in_bytes);

        // calculate strings block size
        remain=file_len-(8+4)-(cnt*8);

        // load strings block
        assert (d3=(char*)malloc (remain));
        assert (read (h, d3, remain)==remain↙
    ↘ );

        printf ("#include <idc.idc>\n\n");
        printf ("static main() {\n");

        for (i=0; i<cnt; i++)
                printf ("\tMakeName(0x%08X, ↙
    ↘ \"%s\");\n", offset + d1[i], &d3[d2[i↙
    ↘ ]]);

        printf ("}\n");

        close (h);
        free (d1); free (d2); free (d3);
};
```

:

```
#include <idc.idc>

static main() {
        MakeName(0x60351000, "_ax_reg");
```

```
        MakeName(0x60351080, "_ax_unreg");
        MakeName(0x603510F0, "_loaddll");
        MakeName(0x60351150, "_wtcsrin0");
        MakeName(0x60351160, "_wtcsrin");
        MakeName(0x603511C0, "_wtcsrfre");
        MakeName(0x603511D0, "_wtclkm");
        MakeName(0x60351370, "_wtcstu");
...
}
```

: beginners.re.

. ?

:

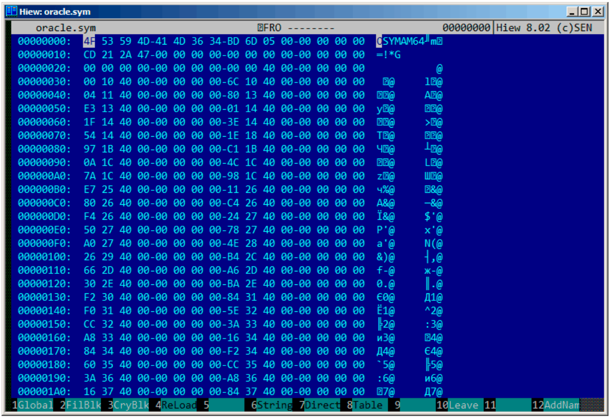

Figura 86.5:

! : GitHub.

 CAPÍTULO 87. ORACLE RDBMS: ARCHIVOS .MSB

Capítulo 87

Oracle RDBMS: Archivos .MSB

.

[1].

ORAUS.MSG :

Listing 87.1: ORAUS.MSG

```
00000, 00000, "normal, successful completion
    "
00001, 00000, "unique constraint (%s.%s) 
    violated"
```

1

```
00017, 00000, "session requested to set ⤦
    ↪ trace event"
00018, 00000, "maximum number of sessions ⤦
    ↪ exceeded"
00019, 00000, "maximum number of session ⤦
    ↪ licenses exceeded"
00020, 00000, "maximum number of processes ⤦
    ↪ (%s) exceeded"
00021, 00000, "session attached to some ⤦
    ↪ other process; cannot switch session"
00022, 00000, "invalid session ID; access ⤦
    ↪ denied"
00023, 00000, "session references process ⤦
    ↪ private memory; cannot detach session"
00024, 00000, "logins from more than one ⤦
    ↪ process not allowed in single-process ⤦
    ↪ mode"
00025, 00000, "failed to allocate %s"
00026, 00000, "missing or invalid session ID⤦
    ↪ "
00027, 00000, "cannot kill current session"
00028, 00000, "your session has been killed"
00029, 00000, "session is not a user session⤦
    ↪ "
00030, 00000, "User session ID does not ⤦
    ↪ exist."
00031, 00000, "session marked for kill"
...
```

..

 CAPÍTULO 87. ORACLE RDBMS: ARCHIVOS .MSB

ORAUS.MSB . :

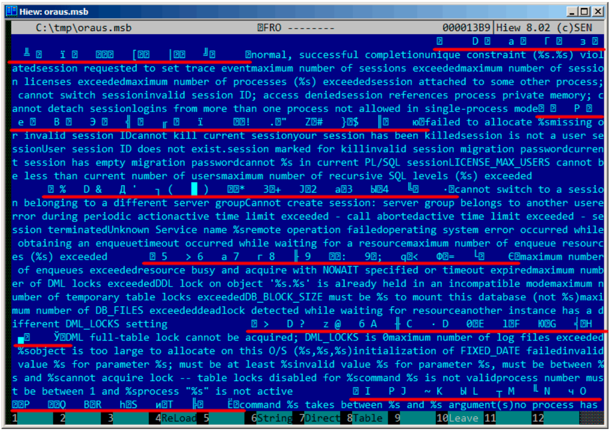

Figura 87.1: Hiew:

. .

 CAPÍTULO 87. ORACLE RDBMS: ARCHIVOS .MSB

:

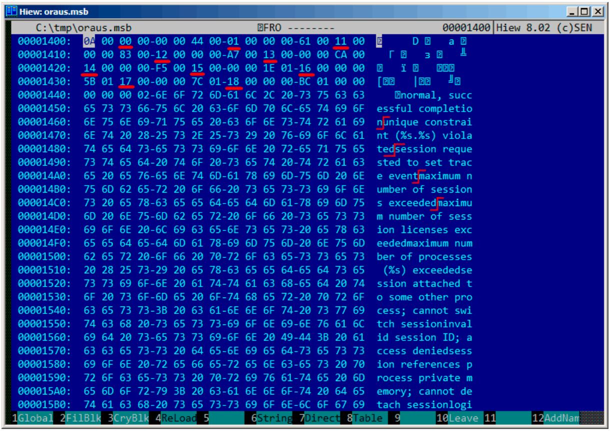

Figura 87.2: Hiew:

. . . . . ORAUS.MSG : 0, 1, 17 (0x11), 18 (0x12), 19 (0x13), 20 (0x14), 21 (0x15), 22 (0x16), 23 (0x17), 24 (0x18)... . .

. . .

«normal, successful completion» 29 (0x1D) . «unique constraint (%s.%s) violated» 34 (0x22) . (0x1D o/y 0x22) .

. Oracle RDBMS ? «normal, successful completion» 0x1444 () 0x44 (). «unique constraint (%s.%s) violated» 0x1461 () 0x61 (). (0x44 y 0x61) ! .

:

CAPÍTULO 87. ORACLE RDBMS: ARCHIVOS .MSB

- 16-;
- 16-;
- 16-.

. ? ! ! .

 CAPÍTULO 87. ORACLE RDBMS: ARCHIVOS .MSB

:

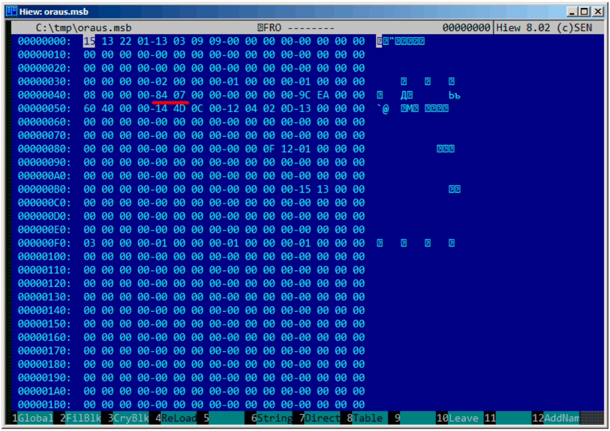

Figura 87.3: Hiew:

. .

 CAPÍTULO 87. ORACLE RDBMS: ARCHIVOS .MSB

:

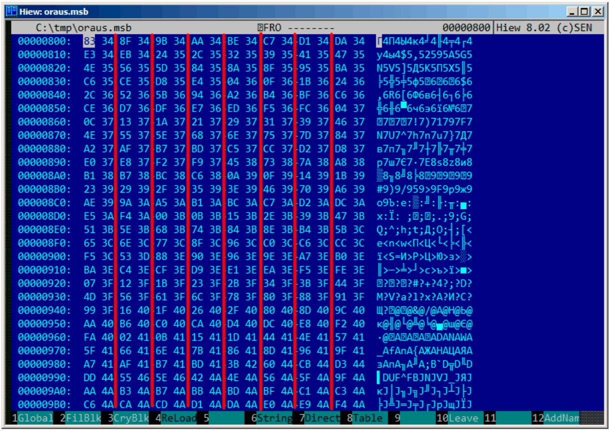

Figura 87.4: Hiew: last_errnos

:

- last_errnos ();
- ;
- ;
- ;
- ;
- ;
- .

 CAPÍTULO 87. ORACLE RDBMS: ARCHIVOS .MSB

: beginners.re.

(Oracle RDBMS 11.1.0.6): beginners.re, beginners.re.

87.1

. . .

Parte X

Capítulo 88

npad

`listing.inc` (MSVC):

```
;; LISTING.INC
;;
;; This file contains assembler macros and ⤦
    ↳ is included by the files created
;; with the -FA compiler switch to be ⤦
    ↳ assembled by MASM (Microsoft Macro
;; Assembler).
;;
;; Copyright (c) 1993-2003, Microsoft ⤦
    ↳ Corporation. All rights reserved.

;; non destructive nops
npad macro size
```

```
if size eq 1
  nop
else
 if size eq 2
   mov edi, edi
 else
  if size eq 3
    ; lea ecx, [ecx+00]
    DB 8DH, 49H, 00H
  else
   if size eq 4
     ; lea esp, [esp+00]
     DB 8DH, 64H, 24H, 00H
   else
    if size eq 5
      add eax, DWORD PTR 0
    else
     if size eq 6
       ; lea ebx, [ebx+00000000]
       DB 8DH, 9BH, 00H, 00H, 00H, 00H
     else
      if size eq 7
        ; lea esp, [esp+00000000]
        DB 8DH, 0A4H, 24H, 00H, 00H, 00H, 00
    ↪ H
      else
       if size eq 8
        ; jmp .+8; .npad 6
        DB 0EBH, 06H, 8DH, 9BH, 00H, 00H, 00
    ↪ H, 00H
       else
        if size eq 9
         ; jmp .+9; .npad 7
```

```
     DB 0EBH, 07H, 8DH, 0A4H, 24H, 00H, ↙
↘ 00H, 00H, 00H
    else
     if size eq 10
      ; jmp .+A; .npad 7; .npad 1
      DB 0EBH, 08H, 8DH, 0A4H, 24H, 00H,↙
↘  00H, 00H, 00H, 90H
     else
      if size eq 11
       ; jmp .+B; .npad 7; .npad 2
       DB 0EBH, 09H, 8DH, 0A4H, 24H, 00H↙
↘ , 00H, 00H, 00H, 8BH, 0FFH
      else
       if size eq 12
        ; jmp .+C; .npad 7; .npad 3
        DB 0EBH, 0AH, 8DH, 0A4H, 24H, 00↙
↘ H, 00H, 00H, 00H, 8DH, 49H, 00H
       else
        if size eq 13
         ; jmp .+D; .npad 7; .npad 4
         DB 0EBH, 0BH, 8DH, 0A4H, 24H, ↙
↘ 00H, 00H, 00H, 00H, 8DH, 64H, 24H, 00H
        else
         if size eq 14
          ; jmp .+E; .npad 7; .npad 5
          DB 0EBH, 0CH, 8DH, 0A4H, 24H, ↙
↘ 00H, 00H, 00H, 00H, 05H, 00H, 00H, 00H↙
↘ , 00H
         else
          if size eq 15
           ; jmp .+F; .npad 7; .npad 6
           DB 0EBH, 0DH, 8DH, 0A4H, 24H,↙
↘  00H, 00H, 00H, 00H, 8DH, 9BH, 00H, 00↙
```

```
    ↳ H, 00H, 00H
                else
                 %out error: unsupported npad ↙
    ↳ size
                 .err
                endif
               endif
              endif
             endif
            endif
           endif
          endif
         endif
        endif
       endif
      endif
     endif
    endif
   endif
  endif
endm
```

Capítulo 89

89.1

. . . . .

.

89.2 x86

:

- . `0x90` (NOP).
- `74 xx` (JZ), . (*jump offset*).
- : `0xEB` JMP.
- RETN (0xC3) . `stdcall` (64.2 on page 1299). `stdcall` RETN (0xC2).

- 0 o 1. `MOV EAX, 0` o `MOV EAX, 1`,. `XOR EAX, EAX` (2 `0x31 0xC0`) o `XOR EAX, EAX / INC EAX` (3 `0x31 0xC0 0x40`).

. ... .

.

(68.2.6 on page 1354) (: Spanish text placeholder.7.12).

 CAPÍTULO 90. COMPILER INTRINSIC

Capítulo 90

Compiler intrinsic

: MSDN.

Capítulo 91

Listing 91.1: libserver11.a

```
.text:08114CF1                              loc_8114CF1
    ↳ : ; CODE XREF: __PGOSF539_kdlimemSer
    ↳ +89A
.text:08114CF1
    ↳   ; __PGOSF539_kdlimemSer+3994
.text:08114CF1 8B 45 08                     mov
    ↳  eax, [ebp+arg_0]
.text:08114CF4 0F B6 50 14                  movzx
    ↳  edx, byte ptr [eax+14h]
.text:08114CF8 F6 C2 01                     test
    ↳  dl, 1
.text:08114CFB 0F 85 17 08 00 00            jnz
    ↳  loc_8115518
.text:08114D01 85 C9                        test
    ↳  ecx, ecx
.text:08114D03 0F 84 8A 00 00 00            jz
    ↳  loc_8114D93
```

```
.text:08114D09 0F 84 09 08 00 00       jz    ↙
    ↘ loc_8115518
.text:08114D0F 8B 53 08                mov   ↙
    ↘ edx, [ebx+8]
.text:08114D12 89 55 FC                mov   ↙
    ↘ [ebp+var_4], edx
.text:08114D15 31 C0                   xor   ↙
    ↘ eax, eax
.text:08114D17 89 45 F4                mov   ↙
    ↘ [ebp+var_C], eax
.text:08114D1A 50                      push  ↙
    ↘ eax
.text:08114D1B 52                      push  ↙
    ↘ edx
.text:08114D1C E8 03 54 00 00          call  ↙
    ↘ len2nbytes
.text:08114D21 83 C4 08                add   ↙
    ↘ esp, 8
```

Listing 91.2:

```
.text:0811A2A5                     loc_811A2A5↙
    ↘ : ; CODE XREF: kdliSerLengths+11C
.text:0811A2A5                                ↙
    ↘  ; kdliSerLengths+1C1
.text:0811A2A5 8B 7D 08                mov   ↙
    ↘ edi, [ebp+arg_0]
.text:0811A2A8 8B 7F 10                mov   ↙
    ↘ edi, [edi+10h]
.text:0811A2AB 0F B6 57 14             movzx ↙
    ↘ edx, byte ptr [edi+14h]
.text:0811A2AF F6 C2 01                test  ↙
    ↘ dl, 1
```

```
.text:0811A2B2 75 3E                        jnz    ↙
    ↳ short loc_811A2F2
.text:0811A2B4 83 E0 01                     and    ↙
    ↳ eax, 1
.text:0811A2B7 74 1F                        jz     ↙
    ↳ short loc_811A2D8
.text:0811A2B9 74 37                        jz     ↙
    ↳ short loc_811A2F2
.text:0811A2BB 6A 00                        push   ↙
    ↳ 0
.text:0811A2BD FF 71 08                     push   ↙
    ↳ dword ptr [ecx+8]
.text:0811A2C0 E8 5F FE FF FF               call   ↙
    ↳ len2nbytes
```

: 19.2.4 on page 595, 39.3 on page 927, 47.7 on page 1010, 18.7 on page 545, 12.4.1 on page 271, 19.5.2 on page 623.

Capítulo 92

OpenMP

OpenMP .

. . «proof of work» [1] ( ).

Bitcoin, «hello, world!_» «hello, world!_<number>» .

0..INT32_MAX-1 (, 0x7FFFFFFE o 2147483646).

:

```
#include <stdio.h>
#include <string.h>
#include <stdlib.h>
#include <time.h>
#include "sha512.h"
```

1

CAPÍTULO 92. OPENMP

```
int found=0;
int32_t checked=0;

int32_t* __min;
int32_t* __max;

time_t start;

#ifdef __GNUC__
#define min(X,Y) ((X) < (Y) ? (X) : (Y))
#define max(X,Y) ((X) > (Y) ? (X) : (Y))
#endif

void check_nonce (int32_t nonce)
{
        uint8_t buf[32];
        struct sha512_ctx ctx;
        uint8_t res[64];

        // update statistics
        int t=omp_get_thread_num();

        if (__min[t]==-1)
                __min[t]=nonce;
        if (__max[t]==-1)
                __max[t]=nonce;

        __min[t]=min(__min[t], nonce);
        __max[t]=max(__max[t], nonce);

        // idle if valid nonce found
        if (found)
```

```
                return;

        memset (buf, 0, sizeof(buf));
        sprintf (buf, "hello, world!_%d", ↙
    ↳ nonce);

        sha512_init_ctx (&ctx);
        sha512_process_bytes (buf, strlen(↙
    ↳ buf), &ctx);
        sha512_finish_ctx (&ctx, &res);
        if (res[0]==0 && res[1]==0 && res↙
    ↳ [2]==0)
        {
                printf ("found (thread %d): ↙
    ↳ [%s]. seconds spent=%d\n", t, buf, ↙
    ↳ time(NULL)-start);
                found=1;
        };
        #pragma omp atomic
        checked++;

        #pragma omp critical
        if ((checked % 100000)==0)
                printf ("checked=%d\n", ↙
    ↳ checked);
};

int main()
{
        int32_t i;
        int threads=omp_get_max_threads();
        printf ("threads=%d\n", threads);
```

```
        __min=(int32_t*)malloc(threads*↙
    ↘ sizeof(int32_t));
        __max=(int32_t*)malloc(threads*↙
    ↘ sizeof(int32_t));
        for (i=0; i<threads; i++)
                __min[i]=__max[i]=-1;

        start=time(NULL);

        #pragma omp parallel for
        for (i=0; i<INT32_MAX; i++)
                check_nonce (i);

        for (i=0; i<threads; i++)
                printf ("__min[%d]=0x%08x ↙
    ↘ __max[%d]=0x%08x\n", i, __min[i], i, ↙
    ↘ __max[i]);

        free(__min); free(__max);
};
```

check_nonce() .

:

```
        #pragma omp parallel for
        for (i=0; i<INT32_MAX; i++)
                check_nonce (i);
```

#pragma check_nonce() INT32_MAX (0x7fffffff o 2147483647). #pragma, [2].

[2] N.B.:

[3] en MSVC 2012:

```
cl openmp_example.c sha512.obj /openmp /O1 /↙
    ↳ Zi /Faopenmp_example.asm
```

GCC:

```
gcc -fopenmp 2.c sha512.c -S -masm=intel
```

92.1 MSVC

MSVC 2012 :

Listing 92.1: MSVC 2012

```
        push    OFFSET _main$omp$1
        push    0
        push    1
        call    __vcomp_fork
        add     esp, 16                          ↙
    ↳                    ; 00000010H
```

vcomp OpenMP vcomp*.dll. .

_mainomp1:

Listing 92.2: MSVC 2012

```
$T1 = -8                                         ↙
    ↳                    ; size = 4
```

[3] sha512.(c|h) y u64.h : http://go.yurichev.com/17324

```
$T2 = -4                                                      ⤦
    ↪                      ; size = 4
_main$omp$1 PROC                                              ⤦
    ↪                      ; COMDAT
        push    ebp
        mov     ebp, esp
        push    ecx
        push    ecx
        push    esi
        lea     eax, DWORD PTR $T2[ebp]
        push    eax
        lea     eax, DWORD PTR $T1[ebp]
        push    eax
        push    1
        push    1
        push    2147483646                                    ⤦
    ↪                      ; 7ffffffeH
        push    0
        call    ⤦
    ↪ __vcomp_for_static_simple_init
        mov     esi, DWORD PTR $T1[ebp]
        add     esp, 24                                       ⤦
    ↪                      ; 00000018H
        jmp     SHORT $LN6@main$omp$1
$LL2@main$omp$1:
        push    esi
        call    _check_nonce
        pop     ecx
        inc     esi
$LN6@main$omp$1:
        cmp     esi, DWORD PTR $T2[ebp]
        jle     SHORT $LL2@main$omp$1
        call    __vcomp_for_static_end
```

```
        pop     esi
        leave
        ret     0
_main$omp$1 ENDP
```

n n . vcomp_for_static_simple_init() for() . $T1 y $T2. 7ffffffeh (o 2147483646) vcomp_for_stati
.

check_nonce() .

check_nonce() .

:

```
threads=4
...
checked=2800000
checked=3000000
checked=3200000
checked=3300000
found (thread 3): [hello, world!_1611446522
    ↳ ]. seconds spent=3
__min[0]=0x00000000 __max[0]=0x1fffffff
__min[1]=0x20000000 __max[1]=0x3fffffff
__min[2]=0x40000000 __max[2]=0x5fffffff
__min[3]=0x60000000 __max[3]=0x7ffffffe
```

:

```
C:\...\sha512sum test
000000
    ↳ f4a8fac5a4ed38794da4c1e39f54279ad5d9bb3c5465
    ↳
```

```
df6e3fe6019f5764fc9975e505a7395fed780fee50eb38dd4c
    ↳  *test
```

task manager : . . . CPU OpenMP .

. .

.

:

```
        #pragma omp atomic
        checked++;

        #pragma omp critical
        if ((checked % 100000)==0)
                printf ("checked=%d\n", ↙
    ↳ checked);
```

Listing 92.3: MSVC 2012

```
        push    edi
        push    OFFSET _checked
        call    __vcomp_atomic_add_i4
; Line 55
        push    OFFSET _$vcomp$critsect$
        call    __vcomp_enter_critsect
        add     esp, 12                                ↙
    ↳                  ; 0000000cH
; Line 56
        mov     ecx, DWORD PTR _checked
        mov     eax, ecx
        cdq
```

```
        mov     esi, 100000                     ↙
    ↘                       ; 000186a0H
        idiv    esi
        test    edx, edx
        jne     SHORT $LN1@check_nonc
; Line 57
        push    ecx
        push    OFFSET ??↙
    ↘ _C@_0M@NPNHLIOO@checked?$DN?$CFd?6?↙
    ↘ $AA@
        call    _printf
        pop     ecx
        pop     ecx
$LN1@check_nonc:
        push    DWORD PTR _$vcomp$critsect$
        call    __vcomp_leave_critsect
        pop     ecx
```

vcomp_atomic_add_i4() vcomp*.dll LOCK XADD[4].

vcomp_enter_critsect() EnterCriticalSection() [5].

92.2 GCC

GCC 4.8.1 , .

Listing 92.4: GCC 4.8.1

[4]: A.6.1 on page 1877
[5]: 68.4 on page 1408

```
        mov     edi, OFFSET FLAT:main.↙
    ↘ _omp_fn.0
        call    GOMP_parallel_start
        mov     edi, 0
        call    main._omp_fn.0
        call    GOMP_parallel_end
```

..

main._omp_fn.0:

Listing 92.5: GCC 4.8.1

```
main._omp_fn.0:
        push    rbp
        mov     rbp, rsp
        push    rbx
        sub     rsp, 40
        mov     QWORD PTR [rbp-40], rdi
        call    omp_get_num_threads
        mov     ebx, eax
        call    omp_get_thread_num
        mov     esi, eax
        mov     eax, 2147483647 ; 0x7FFFFFFF
        cdq
        idiv    ebx
        mov     ecx, eax
        mov     eax, 2147483647 ; 0x7FFFFFFF
        cdq
        idiv    ebx
        mov     eax, edx
        cmp     esi, eax
        jl      .L15
```

```
.L18:
        imul    esi, ecx
        mov     edx, esi
        add     eax, edx
        lea     ebx, [rax+rcx]
        cmp     eax, ebx
        jge     .L14
        mov     DWORD PTR [rbp-20], eax
.L17:
        mov     eax, DWORD PTR [rbp-20]
        mov     edi, eax
        call    check_nonce
        add     DWORD PTR [rbp-20], 1
        cmp     DWORD PTR [rbp-20], ebx
        jl      .L17
        jmp     .L14
.L15:
        mov     eax, 0
        add     ecx, 1
        jmp     .L18
.L14:
        add     rsp, 40
        pop     rbx
        pop     rbp
        ret
```

omp_get_num_threads() y omp_get_thread_num() ,. check_nonce().

GCC LOCK ADD :

Listing 92.6: GCC 4.8.1

```
        lock add        DWORD PTR checked[
    ↳ rip], 1
        call    GOMP_critical_start
        mov     ecx, DWORD PTR checked[rip]
        mov     edx, 351843721
        mov     eax, ecx
        imul    edx
        sar     edx, 13
        mov     eax, ecx
        sar     eax, 31
        sub     edx, eax
        mov     eax, edx
        imul    eax, eax, 100000
        sub     ecx, eax
        mov     eax, ecx
        test    eax, eax
        jne     .L7
        mov     eax, DWORD PTR checked[rip]
        mov     esi, eax
        mov     edi, OFFSET FLAT:.LC2 ; "
    ↳ checked=%d\n"
        mov     eax, 0
        call    printf
.L7:
        call    GOMP_critical_end
```

GOMP GNU OpenMP. vcomp*.dll, : GitHub.

Capítulo 93

Itanium

.

:

Listing 93.1: Linux kernel 3.2.0.4

```
#define TEA_ROUNDS              32
#define TEA_DELTA               0x9e3779b9

static void tea_encrypt(struct crypto_tfm *
    ↘ tfm, u8 *dst, const u8 *src)
{
        u32 y, z, n, sum = 0;
        u32 k0, k1, k2, k3;
        struct tea_ctx *ctx = crypto_tfm_ctx
    ↘ (tfm);
```

```
        const __le32 *in = (const __le32 *)↙
    ↘ src;
        __le32 *out = (__le32 *)dst;

        y = le32_to_cpu(in[0]);
        z = le32_to_cpu(in[1]);

        k0 = ctx->KEY[0];
        k1 = ctx->KEY[1];
        k2 = ctx->KEY[2];
        k3 = ctx->KEY[3];

        n = TEA_ROUNDS;

        while (n-- > 0) {
                sum += TEA_DELTA;
                y += ((z << 4) + k0) ^ (z + ↙
    ↘ sum) ^ ((z >> 5) + k1);
                z += ((y << 4) + k2) ^ (y + ↙
    ↘ sum) ^ ((y >> 5) + k3);
        }

        out[0] = cpu_to_le32(y);
        out[1] = cpu_to_le32(z);
}
```

:

Listing 93.2: Linux Kernel 3.2.0.4 Itanium 2 (McKinley)

```
0090|                                      ↙
    ↘ tea_encrypt:
0090|08 80 80 41 00 21                     ↙
```

```
    ↳ adds r16 = 96, r32                    // ptr↙
    ↳  to ctx->KEY[2]
0096|80 C0 82 00 42 00                         ↙
    ↳ adds r8 = 88, r32                     // ptr↙
    ↳  to ctx->KEY[0]
009C|00 00 04 00                               ↙
    ↳ nop.i 0
00A0|09 18 70 41 00 21                         ↙
    ↳ adds r3 = 92, r32                     // ptr↙
    ↳  to ctx->KEY[1]
00A6|F0 20 88 20 28 00                         ↙
    ↳ ld4 r15 = [r34], 4                    // ↙
    ↳ load z
00AC|44 06 01 84                               ↙
    ↳ adds r32 = 100, r32;;                 // ptr↙
    ↳  to ctx->KEY[3]
00B0|08 98 00 20 10 10                         ↙
    ↳ ld4 r19 = [r16]                       // r19↙
    ↳ =k2
00B6|00 01 00 00 42 40                         ↙
    ↳ mov r16 = r0                          // r0 ↙
    ↳ always contain zero
00BC|00 08 CA 00                               ↙
    ↳ mov.i r2 = ar.lc                      // ↙
    ↳ save lc register
00C0|05 70 00 44 10 10 9E FF FF FF 7F 20       ↙
    ↳ ld4 r14 = [r34]                       // ↙
    ↳ load y
00CC|92 F3 CE 6B                               ↙
    ↳ movl r17 = 0xFFFFFFFF9E3779B9;; // ↙
    ↳ TEA_DELTA
00D0|08 00 00 00 01 00                         ↙
    ↳ nop.m 0
```

```
00D6|50 01 20 20 20 00                                    ↙
    ↘ ld4 r21 = [r8]                              // r21↙
    ↘ =k0
00DC|F0 09 2A 00                                          ↙
    ↘ mov.i ar.lc = 31                            // ↙
    ↘ TEA_ROUNDS is 32
00E0|0A A0 00 06 10 10                                    ↙
    ↘ ld4 r20 = [r3];;                            // r20↙
    ↘ =k1
00E6|20 01 80 20 20 00                                    ↙
    ↘ ld4 r18 = [r32]                             // r18↙
    ↘ =k3
00EC|00 00 04 00                                          ↙
    ↘ nop.i 0
00F0|
00F0|                                                 ↙
    ↘ loc_F0:
00F0|09 80 40 22 00 20                                    ↙
    ↘ add r16 = r16, r17                          // r16↙
    ↘ =sum, r17=TEA_DELTA
00F6|D0 71 54 26 40 80                                    ↙
    ↘ shladd r29 = r14, 4, r21                    // r14↙
    ↘ =y, r21=k0
00FC|A3 70 68 52                                          ↙
    ↘ extr.u r28 = r14, 5, 27;;
0100|03 F0 40 1C 00 20                                    ↙
    ↘ add r30 = r16, r14
0106|B0 E1 50 00 40 40                                    ↙
    ↘ add r27 = r28, r20;;                        // r20↙
    ↘ =k1
010C|D3 F1 3C 80                                          ↙
    ↘ xor r26 = r29, r30;;
0110|0B C8 6C 34 0F 20                                    ↙
```

```
    ↳ xor r25 = r27, r26;;
0116|F0 78 64 00 40 00                                    ↙
    ↳ add r15 = r15, r25                         // r15↙
    ↳ =z
011C|00 00 04 00                                          ↙
    ↳ nop.i 0;;
0120|00 00 00 00 01 00                                    ↙
    ↳ nop.m 0
0126|80 51 3C 34 29 60                                    ↙
    ↳ extr.u r24 = r15, 5, 27
012C|F1 98 4C 80                                          ↙
    ↳ shladd r11 = r15, 4, r19                   // r19↙
    ↳ =k2
0130|0B B8 3C 20 00 20                                    ↙
    ↳ add r23 = r15, r16;;
0136|A0 C0 48 00 40 00                                    ↙
    ↳ add r10 = r24, r18                         // r18↙
    ↳ =k3
013C|00 00 04 00                                          ↙
    ↳ nop.i 0;;
0140|0B 48 28 16 0F 20                                    ↙
    ↳ xor r9 = r10, r11;;
0146|60 B9 24 1E 40 00                                    ↙
    ↳ xor r22 = r23, r9
014C|00 00 04 00                                          ↙
    ↳ nop.i 0;;
0150|11 00 00 00 01 00                                    ↙
    ↳ nop.m 0
0156|E0 70 58 00 40 A0                                    ↙
    ↳ add r14 = r14, r22
015C|A0 FF FF 48                                          ↙
    ↳ br.cloop.sptk.few loc_F0;;
0160|09 20 3C 42 90 15                                    ↙
```

```
     ↳ st4 [r33] = r15, 4                     // ↙
     ↳ store z
0166|00 00 00 02 00 00                          ↙
     ↳ nop.m 0
016C|20 08 AA 00                                ↙
     ↳ mov.i ar.lc = r2;;                     // ↙
     ↳ restore lc register
0170|11 00 38 42 90 11                          ↙
     ↳ st4 [r33] = r14                        // ↙
     ↳ store y
0176|00 00 00 02 00 80                          ↙
     ↳ nop.i 0
017C|08 00 84 00                                ↙
     ↳ br.ret.sptk.many b0;;
```

. .

.

. .

.

. . .

.

. [b0-cc].

We also see a stop bit at 10c. . . . . .

:

http://go.yurichev.com/17322.

: [Bur], [haq].

CAPÍTULO 93. ITANIUM
:
MSDN.

Capítulo 94

(78.3 on page 1513 o 53.5 on page 1170), .

8086/8088 .

. 64KB. . . .

$$real\_address = (segment\_register \ll 4) + address\_register$$

.

64KB.

Ejemplos ampliamente populares incluyen DOS/4GW (el videojuego DOOM fue compilado para él), Phar Lap, PMODE.

.

Capítulo 95

95.1 Profile-guided optimization

.

.

.

.

:

Listing 95.1: orageneric11.dll (win32)

```
                 public _skgfsync
_skgfsync        proc near

; address 0x6030D86A

                 db      66h
```

```
                nop
                push    ebp
                mov     ebp, esp
                mov     edx, [ebp+0Ch]
                test    edx, edx
                jz      short loc_6030D884
                mov     eax, [edx+30h]
                test    eax, 400h
                jnz     __VInfreq__skgfsync ⤦
    ↳ ; write to log
continue:
                mov     eax, [ebp+8]
                mov     edx, [ebp+10h]
                mov     dword ptr [eax], 0
                lea     eax, [edx+0Fh]
                and     eax, 0FFFFFFFCh
                mov     ecx, [eax]
                cmp     ecx, 45726963h
                jnz     error                  ⤦
    ↳ ; exit with error
                mov     esp, ebp
                pop     ebp
                retn
_skgfsync       endp

...

; address 0x60B953F0

__VInfreq__skgfsync:
                mov     eax, [edx]
                test    eax, eax
                jz      continue
```

```
                mov     ecx, [ebp+10h]
                push    ecx
                mov     ecx, [ebp+8]
                push    edx
                push    ecx
                push    offset ... ; "⤦
    ↳ skgfsync(se=0x%x, ctx=0x%x, iov=0x%x)\⤦
    ↳ n"
                push    dword ptr [edx+4]
                call    dword ptr [eax] ; ⤦
    ↳ write to log
                add     esp, 14h
                jmp     continue

error:
                mov     edx, [ebp+8]
                mov     dword ptr [edx], 69⤦
    ↳ AAh ; 27050 "function called with ⤦
    ↳ invalid FIB/IOV structure"
                mov     eax, [eax]
                mov     [edx+4], eax
                mov     dword ptr [edx+8], 0⤦
    ↳ FA4h ; 4004
                mov     esp, ebp
                pop     ebp
                retn
; END OF FUNCTION CHUNK FOR _skgfsync
```

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 CAPÍTULO 95.

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Parte XI

Capítulo 96

96.1 Windows

[RA09].

96.2 C/C++

[ISO13].

96.3 x86 / x86-64

[Int13], [AMD13a]

96.4 ARM

: http://go.yurichev.com/17024

96.5

[Sch94]

Capítulo 97

97.1 Windows

- Microsoft: Raymond Chen
- nynaeve.net

Capítulo 98

: reddit.com/r/ReverseEngineering/ y reddit.com/r/remath ( ).

Stack Exchange : reverseengineering.stackexchange.com.

##re FreeNode[1].

[1] freenode.net

Parte XII

Ejercicios

1)

2) .

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CAPÍTULO 99.   1

Capítulo 99

1

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99.1 Ejercicio 1.4

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- win32 (beginners.re)
- Linux x86 (beginners.re)
- Mac OS X (beginners.re)

CAPÍTULO 99. 1

- MIPS (beginners.re)

Capítulo 100

2

100.1 Ejercicio 2.1

100.1.1 Con optimización MSVC 2010 x86

```
__real@3fe0000000000000 DQ 03↙
    ↘ fe0000000000000r
__real@3f50624dd2f1a9fc DQ 03↙
    ↘ f50624dd2f1a9fcr

_g$ = 8
tv132 = 16
_x$ = 16
f1 PROC
        fld      QWORD PTR _x$[esp-4]
```

```
        fld     QWORD PTR ⤦
    ↳ __real@3f50624dd2f1a9fc
        fld     QWORD PTR ⤦
    ↳ __real@3fe0000000000000
        fld     QWORD PTR _g$[esp-4]
$LN2@f1:
        fld     ST(0)
        fmul    ST(0), ST(1)
        fsub    ST(0), ST(4)
        call    __ftol2_sse
        cdq
        xor     eax, edx
        sub     eax, edx
        mov     DWORD PTR tv132[esp-4], eax
        fild    DWORD PTR tv132[esp-4]
        fcomp   ST(3)
        fnstsw  ax
        test    ah, 5
        jnp     SHORT $LN19@f1
        fld     ST(3)
        fdiv    ST(0), ST(1)
        faddp   ST(1), ST(0)
        fmul    ST(0), ST(1)
        jmp     SHORT $LN2@f1
$LN19@f1:
        fstp    ST(3)
        fstp    ST(1)
        fstp    ST(0)
        ret     0
f1 ENDP

__real@3ff0000000000000 DQ 03⤦
    ↳ ff0000000000000r
```

```
_x$ = 8
f2 PROC
        fld     QWORD PTR _x$[esp-4]
        sub     esp, 16
        fstp    QWORD PTR [esp+8]
        fld1
        fstp    QWORD PTR [esp]
        call    f1
        add     esp, 16
        ret     0
f2 ENDP
```

100.1.2 Con optimización MSVC 2012 x64

```
__real@3fe0000000000000 DQ 03↙
    ↘ fe0000000000000r
__real@3f50624dd2f1a9fc DQ 03↙
    ↘ f50624dd2f1a9fcr
__real@3ff0000000000000 DQ 03↙
    ↘ ff0000000000000r

x$ = 8
f       PROC
        movsdx  xmm2, QWORD PTR ↙
    ↘ __real@3ff0000000000000
        movsdx  xmm5, QWORD PTR ↙
    ↘ __real@3f50624dd2f1a9fc
        movsdx  xmm4, QWORD PTR ↙
    ↘ __real@3fe0000000000000
        movapd  xmm3, xmm0
        npad    4
```

```
$LL4@f:
        movapd  xmm1, xmm2
        mulsd   xmm1, xmm2
        subsd   xmm1, xmm3
        cvttsd2si eax, xmm1
        cdq
        xor     eax, edx
        sub     eax, edx
        movd    xmm0, eax
        cvtdq2pd xmm0, xmm0
        comisd  xmm5, xmm0
        ja      SHORT $LN18@f
        movapd  xmm0, xmm3
        divsd   xmm0, xmm2
        addsd   xmm0, xmm2
        movapd  xmm2, xmm0
        mulsd   xmm2, xmm4
        jmp     SHORT $LL4@f
$LN18@f:
        movapd  xmm0, xmm2
        ret     0
f       ENDP
```

100.2 Ejercicio 2.4

100.2.1 Con optimización MSVC 2010

```
PUBLIC    _f
_TEXT    SEGMENT
_arg1$ = 8              ; size = 4
_arg2$ = 12             ; size = 4
```

```
_f    PROC
    push   esi
    mov    esi, DWORD PTR _arg1$[esp]
    push   edi
    mov    edi, DWORD PTR _arg2$[esp+4]
    cmp    BYTE PTR [edi], 0
    mov    eax, esi
    je     SHORT $LN7@f
    mov    dl, BYTE PTR [esi]
    push   ebx
    test   dl, dl
    je     SHORT $LN4@f
    sub    esi, edi
    npad   6 ; align next label
$LL5@f:
    mov    ecx, edi
    test   dl, dl
    je     SHORT $LN2@f
$LL3@f:
    mov    dl, BYTE PTR [ecx]
    test   dl, dl
    je     SHORT $LN14@f
    movsx  ebx, BYTE PTR [esi+ecx]
    movsx  edx, dl
    sub    ebx, edx
    jne    SHORT $LN2@f
    inc    ecx
    cmp    BYTE PTR [esi+ecx], bl
    jne    SHORT $LL3@f
$LN2@f:
    cmp    BYTE PTR [ecx], 0
    je     SHORT $LN14@f
    mov    dl, BYTE PTR [eax+1]
```

```
    inc     eax
    inc     esi
    test    dl, dl
    jne     SHORT $LL5@f
    xor     eax, eax
    pop     ebx
    pop     edi
    pop     esi
    ret     0
_f    ENDP
_TEXT    ENDS
END
```

100.2.2 GCC 4.4.1

```
                public f
f               proc near

var_C           = dword ptr -0Ch
var_8           = dword ptr -8
var_4           = dword ptr -4
arg_0           = dword ptr  8
arg_4           = dword ptr  0Ch

                push    ebp
                mov     ebp, esp
                sub     esp, 10h
                mov     eax, [ebp+arg_0]
                mov     [ebp+var_4], eax
                mov     eax, [ebp+arg_4]
                movzx   eax, byte ptr [eax]
                test    al, al
```

```
                jnz     short loc_8048443
                mov     eax, [ebp+arg_0]
                jmp     short locret_8048453

loc_80483F4:
                mov     eax, [ebp+var_4]
                mov     [ebp+var_8], eax
                mov     eax, [ebp+arg_4]
                mov     [ebp+var_C], eax
                jmp     short loc_804840A

loc_8048402:
                add     [ebp+var_8], 1
                add     [ebp+var_C], 1

loc_804840A:
                mov     eax, [ebp+var_8]
                movzx   eax, byte ptr [eax]
                test    al, al
                jz      short loc_804842E
                mov     eax, [ebp+var_C]
                movzx   eax, byte ptr [eax]
                test    al, al
                jz      short loc_804842E
                mov     eax, [ebp+var_8]
                movzx   edx, byte ptr [eax]
                mov     eax, [ebp+var_C]
                movzx   eax, byte ptr [eax]
                cmp     dl, al
                jz      short loc_8048402

loc_804842E:
                mov     eax, [ebp+var_C]
```

```
                movzx   eax, byte ptr [eax]
                test    al, al
                jnz     short loc_804843D
                mov     eax, [ebp+var_4]
                jmp     short locret_8048453

loc_804843D:
                add     [ebp+var_4], 1
                jmp     short loc_8048444



loc_8048443:
                nop

loc_8048444:
                mov     eax, [ebp+var_4]
                movzx   eax, byte ptr [eax]
                test    al, al
                jnz     short loc_80483F4
                mov     eax, 0

locret_8048453:
                leave
                retn
f               endp
```

100.2.3 Con optimización Keil (Modo ARM)

```
        PUSH     {r4,lr}
        LDRB     r2,[r1,#0]
        CMP      r2,#0
        POPEQ    {r4,pc}
```

```
        B       |L0.80|
|L0.20|
        LDRB    r12,[r3,#0]
        CMP     r12,#0
        BEQ     |L0.64|
        LDRB    r4,[r2,#0]
        CMP     r4,#0
        POPEQ   {r4,pc}
        CMP     r12,r4
        ADDEQ   r3,r3,#1
        ADDEQ   r2,r2,#1
        BEQ     |L0.20|
        B       |L0.76|
|L0.64|
        LDRB    r2,[r2,#0]
        CMP     r2,#0
        POPEQ   {r4,pc}
|L0.76|
        ADD     r0,r0,#1
|L0.80|
        LDRB    r2,[r0,#0]
        CMP     r2,#0
        MOVNE   r3,r0
        MOVNE   r2,r1
        MOVEQ   r0,#0
        BNE     |L0.20|
        POP     {r4,pc}
```

100.2.4 Con optimización Keil (Modo Thumb)

```
        PUSH    {r4,r5,lr}
        LDRB    r2,[r1,#0]
```

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```
        CMP     r2,#0
        BEQ     |L0.54|
        B       |L0.46|
|L0.10|
        MOVS    r3,r0
        MOVS    r2,r1
        B       |L0.20|
|L0.16|
        ADDS    r3,r3,#1
        ADDS    r2,r2,#1
|L0.20|
        LDRB    r4,[r3,#0]
        CMP     r4,#0
        BEQ     |L0.38|
        LDRB    r5,[r2,#0]
        CMP     r5,#0
        BEQ     |L0.54|
        CMP     r4,r5
        BEQ     |L0.16|
        B       |L0.44|
|L0.38|
        LDRB    r2,[r2,#0]
        CMP     r2,#0
        BEQ     |L0.54|
|L0.44|
        ADDS    r0,r0,#1
|L0.46|
        LDRB    r2,[r0,#0]
        CMP     r2,#0
        BNE     |L0.10|
        MOVS    r0,#0
|L0.54|
        POP     {r4,r5,pc}
```

100.2.5 Con optimización GCC 4.9.1 (ARM64)

Listing 100.1: Con optimización GCC 4.9.1 (ARM64)

```
func:
        ldrb    w6, [x1]
        mov     x2, x0
        cbz     w6, .L2
        ldrb    w2, [x0]
        cbz     w2, .L24
.L17:
        ldrb    w2, [x0]
        cbz     w2, .L5
        cmp     w6, w2
        mov     x5, x0
        mov     x2, x1
        beq     .L18
        b       .L5
.L4:
        ldrb    w4, [x2]
        cmp     w3, w4
        cbz     w4, .L8
        bne     .L8
.L18:
        ldrb    w3, [x5,1]!
        add     x2, x2, 1
        cbnz    w3, .L4
.L8:
        ldrb    w2, [x2]
        cbz     w2, .L27
```

```
.L5:
        ldrb    w2, [x0,1]!
        cbnz    w2, .L17
.L24:
        mov     x2, 0
.L2:
        mov     x0, x2
        ret
.L27:
        mov     x2, x0
        mov     x0, x2
        ret
```

100.2.6 Con optimización GCC 4.4.5 (MIPS)

Listing 100.2: Con optimización GCC 4.4.5 (MIPS) (IDA)

```
f:
                lb      $v1, 0($a1)
                or      $at, $zero
                bnez    $v1, loc_18
                move    $v0, $a0

locret_10:                                  # ↙
    ↘ CODE XREF: f+50
                                            # f↙
    ↘ +78
                jr      $ra
                or      $at, $zero

loc_18:                                     # ↙
    ↘ CODE XREF: f+8
```

```
                lb      $a0, 0($a0)
                or      $at, $zero
                beqz    $a0, locret_94
                move    $a2, $v0

loc_28:                                  # ↙
    ↳ CODE XREF: f+8C
                lb      $a0, 0($a2)
                or      $at, $zero
                beqz    $a0, loc_80
                or      $at, $zero
                bne     $v1, $a0, loc_80
                move    $a3, $a1
                b       loc_60
                addiu   $a2, 1

loc_48:                                  # ↙
    ↳ CODE XREF: f+68
                lb      $t1, 0($a3)
                or      $at, $zero
                beqz    $t1, locret_10
                or      $at, $zero
                bne     $t0, $t1, loc_80
                addiu   $a2, 1

loc_60:                                  # ↙
    ↳ CODE XREF: f+40
                lb      $t0, 0($a2)
                or      $at, $zero
                bnez    $t0, loc_48
                addiu   $a3, 1
                lb      $a0, 0($a3)
                or      $at, $zero
```

```
                beqz    $a0, locret_10
                or      $at, $zero

loc_80:                                     # ⤦
    ↳ CODE XREF: f+30
                                            # f⤦
    ↳ +38 ...
                addiu   $v0, 1
                lb      $a0, 0($v0)
                or      $at, $zero
                bnez    $a0, loc_28
                move    $a2, $v0

locret_94:                                  # ⤦
    ↳ CODE XREF: f+20
                jr      $ra
                move    $v0, $zero
```

100.3 Ejercicio 2.6

100.3.1 Con optimización MSVC 2010

```
PUBLIC    _f
; Function compile flags: /Ogtpy
_TEXT    SEGMENT
_k0$ = -12              ; size = 4
_k3$ = -8               ; size = 4
_k2$ = -4               ; size = 4
_v$ = 8                 ; size = 4
_k1$ = 12               ; size = 4
_k$ = 12                ; size = 4
```

```
_f     PROC
    sub     esp, 12    ; 0000000cH
    mov     ecx, DWORD PTR _v$[esp+8]
    mov     eax, DWORD PTR [ecx]
    mov     ecx, DWORD PTR [ecx+4]
    push    ebx
    push    esi
    mov     esi, DWORD PTR _k$[esp+16]
    push    edi
    mov     edi, DWORD PTR [esi]
    mov     DWORD PTR _k0$[esp+24], edi
    mov     edi, DWORD PTR [esi+4]
    mov     DWORD PTR _k1$[esp+20], edi
    mov     edi, DWORD PTR [esi+8]
    mov     esi, DWORD PTR [esi+12]
    xor     edx, edx
    mov     DWORD PTR _k2$[esp+24], edi
    mov     DWORD PTR _k3$[esp+24], esi
    lea     edi, DWORD PTR [edx+32]
$LL8@f:
    mov     esi, ecx
    shr     esi, 5
    add     esi, DWORD PTR _k1$[esp+20]
    mov     ebx, ecx
    shl     ebx, 4
    add     ebx, DWORD PTR _k0$[esp+24]
    sub     edx, 1640531527    ; 61c88647H
    xor     esi, ebx
    lea     ebx, DWORD PTR [edx+ecx]
    xor     esi, ebx
    add     eax, esi
    mov     esi, eax
    shr     esi, 5
```

```
    add     esi, DWORD PTR _k3$[esp+24]
    mov     ebx, eax
    shl     ebx, 4
    add     ebx, DWORD PTR _k2$[esp+24]
    xor     esi, ebx
    lea     ebx, DWORD PTR [edx+eax]
    xor     esi, ebx
    add     ecx, esi
    dec     edi
    jne     SHORT $LL8@f
    mov     edx, DWORD PTR _v$[esp+20]
    pop     edi
    pop     esi
    mov     DWORD PTR [edx], eax
    mov     DWORD PTR [edx+4], ecx
    pop     ebx
    add     esp, 12                         ; ↙
    ↘ 0000000cH
    ret     0
_f    ENDP
```

100.3.2 Con optimización Keil (Modo ARM)

```
        PUSH     {r4-r10,lr}
        ADD      r5,r1,#8
        LDM      r5,{r5,r7}
        LDR      r2,[r0,#4]
        LDR      r3,[r0,#0]
        LDR      r4,|L0.116|
        LDR      r6,[r1,#4]
        LDR      r8,[r1,#0]
        MOV      r12,#0
```

```
        MOV      r1,r12
|L0.40|
        ADD      r12,r12,r4
        ADD      r9,r8,r2,LSL #4
        ADD      r10,r2,r12
        EOR      r9,r9,r10
        ADD      r10,r6,r2,LSR #5
        EOR      r9,r9,r10
        ADD      r3,r3,r9
        ADD      r9,r5,r3,LSL #4
        ADD      r10,r3,r12
        EOR      r9,r9,r10
        ADD      r10,r7,r3,LSR #5
        EOR      r9,r9,r10
        ADD      r1,r1,#1
        CMP      r1,#0x20
        ADD      r2,r2,r9
        STRCS    r2,[r0,#4]
        STRCS    r3,[r0,#0]
        BCC      |L0.40|
        POP      {r4-r10,pc}

|L0.116|
        DCD      0x9e3779b9
```

100.3.3 Con optimización Keil (Modo Thumb)

```
        PUSH     {r1-r7,lr}
        LDR      r5,|L0.84|
        LDR      r3,[r0,#0]
        LDR      r2,[r0,#4]
        STR      r5,[sp,#8]
```

```
        MOVS     r6,r1
        LDM      r6,{r6,r7}
        LDR      r5,[r1,#8]
        STR      r6,[sp,#4]
        LDR      r6,[r1,#0xc]
        MOVS     r4,#0
        MOVS     r1,r4
        MOV      lr,r5
        MOV      r12,r6
        STR      r7,[sp,#0]
|L0.30|
        LDR      r5,[sp,#8]
        LSLS     r6,r2,#4
        ADDS     r4,r4,r5
        LDR      r5,[sp,#4]
        LSRS     r7,r2,#5
        ADDS     r5,r6,r5
        ADDS     r6,r2,r4
        EORS     r5,r5,r6
        LDR      r6,[sp,#0]
        ADDS     r1,r1,#1
        ADDS     r6,r7,r6
        EORS     r5,r5,r6
        ADDS     r3,r5,r3
        LSLS     r5,r3,#4
        ADDS     r6,r3,r4
        ADD      r5,r5,lr
        EORS     r5,r5,r6
        LSRS     r6,r3,#5
        ADD      r6,r6,r12
        EORS     r5,r5,r6
        ADDS     r2,r5,r2
        CMP      r1,#0x20
```

```
        BCC      |L0.30|
        STR      r3,[r0,#0]
        STR      r2,[r0,#4]
        POP      {r1-r7,pc}

|L0.84|
        DCD      0x9e3779b9
```

100.3.4 Con optimización GCC 4.9.1 (ARM64)

Listing 100.3: Con optimización GCC 4.9.1 (ARM64)

```
f:
        ldr     w3, [x0]
        mov     w4, 0
        ldr     w2, [x0,4]
        ldr     w10, [x1]
        ldr     w9, [x1,4]
        ldr     w8, [x1,8]
        ldr     w7, [x1,12]
.L2:
        mov     w5, 31161
        add     w6, w10, w2, lsl 4
        movk    w5, 0x9e37, lsl 16
        add     w1, w9, w2, lsr 5
        add     w4, w4, w5
        eor     w1, w6, w1
        add     w5, w2, w4
        mov     w6, 14112
        eor     w1, w1, w5
        movk    w6, 0xc6ef, lsl 16
        add     w3, w3, w1
```

```
        cmp     w4, w6
        add     w5, w3, w4
        add     w6, w8, w3, lsl 4
        add     w1, w7, w3, lsr 5
        eor     w1, w6, w1
        eor     w1, w1, w5
        add     w2, w2, w1
        bne     .L2
        str     w3, [x0]
        str     w2, [x0,4]
        ret
```

100.3.5 Con optimización GCC 4.4.5 (MIPS)

Listing 100.4: Con optimización GCC 4.4.5 (MIPS) (IDA)

```
f:
                lui     $t2, 0x9E37
                lui     $t1, 0xC6EF
                lw      $v0, 0($a0)
                lw      $v1, 4($a0)
                lw      $t6, 0xC($a1)
                lw      $t5, 0($a1)
                lw      $t4, 4($a1)
                lw      $t3, 8($a1)
                li      $t2, 0x9E3779B9
                li      $t1, 0xC6EF3720
                move    $a1, $zero

loc_2C:                                     # ↙
    ↳ CODE XREF: f+6C
                addu    $a1, $t2
```

```
                sll     $a2, $v1, 4
                addu    $t0, $a1, $v1
                srl     $a3, $v1, 5
                addu    $a2, $t5
                addu    $a3, $t4
                xor     $a2, $t0, $a2
                xor     $a2, $a3
                addu    $v0, $a2
                sll     $a3, $v0, 4
                srl     $a2, $v0, 5
                addu    $a3, $t3
                addu    $a2, $t6
                xor     $a2, $a3, $a2
                addu    $a3, $v0, $a1
                xor     $a2, $a3
                bne     $a1, $t1, loc_2C
                addu    $v1, $a2
                sw      $v1, 4($a0)
                jr      $ra
                sw      $v0, 0($a0)
```

100.4 Ejercicio 2.13

. ?

100.4.1 Con optimización MSVC 2012

```
_in$ = 8                                                        ↙
    ↘                       ; size = 2
_f      PROC
```

```
        movzx   ecx, WORD PTR _in$[esp-4]
        lea     eax, DWORD PTR [ecx*4]
        xor     eax, ecx
        add     eax, eax
        xor     eax, ecx
        shl     eax, 2
        xor     eax, ecx
        and     eax, 32                              ↙
    ↘                      ; 00000020H
        shl     eax, 10                              ↙
    ↘                      ; 0000000aH
        shr     ecx, 1
        or      eax, ecx
        ret     0
_f      ENDP
```

100.4.2 Keil (Modo ARM)

```
f PROC
        EOR      r1,r0,r0,LSR #2
        EOR      r1,r1,r0,LSR #3
        EOR      r1,r1,r0,LSR #5
        AND      r1,r1,#1
        LSR      r0,r0,#1
        ORR      r0,r0,r1,LSL #15
        BX       lr
        ENDP
```

100.4.3 Keil (Modo Thumb)

```
f PROC
        LSRS     r1,r0,#2
        EORS     r1,r1,r0
        LSRS     r2,r0,#3
        EORS     r1,r1,r2
        LSRS     r2,r0,#5
        EORS     r1,r1,r2
        LSLS     r1,r1,#31
        LSRS     r0,r0,#1
        LSRS     r1,r1,#16
        ORRS     r0,r0,r1
        BX       lr
        ENDP
```

100.4.4 Con optimización GCC 4.9.1 (ARM64)

```
f:
        uxth    w1, w0
        lsr     w2, w1, 3
        lsr     w0, w1, 1
        eor     w2, w2, w1, lsr 2
        eor     w2, w1, w2
        eor     w1, w2, w1, lsr 5
        and     w1, w1, 1
        orr     w0, w0, w1, lsl 15
        ret
```

100.4.5 Con optimización GCC 4.4.5 (MIPS)

Listing 100.5: Con optimización GCC 4.4.5 (MIPS) (IDA)

```
f:
                andi    $a0, 0xFFFF
                srl     $v1, $a0, 2
                srl     $v0, $a0, 3
                xor     $v0, $v1, $v0
                xor     $v0, $a0, $v0
                srl     $v1, $a0, 5
                xor     $v0, $v1
                andi    $v0, 1
                srl     $a0, 1
                sll     $v0, 15
                jr      $ra
                or      $v0, $a0
```

100.5 Ejercicio 2.14

100.5.1 MSVC 2012

```
_rt$1 = -4                                        ↙
    ↘                   ; size = 4
_rt$2 = 8                                         ↙
    ↘                   ; size = 4
_x$ = 8                                           ↙
    ↘                   ; size = 4
_y$ = 12                                          ↙
    ↘                   ; size = 4
?f@@YAIII@Z PROC                                  ↙
    ↘                   ; f
        push    ecx
```

```
        push    esi
        mov     esi, DWORD PTR _x$[esp+4]
        test    esi, esi
        jne     SHORT $LN7@f
        mov     eax, DWORD PTR _y$[esp+4]
        pop     esi
        pop     ecx
        ret     0
$LN7@f:
        mov     edx, DWORD PTR _y$[esp+4]
        mov     eax, esi
        test    edx, edx
        je      SHORT $LN8@f
        or      eax, edx
        push    edi
        bsf     edi, eax
        bsf     eax, esi
        mov     ecx, eax
        mov     DWORD PTR _rt$1[esp+12], eax
        bsf     eax, edx
        shr     esi, cl
        mov     ecx, eax
        shr     edx, cl
        mov     DWORD PTR _rt$2[esp+8], eax
        cmp     esi, edx
        je      SHORT $LN22@f
$LN23@f:
        jbe     SHORT $LN2@f
        xor     esi, edx
        xor     edx, esi
        xor     esi, edx
$LN2@f:
        cmp     esi, 1
```

```
        je      SHORT $LN22@f
        sub     edx, esi
        bsf     eax, edx
        mov     ecx, eax
        shr     edx, cl
        mov     DWORD PTR _rt$2[esp+8], eax
        cmp     esi, edx
        jne     SHORT $LN23@f
$LN22@f:
        mov     ecx, edi
        shl     esi, cl
        pop     edi
        mov     eax, esi
$LN8@f:
        pop     esi
        pop     ecx
        ret     0
?f@@YAIII@Z ENDP
```

100.5.2 Keil (Modo ARM)

```
||f1|| PROC
        CMP      r0,#0
        RSB      r1,r0,#0
        AND      r0,r0,r1
        CLZ      r0,r0
        RSBNE    r0,r0,#0x1f
        BX       lr
        ENDP

f PROC
        MOVS     r2,r0
```

```
        MOV     r3,r1
        MOVEQ   r0,r1
        CMPNE   r3,#0
        PUSH    {lr}
        POPEQ   {pc}
        ORR     r0,r2,r3
        BL      ||f1||
        MOV     r12,r0
        MOV     r0,r2
        BL      ||f1||
        LSR     r2,r2,r0
|L0.196|
        MOV     r0,r3
        BL      ||f1||
        LSR     r0,r3,r0
        CMP     r2,r0
        EORHI   r1,r2,r0
        EORHI   r0,r0,r1
        EORHI   r2,r1,r0
        BEQ     |L0.240|
        CMP     r2,#1
        SUBNE   r3,r0,r2
        BNE     |L0.196|
|L0.240|
        LSL     r0,r2,r12
        POP     {pc}
        ENDP
```

100.5.3 GCC 4.6.3 for Raspberry Pi (Modo ARM)

```
f:
        subs    r3, r0, #0
```

```
        beq     .L162
        cmp     r1, #0
        moveq   r1, r3
        beq     .L162
        orr     r2, r1, r3
        rsb     ip, r2, #0
        and     ip, ip, r2
        cmp     r2, #0
        rsb     r2, r3, #0
        and     r2, r2, r3
        clz     r2, r2
        rsb     r2, r2, #31
        clz     ip, ip
        rsbne   ip, ip, #31
        mov     r3, r3, lsr r2
        b       .L169
.L171:
        eorhi   r1, r1, r2
        eorhi   r3, r1, r2
        cmp     r3, #1
        rsb     r1, r3, r1
        beq     .L167
.L169:
        rsb     r0, r1, #0
        and     r0, r0, r1
        cmp     r1, #0
        clz     r0, r0
        mov     r2, r0
        rsbne   r2, r0, #31
        mov     r1, r1, lsr r2
        cmp     r3, r1
        eor     r2, r1, r3
        bne     .L171
```

```
.L167:
        mov     r1, r3, asl ip
.L162:
        mov     r0, r1
        bx      lr
```

100.5.4 Con optimización GCC 4.9.1 (ARM64)

Listing 100.6: Con optimización GCC 4.9.1 (ARM64)

```
f:
        mov     w3, w0
        mov     w0, w1
        cbz     w3, .L8
        mov     w0, w3
        cbz     w1, .L8
        mov     w6, 31
        orr     w5, w3, w1
        neg     w2, w3
        neg     w7, w5
        and     w2, w2, w3
        clz     w2, w2
        sub     w2, w6, w2
        and     w5, w7, w5
        mov     w4, w6
        clz     w5, w5
        lsr     w0, w3, w2
        sub     w5, w6, w5
        b       .L13
.L22:
        bls     .L12
        eor     w1, w1, w2
```

```
        eor     w0, w1, w2
.L12:
        cmp     w0, 1
        sub     w1, w1, w0
        beq     .L11
.L13:
        neg     w2, w1
        cmp     w1, wzr
        and     w2, w2, w1
        clz     w2, w2
        sub     w3, w4, w2
        csel    w2, w3, w2, ne
        lsr     w1, w1, w2
        cmp     w0, w1
        eor     w2, w1, w0
        bne     .L22
.L11:
        lsl     w0, w0, w5
.L8:
        ret
```

100.5.5 Con optimización GCC 4.4.5 (MIPS)

Listing 100.7: Con optimización GCC 4.4.5 (MIPS) (IDA)

```
f:

var_20          = -0x20
var_18          = -0x18
var_14          = -0x14
var_10          = -0x10
var_C           = -0xC
```

```
var_8           = -8
var_4           = -4

                lui     $gp, (__gnu_local_gp
    ↳ >> 16)
                addiu   $sp, -0x30
                la      $gp, (__gnu_local_gp
    ↳ & 0xFFFF)
                sw      $ra, 0x30+var_4($sp)
                sw      $s4, 0x30+var_8($sp)
                sw      $s3, 0x30+var_C($sp)
                sw      $s2, 0x30+var_10($sp
    ↳ )
                sw      $s1, 0x30+var_14($sp
    ↳ )
                sw      $s0, 0x30+var_18($sp
    ↳ )
                sw      $gp, 0x30+var_20($sp
    ↳ )
                move    $s0, $a0
                beqz    $a0, loc_154
                move    $s1, $a1
                bnez    $a1, loc_178
                or      $s2, $a1, $a0
                move    $s1, $a0

loc_154:                                  #
    ↳ CODE XREF: f+2C
                lw      $ra, 0x30+var_4($sp)
                move    $v0, $s1
                lw      $s4, 0x30+var_8($sp)
                lw      $s3, 0x30+var_C($sp)
```

```
                lw      $s2, 0x30+var_10($sp
    ↳ )
                lw      $s1, 0x30+var_14($sp
    ↳ )
                lw      $s0, 0x30+var_18($sp
    ↳ )
                jr      $ra
                addiu   $sp, 0x30

loc_178:                                 # 
    ↳ CODE XREF: f+34
                lw      $t9, (__clzsi2 & 0
    ↳ xFFFF)($gp)
                negu    $a0, $s2
                jalr    $t9
                and     $a0, $s2
                lw      $gp, 0x30+var_20($sp
    ↳ )
                bnez    $s2, loc_20C
                li      $s4, 0x1F
                move    $s4, $v0

loc_198:                                 # 
    ↳ CODE XREF: f:loc_20C
                lw      $t9, (__clzsi2 & 0
    ↳ xFFFF)($gp)
                negu    $a0, $s0
                jalr    $t9
                and     $a0, $s0
                nor     $v0, $zero, $v0
                lw      $gp, 0x30+var_20($sp
    ↳ )
                srlv    $s0, $v0
```

```
                li      $s3, 0x1F
                li      $s2, 1

loc_1BC:                                   # ↙
    ↳ CODE XREF: f+F0
                lw      $t9, (__clzsi2 & 0↙
    ↳ xFFFF)($gp)
                negu    $a0, $s1
                jalr    $t9
                and     $a0, $s1
                lw      $gp, 0x30+var_20($sp↙
    ↳ )
                beqz    $s1, loc_1DC
                or      $at, $zero
                subu    $v0, $s3, $v0

loc_1DC:                                   # ↙
    ↳ CODE XREF: f+BC
                srlv    $s1, $v0
                xor     $v1, $s1, $s0
                beq     $s0, $s1, loc_214
                sltu    $v0, $s1, $s0
                beqz    $v0, loc_1FC
                or      $at, $zero
                xor     $s1, $v1
                xor     $s0, $s1, $v1

loc_1FC:                                   # ↙
    ↳ CODE XREF: f+D8
                beq     $s0, $s2, loc_214
                subu    $s1, $s0
                b       loc_1BC
                or      $at, $zero
```

```
loc_20C:                                        # ↙
    ↳ CODE XREF: f+78
                b       loc_198
                subu    $s4, $v0

loc_214:                                        # ↙
    ↳ CODE XREF: f+D0
                                                # f↙
    ↳ :loc_1FC
                lw      $ra, 0x30+var_4($sp)
                sllv    $s1, $s0, $s4
                move    $v0, $s1
                lw      $s4, 0x30+var_8($sp)
                lw      $s3, 0x30+var_C($sp)
                lw      $s2, 0x30+var_10($sp↙
    ↳ )
                lw      $s1, 0x30+var_14($sp↙
    ↳ )
                lw      $s0, 0x30+var_18($sp↙
    ↳ )
                jr      $ra
                addiu   $sp, 0x30
```

100.6 Ejercicio 2.15

: 27 on page 830.

100.6.1 Con optimización MSVC 2012 x64

```
__real@412e848000000000 DQ 0412↙
    ↘ e848000000r    ; 1e+006
__real@4010000000000000 DQ 04010000000000000↙
    ↘ r    ; 4
__real@4008000000000000 DQ 04008000000000000↙
    ↘ r    ; 3
__real@3f800000 DD 03f800000r                ↙
    ↘          ; 1

tmp$1 = 8
tmp$2 = 8
f       PROC
        movsdx  xmm3, QWORD PTR ↙
    ↘ __real@4008000000000000
        movss   xmm4, DWORD PTR ↙
    ↘ __real@3f800000
        mov     edx, DWORD PTR ?RNG_state@↙
    ↘ ?1??get_rand@@9@9
        xor     ecx, ecx
        mov     r8d, 200000                  ↙
    ↘                    ; 00030d40H
        npad    2 ; align next label
$LL4@f:
        imul    edx, 1664525                 ↙
    ↘                    ; 0019660dH
        add     edx, 1013904223              ↙
    ↘                    ; 3c6ef35fH
        mov     eax, edx
        and     eax, 8388607                 ↙
    ↘                    ; 007fffffH
        imul    edx, 1664525                 ↙
    ↘                    ; 0019660dH
```

```
        bts      eax, 30
        add      edx, 1013904223                    ↙
   ↘                      ; 3c6ef35fH
        mov      DWORD PTR tmp$2[rsp], eax
        mov      eax, edx
        and      eax, 8388607                       ↙
   ↘                      ; 007fffffH
        bts      eax, 30
        movss    xmm0, DWORD PTR tmp$2[rsp]
        mov      DWORD PTR tmp$1[rsp], eax
        cvtps2pd xmm0, xmm0
        subsd    xmm0, xmm3
        cvtpd2ps xmm2, xmm0
        movss    xmm0, DWORD PTR tmp$1[rsp]
        cvtps2pd xmm0, xmm0
        mulss    xmm2, xmm2
        subsd    xmm0, xmm3
        cvtpd2ps xmm1, xmm0
        mulss    xmm1, xmm1
        addss    xmm1, xmm2
        comiss   xmm4, xmm1
        jbe      SHORT $LN3@f
        inc      ecx
$LN3@f:
        imul     edx, 1664525                       ↙
   ↘                      ; 0019660dH
        add      edx, 1013904223                    ↙
   ↘                      ; 3c6ef35fH
        mov      eax, edx
        and      eax, 8388607                       ↙
   ↘                      ; 007fffffH
        imul     edx, 1664525                       ↙
   ↘                      ; 0019660dH
```

```
    bts     eax, 30
    add     edx, 1013904223                 ↙
↘                       ; 3c6ef35fH
    mov     DWORD PTR tmp$2[rsp], eax
    mov     eax, edx
    and     eax, 8388607                    ↙
↘                       ; 007fffffH
    bts     eax, 30
    movss   xmm0, DWORD PTR tmp$2[rsp]
    mov     DWORD PTR tmp$1[rsp], eax
    cvtps2pd xmm0, xmm0
    subsd   xmm0, xmm3
    cvtpd2ps xmm2, xmm0
    movss   xmm0, DWORD PTR tmp$1[rsp]
    cvtps2pd xmm0, xmm0
    mulss   xmm2, xmm2
    subsd   xmm0, xmm3
    cvtpd2ps xmm1, xmm0
    mulss   xmm1, xmm1
    addss   xmm1, xmm2
    comiss  xmm4, xmm1
    jbe     SHORT $LN15@f
    inc     ecx
$LN15@f:
    imul    edx, 1664525                    ↙
↘                       ; 0019660dH
    add     edx, 1013904223                 ↙
↘                       ; 3c6ef35fH
    mov     eax, edx
    and     eax, 8388607                    ↙
↘                       ; 007fffffH
    imul    edx, 1664525                    ↙
↘                       ; 0019660dH
```

```
        bts     eax, 30
        add     edx, 1013904223                    ↙
    ↘                       ; 3c6ef35fH
        mov     DWORD PTR tmp$2[rsp], eax
        mov     eax, edx
        and     eax, 8388607                       ↙
    ↘                       ; 007fffffH
        bts     eax, 30
        movss   xmm0, DWORD PTR tmp$2[rsp]
        mov     DWORD PTR tmp$1[rsp], eax
        cvtps2pd xmm0, xmm0
        subsd   xmm0, xmm3
        cvtpd2ps xmm2, xmm0
        movss   xmm0, DWORD PTR tmp$1[rsp]
        cvtps2pd xmm0, xmm0
        mulss   xmm2, xmm2
        subsd   xmm0, xmm3
        cvtpd2ps xmm1, xmm0
        mulss   xmm1, xmm1
        addss   xmm1, xmm2
        comiss  xmm4, xmm1
        jbe     SHORT $LN16@f
        inc     ecx
$LN16@f:
        imul    edx, 1664525                       ↙
    ↘                       ; 0019660dH
        add     edx, 1013904223                    ↙
    ↘                       ; 3c6ef35fH
        mov     eax, edx
        and     eax, 8388607                       ↙
    ↘                       ; 007fffffH
        imul    edx, 1664525                       ↙
    ↘                       ; 0019660dH
```

```
        bts     eax, 30
        add     edx, 1013904223                    ↙
    ↘                   ; 3c6ef35fH
        mov     DWORD PTR tmp$2[rsp], eax
        mov     eax, edx
        and     eax, 8388607                       ↙
    ↘                   ; 007fffffH
        bts     eax, 30
        movss   xmm0, DWORD PTR tmp$2[rsp]
        mov     DWORD PTR tmp$1[rsp], eax
        cvtps2pd xmm0, xmm0
        subsd   xmm0, xmm3
        cvtpd2ps xmm2, xmm0
        movss   xmm0, DWORD PTR tmp$1[rsp]
        cvtps2pd xmm0, xmm0
        mulss   xmm2, xmm2
        subsd   xmm0, xmm3
        cvtpd2ps xmm1, xmm0
        mulss   xmm1, xmm1
        addss   xmm1, xmm2
        comiss  xmm4, xmm1
        jbe     SHORT $LN17@f
        inc     ecx
$LN17@f:
        imul    edx, 1664525                       ↙
    ↘                   ; 0019660dH
        add     edx, 1013904223                    ↙
    ↘                   ; 3c6ef35fH
        mov     eax, edx
        and     eax, 8388607                       ↙
    ↘                   ; 007fffffH
        imul    edx, 1664525                       ↙
    ↘                   ; 0019660dH
```

```
        bts      eax, 30
        add      edx, 1013904223                ↙
    ↘                     ; 3c6ef35fH
        mov      DWORD PTR tmp$2[rsp], eax
        mov      eax, edx
        and      eax, 8388607                   ↙
    ↘                     ; 007fffffH
        bts      eax, 30
        movss    xmm0, DWORD PTR tmp$2[rsp]
        mov      DWORD PTR tmp$1[rsp], eax
        cvtps2pd xmm0, xmm0
        subsd    xmm0, xmm3
        cvtpd2ps xmm2, xmm0
        movss    xmm0, DWORD PTR tmp$1[rsp]
        cvtps2pd xmm0, xmm0
        mulss    xmm2, xmm2
        subsd    xmm0, xmm3
        cvtpd2ps xmm1, xmm0
        mulss    xmm1, xmm1
        addss    xmm1, xmm2
        comiss   xmm4, xmm1
        jbe      SHORT $LN18@f
        inc      ecx
$LN18@f:
        dec      r8
        jne      $LL4@f
        movd     xmm0, ecx
        mov      DWORD PTR ?RNG_state@?1??↙
    ↘ get_rand@@9@9, edx
        cvtdq2ps xmm0, xmm0
        cvtps2pd xmm1, xmm0
        mulsd    xmm1, QWORD PTR ↙
    ↘ __real@4010000000000000
```

```
        divsd   xmm1, QWORD PTR ↙
    ↘ __real@412e848000000000
        cvtpd2ps xmm0, xmm1
        ret     0
f       ENDP
```

100.6.2 Con optimización GCC 4.4.6 x64

```
f1:
        mov     eax, DWORD PTR v1.2084[rip]
        imul    eax, eax, 1664525
        add     eax, 1013904223
        mov     DWORD PTR v1.2084[rip], eax
        and     eax, 8388607
        or      eax, 1073741824
        mov     DWORD PTR [rsp-4], eax
        movss   xmm0, DWORD PTR [rsp-4]
        subss   xmm0, DWORD PTR .LC0[rip]
        ret
f:
        push    rbp
        xor     ebp, ebp
        push    rbx
        xor     ebx, ebx
        sub     rsp, 16
.L6:
        xor     eax, eax
        call    f1
        xor     eax, eax
        movss   DWORD PTR [rsp], xmm0
        call    f1
        movss   xmm1, DWORD PTR [rsp]
```

```
        mulss   xmm0, xmm0
        mulss   xmm1, xmm1
        lea     eax, [rbx+1]
        addss   xmm1, xmm0
        movss   xmm0, DWORD PTR .LC1[rip]
        ucomiss xmm0, xmm1
        cmova   ebx, eax
        add     ebp, 1
        cmp     ebp, 1000000
        jne     .L6
        cvtsi2ss        xmm0, ebx
        unpcklps        xmm0, xmm0
        cvtps2pd        xmm0, xmm0
        mulsd   xmm0, QWORD PTR .LC2[rip]
        divsd   xmm0, QWORD PTR .LC3[rip]
        add     rsp, 16
        pop     rbx
        pop     rbp
        unpcklpd        xmm0, xmm0
        cvtpd2ps        xmm0, xmm0
        ret
v1.2084:
        .long   305419896
.LC0:
        .long   1077936128
.LC1:
        .long   1065353216
.LC2:
        .long   0
        .long   1074790400
.LC3:
        .long   0
        .long   1093567616
```

100.6.3 Con optimización GCC 4.8.1 x86

```
f1:
        sub     esp, 4
        imul    eax, DWORD PTR v1.2023, ⤦
    ↪ 1664525
        add     eax, 1013904223
        mov     DWORD PTR v1.2023, eax
        and     eax, 8388607
        or      eax, 1073741824
        mov     DWORD PTR [esp], eax
        fld     DWORD PTR [esp]
        fsub    DWORD PTR .LC0
        add     esp, 4
        ret
f:
        push    esi
        mov     esi, 1000000
        push    ebx
        xor     ebx, ebx
        sub     esp, 16
.L7:
        call    f1
        fstp    DWORD PTR [esp]
        call    f1
        lea     eax, [ebx+1]
        fld     DWORD PTR [esp]
        fmul    st, st(0)
        fxch    st(1)
        fmul    st, st(0)
```

```
        faddp   st(1), st
        fld1
        fucomip st, st(1)
        fstp    st(0)
        cmova   ebx, eax
        sub     esi, 1
        jne     .L7
        mov     DWORD PTR [esp+4], ebx
        fild    DWORD PTR [esp+4]
        fmul    DWORD PTR .LC3
        fdiv    DWORD PTR .LC4
        fstp    DWORD PTR [esp+8]
        fld     DWORD PTR [esp+8]
        add     esp, 16
        pop     ebx
        pop     esi
        ret

v1.2023:
        .long   305419896
.LC0:
        .long   1077936128
.LC3:
        .long   1082130432
.LC4:
        .long   1232348160
```

100.6.4 Keil (Modo ARM):

```
f1      PROC
        LDR      r1,|L0.184|
        LDR      r0,[r1,#0]  ; v1
```

```
        LDR       r2,|L0.188|
        VMOV.F32 s1,#3.00000000
        MUL       r0,r0,r2
        LDR       r2,|L0.192|
        ADD       r0,r0,r2
        STR       r0,[r1,#0]  ; v1
        BFC       r0,#23,#9
        ORR       r0,r0,#0x40000000
        VMOV      s0,r0
        VSUB.F32 s0,s0,s1
        BX        lr
        ENDP

f       PROC
        PUSH      {r4,r5,lr}
        MOV       r4,#0
        LDR       r5,|L0.196|
        MOV       r3,r4
|L0.68|
        BL        f1
        VMOV.F32 s2,s0
        BL        f1
        VMOV.F32 s1,s2
        ADD       r3,r3,#1
        VMUL.F32 s1,s1,s1
        VMLA.F32 s1,s0,s0
        VMOV      r0,s1
        CMP       r0,#0x3f800000
        ADDLT     r4,r4,#1
        CMP       r3,r5
        BLT       |L0.68|
        VMOV      s0,r4
        VMOV.F64 d1,#4.00000000
```

```
        VCVT.F32.S32 s0,s0
        VCVT.F64.F32 d0,s0
        VMUL.F64 d0,d0,d1
        VLDR     d1,|L0.200|
        VDIV.F64 d2,d0,d1
        VCVT.F32.F64 s0,d2
        POP      {r4,r5,pc}
        ENDP

|L0.184|
        DCD      ||.data||
|L0.188|
        DCD      0x0019660d
|L0.192|
        DCD      0x3c6ef35f
|L0.196|
        DCD      0x000f4240
|L0.200|
        DCFD     0x412e848000000000 ; ↙
    ↘ 1000000

        DCD      0x00000000
        AREA ||.data||, DATA, ALIGN=2
v1
        DCD      0x12345678
```

100.6.5 Con optimización GCC 4.9.1 (ARM64)

Listing 100.8: Con optimización GCC 4.9.1 (ARM64)

```
f1:
        adrp    x2, .LANCHOR0
```

```
        mov     w3, 26125
        mov     w0, 62303
        movk    w3, 0x19, lsl 16
        movk    w0, 0x3c6e, lsl 16
        ldr     w1, [x2,#:lo12:.LANCHOR0]
        fmov    s0, 3.0e+0
        madd    w0, w1, w3, w0
        str     w0, [x2,#:lo12:.LANCHOR0]
        and     w0, w0, 8388607
        orr     w0, w0, 1073741824
        fmov    s1, w0
        fsub    s0, s1, s0
        ret
main_function:
        adrp    x7, .LANCHOR0
        mov     w3, 16960
        movk    w3, 0xf, lsl 16
        mov     w2, 0
        fmov    s2, 3.0e+0
        ldr     w1, [x7,#:lo12:.LANCHOR0]
        fmov    s3, 1.0e+0
.L5:
        mov     w6, 26125
        mov     w0, 62303
        movk    w6, 0x19, lsl 16
        movk    w0, 0x3c6e, lsl 16
        mov     w5, 26125
        mov     w4, 62303
        madd    w1, w1, w6, w0
        movk    w5, 0x19, lsl 16
        movk    w4, 0x3c6e, lsl 16
        and     w0, w1, 8388607
        add     w6, w2, 1
```

```
        orr     w0, w0, 1073741824
        madd    w1, w1, w5, w4
        fmov    s0, w0
        and     w0, w1, 8388607
        orr     w0, w0, 1073741824
        fmov    s1, w0
        fsub    s0, s0, s2
        fsub    s1, s1, s2
        fmul    s1, s1, s1
        fmadd   s0, s0, s0, s1
        fcmp    s0, s3
        csel    w2, w2, w6, pl
        subs    w3, w3, #1
        bne     .L5
        scvtf   s0, w2
        str     w1, [x7,#:lo12:.LANCHOR0]
        fmov    d1, 4.0e+0
        fcvt    d0, s0
        fmul    d0, d0, d1
        ldr     d1, .LC0
        fdiv    d0, d0, d1
        fcvt    s0, d0
        ret
.LC0:
        .word   0
        .word   1093567616
v1:
        .word   1013904223
v2:
        .word   1664525
.LANCHOR0 = . + 0
v3.3095:
        .word   305419896
```

100.6.6 Con optimización GCC 4.4.5 (MIPS)

Listing 100.9: Con optimización GCC 4.4.5 (MIPS) (IDA)

```
f1:
        mov     eax, DWORD PTR v1.2084[rip]
        imul    eax, eax, 1664525
        add     eax, 1013904223
        mov     DWORD PTR v1.2084[rip], eax
        and     eax, 8388607
        or      eax, 1073741824
        mov     DWORD PTR [rsp-4], eax
        movss   xmm0, DWORD PTR [rsp-4]
        subss   xmm0, DWORD PTR .LC0[rip]
        ret
f:
        push    rbp
        xor     ebp, ebp
        push    rbx
        xor     ebx, ebx
        sub     rsp, 16
.L6:
        xor     eax, eax
        call    f1
        xor     eax, eax
        movss   DWORD PTR [rsp], xmm0
        call    f1
        movss   xmm1, DWORD PTR [rsp]
        mulss   xmm0, xmm0
        mulss   xmm1, xmm1
```

```
        lea     eax, [rbx+1]
        addss   xmm1, xmm0
        movss   xmm0, DWORD PTR .LC1[rip]
        ucomiss xmm0, xmm1
        cmova   ebx, eax
        add     ebp, 1
        cmp     ebp, 1000000
        jne     .L6
        cvtsi2ss        xmm0, ebx
        unpcklps        xmm0, xmm0
        cvtps2pd        xmm0, xmm0
        mulsd   xmm0, QWORD PTR .LC2[rip]
        divsd   xmm0, QWORD PTR .LC3[rip]
        add     rsp, 16
        pop     rbx
        pop     rbp
        unpcklpd        xmm0, xmm0
        cvtpd2ps        xmm0, xmm0
        ret
v1.2084:
        .long   305419896
.LC0:
        .long   1077936128
.LC1:
        .long   1065353216
.LC2:
        .long   0
        .long   1074790400
.LC3:
        .long   0
        .long   1093567616
```

100.7 Ejercicio 2.16

100.7.1 Con optimización MSVC 2012 x64

```
m$ = 48
n$ = 56
f       PROC
$LN14:
        push    rbx
        sub     rsp, 32
        mov     eax, edx
        mov     ebx, ecx
        test    ecx, ecx
        je      SHORT $LN11@f
$LL5@f:
        test    eax, eax
        jne     SHORT $LN1@f
        mov     eax, 1
        jmp     SHORT $LN12@f
$LN1@f:
        lea     edx, DWORD PTR [rax-1]
        mov     ecx, ebx
        call    f
$LN12@f:
        dec     ebx
        test    ebx, ebx
        jne     SHORT $LL5@f
$LN11@f:
        inc     eax
        add     rsp, 32
        pop     rbx
        ret     0
f       ENDP
```

100.7.2 Con optimización Keil (Modo ARM)

```
f PROC
        PUSH     {r4,lr}
        MOVS     r4,r0
        ADDEQ    r0,r1,#1
        POPEQ    {r4,pc}
        CMP      r1,#0
        MOVEQ    r1,#1
        SUBEQ    r0,r0,#1
        BEQ      |L0.48|
        SUB      r1,r1,#1
        BL       f
        MOV      r1,r0
        SUB      r0,r4,#1
|L0.48|
        POP      {r4,lr}
        B        f
        ENDP
```

100.7.3 Con optimización Keil (Modo Thumb)

```
f PROC
        PUSH     {r4,lr}
        MOVS     r4,r0
        BEQ      |L0.26|
        CMP      r1,#0
        BEQ      |L0.30|
```

```
        SUBS      r1,r1,#1
        BL        f
        MOVS      r1,r0
|L0.18|
        SUBS      r0,r4,#1
        BL        f
        POP       {r4,pc}
|L0.26|
        ADDS      r0,r1,#1
        POP       {r4,pc}
|L0.30|
        MOVS      r1,#1
        B         |L0.18|
        ENDP
```

100.7.4 Sin optimización GCC 4.9.1 (ARM64)

Listing 100.10: Sin optimización GCC 4.9.1 (ARM64)

```
f:
        stp     x29, x30, [sp, -48]!
        add     x29, sp, 0
        str     x19, [sp,16]
        str     w0, [x29,44]
        str     w1, [x29,40]
        ldr     w0, [x29,44]
        cmp     w0, wzr
        bne     .L2
        ldr     w0, [x29,40]
        add     w0, w0, 1
        b       .L3
.L2:
```

```
        ldr     w0, [x29,40]
        cmp     w0, wzr
        bne     .L4
        ldr     w0, [x29,44]
        sub     w0, w0, #1
        mov     w1, 1
        bl      ack
        b       .L3
.L4:
        ldr     w0, [x29,44]
        sub     w19, w0, #1
        ldr     w0, [x29,40]
        sub     w1, w0, #1
        ldr     w0, [x29,44]
        bl      ack
        mov     w1, w0
        mov     w0, w19
        bl      ack
.L3:
        ldr     x19, [sp,16]
        ldp     x29, x30, [sp], 48
        ret
```

100.7.5 Con optimización GCC 4.9.1 (ARM64)

Listing 100.11: Con optimización GCC 4.9.1 (ARM64)

```
ack:
        stp     x29, x30, [sp, -160]!
        add     x29, sp, 0
        stp     d8, d9, [sp,96]
        stp     x19, x20, [sp,16]
```

```
        stp     d10, d11, [sp,112]
        stp     x21, x22, [sp,32]
        stp     d12, d13, [sp,128]
        stp     x23, x24, [sp,48]
        stp     d14, d15, [sp,144]
        stp     x25, x26, [sp,64]
        stp     x27, x28, [sp,80]
        cbz     w0, .L2
        sub     w0, w0, #1
        fmov    s10, w0
        b       .L4
.L46:
        fmov    w0, s10
        mov     w1, 1
        sub     w0, w0, #1
        fmov    s10, w0
        fmov    w0, s13
        cbz     w0, .L2
.L4:
        fmov    s13, s10
        cbz     w1, .L46
        sub     w1, w1, #1
        fmov    s11, s10
        b       .L7
.L48:
        fmov    w0, s11
        mov     w1, 1
        sub     w0, w0, #1
        fmov    s11, w0
        fmov    w0, s14
        cbz     w0, .L47
.L7:
        fmov    s14, s11
```

```
        cbz     w1, .L48
        sub     w1, w1, #1
        fmov    s12, s11
        b       .L10
.L50:
        fmov    w0, s12
        mov     w1, 1
        sub     w0, w0, #1
        fmov    s12, w0
        fmov    w0, s15
        cbz     w0, .L49
.L10:
        fmov    s15, s12
        cbz     w1, .L50
        sub     w1, w1, #1
        fmov    s8, s12
        b       .L13
.L52:
        fmov    w0, s8
        mov     w1, 1
        sub     w0, w0, #1
        fmov    s8, w0
        fmov    w0, s9
        cbz     w0, .L51
.L13:
        fmov    s9, s8
        cbz     w1, .L52
        sub     w1, w1, #1
        fmov    w22, s8
        b       .L16
.L54:
        mov     w1, 1
        sub     w22, w22, #1
```

```
        cbz     w28, .L53
.L16:
        mov     w28, w22
        cbz     w1, .L54
        sub     w1, w1, #1
        mov     w21, w22
        b       .L19
.L56:
        mov     w1, 1
        sub     w21, w21, #1
        cbz     w24, .L55
.L19:
        mov     w24, w21
        cbz     w1, .L56
        sub     w1, w1, #1
        mov     w20, w21
        b       .L22
.L58:
        mov     w1, 1
        sub     w20, w20, #1
        cbz     w25, .L57
.L22:
        mov     w25, w20
        cbz     w1, .L58
        sub     w1, w1, #1
        mov     w26, w20
        b       .L25
.L60:
        mov     w1, 1
        sub     w26, w26, #1
        cbz     w27, .L59
.L25:
        mov     w27, w26
```

```
        cbz     w1, .L60
        sub     w1, w1, #1
        mov     w19, w26
        b       .L28
.L62:
        mov     w23, w19
        mov     w1, 1
        sub     w19, w19, #1
        cbz     w23, .L61
.L28:
        add     w0, w19, 1
        cbz     w1, .L62
        sub     w1, w1, #1
        mov     w23, w19
        sub     w19, w19, #1
        bl      ack
        mov     w1, w0
        cbnz    w23, .L28
.L61:
        add     w1, w1, 1
        sub     w26, w26, #1
        cbnz    w27, .L25
.L59:
        add     w1, w1, 1
        sub     w20, w20, #1
        cbnz    w25, .L22
.L57:
        add     w1, w1, 1
        sub     w21, w21, #1
        cbnz    w24, .L19
.L55:
        add     w1, w1, 1
        sub     w22, w22, #1
```

```
        cbnz    w28, .L16
.L53:
        fmov    w0, s8
        add     w1, w1, 1
        sub     w0, w0, #1
        fmov    s8, w0
        fmov    w0, s9
        cbnz    w0, .L13
.L51:
        fmov    w0, s12
        add     w1, w1, 1
        sub     w0, w0, #1
        fmov    s12, w0
        fmov    w0, s15
        cbnz    w0, .L10
.L49:
        fmov    w0, s11
        add     w1, w1, 1
        sub     w0, w0, #1
        fmov    s11, w0
        fmov    w0, s14
        cbnz    w0, .L7
.L47:
        fmov    w0, s10
        add     w1, w1, 1
        sub     w0, w0, #1
        fmov    s10, w0
        fmov    w0, s13
        cbnz    w0, .L4
.L2:
        add     w0, w1, 1
        ldp     d8, d9, [sp,96]
        ldp     x19, x20, [sp,16]
```

```
        ldp     d10, d11, [sp,112]
        ldp     x21, x22, [sp,32]
        ldp     d12, d13, [sp,128]
        ldp     x23, x24, [sp,48]
        ldp     d14, d15, [sp,144]
        ldp     x25, x26, [sp,64]
        ldp     x27, x28, [sp,80]
        ldp     x29, x30, [sp], 160
        ret
```

100.7.6 Sin optimización GCC 4.4.5 (MIPS)

Listing 100.12: Sin optimización GCC 4.4.5 (MIPS) (IDA)

```
f:                                      # ↙
    ↘ CODE XREF: f+64
                                          # f↙
    ↘ +94 ...

var_C           = -0xC
var_8           = -8
var_4           = -4
arg_0           =  0
arg_4           =  4

                addiu   $sp, -0x28
                sw      $ra, 0x28+var_4($sp)
                sw      $fp, 0x28+var_8($sp)
                sw      $s0, 0x28+var_C($sp)
                move    $fp, $sp
                sw      $a0, 0x28+arg_0($fp)
                sw      $a1, 0x28+arg_4($fp)
```

```
                lw      $v0, 0x28+arg_0($fp)
                or      $at, $zero
                bnez    $v0, loc_40
                or      $at, $zero
                lw      $v0, 0x28+arg_4($fp)
                or      $at, $zero
                addiu   $v0, 1
                b       loc_AC
                or      $at, $zero

loc_40:                                          # ↙
    ↳ CODE XREF: f+24
                lw      $v0, 0x28+arg_4($fp)
                or      $at, $zero
                bnez    $v0, loc_74
                or      $at, $zero
                lw      $v0, 0x28+arg_0($fp)
                or      $at, $zero
                addiu   $v0, -1
                move    $a0, $v0
                li      $a1, 1
                jal     f
                or      $at, $zero
                b       loc_AC
                or      $at, $zero

loc_74:                                          # ↙
    ↳ CODE XREF: f+48
                lw      $v0, 0x28+arg_0($fp)
                or      $at, $zero
                addiu   $s0, $v0, -1
                lw      $v0, 0x28+arg_4($fp)
                or      $at, $zero
```

CAPÍTULO 100. 2

```
                addiu   $v0, -1
                lw      $a0, 0x28+arg_0($fp)
                move    $a1, $v0
                jal     f
                or      $at, $zero
                move    $a0, $s0
                move    $a1, $v0
                jal     f
                or      $at, $zero

loc_AC:                                     # ↙
    ↪ CODE XREF: f+38
                                            # f↙
    ↪ +6C
                move    $sp, $fp
                lw      $ra, 0x28+var_4($sp)
                lw      $fp, 0x28+var_8($sp)
                lw      $s0, 0x28+var_C($sp)
                addiu   $sp, 0x28
                jr      $ra
                or      $at, $zero
```

100.8 Ejercicio 2.17

. ?

:

- Linux x64 (beginners.re)
- Mac OS X (beginners.re)

 CAPÍTULO 100. 2

- Linux MIPS (beginners.re)
- Win32 (beginners.re)
- Win64 (beginners.re)

MSVC 2012 redist.

100.9 Ejercicio 2.18

. .

. .

.

- Win32 (beginners.re)
- Linux x86 (beginners.re)
- Mac OS X (beginners.re)
- Linux MIPS (beginners.re)

100.10 Ejercicio 2.19

2.18.

- Win32 (beginners.re)
- Linux x86 (beginners.re)
- Mac OS X (beginners.re)
- Linux MIPS (beginners.re)

CAPÍTULO 100. 2

100.11 Ejercicio 2.20

?

:

- Linux x64 (beginners.re)
- Mac OS X (beginners.re)
- Linux ARM Raspberry Pi (beginners.re)
- Linux MIPS (beginners.re)
- Win64 (beginners.re)

 CAPÍTULO 101. 3

Capítulo 101

3

.

101.1 Ejercicio 3.2

. .

- Windows x86 (beginners.re)
- Linux x86 (beginners.re)
- Mac OS X (x64) (beginners.re)
- Linux MIPS (beginners.re)

CAPÍTULO 101. 3

101.2 Ejercicio 3.3

...

- Windows x86 (beginners.re)
- Linux x86 (beginners.re)
- Mac OS X (x64) (beginners.re)
- Linux MIPS (beginners.re)

101.3 Ejercicio 3.4

...

.

- Windows x86 (beginners.re)
- : http://go.yurichev.com/17197

101.4 Ejercicio 3.5

. .

:

- .
- .
- Windows x86 (beginners.re)

CAPÍTULO 101. 3

- Linux x86 (beginners.re)
- Mac OS X (x64) (beginners.re)
- Linux MIPS (beginners.re)
- : beginners.re

101.5 Ejercicio 3.6

Spanish text placeholder. . .

- Windows x86 (beginners.re)
- Linux x86 (beginners.re)
- Mac OS X (x64) (beginners.re)

101.6 Ejercicio 3.8

. . .
: beginners.re
: beginners.re
: beginners.re.

. .

- Windows x86 (beginners.re)
- Linux x86 (beginners.re)
- Mac OS X (x64) (beginners.re)

CAPÍTULO 102. CRACKME / KEYGENME

Capítulo 102

crackme / keygenme

http://go.yurichev.com/17315

Capítulo 103

: <dennis(a)yurichev.com>

Spanish text placeholder.

GitHub.

APÉNDICE A. X86

Apéndice A

x86

A.1

byte 8-Spanish text placeholder. . : AL/BL/CL/DL/AH/BH/CI

word 16-Spanish text placeholder. . : AX/BX/CX/DX/SI/DI/

double word («dword») 32-Spanish text placeholder. . (x86) (x64). .

quad word («qword») 64-Spanish text placeholder. . .

tbyte (10) 80-Spanish text placeholder o 10 (IEEE 754 FPU).

paragraph (16) .

 APÉNDICE A. X86

Windows API[1].

A.2

. . .

.

N.B.: .

A.2.1 RAX/EAX/AX/AL

Spanish text placeholder		
RAX[x64]		
	EAX	
	AX	
	AH	AL

AKA . .

A.2.2 RBX/EBX/BX/BL

Spanish text placeholder		
RBX[x64]		
	EBX	
	BX	
	BH	BL

[1] Application programming interface

 APÉNDICE A. X86

A.2.3 RCX/ECX/CX/CL

Spanish text placeholder		
RCX^{x64}		
	ECX	
	CX	
	CH	CL

AKA : (SHL/SHR/RxL/RxR).

A.2.4 RDX/EDX/DX/DL

Spanish text placeholder		
RDX^{x64}		
	EDX	
	DX	
	DH	DL

A.2.5 RSI/ESI/SI/SIL

Spanish text placeholder	
RSI^{x64}	
	ESI
	SI
	SIL^{x64}

AKA «source index». REP MOVSx, REP CMPSx.

A.2.6 RDI/EDI/DI/DIL

Spanish text placeholder	
RDI^{x64}	
	EDI
	DI
	DIL^{x64}

AKA «destination index». REP MOVSx, REP STOSx.

A.2.7 R8/R8D/R8W/R8L

Spanish text placeholder	
R8	
	R8D
	R8W
	R8L

A.2.8 R9/R9D/R9W/R9L

Spanish text placeholder	
R9	
	R9D
	R9W
	R9L

A.2.9 R10/R10D/R10W/R10L

Spanish text placeholder	
R10	
	R10D
	R10W
	R10L

A.2.10 R11/R11D/R11W/R11L

Spanish text placeholder	
R11	
	R11D
	R11W
	R11L

A.2.11 R12/R12D/R12W/R12L

Spanish text placeholder	
R12	
	R12D
	R12W
	R12L

A.2.12 R13/R13D/R13W/R13L

Spanish text placeholder	
R13	
	R13D
	R13W
	R13L

A.2.13 R14/R14D/R14W/R14L

Spanish text placeholder	
R14	
	R14D
	R14W
	R14L

A.2.14 R15/R15D/R15W/R15L

Spanish text placeholder	
R15	
	R15D
	R15W
	R15L

A.2.15 RSP/ESP/SP/SPL

Spanish text placeholder	
RSP	
	ESP
	SP
	SPL

AKA stack pointer. .

A.2.16 RBP/EBP/BP/BPL

Spanish text placeholder	
RBP	
	EBP
	BP
	BPL

AKA frame pointer. : (7.1.2 on page 135).

A.2.17 RIP/EIP/IP

Spanish text placeholder	
RIP^{x64}	
	EIP
	IP

AKA «instruction pointer» [2]. :

```
MOV EAX, ...
```

[2] «program counter»

```
JMP EAX
```

:

```
PUSH value
RET
```

A.2.18 CS/DS/ES/SS/FS/GS

(CS), (DS), (SS).

FS en win32 TLS, . .
(94 on page 1776).

A.2.19

AKA EFLAGS.

 APÉNDICE A. X86

()	()	
0 (1)	CF (Carry)	CLC
2 (4)	PF (Parity)	(17
4 (0x10)	AF (Adjust)	
6 (0x40)	ZF (Zero)	0 0.
7 (0x80)	SF (Sign)	
8 (0x100)	TF (Trap)	. .
9 (0x200)	IF (Interrupt enable)	. CLI
10 (0x400)	DF (Direction)	REP CLD
11 (0x800)	OF (Overflow)	
12, 13 (0x3000)	IOPL (I/O privilege level)[80286]	
14 (0x4000)	NT (Nested task)[80286]	
16 (0x10000)	RF (Resume)[80386]	. .
17 (0x20000)	VM (Virtual 8086 mode)[80386]	
18 (0x40000)	AC (Alignment check)[80486]	
19 (0x80000)	VIF (Virtual interrupt)[Pentium]	
20 (0x100000)	VIP (Virtual interrupt pending)[Pentium]	
21 (0x200000)	ID (Identification)[Pentium]	

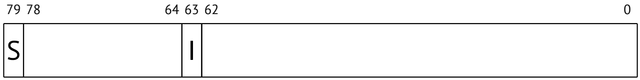
 APÉNDICE A. X86

.

A.3 FPU registros

8 80-: ST(0)-ST(7). N.B.: IDA ST(0) ST. IEEE 754.

:



(S, I)

A.3.1

FPU.

0	IM (Invalid operation Mask)	
1	DM (Denormalized operand Mask)	
2	ZM (Zero divide Mask)	
3	OM (Overflow Mask)	
4	UM (Underflow Mask)	
5	PM (Precision Mask)	
7	IEM (Interrupt Enable Mask)	
8, 9	PC (Precision Control)	00 – 24 (REA 10 – 53 (REA 11 – 64 (REA
10, 11	RC (Rounding Control)	00 – 01 – $-\infty$ 10 – $+\infty$ 11 – 0
12	IC (Infinity Control)	0 – () $+\infty$ y – 1 – $+\infty$ y $-\infty$

PM, UM, OM, ZM, DM, IM .

A.3.2

.

15	B (Busy)	(1) (0)
14	C3	
13, 12, 11	TOP	
10	C2	
9	C1	
8	C0	
7	IR (Interrupt Request)	
6	SF (Stack Fault)	
5	P (Precision)	
4	U (Underflow)	
3	O (Overflow)	
2	Z (Zero)	
1	D (Denormalized)	
0	I (Invalid operation)	

SF, P, U, O, Z, D, I .

C3, C2, C1, C0 : (17.7.1 on page 437).

N.B.: .

A.3.3 Tag Word

.

15, 14	Tag(7)
13, 12	Tag(6)
11, 10	Tag(5)
9, 8	Tag(4)
7, 6	Tag(3)
5, 4	Tag(2)
3, 2	Tag(1)
1, 0	Tag(0)

:

- 00 –
- 01 –
- 10 – (NAN[3], ∞, o)
- 11 –

A.4 SIMD registros

A.4.1 MMX registros

8 64-: MM0..MM7.

A.4.2 SSE y AVX registros

SSE: 8 128-: XMM0..XMM7. x86-64 : XMM8..XMM15.

AVX .

[3] Not a Number

APÉNDICE A. X86

A.5

hardware breakpoints.

- DR0 — #1
- DR1 — #2
- DR2 — #3
- DR3 — #4
- DR6 —
- DR7 —

A.5.1 DR6

()	
0 (1)	B0 —
1 (2)	B1 —
2 (4)	B2 —
3 (8)	B3 —
13 (0x2000)	BD —
14 (0x4000)	BS — . .
15 (0x8000)	BT (task switch flag)

N.B. . (A.2.19 on page 1869).

A.5.2 DR7

.

()	
0 (1)	L0 — #1
1 (2)	G0 — #1
2 (4)	L1 — #2
3 (8)	G1 — #2
4 (0x10)	L2 — #3
5 (0x20)	G2 — #3
6 (0x40)	L3 — #4
7 (0x80)	G3 — #4
8 (0x100)	LE —
9 (0x200)	GE —
13 (0x2000)	GD —
16,17 (0x30000)	#1: R/W —
18,19 (0xC0000)	#1: LEN —
20,21 (0x300000)	#2: R/W —
22,23 (0xC00000)	#2: LEN —
24,25 (0x3000000)	#3: R/W —
26,27 (0xC000000)	#3: LEN —
28,29 (0x30000000)	#4: R/W —
30,31 (0xC0000000)	#4: LEN —

(R/W):

- 00 —
- 01 —

- 10 –
- 11 –

N.B.: .

(LEN):

- 00 –
- 01 –
- 10 –
- 11 –

A.6

(90 on page 1753).

. [Int13] o [AMD13a] .

(89.2 on page 1751).

A.6.1

LOCK . . . ADD, AND, BTR, BTS, CMPXCHG, OR, XADD, XOR. (68.4 on page 1408).

REP MOVSx y STOSx: . MOVSx (A.6.2 on page 1882) y STOSx (A.6.2 on page 1885).

.

REPE/REPNE (AKA REPZ/REPNZ) CMPSx y SCASx: . .

CMPSx (A.6.3 on page 1887) y SCASx (A.6.2 on page 1884).

.

A.6.2

.

ADC (*add with carry*)

```
;
;
ADD val1.lo, val2.lo
ADC val1.hi, val2.hi ;
```

: 24 on page 769.

ADD

AND

CALL :`PUSH address_after_CALL_instruction; JMP label`

CMP

DEC decrement. .

IMUL

INC increment. .

JCXZ, JECXZ, JRCXZ (M)

JMP . jump offset.

Jcc (cccondition code)

. . jump offset.

JAE AKA JNC: : CF=0

JA AKA JNBE: : CF=0 y ZF=0

JBE : CF=1 o ZF=1

JB AKA JC: : CF=1

JC AKA JB:

JE AKA JZ: : ZF=1

JGE : SF=OF

JG : ZF=0 y SF=OF

JLE : ZF=1 o SF≠OF

JL : SF≠OF

JNAE AKA JC: CF=1

JNA CF=1 y ZF=1

JNBE : CF=0 y ZF=0

JNB AKA JNC: : CF=0

JNC AKA JAE: JNB.

JNE AKA JNZ: : ZF=0

 APÉNDICE A. X86

JNGE : SF≠OF

JNG : ZF=1 o SF≠OF

JNLE : ZF=0 y SF=OF

JNL : SF=OF

JNO : OF=0

JNS

JNZ AKA JNE: : ZF=0

JO : OF=1

JPO (Jump Parity Odd)

JP AKA JPE:

JS

JZ AKA JE: : ZF=1

LAHF :

7	6		4		2		0
SF	ZF		AF		PF		CF

LEA **LEAVE** (*Load Effective Address*)

Spanish text placeholder

Spanish text placeholder

Spanish text placeholder.

 APÉNDICE A. X86

```
int f(int a, int b)
{
        return a*8+b;
};
```

Listing A.1: Con optimización MSVC 2010

```
_a$ = 8                                       ↙
    ↘                                     ; size = ↙
    ↘ 4
_b$ = 12                                      ↙
    ↘                                     ; size = ↙
    ↘ 4
_f      PROC
        mov     eax, DWORD PTR _b$[esp↙
    ↘ -4]
        mov     ecx, DWORD PTR _a$[esp↙
    ↘ -4]
        lea     eax, DWORD PTR [eax+ecx↙
    ↘ *8]
        ret     0
_f      ENDP
```

Intel C++ :

```
int f1(int a)
{
        return a*13;
};
```

Listing A.2: Intel C++ 2011

```
_f1     PROC NEAR
        mov     ecx, DWORD PTR [4+esp↙
    ↳ ]       ; ecx = a
        lea     edx, DWORD PTR [ecx+↙
    ↳ ecx*8]  ; edx = a*9
        lea     eax, DWORD PTR [edx+↙
    ↳ ecx*4]  ; eax = a*9 + a*4 = a*13
        ret
```

MOVSB/MOVSW/MOVSD/MOVSQ / 16-/ 32-/ 64- SI/ESI/RSI DI/EDI/RDI.

memcpy() . .

memcpy(EDI, ESI, 15):

```

;
CLD            ;
MOV ECX, 3
REP MOVSD      ;
MOVSW          ;
MOVSB          ;
```

( ).

MOVSX : (15.1.1 on page 378)

MOVZX : (15.1.1 on page 379)

MOV .

MUL

 APÉNDICE A. X86

NEG : $op = -op$

NOP NOP. `XCHG EAX,EAX`. . 6.1.1 on page 91.

: (88 on page 1747).

NOP . NOP .

NOT `op1`: $op1 = \neg op1$.

OR

POP : `value=SS:[ESP]; ESP=ESP+4 (o 8)`

PUSH : `ESP=ESP-4 (o 8); SS:[ESP]=value`

RET : `POP tmp; JMP tmp`.

, `RET` `RETN` («return near») (94 on page 1776), `RETF` («return far»).

: `POP tmp; ADD ESP op1; JMP tmp`. `RET` : 64.2 on page 1299.

SAHF :

7	6		4		2		0
SF	ZF		AF		PF		CF

SBB (*subtraction with borrow*)

```
;
;
SUB val1.lo, val2.lo
SBB val1.hi, val2.hi ;
```

: 24 on page 769.

SCASB/SCASW/SCASD/SCASQ (M) / 16-/ 32-/ 64- AX/EAX/RAX DI/EDI/RDI. .

. .

:

:

```
lea     edi, string
mov     ecx, 0FFFFFFFFh ;
xor     eax, eax        ;
repne scasb
add     edi, 0FFFFFFFFh ;

;

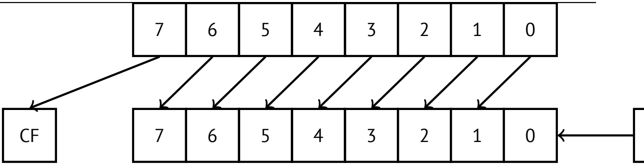
;
;  ECX = -1-strlen

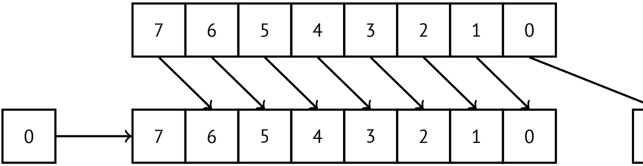
not     ecx
dec     ecx

;
```

SHL

SHR :





2^n. : 19 on page 577.

SHRD op1, op2, op3: .

: 24 on page 769.

STOSB/STOSW/STOSD/STOSQ / 16-/ 32-/ 64- AX/EAX/-RAX DI/EDI/RDI.

: memset() . .

memset(EDI, 0xAA, 15):

```
;
CLD                    ;
MOV EAX, 0AAAAAAAAh
MOV ECX, 3
REP STOSD              ;
STOSW                  ;
STOSB                  ;
```

( ).

SUB

TEST : 19 on page 577

XCHG

XOR op1, op2: XOR[4] . $op1 = op1 \oplus op2$.

0	0	0
0	1	1
1	0	1
1	1	0

A.6.3

BSF *bit scan forward*, : 25.2 on page 814

BSR *bit scan reverse*

BSWAP *(byte swap)*, .

BTC bit test and complement

BTR bit test and reset

BTS bit test and set

BT bit test

CBW/CWD/CWDE/CDQ/CDQE :

CBW

[4] eXclusive OR

CWD

CWDE

CDQ

CDQE (x64)

24.5 on page 788.

> The process of stretching numbers by extending the sign bit is called sign extension. The 8086 provides instructions (Fig. 3.29) to facilitate the task of sign extension. These instructions were initially named SEX (sign extend) but were later renamed to the more conservative CBW (convert byte to word) and CWD (convert word to double word).

CLD .

CLI (M)

CMC (M)

CMOVcc MOV: . (A.6.2 on page 1879).

CMPSB/CMPSW/CMPSD/CMPSQ (M) / 16-/ 32-/ 64- SI/ESI/RSI DI/EDI/RDI. .

.

memcmp() .

(WRK v1.2):

Listing A.3: base\ntos\rtl\i386\movemem.asm

```
; ULONG
; RtlCompareMemory (
;    IN PVOID Source1,
;    IN PVOID Source2,
;    IN ULONG Length
;    )
;
; Routine Description:
;
;    This function compares two blocks ↙
   ↳ of memory and returns the number
;    of bytes that compared equal.
;
; Arguments:
;
;    Source1 (esp+4) - Supplies a ↙
   ↳ pointer to the first block of ↙
   ↳ memory to
;       compare.
;
;    Source2 (esp+8) - Supplies a ↙
   ↳ pointer to the second block of ↙
   ↳ memory to
;       compare.
;
;    Length (esp+12) - Supplies the ↙
   ↳ Length, in bytes, of the memory ↙
   ↳ to be
;       compared.
;
```

```
; Return Value:
;
;    The number of bytes that compared ↙
    ↳ equal is returned as the function
;    value. If all bytes compared equal↙
    ↳ , then the length of the original
;    block of memory is returned.
;
;--

RcmSource1      equ     [esp+12]
RcmSource2      equ     [esp+16]
RcmLength       equ     [esp+20]

CODE_ALIGNMENT
cPublicProc _RtlCompareMemory,3
cPublicFpo 3,0

        push    esi                     ↙
    ↳  ; save registers
        push    edi                     ↙
    ↳  ;
        cld                             ↙
    ↳  ; clear direction
        mov     esi,RcmSource1          ↙
    ↳  ; (esi) -> first block to ↙
    ↳ compare
        mov     edi,RcmSource2          ↙
    ↳  ; (edi) -> second block to ↙
    ↳ compare

;
;   Compare dwords, if any.
```

```
;

rcm10:  mov     ecx,RcmLength                 ↙
    ↳  ; (ecx) = length in bytes
        shr     ecx,2                         ↙
    ↳  ; (ecx) = length in dwords
        jz      rcm20                         ↙
    ↳  ; no dwords, try bytes
        repe    cmpsd                         ↙
    ↳  ; compare dwords
        jnz     rcm40                         ↙
    ↳  ; mismatch, go find byte

;
;   Compare residual bytes, if any.
;

rcm20:  mov     ecx,RcmLength                 ↙
    ↳  ; (ecx) = length in bytes
        and     ecx,3                         ↙
    ↳  ; (ecx) = length mod 4
        jz      rcm30                         ↙
    ↳  ; 0 odd bytes, go do dwords
        repe    cmpsb                         ↙
    ↳  ; compare odd bytes
        jnz     rcm50                         ↙
    ↳  ; mismatch, go report how far we↙
    ↳  got

;
;   All bytes in the block match.
;
```

```
rcm30:  mov     eax,RcmLength                ↙
    ↳ ; set number of matching bytes
        pop     edi                          ↙
    ↳ ; restore registers
        pop     esi                          ↙
    ↳ ;
        stdRET  _RtlCompareMemory

;
;   When we come to rcm40, esi (and edi↙
    ↳ ) points to the dword after the
;   one which caused the mismatch.  ↙
    ↳ Back up 1 dword and find the byte↙
    ↳ .
;   Since we know the dword didn't ↙
    ↳ match, we can assume one byte won↙
    ↳ 't.
;

rcm40:  sub     esi,4                        ↙
    ↳ ; back up
        sub     edi,4                        ↙
    ↳ ; back up
        mov     ecx,5                        ↙
    ↳ ; ensure that ecx doesn't count ↙
    ↳ out
        repe    cmpsb                        ↙
    ↳ ; find mismatch byte

;
;   When we come to rcm50, esi points ↙
    ↳ to the byte after the one that
;   did not match, which is TWO after ↙
```

```
    ↳ the last byte that did match.
;

rcm50:  dec     esi                                ↙
    ↳  ; back up
        sub     esi,RcmSource1                     ↙
    ↳  ; compute bytes that matched
        mov     eax,esi                            ↙
    ↳  ;
        pop     edi                                ↙
    ↳  ; restore registers
        pop     esi                                ↙
    ↳  ;
        stdRET  _RtlCompareMemory

stdENDP _RtlCompareMemory
```

N.B.: .

CPUID . : (21.6.1 on page 707).

DIV

IDIV

INT (M): `INT x` `PUSHF; CALL dword ptr [x*4]` . . . .

. : [Bro] .

.

INT 3 (M): `INT`, (0xCC), . `0xCC` .

Windows NT, `EXCEPTION_BREAKPOINT` . . . MSVS[5] . reverse engineering, .

MSVC compiler intrinsic : `__debugbreak()`[6].

kernel32.dll `DebugBreak()`[7], `INT 3`.

IN (M) . (78.3 on page 1513).

IRET : . `POP tmp; POPF; JMP tmp`.

LOOP (M) CX/ECX/RCX, .

OUT (M) . (78.3 on page 1513).

POPA (M) (R|E)DI, (R|E)SI, (R|E)BP, (R|E)BX, (R|E)DX, (R|E)CX, (R|E)AX .

POPCNT population count. . AKA «hamming weight». AKA «NSA instruction» :

> This branch of cryptography is fast-paced and very politically charged. Most designs are secret; a majority of military encryptions systems in use today are based on LFSRs. In fact, most Cray computers (Cray 1, Cray X-MP, Cray Y-MP) have a rather curious instruction generally known as "population count." It counts the 1 bits in a

[5] Microsoft Visual Studio
[6] MSDN
[7] MSDN

> register and can be used both to efficiently calculate the Hamming distance between two binary words and to implement a vectorized version of a LFSR. I've heard this called the canonical NSA instruction, demanded by almost all computer contracts.

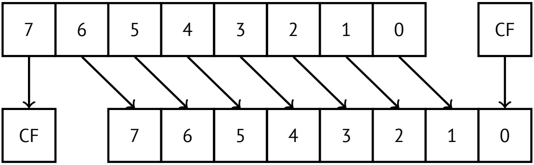
[Sch94]

POPF (AKA)

PUSHA (M) (R|E)AX, (R|E)CX, (R|E)DX, (R|E)BX, (R|E)BP, (R|E)SI, (R|E)DI .

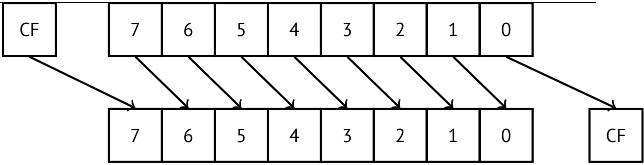
PUSHF (AKA)

RCL (M) :



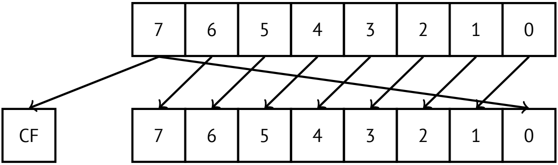
RCR (M) :

 APÉNDICE A. X86

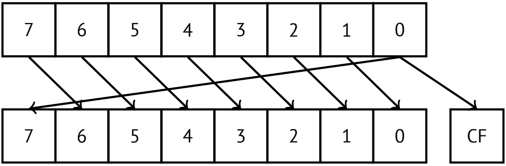


ROL/ROR (M)

ROL: :



ROR: :

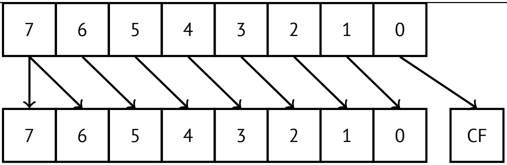


.

(compiler intrinsics) _*rotl()* y _*rotr()*[8], .

SAL , SHL

[8] MSDN



SAR

MSB.

SETcc op: . (A.6.2 on page 1879).

STC (M)

STD (M) . `ntoskrnl.exe` .

STI (M)

SYSCALL (AMD) (66 on page 1325)

SYSENTER (Intel) (66 on page 1325)

UD2 (M)

A.6.4

.

-P

FABS

FADD op: ST(0)=op+ST(0)

FADD ST(0), ST(i): ST(0)=ST(0)+ST(i)

FADDP ST(1)=ST(0)+ST(1);

FCHS ST(0)=-ST(0)

FCOM ST(0) ST(1)

FCOM op: ST(0) op

FCOMP ST(0) ST(1);

FCOMPP ST(0) ST(1);

FDIVR op: ST(0)=op/ST(0)

FDIVR ST(i), ST(j): ST(i)=ST(j)/ST(i)

FDIVRP op: ST(0)=op/ST(0);

FDIVRP ST(i), ST(j): ST(i)=ST(j)/ST(i);

FDIV op: ST(0)=ST(0)/op

FDIV ST(i), ST(j): ST(i)=ST(i)/ST(j)

FDIVP ST(1)=ST(0)/ST(1);

FILD op: .

FIST op: ST(0) op

FISTP op: ST(0) op;

FLD1

FLDCW op: FPU control word (A.3 on page 1871) 16-bit op.

FLDZ

FLD op: .

FMUL op: ST(0)=ST(0)*op

FMUL ST(i), ST(j): ST(i)=ST(i)*ST(j)

FMULP op: ST(0)=ST(0)*op;

FMULP ST(i), ST(j): ST(i)=ST(i)*ST(j);

FSINCOS : tmp=ST(0); ST(1)=sin(tmp); ST(0)=cos(tmp)

FSQRT : $ST(0) = \sqrt{ST(0)}$

FSTCW op: FPU control word (A.3 on page 1871) 16-bit op .

FNSTCW op: FPU control word (A.3 on page 1871) 16-bit op.

FSTSW op: FPU status word (A.3.2 on page 1872) 16-bit op .

FNSTSW op: FPU status word (A.3.2 on page 1872) 16-bit op.

FST op: ST(0) op

FSTP op: ST(0) op;

FSUBR op: ST(0)=op-ST(0)

FSUBR ST(0), ST(i): ST(0)=ST(i)-ST(0)

FSUBRP ST(1)=ST(0)-ST(1);

FSUB op: ST(0)=ST(0)-op

FSUB ST(0), ST(i): ST(0)=ST(0)-ST(i)

FSUBP ST(1)=ST(1)-ST(0);

FUCOM ST(i): ST(0) y ST(i)

FUCOM ST(0) y ST(1)

FUCOMP ST(0) y ST(1); .

FUCOMPP ST(0) y ST(1); .

.

FXCH ST(i)

FXCH

A.6.5

().

. : 82.1 on page 1680.

ASCII		x86
0	30	XOR
1	31	XOR
2	32	XOR
3	33	XOR
4	34	XOR
5	35	XOR
7	37	AAA
8	38	CMP
9	39	CMP
:	3a	CMP
;	3b	CMP
<	3c	CMP

=	3d	CMP
?	3f	AAS
@	40	INC
A	41	INC
B	42	INC
C	43	INC
D	44	INC
E	45	INC
F	46	INC
G	47	INC
H	48	DEC
I	49	DEC
J	4a	DEC
K	4b	DEC
L	4c	DEC
M	4d	DEC
N	4e	DEC
O	4f	DEC
P	50	PUSH
Q	51	PUSH
R	52	PUSH
S	53	PUSH
T	54	PUSH
U	55	PUSH
V	56	PUSH
W	57	PUSH
X	58	POP
Y	59	POP
Z	5a	POP
[	5b	POP

APÉNDICE A. X86

\	5c	POP
]	5d	POP
^	5e	POP
_	5f	POP
`	60	PUSHA
a	61	POPA
f	66	
g	67	
h	68	PUSH
i	69	IMUL
j	6a	PUSH
k	6b	IMUL
p	70	JO
q	71	JNO
r	72	JB
s	73	JAE
t	74	JE
u	75	JNE
v	76	JBE
w	77	JA
x	78	JS
y	79	JNS
z	7a	JP

: AAA, AAS, CMP, DEC, IMUL, INC, JA, JAE, JB, JBE, JE, JNE, JNO, JNS, JO, JP, JS, POP, POPA, PUSH, PUSHA, XOR.

Apéndice B

ARM

B.1

ARM CPU, .

byte 8-Spanish text placeholder. .

halfword 16-Spanish text placeholder. .

word 32-Spanish text placeholder. .

doubleword 64-Spanish text placeholder.

quadword 128-Spanish text placeholder.

B.2

- ARMv4: .
- ARMv6: iPhone 1st gen., iPhone 3G (Samsung 32-bit RISC ARM 1176JZ(F)-S Thumb-2)
- ARMv7: Thumb-2 (2003). iPhone 3GS, iPhone 4, iPad 1st gen. (ARM Cortex-A8), iPad 2 (Cortex-A9), iPad 3rd gen.
- ARMv7s: . iPhone 5, iPhone 5c, iPad 4th gen. (Apple A6).
- ARMv8: 64-, AKA ARM64 AKA AArch64. iPhone 5S, iPad Air (Apple A7).

B.3 32- ARM (AArch32)

B.3.1

- R0
- R1...R12GPRs
- R13AKA SP (stack pointer)
- R14AKA LR (link register)
- R15AKA PC (program counter)

R0-R3 .

 APÉNDICE B. ARM

B.3.2 Current Program Status Register (CPSR)

0..4	Mprocessor mode
5	TThumb state
6	FFIQ disable
7	IIRQ disable
8	Aimprecise data abort disable
9	Edata endianness
10..15, 25, 26	ITif-then state
16..19	GEgreater-than-or-equal-to
20..23	DNMdo not modify
24	JJava state
27	Qsticky overflow
28	Voverflow
29	Ccarry/borrow/extend
30	Zzero bit
31	Nnegative/less than

B.3.3

<table>
<tr><td>0..31 bits</td><td>32..64</td><td>65..96</td><td>97..127</td></tr>
<tr><td colspan="4">Q0 128 bits</td></tr>
<tr><td colspan="2">D0 64 bits</td><td colspan="2">D1</td></tr>
<tr><td>S0 32 bits</td><td>S1</td><td>S2</td><td>S3</td></tr>
</table>

.

.

.

.

.

.

VFPv3 (NEON o «Advanced SIMD») .

NEON o «Advanced SIMD» .

B.4 64- ARM (AArch64)

B.4.1

AArch32.

- X0
- X0…X7.
- X8
- X9…X15.
- X16
- X17
- X18
- X19…X29.
- X29 FP (GCC)
- X30«Procedure Link Register» AKA LR (link register).
- X31 AKA XZR o «Zero Register». WZR.

- SP, .

: [ARM13c].

Spanish text placeholder	Spanish text placeholder
X0	
	W0

B.5

ADD ADDS .

B.5.1

EQ		Z == 1
NE		Z == 0
CS AKA HS (Higher or Same)	/	C == 1
CC AKA LO (LOwer)	/	C == 0
MI	/	N == 1
PL	/	N == 0
VS		V == 1
VC		V == 0
HI	/	C == 1 and Z == 0
LS	/	C == 0 or Z == 1
GE	/	N == V
LT	/	N != V
GT	/	Z == 0 and N == V
LE	/	Z == 1 or N != V
None / AL		

Apéndice C

MIPS

C.1 Registros

( )

C.1.1 GPR

$0	$ZERO	(NOP).
$1	$AT	.
$2 ...$3	$V0 ...$V1	.
$4 ...$7	$A0 ...$A3	.
$8 ...$15	$T0 ...$T7	.
$16 ...$23	$S0 ...$S7	*.
$24 ...$25	$T8 ...$T9	.
$26 ...$27	$K0 ...$K1	.
$28	$GP	**.
$29	$SP	SP*.
$30	$FP	FP*.
$31	$RA	RA.
n/a	PC	PC.
n/a	HI	***.
n/a	LO	***.

C.1.2

$F0..$F1	.
$F2..$F3	.
$F4..$F11	.
$F12..$F15	.
$F16..$F19	.
$F20..$F31	*.

*Callee .

**Callee ().
*** MFHI y MFLO.

C.2 Instrucciones

:

- ```
  instruction destination, source1,
      source2
  ```

  ```
  instruction destination/source1,
      source2
  ```

-

-

C.2.1
  ```

Apéndice D

__divdi3	
__moddi3	
__udivdi3	
__umoddi3	

Apéndice E

`ll` «long long», .

`__alldiv`	
`__allmul`	
`__allrem`	
`__allshl`	
`__allshr`	
`__aulldiv`	
`__aullrem`	
`__aullshr`	

.

`VC/crt/src/intel/*.asm`.

Apéndice F

Cheatsheets

F.1 IDA

Cheatsheet de teclas de acceso rápido:

 APÉNDICE F. CHEATSHEETS

Space	
C	
D	
A	
*	
U	
O	
H	
R	
B	
Q	
N	
?	
G	
:	
Ctrl-X	
X	Spanish text placeholder.
Alt-I	
Ctrl-I	
Alt-B	
Ctrl-B	
Alt-T	
Ctrl-T	
Alt-P	
Enter	
Esc	
Num -	
Num +	

. .

F.2 OllyDbg

Cheatsheet de teclas de acceso rápido:

F7	
F8	pasar por encima
F9	
Ctrl-F2	

F.3 MSVC

.

/O1	
/Ob0	
/Ox	
/GS-	
/Fa(file)	
/Zi	
/Zp(n)	
/MD	`MSVCR*.DLL`

: 55.1 on page 1261.

F.4 GCC

-Os	
-O3	
-regparm=	
-o file	
-g	
-S	
-masm=intel	
-fno-inline	

F.5 GDB

:

APÉNDICE F. CHEATSHEETS

break filename.c:number	
break function	
break *address	
b	—”—
p variable	
run	
r	—”—
cont	
c	—”—
bt	
set disassembly-flavor intel	
disas	disassemble current function
disas function	
disas function,+50	disassemble portion
disas $eip,+0x10	—”—
disas/r	
info registers	
info float	
info locals	
x/w ...	
x/w $rdi	
x/10w ...	
x/s ...	
x/i ...	
x/10c ...	
x/b ...	
x/h ...	
x/g ...	
finish	
next	

Apéndice G

G.1

G.1.1

Ejercicio #1

Ejercicio: 3.7.1 on page 59.

```
MessageBeep (0xFFFFFFFF); // A simple beep. ↙
    ↳ If the sound card is not available, ↙
    ↳ the sound is generated using the ↙
    ↳ speaker.
```

Ejercicio #2

Ejercicio: 3.7.2 on page 60.

```
#include <unistd.h>

int main()
{
        sleep(2);
};
```

G.1.2

Ejercicio #1

Ejercicio: 5.5.1 on page 78.

, RA y argc. .

: RA, argc argv[].

GCC 4.8.x .

Ejercicio #2

Ejercicio: 5.5.2 on page 79.

.

```
#include <stdio.h>
#include <time.h>

int main()
{
        printf ("%d\n", time(NULL));
};
```

G.1.3

Ejercicio #1

Ejercicio: 7.4.1 on page 185.

(“read only data”):

```
$ objdump -s 1

...

Contents of section .rodata:
 400600 01000200 52657375 6c743a20 25730a00 ⤦
     ↳  ....Result: %s..
 400610 48656c6c 6f2c2077 6f726c64 210a00   ⤦
     ↳  Hello, world!..
```

```
C:\...>objdump -s 1.exe

...

Contents of section .data:
 40b000 476f6f64 62796521 00000000 52657375 ⤦
     ↳  Goodbye!....Resu
 40b010 6c743a20 25730a00 48656c6c 6f2c2077 ⤦
     ↳  lt: %s..Hello, w
 40b020 6f726c64 210a0000 00000000 00000000 ⤦
     ↳  orld!...........
 40b030 01000000 00000000 c0cb4000 00000000 ⤦
     ↳  ..........@.....

...
```

```
C:\...>objdump -x 1.exe

...

Sections:
Idx Name          Size      VMA       LMA      ↙
    ↳     File off  Algn
  0 .text         00006d2a  00401000  ↙
    ↳ 00401000  00000400  2**2
                  CONTENTS, ALLOC, LOAD, ↙
    ↳ READONLY, CODE
  1 .rdata        00002262  00408000  ↙
    ↳ 00408000  00007200  2**2
                  CONTENTS, ALLOC, LOAD, ↙
    ↳ READONLY, DATA
  2 .data         00000e00  0040b000  0040↙
    ↳ b000  00009600  2**2
                  CONTENTS, ALLOC, LOAD, ↙
    ↳ DATA
  3 .reloc        00000b98  0040e000  0040↙
    ↳ e000  0000a400  2**2
                  CONTENTS, ALLOC, LOAD, ↙
    ↳ READONLY, DATA
```

G.1.4

G.1.5 Ejercicio #1

Ejercicio: 13.5.1 on page 337.

: `printf()`.

G.1.6

G.1.7 Ejercicio #3

Ejercicio: 14.4.3 on page 366.

```
#include <stdio.h>

int main()
{
        int i;
        for (i=100; i>0; i--)
                printf ("%d\n", i);
};
```

G.1.8 Ejercicio #4

Ejercicio: 14.4.4 on page 371.

```
#include <stdio.h>

int main()
{
        int i;
        for (i=1; i<100; i=i+3)
                printf ("%d\n", i);
};
```

G.1.9

Ejercicio #1

Ejercicio: 15.2.1 on page 392.

```
int f(char *s)
{
        int rt=0;
        for (;*s;s++)
        {
                if (*s==' ')
                        rt++;
        };
        return rt;
};
```

G.1.10

Ejercicio #2

Ejercicio: 16.3.1 on page 409.

```
int f(int a)
{
        return a*7;
};
```

G.1.11

Ejercicio #1

Ejercicio: 17.10.2 on page 479.

```
double f(double a1, double a2, double a3, ↙
    ↳ double a4, double a5)
{
        return (a1+a2+a3+a4+a5) / 5;
};
```

G.1.12

Ejercicio #1

Ejercicio: 18.9.1 on page 548.

:

```
#define M    100
#define N    200

void s(double *a, double *b, double *c)
{
  for(int i=0;i<N;i++)
    for(int j=0;j<M;j++)
      *(c+i*M+j)=*(a+i*M+j) + *(b+i*M+j);
};
```

Ejercicio #2

Ejercicio: 18.9.2 on page 555.

:

```
#define M     100
#define N     200
#define P     300

void m(double *a, double *b, double *c)
{
  for(int i=0;i<M;i++)
    for(int j=0;j<P;j++)
    {
      *(c+i*M+j)=0;
      for (int k=0;k<N;k++) *(c+i*M+j)+=*(a+
    i*M+j) * *(b+i*M+j);
    }
};
```

Ejercicio #3

Ejercicio: 18.9.3 on page 564.

```
double f(double array[50][120], int x, int y
    )
{
        return array[x][y];
};
```

Ejercicio #4

Ejercicio: 18.9.4 on page 566.

```
int f(int array[50][60][80], int x, int y, ↙
    ↳ int z)
{
        return array[x][y][z];
};
```

Ejercicio #5

Ejercicio: 18.9.5 on page 569.

```
int tbl[10][10];

int main()
{
        int x, y;
        for (x=0; x<10; x++)
                for (y=0; y<10; y++)
                        tbl[x][y]=x*y;
};
```

G.1.13

Ejercicio #1

Ejercicio: 19.7.1 on page 634. endianness .

```
unsigned int f(unsigned int a)
{
        return ((a>>24)&0xff) | ((a<<8)&0↙
    ↳ xff0000) | ((a>>8)&0xff00) | ((a<<24)↙
    ↳ &0xff000000);
```

```
};
```

Ejercicio #2

Ejercicio: 19.7.2 on page 637.

```
#include <stdio.h>

unsigned int f(unsigned int a)
{
        int i=0;
        int j=1;
        unsigned int rt=0;
        for (;i<=28; i+=4, j*=10)
                rt+=((a>>i)&0xF) * j;
        return rt;
};

int main()
{
        // test
        printf ("%d\n", f(0x12345678));
        printf ("%d\n", f(0x1234567));
        printf ("%d\n", f(0x123456));
        printf ("%d\n", f(0x12345));
        printf ("%d\n", f(0x1234));
        printf ("%d\n", f(0x123));
        printf ("%d\n", f(0x12));
        printf ("%d\n", f(0x1));
};
```

Ejercicio #3

Ejercicio: 19.7.3 on page 642.

```
#include <windows.h>

int main()
{
        MessageBox(NULL, "hello, world!", "
    ↳ caption",
                MB_TOPMOST | 
    ↳ MB_ICONINFORMATION | MB_HELP | 
    ↳ MB_YESNOCANCEL);
};
```

Ejercicio #4

Ejercicio: 19.7.4 on page 643.

```
#include <stdio.h>
#include <stdint.h>

// source code taken from
// http://www4.wittenberg.edu/academics/
    ↳ mathcomp/shelburne/comp255/notes/
    ↳ binarymultiplication.pdf

uint64_t mult (uint32_t m, uint32_t n)
{
    uint64_t p = 0; // initialize product p 
    ↳ to 0
    while (n != 0) // while multiplier n is 
    ↳ not 0
```

```
    {
        if (n & 1) // test LSB of multiplier
            p = p + m; // if 1 then add ↙
    ↳ multiplicand m
        m = m << 1; // left shift ↙
    ↳ multiplicand
        n = n >> 1; // right shift ↙
    ↳ multiplier
    }
    return p;
}

int main()
{
    printf ("%d\n", mult (2, 7));
    printf ("%d\n", mult (3, 11));
    printf ("%d\n", mult (4, 111));
};
```

G.1.14

Ejercicio #1

Ejercicio: 21.7.1 on page 720.

.

```
#include <sys/types.h>
#include <sys/stat.h>
#include <time.h>
#include <stdio.h>
#include <stdlib.h>
```

```
int main(int argc, char *argv[])
{
    struct stat sb;

    if (argc != 2)
    {
        fprintf(stderr, "Usage: %s <pathname
    ↪ >\n", argv[0]);
        return 0;
    }

    if (stat(argv[1], &sb) == -1)
    {
        // error
        return 0;
    }

    printf("%ld\n",(long) sb.st_uid);
}
```

Ejercicio #1

Ejercicio: 21.7.2 on page 720.

```
#include <stdio.h>

struct some_struct
{
        int a;
        unsigned int b;
        float f;
```

```
        double d;
        char c;
        unsigned char uc;
};

void f(struct some_struct *s)
{
        if (s->a > 1000)
        {
                if (s->b > 10)
                {
                        printf ("%f\n", s->f
    ↳  * 444 + s->d * 123);
                        printf ("%c, %d\n", 
    ↳ s->c, s->uc);
                }
                else
                {
                        printf ("error #2\n
    ↳ ");
                };
        }
        else
        {
                printf ("error #1\n");
        };
};
```

G.1.15

Ejercicio #1

Ejercicio: 50.5.1 on page 1037.

: beginners.re.

G.1.16

Ejercicio #1

Ejercicio: 41.6 on page 949.

```
int f(int a)
{
        return a/661;
};
```

G.2 1

G.2.1 Ejercicio 1.1

.

G.2.2 Ejercicio 1.4

: beginners.re

G.3 2

G.3.1 Ejercicio 2.1

Spanish text placeholder: beginners.re

G.3.2 Ejercicio 2.4

: strstr().

:

```
char * strstr (
        const char * str1,
        const char * str2
        )
{
        char *cp = (char *) str1;
        char *s1, *s2;

        if ( !*str2 )
            return((char *)str1);

        while (*cp)
        {
                s1 = cp;
                s2 = (char *) str2;

                while ( *s1 && *s2 && !(*s1
    ↪ -*s2) )
                        s1++, s2++;

                if (!*s2)
```

```
                    return(cp);

                cp++;
        }

        return(NULL);

}
```

G.3.3 Ejercicio 2.6

.

(wikipedia):

```
void f (unsigned int* v, unsigned int* k) {
    unsigned int v0=v[0], v1=v[1], sum=0, i;↙
    ↪             /* set up */
    unsigned int delta=0x9e3779b9;           ↙
    ↪             /* a key schedule constant ↙
    ↪ */
    unsigned int k0=k[0], k1=k[1], k2=k[2], ↙
    ↪ k3=k[3];    /* cache key */
    for (i=0; i < 32; i++) {                 ↙
    ↪         /* basic cycle start */
        sum += delta;
        v0 += ((v1<<4) + k0) ^ (v1 + sum) ^ ↙
    ↪ ((v1>>5) + k1);
        v1 += ((v0<<4) + k2) ^ (v0 + sum) ^ ↙
    ↪ ((v0>>5) + k3);
    }                                        ↙
    ↪         /* end cycle */
```

```
    v[0]=v0; v[1]=v1;
}
```

G.3.4 Ejercicio 2.13

linear feedback shift register [1].

Spanish text placeholder: beginners.re

G.3.5 Ejercicio 2.14

.

Spanish text placeholder: beginners.re

G.3.6 Ejercicio 2.15

.

Spanish text placeholder: beginners.re

G.3.7 Ejercicio 2.16

[2].

```
int ack (int m, int n)
{
        if (m==0)
                return n+1;
```

[1] wikipedia
[2] wikipedia

```
        if (n==0)
                return ack (m-1, 1);
        return ack(m-1, ack (m, n-1));
};
```

G.3.8 Ejercicio 2.17

:
wikipedia.

Spanish text placeholder: beginners.re

G.3.9 Ejercicio 2.18

Spanish text placeholder: beginners.re

G.3.10 Ejercicio 2.19

Spanish text placeholder: beginners.re

G.3.11 Ejercicio 2.20

: the On-Line Encyclopedia of Integer Sequences (OEIS)) .

: *hailstone numbers*, [3].

Spanish text placeholder: beginners.re

3

G.4 3

G.4.1 Ejercicio 3.2

.

:
beginners.re

G.4.2 Ejercicio 3.3

:
beginners.re

G.4.3 Ejercicio 3.4

: beginners.re

G.4.4 Ejercicio 3.5

.

.

:
beginners.re

G.4.5 Ejercicio 3.6

:
beginners.re

G.4.6 Ejercicio 3.8

:
beginners.re

G.5

G.5.1 «Buscaminas (Windows XP)»

: 76 on page 1436.

:

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RVA Relative Virtual Address 1348

VA Virtual Address 1348

OEP Original Entry Point 1347

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CPU Central processing unit xlvii

FPU Floating-point unit 476

RISC Reduced instruction set computing 7

GUI Graphical user interface 1351

RTTI Run-time type information 1295

BSS Block Started by Symbol 1352

SIMD Single instruction, multiple data 790

BSOD Black Screen of Death 1325

ISA Instruction Set Architecture vi

ELF Formato de archivo ejecutable usado ampliamente en sistemas *NIX, incluido Linux xliii

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STMED Store Multiple Empty Descending () 63

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